

- H Op Amp History**
- ◆ 1 Op Amp Basics**
 - 1 Introduction**
 - 2 Op Amp Topologies**
 - 3 Op Amp Structures**
 - 4 Op Amp Specifications**
 - 5 Precision Op Amps**
 - 6 High Speed Op Amps**
- 2 Specialty Amplifiers**
- 3 Using Op Amps with Data Converters**
- 4 Sensor Signal Conditioning**
- 5 Analog Filters**
- 6 Signal Amplifiers**
- 7 Hardware and Housekeeping Techniques**

■ OP AMP APPLICATIONS

CHAPTER 1: OP AMP BASICS

James Bryant, Walt Jung, Walt Kester

Within Chapter 1, discussions are focused on the basic aspects of op amps. After a brief introductory section, this begins with the fundamental *topology* differences between the two broadest classes of op amps, those using *voltage feedback* and *current feedback*. These two amplifier types are distinguished more by the nature of their internal circuit topologies than anything else. The voltage feedback op amp topology is the classic structure, having been used since the earliest vacuum tube based op amps of the 1940 and 1950's, through the first IC versions of the 1960's, and includes most op amp models produced today. The more recent IC variation of the current feedback amplifier has come into popularity in the mid-to-late 1980's, when higher speed IC op amps were developed. Factors distinguishing these two op amp types are discussed at some length.

Details of op amp input and output *structures* are also covered in this chapter, with emphasis how such factors potentially impact application performance. In some senses, it is logical to categorize op amp types into performance and/or application classes, a process that works to some degree, but not altogether.

In practice, once past those obvious application distinctions such as "high speed" versus "precision," or "single" versus "dual supply," neat categorization breaks down. This is simply the way the analog world works. There is much crossover between various classes, i.e., a high speed op amp can be either single or dual-supply, or it may even fit as a precision type. A low power op amp may be precision, but it need not necessarily be single-supply, and so on. Other distinction categories could include the input stage type, such as FET input (further divided into JFET or MOS, which in turn are further divided into NFET or PFET and PMOS and NMOS, respectively), or bipolar (further divided into NPN or PNP). Then, all of these categories could be further described in terms of the type of input (or output) stage used.

So, it should be obvious that categories of op amps are like an infinite set of analog gray scales; *they don't always fit neatly into pigeonholes, and we shouldn't expect them to*. Nevertheless, it is still very useful to appreciate many of the aspects of op amp design that go into the various structures, as these differences directly influence the optimum op amp choice for an application. Thus structure differences are application drivers, since we choose an op amp to suit the nature of the application, for example single-supply.

In this chapter various op amp performance *specifications* are also discussed, along with those specification differences that occur between the broad distinctions of voltage or current feedback topologies, as well as the more detailed context of individual structures. Obviously, op amp specifications are also application drivers, in fact they are the most important, since they will determine system performance. We choose the best op amp to fit the application, based on for the required bias current, bandwidth, distortion, etc.

▣ OP AMP APPLICATIONS

SECTION 1: INTRODUCTION

Walt Jung

As a precursor to more detailed sections following, this introductory chapter portion considers the most basic points of op amp operation. These initial discussions are oriented around the more fundamental levels of op amp applications. They include: *Ideal Op Amp Attributes*, *Standard Op Amp Feedback Hookups*, *The Non-Ideal Op Amp*, *Op amp Common Mode Dynamic Range(s)*, the various *Functionality Differences of Single and Dual-Supply Operation*, and the *Device Selection* process.

Before op amp applications can be developed, some first requirements are in order. These include an understanding of how the fundamental op amp operating modes differ, and whether dual-supply or single-supply device functionality better suits the system under consideration. Given this, then device selection can begin and an application developed.

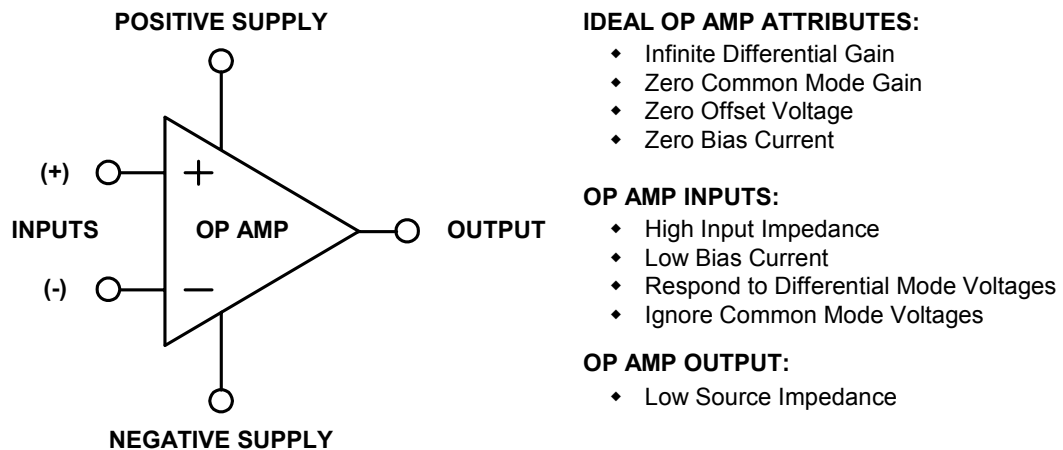


Figure 1-1: The ideal op amp and its attributes

First, an *operational amplifier* (hereafter simply op amp) is a differential input, single ended output amplifier, as shown symbolically in Figure 1-1. This device is an amplifier intended for use with *external feedback elements*, where these elements determine the resultant function, or *operation*. This gives rise to the name "operational amplifier," denoting an amplifier that, by virtue of different feedback hookups, can perform a variety of operations.¹ At this point, note that there is no need for concern with any actual technology to implement the amplifier. Attention is focused more on the behavioral nature of this building block device.

An op amp processes small, differential mode signals appearing between its two inputs, developing a single ended output signal referred to a power supply common terminal. Summaries of the various ideal op amp attributes are given in the figure. While real op amps will depart from these ideal attributes, it is very helpful for first-level understanding of op amp behavior to consider these features. Further, although these initial discussions

¹ The actual *naming* of the operational amplifier occurred in the classic Ragazinni, et al paper of 1947 (see Reference 1). However analog computations using op amps as we know them today began with the work of the Clarence Lovell led group at Bell Labs, around 1940 (acknowledged generally in the Ragazinni paper).

■ OP AMP APPLICATIONS

talk in idealistic terms, they are also flavored by pointed mention of typical "real world" specifications— for a beginning perspective.

It is also worth noting that this op amp is shown with five terminals, a number that happens to be a minimum for real devices. While some single op amps may have more than five terminals (to support such functions as frequency compensation, for example), none will ever have less. By contrast, those elusive ideal op amps don't require power, and symbolically function with just four pins.¹

Ideal Op Amp Attributes

An ideal op amp has infinite gain for *differential* input signals. In practice, real devices will have quite high gain (also called *open-loop gain*) but this gain won't necessarily be precisely known. In terms of specifications, gain is measured in terms of V_{OUT}/V_{IN} , and is given in V/V, the dimensionless numeric gain. More often however, gain is expressed in decibel terms (dB), which is mathematically $\text{dB} = 20 \bullet \log(\text{numeric gain})$. For example, a numeric gain of 1 million (10^6 V/V) is equivalent to a 120 dB gain. Gains of 100-130 dB are common for precision op amps, while high speed devices may have gains in the 60-70 dB range.

Also, an ideal op amp has zero gain for signals *common* to both inputs, that is, *common mode* (CM) signals. Or, stated in terms of the rejection for these common mode signals, an ideal op amp has infinite *CM rejection* (CMR). In practice, real op amps can have CMR specifications of up to 130 dB for precision devices, or as low as 60-70 dB for some high speed devices.

The ideal op amp also has zero *offset voltage* ($V_{OS}=0$), and draws zero *bias current* ($I_B=0$) at both inputs. Within real devices, actual offset voltages can be as low as a μV or less, or as high as several mV. Bias currents can be as low as a few fA, or as high as several μA . This extremely wide range of specifications reflects the different input structures used within various devices, and is covered in more detail later in this chapter.

The attribute headings within Figure 1-1 for INPUTS and OUTPUT summarize the above concepts in more succinct terms. In practical terms, another important attribute is the concept of *low source impedance*, at the output. As will be seen later, low source impedance enables higher useful gain levels within circuits.

To summarize these idealized attributes for a signal-processing amplifier, some of the traits might at first seem strange. However, it is critically important to reiterate that op amps simply are never intended for use without overall feedback! In fact, as noted, the connection of a suitable *external* feedback loop defines the *closed-loop* amplifier's gain and frequency response characteristics.

Note also that all real op amps have a positive and negative power supply terminal, but rarely (if ever) will they have a separate ground connection. In practice, the op amp

¹ Such an op amp generates its own power, has two input pins, an output pin, and an output common pin.

output voltage becomes referred to a power supply common point. *Note: This key point is further clarified with the consideration of typically used op amp feedback networks.*

The basic op amp hookup of Figure 1-2 below applies a signal to the (+) input, and a (generalized) network delivers a fraction of the output voltage to the (−) input terminal. This constitutes *feedback*, with the op amp operating in *closed-loop* fashion. The feedback network (shown here in general form) can be resistive or reactive, linear or non-linear, or any combination of these. More detailed analysis will show that the circuit gain characteristic as a whole follows the inverse of the feedback network transfer function.

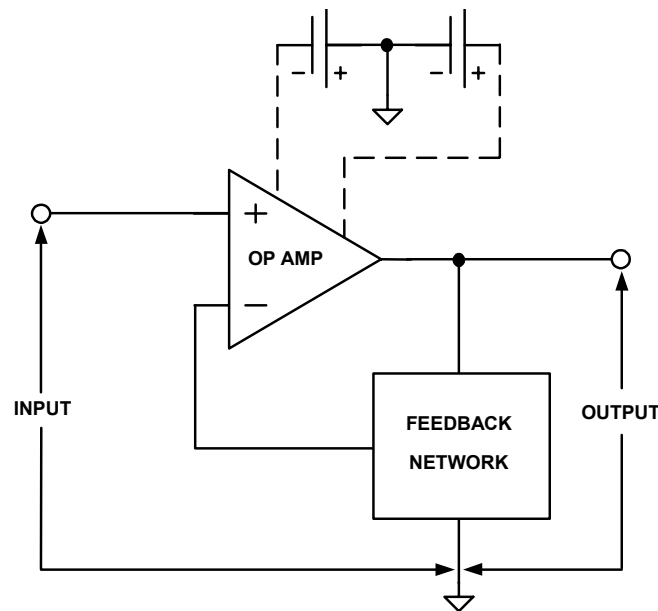


Figure 1-2: A generalized op amp circuit with feedback applied

The concept of feedback is both an essential and salient point concerning op amp use. With feedback, the net closed-loop gain characteristics of a stage such as Fig. 1-2 become primarily dependent upon a set of *external components* (usually passive). Thus behavior is less dependent upon the relatively unstable amplifier open-loop characteristics.

Note that within Figure 1-2, the input signal is applied between the op amp (+) input and a *common or reference point*, as denoted by the ground symbol. It is important to note that this reference point is also common to the output and feedback network. By definition, the op amp stage's output signal appears between the output terminal/feedback network input, and this common ground. This single relevant fact answers the "Where is the op amp grounded?" question so often asked by those new to the craft. The answer is simply that it is grounded *indirectly*, by virtue of the commonality of its input, the feedback network, and the power supply, as is shown within Fig. 1-2.

To emphasize how the input/output signals are referenced to the power supply, dual supply connections are shown dotted, with the $\pm$ power supply midpoint common to the input/output signal ground. But do note, while all op amp application circuits may not show full details of the power supply connections, every *real* circuit will always use power supplies!

■ OP AMP APPLICATIONS

Standard Op Amp Feedback Hookups

Virtually all op amp feedback connections can be categorized into just a few basic types. These include the two most often used, *non-inverting* and *inverting* voltage gain stages, plus a related *differential* gain stage. Having discussed above just the attributes of the ideal op amp, at this point it is possible to conceptually build basic gain stages. Using the concepts of infinite gain, zero input offset voltage, zero bias current, etc., standard op amp feedback hookups can be devised. For brevity, a full mathematical development of these concepts isn't included below (but this follows in a subsequent section). The end-of-section references also include such developments.

The Non-inverting Op Amp Stage

The op amp non-inverting gain stage, also known as a *voltage follower with gain*, or simply *voltage follower*, is shown below in Figure 1-3.

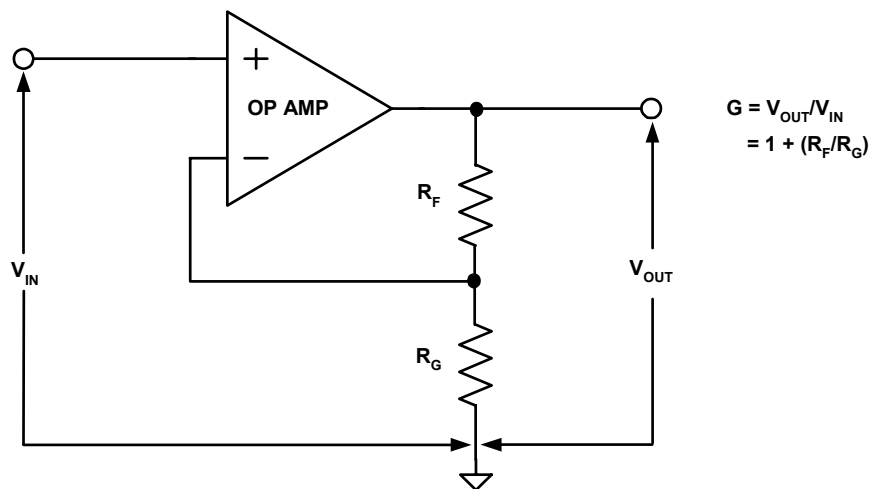


Figure 1-3: The non-inverting op amp stage (voltage follower)

This op amp stage processes the input V_{IN} by a gain of G , so a generalized expression for gain is:

$$G = \frac{V_{OUT}}{V_{IN}} \quad \text{Eq. 1-1}$$

Feedback network resistances R_F and R_G set the stage gain of the follower. For an ideal op amp, the gain of this stage is:

$$G = \frac{R_F + R_G}{R_G} \quad \text{Eq. 1-2}$$

For clarity, these expressions are also included in the figure. Comparison of this figure and the more general Figure 1-2 shows R_F and R_G here as a simple feedback network, returning a fraction of V_{OUT} to the op amp (–) input (*note that some texts may show the more general symbols Z_F and Z_G for these feedback components— both are correct, depending upon the specific circumstances*).

In fact, we can make some useful general points about the network $R_F - R_G$. We will define the transfer expression of the network as seen from the top of R_F to the output across R_G as β . Note that this usage is a general feedback network transfer term, *not* to be confused with bipolar transistor forward gain. β can be expressed mathematically as:

$$\beta = \frac{R_G}{R_F + R_G} \quad \text{Eq. 1-3}$$

So, the feedback network returns a fraction of V_{OUT} to the op amp (–) input. Considering the ideal principles of zero offset and infinite gain, this allows some deductions on gain to be made. The voltage at the (–) input is forced by the op amp's feedback action to be equal to that seen at the (+) input, V_{IN} . Given this relationship, it is relatively easy to work out the ideal gain of this stage, which in fact turns out to be simply the inverse of β . This is apparent from a comparison of equations 1-2 and 1-3.

Thus an ideal non-inverting op amp stage gain is simply equal to $1/\beta$, or:

$$G = \frac{1}{\beta} \quad \text{Eq. 1-4}$$

This non-inverting gain configuration is one of the most useful of all op amp stages, for several reasons. Because V_{IN} sees the op amp's high impedance (+) input, it provides an ideal interface to the driving source. Gain can easily be adjusted over a wide range via R_F and R_G , with virtually no source interaction.

A key point is the interesting relationship concerning R_F and R_G . Note that to satisfy the conditions of Equation 1-2, only their *ratio* is of concern. In practice this means that stable gain conditions can exist over a range of actual $R_F - R_G$ values, so long as they provide the same ratio.

If R_F is taken to zero and R_G open, the stage gain becomes unity, and V_{OUT} is then exactly equal to V_{IN} . This special non-inverting gain case is also called a *unity gain follower*, a stage commonly used for buffering a source.

Note that this op amp example shows only a simple resistive case of feedback. As mentioned, the feedback can also be reactive, i.e., Z_F , to include capacitors and/or inductors. In all cases however, it must include a DC path, if we are to assume the op amp is being biased by the feedback (which is usually the case).

To summarize some key points on op amp feedback stages, we paraphrase from Reference 2 the following statements, which will always be found useful:

The summing point idiom is probably the most used phrase of the aspiring analog artificer, yet the least appreciated. In general, the inverting (–) input is called the summing point, while the non-inverting (+) input is represented as the reference terminal. However, a vital concept is the fact that, within linear op amp applications, the inverting input (or summing point) assumes the same absolute potential as the non-inverting input or reference (within the gain error of the amplifier). In short, the amplifier tries to servo its own summing point to the reference.

■ OP AMP APPLICATIONS

The Inverting Op Amp Stage

The op amp inverting gain stage, also known simply as the *inverter*, is shown in Figure 1-4. As can be noted by comparison of Figures 1-3 and 1-4, the inverter can be viewed as similar to a follower, but with a transposition of the input voltage V_{IN} . In the inverter the signal is applied to R_G of the feedback network, and the op amp (+) input is grounded.

The feedback network resistances, R_F and R_G set the stage gain of the inverter. For an ideal op amp, the gain of this stage is:

$$G = -\frac{R_F}{R_G} \quad \text{Eq. 1-5}$$

For clarity, these expressions are again included in the figure. Note that a major difference between this stage and the non-inverting counterpart is the input-to-output sign reversal, denoted by the minus sign in Equation 1-5. Like the follower stage, applying ideal op amp principles and some basic algebra can derive the gain expression of Eq. 1-5.

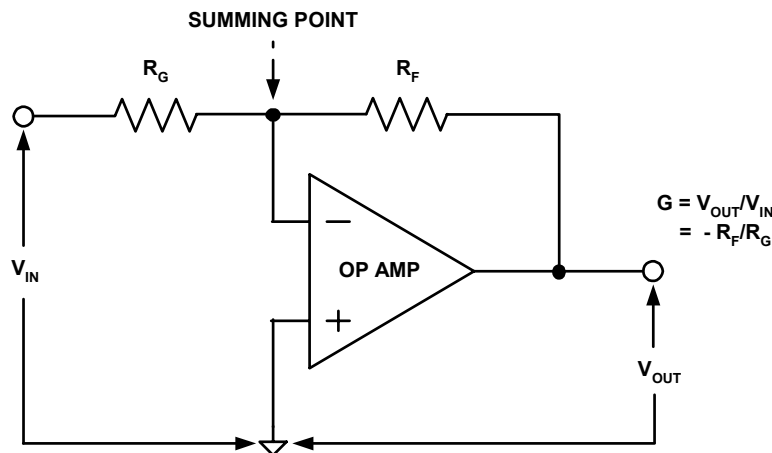


Figure 1-4: The inverting op amp stage (inverter)

The inverting configuration is also one of the more useful op amp stages. Unlike a non-inverting stage however, the inverter presents a relatively low impedance input for V_{IN} , i.e., the value of R_G . This factor provides a finite load to the source. While the stage gain can in theory be adjusted over a wide range via R_F and R_G , there is a practical limitation imposed at high gain, when R_G becomes relatively low. If R_F is zero, the gain becomes zero. R_F can also be made variable, in which case the gain is linearly variable over the dynamic range of the element used for R_F . As with the follower gain stage, the gain is ratio dependent, and is relatively insensitive to the exact R_F and R_G values.

The inverter's gain behavior, due to the principles of infinite op amp gain, zero input offset, and zero bias current, gives rise to an effective node of zero voltage at the (–) input. The input and feedback currents sum at this point, which logically results in the term *summing point*. It is also called a *virtual ground*, because of the fact it will be at the same potential as the grounded reference input.

Note that, technically speaking, *all* op amp feedback circuits have a summing point, whether they are inverters, followers, or a hybrid combination. The summing point is always the feedback junction at the (–) input node, as shown in Fig. 1-4. However in follower type circuits this point isn't a virtual ground, since it follows the (+) input.

A special gain case for the inverter occurs when $R_F = R_G$, which is also called a *unity gain inverter*. This form of inverter is commonly used for generating complementary V_{OUT} signals, i.e., $V_{OUT} = -V_{IN}$. In such cases it is usually desirable to match R_F to R_G accurately, which can readily be done by using a well-specified matched resistor pair.

A variation of the inverter is the *inverting summer*, a case similar to Figure 1-4, but with input resistors R_{G2} , R_{G3} , etc (not shown). For a summer individual input resistors are connected to additional sources V_{IN2} , V_{IN3} , etc., with their common node connected to the summing point. This configuration, called a *summing amplifier*, allows linear input current summation in R_F .¹ V_{OUT} is proportional to an inverse sum of input currents.

The Differential Op Amp Stage

The op amp differential gain stage (also known as a *differential amplifier*, or *subtractor*) is shown in Figure 1-5.

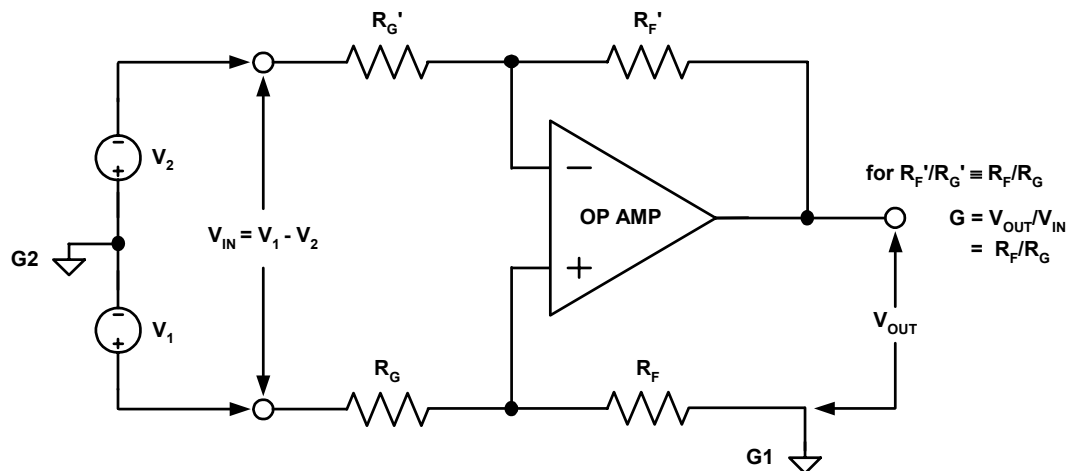


Figure 1-5: The differential amplifier stage (subtractor)

Paired input and feedback network resistances set the gain of this stage. These resistors, $R_F - R_G$ and $R_F' - R_G'$, *must be matched as noted*, for proper operation. Calculation of individual gains for inputs V_1 and V_2 and their linear combination derives the stage gain.

Note that the stage is intended to amplify the *difference* of voltages V_1 and V_2 , so the net input is $V_{IN} = V_1 - V_2$. The general gain expression is then:

$$G = \frac{V_{OUT}}{V_1 - V_2} \quad \text{Eq. 1-6}$$

¹ The very first general-purpose op amp circuit is described by Karl Swartzel in Reference 3, and is titled "Summing Amplifier". This amplifier became a basic building block of the M9 gun director computer and fire control system used by Allied Forces in World War II. It also influenced many vacuum tube op amp designs that followed over the next two decades.

■ OP AMP APPLICATIONS

For an ideal op amp and the resistor ratios matched as noted, the gain of this differential stage from V_{IN} to V_{OUT} is:

$$G = \frac{R_F}{R_G} \quad \text{Eq. 1-7}$$

The great fundamental utility that an op amp stage such as this allows is the property of rejecting voltages *common* to V_1 - V_2 , i.e., common-mode (CM) voltages. For example, if noise voltages appear between grounds G1 and G2, the noise will be suppressed by the common-mode rejection (CMR) of the differential amp. The CMR is however only as good as the matching of the resistor ratios allows, so in practical terms it implies precisely trimmed resistor ratios are necessary. Another disadvantage of this stage is that the resistor networks load the V_1 - V_2 sources, potentially leading to additional errors.

The Non-Ideal Op Amp— Static Errors Due to Finite Amplifier Gain

One of the most distinguishing features of op amps is their staggering magnitude of DC voltage gain. Even the least expensive devices have typical voltage gains of 100,000 (100dB), while the highest performance precision bipolar and chopper stabilized units can have gains as high 10,000,000 (140dB), or more. Negative feedback applied around this much voltage gain readily accomplishes the virtues of closed-loop performance, making the circuit dependent only on the feedback components.

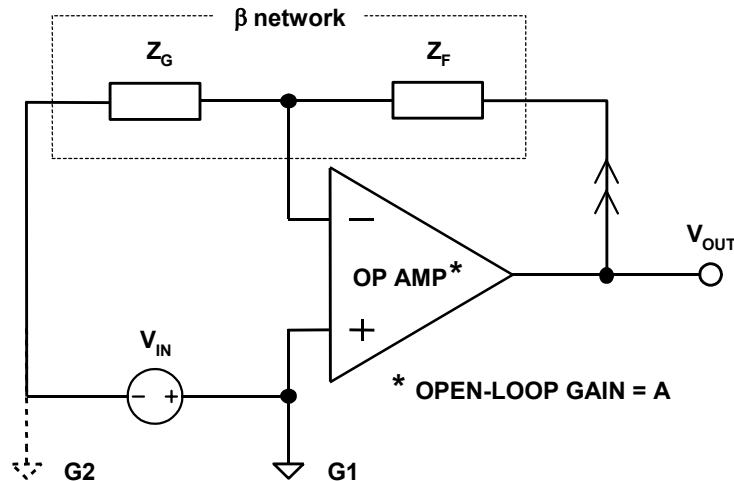


Figure 1-6: Non-ideal op amp stage for gain error analysis

As noted above in the discussion of ideal op amp attributes, the behavioral assumptions follow from the fact that negative feedback, coupled with high open-loop gain, constrains the amplifier input error voltage (and consequently the error current) to infinitesimal values. The higher this gain, the more valid these assumptions become.

But in reality, op amps *do* have finite gain, and errors exist in practical circuits. The op amp gain stage of Figure 1-6 will be used to illustrate how these errors impact performance. In this circuit the op amp is ideal except for the finite open-loop DC voltage gain, A , which is usually stated as A_{VOL} .

Noise Gain (NG)

The first aid to analyzing op amps circuits is to differentiate between *noise gain* and *signal gain*. We have already discussed the differences between non-inverting and inverting stages as to their signal gains, which are summarized in Equations 1-2 and 1-4, respectively. But, as can be noticed from Fig. 1-6, the difference between an inverting and non-inverting stage can be as simple as just where the reference ground is placed. For a ground at point G1, the stage is an inverter; conversely, if the ground is placed at point G2 (with no G1) the stage is non-inverting.

Note however that in terms of the feedback path, *there are no real differences*. To make things more general, the resistive feedback components previously shown are replaced here with the more general symbols Z_F and Z_G , otherwise they function as before. The feedback attenuation, β , is the same for both the inverting and non-inverting stages:

$$\beta = \frac{Z_G}{Z_G + Z_F} \quad \text{Eq. 1-8}$$

Noise gain can now be simply defined as: *The inverse of the net feedback attenuation from the amplifier output to the feedback input*. In other words, the inverse of the β network transfer function. This can ultimately be extended to include frequency dependence (covered later in this chapter). Noise gain can be abbreviated as NG.

As noted, the inverse of β is the ideal non-inverting op amp stage gain. Including the effects of finite op amp gain, a modified gain expression for the non-inverting stage is:

$$G_{CL} = \frac{1}{\beta} \cdot \left[\frac{1}{1 + \frac{1}{A_{VOL}\beta}} \right] \quad \text{Eq. 1-9}$$

where G_{CL} is the finite-gain stage's closed-loop gain, and A_{VOL} is the op amp open-loop voltage gain for loaded conditions.

It is important to note that this expression is identical to the ideal gain expression of Eq. 1-4, with the addition of the bracketed multiplier on the right side. Note also that this right-most term becomes closer and closer to unity, as A_{VOL} approaches infinity. Accordingly, it is known in some textbooks as the *error multiplier* term, when the expression is shown in this form. ¹

It may seem logical here to develop another finite gain error expression for an inverting amplifier, but in actuality there is no need. Both inverting and non-inverting gain stages have a common feedback basis, which is the noise gain. So Eq. 1-9 will suffice for gain error analysis for both stages. Simply use the β factor as it applies to the specific case.

¹ Some early discussions of this finite gain error appear in References 4 and 5. Terman uses the open-loop gain symbol of A , as we do today. West uses Harold Black's original notation of μ for open-loop gain. The form of Eq. 1-9 is identical to Terman's (or to West's, substituting μ for A).

■ OP AMP APPLICATIONS

It is useful to note some assumptions associated with the rightmost error multiplier term of Eq. 1-9. For $A_{VOL}\beta \gg 1$, one assumption is:

$$\frac{1}{1 + \frac{1}{A_{VOL}\beta}} \approx 1 - \frac{1}{A_{VOL}\beta} \quad \text{Eq. 1-10}$$

This in turn leads to an estimation of the percentage error, ε , due to finite gain A_{VOL} :

$$\varepsilon(\%) \approx \frac{100}{A_{VOL}\beta} \quad \text{Eq. 1-11}$$

Gain Stability

The closed-loop gain error predicted by these equations isn't in itself tremendously important, since the ratio Z_F/Z_G could always be adjusted to compensate for this error. But note however that closed-loop gain *stability* is a very important consideration in most applications. Closed-loop gain instability is produced primarily by variations in open-loop gain due to changes in temperature, loading, etc.

$$\frac{\Delta G_{CL}}{G_{CL}} \approx \frac{\Delta A_{VOL}}{A_{VOL}} \cdot \frac{1}{A_{VOL}\beta} \quad \text{Eq. 1-12}$$

From Eq. 1-12, any variation in open-loop gain (ΔA_{VOL}) is reduced by the factor $A_{VOL}\beta$, insofar as the effect on closed-loop gain. This improvement in closed-loop gain stability is one of the important benefits of negative feedback.

Loop gain

The product $A_{VOL}\beta$ which occurs in the above equations, is called *loop gain*, a well-known term in feedback theory. The improvement in closed-loop performance due to negative feedback is, in nearly every case, proportional to loop gain.

The term "loop gain" comes from the method of measurement. This is done by breaking the closed feedback loop at the op amp output, and measuring the total gain around the loop. In Fig. 1-6 for example, this could be done between the amplifier output and the feedback path (see arrows). Approximately, closed-loop output impedance, linearity, and gain instability errors reduce by the factor $A_{VOL}\beta$, with the use of negative feedback.

Another useful approximation is developed as follows. A rearrangement of Eq. 1-9 is:

$$\frac{A_{VOL}}{G_{CL}} = 1 + A_{VOL}\beta \quad \text{Eq. 1-13}$$

So, for high values of $A_{VOL}\beta$,

$$\frac{A_{VOL}}{G_{CL}} \approx A_{VOL}\beta \quad \text{Eq. 1-14}$$

Consequently, in a given feedback circuit the loop gain, $A_{VOL}\beta$, is approximately the numeric ratio (or difference, in dB) of the amplifier open-loop gain to the circuit closed-loop gain.

This loop gain discussion emphasizes that indeed, loop gain is a very significant factor in predicting the performance of closed-loop operational amplifier circuits. The open-loop gain required to obtain an adequate amount of loop gain will, of course, depend on the desired closed-loop gain.

For example, using Equation 1-14, an amplifier with $A_{VOL} = 20,000$ will have an $A_{VOL}\beta \approx 2000$ for a closed-loop gain of 10, but the loop gain will be only 20 for a closed-loop gain of 1000. The first situation implies an amplifier-related gain error the order of $\approx 0.05\%$, while the second would result in about 5% error. Obviously, the higher the required gain, the greater will be the required open-loop gain to support an $A_{VOL}\beta$ for a given accuracy.

Frequency Dependence of Loop Gain

Thus far, it has been assumed that amplifier open-loop gain is independent of frequency. Unfortunately, this isn't the case at all. Leaving the discussion of the effect of open-loop response on bandwidth and dynamic errors until later, let us now investigate the general effect of frequency response on loop gain and static errors.

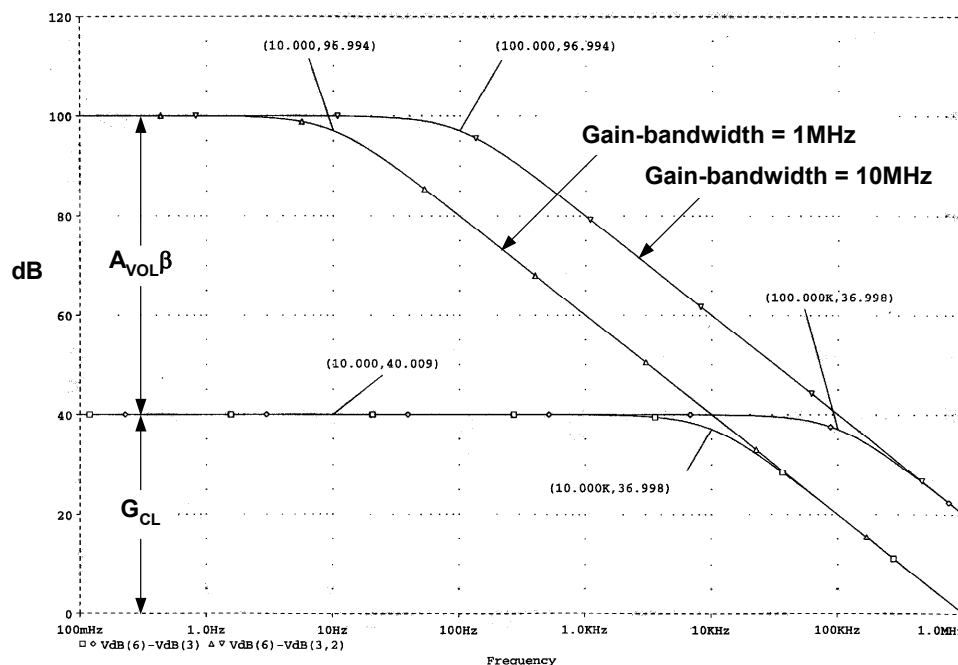


Figure 1-7: Op amp closed-loop gain and loop gain interactions with typical open-loop responses

The open-loop frequency response for a typical operational amplifier with superimposed closed-loop amplifier response for a gain of 100 (40dB), illustrates graphically these results, in Figure 1-7. In these Bode plots, subtraction on a logarithmic scale is equivalent

■ OP AMP APPLICATIONS

to normal division of numeric data.¹ Today, op amp open-loop gain and loop gain parameters are typically given in dB terms, thus this display method is convenient.

A few key points evolve from this graphic figure, which is a simulation involving two hypothetical op amps, both with a DC/low frequency gain of 100dB (100kV/V). The first has a gain-bandwidth of 1MHz, while the gain-bandwidth of the second is 10MHz.

- The open-loop gain A_{VOL} for the two op amps is noted by the two curves marked 1 and 10MHz, respectively. Note that each has a -3 dB corner frequency associated with it, above which the open-loop gain falls at 6dB/octave. These corner frequencies are marked at 10 and 100Hz, respectively, for the two op amps.
- At any frequency on the open-loop gain curve, the numeric product of gain A_{VOL} and frequency, f , is a constant (10,000V/V at 100Hz equates to 1MHz). This, by definition, is characteristic of a constant gain-bandwidth product amplifier. All *voltage feedback* op amps behave in this manner.
- $A_{VOL}\beta$ in dB is the difference between open-loop gain and closed-loop gain, as plotted on log-log scales. At the lower frequency point marked, $A_{VOL}\beta$ is thus 60dB.
- $A_{VOL}\beta$ decreases with increasing frequency, due to the decrease of A_{VOL} above the open-loop corner frequency. At 100Hz for example, the 1MHz gain-bandwidth amplifier shows an $A_{VOL}\beta$ of only $80-40 = 40$ dB.
- $A_{VOL}\beta$ also decreases for higher values of closed-loop gain. Other, higher closed-loop gain examples (not shown) would decrease $A_{VOL}\beta$ to less than 60dB at low frequencies.
- G_{CL} depends primarily on the ratio of the feedback components, Z_F and Z_G , and is relatively independent of A_{VOL} (apart from the errors discussed above, which are inversely proportional to $A_{VOL}\beta$). In this example $1/\beta$ is 100, or 40dB, and is so marked at 10Hz. Note that G_{CL} is flat with increasing frequency, up until that frequency where G_{CL} intersects the open-loop gain curve, and $A_{VOL}\beta$ drops to zero.
- At this point where the closed-loop and open-loop curves intersect, the loop gain is by definition zero, which implies that beyond this point there is no negative feedback. Consequently, closed-loop gain is equal to open-loop gain for further increases in frequency.
- Note that the 10MHz gain-bandwidth op amp allows a $10\times$ increase in closed-loop bandwidth, as can be noted from the -3 dB frequencies; that is 100kHz versus 10kHz for the 10MHz versus the 1MHz gain-bandwidth op amp.

¹ The log-log displays of amplifier gain (and phase) versus frequency are called *Bode* plots. This graphic technique for display of feedback amplifier characteristics, plus definitions for feedback amplifier stability were pioneered by Hendrick W. Bode of Bell Labs (see Reference 6).

Fig. 1-7 illustrates that the high open-loop gain figures typically quoted for op amps can be somewhat misleading. As noted, beyond a few Hz, the open-loop gain falls at 6dB/octave. Consequently, closed-loop gain stability, output impedance, linearity and other parameters dependent upon loop gain are degraded at higher frequencies. One of the reasons for having DC gain as high as 100dB and bandwidth as wide as several MHz, is to obtain adequate loop gain at frequencies even as low as 100Hz.

A direct approach to improving loop gain at high frequencies other than by increasing open-loop gain is to increase the amplifier open-loop bandwidth. Figure 1-7 shows this in terms of two simple examples. It should be borne in mind however that op amp gain-bandwidths available today extend to the hundreds of MHz, allowing video and high-speed communications circuits to fully exploit the virtues of feedback.

Op amp Common Mode Dynamic Range(s)

As a point of departure from the idealized circuits above, some practical basic points are now considered. Among the most evident of these is the allowable input and output dynamic ranges afforded in a real op amp. This obviously varies with not only the specific device, but also the supply voltage. While we can always optimize this performance point with device selection, more fundamental considerations come first.

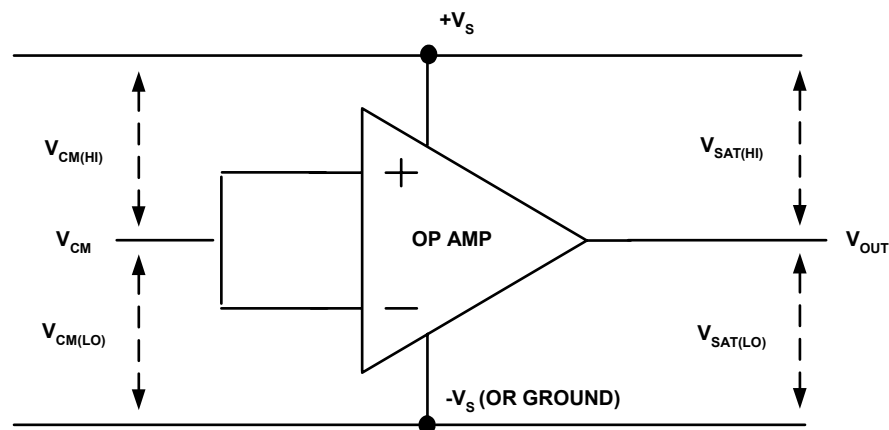


Figure 1-8: *Op amp input and output common mode ranges*

Any real op amp will have a finite voltage range of operation, at both input and output. In modern system designs, supply voltages are dropping rapidly, and 3-5V total supply voltages are now common. This is a far cry from supply systems of the past, which were typically $\pm 15V$ (30V total). Obviously, if designs are to accommodate a 3-5V supply, careful consideration must be given to maximizing dynamic range, by choosing a correct device. Choosing a device will be in terms of exact specifications, but first and foremost it should be in terms of the basic topologies used within it.

Output Dynamic Range

Figure 1-8 above is a general illustration of the limitations imposed by input and output dynamic ranges of an op amp, related to both supply rails. Any op amp will always be powered by two supply potentials, indicated by the positive rail, $+V_S$, and the negative

■ OP AMP APPLICATIONS

rail, $-V_S$. We will define the op amp's input and output CM range in terms of how closely it can approach these two rail voltage limits.

At the output, V_{OUT} has two rail-imposed limits, one high or close to $+V_S$, and one low, or close to $-V_S$. Going high, it can range from an upper saturation limit of $+V_S - V_{SAT(HI)}$ as a positive maximum. For example if $+V_S$ is 5V, and $V_{SAT(HI)}$ is 100mV, the upper V_{OUT} limit or positive maximum is 4.9V. Similarly, going low it can range from a lower saturation limit of $-V_S + V_{SAT(LO)}$. So, if $-V_S$ is ground (0V) and $V_{SAT(LO)}$ is 50mV, the lower limit of V_{OUT} is simply 50mV.

Obviously, the internal design of a given op amp will impact this output CM dynamic range, since, when so necessary, the device itself must be designed to minimize both $V_{SAT(HI)}$ and $V_{SAT(LO)}$, so as to maximize the output dynamic range. Certain types of op amp structures are so designed, and these are generally associated with designs expressly for *single-supply* systems. This is covered in detail later within the chapter.

Input Dynamic Range

At the input, the CM range useful for V_{IN} also has two rail-imposed limits, one high or close to $+V_S$, and one low, or close to $-V_S$. Going high, it can range from an upper CM limit of $+V_S - V_{CM(HI)}$ as a positive maximum. For example, again using the $+V_S = 5V$ example case, if $V_{CM(HI)}$ is 1V, the upper V_{IN} limit or positive CM maximum is $+V_S - V_{CM(HI)}$, or 4V.

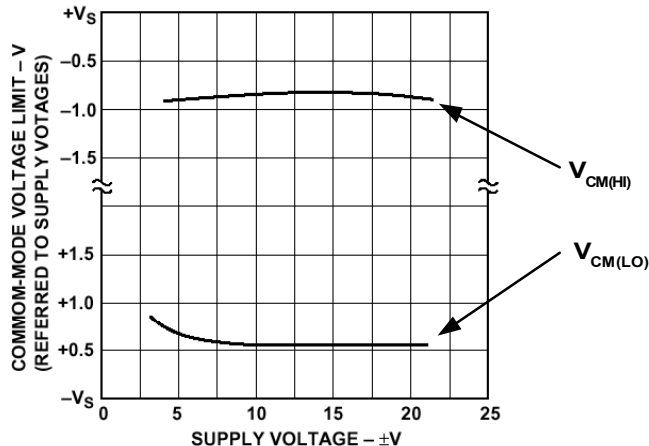


Figure 1-9: A graphical display of op amp input common mode range

Figure 1-9 above illustrates by way of a hypothetical op amp's data how $V_{CM(HI)}$ could be specified, as shown in the upper curve. This particular op amp would operate for V_{CM} inputs *lower* than the curve shown.

In practice the input CM range of real op amps is typically specified as a *range of voltages*, not necessarily referenced to $+V_S$ or $-V_S$. For example, a typical $\pm 15V$ operated dual supply op amp would be specified for an operating CM range of $\pm 13V$. Going low, there will also be a lower CM limit. This can be generally expressed as $-V_S + V_{CM(LO)}$, which would appear in a graph such as Fig. 1-9 as the lower curve, for $V_{CM(LO)}$. If this were again a $\pm 15V$ part, this could represent typical performance.

To use a single-supply example, for the $-V_S = 0V$ case, if $V_{CM(LO)}$ is 100mV, the lower CM limit will be $0V + 0.1V$, or simply 0.1V. Although this example illustrates a lower CM range within 100mV of $-V_S$, it is actually much more typical to see single-supply devices with lower or upper CM ranges, which *include* the supply rail.

In other words, $V_{CM(LO)}$ or $V_{CM(HI)}$ is 0V. There are also single-supply devices with CM ranges that include *both* rails. More often than not however, single-supply devices will not offer graphical data such as Fig. 1-9 for CM limits, but will simply cover performance with a tabular range of specified voltage.

Functionality Differences of Dual-Supply and Single-Supply Devices

There are two major classes of op amps, the choice of which determines how well the selected part will function in a given system. Traditionally, many op amps have been designed to operate on a dual power supply system, which has typically been $\pm 15V$. This custom has been prevalent since the earliest IC op amps days, dating back to the mid-sixties. Such devices can accommodate input/output ranges of $\pm 10V$ (or slightly more), but when operated on supplies of appreciably lower voltage, for example $\pm 5V$ or less, they suffer either loss of performance, or simply don't operate at all. This type of device is referenced here as a *dual-supply* op amp design. This moniker indicates that it performs optimally on dual voltage systems only, typically $\pm 15V$. It may or may not also work at appreciably lower voltages.

PERFORMANCE PARAMETER	DUAL SUPPLY	SINGLE SUPPLY
♦ SUPPLY LIMITATIONS	Best >10V, Limited <10V	Best <10V, Limited >10V
♦ OUTPUT V RANGE	- Limited	+ Greatest
♦ INPUT V RANGE	- Limited	+ Greatest
♦ TOTAL DYNAMIC RANGE	+ Greatest	- Least
♦ V & I OUTPUT	+ Greater	- Less
♦ PRECISION	+ Greatest	- Less (growing)
♦ LOAD IMMUNITY	+ Greatest	- Least
♦ VARIETY AVAILABLE	+ Greater	- Less (growing)

Figure 1-10: Comparison of relative functional performance differences between single and dual-supply op amps

Figure 1-10 above illustrates in a broad overview the relative functional performance differences that distinguish the dual-supply vs. single-supply op amp classes. This table is arranged to illustrate various general performance parameters, with an emphasis on the contrast between single and dual-supply devices. Which particular performance area is more critical will determine what type of device will be the better system choice.

■ OP AMP APPLICATIONS

More recently, with increasing design attention to lower overall system power and the use of single rail power, the single-supply op amp has come into vogue. This has not been without good reason, as the virtues of using single supply rails can be quite compelling. A review of Fig. 1-10 illustrates key points of the dual vs. single supply op amp question.

In terms of *supply voltage limitations*, there is a crossover region in terms of overall utility, which occurs around 10V of total supply voltage.

For example, single-supply devices tend to excel in terms of their *input and output voltage dynamic ranges*. Note that in Figure 1-10 a maximum range is stated as a % of available supply. Single-supply parts operate better in this regard, because they are internally designed to maximize these respective ranges. For example, it is not unusual for a device operating from 5V to swing 4.8V at the output, and so on.

But, rather interestingly, such devices are also usually restricted to lower supply ranges (only), so their upper dynamic range in absolute terms is actually more limited. For example, a traditional $\pm 15\text{V}$ dual-supply device can typically swing 20Vp-p, or more than four times that of a 5V single-supply part. If the *total dynamic range* is considered (assuming an identical input noise), the dual-supply operated part will have 4 times (or 12 dB) greater dynamic range than that of the 5V operated part. Or, stated in another way, the input errors of a real part such as noise, drift etc., become 4 times more critical (relatively speaking), when the output dynamic range is reduced by a factor of 4. Note that these comparisons do not involve any actual device specifications, *they are simply system-based observations*. Device specifications are covered later in this chapter.

In terms of total *voltage and current output*, dual-supply parts tend to offer more in absolute terms, since single-supply parts usually are usually designed not just for low operating voltage ranges, but also more modest current outputs.

In terms of *precision*, the dual-supply op amp has been long favored by designers for highest overall precision. However, this status quo is now beginning to be challenged, by such single-supply parts as the truly excellent chopper-stabilized op amps. With more and more new op amps being designed for single-supply use, high precision is likely to become an ever-increasing strength of this category.

Load immunity is often an application problem with single-supply parts, as many of them use common-emitter or common-source output stages, to maximize signal swing. Such stages are typically much more load sensitive than the classic common-collector stages generally used in dual-supply op amps.

There are now a greater *variety* of dual-supply op amps available. However, this is at least in part due to the ~30-year head start they have been enjoying. Currently, new op amp designs are increasingly oriented around one or more aspects of single-supply compatibility, with strong trends toward lower supply voltages, smaller packages, etc.

Device Selection Drivers

As the op amp design process is begun, it is useful to keep in mind the fact that there are several selection *drivers*, which can dictate priorities. This is illustrated by Figure 1-11.

Actually, any *single* heading along the top of this chart can in fact be the dominant selection driver, and take precedence over all of the others. In the early days of op amp design, when such things as supply range, package type etc. were fairly narrow in spread, performance was usually the major driver. Of course, it is still very much so, and will always be. But, today's systems are much more compact and lower in power, so things like package type, size, supply range, and multiple devices can often be major drivers of selection. As one example, if the only available supply voltage is 3V, you look at 3V compatible devices first, and then fill other performance parameters as you can.

FUNCTION	PERFORM	PACKAGE	MARKET
Single, Dual, Quad	Precision	Type	Cost
Single Or Dual Supply	Speed	Size	Availability
Supply Voltage	Distortion, Noise	Footprint	
	Low Bias Current		
	Power		

Figure 1-11: Some op amp selection drivers

As another example, one coming from another perspective, sometimes all-out performance can drive everything else. An ultra low, non-negotiable input current requirement can drive not only the type of amplifier, but also its package (a FET input device in a glass-sealed hermetic package may be optimum). Then, everything else follows from there. Similarly, high power output may demand a package capable of several watts dissipation, in which case you find the power handling device and package first, and then proceed accordingly.

At this point, the concept of these "selection drivers" is still quite general. The following sections of the chapter introduce device types, which supplements this with further details of a realistic selection process.

REFERENCES: INTRODUCTION

1. John R. Ragazzini, Robert H. Randall and Frederick A. Russell, "Analysis of Problems in Dynamics by Electronic Circuits," **Proceedings of the IRE**, Vol. 35, May 1947, pp. 444-452.
2. Walter Borlase, **An Introduction to Operational Amplifiers (Parts 1-3)**, September 1971, Analog Devices Seminar Notes, Analog Devices, Inc.
3. Karl D. Swartzel, Jr. "Summing Amplifier," **US Patent 2,401,779**, filed May 1, 1941, issued June 11, 1946.
4. Frederick E. Terman, "Feedback Amplifier Design," **Electronics**, Vol. 10, No. 1, January 1937, pp. 12-15, 50.
5. Julian M. West, "Wave Amplifying System," **US Patent 2,196,844**, filed April 26, 1939, issued April 9, 1940.
6. Hendrick W. Bode, "Relations Between Attenuation and Phase In Feedback Amplifier Design," **Bell System Technical Journal**, Vol. 19, No. 3, July, 1940. See also: "Amplifier," **US Patent 2,173,178**, filed June 22, 1937, issued July 12, 1938
7. Ray Stata, "Operational Amplifiers-Parts I and II," **Electromechanical Design**, September, November 1965.
8. Dan Sheingold, Ed., **Applications Manual for Operational Amplifiers for Modeling, Measuring, Manipulating, and Much Else**, George A. Philbrick Researches, Inc., Boston, MA, 1965. See also **Applications Manual for Operational Amplifiers for Modeling, Measuring, Manipulating, and Much Else, 2nd Ed.**, Philbrick/Nexus Research, Dedham, MA, 1966, 1984.
9. Walter G. Jung, **IC Op Amp Cookbook, 3rd Ed.**, Prentice-Hall PTR, 1986, 1997, ISBN: 0-13-889601-1.
10. Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, Inc., 1995, ISBN: 0-916550-15-X.
11. Sergio Franco, **Design With Operational Amplifiers and Analog Integrated Circuits, 2nd Ed.** (Sections 1.2 – 1.4), McGraw-Hill, 1998, ISBN: 0-07-021857-9

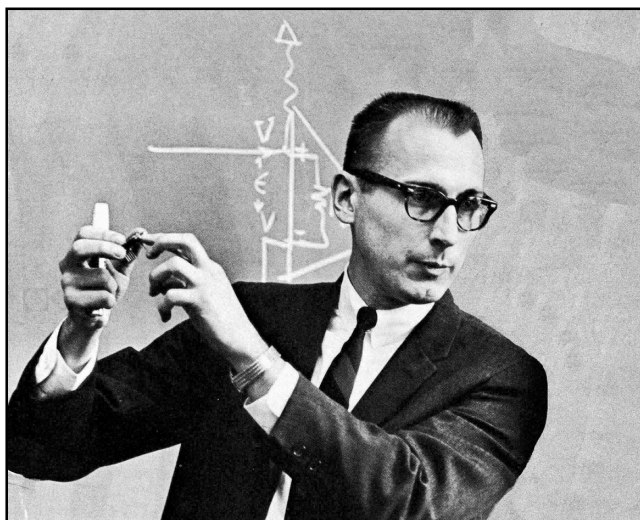
ACKNOWLEDGEMENTS:

Portions of this section were adapted from Ray Stata's "Operational Amplifiers - Part I," **Electromechanical Design**, September, 1965.

Helpful comments on this section were received from Bob Marwin, Dan Sheingold, Ray Stata, and Scott Wayne.

Classic Cameo

Ray Stata Publications Establish ADI Applications Work



In January of 1965 Analog Devices Incorporated (ADI) was founded by Matt Lorber and Ray Stata. Operating initially from Cambridge, MA, modular op amps were the young ADI's primary product. In those days, Ray Stata did more than administrative tasks. He served in sales and marketing roles, and wrote many op amp applications articles. Even today, some of these are still available to ADI customers.

One very early article set was a two part series done for **Electromechanical Design**, which focused on clear, down-to-earth explanation of op amp principles.¹

A second article for the new ADI publication **Analog Dialogue**, was entitled "Operational Integrators," and outlined various errors that plague integrators (including capacitor errors).²

A third impact article was also done for **Analog Dialogue**, titled "User's Guide to Applying and Measuring Operational Amplifier Specifications".³ As the title denotes, this was a comprehensive guide to aid the understanding of op amp specifications, and also showed how to test them.

Ray authored an Applications Manual for the 201, 202, 203 and 210 series of chopper op amps.⁴

Ray was also part of the **EEE** "Speaks Out" series of article-interviews, where he outlined some of the subtle ways that op amp specs and behavior can trap unwary users (above photo from that article).⁵

So, although ADI today makes many other products, those early op amps were the company's roots.

¹ Ray Stata, "Operational Amplifiers - Parts I and II," **Electromechanical Design**, Sept., Nov., 1965.

² Ray Stata, "Operational Integrators," **Analog Dialogue**, Vol. 1, No. 1, April, 1967. See also ADI AN357

³ Ray Stata, "User's Guide to Applying and Measuring Operational Amplifier Specifications," **Analog Dialogue**, Vol. 1, No. 3, September 1967. See also ADI AN356.

⁴ Ray Stata, **Applications Manual for 201, 202, 203 and 210 Chopper Op Amps**, ADI, 1967.

⁵ "Ray Stata Speaks Out on 'What's Wrong with Op Amp Specs'," **EEE**, July 1968.

▣ OP AMP APPLICATIONS

NOTES:

SECTION 1-2: OP AMP TOPOLOGIES

Walt Kester, Walt Jung, James Bryant

The previous section examined op amps without regard to their internal circuitry. In this section the two basic op amp topologies—voltage feedback (VFB) and current feedback (CFB)— are discussed in more detail, leading up to a detailed discussion of the actual circuit structures in Section 1-3.

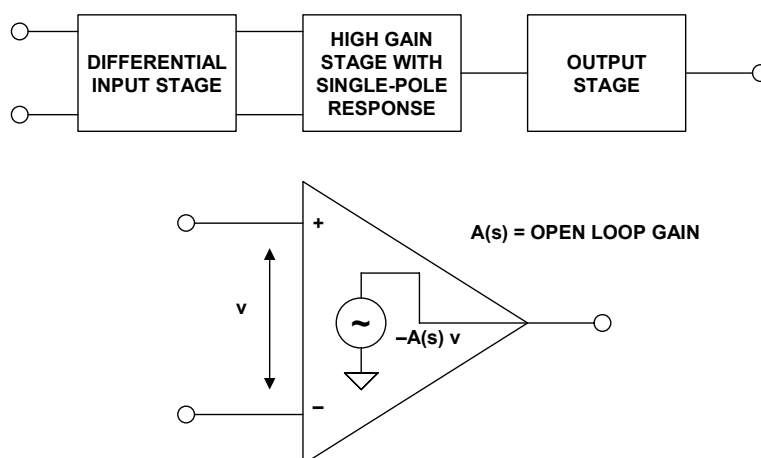


Figure 1-12: Voltage feedback (VFB) op amp

Although not explicitly stated, the previous section focused on the voltage feedback op amp and the related equations. In order to reiterate, the basic voltage feedback op amp is repeated here in Figure 1-12 above (without the feedback network) and in Figure 1-13 below (with the feedback network).

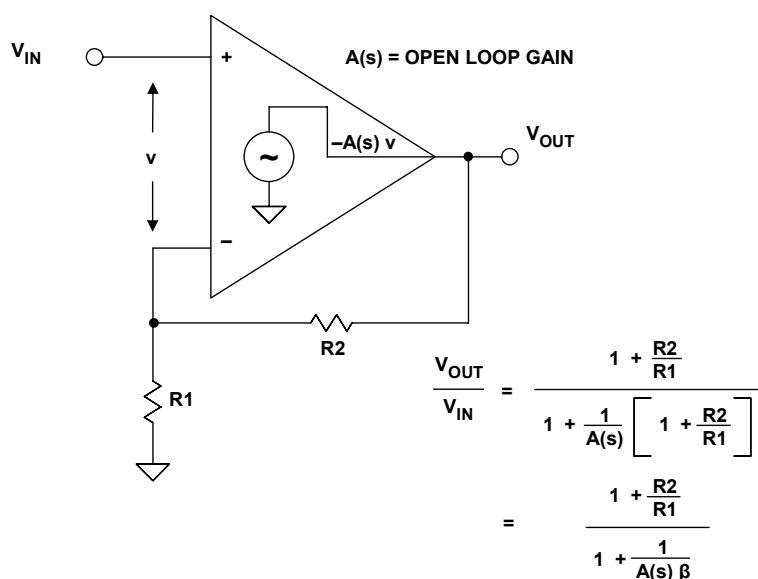


Figure 1-13: Voltage feedback op amp with feedback network connected

It is important to note that the error signal developed because of the feedback network and the finite open-loop gain $A(s)$ is in fact a small voltage, v .

■ OP AMP APPLICATIONS

Current Feedback Amplifier Basics

The basic *current feedback* amplifier topology is shown in Figure 1-14 below. Notice that within the model, a unity gain buffer connects the non-inverting input to the inverting input. In the ideal case, the output impedance of this buffer is zero ($R_O = 0$), and the error signal is a small current, i , which flows into the inverting input. The error current, i , is mirrored into a high impedance, $T(s)$, and the voltage developed across $T(s)$ is equal to $T(s) \cdot i$. (The quantity $T(s)$ is generally referred to as the *open-loop transimpedance gain*.)

This voltage is then buffered, and is connected to the op amp output. If R_O is assumed to be zero, it is easy to derive the expression for the closed-loop gain, V_{OUT}/V_{IN} , in terms of the R_1 - R_2 feedback network and the *open-loop transimpedance gain*, $T(s)$. The equation can also be derived quite easily for a finite R_O , and Fig. 1-14 gives both expressions.

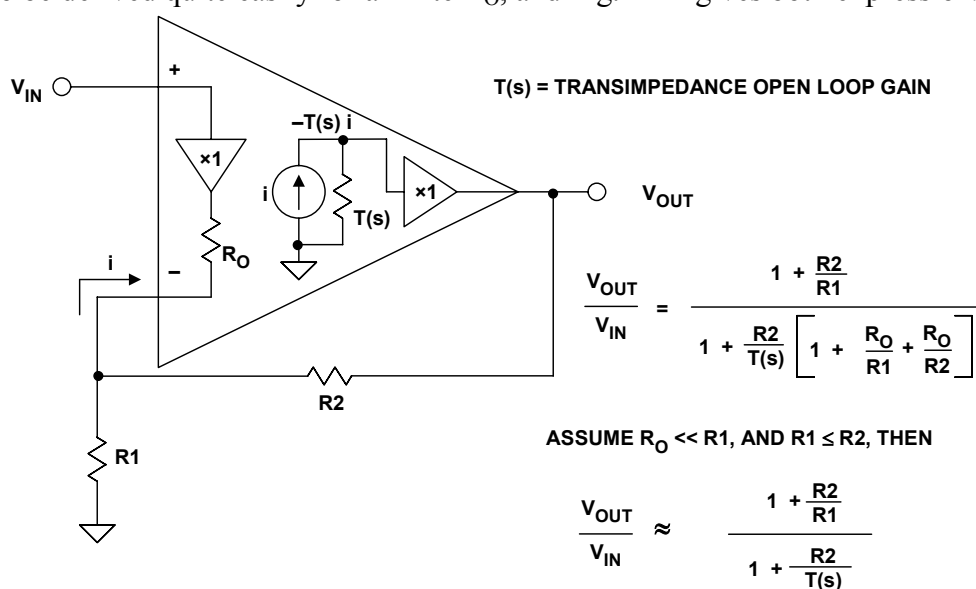


Figure 1-14: Current feedback (CFB) op amp topology

At this point it should be noted that current feedback op amps are often called *transimpedance* op amps, because the *open-loop* transfer function is in fact an impedance as described above. However, the term *transimpedance amplifier* is often applied to more general circuits such as current-to-voltage (I/V) converters, where either CFB or VFB op amps can be used. Therefore, some caution is warranted when the term *transimpedance* is encountered in a given application. On the other hand, the term *current feedback op amp* is rarely confused and is the preferred nomenclature when referring to op amp topology.

From this simple model, several important CFB op amp characteristics can be deduced.

- Unlike VFB op amps, CFB op amps *do not have balanced inputs*. Instead, the non-inverting input is high impedance, and the inverting input is low impedance.
- The open-loop gain of CFB op amps is measured in units of Ω (transimpedance gain) rather than V/V as for VFB op amps.

- For a fixed value feedback resistor R_2 , the closed-loop gain of a CFB can be varied by changing R_1 , without significantly affecting the closed-loop bandwidth. This can be seen by examining the simplified equation in Fig. 1-14. The denominator determines the overall frequency response; and if R_2 is constant, then R_1 of the numerator can be changed (thereby changing the gain) without affecting the denominator—hence the bandwidth remains relatively constant.

The CFB topology is primarily used where the ultimate in high speed and low distortion is required. The fundamental concept is based on the fact that in bipolar transistor circuits currents can be switched faster than voltages, all other things being equal. A more detailed discussion of CFB op amp AC characteristics can be found in Section 1-5.

Figure 1-15 below shows a simplified schematic of an early IC CFB op amp, the AD846—introduced by Analog Devices in 1988 (see Reference 1). Notice that full advantage is taken of the complementary bipolar (CB) process which provides well matched high f_t PNP and NPN transistors.

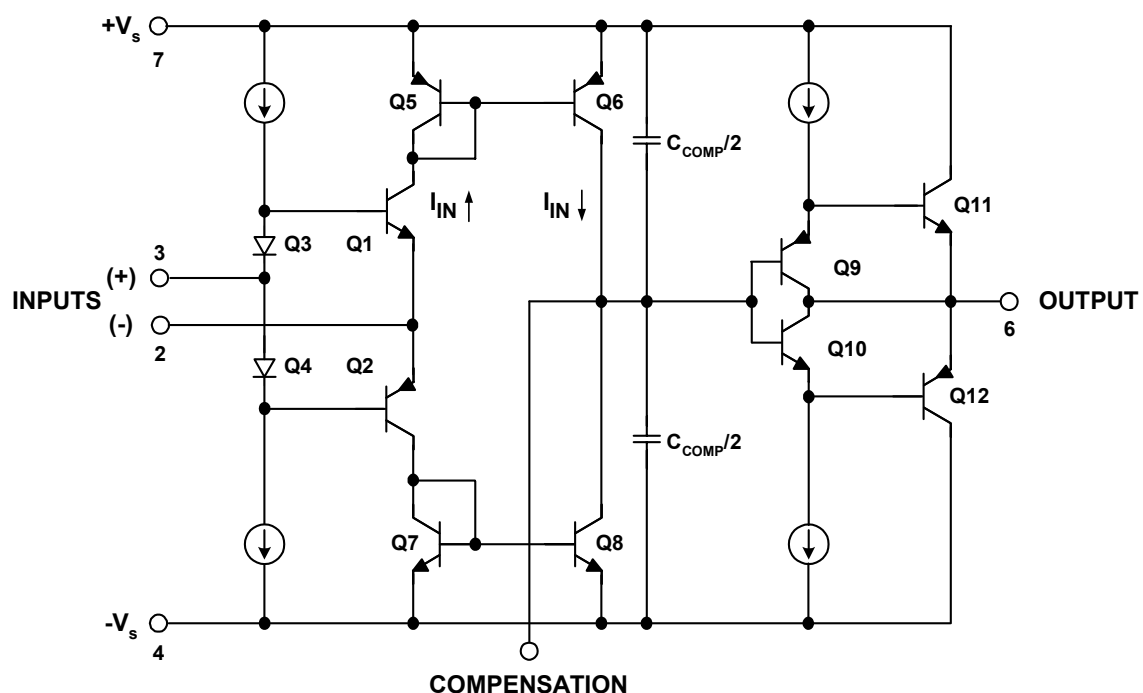


Figure 1-15: AD846 current feedback op amp (1988)

Transistors Q1-Q2 buffer the non-inverting input (pin 3) and drive the inverting input (pin 2). Q5-Q6 and Q7-Q8 act as current mirrors that drive the high impedance node. The C_{COMP} capacitor provides the dominant pole compensation; and Q9, Q10, Q11, and Q12 comprise the output buffer. In order to take full advantage of the CFB architecture, a high speed complementary bipolar (CB) IC process is required. With modern IC processes, this is readily achievable, allowing direct coupling in the signal path of the amplifier.

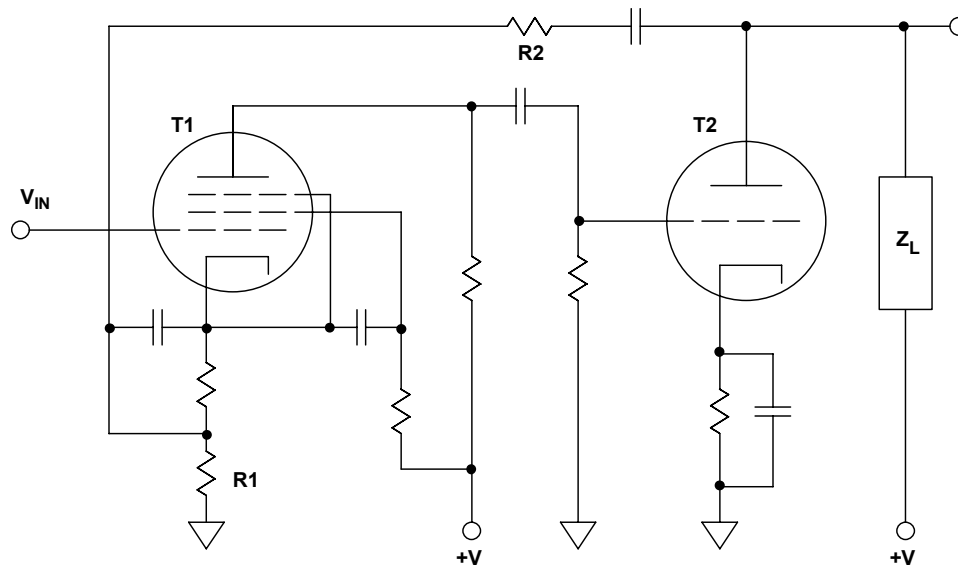
However, the basic concept of current feedback can be traced all the way back to early vacuum tube feedback circuitry, which used negative feedback to the input tube cathode. This use of the cathode for feedback would be analogous to the CFB op amp's low impedance (-) input, in Fig. 1-15.

■ OP AMP APPLICATIONS

Current Feedback Using Vacuum Tubes

Figure 1-16 below is an adaptation from a 1937 article on feedback amplifiers by Frederick E. Terman (see Reference 2). Notice that the AC-coupled R2 feedback resistor for this two-stage amplifier is connected to the low impedance cathode of T1, the pentode vacuum tube input stage. Similar examples of early tube circuits using cathode feedback can be found in Reference 3.

DC-coupled op amp design using vacuum tubes was difficult for numerous reasons. One reason was a lack of suitable level shifters. Multi-stage op amps either required extremely high supply voltages or suffered gain loss because of resistive level shifters. In a 1941 article, Stewart E. Miller describes how to use gas discharge tubes as level shifters in several vacuum tube amplifier circuits (see Reference 4). A circuit of particular interest is shown in Figure 1-17 (opposite).



Adapted from: Frederick E. Terman, "Feedback Amplifier Design," *Electronics*, January 1937, pp. 12-15, 50.

Figure 1-16: A 1937 vacuum tube feedback circuit designed by Frederick E. Terman, using current feedback to the low impedance input cathode (adapted from Reference 2)

In the Fig. 1-17 reproduction of Miller's circuit, the R2 feedback resistor and the R1 gain setting resistor are labeled for clarity, and it can be seen that feedback is to the low impedance cathode of the input tube. The author suggests that the closed-loop gain of the amplifier can be adjusted from 72dB-102dB, by varying the R1 gain-setting resistor from 37.4Ω to 1.04Ω.

What is really interesting about the Miller circuit is its frequency response, which is reproduced in Figure 1-18 (opposite). Notice that the closed-loop bandwidth is nearly independent of the gain setting, and the circuit certainly does not exhibit a constant gain-bandwidth product as would be expected for a traditional VFB op amp.

For a gain of 72dB, the bandwidth is about 30kHz, and for a gain of 102dB (30dB increase), the bandwidth only drops to ~15kHz. With a 72dB gain at 30kHz VFB op amp, bandwidth would be expected to drop 5 octaves to ~0.9kHz for 102dB of gain.

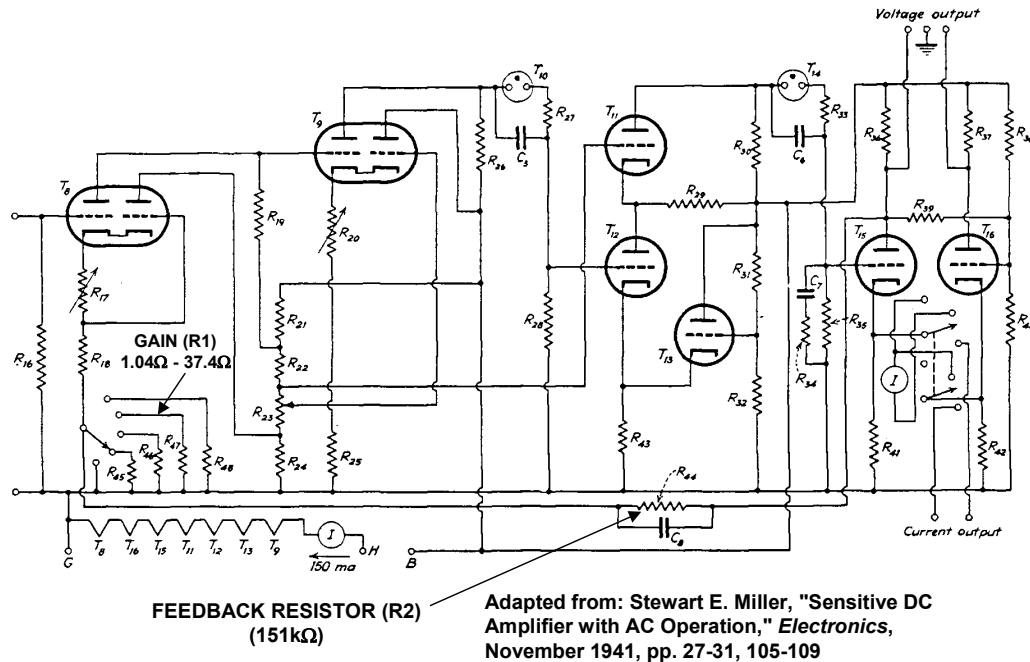
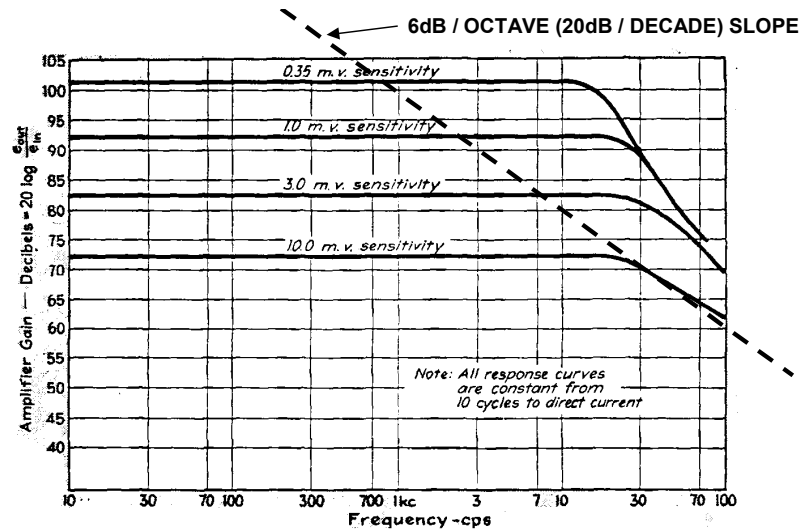


Figure 1-17: A 1941 vacuum tube feedback circuit using current feedback

To clarify this point on bandwidth, a standard VFB op amp 6dB/octave (20dB/decade) slope has been added to Fig. 1-18 for reference.



Adapted from: Stewart E. Miller, "Sensitive DC Amplifier with AC Operation," *Electronics*, November 1941, pp. 27-31, 105-109

Figure 1-18: A 1941 feedback circuit shows characteristic CFB gain-bandwidth relationship

Although there is no mention of the significance of this within the text of the actual article, it nevertheless illustrates a popular application of CFB behavior, in the design of high speed programmable gain amplifiers with relatively constant bandwidth.

■ OP AMP APPLICATIONS

When transistor circuits ultimately replaced vacuum tube circuits between the late 1950s and the mid-1960s, the current feedback architecture became popular for certain high speed op amps. Figure 1-19 below shows a fast-settling op amp designed at Bell Labs in 1965, for use as a building block in high speed A/D converters (see Reference 5).

The circuit shown is a composite amplifier containing a high speed AC amplifier (shown inside the dotted outline) and a separate DC servo amplifier loop (not shown). The feedback resistor R2 is AC coupled to the low-impedance emitter of transistor Q1. The circuit design was somewhat awkward because of the lack of good high frequency PNP transistors, and it also required zener diode level shifters, and non-standard supplies.

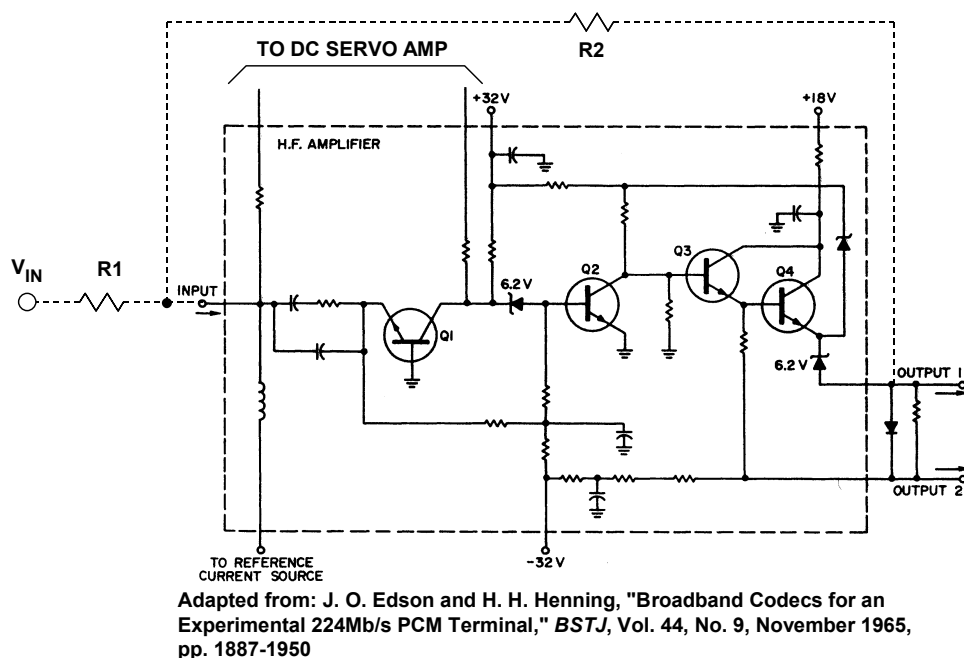


Figure 1-19: A 1965 solid state current feedback op amp design from Bell Labs

Hybrid circuit manufacturing technology, which was well established by the 1980s, allowed the use of fast, relatively well-matched NPN and PNP transistors, to realize CFB op amps. The Analog Devices' AD9610 and AD9611 hybrids were good examples of these devices introduced in the mid-1980s.

With the development of high speed complementary bipolar IC processes in the 1980s (see Reference 6) it became possible to realize completely DC-coupled current feedback op amps using PNP and NPN transistors such as the Analog Devices' AD846, introduced in 1988 (Fig. 1-15, again). Device matching and clever circuit design techniques give these modern IC CFB op amps excellent AC and DC performance without a requirement for separate level shifters, awkward supply voltages, or separate DC servo loops.

Various patents have been issued for these types of designs (see References 7 and 8, for example), but it should be remembered that the fundamental concepts were established decades earlier.

REFERENCES: OP AMP TOPOLOGIES

1. Wyn Palmer, Barry Hilton, "A 500V/ μ s 12 Bit Transimpedance Amplifier," **ISSCC Digest**, February 1987, pp. 176-177, 386.
2. Frederick E. Terman, "Feedback Amplifier Design," **Electronics**, January 1937, pp. 12-15, 50.
3. Edward L. Ginzton, "DC Amplifier Design Techniques," **Electronics**, March 1944, pp. 98-102.
4. Stewart E. Miller, "Sensitive DC Amplifier with AC Operation," **Electronics**, November 1941, pp. 27-31, 105-109.
5. J. O. Edson and H. H. Henning, "Broadband Codecs for an Experimental 224Mb/s PCM Terminal," **Bell System Technical Journal**, Vol. 44, No. 9, November 1965, pp. 1887-1950.
6. "Op Amps Combine Superb DC Precision and Fast Settling," **Analog Dialogue**, Vol. 22, No. 2, pp. 12-15.
7. David A. Nelson, "Settling Time Reduction in Wide-Band Direct-Coupled Transistor Amplifiers," **US Patent 4,502,020**, Filed October 26, 1983, Issued February 26, 1985.
8. Royal A. Gosser, "DC-Coupled Transimpedance Amplifier," **US Patent 4,970,470**, Filed October 10, 1989, Issued November 13, 1990.

▣ OP AMP APPLICATIONS

NOTES:

SECTION 1-3: OP AMP STRUCTURES

Walt Kester, Walt Jung, James Bryant

This section describes op amps in terms of their *structures*, and Section 1-4 discusses op amp *specifications*. It is hard to decide which to discuss first, since discussion of specifications, to be useful, entails reference to structures, and discussion of structures likewise requires reference to the performance feature that they are intended to optimize.

Since the majority of readers will have at least some familiarity with operational amplifiers and their specifications, we shall discuss structures first, and assume that readers will have at least a first-order idea of the definitions of the various specifications. Where this assumption proves ill-founded, the reader should look ahead to the next section to verify any definitions required.

Because single-supply devices permeate practically all modern system designs, the related design issues are integrated into the following op amp structural discussions.

Single-Supply Op Amp Issues

Over the last several years, single-supply operation has become an increasingly important requirement because of market demands. Automotive, set-top box, camera/cam-corder, PC, and laptop computer applications are demanding IC vendors to supply an array of linear devices that operate on a single-supply rail, with the same performance of dual supply parts. Power consumption is now a key parameter for line or battery operated systems, and in some instances, more important than cost. This makes low-voltage/low supply current operation critical; at the same time, however, accuracy and precision requirements have forced IC manufacturers to meet the challenge of "doing more with less" in their amplifier designs.

In a single-supply application, the most immediate effect on the performance of an amplifier is the reduced input and output signal range. As a result of these lower input and output signal excursions, amplifier circuits become more sensitive to internal and external error sources. Precision amplifier offset voltages on the order of 0.1mV are less than a 0.04 LSB error source in a 12-bit, 10V full-scale system. In a single-supply system, however, a "rail-to-rail" precision amplifier with an offset voltage of 1mV represents a 0.8LSB error in a 5V fullscale system (or 1.6LSB for 2.5V fullscale).

To keep battery current drain low, larger resistors are usually used around the op amp. Since the bias current flows through these larger resistors, they can generate offset errors equal to or greater than the amplifier's own offset voltage.

Gain accuracy in some low voltage single-supply devices is also reduced, so device selection needs careful consideration. Many amplifiers with ~120dB open-loop gains typically operate on dual supplies—for example OP07 types. However, many single-supply/rail-to-rail amplifiers for precision applications typically have open-loop gains between 25,000 and 30,000 under light loading ($>10\text{k}\Omega$). Selected devices, like the

■ OP AMP APPLICATIONS

OP113/213/413 family, do have high open-loop gains ($>120\text{dB}$), for use in demanding applications. Another example would be the AD855x chopper-stabilized op amp series.

Many trade-offs are possible in the design of a single-supply amplifier circuit— speed versus power, noise versus power, precision versus speed and power, etc. Even if the noise floor remains constant (highly unlikely), the signal-to-noise ratio will drop as the signal amplitude decreases.

Besides these limitations, many other design considerations that are otherwise minor issues in dual-supply amplifiers now become important. For example, signal-to-noise (SNR) performance degrades as a result of reduced signal swing. "Ground reference" is no longer a simple choice, as one reference voltage may work for some devices, but not others. Amplifier voltage noise increases as operating supply current drops, and bandwidth decreases. Achieving adequate bandwidth and required precision with a somewhat limited selection of amplifiers presents significant system design challenges in single-supply, low-power applications.

Most circuit designers take "ground" reference for granted. Many analog circuits scale their input and output ranges about a ground reference. In dual-supply applications, a reference that splits the supplies (0V) is very convenient, as there is equal supply headroom in each direction, and 0V is generally the voltage on the low impedance ground plane.

In single-supply/rail-to-rail circuits, however, the ground reference can be chosen anywhere within the supply range of the circuit, since there is no standard to follow. The choice of ground reference depends on the type of signals processed and the amplifier characteristics. For example, choosing the negative rail as the ground reference may optimize the dynamic range of an op amp whose output is designed to swing to 0V. On the other hand, the signal may require level shifting in order to be compatible with the input of other devices (such as ADCs) that are not designed to operate at 0V input.

Very early single-supply "zero-in, zero-out" amplifiers were designed on bipolar processes, which optimized the performance of the NPN transistors. The PNP transistors were either lateral or substrate PNPs with much less bandwidth than the NPNs. Fully complementary processes are now required for the new breed of single-supply/rail-to-rail operational amplifiers. These new amplifier designs don't use lateral or substrate PNP transistors within the signal path, but incorporate parallel NPN and PNP input stages to accommodate input signal swings from ground to the positive supply rail. Furthermore, rail-to-rail output stages are designed with bipolar NPN and PNP common-emitter, or N-channel/P-channel common-source amplifiers whose collector-emitter saturation voltage or drain-source channel on-resistance determine output signal swing, as a function of the load current.

The characteristics of a single-supply amplifier input stage (common-mode rejection, input offset voltage and its temperature coefficient, and noise) are critical in precision, low-voltage applications. Rail-rail input operational amplifiers must resolve small signals, whether their inputs are at ground, or in some cases near the amplifier's positive supply. Amplifiers having a minimum of 60dB common-mode rejection over the entire

input common-mode voltage range from 0V to the positive supply are good candidates. It is not necessary that amplifiers maintain common-mode rejection for signals beyond the supply voltages. *But, what is required is that they do not self-destruct for momentary overvoltage conditions!* Furthermore, amplifiers that have offset voltages less than 1mV and offset voltage drifts less than $2\mu\text{V}/^\circ\text{C}$ are also very good candidates for precision applications. Since *input* signal dynamic range and SNR are equally if not more important than *output* dynamic range and SNR, precision single-supply/rail-to-rail operational amplifiers should have noise levels referred-to-input (RTI) less than $5\mu\text{Vp-p}$ in the 0.1Hz to 10Hz band.

The need for rail-to-rail amplifier output stages is also driven by the need to maintain wide dynamic range in low-supply voltage applications. A single-supply/rail-to-rail amplifier should have output voltage swings that are within at least 100mV of either supply rail (under a nominal load). The output voltage swing is very dependent on output stage topology and load current.

- ◆ **Single Supply Offers:**
 - Lower Power
 - Battery Operated Portable Equipment
 - Requires Only One Voltage

- ◆ **Design Tradeoffs:**
 - Reduced Signal Swing Increases Sensitivity to Errors Caused by Offset Voltage, Bias Current, Finite Open-Loop Gain, Noise, etc.
 - Must Usually Share Noisy Digital Supply
 - Rail-to-Rail Input and Output Needed to Increase Signal Swing
 - Precision Less than the best Dual Supply Op Amps but not Required for All Applications
 - Many Op Amps Specified for Single Supply, but do not have Rail-to-Rail Inputs or Outputs

Figure 1-20: Single-supply op amp design issues

Generally, the voltage swing of a good rail-to-rail output stage should maintain its rated swing for loads down to 10k Ω . The smaller the V_{OL} and the larger the V_{OH} , the better. System parameters, such as "zero-scale" or "full-scale" output voltage, should be determined by an amplifier's V_{OL} (for zero-scale) and V_{OH} (for full-scale).

Since the majority of single-supply data acquisition systems require at least 12- to 14-bit performance, amplifiers which exhibit an open-loop gain greater than 30,000 for all loading conditions are good choices in precision applications. Single-supply op amp design issues are summarized in Figure 1-20 above.

■ OP AMP APPLICATIONS

Op Amp Input Stages

It is extremely important to understand input and output structures of op amps in order to properly design the required interfaces. For ease of discussion, the two can be examined separately, as there is no particular reason to relate them at this point.

Bipolar Input Stages

The very common and basic bipolar input stage of Figure 1-21 below consists of a "long-tailed pair" built with bipolar transistors. It has a number of advantages: it is simple, has very low offset, the bias currents in the inverting and non-inverting inputs are well-matched and do not vary greatly with temperature. In addition, minimizing the initial offset voltage of a bipolar op amp by laser trimming also minimizes its drift over temperature. This architecture was used in the very earliest monolithic op amps such as the $\mu A709$. It is also used with modern high speed types, like the AD829 and AD8021. Although NPN bipolars are shown, the concept also applies with the use of PNP bipolars.

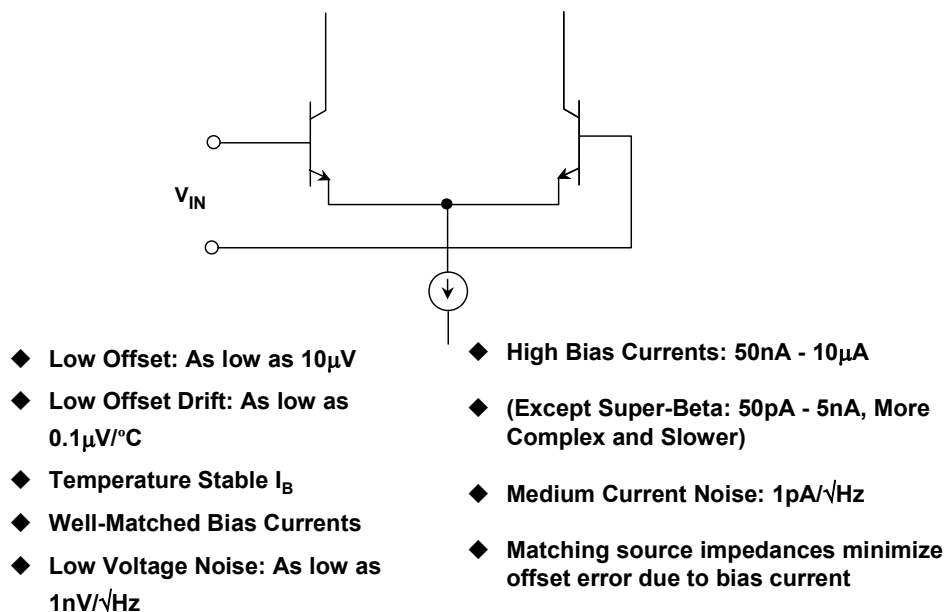


Figure 1-21: A bipolar transistor input stage

The AD829, introduced in 1990, is shown in Figure 1-22 (opposite). This op amp uses a bipolar differential input stage, Q1-Q2, which drives a "folded cascode" gain stage which consists of a fast pair of PNP transistors, Q3-Q4 (see Reference 1). These PNPs drive a current mirror that provides the differential-to-single-ended conversion. The output stage is a two-stage complementary emitter follower.

The AD829 is a wideband video amplifier with a 750MHz uncompensated gain-bandwidth product, and it operates on $\pm 5V$ to $\pm 15V$ supplies. For added flexibility, the AD829 provides access to the internal compensation node (C_{COMP}). This allows the user to customize frequency response characteristics for a particular application where the closed-loop gain is less than 20. The RC network connected between the output and the high impedance node helps maintain stability, when driving capacitive loads.

Input bias current is $7\mu\text{A}$ maximum at $+25^\circ\text{C}$, input voltage noise is $1.7\text{nV}/\sqrt{\text{Hz}}$, and input current noise is $1.5\text{pA}/\sqrt{\text{Hz}}$. Laser wafer trimming reduces the input offset voltage to 0.5mV maximum for the "A" grade. Typical input offset voltage drift is $0.3\mu\text{V}/^\circ\text{C}$.

In an op amp input circuit such as Fig. 1-22, the input bias current is the base current of the transistors comprising the long-tailed pair, Q1-Q2. It can be quite high, especially in high speed amplifiers, because the collector currents are high. It is typically $\sim 3\mu\text{A}$, for the AD829. In amplifiers where the bias current is *uncompensated* (as true in this case), the bias current will be equal to $\frac{1}{2}$ the Q1-Q2 emitter current, divided by the H_{FE} .

The bias current of a simple bipolar input stage can be reduced by a couple of measures. One is by means of *bias current compensation*, to be described further below.

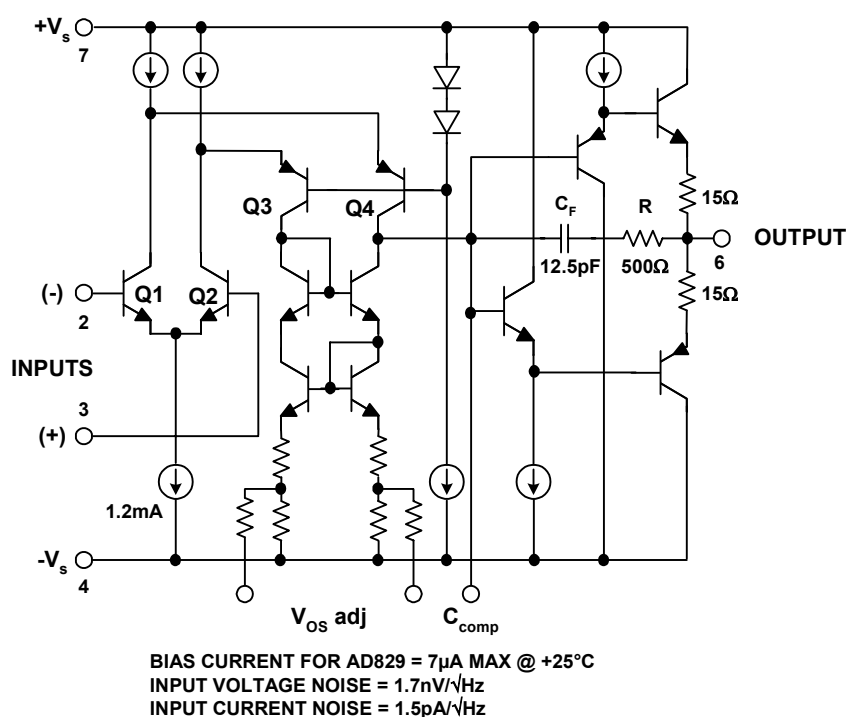


Figure 1-22: AD829 op amp simplified schematic

Another method of reducing bias current is by the use of *super-beta* transistors for Q1-Q2. Super-beta transistors are specially processed devices with a very narrow base region. They typically have a current gain of thousands or tens of thousands (rather than the more usual hundreds). Op amps with super-beta input stages have much lower bias currents, but they also have more limited frequency response.

Since the breakdown voltages of super-beta devices are quite low, they also require additional circuitry to protect the input stage from damage caused by over-voltage (for example, they wouldn't operate in the circuit of Fig. 1-22).

Some examples of super-beta input bipolar op amps are the AD704/705/706 series, and the OP97/297/497 series (single, dual, quad). These devices have typical 25°C bias currents of 100pA or less.

■ OP AMP APPLICATIONS

Bias Current Compensated Bipolar Input Stage

A simple bipolar input stage such as used in Fig. 1-22 exhibits high bias current because the currents seen externally are in fact the base currents of the two input transistors.

By providing this necessary bias currents via an internal current source, as in Figure 1-23 below, the only *external* current then flowing in the input terminals is the difference current between the base current and the current source, which can be quite small.

Most modern precision op amps use some means of internal bias current compensation, examples would be the familiar OP07 and OP27 series.

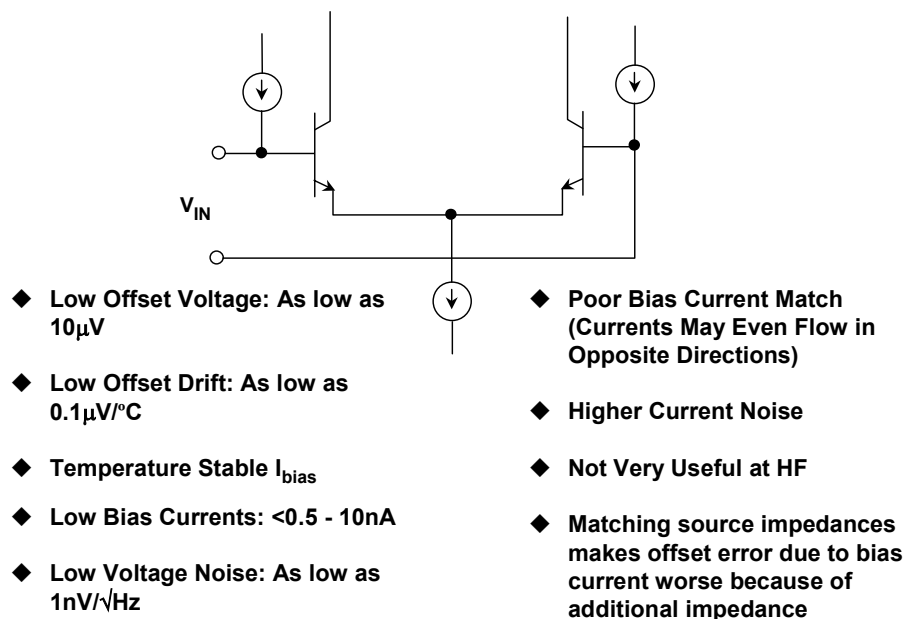


Figure 1-23: A bias current compensated bipolar input stage

The well-known OP27 op amp family is good example of bias compensated op amps (see References 2 and 3). The simplified schematic of the OP27, shown in Figure 1-24 (opposite), shows that the multiple-collector transistor Q6 provides the bias current compensation for the input transistors Q1 and Q2. The "G" grade of the OP27 has a maximum input bias current specification of $\pm 80\text{nA}$ at 25°C . Input voltage noise is $3\text{nV}/\sqrt{\text{Hz}}$, and input current noise is $0.4\text{pA}/\sqrt{\text{Hz}}$. Offset voltage trimming by "Zener-zapping" reduces the input offset voltage of the OP27 to $50\mu\text{V}$ maximum at $+25^\circ\text{C}$ for the "E" grade device (see Reference 4 for details of this trim method).

Bias current compensated input stages have many of the good features of the simple bipolar input stage, namely: low voltage noise, low offset, and low drift. Additionally, they have low bias current which is fairly stable with temperature. However, their current noise is not very good, and their bias current matching is poor.

These latter two undesired side effects result from the external bias current being the *difference* between the compensating current source and the input transistor base current. Both of these currents inevitably have noise. Since they are uncorrelated, the two noises add in a root-sum-of-squares fashion (even though the DC currents subtract).

Since the resulting external bias current is the difference between two nearly equal currents, there is no reason why the net current should have a defined polarity. As a result, the bias currents of a bias-compensated op amp may not only be mismatched, they can actually flow in opposite directions! In most applications this isn't important, but in some it can have unexpected effects (for example the droop of a sample-and-hold (SHA) built with a bias-compensated op amp may have either polarity).

In many cases, the bias current compensation feature is not mentioned on an op amp data sheet, and a simplified schematic isn't supplied. It is easy to determine if bias current compensation is used by examining the bias current specification. If the bias current is specified as a " $\pm$ " value, the op amp is most likely compensated for bias current.

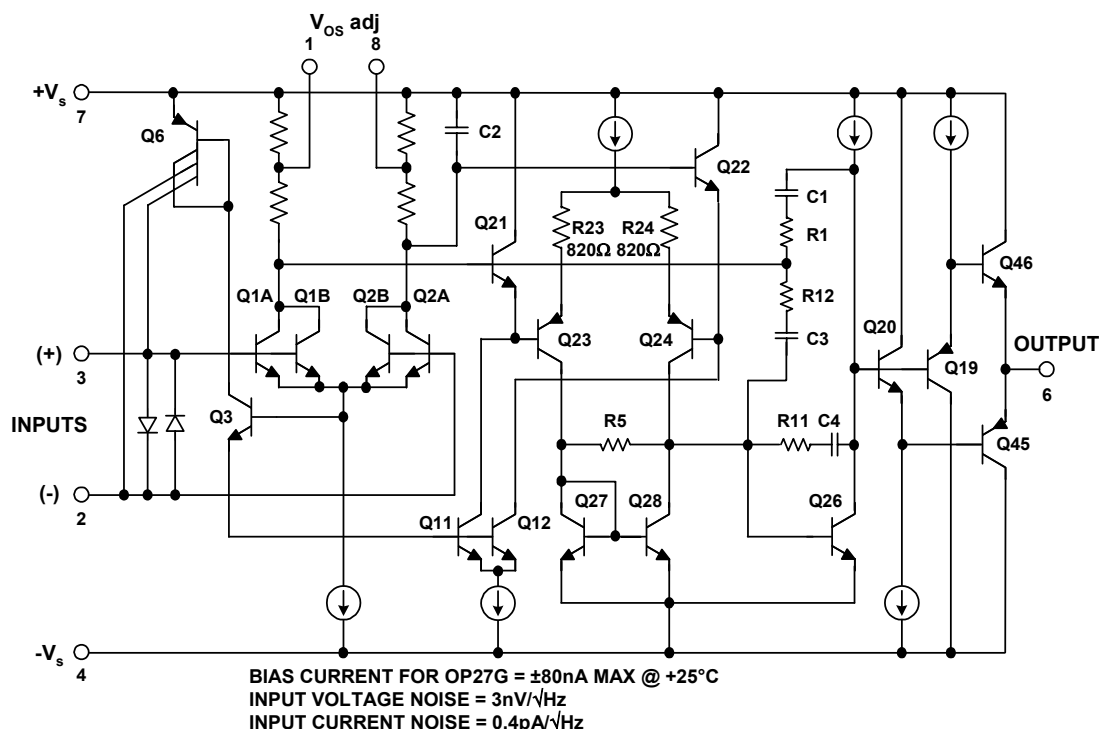


Figure 1-24: OP27 op amp uses bias current compensated input stage

Note that this can easily be verified, by examining the *offset current* specification (the difference in the bias currents). If internal bias current compensation exists, the offset current will be of the same magnitude as the bias current. Without bias current compensation, the offset current will generally be at least a factor of 10 smaller than the bias current. Note that these relationships generally hold, regardless of the exact magnitude of the bias currents.

It is also a well-known fact that, within an op amp application circuit, the effects of bias current on the output offset voltage of an op amp can often be cancelled by making the source resistances at the two inputs equal. But, there is an important caveat here. The validity of this practice only holds true for bipolar input op amps *without* bias current compensation, that is, where the input currents are well matched. In a case of an op amp using internal bias current compensation, adding an extra resistance to either input will usually make the output offset worse!

■ OP AMP APPLICATIONS

Bias Current Compensated Super-Beta Bipolar Input Stage

As mentioned above, the OP97/297/497-series are high performance super-beta op amps, which also use input bias current compensation. As a result, their input bias currents are $\pm 150\text{pA}$ max at 25°C . Note that in this case the " $\pm$ " prefix to the bias current magnitude indicates that the amplifier uses internal bias current compensation.

A simplified schematic of an OP97 (or $\frac{1}{4}$ of the OP497) is shown in Figure 1-25 below. Note that the Q1-Q2 super-beta pair is protected against large destructive differential input voltages, by the use of both back-to-back diodes, and series current-limiting resistors. Note also that Q1-Q2 super-beta pair is also protected against excessive collector voltage, by an elaborate bias and bootstrapping network.

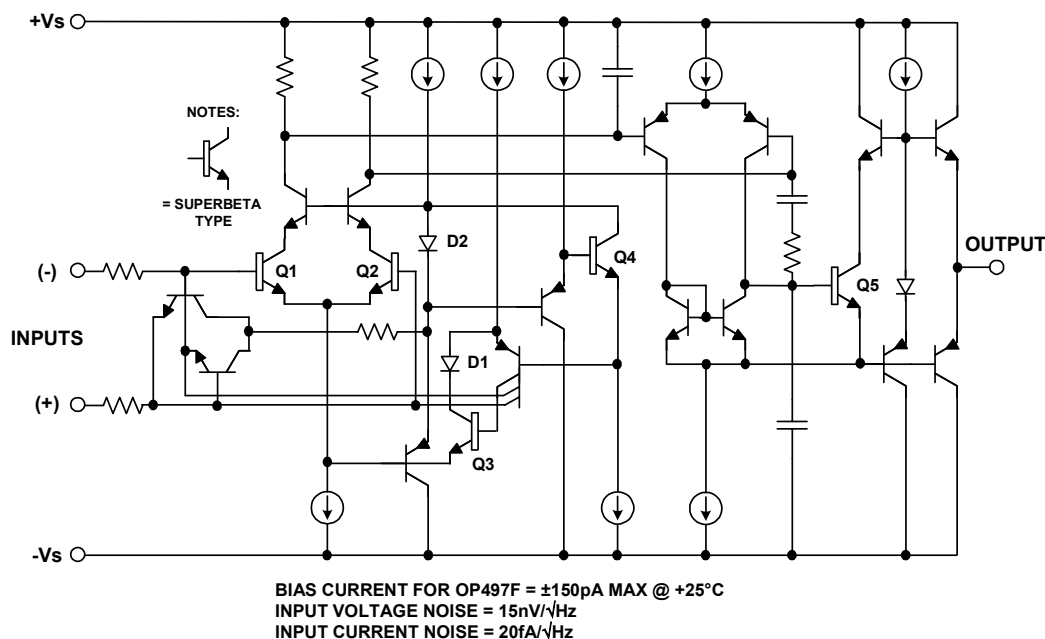


Figure 1-25: The OP97, OP297 and OP497 op amp series uses super-beta input stage transistors and bias current compensation

As a result of these clamping and protection circuits, the input common-mode voltage of this op amp series can safely vary over the full range of the supply voltages used.

FET Input Stages

Field-Effect Transistors (FETs) have much higher input impedance than do bipolar junction transistors (BJTs) and would therefore seem to be ideal devices for op amp input stages. However, they cannot be manufactured on all bipolar IC processes, and when a process does allow their manufacture, they often have their own problems.

FETs have high input impedance, low bias current, and good high frequency performance (in an op amp, the lower g_m of the FET devices allows higher tail currents, thereby increasing the maximum slew rate). FETs also have much lower current noise.

On the other hand, the input offset voltage of FET long-tailed pairs, however, is not as good as the offset of corresponding BJTs, and trimming for minimum offset does not simultaneously minimize drift. A separate trim is needed for drift, and as a result, offset and drift in a JFET op amp, while good, aren't as good as the best BJT ones. A simplified trim procedure for an FET input op amp stage is shown in Figure 1-26 below.

It is possible to make JFET op amps with very low voltage noise, but the devices involved are very large and have quite high input capacitance, which varies with input voltage, and so a trade-off is involved between voltage noise and input capacitance.

The bias current of an FET op amp is the leakage current of the gate diffusion (or the leakage of the gate protection diode, which has similar characteristics for a MOSFET). Such leakage currents double with every 10°C increase in chip temperature so that a FET op amp bias current is *one thousand times greater* at 125°C than at 25°C. Obviously this can be important when choosing between a bipolar or FET input op amp, especially in high temperature applications where bipolar op amp input bias current actually decreases.

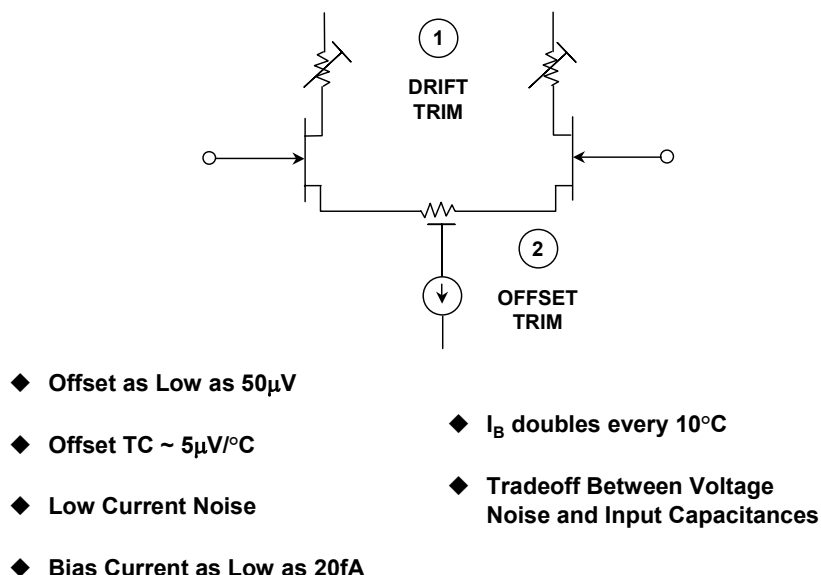


Figure 1-26: Junction field effect transistor (JFET) input op amp stage showing offset and drift trims

Thus far, we have spoken generally of all kinds of FETs, that is junction (JFETs) and MOS (MOSFETs). In practice, combined bipolar/JFET technology op amps (i.e., BiFET) achieve better performance than op amps using purely MOSFET or CMOS technology. While ADI and others make high performance op amps with MOS or CMOS input stages, in general these op amps have worse offset and drift, voltage noise, high-frequency performance than the bipolar counterparts. The power consumption is usually somewhat lower than that of bipolar op amps with comparable, or even better, performance.

JFET devices require more headroom than do BJTs, since their pinchoff voltage is typically greater than a BJT's base-emitter voltage. Consequently, they are more difficult to operate at very low power supply voltages (1-2V). In this respect, CMOS has the advantage of requiring less headroom than JFETs.

■ OP AMP APPLICATIONS

Rail-Rail Input Stages

Today, there is common demand for op amps with input CM voltage that includes *both* supply rails, i.e., *rail-to-rail* CM operation. While such a feature is undoubtedly useful in some applications, engineers should recognize that there are still relatively few applications where it is absolutely essential. These applications should be distinguished from the many more applications where a CM range *close* to the supplies, or one that includes *one* supply is necessary, but true input rail-to-rail operation is not.

In many single-supply applications, it is required that the input CM voltage range extend to one of the supply rails (usually ground). High-side or low-side current-sensing applications are examples of this. Many amplifiers can handle 0V CM inputs, and they are easily designed using PNP differential pairs (or N-channel JFET pairs) as shown in Figure 1-27 below. The input CM range of such an op amp generally extends from about 200mV below the negative rail ($-V_S$ or ground), to about 1-2V of the positive rail, $+V_S$.

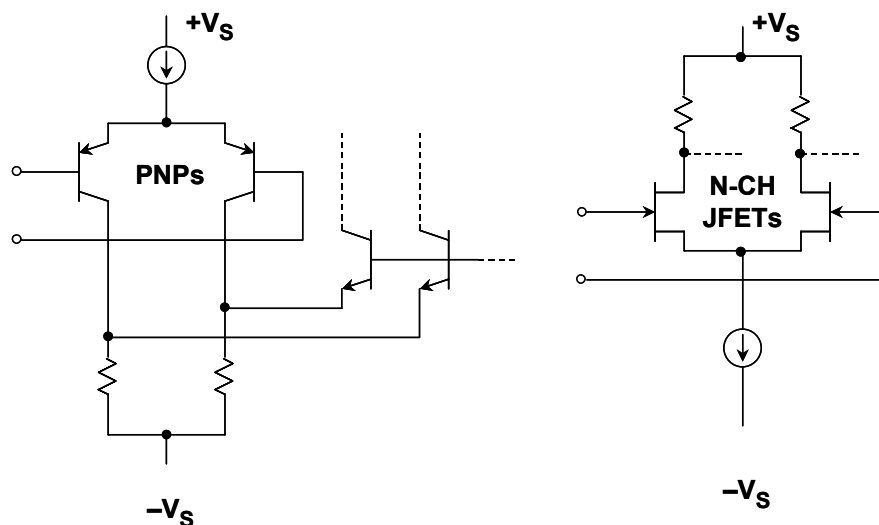


Figure 1-27: PNP or N-channel JFET stages allow CM inputs to the negative rail

An input stage could also be designed with NPN transistors (or P-channel JFETs), in which case the input CM range would include the positive rail, and go to within about 1-2V of the negative rail. This requirement typically occurs in applications such as high-side current sensing. The OP282/OP482 input stage uses a P-channel JFET input pair whose input CM range includes the positive rail, making it suitable for high-side sensing.

The AD823 is a dual 16MHz ($G = +1$) op amp with an N-channel JFET input stage (as in Fig. 1-27 right). A simplified schematic of the AD823 is shown in Figure 1-28 (opposite). This device can operate on single-supply voltages from +3 to +36V. This range also allows operation on traditional $\pm 5V$, or $\pm 15V$ dual supplies if desired. Similar devices in a related (but lower power) family include the AD820, the AD822, and the AD824.

The AD823 JFET input stage allows the input common-mode voltage to range from 200mV below the negative supply to within about 1.5V of the positive supply. Input offset voltage is 0.8mV maximum at $+25^\circ\text{C}$, input bias current is 25pA maximum at $+25^\circ\text{C}$, offset voltage drift is $2\mu\text{V}/^\circ\text{C}$, and input voltage noise is $16\text{nV}/\sqrt{\text{Hz}}$. Current noise

is only $1\text{fA}/\sqrt{\text{Hz}}$. The AD823 is laser wafer trimmed for both offset voltage and offset voltage drift as described above.

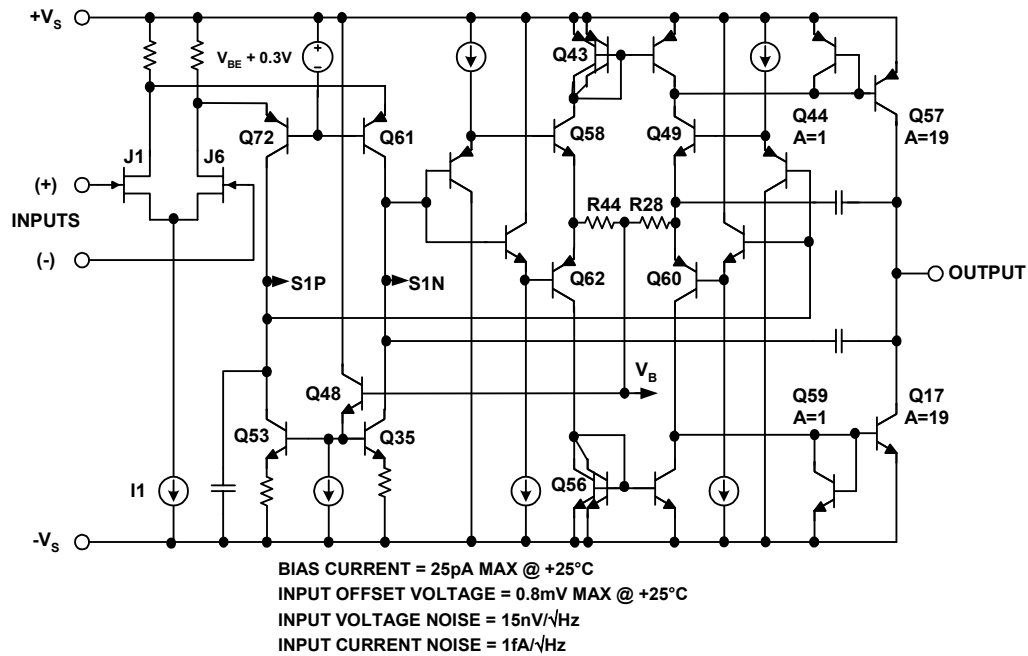


Figure 1-28: AD823 JFET input op amp simplified schematic

A simplified diagram of a true rail-to-rail input stage is shown in Figure 1-29 below. Note that this requires use of *two* long-tailed pairs, one of PNP bipolar transistors Q1-Q2, the other of NPN transistors Q3-Q4. Similar input stages can also be made with CMOS pairs.

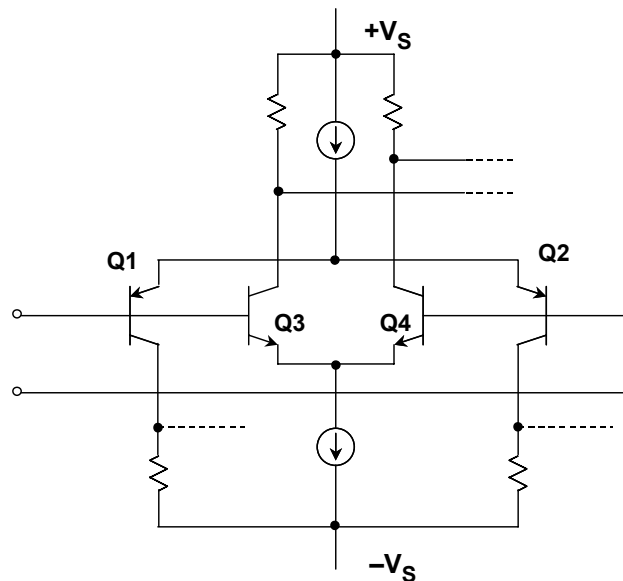


Figure 1-29: A true rail-to-rail bipolar transistor input stage

It should be noted that these two pairs will exhibit *different* offsets and bias currents, so when the applied CM voltage changes, the amplifier input offset voltage and input bias current does also. In fact, when both current sources remain active throughout most of the entire input common-mode range, amplifier input offset voltage is the *average* offset voltage of the two pairs. In those designs where the current sources are alternatively

■ OP AMP APPLICATIONS

switched off at some point along the input common-mode voltage, amplifier input offset voltage is dominated by the PNP pair offset voltage for signals near the negative supply, and by the NPN pair offset voltage for signals near the positive supply. As noted, a true rail-to-rail input stage can also be constructed from CMOS transistors, for example as in the case of the CMOS AD8531/8532/8534 op amp family.

Amplifier input bias current, a function of transistor current gain, is also a function of the applied input common-mode voltage. The result is relatively poor common-mode rejection (CMR), and a changing common-mode input impedance over the CM input voltage range, compared to familiar dual-supply devices. These specifications should be considered carefully when choosing a rail-to-rail input op amp, especially for a non-inverting configuration. Input offset voltage, input bias current, and even CMR may be quite good over *part* of the common-mode range, but much worse in the region where operation shifts between the NPN and PNP devices, and vice versa.

True rail-to-rail amplifier input stage designs must transition from one differential pair to the other differential pair, somewhere along the input CM voltage range. Some devices like the OP191/291/491 family and the OP279 have a common-mode crossover threshold at approximately 1V below the positive supply (where signals do not often occur). The PNP differential input stage is active from about 200mV below the negative supply to within about 1V of the positive supply. Over this common-mode range, amplifier input offset voltage, input bias current, CMR, input noise voltage/current are primarily determined by the characteristics of the PNP differential pair. At the crossover threshold, however, amplifier input offset voltage becomes the average offset voltage of the NPN/PNP pairs and can change rapidly.

Also, as noted previously, amplifier bias currents are dominated by the PNP differential pair over most of the input common-mode range, and change polarity and magnitude at the crossover threshold when the NPN differential pair becomes active.

Op amps like the OP184/284/484 family, shown in Figure 1-30 opposite, utilize a rail-to-rail input stage design where both NPN and PNP transistor pairs are active throughout most of the entire input CM voltage range. With this approach to biasing, there is no CM crossover threshold. Amplifier input offset voltage is the average offset voltage of the NPN and the PNP stages, and offset voltage exhibits a smooth transition throughout the entire input CM range, due to careful laser trimming of input stage resistors.

In the same manner, through careful input stage current balancing and input transistor design, the OP184 family input bias currents also exhibit a smooth transition throughout the entire CM input voltage range. The exception occurs at the very extremes of the input range, where amplifier offset voltages and bias currents increase sharply, due to the slight forward-biasing of parasitic p-n junctions. This occurs for input voltages within approximately 1V of either supply rail.

When *both* differential pairs are active throughout most of the entire input common-mode range, amplifier transient response is faster through the middle of the common-mode range by as much as a factor of 2 for bipolar input stages and by a factor of $\sqrt{2}$ for JFET input stages. This is due to the higher transconductance of two operating input stages.

Input stage g_m determines the slew rate and the unity-gain crossover frequency of the amplifier, hence response time degrades slightly at the extremes of the input common-mode range when either the PNP stage (signals approaching the positive supply rail) or the NPN stage (signals approaching the negative supply rail) are forced into cutoff. The thresholds at which the transconductance changes occur are approximately within 1V of either supply rail, and the behavior is similar to that of the input bias currents.

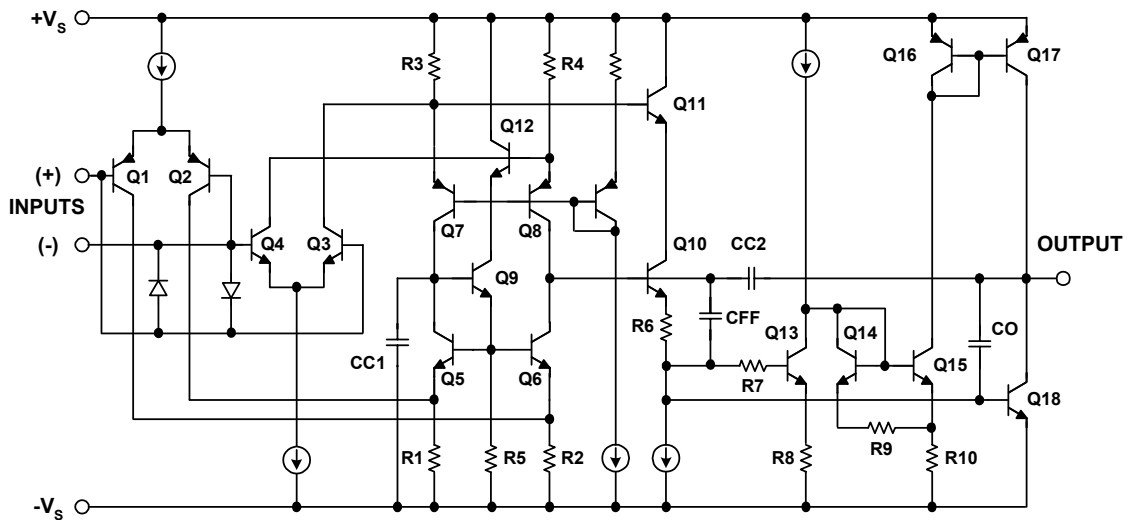


Figure 1-30: OP284 op amp simplified schematic shows true rail-to-rail input stage

In light of the many quirks of true rail-to-rail op amp input stages, applications which do require true rail-to-rail inputs should be carefully evaluated, and an amplifier chosen to ensure that its input offset voltage, input bias current, common-mode rejection, and noise (voltage and current) are suitable.

Don't forget Input Overvoltage Considerations

In order to achieve the performance levels required, it is sometimes not possible to provide complete overdrive protection within IC op amps. Although most op amps have some type of input protection, care must still be taken to prevent possible damage against both CM and differential voltage stress.

This is most likely to occur when the input signal comes from an external sensor, for example. Rather than present a cursory discussion of this topic here, the reader is instead referred to Chapter 7, Section 7-4 for a detailed examination of this important issue.

■ OP AMP APPLICATIONS

Output Stages

The earliest IC op amp output stages were NPN emitter followers with NPN current sources or resistive pull-downs, as shown in Figure 1-31A below. Naturally, the slew rates were greater for positive-going than they were for negative-going signals.

While all modern op amps have push-pull output stages of some sort, many are still asymmetrical, and have a greater slew rate in one direction than the other. Asymmetry tends to introduce distortion on AC signals and generally results from the use of IC processes with faster NPN than PNP transistors. It may also result in an ability of the output to approach one supply more closely than the other in terms of saturation voltage.

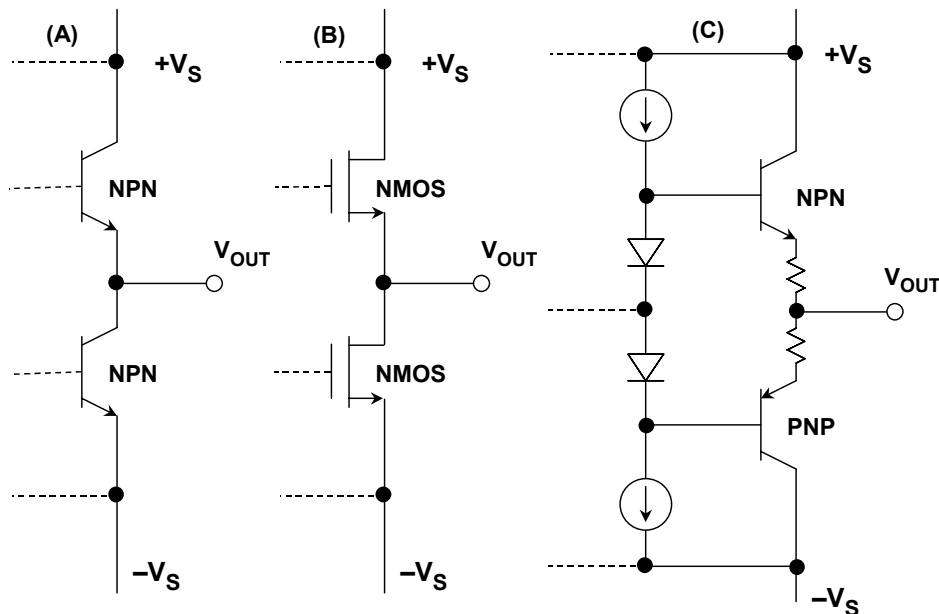


Figure 1-31: Some traditional op amp output stages

In many applications, the output is required to swing only to one rail, usually the negative rail (i.e., ground in single-supply systems). A pulldown resistor to the negative rail will allow the output to approach that rail (provided the load impedance is high enough, or is also grounded to that rail), but only slowly. Using an FET current source instead of a resistor can speed things up, but this adds complexity, as shown in Fig. 1-31B.

With modern complementary bipolar (CB) processes, well matched high speed PNP and NPN transistors are readily available. The complementary emitter follower output stage shown in Fig. 1-31C has many advantages, but the most outstanding one is the low output impedance. However, the output voltage of this stage can only swing within about one V_{BE} drop of either rail. Therefore an output swing of +1V to +4V is typical of such a stage, when operated on a single +5V supply.

The complementary common-emitter/common-source output stages shown in Figure 1-32A and B (opposite) allow the op amp output voltage to swing much closer to the rails, but these stages have much higher open-loop output impedance than do the emitter follower-based stages of Fig. 1-31C.

In practice, however, the amplifier's high open-loop gain and the applied feedback can still produce an application with low output impedance (particularly at frequencies below 10Hz). What should be carefully evaluated with this type of output stage is the loop gain within the application, with the load in place. Typically, the op amp will be specified for a minimum gain with a load resistance of 10k Ω (or more). Care should be taken that the application loading doesn't drop lower than the rated load, or gain accuracy may be lost.

It should also be noted these output stages can cause the op amp to be more sensitive to capacitive loading than the emitter-follower type. Again, this will be noted on the device data sheet, which will indicate a maximum of capacitive loading before overshoot or instability will be noted.

The complementary common emitter output stage using BJTs (Fig. 1-32A) cannot swing completely to the rails, but only to within the transistor saturation voltage (V_{CESAT}) of the rails. For small amounts of load current (less than 100 μ A), the saturation voltage may be as low as 5 to 10mV, but for higher load currents, the saturation voltage can increase to several hundred mV (for example, 500mV at 50mA).

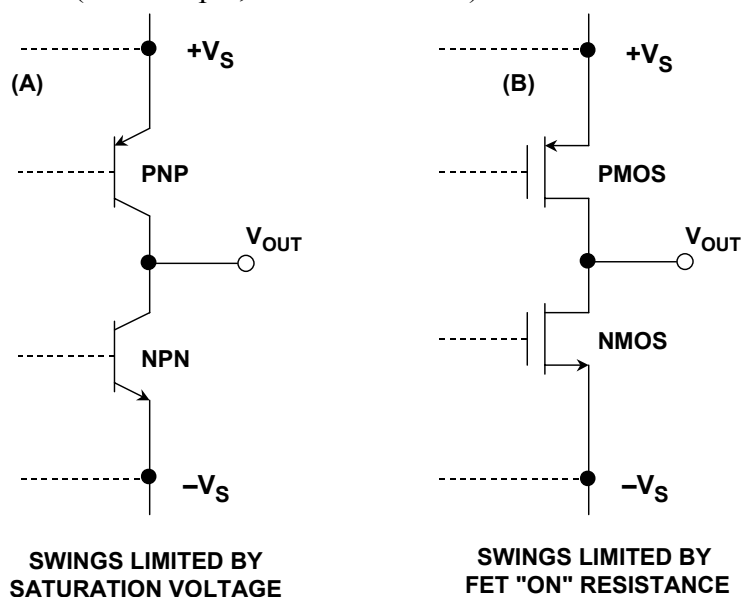


Figure 1-32: "Almost" rail-to-rail output structures

On the other hand, an output stage constructed of CMOS FETs (Fig. 1-32B) can provide nearly true rail-to-rail performance, but only under no-load conditions. If the op amp output must source or sink substantial current, the output voltage swing will be reduced by the $I \times R$ drop across the FETs internal "on" resistance. Typically this resistance will be on the order of 100 Ω for precision amplifiers, but it can be less than 10 Ω for high current drive CMOS amplifiers.

For the above basic reasons, it should be apparent that there is no such thing as a *true* rail-to-rail output stage, hence the caption of Fig. 1-32 ("Almost" Rail-to-Rail Output Structures). The best any op amp output stage can do is an almost rail-to-rail swing, when it is lightly loaded.

■ OP AMP APPLICATIONS

Op amps built on foundry CMOS processes have a primary advantage of low cost. Also, it is relatively straightforward to design rail-to-rail input and output stages with these CMOS devices, which will operate on low supply voltages.

Figure 1-33 below shows a simplified schematic of the AD8531/8532/8534 (single/dual/quad) op amp, which is typical of these design types. The AD8531/8532/8534 operates on a single 2.7V to 6.0V supply and can drive 250mA. Input offset voltage is 25mV maximum at +25°C, and voltage noise is 45nV/ $\sqrt{\text{Hz}}$.

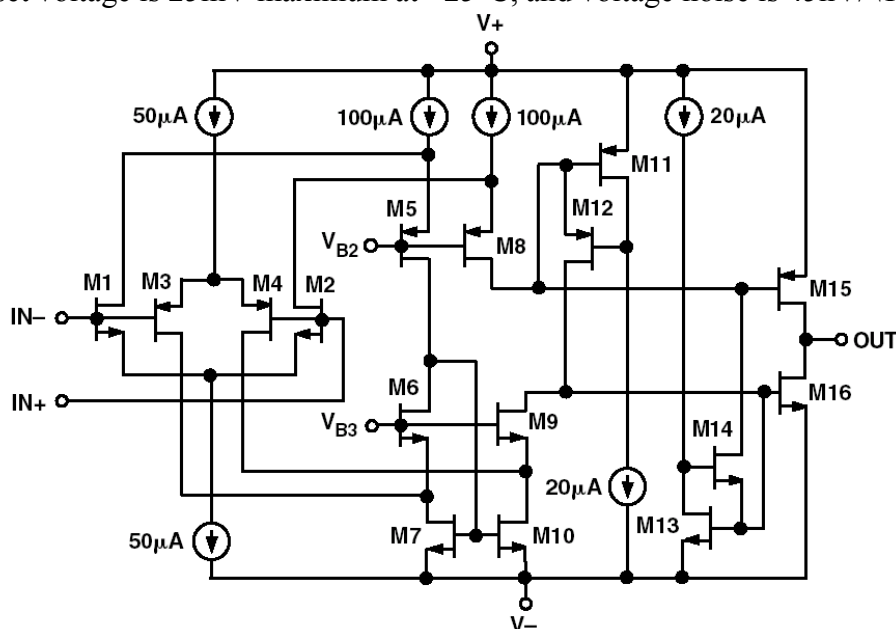


Figure 1-33: AD8531/8532/8534 CMOS rail-to-rail op amp simplified schematic

This type of op amp is simple and cost effective, and the lack of high DC precision is often no disadvantage. To the contrary, the high output drive available can be an overriding plus, particularly in AC-coupled applications.

Output Stage Surge Protection

Most low speed, high precision op amps generally have output stages which are protected against short circuits to ground or to either supply. Their output current is limited to a little more than 10mA. This has the additional advantage that it minimizes self-heating of the chip (and thus minimizes DC errors due to chip temperature differentials).

If an op amp is required to deliver both high precision and a large output current, it is advisable to use a separate output stage (within the loop) to minimize self-heating of the precision op amp. A simple buffer amplifier such as the BUF04, or a section of a non-precision op amp can be used.

Note that high speed op amps cannot have output currents limited to low values, as it would affect their slew rate and load drive ability. Thus most high speed op amps will source/sink between 50-100mA. Although many high speed op amps have internal protection for *momentary* shorts, their junction temperatures can be exceeded with sustained shorts. The user needs to be wary, and consult the specific device ratings.

Offset Voltage Trim Processes

The AD860x CMOS op amp family exploits the advantages of digital technology, so as to minimize the offset voltage normally associated with CMOS amplifiers. Offset voltage trimming is done after the devices are packaged. A digital code is entered into the device to adjust the offset voltage to less than 1mV, depending upon the grade. Wafer testing is not required, and the patented ADI technique called DigiTrim™ requires no extra pins to accomplish the function. These devices have rail-to-rail inputs and outputs (similar to Fig. 1-33), and the NMOS and PMOS parallel input stages are trimmed separately using DigiTrim to minimize the offset voltage in both pairs. A functional diagram of the AD8602 DigiTrim op amp is shown in Figure 1-34 below.

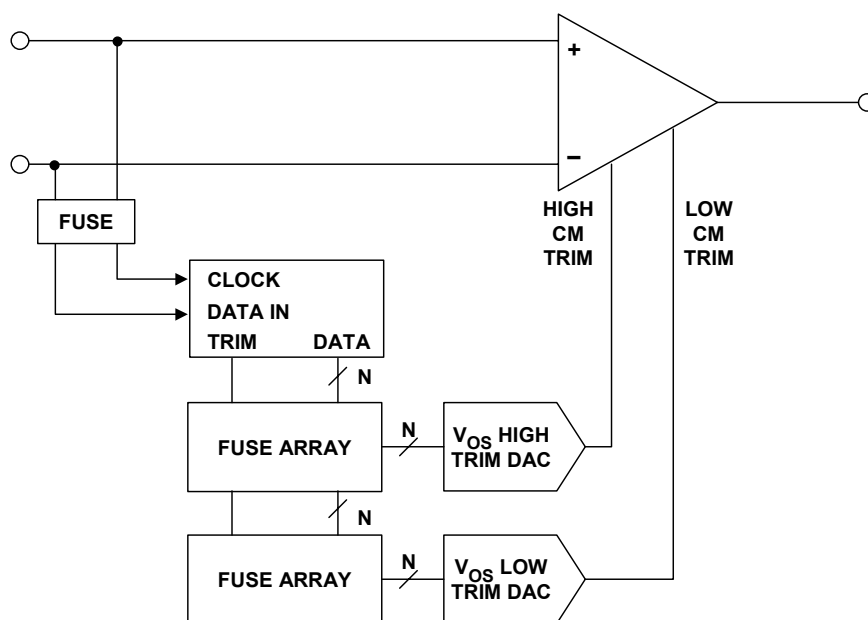


Figure 1-34: AD8602 (1/2) CMOS op amp showing DigiTrim™

DigiTrim adjusts the offset voltage by programming digitally weighted current sources. The trim information is entered through existing pins using a special digital sequence. The adjustment values can be temporarily programmed, evaluated, and readjusted for optimum accuracy before permanent adjustment is performed. After the trim is completed, the trim circuit is locked out to prevent the possibility of any accidental re-trimming by the end user.

The physical trimming, achieved by blowing polysilicon fuses, is very reliable. No extra pads or pins are required, and no special test equipment is needed to perform the trimming. The trims can be done after packaging so that assembly-related shifts can be eliminated. No testing is required at the wafer level because of high die yields.

The first devices to use this new technique are the Analog Devices' AD8601/02/04 (single, dual, quad) rail-to-rail CMOS amplifiers. The offset is trimmed for both high and low common-mode conditions so that the offset voltage is under 500μV over the full common-mode input voltage range. The bandwidth of the op amps is 8MHz, slew rate is 5V/μs, and supply current is only 640μA per amplifier.

■ OP AMP APPLICATIONS

At this point it is useful to review the other popular trim methods. Analog Devices pioneered the use of thin film resistors and laser wafer trimming for precision amplifiers, references, data converters, and other linear ICs (see Reference 5). Up to 16-bit accuracy can be achieved with trimming, and the thin film resistors themselves are very stable with temperature and can add to the thermal stability and accuracy of a device, even without trimming. Thin film deposition and patterning are processes that must be tightly controlled. The laser trimming systems are also quite expensive. In-package trimming is not possible, so assembly-related shifts cannot be easily compensated. Nevertheless, thin film trimming at the wafer level provides continuous fine trim resolution in precision integrated circuits where high accuracy and stability are required.

Zener zapping uses a voltage to create a metallic short circuit across the base-emitter junction of a transistor to remove a circuit element (see References 4 and 6). The base-emitter junction is commonly referred to as a zener, although the mechanism is actually avalanche breakdown of the junction. During the avalanche breakdown across the base-emitter junction, the very high current densities and localized heating generate rapid metal migration between the base and emitter connections, leading to a metallic short across the junction. With proper biasing (current, voltage, and time), this short will have a very low resistance value. If a series of these base-emitter junctions are arranged in parallel with a string of resistors, zapping selected junctions will short out portions of the resistor string, thereby adjusting the total resistance value.

It is possible to perform zener zap trimming in the packaged IC to compensate for assembly-related shifts in the offset voltage. However, trimming in the package requires extra package pins. Alternately, trimming at the wafer level requires additional probe pads. Probe pads do not scale effectively as the process features shrink. So, the die area required for trimming is relatively constant regardless of the process geometries. Some form of bipolar transistor is required for the trim structures, therefore a purely MOS-based process may not have zener zap capability. The nature of the trims is discrete since each zap removes a predefined resistance value. Increasing trim resolution requires additional transistors and pads or pins, which rapidly increase the total die area and/or package cost. This technique is most cost-effective for fairly large-geometry processes where the trim structures and probe pads make up a relatively small percentage of the overall die area.

It was in the process of creating the industry standard OP07 in 1975 that Precision Monolithics Incorporated pioneered zener zap trimming (Reference 6, again). The OP07 and other similar parts must be able to operate from over $\pm 15\text{V}$ supplies. As a result, they utilize relatively large device geometries to support the high voltage requirements, and extra probe pads don't significantly increase die area.

Link trimming is the cutting of metal or poly-silicon links to remove a connection. In link trimming, either a laser or a high current is used to destroy a "shorted" connection across a parallel resistive element. Removing the connection increases the effective resistance of the combined element(s). Laser cutting works similar to laser trimming of thin films. The high local heat from the laser beam causes material changes that lead to a non-conductive area, effectively cutting a metal or conductive polysilicon connector.

The high-current link trim method works as an inverse to zener zapping—the conductive connection is destroyed, rather than created by a zener-zap.

Link trim structures tend to be somewhat more compact than laser trimmed resistor structures. No special processes are required in general, although the process may have to be tailored to the laser characteristics if laser cutting is used. With the high-current trimming method, testing at the wafer level may not be required if die yields are good. The laser cutting scheme doesn't require extra contact pads, but the trim structures don't scale with the process feature sizes. Laser cutting of links cannot be performed in the package, and requires additional probe pads on the die. In addition, it can require extra package pins for in-package high-current trims. Like zener zapping, link trimming is discrete. Resolution improvements require additional structures, increasing area and cost.

PROCESS	TRIMMED AT:	SPECIAL PROCESSING	RESOLUTION
DigiTrim™	Wafer or Final Test	None	Discrete
Laser Trim	Wafer	Thin Film Resistor	Continuous
Zener Zap Trim	Wafer	None	Discrete
Link Trim	Wafer	Thin Film or Poly Resistor	Discrete
EEPROM Trim	Wafer or Final Test	EEPROM	Discrete

Figure 1-35: Summary of ADI trim processes

EEPROM trimming utilizes special, non-volatile digital memory to store trim data. The stored data bits control adjustment currents through on-chip D/A converters. Memory cells and D/A converters scale with the process feature size. In-package trimming and even trimming in the customer's system is possible so that assembly-related shifts can be trimmed out. Testing at the wafer level is not required if yields are reasonable. No special hardware is required for the trimming beyond the normal mixed-signal tester system, although test software development may be more complicated. Since the trims can be overwritten, it is possible to periodically reprogram the system to account for long-term drifts or to modify system characteristics for new requirements. The number of reprogram cycles possible depends on the process, and is finite. Most EEPROM processes provide enough rewrite cycles to handle routine re-calibration.

This trim method does require special processing. Stored trim data can be lost under certain conditions, especially at high operating temperatures. At least one extra digital contact pad/package pin is required to input the trim data to the on-chip memory. This technique is only available on MOS-based processes due to the very thin oxide requirements. The biggest drawback is that the on-chip D/A converters are large—often larger than the amplifier circuits they are adjusting. For this reason, EEPROM trimming is mostly used for data converter or system-level products where the trim D/A converters represent a much smaller percentage of the overall die area.

Figure 1-35 above summarizes the key features of each ADI trim method. It can be seen from that all trim methods have their respective places in producing high performance linear integrated circuits.

■ OP AMP APPLICATIONS

Op Amp Process Technologies

The wide variety of op amp processes is shown in Figure 1-36 below. The early 1960's op amps used standard NPN-based bipolar processes. The PNP transistors of these processes were extremely slow and were used primarily for current sources and level shifting.

The ability to produce matching high speed PNP transistors on a bipolar process added great flexibility to op amp circuit designs. The first p-epi complementary bipolar (CB) process was introduced by ADI in the mid-1980s. The f_t s of the PNP and NPN transistors were approximately 700MHz and 900MHz, respectively, and had 30V breakdowns. Since its original introduction in 1985, several additional CB processes have been developed at ADI designed for higher speeds and lower breakdowns. For example, a current 5V CB process has 9GHz PNPs and 16GHz NPNs. These CB processes are used in today's precision op amps, as well as those requiring wide bandwidths.

- ◆ **BIPOLAR (NPN-BASED):** This is Where it All Started!!
- ◆ **COMPLEMENTARY BIPOLAR (CB):** Rail-to-Rail, Precision, High Speed
- ◆ **BIPOLAR + JFET (BiFET):** High Input Impedance, High Speed
- ◆ **COMPLEMENTARY BIPOLAR + JFET (CBFET):** High Input Impedance, Rail-to-Rail Output, High Speed
- ◆ **COMPLEMENTARY MOSFET (CMOS):** Low Cost Op Amps
(ADI DigiTrim™ Minimizes Offset Voltage and Drift in CMOS op amps)
- ◆ **BIPOLAR (NPN) + CMOS (BiCMOS):** Bipolar Input Stage adds Linearity, Low Power, Rail-to-Rail Output
- ◆ **COMPLEMENTARY BIPOLAR + CMOS (CBCMOS):** Rail-to-Rail Inputs, Rail-to-Rail Outputs, Good Linearity, Low Power, Higher Cost

Figure 1-36: Op amp process technology summary

The JFETs available on the Analog Devices' complementary bipolar processes allow high input impedance op amps to be designed suitable for such applications as photodiode or electrometer preamplifiers. These processes are sometimes designated as *CBFET*.

CMOS op amps, generally have higher offset voltages and offset voltage drift than trimmed bipolar or BiFET op amps, however the Analog Devices' DigiTrim™ process described above yields low offset voltage, while keeping costs low. Voltage noise for CMOS op amps tends to be larger, however, the input bias current is very low. They offer low power and cost (foundry CMOS processes are typically used).

The addition of bipolar or complementary devices to a CMOS process (BiMOS or CBCMOS) adds greater flexibility, better linearity, and low power as well as additional cost. The bipolar devices are typically used for the input stage to provide good gain and linearity, and CMOS devices for the rail-to-rail output stage.

In summary, there is no single IC process which is optimum for all op amps. Process selection and the resulting op amp design depends on the targeted applications and ultimately should be transparent to the customer.

REFERENCES: OP AMP STRUCTURES

1. "Video Op Amp," **Analog Dialogue**, Vol. 24, No. 3, pp. 19.
2. George Erdi, "Amplifier Techniques for Combining Low Noise, Precision, and High-Speed Performance," **IEEE Journal of Solid-State Circuits**, Vol. SC-16, December, 1981 pp. 653-661.
3. George Erdi, Tom Schwartz, Scott Bernardi, and Walt Jung, "Op Amps Tackle Noise-and for Once, Noise Loses," **Electronic Design**, December 12, 1980.
4. George Erdi, "A Precision Trim Technique for Monolithic Analog Circuits," **IEEE Journal of Solid-State Circuits**, Vol. SC-10, December, 1975 pp. 412-416.
5. Richard Wagner, "Laser-Trimming on the Wafer," **Analog Dialogue**, Vol. 9, No. 3, pp. 3-5.
6. Donn Soderquist, George Erdi, "The OP-07 Ultra-Low Offset Voltage Op Amp," **Precision Monolithics AN-13**, December, 1975.
7. Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.
8. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6 (available for download at <http://www.analog.com>).
9. Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8. (available for download at <http://www.analog.com>)..

ACKNOWLEDGEMENTS:

Helpful comments on various op amp schematics were received from op amp designers Derek Bowers, Jim Butler, JoAnn Close, and Scott Wurcer.

NOTES:

SECTION 1-4: OP AMP SPECIFICATIONS

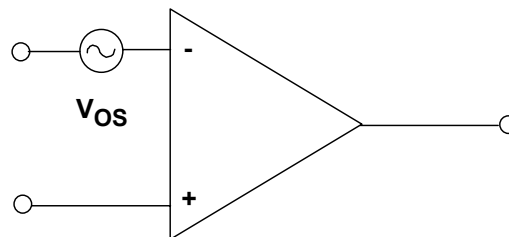
Walt Kester, Walt Jung, James Bryant

In this section, basic op amp specifications are discussed. The importance of any given specification depends of course upon the application. For instance offset voltage, offset voltage drift, and open-loop gain are very critical in precision sensor signal conditioning circuits, but not as important in high speed applications where bandwidth, slew rate, and distortion are the key specifications.

Most op amp specifications are largely topology independent. However, although voltage feedback and current feedback op amps have similar error terms and specifications, the application of each part warrants discussing some of the specifications separately. In the following discussions, this will be done where significant differences exist.

Input Offset Voltage, V_{OS}

Ideally, if both inputs of an op amp are at exactly the same voltage, then the output should be at zero volts. In practice, a small differential voltage must be applied to the inputs to force the output to zero. This is known as the *input offset voltage*, V_{OS} .



- ◆ **Offset Voltage:** The differential voltage which must be applied to the input of an op amp to produce zero output.
- ◆ **Ranges:**
 - Chopper Stabilized Op Amps: $<1\mu\text{V}$
 - General Purpose Precision Op Amps: $50\text{-}500\mu\text{V}$
 - Best Bipolar Op Amps: $10\text{-}25\mu\text{V}$
 - Best FET Op Amps: $100\text{-}1,000\mu\text{V}$
 - High Speed Op Amps: $100\text{-}2,000\mu\text{V}$
 - Untrimmed CMOS Op Amps: $5,000\text{-}50,000\mu\text{V}$
 - DigiTrim™ CMOS Op Amps: $<1,000\mu\text{V}$

Figure 1-37: Input offset voltage

Input offset voltage is modeled as a voltage source, V_{OS} , in series with the inverting input terminal of the op amp as shown in Figure 1-37 above. The corresponding output offset voltage (due to V_{OS}) is obtained by multiplying the input offset voltage by the DC noise gain of the circuit (see Fig. 1-3 and Eq. 1-2, again).

■ OP AMP APPLICATIONS

Chopper stabilized op amps have a V_{OS} which is less than $1\mu\text{V}$ (AD8551 series), and the best precision bipolar op amps (super-beta or bias stabilized) can have offsets as low as $25\mu\text{V}$ (OP177F). The very best trimmed FET types have about $100\mu\text{V}$ of offset (AD8610B), and untrimmed CMOS op amps can range from 5 to 50mV . However, the ADI DigiTrim™ CMOS op amps have offset voltages less than 1mV (AD8605). Generally speaking, "precision" op amps will have $V_{OS} < 0.5\text{mV}$, although some high speed amplifiers may be a little worse than this.

Measuring input offset voltages of a few microvolts requires that the test circuit does not introduce more error than the offset voltage itself. Figure 1-38 below shows a standard circuit for measuring offset voltage. The circuit amplifies the input offset voltage by the noise gain of 1001. The measurement is made at the amplifier output using an accurate digital voltmeter. The offset referred to the input (RTI) is calculated by dividing the output voltage by the noise gain. The small source resistance seen by the inputs results in negligible bias current contribution to the measured offset voltage. For example, 2nA bias current flowing through the 10Ω resistor produces a $0.02\mu\text{V}$ error referred to the input.

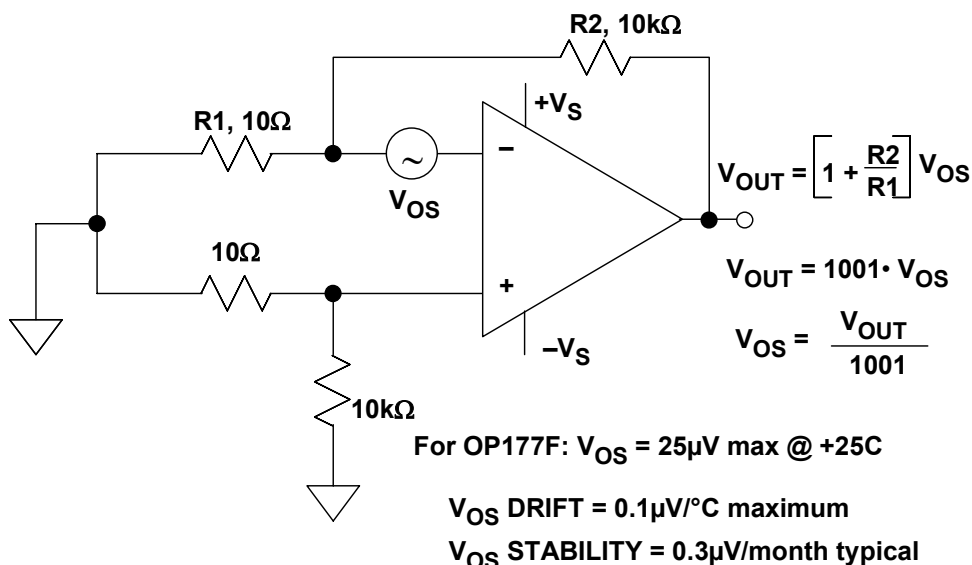


Figure 1-38: Measuring input offset voltage

As simple as this circuit looks, it can give inaccurate results when testing precision op amps, unless care is taken in implementation. The largest potential error source comes from parasitic thermocouple junctions, formed where two different metals are joined. This thermocouple voltage can range from $2\mu\text{V}/^\circ\text{C}$ to more than $40\mu\text{V}/^\circ\text{C}$. Note that in this circuit additional "dummy" resistors have been added to the non-inverting input, in order to exactly match/balance the thermocouple junctions in the inverting input path.

The accuracy of the measurement also depends on the mechanical layout of the components and exactly how they are placed on the PC board. Keep in mind that the two connections of a component such as a resistor create two equal, but opposite polarity thermoelectric voltages (assuming they are connected to the same metal, such as the copper trace on a PC board). These will cancel each other, *assuming both are at exactly the same temperature*. Clean connections and short lead lengths help to minimize

temperature gradients and increase the accuracy of the measurement. Note— see the Chapter 7 discussions on this general topic for more detail.

In the test circuit, airflow should be minimal so that all the thermocouple junctions stabilize at the same temperature. In some cases, the circuit should be placed in a small closed container to eliminate the effects of external air currents. The circuit should be placed flat on a surface so that convection currents flow up and off the top of the board, not across the components, as would be the case if the board were mounted vertically.

Measuring the offset voltage shift over temperature is an even more demanding challenge. Placing the printed circuit board containing the amplifier being tested in a small box or plastic bag with foam insulation prevents the temperature chamber air current from causing thermal gradients across the parasitic thermocouples. If cold testing is required, a dry nitrogen purge is recommended. Localized temperature cycling of the amplifier itself using a Thermostream-type heater/cooler may be an alternative, however these units tend to generate quite a bit of airflow that can be troublesome.

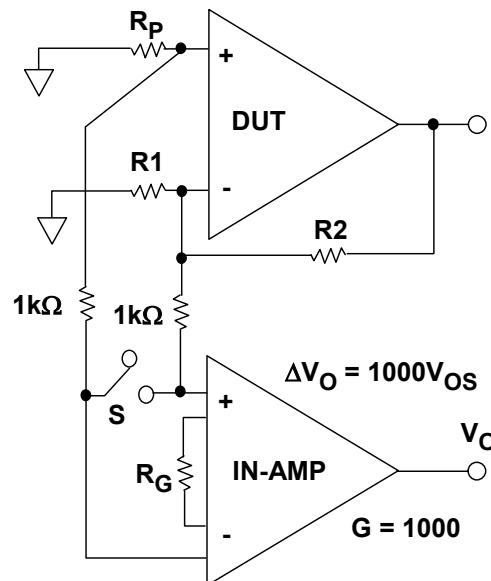


Figure 1-39: Alternate input offset voltage measurement using an in-amp

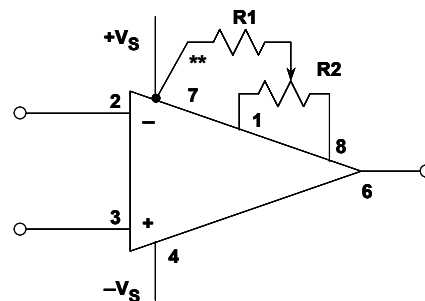
Generally, the test circuit of Fig. 1-38 can be made to work for many amplifiers. Low absolute values for the small resistors (such as 10Ω) will minimize bias current induced errors. An alternate V_{OS} measurement method is shown in Fig. 1-39, and is suitable for cases of high and/or unequal bias currents (as in the case of current feedback op amps).

In this measurement method, an in-amp is connected to the op amp input terminals through isolation resistors, and provides the gain for the measurement. The offset voltage of the in-amp (measured with S closed) must then be subtracted from the final V_{OS} measurement. Also, the circuit shown below in Figure 1-44 for measuring input bias currents can also be used to measure input offset voltage independent of bias currents.

■ OP AMP APPLICATIONS

Offset Adjustment (Internal Method)

Many single op amps have pins available for optional offset null. To make use of this feature, two pins are joined by a potentiometer, and the wiper goes to one of the supplies through a resistor, as shown generally in Figure 1-40 below. Note that if the wiper is accidentally connected to the wrong supply, the op amp will probably be destroyed— this is a common problem, when one op amp type is replaced by another. The range of offset adjustment in a well-designed op amp is no more than two or three times the maximum V_{OS} of the lowest grade device, in order to minimize sensitivity. Nevertheless, the voltage gain of an op amp at its offset adjustment pins may actually be greater than the gain at its signal inputs! It is therefore very important to keep these pins noise-free. Note that it is *never* advisable to use long leads from an op amp to a remote nulling potentiometer.



- ◆ ** Wiper connection may be to either $+V_S$ or $-V_S$ depending on op amp
- ◆ R values depend on op amp. Consult data sheet
- ◆ Use to null out input offset voltage, not system offsets!
- ◆ There may be high gain from offset pins to output — Keep them quiet!
- ◆ Nulling offset causes increase in offset temperature coefficient, approximately $4\mu\text{V}/^\circ\text{C}$ for 1mV offset null for FET inputs

Figure 1-40: Offset adjustment pins

As was mentioned above, the offset drift of an op amp with temperature will vary with the setting of its offset adjustment. The internal adjustment terminals should therefore be used only to adjust the op amp's own offset, *not to correct any system offset errors*, since doing so would be at the expense of increased temperature drift. The drift penalty for a FET input op amp is in the order of $4\mu\text{V}/^\circ\text{C}$ for each millivolt of nulled offset voltage. It is generally better to control offset voltage by proper device/grade selection.

Offset Adjustment (External Methods)

If an op amp doesn't have offset adjustment pins (popular duals and all quads do not), and it is still necessary to adjust the amplifier and system offsets, an external method can be used. This method is also most useful if the offset adjustment is to be done with a system programmable voltage, such as a DAC.

With an inverting op amp configuration, injecting current into the inverting input is the simplest method, as shown in Figure 1-41A (opposite). The disadvantage of this method is that there is some increase in noise gain possible, due to the parallel path of R_3 and the

potentiometer resistance. The resulting increase in noise gain may be reduced by making $\pm V_R$ large enough so that the R_3 value is much greater than $R_1 \parallel R_2$. Note that if the power supplies are stable and noise-free, they can be used as $\pm V_R$.

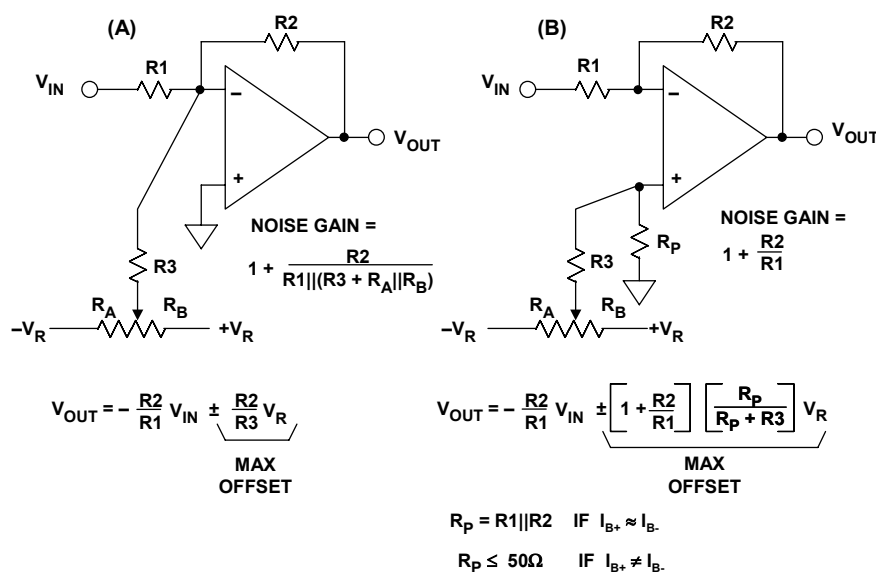


Figure 1-41: Inverting op amp external offset trim methods

Fig. 1-41B shows how to implement offset trim by injecting a small offset voltage into the non-inverting input. This circuit is preferred over 1-41A, as it results in no noise gain increase (but it requires adding R_P). If the op amp has matched input bias currents, then R_P should equal $R_1 \parallel R_2$ (to minimize the added offset voltage). Otherwise, R_P should be less than 50Ω . For higher values, it may be advisable to bypass R_P at high frequencies.

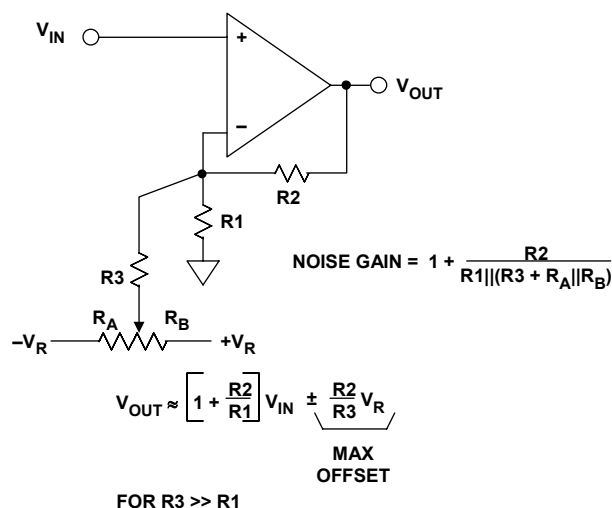


Figure 1-42: Non-inverting op amp external offset trim methods

The circuit shown in Figure 1-42 above can be used to inject a small offset voltage when using an op amp in the non-inverting mode. This circuit works well for small offsets, where R_3 can be made much greater than R_1 . Note that otherwise, the signal gain might be affected as the offset potentiometer is adjusted. The gain may be stabilized, however, if R_3 is connected to a fixed low impedance reference voltage sources, $\pm V_R$.

■ OP AMP APPLICATIONS

Input Offset Voltage Drift and Aging Effects

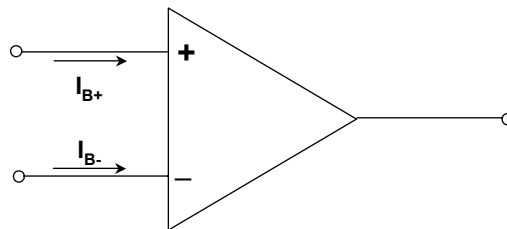
Input offset voltage varies with temperature, and its temperature coefficient is known as TCV_{OS} , or more commonly, *drift*. As we have mentioned, offset drift is affected by offset adjustments to the op amp, but when it has been minimized, it may be as low as $0.1\mu\text{V}/^\circ\text{C}$ (typical value for OP177F). More typical drift values for a range of general purpose precision op amps lie in the range $1\text{--}10\mu\text{V}/^\circ\text{C}$. Most op amps have a specified value of TCV_{OS} , but some, instead, have a second value of maximum V_{OS} that is guaranteed over the operating temperature range. Such a specification is less useful, because there is no guarantee that TCV_{OS} is constant or monotonic.

The offset voltage also changes as time passes, or *ages*. Aging is generally specified in $\mu\text{V}/\text{month}$ or $\mu\text{V}/1000$ hours, but this can be misleading. Since aging is a "drunkard's walk" phenomenon it is proportional to the *square root* of the elapsed time. An aging rate of $1\mu\text{V}/1000$ hour therefore becomes about $3\mu\text{V}/\text{year}$ (not $9\mu\text{V}/\text{year}$).

Long-term stability of the OP177F is approximately $0.3\mu\text{V}/\text{month}$. This refers to a time period *after* the first 30 days of operation. Excluding the initial hour of operation, changes in the offset voltage of these devices during the first 30 days of operation are typically less than $2\mu\text{V}$.

Input Bias Current, I_B

Ideally, no current flows into the input terminals of an op amp. In practice, there is always two *input bias currents*, I_{B+} and I_{B-} . (see Figure 1-43 below).



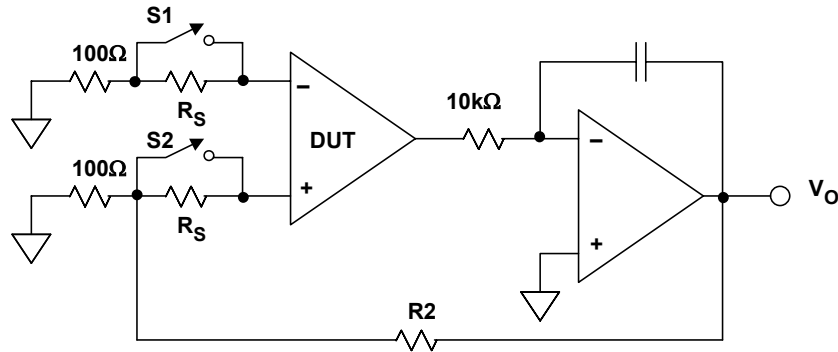
- ◆ A very variable parameter!
- ◆ I_B can vary from 60 fA (1 electron every $3\text{ }\mu\text{s}$) to many μA , depending on the device.
- ◆ Some structures have well-matched I_B , others do not.
- ◆ Some structures' I_B varies little with temperature, but a FET op amp's I_B doubles with every 10°C rise in temperature.
- ◆ Some structures have I_B which may flow in either direction.

Figure 1-43: Op amp input bias current

Values of I_B range from 60 fA (about one electron every three microseconds) in the AD549 electrometer, to tens of microamperes in some high speed op amps. Op amps with simple input structures using BJT or FET long-tailed pair have bias currents that flow in one direction. More complex input structures (bias-compensated and current feedback op amps) may have bias currents that are the difference between two or more internal current sources, and may flow in either direction.

Bias current is a problem to the op amp user because it flows in external impedances and produces voltages, which add to system errors. Consider a non-inverting unity gain buffer driven from a source impedance of $1\text{M}\Omega$. If I_B is 10nA , it will introduce an additional 10mV of error. This degree of error is not trivial in any system.

Or, if the designer simply forgets about I_B and uses capacitive coupling, the circuit won't work— at all! Or, if I_B is low enough, it may work momentarily while the capacitor charges, giving even more misleading results. The moral here is not to neglect the effects of I_B , in any op amp circuit. The same admonition goes for in-amp circuits.



$R_S \gg 100\Omega$ (100kΩ TO 1GΩ)

S1 CLOSED TO TEST I_{B+}

S2 CLOSED TO TEST I_{B-}

BOTH CLOSED TO TEST V_{OS}

BOTH OPEN TO TEST I_{OS}

$$V_O = \left[1 + \frac{R_2}{100} \right] V_{OS} + \left[1 + \frac{R_2}{100} \right] I_{B+} R_S - \left[1 + \frac{R_2}{100} \right] I_{B-} R_S$$

Figure 1-44: Measuring input bias current

Input bias current (or input offset voltage) may be measured using the test circuit of Figure 1-44 above. To measure I_B , a large resistance, R_S , is inserted in series with the input under test, creating an apparent additional offset voltage equal to $I_B \times R_S$. If the actual V_{OS} has previously been measured and recorded, the change in apparent V_{OS} due to the change in R_S can be determined, and I_B is then easily computed. This yields values for I_{B+} and I_{B-} . The rated value of I_B is the average of the two currents, or $I_B = (I_{B+} + I_{B-})/2$.

The *input offset current*, I_{OS} , may also be calculated, by taking the difference between I_{B-} and I_{B+} , or $I_{OS} = I_{B+} - I_{B-}$. Typical useful R_S values vary from $100\text{k}\Omega$ for bipolar op amps to $1000\text{M}\Omega$ for some FET input devices.

Note also that I_{OS} is only meaningful where the two individual bias currents are fundamentally reasonably well-matched, to begin with. This is true for most VFB op amps. However, it wouldn't for example be meaningful to speak of I_{OS} for a CFB op amp, as the currents are radically un-matched.

■ OP AMP APPLICATIONS

Extremely low input bias currents must be measured by integration techniques. The bias current in question is used to charge a capacitor, and the rate of voltage change is measured. If the capacitor and general circuit leakage is negligible (this is very difficult for currents under 10fA), the current may be calculated directly from the rate of change of the output of the test circuit. Figure 1-45 below illustrates the general concept. With one switch open and the opposite closed, either I_{B+} or I_{B-} is measured.

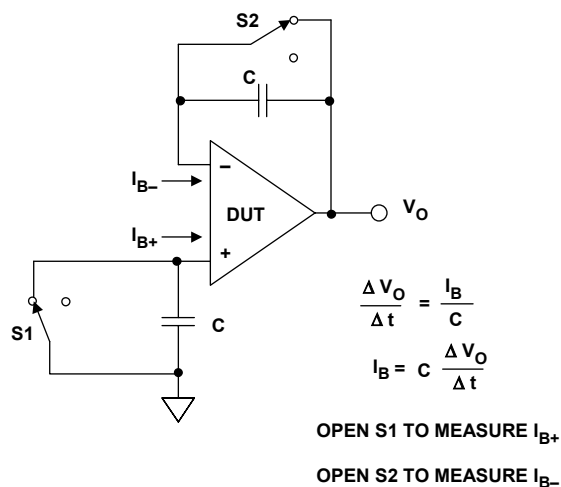


Figure 1-45: Measuring very low bias currents

It should be obvious that only a premium capacitor dielectric can be used for C, for example Teflon or polypropylene types.

Canceling the Effects of Bias Current (External to the Op Amp)

When the bias currents of an op amp are well matched (the case with simple bipolar op amps, but *not* internally bias compensated ones, as noted previously), a bias compensation resistor, R_3 , ($R_3 = R_1 \parallel R_2$) introduces a voltage drop in the non-inverting input to match and thus compensate the drop in the parallel combination of R_1 and R_2 in the inverting input. This minimizes additional offset voltage error, as in Figure 1-46.

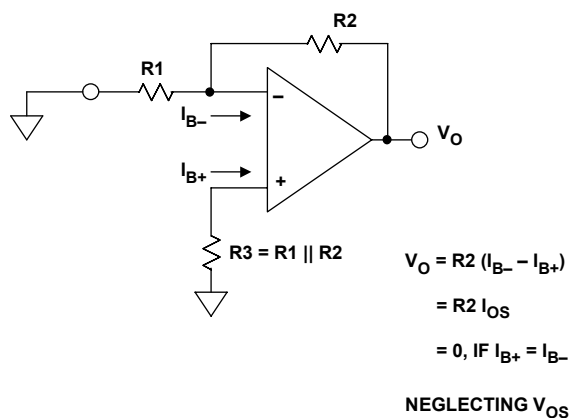


Figure 1-46: Canceling the effects of input bias current within an application

Note that if R_3 is more than 1k Ω or so, it should be bypassed with a capacitor to prevent noise pickup. Also note that this form of bias cancellation is useless where bias currents are not well-matched, and will, in fact, make matters worse.

Calculating Total Output Offset Error Due to I_B and V_{OS}

The equations shown in Figure 1-47 below are useful in referring all the offset voltage and induced offset voltage from bias current errors to the either the input (RTI) or the output (RTO) of the op amp. The choice of RTI or RTO is a matter of preference.

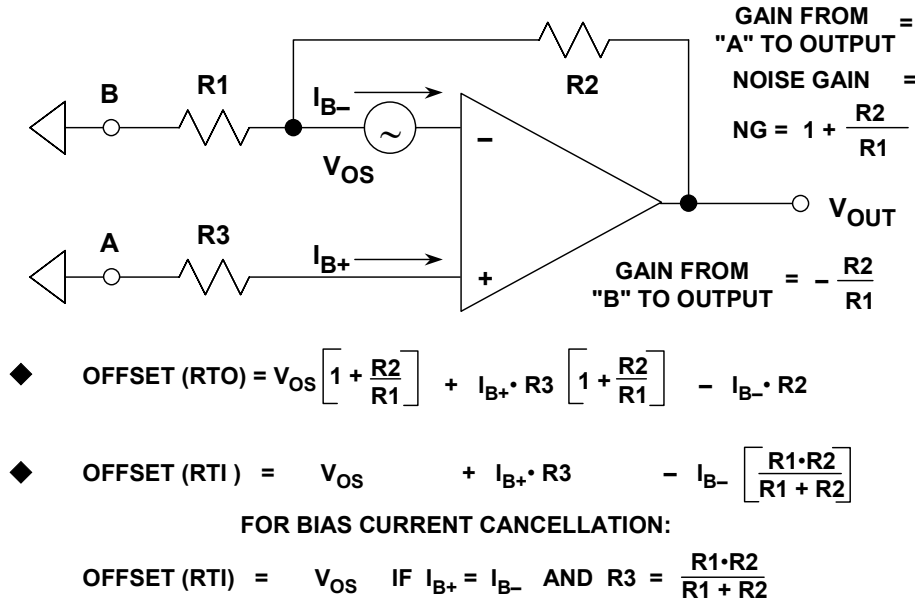


Figure 1-47: Op amp total offset voltage model

The RTI value is useful in comparing the cumulative op amp offset error to the input signal. The RTO value is more useful if the op amp drives additional circuitry, to compare the net errors with that of the next stage.

In any case, the RTO value is simply obtained by multiplying the RTI value by the stage noise gain, which is $1 + R2/R1$.

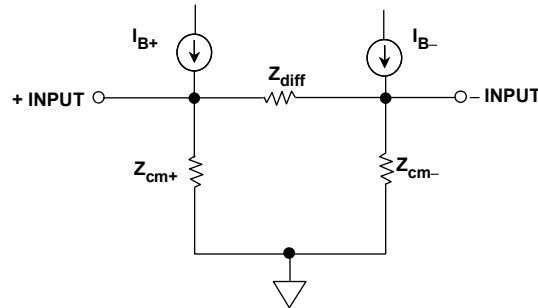
Before departing the topic of offset errors, some simple rules towards minimization might bear repetition:

- Keep input/feedback resistance values low, to minimize offset voltage due to bias current effects.
- Use a bias compensation resistance with VFB op amps *not* using internal bias compensation. Bypass this resistance, for lowest noise pickup.
- If a VFB op amp *does* use internal bias current compensation, *don't* use the compensation resistance.
- When necessary, use *external* offset trim networks, for lowest induced drift.
- Select an appropriate precision op amp specified for low offset and drift, as opposed to trimming.
- For high performance, low drift circuitry, watch out for thermocouple effects and used balanced, low thermal error layouts.

■ OP AMP APPLICATIONS

Input Impedance

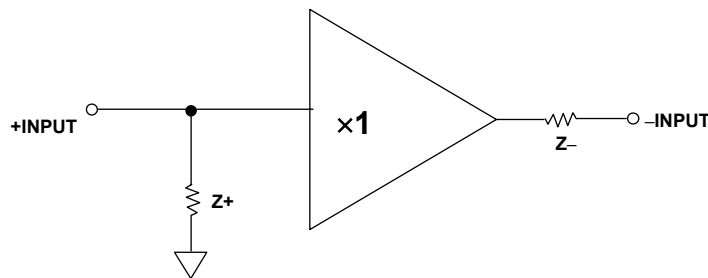
VFB op amps normally have both differential and common-mode input impedances specified. Current feedback op amps normally specify the impedance to ground at each input. Different models may be used for different voltage feedback op amps, but in the absence of other information, it is usually safe to use the model in Figure 1-48 below. In this model the bias currents flow into the inputs from infinite impedance current sources.



- ◆ Z_{cm+} and Z_{cm-} are the common-mode input impedance. The figure on the data sheet is for one, not both, but they are approximately equal. Z_{diff} is the differential input impedance.
- ◆ They are high resistance ($10^5 - 10^{12}\Omega$) in parallel with a small shunt capacitance (sometimes as high as 25pF).
- ◆ In most practical circuits, Z_{cm-} is swamped by negative feedback.

Figure 1-48: Input impedance (voltage feedback op amp)

The common-mode input impedance data sheet specification (Z_{cm+} and Z_{cm-}) is the impedance from either input to ground (NOT from both to ground). The differential input impedance (Z_{diff}) is the impedance between the two inputs. These impedances are usually resistive and high (10^5 - $10^{12}\Omega$) with some shunt capacitance (generally a few pF, sometimes 20-25 pF). In most op amp circuits, the inverting input impedance is reduced to a very low value by negative feedback, and only Z_{cm+} and Z_{diff} are of importance.



- ◆ $Z+$ is high resistance ($10^5 - 10^9\Omega$) with little shunt capacitance.
- ◆ $Z-$ is low and may be reactive (L or C). The resistive component is 10-100 Ω .

Figure 1-49: Input impedance (current feedback op amp)

A current feedback op amp is even more simple, as shown in Figure 1-49 above. $Z+$ is resistive, generally with some shunt capacitance, and high (10^5 - $10^9\Omega$) while $Z-$ is reactive (L or C, depending on the device) but has a resistive component of 10-100 Ω , varying from type to type.

Manipulating Op Amp Noise Gain and Signal Gain

Consider an op amp and two resistors, R1 and R2, arranged as shown in the series of figures of Figure 1-50 below. Note that R1 and R2 need not be resistors; they could also be complex impedances, Z1 and Z2.

If we ground R1 and apply a signal to the non-inverting input, we see a signal gain of $1 + R2/R1$, as in Fig. 1-50A. If we ground the non-inverting input and apply the signal to R1, we see a signal gain of $-R2/R1$, as in Fig. 1-50B. In both cases, the voltage noise of the op amp itself (as well as the input offset voltage) sees a gain of $1 + R2/R1$, i.e., the *noise gain* of the op amp, as discussed earlier in this chapter.

This discussion is aimed at making the point that a stage's noise gain and signal gain *need not necessarily be equal*. Some times it can be to the user's advantage to manipulate them, so as to be somewhat independent of one another.

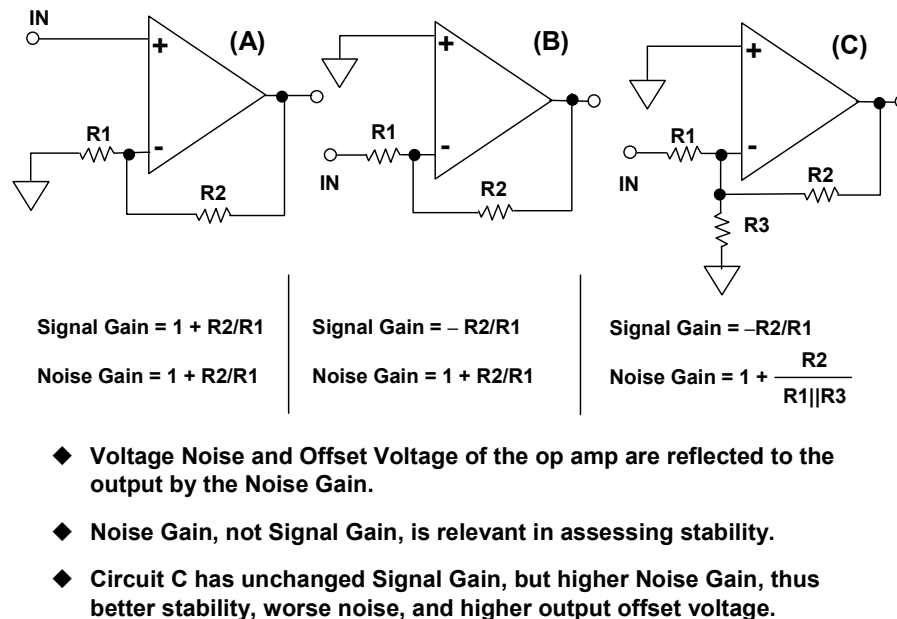


Figure 1-50: Manipulating op amp noise gain and signal gain

But, importantly, *it is the noise gain that is relevant in assessing stability*. It is sometimes possible to alter the noise gain, while leaving signal gain unaffected. When this is done, a marginally stable op amp stage can sometimes be made stable, with the same signal gain.

For example, consider the inverting amplifier of Fig. 1-50B. If we add a third resistor to Fig. 1-50B, it becomes Fig. 1-50C. This dummy resistor R3, from the inverting input to ground, increases the noise gain to $1 + R2/(R1 \parallel R3)$. But, note the signal gain is unaffected; that is it is still $-R2/R1$.

This provides a means of stabilizing an unstable inverting amplifier— at a cost of worse signal-to-noise ratio, less loop gain, and increased sensitivity to input offset voltage. Nevertheless, it is still a sometimes useful trick.

■ OP AMP APPLICATIONS

Open-Loop Gain And Open-Loop Gain Nonlinearity

Open-loop voltage gain, usually called A_{VOL} (sometimes simply A_V), for most VFB op amps is quite high. Common values are 100,000 to 1,000,000, and 10 or 100 times these figures for high precision parts. Some fast op amps have appreciably lower open-loop gain, but gains of less than a few thousand are unsatisfactory for high accuracy use. Note also that open-loop gain isn't highly stable with temperature, and can vary quite widely from device to device of the same type, so it is important that it be reasonably high.

Since a voltage feedback op amp operates as voltage in/voltage out, its open-loop gain is a dimensionless ratio, so no unit is necessary. However, data sheets sometimes express gain in V/mV or V/ μ V instead of V/V, for the convenience of using smaller numbers. Or, voltage gain can also be expressed in dB terms, as gain in dB = $20 \times \log A_{VOL}$. Thus an open-loop gain of 1V/ μ V is equivalent to 120dB, and so on.

CFB op amps have a current input and a voltage output, so their open-loop *transimpedance gain* is expressed in volts per ampere or ohms (or k Ω or M Ω). Values usually lie between hundreds of k Ω and tens of M Ω .

From basic feedback theory, it is understood that in order to maintain accuracy, a precision amplifier's DC open-loop gain, A_{VOL} , should be high. This can be seen by examining the closed-loop gain equation, including errors due to finite gain. The expression for closed loop gain with a finite gain error is:

$$G_{CL} = \frac{1}{\beta} \cdot \left[\frac{1}{1 + \frac{1}{A_{VOL}\beta}} \right] \quad \text{Eq. 1-15A}$$

Since noise gain is equal to $1/\beta$, there are alternate forms of this expression. Combining the two right side terms and using the NG expression, an alternate one is:

$$G_{CL} = \frac{NG}{1 + \frac{NG}{A_{VOL}}} \quad \text{Eq. 1-15B}$$

Equations 1-15A and 1-15B are equivalent, and either can be used. As previously discussed, noise gain (NG) is simply the gain seen by a small voltage source in series with the op amp input, and is also the ideal amplifier signal gain in the non-inverting mode. If A_{VOL} in equations 1-15A and 1-15B is infinite, the closed-loop gain becomes exactly equal to the noise gain, $1/\beta$.

However, for $NG \ll A_{VOL}$ and finite A_{VOL} , there is a closed-loop gain error estimation:

$$\text{Closed loop error(\%)} \approx \frac{NG}{A_{VOL}} \cdot 100 \quad \text{Eq. 1-16}$$

Note that the expression of Eq. 1-16 is equivalent to the earlier mentioned Eq. 1-11, when $1/\beta$ is substituted for NG. Again, either form can be used, at the user's discretion.

Notice from Eq. 1-16 that the percent gain error is directly proportional to the noise gain, therefore the effects of finite A_{VOL} are less for low gain. Some examples illustrate key points about these gain relationships.

In Figure 1-51 below, the first example for a NG of 1000 shows that for an open-loop gain of 2 million, the closed-loop gain error is about 0.05%. Note that if the open-loop gain stays constant over temperature and for various output loads and voltages, the 0.05% gain error can easily be calibrated out of the measurement, and then there is then no overall system gain error. If, however, the open-loop gain *changes*, the resulting closed-loop gain will also change. This introduces a *gain uncertainty*. In the second example, A_{VOL} drops to 300,000, which produces a gain error of 0.33%. This situation introduces a *gain uncertainty* of 0.28% in the closed-loop gain. In most applications, when using a good amplifier, the gain resistors of the circuit will be the largest source of absolute gain error, but it should be noted that gain uncertainty cannot be removed by calibration.

◆ "IDEAL" CLOSED LOOP GAIN = $1/\beta$ = NOISE GAIN (NG)

◆ ACTUAL CLOSED LOOP GAIN = $\frac{1}{\beta} \cdot \frac{1}{1 + \frac{1}{A_{VOL}\beta}} = \frac{NG}{1 + \frac{NG}{A_{VOL}}}$

◆ CLOSED LOOP GAIN ERROR (%) $\approx \frac{NG}{A_{VOL}} \times 100 \quad (NG \ll A_{VOL})$

◆ Ex. 1: Assume $A_{VOL} = 2,000,000$, NG = 1,000
% GAIN ERROR $\approx 0.05\%$

◆ Ex. 2: Assume A_{VOL} Drops to 300,000
% GAIN ERROR $\approx 0.33\%$

◆ CLOSED LOOP GAIN UNCERTAINTY
 $= 0.33\% - 0.05\% = 0.28\%$

Figure 1-51: *Changes in open-loop gain cause closed-loop gain uncertainty*

Changes in the output voltage level and output loading are the most common causes of changes in the open-loop gain of op amps. A change in open-loop gain with signal level produces a *nonlinearity* in the closed-loop gain transfer function, which also cannot be removed during system calibration. Most op amps have fixed loads, so A_{VOL} changes with load are not generally important. However, the sensitivity of A_{VOL} to output signal level may increase for higher load currents.

The severity of this non-linearity varies widely from one device type to another, and generally isn't specified on the data sheet. The minimum A_{VOL} is always specified, and choosing an op amp with a high A_{VOL} will minimize the probability of gain nonlinearity errors. Gain nonlinearity can come from many sources, depending on the design of the op amp. One common source is thermal feedback (for example, from a hot output stage back to the input stage). If temperature shift is the sole cause of the nonlinearity error, it can be assumed that minimizing the output loading will help. To verify this, the nonlinearity is measured with no load, and then compared to the loaded condition.

► OP AMP APPLICATIONS

An oscilloscope X-Y display test circuit for measuring DC open-loop gain nonlinearity is shown in Figure 1-52 below. The same precautions previously discussed relating to the offset voltage test circuit must also be observed in this circuit. The amplifier is configured for a signal gain of -1 . The open-loop gain is defined as the change in output voltage divided by the change in the input offset voltage. However, for large values of A_{VOL} , the actual offset may change only a few microvolts over the entire output voltage swing. Therefore the divider consisting of the 10Ω resistor and R_G ($1M\Omega$) forces the node voltage V_Y to be :

$$V_Y = \left[1 + \frac{R_G}{10\Omega} \right] V_{OS} = 100,001 \bullet V_{OS}. \quad \text{Eq. 1-17}$$

The value of R_G is chosen to give measurable voltages at V_Y depending on the expected values of V_{OS} .

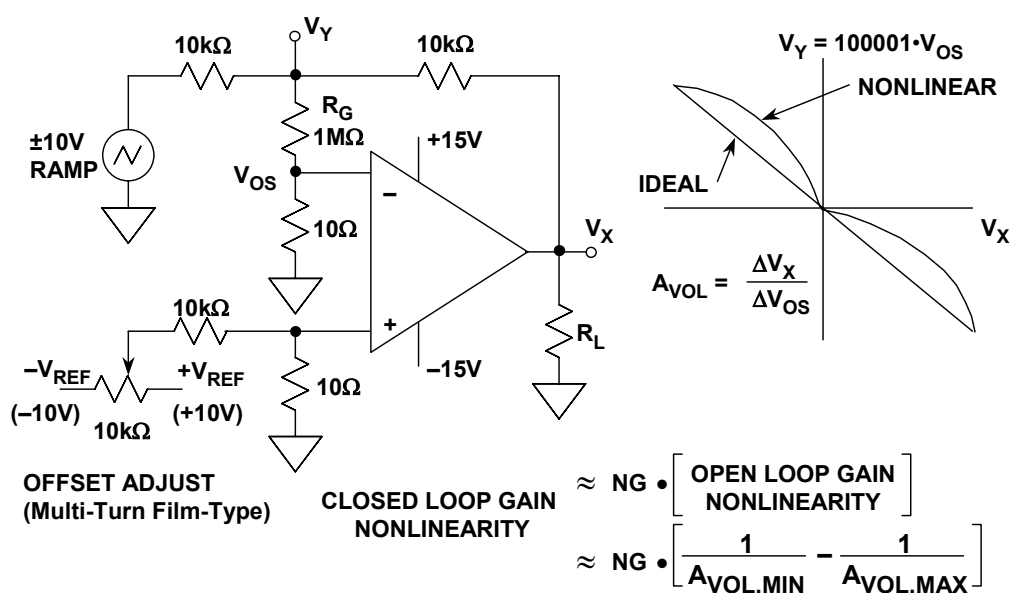


Figure 1-52: Circuit measures open-loop gain nonlinearity

The $\pm 10\text{V}$ ramp generator output is multiplied by the signal gain, -1 , and forces the op amp output voltage V_x to swing from $+10\text{V}$ to -10V . Because of the gain factor applied to the offset voltage, the offset adjust potentiometer is added to allow the initial output offset to be set to zero. The resistor values chosen will null an input offset voltage of up to $\pm 10\text{mV}$. Stable 10V voltage references such as the AD688 should be used at each end of the potentiometer to prevent output drift. *Also, note that the ramp generator frequency must be quite low, probably no more than a fraction of 1Hz because of the low corner frequency of the open-loop gain (0.1Hz for the OP177).*

The plot on the right-hand side of Fig. 1-52 shows V_Y plotted against V_X . If there is no gain nonlinearity the graph will have a constant slope, and A_{VOL} is calculated as follows:

$$A_{vol} = \frac{\Delta V_x}{\Delta V_{OS}} = \left[1 + \frac{R_G}{10\Omega} \right] \left[\frac{\Delta V_x}{\Delta V_y} \right] = 100,001 \bullet \left[\frac{\Delta V_x}{\Delta V_y} \right]. \quad \text{Eq. 1-18}$$

If there is nonlinearity, A_{VOL} will vary dynamically as the output signal changes. The approximate open-loop gain nonlinearity is calculated based on the maximum and minimum values of A_{VOL} over the output voltage range:

$$\text{Open -Loop Gain Nonlinearity} = \frac{1}{A_{VOL,MIN}} - \frac{1}{A_{VOL,MAX}} \quad \text{Eq. 1-19}$$

The closed-loop gain nonlinearity is obtained by multiplying the open-loop gain nonlinearity by the noise gain, NG:

$$\text{Closed -Loop Gain Nonlinearity} \approx NG \cdot \left[\frac{1}{A_{VOL,MIN}} - \frac{1}{A_{VOL,MAX}} \right] \quad \text{Eq. 1-20}$$

In an ideal case, the plot of V_{OS} versus V_X would have a constant slope, and the reciprocal of the slope is the open-loop gain, A_{VOL} . A horizontal line with zero slope would indicate infinite open-loop gain. In an actual op amp, the slope may change across the output range because of nonlinearity, thermal feedback, etc. In fact, the slope can even change sign.

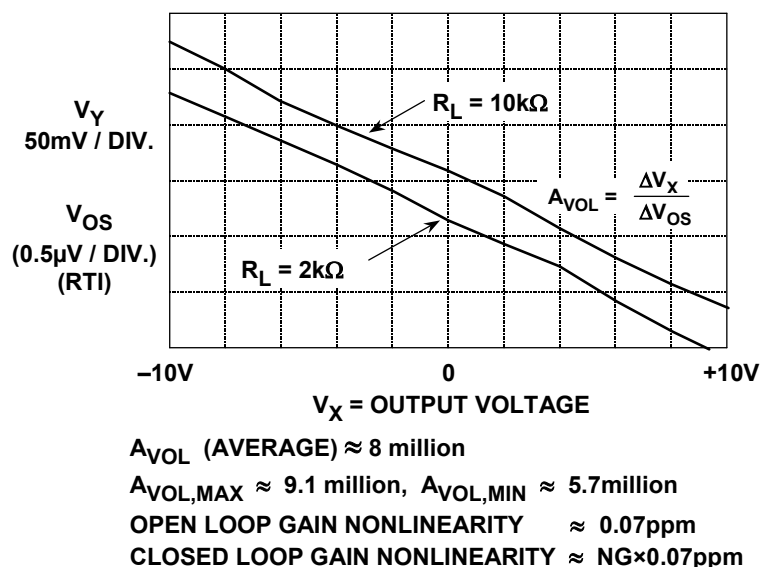


Figure 1-53: OP177 gain nonlinearity

Figure 1-53 above shows the V_Y (and V_{OS}) versus V_X plot for an OP177 precision op amp. The plot is shown for two different loads, $2k\Omega$ and $10k\Omega$. The reciprocal of the slope is calculated based on the end points, and the average A_{VOL} is about 8 million. The maximum and minimum values of A_{VOL} across the output voltage range are measured to be approximately 9.1 million, and 5.7 million, respectively. This corresponds to an open-loop gain nonlinearity of about 0.07ppm. Thus, for a noise gain of 100, the corresponding closed-loop gain nonlinearity is about 7ppm.

These nonlinearity measurements are, of course, most applicable to high precision DC circuits. But they are also applicable to wider bandwidth applications, such as audio. The X-Y display technique of Fig. 1-52 will easily show crossover distortion in a poorly designed op amp output stage, for example.

■ OP AMP APPLICATIONS

Op Amp Frequency Response

There are a number of issues to consider when discussing the frequency response of op amps. Some are relevant to both voltage and current feedback op amp types, some apply to one or the other, but not to both. Issues which vary with type are usually related to small-signal performance, while large-signal issues mostly apply to both.

A good working definition of "large-signal" is where the output voltage swing/frequency limit is set by the slew rate measured at the output stage, rather than the pole(s) of the small signal response. We shall therefore consider large signal parameters applying to both types of op amp before we consider those parameters where they differ.

Frequency Response— Slew Rate and Full-Power Bandwidth

The slew rate (SR) of an amplifier is the maximum rate of change of voltage at its output. It is expressed in V/s (or, more probably, V/ μ s). We have mentioned earlier why op amps might have different slew rates during positive and negative going transitions, but for this analysis we shall assume that good fast op amps have reasonably symmetrical slew rates.

If we consider a sine wave signal with a peak-to-peak amplitude of $2V_p$ and of a frequency f , the expression for the output voltage is:

$$V(t) = V_p \sin 2\pi ft. \quad \text{Eq. 1-21}$$

This sine wave signal has a maximum rate-of-change (slope) at the zero crossing. This maximum rate-of-change is:

$$\left. \frac{dv}{dt} \right|_{\max} = 2\pi f V_p. \quad \text{Eq. 1-22}$$

To reproduce this signal without distortion, an amplifier must be able to respond in terms of its output voltage at this rate (or faster). When an amplifier reaches its maximum output rate-of-change, or *slew rate*, it is said to be *slew limiting* (sometimes also called rate limiting). So, we can see that the maximum signal frequency at which slew limiting *does not* occur is directly proportional to the signal slope, and inversely proportional to the amplitude of the signal. This allows us to define the *full-power bandwidth* (FPBW) of an op amp, which is the maximum frequency at which slew limiting doesn't occur for rated voltage output. It is calculated by letting $2V_p$ in Eq. 1-22 equal the maximum peak-to-peak swing of the amplifier, dV/dt equal the amplifier slew rate, and solving for f :

$$\text{FPBW} = \text{Slew Rate}/2\pi V_p \quad \text{Eq. 1-23}$$

It is important to realize that both slew rate and full-power bandwidth can also depend somewhat on the power supply voltage being used, and the load the amplifier is driving (particularly if it is capacitive).

The key issues regarding slew rate and full-power bandwidth are summarized in Figure 1-54 below. As a point of reference, an op amp with a 1V peak output swing reproducing a 1MHz sine wave must have a minimum SR of 6.28V/μs.

◆ **Slew Rate = Maximum rate at which the output voltage of an op amp can change**

◆ **Ranges: A few volts/μs to several thousand volts/μs**

◆ **For a sinewave, $V_{out} = V_p \sin 2\pi ft$**

$$dV/dt = 2\pi f V_p \cos 2\pi ft$$

$$(dV/dt)_{max} = 2\pi f V_p$$

◆ **If $2V_p$ = full output span of op amp, then**

$$\text{Slew Rate} = (dV/dt)_{max} = 2\pi \cdot \text{FPBW} \cdot V_p$$

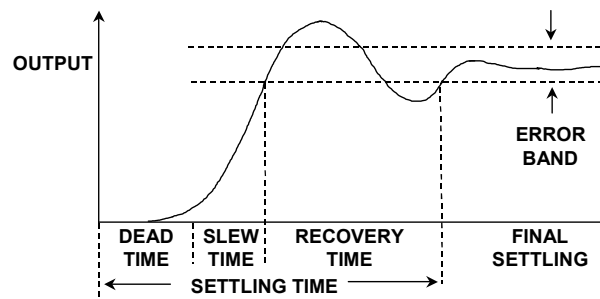
$$\text{FPBW} = \text{Slew Rate} / 2\pi V_p$$

Figure 1-54: Slew rate and full-power bandwidth

Realistically, for a practical circuit the designer would choose an op amp with a SR in excess of this figure, since real op amps show increasing distortion prior to reaching the slew limit point.

Frequency Response— Settling Time

The *settling time* of an amplifier is defined as the time it takes the output to respond to a step change of input and *come into, and remain within a defined error band*, as measured relative to the 50% point of the input pulse, as shown in Figure 1-55 below.



◆ **Error band is usually defined to be a percentage of the step 0.1%, 0.05%, 0.01%, etc.**

◆ **Settling time is non-linear; it may take 30 times as long to settle to 0.01% as to 0.1%.**

◆ **Manufacturers often choose an error band which makes the op amp look good.**

Figure 1-55: Settling time

Unlike a DAC device, there is no natural error band for an op amp (a DAC naturally has an error band of 1 LSB, or perhaps ±1LSB). So, one must be chosen and defined, along with other definitions, such as the step size (1V, 5V, 10V, etc.). What is chosen will

■ OP AMP APPLICATIONS

depend on the performance of the op amp, but since the value chosen will vary from device to device, comparisons are often difficult. This is true because settling is not linear, and many different time constants may be involved. Examples are early op amps using dielectrically isolated (DI) processes. These had very fast settling to 1% of full-scale, but they took almost forever to settle to 10-bits (0.1 %). Similarly, some very high precision op amps have thermal effects that cause settling to 0.001% or better to take tens of ms, although they will settle to 0.025% in a few μ s.

It should also be noted that thermal effects can cause significant differences between short-term settling time (generally measured in nanoseconds) and long-term settling time (generally measured in microseconds or milliseconds). In many AC applications, long-term settling time is not important; but if it is, it must be measured on a much different time scale than short-term settling time.

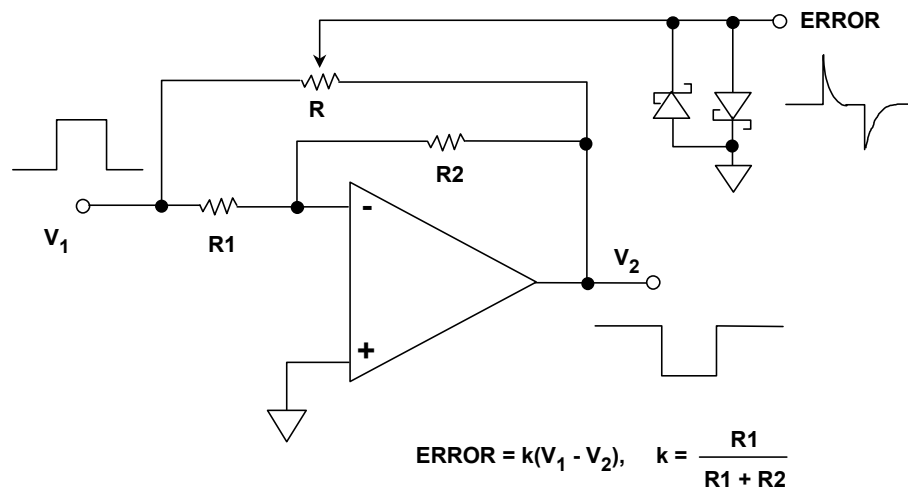


Figure 1-56: Measuring settling time using a "false summing node"

Measuring fast settling time to high accuracy is very difficult. Great care is required in order to generate fast, highly accurate, low noise, flat top pulses. Large amplitude step voltages will overdrive many oscilloscope front ends, when the input scaling is set for high sensitivity.

The example test setup shown in Figure 1-56 above is useful in making settling time measurements on op amps operating in the inverting mode. The signal at the "false summing node" represents the difference between the output and the input signal, multiplied by the constant k , i.e. the $ERROR$ signal.

There are many subtleties involved towards making this setup work reliably. The resistances should be low in value, to minimize parasitic time constants. The back-back Schottky diode clamps help prevent scope overdrive, and allow high sensitivity. If $R_1 = R_2$, then $k = 0.5$. Thus the error band at the $ERROR$ output will be 5mV for 0.1% settling with a 10V input step.

In some cases, a second (very fast) amplifier stage may be used after the false summing node, to increase the signal level. In any case, testing of settling time must be done with a test setup identical to that used by the op amp manufacturer, to ensure validity.

Many modern digitizing oscilloscopes are insensitive to input overdrive and can be used to measure the ERROR waveform directly—this must be verified for each oscilloscope by examining the operating manual carefully. Note that a direct measurement allows measurements of settling time in both the inverting and non-inverting modes. An example of the output step response to a flat pulse input for the AD8039 op amp is shown in Figure 1-57 below. Notice that the settling time to 0.1% is approximately 18ns.

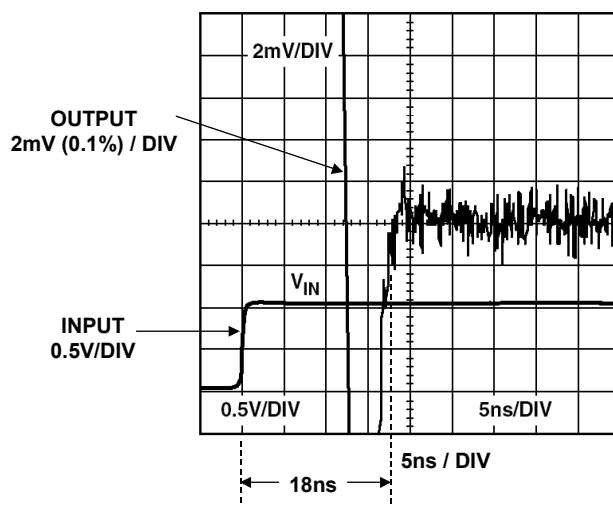


Figure 1-57: AD8039 $G = +2$ settling time measured directly

In making settling time measurements of this type, it is also imperative to use a pulse generator source capable of generating a pulse of sufficient flatness. In other words, if the op amp under test has a settling time of 20ns to 0.1%, the applied pulse should settle to better than 0.05% in less than 5ns.

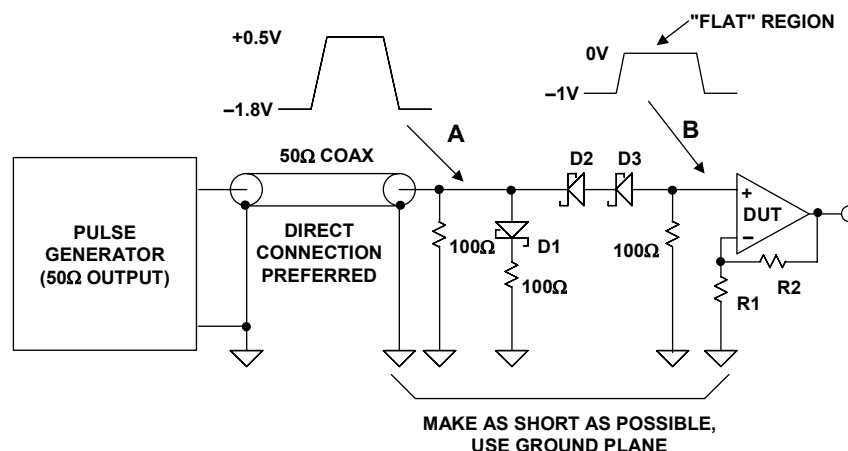


Figure 1-58: A simple flat pulse generator

This type of generator can be expensive, but a simple circuit as shown in Figure 1-58 can be used with a reasonably flat generator to ensure a flat pulse output.

The circuit of Fig. 1-58 works best if low capacitance Schottky diodes are used for D1-D2-D3, and the lead lengths on all the connections are minimized. A short length of 50Ω coax can be used to connect the pulse generator to the circuit, however best results are obtained if the test fixture is connected directly to the output of the generator. The pulse generator is adjusted to output a positive-going pulse at "A" which rises from

■ OP AMP APPLICATIONS

approximately -1.8V to $+0.5\text{V}$ in less than 5ns (assuming the settling time of the DUT is in the order of 20ns). Shorter rise times may generate ringing, and longer rise times can degrade the DUT settling time; therefore some optimization is required in the actual circuit to get best performance. When the pulse generator output "A" goes above 0V , D1 begins to conduct, and D2/D3 are reversed biased. The " 0V " region of the signal "B" at the input of the DUT is flat "by definition"—neglecting the leakage current and stray capacitance of the D2-D3 series combination. The D1 diode and its 100Ω resistor help maintain an approximate 50Ω termination during the time the pulse at "A" is positive.

Frequency Response— Voltage Feedback Op amps, Gain-Bandwidth Product

The open-loop frequency response of a voltage feedback op amp is shown in Figure 1-59 below. There are two possibilities: Fig. 1-59A shows the most common, where a high DC gain drops at 6dB/octave from quite a low frequency down to unity gain. This is a classic single pole response. By contrast, the amplifier in Fig. 1-59B has two poles in its response—gain drops at 6dB/octave for a while, and then drops at 12dB/octave . The amplifier in Fig. 1-59A is known as an *unconditionally stable* or *fully compensated* type and may be used with a noise gain of unity. This type of amplifier is stable with 100% feedback (including capacitance) from output to inverting input.

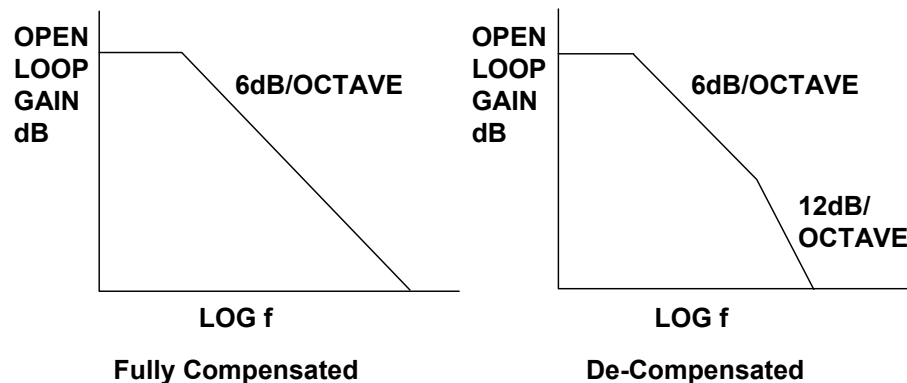


Figure 1-59: Frequency response of voltage feedback op amps

Compare this to the amplifier in Fig 1-59B. If this op amp is used with a noise gain that is lower than the gain at which the slope of the response increases from 6 to 12dB/octave , the phase shift in the feedback will be too great, and it will oscillate. Amplifiers of this type are characterized as "stable at gains $\geq X$ " where X is the gain at the frequency where the $6\text{dB}/12\text{dB}$ transition occurs. Note that here it is, of course, the *noise gain* that is being referenced. The gain level for stability might be between 2 and 25 , typically quoted behavior might be "gain-of-five-stable," etc. These *decompensated* op amps do have higher gain-bandwidth products than fully compensated amplifiers, all other things being equal. So, they are useful, despite the slightly greater complication of designing with them. But, unlike their fully compensated op amp relatives, a decompensated op amp can never be used with direct capacitive feedback from output to inverting input.

The 6dB/octave slope of the response of both types means that over the range of frequencies where this slope occurs, *the product of the closed-loop gain and the 3dB closed-loop bandwidth at that gain is a constant*—this is known as the *gain-bandwidth product* (GBW) and is a figure of merit for an amplifier.

For example, if an op amp has a GBW product of X MHz, then its closed-loop bandwidth at a noise gain of 1 will be X MHz, at a noise gain of 2 it will be X/2 MHz, and at a noise gain of Y it will be X/Y MHz (see Figure 1-60 below). Notice that the *closed-loop* bandwidth is the frequency at which the noise gain plateau intersects the open-loop gain.

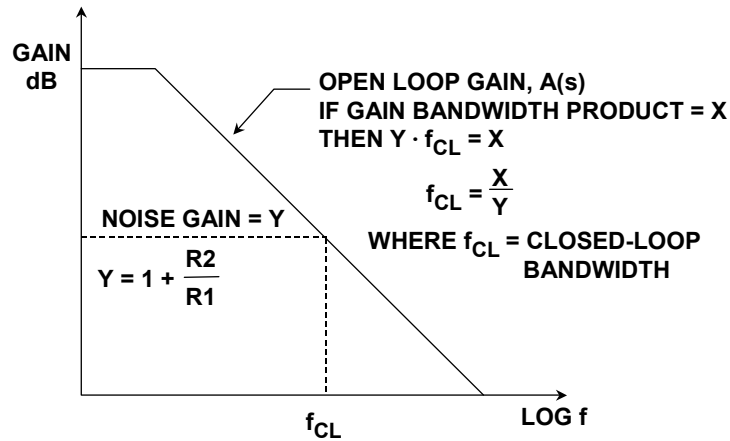


Figure 1-60: Gain-bandwidth product for voltage feedback op amps

In the above example, it was assumed that the feedback elements were resistive. This is not usually the case, especially when the op amp requires a feedback capacitor for stability.

Figure 1-61 below shows a typical example where there is capacitance, C1, on the inverting input of the op amp. This capacitance is the sum of the op amp internal capacitance, plus any external capacitance that may exist. This always-present capacitance introduces a pole in the noise gain transfer function.

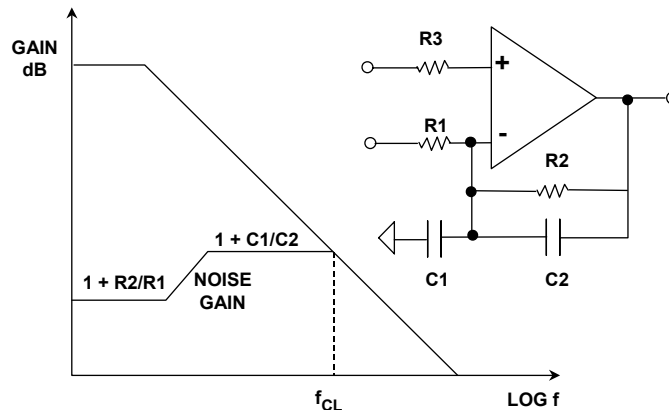


Figure 1-61: Bode plot showing noise gain for voltage feedback op amp with resistive and reactive feedback elements

The net slope of the noise gain curve and the open-loop gain curve, *at the point of intersection*, determines system stability. For unconditional stability, the noise gain must intersect the open-loop gain with a net slope of less than 12dB/octave (20dB per decade). Adding the feedback capacitor, C2, introduces a zero in the noise gain transfer function, which stabilizes the circuit. Notice that in Fig. 1-61 the closed-loop bandwidth, f_{cl} , is the frequency at which the noise gain intersects the open-loop gain.

■ OP AMP APPLICATIONS

The Bode plot of the noise gain is a very useful tool in analyzing op amp stability. Constructing the Bode plot is a relatively simple matter. Although it is outside the scope of this section to carry the discussion of noise gain and stability further, the reader is referred to Reference 1 for an excellent treatment of constructing and analyzing Bode plots. Second-order systems related to noise analysis are discussed later in this section.

Frequency Response— Current Feedback Op amps

Current feedback op amps do not behave in the same way as voltage feedback types. They are not stable with capacitive feedback, nor are they so with a short circuit from output to inverting input. With a CFB op amp, *there is generally an optimum feedback resistance for maximum bandwidth*. Note that the value of this resistance may vary with supply voltage— consult the device data sheet. If the feedback resistance is increased, the bandwidth is reduced. Conversely, if it is reduced, bandwidth increases, and the amplifier may become unstable.

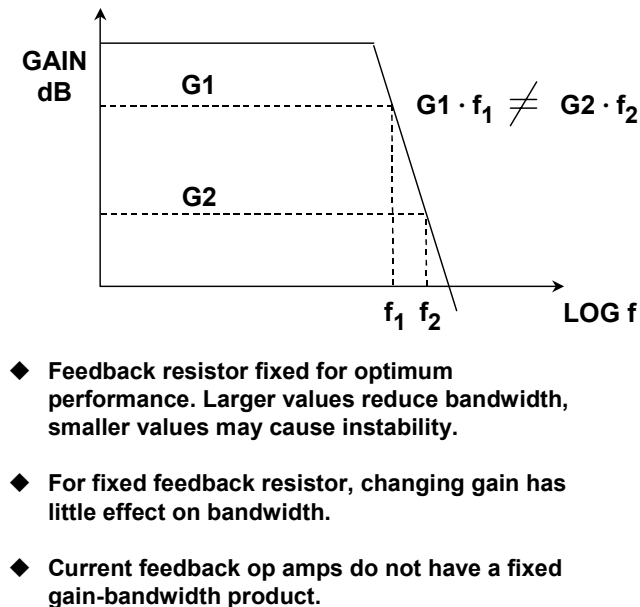


Figure 1-62: Frequency response for current feedback op amps

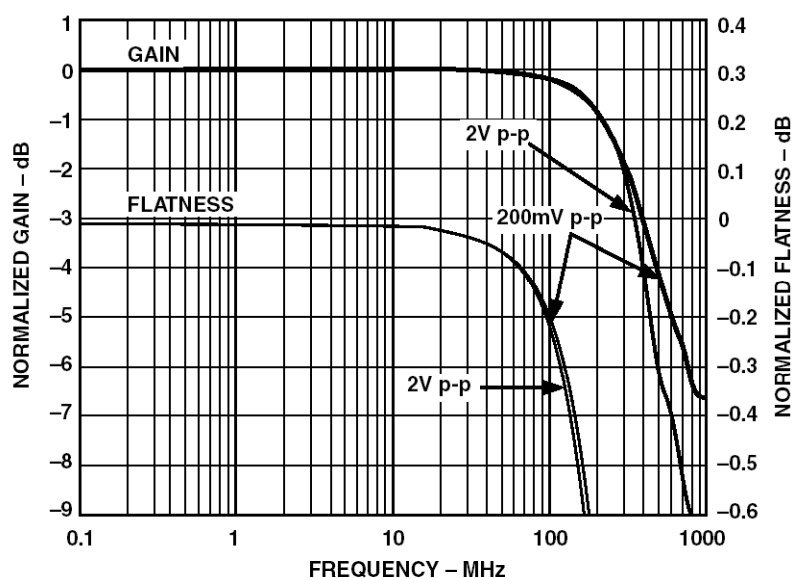
In a CFB op amp, for a given value of feedback resistance (R2), *the closed-loop bandwidth is largely unaffected by the noise gain*, as shown in Figure 1-62 above. Thus it is not correct to refer to gain-bandwidth product, for a CFB amplifier, because of the fact that it is not constant. Gain is manipulated in a CFB op amp application by choosing the correct feedback resistor for the device (R2), and then selecting the bottom resistor (R1) to yield the desired closed loop gain. The gain relationship of R2 and R1 is identical to the case of a VFB op amp (Fig. 1-14, again).

Typically, CFB op amp data sheets will provide a table of recommended resistor values, which provide maximum bandwidth for the device, over a range of both gain and supply voltage. It simplifies the design process considerably to use these tables.

Bandwidth Flatness

In demanding applications such as professional video, it is desirable to maintain a relatively flat bandwidth and linear phase up to some maximum specified frequency, and simply specifying the 3dB bandwidth isn't enough. In particular, it is customary to specify the *0.1dB bandwidth*, or *0.1dB bandwidth flatness*. This means there is no more than 0.1dB ripple up to a specified 0.1dB bandwidth frequency.

Video buffer amplifiers generally have both the 3dB and the 0.1dB bandwidth specified. Figure 1-63 below shows the frequency response of the AD8075 triple video buffer.



3dB BANDWIDTH $\approx$ 400MHz, 0.1dB BANDWIDTH $\approx$ 65MHz

Figure 1-63: 3dB and 0.1dB bandwidth for the AD8075, $G = 2$,
triple video buffer, $R_L = 150\Omega$

Note that the 3dB bandwidth is approximately 400MHz. This can be determined from the response labeled "GAIN" in the graph, and the corresponding gain scale is shown on the left-hand vertical axis (at a scaling of 1dB/division).

The response scale for "FLATNESS" is on the right-hand vertical axis, at a scaling of 0.1dB/division in this case. This allows the 0.1dB bandwidth to be determined, which is about 65MHz in this case. There is the general point to be noted here, and that is the major difference in the applicable bandwidth between the 3dB and 0.1dB criteria. It requires a 400MHz bandwidth amplifier (as conventionally measured) to provide the 65MHz 0.1dB flatness rating.

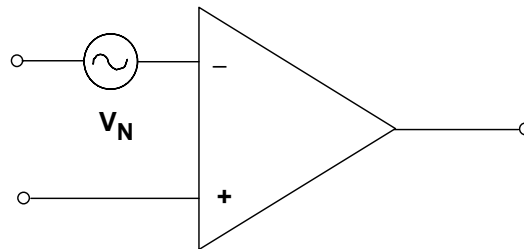
It should be noted that these specifications hold true when driving a 75Ω source and load terminated cable, which represents a resistive load of 150Ω . Any capacitive loading at the amplifier output will cause peaking in the frequency response, and must be avoided.

■ OP AMP APPLICATIONS

Operational Amplifier Noise

This section discusses the noise generated within op amps, not the external noise which they may pick up. External noise is also important, and is discussed in detail in Chapter 7, but in this section we are concerned solely with internal noise.

There are three noise sources in an op amp: a voltage noise which appears differentially across the two inputs, and two current noise sources, one in each input. The simple voltage noise op amp model is shown in Figure 1-64 below. The three noise sources are effectively uncorrelated (independent of each other). There is a slight correlation between the two noise currents, but it is too small to need consideration in practical noise analyses. In addition to these three internal noise sources, it is necessary to consider the Johnson noise of the external gain setting resistors that are used with the op amp.



- ◆ **Input Voltage Noise is bandwidth dependent and measured in $\text{nV}/\sqrt{\text{Hz}}$ (noise spectral density)**
- ◆ **Normal Ranges are $1\text{nV}/\sqrt{\text{Hz}}$ to $20\text{nV}/\sqrt{\text{Hz}}$**

Figure 1-64: Input voltage noise

All resistors have a Johnson noise of $\sqrt{4kTB}$, where k is Boltzmann's Constant ($1.38 \times 10^{-23} \text{J/K}$), T is the absolute temperature, B is the bandwidth, and R is the resistance. Note that this is an intrinsic property—it is not possible to obtain resistors that do not have Johnson noise. The simple model is shown in Figure 1-65 below.



- ◆ **ALL resistors have a voltage noise of $V_{NR} = \sqrt{4kTB}$**
- ◆ **T = Absolute Temperature = $T(^{\circ}\text{C}) + 273.15$**
- ◆ **B = Bandwidth (Hz)**
- ◆ **k = Boltzmann's Constant ($1.38 \times 10^{-23} \text{J/K}$)**
- ◆ **A 1000Ω resistor generates $4\text{nV} / \sqrt{\text{Hz}}$ @ 25°C**

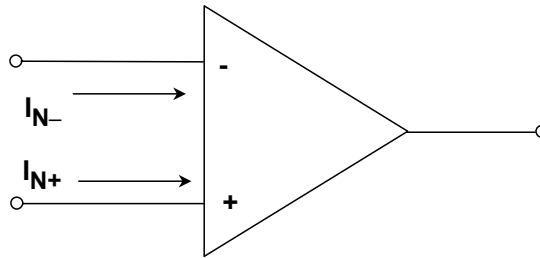
Figure 1-65: Johnson noise of resistors

Uncorrelated noise voltages add in a "root-sum-of-squares" manner; i.e., noise voltages V_1 , V_2 , V_3 give a summed result of $\sqrt{V_1^2 + V_2^2 + V_3^2}$. Noise powers, of course, add

normally. Thus, any noise voltage that is more than 3 to 5 times any of the others is dominant, and the others may generally be ignored. This simplifies noise assessment.

The voltage noise of different op amps may vary from under $1\text{nV}/\sqrt{\text{Hz}}$ to $20\text{nV}/\sqrt{\text{Hz}}$, or even more. Bipolar op amps tend to have lower voltage noise than JFET ones, although it is possible to make JFET op amps with low voltage noise (such as the AD743/AD745), at the cost of large input devices, and hence large input capacitance. Voltage noise is specified on the data sheet, and it isn't possible to predict it from other parameters.

Current noise can vary much more widely, dependent upon the input structure. It ranges from around $0.1\text{fA}/\sqrt{\text{Hz}}$ (in JFET electrometer op amps) to several $\text{pA}/\sqrt{\text{Hz}}$ (in high speed bipolar op amps). It isn't always specified on data sheets, but may be calculated in cases like simple BJT or JFETs, where all the bias current flows in the input junction, because in these cases it is simply the Schottky (or shot) noise of the bias current.



- ◆ **Normal Ranges: $0.1\text{fA}/\sqrt{\text{Hz}}$ to $10\text{pA}/\sqrt{\text{Hz}}$**
- ◆ **In Voltage Feedback op amps the current noise in the inverting and non-inverting inputs is uncorrelated (effectively) but roughly equal in magnitude.**
- ◆ **In simple BJT and JFET input stages, the current noise is the shot noise of the bias current and may be calculated from the bias current.**
- ◆ **In bias-compensated input stages and in current feedback op amps, the current noise cannot be calculated.**
- ◆ **The current noise in the two inputs of a current feedback op amp may be quite different. They may not even have the same $1/f$ corner.**

Figure 1-66: Input current noise

Shot noise spectral density is simply $\sqrt{(2I_{\text{BQ}})}/\sqrt{\text{Hz}}$, where I_{B} is the bias current (in amps) and q is the charge on an electron ($1.6 \times 10^{-19} \text{ C}$). It can't be calculated for bias-compensated or current feedback op amps, where the external bias current is the *difference* of two internal currents. A simple current noise model is shown in Figure 1-66.

Current noise is only important when it flows in an impedance, and thus generates a noise voltage. Maintaining relatively low impedances at the input of an op amp circuit contributes markedly to minimizing the effects of current noise (just as doing the same thing also aids in minimizing offset voltage)

It is logical therefore, that the optimum choice of a low noise op amp depends on the impedances around it. This will be illustrated with the aid of some impedance examples, immediately below.

■ OP AMP APPLICATIONS

Consider for example an OP27, an op amp with low voltage noise ($3\text{nV}/\sqrt{\text{Hz}}$), but quite high current noise ($1\text{pA}/\sqrt{\text{Hz}}$). With zero source impedance, the voltage noise will dominate as shown in Figure 1-67 below (left column). With a source resistance of $3\text{k}\Omega$ (center column), the current noise of $1\text{pA}/\sqrt{\text{Hz}}$ flowing in $3\text{k}\Omega$ will equal the voltage noise, but the Johnson noise of the $3\text{k}\Omega$ resistor is $7\text{nV}/\sqrt{\text{Hz}}$ and is dominant. With a source resistance of $300\text{k}\Omega$ (right column), the current noise portion increases $100\times$ to $300\text{nV}/\sqrt{\text{Hz}}$, voltage noise continues unchanged, and the Johnson noise (which is proportional to the resistance *square root*) increases tenfold. Current noise dominates.

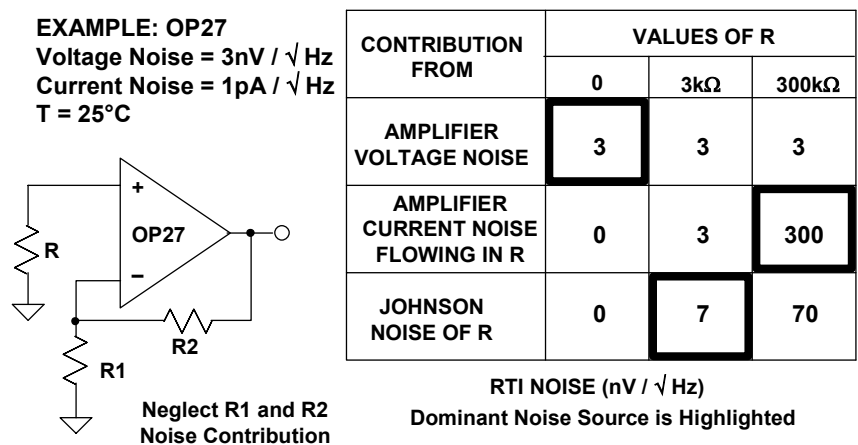


Figure 1-67: Different noise sources dominate at different source impedances

The above example shows that the choice of a low noise op amp depends on the source impedance of the signal, and at high impedances, current noise always dominates.

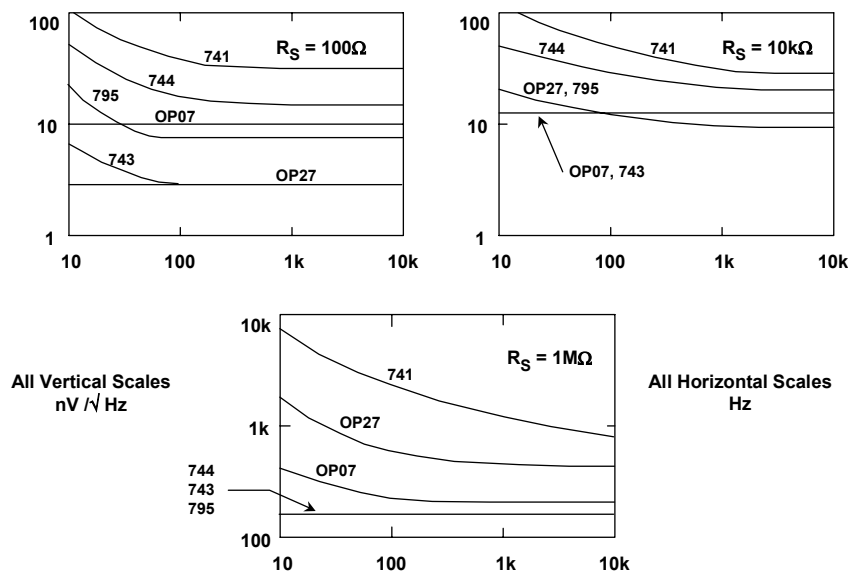
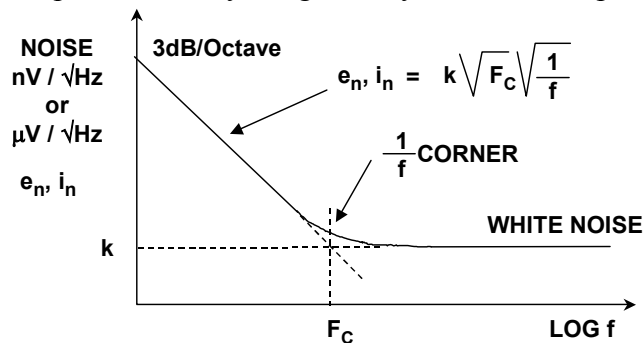


Figure 1-68: Different amplifiers are best at different source impedances

From Figure 1-68 above, it should be apparent that different amplifiers are best at different source impedances. For low impedance circuits, low voltage noise amplifiers such as the OP27 will be the obvious choice, since they are inexpensive, and their comparatively large current noise will not affect the application. At medium resistances, the Johnson noise of resistors is dominant, while at very high source resistance, we must choose an op amp with the smallest possible current noise, such as the AD549 or AD795.

Until recently, BiFET amplifiers tended to have comparatively high voltage noise (though very low current noise), and were thus more suitable for low noise applications in high rather than low impedance circuitry. The AD795, AD743, and AD745 have very low values of both voltage and current noise. The AD795 specifications at 10kHz are 10nV/ $\sqrt{\text{Hz}}$ and 0.6fA/ $\sqrt{\text{Hz}}$, and the AD743/AD745 specifications at 10kHz are 2.9nV/ $\sqrt{\text{Hz}}$ and 6.9fA/ $\sqrt{\text{Hz}}$. These make possible the design of low-noise amplifier circuits that have low noise over a wide range of source impedances.

The *noise figure* of an amplifier is the amount (in dB) by which the noise of the amplifier exceeds the noise of a perfect noise-free amplifier in the same environment. The concept is useful in RF and TV applications, where 50 Ω and 75 Ω transmission lines and terminations are ubiquitous, but is useless for an op amp that is used in a wide range of electronic environments. Noise figure related to communications applications is discussed in more detail in Chapter 6 (Section 6-4). Voltage noise spectral density and current noise spectral density are generally more useful specifications in most cases.



- ◆ **1/f Corner Frequency is a figure of merit for op amp noise performance (the lower the better)**
- ◆ **Typical Ranges: 2Hz to 2kHz**
- ◆ **Voltage Noise and Current Noise do not necessarily have the same 1/f corner frequency**

Figure 1-69: Frequency characteristic of op amp noise

So far, we have assumed that noise is *white* (i.e., its spectral density does not vary with frequency). This is true over most of an op amp's frequency range, but at low frequencies the noise spectral density rises at 3dB/octave, as shown in Figure 1-69 above. The power spectral density in this region is inversely proportional to frequency, and therefore the voltage noise spectral density is inversely proportional to the square root of the frequency. For this reason, this noise is commonly referred to as *1/f noise*. Note however, that some textbooks still use the older term *flicker noise*.

The frequency at which this noise starts to rise is known as the *1/f corner frequency* (F_C) and is a figure of merit—the lower it is, the better. The 1/f corner frequencies are not necessarily the same for the voltage noise and the current noise of a particular amplifier, and a current feedback op amp may have three 1/f corners: for its voltage noise, its inverting input current noise, and its non-inverting input current noise.

■ OP AMP APPLICATIONS

The general equation which describes the voltage or current noise spectral density in the 1/f region is

$$e_n, i_n = k\sqrt{F_C} \sqrt{\frac{1}{f}}, \quad \text{Eq. 1-24}$$

where k is the level of the "white" current or voltage noise level, and F_C is the 1/f corner frequency.

The best low frequency low noise amplifiers have corner frequencies in the range 1-10Hz, while JFET devices and more general purpose op amps have values in the range to 100Hz. Very fast amplifiers, however, may make compromises in processing to achieve high speed which result in quite poor 1/f corners of several hundred Hz or even 1-2kHz. This is generally unimportant in the wideband applications for which they were intended, but may affect their use at audio frequencies, particularly for equalized circuits.

Popcorn Noise

Popcorn noise is so-called because when played through an audio system, it sounds like cooking popcorn. It consists of random step changes of offset voltage that take place at random intervals in the 10+ millisecond timeframe. Such noise results from high levels of contamination and crystal lattice dislocation at the surface of the silicon chip, which in turn results from inappropriate processing techniques or poor quality raw materials.

When monolithic op amps were first introduced in the 1960s, popcorn noise was a dominant noise source. Today, however, the causes of popcorn noise are well understood, raw material purity is high, contamination is low, and production tests for it are reliable so that no op amp manufacturer should have any difficulty in shipping products that are substantially free of popcorn noise. For this reason, it is not even mentioned in most modern op amp textbooks.

RMS Noise Considerations

As was discussed above, noise spectral density is a function of frequency. In order to obtain the RMS noise, the noise spectral density curve must be integrated over the bandwidth of interest.

In the 1/f region, the RMS noise in the bandwidth F_L to F_C is given by

$$v_{n,rms}(F_L, F_C) = v_{nw} \sqrt{F_C} \sqrt{\int_{F_L}^{F_C} \frac{1}{f} df} = v_{nw} \sqrt{F_C \ln \left[\frac{F_C}{F_L} \right]} \quad \text{Eq. 1-25}$$

where v_{nw} is the voltage noise spectral density in the "white" region, F_L is the lowest frequency of interest in the 1/f region, and F_C is the 1/f corner frequency.

The next region of interest is the "white" noise area which extends from F_C to F_H .

The RMS noise in this bandwidth is given by

$$v_{n,rms}(F_C, F_H) = v_{nw} \sqrt{F_H - F_C} \quad \text{Eq. 1-26}$$

Eq. 1-25 and 1-26 can be combined to yield the total RMS noise from F_L to F_H :

$$v_{n,rms}(F_L, F_H) = v_{nw} \sqrt{F_C \ln \left[\frac{F_C}{F_L} \right] + (F_H - F_C)} \quad \text{Eq. 1-27}$$

In many cases, the low frequency p-p noise is specified in a 0.1 to 10Hz bandwidth, measured with a 0.1 to 10Hz bandpass filter between op amp and measuring device.

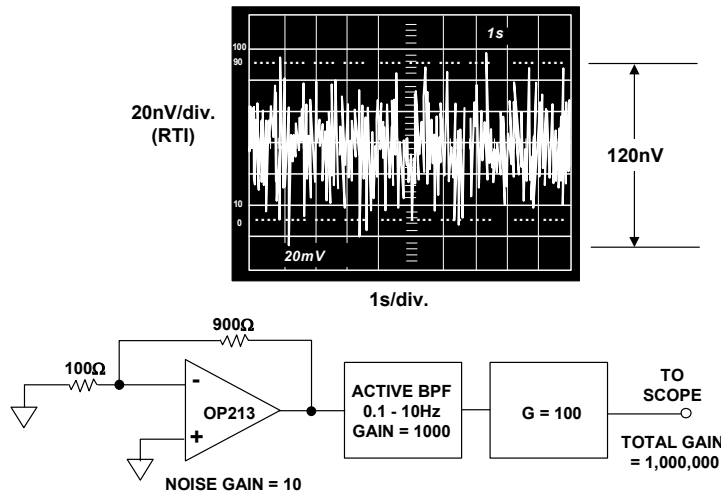
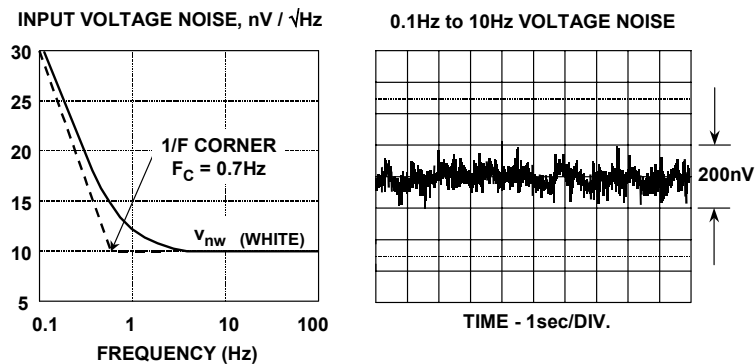


Figure 1-70: The peak-to-peak noise in the 0.1Hz to 10Hz bandwidth for the OP213 is less than 120nV

The measurement is often presented as a scope photo with a time scale of 1s/div, as is shown in Figure 1-70 above for the OP213.



$$\begin{aligned} \blacklozenge \quad v_{n,rms}(F_L, F_H) &= v_{nw} \sqrt{F_C \ln \left[\frac{F_C}{F_L} \right] + (F_H - F_C)} \\ \blacklozenge \quad \text{For } F_L &= 0.1\text{Hz}, F_H = 10\text{Hz}, v_{nw} = 10\text{nV}/\sqrt{\text{Hz}}, F_C = 0.7\text{Hz}: \\ \blacklozenge \quad v_{n,rms} &= 33\text{nV} \\ \blacklozenge \quad v_{n,pp} &= 6.6 \times 33\text{nV} = 218\text{nV} \end{aligned}$$

Figure 1-71: Input voltage noise for the OP177

It is possible to relate the 1/f noise measured in the 0.1 to 10Hz bandwidth to the voltage noise spectral density. Figure 1-71 above shows the OP177 input voltage noise spectral

■ OP AMP APPLICATIONS

density on the left-hand side of the diagram, and the 0.1 to 10Hz peak-to-peak noise scope photo on the right-hand side. Equation 1-26 can be used to calculate the total RMS noise in the bandwidth 0.1 to 10Hz by letting $F_L = 0.1\text{Hz}$, $F_H = 10\text{Hz}$, $F_C = 0.7\text{Hz}$, $v_{nw} = 10\text{nV}/\sqrt{\text{Hz}}$. The value works out to be about 33nV RMS, or 218nV peak-to-peak (obtained by multiplying the RMS value by 6.6— see the following discussion). This compares well to the value of 200nV as measured from the scope photo.

It should be noted that at higher frequencies, the term in the equation containing the natural logarithm becomes insignificant, and the expression for the RMS noise becomes:

$$V_{n,\text{rms}}(F_H, F_L) \approx v_{nw} \sqrt{F_H - F_L} . \quad \text{Eq. 1-28}$$

And, if $F_H \gg F_L$,

$$V_{n,\text{rms}}(F_H) \approx v_{nw} \sqrt{F_H} . \quad \text{Eq. 1-29}$$

However, some op amps (such as the OP07 and OP27) have voltage noise characteristics that increase slightly at high frequencies. The voltage noise versus frequency curve for op amps should therefore be examined carefully for flatness when calculating high frequency noise using this approximation.

At very low frequencies when operating exclusively in the 1/f region, $F_C \gg (F_H - F_L)$, and the expression for the RMS noise reduces to:

$$V_{n,\text{rms}}(F_H, F_L) \approx v_{nw} \sqrt{F_C \ln \left[\frac{F_H}{F_L} \right]} . \quad \text{Eq. 1-30}$$

Note that there is no way of reducing this 1/f noise by filtering if operation extends to DC. Making $F_H = 0.1\text{Hz}$ and $F_L = 0.001$ still yields an RMS 1/f noise of about 18nV RMS, or 119nV peak-to-peak.

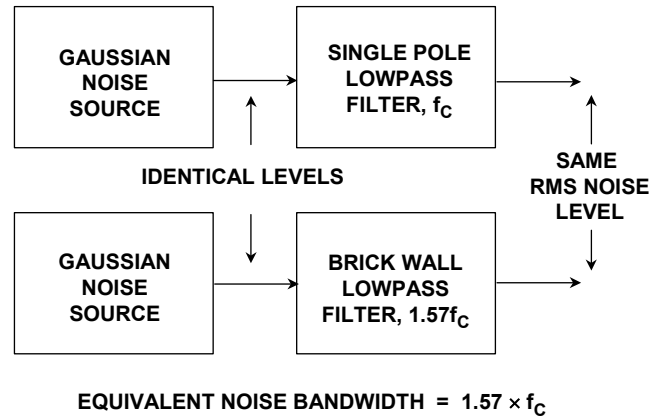


Figure 1-72: Equivalent noise bandwidth

The point is that averaging results of a large number of measurements over a long period of time has practically no effect on the RMS value of the 1/f noise. A method of reducing it further is to use a chopper stabilized op amp, to remove the low frequency noise.

In practice, it is virtually impossible to measure noise within specific frequency limits with no contribution from outside those limits, since practical filters have finite rolloff

characteristics. Fortunately, measurement error introduced by a single pole lowpass filter is readily computed. The noise in the spectrum above the single pole filter cutoff frequency, f_c , extends the corner frequency to $1.57f_c$. Similarly, a two pole filter has an apparent corner frequency of approximately $1.2f_c$. The error correction factor is usually negligible for filters having more than two poles. The net bandwidth after the correction is referred to as the filter *equivalent noise bandwidth* (see Figure 1-72 opposite).

It is often desirable to convert RMS noise measurements into peak-to-peak. In order to do this, one must have some understanding of the statistical nature of noise. For Gaussian noise and a given value of RMS noise, statistics tell us that the chance of a particular peak-to-peak value being exceeded decreases sharply as that value increases— but this probability never becomes zero.

Thus, for a given RMS noise, it is possible to predict the percentage of time that a given peak-to-peak value will be exceeded, but it is not possible to give a peak-to-peak value which will never be exceeded as shown in Figure 1-73 below.

Nominal Peak-to-Peak	% of the Time Noise will Exceed Nominal Peak-to-Peak Value
2 × rms	32%
3 × rms	13%
4 × rms	4.6%
5 × rms	1.2%
6 × rms	0.27%
6.6 × rms**	0.10%
7 × rms	0.046%
8 × rms	0.006%

****Most often used conversion factor is 6.6**

Figure 1-73: RMS to peak-to-peak ratios

Peak-to-peak noise specifications, therefore, must always be written with a time limit. A suitable one is 6.6 times the RMS value, which is exceeded only 0.1% of the time.

Total Output Noise Calculations

We have already pointed out that any noise source which produces less than one third to one fifth of the noise of some greater source can be ignored, with little error. When so doing, both noise voltages must be measured at the same point in the circuit. To analyze the noise performance of an op amp circuit, we must assess the noise contributions of each part of the circuit, and determine which are significant. To simplify the following calculations, we shall work with noise spectral densities, rather than actual voltages, to

■ OP AMP APPLICATIONS

leave bandwidth out of the expressions (the noise spectral density, which is generally expressed in $\text{nV}/\sqrt{\text{Hz}}$, is equivalent to the noise in a 1 Hz bandwidth).

If we consider the circuit in Figure 1-74 below, which is an amplifier consisting of an op amp and three resistors (R_3 represents the source resistance at node A), we can find six separate noise sources: the Johnson noise of the three resistors, the op amp voltage noise, and the current noise in each input of the op amp. Each source has its own contribution to the noise at the amplifier output. Noise is generally specified RTI, or *referred to the input*, but it is often simpler to calculate the noise referred to the output (RTO) and then divide it by the *noise gain* (not the *signal gain*) of the amplifier to obtain the RTI noise.

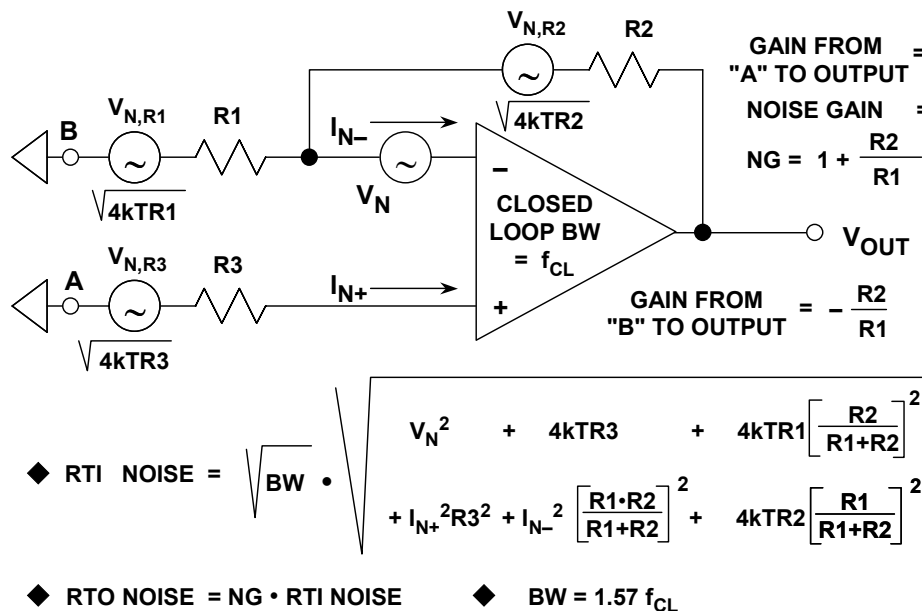


Figure 1-74: Op amp noise model for single pole system

Figure 1-75 (opposite) is a detailed analysis of how each of the noise sources in Fig. 1-74 is reflected to the output of the op amp. Some further discussion regarding the effect of the current noise at the inverting input is warranted. This current, I_{N-} , does not flow in R_1 , as might be expected—the negative feedback around the amplifier works to keep the potential at the inverting input unchanged, so that a current flowing from that pin is forced, by negative feedback, to flow in R_2 only, resulting in a voltage at the output of $I_{N-} R_2$. We could equally well consider the voltage caused by I_{N-} flowing in the parallel combination of R_1 and R_2 and then amplified by the noise gain of the amplifier, but the results are identical—only the calculations are more involved.

Notice that the Johnson noise voltage associated with the three resistors has been included in the expressions of Fig. 1-75. All resistors have a Johnson noise of $\sqrt{(4kTBR)}$, where k is Boltzmann's Constant (1.38×10^{-23} J/K), T is the absolute temperature, B is the bandwidth in Hz, and R is the resistance in Ω . A simple relationship which is easy to remember is that a 1000Ω resistor generates a Johnson noise of $4\text{nV}/\sqrt{\text{Hz}}$ at 25°C .

The analysis so far assumes that the feedback network is purely resistive and that the noise gain versus frequency is flat. This applies to most applications, but if the feedback network contains reactive elements (usually capacitors) the noise gain is not constant

over the bandwidth of interest, and more complex techniques must be used to calculate the total noise (see in particular, Reference 2 and Chapter 4, Section 4-4 of this book).

NOISE SOURCE EXPRESSED AS A VOLTAGE	MULTIPLY BY THIS FACTOR TO REFER TO THE OP AMP OUTPUT
Johnson noise in R3: $\sqrt{(4kTR3)}$	Noise Gain = $1 + R2/R1$
Non-inverting input current noise flowing in R3: $I_{N+}R3$	Noise Gain = $1 + R2/R1$
Input voltage noise: V_N	Noise Gain = $1 + R2/R1$
Johnson noise in R1: $\sqrt{(4kTR1)}$	$-R2/R1$ (Gain from input of R1 to output)
Johnson noise in R2: $\sqrt{(4kTR2)}$	1
Inverting input current noise flowing in R2: $I_{N-}R2$	1

Figure 1-75: Noise sources referred to the output (RTO)

The circuit shown in Figure 1-76 below represents a second-order system, where capacitor C1 represents the source capacitance, stray capacitance on the inverting input, the input capacitance of the op amp, or any combination of these. C1 causes a breakpoint in the noise gain, and C2 is the capacitor that must be added to obtain stability.

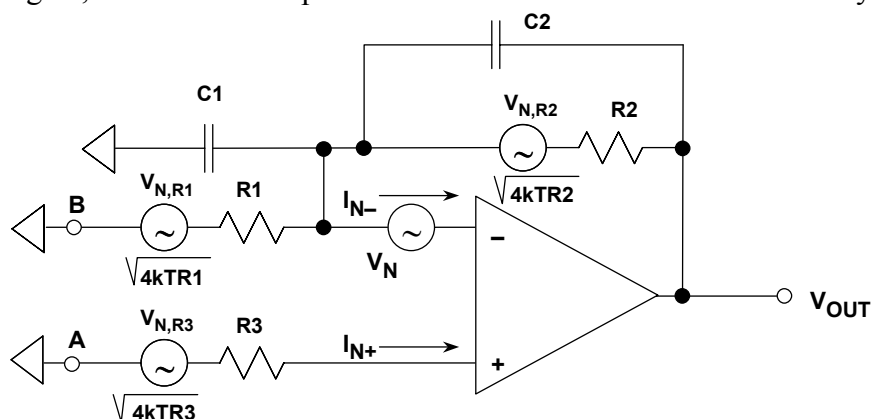


Figure 1-76: Op amp noise model with reactive elements (second-order system)

Because of C1 and C2, the noise gain is a function of frequency, and has peaking at the higher frequencies (assuming C2 is selected to make the second-order system critically damped). Textbooks state that a flat noise gain can be achieved if one simply makes $R1C1 = R2C2$.

But in the case of current-to-voltage converters, however, R1 is typically a high impedance, and the method doesn't work. Maximizing the signal bandwidth in these situations is somewhat complex and is treated in detail in Section 1-6 of this chapter and in Chapter 4, Section 4-4 of this book.

■ OP AMP APPLICATIONS

A DC signal applied to input A (B being grounded) sees a gain of $1 + R_2/R_1$, the low frequency noise gain. At higher frequencies, the gain from input A to the output becomes $1 + C_1/C_2$ (the high frequency noise gain).

The closed-loop bandwidth f_{cl} is the point at which the noise gain intersects the open-loop gain. A DC signal applied to B (A being grounded) sees a gain of $-R_2/R_1$, with a high frequency cutoff determined by R_2 - C_2 . Bandwidth from B to the output is $1/2\pi R_2 C_2$.

The current noise of the non-inverting input, I_{N+} , flows in R_3 and gives rise to a noise voltage of $I_{N+}R_3$, which is amplified by the frequency-dependent noise gain, as are the op amp noise voltage, V_N , and the Johnson noise of R_3 , which is $\sqrt{4kTR_3}$. The Johnson noise of R_1 is amplified by $-R_2/R_1$ over a bandwidth of $1/2\pi R_2 C_2$, and the Johnson noise of R_2 is not amplified at all but is connected directly to the output over a bandwidth of $1/2\pi R_2 C_2$. The current noise of the inverting input, I_{N-} , flows in R_2 only, resulting in a voltage at the amplifier output of $I_{N-}R_2$ over a bandwidth of $1/2\pi R_2 C_2$.

If we consider these six noise contributions, we see that if R_1 , R_2 , and R_3 are low, then the effect of current noise and Johnson noise will be minimized, and the dominant noise will be the op amp's voltage noise. As we increase resistance, both Johnson noise and the voltage noise produced by noise currents will rise.

If noise currents are low, then Johnson noise will take over from voltage noise as the dominant contributor. Johnson noise, however, rises with the square root of the resistance, while the current noise voltage rises linearly with resistance, so ultimately, as the resistance continues to rise, the voltage due to noise currents will become dominant.

These noise contributions we have analyzed are not affected by whether the input is connected to node A or node B (the other being grounded or connected to some other low-impedance voltage source), which is why the non-inverting gain ($1 + Z_2/Z_1$), which is seen by the voltage noise of the op amp, V_N , is known as the "noise gain".

Calculating the total output RMS noise of the second-order op amp system requires multiplying each of the six noise voltages by the appropriate gain and integrating over the appropriate frequency as shown in Figure 1-77 (opposite).

The root-sum-square of all the output contributions then represents the total RMS output noise. Fortunately, this cumbersome exercise may be greatly simplified in most cases by making the appropriate assumptions and identifying the chief contributors.

NOISE SOURCE EXPRESSED AS A VOLTAGE	MULTIPLY BY THIS FACTOR TO REFER TO OUTPUT	INTEGRATION BANDWIDTH
Johnson noise in R3: $\sqrt{4kTR3}$	Noise Gain as a function of frequency	Closed-Loop BW
Non-inverting input current noise flowing in R3: $I_{N+}R3$	Noise Gain as a function of frequency	Closed-Loop BW
Input voltage noise: V_N	Noise Gain as a function of frequency	Closed-Loop BW
Johnson noise in R1: $\sqrt{4kTR1}$	$-R2/R1$ (Gain from B to output)	$1/2\pi R2C2$
Johnson noise in R2: $\sqrt{4kTR2}$	1	$1/2\pi R2C2$
Inverting input current noise flowing in R2: $I_{N-}R2$	1	$1/2\pi R2C2$

Figure 1-77: Noise sources referred to the output for a second-order system

Although shown before, the noise gain for a typical second-order system is repeated in Figure 1-78 below. It is quite easy to perform the voltage noise integration in two steps, but notice that because of peaking, the majority of the output noise due to the input voltage noise will be determined by the high frequency portion where the noise gain is $1 + C1/C2$. This type of response is typical of second-order systems.

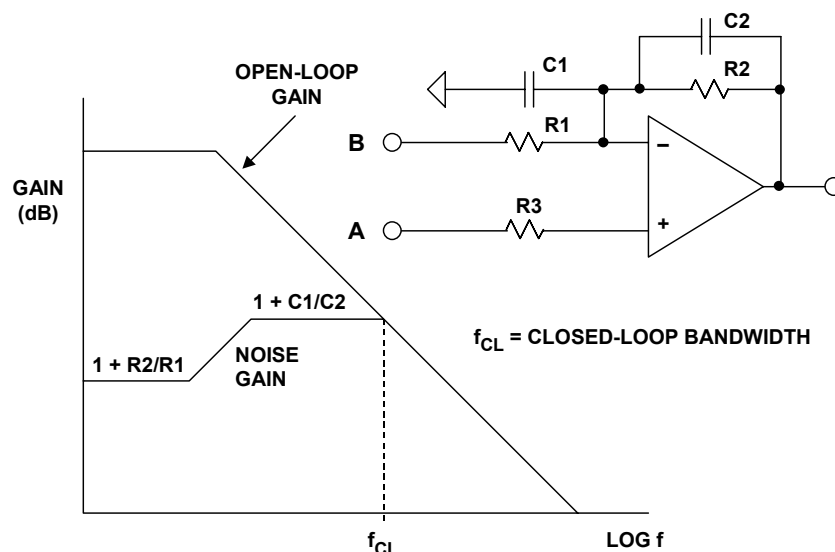


Figure 1-78: Noise gain of a typical second-order system

The noise due to the inverting input current noise, R1, and R2 is only integrated over the bandwidth $1/2\pi R2C2$.

■ OP AMP APPLICATIONS

Op Amp Distortion

Dynamic range of an op amp may be defined in several ways. The most common ways are to specify *harmonic distortion*, *total harmonic distortion* (THD), or *total harmonic distortion plus noise* (THD + N).

Other specifications related specifically to communications systems such as *intermodulation distortion* (IMD), *intercept points* (IP), *spurious free dynamic range* (SFDR), *multitone power ratio* (MTPR) and others are covered thoroughly in Chapter 6, Section 6-4. In this section, only harmonic distortion, THD, and THD + N will be covered.

The distortion components which makes up total harmonic distortion is usually calculated by taking the root sum of the squares of the first five or six harmonics of the fundamental. In many practical situations, however, there is negligible error if only the second and third harmonics are included. The definition of THD and THD + N is shown in Figure 1-79 below.

- ◆ V_s = Signal Amplitude (RMS Volts)
- ◆ V_2 = Second Harmonic Amplitude (RMS Volts)
- ◆ V_n = nth Harmonic Amplitude (RMS Volts)
- ◆ V_{noise} = RMS value of noise over measurement bandwidth

- ◆
$$\text{THD} + \text{N} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots + V_n^2 + V_{\text{noise}}^2}}{V_s}$$
- ◆
$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \dots + V_n^2}}{V_s}$$

Figure 1-79: Definitions of THD and THD + N

It is important to note that the THD measurement does not include noise terms, while THD + N does. The noise in the THD + N measurement must be integrated over the measurement bandwidth. In audio applications, the bandwidth is normally chosen to be around 100kHz. In narrow-band applications, the level of the noise may be reduced by filtering.

On the other hand, harmonics and intermodulation products which fall within the measurement bandwidth cannot be filtered, and therefore may limit the system dynamic range.

Common-Mode Rejection Ratio (CMRR), Power Supply Rejection Ratio (PSRR)

If a signal is applied equally to both inputs of an op amp, so that the differential input voltage is unaffected, the output should not be affected. In practice, changes in common-mode voltage will produce changes in output. The op amp *common-mode rejection ratio* (CMRR) is the ratio of the common-mode gain to differential-mode gain. For example, if a differential input change of Y volts produces a change of 1V at the output, and a common-mode change of X volts produces a similar change of 1V, then the CMRR is X/Y. When the common-mode rejection ratio is expressed in dB, it is generally referred to as common-mode rejection (CMR). Typical LF CMR values are between 70 and 120dB, but at higher frequencies, CMR deteriorates. Many op amp data sheets show a plot of CMR versus frequency, as shown in Figure 1-80 for an OP177 op amp.

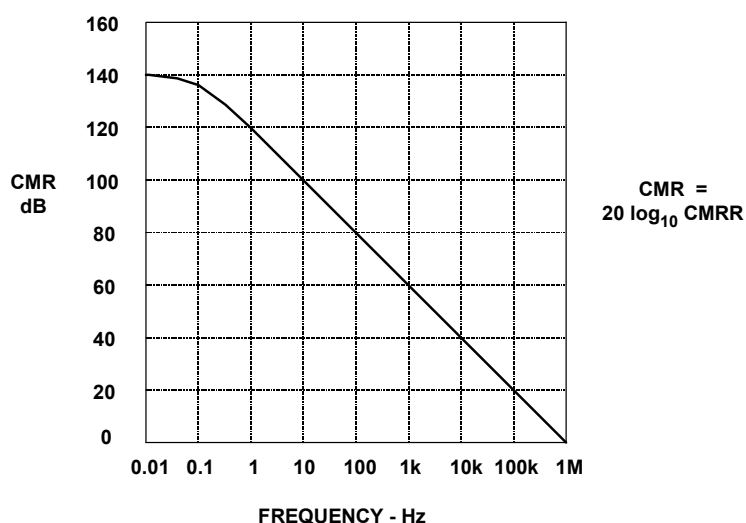


Figure 1-80: OP177 common-mode rejection (CMR)

CMRR produces a corresponding output offset voltage error in op amps configured in the non-inverting mode as shown in Figure 1-81 below.

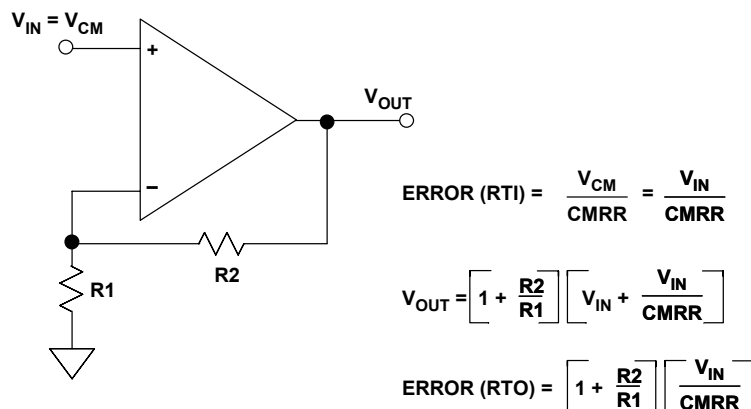


Figure 1-81: Calculating offset error due to common-mode rejection ratio (CMRR)

Note inverting mode operating op amps will have negligible CMRR error, as both inputs are held at a ground (or virtual ground), i.e., there is no CM dynamic voltage.

■ OP AMP APPLICATIONS

Common-mode rejection ratio can be measured in several ways. The method shown in Figure 1-82 below uses four precision resistors to configure the op amp as a differential amplifier, a signal is applied to both inputs, and the change in output is measured—an amplifier with infinite CMRR would have no change in output. The disadvantage inherent in this circuit is that the ratio match of the resistors is as important as the CMRR of the op amp. A mismatch of 0.1% between resistor pairs will result in a CMR of only 66dB—no matter how good the op amp! Since most op amps have a LF CMR of between 80 and 120dB, it is clear that this circuit is only marginally useful for measuring CMRR (although it does an excellent job in measuring the matching of the resistors!).

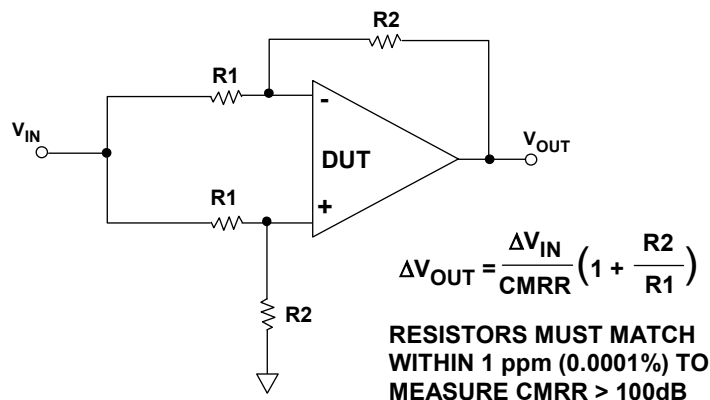


Figure 1-82: Simple common-mode rejection ratio (CMRR) test circuit

The slightly more complex circuit shown in Figure 1-83 below measures CMRR without requiring accurately matched resistors. In this circuit, the common-mode voltage is changed by switching the power supply voltages. (This is easy to implement in a test facility, and the same circuit with different supply voltage connections can be used to measure power supply rejection ratio).

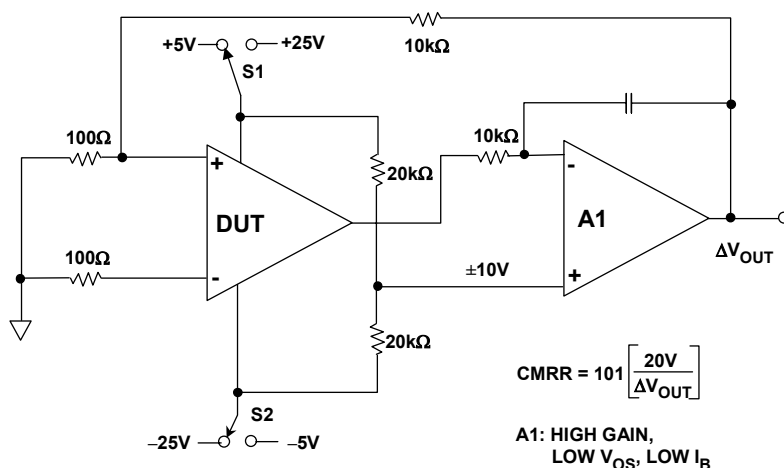


Figure 1-83: CMRR test circuit does not require precision resistors

The power supply values shown in the circuit are for a $\pm 15V$ DUT op amp, with a common-mode voltage range of $\pm 10V$. Other supplies and common-mode ranges can also be accommodated by changing voltages, as appropriate. The integrating amplifier A1 should have high gain, low V_{OS} and low I_B , such as an OP97 family device.

If the supply of an op amp changes, its output should not, but it does. The specification of *power supply rejection ratio* or PSRR is defined similarly to the definition of CMRR. If a change of X volts in the supply produces the same output change as a differential input change of Y volts, then the PSRR on that supply is X/Y. The definition of PSRR assumes that both supplies are altered equally in opposite directions—otherwise the change will introduce a common-mode change as well as a supply change, and the analysis becomes considerably more complex. It is this effect which causes apparent differences in PSRR between the positive and negative supplies.

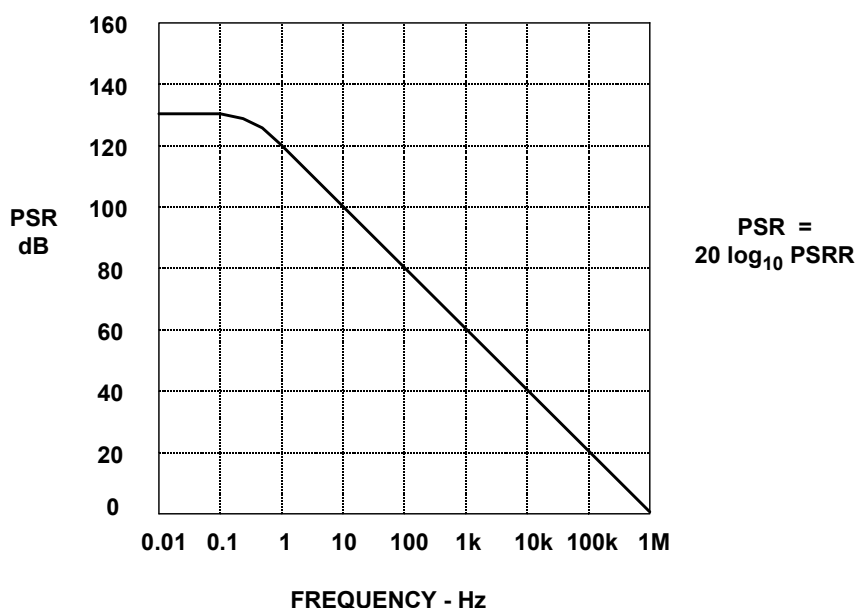


Figure 1-84: OP177 power supply rejection

Typical PSR for the OP177 is shown in Figure 1-84 above.

The test setup used to measure CMRR may be modified to measure PSRR as shown in Figure 1-85 below.

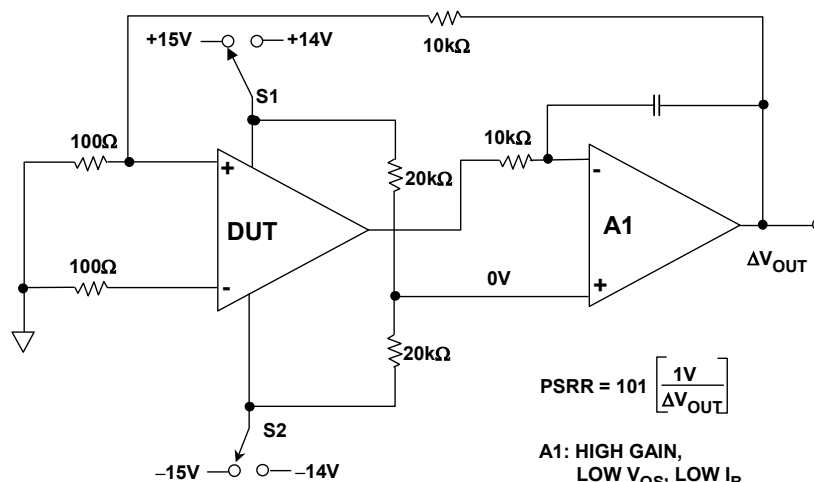


Figure 1-85: Test setup for measuring power supply rejection ratio (PSRR)

The voltages are chosen for a symmetrical power supply change of 1V. Other values may be used where appropriate.

■ OP AMP APPLICATIONS

Power Supplies and Decoupling

Because op amp PSRR is frequency dependent, op amp power supplies must be well decoupled. At low frequencies, several devices may share a 10-50 μ F capacitor on each supply, provided it is no more than 10cm (PC track distance) from any of them.

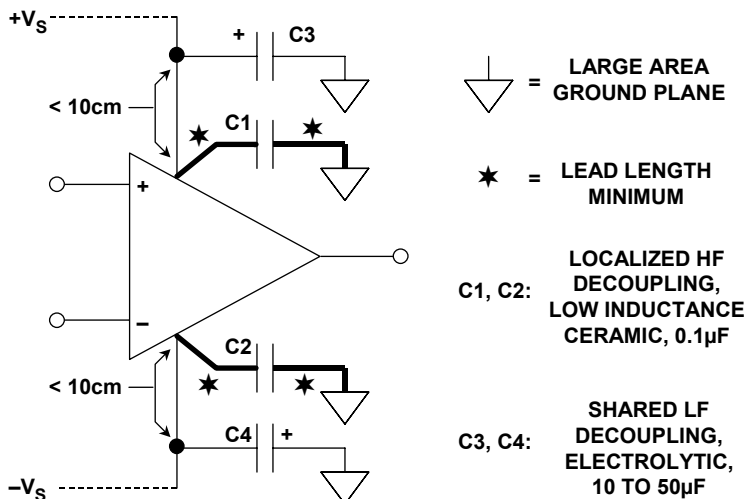


Figure 1-86: Proper low and high-frequency decoupling techniques for op amps

At high frequencies, each IC should have the supply leads decoupled by a low inductance 0.1 μ F (or so) capacitor with short leads/PC tracks. These capacitors must also provide a return path for HF currents in the op amp load. Typical decoupling circuits are shown in Figure 1-86 above. Further bypassing and decoupling information is found Chapter 7.

Power Supplies and Power Dissipation

Op amps have no ground terminal. Specifications of power supply are quite often in the form $\pm X$ Volts, but in fact it might equally be expressed as $2X$ Volts. What is important is where the CM and output ranges lie relative to the supplies. This information may be provided in tabular form or as a graph.

Often data sheets will advise that an op amp will work over a range of supplies (from +3 to ± 16.5 V for example), and will then give parameters at several values of supply, so that users may extrapolate. If the minimum supply is quite high, it is usually because the device uses a structure requiring a threshold voltage to function (zener diode).

Data sheets also give current consumption. Any current flowing into one supply pin will flow out of the other or out of the output terminal. When the output is open circuit, the dissipation is easily calculated from the supply voltage and current. When current flows in a load, it is easiest to calculate the total dissipation (remember that if the load is grounded to the center rail the load current flows from a supply to ground, not between supplies), and then subtract the load dissipation to obtain the device dissipation. Data sheets normally give details of thermal resistances and maximum junction temperature ratings, from which dissipation limits may be calculated knowing conditions. Details of further considerations relating to power dissipation, heatsinking, etc., can be found in Chapter 7, Section 7-5.

REFERENCES: OP AMP SPECIFICATIONS

1. James L. Melsa and Donald G. Schultz, **Linear Control Systems**, McGraw-Hill, 1969, pp. 196-220, ISBN: 0-07-041481-5
2. Lewis Smith and Dan Sheingold, "Noise and Operational Amplifier Circuits," **Analog Dialogue**, Vol. 3, No.1, pp. 1, 5-16. See also: **Analog Dialogue 25th Anniversary Issue**, pp. 19-31, 1991.
3. Thomas M. Frederiksen, **Intuitive Operational Amplifiers**, McGraw-Hill, 1988., ISBN: 0-07-021966-4
4. Walter G. Jung, **IC Op Amp Cookbook, 3rd Ed.**, Prentice-Hall PTR, 1986, 1997, ISBN: 0-13-889601-1.
5. J. K. Roberge, **Operational Amplifiers-Theory and Practice**, John Wiley, 1975, ISBN: 0-471-72585-4.
6. D. Stout, M. Kaufman, **Handbook of Operational Amplifier Circuit Design**, New York, McGraw-Hill, 1976, ISBN: 0-07-061797-X.
7. J. Dostal, **Operational Amplifiers**, Elsevier Scientific Publishing, New York, 1981, ISBN: 0-444-99760-1.
8. Paul R. Gray and Robert G. Meyer, **Analysis and Design of Analog Integrated Circuits, 3rd Edition**, John Wiley, 1993, ISBN: 0-471-57495-3.
9. Sergio Franco, **Design With Operational Amplifiers and Analog Integrated Circuits, 2nd Ed.**, McGraw-Hill, 1998, ISBN: 0-07-021857-9
10. Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, Inc., 1995, ISBN: 0-916550-15-X.
11. Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8, (available for download at <http://www.analog.com>).
12. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6, (available for download at <http://www.analog.com>).

NOTES:

SECTION 1-5: PRECISION OP AMPS

Walt Kester, Walt Jung

This section examines in more detail some of the issues relating to amplifiers for use in precision signal conditioning applications. Although the OP177 op amp is used for the "gold standard" for precision in these discussions, more recent product introductions such as the rail-to-rail output OP777, OP727, and OP747, along with the OP1177, OP2177, and OP4177 offer nearly as good performance in smaller packages.

Precision op amp open-loop gains greater than 1 million are available, along with common-mode and power supply rejection ratios of the same magnitude. Offset voltages of less than $25\mu\text{V}$ and offset drift less than $0.1\mu\text{V}/^\circ\text{C}$ are available in dual supply op amps such as the OP177, however, the performance in single-supply precision bipolar op amps may sometimes fall short of this performance. This is the tradeoff that must sometimes be made in low power, low voltage applications. On the other hand, however, modern chopper stabilized op amps provide offsets and offset voltage drifts which cannot be distinguished from noise, and these devices operate on single supplies and provide rail-to-rail inputs and outputs. They too come with their own set of problems that are discussed later within this section.

It is important to understand that DC open-loop gain, offset voltage, power supply rejection (PSR), and common-mode rejection (CMR) alone shouldn't be the only considerations in selecting precision amplifiers. The AC performance of the amplifier is also important, even at "low" frequencies. Open-loop gain, PSR, and CMR all have relatively low corner frequencies, and therefore what may be considered "low" frequency may actually fall above these corner frequencies, increasing errors above the value predicted solely by the DC parameters. For example, an amplifier having a DC open-loop gain of 10 million and a unity-gain crossover frequency of 1MHz has a corresponding corner frequency of 0.1Hz! One must therefore consider the open-loop gain at the actual *signal* frequency. The relationship between the single-pole unity-gain crossover frequency, f_u , the signal frequency, f_{sig} , and the open-loop gain $A_{\text{VOL}(f_{\text{sig}})}$ (measured at the signal frequency) is given by:

$$A_{\text{VOL}(f_{\text{sig}})} = \frac{f_u}{f_{\text{sig}}} . \quad \text{Eq. 1-31}$$

In the example above, the open-loop gain is 10 at 100kHz, and 100,000 at 10Hz. Note that the constant gain-bandwidth product concept only holds true for VFB op amps. It doesn't apply to CFB op amps, but then they are rarely used in precision applications.

Loss of open-loop gain at the frequency of interest can introduce distortion, especially at audio frequencies. Loss of CMR or PSR at the line frequency or harmonics thereof can also introduce errors.

■ OP AMP APPLICATIONS

The challenge of selecting the right amplifier for a particular signal conditioning application has been complicated by the sheer proliferation of various types of amplifiers in various processes (Bipolar, Complementary Bipolar, BiFET, CMOS, BiCMOS, etc.) and architectures (traditional op amps, instrumentation amplifiers, chopper amplifiers, isolation amplifiers, etc.)

In addition, a wide selection of precision amplifiers are now available which operate on single-supply voltages which complicates the design process even further because of the reduced signal swings and voltage input and output restrictions. Offset voltage and noise are now a more significant portion of the input signal.

◆ Input Offset Voltage	<100 μ V
◆ Input Offset Voltage Drift	<1 μ V/ $^{\circ}$ C
◆ Input Bias Current	<2nA
◆ Input Offset Current	<2nA
◆ DC Open Loop Gain	>1,000,000
◆ Unity Gain Bandwidth Product, f_u	500kHz - 5MHz
◆ Always Check Open Loop Gain at Signal Frequency!	
◆ 1/f (0.1Hz to 10Hz) Noise	<1 μ V p-p
◆ Wideband Noise	<10nV/ $\sqrt{\text{Hz}}$
◆ CMR, PSR	>100dB
◆ Tradeoffs:	
● Single supply operation	
● Low supply currents	

Figure 1-87: Precision op amp characteristics

Selection guides and parametric search engines, which can simplify this process somewhat, are available on the world-wide-web (<http://www.analog.com>) as well as on CDROM. Some general attributes of precision op amps are summarized in Figure 1-87.

Precision Op Amp Amplifier DC Error Budget Analysis

In order to develop a concept for the magnitudes of the various errors in a high precision op amp circuit, a simple room temperature analysis for the OP177F is shown on the opposite page, in Figure 1-88. The amplifier is connected in the inverting mode with a signal gain of 100. The key data sheet specifications are also shown in the diagram. We assume an input signal of 100mV fullscale which corresponds to an output signal of 10V. The various error sources are normalized to fullscale and expressed in parts per million (ppm). Note: parts per million (ppm) error = fractional error $\times 10^6$ = % error $\times 10^4$.

Note that the offset errors due to V_{OS} and I_{OS} and the gain error due to finite A_{VOL} can be removed with a system calibration. However, the error due to open-loop gain nonlinearity cannot be removed with calibration and produces a relative accuracy error, often called *resolution error*.

A second contributor to resolution error is the 1/f noise. This noise is always present and adds to the uncertainty of the measurement. The overall relative accuracy of the circuit at room temperature is 9ppm, equivalent to ~17 bits of resolution.

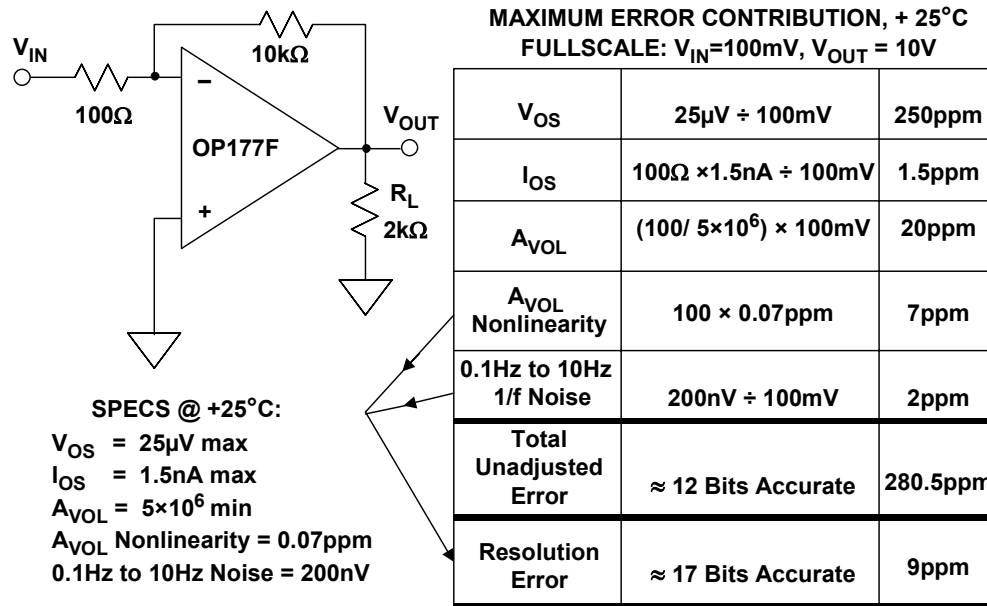


Figure 1-88: Precision op amp (OP177F) DC error budget analysis

It is also useful to compare the performance of a number of single-supply op amps to that of the "gold standard" OP177, and this is done in Figure 1-89 below for some representative devices.

LISTED IN ORDER OF INCREASING SUPPLY CURRENT

PART NO.	V_{OS} max	V_{OS} TC	A_{VOL} min	NOISE (1kHz)	INPUT	OUTPUT	$I_{SY}/\text{AMP MAX}$
OP293	250μV	2μV/°C	200k	5nV/√Hz	0, 4V	5mV, 4V	20μA
OP196/296/496	300μV	2μV/°C	150k	26nV/√Hz	R/R	"R/R"	60μA
OP777	100μV	1.3μV/°C	300k	15nV/√Hz	0, 4V	"R/R"	270μA
OP191/291/491	700μV	5μV/°C	25k	35nV/√Hz	R/R	"R/R"	350μA
*AD820/822/824	1000μV	20μV/°C	500k	16nV/√Hz	0, 4V	"R/R"	800μA
**AD8601/2/4	600μV	2μV/°C	20k	33nV/√Hz	R/R	"R/R"	1000μA
OP184/284/484	150μV	2μV/°C	50k	3.9nV/√Hz	R/R	"R/R"	1350μA
OP113/213/413	175μV	4μV/°C	2M	4.7nV/√Hz	0, 4V	5mV, 4V	3000μA
OP177F (±15V)	25μV	0.1μV/°C	5M	10nV/√Hz	N/A	N/A	2000μA

*JFET INPUT **CMOS

NOTE: Unless Otherwise Stated
Specifications are Typical @ +25°C
 $V_S = +5\text{V}$

Figure 1-89: Precision single-supply op amp performance characteristics

Note that the Fig. 1-89 amplifier list does *not* include the category of chopper op amps, which excel in many of the categories. These are covered separately, immediately below.

Chopper Stabilized Amplifiers

For the lowest offset and drift performance, chopper-stabilized amplifiers may be the only solution. The best bipolar amplifiers offer offset voltages of $25\mu\text{V}$ and $0.1\mu\text{V}/^\circ\text{C}$ drift. Offset voltages less than $5\mu\text{V}$ with practically no measurable offset drift are obtainable with choppers, albeit with some penalties.

A basic chopper amplifier circuit is shown in Figure 1-90 below. When the switches are in the "Z" (auto-zero) position, capacitors C2 and C3 are charged to the amplifier input and output offset voltage, respectively. When the switches are in the "S" (sample) position, V_{IN} is connected to V_{OUT} through the path comprised of R1, R2, C2, the amplifier, C3, and R3. The chopping frequency is usually between a few hundred Hz and several kHz, and it should be noted that because this is a sampling system, the input frequency must be much less than one-half the chopping frequency in order to prevent errors due to aliasing. The R1-C1 combination serves as an antialiasing filter. It is also assumed that after a steady state condition is reached, there is only a minimal amount of charge transferred during the switching cycles. The output capacitor, C4, and the load, R_L , must be chosen such that there is minimal V_{OUT} droop during the auto-zero cycle.

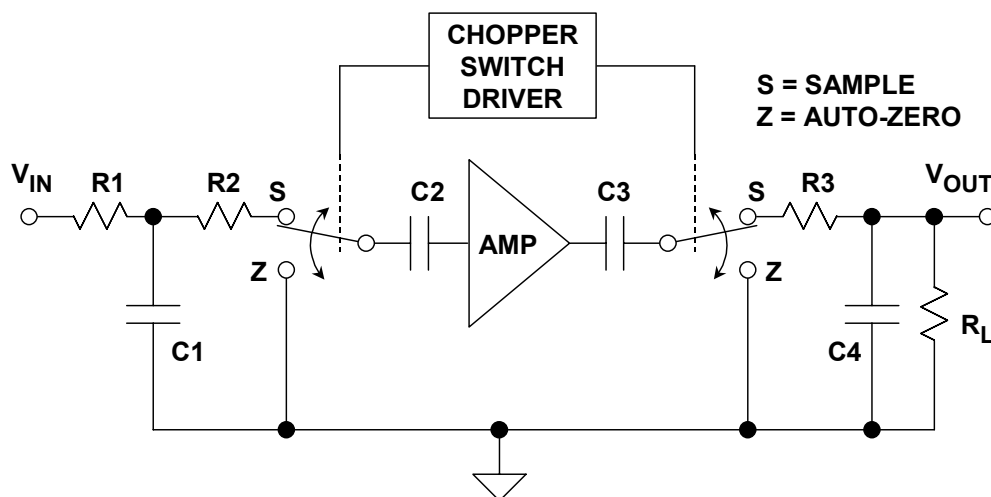


Figure 1-90: Classic chopper amplifier

The basic chopper amplifier of Fig. 1-90 can pass only very low frequencies because of the input filtering required to prevent aliasing. In contrast to this, the *chopper-stabilized* architecture shown in Figure 1-91 (opposite) is most often used in chopper amplifier implementations. In this circuit, A1 is the *main* amplifier, and A2 is the *nulling* amplifier. In the sample mode (switches in "S" position), the nulling amplifier, A2, monitors the input offset voltage of A1 and drives its output to zero by applying a suitable correcting voltage at A1's null pin. Note, however, that A2 also has an input offset voltage, so it must correct its own error before attempting to null A1's offset. This is achieved in the auto-zero mode (switches in "Z" position) by momentarily disconnecting A2 from A1, shorting its inputs together, and coupling its output to its own null pin. During the auto-zero mode, the correction voltage for A1 is momentarily held by C1. Similarly, C2 holds the correction voltage for A2 during the sample mode. In modern IC chopper-stabilized op amps, the storage capacitors C1 and C2 are on-chip.

Note in this architecture that *the input signal is always connected to the output, through A1*. The bandwidth of A1 thus determines the overall signal bandwidth, and the input signal is not limited to less than one-half the chopping frequency as in the case of the traditional chopper amplifier architecture. However, the switching action does produce small transients at the chopping frequency, that can mix with the input signal frequency and produce intermodulation distortion.

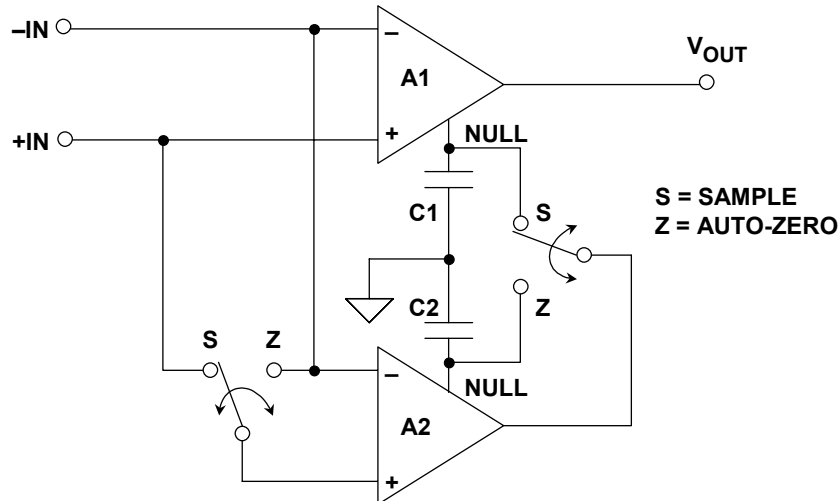


Figure 1-91: Modern chopper stabilized op amp

A patented spread-spectrum technique is used in the AD8571/72/74 series of single-supply chopper-stabilized op amps, to virtually eliminate intermodulation effects.

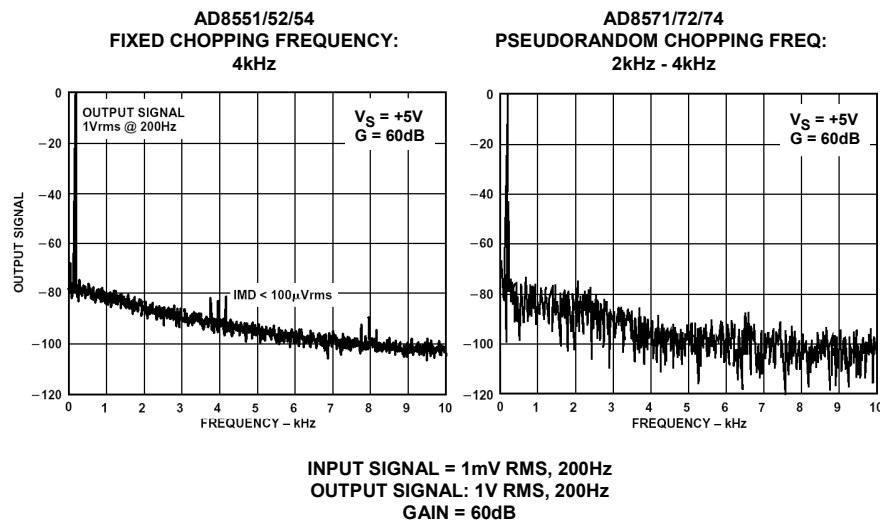


Figure 1-92: Intermodulation product: fixed versus pseudorandom chopping frequency

These devices use a pseudorandom chopping frequency swept between 2kHz and 4kHz. Figure 1-92 above compares the intermodulation distortion of a traditional chopper stabilized op amp (AD8551/52/54, left) that uses a fixed 4kHz chopping frequency to that of the AD8571/72/74 (right) that uses the pseudorandom chopping frequency.

■ OP AMP APPLICATIONS

A comparison between fixed and pseudorandom chopping on the voltage noise is shown in Figure 1-93 below. Notice for the fixed chopping frequency, there are distinct peaks in the noise spectrum at the odd harmonics of 4kHz, whereas with pseudorandom chopping, the spectrum is much more uniform, although the average noise level is higher.

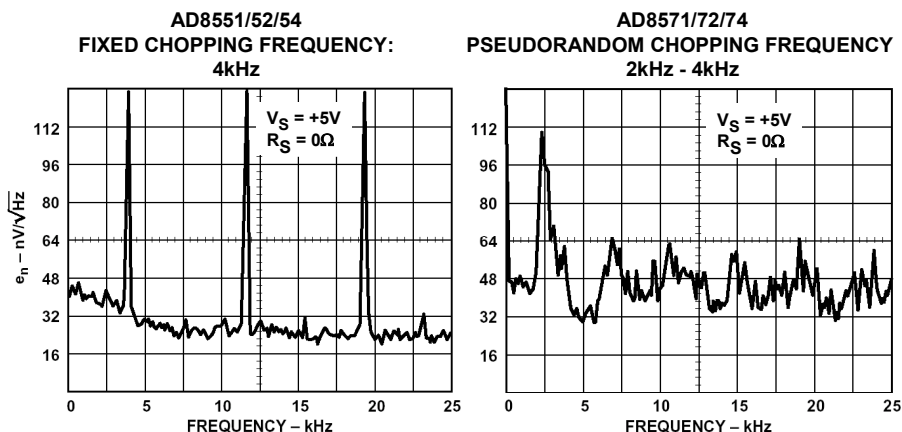


Figure 1-93: Voltage noise spectral density comparison: fixed versus pseudorandom chopping frequency

The AD8571/8572/8574 family of chopper-stabilized op amps offers rail-to-rail input and output single-supply operation, low offset voltage, and low offset drift. As discussed above, the pseudorandom chopping frequency minimizes intermodulation distortion with the input signal. The storage capacitors are internal to the IC, and no external capacitors other than standard decoupling capacitors are required. Key specifications for the devices are given in Figure 1-94 below.

- ◆ Single Supply: +2.7V to +5V
- ◆ 1 μV Typical Input Offset Voltage
- ◆ 0.005 $\mu\text{V}/^\circ\text{C}$ Typical Input Offset Voltage Drift
- ◆ 130dB CMR, PSR
- ◆ 750 μA Supply Current / Op Amp
- ◆ 50 μs Overload Recovery Time
- ◆ 50nV/ $\sqrt{\text{Hz}}$ Input Voltage Noise
- ◆ Pseudorandom Chopping Frequency
- ◆ 1.5MHz Gain-Bandwidth Product
- ◆ Single (AD8571), Dual (AD8572) and Quad (AD8574)

Figure 1-94: AD8571/72/74 chopper stabilized rail-to-rail input/output amplifiers

It should be noted that extreme care must be taken when applying all of the chopper stabilized devices. This is because in order to fully realize the full offset and drift performance inherent to the parts, parasitic thermocouple effects in external circuitry must be avoided. See Chapter 4, Section 4-5 for a general discussion of thermocouples, and Chapter 7, Section 7-1 related to passive components.

Noise Considerations for Chopper-Stabilized Op Amps

It is interesting to consider the effects of a chopper amplifier on low frequency $1/f$ noise. If the chopping frequency is considerably higher than the $1/f$ corner frequency of the input noise, the chopper-stabilized amplifier continuously nulls out the $1/f$ noise on a sample-by-sample basis. Theoretically, a chopper op amp therefore has no $1/f$ noise. However, the chopping action produces wideband noise which is generally much worse than that of a precision bipolar op amp.

Figure 1-95 below shows the noise of a precision bipolar amplifier (OP177) versus that of the AD8571/72/74 chopper-stabilized op amp. The peak-to-peak noise in various bandwidths is calculated for each in the table below the graphs.

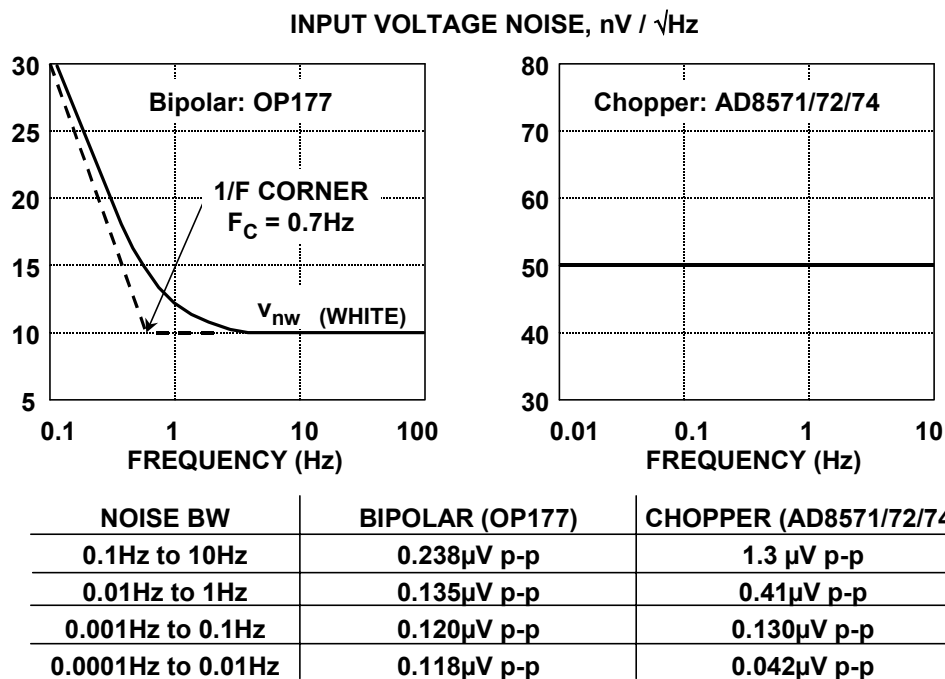


Figure 1-95: Noise: bipolar versus chopper stabilized op amp

Note from the data that as the frequency is lowered, the chopper amplifier noise continues to drop, while the bipolar amplifier noise approaches a limit determined by the $1/f$ corner frequency and its white noise. Notice that only at very low frequencies ($<0.01\text{Hz}$) is the chopper noise performance superior to that of the bipolar op amp.

In order to take advantage of the chopper op amp's lack of $1/f$ noise, much filtering is required—otherwise the total noise of a chopper will always be worse than a good bipolar op amp. Choppers should therefore be selected on the basis of their low offset and drift—not because of their lack of $1/f$ noise.

■ OP AMP APPLICATIONS

REFERENCES: PRECISION OP AMPS

1. Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, 1992, ISBN: 0-916550-10-9.
2. Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
3. Data Sheet for **AD8551/AD8552/AD8554 Zero-Drift, Single-Supply, Rail-to-Rail Input/Output Operational Amplifiers**, <http://www.analog.com>
4. Data Sheet for **AD8571/AD8572/AD8574 Zero-Drift, Single-Supply, Rail-to-Rail Input/Output Operational Amplifiers**, <http://www.analog.com>
5. Data Sheet for **OP777/OP727/OP747 Precision Micropower Single-Supply Operational Amplifiers**, <http://www.analog.com>
6. Data Sheet for **OP1177/OP2177/OP4177 Precision Low Noise, Low Input Bias Current Operational Amplifiers**, <http://www.analog.com>

SECTION 1-6: HIGH SPEED OP AMPS

Walt Kester

Introduction

High speed analog signal processing applications, such as video and communications, require op amps which have wide bandwidth, fast settling time, low distortion and noise, high output current, good DC performance, and operate at low supply voltages. These devices are widely used as gain blocks, cable drivers, ADC pre-amps, current-to-voltage converters, etc. Achieving higher bandwidths for less power is extremely critical in today's portable and battery-operated communications equipment. The rapid progress made over the last few years in high speed linear circuits has hinged not only on the development of IC processes but also on innovative circuit topologies.

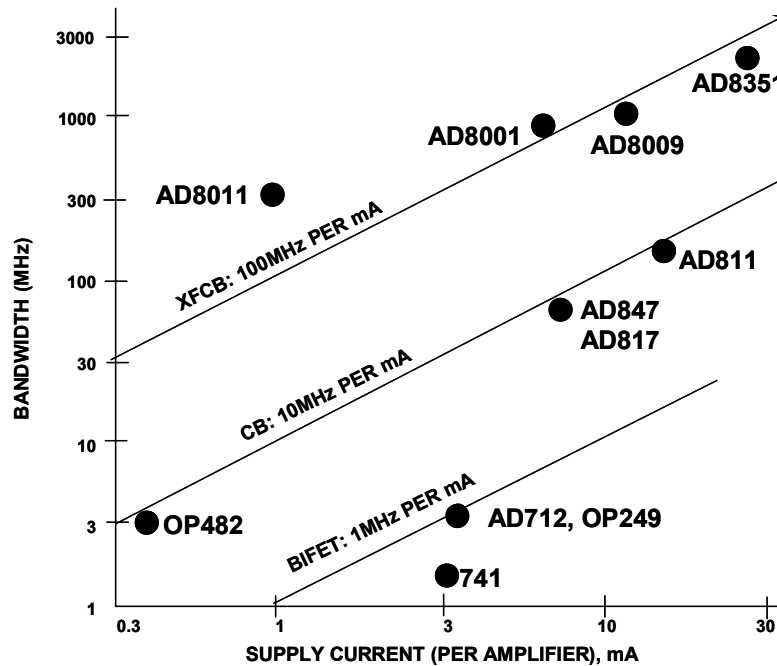


Figure 1-96: Amplifier bandwidth versus supply current for Analog Devices' processes

The evolution of high speed processes using amplifier bandwidth as a function of supply current as a figure of merit is shown in Figure 1-96 above. (In the case of duals, triples, and quads, the current per amplifier is used). Analog Devices BiFET process, which produced the AD712 (3MHz bandwidth, 3mA current, yields about 1MHz per mA.

The CB (Complementary Bipolar) process (AD817, AD847, AD811, etc.) yields about 10MHz/mA of supply current. The f_s of the CB process PNP transistors are about 700MHz, and the NPN's about 900MHz. The CB process at Analog Devices was introduced in 1985.

■ OP AMP APPLICATIONS

The next complementary bipolar process from Analog Devices is a high speed dielectrically isolated process called "XFCB" (eXtra Fast Complementary Bipolar) which was introduced in 1992. This process yields 3GHz PNPs and 5GHz matching NPNs, and coupled with innovative circuit topologies allows op amps to achieve new levels of cost-effective performance at astonishing low quiescent currents. The approximate figure of merit for this process is typically 100MHz/mA, although the AD8011 op amp is capable of 300MHz bandwidth on 1mA of supply current due to its unique two-stage current-feedback architecture described later in this section.

Even faster CB processes have been developed at Analog Devices for low voltage supply products such as "XFCB 1.5" (5GHz PNP, 9GHz NPN), and "XFCB 2" (9GHz PNP, 16GHz NPN). The AD8351 differential low distortion RF amplifier (shown on Fig. 1-96) is fabricated on "XFCB 1.5" and has a bandwidth of 2GHz for a gain of 12dB. It is expected that newer complementary bipolar processes will be optimized for higher f_{ts} .

In order to select intelligently the correct high speed op amp for a given application, an understanding of the various op amp topologies as well as the tradeoffs between them is required. The two most widely used topologies are voltage feedback (VFB) and current feedback (CFB). An overview of these topologies has been presented in a previous section, but the following discussion treats the frequency-related aspects of the two topologies in considerably more detail.

Voltage Feedback (VFB) Op Amps

A voltage feedback (VFB) op amp is distinguished from a current feedback (CFB) op amp by circuit topology. The VFB op amp is certainly the most popular in low frequency applications, but the CFB op amp has some advantages at high frequencies. We will discuss CFB in detail later, but first the more traditional VFB architecture.

Early IC voltage feedback op amps were made on "all NPN" processes. These processes were optimized for NPN transistors—the "lateral" PNP transistors had relatively poor performance. Some examples of these early VFB op amps which used these poor quality PNPs include the 709, the LM101 and the 741 (see Chapter H: "Op Amp History").

Lateral PNPs were generally only used as current sources, level shifters, or for other non-critical functions. A simplified diagram of a typical VFB op amp manufactured on such a process is shown in Figure 1-97 (opposite).

The input stage is a differential pair (sometimes called a *long-tailed pair*) consisting of either a bipolar pair (Q1, Q2) or a FET pair. This " g_m " (transconductance) stage converts the small-signal differential input voltage, v_i , into a current, i_i , and its transfer function is measured in units of conductance, $1/\Omega$, (or mhos). The small-signal emitter resistance, r_e , is approximately equal to the reciprocal of the small-signal g_m .

The formula for the small-signal g_m of a single bipolar transistor is given by the following equation:

$$g_m = \frac{1}{r_e} = \frac{q}{kT} (I_C) = \frac{q}{kT} \left(\frac{I_T}{2} \right), \text{ or} \quad \text{Eq. 1-32}$$

$$g_m \approx \left(\frac{1}{26\text{mV}} \right) \left(\frac{I_T}{2} \right). \quad \text{Eq. 1-33}$$

where I_T is the differential pair tail current, I_C is the collector quiescent bias current ($I_C = I_T/2$), q is the electron charge, k is Boltzmann's constant, and T is absolute temperature. At $+25^\circ\text{C}$, $V_T = kT/q = 26\text{mV}$ (often called the *thermal voltage*, V_T).

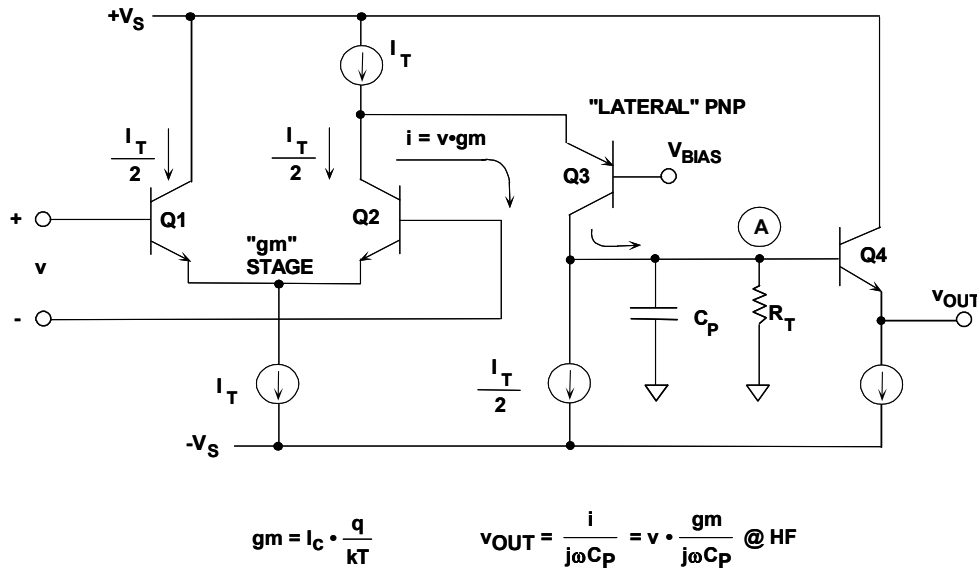


Figure 1-97: Voltage feedback (VFB) op amp designed on an "all NPN" IC process

As we will see shortly, the amplifier unity gain-bandwidth product, f_u , is equal to $g_m/2\pi C_P$, where the capacitance C_P is used to set the dominant pole frequency. For this reason, the tail current, I_T , is made proportional to absolute temperature (PTAT). This current tracks the variation in r_e with temperature thereby making g_m independent of temperature. It is relatively easy to make C_P reasonably constant over temperature.

The Q_2 collector output of the g_m stage drives the emitter of a lateral PNP transistor (Q_3). It is important to note that Q_3 is not used to amplify the signal, only to level shift, i.e., the signal current variation in the collector of Q_2 appears at the collector of Q_3 . The collector current of Q_3 develops a voltage across high impedance node A, and C_P sets the dominant pole of the amplifier. Emitter follower Q_4 provides a low impedance output.

The effective load at the high impedance node A can be represented by a resistance, R_T , in parallel with the dominant pole capacitance, C_P . The small-signal output voltage, v_{out} , is equal to the small-signal current, i , multiplied by the impedance of the parallel combination of R_T and C_P .

■ OP AMP APPLICATIONS

Figure 1-98 below shows a simple model for the single-stage amplifier and the corresponding Bode plot. The Bode plot is conveniently constructed on a log-log scale.

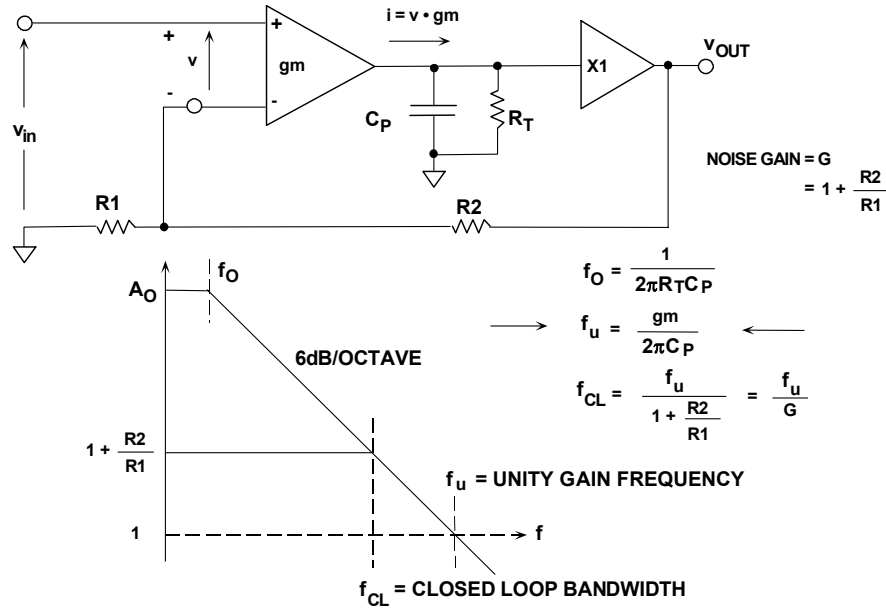


Figure 1-98: Model and Bode plot for a VFB op amp

The low frequency breakpoint, f_O , is given by:

$$f_O = \frac{1}{2\pi R_T C_P} \quad \text{Eq. 1-34}$$

Note that the high frequency response is determined solely by g_m and C_P :

$$v_{out} = v \cdot \frac{g_m}{j\omega C_P} \quad \text{Eq. 1-35}$$

The unity gain-bandwidth frequency, f_u , occurs where $|v_{out}| = |v|$. Letting $\omega = 2\pi f_u$ and $|v_{out}| = |v|$, Eq. 1-35 can be solved for f_u ,

$$f_u = \frac{g_m}{2\pi C_P} \quad \text{Eq. 1-36}$$

We can use feedback theory to derive the closed-loop relationship between the circuit's signal input voltage, v_{in} , and its output voltage, v_{out} :

$$\frac{v_{out}}{v_{in}} = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{j\omega C_P}{g_m} \left(1 + \frac{R_2}{R_1}\right)} \quad \text{Eq. 1-37}$$

At the op amp 3dB closed-loop bandwidth frequency, f_{cl} , the following is true:

$$\frac{2\pi f_{cl} C_P}{g_m} \left(1 + \frac{R_2}{R_1} \right) = 1, \text{ and hence} \quad \text{Eq. 1-38}$$

$$f_{cl} = \frac{g_m}{2\pi C_P} \left(\frac{1}{1 + \frac{R_2}{R_1}} \right), \text{ or} \quad \text{Eq. 1-39}$$

$$f_{cl} = \frac{f_u}{1 + \frac{R_2}{R_1}}. \quad \text{Eq. 1-40}$$

This demonstrates the fundamental property of VFB op amps: *The closed-loop bandwidth multiplied by the closed-loop gain is a constant*, i.e., the VFB op amp exhibits a constant gain-bandwidth product over most of the usable frequency range.

As noted previously, some VFB op amps (called *de-compensated*) are not stable at unity gain, but designed to be operated at some minimum (higher) amount of closed-loop gain. However, even for these op amps, the gain-bandwidth product is still relatively constant over the region of stability.

Now, consider the following typical example: $I_T = 100\mu\text{A}$, $C_P = 2\text{pF}$. We find that:

$$g_m = \frac{I_T / 2}{V_T} = \frac{50\mu\text{A}}{26\text{mV}} = \frac{1}{520\Omega} \quad \text{Eq. 1-41}$$

$$f_u = \frac{g_m}{2\pi C_P} = \frac{1}{2\pi(520)(2 \cdot 10^{-12})} = 153\text{MHz}. \quad \text{Eq. 1-42}$$

Now, we must consider the large-signal response of the circuit. The slew-rate, SR, is simply the total available charging current, $I_T/2$, divided by the dominant pole capacitance, C_P . For the example under consideration,

$$I = C \frac{dv}{dt}, \frac{dv}{dt} = \text{SR}, \text{SR} = \frac{I}{C} \quad \text{Eqs. 1-43}$$

$$\text{SR} = \frac{I_T / 2}{C_P} = \frac{50\mu\text{A}}{2\text{pF}} = 25\text{V} / \mu\text{s}. \quad \text{Eq. 1-44}$$

The full-power bandwidth (FPBW) of the op amp can now be calculated from the formula:

$$\text{FPBW} = \frac{\text{SR}}{2\pi A} = \frac{25\text{V} / \mu\text{s}}{2\pi \cdot 1\text{V}} = 4\text{MHz}, \quad \text{Eq. 1-45}$$

■ OP AMP APPLICATIONS

where A is the peak amplitude of the output signal. If we assume a 2V peak-to-peak output sinewave (certainly a reasonable assumption for high speed applications), then we obtain a FPBW of only 4MHz, even though the small-signal unity gain-bandwidth product is 153MHz! For a 2V p-p output sinewave, distortion will begin to occur much lower than the actual FPBW frequency. We must increase the SR by a factor of about 40 in order for the FPBW to equal 153MHz. The only way to do this is to increase the tail current, I_T , of the input differential pair by the same factor. This implies a bias current of 4mA in order to achieve a FPBW of 160MHz. We are assuming that C_P is a fixed value of 2pF and cannot be lowered by design. These calculations are summarized below in Figure 1-99.

- ◆ Assume that $I_T = 100\mu\text{A}$, $C_P = 2\text{pF}$

$$g_m = \frac{I_C}{V_T} = \frac{50\mu\text{A}}{26\text{mV}} = \frac{1}{520\Omega}$$

$$f_u = \frac{g_m}{2\pi C_P} = 153\text{MHz}$$

- ◆ Slew Rate = SR =

BUT FOR 2V PEAK-PEAK OUTPUT ($A = 1\text{V}$)

$$\text{FPBW} = \frac{\text{SR}}{2\pi A} = 4\text{MHz}$$

- ◆ Must increase I_T to 4mA to get FPBW = 160MHz!!

- ◆ Reduce g_m by adding emitter degeneration resistors

Figure 1-99: VFB op amp bandwidth and slew rate calculations

In practice, the FPBW of the op amp should be approximately 5 to 10 times the maximum output frequency in order to achieve acceptable distortion performance (typically 55-80dBc @ 5 to 20MHz, but actual system requirements vary widely).

Notice, however, that increasing the tail current causes a proportional increase in g_m and hence f_u . In order to prevent possible instability due to the large increase in f_u , g_m can be reduced by inserting resistors in series with the emitters of Q1 and Q2 (this technique, called *emitter degeneration*, also serves to linearize the g_m transfer function and thus also lowers distortion).

This analysis points out that a major inefficiency of conventional bipolar voltage feedback op amps is their inability to achieve high slew rates without proportional increases in quiescent current (assuming that C_P is fixed, and has a reasonable minimum value of 2 or 3pF).

This of course is not meant to say that high speed op amps designed using this architecture are deficient, just that there are circuit design techniques available which allow equivalent performance at much lower quiescent currents. This is extremely important in portable battery operated equipment where every milliwatt of power dissipation is critical.

VFB Op Amps Designed on Complementary Bipolar Processes

With the advent of complementary bipolar (CB) processes having high quality PNP transistors as well as NPNs, VFB op amp configurations such as the one shown in the simplified diagram in Figure 1-100 below became popular.

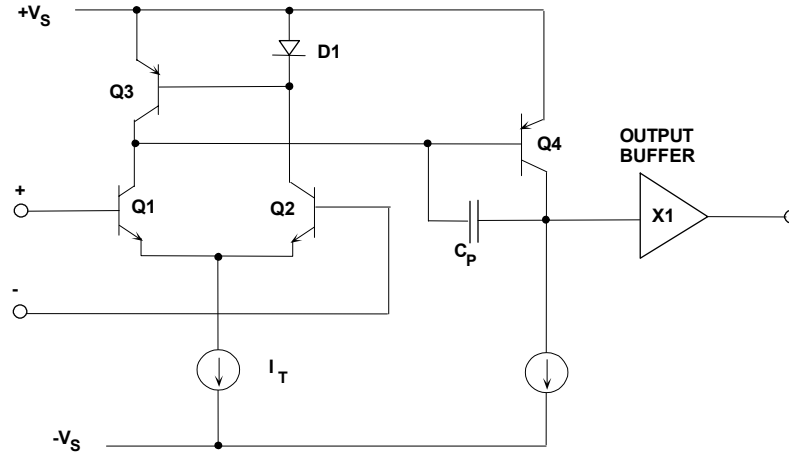


Figure 1-100: VFB op amp using two gain stages

Notice that the input differential pair (Q1, Q2) is loaded by a current mirror (Q3 and D1). We show D1 as a diode for simplicity, but it is actually a diode-connected PNP transistor (matched to Q3) with the base and collector connected to each other. This simplification will be used in many of the circuit diagrams to follow in this section. The common emitter transistor, Q4, provides a *second* voltage gain stage.

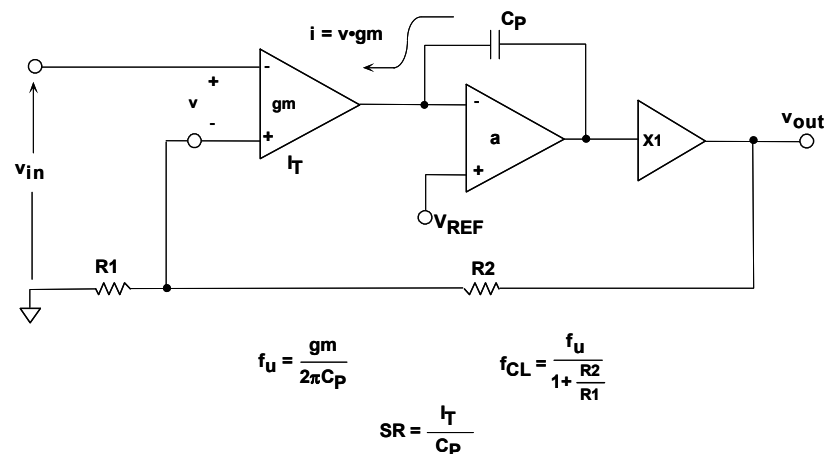


Figure 1-101: Model for two stage VFB op amp

Since the PNP transistors are fabricated on a complementary bipolar process, they are high quality and matched to the NPNs, and therefore suitable for voltage gain. The dominant pole of the Fig. 1-100 amplifier is set by C_P , and the combination of the gain stage, Q4 and local feedback capacitor C_P is often referred to as a *Miller Integrator*. The unity-gain output buffer is usually a complementary emitter follower.

A model for this two-stage VFB op amp is shown in Figure 1-101 above. Notice that the unity gain-bandwidth frequency, f_u , is still determined by the input stage g_m and the

■ OP AMP APPLICATIONS

dominant pole capacitance, C_P . The second gain stage increases the DC open-loop gain, but maximum slew rate is still limited by the input stage tail current as: $SR = I_T/C_P$.

A two-stage amplifier topology such as this is widely used throughout the IC industry in VFB op amps, both precision and high speed. It can be recalled that a similar topology with a dual FET input stage was used in the early high speed, fast settling FET modular op amps (see Chapter H: "Op Amp History").

Another popular VFB op amp architecture is the *folded cascode* as shown in Figure 1-102 below. An industry-standard video amplifier family (the AD847) is based on this architecture. This circuit also takes advantage of the fast PNPs available on a CB process. The differential signal currents in the collectors of Q1 and Q2 are fed to the emitters of a PNP cascode transistor pair (hence the term *folded cascode*). The collectors of Q3 and Q4 are loaded with the current mirror, D1 and Q5, and voltage gain is developed at the Q4-Q5 node. This single-stage architecture uses the junction capacitance at the high-impedance node for compensation (C_{STRAY}). Some variations of the design bring this node to an external pin so that additional external capacitance can be added if desired.

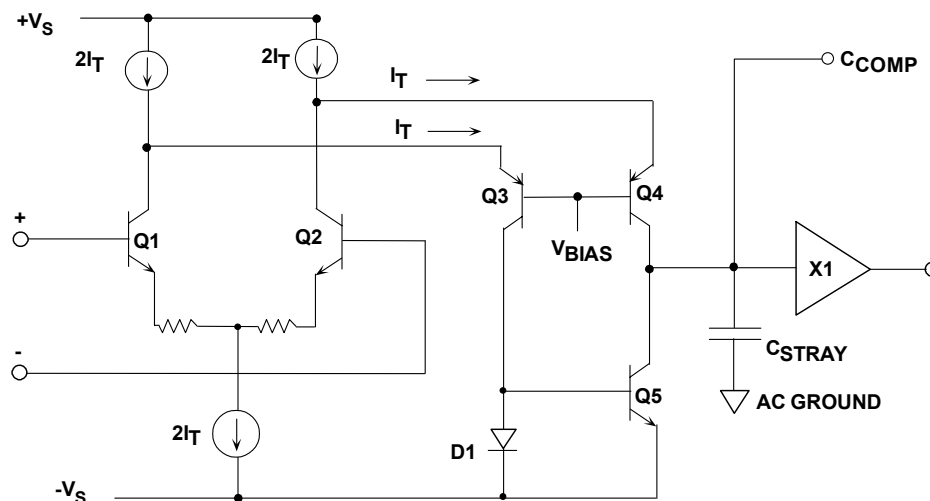


Figure 1-102: AD847-family folded cascode simplified circuit

With no emitter degeneration resistors in Q1 and Q2, and no additional external compensating capacitance, this circuit is only stable for high closed-loop gains. However, unity-gain compensated versions of this family are available which have the appropriate amount of emitter degeneration.

The availability of JFETs on a CB process allows not only low input bias current but also improvements in the slew rate tradeoff, which must be made between g_m and I_T found in bipolar input stages. Figure 1-103 (opposite) shows a simplified diagram of the AD845 16MHz op amp. JFETs have a much lower g_m per mA of tail current than a bipolar transistor. This lower g_m of the FET allows the input tail current (hence the slew rate) to be increased, without having to increase C_P to maintain stability.

The unusual thing about this seemingly poor performance of the JFET is that it is exactly what is needed for a fast, high SR input stage. For a typical JFET, the value of g_m is approximately $I_S/1V$ (I_S is the source current), rather than $I_C/26mV$ for a bipolar transistor,

i.e., the FET g_m is about 40 times lower. This allows much higher tail currents (and higher slew rates) for a given g_m when JFETs are used as the input stage.

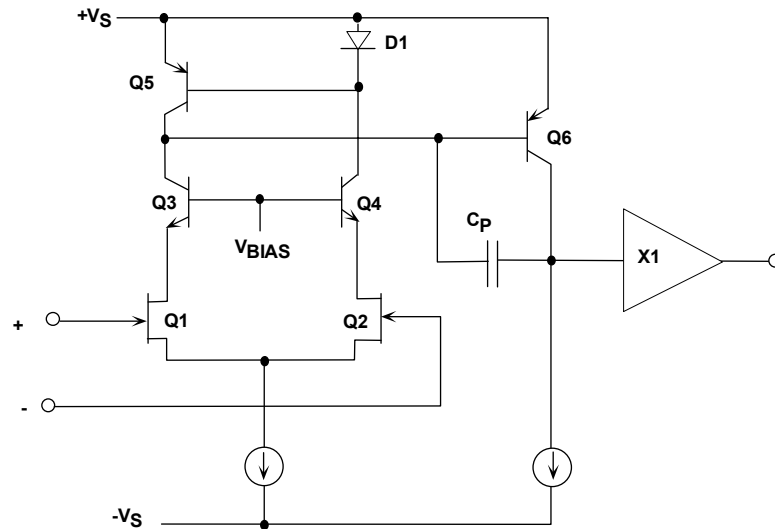


Figure 1-103: AD845 BiFET 16MHz op amp simplified circuit

A New VFB Op Amp Architecture for "Current-on-Demand" Performance, Lower Power, and Improved Slew Rate

Until recently, op amp designers had to make the above tradeoffs between the input g_m stage quiescent current and the slew-rate and distortion performance. ADI has patented a circuit core which supplies *current-on-demand*, to charge and discharge the dominant pole capacitor, C_p , while allowing the quiescent current to be small. The additional current is proportional to the fast slewing input signal and adds to the quiescent current.

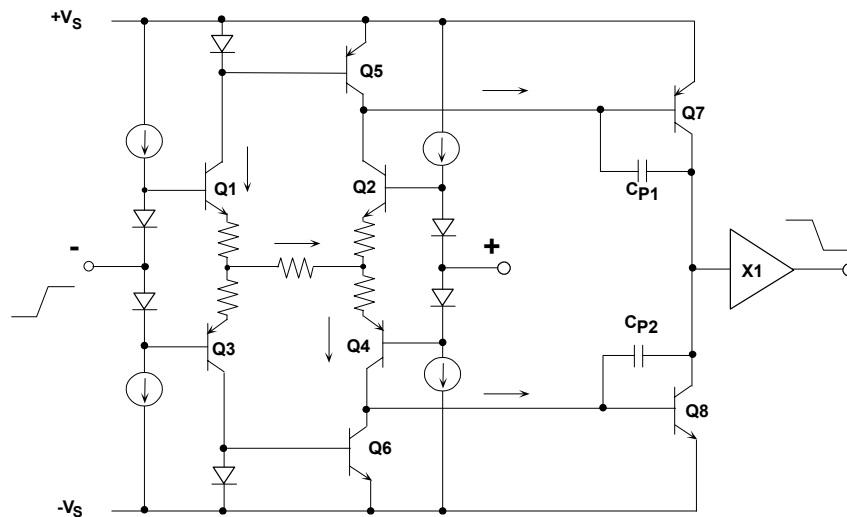


Figure 1-104: "Quad-Core" VFB g_m stage for current-on-demand

A simplified diagram of the basic core cell is shown in Figure 1-104 above. The *quad-core* (g_m stage) consists of transistors Q1, Q2, Q3, and Q4 with their emitters connected together as shown. Consider a positive step voltage on the inverting input. This voltage produces a proportional current in Q1 that is mirrored into C_{p1} by Q5. The current through Q1 also flows through Q4 and C_{p2} .

■ OP AMP APPLICATIONS

At the dynamic range limit, Q2 and Q3 are correspondingly turned off. Notice that the charging and discharging current for C_{P1} and C_{P2} is not limited by the quad core bias current. In practice, however, small current-limiting resistors are required forming an "H" resistor network as shown. Q7 and Q8 form the second gain stage (driven differentially from the collectors of Q5 and Q6), and the output is buffered by a unity-gain complementary emitter follower (X1).

The quad core configuration is patented (see Reference 1), as well as the circuits that establish the quiescent bias currents (not shown in Fig. 1-104). A number of new VFB op amps using this proprietary configuration have been released and have unsurpassed high frequency low distortion performance, bandwidth, and slew rate at the indicated quiescent current levels as shown in Figure 1-105 below.

LISTED IN ORDER OF DECREASING SUPPLY CURRENT

PART #	I_{SY} / AMP	BANDWIDTH	SLEWRATE	DISTORTION
AD9631/32 (1)	17mA	320MHz	1300V/ μ s	-72dBc@20MHz
AD8074/75 (3)	8mA	600MHz	1600V/ μ s	-62dBc@20MHz
AD8047/48 (1)	5.8mA	250MHz	750V/ μ s	-66dBc@5MHz
AD8041 (1)	5.2mA	160MHz	160V/ μ s	-69dBc@10MHz
AD8042 (2)	5.2mA	160MHz	200V/ μ s	-64dBc@10MHz
AD8044 (3)	2.8mA	150MHz	150V/ μ s	-75dBc@5MHz
AD8039 (2)	1.5mA	300MHz	425V/ μ s	-65dBc@5MHz
AD8031 (1)	0.75mA	80MHz	30V/ μ s	-62dBc@1MHz
AD8032 (2)	0.75mA	80MHz	30V/ μ s	-72dBc@1MHz

Number in () indicates single, dual, triple, or quad

Figure 1-105: High speed VFB op amps

The AD9631, AD8074, and AD8047 are optimized for a gain of +1, while the AD9632, AD8075, and AD8048 for a gain of +2.

The same quad-core architecture is used as the *second* stage of the AD8041 rail-to-rail output, zero-volt input single-supply op amp. The input stage is a differential PNP pair which allows the input common-mode signal to go about 200mV below the negative supply rail. The AD8042 and AD8044 are dual and quad versions of the AD8041.

Current Feedback (CFB) Op Amps

We will now examine in more detail the current feedback (CFB) op amp topology which is very popular in high speed op amps. As mentioned previously, the circuit concepts were introduced decades ago, however modern high speed complementary bipolar processes are required to take full advantage of the architecture.

It has long been known that in bipolar transistor circuits, currents can be switched faster than voltages, other things being equal. This forms the basis of non-saturating emitter-coupled logic (ECL) and devices such as current-output DACs. Maintaining low impedances at the current switching nodes helps to minimize the effects of stray capacitance, one of the largest detriments to high speed operation. The current mirror is a good example of how currents can be switched with a minimum amount of delay.

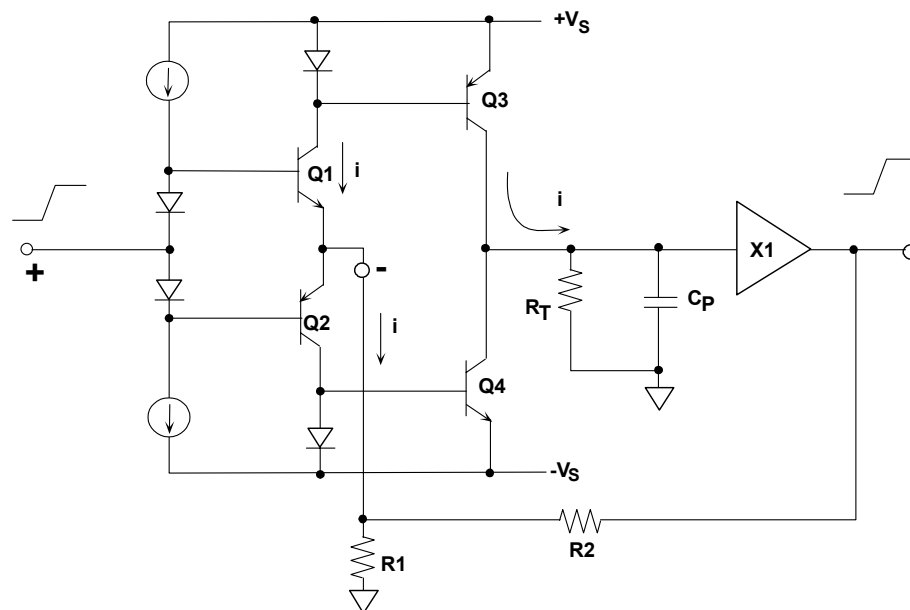


Figure 1-106: Simplified current feedback (CFB) op amp

The current feedback op amp topology is simply an application of these fundamental principles of current steering. A simplified CFB op amp is shown in Figure 1-106 above. The non-inverting input is high impedance and is buffered directly to the inverting input through the complementary emitter follower buffers Q1 and Q2. Note that the inverting input impedance is very low (typically 10 to 100 Ω), because of the low emitter resistance (ideally, would be zero). This is a fundamental difference between a CFB and a VFB op amp, and also a feature that gives the CFB op amp some unique advantages.

The collector outputs of Q1 and Q2 drive current mirrors, which mirror the inverting input current to the high impedance node, modeled by R_T and C_P . The high impedance node is buffered by a complementary unity gain emitter follower. Feedback from the output to the inverting input acts to force the inverting input *current* to zero, hence the term *Current Feedback*. Note that in a ideal case, for zero inverting input impedance, no small signal voltage can exist at this node, only small-signal current.

■ OP AMP APPLICATIONS

Now, consider a positive step voltage applied to the non-inverting input of the CFB op amp. Q1 immediately sources a proportional current into the external feedback resistors creating an *error* current, which is mirrored to the high impedance node by Q3. The voltage developed at the high impedance node is equal to this current multiplied by the equivalent impedance. This is where the term *transimpedance op amp* originated, since the transfer function is an impedance, rather than a unitless voltage ratio as in a traditional VFB op amp.

Note also that the error current delivered to the high impedance node is not limited by the input stage tail current. In other words, unlike a conventional VFB op amp, *there is no slew-rate limitation in an ideal CFB op amp*. The current mirrors supply *current-on-demand* from the power supplies. The negative feedback loop then forces the output voltage to a value that reduces the inverting input error current to zero.

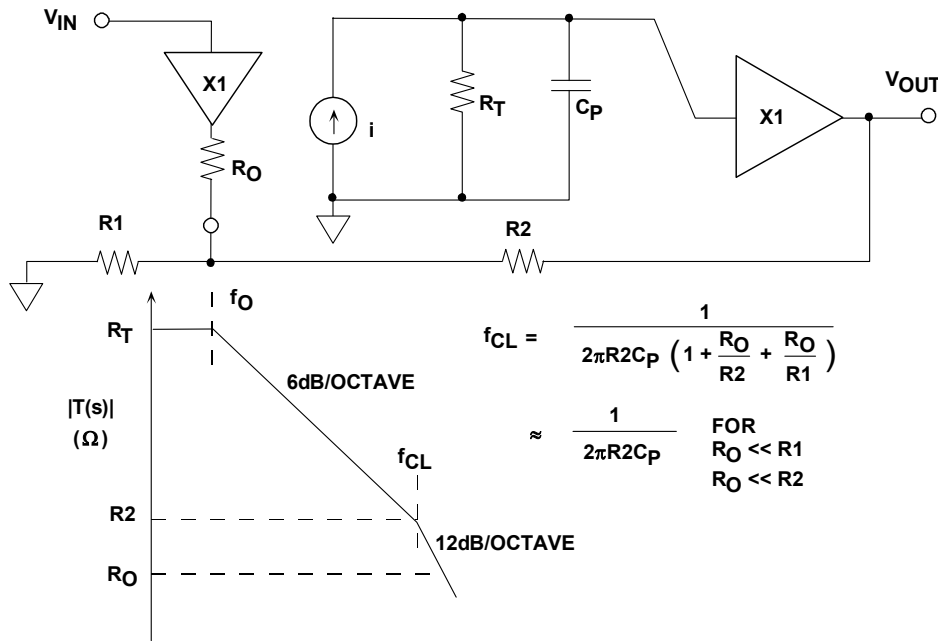


Figure 1-107: CFB op amp model and Bode plot

The model for a CFB op amp is shown in Figure 1-107 above, along with the corresponding Bode plot. The Bode plot is plotted on a log-log scale, and the open-loop gain is expressed as a transimpedance, $T(s)$, with units of ohms.

The finite output impedance of the input buffer is modeled by R_O . The input error current is i . By applying the principles of negative feedback, we can derive the expression for the op amp transfer function:

$$\frac{v_{out}}{v_{in}} = \frac{1 + \frac{R_2}{R_1}}{1 + j\omega C_P R_2 \left(1 + \frac{R_O}{R_2} + \frac{R_O}{R_1}\right)} \quad \text{Eq. 1-46}$$

At the op amp 3dB closed-loop bandwidth frequency, f_{cl} , the following is true:

$$2\pi f_{cl} C_P R_2 \left(1 + \frac{R_o}{R_2} + \frac{R_o}{R_1} \right) = 1. \quad \text{Eq. 1-47}$$

Solving for f_{cl} :

$$f_{cl} = \frac{1}{2\pi C_P R_2 \left(1 + \frac{R_o}{R_2} + \frac{R_o}{R_1} \right)}. \quad \text{Eq. 1-48}$$

For the condition $R_o \ll R_2$ and R_1 , the equation simply reduces to:

$$f_{cl} = \frac{1}{2\pi C_P R_2}. \quad \text{Eq. 1-49}$$

Examination of this equation quickly reveals that *the closed-loop bandwidth of a CFB op amp is determined by the internal dominant pole capacitor, C_P , and the external feedback resistor R_2 , and is independent of the gain-setting resistor, R_1 .* This ability to maintain constant bandwidth independent of gain makes CFB op amps ideally suited for wideband programmable gain amplifiers.

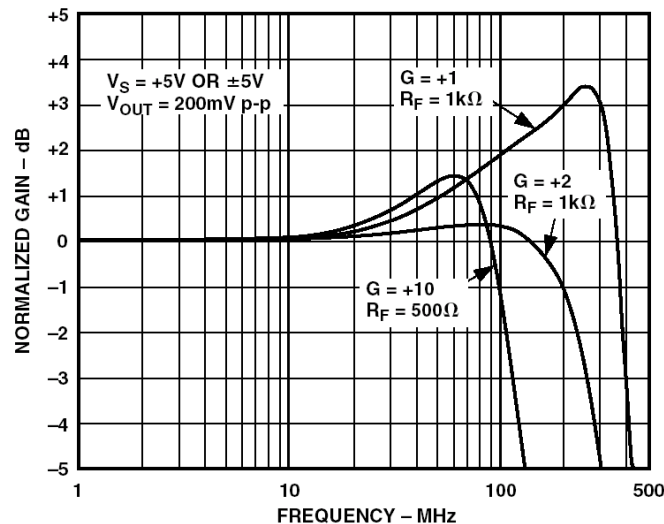


Figure 1-108: AD8011 frequency response, $G = +1, +2, +10$

Because the closed-loop bandwidth is inversely proportional to the external feedback resistor, R_2 , a CFB op amp is usually optimized for a specific R_2 . Increasing R_2 from its optimum value lowers the bandwidth, and decreasing it may lead to oscillation and instability because of high frequency parasitic poles.

The frequency response of the AD8011 CFB op amp is shown in Figure 1-108 above for various closed-loop values of gain (+1, +2, and +10). Note that even at a gain of +10, the closed-loop bandwidth is still greater than 100MHz. The peaking which occurs at a gain of +1 is typical of wideband CFB op amps used in the non-inverting mode, and is due primarily to stray capacitance at the inverting input. This peaking can be reduced by sacrificing bandwidth, by using a slightly larger feedback resistor.

■ OP AMP APPLICATIONS

The AD8011 CFB op amp (introduced in 1995) still represents state-of-the-art performance, and key specifications are shown in Figure 1-109 below.

- ◆ 1mA Power Supply Current (+5V or $\pm 5V$)
- ◆ 300MHz Bandwidth ($G = +1$)
- ◆ 2000 V/ μ s Slew Rate
- ◆ 29ns Settling Time to 0.1%
- ◆ Video Specifications ($G = +2$)
 - Differential Gain Error 0.02%
 - Differential Phase Error 0.06°
 - 25MHz 0.1dB Bandwidth
- ◆ Distortion
 - 70dBc @ 5MHz
 - 62dBc @ 20MHz
- ◆ Fully Specified for $\pm 5V$ or +5V Operation

Figure 1-109: AD8011 key specifications

Traditional current feedback op amps have been limited to a single gain stage, using current-mirrors. The AD8011 (and also others in this family) unlike traditional CFB op amps, use a *two-stage gain configuration*, as shown in Figure 1-110 below.

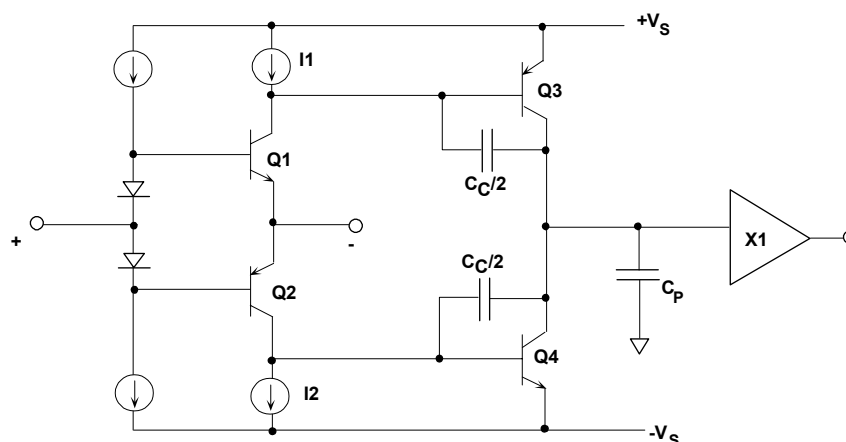


Figure 1-110: Simplified two-stage current feedback op amp

Until the advent of the AD8011, fully complementary two-gain stage CFB op amps had been impractical because of their high power dissipation. The AD8011 employs a patented (see Reference 2) second gain stage consisting of a pair of complementary amplifiers (Q3 and Q4). Note that they are not connected as current mirrors but as grounded-emitter gain stages. The detailed design of current sources (I1 and I2), and their respective bias circuits are the key to the success of the two-stage CFB circuit; they keep the amplifier's quiescent power low, yet are capable of supplying *current-on-demand* for wide current excursions required during fast slewing.

A further advantage of the two-stage amplifier is the higher overall bandwidth (for the same power), which means lower signal distortion and the ability to drive heavier external loads.

Thus far, we have learned several key features of CFB op amps. The most important is that for a given complementary bipolar IC process, *CFB generally yields higher FPBW (hence lower distortion) than VFB for the same amount of quiescent supply current*. This is because there is practically no slew-rate limiting in CFB. Because of this, the full power bandwidth and the small signal bandwidth are approximately the same.

LISTED IN ORDER OF DECREASING SUPPLY CURRENT

PART #	I _{sy} / AMP	BANDWIDTH	SLEWRATE	DISTORTION
AD8009 (1)	14mA	1000MHz	5500V/μs	-80dBc@5MHz
AD8023 (3)	10mA	250MHz	1200V/μs	ΔG=0.06%, Δφ=0.02°
AD8001 (1)	5.0mA	600MHz	1200V/μs	-65dBc@5MHz
AD8002 (2)	5.0mA	600MHz	1200V/μs	-65dBc@5MHz
AD8004 (4)	3.5mA	250MHz	3000V/μs	-78dBc@5MHz
AD8013 (3)	4.0mA	140MHz	1000V/μs	ΔG=0.02%, Δφ=0.06°
AD8072 (2)	3.5mA	100MHz	500V/μs	ΔG=0.05%, Δφ=0.1°
AD8073 (3)	3.5mA	100MHz	500V/μs	ΔG=0.05%, Δφ=0.1°
AD8012 (2)	1.7mA	350MHz	2250V/μs	-66dBc@5MHz
AD8014 (1)	1.2mA	400MHz	4000V/μs	-70dBc@5MHz
AD8011 (1)	1.0mA	300MHz	5500V/μs	-70dBc@5MHz
AD8005 (1)	0.4mA	270MHz	1500/μs	-53dBc@5MHz

Number in () Indicates Single, Dual, Triple, or Quad

Figure 1-111: Performance of selected CFB op amps

The second important feature is that the *inverting input impedance of a CFB op amp is very low*. This is advantageous when using the op amp in the inverting mode as an I/V converter, because there is less sensitivity to inverting input capacitance than with VFB.

The third feature is that *the closed-loop bandwidth of a CFB op amp is determined by the value of the internal C_p capacitor and the external feedback resistor R₂ and is relatively independent of the gain-setting resistor R₁*.

The performance for a selected group of current feedback op amps is shown in Figure 1-111 above. Note that the op amps are listed in order of decreasing power supply current.

- ◆ **CFB yields higher FPBW and lower distortion than VFB for the same process and power dissipation**
- ◆ **Inverting input impedance of a CFB op amp is low, non-inverting input impedance is high**
- ◆ **Closed-loop bandwidth of a CFB op amp is determined by the internal dominant-pole capacitance and the external feedback resistor, independent of the gain-setting resistor**

Figure 1-112: Summary: CFB op amps

Figure 1-112 above summarizes the general characteristics of CFB op amps.

■ OP AMP APPLICATIONS

Effects of Feedback Capacitance in Op Amps

It is quite common to use a capacitor in the feedback loop of a VFB op amp, to shape the frequency response as in a simple single-pole lowpass filter shown in Figure 1-113A below. The resulting noise gain is plotted on a Bode plot to analyze stability and phase margin. Stability of the system is determined by the net slope of the noise gain and the open-loop gain where they intersect.

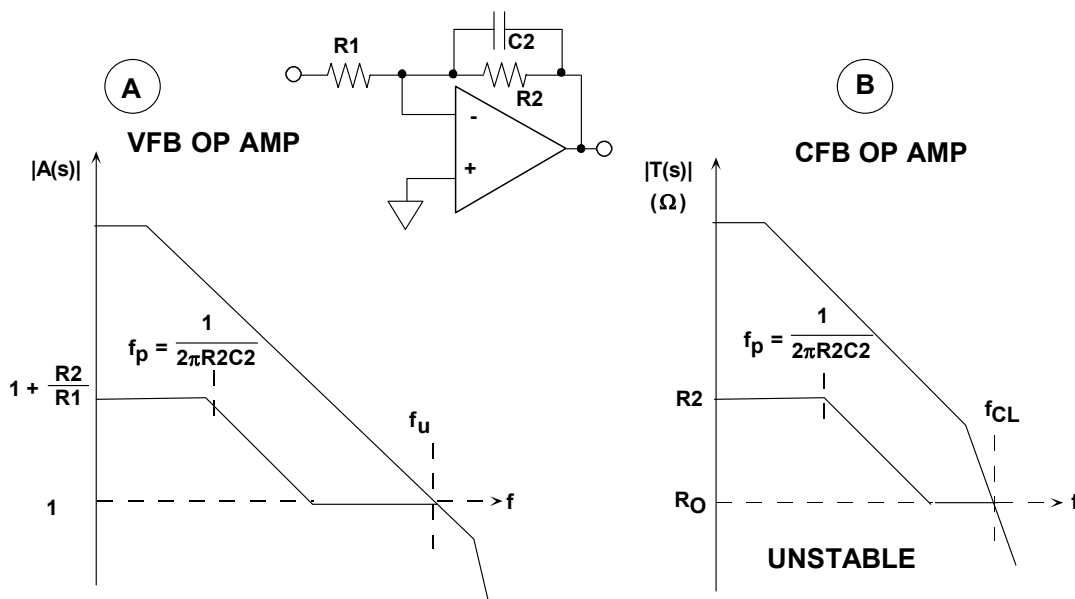


Figure 1-113: Noise gain stability analysis for VFB and CFB op amps with feedback capacitor

For unconditional stability, the noise gain plot must intersect the open-loop response with a net slope of less than 12dB/octave. In this case, the net slope where they intersect is 6dB/octave, indicating a stable condition. Notice for the case drawn in Fig. 1-113A, the second pole in the frequency response occurs at a considerably higher frequency than f_u .

In the case of the CFB op amp (Fig. 1-113B), the same analysis is used, except that the open-loop transimpedance gain, $T(s)$, is used to construct the Bode plot.

The definition of *noise gain* (for the purposes of stability analysis) for a CFB op amp, however, must be redefined in terms of a *current* noise source attached to the inverting input as shown in Figure 1-114 on the following page. This current is reflected to the output by an impedance, which we define to be the "current noise gain" of a CFB op amp:

$$\text{"CURRENT NOISE GAIN"} \equiv R_o + Z_2 \left(1 + \frac{R_o}{Z_1} \right). \quad \text{Eq. 1-50}$$

Now, return to Fig. 1-113B, and observe the CFB *current noise gain* plot. At low frequencies, the CFB current noise gain is simply R_2 (making the assumption that R_O is much less than Z_1 or Z_2). The first pole is determined by R_2 and C_2 . As the frequency continues to increase, C_2 becomes a short circuit, and all the inverting input current flows through R_O (again refer to Fig. 1-114).

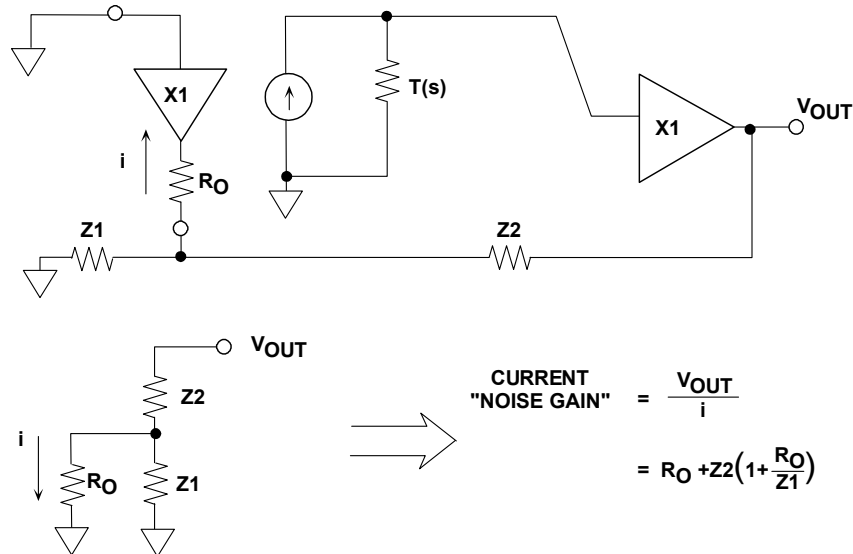


Figure 1-114: Current "noise gain" definition for CFB op amp for use in stability analysis

A CFB op amp is normally optimized for best performance for a fixed feedback resistor, R_2 . Additional poles in the transimpedance gain, $T(s)$, occur at frequencies above the closed-loop bandwidth, f_{cl} , (set by R_2). Note that the intersection of the CFB current noise gain with the open-loop $T(s)$ occurs where the slope of the $T(s)$ function is 12dB/octave. This indicates instability and possible oscillation.

It is for this reason that *CFB op amps are not suitable in configurations which require capacitance in the feedback loop*, such as simple active integrators or lowpass filters.

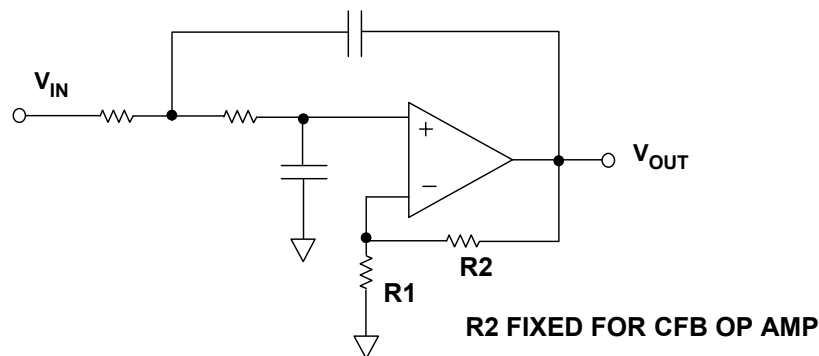


Figure 1-115: The Sallen-Key filter configuration

They can, however, be used in certain active filters such as the Sallen-Key configuration shown in Figure 1-115 above, which do not require capacitance in the feedback network.

■ OP AMP APPLICATIONS

On the other hand, VFB op amps, do make very flexible active filters. A multiple feedback 20MHz lowpass filter example using an AD8048 op amp is shown below in Figure 1-116.

In general, an active filter amplifier should have a bandwidth that is at least ten times the bandwidth of the filter, if problems due to phase shift of the amplifier are to be avoided. (The AD8048 has a bandwidth of over 200MHz in this configuration).

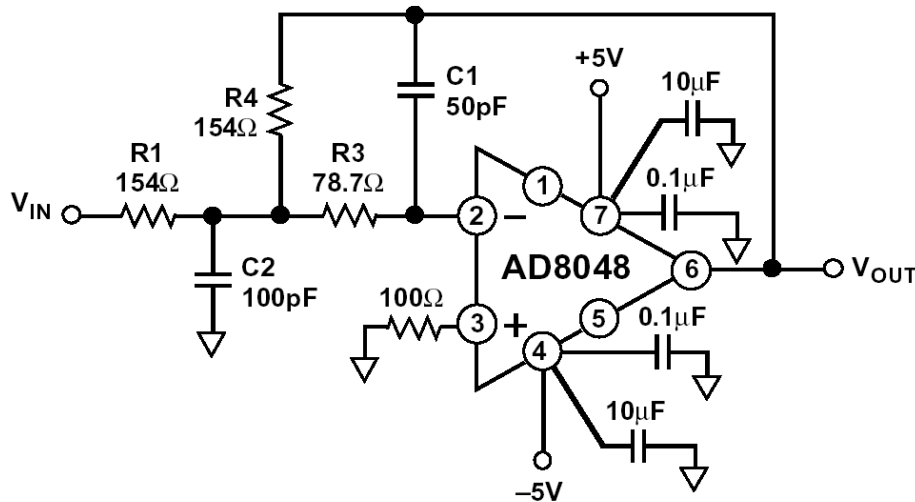


Figure 1-116: Multiple feedback 20MHz lowpass filter using the AD8048 VFB op amp

Design details of the filter design can be found on the AD8048 data sheet. Further discussions on active filter design are included in Chapter 5 of this book.

High Speed Current-to-Voltage Converters, and the Effects of Inverting Input Capacitance

Fast op amps are useful as current-to-voltage converters in such applications as high speed photodiode preamplifiers and current-output DAC buffers. A typical application using a VFB op amp as an I/V converter is shown in Figure 1-117 on the following page.

The net input capacitance, $C1$, forms a pole at a frequency f_p in the noise gain transfer function as shown in the Bode plot, and is given by:

$$f_p = \frac{1}{2\pi R_2 C_1} \quad \text{Eq. 1-51}$$

If left uncompensated, the phase shift at the frequency of intersection, f_x , will cause instability and oscillation. Introducing a zero at f_x by adding feedback capacitor $C2$ stabilizes the circuit and yields a phase margin of about 45° .

The location of the zero is given by:

$$f_x = \frac{1}{2\pi R_2 C_2} \quad \text{Eq. 1-52}$$

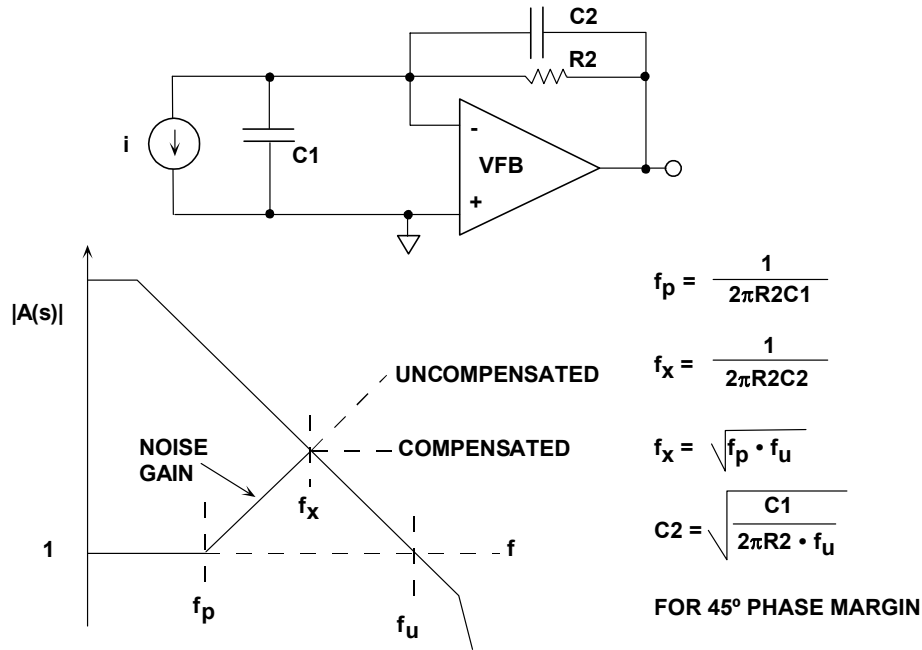


Figure 1-117: Compensating for input capacitance in a current-to-voltage converter using VFB op amp

Although the addition of C_2 actually decreases the pole frequency slightly, this effect is negligible if $C_2 \ll C_1$. The frequency f_x is the geometric mean of f_p and the unity-gain bandwidth frequency of the op amp, f_u ,

$$f_x = \sqrt{f_p \cdot f_u} \quad \text{Eq. 1-53}$$

Combining Eq. 1-52 and Eq. 1-53 and solving for C_2 yields:

$$C_2 = \sqrt{\frac{C_1}{2\pi R_2 \cdot f_u}} \quad \text{Eq. 1-54}$$

This value of C_2 will yield a phase margin of about 45°. Increasing the capacitor by a factor of 2 increases the phase margin to about 65° (see Reference 3).

In practice, the optimum value of C_2 may be optimized experimentally by varying it slightly, to optimize the output pulse response.

■ OP AMP APPLICATIONS

A similar analysis can be applied to a CFB op amp as shown in Figure 1-118 below. In this case, however, the low inverting input impedance, R_O , greatly reduces the sensitivity to input capacitance. In fact, an ideal CFB with zero input impedance would be totally insensitive to any amount of input capacitance!

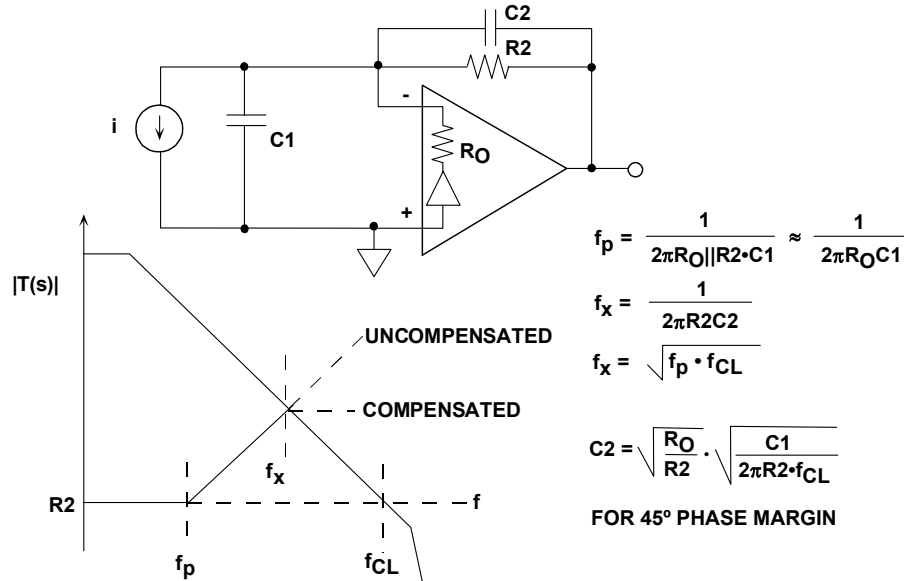


Figure 1-118: Current-to-voltage converter using a CFB op amp

The pole caused by C_1 occurs at a frequency f_p :

$$f_p = \frac{1}{2\pi(R_O || R_2)C_1} \approx \frac{1}{2\pi R_O C_1} \quad \text{Eq. 1-55}$$

This pole frequency will generally be much higher than the case for a VFB op amp, and the pole can be ignored completely if it occurs at a frequency greater than the closed-loop bandwidth of the op amp.

We next introduce a compensating zero at the frequency f_x by inserting the capacitor C_2 :

$$f_x = \frac{1}{2\pi R_2 C_2} \quad \text{Eq. 1-56}$$

As in the case for VFB, f_x is the geometric mean of f_p and f_{cl} :

$$f_x = \sqrt{f_p \cdot f_{cl}} \quad \text{Eq. 1-57}$$

Combining Eq. 1-56 and Eq. 1-57 and solving for C_2 yields:

$$C_2 = \sqrt{\frac{R_O}{R_2}} \cdot \sqrt{\frac{C_1}{2\pi R_2 \cdot f_{cl}}} \quad \text{Eq. 1-58}$$

There is a significant advantage in using a CFB op amp in this configuration as can be seen by comparing Eq. 1-58 with the similar equation for C_2 required for a VFB op amp, Eq. 1-54. If the unity-gain bandwidth product of the VFB is equal to the closed-loop bandwidth of the CFB (at the optimum R_2), then the size of the CFB compensation capacitor, C_2 , is reduced by a factor of $\sqrt{(R_2/R_O)}$.

A comparison in an actual application is shown in Figure 1-119 below. The full scale output current of the DAC is 4mA, the net capacitance at the inverting input of the op amp is 20pF, and the feedback resistor is 500Ω. In the case of the VFB op amp, the pole due to C1 occurs at 16MHz. A compensating capacitor of 5.6pF is required for 45° of phase margin, and the signal bandwidth is 57MHz.

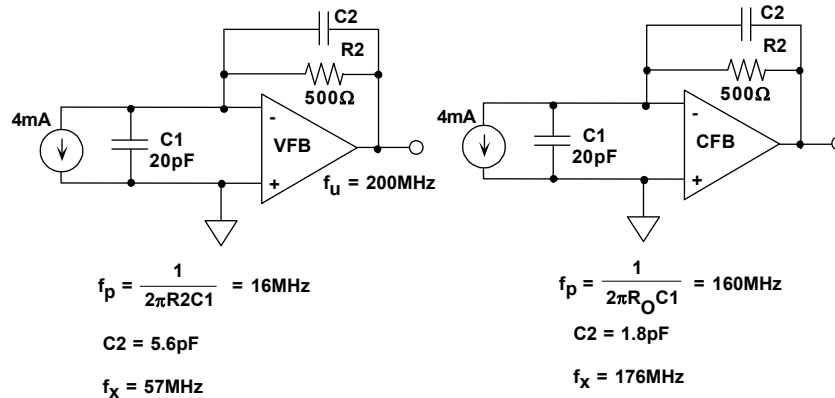


Figure 1-119: CFB op amp is relatively insensitive to input capacitance when used as an I/V converter

For the CFB op amp, however, because of the low inverting input impedance ($R_O = 50\Omega$), the pole occurs at 160MHz, the required compensation capacitor is about 1.8pF, and the corresponding signal bandwidth is 176MHz. In practice, the pole frequency is so close to the closed-loop bandwidth of the op amp that it could probably be left uncompensated.

It should be noted that a CFB op amp's relative insensitivity to inverting input capacitance is when it is used in the inverting mode. In the non-inverting mode, however, even a few picofarads of stray capacitance on the inverting input can cause significant gain-peaking and potential instability.

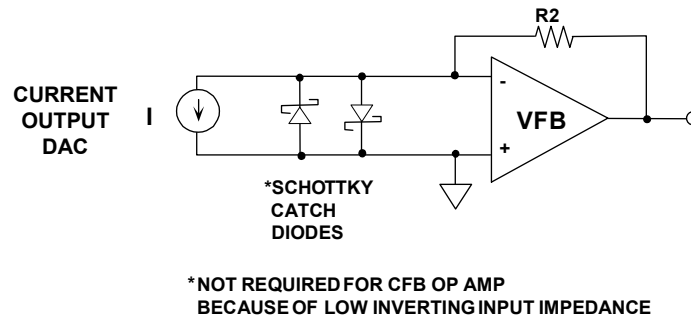


Figure 1-120: Low inverting input impedance of CFB op amp helps reduce effects of fast DAC transients

Another advantage of the low inverting input impedance of the CFB op amp is when it is used as an I/V converter to buffer the output of a high speed current output DAC. When a step function current (or DAC switching glitch) is applied to the inverting input of a VFB op amp, it can produce a large voltage transient until the signal can propagate through the op amp to its output and negative feedback is regained. Back-to-back Schottky diodes are often used to limit this voltage swing as shown in Figure 1-120 above. These diodes must be low capacitance, small geometry devices because their capacitance adds to the total input capacitance.

■ OP AMP APPLICATIONS

A CFB op amp, on the other hand, presents a low impedance (R_O) to fast switching currents even before the feedback loop is closed, thereby limiting the voltage excursion without the requirement of the external diodes. This greatly improves the settling time of the I/V converter.

Noise Comparisons between VFB and CFB Op Amps

In most applications of high speed op amps, it is the total output RMS noise that is generally of interest. Because of the high bandwidths involved, the chief contributor to the output RMS noise is therefore the white noise, and the $1/f$ noise is negligible.

Typical high speed op amps with bandwidths greater than 150MHz or so, and bipolar VFB input stages have input voltage noises ranging from about 2 to 20nV/ $\sqrt{\text{Hz}}$.

For a VFB op amp, the inverting and non-inverting input current noise are typically equal, and almost always uncorrelated. Typical values for wideband VFB op amps range from 0.5pA/ $\sqrt{\text{Hz}}$ to 5pA/ $\sqrt{\text{Hz}}$. The input current noise of a bipolar input stage is increased when input bias-current compensation generators are added, because their current noise is not correlated, and therefore adds (in an RSS manner) to the intrinsic current noise of the bipolar stage. However, bias current compensation is rarely used in high speed op amps.

The input voltage noise in CFB op amps tends to be lower than for VFB op amps having the same approximate bandwidth. This is because the input stage in a CFB op amp is usually operated at a higher current, thereby reducing the emitter resistance and hence the voltage noise. Typical values for CFB op amps range from about 1 to 5nV/ $\sqrt{\text{Hz}}$.

The input current noise of CFB op amps tends to be larger than for VFB op amps because of the generally higher bias current levels. The inverting and non-inverting current noise of a CFB op amp is usually different because of the unique input architecture, and are specified separately. In most cases, the inverting input current noise is the larger of the two. Typical input current noise for CFB op amps ranges from 5 to 40pA/ $\sqrt{\text{Hz}}$. This can often be dominant, except in cases of very high gain, when R_1 is small.

The noise sources which dominate the output noise are highly dependent on the closed-loop gain of the op amp and the values of the feedback and feedforward resistors. For high values of closed-loop gain, the op amp voltage noise will tend to be the chief contributor to the output noise. At low gains, the effects of the input current noise must also be considered, and may dominate, especially in the case of a CFB op amp.

Feedforward/feedback resistors in high speed op amp circuits may range from less than 100 Ω to more than 1k Ω , so it is difficult to generalize about their contribution to the total output noise without knowing the specific values and the closed-loop gain.

The best way to make the noise calculations is to write a simple computer program that performs the calculations automatically, and include all the noise sources. The equation previously discussed can be used for this purpose (see Fig. 1-74, again). In most high speed op amp applications, the source impedance noise can often be neglected for source impedances of 100Ω or less.

- ◆ **Voltage Feedback Op Amps:**
 - **Voltage Noise:** 2 to $20\text{nV}/\sqrt{\text{Hz}}$
 - **Current Noise:** 0.5 to $5\text{pA}/\sqrt{\text{Hz}}$
- ◆ **Current Feedback Op Amps:**
 - **Voltage Noise:** 1 to $5\text{nV}/\sqrt{\text{Hz}}$
 - **Current Noise:** 5 to $40\text{pA}/\sqrt{\text{Hz}}$
- ◆ **Noise Contribution from Source Negligible if $< 100\Omega$**
- ◆ **Voltage Noise Usually Dominates at High Gains**
- ◆ **Reflect Noise Sources to Output and Combine (RSS)**
- ◆ **Errors Will Result if there is Significant High Frequency Peaking**

Figure 1-121: High speed op amp noise summary

Figure 1-121 above summarizes the noise characteristics of high speed op amps

DC Characteristics of High Speed Op Amps

High speed op amps are optimized for bandwidth and settling time, not for precision DC characteristics as found in lower frequency precision op amps. In spite of this, however, high speed op amps do have reasonably good DC performance.

Input offset voltages of high speed bipolar input op amps are rarely trimmed, since offset voltage matching of the input stage is excellent, typically ranging from 1 to 3mV , with offset temperature coefficients of 5 to $15\mu\text{V}/^\circ\text{C}$.

Input bias currents on VFB op amps (with no input bias current compensation circuits) are approximately equal for (+) and (–) inputs, and can range from 1 to $5\mu\text{A}$. The output offset voltage due to the input bias currents can be nulled by making the effective source resistance, R_3 , equal to the parallel combination of R_1 and R_2 .

As previously discussed, this scheme will not work with bias-current compensated VFB op amps which have additional current generators on their inputs. In this case, the net input bias currents are not necessarily equal or of the same polarity.

■ OP AMP APPLICATIONS

CFB op amps generally have unequal and uncorrelated input bias currents because the (+) and (–) inputs have completely different architectures. For this reason, external bias current cancellation schemes are also ineffective. CFB input bias currents range from 5 to 15 μ A, being generally higher at the inverting input.

◆ **High Speed Bipolar Op Amp Input Offset Voltage:**

- **Ranges from 1 to 3mV for VFB and CFB**
- **Offset TC Ranges from 5 to 15 μ V/°C**

◆ **High Speed Bipolar Op Amp Input Bias Current:**

- **For VFB Ranges from 1 to 5 μ A**
- **For CFB Ranges from 5 to 15 μ A**

◆ **Bias Current Cancellation Doesn't Work for:**

- **Bias Current Compensated Op Amps**
- **Current Feedback Op Amps**

Figure 1-122: High speed op amp offset voltage summary

Figure 1-122 above summarizes the offset considerations for high speed op amps.

REFERENCES: HIGH SPEED OP AMPS

1. Roy Gosser, "Wide-Band Transconductance Generator," **US Patent 5,150,074**, filed May 3, 1991, issued September 22, 1992.
2. Roy Gosser, DC-Coupled Transimpedance Amplifier, **US Patent 4,970,470**, filed October 10, 1989, issued November 13, 1990.
3. James L. Melsa and Donald G. Schultz, **Linear Control Systems**, McGraw-Hill, 1969, pp. 196-220, ISBN: 0-07-041481-5
4. Thomas M. Frederiksen, **Intuitive Operational Amplifiers**, McGraw-Hill, 1988., ISBN: 0-07-021966-4
5. Sergio Franco, **Design With Operational Amplifiers and Analog Integrated Circuits, 2nd Ed.**, McGraw-Hill, 1998, ISBN: 0-07-021857-9
6. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6, (available for download at <http://www.analog.com>).
7. Data sheet for **AD8011 300 MHz, 1 mA Current Feedback Amplifier**, <http://www.analog.com>

NOTES:

- H Op Amp History**
 - 1 Op Amp Basics**
 - ◆ 2 Specialty Amplifiers**
 - 1 Instrumentation Amplifiers**
 - 2 Programmable Gain Amplifiers**
 - 3 Isolation Amplifiers**
 - 3 Using Op Amps with Data Converters**
 - 4 Sensor Signal Conditioning**
 - 5 Analog Filters**
 - 6 Signal Amplifiers**
 - 7 Hardware and Housekeeping Techniques**

▣ OP AMP APPLICATIONS

CHAPTER 2: SPECIALTY AMPLIFIERS

Walt Kester, Walt Jung, James Bryant

This chapter of the book discusses several popular types of *specialty* amplifiers, or amplifiers that are based in some way on op amp techniques. However, in an overall applications sense, they are not generally used as universally as op amps. Examples of specialty amplifiers include instrumentation amplifiers of various configurations, programmable gain amplifiers (PGAs), isolation amplifiers, and difference amplifiers.

Other types of amplifiers, for example such types as audio and video amplifiers, cable drivers, high-speed variable gain amplifiers (VGAs), and various communications-related amplifiers might also be viewed as specialty amplifiers. However these applications are more suitably covered in Chapter 6, within the various signal amplification sections.

SECTION 2-1: INSTRUMENTATION AMPLIFIERS

Walt Kester, Walt Jung

Probably the most popular among all of the specialty amplifiers is the *instrumentation amplifier* (hereafter called simply an *in-amp*). The in-amp is widely used in many industrial and measurement applications where DC precision and gain accuracy must be maintained within a noisy environment, and where large common-mode signals (usually at the AC power line frequency) are present.

Op amp/In-Amp Functionality Differences

An in-amp is unlike an op amp in a number of very important ways. As already discussed, an op amp is a general-purpose gain block— user-configurable in myriad ways using external feedback components of R, C, and, (sometimes) L. The final configuration and circuit function using an op amp is truly whatever you make of it.

In contrast to this, an in-amp is a more constrained device in terms of functioning, and also the allowable range(s) of operating gain. And, in many ways, it is better suited to its task than would be an op amp— even though, ironically, an in-amp may actually be composed of a number of op amps within it! People also often confuse in-amps as to their function, calling them "op amps". But the converse is seldom (if ever) true. It should be understood that an in-amp is *not* just a special type op amp; the function of the two devices is actually fundamentally different.

Perhaps a good way to differentiate the two devices is to remember that an op amp can be programmed to do almost anything, by virtue of its feedback flexibility. In contrast to this, an in-amp *cannot* be programmed to do just anything. It can *only* be programmed for gain, and then over a specific range. An op amp is configured via a number of external components, while an in-amp is configured by either one resistor, or by pin-selectable taps for its working gain.

In-Amp Definitions

An in-amp is a *precision* closed-loop gain block. It has a pair of differential input terminals, and a single-ended output that works with respect to a reference or common terminal, as shown in Figure 2-1 below. The input impedances are balanced and high in value, typically $\geq 10^9 \Omega$. Again, unlike an op amp, an in-amp uses an internal feedback resistor network, plus one (usually) gain set resistance, R_G . Also unlike an op amp is the fact that the internal resistance network and R_G are *isolated* from the signal input terminals. In-amp gain can also be preset via an internal R_G by pin selection, (again isolated from the signal inputs). Typical in-amp gains range from 1 to 1,000.

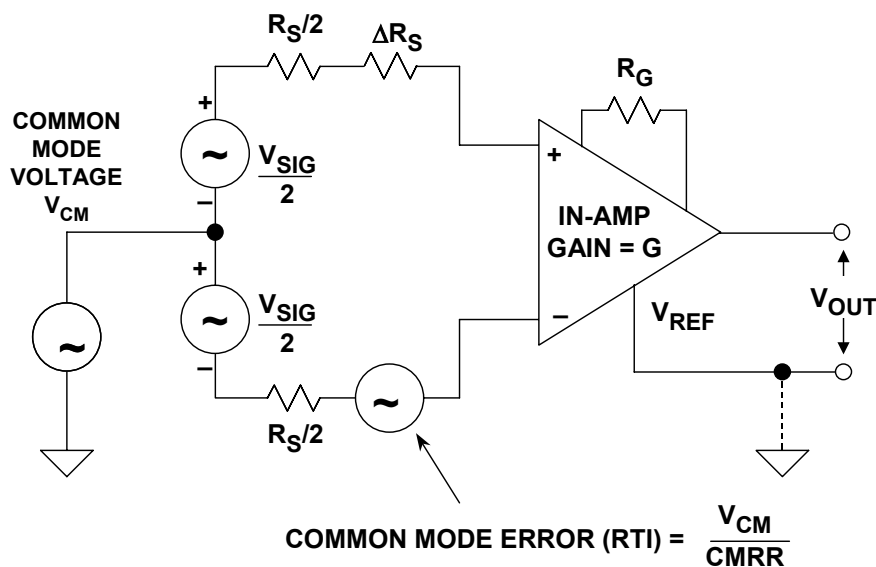


Figure 2-1: The generic instrumentation amplifier (in-amp)

The in-amp develops an output voltage which is referenced to a pin usually designated REFERENCE, or V_{REF} . In many applications, this pin is connected to circuit ground, but it can be connected to other voltages, as long as they lie within a rated compliance range. This feature is especially useful in single-supply applications, where the output voltage is usually referenced to mid-supply (i.e., +2.5V in the case of a +5V supply).

In order to be effective, an in-amp needs to be able to amplify microvolt-level signals, while simultaneously rejecting volts of *common mode* (CM) signal at its inputs. This requires that in-amps have very high *common mode rejection* (CMR). Typical values of in-amp CMR are from 70 to over 100dB, with CMR usually improving at higher gains.

It is important to note that a CMR specification for DC inputs alone isn't sufficient in most practical applications. In industrial applications, the most common cause of external interference is 50/60Hz AC power-related noise (including harmonics). In differential measurements, this type of interference tends to be induced equally onto both in-amp inputs, so the interference appears as a CM input signal. Therefore, specifying CMR over frequency is just as important as specifying its DC value. Note that imbalance in the two source impedances can degrade the CMR of some in-amps. Analog Devices fully specifies in-amp CMR at 50/60Hz, with a source impedance imbalance of 1k Ω .

Subtractor or Difference Amplifiers

A simple subtractor or difference amplifier can be constructed with four resistors and an op amp, as shown in Figure 2-2 below. It should be noted that this is *not* a true in-amp (based on the previously discussed criteria), but it is often used in applications where a simple differential to single-ended conversion is required. Because of its popularity, this circuit will be examined in more detail, in order to understand its fundamental limitations before discussing true in-amp architectures.

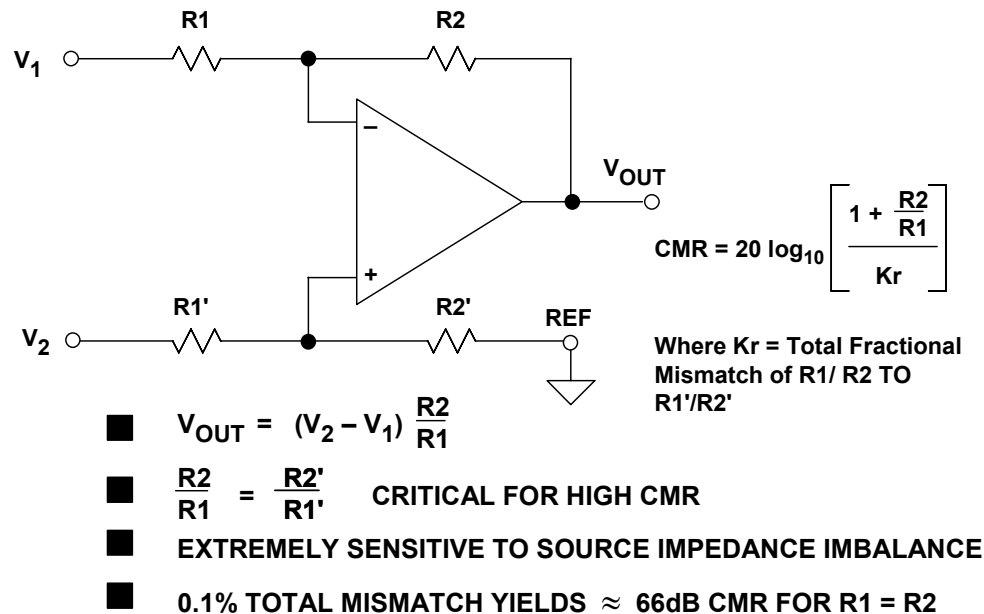


Figure 2-2: Op amp subtractor or difference amplifier

There are several fundamental problems with this simple circuit. First, the input impedance seen by V_1 and V_2 isn't balanced. The input impedance seen by V_1 is $R1$, but the input impedance seen by V_2 is $R1' + R2'$. The configuration can also be quite problematic in terms of CMR, since even a small source impedance imbalance will degrade the workable CMR. This problem can be solved with well-matched open-loop buffers in series with each input (for example, using a precision dual op amp). But, this adds complexity to a simple circuit, and may introduce offset drift and non-linearity.

The second problem with this circuit is that the *CMR is primarily determined by the resistor ratio matching*, not the op amp. The resistor ratios $R1/R2$ and $R1'/R2'$ must match extremely well to reject common mode noise—at least as well as a typical op amp CMR of $\geq 100\text{dB}$. Note also that the *absolute* resistor values are relatively unimportant.

Picking four 1% resistors from a single batch may yield a net ratio matching of 0.1%, which will achieve a CMR of 66dB (assuming $R1 = R2$). But if one resistor differs from the rest by 1%, the CMR will drop to only 46dB. Clearly, very limited performance is possible using ordinary discrete resistors in this circuit (without resorting to hand matching). This is because the best standard off-the-shelf RNC/RNR style resistor tolerances are on the order of 0.1% (see Reference 1).

■ OP AMP APPLICATIONS

In general, the worst case CMR for a circuit of this type is given by the following equation (see References 2 and 3):

$$\text{CMR(dB)} = 20 \log \left[\frac{1 + R2/R1}{4K_r} \right], \quad \text{Eq. 2-1}$$

where K_r is the *individual* resistor tolerance in fractional form, for the case where 4 discrete resistors are used. This equation shows that the worst case CMR for a tolerance build-up for 4 unselected same-nominal-value 1% resistors to be no better than 34dB.

A single resistor network with a net matching tolerance of K_r would probably be used for this circuit, in which case the expression would be as noted in the figure, or:

$$\text{CMR(dB)} = 20 \log \left[\frac{1 + R2/R1}{K_r} \right] \quad \text{Eq. 2-2}$$

A net matching tolerance of 0.1% in the resistor ratios therefore yields a worst case DC CMR of 66dB using Equation 2-2, and assuming $R1 = R2$. Note that either case assumes a significantly higher amplifier CMR (i.e., >100dB). Clearly for high CMR, such circuits need four single-substrate resistors, with very high absolute and TC matching. Such networks using thick/thin-film technology are available from companies such as Caddock and Vishay, in ratio matches of 0.01% or better.

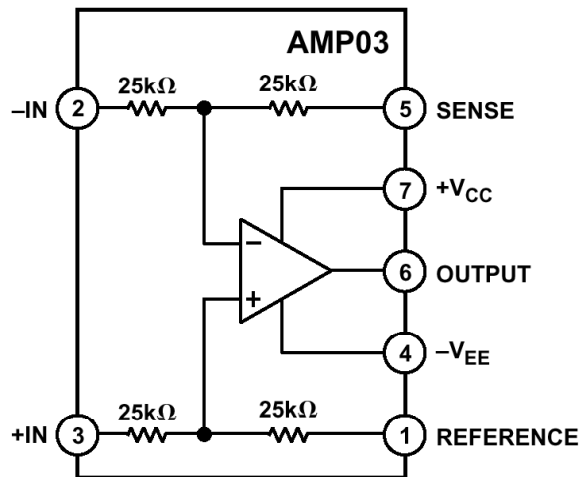


Figure 2-3: AMP03 precision difference amplifier

In implementing the simple difference amplifier, rather than incurring the higher costs and PCB real estate limitations of a precision op amp plus a separate resistor network, it is usually better to seek out a completely monolithic solution. The AMP03 is just such a precision difference amplifier, which includes an on-chip laser trimmed precision thin film resistor network. It is shown in Figure 2-3 above. The typical CMR of the AMP03F is 100dB, and the small-signal bandwidth is 3MHz.

There are several devices related to the AMP03 in function. These are namely the SSM2141 and SSM2143 difference amplifiers. These sister parts are designed for audio

line receivers (see Figure 2-4 below). They have low distortion, and high (pre-trimmed) CMR. The net gains of the SSM2141 and SSM2143 are unity and 0.5, respectively. They are designed to be used with balanced 600 Ω audio sources (see the related discussions on these devices in the Audio Amplifiers section of Chapter 6).

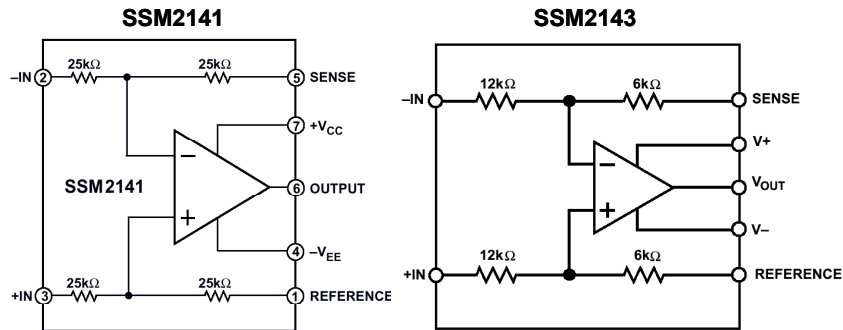


Figure 2-4: SSM2141 and SSM2143 difference amplifiers (audio line receivers)

Another interesting variation on the simple difference amplifier is found in the AD629 difference amplifier, optimized for high common-mode input voltages. A typical current-sensing application is shown in Figure 2-5 below. The AD629 is a differential-to-single-ended amplifier with a gain of unity. It can handle a common-mode voltage of $\pm 270\text{V}$ with supply voltages of $\pm 15\text{V}$, with a small signal bandwidth of 500kHz.

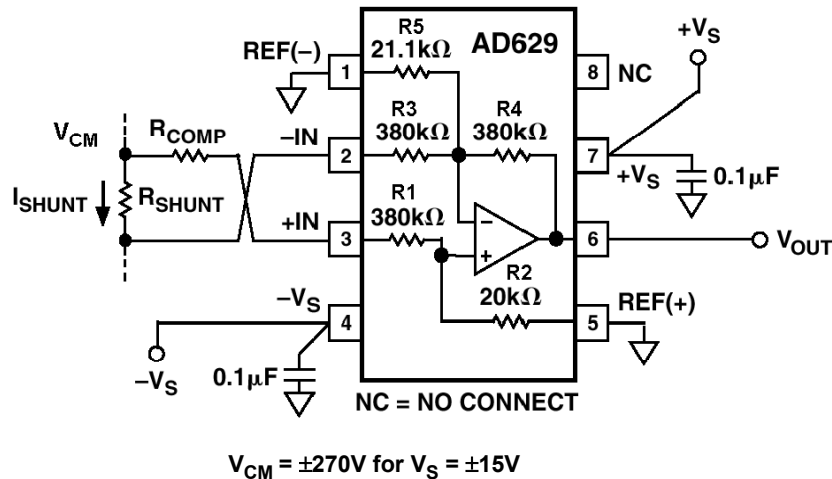


Figure 2-5: High common-mode current sensing using the AD629 difference amplifier

The high common-mode voltage range is obtained by attenuating the non-inverting input (pin 3) by a factor of 20 times, using the R1–R2 divider network. On the inverting input, resistor R5 is chosen such that $R5 \parallel R3$ equals resistor R2. The noise gain of the circuit is equal to $20 [1 + R4/(R3 \parallel R5)]$, thereby providing unity gain for differential input voltages. Laser wafer trimming of the R1–R5 thin film resistors yields a minimum CMR of 86dB @ 500Hz for the AD629B. Within an application, it is good practice to maintain balanced source impedances on both inputs, so dummy resistor R_{COMP} is chosen to equal the value of the shunt sensing resistor R_{SHUNT} .

OP AMP APPLICATIONS

David Birt (see Reference 4) of the BBC has analyzed the simple line receiver topology in terms of loading presented to the source, and presented a modified and balanced form, shown as Figure 2-6, below. Here stage U1 uses a 4 resistor network identical to that of Figure 2-2, while feedback from the added unity gain inverter U2 drives the previously grounded R2' reference terminal. This has two overall effects; the input currents in the $\pm$ input legs become equal in magnitude, and the gain of the stage is halved.

Compared to Fig. 2-2, and for like resistor ratios, the Fig. 2-6 gain from V_{IN} to V_{OUT} is $\frac{1}{2}$, or a gain of -6dB (0.5) as shown. However the new circuit form also offers a complementary output from U2, $-V_{OUT}$.

The common-mode range of this circuit is the same as for Fig. 2-2, but the CMR is about doubled with all resistors nominally equal (as measured to a single output). The inverter resistor ratio $R3/R4$ affects output balance, but not CMR. Like Fig. 2-2, the gain of this circuit is not easily changed, as it involves precise resistor ratios.

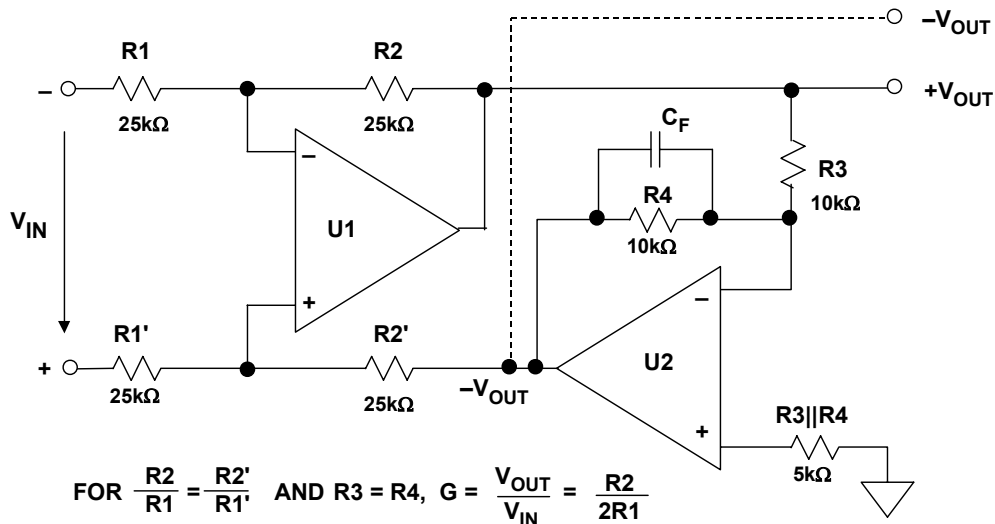


Figure 2-6: Balanced difference amplifier using push-pull feedback path

Because of the two feedback paths, this circuit holds the inputs of U1 at a null for differential input signals. However CM signals are seen by U1, and the CM range of the circuit is $[1+(R2'/R1')]\times V_{CM(U1)}$. Differential input resistance is $R1+R1'$.

As can be noted from Fig. 2-6, this circuit can be broken into a simple line receiver (left), plus an inverter (right). Thus existing line receivers like Fig. 2-2 can be converted to the fully balanced topology, by simply adding an appropriate inverter, U2. This of course not only balances the *input* currents, but it also provides a balanced *output* signal.

For example, the SSM2141 line receiver and the OP275 are a good combination for implementing this approach (see Reference 5, and the further discussions on these circuits in the Audio Amplifiers section of Chapter 6).

In-amp Configurations

The simple difference amplifier circuits described above are quite useful (especially at higher frequencies) but lack the performance required for most precision applications. In many cases, true in-amps are more suitable, because of their balanced and high input impedance, as well as their high common-mode rejection.

Two Op Amp In-Amps

As noted initially, in-amps are based on op amps, and there are two basic configurations that are extremely popular. The first is based on two op amps, and the second on three op amps. The circuit shown in Figure 2-7 is referred to as the *two op amp in-amp*. Dual IC op amps are used in most cases for good matching, such as the OP297 or the OP284. The resistors are usually a thin film laser trimmed array on the same chip. The in-amp gain can be easily set with an external resistor, R_G . Without R_G , the gain is simply $1 + R_2/R_1$. In a practical application, the R_2/R_1 ratio is chosen for the desired minimum in-amp gain.

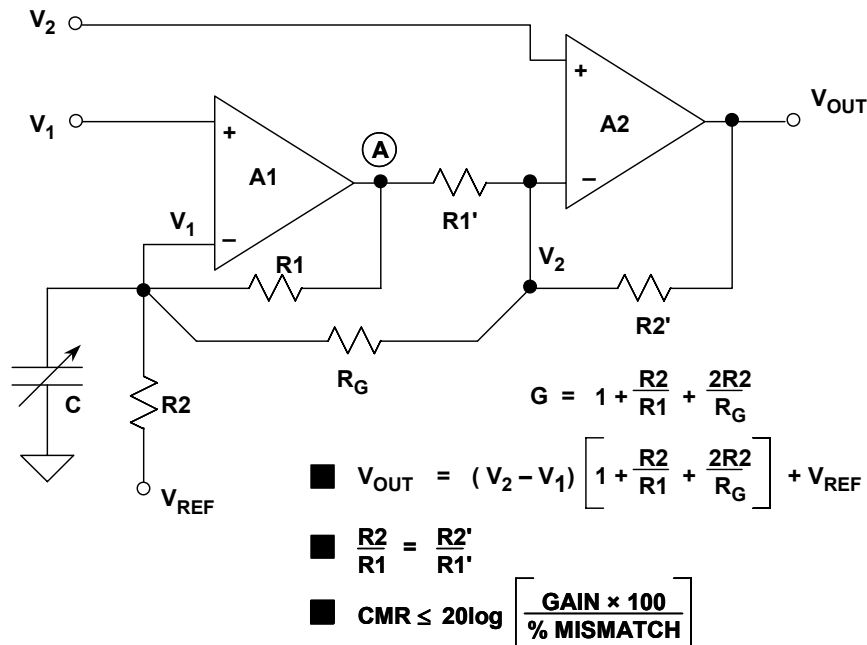


Figure 2-7: The two op amp instrumentation amplifier

The input impedance of the two op amp in-amp is inherently high, permitting the impedance of the signal sources to be high and unbalanced. The DC common mode rejection is limited by the matching of R_1/R_2 to R_1'/R_2' . If there is a mismatch in any of the four resistors, the DC common mode rejection is limited to:

$$CMR \leq 20 \log \left[\frac{GAIN \times 100}{\% \text{ MISMATCH}} \right]. \quad \text{Eq. 2-3}$$

Notice that the net CMR of the circuit increases proportionally with the working gain of the in-amp, an effective aid to high performance at higher gains.

OP AMP APPLICATIONS

IC in-amps are particularly well suited to meeting the combined needs of ratio matching and temperature tracking of the gain-setting resistors. While thin film resistors fabricated on silicon have an initial tolerance of up to $\pm 20\%$, laser trimming during production allows the ratio error between the resistors to be reduced to 0.01% (100ppm). Furthermore, the tracking between the temperature coefficients of the thin film resistors is inherently low and is typically less than 3ppm/ $^{\circ}\text{C}$ (0.0003%/ $^{\circ}\text{C}$).

When dual supplies are used, V_{REF} is normally connected directly to ground. In single supply applications, V_{REF} is usually connected to a low impedance voltage source equal to one-half the supply voltage. The gain from V_{REF} to node "A" is $R1/R2$, and the gain from node "A" to the output is $R2'/R1'$. This makes the gain from V_{REF} to the output equal to unity, assuming perfect ratio matching. Note that it is critical that the source impedance seen by V_{REF} be low, otherwise CMR will be degraded.

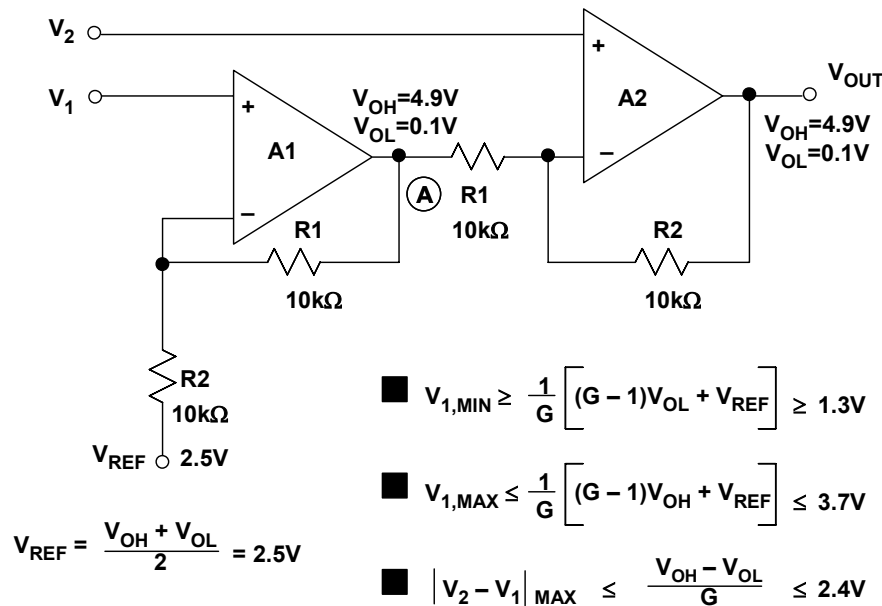


Figure 2-8: Two op amp in-amp single-supply restrictions for $V_s = +5\text{V}$, $G = 2$

One major disadvantage of the two op amp in-amp design is that common mode voltage input range must be traded off against gain. The amplifier A1 must amplify the signal at V_1 by $1 + R1/R2$. If $R1 \gg R2$ (a low gain example in Figure 2-7), A1 will saturate if the V_1 common mode signal is too high, leaving no A1 headroom to amplify the wanted differential signal. For high gains ($R1 \ll R2$), there is correspondingly more headroom at node "A", allowing larger common mode input voltages.

The AC common mode rejection of this configuration is generally poor because the signal path from V_1 to V_{OUT} has the additional phase shift of A1. In addition, the two amplifiers are operating at different closed-loop gains (and thus at different bandwidths). The use of a small trim capacitor "C" as shown in Fig. 2-7 can improve the AC CMR somewhat.

A low gain ($G = 2$) single supply two op amp in-amp configuration results when R_G is not used, and is shown above in Figure 2-8. The input common mode and differential signals must be limited to values which prevent saturation of either A1 or A2. In the example, the

op amps remain linear to within 0.1V of the supply rails, and their upper and lower output limits are designated V_{OH} and V_{OL} , respectively. These saturation voltage limits would be typical for a single-supply, rail-rail output op amp (such as the AD822, for example).

Using the Fig. 2-8 equations, the voltage at V_1 must fall between 1.3V and 2.4V to prevent A1 from saturating. Notice that V_{REF} is connected to the average of V_{OH} and V_{OL} (2.5V). This allows for bipolar differential input signals with V_{OUT} referenced to +2.5V.

A high gain ($G = 100$) single supply two op amp in-amp configuration is shown below in Figure 2-9. Using the same equations, note that voltage at V_1 can now swing between 0.124V and 4.876V. V_{REF} is again 2.5V, to allow for bipolar input and output signals.

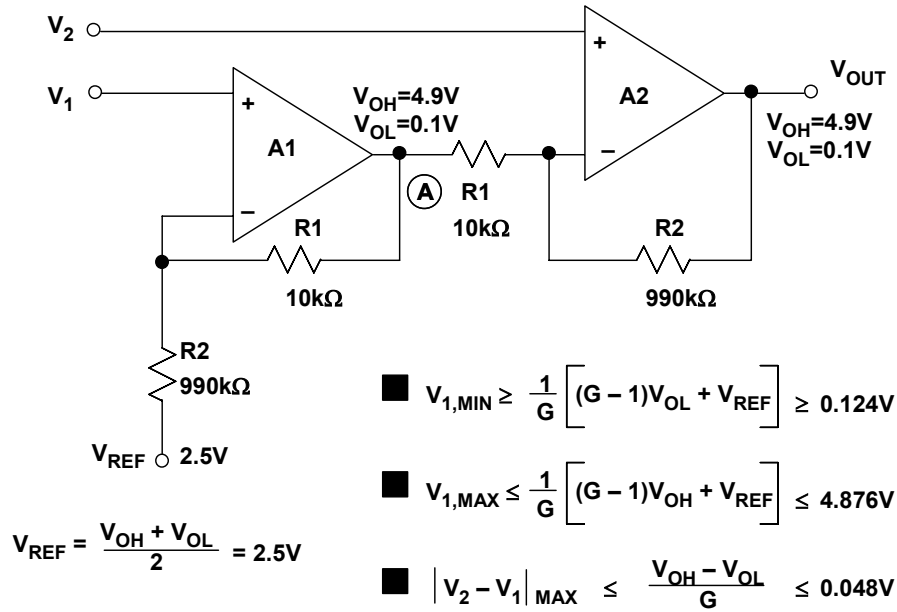


Figure 2-9: Two op amp in-amp single-supply restrictions for $V_s = +5V$, $G = 100$

All of these discussions show that the conventional two op amp in-amp architecture is fundamentally limited, when operating from a single power supply. These limitations can be viewed in one sense as a restraint on the allowable input CM range for a given gain. Or, alternately, it can be viewed as limitation on the allowable gain range, for a given CM input voltage.

Nevertheless, there are ample cases where a combination of gain and CM voltage cannot be supported by the basic two op amp structures of Figs. 2-7 through 2-9, even with perfect amplifiers (i.e., zero output saturation voltage to both rails).

In summary, regardless of gain, the basic structure of the common two op amp in-amp does not allow for CM input voltages of zero when operated on a single supply. The only route to removing these restrictions for single supply operation is to modify the in-amp architecture.

OP AMP APPLICATIONS

The AD627 Single-Supply Two Op Amp In-Amp

The above-mentioned CM limitations can be overcome with some key modifications to the basic two op amp in-amp architecture. These modifications are implemented in the circuit shown in Figure 2-10 below, which represents the AD627 in-amp architecture.

In this circuit, each of the two op amps is composed of a PNP common emitter input stage and a gain stage, designated Q1/A1, and Q2/A2, respectively. The PNP transistors not only provide gain but also level shift the input signal positive by about 0.5V, thereby allowing the common mode input voltage to go to 0.1V below the negative supply rail. The maximum positive input voltage allowed is 1V less than the positive supply rail.

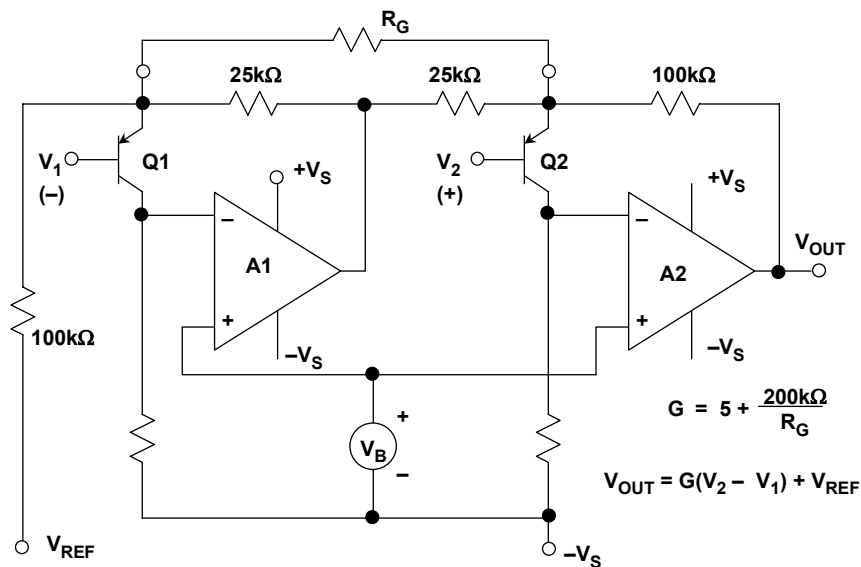


Figure 2-10: The AD627 in-amp architecture

The AD627 in-amp delivers rail-to-rail output swing, and operates over a wide supply voltage range (+2.7V to ±18V). Without the external gain setting resistor R_G , the in-amp gain is a minimum of 5. Gains up to 1000 can be set with the addition of this external resistor. Common mode rejection of the AD627B at 60Hz with a 1kΩ source imbalance is 85dB when operating on a single +3V supply and $G = 5$.

Even though the AD627 is a two op amp in-amp, it is worthwhile noting that it is not subject to the same CM frequency response limitations as the basic circuit of Fig. 2-7. A patented circuit keeps the AD627 CMR flat out to a much higher frequency than would otherwise be achievable with a conventional discrete two op amp in-amp.

The AD627 data sheet has a detailed discussion of allowable input/output voltage ranges as a function of gain and power supply voltages (see Reference 7). In addition, interactive design tools are available on the ADI web site, which perform calculations relating these parameters for a number of in-amps, including the AD627.

Key specifications for the AD627 are summarized in Figure 2-11 below. Although it has been designed as a low power, single-supply device, the AD627 is capable of operating on traditional higher voltage supplies such as $\pm 15\text{V}$, with excellent performance.

- ◆ Wide Supply Range : $+2.7\text{V}$ to $\pm 18\text{V}$
- ◆ Input Voltage Range: $-V_S - 0.1\text{V}$ to $+V_S - 1\text{V}$
- ◆ $85\mu\text{A}$ Supply Current
- ◆ Gain Range: 5 to 1000
- ◆ $75\mu\text{V}$ Maximum Input Offset Voltage (AD627B)
- ◆ $10\text{ppm}/^\circ\text{C}$ Maximum Offset Voltage TC (AD627B)
- ◆ 10ppm Gain Nonlinearity
- ◆ 85dB CMR @ 60Hz , $1\text{k}\Omega$ Source Imbalance ($G = 5$)
- ◆ $3\mu\text{V}$ p-p 0.1Hz to 10Hz Input Voltage Noise ($G = 5$)

Figure 2-11: AD627 in-amp key specifications

Three Op Amp In-Amps

A second popular in-amp architecture is based on three op amps, and is shown below in Figure 2-12. This circuit is typically referred to as the *three op amp in-amp*.

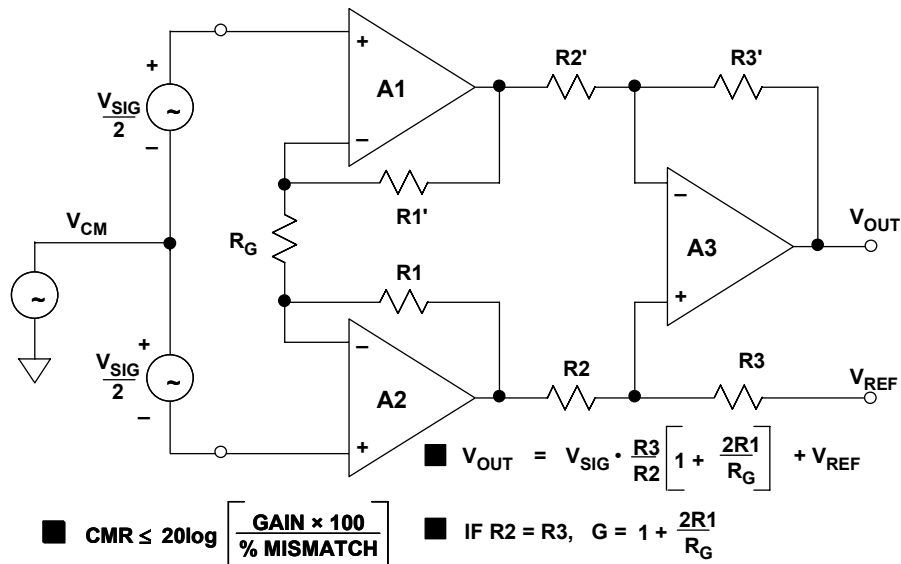


Figure 2-12: The three op amp in-amp

Resistor R_G sets the overall gain of this amplifier. It may be internal, external, or (software or pin-strap) programmable, depending upon the in-amp. In this configuration, CMR depends upon the ratio matching of $R3/R2$ to $R3'/R2'$. Furthermore, common mode signals are only amplified by a factor of 1 regardless of gain (no common mode voltage will appear across R_G , hence, no common mode current will flow in it because the input terminals of an op amp will have no significant potential difference between them).

OP AMP APPLICATIONS

As a result of the high ratio of differential to CM gain in A1-A2, CMR of this in-amp theoretically increases in proportion to gain. Large common mode signals (within the A1-A2 op amp headroom limits) may be handled at all gains. Finally, because of the symmetry of this configuration, common mode errors in the input amplifiers, if they track, tend to be canceled out by the subtractor output stage. These features explain the popularity of this three op amp in-amp configuration— it is capable of delivering the highest performance.

The classic three op amp configuration has been used in a number of monolithic IC in-amps (see References 8 and 9). Besides offering excellent matching between the three internal op amps, thin film laser trimmed resistors provide excellent ratio matching and gain accuracy at much lower cost than using discrete precision op amps and resistor networks. The AD620 (see Reference 10) is an excellent example of monolithic IC in-amp technology. A simplified device schematic is shown in Figure 2-13 below.

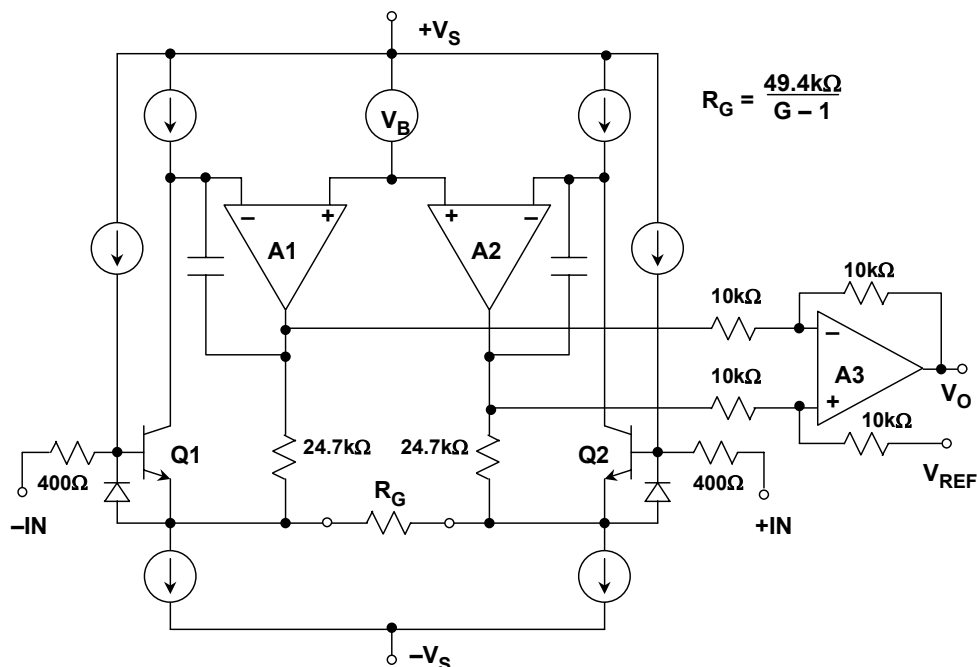


Figure 2-13: The AD620 in-amp simplified schematic

The AD620 is a highly popular in-amp and is specified for power supply voltages from $\pm 2.3\text{V}$ to $\pm 18\text{V}$. Input voltage noise is only $9\text{nV}/\sqrt{\text{Hz}}$ @ 1kHz . Maximum input bias current is only 1nA , due to the use of superbeta transistors for Q1-Q2.

Overvoltage protection is provided by the internal 400Ω thin-film current-limit resistors in conjunction with the diodes connected from the emitter-to-base of Q1 and Q2. The gain G is set with a single external R_G resistor, as noted by equation 2-4 below.

$$G = (49.4\text{k}\Omega/R_G) + 1 \quad \text{Eq. 2-4}$$

As can be noted from this expression and Fig. 2-13, the AD620 internal resistors are trimmed so that standard 1% or 0.1% resistors can be used to set gain to popular values.

As is true in the case of the two op amp in-amp configuration, single supply operation of the three op amp in-amp requires an understanding of the internal node voltages. Figure 2-14 below shows a generalized diagram of the in-amp operating on a single +5V supply. The maximum and minimum allowable output voltages of the individual op amps are designated V_{OH} (maximum high output) and V_{OL} (minimum low output) respectively.

Note that the gain from the common mode voltage to the outputs of A1 and A2 is unity. It can be stated that *the sum of the common mode voltage and the signal voltage at these outputs must fall within the amplifier output voltage range*. Obviously this configuration cannot handle input common mode voltages of either zero volts or +5V, because of saturation of A1 and A2. As in the case of the two op amp in-amp, the output reference is positioned halfway between V_{OH} and V_{OL} to allow for bipolar differential input signals.

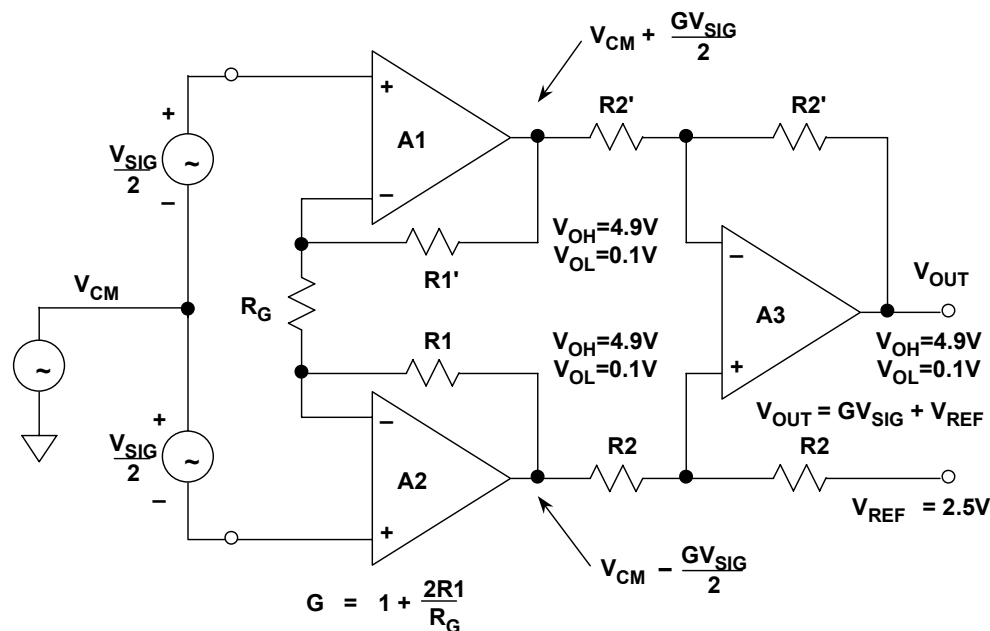


Figure 2-14: Three op amp in-amp single +5V supply restrictions

While there are a number of good single-supply in-amps, such as the AD627 discussed above, the highest performance devices are still among those specified for traditional dual-supply operation, i.e., the just-discussed AD620. For certain applications, even such devices as the AD620, which has been designed for dual supply operation, can be used with full precision on a single-supply power system.

Precision Single-Supply Composite In-Amp

One way to achieve both high precision and single-supply operation takes advantage of the fact that many popular sensors (e.g. strain gauges) provide an output signal which is inherently centered around an approximate mid-point of the supply voltage (and/or the reference voltage). Taking advantage of this basic point allows the inputs of a signal conditioning in-amp to be biased at "mid-supply". As a consequence of this step, the inputs needn't operate near ground or the positive supply voltage, and the in-amp can still be used with all its precision.

OP AMP APPLICATIONS

Under these conditions, an AD620 dual-supply in-amp referenced to the supply mid-point followed by an rail-to-rail op amp output gain stage provides very high DC precision. Figure 2-15 below illustrates one such high-performance in-amp, which operates on a single +5V supply.

This circuit uses the AD620 as a low-cost precision in-amp for the input stage, along with an AD822 JFET-input dual rail-to-rail output op amp for the output stage, comprised of A1 and A2. The output stage operates at a fixed gain of 3, with overall gain set by R_G .

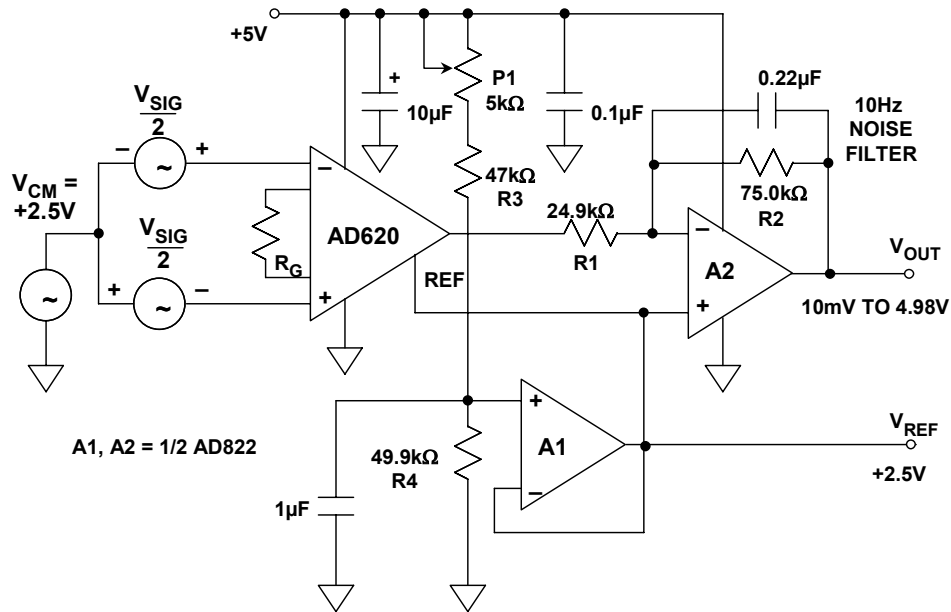


Figure 2-15: A precision single-supply composite in-amp with rail-to-rail output

In this circuit, R3 and R4 form a voltage divider which splits the supply voltage nominally in half to +2.5V, with fine adjustment provided by a trimming potentiometer, P1. This voltage is applied to the input of A1, an AD822 voltage follower, which buffers it and provides a low-impedance source needed to drive the AD620's reference pin as well as providing the output reference voltage V_{REF} . *Note that this feature allows a bipolar V_{OUT} to be measured with respect to this +2.5V reference (not to GND).* This is despite the fact that the entire circuit operates from a single (unipolar) supply.

The other half of the AD822 is connected as a gain-of-3 inverter, so that it can output $\pm 2.5V$, "rail-to-rail," with only $\pm 0.83V$ required of the AD620. This output voltage level of the AD620 is well within the AD620's capability, thus ensuring high linearity for the front end.

The general gain expression for this composite in-amp is the product of the gain of the AD620 stage, and the gain of inverting amplifier:

$$\text{GAIN} = \left(\frac{49.4\text{k}\Omega}{R_G} + 1 \right) \left(\frac{R_2}{R_1} \right). \quad \text{Eq. 2-5}$$

For this example, an overall gain of 10 is realized with $R_G = 21.5k\Omega$ (closest standard value). The table shown in Figure 2-16 below summarizes various R_G gain values, and the resulting performance for gains ranging from 10 to 1000.

In this application, the allowable input voltage on either input to the AD620 must lie between +2V and +3.5V in order to maintain linearity. For example, at an overall circuit gain of 10, the common mode input voltage range spans 2.25V to 3.25V, allowing room for the $\pm 0.25V$ full-scale differential input voltage required to drive the output $\pm 2.5V$ about V_{REF} .

CIRCUIT GAIN	R_G (Ω)	V_{OS}, RTI (μV)	TC V_{OS}, RTI ($\mu V/^{\circ}C$)	NONLINEARITY (ppm) *	BANDWIDTH (kHz)**
10	21.5k	1000	1000	< 50	600
30	5.49k	430	430	< 50	600
100	1.53k	215	215	< 50	300
300	499	150	150	< 50	120
1000	149	150	150	< 50	30

* Nonlinearity Measured Over Output Range: $0.1V < V_{OUT} < 4.90V$

** Without 10Hz Noise Filter

Figure 2-16: Performance summary of the +5V single-supply AD620/AD822 composite in-amp

The inverting configuration was chosen for the output buffer to facilitate system output offset voltage adjustment by summing currents into the A2 stage buffer's feedback summing node. These offset currents can be provided by an external DAC, or from a resistor connected to a reference voltage.

The AD822 rail-to-rail output stage exhibits a very clean transient response (not shown) and a small-signal bandwidth over 100kHz for gain configurations up to 300. Note that excellent linearity is maintained over 0.1V to 4.9V V_{OUT} .

To reduce the effects of unwanted noise pickup, a filter capacitor is recommended across A2's feedback resistance to limit the circuit bandwidth to the frequencies of interest. This capacitor forms a 1st order low pass filter with R2. The corner frequency is 10Hz as shown, but this may be easily modified. The capacitor should be a high quality film type, such as polypropylene.

■ OP AMP APPLICATIONS

The AD623 In-Amp

Like the two op amp in-amp counterparts discussed previously, three op amp in-amps require special design attention for wide CM range inputs on single power supplies. The AD623 single supply in-amp configuration (see Reference 11), shown below in Figure 2-17 offers an attractive solution. In this device PNP emitter follower level shifters Q1 and Q2 allow the input signal to go 150mV below the negative supply, and to within 1.5V of the positive supply. The AD623 is fully specified for both single power supplies between +3V and +12V, and dual supplies between $\pm 2.5\text{V}$ and $\pm 6\text{V}$.

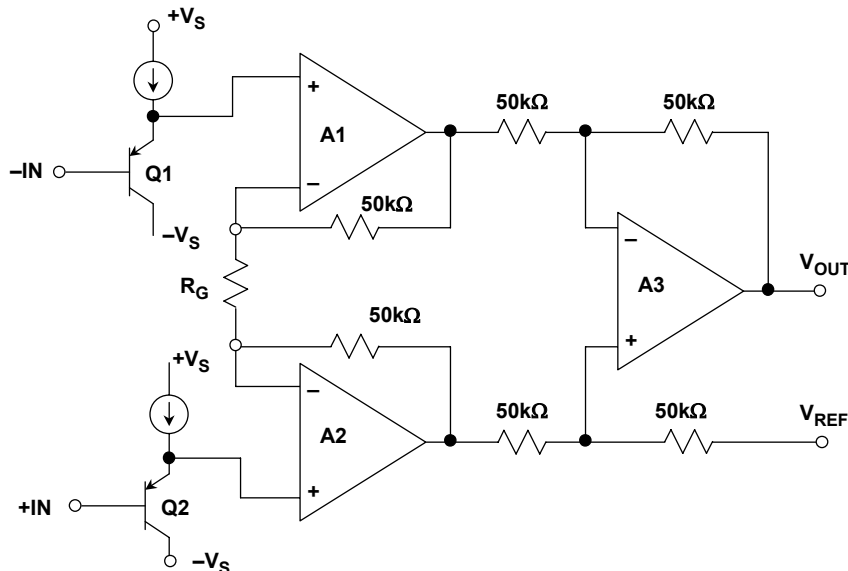


Figure 2-17: AD623 single-supply in-amp architecture

The AD623 data sheet (Reference 11, again) contains excellent discussions and data on allowable input/output voltage ranges as a function of gain and power supply voltages. In addition, interactive design tools are available on the ADI web site which perform calculations relating these parameters for a number of in-amps, including the AD623.

- ◆ **Wide Supply Range: +3V to $\pm 6\text{V}$**
- ◆ **Input Voltage Range: $-V_S - 0.15\text{V}$ to $+V_S - 1.5\text{V}$**
- ◆ **575 μA Maximum Supply Current**
- ◆ **Gain Range: 1 to 1000**
- ◆ **100 μV Maximum Input Offset Voltage (AD623B)**
- ◆ **1 $\mu\text{V}/^\circ\text{C}$ Maximum Offset Voltage TC (AD623B)**
- ◆ **50ppm Gain Nonlinearity**
- ◆ **105dB CMR @ 60Hz, 1k Ω Source Imbalance, $G \geq 100$**
- ◆ **3 μV p-p 0.1Hz to 10Hz Input Voltage Noise ($G = 1$)**

Figure 2-18: AD623 in-amp key specifications

The key specifications of the AD623 are summarized in Figure 2-18, above.

In-Amp DC Error Sources

The DC and noise specifications for in-amps differ slightly from conventional op amps, so some discussion is required in order to fully understand the error sources.

The gain of an in-amp is usually set by a single resistor. If the resistor is external to the in-amp, its value is either calculated from a formula or chosen from a table on the data sheet, depending on the desired gain.

Absolute value laser wafer trimming allows the user to program gain accurately with this single resistor. The absolute accuracy and temperature coefficient of this resistor directly affects the in-amp gain accuracy and drift. Since the external resistor will never exactly match the internal thin film resistor tempcos, a low TC ($<25\text{ppm}/^{\circ}\text{C}$) metal film resistor should be chosen, preferably with a 0.1% or better accuracy.

Often specified as having a gain range of 1 to 1000, or 1 to 10,000, many in-amps will work at higher gains, but the manufacturer will not guarantee a specific level of performance at these high gains. In practice, as the gain-setting resistor becomes smaller, any errors due to the resistance of the metal runs and bond wires become significant. These errors, along with an increase in noise and drift, may make higher single-stage gains impractical. In addition, input offset voltages can become quite sizable when reflected to output at high gains. For instance, a 0.5mV input offset voltage becomes 5V at the output for a gain of 10,000. For high gains, the best practice is to use an in-amp as a preamplifier, then use a post amplifier for further amplification.

In a *pin-programmable-gain* in-amp such as the AD621, the gain-set resistors are internal, well matched, and the device gain accuracy and gain drift specifications include their effects. The AD621 is otherwise generally similar to the externally gain-programmed AD620.

The *gain error* specification is the maximum deviation from the gain equation. Monolithic in-amps such as the AD624C have very low factory trimmed gain errors, with its maximum error of 0.02% at $G = 1$ and 0.25% at $G = 500$ being typical for this high quality in-amp. Notice that the gain error increases with increasing gain. Although externally connected gain networks allow the user to set the gain exactly, the temperature coefficients of the external resistors and the temperature differences between individual resistors within the network all contribute to the overall gain error. If the data is eventually digitized and presented to a digital processor, it may be possible to correct for gain errors by measuring a known reference voltage and then multiplying by a constant.

Nonlinearity is defined as the maximum deviation from a straight line on the plot of output versus input. The straight line is drawn between the end-points of the actual transfer function. Gain nonlinearity in a high quality in-amp is usually 0.01% (100ppm) or less, and is relatively insensitive to gain over the recommended gain range.

■ OP AMP APPLICATIONS

The total *input offset voltage* of an in-amp consists of two components (see Figure 2-19 below). *Input offset voltage*, V_{OSI} , is the input offset component that is reflected to the output of the in-amp by the gain G . *Output offset voltage*, V_{OSO} , is independent of gain.

At low gains, output offset voltage is dominant, while at high gains input offset dominates. The output offset voltage drift is normally specified as drift at $G=1$ (where input effects are insignificant), while input offset voltage drift is given by a drift specification at a high gain (where output offset effects are negligible).

The total output offset error, referred to the input (RTI), is equal to $V_{OSI} + V_{OSO}/G$. In-amp data sheets may specify V_{OSI} and V_{OSO} separately, or give the total RTI input offset voltage for different values of gain.

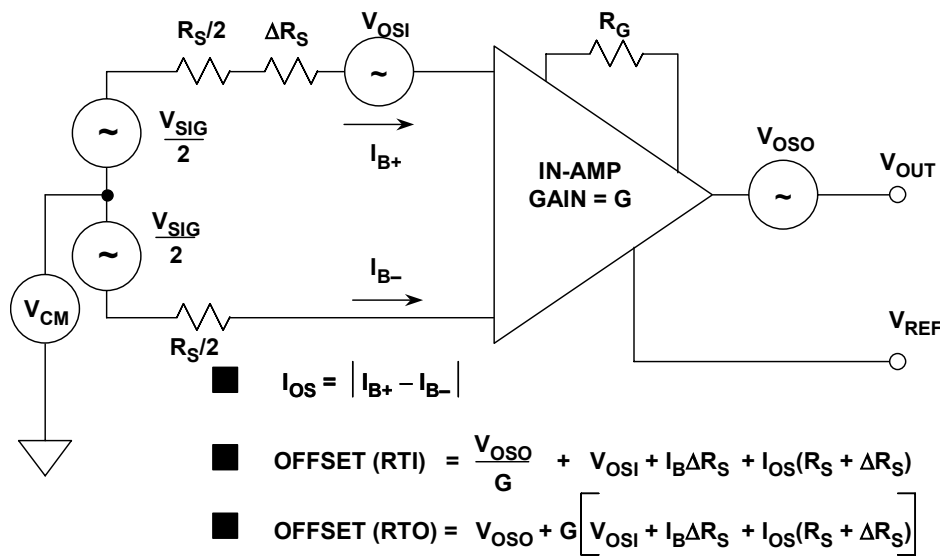


Figure 2-19: In-amp offset voltage model

Input bias currents may also produce offset errors in in-amp circuits (Fig. 2-19, again). If the source resistance, R_S , is unbalanced by an amount, ΔR_S , (often the case in bridge circuits), then there is an additional input offset voltage error due to the bias current, equal to $I_B \Delta R_S$ (assuming that $I_{B+} \approx I_{B-} = I_B$). This error is reflected to the output, scaled by the gain G .

The input offset current, I_{OS} , creates an input offset voltage error across the source resistance, $R_S + \Delta R_S$, equal to $I_{OS}(R_S + \Delta R_S)$, which is also reflected to the output by the gain, G .

In-amp *common mode error* is a function of both gain and frequency. Analog Devices specifies in-amp CMR for a $1\text{k}\Omega$ source impedance unbalance at a frequency of 60Hz. The RTI common mode error is obtained by dividing the common mode voltage, V_{CM} , by the common mode rejection ratio, CMRR.

Figure 2-20 below shows the CMR for the AD620 in-amp as a function of frequency, with a $1\text{k}\Omega$ source impedance imbalance.

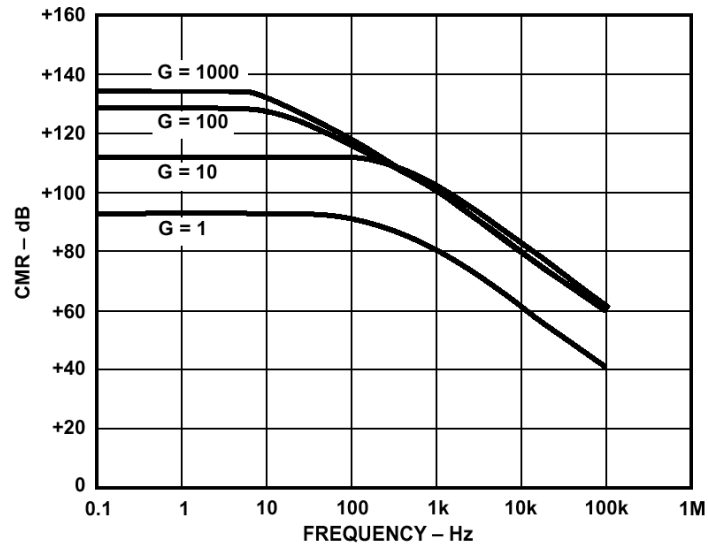


Figure 2-20: AD620 In-amp common-mode rejection (CMR) versus frequency for $1\text{k}\Omega$ source impedance imbalance

Power supply rejection (PSR) is also a function of gain and frequency. For in-amps, it is customary to specify the sensitivity to each power supply separately, as shown in Figure 2-21 below for the AD620. The RTI power supply rejection error is obtained by dividing the power supply deviation from nominal by the power supply rejection ratio, PSRR.

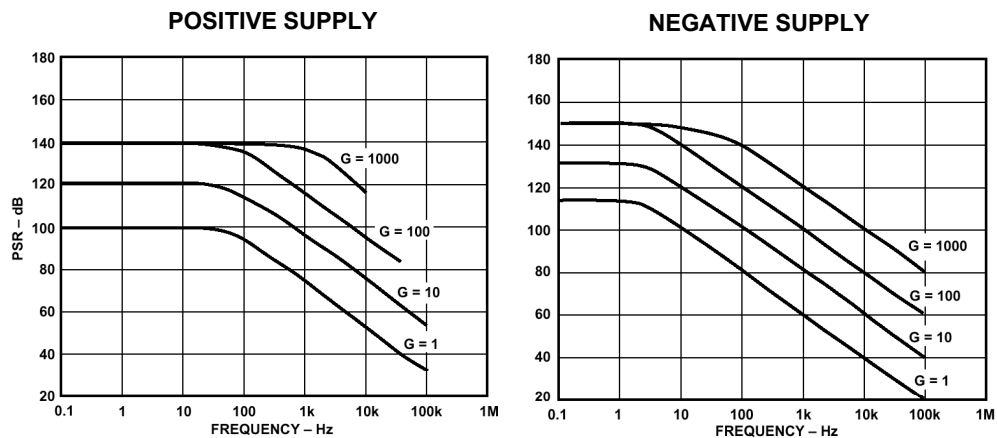


Figure 2-21: AD620 in-amp power supply rejection (PSR) versus frequency

Because of the relatively poor PSR at high frequencies, decoupling capacitors are required on both power pins to an in-amp. Low inductance ceramic capacitors (0.01 to $0.1\mu\text{F}$) are appropriate for high frequencies. Low ESR electrolytic capacitors should also be located at several points on the PC board for low frequency decoupling.

Note that these decoupling requirements apply to all linear devices, including op amps and data converters. Further details on power supply decoupling are found in Chapter 7.

■ OP AMP APPLICATIONS

Now that all DC error sources have been accounted for, a worst case DC error budget can be calculated by reflecting all the sources to the in-amp input, as is illustrated by the table of Figure 2-22, below.

ERROR SOURCE	RTI VALUE
Gain Accuracy (ppm)	Gain Accuracy $\times$ FS Input
Gain Nonlinearity (ppm)	Gain Nonlinearity $\times$ FS Input
Input Offset Voltage, V_{OSI}	V_{OSI}
Output Offset Voltage, V_{OSO}	$V_{OSO} \div G$
Input Bias Current, I_B , Flowing in ΔR_S	$I_B \Delta R_S$
Input Offset Current, I_{OS} , Flowing in R_S	$I_{OS}(R_S + \Delta R_S)$
Common Mode Input Voltage, V_{CM}	$V_{CM} \div CMRR$
Power Supply Variation, ΔV_S	$\Delta V_S \div PSRR$

Figure 2-22: In-amp DC errors referred to the input (RTI)

It should be noted that the DC errors can be referred to the in-amp output (RTO), by simply multiplying the RTI error by the in-amp gain.

In-Amp Noise Sources

Since in-amps are primarily used to amplify small precision signals, it is important to understand the effects of all the associated noise sources. The in-amp noise model is shown in Figure 2-23, below.

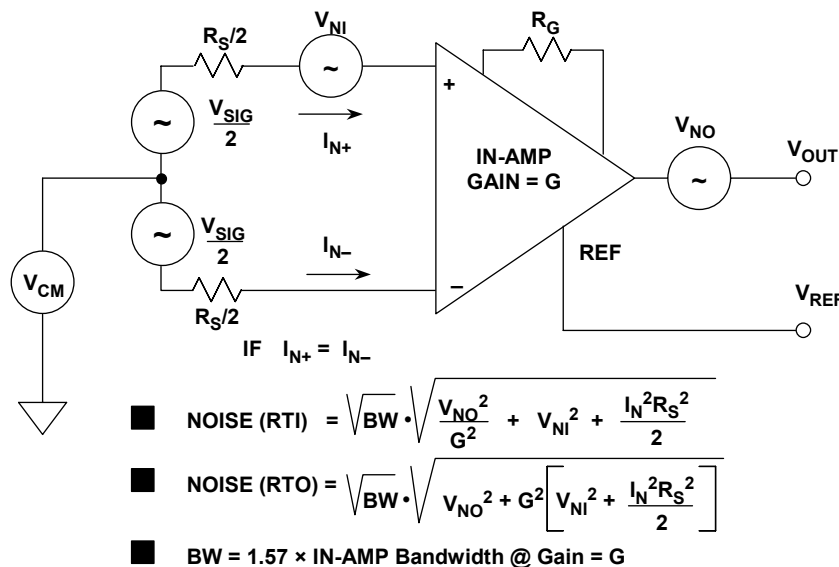


Figure 2-23: In-amp noise model

There are two sources of input voltage noise. The first is represented as a noise source, V_{NI} , in series with the input, as in a conventional op amp circuit. This noise is reflected to the output by the in-amp gain, G . The second noise source is the output noise, V_{NO} , represented as a noise voltage in series with the in-amp output. The output noise, shown here referred to V_{OUT} , can be referred to the input by dividing by the gain, G .

There are also two noise sources associated with the input noise currents I_{N+} and I_{N-} . Even though I_{N+} and I_{N-} are usually equal ($I_{N+} \approx I_{N-} = I_N$), they are *uncorrelated*, and therefore, the noise they each create must be summed in a root-sum-squares (RSS) fashion. I_{N+} flows through one half of R_S , and I_{N-} the other half. This generates two noise voltages, each having an amplitude, $I_N R_S / 2$. Each of these two noise sources is reflected to the output by the in-amp gain, G .

The total output noise is calculated by combining all four noise sources in an RSS manner:

$$\text{NOISE (RTO)} = \sqrt{BW} \sqrt{V_{NO}^2 + G^2 \left(V_{NI}^2 + \frac{I_{N+}^2 R_S^2}{4} + \frac{I_{N-}^2 R_S^2}{4} \right)} \quad \text{Eq. 2-6}$$

If $I_{N+} = I_{N-} = I_N$,

$$\text{NOISE (RTO)} = \sqrt{BW} \sqrt{V_{NO}^2 + G^2 \left(V_{NI}^2 + \frac{I_N^2 R_S^2}{2} \right)} \quad \text{Eq. 2-7}$$

The total noise, referred to the input (RTI) is simply the above expression divided by the in-amp gain, G :

$$\text{NOISE (RTI)} = \sqrt{BW} \sqrt{\frac{V_{NO}^2}{G^2} + \left(V_{NI}^2 + \frac{I_N^2 R_S^2}{2} \right)} \quad \text{Eq. 2-8}$$

In-amp data sheets often present the total voltage noise RTI as a function of gain. This noise spectral density includes both the input (V_{NI}) and output (V_{NO}) noise contributions. The input current noise spectral density is specified separately.

As in the case of op amps, the total in-amp noise RTI must be integrated over the applicable in-amp closed-loop bandwidth to compute an RMS value. The bandwidth may be determined from data sheet curves that show frequency response as a function of gain.

Regarding this bandwidth, some care must be taken in computing it, as it is often *not* constant bandwidth product relationship, as is true with VFB op amps. In the case of the AD620 in-amp family for example, the gain-bandwidth pattern is more like that of a CFB op amp. In such cases, the safest way to predict the bandwidth at a given gain is to use the curves supplied within the data sheet.

In-Amp Bridge Amplifier Error Budget Analysis

It is important to understand in-amp error sources in a typical application. Figure 2-24 below shows a 350Ω load cell with a fullscale output of 100mV when excited with a 10V source. The AD620 is configured for a gain of 100 using the external 499Ω gain-setting resistor. The table shows how each error source contributes to a total unadjusted error of 2145ppm . Note however that the gain, offset, and CMR errors can all be removed with a system calibration. The remaining errors— gain nonlinearity and 0.1Hz to 10Hz noise - cannot be removed with calibration and ultimately limit the system resolution to 42.8ppm (approximately 14-bit accuracy).

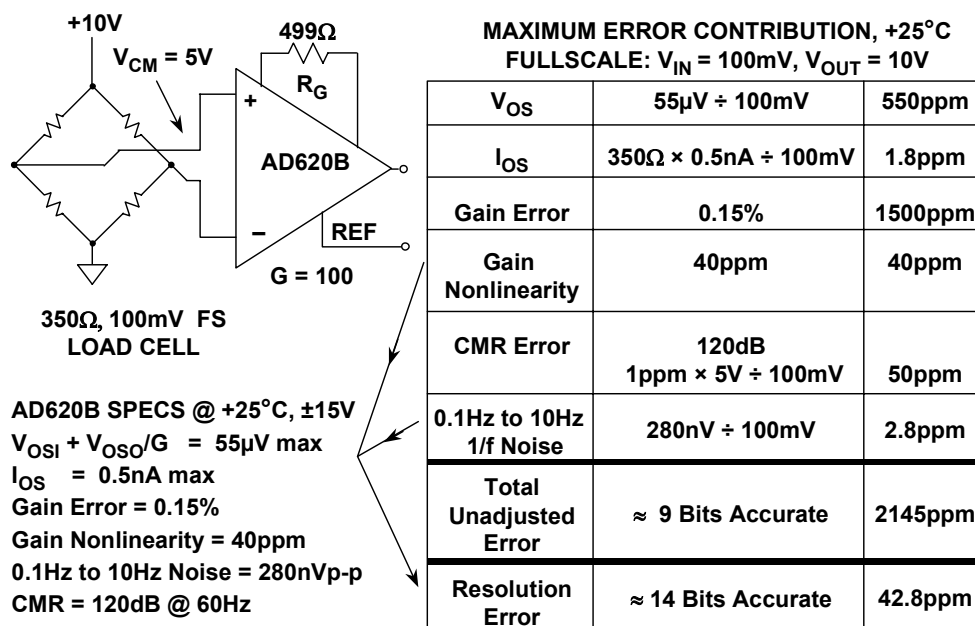


Figure 2-24: AD620B bridge amplifier DC error budget

This example is of course just an illustration, but should be useful towards the importance of addressing performance-limiting errors such as gain nonlinearity and LF noise.

In-Amp Performance Tables

Figure 2-25 opposite shows a selection of precision in-amps designed primarily for operation on dual supplies. It should be noted that the AD620 is capable of single +5V supply operation (see Figure 2-15), but neither its input nor its output are capable of rail-to-rail swings.

These tables allow at-a-glance inspection of key errors, which can be critical towards getting the most performance from a system. From Fig. 2-25 for example, it can be noted that the use of an AD621 in lieu of the AD620B in the gain-of-100 bridge circuit of Fig. 2-24 allows reduction of the gain nonlinearity component of error by a factor of 4 times.

It is also important to separate out errors that can be calibrated out as mentioned above, and those that can only be minimized by device specification improvements. Comparison

of the AD620B and the AD622 specifications for example shows a higher V_{OS} for the latter. But, since V_{OS} can be calibrated out, the fact that it is higher for the AD622 isn't material to this particular application. The gain nonlinearity between the AD620 and AD622 is the same, so in an auto-cal system, they would likely perform comparably. On the other hand, the AD621B would be preferable for its lower gain nonlinearity, as noted.

	Gain Accuracy *	Gain Nonlinearity	V_{OS} Max	V_{OS} TC	CMR Min	0.1Hz to 10Hz p-p Noise
AD524C	0.5% / P	100ppm	50 μ V	0.5 μ V/ $^{\circ}$ C	120dB	0.3 μ V
AD620B	0.5% / R	40ppm	50 μ V	0.6 μ V/ $^{\circ}$ C	120dB	0.28 μ V
AD621B ¹	0.05% / P	10ppm	50 μ V	1.6 μ V/ $^{\circ}$ C	100dB	0.28 μ V
AD622	0.5% / R	40ppm	125 μ V	1 μ V/ $^{\circ}$ C	103dB	0.3 μ V
AD624C ²	0.25% / R	50ppm	25 μ V	0.25 μ V/ $^{\circ}$ C	130dB	0.2 μ V
AD625C	0.02% / R	50ppm	25 μ V	0.25 μ V/ $^{\circ}$ C	125dB	0.2 μ V
AMP01A	0.6% / R	50ppm	50 μ V	0.3 μ V/ $^{\circ}$ C	125dB	0.12 μ V
AMP02E	0.5% / R	60ppm	100 μ V	2 μ V/ $^{\circ}$ C	115dB	0.4 μ V

* / P = Pin Programmable ¹ G = 100
* / R = Resistor Programmable ² G = 500

Figure 2-25: Precision in-amps: data for $V_S = \pm 15V$, $G = 1000$

In-amps specifically designed for single supply operation are also shown, in Figure 2-26 below. It should be noted that although the specifications in this figure are given for a single +5V supply, all of the amplifiers are also capable of dual supply operation and are specified for both dual and single supply operation on their data sheets. In addition, the AD623 and AD627 will operate on a single +3V supply

	Gain Accuracy *	Gain Nonlinearity	V_{OS} Max	V_{OS} TC	CMR Min	0.1Hz to 10Hz p-p Noise	Supply Current
AD623B	0.5% / R	50ppm	100 μ V	1 μ V/ $^{\circ}$ C	105dB	1.5 μ V	575 μ A
AD627B	0.35% / R	10ppm	75 μ V	1 μ V/ $^{\circ}$ C	85dB	1.5 μ V	85 μ A
AMP04E	0.4% / R	250ppm	150 μ V	3 μ V/ $^{\circ}$ C	90dB	0.7 μ V	290 μ A
AD626B ¹	0.6% / P	200ppm	2.5mV	6 μ V/ $^{\circ}$ C	80dB	2 μ V	700 μ A

* / P = Pin Programmable ¹ Differential Amplifier, G = 100
* / R = Resistor Programmable

Figure 2-26: Single-supply in-amps: data for $V_S = +5V$, $G = 1000$

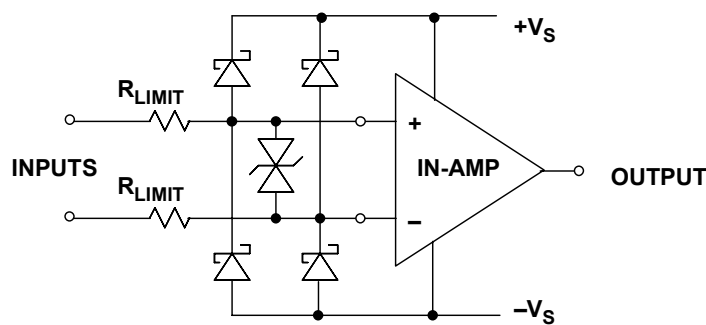
Note that the AD626 is not a true in-amp, but is in fact a differential amplifier with a thin-film input attenuator that allows the common mode voltage to exceed the supply voltages. This device is designed primarily for high and low-side current-sensing applications. It will also operate on a single +3V supply.

■ OP AMP APPLICATIONS

In-Amp Input Overvoltage Protection

As interface amplifiers for data acquisition systems, in-amps are often subjected to input overloads, i.e., voltage levels in excess of the full scale for the selected gain range. The manufacturer's "absolute maximum" input ratings for the device should be closely observed. As with op amps, many in-amps have absolute maximum input voltage specifications equal to $\pm V_S$.

In some cases, external series resistors (for current limiting) and diode clamps may be used to prevent overload, if necessary (see Figure 2-27, below). Some in-amps have built-in overload protection circuits in the form of series resistors. For example, the AD620 series have thin film resistors, and the substrate isolation they provide allows input voltages that can exceed the supplies. Other devices use series-protection FETs, for example the AMP02 and the AD524, because they act as a low impedance during normal operation, and a high impedance during overvoltage fault conditions. In any instance however, there are always finite safe limits to applied overvoltage (Fig. 2-27, again).



- ◆ **Always Observe Absolute Maximum Data Sheet Specs!**
- ◆ **Schottky Diode Clamps to the Supply Rails Will Limit Input to Approximately $\pm V_S \pm 0.3V$, TVSs Limit Differential Voltage**
- ◆ **External Resistors (or Internal Thin-Film Resistors) Can Limit Input Current, but will Increase Noise**
- ◆ **Some In-Amps Have Series-Protection Input FETs for Lower Noise and Higher Input Over-Voltages (up to $\pm 60V$, Depending on Device)**

Figure 2-27: In-amp input overvoltage considerations

In some instances, an additional Transient Voltage Suppressor (TVS) may be required across the input pins to limit the maximum differential input voltage. This is especially applicable to three op amp in-amps operating at high gain with low values of R_G .

A more detailed discussion of input overvoltage and EMI/RFI protection can be found in Chapter 7 of this book.

In-Amp Applications

Some representative in-amp applications round out this section, illustrating how the characteristics lend utility and efficiency to a range of circuits.

In-Amp Bridge Amplifier

In-amps are widely used as precision signal conditioning elements. A popular application is a bridge amplifier, shown below in Figure 2-28. The in-amp is ideally suited for this application because the bridge output is fundamentally balanced, and the in-amp presents it with a truly balanced high impedance load. The nominal resistor values in the bridge can range from 100Ω to several kΩ, but 350Ω is popular for most precision load cells.

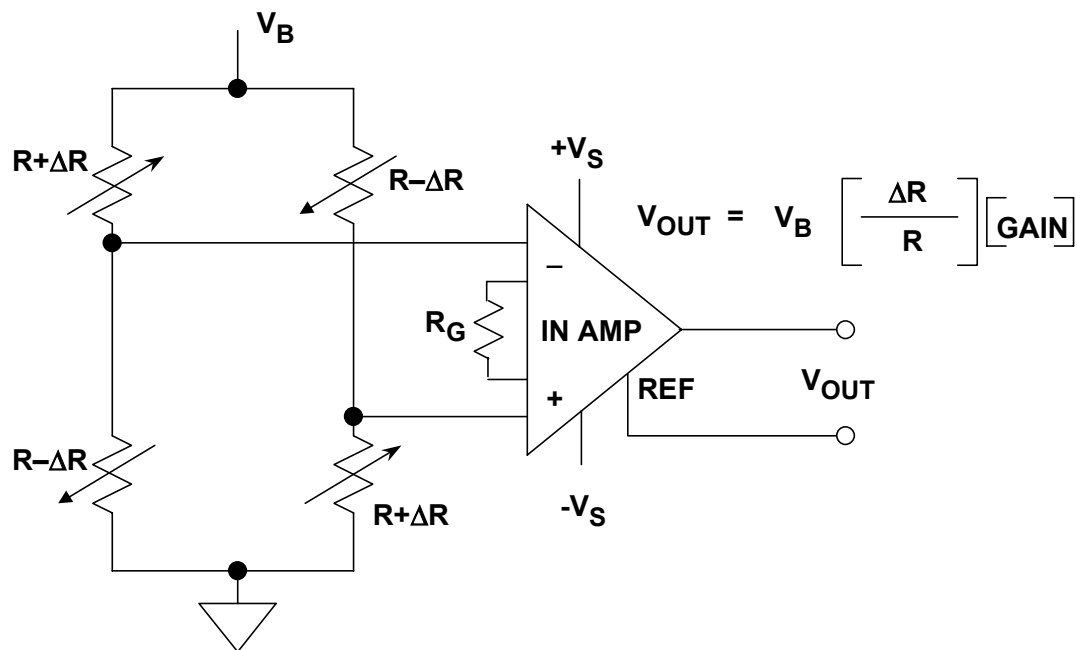


Figure 2-28: Generalized bridge amplifier using an in-amp

Full scale output voltages from a typical bridge circuit can range from approximately 10mV to several hundred mV. Typical in-amp gains in the order of 100 to 1000 are therefore ideally suited for amplifying these small voltages to levels compatible with popular analog-to-digital converter (ADC) input voltage ranges (usually 1V to 10V full scale).

In addition, the in-amp's high CMR at power line frequencies allows common-mode noise to be rejected, when the bridge must be located remotely from the in-amp.

Note that a much more thorough discussion of bridge applications can be found in Chapter 4 of this book.

■ OP AMP APPLICATIONS

In-Amp A/D Interface

Interfacing bipolar signals to single supply ADCs presents a challenge. The bipolar signal must be amplified and level-shifted into the input range of the ADC. Figure 2-29 below shows how this translation can be achieved using the AD623 in-amp, when interfacing a bridge circuit to the AD7776 10-bit, 2.5 μ s ADC.

The bridge circuit is excited by a +5V supply. The full scale output from the bridge ($\pm 10\text{mV}$) therefore has a common-mode voltage of +2.5V. The AD623 removes the common-mode component, and amplifies the bridge output by a factor of 100 ($R_G = 1.02\text{k}\Omega$).

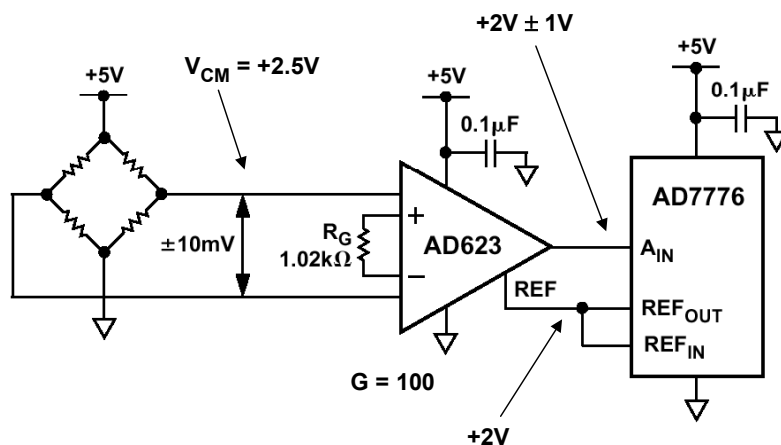


Figure 2-29: Single-supply data acquisition system

This results in an output signal swing of $\pm 1\text{V}$. This signal is level shifted by connecting the REF pin of the AD623 to the +2V REF_{OUT} of the AD7776 ADC. This sets the common-mode output voltage of the AD623 to +2V, and the resulting signal into the ADC is +2V $\pm 1\text{V}$, corresponding to the input range of the AD7776.

In-Amp Driven Current Source

Figure 2-30 (opposite) shows a precision voltage controlled current source using an in-amp. The input voltage V_{IN} develops an output voltage V_{OUT} equal to GV_{IN} between the output pin of the AD620 and the REF pin. With the connections shown, V_{OUT} is also applied across sense resistor R_{SENSE} , thus developing a load current of $V_{\text{OUT}}/R_{\text{SENSE}}$. The OP97 acts as a unity gain buffer to isolate the load from the 20k Ω impedance of the REF pin of the AD620. In this circuit the input voltage can be floating with respect to the load ground (as long as there exists a path for the in-amp bias currents). The high CMR of the in-amp allows high accuracy to be achieved for the load current, despite CM voltages.

The circuit will work for both large and small values of G in the AD620. The most simple form would be to let $G = 1$ with R_G open. In this case, $V_{\text{OUT}} = V_{\text{IN}}$, and I_{LOAD} is proportional to V_{IN} . But the gain factor of the in-amp can be readily used to scale almost any input voltage to a desired current level.

The output load voltage compliance is typically $\pm 10\text{V}$ when operating on $\pm 15\text{V}$ power supplies, and load currents up to $\pm 15\text{mA}$ are allowable, limited by the AD620's drive. A typical operating condition might be a full scale load current of 10mA , a full scale $V_{\text{OUT}} = 0.5\text{V}$, and $R_{\text{SENSE}} = 50\Omega$.

For small values of R_{SENSE} , the OP97 buffer could possibly be eliminated provided the resulting error incurred by the loading effect of the AD620 REF pin is acceptable. In this case the load and bottom R_{SENSE} node would be connected directly to the in-amp REF pin.

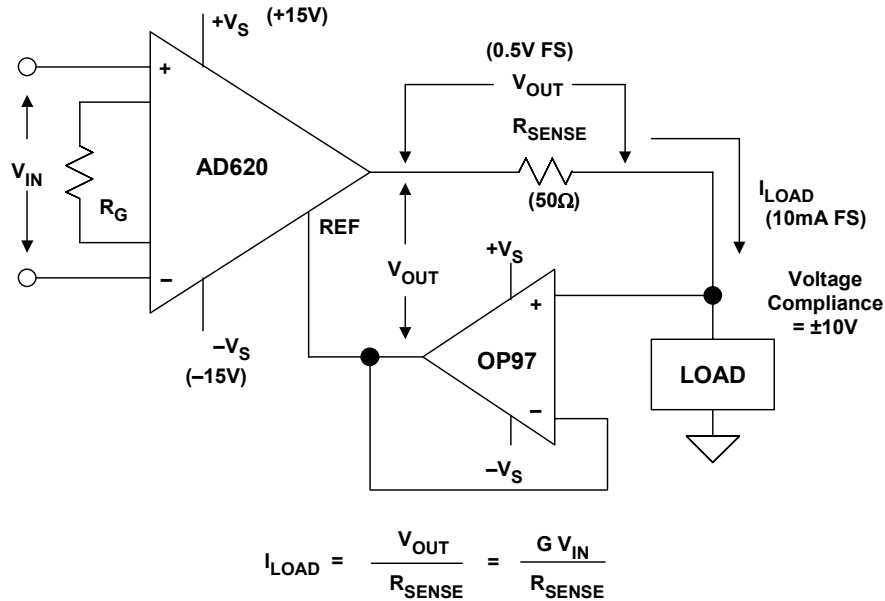


Figure 2-30: Precision voltage controlled current source using an in-amp

Many other useful variation of the basic circuit exist, and can easily be added. For currents of up to 50mA , a unity-gain, low offset buffer can be added between the AD620 output and the top of R_{SENSE} . This will remove all load current from the AD620, allowing it to operate with greatest linearity.

The circuit is also very useful at very small currents. It will work well with the OP97 down to around one μA , before bias current of the op amp becomes a performance limitation. For even lower currents, a precision JFET op amp such as the AD8610 can easily be substituted. This step will allow precise low-level currents, down to below one nA . Note that the AD8610 must be operated on supplies of $\pm 13\text{V}$ or less, but this isn't necessarily a problem (the AD620 will still operate well on supplies as low as $\pm 2.5\text{V}$).

A factor that may not be obvious is that the output current capability of this current source is bi-lateral, as it is shown. This makes this form of current source a great advantage over a Howland type current source, which are always problematic with the numerous resistors required, which must be well-matched and stable for good performance. In contrast, the current source of Fig. 2-30 is clean and efficient, requires no matched resistors, and is precise over very wide current ranges.

▶ OP AMP APPLICATIONS

In-Amp Remote Load Driver

Often remote loads present a problem in driving, when high accuracy must be maintained at the load end. For this type of requirement, an in-amp (or simple differential amplifier) with separate SENSE/FORCE terminals can serve very well, providing a complete solution in one IC. Most of the more popular in-amps available today have removed the separate SENSE/FORCE connections, due to the pin limitations of an 8 pin package (AD620, etc.). However, many classic in-amps such as the AMP01 do have access to the SENSE/FORCE pins, and can perform remote sensing, as shown in Figure 2-31 below.

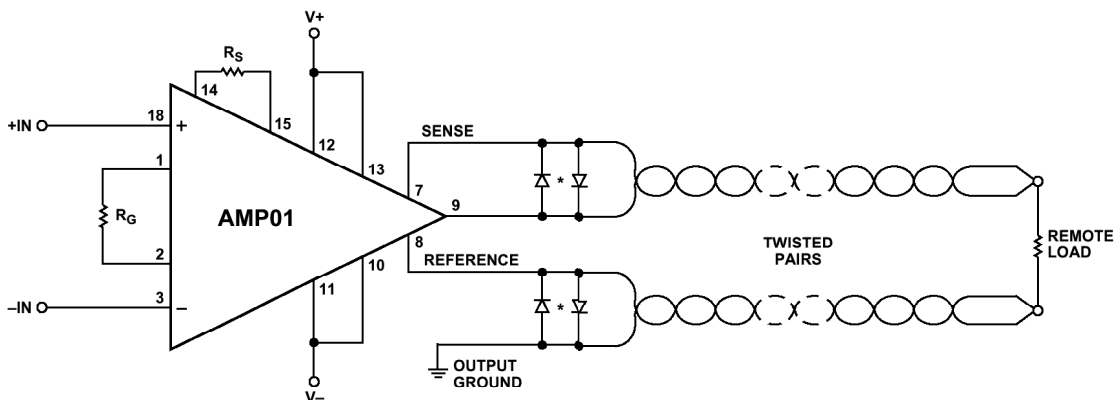


Figure 2-31: Precision in-amp remote load driver using FORCE/SENSE connections

In this circuit a quad cable composed of two twisted pairs is used. One pair is dedicated to the load HIGH side, the other to the LO side. At the remote end, the load is connected as shown, with each twisted pair terminated at one end of the load.

Although the full load current still flows in the FORCE (AMP01 pin 9) and OUTPUT GROUND connections, the resulting drop does not create an error, since the remote sensing of the second lead of each pair is returned back to the driver, and carries comparatively very little current. The reverse-parallel connected diodes are optional, and perform a "safety-valve" function, in case a sense line becomes open-circuit (100Ω resistors might also be used).

The AMP01 is valuable to this function not simply because of the SENSE/FORCE capability, but because it also is capable of 50mA output currents, and is stable with the capacitive loading presented by a cable. Alternately, a precision differential amplifier like the AMP03 can also be used, at lower current levels.

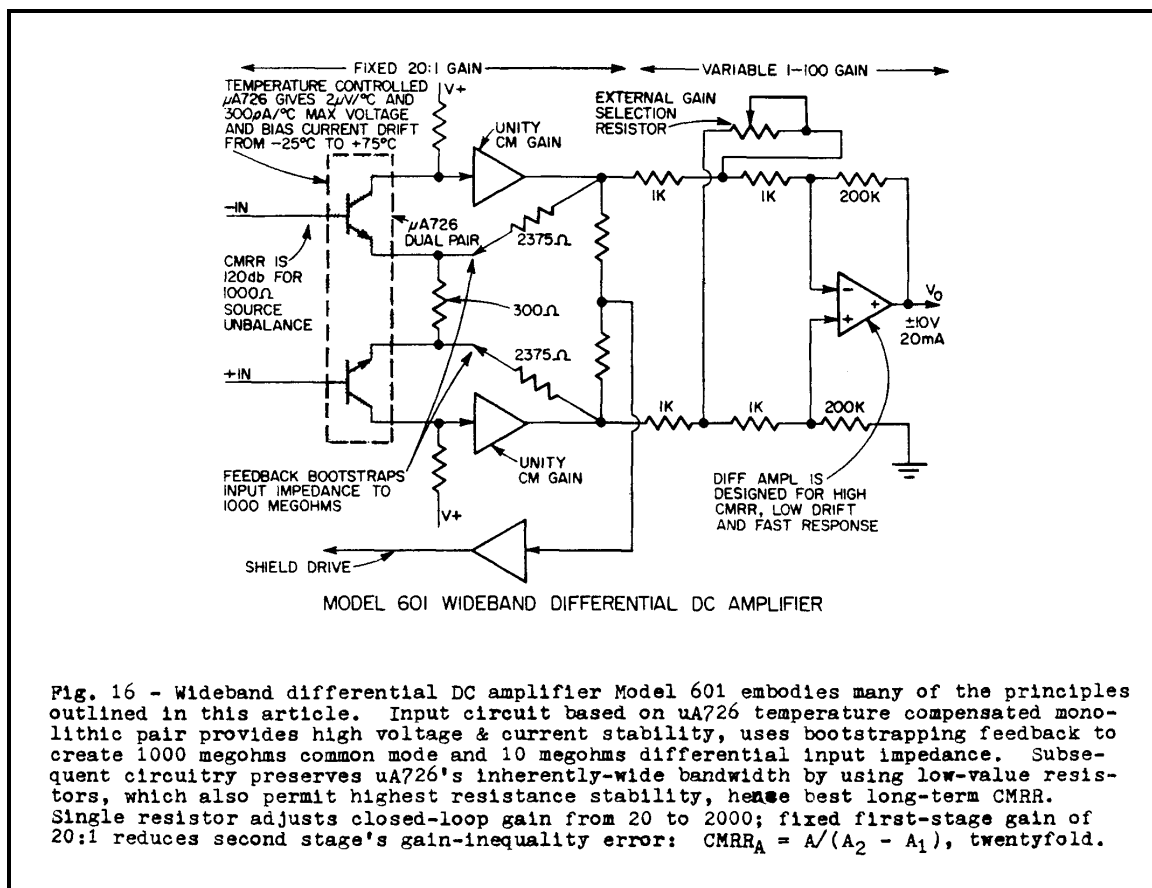
For additional in-amp background and reference material, see References 12 through 15.

REFERENCES: INSTRUMENTATION AMPLIFIERS

1. MIL-PRF-55182G, "Resistors, Fixed, Film, Nonestablished Reliability, Established Reliability, And Space Level, General Specification For," June 9, 1997.
2. Robert Demrow, "Narrowing the Margin of Error," **Electronics**, April 15, 1968, pp. 108-117.
3. Robert Demrow, "Evolution from Operational Amplifier to Data Amplifier," **Analog Devices Application Note**, September, 1968.
4. David Birt, "Electronically Balanced Analogue-Line Interfaces," **Proceedings of Institute of Acoustics Conference**, Windermere, U.K., Nov. 1990.
5. Walt Jung, "Op Amps in Line-Driver and Receiver Circuits, Part 1," **Analog Dialogue**, Vol. 26 No. 2, 1992.
6. W. Jung, A. Garcia, "Op Amps in Line-Driver and Receiver Circuits, Part 2," **Analog Dialogue**, Vol. 27, No. 1, 1993.
7. Data sheet for **AD627 Micropower, Single and Dual Supply Rail-to-Rail Instrumentation Amplifier**, <http://www.analog.com>
8. S. Wurcer, L. Counts, "A Programmable Instrumentation Amplifier for 12-Bit Resolution Systems," **IEEE Journal of Solid State Circuits**, Vol. SC-17 #6, Dec. 1982, pp. 1102-1111.
9. L. Counts, S. Wurcer, "Instrumentation Amplifier Nears Input Noise Floor," **Electronic Design**, June 10, 1982, pp. 177.
10. Data sheet for **AD620 Low Cost, Low Power Instrumentation Amplifier**, <http://www.analog.com>
11. Data sheet for **AD623 Single Supply, Rail-to-Rail, Low Cost Instrumentation Amplifier**, <http://www.analog.com>
12. Charles Kitchen and Lew Counts, **A Designer's Guide to Instrumentation Amplifiers**, Analog Devices, 2000.
13. Sections 2, 3, 4, of Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, Inc., 1999, ISBN: 0-916550-20-6.
14. Walter Borlase, "Application/Analysis of the AD520 Monolithic Data Amplifier," **Analog Devices Application Note**, 1972.
15. Jeff Riskin, "A User's Guide to IC Instrumentation Amplifiers, **Analog Devices AN244**, January, 1978.

Classic Cameo

Robert Demrow's "Evolution from Operational Amplifier to Data Amplifier"



As applications engineering manager in the early years of ADI, Robert Demrow published numerous articles and application notes. It is a testament to the quality of these articles that most of them are still germane today— due in no small part to their lucid outlining of fundamental principles.

Demrow's 1968 application note, "Evolution from Operational Amplifier to Data Amplifier" outlined the relevant amplifier operating principles for retrieving analog signals from a noisy environment. It also introduced the ADI Model 601 *data amplifier* (above). Of course, a data amplifier is what we know today as an instrumentation amplifier. Within his Figure 16 can be seen several key operating principles: 1) dual high impedance inputs, as necessary for high CMR, 2) the use of a precision bipolar transistor differential pair front end, for low offset and drift (the $\mu A726$), ¹ 3) a balanced, three amplifier stage topology.

It is interesting to note that some more popular IC in-amps of 2002 utilize many of the same principles—for example, the AD620 family. Back in 1968, Robert Demrow outlined a host of sound design concepts, leading the way to later solid-state developments, and the completely monolithic in-amp ICs of today.

¹ "The $\mu A726$ Temperature-Stabilized Transistor Pair", Chapter 8 within James N. Giles, Editor, **Fairchild Semiconductor Linear Integrated Circuits Handbook**, Fairchild Semiconductor, 1967.

SECTION 2-2: PROGRAMMABLE GAIN AMPLIFIERS

Walt Kester, James Bryant

Most data acquisition systems with wide dynamic range need some method of adjusting the input signal level to the analog-to-digital-converter (ADC). Typical ADC full scale input voltage ranges lie between 2V and 10V. To achieve the rated precision of the converter, the maximum input signal should be fairly near its full scale voltage.

Transducers however, have a very wide range of output voltages. High gain is needed for a small sensor voltage, but with a large output, a high gain will cause the amplifier or ADC to saturate. So, some type of predictably controllable gain device is needed. Amplifiers with programmable gain have a variety of applications, and Figure 2-32 below lists some of them.

- ◆ **Instrumentation**
- ◆ **Photodiode circuits**
- ◆ **Ultrasound preamplifiers**
- ◆ **Sonar**
- ◆ **Wide dynamic range sensors**
- ◆ **Driving ADCs (some ADCs have on-chip PGAs)**
- ◆ **Automatic gain control (AGC) loops**

Figure 2-32: Programmable gain amplifier (PGA) applications

Such a device has a gain that is controlled by a DC voltage or, more commonly, a digital input. This device is known as a *programmable gain amplifier*, or PGA. Typical PGAs may be configured either for selectable *decade gains* such as 10, 100, 1000, etc., or they might also be configured for *binary gains* such as 1, 2, 4, 8, etc. It is a function of the end system of course, which type might be the more desirable.

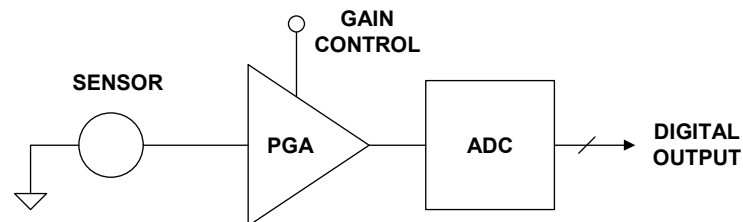
It should be noted that a factor common to the above application examples is that the different types of signals being handled is diverse. Some may require wide bandwidth, others very low noise, from either high or low impedance sources. The inputs may be either single-ended, or they may be differential, crossing over into the realm of the just-discussed in-amps.

The output from the PGA may be required to drive some defined input range of an ADC, or it may be part of a smaller sub-system, such as an AGC or gain-ranging loop. The circuits following fall into a range of categories addressing some of these requirements.

■ OP AMP APPLICATIONS

A PGA is usually located between a sensor and its ADC, as shown in Figure 2-33 below. Additional signal conditioning may take place before or after the PGA, depending on the application. For example, a photodiode needs a current-to-voltage converter between it and the PGA. In most other systems, it is better to place the gain first, and condition a larger signal. This reduces errors introduced by the signal conditioning circuitry.

To understand the benefits of variable gain, assume an ideal PGA with two settings, gains of one and two. The dynamic range of the system is increased by 6dB. Increasing the gain to a maximum four results in a 12dB increase in dynamic range. If the LSB of an ADC is equivalent to 10mV of input voltage, the ADC cannot resolve smaller signals, but when the gain of the PGA is increased to two, input signals of 5mV may be resolved.



- ◆ Used to increase the dynamic range of the system
- ◆ A PGA with a gain of 1 to 2 theoretically increases the dynamic range by 6dB.
- ◆ A gain of 1 to 4 gives a 12dB increase, etc.

Figure 2-33: PGAs in data acquisition systems

Thus, a central processor can combine PGA gain information with the digital output of the ADC to increase its resolution by one bit. Essentially, this is the same as adding additional resolution to the ADC. In fact, a number of ADCs now have on-chip PGAs for increased dynamic range (AD77XX-series, for example, covered later).

PGA Design Issues

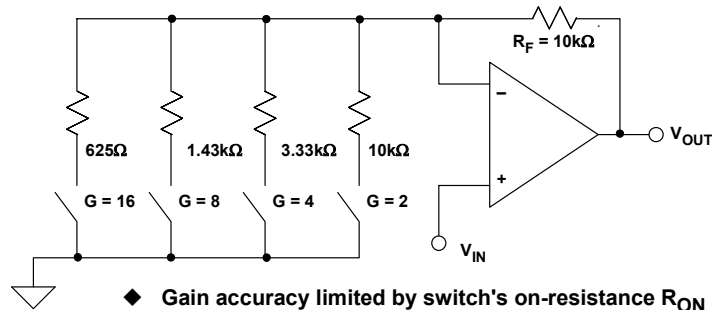
- ◆ How to switch the gain
- ◆ Effects of the switch on-resistance (R_{ON})
- ◆ Gain accuracy
- ◆ Gain linearity
- ◆ Bandwidth versus frequency versus gain
- ◆ DC offset
- ◆ Gain and offset drift over temperature
- ◆ Settling time after switching gain

Figure 2-34: PGA design issues

In practice, PGAs aren't ideal, and their error sources must be studied and dealt with. A number of the various PGA design issues are summarized in Figure 2-34, above. A fundamental PGA design problem is programming gain accurately. Electromechanical relays have minimal on-resistance (R_{ON}), but are unsuitable for gain switching—slow,

large, and expensive. CMOS switches are small, but they have voltage/temperature dependent R_{ON} , as well as stray capacitance, which may affect PGA AC parameters.

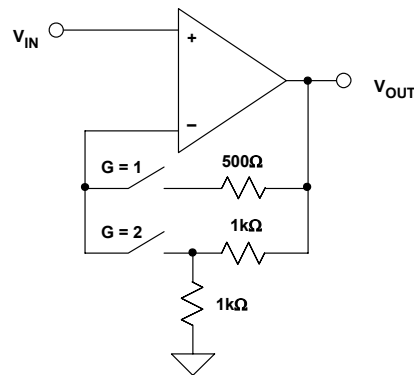
To understand R_{ON} 's effect on performance, consider Figure 2-35 below, a poor PGA design. A non-inverting op amp has 4 different gain-set resistors, each grounded by a switch, with an R_{ON} of 100 Ω -500 Ω . Even with R_{ON} as low as 25 Ω , the gain of 16 error would be 2.4%, worse than 8-bits! R_{ON} also changes over temperature, and switch-switch.



- ◆ Gain accuracy limited by switch's on-resistance R_{ON} and R_{ON} modulation
- ◆ R_{ON} typically 100 - 500 Ω for CMOS or JFET switch
- ◆ Even for $R_{ON} = 25\Omega$, there is a 2.4% gain error for $G = 16$
- ◆ R_{ON} drift over temperature limits accuracy
- ◆ Must use very low R_{ON} switches (relays)

Figure 2-35: A poorly designed PGA

To attempt "fixing" this design, the resistors might be increased, but noise and offset could then be a problem. The only way to accuracy with this circuit is to use relays, with virtually no R_{ON} . Only then will the few m Ω of relay R_{ON} be a small error vis-à-vis 625 Ω .



- ◆ R_{ON} is not in series with gain setting resistors
- ◆ R_{ON} is small compared to input impedance
- ◆ Only slight offset errors occur due to bias current flowing through the switches

Figure 2-36: Alternate PGA configuration minimizes the effects of R_{ON}

It is much better to use a circuit *insensitive* to R_{ON} ! In Figure 2-36 above, the switch is placed in series with the inverting input of an op amp. Since the op amp input impedance is very large, the switch R_{ON} is now irrelevant, and gain is now determined solely by the external resistors. Note— R_{ON} may add a small offset error if op amp bias current is high (if this is the case, it can readily be compensated with an equivalent resistance at V_{IN}).

■ OP AMP APPLICATIONS

PGA Applications

The following section illustrates several PGA circuits using the above and other concepts.

AD526 Software Programmable PGA

The AD526 amplifier uses the just-described PGA architecture, integrating it onto a single chip, as diagrammed in Figure 2-37 below (see References 1 and 2). The AD526 has 5 binary gain settings from 1 to 16, and its internal JFET switches are connected to the inverting input of the amplifier as in Fig. 2-37. The gain resistors are laser trimmed, providing a maximum gain error of only 0.02%, and a linearity of 0.001%. The use of the FORCE/SENSE terminals connected at the load ensures highest accuracy (it also allows the use of an optional unity-gain buffer, for low impedance loads).

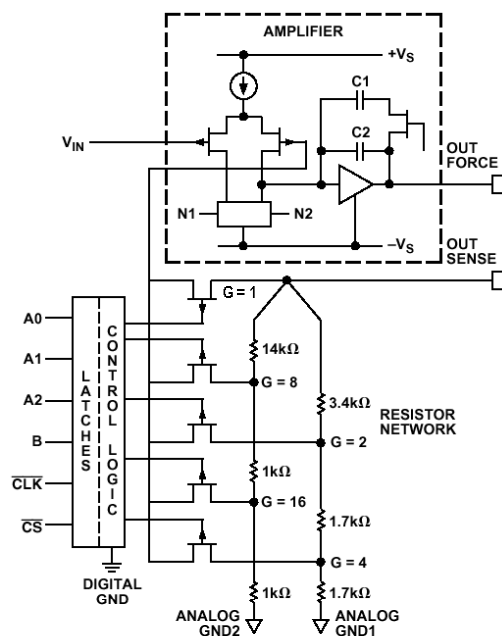


Figure 2-37: AD526 software programmable PGA simplified schematic

Functionally speaking, the AD526 is a programmable, precision, non-inverting op amp gain stage, logic programmable over a range of 1 to 16 times V_{IN} . It typically operates from a $\pm 15V$ power supply, and has $\pm 10V$ output range (like a conventional op amp).

- ◆ Software programmable binary gains from 1 to 16
- ◆ Low bias current JFET input stage
- ◆ Worst case gain error: 0.02% (12-bit performance)
- ◆ Maximum gain nonlinearity: 0.001%
- ◆ Gain change settling time: $5.6\mu s$ ($G = 16$)
- ◆ Small signal bandwidth: 4MHz ($G = 1$), 0.35MHz ($G = 16$)
- ◆ Latched TTL compatible control inputs

Figure 2-38: AD526 PGA key specifications

The key specifications for the AD526 are summarized in Figure 2-38 above.

Low Noise PGA

This same design concepts can be used to build a low noise PGA as shown in Figure 2-39 below. It uses a single op amp, a quad switch, and precision resistors. The lower noise AD797 replaces the JFET input op amp of the AD526, but almost any voltage feedback op amp could be used in this circuit. The ADG412 was picked for its R_{ON} of 35Ω .

The resistors were chosen to give decade gains of 1, 10, 100 and 1000, but if other gains are required, the resistor values may easily be altered. Ideally, a single trimmed resistor network should be used both for initial gain accuracy and for low drift over temperature. The 20pF feedback capacitor ensures stability and holds the output voltage when the gain is switched. The control signal to the switches turns one switch off a few nanoseconds before the second switch turns on. During this break, the op amp is open-loop. Without the capacitor the output would start slewing. Instead, the capacitor holds the output voltage during switching. Since the time that both switches are open is very short, only 20pF is needed. For slower switches, a larger capacitor may be necessary.

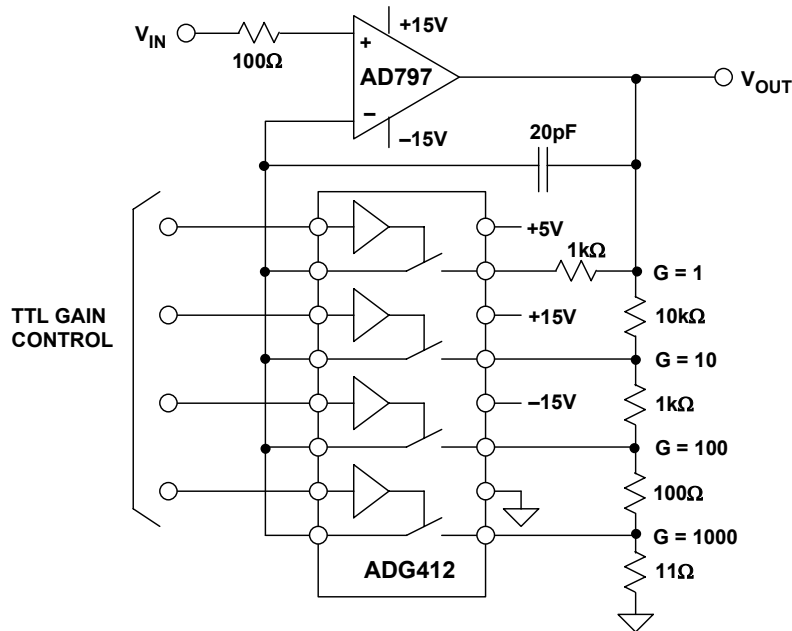


Figure 2-39: A very low noise PGA using the AD797 and the ADG412

The PGA's input voltage noise spectral density at a gain of 1000 is only $1.65\text{nV}/\sqrt{\text{Hz}}$ at 1kHz, only slightly higher than the noise performance of the AD797 alone. The increase is due to the ADG412 noise, and the current noise of the AD797 flowing through R_{ON} .

The accuracy of the PGA is important in determining the overall accuracy of a system. The AD797 has a bias current of $0.9\mu\text{A}$, which, flowing in a 35Ω R_{ON} , results in an additional offset error of $31.5\mu\text{V}$. Combined with the AD797 offset, the total V_{OS} becomes $71.5\mu\text{V}$ (max). Offset temperature drift is affected by the change in bias current and R_{ON} . Calculations show that the total temperature coefficient increases from $0.6\mu\text{V}/^\circ\text{C}$ to $1.6\mu\text{V}/^\circ\text{C}$. Note that while these errors are small (and may not matter in the end) it is still important to be aware of them.

■ OP AMP APPLICATIONS

In practice, circuit accuracy and gain TC will be determined by the external resistors. Input characteristics such as common mode range and input bias current are determined solely by the AD797. A performance summary is shown below in Figure 2-40.

- ◆ R_{ON} adds additional input offset and drift:
 - $\Delta V_{OS} = I_b R_{ON} = (0.9\mu A)(35\Omega) = 31.5\mu V$ (max)
 - Total $V_{OS} = 40\mu V + 31.5\mu V = 71.5\mu V$ (max)
- ◆ Temperature drift due to R_{ON} :
 - At $+85^\circ C$, $\Delta V_{OS} = (2\mu A)(45\Omega) = 90\mu V$ (max)
- ◆ Temperature coefficient total:
 - $\Delta V_{OS} / \Delta T = 0.6\mu V/^\circ C + 1.0\mu V/^\circ C = 1.6\mu V/^\circ C$ (max)
 - Note: $0.6\mu V/^\circ C$ is due to the AD797B
- ◆ RTI Noise : $1.65nV/\sqrt{Hz}$ @ 1kHz, $G = 1000$
- ◆ Gain switching time $< 1\mu s$, $G = 10$

Figure 2-40: AD797/ADG412 PGA performance summary

DAC Programmed PGA

Another PGA configuration uses a DAC in the feedback loop of an op amp to adjust the gain under digital control, as shown in Figure 2-41 below. The digital code of the DAC controls its attenuation with respect to its reference input V_{REF} , acting functionally similar to a potentiometer. Attenuating the feedback signal increases the closed-loop gain.

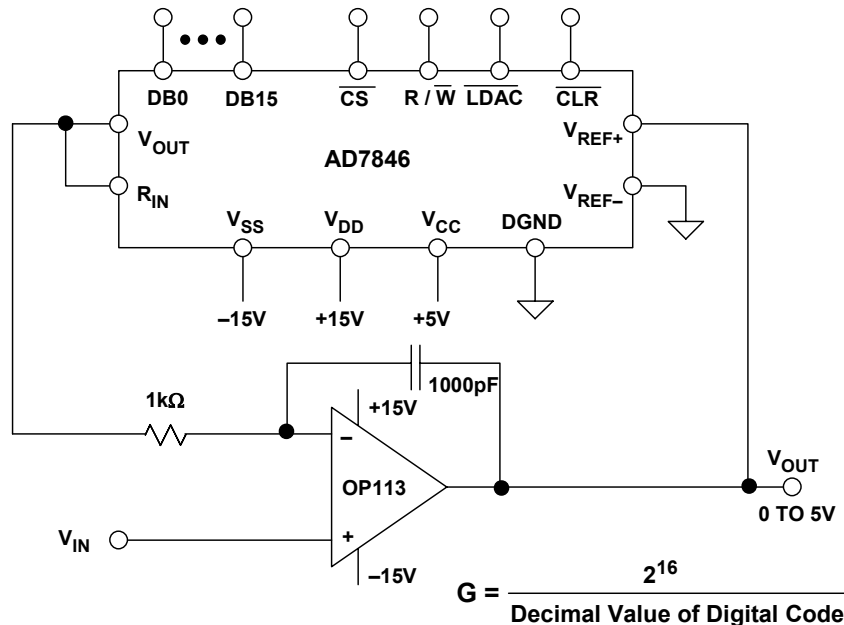


Figure 2-41: Binary gain PGA using a DAC in the feedback path of an op amp

A noninverting PGA of this type requires a multiplying DAC with a voltage mode output. Note that a multiplying DAC is a DAC with a wide reference voltage range, *which includes zero*. For most applications of the PGA, the reference input must be capable of handling bipolar signals. The AD7846 is a 16-bit converter that meets these requirements. In this application, it is used in the standard 2-quadrant multiplying mode.

The OP113 is a low drift, low noise amplifier, but the choice of the amplifier is flexible, and depends on the intended application. The input voltage range depends on the output swing of the AD7846, which is 3V less than the positive supply, and 4V above the negative supply. A 1000pF capacitor is used in the feedback loop for stability.

The gain of the circuit is set by adjusting the digital inputs of the DAC, according to the equation given in Fig. 2-41. D_{0-15} represents the decimal value of the digital code. For example, if all the bits were set high, the gain would be $65,536/65,535 = 1.000015$. If the 8 least significant bits are set high and the rest low, the gain would be $65,536/255 = 257$.

The bandwidth of the circuit is a fairly high 4MHz for a gain of +1. However, this does reduce with gain, and for a gain of 256, the bandwidth is only 600Hz. If the gain-bandwidth product were constant, the bandwidth in a gain of 256 should be 15.6kHz; but the internal capacitance of the DAC reduces the bandwidth to 600Hz.

Performance characteristics of this binary PGA are summarized in Figure 2-42 below.

- ◆ **Gain Accuracy:**
 - 0.003% (G = +1)
 - 0.1% (G = +256)
- ◆ **Nonlinearity: 0.001% (G = 1)**
- ◆ **Offset: 100μV**
- ◆ **Noise: 50nV/√Hz**
- ◆ **Bandwidth:**
 - 4MHz (G = +1)
 - 600Hz (G = +256)

Figure 2-42: Binary gain PGA performance

The gain accuracy of the circuit is determined by the resolution of the DAC and the gain setting. At a gain of 1, all bits are on, and the accuracy is determined by the DNL specification of the DAC, which is ± 1 LSB maximum. Thus, the gain accuracy is equivalent to 1LSB in a 16-bit system, or 0.003%.

However, as the gain is increased, fewer of the bits are on. For a gain of 256, only bit 8 is turned on. The gain accuracy is still dependent on the ± 1 LSB of DNL, but now that is compared to only the lowest 8 bits. Thus, the gain accuracy is reduced to 1 LSB in a 8-bit system, or 0.4%. If the gain is increased above 256, the gain accuracy is reduced further. The designer must determine an acceptable level of accuracy. In this particular circuit, the gain was limited to 256.

■ OP AMP APPLICATIONS

Differential Input PGAs

There are often applications where a PGA with differential inputs is needed, instead of the single ended types discussed so far. The AD625 combines an instrumentation amplifier topology similar to the AD620 with external gain switching capabilities to accomplish 12-bit gain accuracy (see Reference 3). An external switch is needed to switch between different gain settings, but its on-resistance does not significantly affect the gain accuracy due to the unique design of the AD625.

The circuit in Figure 2-43 below uses an ADG409 CMOS switch to switch the connections to an external gain-setting resistor network. In the example shown, resistors were chosen for gains of 1, 4, 16, and 64. Other features of the AD625 are 0.001% non-linearity, wide bandwidth, and very low input noise.

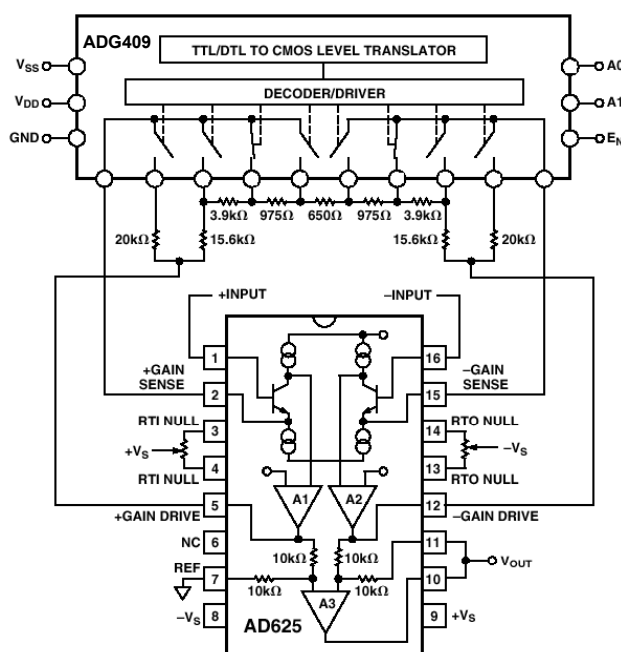


Figure 2-43: A software programmable gain amplifier

The AD625 is uniquely designed so that the on-resistance of the switches does not introduce significant error in the circuit. This can be understood by considering the simplified AD625 circuit shown Figure 2-44 (opposite). The voltages shown are for an input of +1mV on +IN and 0V on -IN. The gain is set to 64 with $R_G = 635\Omega$ and the two resistors, $R_F = 20k\Omega$.

Since transistors Q1 and Q2 have $50\mu A$ current sources in both their emitters and their collectors, negative feedback around A1 and A2 respectively will ensure that no net current flows through either *gain sense* pin into either emitter. Since no current flows in the gain sense pins, no current flows in the gain setting switches, and their R_{ON} does not affect either gain or offset. In real life there will be minor mismatches, but the errors are well under the 12-bit level.

SPECIALTY AMPLIFIERS PROGRAMMABLE GAIN AMPLIFIERS

The differential gain between the inputs and the A1-A2 outputs is $2R_F/R_G + 1$. The unity-gain difference amplifier and matched resistors removes CM voltage, and drives the output.

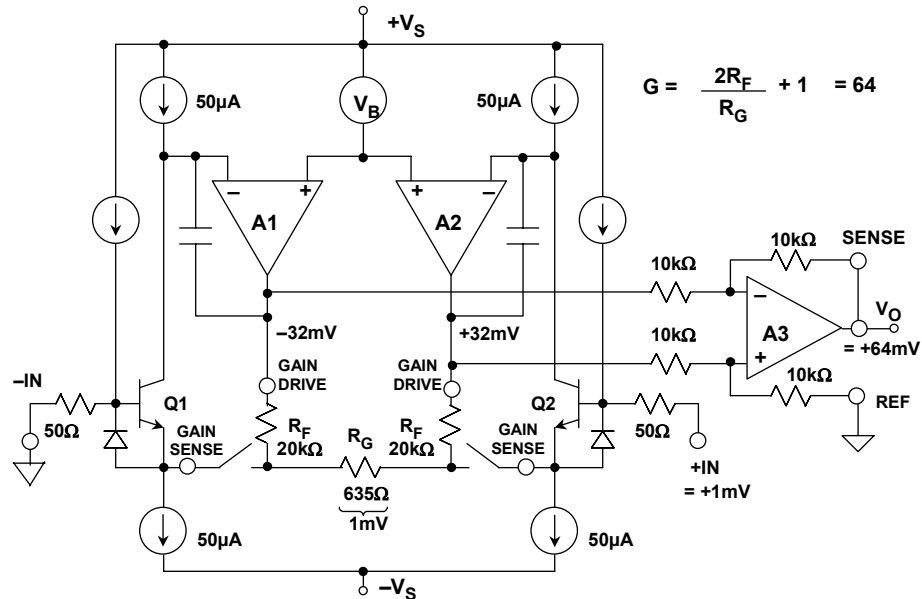


Figure 2-44: AD625 details showing external switches and gain-setting resistors for $R_G = 635\Omega$, $+IN = 1mV$, $-IN = 0V$

Non-inverting PGA circuits using an op amp are easily adaptable to single supply operation, but when differential inputs are desired, a single-supply in-amp such as the AD623, AD627, or the AMP04 should be used. The AMP04 is used with an external CMOS switch in the single supply in-amp PGA shown in Figure 2-45 below.

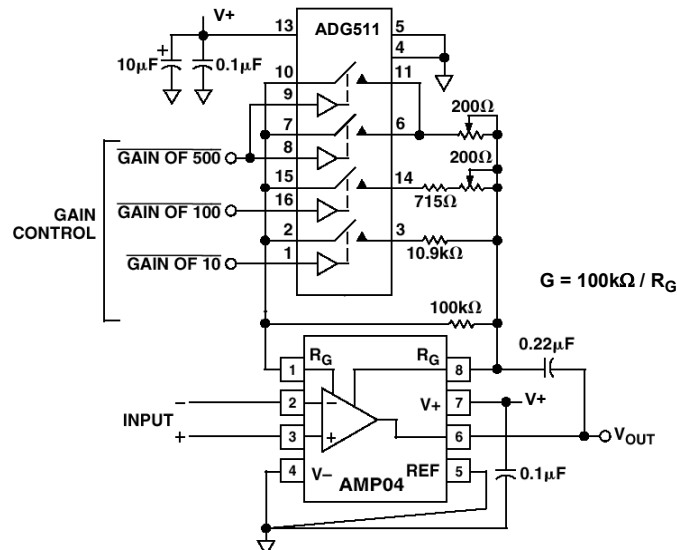


Figure 2-45: Single supply instrumentation PGA using the AMP04 in-amp and the ADG511 switch

This circuit has selectable gains of 1, 10, 100, and 500, which are controlled by an ADG511. The ADG511 was chosen as a single supply switch with a low R_{ON} of 45Ω . A disadvantage of this circuit is that the gain of this circuit is dependent on the R_{ON} of the

■ OP AMP APPLICATIONS

switches. Trimming is required at the higher gains to achieve accuracy. At a gain of 500, two switches are used in parallel, but their resistance causes a 10% gain error in the absence of adjustment.

ADC With Onboard PGA

Certain ADCs (such as the AD77XX measurement series) have built in PGAs and other conditioning circuitry. Circuit design with these devices is much easier, because an external PGA and its control logic are not needed. Furthermore, all the errors of the PGA are included in the specifications of the ADC, making error calculations simple.

The PGA gain is controlled over the common ADC serial interface, and the gain setting is factored into the conversion, saving additional calculations to determine input voltage.

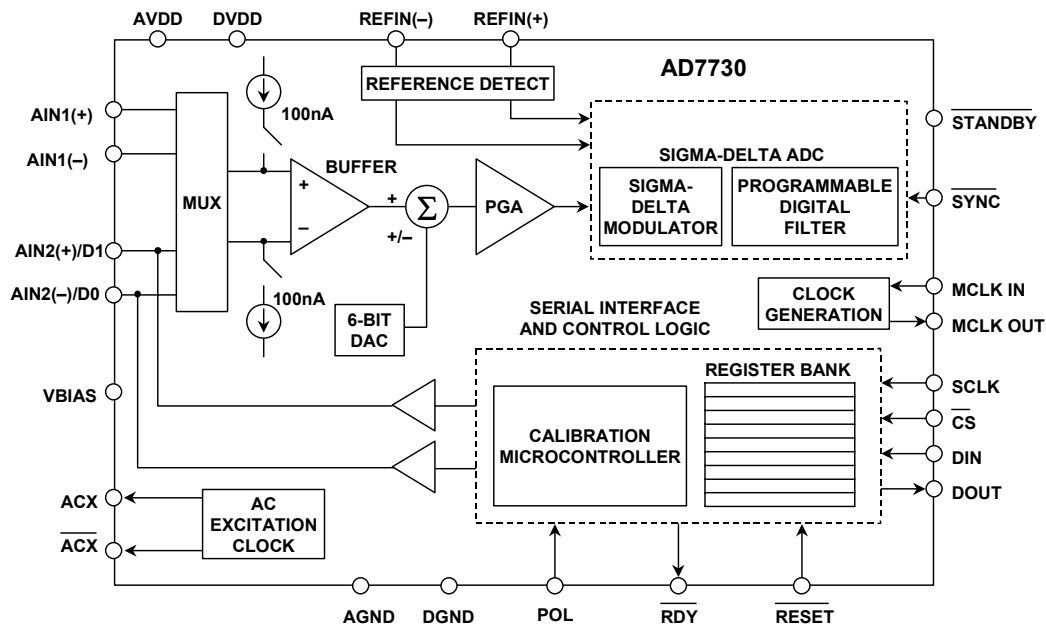


Figure 2-46: AD7730 sigma-delta measurement ADC with on-chip PGA

This combination of ADC and PGA is very powerful and enables the realization of a highly accurate system, with a minimum of circuit design. As an example, Figure 2-46 above shows a simplified diagram of the AD7730 sigma-delta measurement ADC which is optimized for digitizing low voltage bridge outputs directly (as low as 10mV full scale) to greater than 16-bits noise free code resolution, without the need for external signal conditioning circuits.

Additional information and background reading on PGAs can be found in References 4 and 5.

REFERENCES: PROGRAMMABLE GAIN AMPLIFIERS

1. John Krehbiel, "Monolithic Software Programmable-Gain Amplifier," **Analog Dialogue**, Vol. 21, No. 2, 1987, pp. 12-13.
2. Data sheet for **AD526 Software Programmable Gain Amplifier**, <http://www.analog.com>
3. Data sheet for **AD625 Programmable Gain Instrumentation Amplifier**, <http://www.analog.com>
4. Chapters 2, 3, 8, Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
5. Chapter 2, Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.

NOTES:

SECTION 2-3: ISOLATION AMPLIFIERS

Walt Kester, James Bryant

Analog Isolation Techniques

There are many applications where it is desirable, or even essential, for a sensor to have no direct ("galvanic") electrical connection with the system to which it is supplying data. This might be in order to avoid the possibility of dangerous voltages or currents from one half of the system doing damage in the other, or to break an intractable ground loop. Such a system is said to be "isolated", and the arrangement that passes a signal without galvanic connections is known as an *isolation barrier*.

The protection of an isolation barrier works in both directions, and may be needed in either, or even in both. The obvious application is where a sensor may accidentally encounter high voltages, and the system it is driving must be protected. Or a sensor may need to be isolated from accidental high voltages arising downstream, in order to protect its environment: examples include the need to prevent the ignition of explosive gases by sparks at sensors and the protection from electric shock of patients whose ECG, EEG or EMG is being monitored. The ECG case is interesting, as protection may be required in *both* directions: the patient must be protected from accidental electric shock, but if the patient's heart should stop, the ECG machine must be protected from the very high voltages (>7.5 kV) applied to the patient by the defibrillator which will be used to attempt to restart it. A summary of applications for isolation amplifiers (both analog and digital) is shown in Figure 2-47 below.

- ◆ **Sensor is at a High Potential Relative to Other Circuitry
(or may become so under Fault Conditions)**
- ◆ **Sensor May Not Carry Dangerous Voltages, Irrespective
of Faults in Other Circuitry
(e.g. Patient Monitoring and Intrinsically Safe Equipment
for use with Explosive Gases)**
- ◆ **To Break Ground Loops**

Figure 2-47: Applications for isolation amplifiers

Just as interference, or *unwanted* information, may be coupled by electric or magnetic fields, or by electromagnetic radiation, these phenomena may be used for the transmission of *wanted* information in the design of isolated systems.

The most common isolation amplifiers use *transformers*, which exploit magnetic fields, and another common type uses small high voltage capacitors, exploiting electric fields. *Optoisolators*, which consist of an LED and a photocell, provide isolation by using light, a form of electromagnetic radiation. Different isolators have differing performance: some are sufficiently linear to pass high accuracy analog signals across an isolation barrier. With others, the signal may need to be converted to digital form before transmission for accuracy is to be maintained (note this is a common V/F converter application).

■ OP AMP APPLICATIONS

Transformers are capable of analog accuracy of 12-16 bits and bandwidths up to several hundred kHz, but their maximum voltage rating rarely exceeds 10kV, and is often much lower. *Capacitively-coupled* isolation amplifiers have lower accuracy, perhaps 12-bits maximum, lower bandwidth, and lower voltage ratings— but they are low cost. Optical isolators are fast and cheap, and can be made with very high voltage ratings (4 -7kV is one of the more common ratings), but they have poor analog domain linearity, and are not usually suitable for direct coupling of precision analog signals.

Linearity and isolation voltage are not the only issues to be considered in the choice of isolation systems. Operating power is of course, essential. Both the input and the output circuitry must be powered, and unless there is a battery on the isolated side of the isolation barrier (which is possible, but rarely convenient), some form of isolated power must be provided. Systems using transformer isolation can easily use a transformer (either the signal transformer or another one) to provide isolated power, but it is impractical to transmit useful amounts of power by capacitive or optical means. Systems using these forms of isolation must make other arrangements to obtain isolated power supplies— this is a powerful consideration in favor of choosing transformer isolated isolation amplifiers: they almost invariably include an isolated power supply.

The isolation amplifier has an input circuit that is galvanically isolated from the power supply and the output circuit. In addition, there is minimal capacitance between the input and the rest of the device. Therefore, there is no possibility for DC current flow, and minimum AC coupling. Isolation amplifiers are intended for applications requiring safe, accurate measurement of low frequency voltage or current (up to about 100kHz) in the presence of high common-mode voltage (to thousands of volts) with high common mode rejection. They are also useful for line-receiving of signals transmitted at high impedance in noisy environments, and for safety in general-purpose measurements, where DC and line-frequency leakage must be maintained at levels well below certain mandated minimums. Principal applications are in electrical environments of the kind associated with medical equipment, conventional and nuclear power plants, automatic test equipment, and industrial process control systems.

AD210 3-Port Isolator

In a basic two-port form of isolator, the output and power circuits are not isolated from one another. A *three-port isolator* (input, power, output) is shown in Figure 2-48 (opposite). Note that in this diagram, the input circuits, output circuits, and power source are all isolated from one another. This figure represents the circuit architecture of a self-contained isolator, the AD210 (see References 1 and 2).

An isolator of this type requires power from a two-terminal DC power supply (PWR, PWR COM). An internal oscillator (50kHz) converts the DC power to AC, which is transformer-coupled to the shielded input section, then converted to DC for the input stage and the auxiliary power output.

SPECIALTY AMPLIFIERS ISOLATION AMPLIFIERS

The AC carrier is also modulated by the input stage amplifier output, transformer-coupled to the output stage, demodulated by a phase-sensitive demodulator (using the carrier as the reference), filtered, and buffered using isolated DC power derived from the carrier.

The AD210 allows the user to select gains from 1 to 100, using external resistors with the input section op amp. Bandwidth is 20kHz, and voltage isolation is 2500V RMS (continuous) and $\pm 3500\text{V}$ peak (continuous).

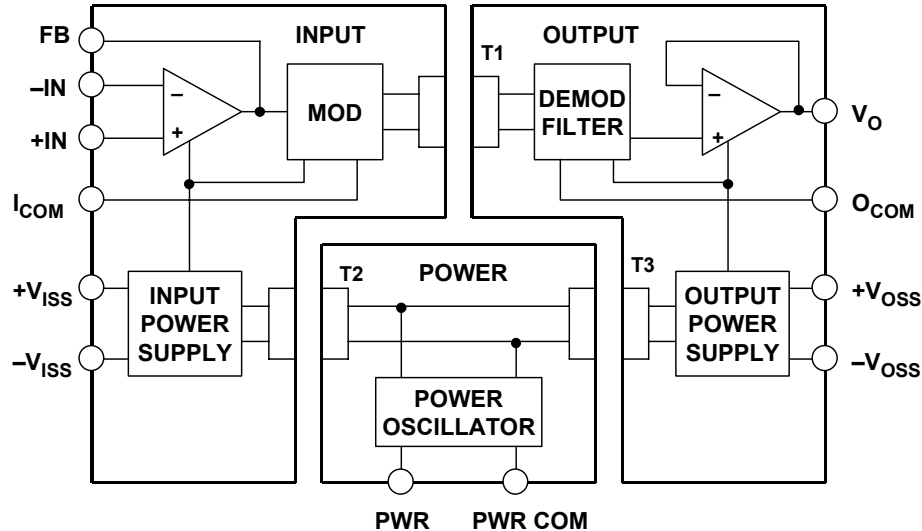


Figure 2-48: AD210 3-port isolation amplifier

The AD210 is a 3-port isolation amplifier, thus the power circuitry is isolated from both the input and the output stages and may therefore be connected to either (or to neither), without change in functionality. It uses transformer isolation to achieve 3500V isolation with 12-bit accuracy.

- ◆ **Transformer Coupled**
- ◆ **High Common Mode Voltage Isolation:**
 - **2500V RMS Continuous**
 - **$\pm 3500\text{V}$ Peak Continuous**
- ◆ **Wide Bandwidth: 20kHz (Full Power)**
- ◆ **0.012% Maximum Linearity Error**
- ◆ **Input Amplifier: Gain 1 to 100**
- ◆ **Isolated Input and Output Power Supplies, $\pm 15\text{V}$, $\pm 5\text{mA}$**

Figure 2-49: AD210 isolation amplifier key specifications

Key specifications for the AD210 are summarized in Figure 2-49 above.

Motor Control Isolation Amplifier

A typical isolation amplifier application using the AD210 is shown in Figure 2-50 below. The AD210 is used with an AD620 instrumentation amplifier in a current-sensing system for motor control. The input of the AD210, being isolated, can be directly connected to a 110 or 230 V power line without protection being necessary. The input section's isolated $\pm 15\text{V}$ powers the AD620, which senses the voltage drop in a small value current sensing resistor. The AD210 input stage op amp is simply connected as a unity-gain follower, which minimizes its error contribution. The 110 or 230V RMS common-mode voltage is ignored by this isolated system.

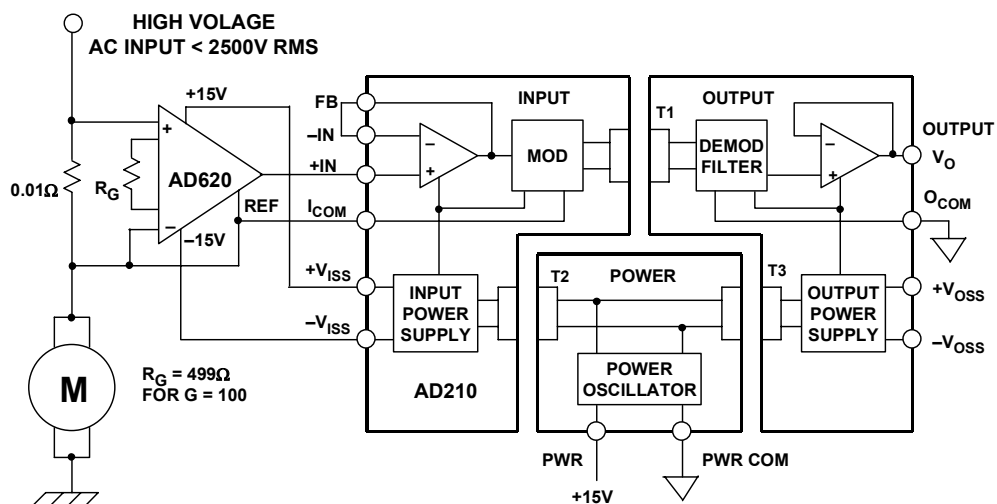


Figure 2-50: Motor control current sensing

Within this system the AD620 preamp is used as the system scaling control point, and will produce and output voltage proportional to motor current, as scaled by the sensing resistor value and gain as set by the AD620's R_G . The AD620 also improves overall system accuracy, as the AD210 V_{OS} is 15mV, versus the AD620's 30 μ V (with less drift also). Note that if higher DC offset and drift are acceptable, the AD620 may be omitted and the AD210 connected at a gain of 100.

Optional Noise Reduction Post Filter

Due to the nature of this type of carrier-operated isolation system, there will be certain operating situations where some residual AC carrier component will be superimposed upon the recovered output DC signal. When this occurs, a low impedance passive RC filter section following the output stage may be used (if the following stage has a high input impedance, i.e., non-loading to this filter). Note that will be the case for many high input impedance sampling ADCs, which appear essentially as a small capacitor. A 150 Ω resistance and 1nF capacitor will provide a corner frequency of about 1kHz. Note also that the capacitor should be a film type for low errors, such as polypropylene.

AD215 Two-Port Isolator

The AD215 is a high speed, two-port isolation amplifier, designed to isolate and amplify wide bandwidth analog signals (see Reference 3). The innovative circuit and transformer design of the AD215 ensures wide-band dynamic characteristics, while preserving DC performance specifications. An AD215 block diagram is shown in Figure 2-51 below.

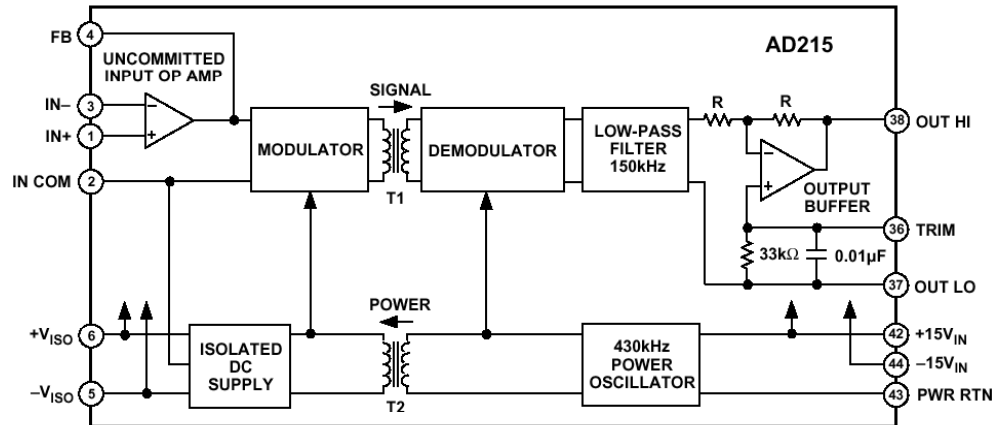


Figure 2-51: AD215 120kHz low distortion 2-port isolation amplifier

The AD215 provides complete galvanic isolation between the input and output of the device, which also includes the user-available front-end isolated bipolar power supply. The functionally complete design, powered by a $\pm 15\text{V}$ DC supply on the output side, eliminates the need for a user supplied isolated DC/DC converter. This permits the designer to minimize circuit overhead and reduce overall system design complexity and component costs.

The design of the AD215 emphasizes maximum flexibility and ease of use in a broad range of applications where fast analog signals must be measured under high common-mode voltage (CMV) conditions.

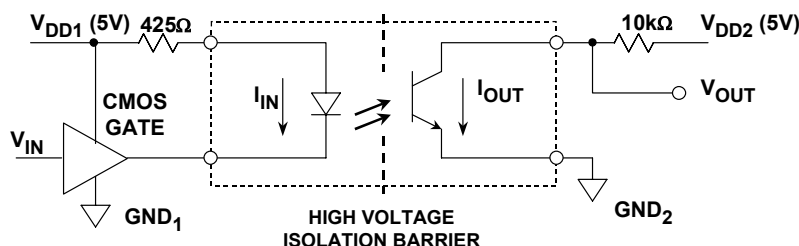
- ◆ Isolation voltage: 1500V rms
- ◆ Full power bandwidth: 120kHz
- ◆ Slew rate: 6V / μs
- ◆ Harmonic distortion: -80dB @ 1kHz
- ◆ 0.005% maximum linearity error
- ◆ Gain range: 1 to 10
- ◆ Isolated input power supply: $\pm 15\text{V}$ @ $\pm 10\text{mA}$

Figure 2-52: AD215 isolation amplifier key specifications

The AD215 has a $\pm 10\text{V}$ input/output range, a specified gain range of 1V/V to 10V/V, a buffered output with offset trim and a user-available isolated front end power supply which produces $\pm 15\text{V}$ DC at $\pm 10\text{mA}$. The key specifications of the AD215 are summarized in Figure 2-52 above.

Digital Isolation Techniques

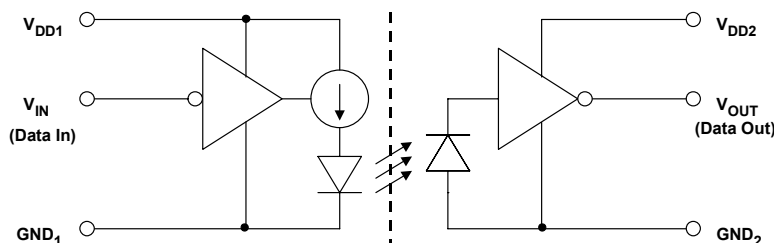
Analog isolation amplifiers find many applications where a high isolation is required, such as in medical instrumentation. Digital isolation techniques provide similar galvanic isolation, and are a reliable method of transmitting digital signals without ground noise.



- ◆ Uses Light for Transmission Over a High Voltage Barrier
- ◆ The LED is the Transmitter, and the Phototransistor is the Receiver
- ◆ High Voltage Isolation: 5000V to 7000V RMS
- ◆ Non-Linear -- Best for Digital or Frequency Information
- ◆ Rise and Fall-times can be 10 to 20μs in Slower Devices
- ◆ Example: Siemens ILQ-1 Quad (<http://www.siemens.com>)

Figure 2-53: Digital isolation using LED / phototransistor optocouplers

Optocouplers (also called optoisolators) are useful and available in a wide variety of styles and packages. A typical optocoupler based on an LED and a phototransistor is shown in Figure 2-53 above. A current of approximately 10mA drives an LED transmitter, with light output is received by a phototransistor. The light produced by the LED saturates the phototransistor. Input/output isolation of 5000V RMS to 7000V RMS is common. Although fine for digital signals, optocouplers are too nonlinear for most analog applications. Also, since the phototransistor is being saturated, response times can range from 10μs to 20μs in slower devices, limiting high speed applications.



- ◆ +5V Supply Voltage
- ◆ 2500V RMS I/O Withstand Voltage
- ◆ Logic Signal Frequency: 12.5MHz Maximum
- ◆ 25MBd Maximum Data Rate
- ◆ 40ns Maximum Propagation Delay
- ◆ 9ns Typical Rise/Fall Time
- ◆ Example: Agilent HCPL-7720
- ◆ (<http://www.semiconductor.agilent.com>)

Figure 2-54: Digital isolation using LED / photodiode optocouplers

A much faster optocoupler architecture is shown in Figure 2-54 above, and is based on an LED and a photodiode. The LED is again driven with a current of approximately 10mA.

This produces a light output sufficient to generate enough current in the receiving photodiode to develop a valid high logic level at the output of the transimpedance amplifier. Speed can vary widely between optocouplers, and the fastest ones have propagation delays of 20ns typical, and 40ns maximum, and can handle data rates up to 25MBd for NRZ data. This corresponds to a maximum square wave operating frequency of 12.5MHz, and a minimum allowable passable pulse width of 40ns.

AD260/AD261 High Speed Logic Isolators

The AD260/AD261 family of digital isolators operates on a principle of transformer-coupled isolation (see Reference 4). They provide isolation for five digital control signals to/from high speed DSPs, microcontrollers, or microprocessors. The AD260 also has a 1.5W transformer for a 3.5kV RMS isolated external DC/DC power supply circuit.

Each line of the AD260 can handle digital signals up to 20MHz (40MBd) with a propagation delay of only 14ns which allows for extremely fast data transmission. Output waveform symmetry is maintained to within ± 1 ns of the input so the AD260 can be used to accurately isolate time-based pulse width modulator (PWM) signals.

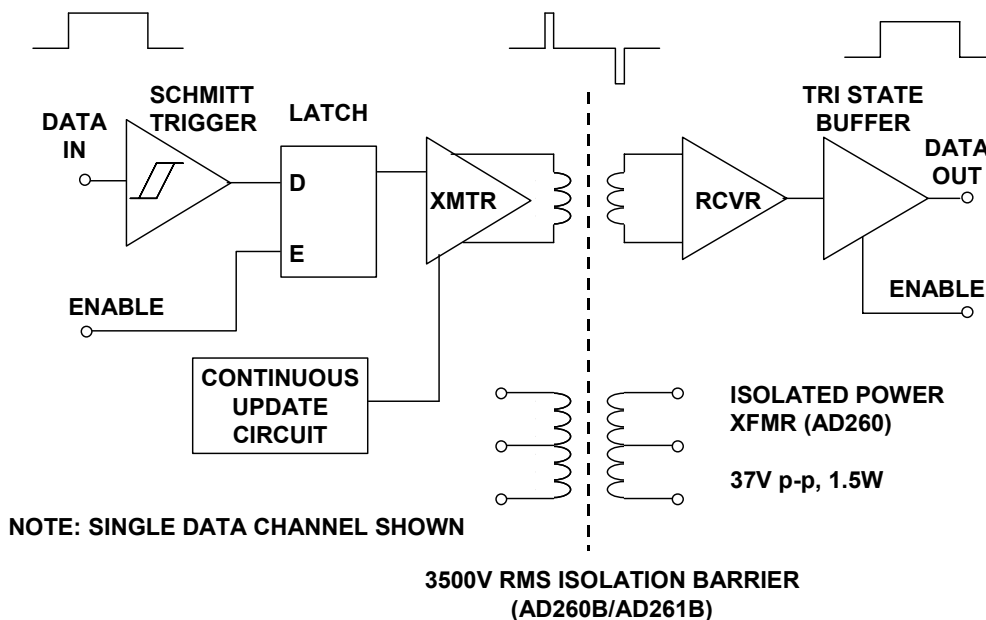


Figure 2-55: AD260/AD261 digital isolators

A simplified schematic of one channel of the AD260/AD261 is shown in Figure 2-55 above. The data input is passed through a schmitt trigger circuit, through a latch, and a special transmitter circuit which differentiates the edges of the digital input signal and drives the primary winding of a proprietary transformer with a "set-high/set-low" signal. The secondary of the isolation transformer drives a receiver with the same "set-hi/set-low" data, which regenerates the original logic waveform. An internal circuit operating in the background interrogates all inputs about every 5 μ s, and in the absence of logic transitions, sends appropriate "set-hi/set-low" data across the interface. Recovery time from a fault condition or at power-up is thus between 5 μ s and 10 μ s.

■ OP AMP APPLICATIONS

The power transformer (available on the AD260) is designed to operate between 150kHz and 250kHz and will easily deliver more than 1W of isolated power when driven push-pull (5V) on the transmitter side. Different transformer taps, rectifier and regulator schemes will provide combinations of $\pm 5\text{V}$, 15V, 24V, or even 30V or higher.

The transformer output voltage when driven with a low voltage-drop drive will be 37V p-p across the entire secondary with a 5V push-pull drive. Figure 2-56 below summarizes the key specifications of the AD260/261 series.

- ◆ **Isolation Test Voltage to 3500V RMS (AD260B/AD261B)**
- ◆ **Five Isolated Digital Lines Available in 6 Input/Output Configurations**
- ◆ **Logic Signal Frequency: 20MHz Max.**
- ◆ **Data Rate: 40MBd Max.**
- ◆ **Isolated Power Transformer: 37V p-p, 1.5W (AD260)**
- ◆ **Waveform Edge Transmission Symmetry: $\pm 1\text{ns}$**
- ◆ **Propagation Delay: 14ns**
- ◆ **Rise and Fall-Times < 5ns**

Figure 2-56: AD260/AD261 digital isolator key specifications

The availability of low cost digital isolators such as those previously discussed solves most system isolation problems in data acquisition systems as shown in Figure 2-57 below. In the upper example, digitizing the signal first, then using digital isolation eliminates the problem of analog isolation amplifiers. While digital isolation can be used with parallel output ADCs provided the bandwidth of the isolator is sufficient, it is more practical with ADCs that have *serial* outputs. This minimizes cost and component count. A 3-wire interface (data, serial clock, framing clock) is all that is required in these cases.

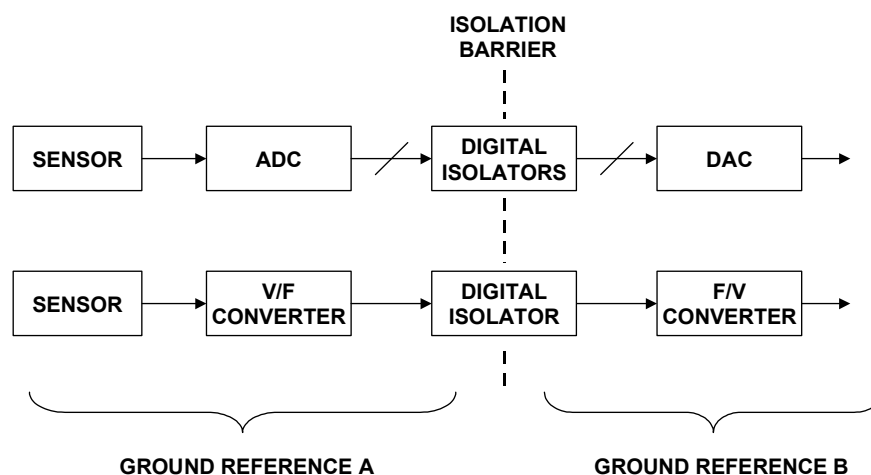


Figure 2-57: Practical application of digital isolation in data acquisition systems

An alternative (lower example) is to use a voltage-to-frequency converter (VFC) as a transmitter and a frequency-to-voltage converter (FVC) as a receiver. In this case, only one digital isolator is required.

REFERENCES: ISOLATION AMPLIFIERS

1. James Conant, "Precision Wideband Three-Port Isolation Amplifier," **Analog Dialogue**, Vol. 20, No. 2, 1986, pp. 14.
2. Data sheet for **AD210 Precision, Wide Bandwidth 3-Port Isolation Amplifier**, <http://www.analog.com>
3. Data sheet for **AD215 120 kHz Bandwidth, Low Distortion, Isolation Amplifier**, <http://www.analog.com>
4. Data sheets for **AD260 and AD261 High Speed Logic Isolators**, <http://www.analog.com>
5. Chapters 2, 3, 8, Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
6. Chapter 2, Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.
7. Chapter 10, Walt Kester, Editor, **Mixed-Signal and DSP Design Techniques**, Analog Devices, 2000, ISBN: 0-916550-23-0.

NOTES:

- H Op Amp History**
 - 1 Op Amp Basics**
 - 2 Specialty Amplifiers**
 - ◆ 3 Using Op Amps with Data Converters**
 - 1 Introduction**
 - 2 ADC/DAC Specifications**
 - 3 Driving ADC Inputs**
 - 4 Driving ADC/DAC Reference Inputs**
 - 5 Buffering DAC Outputs**
 - 4 Sensor Signal Conditioning**
 - 5 Analog Filters**
 - 6 Signal Amplifiers**
 - 7 Hardware and Housekeeping Techniques**

▣ OP AMP APPLICATIONS

CHAPTER 3: USING OP AMPS WITH DATA CONVERTERS

Walt Kester, James Bryant, Paul Hendriks

SECTION 3-1: INTRODUCTION

Walt Kester

This chapter of the book deals with data conversion and associated signal conditioning circuitry involving the use of op amps. Data conversion is a very broad topic, and this chapter will provide only enough background material so that the reader can make intelligent decisions regarding op amp selection. There is much more material on the subject available in the references (see References 1-5).

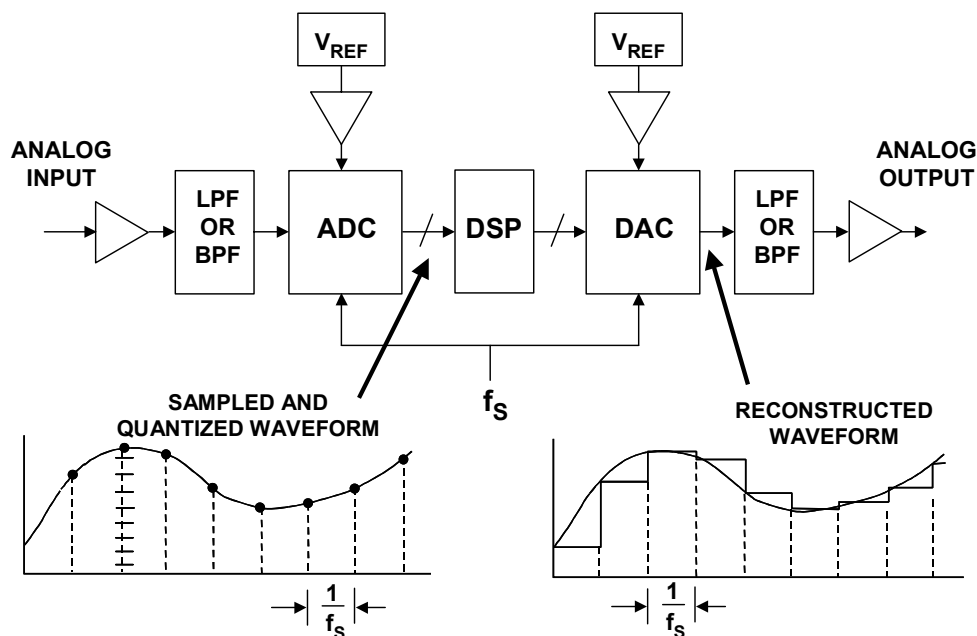


Figure 3-1: Typical sampled data system showing potential amplifier applications

Figure 3-1 above shows a generalized sampled data system and some possible applications of op amps. The analog input signal is first buffered and filtered before it is applied to the analog-to-digital converter (ADC). The buffer may or may not be required, depending upon the input structure of the ADC. For example, some ADCs (such as switched capacitor) generate transient currents at their inputs due to the internal conversion architecture, and these currents must be isolated from the signal source. A suitable buffer amplifier provides a low impedance drive and absorbs these currents. In some cases, an op amp is required to provide the appropriate gain and offset to match the signal to the input range of the ADC.

■ OP AMP APPLICATIONS

Another key component in a sampled data system is the anti-aliasing filter which removes signals that fall outside the Nyquist bandwidth, $f_s/2$. Normally this filter is a lowpass filter, but it can be a bandpass filter in certain undersampling applications. If the op amp buffer is required, it may be located before or after the filter, depending on system considerations. In fact, the filter itself may be an active one, in which case the buffering function can be performed by the actual output amplifier of the filter. More discussions regarding active filters can be found in Chapter 5 of this book.

After the signal is buffered and filtered, it is applied to the ADC. The full scale input voltage range of the ADC is generally determined by a voltage reference, V_{REF} . Some ADCs have this function on chip, while others require an external reference. If an external reference is required, its output may require buffering using an appropriate op amp. The reference input to the ADC may be connected to an internal switched capacitor network, causing transient currents to be generated at that node (similar to the analog input of such converters). Therefore some references may require a buffer to isolate these transient currents from the actual reference output. Other references may have internal buffers that are sufficient, and no additional buffering is needed in those cases.

- ◆ **Gain setting**
- ◆ **Level shifting**
- ◆ **Buffering ADC transients from signal source**
- ◆ **Buffering voltage reference outputs**
- ◆ **Buffering DAC outputs**
- ◆ **Active anti-aliasing filter before ADC**
- ◆ **Active anti-imaging filter after DAC**

Figure 3-2: Data converter amplifier applications

The output of the ADC is then processed digitally by an appropriate processor, shown in the diagram as a digital signal processor (DSP). DSPs are processors that are optimized to perform fast repetitive arithmetic, as required in digital filters or fast Fourier transform (FFT) algorithms. The DSP output then drives a digital-to-analog converter (DAC) which converts the digital signal back into an analog signal.

The DAC analog output must be filtered to remove the unwanted image frequencies caused by the sampling process, and further buffering may be required to provide the proper signal amplitude and offset. The output filter is generally placed between the DAC and the buffer amplifier, but their positions can be reversed in certain applications. It is also possible to combine the filtering and buffering function if an active filter is used to condition the DAC output.

Trends in Data Converters

It is useful to examine a few general trends in data converters, to better understand any associated op amp requirements. Converter performance is first and foremost; and maintaining that performance in a system application is extremely important. In low frequency measurement applications (10Hz bandwidth signals or lower), sigma-delta ADCs with resolutions up to 24 bits are now quite common. These converters generally have automatic or factory calibration features to maintain required gain and offset accuracy. In higher frequency signal processing, ADCs must have wide dynamic range (low distortion and noise), high sampling frequencies, and generally excellent AC specifications.

In addition to sheer performance, other characteristics such as low power, single supply operation, low cost, and small surface mount packages also drive the data conversion market. These requirements result in application problems because of reduced signal swings, increased sensitivity to noise, etc. In addition, many data converters are now produced on low-cost foundry CMOS processes which generally make on-chip amplifier design more difficult and therefore less likely to be incorporated on-chip.

- ◆ **Higher sampling rates, higher resolution, higher AC performance**
- ◆ **Single supply operation (e.g., +5V, +3V)**
- ◆ **Lower power**
- ◆ **Smaller input/output signal swings**
- ◆ **Maximize usage of low cost foundry CMOS processes**
- ◆ **Smaller packages**
- ◆ **Surface mount technology**

Figure 3-3: *Some general trends in data converters*

As has been mentioned previously, the analog input to a CMOS ADC is usually connected directly to a switched-capacitor sample-and-hold (SHA), which generates transient currents that must be buffered from the signal source. On the other hand, data converters fabricated on Bi-CMOS or bipolar processes are more likely to have internal buffering, but generally have higher cost and power than their CMOS counterparts.

It should be clear by now that selecting an appropriate op amp for a data converter application is highly dependent on the particular converter under consideration. Generalizations are difficult, but some meaningful guidelines can be followed.

The most obvious requirement for a data converter buffer amplifier is that it not degrade the DC or AC performance of the converter. One might assume that a careful reading of the op amp datasheets would assist in the selection process: simply lay the data converter and the op amp datasheets side by side, and compare each critical performance specification. It is true that this method will provide some degree of success, however in order to perform an accurate comparison, the op amp must be specified under the exact operating conditions required by the data converter application. Such factors as gain, gain setting resistor values, source impedance, output load, input and output signal amplitude, input and output common-mode (CM) level, power supply voltage, etc., all affect op amp performance.

■ OP AMP APPLICATIONS

It is highly unlikely that even a well written op amp datasheet will provide an exact match to the operating conditions required in the data converter application. Extrapolation of specified performance to fit the exact operating conditions can give erroneous results. Also, the op amp may be subjected to transient currents from the data converter, and the corresponding effects on op amp performance are rarely found on datasheets.

Converter datasheets themselves can be a good source for recommended op amps and other application circuits. However this information can become obsolete as newer op amps are introduced after the converter's initial release.

Analog Devices and other op amp manufacturers today have on-line websites featuring parametric search engines, which facilitate part selection (see Reference 1). For instance, the first search might be for minimum power supply voltage, e.g., +3V. The next search might be for bandwidth, and further searches on relevant specifications will narrow the selection of op amps even further. Figure 3-4 below summarizes the selection process.

- ◆ **The amplifier should not degrade the performance of the ADC/DAC**
- ◆ **AC specifications are usually the most important**
 - Noise
 - Bandwidth
 - Distortion
- ◆ **Selection based on op amp data sheet specifications difficult due to varying conditions in actual application circuit with ADC/DAC:**
 - Power supply voltage
 - Signal range (differential and common-mode)
 - Loading (static and dynamic)
 - Gain
- ◆ **Parametric search engines may be useful**
- ◆ **ADC/DAC data sheets often recommend op amps (but may not include newly released products)**

Figure 3-4: General amplifier selection criteria

While not necessarily suitable for the final selection, this process can narrow the search to a manageable number of op amps whose individual datasheets can be retrieved, then reviewed in detail before final selection.

From the discussion thus far, it should be obvious that in order to design a proper interface, an understanding of both op amps and data converters is required. References 2-6 provide background material on data converters.

The next section of this chapter addresses key data converter performance specifications without going into the detailed operation of converters themselves. The remainder of the chapter shows a number of specific applications of op amps with various data converters.

REFERENCES: INTRODUCTION

1. ADI website, at <http://www.analog.com>
2. Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8.
3. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6.
4. Chapters 3, 8, Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
5. Chapters 2, 3, 4, Walt Kester, Editor, **Mixed-Signal and DSP Design Techniques**, Analog Devices, 2000, ISBN: 0-916550-23-0.
6. Chapters 4, 5, Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.

▣ OP AMP APPLICATIONS

NOTES:

SECTION 3-2: ADC/DAC SPECIFICATIONS

Walt Kester

ADC and DAC Static Transfer Functions and DC Errors

The most important thing to remember about both DACs and ADCs is that either the input or output is digital, and therefore the signal is *quantized*. That is, an N-bit word represents one of 2^N possible states, and therefore an N-bit DAC (with a fixed reference) can have only 2^N possible analog outputs, and an N-bit ADC can have only 2^N possible digital outputs. The analog signals will generally be voltages or currents.

The resolution of data converters may be expressed in several different ways: the weight of the Least Significant Bit (LSB), parts per million of full scale (ppm FS), millivolts (mV), etc. It is common that different devices (even from the same manufacturer) will be specified differently, so converter users must learn to translate between the different types of specifications if they are to compare devices successfully. The size of the least significant bit for various resolutions is shown in Figure 3-5 below.

RESOLUTION N	2^N	VOLTAGE (10V FS)	ppm FS	% FS	dB FS
2-bit	4	2.5 V	250,000	25	-12
4-bit	16	625 mV	62,500	6.25	-24
6-bit	64	156 mV	15,625	1.56	-36
8-bit	256	39.1 mV	3,906	0.39	-48
10-bit	1,024	9.77 mV (10 mV)	977	0.098	-60
12-bit	4,096	2.44 mV	244	0.024	-72
14-bit	16,384	610 μ V	61	0.0061	-84
16-bit	65,536	153 μ V	15	0.0015	-96
18-bit	262,144	38 μ V	4	0.0004	-108
20-bit	1,048,576	9.54 μ V (10 μ V)	1	0.0001	-120
22-bit	4,194,304	2.38 μ V	0.24	0.000024	-132
24-bit	16,777,216	596 nV*	0.06	0.000006	-144

*600nV is the Johnson Noise in a 10kHz BW of a 2.2k Ω Resistor @ 25°C

Remember: 10-bits and 10V FS yields an LSB of 10mV, 1000ppm, or 0.1%.
All other values may be calculated by powers of 2.

Figure 3-5: Quantization: the size of a least significant bit (LSB)

As noted above (and obvious from this table), the LSB scaling for a given converter resolution can be expressed in various ways. While it is convenient to relate this to a full scale of 10V, as in the figure, other full scale levels can be easily extrapolated.

Before we can consider op amp applications with data converters, it is necessary to consider the performance to be expected, and the specifications that are important when operating with data converters. The following sections will consider the definition of errors and specifications used for data converters.

OP AMP APPLICATIONS

The first applications of data converters were in measurement and control, where the exact timing of the conversion was usually unimportant, and the data rate was slow. In such applications, the DC specifications of converters are important, but timing and AC specifications are not. Today many, if not most, converters are used in *sampling* and *reconstruction* systems where AC specifications are critical (and DC ones may not be).

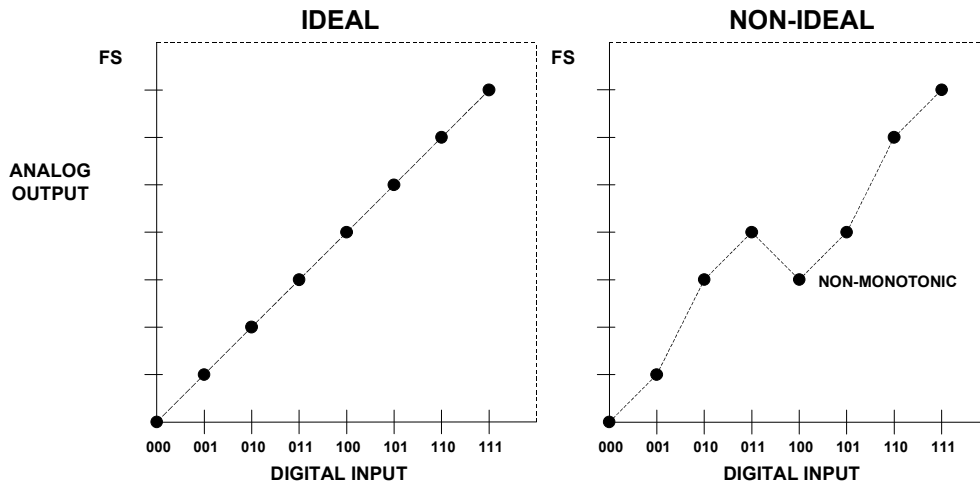


Figure 3-6: DAC transfer functions

Figure 3-6 above shows the transfer characteristics for a 3-bit unipolar ideal and non-ideal DAC. In a DAC, both the input and output are quantized, and the graph consists of eight points— while it is reasonable to discuss a line through these points, it is critical to remember that the actual transfer characteristic is *not* a line, but a series of discrete points.

Similarly, Figure 3-7 below shows the transfer characteristics for a 3-bit unipolar ideal and non-ideal ADC. Note that the input to an ADC is analog and is therefore *not* quantized, but its output *is* quantized.

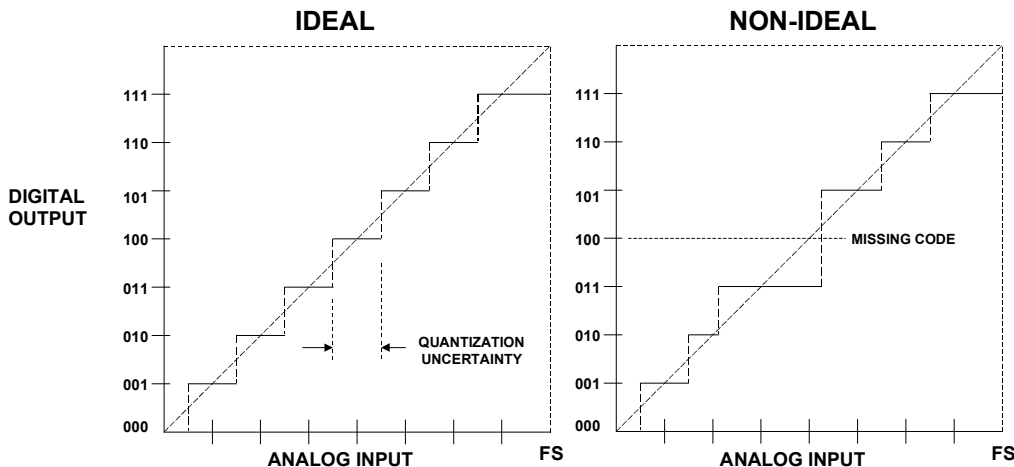


Figure 3-7: ADC transfer functions

The ADC transfer characteristic therefore consists of eight horizontal steps (when considering the offset, gain and linearity of an ADC we consider the line joining the midpoints of these steps).

USING OP AMPS WITH DATA CONVERTERS

ADC/DAC SPECIFICATIONS

The (ideal) ADC transitions take place at $\frac{1}{2}$ LSB above zero, and thereafter every LSB, until $1\frac{1}{2}$ LSB below analog full scale. Since the analog input to an ADC can take any value, but the digital output is quantized, there may be a difference of up to $\frac{1}{2}$ LSB between the actual analog input and the exact value of the digital output. This is known as the quantization error or quantization uncertainty as shown in Figure 3-7. In AC (sampling) applications this quantization error gives rise to quantization noise which will be discussed shortly.

The *integral linearity* error of a converter is analogous to the linearity error of an amplifier, and is defined as the maximum deviation of the actual transfer characteristic of the converter from a straight line. It is generally expressed as a percentage of full scale (but may be given in LSBs). There are two common ways of choosing the straight line: *end point* and *best straight line*.

In the end point system, the deviation is measured from the straight line through the origin and the full scale point (after gain adjustment). This is the most useful integral linearity measurement for measurement and control applications of data converters (since error budgets depend on deviation from the ideal transfer characteristic, not from some arbitrary "best fit"), and is the one normally adopted by Analog Devices, Inc.

The best straight line, however, does give a better prediction of distortion in AC applications, and also gives a lower value of "linearity error" on a datasheet. The best fit straight line is drawn through the transfer characteristic of the device using standard curve fitting techniques, and the maximum deviation is measured from this line. In general, the integral linearity error measured in this way is only 50% of the value measured by end point methods. This makes the method good for producing impressive datasheets, but it is less useful for error budget analysis. For AC applications, it is even better to specify distortion than DC linearity, so it is rarely necessary to use the best straight line method to define converter linearity.

The other type of converter non-linearity is *differential non-linearity* (DNL). This relates to the linearity of the code transitions of the converter. In the ideal case, a change of 1 LSB in digital code corresponds to a change of exactly 1 LSB of analog signal. In a DAC, a change of 1 LSB in digital code produces exactly 1 LSB change of analog output, while in an ADC there should be exactly 1 LSB change of analog input to move from one digital transition to the next.

Where the change in analog signal corresponding to 1 LSB digital change is more or less than 1 LSB, there is said to be a DNL error. The DNL error of a converter is normally defined as the maximum value of DNL to be found at any transition.

If the DNL of a DAC is less than -1 LSB at any transition (Figure 3-6, again), the DAC is *non-monotonic*; i.e., its transfer characteristic contains one or more localized maxima or minima. A DNL greater than $+1$ LSB does not cause non-monotonicity, but is still undesirable. In many DAC applications (especially closed-loop systems where non-monotonicity can change negative feedback to positive feedback), it is critically important that DACs are monotonic. DAC monotonicity is often explicitly specified on datasheets,

■ OP AMP APPLICATIONS

but if the DNL is guaranteed to be less than 1 LSB (i.e., $|DNL| \leq 1\text{LSB}$) then the device must be monotonic, even without an explicit guarantee.

ADCs can be non-monotonic, but a more common result of excess DNL in ADCs is *missing codes* (Figure 3-7, again). Missing codes (or non-monotonicity) in an ADC are as objectionable as non-monotonicity in a DAC. Again, they result from $DNL > 1\text{LSB}$.

Quantization Noise in Data Converters

The only errors (DC or AC) associated with an ideal N-bit ADC are those related to the sampling and quantization processes. The maximum error an ideal ADC makes when digitizing a DC input signal is $\pm 1/2\text{LSB}$. Any AC signal applied to an ideal N-bit ADC will produce quantization noise whose rms value (measured over the Nyquist bandwidth, DC to $f_s/2$) is approximately equal to the weight of the least significant bit (LSB), q , divided by $\sqrt{12}$. This assumes that the signal is at least a few LSBs in amplitude so that the ADC output always changes state. The quantization error signal from a linear ramp input is approximated as a sawtooth waveform with a peak-to-peak amplitude equal to q , and its rms value is therefore $q/\sqrt{12}$ (see Figure 3-8 below).

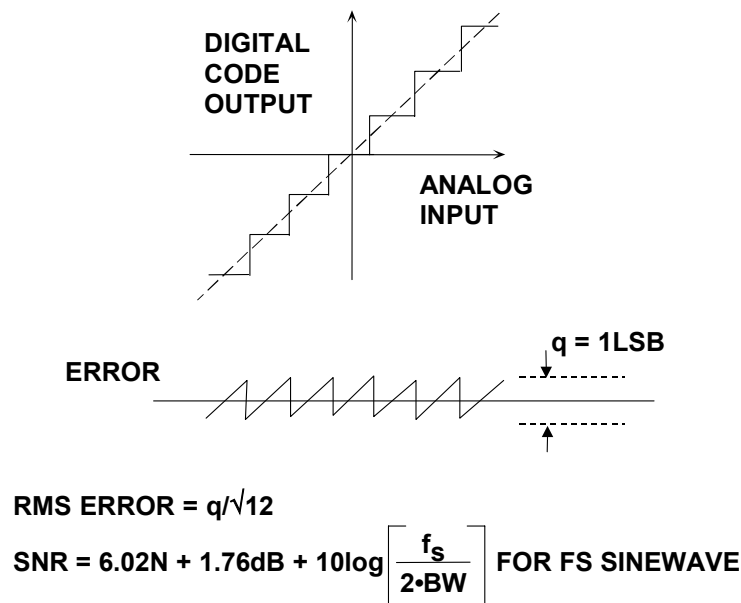


Figure 3-8: Ideal N-bit ADC quantization noise

It can be shown that the ratio of the rms value of a full scale sinewave to the rms value of the quantization noise (expressed in dB) is:

$$\text{SNR} = 6.02N + 1.76\text{dB}, \quad \text{Eq. 3-1}$$

where N is the number of bits in the ideal ADC. *Note— this equation is only valid if the noise is measured over the entire Nyquist bandwidth from DC to $f_s/2$. If the signal bandwidth, BW, is less than $f_s/2$, then the SNR within the signal bandwidth BW is increased because the amount of quantization noise within the signal bandwidth is less.*

The correct expression for this condition is given by:

$$\text{SNR} = 6.02N + 1.76\text{dB} + 10 \log\left(\frac{f_s}{2 \cdot \text{BW}}\right). \quad \text{Eq. 3-2}$$

The above equation reflects the condition called *oversampling*, where the sampling frequency is higher than twice the signal bandwidth. The correction term is often called *processing gain*. Notice that for a given signal bandwidth, doubling the sampling frequency increases the SNR by 3dB.

ADC Input-Referred Noise

The internal circuits of an ADC produce a certain amount of wideband rms noise due to thermal and kT/C effects. This noise is present even for DC input signals, and accounts for the fact that the output of most wideband (or high resolution) ADCs is a distribution of codes, centered around the nominal DC input value, as is shown in Figure 3-9 below.

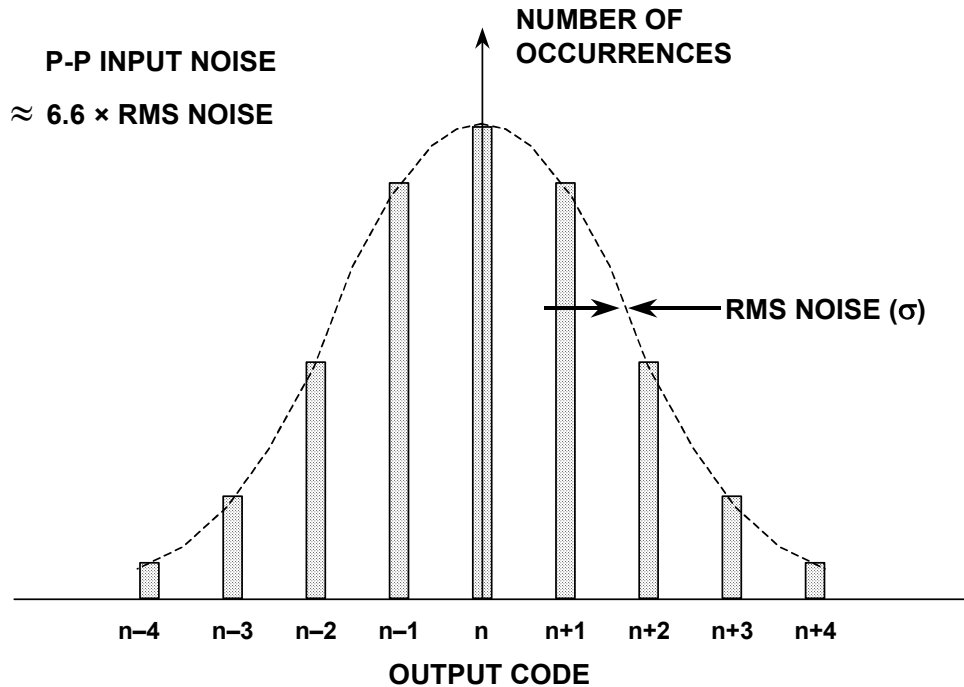


Figure 3-9: Effect of ADC input-referred noise on "grounded input" histogram

To measure its value, the input of the ADC is grounded, and a large number of output samples are collected and plotted as a histogram (sometimes referred to as a *grounded-input* histogram).

Since the noise is approximately Gaussian, the standard deviation (σ) of the histogram is easily calculated, corresponding to the effective input rms noise. It is common practice to express this rms noise in terms of LSBs, although it can be expressed as an rms voltage.

Calculating Op Amp Output Noise and Comparing it with ADC Input-Referred Noise

In precision measurement applications utilizing 16- to 24-bit sigma-delta ADCs operating on low frequency (<20Hz, e.g.) signals, it is generally undesirable to use a drive amplifier in front of the ADC because of the increased noise due to the amplifier itself. If an op amp is required, however, the op amp output 1/f noise should be compared to the input-referred ADC noise. The 1/f noise is usually specified as a peak-to-peak value measured over the 0.1Hz to 10Hz bandwidth and referred to the op amp input (see Chapter 1 of this book). Op amps such as the OP177 and the AD707 (input voltage noise 350nV p-p) or the AD797 (input voltage noise 50nV p-p) are appropriate for high resolution measurement applications if required.

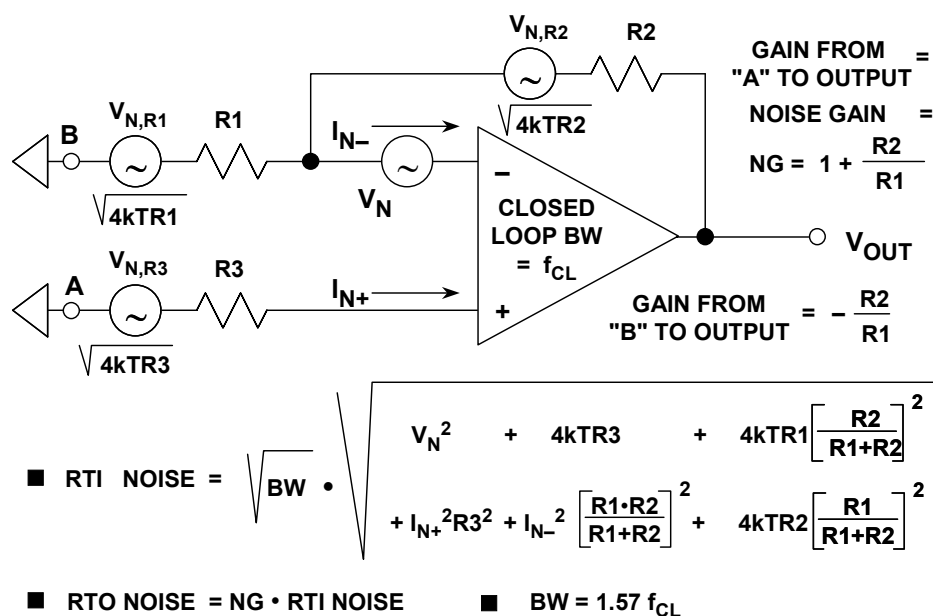


Figure 3-10: Op amp noise model for a first-order circuit with resistive feedback

The general model for calculating the referred-to-input (RTI) or referred-to-output (RTO) noise of an op amp is shown in Figure 3-10 above. This model shows all possible noise sources. The results using this model are relatively accurate, provided there is less than 1dB gain peaking in the closed loop frequency response. For higher frequency applications, 1/f noise can be neglected, because the dominant contributor is white noise.

An example of a practical noise calculation is shown in Figure 3-11 (opposite). In this circuit, a wideband, low distortion amplifier (AD9632) drives a 12-bit, 25MSPS ADC (AD9225). The input voltage noise spectral density of the AD9632 (4.3nV/√Hz) dominates the op amp noise because of the low gain and the low values of the external feedback resistors. The noise at the output of the AD9632 is obtained by multiplying the input voltage noise spectral density by the noise gain of 2. To obtain the rms noise, the noise spectral density is multiplied by the equivalent noise bandwidth of 50MHz which is set by the single-pole lowpass filter placed between the op amp and the ADC input.

USING OP AMPS WITH DATA CONVERTERS ADC/DAC SPECIFICATIONS

Note that the closed-loop bandwidth of the AD9632 is 250MHz, and the input bandwidth of the AD9225 is 105MHz. With no filter, the output noise of the AD9632 would be integrated over the full 105MHz ADC input bandwidth.

However, the sampling frequency of the ADC is 25MSPS, thereby implying that signals above 12.5MHz are not of interest, assuming Nyquist operation (as opposed to undersampling applications where the input signal can be greater than the Nyquist frequency, $f_s/2$). The addition of this simple filter significantly reduces noise effects.

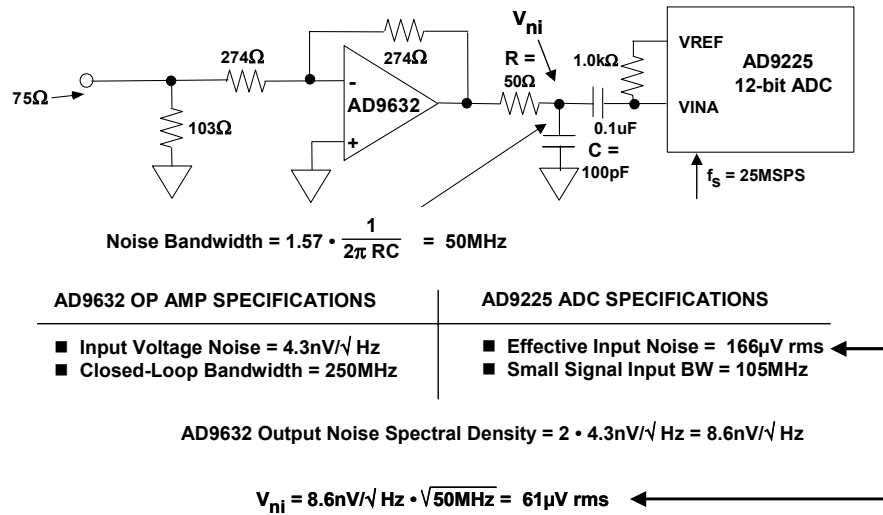


Figure 3-11: Noise calculations for the AD9632 op amp driving the AD9225 12-bit, 25MSPS ADC

The noise at the output of the lowpass filter is calculated as approximately 61μV rms which is less than half the effective input noise of the AD9225, 166μV rms. Without the filter, the noise from the op amp would be about 110μV rms (integrating over the full equivalent ADC input noise bandwidth of $1.57 \times 105\text{MHz} = 165\text{MHz}$).

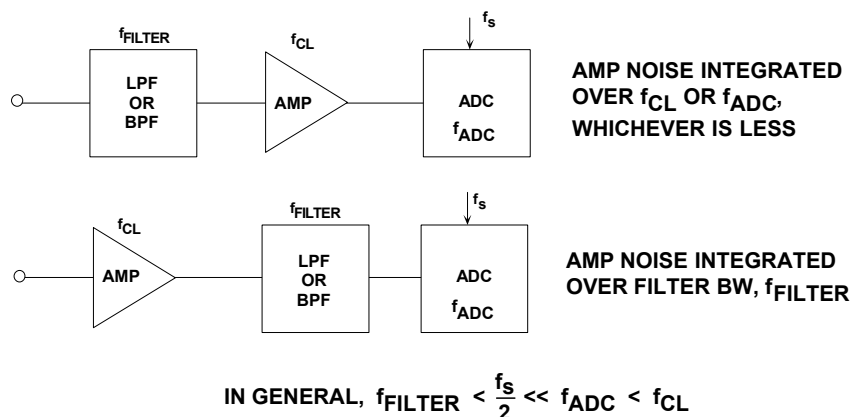


Figure 3-12: Proper positioning of the antialiasing filter will reduce the effects of op amp noise

This serves to illustrate the general concept shown in Figure 3-12 above. In most high speed system applications a passive antialiasing filter (either lowpass for baseband sampling, or bandpass for undersampling) is required, and placing this filter between the op amp and the ADC will serve to reduce the noise due to the op amp.

Quantifying and Measuring Converter Dynamic Performance

There are various ways to characterize the AC performance of ADCs. In the early years of ADC technology (over 30 years ago) there was little standardization with respect to AC specifications, and measurement equipment and techniques were not well understood or available. Over nearly a 30 year period, manufacturers and customers have learned more about measuring the dynamic performance of converters, and the specifications shown in Figure 3-13 below represent the most popular ones used today.

- ◆ **Signal-to-Noise-and-Distortion Ratio (SINAD, or S/N +D)**
- ◆ **Effective Number of Bits (ENOB)**
- ◆ **Signal-to-Noise Ratio (SNR)**
- ◆ **Analog Bandwidth (Full-Power, Small-Signal)**
- ◆ **Harmonic Distortion**
- ◆ **Worst Harmonic**
- ◆ **Total Harmonic Distortion (THD)**
- ◆ **Total Harmonic Distortion Plus Noise (THD + N)**
- ◆ **Spurious Free Dynamic Range (SFDR)**
- ◆ **Two-Tone Intermodulation Distortion**
- ◆ **Multi-tone Intermodulation Distortion**

Figure 3-13: Quantifying ADC dynamic performance

Practically all the specifications represent the converter's performance in the frequency domain, and all are related to noise and distortion in one manner or another.

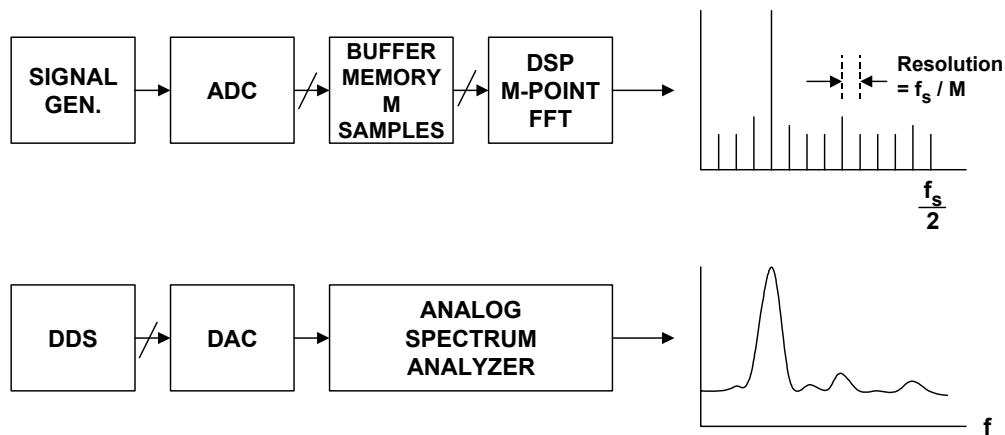


Figure 3-14: Measuring ADC / DAC dynamic performance

ADC outputs are analyzed using fast Fourier transform (FFT) techniques, and DAC outputs are analyzed using conventional analog spectrum analyzers, as shown in Figure 3-14 above. In the case of an ADC, the input signal is an analog sinewave, and in the case of a DAC, the input is a digital sinewave generated by a direct digital synthesis (DDS) system.

Signal-to-Noise-and-Distortion Ratio (SINAD), Signal-to-Noise Ratio (SNR), and Effective Number of Bits (ENOB)

SINAD and SNR deserve careful attention (see Figure 3-15 below), because there is still some variation between ADC manufacturers as to their precise meaning. *Signal-to-noise-and Distortion* (SINAD, or S/N+D) is the ratio of the rms signal amplitude to the mean value of the root-sum-square (RSS) of all other spectral components, *including harmonics*, but excluding DC. SINAD is a good indication of the overall dynamic performance of an ADC as a function of input frequency, because it includes all components which make up noise (including thermal noise) and distortion. It is often plotted for various input amplitudes. SINAD is equal to THD+N if the bandwidth for the noise measurement is the same.

◆ **SINAD (Signal-to-Noise-and-Distortion Ratio):**

- The ratio of the rms signal amplitude to the mean value of the root-sum-squares (RSS) of all other spectral components, including harmonics, but excluding DC.

◆ **ENOB (Effective Number of Bits):**

$$\text{ENOB} = \frac{\text{SINAD} - 1.76\text{dB}}{6.02}$$

◆ **SNR (Signal-to-Noise Ratio, or Signal-to-Noise Ratio Without Harmonics):**

- The ratio of the rms signal amplitude to the mean value of the root-sum-squares (RSS) of all other spectral components, excluding the first 5 harmonics and DC

Figure 3-15: SINAD, ENOB, and SNR

The SINAD plot shows where the AC performance of the ADC degrades due to high-frequency distortion, and is usually plotted for frequencies well above the Nyquist frequency so that performance in undersampling applications can be evaluated.

SINAD is often converted to *effective-number-of-bits* (ENOB) using the relationship for the theoretical SNR of an ideal N-bit ADC: $\text{SNR} = 6.02N + 1.76\text{dB}$. The equation is solved for N, and the value of SINAD is substituted for SNR:

$$\text{ENOB} = \frac{\text{SINAD} - 1.76\text{dB}}{6.02} \quad \text{Eq. 3-3}$$

Signal-to-noise ratio (SNR, or *SNR-without-harmonics*) is calculated the same as SINAD except that the signal harmonics are excluded from the calculation, leaving only the noise terms. In practice, it is only necessary to exclude the first 5 harmonics since they dominate. The SNR plot will degrade at high frequencies, but not as rapidly as SINAD because of the exclusion of the harmonic terms.

■ OP AMP APPLICATIONS

Many current ADC datasheets somewhat loosely refer to SINAD as SNR, so the design engineer must be careful when interpreting these specifications.

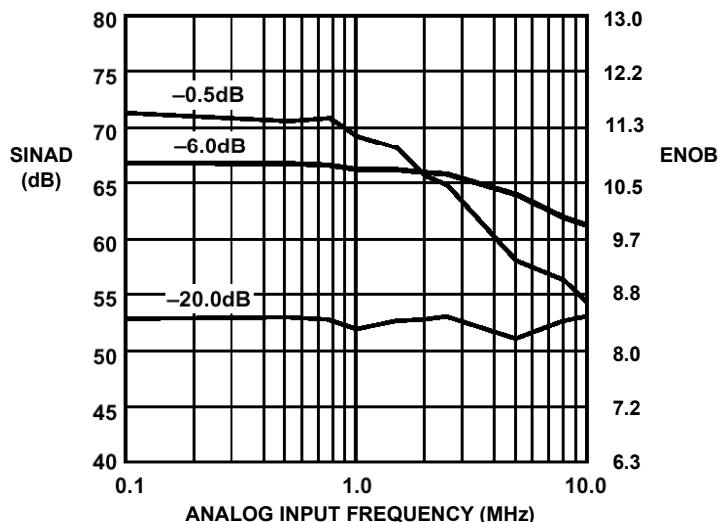


Figure 3-16: AD9220 12-bit, 10MSPS ADC SINAD and ENOB for various input signal levels

A SINAD/ENOB plot for the AD9220 12-bit, 10MSPS ADC is shown in Figure 3-16.

Analog Bandwidth

The analog bandwidth of an ADC is that frequency at which the spectral output of the *fundamental* swept frequency (as determined by the FFT analysis) is reduced by 3dB. It may be specified for either a small signal bandwidth (SSBW), or a full scale signal (FPBW- full power bandwidth), so there can be a wide variation in specifications between manufacturers.

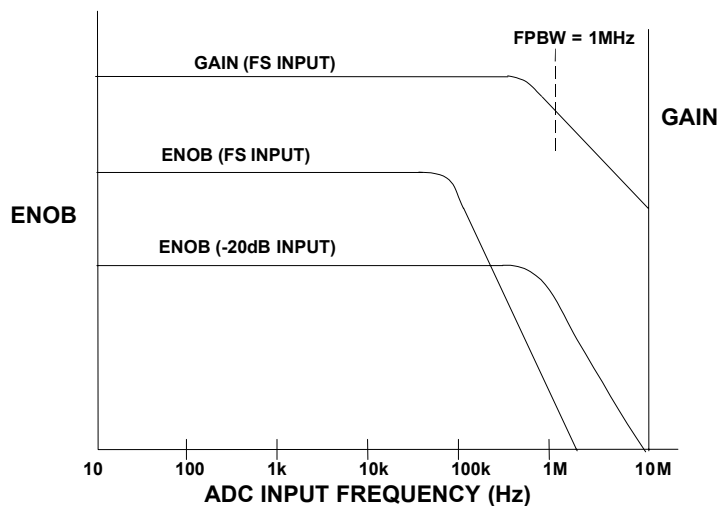


Figure 3-17: ADC Gain (bandwidth) and ENOB versus frequency shows importance of ENOB specification

Like an amplifier, the analog bandwidth specification of a converter does not imply that the ADC maintains good distortion performance up to its bandwidth frequency. In fact,

the SINAD (or ENOB) of most ADCs will begin to degrade considerably before the input frequency approaches the actual 3dB bandwidth frequency. Figure 3-17 (opposite) shows ENOB and full scale frequency response of an ADC with a FPBW of 1MHz, however, the ENOB begins to drop rapidly above 100kHz.

Harmonic Distortion, Worst Harmonic, Total Harmonic Distortion (THD), Total Harmonic Distortion Plus Noise (THD + N)

There are a number of ways to quantify the distortion of an ADC. An FFT analysis can be used to measure the amplitude of the various harmonics of a signal. The harmonics of the input signal can be distinguished from other distortion products by their location in the frequency spectrum. Figure 3-18 below shows a 7MHz input signal sampled at 20MSPS and the location of the first 9 harmonics.

Aliased harmonics of f_a fall at frequencies equal to $|\pm Kf_s \pm nf_a|$, where n is the order of the harmonic, and $K = 0, 1, 2, 3, \dots$. The second and third harmonics are generally the only ones specified on a datasheet because they tend to be the largest, although some datasheets may specify the value of the *worst* harmonic.

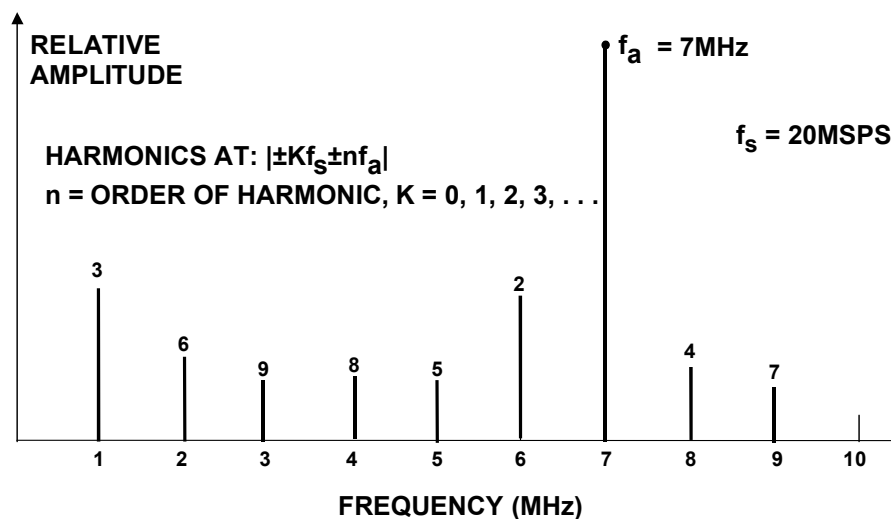


Figure 3-18: Location of harmonic distortion products: Input signal = 7MHz, sampling rate = 20MSPS

Harmonic distortion is normally specified in dBc (decibels below *carrier*), although in audio applications it may be specified as a percentage. Harmonic distortion is generally specified with an input signal near full scale (generally 0.5 to 1dB below full scale to avoid clipping), but it can be specified at any level. For signals much lower than full scale, other distortion products due to the DNL of the converter (not direct harmonics) may limit performance.

Total harmonic distortion (THD) is the ratio of the rms value of the fundamental signal to the mean value of the root-sum-square of its harmonics (generally, only the first 5 are significant). THD of an ADC is also generally specified with the input signal close to full scale, although it can be specified at any level.

■ OP AMP APPLICATIONS

Total harmonic distortion plus noise (THD+ N) is the ratio of the rms value of the fundamental signal to the mean value of the root-sum-square of its harmonics plus all noise components (excluding DC). The bandwidth over which the noise is measured must be specified. In the case of an FFT, the bandwidth is DC to $f_s/2$. (If the bandwidth of the measurement is DC to $f_s/2$, THD+N is equal to SINAD.

Spurious Free Dynamic Range (SFDR)

Probably the most significant specification for an ADC used in a communications application is its spurious free dynamic range (SFDR). The SFDR specification is to ADCs what the third order intercept specification is to mixers and LNAs.

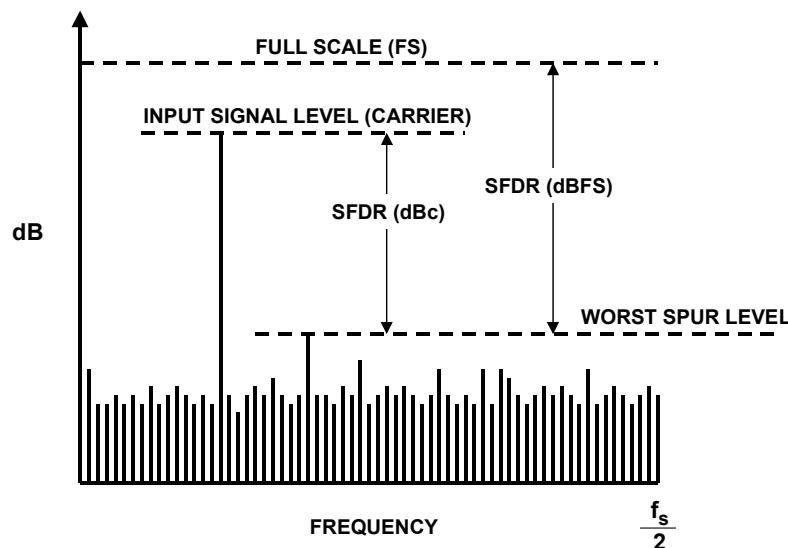


Figure 3-19: Spurious free dynamic range (SFDR)

SFDR of an ADC is defined as the ratio of the rms signal amplitude to the rms value of the *peak spurious spectral content* (measured over the entire first Nyquist zone, DC to $f_s/2$). SFDR is generally plotted as a function of signal amplitude and may be expressed relative to the signal amplitude (dBc) or the ADC full scale (dBFS) as shown in Figure 3-19 above.

For a signal near full scale, the peak spectral spur is generally determined by one of the first few harmonics of the fundamental. However, as the signal falls several dB below full scale, other spurs generally occur which are not direct harmonics of the input signal. This is because of the differential non-linearity of the ADC transfer function as discussed earlier. Therefore, SFDR considers *all* sources of distortion, regardless of their origin.

Two Tone Intermodulation Distortion (IMD)

Two tone IMD is measured by applying two spectrally pure sinewaves to the ADC at frequencies f_1 and f_2 , usually relatively close together. The amplitude of each tone is set slightly more than 6dB below full scale so that the ADC does not clip when the two tones add in-phase. Second and third-order product locations are shown in Figure 3-20 below.

Notice that the second-order products fall at frequencies which can be removed by digital filters. However, the third-order products $2f_2 - f_1$ and $2f_1 - f_2$ are close to the original signals, and are almost impossible to filter. Unless otherwise specified, two-tone IMD refers to these third-order products. The value of the IMD product is expressed in dBc relative to the value of *either* of the two original tones, and not to their sum.

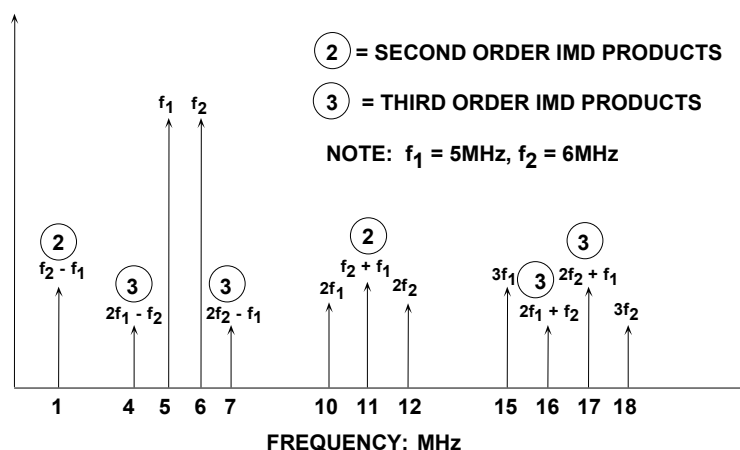


Figure 3-20: Second and third-order intermodulation products for $f_1 = 5\text{MHz}$, $f_2 = 6\text{MHz}$

Note, however, that if the two tones are close to $f_s/4$, then the aliased third harmonics of the fundamentals can make the identification of the actual $2f_2 - f_1$ and $2f_1 - f_2$ products difficult. This is because the third harmonic of $f_s/4$ is $3f_s/4$, and the alias occurs at $f_s - 3f_s/4 = f_s/4$. Similarly, if the two tones are close to $f_s/3$, the aliased second harmonics may interfere with the measurement. The same reasoning applies here; the second harmonic of $f_s/3$ is $2f_s/3$, and its alias occurs at $f_s - 2f_s/3 = f_s/3$.

The concept of *second and third-order intercept points* is not valid for an ADC, because the distortion products don't vary predictably (as a function of signal amplitude). The ADC doesn't gradually begin to compress signals approaching full scale (there is no 1dB compression point); it acts as a *hard limiter* as soon as the signal exceeds the input range, producing extreme distortion due to clipping. Conversely, for signals much below full scale, the distortion floor remains relatively constant and is independent of signal level.

Multi-tone SFDR is often measured in communications applications. The larger number of tones more closely simulates the wideband frequency spectrum of cellular telephone systems such as AMPS or GSM. High SFDR increases the receiver's ability to capture small signals in the presence of large ones, and prevent the small signals from being masked by the intermodulation products of the larger ones.

REFERENCES: ADC/DAC SPECIFICATIONS

1. Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8.
2. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6.
3. Chapters 3, 8, Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
4. Chapters 2, 3, 4, Walt Kester, Editor, **Mixed-Signal and DSP Design Techniques**, Analog Devices, 2000, ISBN: 0-916550-23-0.
5. Chapters 4, 5, Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.

SECTION 3-3: DRIVING ADC INPUTS

Walt Kester, Paul Hendriks

Introduction

Op amps are often used as drivers for ADCs to provide the gain and level shifting required for the input signal to match the input range of the ADC. An op amp may be required because of the antialiasing filter impedance matching requirements. In some cases, the antialiasing filter may be an active filter and include op amps as part of the filter itself. Some ADCs also generate transient currents on their inputs due to the conversion process, and these must be isolated from the signal source with an op amp. This section examines these and other issues involved in driving high performance ADCs.

To begin with, one shouldn't necessarily assume that a driver op amp is always required. Some converters have relatively benign inputs and are designed to interface directly to the signal source. There is practically no industry standardization regarding ADC input structures, and therefore each ADC must be carefully examined on its own merits before designing the input interface circuitry. In some applications, transformer drive may be preferable.

- ◆ **Minimize degradation of ADC / DAC performance specifications**
- ◆ **Fast settling to ADC/DAC transient**
- ◆ **High bandwidth**
- ◆ **Low noise**
- ◆ **Low distortion**
- ◆ **Low power**

- ◆ **Note: Op amp performance must be measured under identical conditions as encountered in ADC / DAC application**
 - Gain setting resistors
 - Input source impedance, output load impedance
 - Input / output signal voltage range
 - Input signal frequency
 - Input / output common-mode level
 - Power supply voltage (single or dual supply)
 - Transient loading

Figure 3-21: General op amp requirements in ADC driver applications

Assuming an op amp is required for one reason or another, the task of its selection is a critical one and not at all straightforward. Figure 3-21 above lists a few of the constraints and variables. *The most important requirement is that the op amp should not significantly degrade the overall DC or AC performance of the ADC.* At first glance, it would appear that a careful comparison of an op amp datasheet with the ADC datasheet would allow an appropriate choice. However, this is rarely the case.

The problem is that the op amp performance specifications must be known for the *exact* circuit configuration used in the ADC driver circuit. Even a very complete datasheet is unlikely to provide all information required, due to the wide range of possible variables.

■ OP AMP APPLICATIONS

Although the op amp and ADC datasheets should definitely be used as a guide in the selection process, it is unlikely that the overall performance of the op amp/ADC combination can be predicted accurately without actually prototyping the circuit, especially in high performance applications.

Various tested application circuits are often recommended on either the op amp or the ADC datasheet, but these can become obsolete quickly as new op amps are released. In most cases, however, the ADC datasheet application section should be used as the primary source for tested interfaces.

Op Amp Specifications Key to ADC Applications

The two most popular applications for ADCs today are in either precision high-resolution measurements or in low distortion high speed systems. Precision measurement applications require ADCs of at least 16 bits of resolution, and sometimes up to 24 bits. Op amps used with these ADCs must be low noise and have excellent DC characteristics. In fact, high resolution measurement ADCs are often designed to interface directly with the transducer, eliminating the need for an op amp entirely.

◆ DC

- Offset, offset drift
- Input bias current
- Open loop gain
- Integral linearity
- 1/f noise (voltage and current)

◆ AC (Highly application dependent!)

- Wideband noise (voltage and current)
- Small and Large Signal Bandwidth
- Harmonic Distortion
- Total Harmonic Distortion (THD)
- Total Harmonic Distortion + Noise (THD + N)
- Spurious Free Dynamic Range (SFDR)
- Third Order Intermodulation Distortion
- Third Order Intercept Point

Figure 3-22: Key op amp specifications

If op amps are required, it is generally relatively straightforward to select one based on well-understood DC specifications, as listed in Figure 3-22 above.

It is much more difficult to provide a complete set of op amp AC specifications because they are highly dependent upon the application circuit. For example, Figure 3-23 (opposite) shows some key specifications taken from the table of specifications on the datasheet for the AD8057/AD8058 high speed, low distortion op amp (see Reference 1). Note that the specifications depend on the supply voltage, the signal level, output loading, etc. It should also be emphasized that it is customary to provide only *typical* AC specifications (as opposed to *maximum* and *minimum* values) for most op amps. In

USING OP AMPS WITH DATA CONVERTERS DRIVING ADC INPUTS

addition, there are restrictions on the input and output CM signal ranges, which are especially important when operating on low voltage dual (or single) supplies.

	$V_S = \pm 5V$	$V_S = +5V$
Input Common Mode Voltage Range	–4.0V to +4.0V	+0.9V to +3.4V
Output Common Mode Voltage Range	–4.0V to +4.0V	+0.9V to +4.1V
Input Voltage Noise	7nV/$\sqrt{\text{Hz}}$	7nV/$\sqrt{\text{Hz}}$
Small Signal Bandwidth	325MHz	300MHz
THD @ 5MHz, $V_O = 2V$ p-p, $R_L = 1k\Omega$	– 85dBc	– 75dBc
THD @ 20MHz, $V_O = 2V$ p-p, $R_L = 1k\Omega$	– 62dBc	– 54dBc

Figure 3-23: AD8057/AD8058 op amp key AC specifications, $G = +1$

Most op amp datasheets contain a section that provides supplemental performance data for various other conditions not explicitly specified in the primary specification tables. For instance, Figure 3-24 below shows the AD8057/AD8058 distortion as a function of frequency for $G = +1$ and $V_S = \pm 5V$. Unless it is otherwise specified, the data represented by these curves should be considered typical (it is usually marked as such).

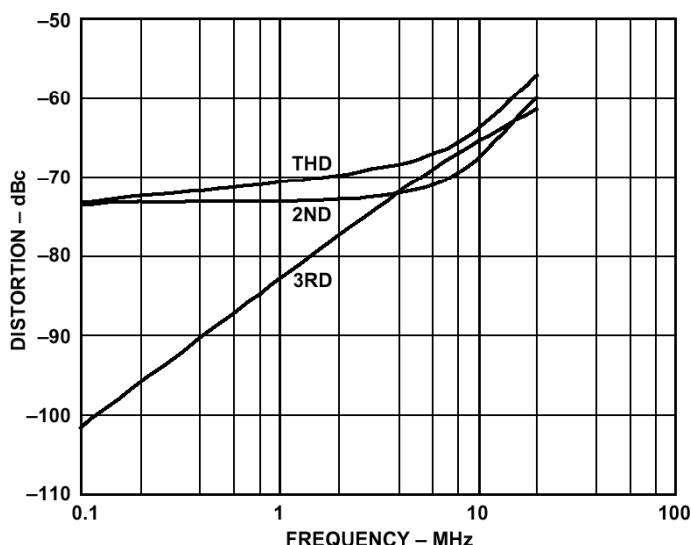


Figure 3-24: AD8057/AD8058 op amp distortion versus frequency
 $G = +1$, $R_L = 150\Omega$, $V_S = \pm 5V$

Note however that the data in both Fig. 3-24 (and also the following Figure 3-25) is given for a DC load of 150Ω . This is a load presented to the op amp in the popular application of driving a source and load-terminated 75Ω cable. Distortion performance is generally better with lighter DC loads, such as 500Ω - 1000Ω (more typical of many ADC inputs), and this data may or may not be found on the datasheet.

■ OP AMP APPLICATIONS

On the other hand, Figure 3-25 below shows distortion as a function of output signal level for a frequencies of 5MHz and 20MHz.

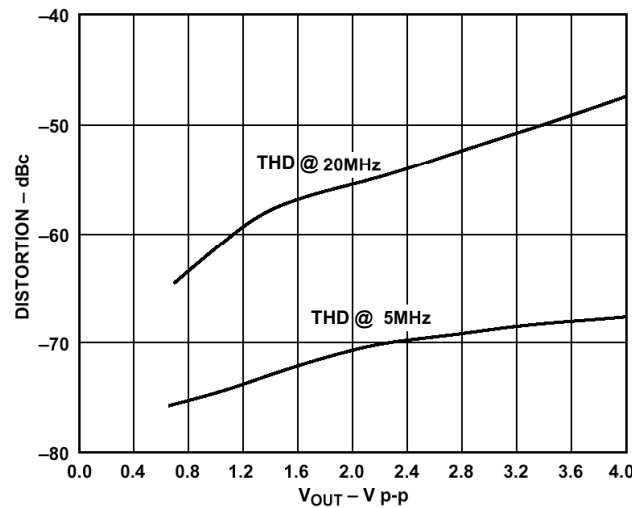


Figure 3-25: AD8057/AD8058 op amp distortion versus output voltage
 $G = +1$, $R_L = 150\Omega$, $V_S = \pm 5V$

Whether or not specifications such as those just described are complete enough to select an op amp for an ADC driver application depends upon the ability to match op amp specifications to the actually required ADC operating conditions. In many cases, these comparisons will at least narrow the op amp selection process. The following sections will examine a number of specific driver circuit examples using various types of ADCs, ranging from high resolution measurement to high-speed, low distortion applications.

Driving High Resolution Sigma-Delta Measurement ADCs

The AD77XX family of ADCs is optimized for high resolution (16–24 bits) low frequency transducer measurement applications. Details of operation can be found in Reference 2, and general characteristics of the family are listed in Figure 3-26 below.

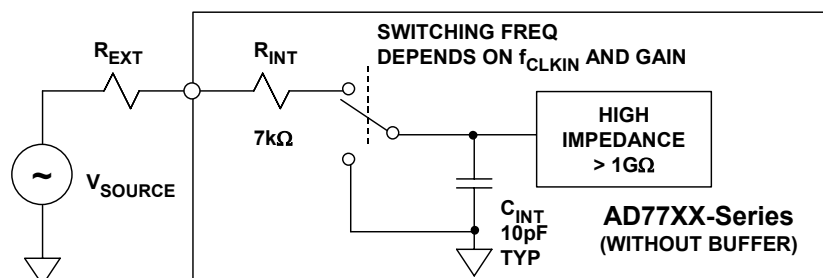
- ◆ **Resolution: 16 - 24 bits**
- ◆ **Input signal bandwidth: <60Hz**
- ◆ **Effective sampling rate: <100Hz**
- ◆ **Generally Sigma-Delta architecture**
- ◆ **Designed to interface directly to sensors (< 1 k Ω) such as bridges with no external buffer amplifier (e.g., AD77XX - series)**
 - On-chip PGA and high resolution ADC eliminates the need for external amplifier
- ◆ **If buffer is used, it should be precision low noise (especially 1/f noise)**
 - OP177
 - AD707
 - AD797

Figure 3-26: High resolution low frequency measurement ADCs

Some members of this family, such as the AD7730, have a high impedance input buffer which isolates the analog inputs from switching transients generated in the front-end programmable gain amplifier (PGA) and the sigma-delta modulator. Therefore, no special

precautions are required in driving the analog inputs. Other members of the AD77XX family, however, either do not have the input buffer, or if one is included on-chip, it can be switched either in or out under program control. Bypassing the buffer offers a slight improvement in noise performance.

The equivalent input circuit of the AD77XX family without an input buffer is shown below in Figure 3-27. The input switch alternates between the 10pF sampling capacitor and ground. The 7k Ω internal resistance, R_{INT} , is the on-resistance of the input multiplexer. The switching frequency is dependent on the frequency of the input clock and also the internal PGA gain. If the converter is working to an accuracy of 20-bits, the 10pF internal capacitor, C_{INT} , must charge to 20-bit accuracy during the time the switch connects the capacitor to the input. This interval is one-half the period of the switching signal (it has a 50% duty cycle). The input RC time constant due to the 7k Ω resistor and the 10pF sampling capacitor is 70ns. If the charge is to achieve 20-bit accuracy, the capacitor must charge for at least 14 time constants, or 980ns. Any external resistance in series with the input will increase this time constant.



- R_{EXT} Increases C_{INT} Charge Time and May Result in Gain Error
- Charge Time Dependent on the Input Sampling Rate and Internal PGA Gain Setting
- Refer to Specific Data Sheet for Allowable Values of R_{EXT} to Maintain Desired Accuracy
- Some AD77XX-Series ADCs Have Internal Buffering Which Isolates Input from Switching Circuits

Figure 3-27: Driving unbuffered AD77XX-series $\Sigma\Delta$ ADC inputs

There are tables on the datasheets for the various AD77XX ADCs, which give the maximum allowable values of R_{EXT} in order to maintain a given level of accuracy. These tables should be consulted if the external source resistance is more than a few k Ω .

Note that for instances where an external op amp buffer is found to be required with this type of converter, guidelines exist for best overall performance. This amplifier should be a precision low-noise bipolar-input type, such as the OP177, AD707, or the AD797.

Op Amp Considerations for Multiplexed Data Acquisition Applications

Multiplexing is a fundamental part of many data acquisition systems. Switches used in multiplexed data acquisition systems are generally CMOS-types shown in Figure 3-28 below. Utilizing P-Channel and N-Channel MOSFET switches in parallel minimizes the change of *on-resistance* (R_{ON}) as a function of signal voltage. On-resistance can vary from less than five to several hundred ohms depending upon the device. Variation in on-resistance as a function of signal level (often called *R_{ON} -modulation*) causes distortion if the multiplexer drives a load, therefore *R_{ON} flatness* is also an important specification.

Because of the effects of non-zero R_{ON} and R_{ON} -modulation, multiplexer outputs should be isolated from the load with a suitable buffer op amp. A separate buffer is not required if the multiplexer drives a high input impedance, such as a PGA, SHA or ADC—but beware! Some SHAs and ADCs draw high frequency pulse current at their sampling rate and cannot tolerate being driven by an unbuffered multiplexer.

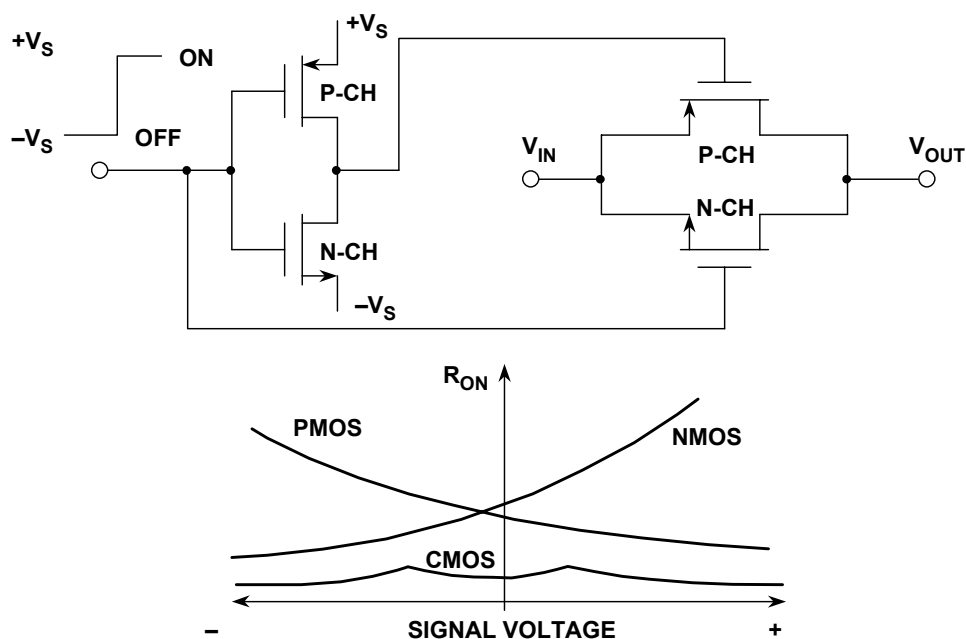


Figure 3-28: Basic CMOS analog switch

Key multiplexer specifications are *switching time*, *on-resistance*, *on-resistance flatness*, and *off-channel isolation*, and *crosstalk*. Multiplexer switching time ranges from less than 20ns to over 1 μ s, R_{ON} from less than 5 Ω to several hundred ohms, and off-channel isolation from 50 to 90dB.

A number of CMOS switches can be connected to form an analog multiplexer, as shown in Figure 3-29 (opposite). The number of input channels typically ranges from 4 to 16, and some multiplexers have internal channel-address decoding logic and registers, while with others, these functions must be performed externally. Unused multiplexer inputs *must* be grounded or severe loss of system accuracy may result. In applications requiring an op amp buffer, it should be noted that when the multiplexer changes channels it is possible to have a full scale step function into the op amp and the ADC which follows it.

Op amp settling time must be fast enough so that conversion errors do not result. It is customary to specify the op amp settling time to 1 LSB, and the allowed time for this settling is generally the reciprocal of the sampling frequency.

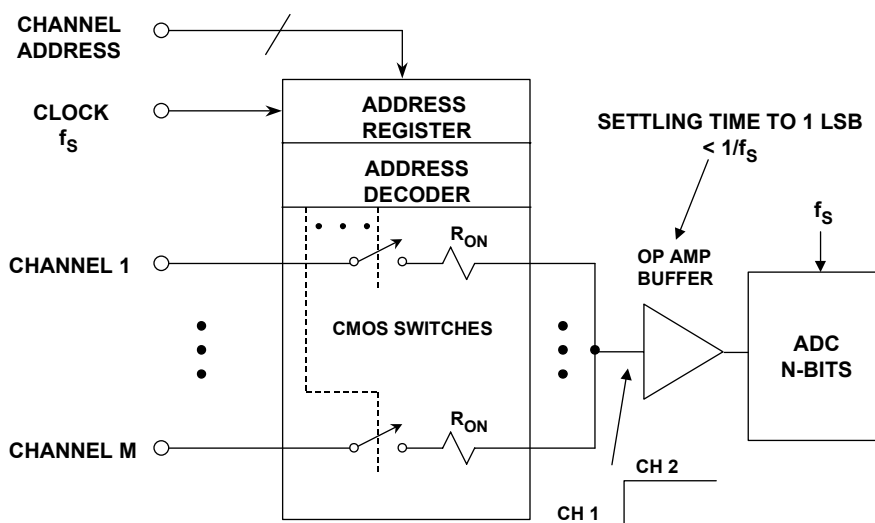


Figure 3-29: Typical multiplexed data acquisition system requires fast settling op amp buffer

Driving Single-Supply Data Acquisition ADCs with Scaled Inputs

The AD789X and AD76XX family of single supply SAR ADCs (as well as the AD974, AD976, and AD977) includes a thin film resistive attenuator and level shifter on the analog input to allow a variety of input range options, both bipolar and unipolar.

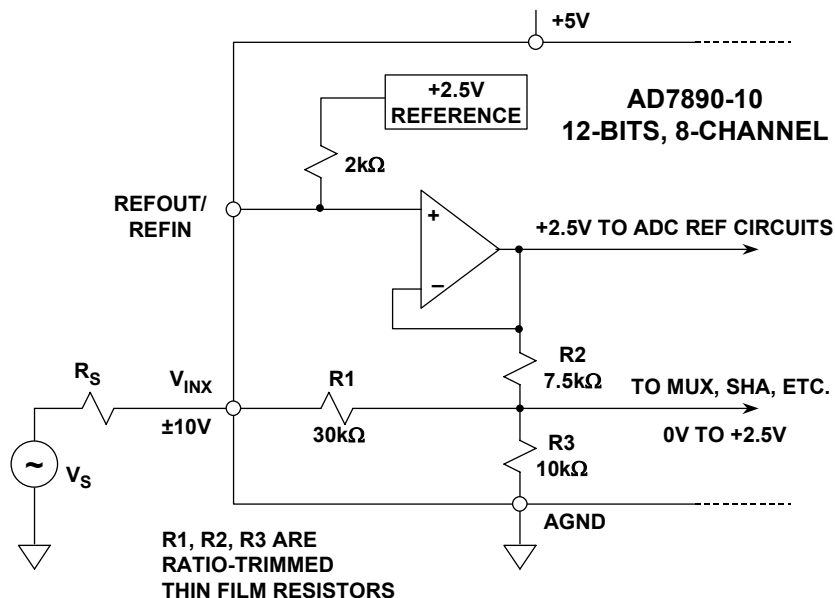


Figure 3-30: Driving single-supply data acquisition ADCs with scaled inputs

A simplified diagram of the input circuit of the AD7890-10 12-bit, 8-channel ADC is shown in Figure 3-30 above. This arrangement allows the converter to digitize a $\pm 10\text{V}$ input while operating on a single $+5\text{V}$ supply.

OP AMP APPLICATIONS

Within the ADC, the R1/R2/R3 thin film network provides attenuation and level shifting to convert the $\pm 10\text{V}$ input to a 0V to $+2.5\text{V}$ signal that is digitized. This type of input requires no special drive circuitry, because R1 isolates the input from the actual converter circuitry that may generate transient currents due to the conversion process. Nevertheless, the external source resistance R_S should be kept reasonably low, to prevent gain errors caused by the $R_S/R1$ divider.

Driving ADCs with Buffered Inputs

Some ADCs have on-chip buffer amplifiers on their analog input to simplify the interface. This feature is most often found on ADCs designed on either bipolar or BiCMOS processes. Conversely, input amplifiers are rarely found on CMOS ADCs because of the inherent difficulty associated with amplifier design in CMOS.

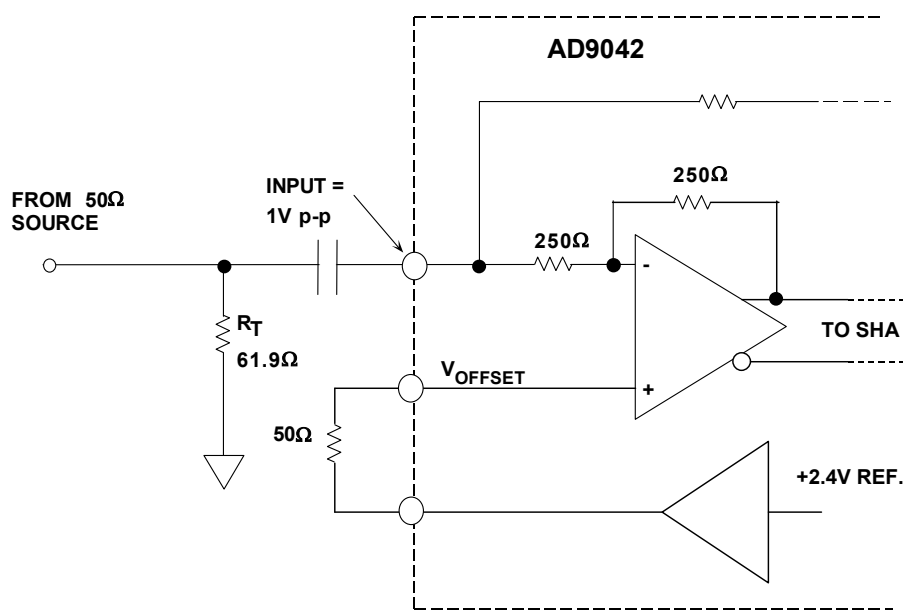


Figure 3-31: AD9042 ADC is designed to be driven directly from 50Ω source for best SFDR

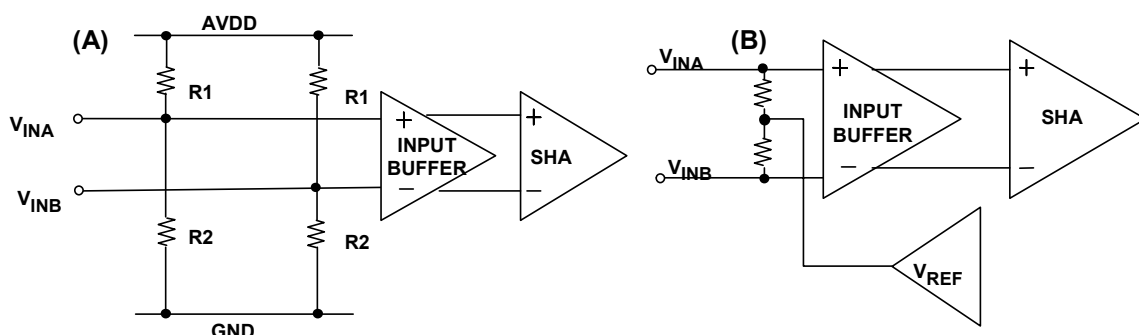
A typical input structure for an ADC with an input buffer is shown in Figure 3-31 above for the AD9042 12-bit, 41MSPS ADC. The effective input impedance is 250Ω, and an external 61.9Ω resistor in parallel with this internal 250Ω provides an effective input termination of 50Ω to the signal source. The circuit shows an AC coupled input. An internal reference voltage of 2.4V sets the input CM voltage of the AD9042.

The input amplifier precedes the ADC sample-and-hold (SHA), and therefore isolates the input from any transients produced by the conversion process. The gain of the amplifier is set such that the input range of the ADC is 1Vp-p. In the case of a single-ended input structure, the input amplifier serves to convert the single-ended signal to a differential one, which allows fully differential circuit design techniques to be used throughout the remainder of the ADC.

Driving Buffered Differential Input ADCs

Figure 3-32 below shows two possible input structures for an ADC with buffered differential inputs. The input CM voltage is set with an internal resistor divider network in Figure 3-32A (left), and by a voltage reference in Figure 3-32B (right).

In single supply ADCs, the CM voltage is usually equal to one-half the power supply voltage, but some ADCs may use other values. Although the input buffers provide for a simplified interface, the fixed CM voltage may limit flexibility in some DC coupled applications.



- ◆ Input buffers typical on BiMOS and bipolar processes
- ◆ Difficult on CMOS
- ◆ Simplified input interface - no transient currents
- ◆ Fixed common-mode level may limit flexibility

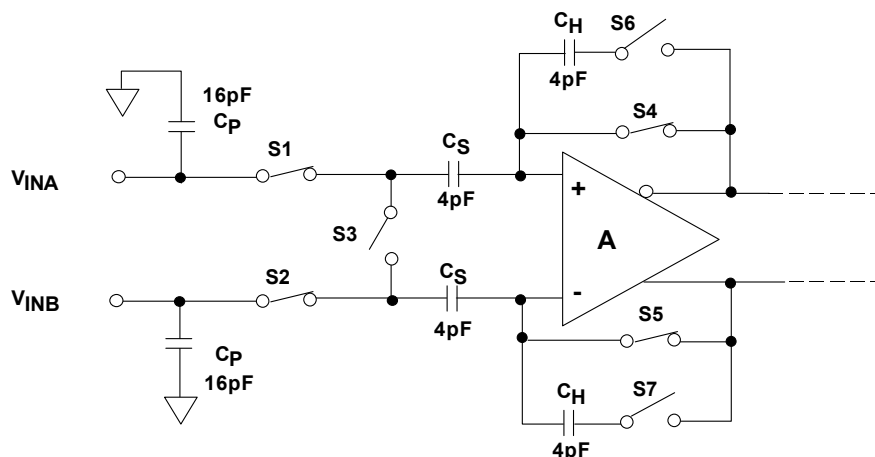
Figure 3-32: Simplified input circuit of typical buffered ADC with differential inputs

It is worthwhile noting that differential ADC inputs offer several advantages over single ended ones. First, many signal sources in communications applications are differential, such as the output of a balanced mixer or an RF transformer. Thus an ADC that accepts differential inputs interfaces easily in such systems. Secondly, maintaining balanced differential transmission in the signal path and within the ADC itself often minimizes even-order distortion products as well as improving CM noise rejection. Third, (and somewhat more subtly), a differential ADC input swing of say, 2Vp-p requires only 1Vp-p from twin driving sources. On low voltage and single-supply systems, this lower absolute level of drive can often make a real difference in the dual amplifier driver distortion, due to practical headroom limitations.

Given all of these points, it behooves the system engineer to operate a differential-capable ADC in the differential mode for best overall performance. This may be true even if a second amplifier need be added for the complementary drive signal, since dual op amps are only slightly more expensive than singles.

Driving CMOS ADCs with Switched Capacitor Inputs

CMOS ADCs are quite popular because of their low power and low cost. The equivalent input circuit of a typical CMOS ADC using a differential sample-and-hold is shown in Figure 3-33 below. While the switches are shown in the *track* mode, note that they open/close at the sampling frequency. The 16pF capacitors represent the effective capacitance of switches S1 and S2, plus the stray input capacitance. The C_S capacitors (4pF) are the sampling capacitors, and the C_H capacitors are the hold capacitors. Although the input circuit is completely differential, this ADC structure can be driven either single-ended or differentially. Optimum performance, however, is generally obtained using a differential transformer or differential op amp drive.



SWITCHES SHOWN IN TRACK MODE

Figure 3-33: Simplified input circuit for a typical switched capacitor CMOS sample-and-hold

In the *track* mode, the differential input voltage is applied to the C_S capacitors. When the circuit enters the *hold* mode, the voltage across the sampling capacitors is transferred to the C_H hold capacitors and buffered by the amplifier A (the switches are controlled by the appropriate sampling clock phases). When the SHA returns to the *track* mode, the input source must charge or discharge the voltage stored on C_S to a new input voltage. This action of charging and discharging C_S , averaged over a period of time and for a given sampling frequency f_s , makes the input impedance appear to have a benign resistive component. However, if this action is analyzed within a sampling period ($1/f_s$), the input impedance is dynamic, and certain input drive source precautions should be observed.

The resistive component to the input impedance can be computed by calculating the average charge that is drawn by C_H from the input drive source. It can be shown that if C_S is allowed to fully charge to the input voltage before switches S1 and S2 are opened that the average current into the input is the same as if there were a resistor equal to $1/(C_S f_s)$ connected between the inputs. Since C_S is only a few picofarads, this resistive component is typically greater than several k Ω for an $f_s = 10\text{MSPS}$.

Over a sampling period, the SHA's input impedance appears as a dynamic load. When the SHA returns to the track mode, the input source should ideally provide the charging current through the R_{ON} of switches S1 and S2 in an exponential manner. The requirement of exponential charging means that the source impedance should be both low and resistive up to and beyond the sampling frequency.

The output impedance of an op amp can be modeled as a series inductor and resistor. When a capacitive load is switched onto the output of the op amp, the output will momentarily change due to its effective high frequency output impedance. As the output recovers, ringing may occur. To remedy this situation, a series resistor can be inserted between the op amp and the SHA input. The optimum value of this resistor is dependent on several factors including the sampling frequency and the op amp selected, but in most applications, a 25 to 100 Ω resistor is optimum.

Single Ended ADC Drive Circuits

Although most CMOS ADC inputs are differential, they can be driven single-ended with some AC performance degradation. An important consideration in CMOS ADC applications are the input switching transients previously discussed.

- ◆ **Hold-to-Sample Mode Transition** - C_S Returned to Source for “recharging”. Transient Consists of Linear, Nonlinear, and Common-Mode Components at Sample Rate .
- ◆ **Sample-to-Hold Mode Transition** - Input Signal Sampled when C_S is disconnected from Source.

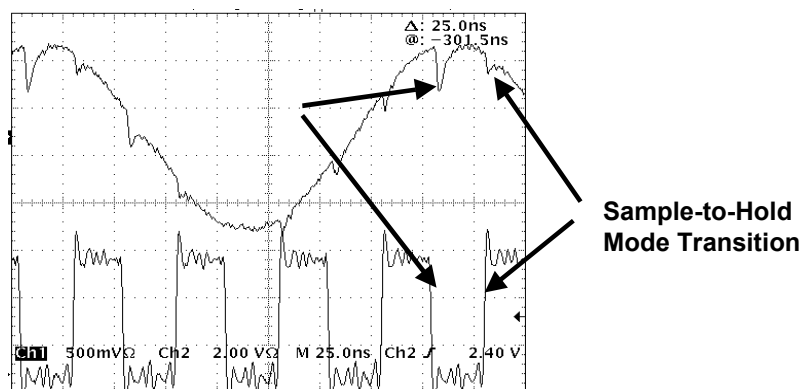


Figure 3-34: Single-ended input transient response of CMOS switched capacitor SHA (AD9225)

For instance, the input switching transient on one of the inputs of the AD9225 12-bit, 25MSPS ADC is shown above in Figure 3-34. This data was taken driving the ADC with an equivalent 50 Ω source impedance. During the sample-to-hold transition, the input signal is sampled when C_S is disconnected from the source. Notice that during the hold-to-sample transition, C_S is reconnected to the source for recharging. The transients consist of linear, nonlinear, and CM components at the sample rate.

In addition to selecting an op amp with sufficient bandwidth and distortion performance, the output should settle to these transients during the sampling interval, $1/f_S$. The general

■ OP AMP APPLICATIONS

circuit shown below in Figure 3-35 is typical for this type of single-ended op amp ADC driver application.

In this circuit, series resistor R_S has a dual purpose. Typically chosen in the range of 25-100 Ω , it limits the peak transient current from the driving op amp. Importantly, it also decouples the driver from the ADC input capacitance (and possible phase margin loss).

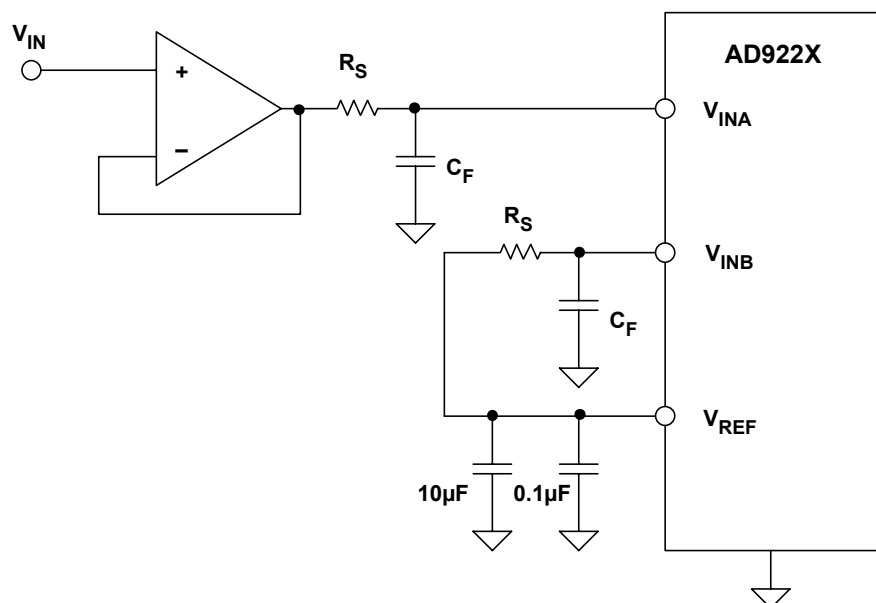


Figure 3-35: Optimizing single-ended switched capacitor ADC input drive circuit

Another feature of the circuit are the dual networks of R_S and C_F . Matching both the DC and AC the source impedance for the ADC's V_{INA} and V_{INB} inputs ensures symmetrical settling of CM transients, for optimum noise and distortion performance. At both inputs, the C_F shunt capacitor acts as a charge reservoir and steers the CM transients to ground.

In addition to the buffering of transients, R_S and C_F also form a lowpass filter for V_{IN} , which limits the output noise of the drive amplifier into the ADC input V_{INA} . The exact values for R_S and C_F are generally optimized within the circuit, and the recommended values given on the ADC datasheet. The ADC data sheet information should also be consulted for the recommended drive op amp for best performance.

To enable best correlation of performance between environments, an ADC evaluation board should be used (if available). This will ensure confidence when the ADC data sheet circuit performance is duplicated. ADI makes evaluation boards available for many of their ADC and DAC devices (plus of course, op amps), and general information on them is contained in Chapter 7 of this book.

Op Amp Gain Setting and Level Shifting in DC Coupled Applications

In DC coupled applications, the drive amplifier must provide the required gain and offset voltage, to match the signal to the input voltage range of the ADC. Figure 3-36 below summarizes various op amp gain and level shifting options. The circuit of Figure 3-36A operates in the non-inverting mode, and uses a low impedance reference voltage, V_{REF} , to offset the output. Gain and offset interact according to the equation:

$$V_{OUT} = [1 + (R2/R1)] \cdot V_{IN} - [(R2/R1) \cdot V_{REF}] \quad \text{Eq. 3-4}$$

The circuit in Figure 3-36B operates in the inverting mode, and the signal gain is independent of the offset. The disadvantage of this circuit is that the addition of $R3$ increases the noise gain, and hence the sensitivity to the op amp input offset voltage and noise. The input/output equation is given by:

$$V_{OUT} = -(R2/R1) \cdot V_{IN} - (R2/R3) \cdot V_{REF} \quad \text{Eq. 3-5}$$

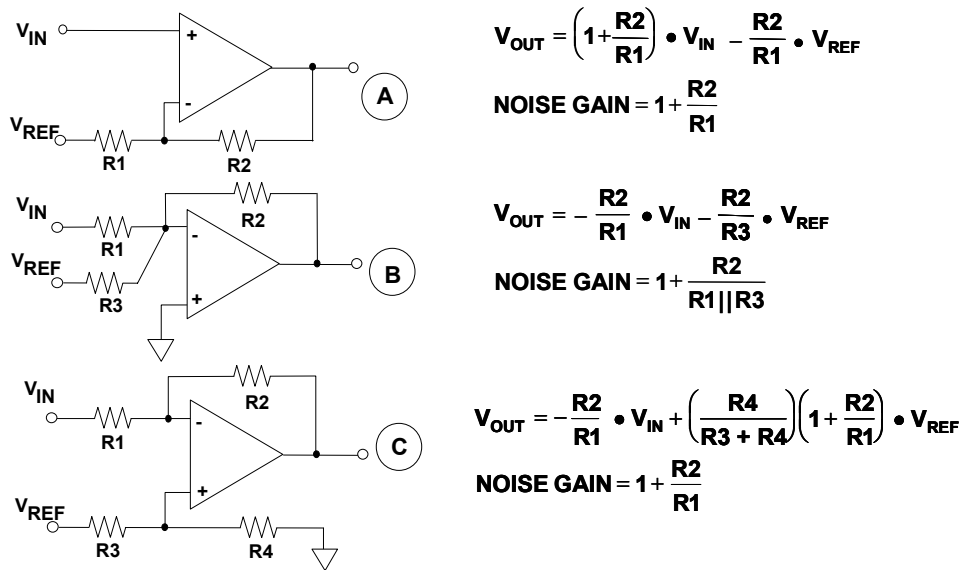


Figure 3-36: Op amp gain and level shifting circuits

The circuit in Figure 3-36C also operates in the inverting mode, and the offset voltage V_{REF} is applied to the non-inverting input without noise gain penalty. This circuit is also attractive for single-supply applications ($V_{REF} > 0$). The input/output equation is given by:

$$V_{OUT} = -(R2/R1) \cdot V_{IN} + [(R4/(R3+R4))(1 + (R2/R1))] \cdot V_{REF} \quad \text{Eq. 3-6}$$

Note that the circuit of Fig. 3-36A is sensitive to the impedance of V_{REF} , unlike the counterparts in B and C. This is due to the fact that the signal current flows into/from V_{REF} , due to V_{IN} operating the op amp over its CM range. In the other two circuits the CM voltages are fixed, and no signal current flows in V_{REF} .

OP AMP APPLICATIONS

A DC coupled single-ended op amp driver for the AD9225 12-bit, 25MSPS ADC is shown in Figure 3-37 below. This circuit interfaces a $\pm 2\text{V}$ input signal to the single-supply ADC, and provides transient current isolation. The ADC input voltage range is 0 to $+4\text{V}$, and a dual supply op amp is required, since the ADC minimum input is 0V .

The non-inverting input of the AD8057 is biased at $+1\text{V}$, which sets the output CM voltage at V_{INA} to $+2\text{V}$ for a bipolar input signal source. Note that the V_{INA} and V_{INB} source impedances are matched for better CM transient cancellation. The 100pF capacitors act as small charge reservoirs for the input transient currents, and also form lowpass noise filters with the 33Ω series resistors.

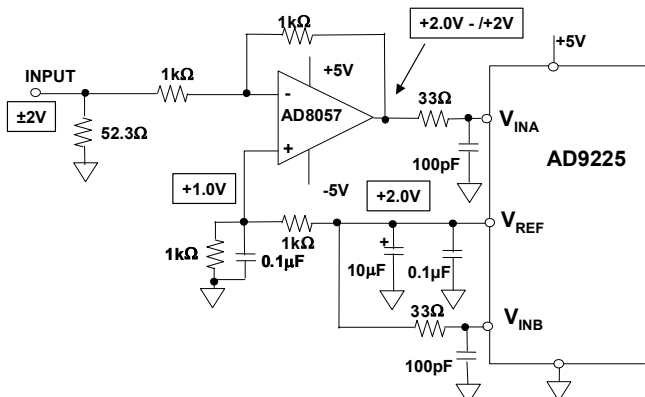


Figure 3-37: DC coupled single-ended level shifter and driver for the AD9225 12-bit, 25MSPS CMOS ADC

A similar level shifter and drive circuit is shown in Figure 3-38 below, operating on a single $+5\text{V}$ supply. In this circuit the bipolar $\pm 1\text{V}$ input signal is interfaced to the input of the ADC whose span is set for 2V about a $+2.5\text{V}$ CM voltage. The AD8041 rail-to-rail output op amp is used. The $+1.25\text{V}$ input CM voltage for the AD8041 is developed by a voltage divider from the external AD780 2.5V reference.

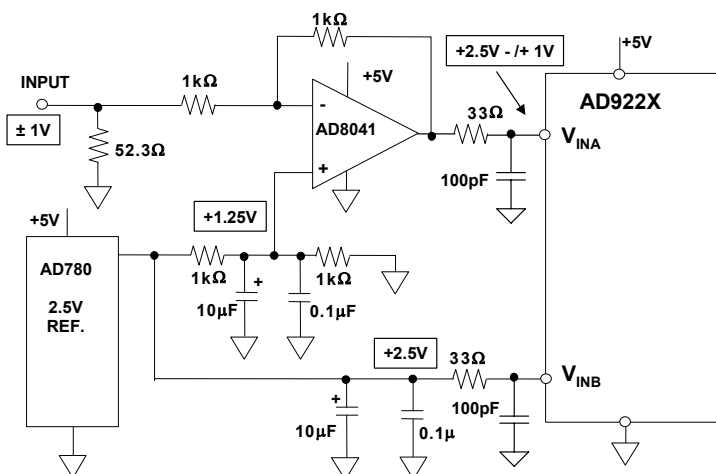


Figure 3-38: DC coupled single-ended, single-supply ADC driver / level shifter using external reference

Note that single-supply circuits of this type must observe op amp input and output CM voltage restrictions, to prevent clipping and excess distortion.

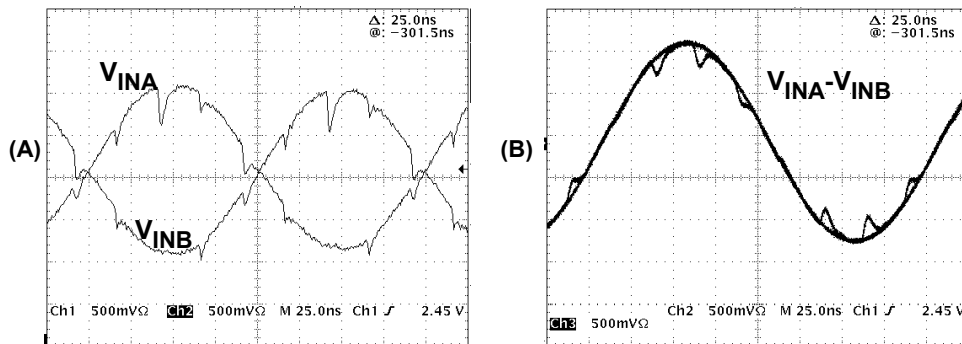
Drivers for Differential Input ADCs

Most high performance ADCs are now being designed with differential inputs. A fully differential ADC design offers the advantages of good CM rejection, reduction in second-order distortion products, and simplified DC trim algorithms. Although they can be driven single-ended as previously described, a fully differential driver usually optimizes overall performance.

- ◆ High common-mode noise rejection
- ◆ Flexible input common-mode voltage levels
- ◆ Reduced input signal swings helps in low voltage, single-supply applications
- ◆ Reduced second-order distortion products
- ◆ Simplified DC trim algorithms because of internal matching
- ◆ Requires high performance differential driver

Figure 3-39: Differential input ADCs offer performance advantages

Waveforms at the two inputs of the AD9225 12-bit, 25MSPS CMOS ADC are shown in Figure 3-40A, designated as V_{INA} and V_{INB} . The balanced source impedance is 50Ω , and the sampling frequency is set for 25MSPS. The diagram clearly shows the switching transients due to the internal ADC switched capacitor sample-and-hold. Figure 3-40B shows the difference between the two waveforms, $V_{INA} - V_{INB}$.



- ◆ Differential charge transient is symmetrical around mid-scale and dominated by linear component
- ◆ Common-mode transients cancel with equal source impedance

Figure 3-40: Differential input transient response of CMOS switched capacitor SHA (AD9225)

Note that the resulting differential charge transients are symmetrical about mid-scale, and that there is a distinct linear component to them. This shows the reduction in the CM transients, and also leads to better distortion performance than would be achievable with a single-ended input.

OP AMP APPLICATIONS

Transformer coupling into a differential input ADC provides excellent CM rejection and low distortion if performance to DC is not required. Figure 3-41 shows a typical circuit. The transformer is a Mini-Circuits RF transformer, model #T4-6T which has an impedance ratio of 4 (turns ratio of 2). The schematic assumes that the signal source has a 50Ω source impedance. The 1:4 impedance ratio requires the 200Ω secondary termination for optimum power transfer and low VSWR. The Mini-Circuits T4-6T has a 1dB bandwidth from 100kHz to 100MHz. The center tap of the transformer provides a convenient means of level shifting the input signal to the optimum CM voltage of the ADC. The AD922X CML (common-mode level) pin is used to provide the +2.5V CM voltage.

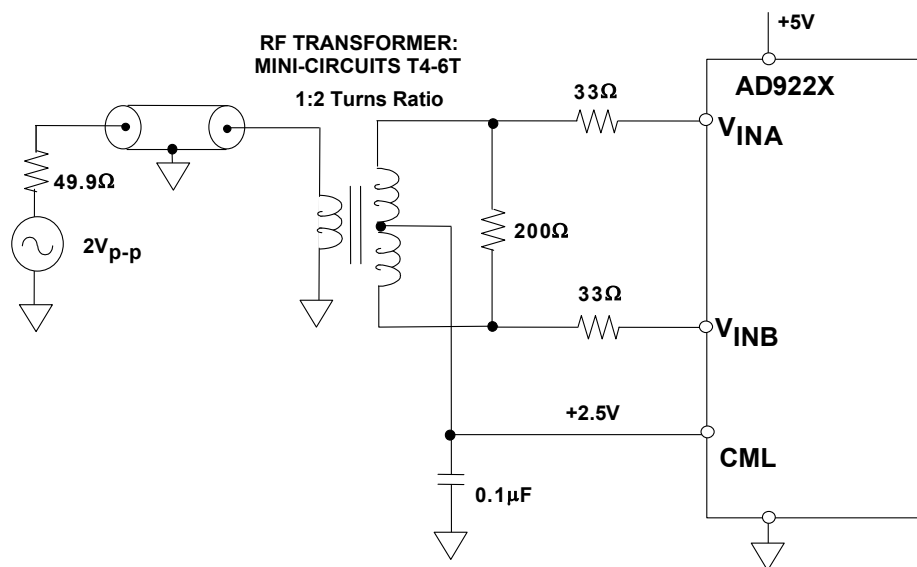


Figure 3-41: Transformer coupling into AD922x ADC

Transformers with other turns ratios may also be selected to optimize the performance for a given application. For example, a given input signal source or amplifier may realize an improvement in distortion performance at reduced output power levels and signal swings. Hence, selecting a transformer with a higher impedance ratio (i.e. Mini-Circuits #T16-6T with a 1:16 impedance ratio, turns ratio 1:4) effectively "steps up" the signal level thus reducing the driving requirements of the signal source.

Note the 33Ω series resistors inserted between the transformer secondary and the ADC input. These values were specifically selected to optimize both the SFDR and the SNR performance of the ADC. They also provide isolation from transients at the ADC inputs. Transients currents are approximately equal on the V_{INA} and V_{INB} inputs, so they are isolated from the primary winding of the transformer by the transformer's CM rejection.

Transformer coupling using a CM voltage of +2.5V provides the maximum SFDR when driving the AD922X-series. By driving the ADC differentially, even-order harmonics are reduced compared with the single-ended circuit.

Driving ADCs with Differential Amplifiers

There are many applications where differential input ADCs cannot be driven with transformers because the frequency response must extend to DC. In these cases, op amps can be used to implement the differential drivers. Figure 3-42 shows how the dual AD8058 op amp can be connected to convert a single-ended bipolar signal to a differential one suitable for driving the AD922X family of ADCs. The input range of the ADC is set for a 2Vp-p signal on each input (4V span), and a CM voltage of +2V.

The A1 amplifier is configured as a non-inverting op amp. The 1k Ω divider resistors level shift the $\pm 1V$ input signal to +1V $\pm 0.5V$ at the non-inverting input of A1. The output of A1 is therefore +2V $\pm 1V$.

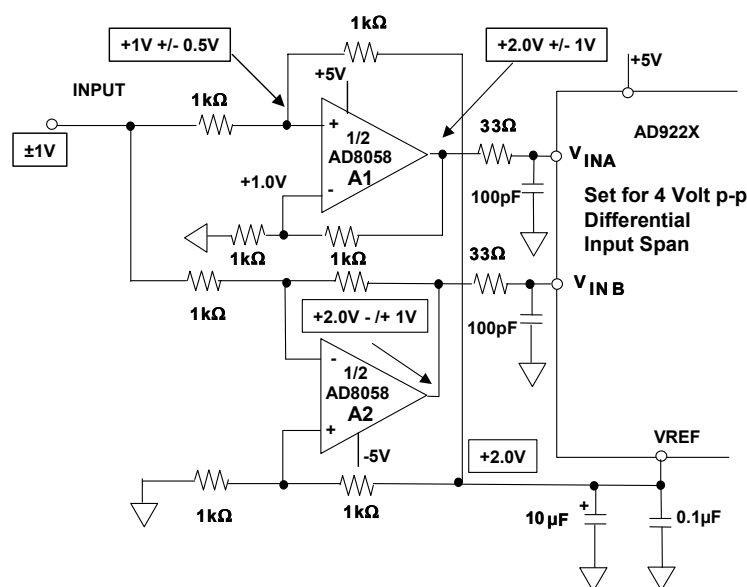


Figure 3-42: Op amp single-ended to differential DC-coupled driver with level shifting.

The A2 op amp inverts the input signal, and the 1k Ω divider resistors establish a +1V CM voltage on its non-inverting input. The output of A2 is therefore +2V $\mp 1V$.

This circuit provides good matching between the two op amps because they are duals on the same die and are both operated at the same noise gain of 2. However, the input voltage noise of the AD8058 is 20nV/ $\sqrt{\text{Hz}}$, and this appears as 40nV/ $\sqrt{\text{Hz}}$ at the output of both A1 and A2 thereby introducing possible SNR degradation in some applications. In the circuit of Fig. 3-42, this is mitigated somewhat by the 100pF input capacitors which not only reduce the input noise but absorb some of the transient currents. It should be noted that because the input CM voltage of A1 can go as low as +0.5V, dual supplies must be used for the op amps.

OP AMP APPLICATIONS

A block diagram of the AD813X family of fully differential amplifiers optimized for ADC driving is shown in Figure 3-43 (see References 3-5). Figure 3-43A shows the details of the internal circuit, and Figure 3-43B shows the equivalent circuit. The gain is set by the external R_F and R_G resistors, and the CM voltage is set by the voltage on the V_{OCM} pin. The internal CM feedback forces the V_{OUT+} and V_{OUT-} outputs to be balanced, i.e., the signals at the two outputs are always equal in amplitude but 180° out of phase per the equation,

$$V_{OCM} = (V_{OUT+} + V_{OUT-}) / 2 . \quad \text{Eq. 3-7}$$

The circuit can be used with either a differential or a single-ended input, and the voltage gain is equal to the ratio of R_F to R_G .

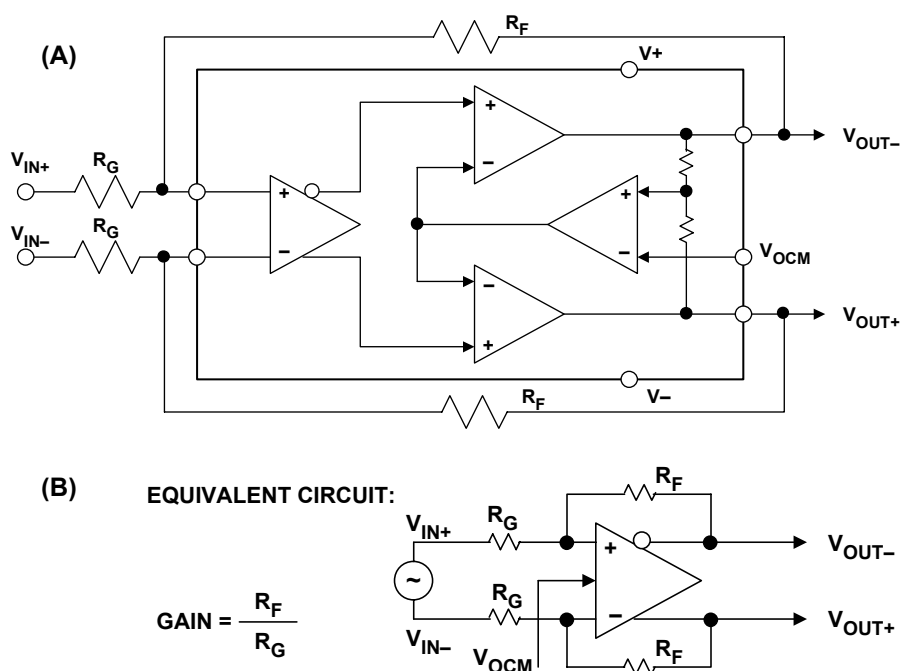


Figure 3-43: AD813X differential ADC driver functional diagram and equivalent circuit

The AD8138 has a 3dB small-signal bandwidth of 320MHz ($G = +1$) and is designed to give low harmonic distortion as an ADC driver. The circuit provides excellent output gain and phase matching, and the balanced structure suppresses even-order harmonics.

Figure 3-44 (opposite) shows the AD8138 driving the AD9203 10-bit, 40MSPS ADC (see Reference 6). This entire circuit operates on a single +3V supply. A 1Vp-p bipolar single-ended input signal produces a 1Vp-p differential signal at the output of the AD8138, centered around a CM voltage of +1.5V (mid-supply).

Each of the differential inputs of the AD8138 swing between +0.625V and +0.875V, and each output swings between +1.25V and +1.75V. These voltages fall within the allowable input and output CM voltage range of the AD8138 operating on a single +3V supply.

USING OP AMPS WITH DATA CONVERTERS DRIVING ADC INPUTS

The circuit as shown operates on a 1Vp-p single-ended bipolar input signal, and the input span of the AD9203 ADC is set for 1Vp-p differential. If the signal input amplitude is increased to 2Vp-p, the span of the AD9203 must be set for 2Vp-p differential. Under these conditions, each of the AD8138 inputs must swing between +0.5V and +1V, and each of the outputs between +1V and +2V.

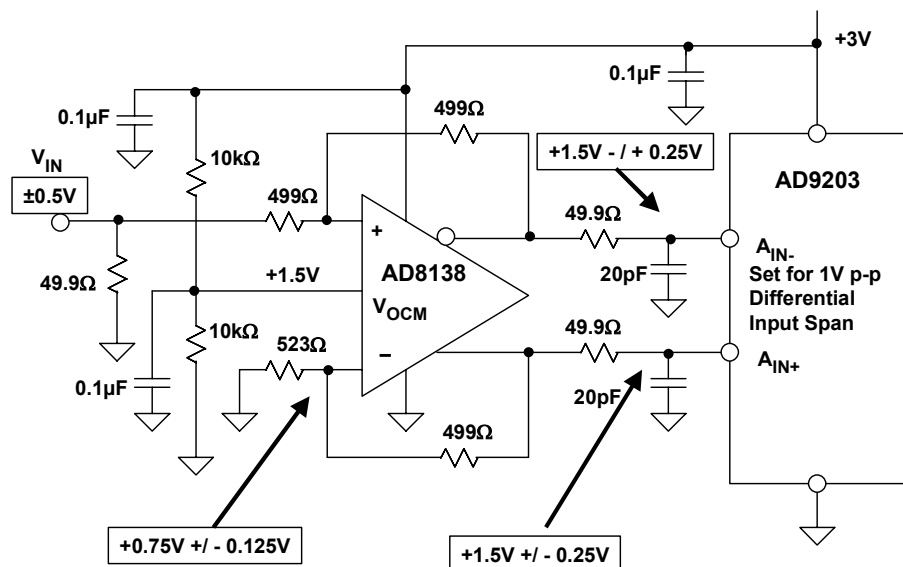


Figure 3-44: AD8138 driving AD9203 10-bit, 40MSPS ADC

As shown in Figure 3-45 below, increasing the amplitude in this manner offers a slight improvement in low frequency SINAD due to the improvement in low frequency SNR.

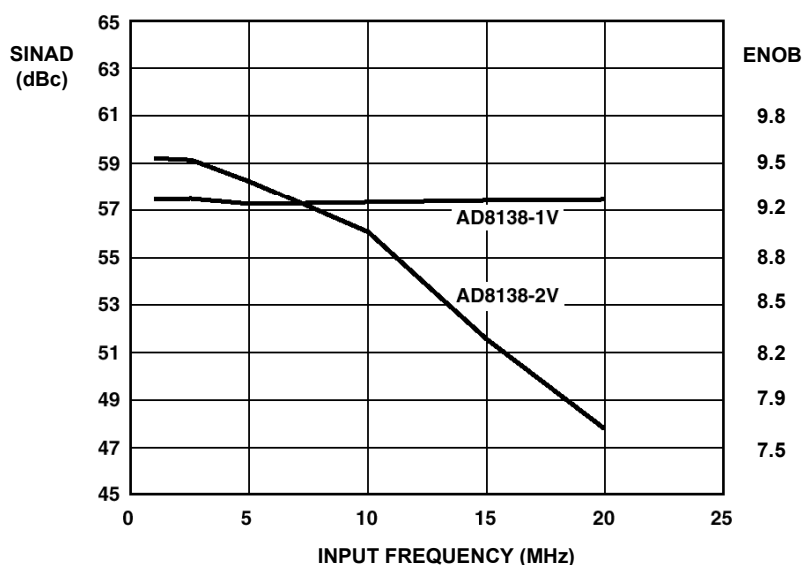


Figure 3-45: SINAD and ENOB for AD9203 12-bit, 40MSPS ADC driven by AD8138 differential amplifier

But at the same time however, a degradation occurs in the high frequency SINAD because of the larger distortion due to the larger signal swings.

Overvoltage Considerations

The input structures of most high performance ADCs are sensitive to overvoltage conditions because of the small geometry devices used in the designs. Although ADC inputs generally have ESD protection diodes connected from the analog input to each supply rail, these diodes are not designed to handle the large currents that can be generated from typical op amp drivers. Two good rules of thumb are to (1) limit the analog input voltage to no more than 0.3V above or below the positive and negative supply voltages, respectively, and (2) limit the analog input current to 5mA maximum in overvoltage conditions.

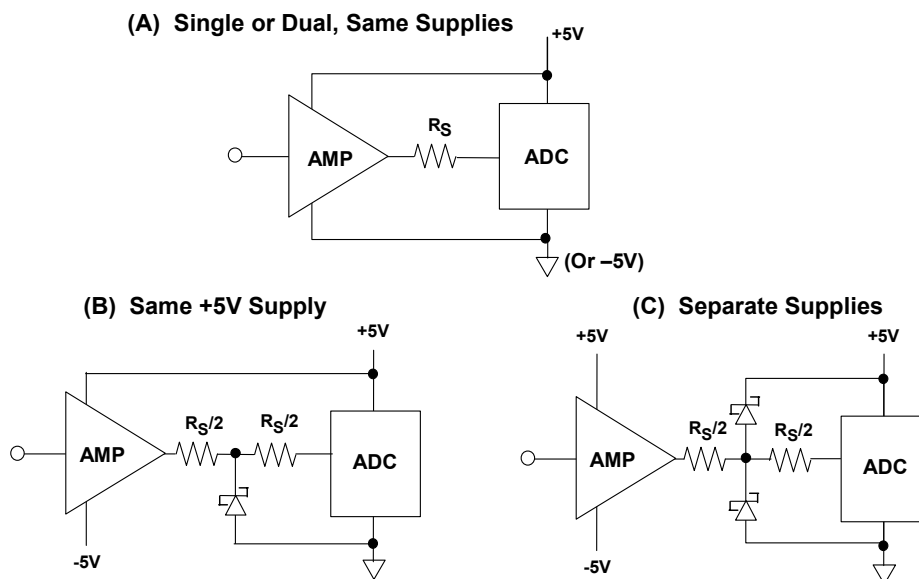


Figure 3-46: ADC input overvoltage protection circuits

Several typical configurations for the drive amp/ADC interface are shown in Figure 3-46 above. In Figure 3-46A, the ADC requires no additional input protection because both the op amp and the ADC are driven directly from the same supply voltages. While the R_S resistor is not required for overvoltage protection, it does serve to isolate the capacitive input of the ADC from the output of the op amp.

Figure 3-46B shows a dual supply op amp driving a single supply ADC, with the +5V supply is shared between the two devices. The diode protects the input of the ADC in case the output of the op amp is driven below ground. A Schottky diode is used because of its low forward voltage drop and its low capacitance. The R_S resistor is split into two equal resistors, and they are chosen to limit the ADC input fault current to 5mA maximum. Note that the R_S resistor in conjunction with the ADC input capacitance forms a lowpass filter. If R_S is made too large, the input bandwidth may be restricted.

Figure 3-46C shows the condition where the op amp and the ADC are driven from separate supplies. Two Schottky diodes are required to protect the ADC under all power supply and signal conditions. As in 3-46A, the $R_S/2$ resistors limit ADC fault current.

REFERENCES: DRIVING ADC INPUTS

1. Data sheet for **AD8057/AD8058 Low Cost, High Performance Voltage Feedback, 325 MHz Amplifiers**, <http://www.analog.com>.
2. Chapter 8 of Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.
3. Data sheet for **AD8131 Low-Cost, High-Speed Differential Driver**, <http://www.analog.com>
4. Data sheet for **AD8132 Low-Cost, High-Speed Differential Amplifier**, <http://www.analog.com>
5. Data sheet for **AD8138 Low Distortion Differential ADC Driver**, <http://www.analog.com>
6. Data Sheet for **AD9203 10-Bit, 40 MSPS, 3V, 74mW A/D Converter**, <http://www.analog.com>
7. Chapters 3-6, Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8.
8. Chapters 4, 5, Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6.
9. Chapters 3, 4, Walt Kester, Editor, **Mixed-Signal and DSP Design Techniques**, Analog Devices, 2000, ISBN: 0-916550-23-0.
10. Chapters 4, 5, Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.

NOTES:

SECTION 4-4: DRIVING ADC/DAC REFERENCE INPUTS

Walt Jung, Walt Kester

It might seem odd to include a section on voltage references in a book devoted primarily to op amp applications, but the relevance will shortly become obvious. Unfortunately, there is little standardization with respect to ADC/DAC voltage references. Some ADCs and DACs have internal references, while others do not. In some cases, the DC accuracy of a converter with an internal reference can often be improved by overriding the internal reference with a more accurate and stable external one.

Although the reference element itself can be either a bandgap, buried zener, or XFET™ (see Reference 1), practically all references have some type of output buffer op amp. The op amp isolates the reference element from the output and also provides drive capability. However, this op amp must obey the general laws relating to op amp stability, and that is what makes the topic of references relevant to the discussion. Figure 3-47 below summarizes voltage reference considerations.

- ◆ **Data converter accuracy determined by the reference, whether internal or external**
- ◆ **Bandgap, buried zener, XFET™ generally have on-chip output buffer op amp**
- ◆ **Transient loading can cause instability and errors**
- ◆ **External decoupling capacitors may cause oscillation**
- ◆ **Output may require external buffer to source and sink current**
- ◆ **Reference voltage noise may limit system resolution**

Figure 3-47: ADC / DAC voltage reference considerations

Note that a reference input to an ADC or DAC is similar to the analog input of an ADC, in that the internal conversion process can inject transient currents at that pin. This requires adequate decoupling to stabilize the reference voltage. Adding such decoupling might introduce instability in some reference types, depending on the output op amp design. Of course, a reference data sheet may not show any details of the output op amp, which leaves the designer in somewhat of a dilemma concerning whether or not it will be stable and free from transient errors. Fortunately, some simple lab tests can exercise a reference circuit for transient errors, and also determine stability for capacitive loading.

Figure 3-48 shows the transients associated with the reference input of a typical successive approximation ADC. The ADC reference voltage input must be stabilized with a sufficiently large decoupling capacitor, in order to prevent conversion errors. The value of the capacitor required as C_B may range from below $1\mu\text{F}$, to as high as $100\mu\text{F}$. This capacitor must of course have a voltage rating greater than the reference voltage. Physically, it will be of minimum size when purchased in a surface mount style.

OP AMP APPLICATIONS

Note that in this case, a $1\mu\text{F}$ capacitor on the reference input is required to reduce the transients to an acceptable level. Note that the capacitor size can be electrically larger for further noise reduction—the tradeoff here is of course cost and PCB real estate. The AD780 will work with capacitors of up to $100\mu\text{F}$.

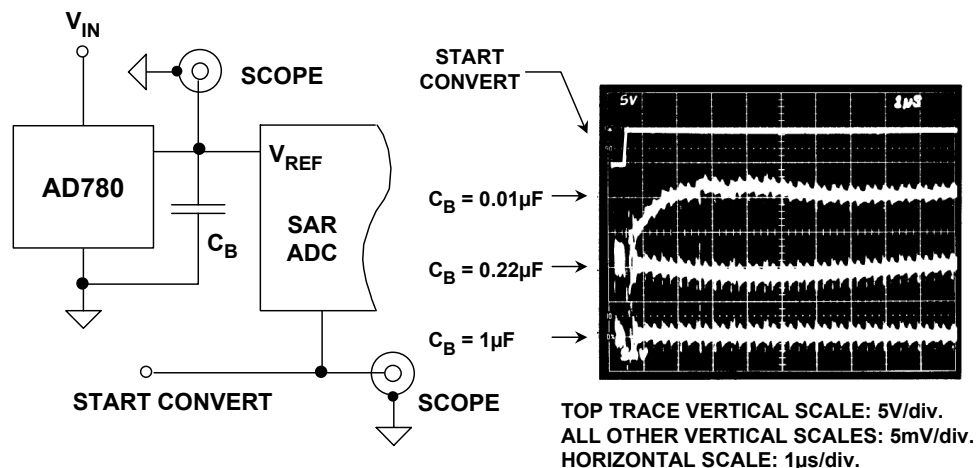


Figure 3-48: Successive approximation ADCs can present a transient load to the reference

A well-designed voltage reference is stable with heavy capacitive decoupling. Unfortunately, some are not, as shown in Figure 3-49 below, where the addition of C_L to the reference output (a $0.01\mu\text{F}$ capacitor) actually increases the amount of transient ringing. Such references are practically useless in data converter applications, because some amount of local decoupling is almost always required at the converter.

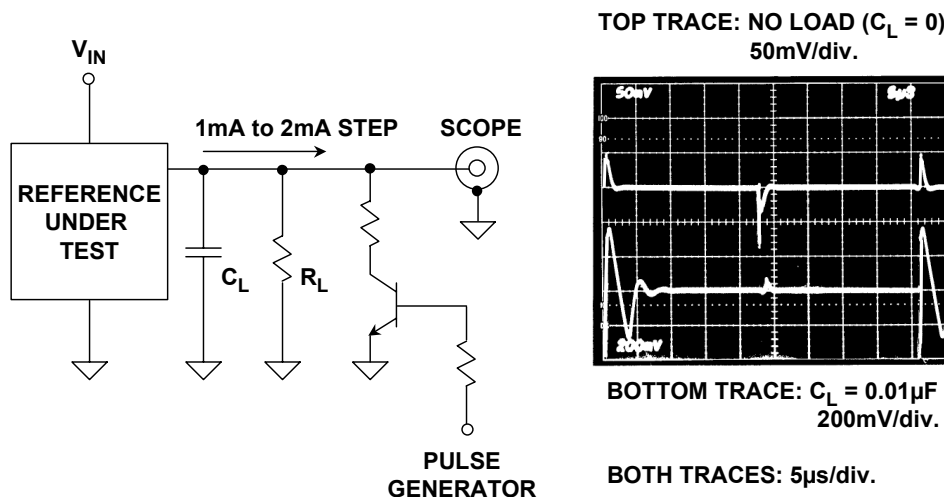


Figure 3-49: Make sure reference is stable with large capacitive loads

A suitable op amp buffer might be added between the reference and the data converter. But, there are many good references available (such as the AD780) which are stable with an output capacitor. This type of reference should be chosen for a data converter application, rather than incurring the further complication and expense of an op amp.

USING OP AMPS WITH DATA CONVERTERS DRIVING ADC/DAC REFERENCE INPUTS

If very low noise levels are required from a reference, an additional low pass filter followed by a low noise op amp can be used to achieve the desired performance. The reference circuit of Figure 3-50 is one such example (see References 2 and 3). This circuit uses external filtering and a precision low-noise op-amp to provide both very low noise and high DC accuracy. Reference U1 is a 2.5, 3.0, 5, or 10V reference with a low noise buffered output. The output of U1 is applied to the R1-C1/C2 noise filter to produce a corner frequency of about 1.7 Hz.

Electrolytic capacitors usually imply DC leakage errors, but the bootstrap connection of C1 causes its applied bias voltage to be only the relatively small drop across R2. This lowers the leakage current through R1 to acceptable levels. Since the filter attenuation is modest below a few Hertz, the reference noise still affects overall performance at low frequencies (i.e., <10 Hz).

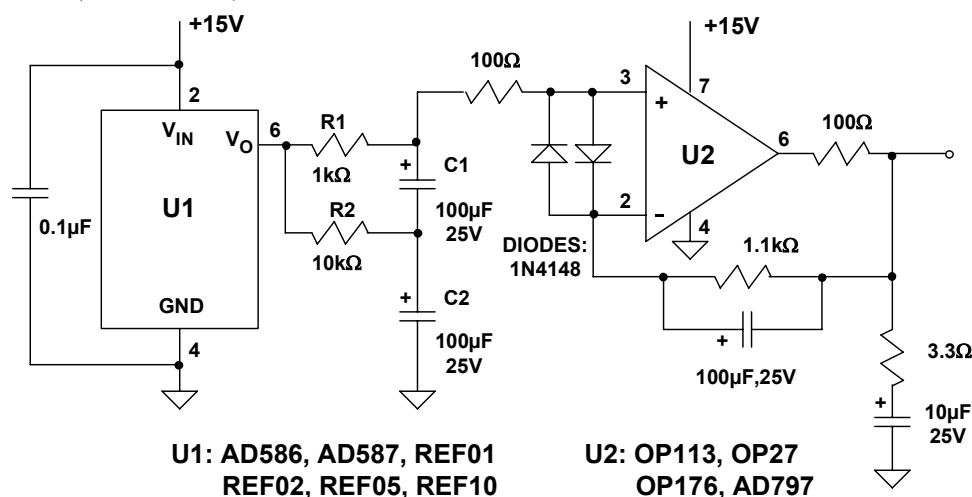


Figure 3-50: Low-noise op amp with filtering yields reference noise performance (1.5 to 5nV/√Hz @ 1kHz)

A precision low noise unity-gain follower, such as the OP113, then buffers the output of the filter. With less than $\pm 150\mu\text{V}$ of offset error and under $1\mu\text{V}/^\circ\text{C}$ drift, the buffer amplifier's DC performance will not seriously affect the accuracy/drift of most references. For example, an ADR292E for U1 will have a typical drift of 3ppm/ $^\circ\text{C}$, equivalent to $7.5\mu\text{V}/^\circ\text{C}$, higher than the buffer amplifier.

Almost any op amp will have a current limit higher than a typical IC reference, so this circuit allows greater current output. It also removes any load related thermal errors that might occur when the reference IC is loaded directly.

Even lower noise op-amps are available, for 5-10V use. The AD797 offers 1kHz noise performance less than $2\text{nV}/\sqrt{\text{Hz}}$ in this circuit, compared to about $5\text{nV}/\sqrt{\text{Hz}}$ for the OP113. With any buffer amplifier, Kelvin sensing can be used at the load point, a technique that eliminates $I \times R$ related output voltage errors.

REFERENCES: DRIVING ADC/DAC REFERENCE INPUTS

1. Chapter 2, Walt Kester, Editor, **Practical Design Techniques for Power and Thermal Management**, Analog Devices, 1998, ISBN: 0-916550-19-2.
2. Walt Jung, "Build an Ultra-Low-Noise Voltage Reference," **Electronic Design Analog Applications Issue**, June 24, 1993.
3. Walt Jung, "Getting the Most from IC Voltage References," **Analog Dialogue**, Vol. 28, No., 1994, pp. 13-21.

SECTION 3-5: BUFFERING DAC OUTPUTS

Walt Kester, Paul Hendriks

General Considerations

Another important op amp application is buffering DAC outputs. Modern IC DACs provide either voltage or current outputs. Figure 3-51 below shows three fundamental configurations, all with the objective of using an op amp for a buffered output voltage.

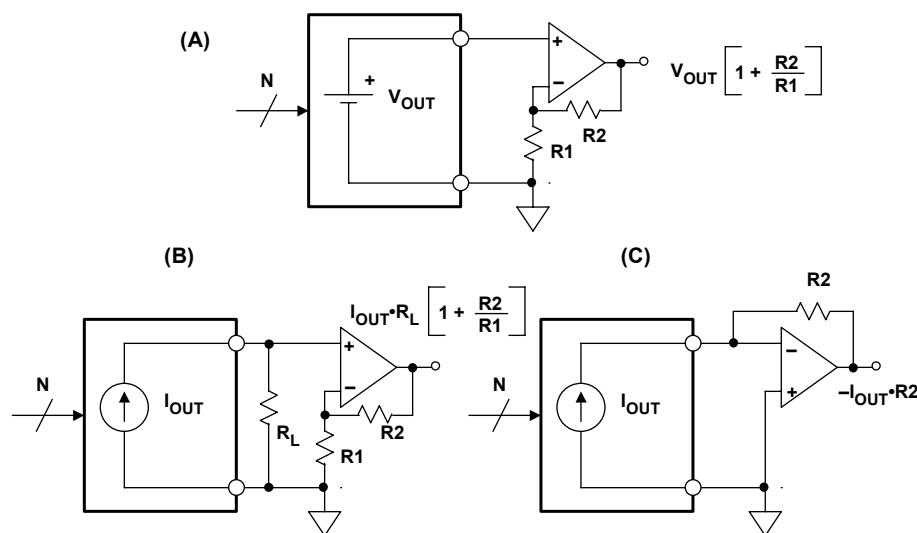


Figure 3-51: Buffering DAC outputs with op amps

Figure 3-51A shows a buffered voltage output DAC. In many cases, the DAC output can be used directly, without additional buffering. If an additional op amp buffer is needed, it is usually configured in a non-inverting mode, with gain determined by R_1 and R_2 .

There are two basic methods for dealing with a current output DAC. In Figure 3-51B, a voltage is simply developed across external load resistor, R_L . An external op amp can be used to buffer and/or amplify this voltage if required. Many DACs supply fullscale currents of 20mA or more, thereby allowing reasonable voltages to be developed across fairly low value load resistors. For instance, fast settling video DACs typically supply nearly 30mA fullscale current, allowing 1V to be developed across a source and load terminated 75 Ω coaxial cable (representing a DC load of 37.5 Ω to the DAC output).

A direct method to convert the output current into a voltage is shown in Figure 3-51C. This circuit is usually called a current-to-voltage converter, or I/V. In this circuit, the DAC output drives the inverting input of an op amp, with the output voltage developed across the R_2 feedback resistor. In this approach the DAC output always operates at virtual ground (which may give a linearity improvement vis-à-vis Fig. 3-51B).

The general selection process for an op amp used as a DAC buffer is similar to that of an ADC buffer. The same basic specifications such as DC accuracy, noise, settling time, bandwidth, distortion, etc., apply to DACs as well as ADCs, and the discussion will not be repeated here. Rather, some specific application examples will be shown.

Differential to Single-Ended Conversion Techniques

A general model of a modern current output DAC is shown in Figure 3-52 below. This model is typical of the AD976X and AD977X TxDAC™ series (see Reference 1). Current output is more popular than voltage output, especially at audio frequencies and above. If the DAC is fabricated on a bipolar or BiCMOS process, it is likely that the output will sink current, and that the output impedance will be less than 500Ω (due to the internal R/2R resistive ladder network). On the other hand, a CMOS DAC is more likely to source output current and have a high output impedance, typically greater than 100kΩ.

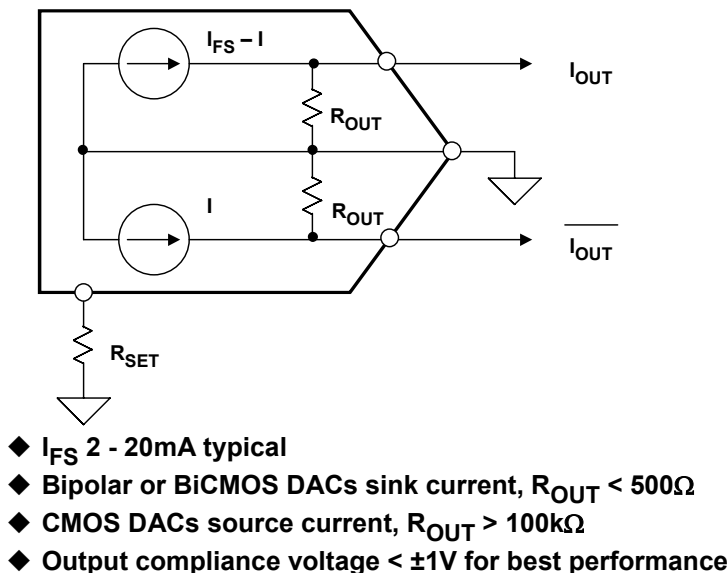


Figure 3-52: Model of high speed DAC output

Another consideration is the output *compliance voltage*— the maximum voltage swing allowed at the output in order for the DAC to maintain its linearity. This voltage is typically 1V to 1.5V, but can vary depending upon the DAC. Best DAC linearity is generally achieved when driving a virtual ground, such as an op amp I/V converter.

Modern current output DACs usually have differential outputs, to achieve high CM rejection and reduce the even-order distortion products. Fullscale output currents in the range of 2mA to 20mA are common.

In most applications, it is desirable to convert the differential output of the DAC into a single-ended signal, suitable for driving a coax line. This can be readily achieved with an RF transformer, provided low frequency response is not required. Figure 3-53 (opposite) shows a typical example of this approach. The high impedance current output of the DAC is terminated differentially with 50Ω, which defines the source impedance to the transformer as 50Ω.

The resulting differential voltage drives the primary of a 1:1 RF transformer, to develop a single-ended voltage at the output of the secondary winding. The output of the 50 Ω LC filter is matched with the 50 Ω load resistor R_L , and a final output voltage of 1Vp-p is developed.

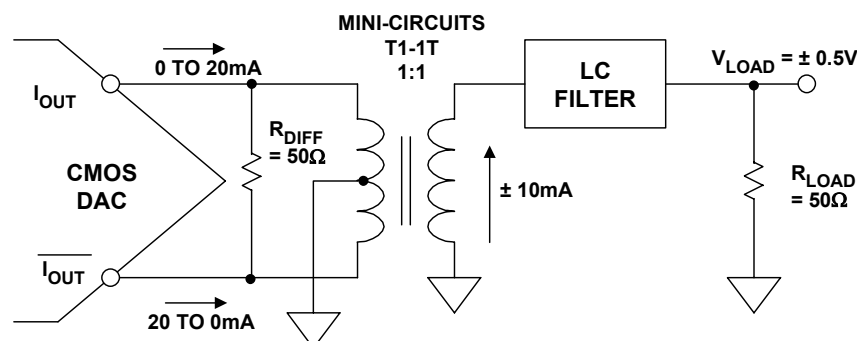


Figure 3-53: Differential transformer coupling

The transformer not only serves to convert the differential output into a single-ended signal, but it also isolates the output of the DAC from the reactive load presented by the LC filter, thereby improving overall distortion performance.

An op amp connected as a differential to single-ended converter can be used to obtain a single-ended output when frequency response to DC is required. In Figure 3-54 below the AD8055 op amp is used to achieve high bandwidth and low distortion (see Reference 2). The current output DAC drives balanced 25 Ω resistive loads, thereby developing an out-of-phase voltage of 0 to +0.5V at each output. The AD8055 is configured for a gain of 2, to develop a final single-ended ground-referenced output voltage of 2Vp-p. Note that because the output signal swings above and below ground, a dual-supply op amp is required.

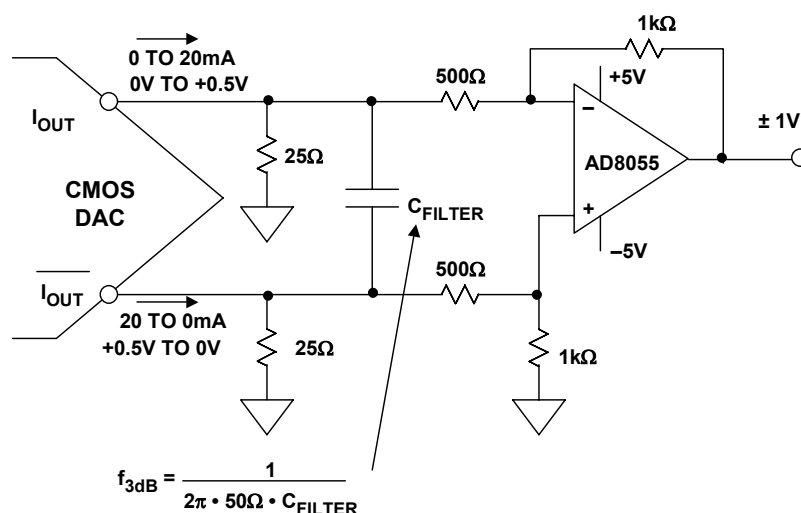


Figure 3-54: Differential DC coupled output using a dual supply op amp

The C_{FILTER} capacitor forms a differential filter with the equivalent 50 Ω differential output impedance. This filter reduces any slew induced distortion of the op amp, and the optimum cutoff frequency of the filter is determined empirically to give the best overall distortion performance.

OP AMP APPLICATIONS

A modified form of the Fig. 3-54 circuit can also be operated on a single supply, provided the CM voltage of the op amp is set to mid-supply (+2.5V). This is shown in Figure 3-55 below. The output voltage is 2Vp-p centered around a CM voltage of +2.5V. This CM voltage can be either developed from the +5V supply using a resistor divider, or directly from a +2.5V voltage reference. If the +5V supply is used as the CM voltage, it must be heavily decoupled to prevent supply noise from being amplified.

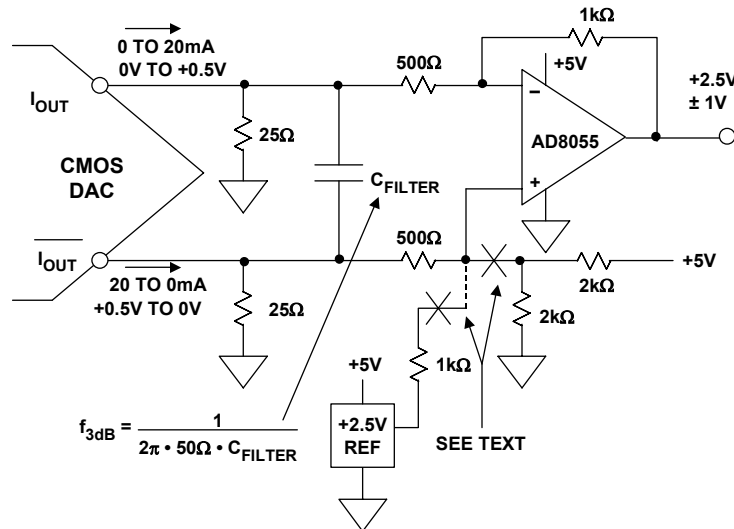


Figure 3-55: Differential DC coupled output using a single-supply op amp

Single-Ended Current-to-Voltage Conversion

Single-ended current-to-voltage conversion is easily performed using a single op amp as an I/V converter, as shown in Figure 3-56 below. The 10mA fullscale DAC current from the AD768 (see Reference 3) develops a 0 to +2V output voltage across the 200Ω R_F .

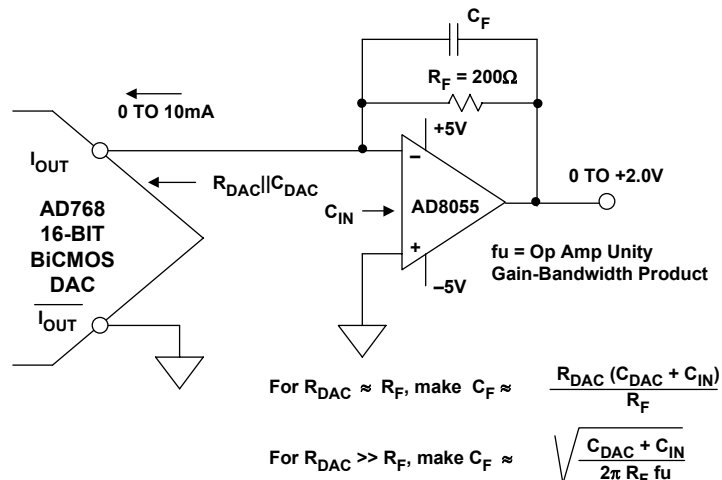


Figure 3-56: Single-ended I/V op amp interface for precision 16-bit AD768 DAC

Driving the virtual ground of the AD8055 op amp minimizes any distortion due to nonlinearity in the DAC output impedance. In fact, most high resolution DACs of this type are factory trimmed using an I/V converter.

It should be recalled, however, that using the single-ended output of the DAC in this manner will cause degradation in the CM rejection and increased second-order distortion products, compared to a differential operating mode.

The C_F feedback capacitor should be optimized for best pulse response in the circuit. The equations given in the diagram should only be used as guidelines. A more detailed analysis of this circuit is given in Reference 6.

Differential Current-to-Differential Voltage Conversion

If a buffered differential voltage output is required from a current output DAC, the AD813X-series of differential amplifiers can be used as shown in Figure 3-57 below.

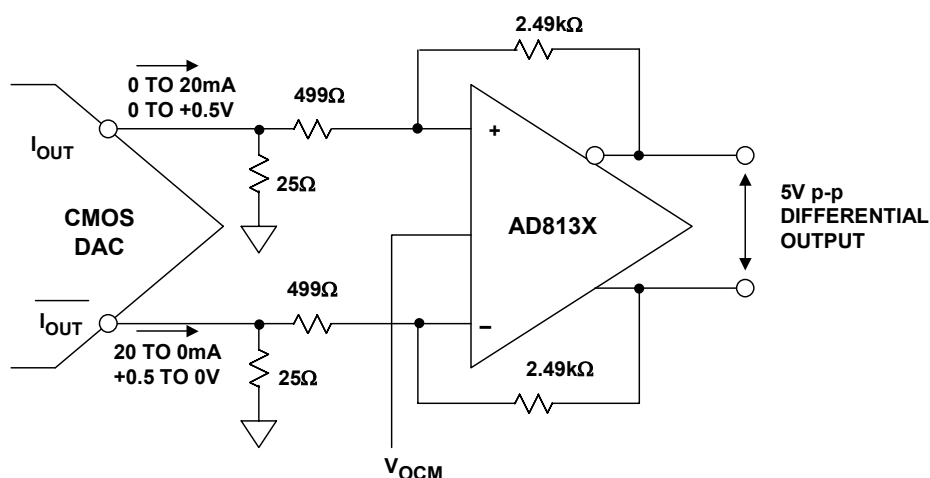


Figure 3-57: Buffering high speed DACs using AD813X differential amplifier

The DAC output current is first converted into a voltage that is developed across the 25Ω resistors. The voltage is amplified by a factor of 5 using the AD813X. This technique is used in lieu of a direct I/V conversion to prevent fast slewing DAC currents from overloading the amplifier and introducing distortion. Care must be taken so that the DAC output voltage is within its compliance rating.

The V_{OCM} input on the AD813X can be used to set a final output CM voltage within the range of the AD813X. If transmission lines are to be driven at the output, adding a pair of 75Ω resistors will allow this.

An Active Lowpass Filter for Audio DAC

Figure 3-58 below shows an active lowpass filter which also serves as a current-to-voltage converter for the AD1853 sigma-delta audio DAC (see Reference 4). The filter is a 4-pole filter with a 3dB cutoff frequency of approximately 75kHz. Because of the high oversampling frequency (24.576MSPS when operating the DAC at a 48KSPS throughput rate), a simple filter is all that is required to remove aliased components above 12MHz).

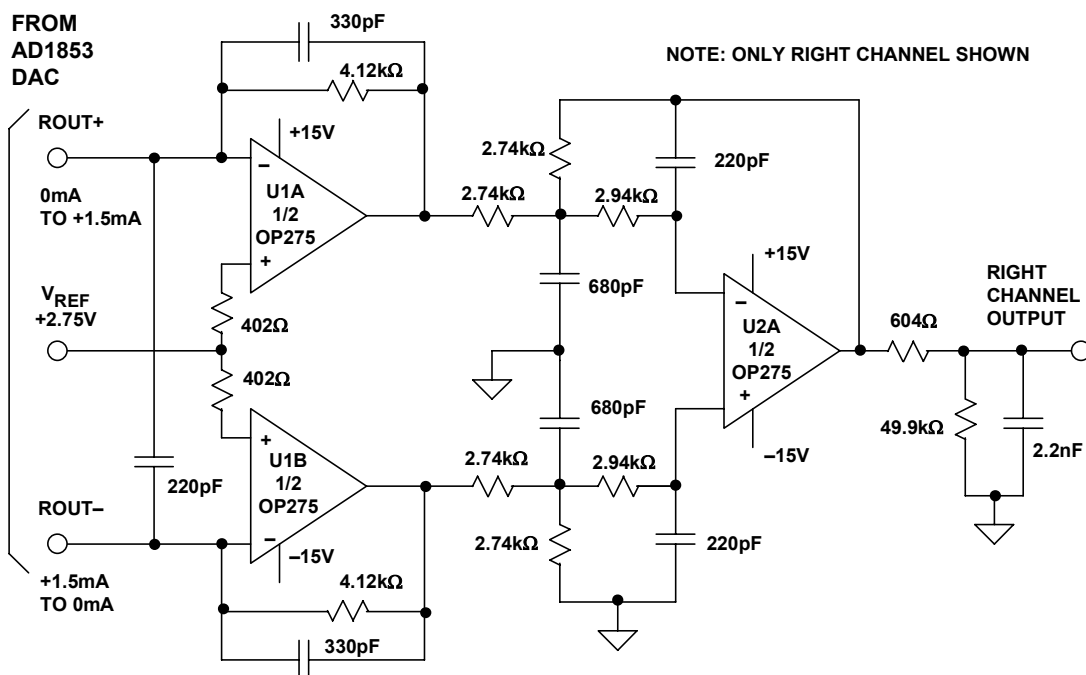


Figure 3-58: A 75kHz 4-pole gaussian active filter for buffering the output of AD1853 stereo DAC

The diagram shows a single channel for the dual channel DAC output. U1A and U1B I/V stages form a 1-pole differential filter, while U2 forms a 2-pole multiple-feedback filter that also performs a differential-to-single-ended conversion.

A final fourth passive pole is formed by the 604Ω resistor and the 2.2nF capacitor across the output. The OP275 op amp was chosen for operation at U1 and U2, and for its quality audio characteristics (see Reference 5).

For further details of active filter designs, see Chapter 5 of this book.

REFERENCES: BUFFERING DAC OUTPUTS

1. Data sheet for **AD9772A 14-Bit, 160 MSPS TxDAC+® with 2x Interpolation Filter**, <http://www.analog.com>
2. Data sheet for **AD8055/AD8056 Low Cost, 300 MHz Voltage Feedback Amplifiers**, <http://www.analog.com>.
3. Data sheet for **AD768 16-Bit, 30 MSPS D/A Converter**, <http://www.analog.com>.
4. Data sheet for **AD1853 Stereo, 24-Bit, 192 kHz, Multibit $\Sigma\Delta$ DAC**, <http://www.analog.com>.
5. Data sheet for **OP275 Dual Bipolar/JFET, Audio Operational Amplifier**, <http://www.analog.com>.
6. Chapters 5, Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, 1999, ISBN: 0-916550-20-6.

NOTES:

- H Op Amp History**
 - 1 Op Amp Basics**
 - 2 Specialty Amplifiers**
 - 3 Using Op Amps with Data Converters**
 - ◆ 4 Sensor Signal Conditioning**
 - 1 Introduction**
 - 2 Bridge Circuits**
 - 3 Strain, Force, Pressure and Flow Measurements**
 - 4 High Impedance Sensors**
 - 5 Temperature Sensors**
- 5 Analog Filters**
- 6 Signal Amplifiers**
- 7 Hardware and Housekeeping Techniques**

▣ OP AMP APPLICATIONS

CHAPTER 4: SENSOR SIGNAL CONDITIONING

*Walt Kester, James Bryant, Walt Jung,
Scott Wurcer, Chuck Kitchin*

SECTION 4-1: INTRODUCTION

Walt Kester

This chapter of the book deals with various sensors and associated signal-conditioning circuitry involving the use of op amps and in-amps. While the topic is generally very broad, the focus is to concentrate on circuit and signal processing applications of sensors rather than the details of the actual sensors themselves.

Strictly speaking, a *sensor* is a device that receives a signal or stimulus and responds with an electrical signal, while a *transducer* is a converter of one type of energy into another. In practice, however, the terms are often used interchangeably.

Sensors and their associated circuits are used to measure various physical properties such as temperature, force, pressure, flow, position, light intensity, etc. These properties act as the stimulus to the sensor, and the sensor output is conditioned and processed to provide the corresponding measurement of the physical property. We will not cover all possible types of sensors, only the most popular ones, and specifically, those that lend themselves to process control and data acquisition systems.

Sensors do not operate by themselves. They are generally part of a larger system consisting of signal conditioners and various analog or digital signal processing circuits. The *system* could be a measurement system, data acquisition system, or process control system, for example.

Sensors may be classified in a number of ways. From a signal-conditioning viewpoint it is useful to classify sensors as either *active* or *passive*. An *active* sensor requires an external source of excitation. Resistor-based sensors such as thermistors, resistance temperature detectors (RTDs), and strain gages are examples of active sensors, because a current must be passed through them and the corresponding voltage measured in order to determine the resistance value. An alternative would be to place the devices in a bridge circuit, however in either case, an external current or voltage is required.

On the other hand, *passive* (or *self-generating*) sensors generate their own electrical output signal without requiring external voltages or currents. Examples of passive sensors are thermocouples and photodiodes which generate thermoelectric voltages and photocurrents, respectively, which are independent of external circuits.

It should be noted that these definitions (*active* vs. *passive*) refer to the need (or lack thereof) of external active circuitry to produce a sensor electrical output signal. It would

■ OP AMP APPLICATIONS

seem equally logical to consider a thermocouple *active*, in the sense that it produces an output voltage without external circuitry, however the convention in the industry is to classify the sensor with respect to the external circuit requirement as defined above.

A logical way to classify sensors is with respect to the physical property the sensor is designed to measure. Thus we have temperature sensors, force sensors, pressure sensors, motion sensors, etc. However, sensors which measure different properties may have the same type of electrical output. For instance, a Resistance Temperature Detector (RTD) is a variable resistance, as is a resistive strain gauge. Both RTDs and strain gages are often placed in bridge circuits, and the conditioning circuits are therefore quite similar. In fact, bridges and their conditioning circuits deserve a detailed discussion. Figure 4-1 below is an overview of basic sensor characteristics.

- ◆ **Sensors:**
Convert a Signal or Stimulus (Representing a Physical Property) into an Electrical Output
- ◆ **Transducers:**
Convert One Type of Energy into Another
- ◆ **The Terms are often Interchanged**
- ◆ **Active Sensors Require an External Source of Excitation:**
RTDs, Strain-Gages
- ◆ **Passive (Self-Generating) Sensors do not:**
Thermocouples, Photodiodes

Figure 4-1: *An overview of sensor characteristics*

The full-scale outputs of most sensors (passive or active) are relatively small voltages, currents, or resistance changes, and therefore their outputs must be properly conditioned before further analog or digital processing can occur. Because of this, an entire class of circuits have evolved, generally referred to as signal-conditioning circuits. Amplification, level translation, galvanic isolation, impedance transformation, linearization, and filtering are fundamental signal-conditioning functions that may be required. Figure 4-2 (opposite) summarizes sensors and their outputs.

Whatever form the conditioning takes, however, the circuitry and performance will be governed by the electrical character of the sensor and its output. Accurate characterization of the sensor in terms of parameters appropriate to the application, e.g., sensitivity, voltage and current levels, linearity, impedances, gain, offset, drift, time constants, maximum electrical ratings, and stray impedances and other important considerations can spell the difference between substandard and successful application of the device, especially where high resolution, precision, or low-level measurements are necessary.

Higher levels of integration now allow ICs to play a significant role in both analog and digital signal conditioning. ADCs specifically designed for measurement applications often contain on-chip programmable-gain amplifiers (PGAs) and other useful circuits, such as current sources for driving RTDs, thereby minimizing the external conditioning circuit requirements.

To some degree or another, most sensor outputs are non-linear with respect to the applied stimulus, and as a result their outputs must often be linearized in order to yield correct measurements. In terms of the design approach choice towards linearization, the designer can take a route along either of two major paths.

PROPERTY	SENSOR	ACTIVE/ PASSIVE	OUTPUT
Temperature	Thermocouple	Passive	Voltage
	Silicon	Active	Voltage/Current
	RTD	Active	Resistance
	Thermistor	Active	Resistance
Force / Pressure	Strain Gage	Active	Resistance
	Piezoelectric	Passive	Voltage
Acceleration	Accelerometer	Active	Capacitance
Position	LVDT	Active	AC Voltage
Light Intensity	Photodiode	Passive	Current

Figure 4-2: Typical sensors and their output formats

Analog is one viable route, and such techniques may be used to perform an “analog domain” linearization function.

However, the recent introduction of high performance ADCs now allows linearization to be done much more efficiently and accurately in software. This “digital domain” approach to linearization eliminates the need for tedious manual calibration using multiple and sometimes interactive analog trim adjustments.

■ OP AMP APPLICATIONS

A quite common application of sensors is within process control systems. One example would be control of a physical property, such as temperature. A sample block diagram of how this might be implemented is illustrated in Figure 4-3 below.

In this system, an output from a temperature sensor is conditioned, transmitted over some distance, received, and then digitized by an ADC. The microcontroller or host computer determines if the temperature is above or below the desired value, and outputs a digital word to the digital-to-analog converter (DAC). The DAC output is conditioned and drives the remotely located *actuator*, in this case - a heater. Notice that the interface between the control center and the remote process is via the industry-standard 4-20mA loop.

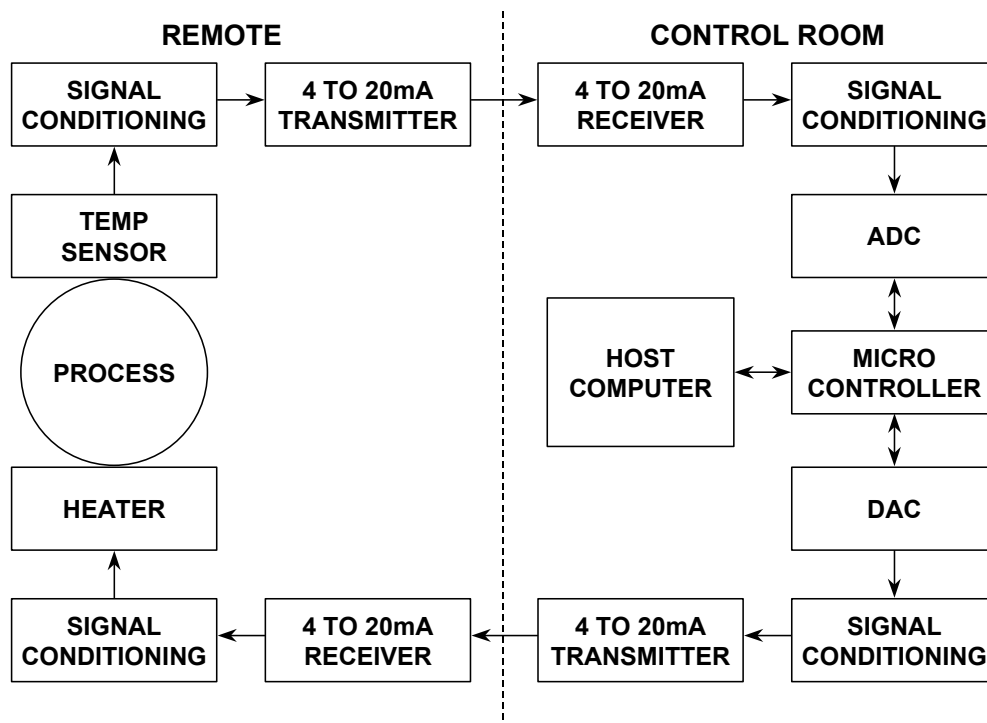


Figure 4-3: A typical industrial process temperature control loop

Digital techniques are becoming more and more popular in processing sensor outputs in data acquisition, process control, and measurement. 8-bit microcontrollers (8051-based, for example) generally have sufficient speed and processing capability for most applications.

By including the A/D conversion and the microcontroller programmability on the sensor itself, a "smart sensor" can be implemented with self-contained calibration and linearization features among others. However, such digital techniques aren't a major focus of this text, so the section references may be consulted for further information.

The remaining sections of the chapter deal with analog signal-conditioning methods for a variety of sensor types.

REFERENCES: INTRODUCTION

1. Walt Kester, Bill Chestnut, and Grayson King, *Smart Sensors*, Chapter 9 of **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, Inc., 1999, ISBN: 0-916550-20-6.
2. Compatibility of Analog Signals for Electronic Industrial Process Instruments, ANSI/ISA-S50.1-1982 (Rev. 1992), <http://www.isa.org>.
3. Editors, "Fieldbuses: Look Before You Leap," **EDN**, November 5, 1998, p. 197.
4. "MicroConverter Technology Backgrounder," Whitepaper, Analog Devices, Inc., <http://www.analog.com>.
5. Scott MacKenzie, **The 8051 Microcontroller, 3rd Ed.**, Prentice-Hall, 1999, ISBN: 0-13-780008-8

NOTES:

SECTION 4-2: BRIDGE CIRCUITS

Walt Kester

An Introduction to Bridges

This section of the chapter, 4-2, discusses more fundamental bridge circuit concepts. To gain greatest appreciation of these ideas, it should be studied along with those sections discussing precision op amps within Chapters 1 and 2. The next section (4-3) focuses on the detailed application circuits relating to strain-gage-based sensors. These sections can be read sequentially if the reader already understands the design issues related to precision op amp applications.

Resistive elements are some of the most common sensors. They are inexpensive, and relatively easy to interface with signal-conditioning circuits. Resistive elements can be made sensitive to temperature, strain (by pressure or by flex), and light. Using these basic elements, many complex physical phenomena can be measured, such as: fluid or mass flow (by sensing the temperature difference between two calibrated resistances), dew-point humidity (by measuring two different temperature points), etc.

◆ Strain Gages	120Ω, 350Ω, 3500Ω
◆ Weigh-Scale Load Cells	350Ω - 3500Ω
◆ Pressure Sensors	350Ω - 3500Ω
◆ Relative Humidity	100kΩ - 10MΩ
◆ Resistance Temperature Devices (RTDs)	100Ω , 1000Ω
◆ Thermistors	100Ω - 10MΩ

Figure 4-4: *Sensor resistances used in bridge circuits span a wide dynamic range*

Sensor element resistance can range from less than 100Ω to several hundred kΩ, depending on the sensor design and the physical environment to be measured. Figure 4-4 indicates the wide range of sensor resistance encountered. For example, RTDs are typically 100Ω or 1000Ω. Thermistors are typically 3500Ω or higher.

Resistive sensors such as RTDs and strain gages produce relatively small percentage changes in resistance, in response to a change in a physical variable such as temperature or force. For example, platinum RTDs have a temperature coefficient of about 0.385%/°C. Thus, in order to accurately resolve temperature to 1°C, the overall measurement accuracy must be much better than 0.385Ω when using a 100Ω RTD.

■ OP AMP APPLICATIONS

Strain gages present a significant measurement challenge because the typical change in resistance over the entire operating range of a strain gage may be less than 1% of the nominal resistance value. Accurately measuring small resistance changes is therefore critical when applying resistive sensors.

A simple method for measuring resistance is to force a constant current through the resistive sensor, and measure the voltage output. This requires both an accurate current source and an accurate means of measuring the voltage. Any change in the current will be interpreted as a resistance change. In addition, the power dissipation in the resistive sensor must be small and in accordance with the manufacturer's recommendations, so that self-heating does not produce errors. As a result, the drive current must be small, which tends to limit the resolution of this approach.

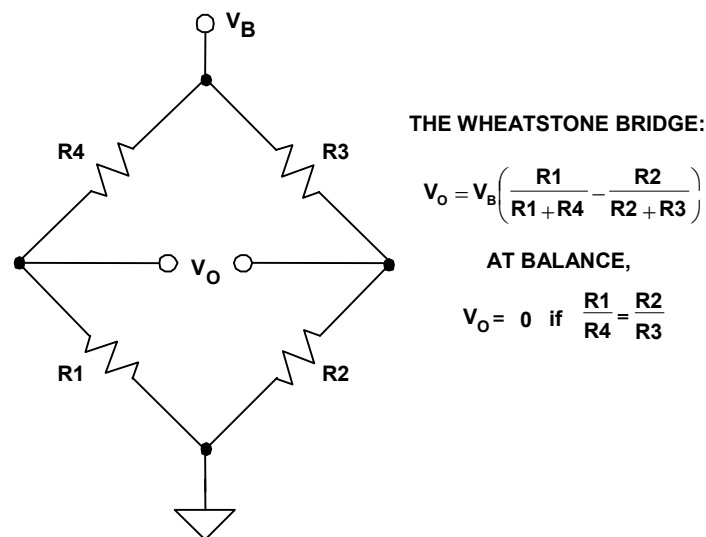


Figure 4-5: The basic Wheatstone bridge produces an output null when the ratios of sidearm resistances match

A *resistance bridge*, shown in Figure 4-5, offers an attractive alternative for measuring small resistance changes accurately. This is a basic Wheatstone bridge (actually developed by S. H. Christie in 1833), and is a prime example. It consists of four resistors connected to form a quadrilateral, a source of excitation voltage V_B (or, alternately, a current) connected across one of the diagonals, and a voltage detector connected across the other diagonal. The detector measures the difference between the outputs of the two voltage dividers connected across the excitation voltage, V_B . The general form of the bridge output V_O is noted in the figure.

There are two principal ways of operating a bridge such as this. One is by operating it as a null detector, where the bridge measures resistance indirectly by comparison with a similar standard resistance. On the other hand, it can be used as a device that reads a resistance difference directly, as a proportional voltage output.

When $R1/R4 = R2/R3$, the resistance bridge is said to be at a *null*, irrespective of the mode of excitation (current or voltage, AC or DC), the magnitude of excitation, the mode of readout (current or voltage), or the impedance of the detector. Therefore, if the ratio of $R2/R3$ is fixed at K , a null is achieved when $R1 = K \cdot R4$. If $R1$ is unknown and $R4$ is an accurately determined variable resistance, the magnitude of $R1$ can be found by adjusting $R4$ until an output null is achieved. Conversely, in sensor-type measurements, $R4$ may be a fixed reference, and a null occurs when the magnitude of the external variable (strain, temperature, etc.) is such that $R1 = K \cdot R4$.

Null measurements are principally used in feedback systems involving electromechanical and/or human elements. Such systems seek to force the active element (strain gage, RTD, thermistor, etc.) to balance the bridge by influencing the parameter being measured.

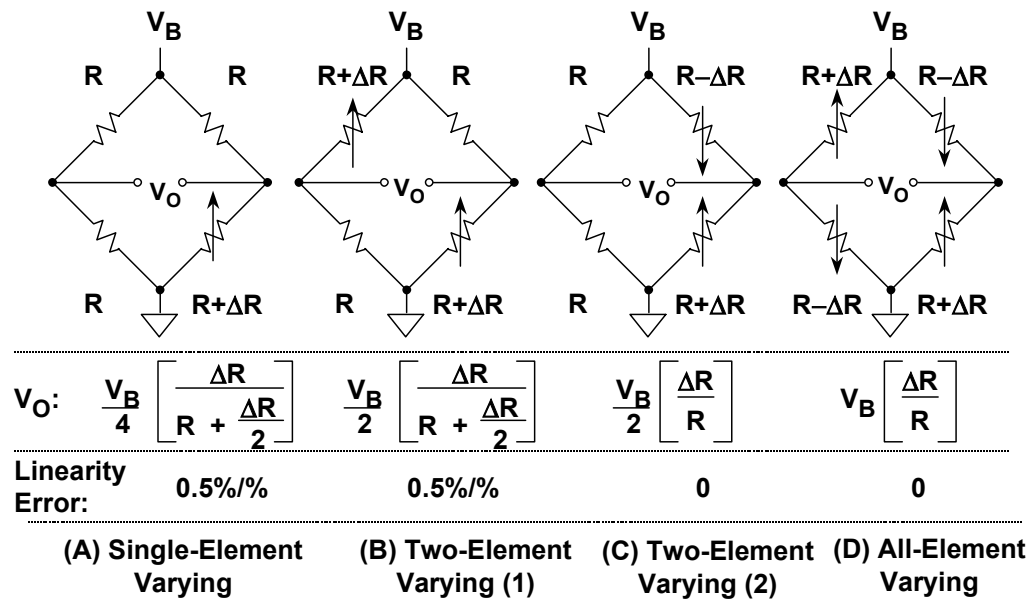


Figure 4-6: The output voltage sensitivity and linearity of constant voltage drive bridge configurations differs according to the number of active elements

For the majority of sensor applications employing bridges, however, the deviation of one or more resistors in a bridge from an initial value is measured as an indication of the magnitude (or a change) in the measured variable. In these cases, the output voltage change is an indication of the resistance change. Because very small resistance changes are common, the output voltage change may be as small as tens of millivolts, even with the excitation voltage $V_B = 10V$ (typical for a load cell application).

In many bridge applications, there may not just be a single variable element, but two, or even four elements, all of which may vary. Figure 4-6 above shows a family of four voltage-driven bridges, those most commonly suited for sensor applications. In the four cases the corresponding equations for V_O relate the bridge output voltage to the excitation voltage and the bridge resistance values. In all cases we assume a constant voltage drive, V_B . Note that since the bridge output is always directly proportional to V_B , the measurement accuracy can be no better than that of the accuracy of the excitation voltage.

■ OP AMP APPLICATIONS

In each case, the value of the fixed bridge resistor “R” is chosen to be equal to the nominal value of the variable resistor(s). The deviation of the variable resistor(s) about the nominal value is assumed to be proportional to the quantity being measured, such as strain (in the case of a strain gage), or temperature (in the case of an RTD).

The *sensitivity* of a bridge is the ratio of the maximum expected change in the output voltage to the excitation voltage. For instance, if $V_B = 10\text{V}$, and the fullscale bridge output is 10mV , then the sensitivity is 1mV/V . For the four cases of Figure 4-6, sensitivity can be said to increase going left-right, or as more elements are made variable.

The *single-element varying* bridge of Figure 4-6A is most suited for temperature sensing using RTDs or thermistors. This configuration is also used with a single resistive strain gage. All the resistances are nominally equal, but one of them (the sensor) is variable by an amount ΔR . As the equation indicates, the relationship between the bridge output and ΔR is not linear. For example, if $R = 100\Omega$ and $\Delta R = 0.1\Omega$ (0.1% change in resistance), the output of the bridge is 2.49875mV for $V_B = 10\text{V}$. The error is $2.50000\text{mV} - 2.49875\text{mV}$, or 0.00125mV . Converting this to a % of fullscale by dividing by 2.5mV yields an end-point linearity error in percent of approximately 0.05%. (Bridge end-point linearity error is calculated as the worst error in % FS from a straight line which connects the origin and the end point at FS, i.e., the FS gain error is not included). If $\Delta R = 1\Omega$, (1% change in resistance), the output of the bridge is 24.8756mV , representing an end-point linearity error of approximately 0.5%. The end-point linearity error of the single-element bridge can be expressed in equation form:

$$\begin{array}{l} \text{Single-Element Varying} \\ \text{Bridge End-Point Linearity Error} \quad \approx \quad \% \text{ Change in Resistance} \div 2 \end{array}$$

It should be noted that the above nonlinearity refers to the nonlinearity of the bridge itself and not the sensor. In practice, most sensors themselves will exhibit a certain specified amount of nonlinearity, which must also be accounted for in the final measurement.

In some applications, the bridge nonlinearity noted above may be acceptable. But, if not, there are various methods available to linearize bridges. Since there is a fixed relationship between the bridge resistance change and its output (shown in the equations), software can be used to remove the linearity error in digital systems. Circuit techniques can also be used to linearize the bridge output directly, and these will be discussed shortly.

There are two cases to consider in the instance of a *two-element varying* bridge. In Case 1 (Figure 4-6B), both of the diagonally opposite elements change in the same direction. An example would be two identical strain gages mounted adjacent to each other, with their axes in parallel.

The nonlinearity for this case, 0.5%/%, the same as that of the single-element varying bridge of Figure 4-6A. However, it is interesting to note the sensitivity is now improved by a factor of 2, vis-à-vis the single-element varying setup. The two-element varying bridge is commonly found in pressure sensors and flow meter systems.

A second case of the two-element varying bridge, Case 2, is shown in Figure 4-6C. This bridge requires two identical elements that vary in *opposite* directions. This could correspond to two identical strain gages: one mounted on top of a flexing surface, and one on the bottom. Note that this configuration is now linear, and like two-element varying Case 1, it has twice the sensitivity of the Figure 4-6A configuration. Another way to view this configuration is to consider the terms $R+\Delta R$ and $R-\Delta R$ as comprising two sections of a linear potentiometer.

The *all-element varying* bridge of Figure 4-6D produces the most signal for a given resistance change, and is inherently linear. It is also an industry-standard configuration for load cells constructed from four identical strain gages. Understandably, it is also one of the most popular bridge configurations.

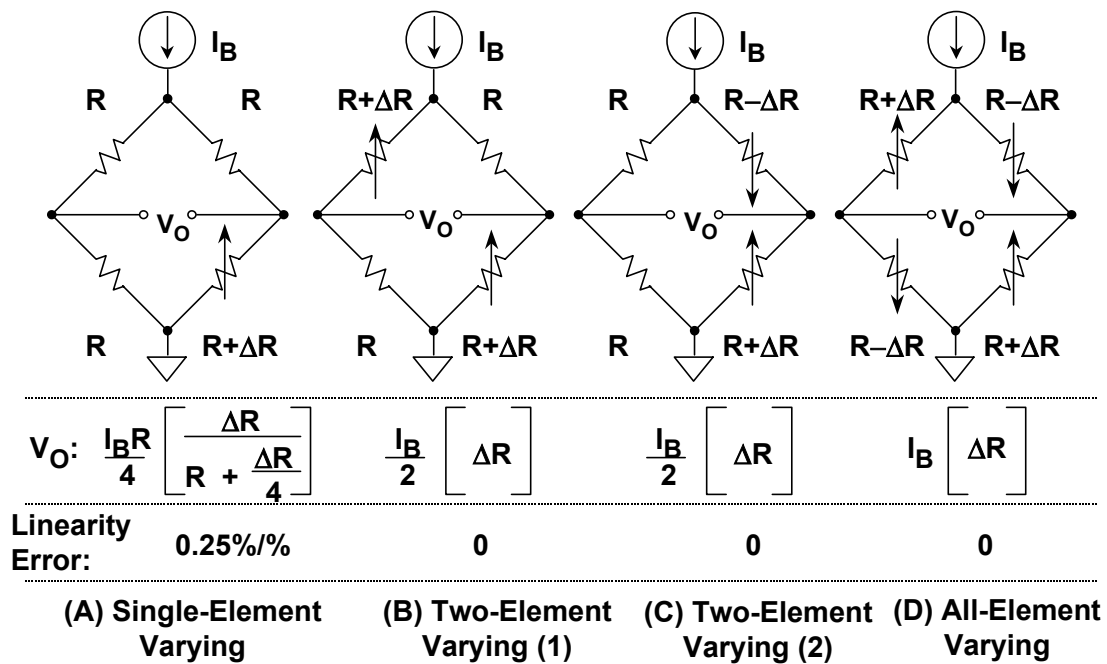


Figure 4-7: The output voltage sensitivity and linearity of constant current drive bridge configurations also differs according to the number of active elements

Bridges may also be driven from constant current sources, as shown in Figure 4-7, for the corresponding cases of single, dual, dual, and four active element(s). As with the voltage-driven bridges, the analogous output expressions are noted, along with the sensitivities.

Current drive, although not as popular as voltage drive, does have advantages when the bridge is located remotely from the source of excitation. One advantage is that the wiring resistance doesn't introduce errors in the measurement; another is simpler, less expensive cabling. Note also that with constant current excitation, all bridge configurations are linear except the single-element varying case of Figure 4-7A.

▣ OP AMP APPLICATIONS

In summary, there are many design issues relating to bridge circuits, as denoted by Figure 4-8 below. After selecting the basic configuration, the excitation method must be determined. The value of the excitation voltage or current must first be determined, as this directly influences sensitivity. Recall that the fullscale bridge output is directly proportional to the excitation voltage (or current). Typical bridge sensitivities are 1mV/V to 10mV/V.

Although large excitation voltages yield proportionally larger fullscale output voltages, they also result in higher bridge power dissipation, and thus raise the possibility of sensor resistor self-heating errors. On the other hand, low values of excitation voltage require more gain in the conditioning circuits, and also increase sensitivity to low level errors such as noise and offset voltages.

- ◆ **Selecting Configuration (1, 2, 4 - Element Varying)**
- ◆ **Selection of Voltage or Current Excitation**
- ◆ **Ratiometric Operation**
- ◆ **Stability of Excitation Voltage or Current**
- ◆ **Bridge Sensitivity: FS Output / Excitation Voltage**
1mV / V to 10mV / V Typical
- ◆ **Fullscale Bridge Outputs: 10mV - 100mV Typical**
- ◆ **Precision, Low Noise Amplification / Conditioning**
Techniques Required
- ◆ **Linearization Techniques May Be Required**
- ◆ **Remote Sensors Present Challenges**

Figure 4-8: *A number of bridge considerations impact design choices*

Regardless of the absolute level, the stability of the excitation voltage or current directly affects the overall accuracy of the bridge output, as is evident from the V_B and I_B terms in the output expressions. Therefore stable references and/or *ratiometric* drive techniques are required, to maintain highest accuracy.

Here, ratiometric simply refers to the use of the bridge drive voltage of a voltage-driven bridge (or a current-proportional voltage, for a current-driven bridge) as the reference input to the ADC that digitizes the amplified bridge output voltage. In this manner the absolute accuracy and stability of the excitation voltage becomes a second order error. Examples to follow illustrate this point further.

Amplifying and Linearizing Bridge Outputs

The output of a single-element varying bridge may be amplified by a single precision op-amp connected as shown in Figure 4-9. Unfortunately this circuit, although attractive because of relative simplicity, has poor overall performance. Its gain predictability and accuracy are poor, and it unbalances the bridge due to loading from R_F and the op amp bias current. The R_F resistors must be carefully chosen and matched to maximize common mode rejection (CMR). Also, it is difficult to maximize the CMR while at the same time allowing different gain options. Gain is dependent upon the bridge resistances and R_F . In addition, the output is nonlinear, as the configuration does nothing to address the intrinsic bridge non-linearity. In summary, the circuit isn't recommended for precision use.

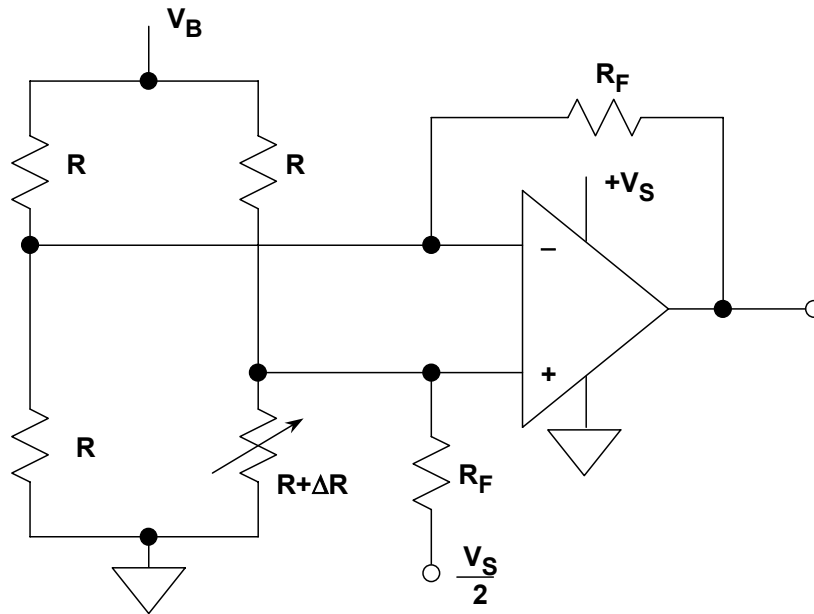


Figure 4-9: Using a single op amp as a bridge amplifier

However, a redeeming feature of this circuit is that it is capable of single supply operation, with a solitary op amp. Note that the R_F resistor connected to the non-inverting input is returned to $V_S/2$ (rather than ground) so that both positive and negative ΔR values can be accommodated, with the bipolar op amp output swing referenced to $V_S/2$.

■ OP AMP APPLICATIONS

A much better approach is to use an *instrumentation amplifier* (in-amp) for the required gain, as shown in Figure 4-10 below. This efficient circuit provides better gain accuracy, with the in-amp gain usually set with a single resistor, R_G . Since the amplifier provides dual, high-impedance loading to the bridge nodes, it does not unbalance or load the bridge. Using modern in-amp devices with gains ranging from 10-1000, excellent common mode rejection and gain accuracy can be achieved with this circuit.

However, due to the intrinsic characteristics of the bridge, the output is still nonlinear (see expression). As noted earlier, this can be corrected in software (assuming that the in-amp output is digitized using an analog-to-digital converter and followed by a microcontroller or microprocessor).

The in-amp can be operated on either dual supplies as shown, or alternately, on a single positive supply. In the figure, this corresponds to $-V_S = 0$. This is a key advantageous point, due the fact that all such bridge circuits bias the in-amp inputs at $V_B/2$, a voltage range typically compatible with amplifier bias requirements. In-amps such as the AD620 family, the AD623, and AD627 can be used in single (or dual) supply bridge applications, provided their restrictions on the gain and input and output voltage swings are observed.

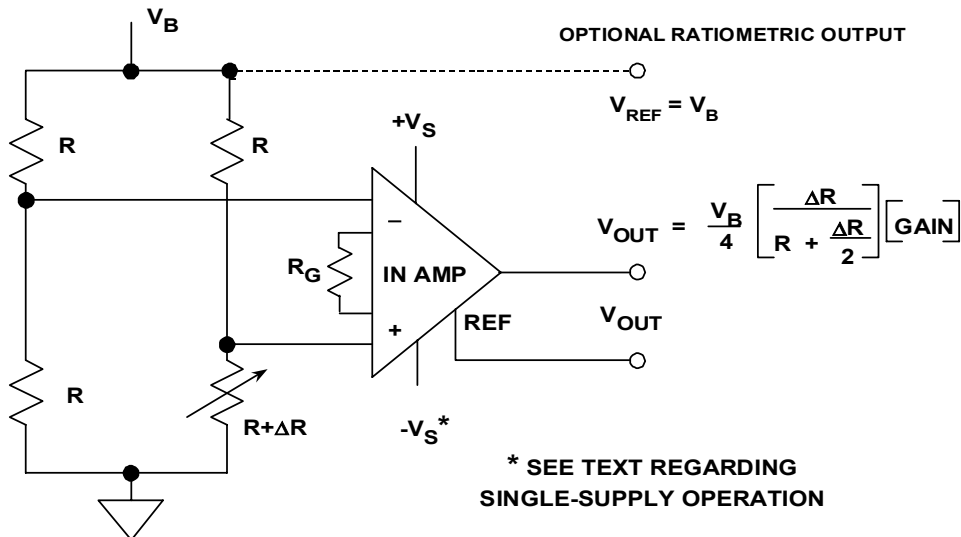


Figure 4-10: A generally preferred method of bridge amplification employs an instrumentation amplifier for stable gain and high CMR

The bridge in this example is voltage driven, by the voltage V_B . This voltage can optionally be used for an ADC reference voltage, in which case it also is an additional output, V_{REF} .

Various techniques are available to linearize bridge outputs, but it is important to distinguish between the linearity of the bridge equation (discussed earlier), and the sensor response linearity to the phenomenon being sensed. For example, if the active sensor element is an RTD, the bridge used to implement the measurement might have perfectly adequate linearity; *yet the output could still be nonlinear*, due to the RTD device's intrinsic nonlinearity. Manufacturers of sensors employing bridges address the

nonlinearity issue in a variety of ways, including keeping the resistive swings in the bridge small, shaping complementary nonlinear response into the active elements of the bridge, using resistive trims for first-order corrections, and others. In the examples which follow, what is being addressed is the linearity error of the bridge configuration itself (as opposed to a sensor element within the bridge).

Figure 4-11 shows a single-element varying active bridge circuit, in which an op amp produces a forced bridge null condition. For this single-element varying case, only the op amp feedback resistance varies, with the remaining three resistances fixed.

As used here, the op amp output provides a buffered, ground referenced, low impedance output for the bridge measurement, effectively suppressing the $V_B/2$ CM bridge component at the op amp inputs.

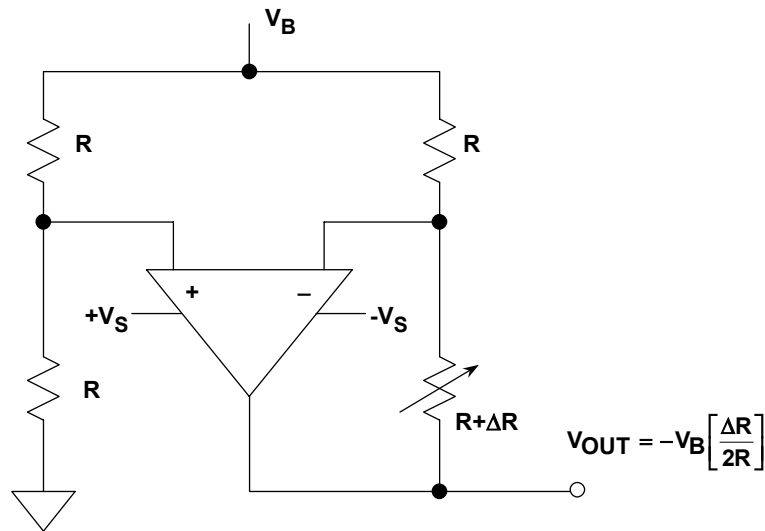


Figure 4-11: *Linearizing a single-element varying bridge (Method 1)*

The circuit works by adding a voltage in series with the variable resistance arm. This voltage is equal in magnitude and opposite in polarity to the incremental voltage across the varying element, and is linear with ΔR . As can be noted, the three constant “R” valued resistances and the op amp operate to drive a constant current in the variable resistance. This is the basic mechanism that produces the linearized output.

This active bridge has a sensitivity gain of two over the standard single-element varying bridge (Figure 4-6A, again). The key point is that the bridge’s incremental resistance/voltage output becomes linear, even for large values of ΔR . However, because of a still relatively small output signal, a second amplifier must usually follow this bridge. Note also that the op amp used in this circuit requires dual supplies, because its output must go negative for conditions where ΔR is positive.

■ OP AMP APPLICATIONS

Another circuit for linearizing a single-element varying bridge is shown in Figure 4-12. The top node of the bridge is excited by the voltage, V_B . The bottom of the bridge is driven in complementary fashion by the left op amp, which maintains a constant current of V_B/R in the varying resistance element, $R + \Delta R$.

Like the circuit of Figure 4-11, the constant current drive for the single-element variable resistance provides the mechanism for linearity improvement. Also, because of the fact that the bridge left-side center node is ground-referenced by the op amp, this configuration effectively suppresses CM voltages. This has the virtue of making the op amp selection somewhat less critical. Of course, performance parameters of high gain, low offset/noise, and high stability are all still needed.

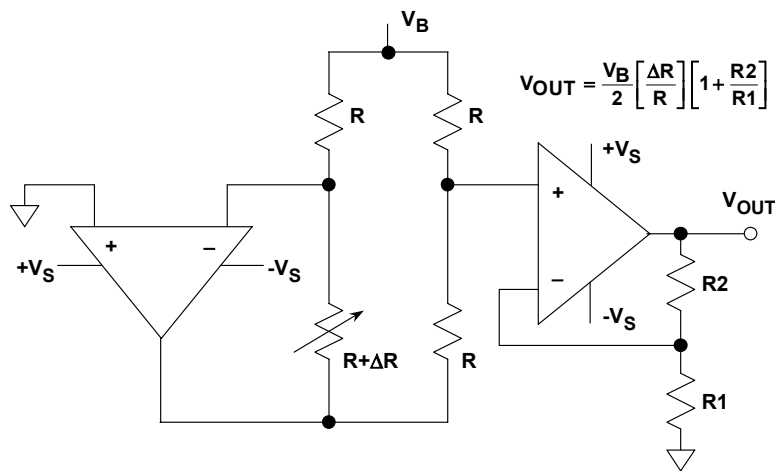


Figure 4-12: *Linearizing a single-element varying bridge (Method 2)*

The output signal is taken from the right-hand leg of the bridge, and is amplified by a second op amp, connected as a non-inverting gain stage. With the scaling freedom provided by the second op amp, the configuration is very flexible. The net output is linear, and has a bridge-output referred sensitivity comparable to the single-element varying circuit of Figure 4-11.

The Fig. 4-12 circuit requires two op amps operating on dual supplies. In addition, paired resistors R1-R2 must be ratio matched and stable types, for overall accurate and stable gain. The circuit can be a practical one using a dual precision op amp, such as an AD708, the OP2177 or the OP213.

A closely related circuit for linearizing a voltage-driven, *two-element* varying bridge can be adapted directly from the basic circuit of Figure 4-11. This form of the circuit, shown in Figure 4-13 (opposite), is identical to the previous single-element varying case, with the exception that the resistance between V_B and the op amp (+) input is now also variable (i.e., both diagonal $R + \Delta R$ resistances vary, in a like manner).

For the same applied voltage V_B , this form of the circuit has twice the sensitivity, which is evident in the output expressions. A dual supply op amp is again required, and additional gain may also be necessary.

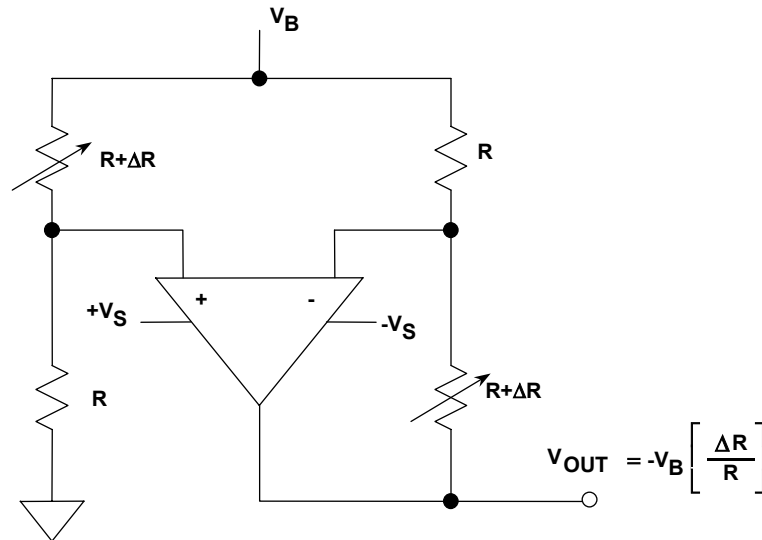


Figure 4-13: Linearizing a two-element varying voltage-driven bridge (Method 1)

The two-element varying bridge circuit shown in Figure 4-14 uses an op amp, a sense resistor, and a voltage reference, set up in a feedback loop containing the sensing bridge. The net effect of the loop is to maintain a constant current through the bridge of $I_B = V_{REF}/R_{SENSE}$. The current through each leg of the bridge remains constant ($I_B/2$) as the resistances change, therefore the output is a linear function of ΔR . An in-amp provides the additional gain.

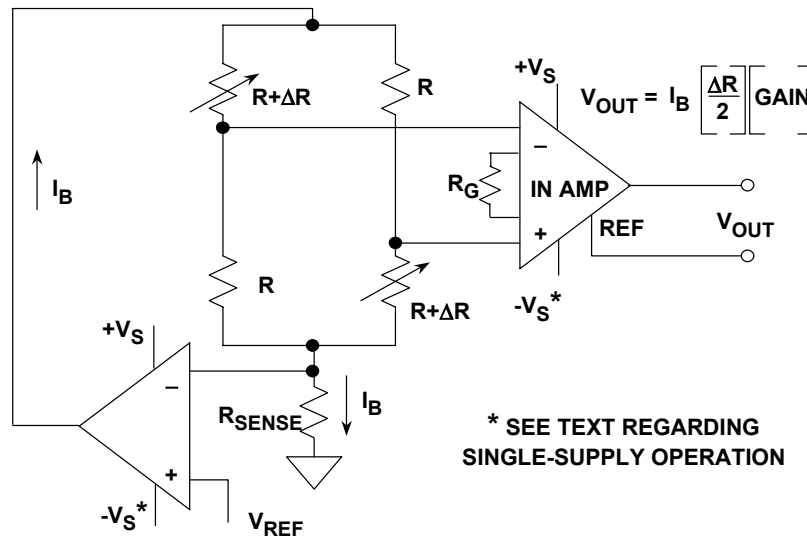


Figure 4-14: Linearizing a two-element varying current-driven bridge (Method 2)

This circuit can be operated on a single supply with the proper choice of amplifiers and signal levels. If ratiometric operation of an ADC is desired, the V_{REF} voltage can be used to drive the ADC.

Driving Remote Bridges

Wiring resistance and noise pickup are the biggest problems associated with remotely located bridges. Figure 4-15 shows a 350Ω strain gage, which is connected to the rest of the bridge circuit by 100 feet of 30 gage twisted pair copper wire. The resistance of the wire at 25°C is $0.105\Omega/\text{ft}$, or 10.5Ω for 100ft. The total lead resistance in series with the 350Ω strain gage is therefore 21Ω . The temperature coefficient of the copper wire is $0.385\%/^\circ\text{C}$. Now we will calculate the gain and offset error in the bridge output due to a $+10^\circ\text{C}$ temperature rise in the cable. These calculations are easy to make, because the bridge output voltage is simply the difference between the output of two voltage dividers, each driven from a $+10\text{V}$ source.

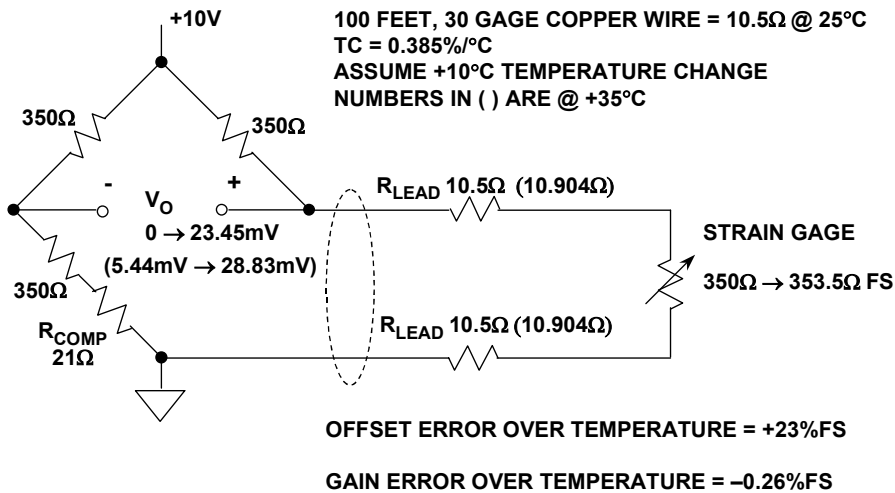


Figure 4-15: Wiring resistance related errors with remote bridge sensor

The fullscale variation of the strain gage resistance (with flex) above its nominal 350Ω value is $+1\%$ ($+3.5\Omega$), corresponding to a fullscale strain gage resistance of 353.5Ω which causes a bridge output voltage of $+23.45\text{mV}$. Notice that the addition of the 21Ω R_{COMP} resistor compensates for the wiring resistance and balances the bridge when the strain gage resistance is 350Ω . Without R_{COMP} , the bridge would have an output offset voltage of 145.63mV for a nominal strain gage resistance of 350Ω . This offset could be compensated for in software just as easily, but for this example, we chose to do it with R_{COMP} .

Assume that the cable temperature increases $+10^\circ\text{C}$ above nominal room temperature. This results in a total lead resistance increase of $+0.404\Omega$ ($10.5\Omega \times 0.00385/^\circ\text{C} \times 10^\circ\text{C}$) in each lead. *Note: The values in parentheses in the diagram indicate the values at $+35^\circ\text{C}$.* The total additional lead resistance (of the two leads) is $+0.808\Omega$. With no strain, this additional lead resistance produces an offset of $+5.44\text{mV}$ in the bridge output. Fullscale strain produces a bridge output of $+28.83\text{mV}$ (a change of $+23.39\text{mV}$ from no strain). Thus the increase in temperature produces an offset voltage error of $+5.44\text{mV}$ ($+23\%$ fullscale) and a gain error of -0.06mV ($23.39\text{mV} - 23.45\text{mV}$), or -0.26% fullscale. Note that these errors are produced solely by the 30 gage wire, and do not include any temperature coefficient errors in the strain gage itself.

SENSOR SIGNAL CONDITIONING BRIDGE CIRCUITS

The effects of wiring resistance on the bridge output can be minimized by the 3-wire connection shown in Figure 4-16. We assume that the bridge output voltage is measured by a high impedance device, therefore there is no current in the sense lead. Note that the sense lead measures the voltage output of a divider: the top half is the bridge resistor plus the lead resistance, and the bottom half is strain gage resistance plus the lead resistance. The nominal sense voltage is therefore independent of the lead resistance. When the strain gage resistance increases to fullscale (353.5Ω), the bridge output increases to $+24.15\text{mV}$.

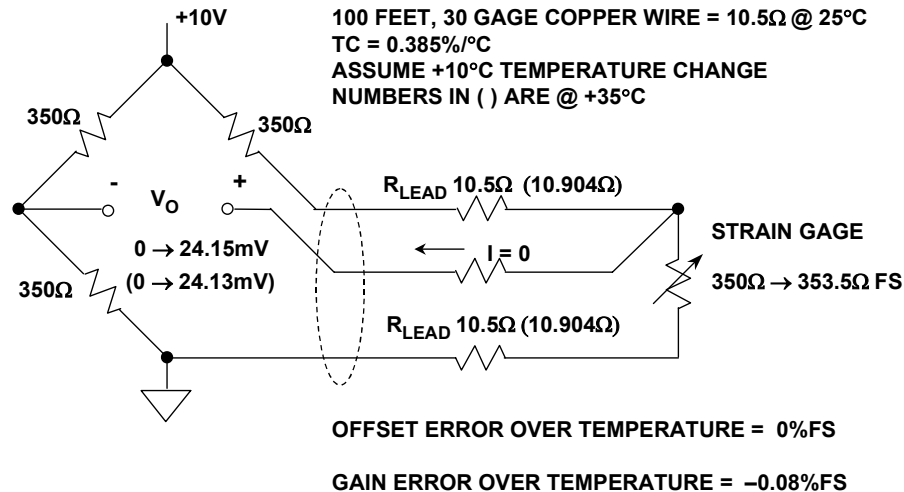


Figure 4-16: Remote bridge wiring resistance errors are reduced with 3-wire sensor connection

Increasing the temperature to $+35^\circ\text{C}$ increases the lead resistance by $+0.404\Omega$ in each half of the divider. The fullscale bridge output voltage decreases to $+24.13\text{mV}$ because of the small loss in sensitivity, but there is no offset error. The gain error due to the temperature increase of $+10^\circ\text{C}$ is therefore only -0.02mV , or -0.08% of fullscale. Compare this to the $+23\%$ fullscale offset error and the -0.26% gain error for the two-wire connection shown in Figure 4-14.

The three-wire method works well for remotely located resistive elements which make up one leg of a single-element varying bridge. However, all-element varying bridges generally are housed in a complete assembly, as in the case of a load cell. When these bridges are remotely located from the conditioning electronics, special techniques must be used to maintain accuracy.

Of particular concern is maintaining the accuracy and stability of the bridge excitation voltage. The bridge output is directly proportional to the excitation voltage, and any drift in the excitation voltage produces a corresponding drift in the output voltage.

■ OP AMP APPLICATIONS

For this reason, most all-element varying bridges (such as load cells) are six-lead assemblies: two leads for the bridge output, two leads for the bridge excitation, and two *sense* leads. To take full advantage of the additional accuracy that these extra leads allow, a method called Kelvin or 4-wire sensing is employed, as shown in Figure 4-17 below.

In this setup the drive voltage V_B is not applied directly to the bridge, but goes instead to the input of the upper precision op amp, which is connected in a feedback loop around the bridge (+) terminal. Although there may be a substantial voltage drop in the +FORCE lead resistance of the remote cable, the op amp will automatically correct for it, since it has a feedback path through the +SENSE lead. The net effect is that the upper node of the remote bridge is maintained at a precise level of V_B (within the capability of the op amp used, of course). A similar situation occurs with the bottom precision op amp, which drives the bridge (-) terminal to a ground level, as established by the op amp input ground reference. Again, the voltage drop in the -FORCE lead is relatively immaterial, because of the sensing at the -SENSE terminal.

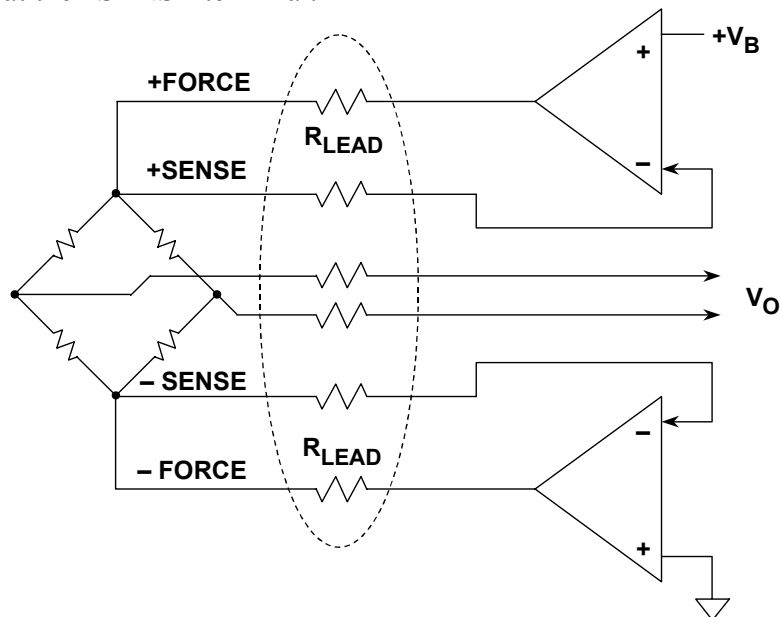


Figure 4-17: A Kelvin sensing system with a 6-wire voltage-driven bridge connection and precision op amps minimizes errors due to wire lead resistances

In both cases, the sense lines go to high impedance op amp inputs, thus there is minimal error due to the bias current induced voltage drop across their lead resistance. The op amps maintain the required excitation voltage at the remote bridge, to make the voltage measured between the (+) and (-) sense leads always equal to V_B .

Note— a subtle point is that the lower op amp will need to operate on dual supplies, since the drive to the -FORCE lead will cause the op amp output to go negative. Because of relatively high current in the bridge ($\sim 30\text{mA}$), current buffering stages at the op amp outputs are likely advisable for this circuit.

Although Kelvin sensing eliminates errors due to voltage drops in the bridge wiring resistance, the basic drive voltage V_B must still be highly stable since it directly affects the bridge output voltage. In addition, the op amps must have low offset, low drift, and low noise. Ratiometric operation can be optionally added, simply by using V_B to drive the ADC reference input.

The constant current excitation method shown in Figure 4-18 below is another method for minimizing the effects of wiring resistance on the measurement accuracy. This system drives a precise current I through the bridge, proportioned as per the expression in the figure. An advantage of the circuit in Fig. 4-18 is that it only uses one amplifier.

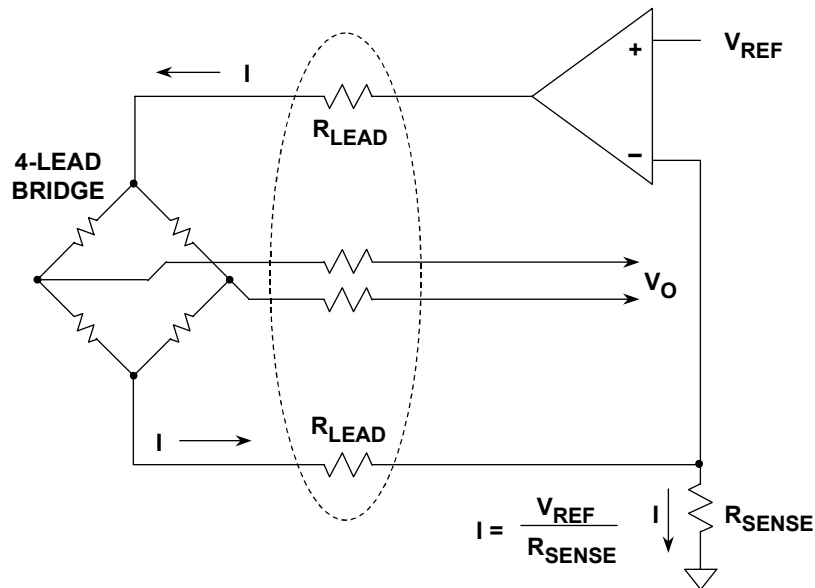


Figure 4-18: A 4-wire current-driven bridge scheme also minimizes errors due to wire lead resistances, plus allows simpler cabling

However, the accuracy of the reference, the sense resistor, and the op amp all influence the overall accuracy. While the precision required of the op amp should be obvious, one thing not necessarily obvious is that it may be required to deliver appreciable current, when I is more than a few mA (which it will be with standard 350Ω bridges). In such cases, current buffering of the op amp is again in order.

Therefore for highest precision with this circuit, a buffer stage is recommended. This can be as simple as a small transistor, since the bridge drive is unidirectional.

■ OP AMP APPLICATIONS

System offset minimization

Maintaining an accuracy of 0.1% or better with a fullscale bridge output voltage of 20mV requires that the sum of all offset errors be less than 20 μ V. Parasitic thermocouples are cases in point, and if not given due attention, they can cause serious temperature drift errors. All dissimilar metal-metal connections generate voltages between a few and tens of microvolts for a 1°C temperature differential, are basic thermocouple fact-of-life.

Fortunately however, within a bridge measurement system the signal connections are differential, therefore this factor can be used to minimize the impact of parasitic thermocouples.

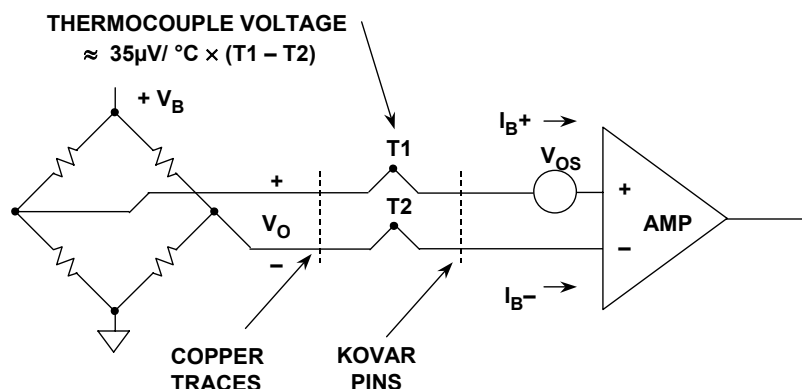


Figure 4-19: Typical sources of offset voltage within bridge measurement systems

Figure 4-19 above shows some typical sources of offset error that are inevitable in a system. Within a differential signal path, only those thermocouple pairs whose junctions are actually at different temperatures will degrade the signal. The diagram shows a typical parasitic junction formed between the copper printed circuit board traces and the kovar pins of an IC amplifier.

This thermocouple voltage is about 35 μ V/°C temperature differential. Note that this package-PC trace thermocouple voltage is significantly less when using a plastic package with a copper lead frame (recommended). Regardless of what package is used, all metal-metal connections along the signal path should be designed so that minimal temperature differences occur between the sides.

The amplifier offset voltage and bias currents are further sources of offset error. The amplifier bias current must flow through the source impedance. Any unbalance in either the source resistances or the bias currents produce offset errors. In addition, the offset voltage and bias currents are a function of temperature.

High performance low offset, low offset drift, low bias current, and low noise precision amplifiers such as the AD707, the OP177 or OP1177 are required. In some cases, chopper-stabilized amplifiers such as the AD8551/AD8552/AD8554 may be a solution.

AC bridge excitation such as that shown in Figure 4-20 below can effectively remove offset voltage effects in series with a bridge output, V_O .

The concept is simple, and can be described as follows. The net bridge output voltage is measured under the two phased-sequence conditions, as shown. A first measurement (top) drives the bridge at the top node with excitation voltage V_B . This yields a first-phase measurement output V_A , where V_A is the sum of the desired bridge output voltage V_O and the net offset error voltage E_{OS} .

In the second measurement (bottom) the polarity of the bridge excitation is then reversed, and a second measurement, V_B , is made. Subtracting V_B from V_A yields $2V_O$, and the offset error term E_{OS} cancels as noted from the mathematical expression in the figure.

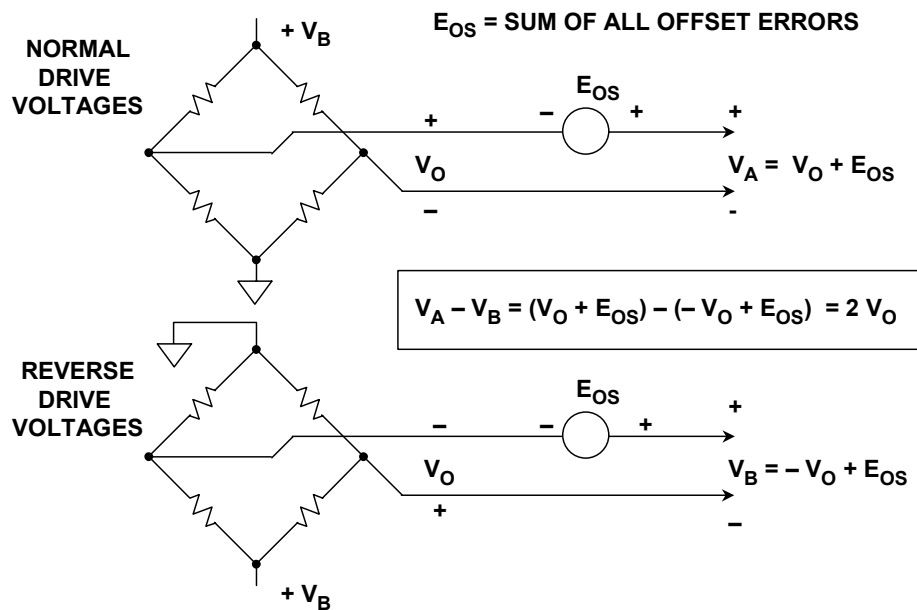


Figure 4-20: AC bridge excitation minimizes system offset voltages

Obviously, a full implementation of this technique requires a highly accurate measurement ADC such as the AD7730 (see Reference 5) as well as a microcontroller to perform the subtraction.

Note that if a ratiometric reference is desired, the ADC must also accommodate the changing polarity of the reference voltage, as well as sense the magnitude. Again, the AD7730 includes this capability.

OP AMP APPLICATIONS

A very powerful combination of bridge circuit techniques is shown in Figure 4-21 below, an example of a high performance ADC. In Fig. 4-21A is shown a basic DC operated ratiometric technique, combined with Kelvin sensing to minimize errors due to wiring resistance, which eliminates the need for an accurate excitation voltage.

The AD7730 measurement ADC can be driven from a single supply voltage of 5V, which in this case is also used to excite the remote bridge. Both the analog input and the reference input to the ADC are high impedance and fully differential. By using the + and – SENSE outputs from the bridge as the differential reference voltage to the ADC, there is no loss in measurement accuracy if the actual bridge excitation voltage varies.

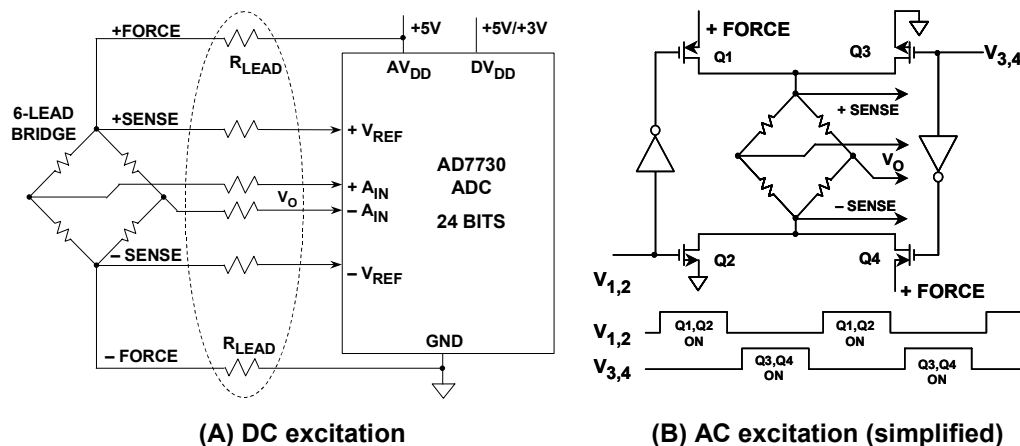


Figure 4-21: Ratiometric DC or AC operation with Kelvin sensing can be implemented using the AD7730 ADC

To implement AC bridge operation of the AD7730, an "H" bridge driver of P-Channel and N-Channel MOSFETs can be configured as shown in Fig. 4-21B (note — dedicated bridge driver chips are available, such as the Micrel MIC4427). This scheme, added to the basic functionality of the AD7730 configuration of 4-21A greatly increases the utility of the offset canceling circuit, as generally outlined in the preceding discussion of Fig. 4-20.

Because of the on-resistance of the H-bridge MOSFETs, Kelvin sensing must also be used in these AC bridge applications. It is also important that the drive signals be non-overlapping, as noted, to prevent excessive MOSFET switching currents. The AD7730 ADC has on-chip circuitry which generates the required non-overlapping drive signals to implement this AC bridge excitation. All that needs adding is the switching bridge as noted in Fig. 4-21B.

The AD7730 is one of a family of sigma-delta ADCs with high resolution (24 bits) and internal programmable gain amplifiers (PGAs) and is ideally suited for bridge applications. These ADCs have self- and system calibration features, which allow offset and gain errors due to the ADC to be minimized. For instance, the AD7730 has an offset drift of 5nV/°C and a gain drift of 2ppm/°C. Offset and gain errors can be reduced to a few microvolts using the system calibration feature.

REFERENCES: BRIDGE CIRCUITS

1. Ramon Pallas-Areny and John G. Webster, **Sensors and Signal Conditioning**, John Wiley, New York, 1991.
2. Dan Sheingold, Editor, **Transducer Interfacing Handbook**, Analog Devices, Inc., 1980, ISBN: 0-916550-05-2.
3. Sections 2, 3, Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, 1992, ISBN: 0-916550-10-9.
4. Sections 1, 6, Walt Kester, Editor, **System Applications Guide**, Analog Devices, 1993, ISBN: 0-916550-13-3.
5. Data sheet for **AD7730 Bridge Transducer ADC**, <http://www.analog.com>

NOTES:

SECTION 4-3: STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

Walt Kester

Strain Gages

The most popular electrical elements used in force measurements include the resistance strain gage, the semiconductor strain gage, and piezoelectric transducers. The strain gage measures force indirectly by measuring the deflection it produces in a calibrated carrier. Pressure can be converted into a force using an appropriate transducer, and strain gage techniques can then be used to measure pressure. Flow rates can be measured using differential pressure measurements, which also make use of strain gage technology. These principles are summarized in Figure 4-22 below.

- ◆ **Strain:** **Strain Gage, PiezoElectric Transducers**
- ◆ **Force:** **Load Cell**
- ◆ **Pressure:** **Diaphragm to Force to Strain Gage**
- ◆ **Flow:** **Differential Pressure Techniques**

Figure 4.22 *Strain gages are directly or indirectly the basis for a variety of physical measurements*

The resistance-based strain gage uses a resistive element which changes in length, hence resistance, as the force applied to the base on which it is mounted causes stretching or compression. It is perhaps the most well known transducer for converting force into an electrical variable.

An *unbonded* strain gage consists of a wire stretched between two points. Force acting upon the wire (area = A, length = L, resistivity = ρ) will cause the wire to elongate or shorten, which will cause the resistance to increase or decrease proportionally according to:

$$R = \rho L/A \quad \text{Eq. 4-1}$$

and,

$$\Delta R/R = GF \cdot \Delta L/L \quad \text{Eq. 4-2}$$

where GF = Gage factor (2.0 to 4.5 for metals, and more than 150 for semiconductors).

In this expression, the dimensionless quantity $\Delta L/L$ is a measure of the force applied to the wire and is expressed in *microstrains* ($1\mu\epsilon = 10^{-6}$ cm/cm) which is the same as parts-per-million (ppm).

OP AMP APPLICATIONS

From equation 4-2, note that larger gage factors result in proportionally larger resistance changes, hence this implies greater strain gage sensitivity. These concepts are summarized in the drawing of Figure 4-23 below.

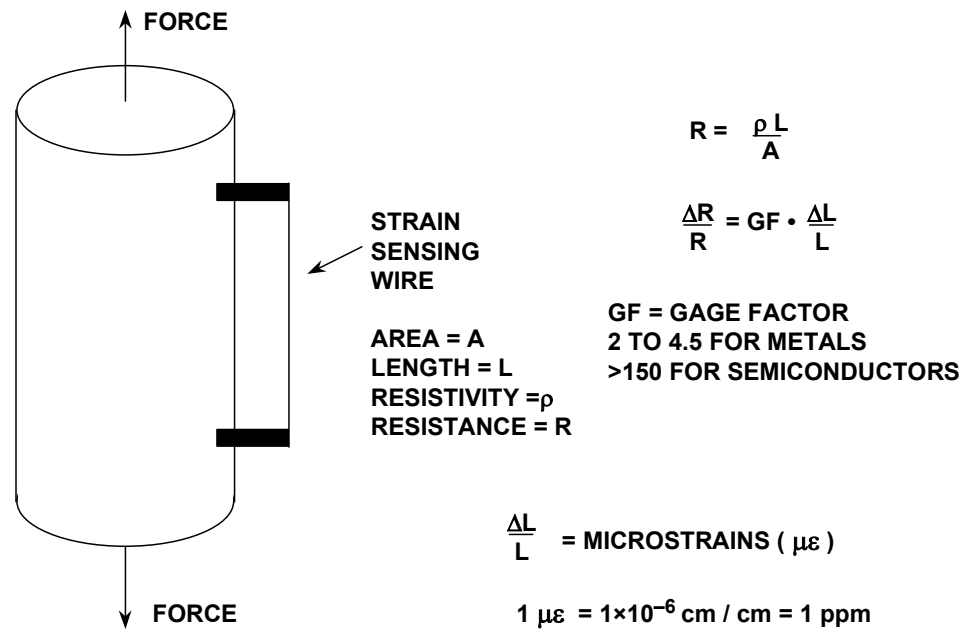


Figure 4-23: Operating principles of a basic unbonded strain gage

A *bonded* strain gage consists of a thin wire or conducting film arranged in a coplanar pattern and cemented to a base or carrier. The basic form of this type of gage is shown in Figure 4-24 below.

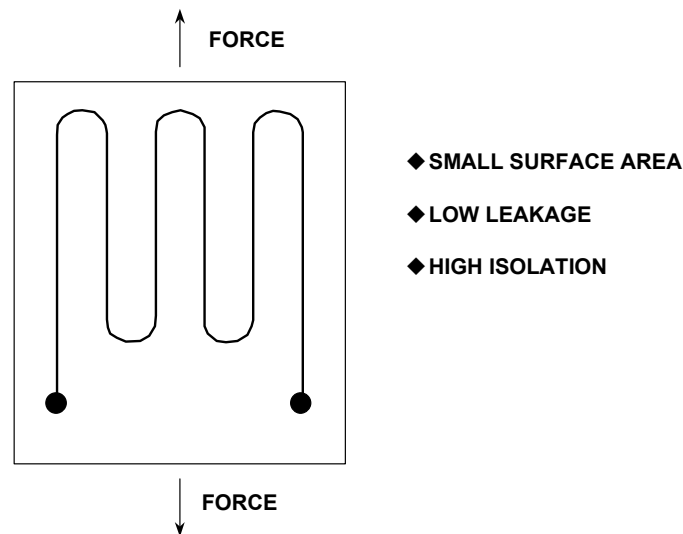


Figure 4-24: A bonded wire strain gage

This strain gage is normally mounted so that as much as possible of the length of the conductor is aligned in the direction of the stress that is being measured, i.e., longitudinally. Lead wires are attached to the base and brought out for interconnection. Bonded devices are considerably more practical and are in much wider use than are the aforementioned unbonded devices.

SENSOR SIGNAL CONDITIONING

STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

Perhaps the most popular version is the *foil-type gage*, produced by photo-etching techniques, and using similar metals to the wire types. Typical alloys are of copper-nickel (Constantan), nickel-chromium (Nichrome), nickel-iron, platinum-tungsten, etc. This strain gage type is shown in Figure 4-25 below.

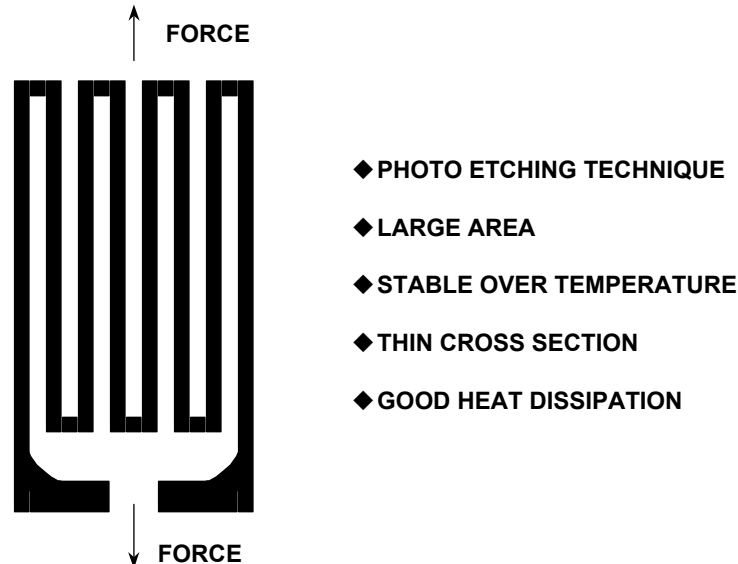


Figure 4-25: A metal foil strain gage

Gages having *wire sensing elements* present a small surface area to the specimen; this reduces leakage currents at high temperatures and permits higher isolation potentials between the sensing element and the specimen. Foil sensing elements, on the other hand, have a large ratio of surface area to cross-sectional area and are more stable under extremes of temperature and prolonged loading. The large surface area and thin cross section also permit the device to follow the specimen temperature and facilitate the dissipation of self-induced heat.

Semiconductor strain gages

Semiconductor strain gages make use of the piezoresistive effect in certain semiconductor materials such as silicon and germanium in order to obtain greater sensitivity and higher-level output.

Semiconductor gages can be produced to have either positive or negative changes when strained. They can be made physically small while still maintaining a high nominal resistance.

Semiconductor strain gage bridges may have 30 times the sensitivity of bridges employing metal films, but are temperature sensitive and difficult to compensate. Their change in resistance with strain is also nonlinear. They are not in as widespread use as the more stable metal-film devices for precision work; however, where sensitivity is important and temperature variations are small, they may have some advantage.

■ OP AMP APPLICATIONS

Instrumentation is similar to that for metal-film bridges but is less critical because of the higher signal levels and decreased transducer accuracy. Figure 4-26 below summarizes the relative performance of metal and semiconductor strain gages.

PARAMETER	METAL STRAIN GAGE	SEMICONDUCTOR STRAIN GAGE
Measurement Range	0.1 to 40,000 $\mu\epsilon$	0.001 to 3000 $\mu\epsilon$
Gage Factor	2.0 to 4.5	50 to 200
Resistance, Ω	120, 350, 600, ..., 5000	1000 to 5000
Resistance Tolerance	0.1% to 0.2%	1% to 2%
Size, mm	0.4 to 150 Standard: 3 to 6	1 to 5

Figure 4-26: A comparison of metal and semiconductor type strain gages

Piezoelectric force transducers are employed where the forces to be measured are dynamic (i.e., continually changing over the period of interest - usually of the order of milliseconds). These devices utilize the effect that changes in charge are produced in certain materials when they are subjected to physical stress. In fact, piezoelectric transducers are *displacement* transducers with quite large charge outputs for very small displacements, but they are invariably used as force transducers on the assumption that in an elastic material, displacement is proportional to force. Piezoelectric devices produce substantial output voltage in instruments such as accelerometers for vibration studies. Piezoelectric sensor output conditioning is discussed within Section 4-4 of this chapter.

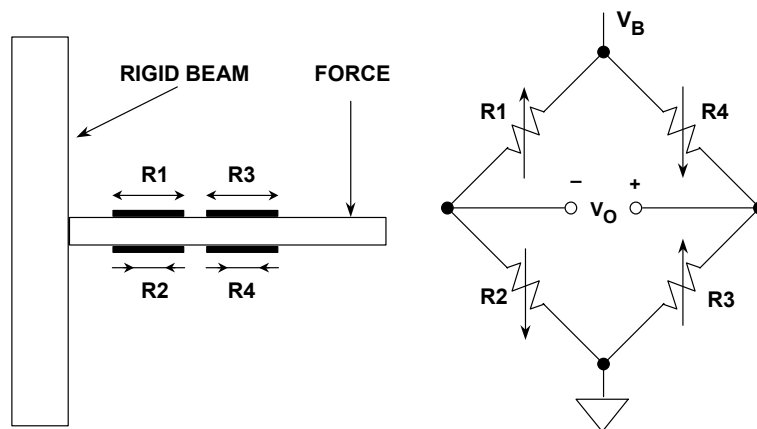


Figure 4-27: A beam force sensor using a strain gage bridge

Strain gages can be used to measure force, as shown in Figure 4-27 above, where a cantilever beam is slightly deflected by the applied force. Four strain gages are used to measure the flex of the beam, two on the top, and two on the bottom. The gages are connected in a four-element bridge configuration. Recall from Section 4-2 that this configuration gives maximum sensitivity and is inherently linear. This configuration also offers first-order correction for temperature drift in the individual strain gages.

SENSOR SIGNAL CONDITIONING

STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

Strain gages are low-impedance devices, consequently they require significant excitation power to obtain reasonable levels of output voltage. A typical strain-gage based load cell bridge will have a 350Ω impedance and is specified as having a sensitivity in a range 3-10 millivolts full scale, per volt of excitation.

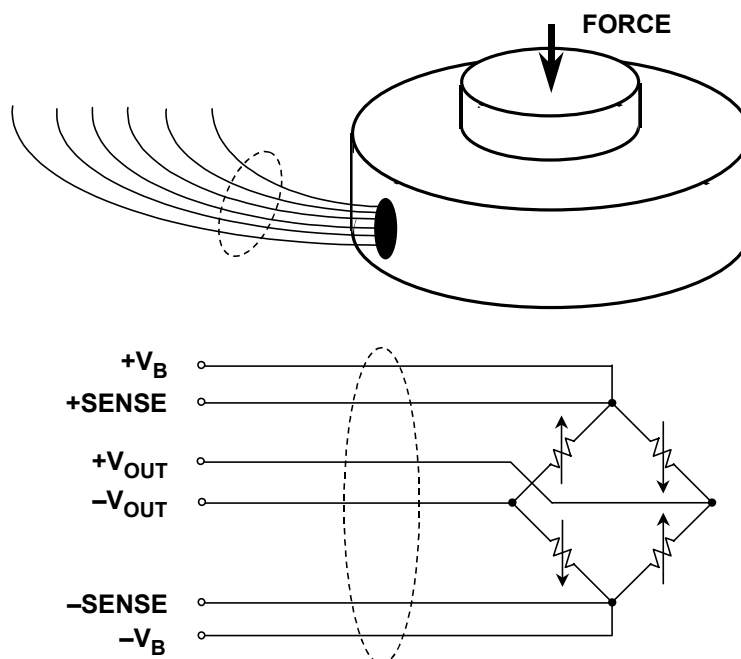


Figure 4-28: A load cell comprised of 4 strain gages is shown in physical (top) and electrical (bottom) representations

The load cell is composed of four individual strain gages arranged as a bridge, as shown in Figure 4-28 above. For a 10V bridge excitation voltage with a rating of 3mV/V , 30 millivolts of signal will be available at full scale loading.

While increasing the drive to the bridge can increase the output, self-heating effects are a significant limitation to this approach— they can cause erroneous readings, or even device destruction. One technique for evading this limitation is to use a low duty cycle pulsed drive signal for the excitation.

Many load cells have the $\pm$ "SENSE" connections as shown, to allow the signal-conditioning electronics to compensate for DC drops in the wires (Kelvin sensing as discussed in section 4-2). This brings the wires to a total of 6 for the fully instrumented bridge. Some load cells may also have additional internal resistors, for temperature compensation purposes.

▣ OP AMP APPLICATIONS

Pressures in liquids and gases are measured electrically by a variety of pressure transducers. A number of mechanical converters (including diaphragms, capsules, bellows, manometer tubes, and Bourdon tubes) are used to measure pressure by measuring an associated length, distance, or displacement, and to measure pressure changes by the motion produced, as shown by Figure 4-29 below.

The output of this mechanical interface is then applied to an electrical converter such as a strain gage, or piezoelectric transducer. Unlike strain gages, piezoelectric pressure transducers are typically used for high-frequency pressure measurements (such as sonar applications, or crystal microphones).

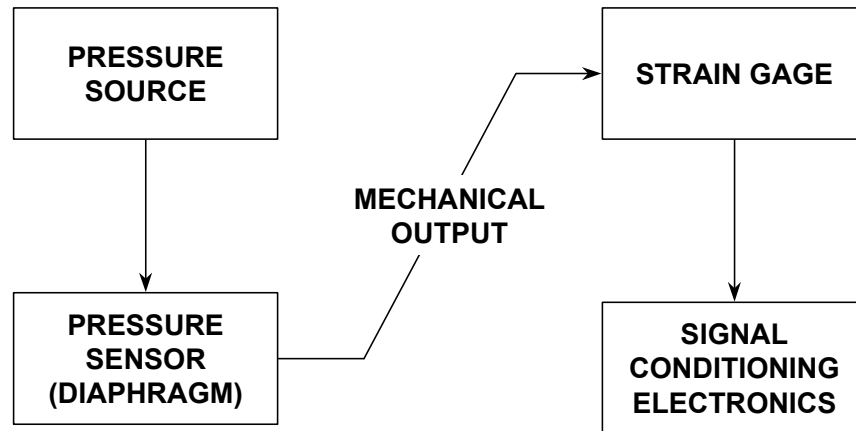


Figure 4-29: *Pressure sensors use strain gages for indirect pressure measurement*

There are many ways of defining flow (mass flow, volume flow, laminar flow, turbulent flow). Usually the *amount* of a substance flowing (mass flow) is the most important, and if the fluid's density is constant, a volume flow measurement is a useful substitute that is generally easier to perform. One commonly used class of transducers, which measures flow rate indirectly, involves the measurement of pressure.

Flow can be derived by taking the differential pressure across two points in a flowing medium - one at a static point and one in the flow stream. *Pitot tubes* are one form of device used to perform this function, where flow rate is obtained by measuring the differential pressure with standard pressure transducers.

Differential pressure can also be used to measure flow rate using the *venturi* effect by placing a restriction in the flow. Although there are a wide variety of physical parameters being sensed, the electronics interface is very often strain gage based.

STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

The remaining discussions of this section deal with applications that apply the bridge and strain gage concepts discussed thus far in general terms.

An example of an all-element varying bridge circuit is a fatigue monitoring strain sensing circuit, as shown in Figure 4-30 below. The full bridge is an integrated unit, which can be attached to the surface on which the strain or flex is to be measured. In order to facilitate remote sensing, current mode bridge drive is used. The remotely located bridge is connected to the conditioning electronics through a 4-wire shielded cable. The OP177 precision op amp servos the bridge current to 10mA, being driven from an AD589 reference voltage of 1.235V. Current buffering of the op amp is employed in the form of the PNP transistor, for lowest op amp self-heating, and highest gain linearity.

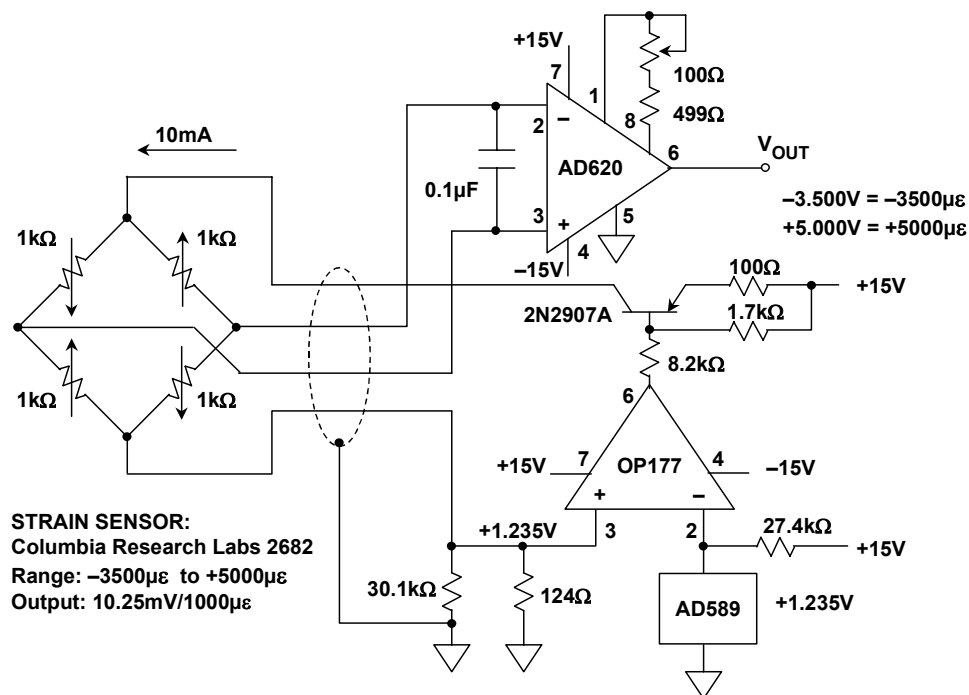


Figure 4-30: A precision strain gage sensor amplifier using a remote current-driven $1k\Omega$ bridge, a buffered precision op amp driver, and a precision in-amp 100X gain stage

The strain gauge produces an output of 10.25mV/1000 $\mu\epsilon$. The signal is amplified by the AD620 in-amp, which is configured for a gain of 100 times, via an effective R_G of 500 Ω . Full-scale voltage calibration is set by adjusting the 100 Ω gain potentiometer such that, for a sensor strain of $-3500\mu\epsilon$, the output reads $-3.500V$; and for a strain of $+5000\mu\epsilon$, the output registers $+5.000V$. The measurement may then be digitized with an ADC which has a 10V fullscale input range.

The 0.1 μ F capacitor across the AD620 input pins serves as an EMI/RFI filter in conjunction with the bridge resistance of 1k Ω . The corner frequency of this filter is approximately 1.6kHz.

■ OP AMP APPLICATIONS

Another example is a load cell amplifier circuit, shown in Figure 4-31. This circuit is more typical of a bridge workhorse application. It interfaces with a typical 350Ω load cell, and can be configured to accommodate typical bridge sensitivities over a range of 3-10mV/V.

A 10.000V bridge excitation is derived from an AD588 10V reference, with an OP177 and 2N2219A used as a buffer. The 2N2219A is within the OP177 feedback loop and supplies the necessary bridge drive current (28.57mA). This insures that the op amp performance will not be compromised. The Kelvin sensing scheme used at the bridge provides for low errors due to wiring resistances, and a precision zener diode reference, the AD588, provides lowest excitation drift and scaling with temperature changes.

To ensure highest linearity is preserved, a low drift instrumentation amplifier is used as the gain stage. This design has a minimum number of critical resistors and amplifiers, making the entire implementation accurate, stable, and cost effective. In addition to low excitation voltage TC, another stability requirement is minimum in-amp gain TC. Both factors are critical towards insuring stable circuit scaling over temperature.

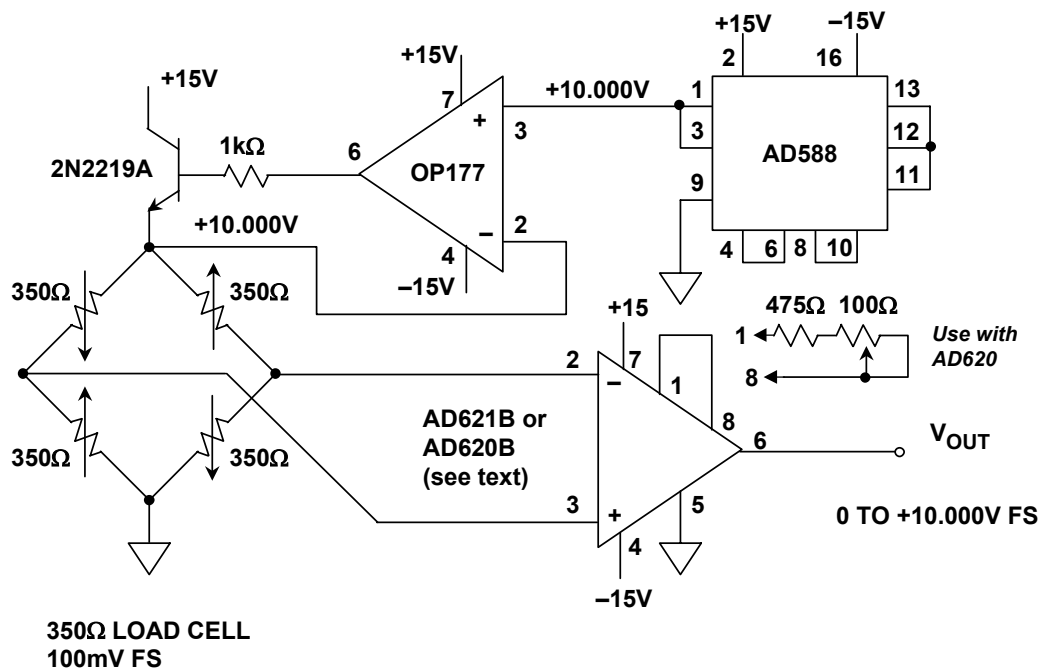


Figure 4-31: A precision 350Ω load cell amplifier, using a buffered voltage-driven configuration with Kelvin sensing and a precision in-amp

With the use of the AD621B in-amp as shown, the scaling is for a precise gain of 100 (as set by the pin 1-8 jumper), for lowest in-amp gain TC. The AD621B is specified for a very low gain TC, only 5ppm/°C. The gain of 100 translates a 100mV fullscale bridge output to a nominal 10V output. Alternately, an AD620B could also be used, with the optional gain network consisting of the fixed 475Ω resistor, and 100Ω potentiometer for gain adjustment. This will provide a 50ppm/°C gain TC for the in-amp, plus the TC of the external parts (which should have low temperature coefficients).

SENSOR SIGNAL CONDITIONING STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

While the lowest TC is provided by the fixed gain AD621 setup, it doesn't allow direct control of overall scaling. To retain the very lowest TC, scaling could be accomplished via a software auto-calibration routine. Alternately, the AD588 and OP177 reference/op amp stage could be configured for a variable excitation voltage (as opposed to a fixed 10.000V as shown). Variable gain in the reference voltage driver will effectively alter the excitation voltage as seen by the bridge, and thus provide flexible overall system scaling. Of course, it is imperative that such a scheme be implemented with low TC resistances.

As shown previously, a precision load cell is usually configured as a 350Ω bridge. Figure 4-32 shows a precision load cell amplifier, within a circuit possessing the advantage of being powered from just a single power supply.

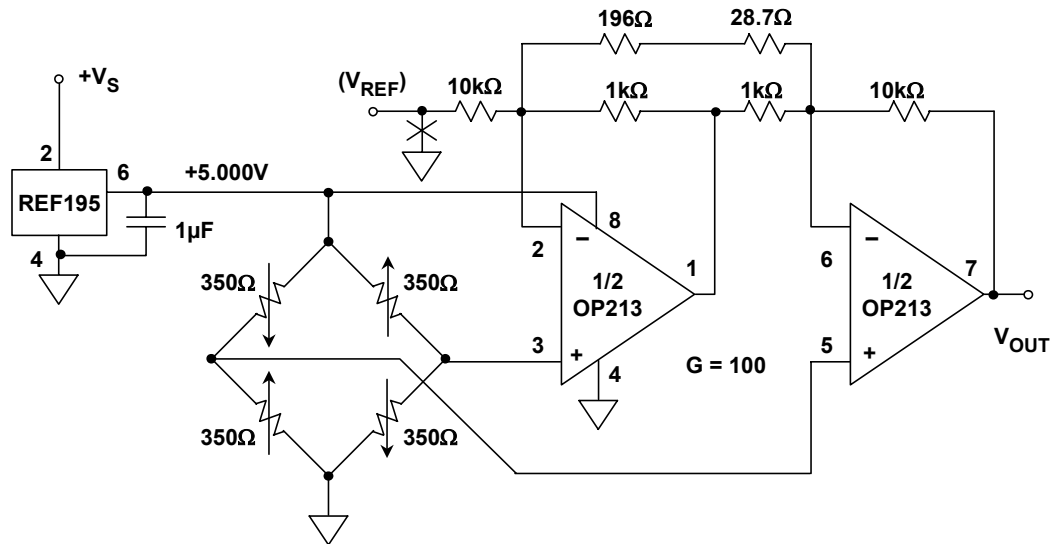


Figure 4-32: a single-supply load cell amplifier

As noted previously, the bridge excitation voltage must be both precise and stable, otherwise it can introduce measurement errors. In this circuit, a precision REF195 5V reference is used as the bridge drive, allowing a TC as low as 5ppm/°C. The REF195 reference can also supply more than 30mA to a load, so it can drive a 350Ω bridge (~14mA) without need of a buffer. The dual OP213 is configured as a gain-of-100, two op amp in-amp. The resistor network sets the gain according to the formula:

$$G = 1 + \frac{10\text{k}\Omega}{1\text{k}\Omega} + \frac{20\text{k}\Omega}{196\Omega + 28.7\Omega} = 100 \quad \text{Eq. 4-3}$$

For optimum CMR, the 10kΩ/1kΩ resistor ratio matching should be precise. Close tolerance resistors (±0.5% or better) should be used, and all resistors should be of the same type.

For a zero volt bridge output signal, the amplifier will swing to within 2.5mV of 0V. This is the minimum output limit of the OP213. Therefore, if an offset adjustment is required, the adjustment should start from a positive voltage at V_{REF} and adjust V_{REF} downward

■ OP AMP APPLICATIONS

until the output (V_{OUT}) stops changing. This is the point where the amplifier limits the swing. Because of the single supply design, the amplifier cannot sense input signals which have negative polarity.

If linearity around or at zero volts input is required, or if negative polarity signals must be processed, the V_{REF} connection can be connected to a stable voltage which is mid-supply (i.e., 2.5V) rather than ground. Note that when V_{REF} is not at ground, the output must be referenced to V_{REF} . An advantage of this type of referencing is that the output is now bipolar, with respect to V_{REF} .

The AD7730 24-bit sigma-delta ADC is ideal for direct conditioning of bridge outputs, and requires no interface circuitry (see Reference 10). A simplified connection diagram was shown in Figure 4.21A (again). The entire circuit operates on a single +5V supply, which also serves as the bridge excitation voltage. Note that the measurement is ratiometric, because the sensed bridge excitation voltage is also used as the ADC reference. Variations in the +5V supply do not affect the accuracy of the measurement.

The AD7730 has an internal programmable gain amplifier which allows a fullscale bridge output of $\pm 10\text{mV}$ to be digitized to 16-bit accuracy. The AD7730 has self and system calibration features which allow offset and gain errors to be minimized with periodic recalibrations.

A "chop" or AC mode option minimizes the offset voltage and drift and operates similarly to a chopper-stabilized amplifier. The effective input voltage noise RTI is approximately 40nV rms, or 264nV peak-to-peak. This corresponds to a resolution of 13 ppm, or approximately 16.5-bits. Gain linearity is also approximately 16-bits.

REFERENCES: STRAIN, FORCE, PRESSURE AND FLOW MEASUREMENTS

1. Ramon Pallas-Areny and John G. Webster, **Sensors and Signal Conditioning**, John Wiley, New York, 1991.
2. Dan Sheingold, Editor, **Transducer Interfacing Handbook**, Analog Devices, Inc., 1980, ISBN: 0-916550-05-2.
3. Sections 2, 3, Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, 1992, ISBN: 0-916550-10-9.
4. Sections 1, 6, Walt Kester, Editor, **System Applications Guide**, Analog Devices, 1993, ISBN: 0-916550-13-3.
5. Harry L. Trietley, **Transducers in Mechanical and Electronic Design**, Marcel Dekker, Inc., 1986.
6. Jacob Fraden, **Handbook of Modern Sensors**, 2nd Ed., Springer-Verlag, New York, NY, 1996.
7. **The Pressure, Strain, and Force Handbook, Vol. 29**, Omega Engineering, One Omega Drive, P.O. Box 4047, Stamford CT, 06907-0047, 1995. <http://www.omega.com>
8. **The Flow and Level Handbook, Vol. 29**, Omega Engineering, One Omega Drive, P.O. Box 4047, Stamford CT, 06907-0047, 1995. (<http://www.omega.com>)
9. Ernest O. Doebelin, **Measurement Systems Applications and Design**, 4th Ed., McGraw-Hill, 1990.
10. Data sheet for AD7730 Bridge Transducer ADC, <http://www.analog.com>

NOTES:

SECTION 4-4: HIGH IMPEDANCE SENSORS

Walt Kester, Scott Wurcer, Chuck Kitchin

Many popular sensors have output impedances greater than several megohms, and thus the associated signal-conditioning circuitry must be carefully designed to meet the challenges of low bias current, low noise, and high gain. Figure 4-33 below lists a few examples of high impedance sensors.

A large portion of this section is devoted to a germane example, the analysis of a photodiode preamplifier. This application demonstrates many of the problems associated with high impedance sensor signal-conditioning circuits, and offers a host of practical solutions that can be applied to practically all such sensors.

- ◆ **Photodiode Preamplifiers**
- ◆ **Piezoelectric Sensors**
- ◆ **Humidity Monitors**
- ◆ **pH Monitors**
- ◆ **Chemical Sensors**
- ◆ **Smoke Detectors**

Figure 4-33: High impedance sensors

Other examples of high impedance sensors to be discussed are piezoelectric sensors and charge-output sensors.

Photodiode Preamplifier Design

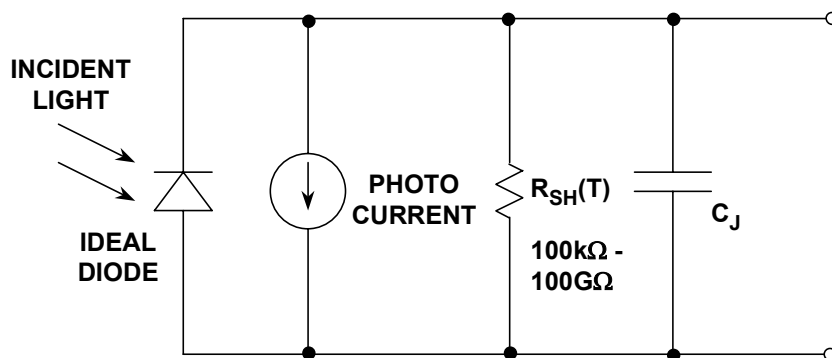
Photodiodes generate a small current that is proportional to the level of illumination. Their applications range from relatively low speed, wide dynamic range circuits to much higher speed circuits. Examples of the types of applications are precision light meters and high-speed fiber optic receivers.

One of the standard methods for specifying photodiode sensitivity is to state its short-circuit photocurrent (I_{SC}) for a given light level from a well-defined light source. The most commonly used source is an incandescent tungsten lamp running at a color temperature of 2850K.

At 100fc (foot-candles) of illumination (approximately the light level on an overcast day), the short-circuit current usually falls in a range of picoamps to hundreds of microamps for small area (less than 1mm^2) diodes.

OP AMP APPLICATIONS

The equivalent circuit for a photodiode is shown in Figure 4-34 below. The short-circuit current is very linear over 6 to 9 decades of light intensity, and is therefore often used as a measure of absolute light levels. The open-circuit forward voltage drop across the photodiode varies logarithmically with light level, but, because of its large temperature coefficient, the diode voltage is seldom used as an accurate measure of light intensity.



NOTE: R_{SH} HALVES EVERY 10°C TEMPERATURE RISE

Figure 4-34: A photodiode equivalent circuit

The shunt resistance R_{SH} is usually on the order of 1000MΩ at room temperature, and decreases by a factor of 2 for every 10°C rise in temperature. Diode capacitance C_J is a function of junction area and the diode bias voltage. A value of 50pF at zero bias is typical for small-area diodes.

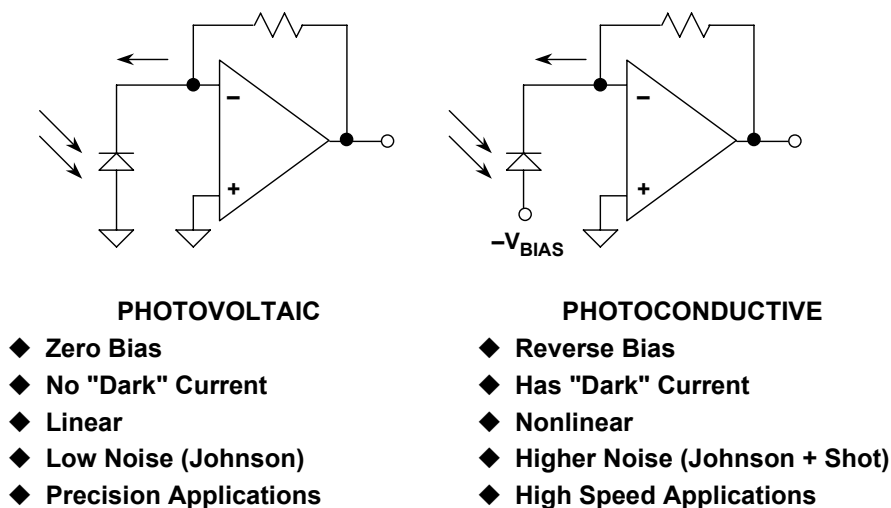


Figure 4-35: Photodiode operating modes

Photodiodes may be operated in either of two basic modes, as shown in Figure 4-35, above. These modes are with zero bias voltage (*photovoltaic* mode, left) or with a reverse-bias voltage (*photoconductive* mode, right).

The most precise linear operation is obtained in the photovoltaic mode, while higher switching speeds are realizable when the diode is operated in the photoconductive mode at the expense of linearity. Under these reverse-bias conditions, a small amount of current called *dark current* will flow— even when there is no illumination.

There is no dark current in the photovoltaic mode. In the photovoltaic mode, the diode noise is basically the thermal noise generated by the shunt resistance. In the photoconductive mode, shot noise due to conduction is an additional source of noise. Photodiodes are usually optimized during the design process for use in either the photovoltaic mode or the photoconductive mode, but not both.

ENVIRONMENT	ILLUMINATION (fc)	SHORT CIRCUIT CURRENT
Direct Sunlight	1000	30 μ A
Overcast Day	100	3 μ A
Twilight	1	0.03 μ A
Full Moonlit Night	0.1	3000pA
Clear Night / No Moon	0.001	30pA

Figure 4-36: Short circuit current versus light intensity for SD-020-12-001 photodiode (photovoltaic operating mode)

Figure 4-36 above shows the photosensitivity for a small photodiode (Silicon Detector Part Number SD-020-12-001). This diode has a basic sensitivity of 0.03 μ A/fc, and was chosen for the design example to follow. As this chart indicates, this photodiode's dynamic range covers six orders of magnitude!

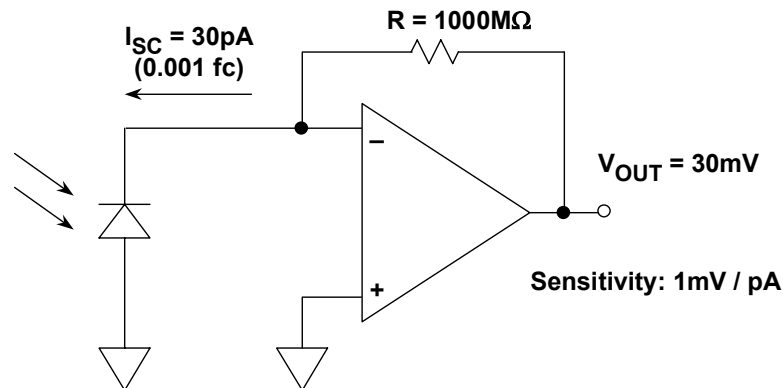


Figure 4-37: A simplified current-to-voltage converter uses a low bias current op amp and a high value feedback resistor

A convenient way to convert the photodiode current into a usable voltage is to use a low bias current op amp, configured as a current-to-voltage converter as shown in Figure 4-37 above. The diode bias is maintained at zero volts by the virtual ground of the op amp, and the short-circuit current is converted into a voltage. At maximum sensitivity, the amplifier must be able to detect a diode current of 30pA. This implies that the feedback resistor must be very large, and the amplifier bias current very small.

■ OP AMP APPLICATIONS

For example, $1000\text{M}\Omega$ will yield a corresponding voltage of 30mV for this amount of current. Larger resistor values are impractical, so we will use $1000\text{M}\Omega$ for the most sensitive range. This will give an output voltage range of 10mV for 10pA of diode current and 10V for 10nA of diode current. This yields a range of 60dB . For higher values of light intensity, the gain of the circuit must be reduced by using a smaller feedback resistor. For this range of maximum sensitivity, we should be able to easily distinguish between the light intensity on a clear, moonless night (0.001fc), and that of a full moon (0.1fc)!

Notice that we have chosen to get as much gain as possible from one stage, rather than cascading two stages. This is in order to maximize the signal-to-noise ratio (SNR). If we halve the feedback resistor value, the signal level decreases by a factor of 2, while the noise due to the feedback resistor ($\sqrt{4kTR \cdot \text{Bandwidth}}$) decreases by only $\sqrt{2}$. This reduces the SNR by 3dB , assuming the closed loop bandwidth remains constant. Later in the analysis, we will find the resistors one of the largest overall output noise contributors.

PART #	V_{OS} , MAX*	TC V_{OS} , TYP	I_B , MAX*	0.1Hz TO 10Hz NOISE, TYP	PACKAGE
AD549K	$250\mu\text{V}$	$5\mu\text{V}/^\circ\text{C}$	100fA	$4\mu\text{V p-p}$	TO-99
AD795JR	$500\mu\text{V}$	$3\mu\text{V}/^\circ\text{C}$	3pA	$1\mu\text{V p-p}$	SOIC
AD820B	$1000\mu\text{V}$	$2\mu\text{V}/^\circ\text{C}$	10pA	$2\mu\text{V p-p}$	SOIC, DIP

* 25°C SPECIFICATION

Figure 4-38: Some JFET input electrometer grade op amps suitable for use in photodiode preamplifiers

To accurately measure photodiode currents in the tens of picoamps range, the bias current of the op amp should be no more than a few picoamps. This narrows the choice considerably. The industry-standard OP07 is an ultra-low offset voltage ($10\mu\text{V}$) bipolar op amp, but its bias current is 4nA (4000pA !). Even super-beta bipolar op amps with bias current compensation (such as the OP97) have bias currents on the order of 100pA at room temperature, but they might be suitable for very high temperature applications, as unlike FET amplifiers, the bias currents do not double for every 10°C increase.

A JFET-input electrometer-grade op amp is chosen for our photodiode preamp, since it must operate only over a limited temperature range. Figure 4-38 above summarizes the performance of several popular "electrometer grade" FET input op amps.

As can be noted from this figure, the 25°C maximum bias current specification ranges from a few pA down to as low as 100fA , and there are a number of packages types from which to choose. As will be seen shortly, the package finally chosen can and will affect the performance of the circuit in terms of the bias current realized within an application. This is due to relative ability to control the inevitable leakage currents in a design's production environment.

Of these devices, the AD549 and AD795 are fabricated on a BiFET process and use P-Channel JFETs as the input stage, as is shown in Figure 4-39 below. The rest of the op amp circuit is designed using bipolar devices. These BiFET op amps are laser trimmed at the wafer level, to minimize offset voltage and offset voltage drift. The offset voltage drift is minimized, by first trimming the input stage for equal currents in the JFET differential pair (DRIFT TRIM RESISTORS). A further trim of the JFET source resistors minimizes the input offset voltage (OFFSET VOLTAGE TRIM RESISTORS).

For these discussions, an AD795JR was selected for the photodiode preamplifier, with key specifications summarized in Fig. 4-38 (again). This allows high circuit performance in an SOIC packaged device.

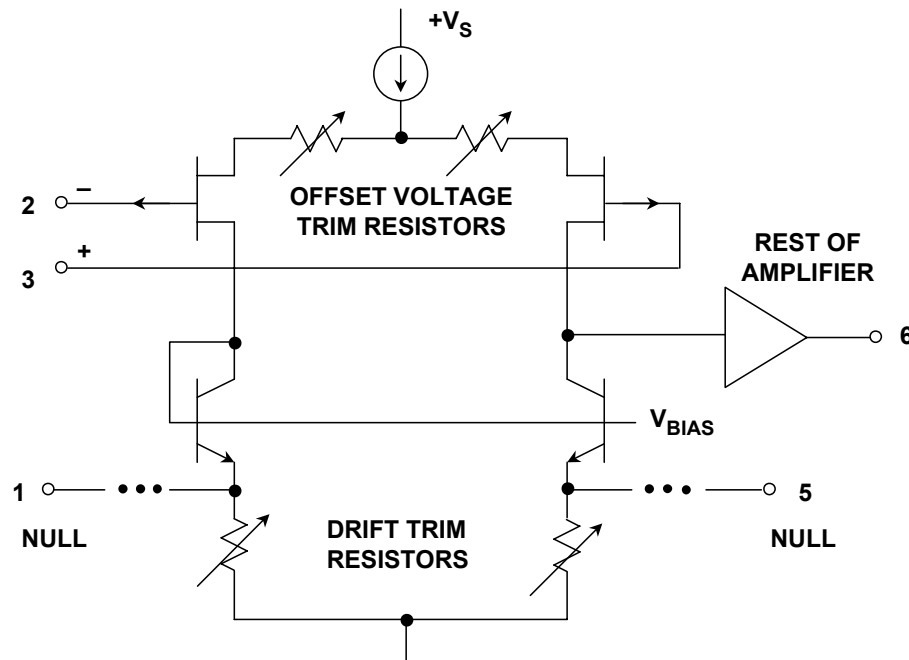


Figure 4-39: JFET input stage op amp with separate trims for offset voltage and drift

Alternately, for even greater performance, the AD549 could be used. The AD549 uses the glass sealed TO-99 package, which allows the very highest performance in terms of low leakage. More on this follows below.

Since the photodiode current is measured in terms of picoamperes, it should be understood that extremely close attention must be given to all potential leakage paths in the actual physical circuit. To put this in some perspective, consider the a simple printed circuit card example with two parallel conductor traces on a high-quality, well-cleaned epoxy-glass PC board 0.05 inches apart and parallel for 1 inch. Such an insulator has a leakage resistance of approximately 10^{11} ohms at $+125^{\circ}\text{C}$. By simple application of Ohm's law, 15 volts of bias between these runs produces a 150pA current— sufficient to mask all signal levels below this current. Obviously then, low-level photodiode circuitry needs to employ all possible means of minimizing such parasitic currents. Unfortunately, they can arise from numerous sources, some of which can be quite subtle in origin.

■ OP AMP APPLICATIONS

Figure 4-40 below illustrates the circuit elements subject to leakage for the photodiode circuit, as enclosed within the dotted lines. The feedback resistor is highly critical, and should be a close tolerance (1%), low TC (50ppm/°C) unit. Typical units suitable for R2 will be manufactured with thin film or metal oxide construction on ceramic or glass, with glass insulation. It should be readily apparent that any shunt conductive paths across this resistor's body can (and will) degrade or lower the net effective resistance, producing scaling errors. It is for this reason that such high value resistors are often glass enclosed, and can require special handling. Some sources of suitable high value resistors are listed in the section references. If used, compensation capacitor C2 should use the lowest loss dielectric possible. Typically this will mean a film type capacitor of Teflon, polypropylene, or polystyrene construction.

All connections to the op amp's summing junction should be kept short, clean, and free from manufacturing process chemicals and residues. In cases where an input cable is used to connect the photodiode to the preamp, it should be kept as short as possible, and should use Teflon or similar low-loss dielectric insulation.

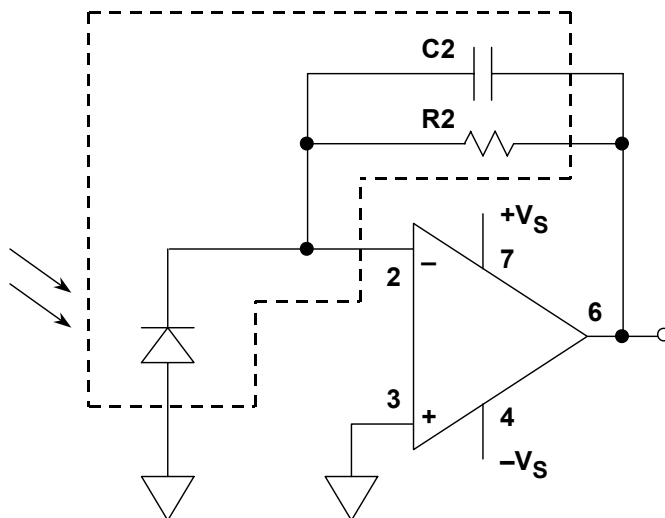


Figure 4-40: Critical leakage paths and components for a photodiode preamplifier circuit are those within the dotted line area

The above considerations deal mainly with the more obvious construction points towards optimizing accuracy and keeping leakage low. However, two of the more difficult leakage sources which can plague this circuit aren't quite as obvious. These are the op amp package-related parasitic leakages, which can occur from all op amp package pins adjacent to the input pin. Consider leakage as a high-value resistance to pin 2.

Although it doesn't show in this particular figure (since pin 1 isn't actively used by the application), *any leakage from package pin 1 can be very relevant*. Note also, that since pin 3 is grounded, it prevents error from leakage between pins 4 and 2. If however, pin 1 has any significant voltage on it (which it does in the case of the offset null pin of the AD820BN DIP device) serious leakage will then occur between pins 1 and 2. The AD795JR SOIC is immune to this leakage, as pin 1 isn't connected internally. These comments serve to illustrate some of the subtleties of these leakage sources.

The situation just described for the AD820BN DIP packaged device is by no means unique, as pin 1 is a standard offset trim pin on many op amps. This circumstance will always tend to leak current into any high impedance source seen at pin 2. There are also cases for follower-connected stages where leakage is just as critical, if not more so. In such cases the leakage goes into pin 3 as a high impedance, typically from pin 4, which is $-V_S$. Fortunately however, there is a highly effective answer to controlling both of these leakage problems, and that is the use of circuit *guard* techniques.

Guarding is used to reduce parasitic leakage currents, by isolating a sensitive amplifier input from large voltage gradients across the PC board. It does this by interposing a conductive barrier or screen between a high voltage source and a sensitive input. The barrier intercepts the leakage which would otherwise flow into the sensitive node, and diverts it away. In physical terms, a guard is a low impedance conductor that completely surrounds an input line or node, and it is biased to a potential equal to the line's voltage.

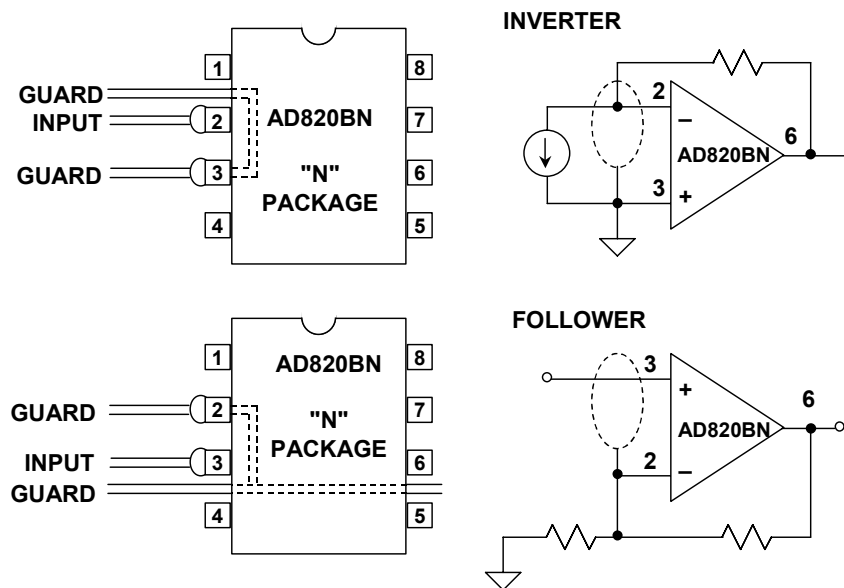


Figure 4-41: Guard techniques for inverting and non-inverting op amp stages using DIP package devices

Note that the low impedance nature of a guard conductor shunts leakage harmlessly away. The biasing of the guard to the same potential as the guarded pin reduces any possibility of leakage between the guard itself and the guarded node. The exact technique for guarding depends on the amplifier's mode of operation, i.e., whether the connection is inverting (like Fig. 4-40), or a non-inverting stage.

Figure 4-41 above shows a PC board layout for guarding the inputs of the AD820 op amp, as operated within either an inverting (top) or a non-inverting gain stage (bottom). This setup uses the DIP ("N") package, and would also be applicable to other devices where relatively high voltages occur at pin 1 or 4. Using a standard 8 pin DIP outline, it can be noted that this package's 0.1" pin spacing allows a PC trace (the guard trace) to pass between adjacent pins. This is the key to implementing effective DIP package guarding—the complete surrounding of the guarded trace with a low impedance trace.

■ OP AMP APPLICATIONS

In the inverting mode (top), note that the pin 3 connected and grounded guard traces surround the op amp inverting input (pin 2), and run parallel to the input trace. This guard would be continued out to and around the source device and feedback connection in the case of a photodiode (or around the input pad, in the case of a cable). In the follower mode (bottom), the guard voltage is the feedback divider tap voltage to pin 2, i.e., the inverting input node of the amplifier. Although the feedback divider impedance isn't as low in absolute terms as a direct ground, it is still quite effective. Even a $1\text{k}\Omega$ or so impedance here will still be many orders of magnitude lower than the pin 3 impedance. In both the inverting and the non-inverting modes, the guard traces should be located on both sides of the PC board, with top and bottom side traces connected with several vias.

Things become slightly more complicated when using guarding techniques with the SOIC surface mount ("R") package, because the 0.05" pin spacing doesn't allow routing of PC board traces between the pins. But, there still is an effective guarding answer, at least for the inverting case. Figure 4-42 below shows the preferred method.

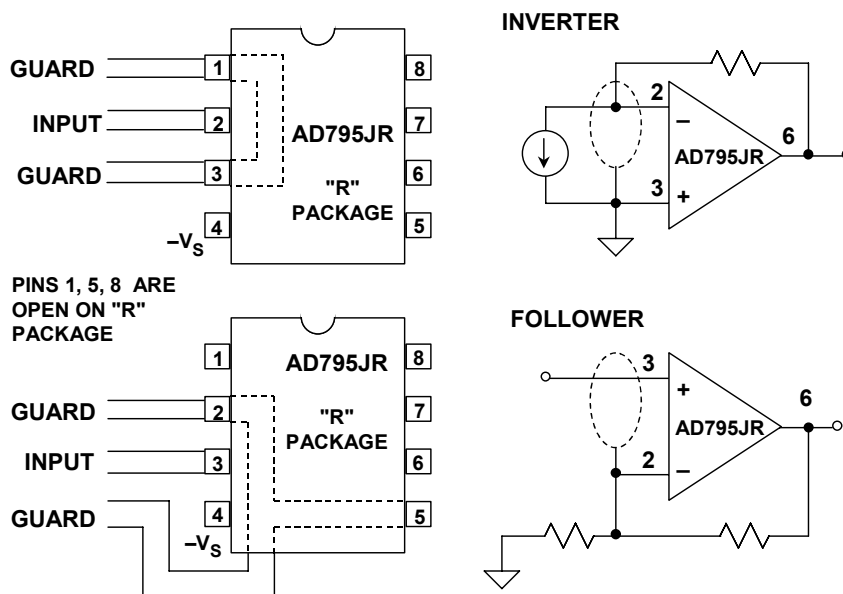


Figure 4-42: Guard techniques for inverting and non-inverting op amp stages using SOIC package devices

In the AD795 SOIC "R" package, pins 1, 5, and 8 are "no-connect" pins and can be used to route signal traces as shown. Thus in the case of the inverting stage (top), guarding is still completely effective, with dummy pin 1 and pin 3 acting as a grounded guard trace.

In the case of the follower stage (bottom), the guard trace must be routed around the $-V_S$ pin, and thus the pin 4 to pin 3 leakage is *not* fully guarded. For this reason, a very high impedance follower stage using an SOIC package op amp isn't generally recommended, as adequate guarding simply isn't possible. An exception to this caveat would apply to the use of a *single-supply* op amp as a follower (for example, the AD820), in which case pin 4 becomes grounded by default, and some degree of intrinsic guarding is established.

For extremely low bias current applications, such as for example with the AD549 with input bias current of 100fA, the high impedance input signal connection of the op amp should be made to a virgin Teflon standoff insulator, as shown in Figure 4-43 below. Note—"virgin" Teflon is a solid piece of new Teflon material that has been machined to shape (as opposed to one welded together from powder or grains).

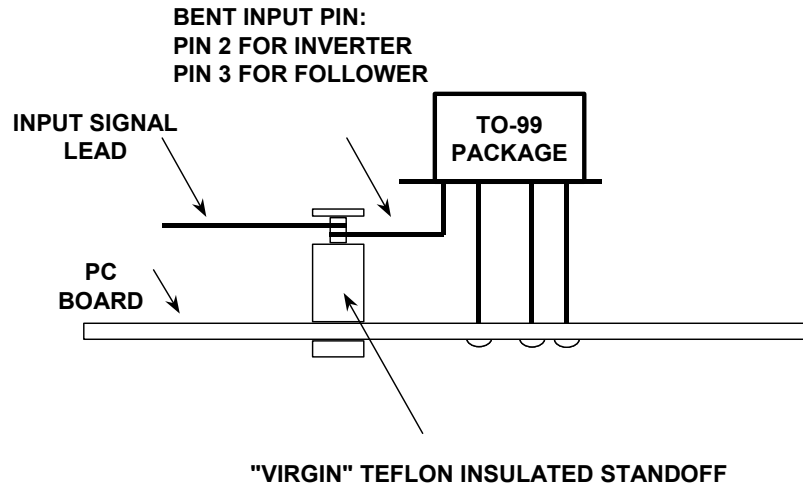


Figure 4-43: Input pin isolation technique using virgin Teflon standoff insulator

If mechanical and manufacturing considerations allow, the sensitive op amp input pin should be soldered directly to the Teflon standoff, rather than going through a PCB hole.

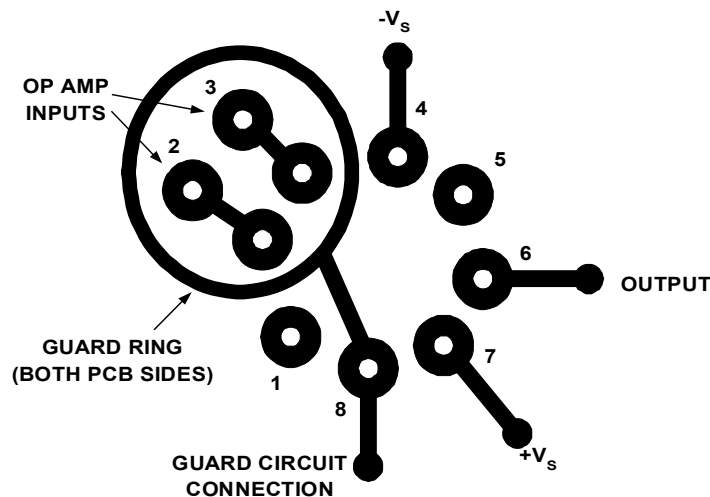


Figure 4-44: TO-99 package devices can use guard rings surrounding input pins 2 and 3 (PCB bottom view shown)

For TO-99 packaged devices, such as the AD549KH, two possible guarding choices present themselves. One method is to employ the device in a scheme like Fig. 4-43, with the sensitive input pin going to the Teflon standoff. Alternately, a round PCB layout scheme that is more amenable to the TO-99 package can be used, as shown above in Figure 4-44.

This scheme uses a guard ring, which completely surrounds the input and feedback nodes, with the ring tied to the device's metal can through the pin 8 connection. The guard ring

■ OP AMP APPLICATIONS

is then also tied to either ground or the feedback divider, as suits the application. This setup can also be further modified, to use the more sensitive of the two inputs going to a *Teflon standoff within the guard ring*, for the ultimate in performance.

Note that in all cases where control of leakage is critical, the PC board itself must be cleaned carefully and then sealed against humidity and dirt using a high quality conformal coating material. In addition to minimizing leakage currents, the entire circuit should be well shielded with a grounded metal shield to prevent stray signal pickup.

Preamplifier Offset Voltage and Drift Analysis

A photodiode preamp offset voltage and bias current model is shown in Figure 4-45. There are two important considerations in this circuit. First, the diode shunt resistance (R_1) is a function of temperature—it halves every time the temperature increases by 10°C . At room temperature ($+25^\circ\text{C}$), $R_1 = 1000\text{M}\Omega$, but at $+70^\circ\text{C}$ it decreases to $43\text{M}\Omega$. This has a drastic impact on the circuit noise gain and hence the output offset voltage. In the example, at $+25^\circ\text{C}$ the DC noise gain is 2, but at $+70^\circ\text{C}$ it increases to 24.

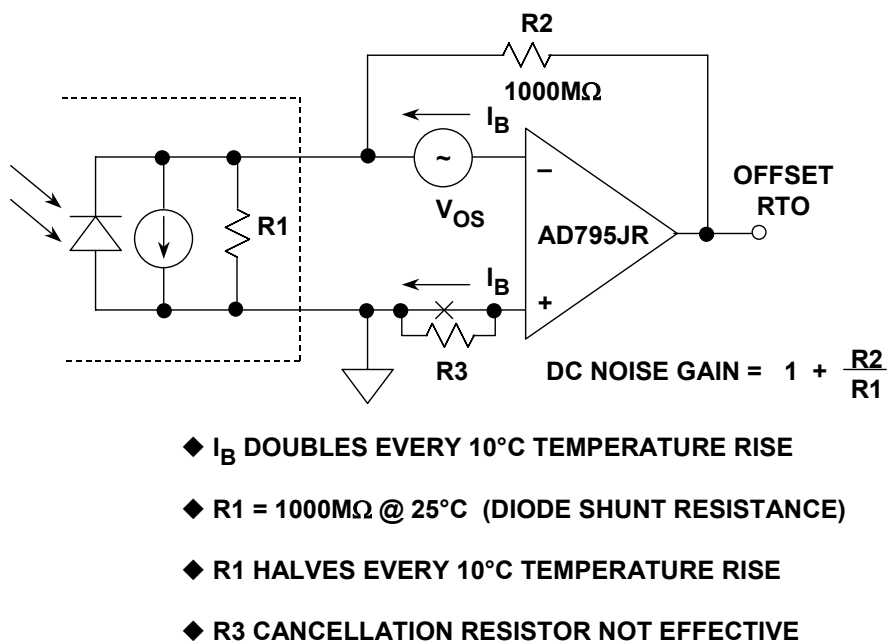


Figure 4-45: AD795JR photodiode preamplifier offset error model

The second circuit difficulty is that the input bias current doubles with every 10°C temperature rise. The bias current produces an output offset error equal to $I_B R_2$. At $+70^\circ\text{C}$ bias current increases to 72pA , compared to 3pA at room temperature. Normally, the addition of a resistor (R_3) between the non-inverting input of the op amp and ground, with a value of $R_1 \parallel R_2$ would yield a first-order cancellation of this effect. However, because R_1 changes with temperature, this method simply isn't effective. In addition, if R_3 is used, the bias current then develops a voltage across it, which in turn would be applied to the photodiode as a parasitic bias. Such a bias would cause the diode response to become nonlinear, thus the use of R_3 is also undesirable from a linearity point-of-view.

The total referred-to-output (RTO) offset voltage preamp errors are summarized in Figure 4-46 below. Notice that at +70°C the total error is 87.2mV. This error is acceptable for the design under consideration. The primary contributor to the error at high temperature is of course the bias current.

There are several steps that can be taken to minimize amplifier temperature rise, and thus offset drift. Operating the amplifier at reduced supply voltages, minimizing the output drive requirements, and heat sinking are some ways to reduce this error. The addition of an external offset nulling circuit would minimize the initial input offset voltage error.

	0°C	25°C	50°C	70°C
V_{OS}	0.575mV	0.500mV	0.575mV	0.635mV
Noise Gain	1.1	2	7	24
V_{OS} Error RTO	0.6mV	1.0mV	4.0mV	15.2mV
I_B	0.6pA	3.0pA	18.0pA	72.0pA
I_B Error RTO	0.6mV	3.0mV	18.0mV	72.0mV
Total Error RTO	1.2mV	4.0mV	22.0mV	87.2mV

Figure 4-46: AD795JR photodiode preamplifier offset error summary

Thermoelectric Voltages as Sources of Input Offset Voltage

As discussed earlier in this chapter, thermoelectric potentials are generated by electrical connections that are made between different metals. For example, the copper PC board electrical contacts to the kovar input pins of a TO-99 IC package can create an offset voltage of up to 40μV/°C, if the two bi-metal junctions so formed are at different temperatures. Even ordinary solders, being composed of alloys different from PCB copper traces, can give rise to thermoelectric voltages. For example, common high tin content lead-tin solder alloys, when used with copper, create thermoelectric voltages on the order of 1 to 3μV/°C (see Reference 8). While some special cadmium-tin solders can reduce this voltage to 0.3μV/°C, cadmium solders aren't in general use for health reasons. Another possible low thermal EMF solder is a low tin alloy such as Sn10Pb90.

The best general solution to minimizing this spurious thermocouple problem is to ensure that the connections to the inverting and non-inverting input pins of the IC are made with the same material, and that the PC board thermal layout is such that these two pins remain at the same temperature. Everything should be balanced from a thermal standpoint. In the case where a Teflon standoff is used as an insulated connection point for the inverting input (as in the case of this preamp), prudence dictates that connections to the non-inverting inputs also be made in a similar manner to minimize possible thermoelectric effects, and in keeping with the principle of thermal symmetry.

Preamplifier AC Design, Bandwidth, and Stability

The key to the preamplifier AC design is an understanding of the circuit noise gain as a function of frequency. Plotting gain versus frequency on a log-log scale makes the analysis relatively simple (see Figure 4-47 below). This type of plot is also referred to as a Bode plot. The noise gain is the gain seen by a small voltage source in series with one of the op amp input terminals. It is also the same as the non-inverting signal gain (the gain from "A" to the output). In the photodiode preamplifier, the signal current from the photodiode passes through the $C2/R2$ network. It is important to distinguish between the signal gain and the noise gain, because it is the noise gain characteristic that determines the net circuit stability, regardless of where the signal is actually applied.

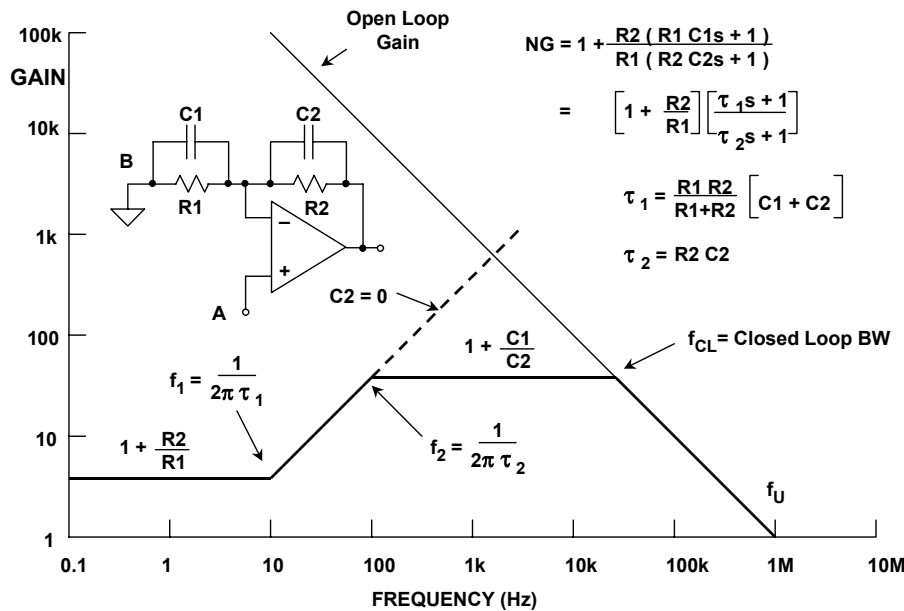


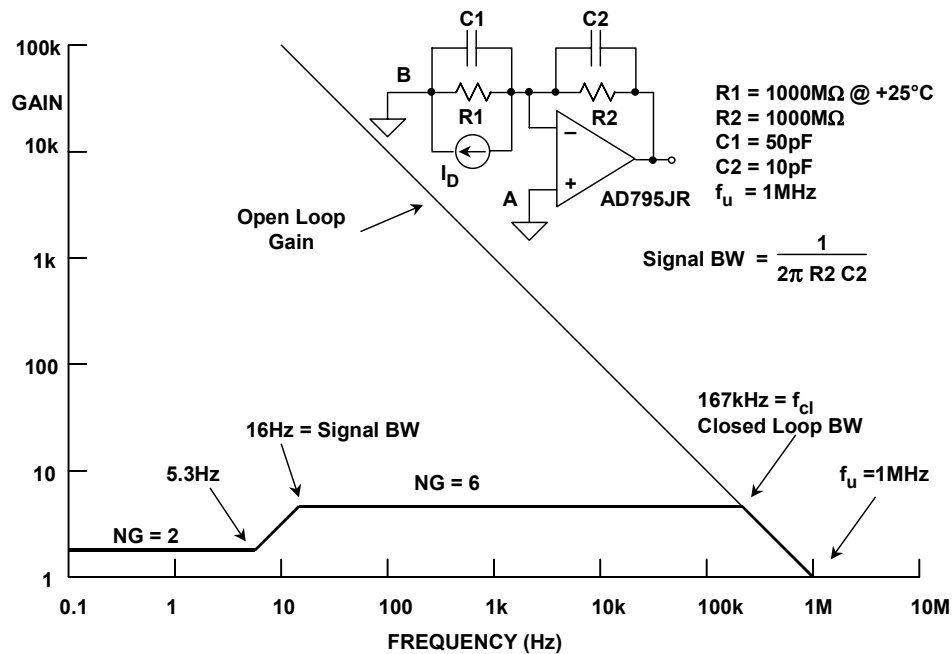
Figure 4-47: A generalized noise gain Bode plot

Note that the net slope between the noise gain and the open loop gain curves, *at the point where they intersect*, determines system stability. For unconditional stability, the noise gain curve must intersect the open loop response with a net slope of less than 12dB/octave (or 20dB per decade). In the figure, the dotted ($C2 = 0$) line shows a noise gain that intersects the open loop gain at a net slope of 12dB/octave, indicating an unstable condition. This is what would occur in the circuit, without a feedback capacitor.

The general equations for determining the break points and gain values in the Bode plot are also given in Fig. 4-47. It is useful to examine these gain characteristics with increasing frequency. At low frequencies, the circuit noise gain is $1 + R2/R1$, as indicated by the lowest frequency shelf (below 10Hz). There are two key time constants in this circuit, τ_1 and τ_2 . The first comes into play as a zero in the noise gain transfer function, which occurs at a frequency of $f_1 = 1/2\pi\tau_1$, where $\tau_1 = R1 \parallel R2 (C1 + C2)$. Stated simply, this frequency falls where the noise gain begins to increase to a new (higher) value from the low frequency gain of $1 + R2/R1$ plateau. In the Fig. 4-47 example f_1 occurs at 10Hz.

Above f_1 , gain increases towards a high frequency gain plateau where the gain is $1 + C1/C2$, which is indicated as the highest frequency shelf (above 100Hz). The second time constant, τ_2 , comes into play as a pole of the transfer function, which occurs at a corner frequency, $f_2 = 1/2\pi\tau_2$, where $\tau_2 = R2C2$. It can also be noted that this is equal to the signal bandwidth, if the signal is applied at point "B".

Plotting the composite noise gain curve on the log-log graph is a simple matter of connecting the f_1 and f_2 breakpoints with a line having a 45° slope, after first sketching the flat low and high frequency gain plateaus. The point at which the high frequency noise gain intersects the op amp open loop gain is called the *closed loop bandwidth*. Notice that the *signal bandwidth* for a signal applied at point "B" is much less, and is



$1/2\pi R2C2$.

Figure 4-48: Noise gain of the AD795 photodiode preamplifier at 25 °C

Figure 4-48 above shows the noise gain plot for the photodiode preamplifier using actual circuit values. The choice of $C2$ determines the actual signal bandwidth and also the phase margin. In the example, a signal bandwidth of 16Hz was chosen. Notice that a smaller value of $C2$ would result in a higher signal bandwidth and a corresponding reduction in phase margin. It is also interesting to note that although the signal bandwidth is only 16Hz, the closed loop bandwidth is 167kHz. This will have important implications with respect to the output noise voltage analysis to follow.

It is important to note that temperature changes do not significantly affect the stability of the circuit. Changes in $R1$ (the photodiode shunt resistance) only affect the low frequency noise gain and the frequency at which the zero in the noise gain response occurs. The high frequency noise gain is determined by the $C1/C2$ ratio.

Photodiode Preamplifier Noise Analysis

To begin a noise analysis, we first consider the AD795 input voltage and current noise spectral densities, as shown in Figure 4-49 below. The AD795 performance is truly impressive for a JFET input op amp: 1 μ V p-p typical 0.1Hz to 10Hz noise, and a 1/f corner frequency of 12Hz, comparing favorably with all but the best bipolar op amps. As shown in the (right) figure, the current noise is much lower than for bipolar op amps, a key factor making it an ideal choice for high impedance applications.

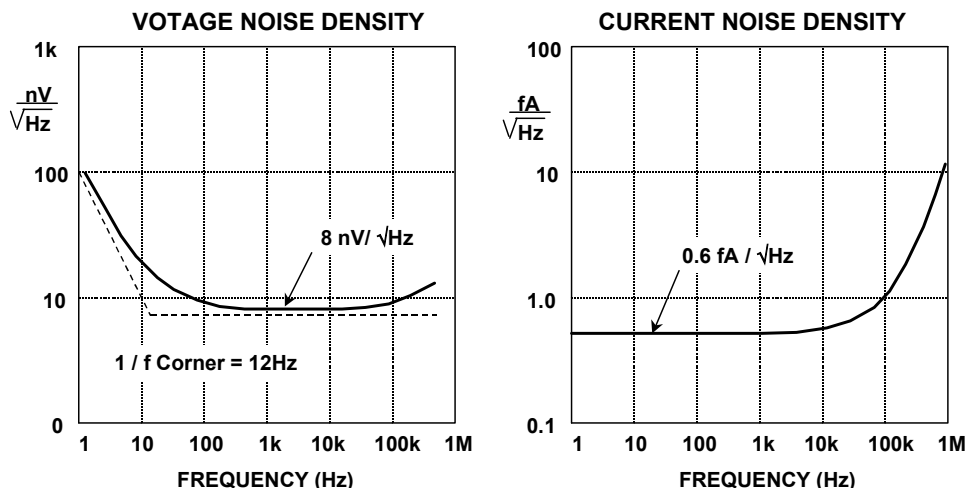


Figure 4-49: AD795 Voltage and current noise density performance

The complete noise model for an op amp is shown in Figure 4-50 (opposite). This model includes the reactive elements C1 and C2. Each individual output noise contributor is calculated by integrating the square of its spectral density over the appropriate frequency bandwidth and then taking the square root, as:

$$\text{RMS OUTPUT NOISE DUE TO } V_1 = \sqrt{\int V_1(f)^2 df} \quad \text{Eq. 4-4}$$

In most cases, this integration can be done by inspection of the graph of the individual spectral densities superimposed on a graph of the noise gain. The total output noise is then obtained by combining the individual components in a root-sum-squares manner. The table in the diagram shows how each individual source is reflected to the output, and the corresponding bandwidth for integration. The factor of 1.57 ($\pi/2$) is required to convert the single pole bandwidth into its equivalent noise bandwidth.

The resistor Johnson noise spectral density V_R is given by:

$$V_R = \sqrt{4kTR} \quad \text{Eq. 4-5}$$

where R is the resistance in ohms, k is Boltzmann's constant (1.38×10^{-23} J/K), and T is the absolute temperature in kelvins.

A simple way to compute this is to remember that the noise spectral density of a $1\text{k}\Omega$ resistor is $4\text{nV}/\sqrt{\text{Hz}}$ at $+25^\circ\text{C}$. The Johnson noise of another resistor value can be found by multiplying by the square root of the ratio of the resistor value to 1000Ω . For example, a $4\text{k}\Omega$ resistor produces a noise density $\sqrt{4}$ times a $1\text{k}\Omega$ resistor, or $8\text{nV}/\sqrt{\text{Hz}}$ (at $+25^\circ\text{C}$).

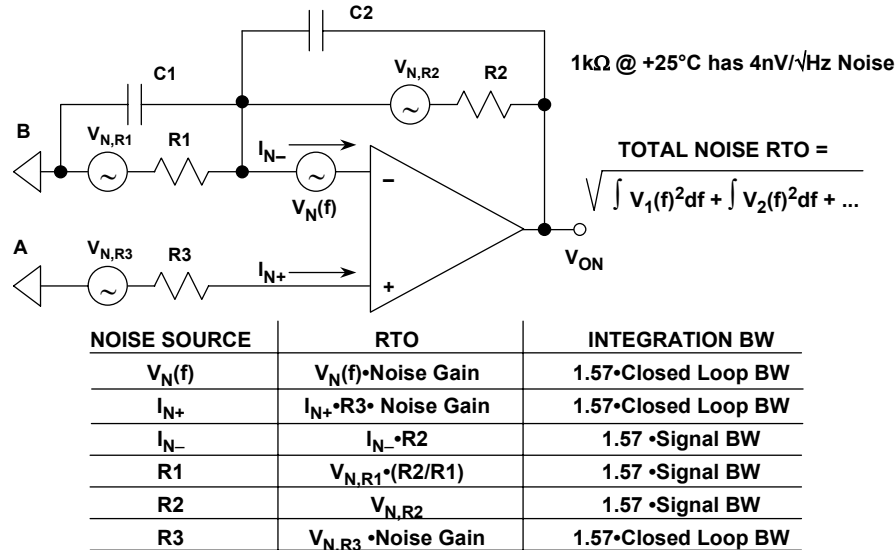


Figure 4-50: A noise model of preamp

Finally, note that Johnson noise is broadband, and its spectral density is constant with frequency.

Input Voltage Noise

In order to obtain the output voltage noise spectral density plot due to the input voltage noise, the input voltage noise spectral density plot is multiplied by the noise gain plot. This is easily accomplished using the Bode plot on a log-log scale. The total RMS output voltage noise due to the input voltage noise is then obtained by integrating the square of the output voltage noise spectral density plot, and then taking the square root. In most cases, this integration may be approximated. A lower frequency limit of 0.01Hz in the $1/f$ region is normally used. If the bandwidth of integration for the input voltage noise is greater than a few hundred Hz, the input voltage noise spectral density may be assumed to be constant. Usually, the value of the input voltage noise spectral density at 1kHz will provide sufficient accuracy.

It is important to note that the input voltage noise contribution must be integrated over the entire closed loop bandwidth of the circuit (the closed loop bandwidth, f_{cl} , is the frequency at which the noise gain intersects the op amp open loop response). This is also true of the other noise contributors which are reflected to the output by the noise gain (namely, the non-inverting input current noise and the non-inverting input resistor noise).

The inverting input noise current flows through the feedback network to produce a noise voltage contribution at the output. The input noise current is approximately constant with frequency, therefore, the integration is accomplished by multiplying the noise current

OP AMP APPLICATIONS

spectral density (measured at 1kHz) by the noise bandwidth which is 1.57 times the signal bandwidth ($1/2\pi R_2 C_2$). The factor of 1.57 ($\pi/2$) arises when single-pole 3dB bandwidth is converted to equivalent noise bandwidth.

Johnson Noise Due to Feedforward Resistor R1

The noise current produced by the feedforward resistor R1 also flows through the feedback network to produce a contribution at the output. The noise bandwidth for integration is also 1.57 times the signal bandwidth.

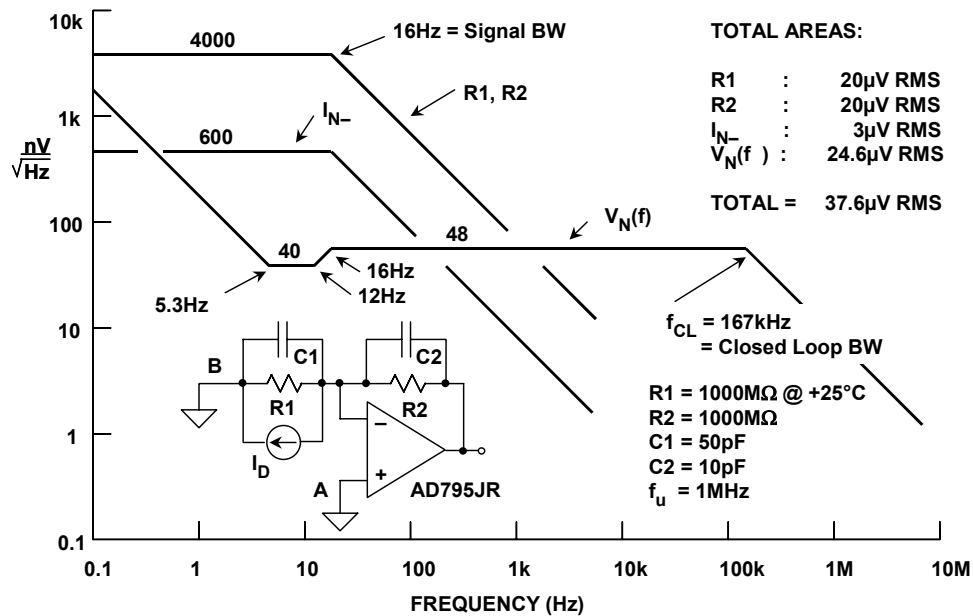


Figure 4-51: Preamp output spectral noise densities ($nV/\sqrt{Hz}$) @25 °C

Non-Inverting Input Current Noise

The non-inverting input current noise, I_{N+} , develops a voltage noise across R3 that is reflected to the output by the noise gain of the circuit. The bandwidth for integration is therefore the closed loop bandwidth of the circuit. However, there is no contribution at the output if R3 = 0 (or, if R3 is used, but it is bypassed with a large capacitor). This will usually be desirable when operating the op amp in the inverting mode.

Johnson Noise Due to Resistor in Non-Inverting Input

The Johnson voltage noise due to R3 is also reflected to the output by the noise gain of the circuit. Again, if R3 is bypassed sufficiently, it makes no significant contribution to the output noise.

Summary of Photodiode Circuit Noise Performance

Figure 4-51 (opposite) shows the output noise spectral densities for each of the contributors at +25°C. As can be noted, there is no contribution due to I_{N+} or R_3 , since the non-inverting input of the op amp is grounded.

Noise Reduction Using Output Filtering

From the above analysis, the largest contributor to the output noise voltage at +25°C is the input voltage noise of the op amp reflected to the output by the noise gain. This contributor is large primarily because the noise gain over which the integration is performed extends to a bandwidth of 167kHz (the intersection of the noise gain curve with the open-loop response of the op amp). If the op amp output is filtered by a single pole low pass filter with a 20Hz cutoff frequency ($\tau = 7.95\text{ms}$), this contribution is reduced to less than $1\mu\text{Vrms}$. The diagram for the final, filtered, optimized photodiode circuit design is shown in Figure 4-52 below.

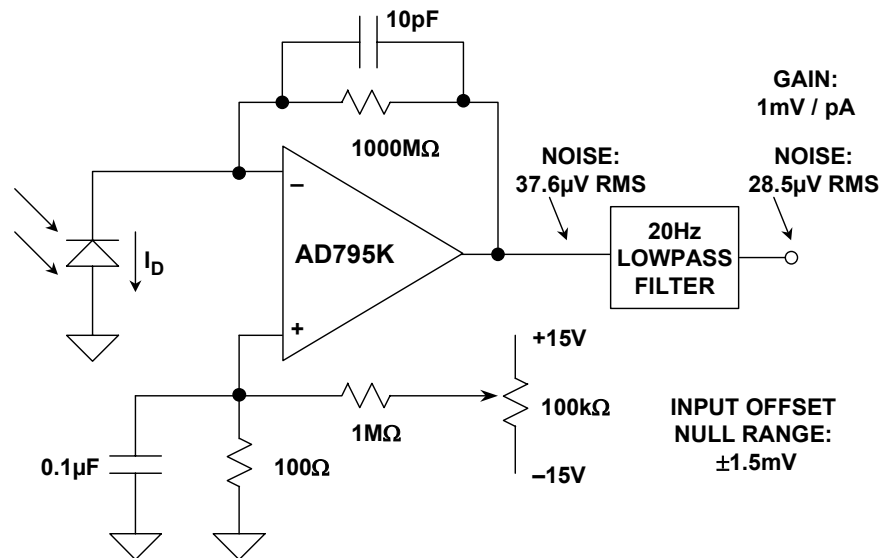


Figure 4-52: AD795K preamp with output filter and offset null option

Notice that the same results would not be achieved simply by increasing the feedback capacitor, C_2 . Increasing C_2 lowers the high frequency noise gain, but the integration bandwidth becomes proportionally higher. Larger values of C_2 may also decrease the signal bandwidth to unacceptable levels.

The addition of the post-filter stage reduces output noise to $28.5\mu\text{Vrms}$; approximately 75% of its former value, and the resistor noise and current noise are now the largest contributors to output noise. Practically, this filter can be either active or passive. Care will need to be taken of course, that the filter circuit does not add any significant noise of its own to the signal. Filter design is discussed in greater detail in Chapter 5 of this book. The final circuit also includes an offset trim arrangement that is capable of nulling op amp offsets of up to $\pm 1.5\text{mV}$.

▣ OP AMP APPLICATIONS

Summary of Circuit Performance

Performance characteristics are summarized in Figure 4-53 below. The total output voltage drift over 0 to +70°C is 87.2mV, corresponding to 87.2pA of diode current. The offset nulling circuit shown on the non-inverting input can be used to null out the room temperature offset. Note that this method is better than using the offset null pins because using the offset null pins will increase the offset voltage TC by about 3μV/°C for each millivolt nulled. In addition, the AD795 SOIC package does not have offset nulling pins.

- ◆ **Output Offset Error (0°C to +70°C) : 87.2mV**
- ◆ **Output Sensitivity: 1mV / pA**
- ◆ **Output Photosensitivity: 30V / foot-candle**
- ◆ **Total Output Noise @ +25°C : 28.5μV RMS**
- ◆ **Total Noise RTI @ +25°C : 44fA RMS, or 26.4pA p-p**
- ◆ **Range with R2 = 1000MΩ : 0.001 to 0.33 foot-candles**
- ◆ **Bandwidth: 16Hz**

Figure 4-53: AD795JR photodiode preamp performance summary

The input sensitivity based on a total output voltage noise of 44μV is obtained by dividing the output voltage noise by the value of the feedback resistor R2. This yields a minimum detectable diode current of 44fA. If a 12 bit ADC is used to digitize the 10V fullscale output, the weight of the least significant bit (LSB) is 2.5mV. The output noise level is much less than this.

Photodiode Circuit Tradeoffs

There are many tradeoffs which could be made in the basic photodiode circuit design we have described. More signal bandwidth can be achieved in exchange for a larger output noise level. Reducing the feedback capacitor C2 to 1pF increases the signal bandwidth to approximately 160Hz. Further reductions in C2 are not practical because the parasitic capacitance is probably in the order of 1 to 2pF. Some small amount of feedback capacitance is also required to maintain stability.

If the circuit is to be operated at higher levels of illumination (greater than approximately 0.3fc), the value of the feedback resistor can be reduced, thereby resulting in further increases in circuit bandwidth and less resistor noise.

If gain-ranging is to be used to measure higher light levels, extreme care must be taken in the design and layout of the additional switching networks, so as to minimize leakage paths and other parasitic effects.

Compensation of a High Speed Photodiode I/V Converter

A classical I/V converter is shown in Figure 4-54 below. Note that it is the same as the previous photodiode preamplifier, if we assume that $R_1 \gg R_2$. The total input capacitance, C_1 , is the sum of the diode capacitance and the op amp input capacitance. Dynamically, this is a classical second-order system, and the following guidelines can be applied in order to determine the proper compensation.

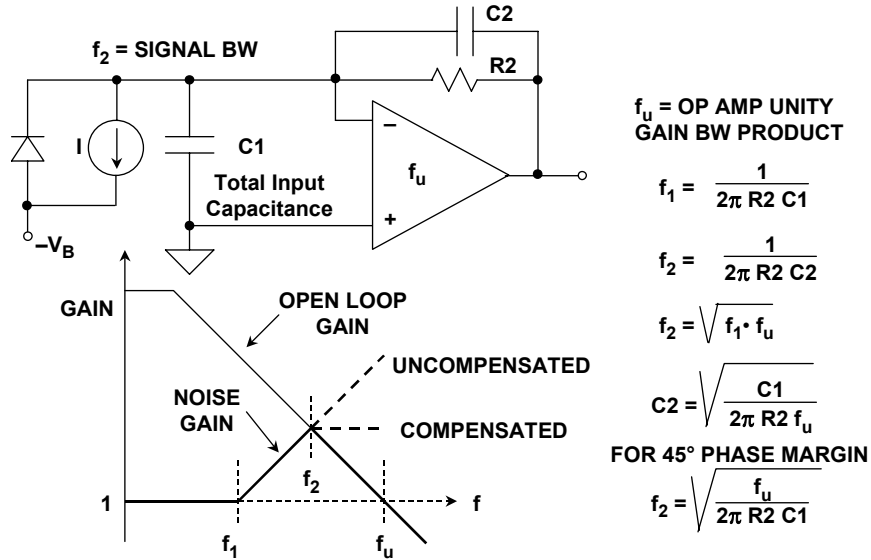


Figure 4-54: Input capacitance compensation for an I/V converter

The net input capacitance, C_1 , forms a zero at a frequency f_1 in the noise gain transfer function as shown in the Bode plot.

$$f_1 = \frac{1}{2\pi R_2 C_1} \quad \text{Eq. 4-6}$$

Note that we are neglecting the effects of the compensation capacitor C_2 and are assuming that it is small relative to C_1 and will not significantly affect the zero frequency f_1 when it is added to the circuit. In most cases, this approximation yields results which are close enough, considering the other variables in the circuit.

If left uncompensated, the phase shift at the frequency of intersection, f_2 , will cause instability and oscillation. Introducing a pole at f_2 by adding the feedback capacitor C_2 stabilizes the circuit and yields a phase margin of about 45 degrees.

$$f_2 = \frac{1}{2\pi R_2 C_2} \quad \text{Eq. 4-7}$$

■ OP AMP APPLICATIONS

Since f_2 is the geometric mean of f_1 and the unity-gain bandwidth frequency of the op amp, f_u ,

$$f_2 = \sqrt{f_1 \cdot f_u} \quad \text{Eq. 4-8}$$

These equations can be combined and solved for C2:

$$C2 = \sqrt{\frac{C1}{2\pi R2 \cdot f_u}} \quad \text{Eq. 4-9}$$

This C2 value yields a phase margin of about 45 degrees; increasing it by a factor of 2 increases phase margin to about 65 degrees. In practice, an optimum C2 value should be determined experimentally, by varying it slightly to optimize the output pulse response.

Op Amp Selection for Wideband Photodiode I/V Converters

The op amp in the high speed photodiode I/V converter should be a wideband FET-input one in order to minimize the effects of input bias current and allow low values of photocurrents to be detected. In addition, if the equation for the 3dB bandwidth, f_2 , is rearranged in terms of f_u , R2, and C1, then

$$f_2 = \sqrt{\frac{f_u}{2\pi R2 C1}} \quad \text{Eq. 4-10}$$

where C1 is the sum of the diode capacitance, C_D , and the op amp input capacitance, C_{IN} . In a high speed application, the diode capacitance will be much smaller than that of the low frequency preamplifier design previously discussed— perhaps as low as a few pF.

By inspection of this equation, it is clear that in order to maximize f_2 , the FET-input op amp should have both a high unity gain-bandwidth product, f_u , and a low input capacitance, C_{IN} . In fact, the ratio of f_u to C_{IN} is a good figure-of-merit when evaluating different op amps for this application.

Figure 4-55 below compares a number of FET-input op amps suitable for photodiode preamps. By inspection, the AD823 op amp has the highest ratio of unity gain-bandwidth product to input capacitance, in addition to relatively low input bias current.

	Unity GBW Product f_u (MHz)	Input Capacitance C_{IN} (pF)	f_u/C_{IN} (MHz/pF)	Input Bias Current I_B (pA)	Voltage Noise @ 10kHz (nV/ $\sqrt{\text{Hz}}$)
AD823	16	1.8	8.9	3	16
AD843	34	6	5.7	600	19
AD744	13	5.5	2.4	100	16
AD845	16	8	2	500	18
OP42	10	6	1.6	100	12
AD745*	20	20	1	250	2.9
AD795	1	1	1	3	8
AD820	1.9	2.8	0.7	10	13
AD743	4.5	20	0.2	250	2.9

*Stable for Noise Gains ≥ 5 , Usually the Case,
Since High Frequency Noise Gain = $1 + C1/C2$,
and $C1$ Usually $\geq 4C2$

Figure 4-55: FET input op amps suitable for high speed photodiode preamps

For these reasons, the AD823 op amp was chosen for the wideband design.

High Speed Photodiode Preamp Design

The HP 5082-4204 PIN Photodiode will be used as an example for our discussion. Its characteristics are listed in Figure 4-56 below. It is typical of many PIN photodiodes.

- ◆ Sensitivity: 350 μ A @ 1mW, 900nm
- ◆ Maximum Linear Output Current: 100 μ A
- ◆ Area: 0.002cm² (0.2mm²)
- ◆ Capacitance: 4pF @ 10V Reverse Bias
- ◆ Shunt Resistance: 10¹¹ Ω
- ◆ Risetime: 10ns
- ◆ Dark Current: 600pA @ 10V Reverse Bias

Figure 4-56: HP 5082-4204 photodiode characteristics

As in most high-speed photodiode applications, the diode will be operated in the reverse-biased or *photoconductive* mode. This greatly lowers the diode junction capacitance, but causes a small amount of *dark current* to flow even when the diode is not illuminated (we will show a circuit which compensates for the dark current error later in the section). This photodiode is linear with illumination up to approximately 50 to 100 μ A of output current.

■ OP AMP APPLICATIONS

The available dynamic range is limited by the total circuit noise, and the diode dark current (assuming no dark current compensation).

Using the circuit shown in Figure 4-57 below, assume that we wish to have a full scale output of 10V for a diode current of $100\mu\text{A}$. This determines the value of the feedback resistor R2 to be $10\text{V}/100\mu\text{A} = 100\text{k}\Omega$.

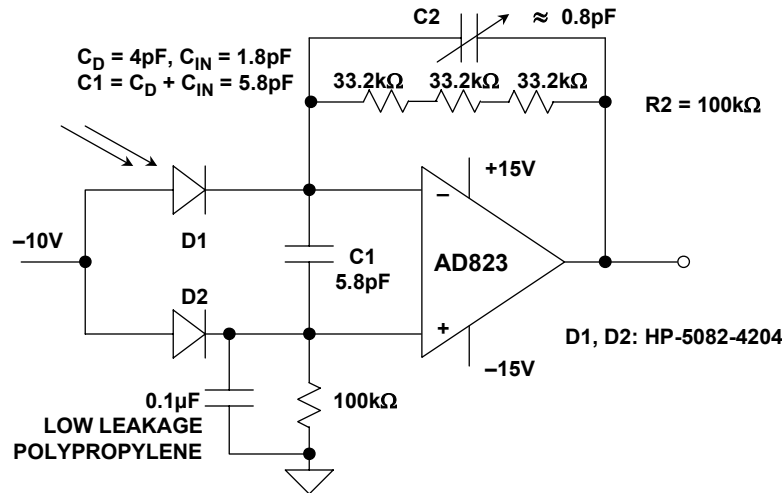


Figure 4-57: 2MHz bandwidth photodiode preamp with dark current compensation

Using the diode capacitance, $C_D=4\text{pF}$, and the AD823 input capacitance, $C_{IN}=1.8\text{pF}$, the value of $C_1 = C_D + C_{IN} = 5.8\text{pF}$. Solving the above equations using $C_1=5.8\text{pF}$, $R_2 = 100\text{k}\Omega$, and $f_u = 16\text{MHz}$, we find that:

$$\begin{aligned}
 f_1 &= 274\text{kHz} \\
 C_2 &= 0.76\text{pF} \\
 f_2 &= 2.1\text{MHz}
 \end{aligned}$$

In the final circuit shown, notice at the $100\text{k}\Omega$ resistor is replaced with three $33.2\text{k}\Omega$ film resistors to minimize stray capacitance. The feedback capacitor, C2, is a variable 1.5pF ceramic and is adjusted in the final circuit for best bandwidth/pulse response. The overall circuit bandwidth is approximately 2MHz.

The full scale output voltage of the preamp for $100\mu\text{A}$ diode current is 10V, and therefore the (uncompensated) error (RTO) due to the photodiode dark current of 600pA is $60\mu\text{V}$.

This dark current error can be effectively canceled using a second photodiode, D2, of the same type. This diode is biased with a voltage identical to D1 and, with nominally matched characteristics, will tend to conduct a similar dark current. In the circuit, this “dummy” dark current drives the $100\text{k}\Omega$ resistance in the non-inverting input of the op amp. This produces a dark current proportional bias voltage, which, due to the CM rejection of the op amp, has an end result of suppressing the dark current effects.

High Speed Photodiode Preamp Noise Analysis

As in most noise analyses, only the key contributors need be identified. Because the noise sources combine in an RSS manner, any single noise source that is at least three or four times as large as any of the others will dominate.

In the case of the wideband photodiode preamp, the dominant sources of output noise are the input voltage noise of the op amp, V_N , and the resistor noise due to R_2 , $V_{N,R2}$ (see Figure 4-58). The input current noise of the FET-input op amp is negligible. The shot noise of the photodiode (caused by the reverse bias) is negligible because of the filtering effect of the shunt capacitance C_1 .

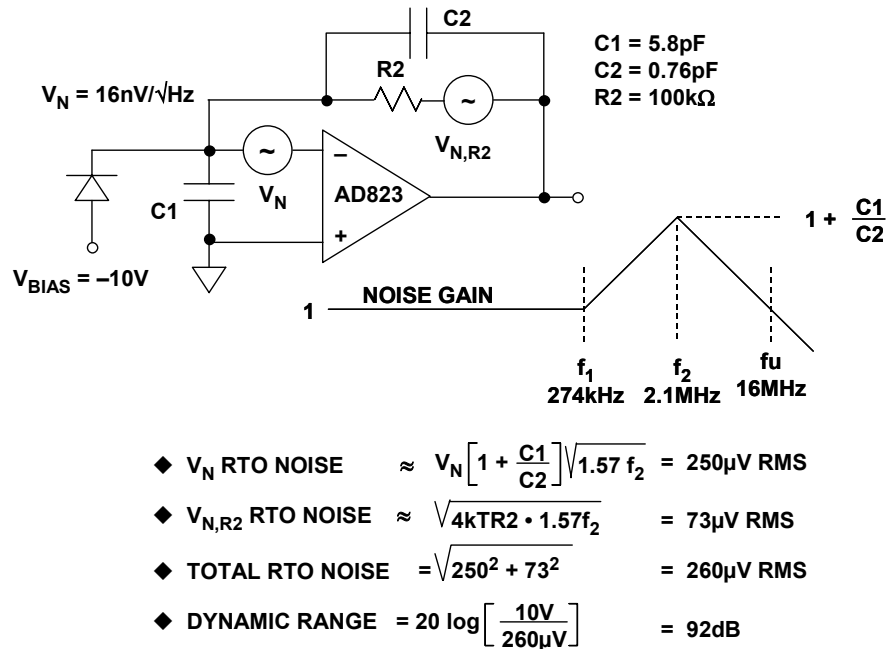


Figure 4-58: Equivalent circuit of preamp for output noise analysis

The resistor noise is easily calculated by knowing that a $1\text{k}\Omega$ resistor generates about $4\text{nV}/\sqrt{\text{Hz}}$, therefore, a $100\text{k}\Omega$ resistor generates $40\text{nV}/\sqrt{\text{Hz}}$. The bandwidth for integration is the signal bandwidth, 2.1MHz , yielding a total output rms noise of:

$$V_{N,R2} \text{ RTO NOISE} = 40 \sqrt{1.57 \cdot 2.1 \cdot 10^6} = 73\mu\text{Vrms}$$

The factor of 1.57 converts the amplifier approximate single-pole bandwidth of 2.1MHz into the *equivalent noise bandwidth*.

The output noise due to the input voltage noise is obtained by multiplying the noise gain by the voltage noise and integrating the entire function over frequency. This would be tedious if done rigorously, but a few reasonable approximations can be made which greatly simplify the math. Obviously, the low frequency $1/f$ noise can be neglected in the case of the wideband circuit. The primary source of output noise is due to the high-

■ OP AMP APPLICATIONS

frequency noise-gain peaking which occurs between f_l and f_u . If we simply assume that the output noise is constant over the entire range of frequencies and use the maximum value for AC noise gain $[1+(C1/C2)]$, then

$$V_N \text{ RTO NOISE} \approx V_N \left(1 + \frac{C1}{C2} \right) \sqrt{1.57 f_2} = 250 \mu V_{rms}$$

The total rms noise referred to the output is then the RSS value of the two components:

$$\text{TOTAL RTO NOISE} = \sqrt{(73)^2 + (250)^2} = 260 \mu V_{rms}$$

The total output dynamic range can be calculated by dividing the 10V full scale output by the total 260 μ Vrms noise, and, converting to dB, yielding approximately 92dB.

High Impedance Charge Output Sensors

High impedance transducers such as piezoelectric sensors, hydrophones, and some accelerometers require an amplifier that converts a transfer of charge into a voltage change. Due to the high DC output impedance of these devices, appropriate buffer amplifiers are required. The basic circuit for an inverting charge sensitive amplifier is shown in Figure 4-59 below.

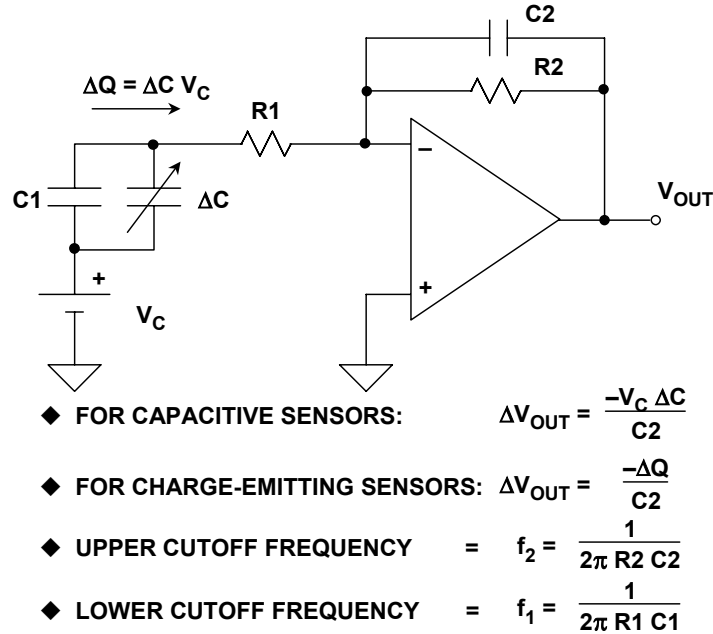


Figure 4-59: Charge amplifier basic principles

There are basically two types of charge transducers: capacitive and charge-emitting. In a capacitive transducer, the voltage across the capacitor (V_C) is held constant. The change in capacitance, ΔC , produces a change in charge, $\Delta Q = \Delta C V_C$. This charge is transferred to the op amp output as a voltage, $\Delta V_{OUT} = -\Delta Q/C2 = -\Delta C V_C/C2$.

Charge-emitting transducers produce an output charge, ΔQ , and their output capacitance remains constant. This charge would normally produce an open-circuit output voltage at the transducer output equal to $\Delta Q/C$. However, since the voltage across the transducer is held constant by the virtual ground of the op amp (R_1 is usually small), the charge is transferred to capacitor C_2 producing an output voltage $\Delta V_{OUT} = -\Delta Q/C_2$.

In an actual application, the charge amplifier only responds to AC inputs. The upper cutoff frequency is given by $f_2 = 1/2\pi R_2 C_2$, and the lower by $f_1 = 1/2\pi R_1 C_1$.

Low Noise Charge Amplifier Circuit Configurations

Figure 4-60 below shows two ways to buffer and amplify the output of a charge output transducer. Both require using an amplifier which has a very high input impedance, such as the AD745. The AD745 provides both low voltage noise and low current noise. This combination makes this device particularly suitable in applications requiring very high charge sensitivity, such as capacitive accelerometers and hydrophones.

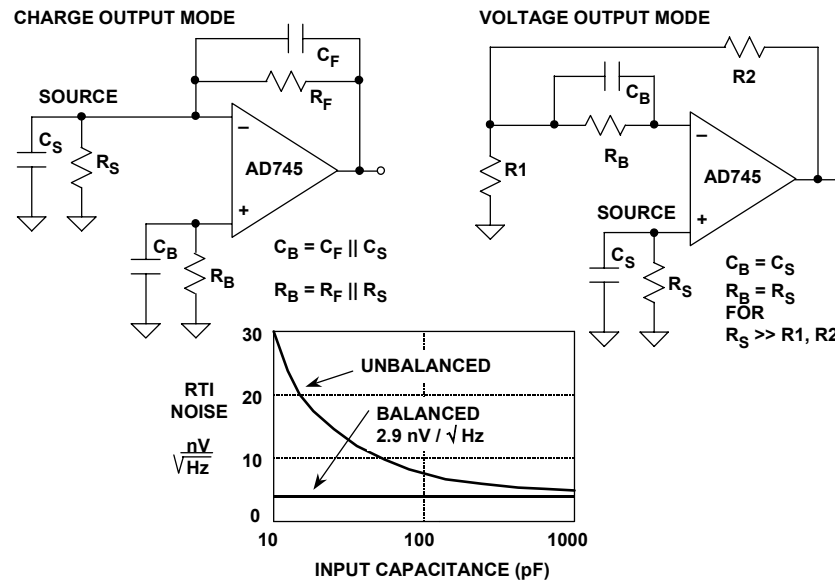


Figure 4-60: Two basic charge amplifier configurations using the AD745 low noise FET op amp

The first (left) circuit in Fig. 4-60 uses the op amp in the inverting mode. Amplification depends on the principle of conservation of charge at the inverting input of the amplifier. The charge on capacitor C_S is transferred to capacitor C_F , thus yielding an output voltage of $\Delta Q/C_F$. The amplifier's input voltage noise will appear at the output amplified by the AC noise gain of the circuit, $1 + C_S/C_F$.

The second (right) circuit shown in Figure 4-60 is simply a high impedance follower with gain. Here the noise gain ($1 + R_2/R_1$) is the same as the gain from the transducer to the output. Resistor R_B , in both circuits, is required as a DC bias current return.

OP AMP APPLICATIONS

To maximize DC performance over temperature, source resistances should be balanced at the amplifier inputs, as represented by the resistor R_B shown in Fig. 4-60 (again). For best noise performance, the source capacitance should also be balanced with the capacitor C_B .

In general, it is good practice to balance the source impedances (both resistive and reactive) as seen by the inputs of precision low noise BiFET amplifiers such as the AD743/AD745. Balancing the resistive components will optimize DC performance over temperature, as balancing mitigates the effects of any bias current errors. Balancing the input capacitance will minimize AC response errors due to the amplifier's non-linear common mode input capacitance, and as shown in Figure 4-60, noise performance will be optimized. In any FET input amplifier, the current noise of the internal bias circuitry can be coupled to the inputs via the gate-to-source capacitances (20pF for the AD743 and AD745) and appears as excess input voltage noise. This noise component is correlated at the inputs, so source impedance matching will tend to cancel out its effect. Fig. 4-60 shows the required external components for both inverting and noninverting configurations. For values of C_B greater than 300pF, there is a diminishing impact on noise, and C_B can then be simply a polyester bypass capacitor of 0.01 μ F or greater.

40dB Gain Piezoelectric Transducer Amplifier Operates on Reduced Supply Voltages for Lower Bias Current

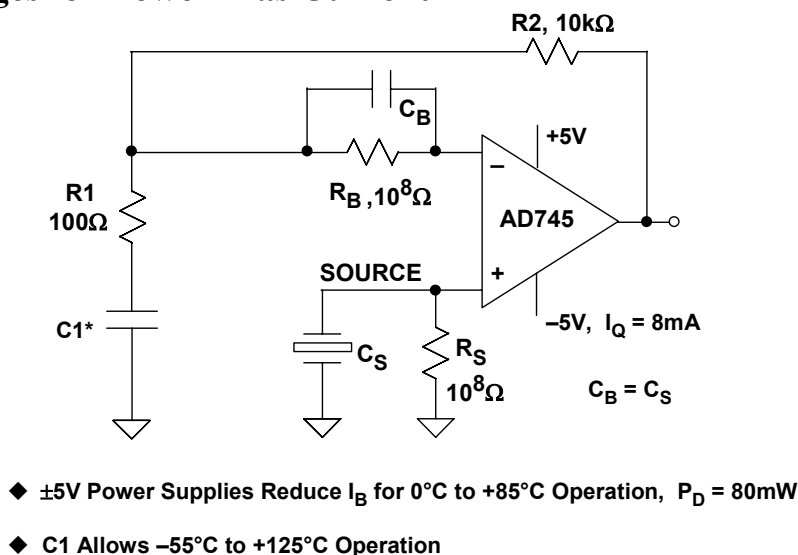


Figure 4-61: A gain-of-100 piezoelectric transducer amplifier

Figure 4-61 above shows a gain-of-100 piezoelectric transducer amplifier connected in the voltage-output mode. Reducing the op amp power supplies to $\pm 5\text{V}$ reduces the effects of bias current in two ways: first, by lowering the total power dissipation and, second, by reducing the basic gate-to-junction leakage current. The addition of a clip-on heat sink such as the Aavid #5801 will further limit the internal junction temperature rise.

With AC coupling capacitor C_1 shorted, the amplifier will operate over a range of 0°C to +85°C. If the optional AC coupling capacitor C_1 is used, the circuit will operate over the entire -55°C to +125°C temperature range, but DC information is lost.

Hydrophones

Interfacing the outputs of highly capacitive transducers such as hydrophones, some accelerometers, and condenser microphones to the outside world presents many design challenges. Previously designers had to use costly hybrid amplifiers consisting of discrete low-noise JFETs in front of conventional op amps to achieve the low levels of voltage and current noise required by these applications. Using AD743/AD745 monolithic ICs, designers can achieve almost the same level of performance of a hybrid approach.

In sonar applications, a piezo-ceramic cylinder is commonly used as the active element in the hydrophone. A typical cylinder has a nominal capacitance of around 6,000pF with a series resistance of 10Ω . The output impedance is typically $10^8\Omega$ or $100M\Omega$.

Since the hydrophone signals of interest are inherently AC with wide dynamic range, noise is an overriding concern for sonar system designs. The noise floor of the hydrophone and its preamplifier together limit the system sensitivity, and thus the overall hydrophone usefulness. Typical hydrophone bandwidths are in the 1kHz to 10kHz range. The AD743 and AD745 op amps, with their low noise voltages of $2.9nV/\sqrt{Hz}$ and high input impedance of $10^{10}\Omega$ (or 10G Ω) are ideal for use as hydrophone amplifiers.

Op Amp Performance: JFET versus Bipolar

The AD743 and AD745 op amps are the first monolithic JFET devices to offer the low input voltage noise comparable to a bipolar op amp, *without* the high input bias currents typically associated with bipolar op amps. Figure 4-62 below shows input voltage noise versus input source resistance of the OP27 (bipolar-input) and the JFET-input AD745 op amps. Note: the noise levels of the AD743 and the AD745 are identical.

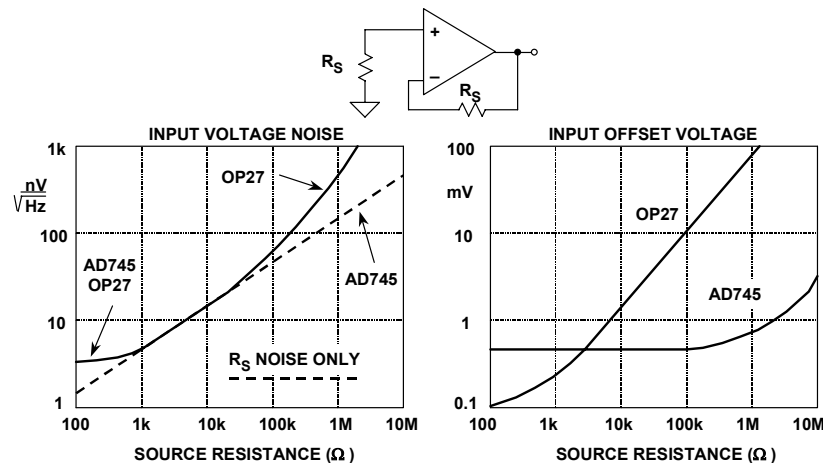


Figure 4-62: Total noise performance comparison, OP27 (bipolar) and AD745 (FET) op amps

From this figure (left), it is clear that at high source impedances, the low current noise of the AD745 provides lower overall noise than a high performance bipolar op amp such as the OP27. It is also important to note that, with the AD745, this noise reduction extends

■ OP AMP APPLICATIONS

down to low source impedances. At high source impedances, the lower DC current errors of the AD745 also reduce errors due to offset and drift, as shown in Fig. 4-62 (right).

The AD743 and AD745 are related, companion amplifiers, which differ in their levels of internal compensation. The AD743 is internally compensated for unity gain stability. The AD745, stable for noise gains of 5 or higher, has a much higher bandwidth and slew rate. This makes the latter device especially useful as a high-gain preamplifier where it provides both high gain and wide bandwidth. The AD743 and AD745 also operate with very low levels of distortion; less than 0.0003% and 0.0002% (at 1kHz), respectively.

A pH Probe Buffer Amplifier

A typical pH probe requires a buffer amplifier to isolate its 10^6 to $10^9 \Omega$ source resistance from external circuitry. Such an amplifier is shown in Figure 4-63 below. The low input current of the AD795JR minimizes the voltage error produced by the bias current and electrode resistance. The use of guarding, shielding, high insulation resistance standoffs, and other such standard picoamp methods previously discussed should be used to minimize leakage. Are all needed to maintain the accuracy of this circuit.

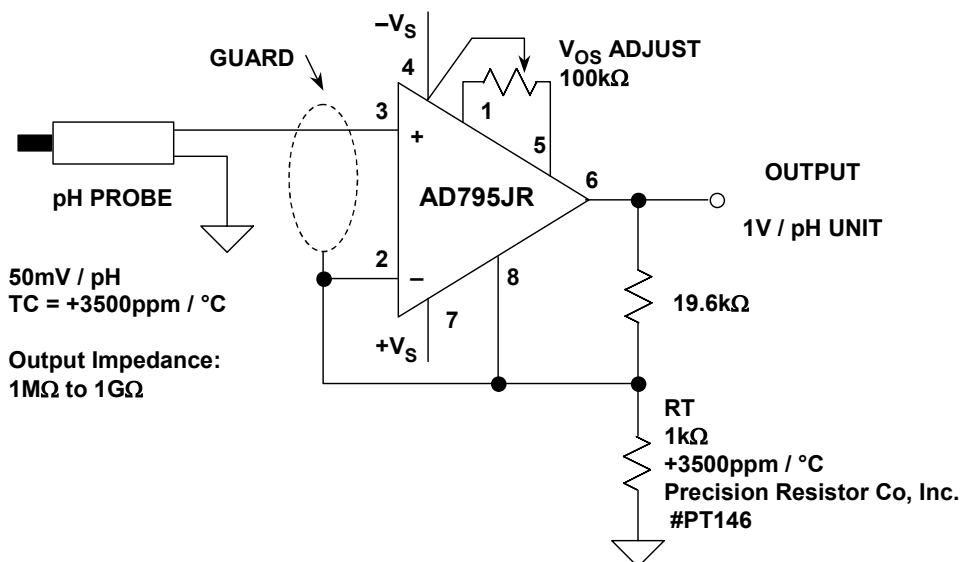


Figure 4-63: A gain of 20x pH probe amplifier allows 1V/pH output scaling

The slope of the pH probe transfer function, 50mV per pH unit at room temperature, has an approximate +3500ppm/°C temperature coefficient. The buffer amplifier shown above as Fig. 4-63 provides a gain of 20, and yields a final output voltage equal to 1volt/pH unit. Temperature compensation is provided by resistor RT, which is a special temperature compensation resistor, 1kΩ, 1%, +3500ppm/°C, #PT146 available from Precision Resistor Co., Inc. (see Reference 15).

REFERENCES: HIGH IMPEDANCE SENSORS

1. Ramon Pallas-Areny and John G. Webster, **Sensors and Signal Conditioning**, John Wiley, New York, 1991.
2. Dan Sheingold, Editor, **Transducer Interfacing Handbook**, Analog Devices, Inc., 1980, ISBN: 0-916550-05-2.
3. Section 3, Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, 1992, ISBN: 0-916550-10-9.
4. Walt Kester, Editor, **System Applications Guide**, Analog Devices, 1993, ISBN: 0-916550-13-3.
5. Walt Kester, Editor, **Linear Design Seminar**, Analog Devices, 1995, ISBN: 0-916550-15-X.
6. Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, 1995, ISBN: 0-916550-16-8.
7. Walt Kester, Editor, **High Speed Design Techniques**, Analog Devices, 1996, ISBN: 0-916550-17-6.
8. Jiri Dostal, section 9.3.1, "Thermoelectric Voltages," **Operational Amplifiers 2nd Ed.**, Butterworth-Heinemann, 1993, p265-268.
9. Optoelectronics Data Book, EG&G Vactec, St. Louis, MO, 1990.
10. Silicon Detector Corporation, Camarillo, CA, Part Number SD-020-12-001 Data Sheet.
11. Photodiode 1991 Catalog, Hamamatsu Photonics, Bridgewater, NJ
12. Lewis Smith and Dan Sheingold, "Noise and Operational Amplifier Circuits," **Analog Dialogue 25th Anniversary Issue**, pp. 19-31, Analog Devices Inc., 1991.
13. James L. Melsa and Donald G. Schultz, **Linear Control Systems**, pp. 196-220, McGraw-Hill, 1969.
14. Jerald G. Graeme, **Photodiode Amplifiers: Op Amp Solutions**, McGraw-Hill, 1995.
15. Precision Resistor Co., Inc., 10601 75th St. N., Largo, FL, 33777-1427, 727 541-5771,
<http://www.precisionresistor.com>
16. **Ohmite Victoreen MAXI-MOX Resistors**, 3601 Howard Street, Skokie, IL 60076, 847 675-2600,
<http://www.ohmite.com/victoreen/>.
17. **Vishay/Dale RNX Resistors**, 2300 Riverside Blvd., Norfolk, NE, 68701-2242, 402 371-0800,
<http://www.vishay.com>.

NOTES:

SECTION 4-5: TEMPERATURE SENSORS

Walt Kester, James Bryant, Walt Jung

Introduction

Temperature measurement is critical in many modern electronic devices, especially expensive laptop computers and other portable devices; their densely packed circuitry dissipates considerable power in the form of heat. Knowledge of system temperature can also be used to control battery charging, as well as to prevent damage to expensive microprocessors.

Compact high power portable equipment often has fan cooling to maintain junction temperatures at proper levels. In order to conserve battery life, the fan should only operate when necessary. Accurate control of the fan requires, in turn, knowledge of critical temperatures from an appropriate temperature sensor.

THERMOCOUPLE	RTD	THERMISTOR	SEMICONDUCTOR
Widest Range: −184°C to +2300°C	Range: −200°C to +850°C	Range: 0°C to +100°C	Range: −55°C to +150°C
High Accuracy and Repeatability	Fair Linearity	Poor Linearity	Linearity: 1°C Accuracy: 1°C
Needs Cold Junction Compensation	Requires Excitation	Requires Excitation	Requires Excitation
Low-Voltage Output	Low Cost	High Sensitivity	10mV/K, 20mV/K, or 1μA/K Typical Output

Figure 4-64: *Some common types of temperature transducers*

Accurate temperature measurements are required in many other measurement systems, for example within process control and instrumentation applications. Some popular types of temperature transducers and their characteristics are indicated in Figure 4-64 above. In most cases, because of low-level and/or nonlinear outputs, the sensor output must be properly conditioned and amplified before further processing can occur.

Except for the semiconductor sensors of the last column, all of the temperature sensors shown have nonlinear transfer functions. In the past, complex analog conditioning circuits were designed to correct for the sensor nonlinearity. These circuits often required manual calibration and precision resistors to achieve the desired accuracy. Today, however, sensor outputs may be digitized directly by high resolution ADCs. Linearization and calibration can then be performed digitally, substantially reducing cost and complexity.

■ OP AMP APPLICATIONS

Resistance Temperature Devices (RTDs) are accurate, but require excitation current and are generally used within bridge circuits such as those described earlier. *Thermistors* have the most sensitivity, but are also the most non-linear. Nevertheless, they are popular in portable applications for measurement of battery and other critical system temperatures.

Modern *semiconductor temperature sensors* offer both high accuracy and linearity over about a -55°C to $+150^{\circ}\text{C}$ operating range. Internal amplifiers can scale output to convenient values, such as $10\text{mV}/^{\circ}\text{C}$. They are also useful in cold-junction compensation circuits for wide temperature range thermocouples. Semiconductor temperature sensors are also integrated into ICs that perform other hardware monitoring functions—an example is the imbedded transistor sensors used within modern μP chips.

Thermocouple Principles and Cold-Junction Compensation

Thermocouples comprise a more common form of temperature measurement. In operation, thermocouples rely on the fact that two dissimilar metals joined together produce a voltage output roughly proportional to temperature. They are small, rugged, relatively inexpensive, and operate over the widest range of contact temperature sensors.

JUNCTION MATERIALS	TYPICAL USEFUL RANGE ($^{\circ}\text{C}$)	NOMINAL SENSITIVITY ($\mu\text{V}/^{\circ}\text{C}$)	ANSI DESIGNATION
Platinum (6%)/Rhodium- Platinum (30%)/Rhodium	38 to 1800	7.7	B
Tungsten (5%)/Rhenium - Tungsten (26%)/Rhenium	0 to 2300	16	C
Chromel - Constantan	0 to 982	76	E
Iron - Constantan	0 to 760	55	J
Chromel - Alumel	-184 to 1260	39	K
Platinum (13%)/Rhodium- Platinum	0 to 1593	11.7	R
Platinum (10%)/Rhodium- Platinum	0 to 1538	10.4	S
Copper-Constantan	-184 to 400	45	T

Figure 4-65: Some common thermocouples and their characteristics

Thermocouples are especially useful for making measurements at extremely high temperatures (up to $+2300^{\circ}\text{C}$), and within hostile environments. Characteristics of some common types are shown in Figure 4-65 above.

However, thermocouples produce only millivolts of output, and thus they typically require precision amplification for further processing. They also require *cold-junction-compensation* (CJC) techniques, to be discussed shortly. They are more linear than many other sensors, and their non-linearity has been well characterized.

The most common metals used for thermocouples are iron, platinum, rhodium, rhenium, tungsten, copper, alumel (composed of nickel and aluminum), chromel (composed of nickel and chromium) and constantan (composed of copper and nickel).

Figure 4-66 below shows the voltage-temperature curves for three commonly used thermocouples, Types J, K, and S, referred to a 0°C fixed-temperature reference junction. Of these, Type J thermocouples are the most sensitive, producing the highest output voltage for a given temperature change, but over a relatively narrow temperature span. On the other hand, Type S thermocouples are the least sensitive, but can operate over a much wider range.

These characteristics are very important considerations when designing thermocouple signal-conditioning circuitry. A main consideration is the fact that virtually any thermocouple employed will have a relatively low output signal, thus they generally require the careful application of stable, high-gain amplifiers.

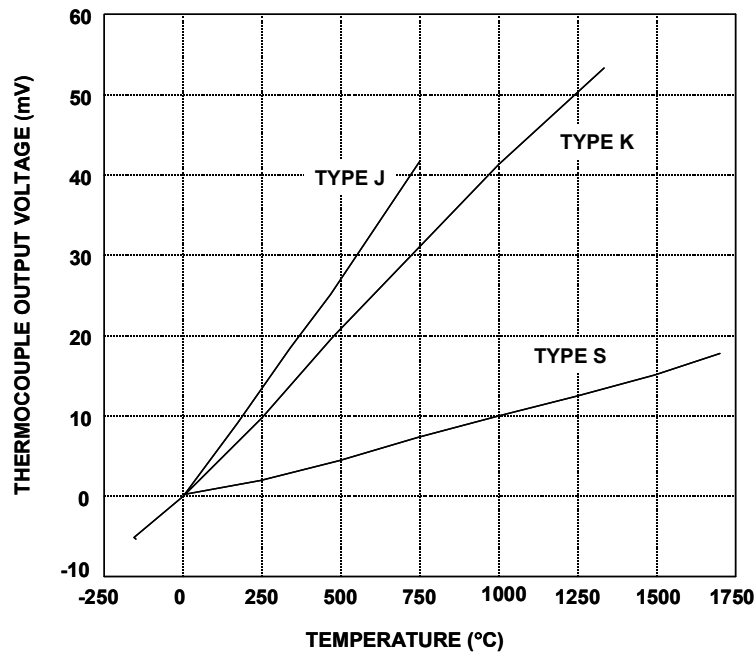


Figure 4-66: Type J, K and S thermocouple output voltage vs. temperature

To understand thermocouple behavior more fully, it is also necessary to consider the non-linearities in their response to temperature differences. As noted, Figure 4-66 shows the relationships between sensing junction temperature and voltage output for a number of thermocouple types (in all cases, the reference *cold* junction is maintained at 0°C).

While close scrutiny of these data may reveal the fact that none of the responses are quite linear, the exact nature of the non-linearity isn't so obvious. What is needed is another perspective on the relationships displayed by these curves, to gain better insight into how the various devices can be best utilized.

■ OP AMP APPLICATIONS

Figure 4-67 below shows how the thermocouple *Seebeck* coefficient varies with sensor junction temperature. The Seebeck coefficient is the *change* of output voltage with *change* of sensor junction temperature (i.e., the first derivative of output with respect to temperature). Note that we are still considering the case where the reference junction is maintained at 0°C.

An ideal linear thermocouple would have a constant Seebeck coefficient with varying temperature, but in practice all thermocouples are non-linear to some degree. In selecting a measurement thermocouple for a particular temperature range, we should therefore choose one whose Seebeck coefficient varies as little as possible over that range.

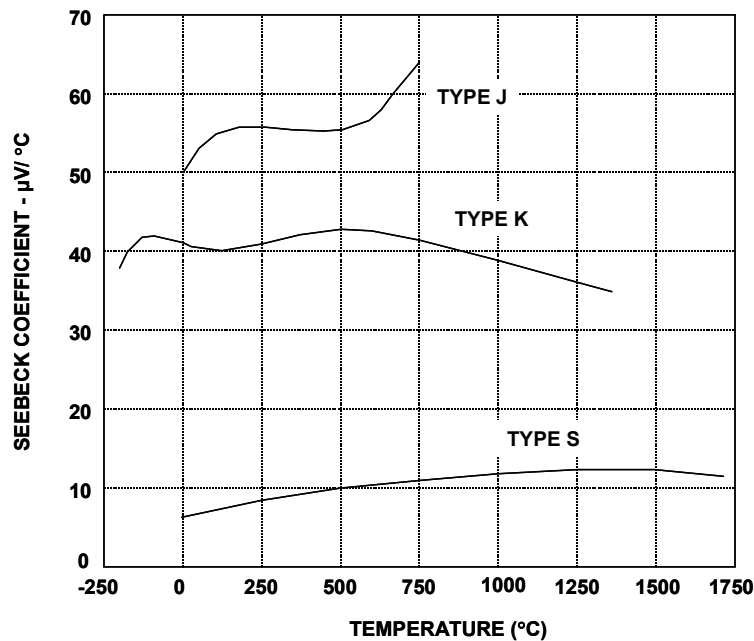


Figure 4-67: Type J, K and S thermocouple Seebeck coefficient vs. temperature

For example, a Type J thermocouple has a nominal Seebeck coefficient of $55\mu\text{V}/^\circ\text{C}$, which varies by less than $1\mu\text{V}/^\circ\text{C}$ between 200 and 500°C , making it ideal for measurements over this range (the flat region of the upper curve within Fig. 4-67).

Presenting these data on thermocouples serves two purposes. First, Fig. 4-66 illustrates the range and sensitivity of the three thermocouple types so that the system designer can, at a glance, determine that a Type S thermocouple has the widest useful temperature range, but a Type J thermocouple is the more sensitive of the three.

Second, the relative stability of the Seebeck coefficient over temperature provides a quick guide to a thermocouple's linearity. Using Fig. 4-67, a system designer can choose a Type K thermocouple for its relatively linear Seebeck coefficient over the range of 400°C to 800°C , or a Type S over the range of 900°C to 1700°C . The behavior of a thermocouple's Seebeck coefficient is important in applications where variations of temperature rather than absolute magnitude are important. These data also indicate what performance is required of the associated signal-conditioning circuitry.

To apply thermocouples successfully we must also understand their basic operating principles. Consider the diagrams shown in Figure 4-68 below.

If we join two dissimilar metals, A and B, at any temperature above absolute zero, there will be a potential difference between them, i.e., their "thermoelectric EMF " or "contact potential", V_1 . This voltage is a function of the temperature of the measurement junction, T_1 , as is noted in Figure 4-68A. If we join two wires of metal A with metal B at two places, two measurement junctions are formed, T_1 and T_2 (Figure 4-68B). If the two junctions are at different temperatures, there will be a net EMF in the circuit, and a current I will flow, as determined by the EMF $V_1 - V_2$, and the total resistance R in the circuit (Fig. 4-68B, again).

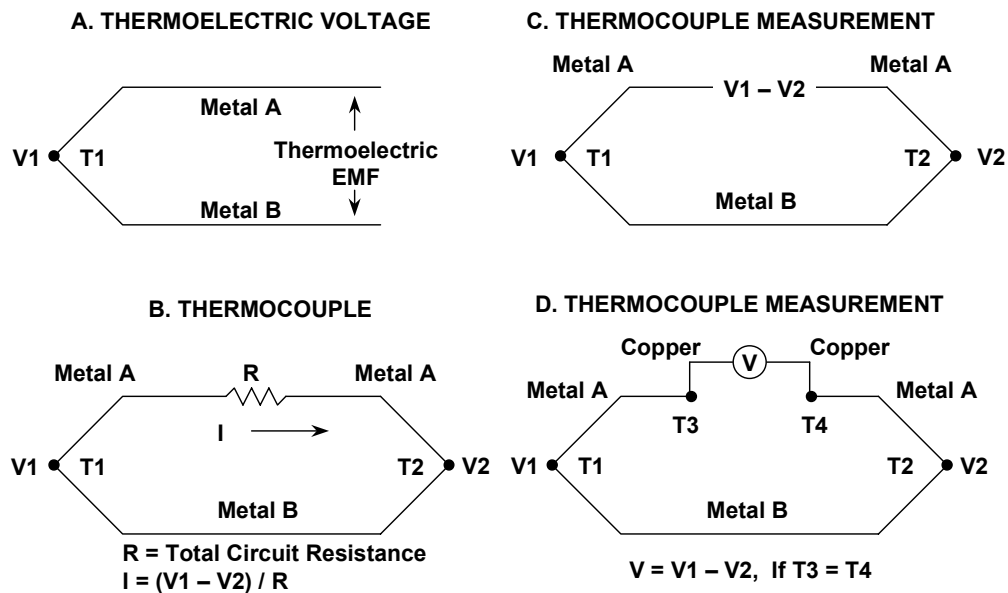


Figure 4-68: Thermocouple basics

If we open the circuit as in Figure 4-68C, the voltage across the break will be equal to the net thermoelectric EMF of the circuit; and if we measure this voltage, we can use it to calculate the temperature difference between the two junctions. We must remember that a thermocouple always measures the temperature *difference* between two junctions, not the absolute temperature at one junction. We can only measure the temperature at the measuring junction if we know the temperature of the other junction. This is the origin of the terms "reference" or "cold" junction.

But of course, it is not so easy to just measure the voltage generated by a thermocouple. *Any wire attached to a thermocouple is also a thermocouple itself*, and if care is not taken, errors can be introduced. Suppose that we attach a voltmeter to the circuit of Fig. 4-68C, as shown in Figure 4-68D. The wires of the voltmeter will form further thermocouple junctions where they are attached (T_3 , T_4). If both these additional junctions are at the same temperature (it does not matter exactly what temperature), then the "Law of Intermediate Metals" states that they will make no net contribution to the total EMF of the system. If they are at different temperatures, they will introduce errors.

■ OP AMP APPLICATIONS

Since every pair of dissimilar metals in contact generates a thermoelectric EMF (including copper/solder, kovar/copper [kovar is the alloy used for IC leadframes] and aluminum/kovar [at the bond inside the IC]), it is obvious that in practical circuits the problem is even more complex. In fact, it is necessary to take extreme care to ensure that both junctions of each junction pair in series with a thermocouple are at the same temperature, *except for the measurement and reference junctions themselves*.

Thermocouples generate a voltage, albeit a very small one, and don't require excitation for this most basic operation. As was shown in Figure 4-68D, however, two junctions are always involved (T_1 , the measurement junction temperature and T_2 , the reference junction temperature). If $T_2 = T_1$, then $V_2 = V_1$, and the output voltage $V = 0$. Thermocouple output voltages are often defined with respect to a reference junction temperature of 0°C (hence the term *cold* or *ice point* junction). In such a system the thermocouple provides a convenient output voltage of 0V at 0°C . Obviously, to maintain system accuracy, the reference junction must remain at a well-defined temperature (but not necessarily 0°C).

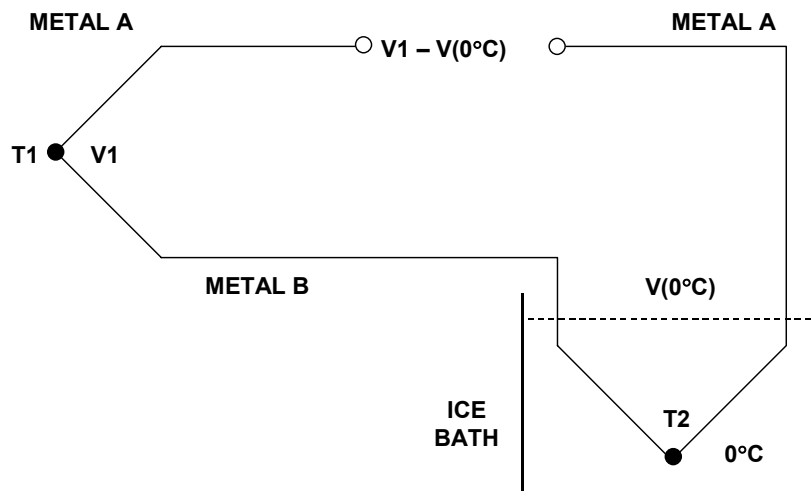


Figure 4-69: A thermocouple cold junction reference system using an ice-point (0°C) T_2 reference

A conceptually simple approach to this need is shown in Figure 4-69 above, the ice-point reference, where junction T_2 is kept at 0°C by virtue of being immersed in an ice water slurry. Although this ice water bath is relatively easy to define conceptually, it is quite inconvenient to maintain.

Today the inconvenient ice/water bath reference is replaced by an electronic equivalent. A temperature sensor of another sort (often a semiconductor sensor, sometimes a thermistor) measures the cold junction temperature and is used to inject a voltage into the thermocouple circuit which compensates for the difference between the actual cold junction temperature and its ideal value (usually 0°C) as shown in Figure 4-70 opposite.

Ideally, the compensation voltage should be an exact match for the difference voltage required, which is why the diagram gives the voltage $V(\text{COMP})$ as $f(T_2)$ (a function of

T_2) rather than KT_2 , where K is a simple constant. In practice, since the cold junction is rarely more than a few tens of degrees from 0°C , and generally varies by little more than $\pm 10^\circ\text{C}$, a linear approximation ($V=KT_2$) to the more complex reality is usually sufficiently accurate and is often used. (Note—the expression for the output voltage of a thermocouple with its measuring junction at $T^\circ\text{C}$ and its reference at 0°C is a polynomial of the form $V = K_1T + K_2T^2 + K_3T^3 + \dots$, but the values of the coefficients K_2 , K_3 , etc. are very small for most common types of thermocouple). References 7 and 8 give the values of these coefficients for a wide range of thermocouples.

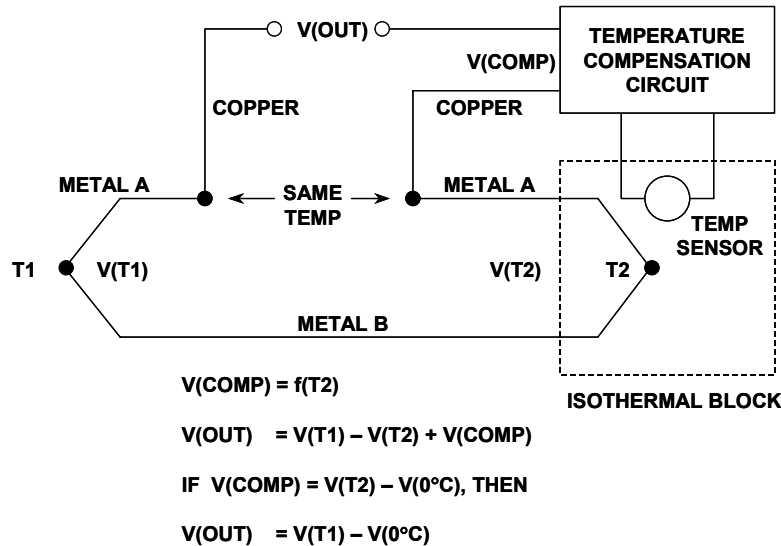


Figure 4-70: A semiconductor temperature sensor can be used to provide cold junction compensation

When electronic cold-junction compensation is used, it is common practice to eliminate the additional thermocouple wire, and terminate the thermocouple leads in the isothermal block, as shown in the arrangement of Figure 4-71 below.

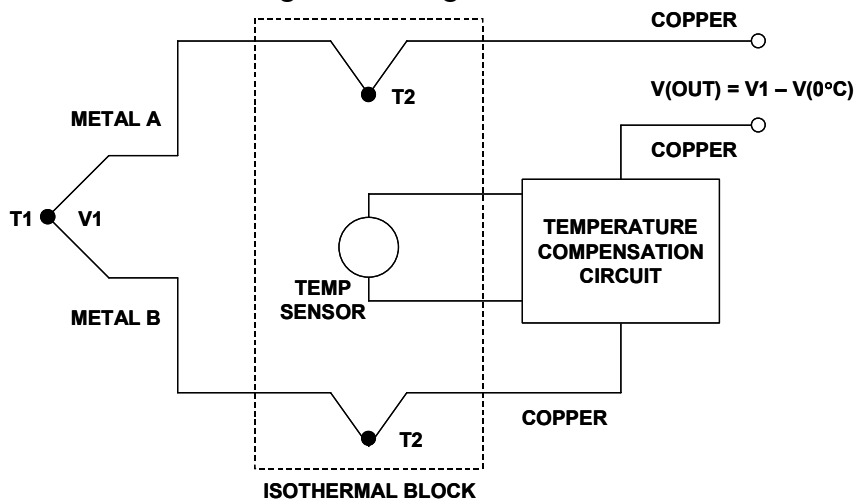


Figure 4-71: Termination of thermocouple leads directly to an isothermal block

In Fig. 4-71, the Metal A-Copper and the Metal B-Copper junctions, if at temperature T_2 , are equivalent to the Metal A-Metal B thermocouple junction in Fig. 4-70.

Type K thermocouple amplifier and cold junction compensator

The circuit in Figure 4-72 below conditions the output of a Type K thermocouple, while providing cold-junction compensation, operating over temperatures between 0°C and 250°C. The circuit operates from single +3.3V to +5.5V supplies with the AD8551, and has been designed to produce a basic output voltage transfer characteristic of 10mV/°C. A Type K thermocouple exhibits a Seebeck coefficient of approximately 40μV/°C; therefore, at the cold junction, the TMP35 voltage output sensor with a temperature coefficient of 10mV/°C is used with divider R1 and R2, to introduce an opposing cold-junction temperature coefficient of -40μV/°C. This prevents the isothermal, cold-junction connection between the circuit's printed circuit board traces and the thermocouple's wires from introducing an error in the measured temperature. This compensation works extremely well for conditioning circuit ambient temperatures of 20°C to 50°C.

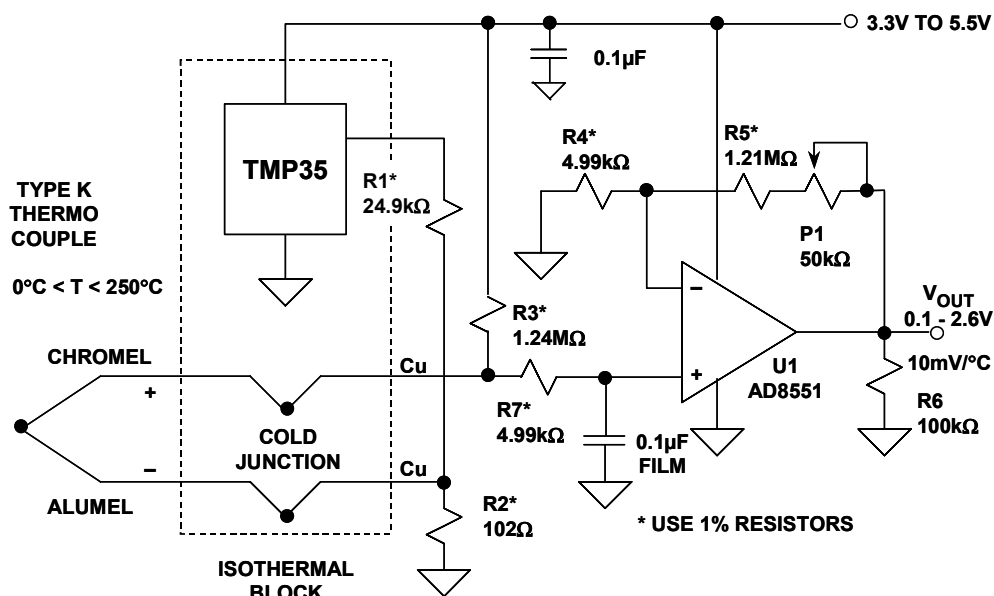


Figure 4-72: Using a TMP35 temperature sensor for cold junction compensation within a Type K thermocouple amplifier-conditioner

Over a 250°C measurement temperature range, the thermocouple produces an output voltage change of ~10mV. Since the circuit's required full-scale output voltage change is 2.5V, the required gain is ~250. Choosing R4 equal to 4.99kΩ sets R5 ~1.24MΩ. With a fixed 1% value for R5 of 1.21MΩ, a 50kΩ potentiometer is used with R5 for fine trim of the full-scale output voltage. The U1 amplifier should be a low drift, very high gain type. A chopper-stabilized AD8551 or an OP777 precision bipolar op amp is suitable for U1.

Both the AD8551 and the OP777 have rail-rail output stages. To extend low range linearity, bias resistor R3 is added to the circuit, supplying an output offset voltage of about 0.1V (for a nominal supply voltage of 5V). Note that this 10°C offset must be subtracted, when making final measurements referenced to the U1 output. Note also that R3 provides a useful open thermocouple detection function, forcing the U1 output to greater than 3V should the thermocouple open. Resistor R7 balances the DC input impedance at the U1 (+) input, and the 0.1μF film capacitor reduces noise coupling.

Single-chip thermocouple signal conditioners

While the construction of thermocouple signal conditioners with op amps and other discrete circuit elements offers great flexibility, this does come at the expense of component count. To achieve a greater level of integration in the thermocouple conditioning function, dedicated thermocouple signal conditioners can be used.

One such solution lies with the AD594 and AD595 (see Reference 9), which are complete, single-chip instrumentation amplifiers and thermocouple cold junction compensators, as shown in Figure 4-73 below. Suitable for use with either Type J (AD594) or Type K (AD595) thermocouples, these two devices combine an ice point reference with a precalibrated scaling amplifier. They provide a high level ($10\text{mV}/^\circ\text{C}$) output working directly from a thermocouple input, without additional precision parts. Pin-strapping options allow the devices to be used as a linear amplifier-compensator, or as a switched output set-point controller with fixed or remote set-point control.

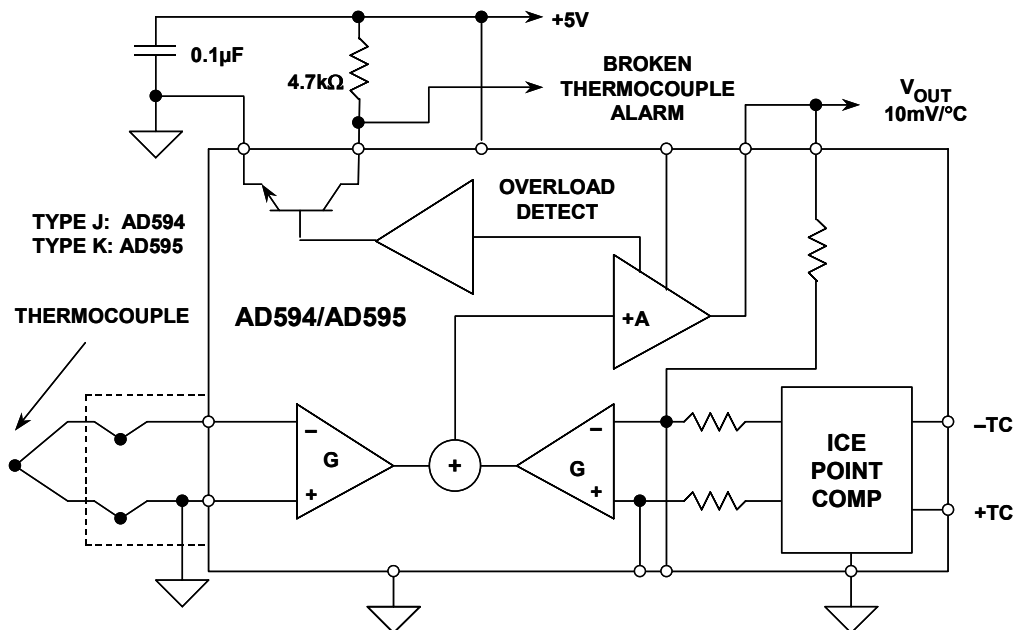


Figure 4-73: Functional diagram of AD594 and AD595 thermocouple signal-conditioning amplifiers

The AD594 and AD595 can be used to amplify the cold junction compensation voltage directly, thereby becoming a stand-alone, $10\text{mV}/^\circ\text{C}$ output Celsius transducer. In such applications it is very important that the IC chip be at the same temperature as the cold junction of the thermocouple; this is usually achieved by keeping the two in close proximity and isolated from any heat sources.

The AD594/AD595 structure includes a flexible thermocouple failure alarm output, which provides broken thermocouple indication. The devices can be powered from either dual or single power supplies (as low as +5V), but the use of a negative supply also allows temperatures below 0°C to be measured. To minimize self-heating, an unloaded AD594/AD595 operates with $160\mu\text{A}$ of supply current, and can deliver $\pm 5\text{mA}$ to a load.

■ OP AMP APPLICATIONS

Although the AD594 is precalibrated by laser wafer trimming to match the characteristics of type J thermocouples, and the AD595 for type K, the temperature transducer voltages and gain control resistors are also made available at the package pins. So, if desired, the circuit can be recalibrated for other thermocouple types with the addition of external resistors. These terminals also allow more precise calibration for both thermocouple and thermometer applications. The AD594/AD595 are available in C and A performance grades, with calibration accuracies of $\pm 1^\circ\text{C}$ and $\pm 3^\circ\text{C}$, respectively. Both are designed to be used with cold junctions between 0 to $+50^\circ\text{C}$.

The 5V powered, single-supply circuit shown of Fig. 4-73 provides a scaled $10\text{mV}/^\circ\text{C}$ output capable of measuring a range of 0 to $+300^\circ\text{C}$. This can be from either a type J thermocouple using the AD594, or a type K with the AD595.

Resistance Temperature Detectors

The Resistance Temperature Detector (RTD), is a sensor whose resistance changes with temperature. Typically built of a platinum (Pt) wire wrapped around a ceramic bobbin, the RTD exhibits behavior which is more accurate and more linear over wide temperature ranges than a thermocouple.

- ◆ **Platinum (Pt) the Most Common**
- ◆ **100 Ω , 1000 Ω Standard Values**
- ◆ **Typical TC = $0.385\% / ^\circ\text{C}$**
= $0.385\Omega / ^\circ\text{C}$ for 100 Ω Pt RTD
- ◆ **Good Linearity - Better than Thermocouple, Easily Compensated**

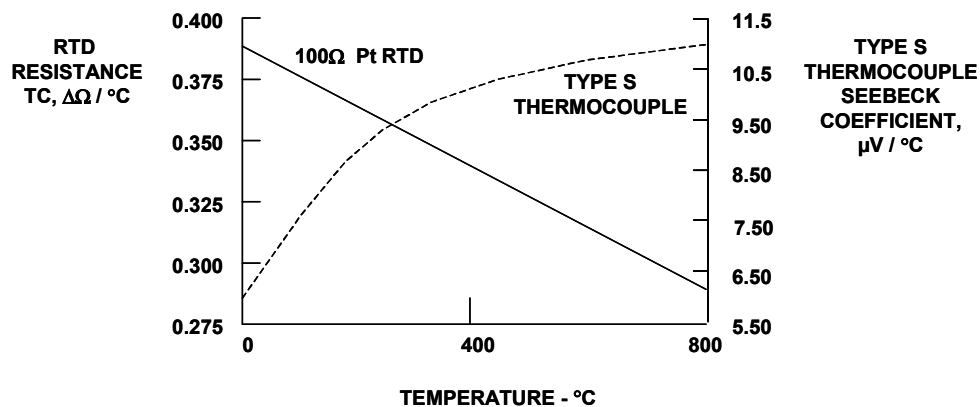


Figure 4-74: Resistance temperature detectors

Figure 4-74 above illustrates the temperature coefficient of a 100 Ω RTD, and the Seebeck coefficient of a Type S thermocouple. Over the entire range (approximately -200°C to $+850^\circ\text{C}$), the RTD is a more linear device. Hence, linearizing an RTD is less complex.

Unlike a thermocouple, however, an RTD is a passive sensor, and a current excitation is required to produce an output voltage. The RTD's low temperature coefficient of

0.385%/°C requires high-performance signal-conditioning circuitry similar to that used by a thermocouple. However, the typical voltage drop seen across an RTD is much larger than a thermocouple's output voltage. A system designer may opt for large value RTDs with higher output, but large-valued RTDs exhibit slow response times. Furthermore, although the cost of RTDs is higher than that of thermocouples, they use copper leads, and thermoelectric effects from terminating junctions do not affect their accuracy. And finally, because their resistance is a function of the absolute temperature, RTDs do not require cold-junction compensation.

Caution must be exercised with the level of current excitation applied to an RTD, because excessive current can cause self-heating. Any self-heating changes the RTD temperature, and therefore results in a measurement error. Hence, careful attention must be paid to the design of the signal-conditioning circuitry so that self-heating errors are kept below 0.5°C. Manufacturers specify self-heating errors for various RTD values and sizes, in both still and moving air. To reduce the error due to self-heating, the minimum current should be used to achieve the required system resolution, and the largest RTD value chosen that results in acceptable response time.

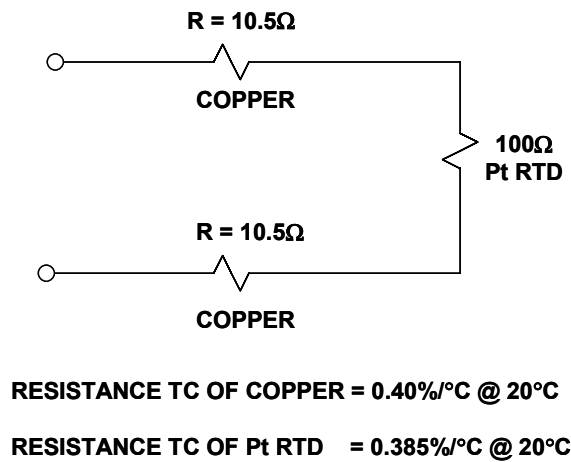


Figure 4-75: A 100 ohm Pt RTD with 100 foot #30 AWG lead wires

Another effect that can produce measurement error is voltage drop in RTD lead wires. This is especially critical with low-value, 2-wire RTDs, because both the temperature coefficient and absolute value of the RTD resistance are small. If the RTD is located a long distance from the signal-conditioning circuitry, the connecting lead resistance can be ohms or tens of ohms. Even this small amount of lead resistance can contribute a significant error to the temperature measurement, as shown above in Figure 4-75.

To illustrate this point, assume that a 100Ω platinum RTD with 30-gauge copper leads is located about 100 feet from a controller's display console. The resistance of 30-gauge copper wire is 0.105Ω/ft, and the two leads of the RTD will contribute a total 21Ω to the network. Uncorrected, this additional resistance will produce a 55°C measurement error! Obviously, the temperature coefficient of the connecting leads can contribute an additional, and possibly significant, error to the measurement.

■ OP AMP APPLICATIONS

To eliminate the effect of the lead resistance, a 4-wire technique is used. In Figure 4-76 below, a 4-wire (Kelvin) connection is made to the RTD. A constant current, I , is applied through the FORCE leads of the RTD, and the voltage across the RTD itself is measured remotely, via the SENSE leads. The measuring device can be a DVM or an in-amp, and high accuracy can be achieved provided that the measuring device exhibits high input impedance and/or low input bias current. Since the SENSE leads don't carry appreciable current, this technique is relatively insensitive to lead wire length. Some major sources of errors in this scheme are the stability of the constant current source, and the input impedance and/or bias currents in the amplifier or DVM, and the associated drift.

RTDs are generally configured in a four-resistor bridge circuit. The bridge output is amplified by an in-amp for further processing. However, high resolution measurement ADCs allow the RTD output to be digitized directly. In this manner, linearization can be performed digitally, thereby easing the analog circuit requirements considerably.

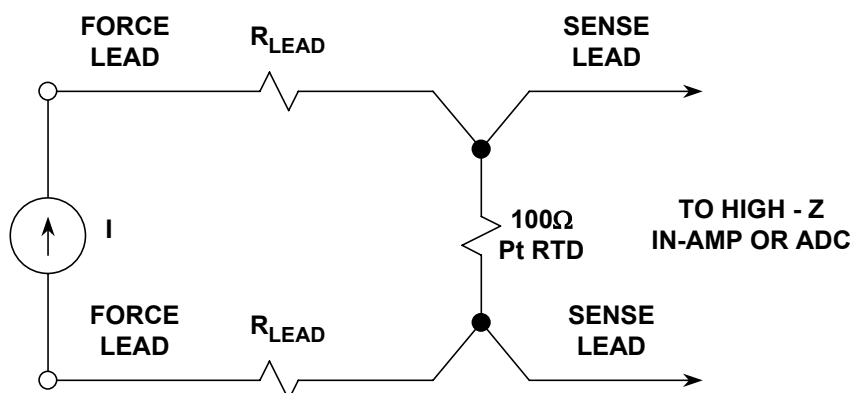


Figure 4-76: Use of Kelvin or 4-wire Pt RTD connections provides high accuracy

For example, an RTD output can be digitized by one of the AD77XX series high resolution ADCs. Figure 4-77 (opposite) shows a 100Ω Pt RTD, driven with a 400μA excitation current source. Note that the 400μA RTD excitation current source also generates a 2.5V reference voltage for the ADC, by virtue of flowing in 6.25kΩ resistor, R_{REF} , with the drop across this resistance being metered by the ADC's V_{REF} (+) and (-) input terminals.

It should be noted that this simple scheme has great benefits (beyond the obvious one of simplicity). Variations in the magnitude of the 400μA excitation current do not affect circuit accuracy, since both the input voltage drop across the RTD as well as the reference voltage across R_{REF} vary ratiometrically with the excitation current. However, it should be noted that the 6.25kΩ resistor must be a stable type with a low temperature coefficient, to avoid errors in the measurement. Either a wirewound resistor, or a very low TC metal film type is most suitable for R_{REF} within this application.

SENSOR SIGNAL CONDITIONING TEMPERATURE SENSORS

In this application, the ADC's high resolution and the gain of 1 to 128 input PGA eliminates the need for any additional conditioning. The high resolution ADC can in fact perform virtually all the conditioning necessary for an RTD, leaving any further processing such as linearization to be performed in the digital domain.

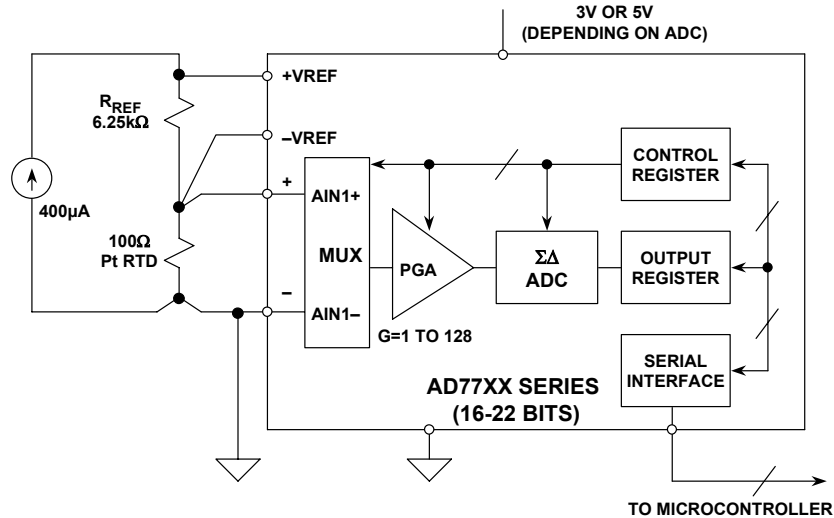


Figure 4-77: A Pt RTD interfaced to the AD77XX series of high resolution ADCs

Thermistors

Similar in general function to RTDs, thermistors are low-cost temperature-sensitive resistors, constructed of solid semiconductor materials which exhibit a positive or negative temperature coefficient. Although positive temperature coefficient devices do exist, the most common thermistors are negative temperature coefficient (NTC) devices.

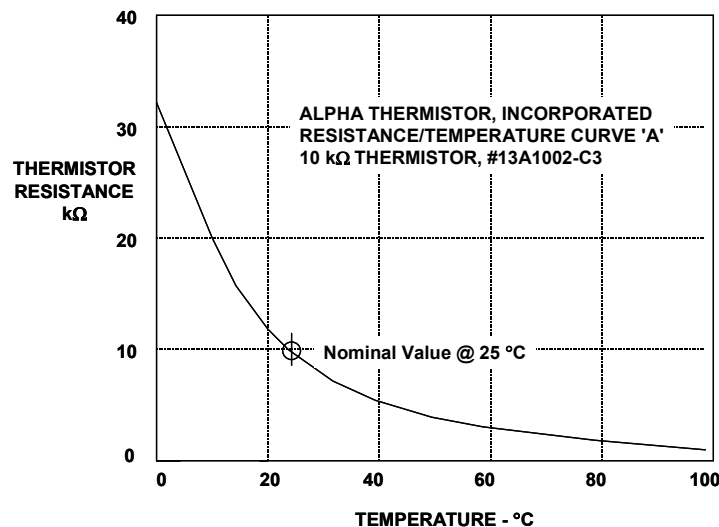


Figure 4-78: Resistance characteristics for a 10kΩ NTC thermistor

Figure 4-78 above shows the resistance-temperature characteristic of a commonly used NTC thermistor. Although the thermistor is the most non-linear of the three temperature sensors discussed, it is also the most sensitive.

■ OP AMP APPLICATIONS

The thermistor's very high sensitivity (typically $-44,000\text{ppm}/^{\circ}\text{C}$ at 25°C) allows it to detect minute temperature variations not readily observable with an RTD or thermocouple. This high sensitivity is a distinct advantage over the RTD, in that 4-wire Kelvin connections to the thermistor aren't needed for lead wire error compensation. To illustrate this point, suppose a $10\text{k}\Omega$ NTC thermistor with a typical 25°C temperature coefficient of $-44,000\text{ppm}/^{\circ}\text{C}$ were substituted for the 100Ω Pt RTD in the example given earlier. The total lead wire resistance of 21Ω would generate less than 0.05°C error in the measurement, using the thermistor in lieu of the RTD. This is roughly a factor of 500 improvement in error sensitivity over an RTD.

However, the thermistor's high sensitivity to temperature does not come without a price. As previously shown in Fig. 4-78, the temperature coefficient of thermistors does not decrease linearly with increasing temperature as with RTDs, and as a result linearization is required for all but the most narrow temperature ranges. Thermistor applications are limited to a few hundred degrees at best, because thermistors are also more susceptible to damage at high temperatures.

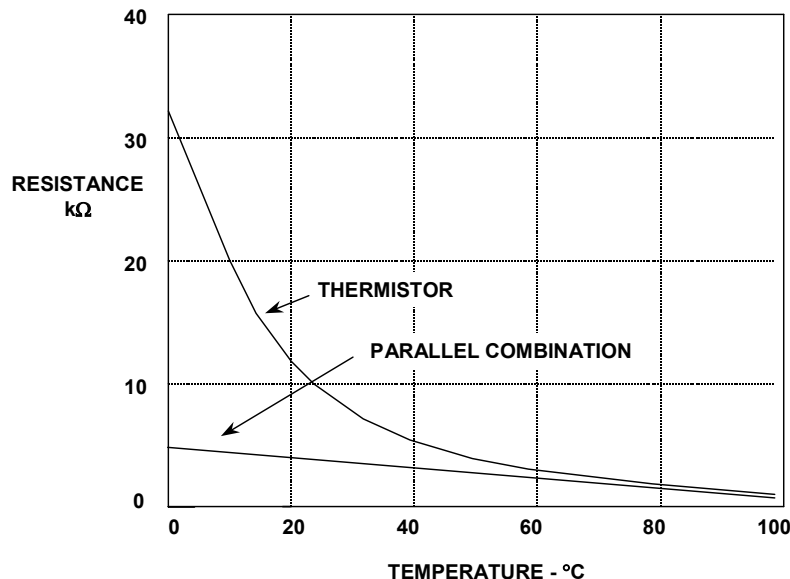


Figure 4-79: Linearization of NTC thermistor using a fixed shunt resistance

Compared to thermocouples and RTDs, thermistors are fragile in construction and require careful mounting procedures to prevent crushing or bond separation. Although a thermistor's response time is short due to its small size, its small thermal mass also makes it very sensitive to self-heating errors.

Thermistors are very inexpensive, highly sensitive temperature sensors. However, we have noted that a thermistor's temperature coefficient can vary, from $-44,000\text{ppm}/^{\circ}\text{C}$ at 25°C , to $-29,000\text{ppm}/^{\circ}\text{C}$ at 100°C . Not only is this non-linearity the largest source of error in a temperature measurement, it also limits useful applications to very narrow temperature ranges without linearization.

As shown in Figure 4-79 above, a parallel resistor combination exhibits a more linear variation with temperature compared to the thermistor itself. This approach to linearizing

a thermistor simply shunts it with a fixed, temperature-stable resistor. Paralleling the thermistor with a fixed resistor increases the linearity significantly. Also, the sensitivity of the combination still is high compared to a thermocouple or RTD. The primary disadvantage of the technique is that linearization is only effective within a narrow range. However, it is possible to use a thermistor over a wide temperature range, if the system designer can tolerate a lower net sensitivity, in order to achieve improved linearity.

R, the value of the fixed shunt resistor, can be calculated from the following equation:

$$R = \frac{RT_2 \cdot (RT_1 + RT_3) - 2 \cdot RT_1 \cdot RT_3}{RT_1 + RT_3 - 2 \cdot RT_2} \quad \text{Eq. 4-11}$$

where RT_1 is the thermistor resistance at T_1 , the lowest temperature in the measurement range, RT_3 is the thermistor resistance at T_3 , the highest temperature in the range, and RT_2 is the thermistor resistance at T_2 , the midpoint, $T_2 = (T_1 + T_3)/2$.

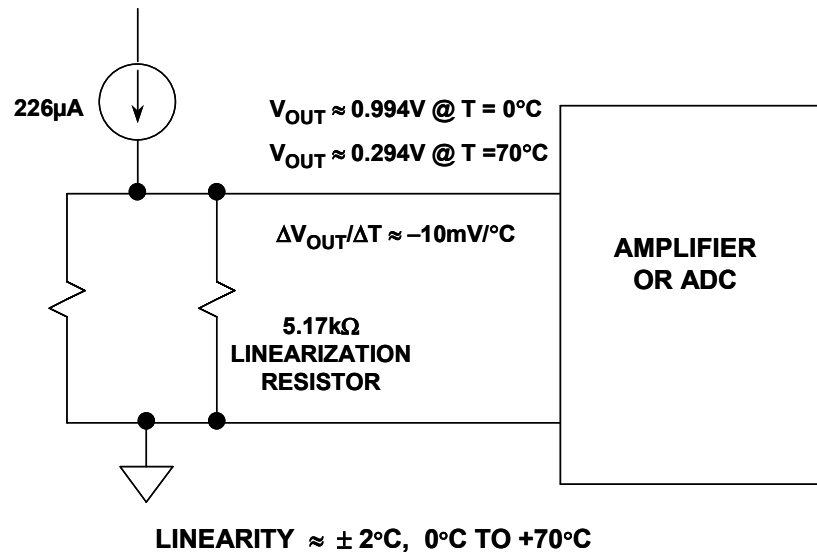


Figure 4-80: Linearized thermistor network with amplifier or ADC

For a typical $10\text{k}\Omega$ NTC thermistor, $RT_1 = 32,650\Omega$ at 0°C , $RT_2 = 6,532\Omega$ at 35°C , and $RT_3 = 1,752\Omega$ at 70°C . This results in a value of $5.17\text{k}\Omega$ for R . The accuracy needed in the associated signal-conditioning circuitry depends on the linearity of the network. For the example given above, the network shows a non-linearity of $-2.3^\circ\text{C}/+2.0^\circ\text{C}$.

The output of the network can be applied to an ADC for digital conversion (with optional linearization) as shown in Figure 4-80 above. Note that the output of the thermistor network has a slope of approximately $-10\text{mV}/^\circ\text{C}$, which implies that an 8 or 10-bit ADC easily has more than sufficient resolution with a full scale range of 1V or less. The further linearization can be applied to the data in the digital domain, if desired.

Semiconductor Temperature Sensors

Modern semiconductor temperature sensors offer high accuracy and high linearity over an operating range of about -55°C to $+150^{\circ}\text{C}$. Internal amplifiers can scale the output to convenient values, such as $10\text{mV}/^{\circ}\text{C}$. They are also useful in cold-junction-compensation circuits for wide temperature range thermocouples.

All semiconductor temperature sensors make use of the relationship between a bipolar junction transistor's (BJT) base-emitter voltage to its collector current:

$$V_{BE} = \frac{kT}{q} \ln\left(\frac{I_C}{I_S}\right) \quad \text{Eq. 4-12}$$

In this expression k is Boltzmann's constant, T is the absolute temperature, q is the charge of an electron, and I_S is a current related to the geometry and the temperature of the transistors. (The equation assumes a voltage of at least a few hundred mV on the collector, and ignores Early effects.)

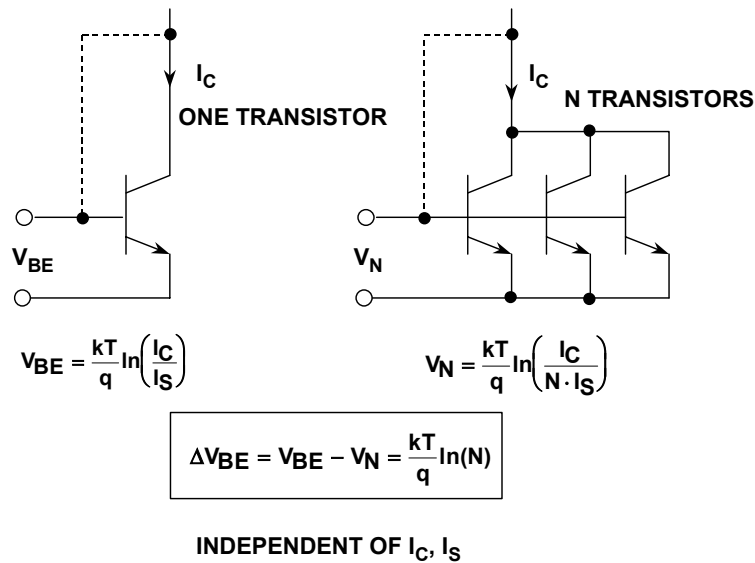


Figure 4-81: The basic relationships for BJT-based semiconductor temperature sensors

If we take ‘ N ’ transistors identical to the first (see Figure 4-81) and allow the total current I_C to be shared equally among them, we find that the new base-emitter voltage applicable to this case is given by the equation

$$V_N = \frac{kT}{q} \ln\left(\frac{I_C}{N \cdot I_S}\right) \quad \text{Eq. 4-13}$$

Neither of these circuits is of much use by itself, because of the strong temperature dependence of I_S . However, if we have equal currents flowing in one BJT, as well as the N similar BJTs, then the expression for the *difference* between the respective base-emitter voltages (or ΔV_{BE}) is proportional to absolute temperature, and it does not contain I_S .

This then leads to a far more useful relationship, developed as follows:

$$\Delta V_{BE} = V_{BE} - V_N = \frac{kT}{q} \ln\left(\frac{I_C}{I_S}\right) - \frac{kT}{q} \ln\left(\frac{I_C}{N \cdot I_S}\right)$$

$$\Delta V_{BE} = V_{BE} - V_N = \frac{kT}{q} \left[\ln\left(\frac{I_C}{I_S}\right) - \ln\left(\frac{I_C}{N \cdot I_S}\right) \right]$$

$$\Delta V_{BE} = V_{BE} - V_N = \frac{kT}{q} \ln \left[\frac{\left(\frac{I_C}{I_S}\right)}{\left(\frac{I_C}{N \cdot I_S}\right)} \right] = \frac{kT}{q} \ln(N)$$

This end result of this algebra is expressed in a single key equation, one worthy of restatement:

$$\Delta V_{BE} = \frac{kT}{q} \ln(N) \quad \text{Eq. 4-14}$$

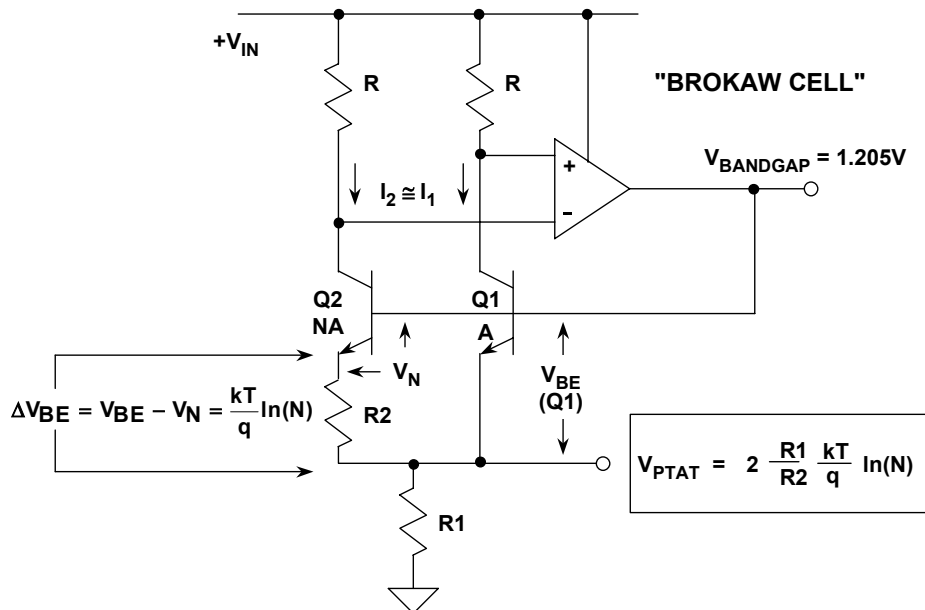


Figure 4-82: The “Brokaw Cell” is both a silicon bandgap voltage based reference as well as a temperature sensor

As one can note, equation 4-14 contains only the transistor emitter area ratio N and T as variables. Since N is fixed within a given design, it can be the basis of a transducer for T measurement. The circuit as shown in Figure 4-82 above implements the above equation, and is popularly known as the “Brokaw Cell”, after its inventor (see Reference 10).

■ OP AMP APPLICATIONS

The voltage $\Delta V_{BE} = V_{BE} - V_N$ appears across resistor R2, as noted. The emitter current in Q2 is therefore $\Delta V_{BE}/R2$. The op amp's servo loop and the two resistors 'R' force an identical current to flow through Q1. The Q1 and Q2 currents are equal, and they are summed, flowing in resistor R1.

The corresponding voltage developed across R1 is V_{PTAT} , a voltage proportional to absolute temperature (PTAT). This is given by:

$$V_{PTAT} = \frac{2R1(V_{BE} - V_N)}{R2} = 2 \frac{R1}{R2} \frac{kT}{q} \ln(N)$$

Within the circuit, a voltage labeled as $V_{BANDGAP}$ appears at the base of Q1, and, as can be noted, is the sum of $V_{BE(Q1)}$ and V_{PTAT} . When this voltage is set by the design to be exactly equal to the bandgap voltage of silicon, it will then become independent of temperature. The voltage $V_{BE(Q1)}$ is complementary to absolute temperature (CTAT), and summing it with a properly proportioned V_{PTAT} from across R1 gives the desired end result; the bandgap voltage becomes constant with respect to temperature. Note that this assumes the proper choice of R1/R2 ratio and N, so as to make the summed voltage equal to $V_{BANDGAP}$, the silicon bandgap voltage (in this instance 1.205V). This circuit has the virtues of dual application because of the above features. It is useful as a basic silicon bandgap temperature sensor, with either direct or scaled use of the voltage V_{PTAT} . It is also widely used a temperature stable reference voltage source, by suitable scaling of $V_{BANDGAP}$ to standard outputs of 2.500, 5.000V, etc.

Current and Voltage Output Temperature Sensors

The concepts used in the bandgap voltage temperature sensor discussion above can also be used as the basis for a variety of IC temperature sensors, with linear, proportional-to-temperature outputs, of either current or voltage.

The AD592 device shown in Figure 4-83 (opposite) is a two-terminal, current output sensor with a scale factor of $1\mu A/K$. This device does not require external calibration, and is available in several accuracy grades. The AD592 is a TO92 packaged version of the original AD590 TO52 metal packaged temperature transducer device (see Reference 11).

The simplest operating mode for current mode temperature sensors is to load them with a precision resistor of 1% or better tolerance, and read the output voltage developed with either an ADC or a scaling amplifier/buffer. Figure 4-84 (opposite, lower) shows this technique with an ADC, as applicable to the AD592. The resistor load R1 converts the basic scaling of the sensor ($1\mu A/K$) into a proportional voltage.

Choice of this resistor determines the overall sensitivity of the temperature sensor, in terms of V/K. For example, with a $1k\Omega$ precision resistor load as shown, the net circuit sensitivity becomes $1mV/K$. With a 5V bias on the temperature sensor as shown, the AD592's full dynamic range is allowed with a $1K\Omega$ load. If a higher value R1 is used, higher bias voltage may be required, as the AD592 requires 4V of operating headroom.

The function just described is a Kelvin-scaled temperature sensor, so the ADC will be required to read the full dynamic range of voltage across R1. With an AD592, this span is from the range corresponding to -25°C (248K) to 105°C (378K), which is 0.248 to 0.378V. A 10-bit, 0.5V scaled ADC can read this range directly with $\approx 0.5^{\circ}\text{C}$ resolution.

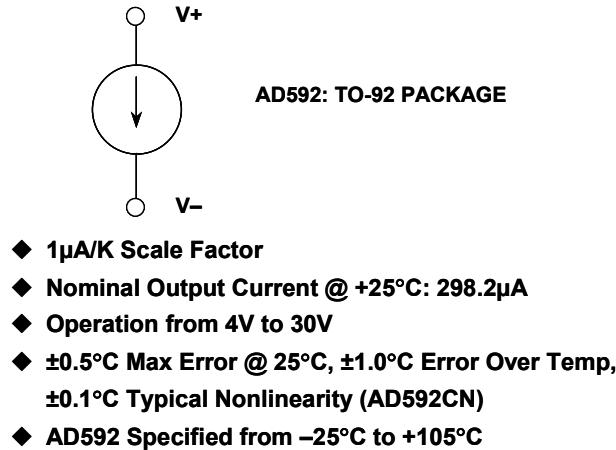


Figure 4-83: Current output absolute temperature sensor

If a centigrade-scaled reading is desired, two options present themselves. For a traditional analog approach, the common terminal of the ADC input can easily be biased with a reference voltage corresponding to 0°C , or 0.273V. Alternately, the 0°C reference can be inserted in the digital domain, with the advantage of no additional hardware requirement.

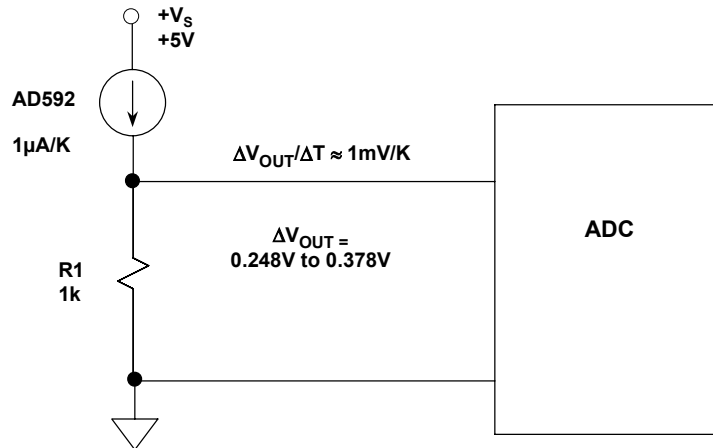


Figure 4-84: Current output temperature sensor driving a resistive load

The AD592 is available in three accuracy grades. The highest grade version (AD592CN) has a maximum error @ 25°C of $\pm 0.5^{\circ}\text{C}$ and $\pm 1.0^{\circ}\text{C}$ error from -25°C to $+105^{\circ}\text{C}$, and a linearity error of $\pm 0.35^{\circ}\text{C}$. The AD592 is available in a TO-92 package.

With regard to stand-alone digital output temperature sensors, it is worthy of note that such devices do exist, that is ADCs with the temperature sensing built in. The AD7816/7817/7818-series ADCs have on-board temperature sensors digitized by a 10-bit 9µs conversion time switched capacitor SAR ADC. The device family offers a variety of input options for flexibility. The similar AD7416/7417/7418 have serial interfaces.

■ OP AMP APPLICATIONS

For a great many temperature sensing applications, a voltage mode output sensor is most appropriate. For this, there are a variety of stand-alone sensors that can be directly applied. In such devices the basic mode of operation is as a three-terminal device, using power input, common, and voltage output pins. In addition, some devices offer an optional shutdown pin.

The TMP35/TMP36 are low voltage (2.7V to 5.5V), SO-8 or TO-92 packaged voltage output temperature sensors with a 10mV/°C scale factor, as shown in Figure 4-85. Supply current is below 50μA, providing very low self-heating (less than 0.1°C in still air).

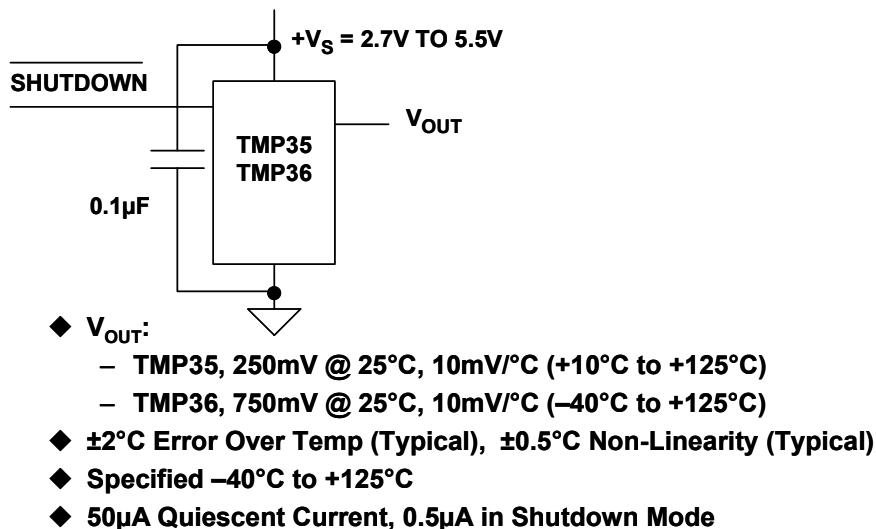


Figure 4-85: *TMP35/36 absolute scaled voltage mode output temperature sensors with shutdown capability*

Output scaling of this device family differs in range and 25°C offset. The TMP35 provides a 250mV output at +25°C, and reads temperature from +10°C to +125°C. The TMP36 is specified from –40°C to +125°C, and provides a 750mV output at 25°C. Both the TMP35 and TMP36 have an output scale factor of +10mV/°C.

An optional shutdown feature is provided for the SO8 package devices, which reduces the standby current to 0.5μA. This pin, when taken to a logic LOW, turns the device OFF, and the output becomes a high impedance state. If shutdown isn't used, the pin should be connected to +V_S.

The power supply pin of these voltage mode sensors should be bypassed to ground with a 0.1μF ceramic capacitor having very short leads (preferably surface mount) and located as close to the power supply pin as possible. Since these temperature sensors operate on very little supply current and could be exposed to very hostile electrical environments, it is also important to minimize the effects of EMI/RFI on these devices. The effect of RFI on these temperature sensors is manifested as abnormal DC shifts in the output voltage due to rectification of the high frequency noise by the internal IC junctions. In those cases where the devices are operated in the presence of high frequency radiated or conducted noise, a large value tantalum electrolytic capacitor (>2.2μF) placed across the 0.1μF ceramic may offer additional noise immunity.

Ratiometric Voltage Output Temperature Sensors

In some cases, it is desirable for the output of a temperature sensor to be *ratiometric* with respect its supply voltage. A series of ADI temperature sensors have been designed to fulfill this need, in the form of the AD2210x series (see references 12–14). Of this series, the AD22103 illustrated in Figure 4-86 below has an output that is ratiometric with regard its nominal 3.3V supply voltage, according to the equation:

$$V_{OUT} = \frac{V_S}{3.3V} \times \left(0.25V + \frac{28mV}{^{\circ}C} \times T_A \right). \quad \text{Eq. 4-15}$$

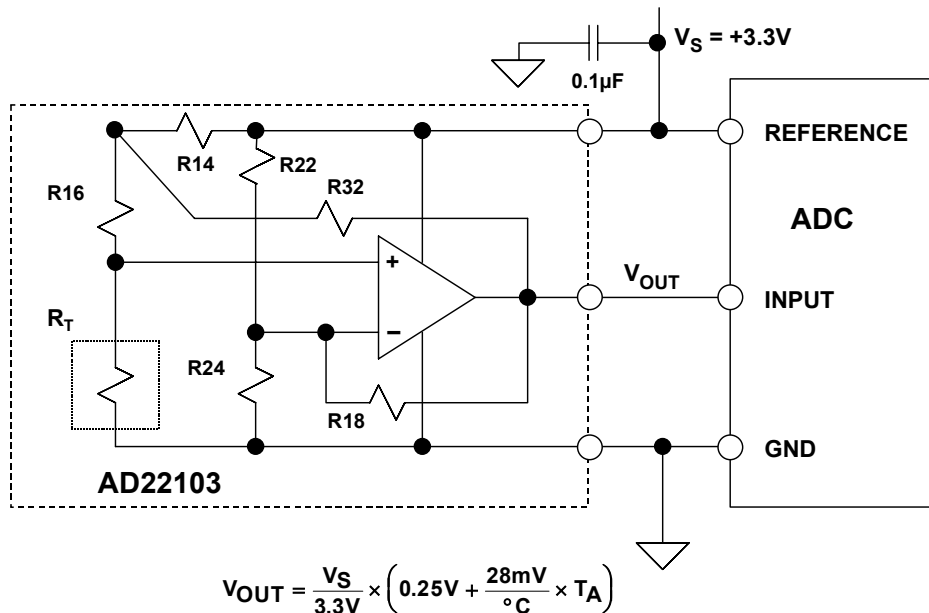


Figure 4-86: Ratiometric voltage output temperature sensors

Note also that the Fig. 4-86 circuit uses the 3.3V AD22103 power supply voltage as the reference input to the ADC. This step eliminates the need for a separate precision voltage reference source. Within a system, this key point potentially can have a positive impact.

Operation of the AD22103 is accomplished with an on-chip temperature sensing resistance R_T , which operates similarly to the RTD types discussed earlier in this section. This resistance is fed from a resistance network comprised of R14, R16 and R32. R14 and R16 provide a current drive component for R_T that is proportional to the supply voltage, a factor that gives the AD22103 a basic sensitivity that is proportional to the supply. R_T , like a classic platinum RTD, exhibits a nonlinear resistance vs. temperature behavior. This nonlinear characteristic of R_T is corrected by a positive feedback loop, composed of R32 along with the Thevenin equivalent of resistances R14 and R16.

Gain scaling for the changing R_T output voltage is provided by the op amp negative feedback loop, R18 and the Thevenin equivalent of resistances R24 and R22. This references the gain network of the op amp to the supply voltage, instead of ground. The various resistance networks around the op amp are actively trimmed at temperature, to calibrate the sensor for its rated offset and scaling.

■ OP AMP APPLICATIONS

The net combination of these factors allows the device to behave in accordance with the relationship of equation 4-15. The AD22103 is specified over a range of 0°C to +100°C, and has an accuracy better than $\pm 2.5^\circ\text{C}$, along with a linearity better than $\pm 0.5\%$ of full scale, i.e., 0.5°C over 100°C .

Since the AD22103 is a single-supply part, the sensing of low temperatures necessarily involves a positive output offset. For example, for 3.3V operation of this example, the output offset is simply the 0.25V term of equation 4-15. Accordingly, the 0 to 100°C temperature span translates to an output swing of 0.25V to 3.05V.

Should it be desired, operation of the AD22103 device is also possible at higher supply voltages. Because of the ratiometric operation feature, this will necessarily involve a change to both the basic sensitivity, as well as the offset. For example, in operating the AD22103 at 5V, the output expression changes to:

$$V_{\text{OUT}} = \frac{V_S}{5V} \times \left(0.378V + \frac{42.42\text{mV}}{^\circ\text{C}} \times T_A \right). \quad \text{Eq. 4-16}$$

However, it should be noted that the fact that the AD22103 is ratiometric does *not* preclude operating the part from a fixed reference voltage. The nominal current drain of the AD22103 is $500\mu\text{A}$, allowing a number of these sensors to be operated from a common reference IC without danger of overload (as well as other analog parts). For example, one such reference family is the REF19x series, which can supply output currents of up to 30mA.

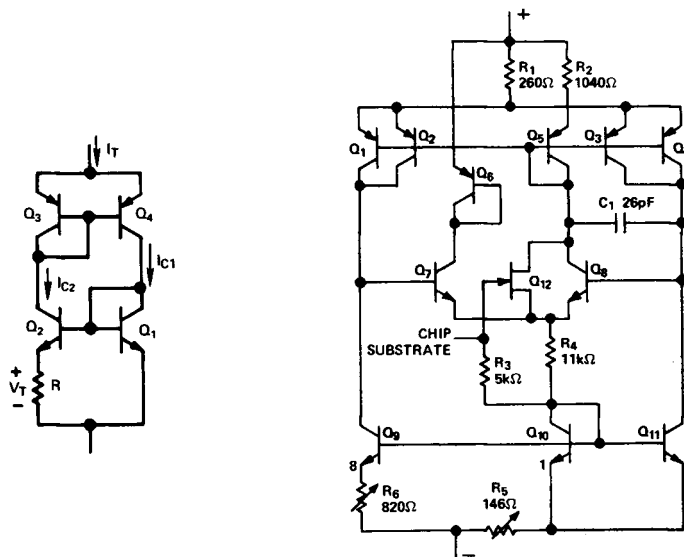
In addition to the above-described AD22103 3.3V part, there is also a companion device, the AD22100. While quite similar to the AD22103 basically, the AD22100 operates from a nominal 5V power supply with reduced sensitivity, allowing operation over a range of -50 to 150°C . Over this range the rated accuracy of the AD22100 is 2% or better, and linearity error is 1% or less.

REFERENCES: TEMPERATURE SENSORS

1. Ramon Pallas-Areny and John G. Webster, **Sensors and Signal Conditioning**, John Wiley, New York, 1991.
2. Dan Sheingold, Editor, **Transducer Interfacing Handbook**, Analog Devices, Inc., 1980, ISBN: 0-916550-05-2.
3. Sections 2, 3, Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, 1992, ISBN: 0-916550-10-9.
4. Sections 1, 6, Walt Kester, Editor, **System Applications Guide**, Analog Devices, 1993, ISBN: 0-916550-13-3.
5. Dan Sheingold, Editor, **Nonlinear Circuits Handbook**, Analog Devices, Inc., 1974.
6. James Wong, "Temperature Measurements Gain from Advances in High-precision Op Amps," **Electronic Design**, May 15, 1986.
7. OMEGA Temperature Measurement Handbook, Omega Instruments, Inc.
8. **Handbook of Chemistry and Physics**, Chemical Rubber Co.
9. Joe Marcin, "Thermocouple Signal Conditioning Using the AD594/AD595," **Analog Devices AN369**.
10. Paul Brokaw, "A Simple Three-Terminal IC Bandgap Voltage Reference," **IEEE Journal of Solid State Circuits**, Vol. SC-9, No. 6, December, 1974, pp. 388-393.
11. Mike Timko, "A Two-Terminal IC Temperature Transducer," **IEEE Journal of Solid State Circuits**, Vol. SC-11, No. 6, December, 1976, pp. 784-788.
12. Data sheet for **AD22103 3.3V Supply, Voltage Output Temperature Sensor with Signal Conditioning**, <http://www.analog.com>
13. Data sheet for **AD22100 Voltage Output Temperature Sensor with Signal Conditioning**, <http://www.analog.com>
14. Adrian P. Brokaw, "Monolithic Ratiometric Temperature Measurement Circuit," **US Patent No. 5,030,849**, filed June 30, 1989, issued July 9, 1991

Classic Cameo

AD590 Two-Terminal IC Temperature Transducer



AD590 basic (left), and complete schematics (right)

Designed by Mike Timko, based on an original Paul Brokaw concept,¹ the AD590^{2,3,4} current mode IC temperature transducer was introduced in 1977. The AD590 established early an elegant method of accurate temperature measurement, based upon fundamental silicon transistor operating principles. It has been in ADI production since introduction, along with such related ICs as the AD592 discussed within this chapter.

The references below discuss operation in great detail, but suffice it to say that in the basic structure (left) a voltage proportional to absolute temperature, V_T , appears across resistor R . This makes the current I_T drawn from an external source proportional to absolute temperature. In the full circuit (right), trimmed thin film resistors implement a calibrated scaling for I_T of $1\mu\text{A/K}$. Additional circuitry is added for startup and for increased accuracy, both with respect to applied voltage as well as against high temperature leakage.

A current-operated transducer such as this is quite convenient to operate, the output being impervious to long lead lengths, and also virtually noise-immune. The low scaling factor also makes the AD590 easy to operate on low voltage supplies without self-heating, yet high output impedance also holds calibration with higher applied voltages. Readout is simply accomplished with a single resistance, making a simple, two-component Kelvin-scaled thermometer possible.

¹ Adrian P. Brokaw, "Digital-to-Analog Converter with Current Source Transistors Operated Accurately at Different Current Densities," **US Patent No. 3,940,760**, filed March 21, 1975, issued Feb. 24, 1976.

² Mike Timko, "A Two-Terminal IC Temperature Transducer," **IEEE Journal of Solid State Circuits**, Vol. SC-11, No. 6, December, 1976, pp. 784-788.

³ Mike Timko, Goodloe Suttler, " $1\mu\text{A/K}$ IC Temperature-to-Current Transducer," **Analog Dialogue**, Vol. 12, No. 1, 1978, pp. 3-5.

⁴ Michael P. Timko, Adrian P. Brokaw, "Integrated Circuit Two Terminal Temperature Transducer," **US Patent No. 4,123,698**, filed July 6, 1976, issued October 31, 1978.

- H Op Amp History**
 - 1 Op Amp Basics**
 - 2 Specialty Amplifiers**
 - 3 Using Op Amps with Data Converters**
 - 4 Sensor Signal Conditioning**
 - 5 Analog Filters**
 - ◆ 6 Signal Amplifiers**
 - 1 Audio Amplifiers**
 - 2 Buffer Amplifiers/Driving Cap Loads**
 - 3 Video Amplifiers**
 - 4 Communications Amplifiers**
 - 5 Amplifier Ideas**
 - 6 Composite Amplifiers**
- 7 Hardware and Housekeeping Techniques**

▣ OP AMP APPLICATIONS

CHAPTER 6: SIGNAL AMPLIFIERS

Walt Jung, Walt Kester

SECTION 6-1: AUDIO AMPLIFIERS

Walt Jung

Audio Preamplifiers

Audio signal preamplifiers (preamps) represent the low-level end of the dynamic range of practical audio circuits using modern IC devices. In general, amplifying stages with input signal levels of 10mV or less fall into the preamp category. This section discusses some basic types of audio preamps, which are:

■ *Microphone*— including preamps for dynamic, electret and phantom powered microphones, using transformer input circuits, operating from dual and single supplies.

■ *Phonograph*— including preamps for moving magnet and moving coil phono cartridges in various topologies, with detailed response analysis and discussion.

In general, when working signals drop to a level of $\approx 1\text{mV}$, the input noise generated by the first system amplifying stage becomes critical for wide dynamic range and good signal-to-noise ratio. For example, if internally generated noise of an input stage is $1\mu\text{V}$ and the input signal voltage 1mV, the best signal-to-noise ratio possible is just 60dB.

In a given application, both the input voltage level and impedance of a source are usually fixed. Thus, for best signal-to-noise ratio, the input noise generated by the first amplifying stage must be minimized when operated from the intended source. This factor has definite implications to the preamp designer, as a "low noise" circuit for low impedances is quite different from one with low noise operating from a high impedance.

Successfully minimizing the input noise of an amplifier requires a full understanding of all the various factors which contribute to total noise. This includes the amplifier itself as well as the external circuit in which it is used, in fact *the total circuit environment must be considered*, both to minimize noise and maximize dynamic range and signal fidelity.

A further design complication is the fact that not only is a basic gain or signal scaling function to be accomplished, but *signal frequency response* may also need to be altered in a predictable manner. Microphone preamps are an example of wideband, flat frequency response, low noise amplifiers. In contrast to this, phonograph preamp circuits not only scale the signal, they also impart a specific frequency response characteristic to it. A major part of the design for the RIAA phono preamps of this section is a systematic analysis process, which can be used to predictably select components for optimum performance in frequency response terms. This leads to very precise functioning, and excellent correlation between a computer based design and measured lab operation.

■ OP AMP APPLICATIONS

Microphone Preamplifiers

The microphone preamplifier (mic preamp) is a basic low level audio amplification requirement. Mic preamps can assume a variety of forms, considering the wide range of possible signal levels, the microphone types, and their impedances. These factors influence the optimum circuit for a specific application. In this section mic preamps are discussed which work with both high and low impedance microphones, both with and without phantom power, and with transformer input stages.

Single-Ended, Single-Supply High-Impedance Mic Preamp

A very simple form of mic preamp is shown in Figure 6-1. This is a non-inverting stage with a single-ended input, most useful with high-impedance microphones such as dynamic and piezoelectric types. As shown it has adjustable gain of 20-40 dB via R_{GAIN} , and is useful with audio sources with 600Ω or greater source impedances.

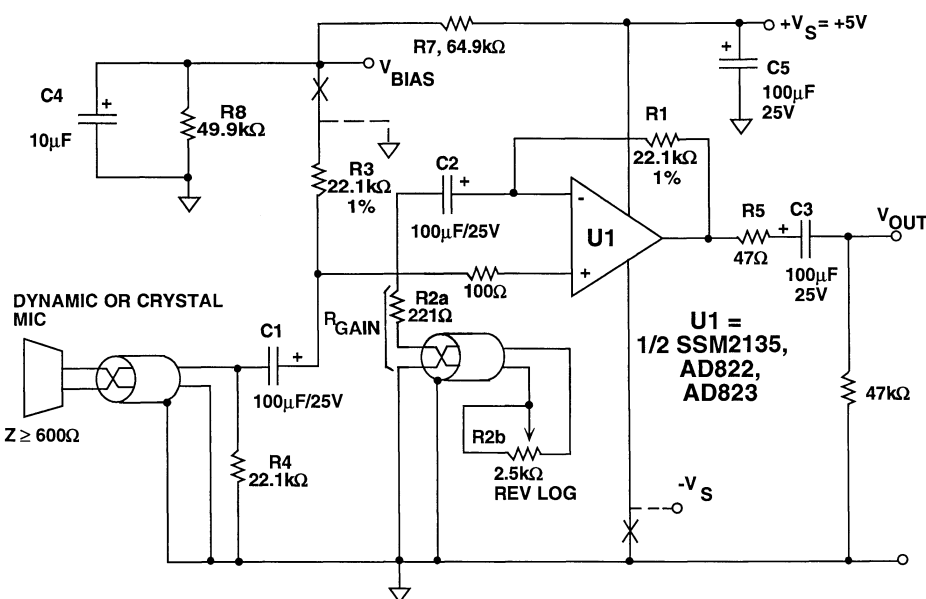


Figure 6-1: A single-ended, single-supply mic preamp

The U1 op amp can greatly affect the overall performance, both in general amplification terms but also in suitability for single supply operation (as shown here). In terms of noise performance, the U1 device should have a low input noise with $\geq 500\Omega$ sources, with the external circuit values adjusted so that the source impedance (microphone) dominates the overall source resistance.

For very low noise on 5V supplies, very few devices are suitable. Among these the dual SSM2135 or the OP213, and AD822/AD823 stand out, and are recommended as first choices. For very low power, minimal quiescent current parts like the AD8541 can be considered. Many other low noise devices can also work well in this circuit for total supply voltages of 10V or more, for example the OP275, and OP270/470 types. The circuit is also easily adapted for dual supply use, as noted below.

In this circuit, gain-determining resistors $R1 \parallel R2$ (where $R2a + R2b = R_{\text{GAIN}}$) are scaled such that their total resistance is less than the expected source impedance, that is $1\text{k}\Omega$ or less. This minimizes the contribution of the gain resistors to input noise, at high gain. As noted, gain of the circuit is adjusted in the feedback path via resistor R_{GAIN} . In a system sense, control of a microphone or other low level channel signal level is preferably done *after* it has undergone some gain, as the case here. R_{GAIN} can of course be a fixed value.

Because of the single supply operation, input/output coupling is via polar capacitors, namely C1, C2, and C3. C4 is a noise filter, and C5 a bypass. For lowest noise in the circuit, the amplifier biasing must also be *noiseless*, that is free from noise added directly or indirectly by the biasing (see Reference 1). Resistors with DC across them should have low excess noise (film types), or be AC-bypassed. Thus R1, R2, R3, R4, R7, and R8 are preferably metal films, with R7-R8 bypassed. A 2.2V bias provided from R7-R8 biases the output of U1 to near mid-supply. If higher supply voltage is used, R7-R8 can be adjusted for maximum output with a particular amplifier. For example, with low bias current, rail-rail output op amps, R7 and R8 should be high, equal values ($\geq 100\text{k}\Omega$).

While the OP213 or SSM2135 for U1 is optimum when operating from lower impedance sources, FET input types such as the AD82x families (or a select CMOS part) is preferable for high impedance sources, such as crystal or ceramic mics. To adapt the circuit for this, R3 and R4 should be $1\text{M}\Omega$ or more, and C1 a $0.1\mu\text{F}$ film capacitor.

Bandwidth using the OP213 or SSM2135 is about 30kHz at maximum gain, or about 20kHz for similar conditions with the AD822 (or AD820). Distortion and noise performance will reflect the U1 device and source impedance. With a shorted input, an SSM2135 measures output noise of about $110\mu\text{V}_{\text{rms}}$ at a gain of 100, with a 1kHz THD+N of 0.022% at 1V_{rms} into a $2\text{k}\Omega$ load. The AD820 measures about $200\mu\text{V}_{\text{rms}}$ with 0.05% THD+N for similar conditions. For both, the figures improve at lower gains.

The circuit of Figure 6-1 is a good one if modest performance and simplicity are required, but requires attention to details. The input cable to the microphone must be shielded, and no longer than required. Similar comments apply to a cable for R_{GAIN} (if remote).

To adapt this circuit for dual supply use, R3 is returned to ground as noted, plus the bias network of R7, R8 and C4 is eliminated. U1 is operated on symmetric supplies ($\pm 5\text{V}$, $\pm 15\text{V}$, etc.), with the $-V_{\text{S}}$ rail bypassed similar to $+V_{\text{S}}$. Coupling caps C1, C2 and C3 are retained, but must be polarized to matched the amplifier used (or non-polar types).

Although microphones with output impedances of less than 600Ω can be used with this circuit, the noise performance will not be optimum. Also, many of these typically require a balanced input interface. Subsequent circuits show methods of optimizing noise with low impedance, balanced output microphones, as suited for professional applications.

■ OP AMP APPLICATIONS

Electret Mic Preamp Interface

A popular mic type for speech recording and other non-critical applications is the electret type. This is a permanently polarized condenser mic, typically with a built in common-source FET amplifier. The amplified output signal is taken from the same single ended lead which supplies the microphone with DC power, typically from a 3-10V DC source.

Figure 6-2 illustrates a basic interface circuit which is useful in powering and scaling the output signal of an electret mic for further use. In this case the scaled output signal from this interface is fed into the LEFT and RIGHT inputs of a +5V supply powered CODEC, for digitization and processing. DC phantom power is fed to the mic capsules by the R_A - C_A - R_B decoupling network from the +5V supply, and the AC output signal is tapped off by C_{IN} - R_2 , and fed to U1. The R_B resistors will vary with different mics and supply voltages, and the values shown are typical. For a quiet mic supply voltage, a filtered/scaled V_R can be generated by the optional U2 connection shown.

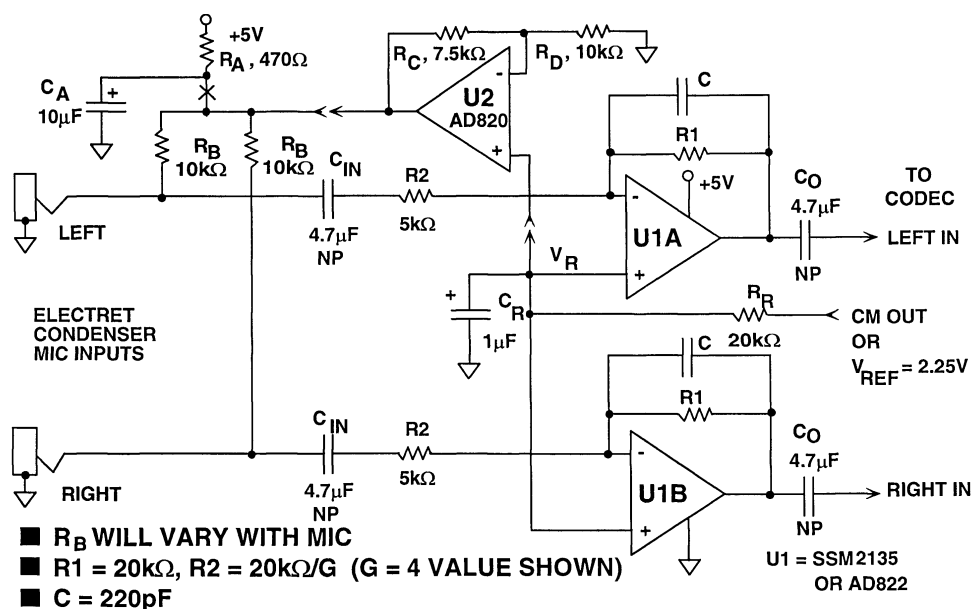


Figure 6-2: An electret mic interface for 5V powered CODECs

The U1 dual scaling amplifier is an SSM2135 or AD822, and is used to normalize the mic signal to either a 1Vrms line level or 100mVrms mic level typically required by CODEC inputs, and also to low pass filter it prior to digitization. With a wide variety of electret mics and operating parameters, some signal level scaling is often required.

The scaling gain is simply R_1/R_2 , and R_2 is selected to provide a gain "G", to yield 0.1Vrms at the mic inputs of the CODEC, with the rated output from the mic. The U1 stages are inverting, so G can be greater or less than unity, i.e., other than 4 as is shown here, to normalize any practical input signal to an optimum CODEC level. The amplifier's low pass corner frequency is set by the time constant R_1 - C , which results in a -3dB point of 36kHz. Bias for the U1 stages is provided from the CODEC, via the reference or CMOUT pins, typically a 2.25-2.5V reference voltage. The low frequency time constants C_{IN} - R_B/R_2 and C_O - $20k\Omega$ are wideband to minimize LF phase shift. These (non-polar) capacitors can be reduced to 1μF or less, for narrowband uses.

Transformer-Coupled Low-Impedance Microphone Preamps

For any op amp, the best noise performance is attained when the characteristic *noise resistance* of the amplifier, R_n , is equal to the source resistance, R_s . Examples of microphone preamps that make use of this factor are discussed in this section. They utilize an input matching transformer to more closely optimize an amplifier to a source impedance which is unequal to the amplifier R_n . A basic circuit operating on this principle is shown in Figure 6-3. In order to select an optimum transformer turns ratio for a given source resistance (R_s), calculate the characteristic R_n of the op amp in use.

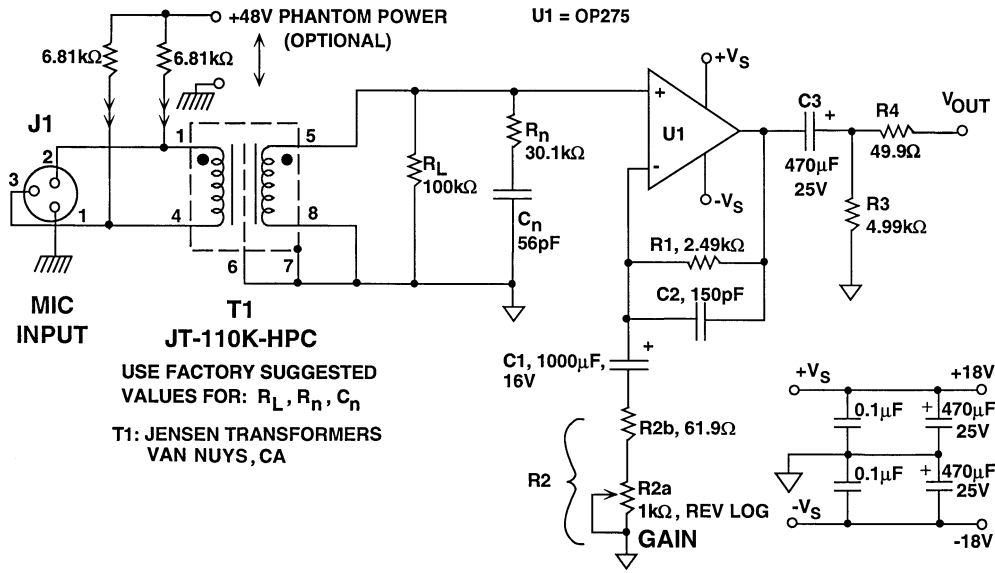


Figure 6-3: Transformer input mic preamplifier with 28 to 50 dB gain

R_n must first be calculated from the op amp's e_n and i_n data as:

$$R_n = \frac{e_n}{i_n} \quad \text{Eq. 6-1}$$

where e_n is in $V/\sqrt{\text{Hz}}$ and i_n is in $A/\sqrt{\text{Hz}}$. A turns ratio for T1 may be calculated as:

$$\frac{N_s}{N_p} = \sqrt{\frac{R_n}{R_s}} \quad \text{Eq. 6-2}$$

where N_s/N_p is the transformer secondary/primary turns ratio. For the OP275 op amp, the values of e_n and i_n are $7\text{nV}/\sqrt{\text{Hz}}$ and $1.5\text{pA}/\sqrt{\text{Hz}}$, respectively; thus,

$$\begin{aligned} R_n &= \frac{e_n}{i_n} = \frac{7 \times 10^{-9}}{1.5 \times 10^{-12}} \\ &= 4.7\text{k}\Omega \end{aligned}$$

■ OP AMP APPLICATIONS

Since both e_n and i_n vary with frequency, R_n will also vary with frequency. Therefore, a value calculated for R_n from the data sheet (such as above) is most accurate at the specified frequency. If the amplifier is to be optimized for a specific frequency, then the e_n and i_n values should be for that frequency. However audio amplifiers are wideband circuits, so latitude is due here. When available, a minimum noise-figure plot for the amplifier will allow graphical determination of the optimum source resistance for noise.

For this case, an optimum transformer turns ratio can be calculated to provide the optimum R_n to the op amp, working from a given R_s . For example, if R_s is 150Ω , then an optimum turns ratio for an OP275 (or other amplifier) with an R_n of $4.7k\Omega$ will be:

$$\frac{N_s}{N_p} = \sqrt{\frac{R_n}{R_s}} = \sqrt{\frac{4.7 \times 10^3}{1.5 \times 10^2}} \\ \approx 5.6$$

Other examples matching these criteria would include OP27 family types.

Transformers are catalogued in fairly narrow and specific impedance ranges, so a unit with a rated secondary impedance in the range of $5k\Omega$ to $10k\Omega$ will be useful (the amplifier minimum noise impedance is reasonably broad). A suitable unit for this purpose is the Jensen JT-110K-HPC. Note that T1 must be adequately shielded and otherwise suitable for operation in low-level environments. The use of the matching transformer allows the circuit to achieve an equivalent input noise (referred to the transformer input) that is only a few decibels above the theoretical limit, or very close to the thermal noise of the source resistance. For example, the thermal noise of a 150Ω resistor in a $20kHz$ noise bandwidth at room temperature is $219nV$. A real circuit has a higher input referred noise, due to the transformer plus op amp noise.

An additional advantage of the transformer lies in the *effective voltage gain* that it provides, due to the step up turns ratio. For a given circuit total numeric gain, G_{total} , this reduces the gain required from the op amp U1, $G_{(U1)}$, to:

$$G_{(U1)} = \frac{G_{total}}{N_s / N_p} \quad \text{Eq. 6-3}$$

Thus, in the composite circuit of Figure 6-3 gain G_{total} is the product of the transformer step up, N_s/N_p , and $(R1 + R2)/R1$, which is $G_{(U1)}$. This has advantages of allowing more amplifier loop gain, thus greater bandwidth and accuracy, lower distortion, etc.

The transformer input example mic preamp stage of Figure 6-3 uses the JT-110K-HPC transformer for T1 with a primary/secondary ratio of about 1/8 ($150\Omega/10k\Omega$). The op amp section has a variable gain of about 3.3-41 times, which, in combination with the 17.8dB transformer gain, yields a composite gain of 28 to 50 dB (26 to 300 times). Transient response of the transformer plus U1 amplifier is excellent. U1 here is $\frac{1}{2}$ an OP275, operating on $\pm 18V$ power. Supplies should be well regulated and decoupled close

to U1, particularly with low impedance loads. *Care should be used to operate U1 below maximum voltage rating.* The OP275 is rated for maximum supplies of $\pm 22\text{V}$.

For best results, passive components should be high-quality, such as 1% metal film resistors, a reverse log taper film pot for R2a, and low ESR capacitors for C1 & C3. Microphone phantom powering (see References 2 and 3) can be used, simply by adding the $\pm 0.1\%$ matched $6.81\text{k}\Omega$ resistors and a 48V DC source, as shown. Close matching of the DC feed resistors is recommended by the transformer manufacturer whenever phantom power is used, to optimize CMR and to minimize the transformer's primary DC current flow (see Reference 4). Note that use of phantom powering has little or no effect on the preamp, since the transformer decouples the CM DC variations at the primary. CMR in an input transformer such as the JT-110K-HPC is typically 85dB or more at 1kHz, and substantially better at lower frequencies.

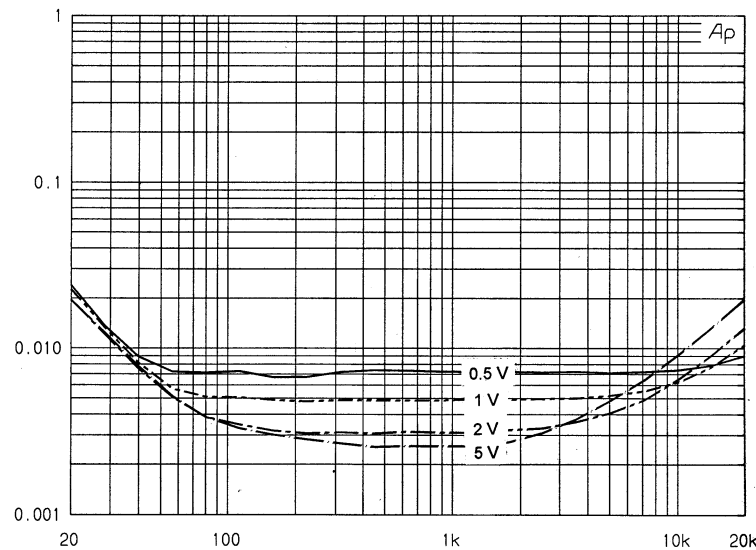


Figure 6-4: Transformer coupled mic preamplifier THD+N (%) versus frequency (Hz) for 35dB gain, outputs of 0.5, 1, 2, and 5Vrms into 600 Ω

THD+N performance versus frequency of this OP275 mic preamp is shown in the family of curves in Figure 6-4. The test conditions are 35dB gain, and successive input sweeps resulting in outputs of 0.5, 1, 2, and 5Vrms into 600 Ω . For these distortion tests as well as most of those following throughout these sections, THD+N frequency sweeps at various levels are used for sensitivity to slewing related distortions (see references 5-7), and output loaded tests are used for sensitivity to load related non-linearities.

For the OP275 data shown in Figure 6-4, there are three interest regions, a sub-100Hz region where distortion is largely transformer-related, a 100Hz-3kHz region where distortion is lowest, and a greater-than-3kHz region where it again rises. For most of the spectrum THD+N is $\leq 0.01\%$ for medium outputs, and slightly higher at high frequencies.

The -3dB bandwidth of this circuit is about 100kHz, and is dominated by the JT-110K-HPC transformer and its termination network, assuming a 150 Ω source impedance. Conversely, for higher or lower source impedances, the bandwidth will lower or rise in proportion, so application of this circuit should take this into account. For example, capacitor microphone capsules with emitter follower outputs appear as a $\approx 15\Omega$ source.

■ OP AMP APPLICATIONS

Very Low Noise Transformer Coupled Mic Preamp

A high performance low noise mic preamp is shown in Figure 6-5, using a lower ratio transformer, the Jensen JT-16A. This transformer has a lower nominal step up ratio of about 2/1, and is optimized for use with lower noise resistance amplifiers such as the AD797. As can be noted from the figure, the general topology is similar to the previous transformer coupled preamp, but some details allow premium levels of performance.

This preamp has a selectable gain feature, using GAIN switch S1 to alter R2 of the feedback network. This varies U1's gain (and thus overall gain) over a range of 20-50dB, making the preamp suitable for a wide range of uses. With the R2 step values shown, gain is selectable in 5dB increments. This ranges from 50dB with R2 (total) = 15Ω, down to 20dB with R2 (total) = 588.5Ω. The transformer provides a fixed gain of about 5.6dB.

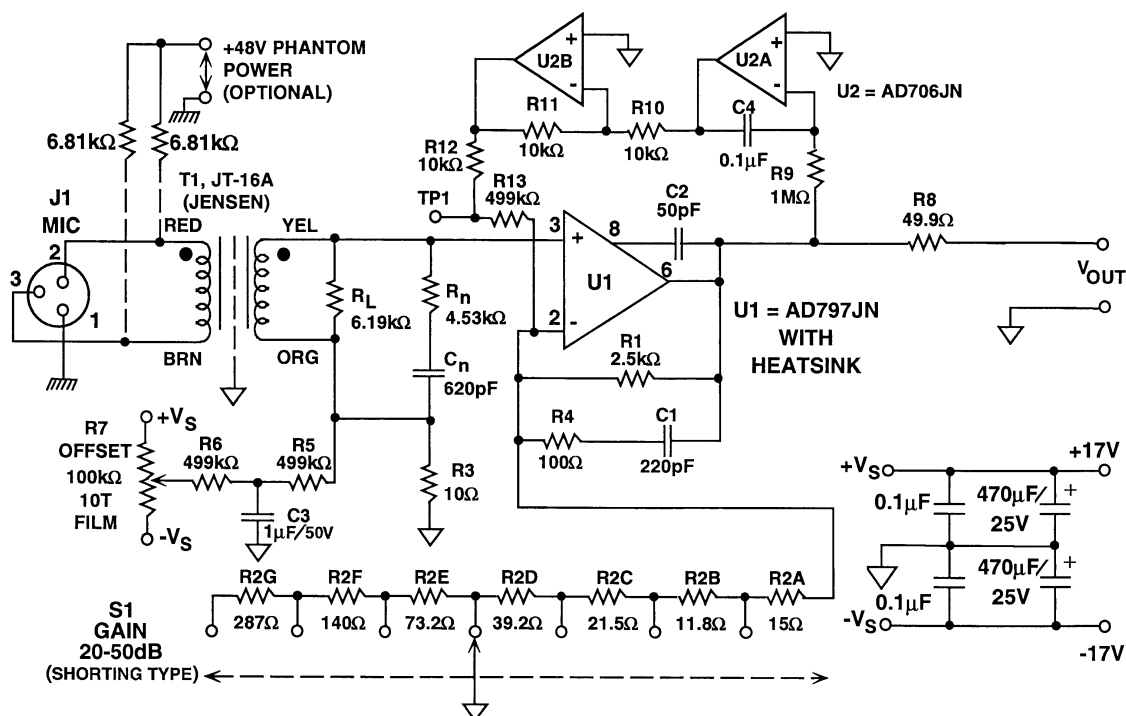


Figure 6-5: Low noise transformer input 20 to 50 dB gain mic preamp

Inasmuch as the AD797 has high precision as well as low distortion audio characteristics, this circuit can be DC coupled quite effectively. This has the worthwhile advantage of eliminating large electrolytic coupling caps in the gain network and in the output coupling between U1 and V_{OUT}. This is accomplished as follows:

The initial device offset of the AD797 is 80μV(max), a factor which allows a relatively simple trim by OFFSET trimmer R7 to null offset. R7 has a range of ±150μV at the AD797 input, with noise well decoupled by C3. With the preamp warmed up well, and working at a mid-range gain setting of 35dB, the offset can be trimmed out. This is best done with the servo temporarily defeated, by grounding test point TP1. Under this condition the V_{OUT} DC level is then trimmed to <1mV, via R7. This nulls out the residual offset of the AD797, and also ensures that the gain-range network sees minimal DC, which minimizes "pops" with gain changes. The offset shift thereafter with gain is only a

few mV, and is of little concern, since the servo circuit of U2A and U2B holds the longer term DC offset to 100 μ V or less, with little gain interaction. Note that for the gain-change scheme to work properly, S1 *must* be a shorting (make-before-break) type.

THD+N performance versus frequency of this mic preamp is shown in Figure 6-6, for conditions of 35dB gain, and successive input sweeps resulting in outputs of 0.5, 1, 2, and 5V_{rms} into 600 Ω . From these data it is essentially clear that the only distortion in the circuit is due to the transformer, which is small and occurs only at the low frequencies. Above 100Hz, the apparent distortion is noise limited, to the highest frequencies.

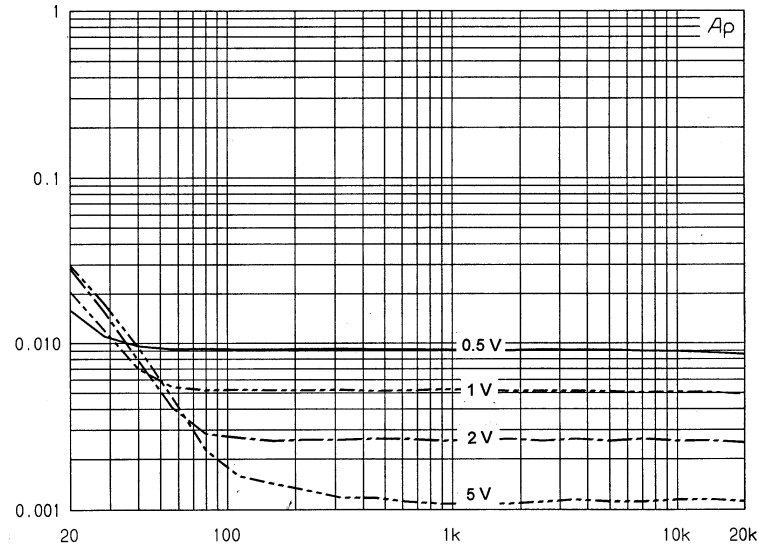


Figure 6-6: Low noise transformer input mic preamp THD+N (%) versus frequency (Hz) for 35dB gain, outputs of 0.5, 1, 2, and 5V_{rms} into 600 Ω

The -3dB bandwidth of this circuit is just under 150kHz, and while this is essentially dominated by the JT-16A transformer and termination, bandwidth does reduce slightly at the highest gain (50dB). Like the previous transformer coupled circuit, this circuit also assumes a 150 Ω source impedance, and similar application caveats apply.

The basic circuit as shown is single-ended with V_{OUT} taken from R8. However, a transformer can be simply added, as an option for driving balanced lines. When this is done, a nickel core type is suggested, for lowest distortion. One type suitable would be a Jensen JT-11-DM (or similar). It is coupled to the U1 output via a 10 Ω resistor.

Just as shown the circuit is suited for local, higher impedance loads of 1k Ω and more. For very high levels of output drive or to drive long lines, a dedicated high current output driver should be used with U1, as generally described in the "Line Drivers" section. This can be most simply implemented by making U1 a composite amplifier, using a AD797 input section plus a follower-type output stage. A good choice for this would be a BUF04 IC, connected between pin 6 of the AD797 and the remaining circuitry. The buffer will isolate the U1 stage, allowing it to operate with highest linearity with difficult loads. Note also that $\pm$ 17V supplies won't be necessary with the AD797 unless extreme voltage swings are required. More conventional ($\pm$ 15V) supplies will minimize the U1 heating.

REFERENCES: MICROPHONE PREAMPLIFIERS

1. C. D. Motchenbacher, F. C. Fitchen, **Low-Noise Electronic Design**, Wiley, New York, 1973, ISBN 0-471-61950-7.
2. G. Bore, "Powering Condenser Microphones," **db**, June 1970.
3. "ANSI Standard 268-15 (Revision 1987, amendments 1989, 1990, 1991)". American National Standards Institute, 11 W. 42nd St., New York, NY, 10036.
4. Steve Hogan, "Standard Mic Input Application," Jensen Transformers Application note JT99-0003, November, 1992.
5. W. Jung, M. Stephens, C. Todd, "Slewing Induced Distortion & Its Effect on Audio Amplifier Performance— With Correlated Measurement/Listening Results," **presented at 57th AES convention**, May 1977, AES preprint # 1252.
6. W. Jung, M. Stephens, C. Todd, "An Overview of SID and TIM," Parts 1-3, **Audio**, June, July, August, 1979.
7. Walter G. Jung, **Audio IC Op Amp Applications, 3d Ed.**, Howard W. Sams & Co., 1987, ISBN 0-672-22452-6.
8. W. Jung, A. Garcia, "A Low Noise Microphone Preamp with a Phantom Power Option," **Analog Devices AN242**, November 1992.
9. Walt Jung, "Audio Preamplifiers, Line Drivers, and Line Receivers," within Chapter 8 of Walt Kester, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3, pp. 8-1 to 8-100.
10. Walt Jung, "Microphone Preamplifiers for Audio," **Analog Dialogue**, Vol. 28, No. 2, 1994, pp. 12-18.

RIAA Phono Preamplifiers

An example of an audio range preamplifier application requiring equalized frequency response is the RIAA phono preamp. While LP record sales have faded with the establishment of new digital media, for completeness equipment is still designed to include phono playback stages. RIAA preamp stages, as amplifiers with predictable, non-flat frequency response, have more general application connotations. The design techniques within this section are specific to RIAA as an example, but they are also applicable to other frequency dependent amplitude designs in general. The techniques are also useful as a study tool, considering the various approaches advanced to optimize the function of high performance gain with predictable equalization (EQ). These last two points make these discussions useful in a much broader sense.

Some RIAA Basics

The RIAA equalization curve (see Reference 1) is shown in Figure 6-7, expressed as it is relative to DC. This curve indicates maximum gain below 50Hz (f_1), with two high-frequency inflection points. Above f_1 , the gain rolls off at 6 dB/octave until a first high-frequency breakpoint is reached at 500Hz (f_2). Gain then remains relatively constant until a second high-frequency breakpoint is reached at about 2.1kHz (f_3), where it again rolls off at 6 dB/octave through the remainder of the audio region and above.

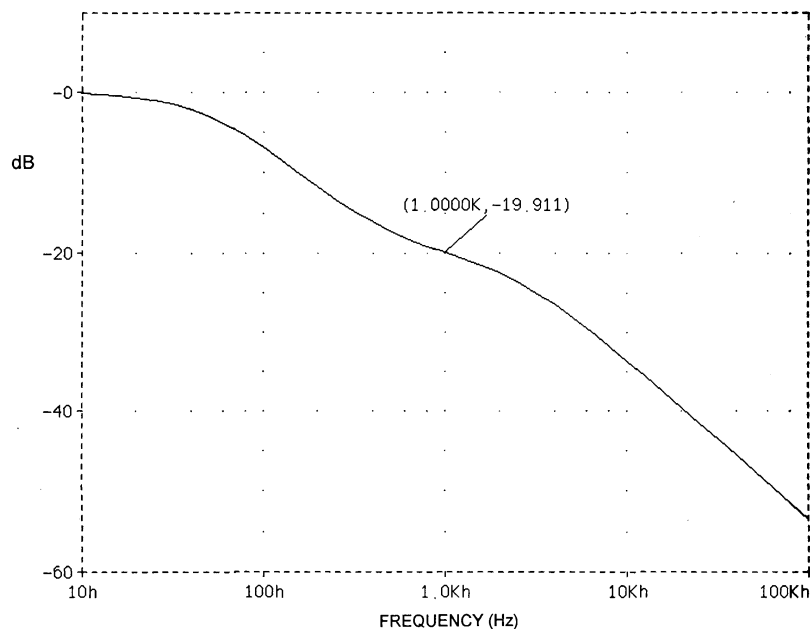


Figure 6-7: *Ideal RIAA de-emphasis (time constants of 3180, 318, 75 μ s)*

Use of a low frequency rolloff (f_0 , not shown) is at the option of the designer. Frequency response can be extended towards DC, or, alternately, rolled off at a low frequency below 50 Hz. When applied, this roll off is popularly called a "rumble" filter, as it reduces turntable/record related low-frequency disturbances, lessening low-frequency driver overload. This rolloff may or may not coincide with a fourth time constant (below).

■ OP AMP APPLICATIONS

However, gain at the frequencies f_1 , f_2 , and f_3 describes the basic RIAA curve. In the standard, this is described in terms of three corresponding time constants, T_1 , T_2 , and T_3 , defined as $3180\mu\text{s}$, $318\mu\text{s}$, and $75\mu\text{s}$, respectively (Reference 1, again). The T_1 - T_3 are here described as they correspond to *ascending frequency*, the reverse of the terminology in Reference 1 (however the time constants themselves are identical). In some literature occasionally one may find the frequencies corresponding to T_1 , T_2 , and T_3 referenced. These exact frequencies can be found simply by the basic relationship of:

$$f = 1/(2 \bullet \pi \bullet T) \quad \text{Eq. 6-4}$$

So, for the three time constants specified, the frequencies are:

$$f_1 = 1/(2 \bullet \pi \bullet T_1) = 1/(2 \bullet \pi \bullet 3180\text{E-6})$$

$$= 50\text{Hz}$$

$$f_2 = 1/(2 \bullet \pi \bullet T_2) = 1/(2 \bullet \pi \bullet 318\text{E-6})$$

$$= 500\text{Hz}$$

$$f_3 = 1/(2 \bullet \pi \bullet T_3) = 1/(2 \bullet \pi \bullet 75\text{E-6})$$

$$= 2122\text{Hz}$$

An IEC amendment to the basic RIAA response adds a fourth time constant of $7950\mu\text{s}$, corresponding to an f_0 of 20Hz when used (see Reference 2). Use of this rolloff has never been standardized in the US, and isn't treated in detail here.

The characteristic gain in dB for an RIAA preamp is generally specified relative to a 1kHz reference frequency. For convenience in evaluating the RIAA curve numerically, Figure 6-8 (opposite) is a complete 10-100kHz relative decibel table for the three basic RIAA time constants. From these data several key points can be observed: If the 1kHz gain is taken as the zero dB reference, frequencies below or above show higher or lower dB levels, respectively (Note 1, column 2). With a DC 0dB reference, it can be noted that the 1kHz gain is 19.91dB below the DC gain (Note 2, column 3).

Expressed in terms of a gain ratio, this means that in an ideal RIAA preamp the 1kHz gain is always 0.101 times the DC gain. The constant 0.101 is unique to all RIAA preamp designs following the above curve, therefore it can be designated as " K_{RIAA} ", or:

$$K_{\text{RIAA}} = 0.101 \quad \text{Eq. 6-5}$$

This constant logically shows up in the various gain expressions of the RIAA preamp designs following. In all examples discussed here (and virtually all RIAA preamps in general), the shape of the standard RIAA curve is fixed, so specifying gain for a given frequency (1kHz) also defines the gain for all other frequencies.

FREQ	VDB(6)(1)	VDB(5)(2)
1.000E+01	1.974E+01	-1.684E-01
1.259E+01	1.965E+01	-2.639E-01
1.585E+01	1.950E+01	-4.109E-01
1.995E+01	1.928E+01	-6.341E-01
2.512E+01	1.895E+01	-9.654E-01
3.162E+01	1.847E+01	-1.443E+00
3.981E+01	1.781E+01	-2.103E+00
5.012E+01	1.694E+01	-2.975E+00
6.310E+01	1.584E+01	-4.067E+00
7.943E+01	1.455E+01	-5.362E+00
1.000E+02	1.309E+01	-6.823E+00
1.259E+02	1.151E+01	-8.398E+00
1.585E+02	9.877E+00	-1.003E+01
1.995E+02	8.236E+00	-1.167E+01
2.512E+02	6.645E+00	-1.327E+01
3.162E+02	5.155E+00	-1.476E+01
3.981E+02	3.810E+00	-1.610E+01
5.012E+02	2.636E+00	-1.727E+01
6.310E+02	1.636E+00	-1.828E+01
7.943E+02	7.763E-01	-1.913E+01
1.000E+03	8.338E-07	-1.991E+01
1.259E+03	-7.682E-01	-2.068E+01
1.585E+03	-1.606E+00	-2.152E+01
1.995E+03	-2.578E+00	-2.249E+01
2.512E+03	-3.726E+00	-2.364E+01
3.162E+03	-5.062E+00	-2.497E+01
3.981E+03	-6.572E+00	-2.648E+01
5.012E+03	-8.227E+00	-2.814E+01
6.310E+03	-9.992E+00	-2.990E+01
7.943E+03	-1.184E+01	-3.175E+01
1.000E+04	-1.373E+01	-3.365E+01
1.259E+04	-1.567E+01	-3.558E+01
1.585E+04	-1.763E+01	-3.754E+01
1.995E+04	-1.960E+01	-3.951E+01
2.512E+04	-2.158E+01	-4.149E+01
3.162E+04	-2.357E+01	-4.348E+01
3.981E+04	-2.557E+01	-4.548E+01
5.012E+04	-2.756E+01	-4.747E+01
6.310E+04	-2.956E+01	-4.947E+01
7.943E+04	-3.156E+01	-5.147E+01
1.000E+05	-3.356E+01	-5.347E+01

Notes:

(1) denotes 1kHz 0dB reference

(2) denotes DC 0dB reference

Figure 6-8: Idealized RIAA frequency response referred to 1kHz and to DC

It can also be noted from the RIAA curve of Figure 6-7 that the gain characteristic continues to fall at higher frequencies. This implies that an amplifier with unity-gain stability for 100% feedback is ultimately required, which can indeed be true, when a standard feedback configuration is used. There are many circuit approaches which can be used to accomplish RIAA phono-playback equalization, however all must satisfy the general frequency response characteristic of Figure 6-7.

■ OP AMP APPLICATIONS

Equalization Networks for RIAA Equalizers

Two equalization networks well suited in practice to RIAA phono reproduction are illustrated in Figure 6-9a and 6-9b, networks N1 and N2. Both networks with values as listed can yield with high accuracy the three standard RIAA time constants of 3180, 318, and 75 μ s as outlined by network theory (see References 3-6). For convenience, both theoretical values for the ideal individual time constants are shown at the left, as well as closest fit standard "no trim" values to the right. Designers can of course, parallel and/or series RC values as may be deemed appropriate, adhering to network theory.

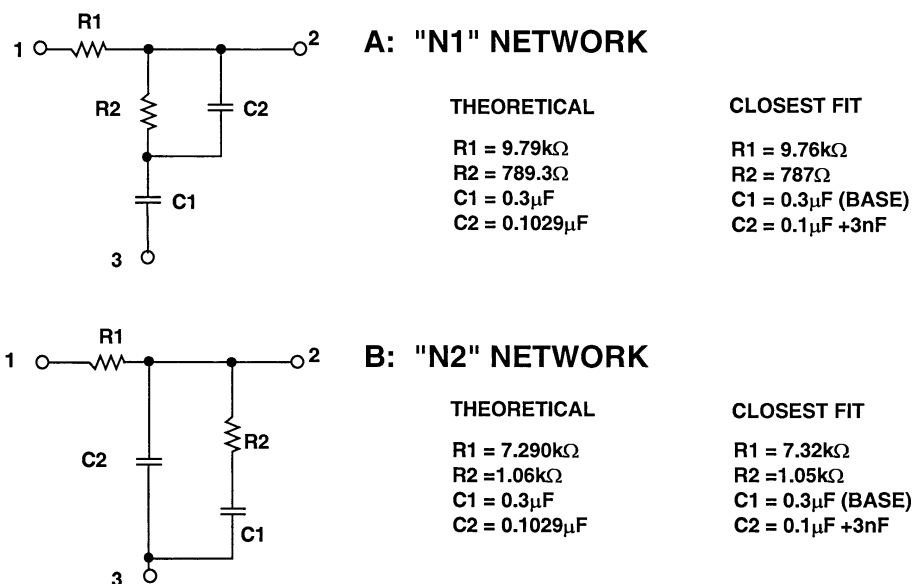


Figure 6-9: Two RIAA EQ networks ($T_1=3180\mu$ s, $T_2=318\mu$ s, $T_3=75\mu$ s)

There are of course an infinite set of possible RC combinations from which to choose network values, but practicality should rule any final selection. A theoretical starting point for a network value selection can begin with *any* component, but in practice the much smaller range of available capacitors suggests their selection first, then resistors, since they have a much broader span of (stock) values. Note that precision film resistors can in fact be obtained (on special order) in virtually *any* value, up to several megohms. The values listed here are those taken as standard from the E96 series.

Very high standards of EQ accuracy are possible, to tolerances of noticeably better than ± 0.1 dB (see for example data from Reference 8, also quoted in 6). In the design process, there are several distinct general aspects of EQ component selection which can impact the ultimate accuracy. These are worth placing in perspective before starting a design.

■ The *selection tolerance* of the component defines how far an ideal (zero manufacturing tolerance) component deviates from the theoretical value. A good design will seek to minimize this error by using either carefully selected standard values, or series and/or shunt combinations, so as to achieve selection tolerance of less than 1%, preferably zero.

■ The *manufacturing tolerance* of the component defines how far an otherwise ideal component deviates from its stated catalog value, such as ± 1 , $\pm 2\%$, etc. This can

obviously be controlled by tighter specifications, but usually at some premium, particularly with capacitors of $\pm 1\%$ or less. Note that a "hidden" premium here can be long delivery times for certain values. Care should be taken to use standard stock values with capacitors— even to the extent that multiple standard values may be preferable (3 times $0.01\mu\text{F}$ for $0.03\mu\text{F}$, as an example).

■ *Topology-related parasitics* must also be given attention, as they can also potentially wreck accuracy. Amplifier gain-bandwidth is one possible source of parasitic EQ error. However, a more likely error source is the parasitic zero associated with active feedback equalizers. If left uncompensated below 100kHz, this alone can be a serious error.

In any event, for high equalization accuracy to be "real", once a basic solid topology is selected, the designer must provide for the qualification of components used, by precise measurement and screening, or tight purchase tolerances. An alternative is iterative trimming against a reference standard such as that of Reference 9, but this isn't suited for production. An example is the data of Reference 8, derived with the network of Reference 9. If used, the utility of such a trim technique lies in the reduction of the equipment accuracy burden. While the comparator used needs to have high *resolution*, the accuracy is transferred to the network comparison standard used.

It should be understood that an appropriately selected high quality network will allow excellent accuracy, for example either N1 or N2 with the "closest fit" (single component) values of exact value yield a broadband error of about $\pm 0.15\text{dB}$. Accuracy about 3 times better than this is achieved with the use of N1 and the composite C2, as noted. The composite C2 is strongly suggested, as without it there is a selection error of about 3%.

It is also strongly recommended that only the highest quality components be employed for use in these networks, for obvious reasons. Regardless of the quality of the remainder of the circuit, it is surely true that the equalization accuracy and fidelity can be no better than the quality of those components used to define the transfer function. Thus only the best available components are used in the N1 (or N2) RC network, selected as follows:

■ *Capacitors*— should have close initial tolerance (1-2%), a low dissipation factor and low dielectric absorption, be non-inductive in construction, and have stably terminated low-loss leads. These criteria in general are best met by capacitors of the Teflon, polypropylene and polystyrene film families, with 1-2% polypropylene types being preferred as the most practical. Types to definitely avoid are the "high K" ceramics. In contrast, "low K" ceramic types, such as "NP0" or "COG" dielectrics, have excellent dissipation factors. See the passive component discussions of Chapter 7 on capacitors, as well as the component-specific references at the end of this section.

■ *Resistors*— should also be close tolerance ($\leq 1\%$), have low non-linearity (low voltage coefficient), be temperature stable, with solid stable terminations and low-loss non-inductive leads. Types which meet these criteria best are the bulk metal foil types and selected thick films, or selected military grade RN55 or RN60 style metal film resistor types. See the passive component discussions of Chapter 7 on resistors, as well as the component-specific references at the end of this section.

■ OP AMP APPLICATIONS

It should be noted also that the specific component values suggested might not be totally optimum from a low impedance, low noise standpoint. But, practicalities will likely deter using appreciably lower ones. For example, one could reduce the input resistance of either network down to say $1\text{k}\Omega$, and thus lower the input referred noise contribution of the network. But, this in turn would necessitate greater drive capability from the amplifier stage, and raise the C values up to $1\text{--}3\mu\text{F}$, where they are large, expensive, and most difficult to obtain. This may be justified for some uses, where performance is the guiding criterion rather than cost effectiveness, or the amplifiers used are sufficiently low in noise to justify such a step. Regardless of the absolute level of impedance used, in any case the components should be adequately shielded against noise pickup, with the outside foils of C1 or C2 connected either to common or a low impedance point.

These very same N1/N2 networks can suffice for both active and passive type equalization. Active (feedback) equalizers use the network simply by returning the input resistor R1 to common, that is jumpering points 1-3, and employing the network as a two-terminal impedance between points 1+3, and 2. Passive equalizers use the same network in a three-terminal mode, placed between two wideband gain blocks.

RIAA Equalizer Topologies

There are of course many different circuit topologies that can be used to realize an RIAA equalizer. Dependent upon the output level of the phono cartridge to be used, the 1kHz gain of the preamp can range from 30 to more than 50dB.

Magnetic phono cartridges in popular use consist of two basic types: moving magnet (MM) and moving coil (MC). The moving magnet types, which are the most familiar, are suitable for the first two circuits described. The moving coil cartridge types are higher performance devices; they are less commonplace but still highly popular.

Functionally, both types of magnetic cartridges perform similarly, and both must be equalized for flat response in accordance with the RIAA characteristic. A big difference in application, however, is the fact that moving magnet types have typical sensitivities of about 1mV of output for each cm/s of recorded velocity. In moving coil types, sensitivity on the order of 0.1mV is more common (for a similar velocity). In application then, a moving coil RIAA preamp must have more gain than one for moving magnets. Typically, 1kHz gains are 40-50 dB for moving coils, but only 30-40 dB for moving magnets.

Noise performance of a moving coil preamp can become a critical performance factor however, because of low-output voltage and low impedance involved—typically this is in the range of just 3-40 Ω . The following circuit examples illustrate techniques that are useful to these requirements.

Actively Equalized RIAA Preamp Topologies

The most familiar RIAA topology is shown in general form in Figure 6-10, and is called an active feedback equalizer, as the network N used to accomplish the EQ is part of an active feedback path (see References 10, 11). In these and all of the following discussions it is assumed that the input from the pickup is appropriately terminated by R_t - C_t , which are selected for flat *cartridge* frequency response driving U1. The following discussions deal with the *amplification* frequency response, given this ideal input signal.

Assuming an adequately high gain amplifier for U1, the gain/frequency characteristics of this circuit are determined largely by the network. The gain of the stage is set by the values of the network N and R_3 , and the U1 output is a low impedance, V_{OUT} . The 1kHz gain of this stage is defined by the RIAA curve and resistors R_1 and R_3 , and is:

$$G = 0.101 \cdot [1 + (R_1/R_3)] \quad \text{Eq. 6-6}$$

where 0.101 is the constant K_{RIAA} . R_1 is within N; R_4 and R_5 are discussed momentarily.

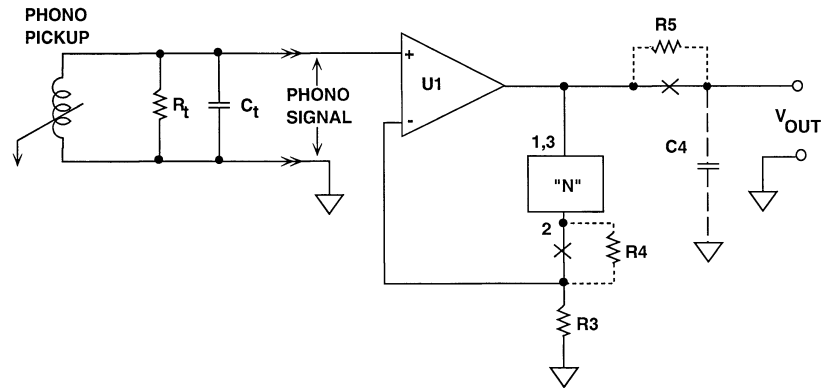


Figure 6-10: Active feedback RIAA equalizer

As noted previously, an ideal RIAA response continues to fall with increasing frequency, and can in fact be less than unity at some high frequency (Fig. 6-7, again). But, the basic U1 topology of Figure 6-10 can't achieve this, as the minimum gain seen at the output of U1 approaches unity at some (high) parasitic zero frequency, where the network equivalent series capacitive impedance of C_1 and C_2 is equal to R_3 . At this zero frequency, the response from U1 simply levels off and ceases to track the RIAA curve.

However, in terms of practical consequence the error created by this zero may or may not be of significance, dependent upon where the zero falls (as determined by gain). If well above audibility (i.e., $\geq 100\text{kHz}$), it will introduce a small equalization error at the upper end of the audio range. For example, if it falls at 100kHz, the 20kHz error is only about 0.3dB. Fortunately, this error is easily compensated by a simple low-pass filter after the amplifier, R_5 - C_4 . The filter time constant is set to match the zero T4, which is:

$$T4 = R_3 \cdot C_{EQUIV} \quad \text{Eq. 6-7}$$

where R_3 is the value required for a specific gain in the design.

■ OP AMP APPLICATIONS

C_{EQUIV} is the series equivalent capacitance of network capacitors C_1 and C_2 , or:

$$C_{EQUIV} = (C_1 \bullet C_2)/(C_1 + C_2) \quad \text{Eq. 6-8}$$

Here the C_{EQUIV} is 7.6nF and R_3 200 Ω , so $T_4 = 1.5\mu s$. The product of R_5 and C_4 are set equal to T_4 , so picking a R_5 value solves for C_4 as:

$$C_4 = T_4/R_5 \quad \text{Eq. 6-9}$$

The 1.5 μs R_5 - C_4 time constant is realized with $R_5 = 499\Omega$ and $C_4 = 3nF$. This design step increases the output impedance, making it more load susceptible. This should be weighed against the added parts and loading. In general, R_5 should be low, i.e., $\leq 1k\Omega$.

In some designs, a resistor R_4 (dotted in Fig. 6-10) may be used with N (for example, for purposes of amplifier stability at a gain higher than unity). With R_4 , T_4 is calculated as:

$$T_4 = (R_3 + R_4) \bullet C_{EQUIV} \quad \text{Eq. 6-10}$$

The R_5 - C_4 product is again chosen to be equal to this T_4 (more on this below).

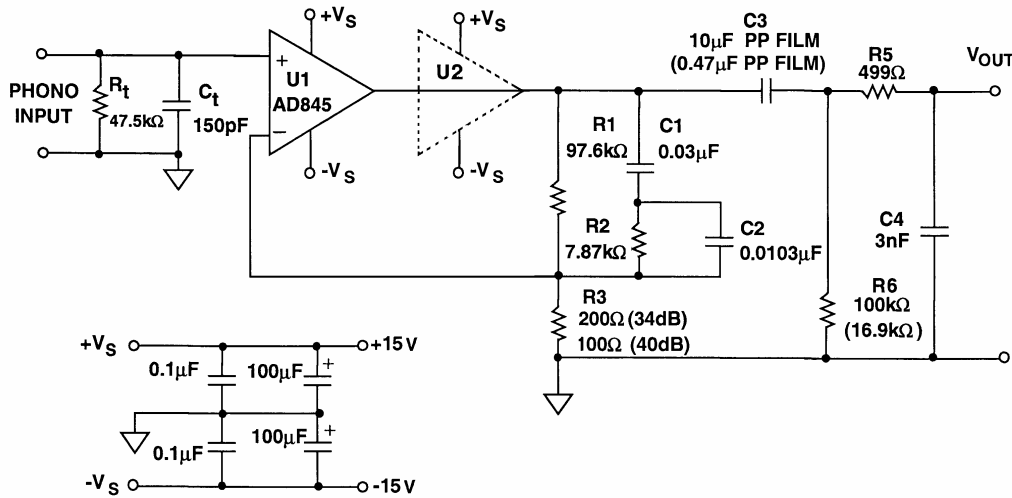


Figure 6-11: A DC-coupled active feedback RIAA moving magnet preamp

The next two schematics illustrate variations of the most popular approach to achieving a simple RIAA phono preamp, using active feedback, as just described. Figure 6-11 above is a high-performance, DC coupled version using precision 1% metal film resistors and 1 or 2% capacitors of polystyrene or polypropylene type. Amplifier U1 provides the gain, and equalization components R_1 - R_2 - C_1 - C_2 form the RIAA network, providing accurate realization with standard component values. N_1 is the network, with 1 and 3 common.

As mentioned, input RC components R_t - C_t terminate the moving magnet cartridge with recommended values (shown as typical). In terms of desired amplifier parameters for optimum performance, they are considerably demanding. For lowest noise from a cartridge's inductive source, the amplifier should have an input voltage noise density of 5nV/ $\sqrt{Hz}$ or less (favoring a bipolar), and an input current noise density of 1pA/ $\sqrt{Hz}$ or less (favoring a FET). In either case, the 1/F noise corner should be as low as possible.

For bipolar-input amplifiers, DC input-bias current can be a potential problem when direct coupling to the cartridge, so in this circuit only a very low input bias current type is suggested. If a bipolar input amplifier is used for U1, it should have an input current of $\ll 100\text{nA}$ for minimum DC offset problems (assuming a typical phono cartridge of $\approx 1\text{k}\Omega$ resistance). Examples are the OP27, OP270 families. FET-input amplifiers generally have negligible bias currents but also tend typically to have higher voltage noise. FET-input types useful for U1 are the AD845 and OP42, even though their voltage noise is not as low as the best of the bipolar devices mentioned. On the plus side, they both have a high output current and slew rate, for low distortion driving the feedback network load (approximately the R_3 value at high frequencies). Of the two, the OP42 has lower noise, the AD845 higher output current and slew rate.

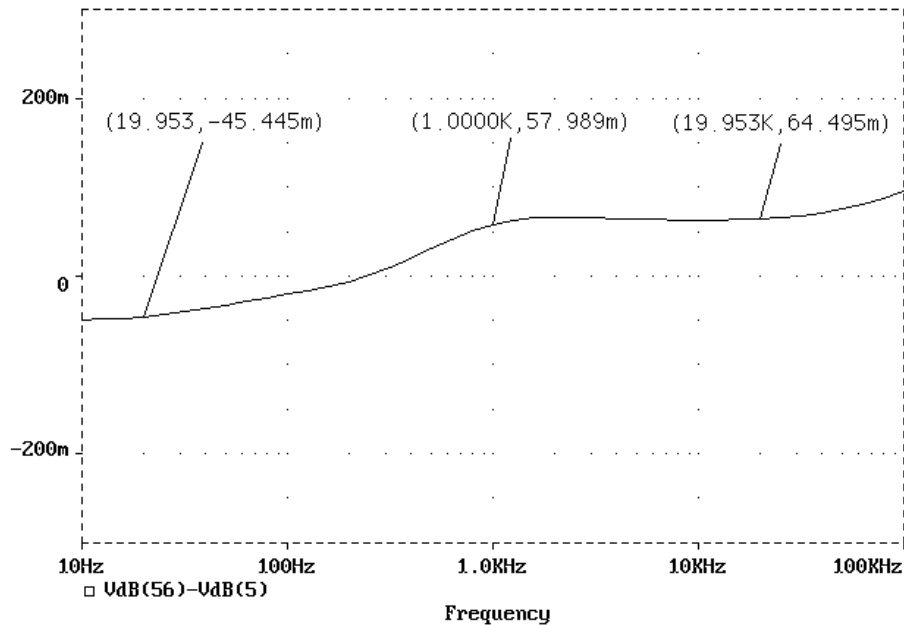


Figure 6-12: *Relative error (B) versus frequency for DC-coupled active feedback RIAA moving magnet preamp, gain of 34dB*

For high gain accuracy at high stage gains, the amplifier should have a high gain-bandwidth product; preferably $>5\text{MHz}$ at audio frequencies. Because of the 100% feedback through the network at high frequencies, the U1 amplifier must be unity-gain-stable. To minimize noise from sources other than the amplifier, gain resistor R_3 is set to a relatively low value, which generates a low voltage noise in relation to the amplifier.

RIAA accuracy is quite good using the stock equalizer values. A PSpice simulation run is shown in Figure 6-12 above for the suggested gain of 34dB. In this expanded scale plot over the 20-20kHz range, the error relative to the 1kHz gain is less than $\pm 0.1\text{dB}$.

As can be noted from Fig. 6-12, the relative amplitude is expanded, to easily show response errors. A perfect response would be a straight line at 0dB, meaning that the circuit under test had exactly the same gain as an ideal RIAA amplifier of the same 1kHz gain. This high sensitivity in the simulation is done via the use of a feature in PSpice allowing the direct entry of Laplace statements (see Reference 10). With this evaluation

■ OP AMP APPLICATIONS

tool, the ideal transfer function of an RIAA equalizer can be readily generated. The key parameters are the three time constants described above, and the ideal DC gain.

The syntax to enable this mode of comparison is contained in the listing of Figure 6-13, which is the PSpice CIR file for the circuit of Fig. 6-11. The Laplace details are all contained within the dotted box, and need only the editing of one value, "ENORM", for gain normalization from one circuit to another (see boldface). In this case ENORM is set to 490.7, to match the ideal R1 and R3 values of Fig. 6-13. When the analysis is run, a difference display of the circuit-under-test and the ideal outputs (i.e., VdB(56)-VdB(5)) shows the relative response (Fig. 6-12, again). Vertical axis scaling is easily adjusted for sensitivity, and is $\pm 300\text{mB}$ as displayed in Fig. 6-12.

```
RIAA34LP: 34 dB gain RIAA preamp with AD845
*
.OPT ACCT LIST NODE OPTS NOPAGE LIBRARY
.AC DEC 10 10 100KHZ
.LIB D:\PS\ADLIB\AD_RELL.LIB
.PRINT AC VDB(5) VDB(56)
.PROBE
VIN 1 0 AC 1E-3
VCC 52 0 +15V
VEE 53 0 -15V
* ----- V(5) = idealized RIAA frequency response -----
*
* Uses Laplace feature of PSpice Analog Behavioral option
* for frequency response reference.
* ENORM = ideal U1 DC gain = 1+(R1/R3) Use ideal values for R1, R3
* T1 - T3 are time constants desired (in  $\mu\text{s}$ ).
* Input = node 1, Laplace Output = node 5
.PARAM ENORM = {490.7}
.PARAM T1 = {3180} ; Reference RIAA constants, do not alter!
.PARAM T2 = {318} ; Reference RIAA constants, do not alter!
.PARAM T3 = {75} ; Reference RIAA constants, do not alter!
*
ERIAA 5 0 LAPLACE {ENORM*V(1)}={ (1+(T2*1E-6)*S)/((1+(T1*1E-6)*S)*(1+(T3*1E-6)*S)) }
RDUMMY5 5 0 1E9
*
* -----
*
* (+) (-) V+ V- OUT
XU3 1 21 52 53 55 AD845
* Active values Theoretical values
R1 55 21 97.6K ; 97.9k
R2 21 8 7.87K ; 7.8931563k
C1 55 8 30NF ; 30nF
C2 21 8 10.3NF ; 10.2881nF
R3 21 0 200 ; 199.9148
C3 55 100 10E-6
R6 100 0 100K
R5 100 56 499
C4 56 0 3.0000E-9
.END
```

Figure 6-13: An example PSpice circuit file which uses the Laplace feature for ideal RIAA response comparison

The 1kHz gain of this circuit can be calculated from Eq. 6-6 above. For the values shown, the gain is just under 50 times ($\approx 34\text{dB}$). Higher gains are possible by decreasing R_3 , but gains $>40\text{dB}$ may show increasing equalization errors, dependent upon amplifier bandwidth. For example, R_3 can be 100Ω for a gain of about 100 times ($\approx 40\text{dB}$). Note that if R_3 is changed to 100Ω , C_4 should also be changed to 1.5nF , to satisfy Eq. 6-9.

Dependent upon the amplifier in use, this circuit is capable of very low distortion over its entire range, generally below 0.01% at levels up to 7Vrms, assuming $\pm 15\text{V}$ supplies. Higher output with $\pm 17\text{V}$ supplies is possible, but will require a heat sink for the AD845. U2 is an optional unity-gain buffer useful with some op amps, particularly at higher gains or with a low-Z network. But this isn't likely to be necessary with U1 an AD845.

For extended low-frequency response, C_3 and R_6 are the large values, with C_3 preferably a polypropylene film type. If applied, the alternate values form a simple 6dB per octave rumble filter with a 20Hz corner. As can be noted from the figure's simplicity, C_3 is the only DC blocking capacitor in the circuit. Since the DC circuit gain is on the order of 54dB, the amplifier used must be a low offset-voltage device, with an offset voltage that is insensitive to the source. Since these preamps are high-gain, low-level circuits ($\geq 50\text{dB}$ of gain at 50/60Hz), supply voltages should be well regulated and noise-free, and reasonable care should be taken with the shielding and conductor routing in their layout.

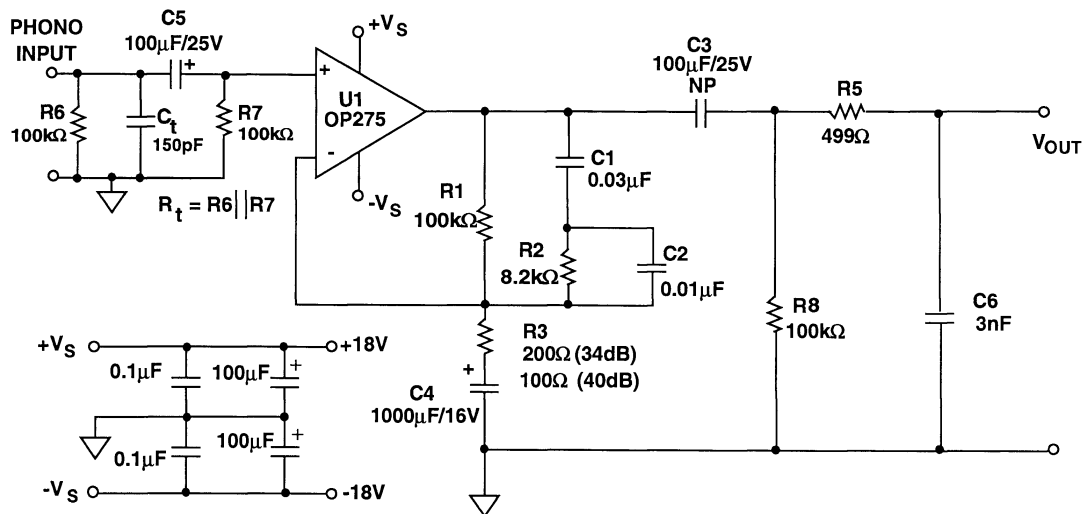


Figure 6-14: An AC-coupled active feedback moving magnet RIAA preamp

Alternately, an inexpensive AC-coupled form of this circuit can be built with higher bias current, low-noise bipolar op amps, for example the OP275, $I_B = 350\text{nA}(\text{max})$, which would tend to make direct coupling to a cartridge difficult. This form of the circuit is shown in Figure 6-14, and can be used with many unity gain stable bipolar op amps.

Here input AC coupling to U1 is added with C_5 , and the cartridge termination resistance R_t is made up of the R_6 - R_7 parallel equivalent. R_3 of the feedback network is AC-grounded via C_4 , a large value electrolytic. These measures reduce the DC offset at the output of U1 to a few mV. Nearest 5% values are also used for the network components, making it easily reproducible and inexpensive. C_3 is a non-polar electrolytic type, and the R_3 - C_4 time constant as shown provides a corner frequency of $\leq 1\text{Hz}$ at the 34dB gain.

Frequency response of this version (not shown) isn't quite as good as that of Figure 6-11, but is still within $\pm 0.2\text{dB}$ over 20-20kHz (neglecting the effects of the low frequency rolloff). If a tighter frequency response is desired, the N1 network values can be adjusted. With a higher rated maximum supply voltage for the OP275, the power supplies of this version can be $\pm 21\text{V}$ if desired, for outputs up to 10Vrms.

■ OP AMP APPLICATIONS

There is another, very useful variation on the actively equalized RIAA topology. This is one that operates at appreciably higher gain and with lower noise, making it suitable for operation with higher output moving coil (MC) cartridges. In this design example, shown below in Figure 6-15, the basic circuit is used is quite similar to that of Fig. 6-11. The lower R_t and C_t values shown are typical for moving coil cartridges. They are of course chosen per the manufacturer's recommendations (in particular the resistance).

To make it suitable for a high-output MC cartridge, a very low-noise FET op amp is used for U1, the AD745. The AD745 is stable at a minimum gain of five times, as opposed to the unity-gain stable op amps of the prior examples. This factor requires a modification to gain resistors R1-R3. This is the inclusion of an extra resistor, R4. With the ratio shown, R3 and R4 form a 5/1 voltage divider for the voltage seen at the bottom of network N (the R1-R2-C2 node). This satisfies U1's gain-of-five stability requirement.

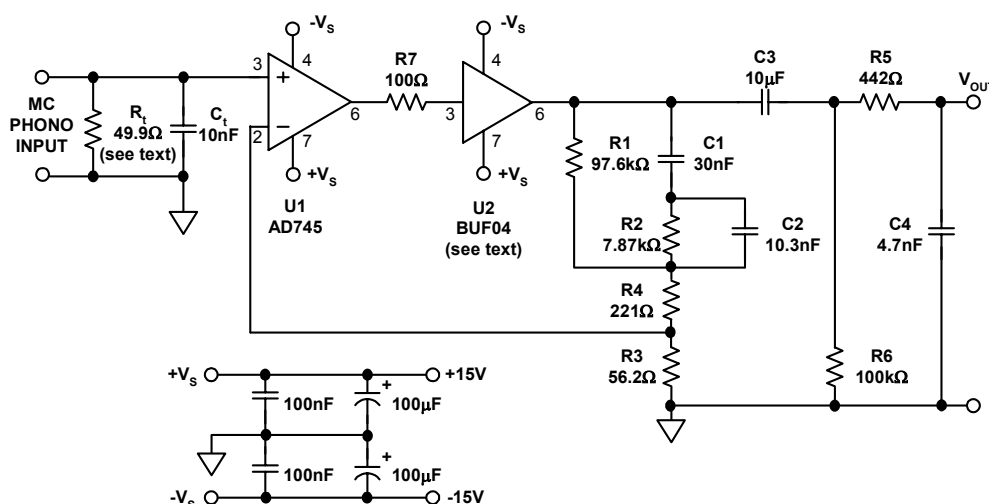


Figure 6-15: A low noise DC-coupled active feedback RIAA moving coil preamp with 45dB of gain

In this gain setup, R3 is still used for the gain adjustment, and R1-R2-C1-C2 still form the basic N1 RIAA network. With R4 used, Eq. 6-9 is used to calculate the T4 time constant. With C4 chosen as a standard value, R5 is then calculated. With these N1 network values and a 45dB 1kHz gain, R3 is 56.2Ω, which is still suitable as a low noise value operating with either an AD745 or an OP37 used for U1.

Some subtle points of circuit operation are worth noting. The DC gain of this circuit is close to 1800, which can result in saturation of U1 if offset isn't sufficiently low. Fortunately, the AD745 has a maximum offset of 1.5mV over temperature, making the output referred offset always less than 3V. While this may limit the maximum output swing some due to asymmetrical clipping, 5Vrms or more of swing should be available operating from ±15V supplies. Coupling capacitor C3 decouples the DC output offset at U2, so any negative consequences of DC-coupling the U1 gain path are minimal.

For minimal loading of U1 and maximum linearity at high gains, the unity-gain buffer amplifier U2 is used, a BUF04. The BUF04 is internally configured for unity-gain operation, and needs no additional components. Note that this buffer is optional, and is not absolutely required. Other buffer amplifiers are discussed later on in this chapter.

This Fig. 6-15 circuit was analyzed with PSpice using the Laplace comparison technique earlier described, and the results are displayed in Figure 6-16. As was true previously, the vertical scaling of this display is very sensitive; $\pm 300\text{mB}$ (or $\pm 0.3\text{dB}$). Thus, placed in context, gain errors relative to 1kHz over 20Hz - 20kHz are extremely small, $\approx 0.1\text{dB}$. Lab measurements of the circuit were also consistent with the simulation. Of course in terms of audible effects, errors of $\pm 0.1\text{dB}$ or less aren't likely to be apparent.

Distortion/noise measurements of the circuit are essentially dominated by noise (as opposed to actual distortion) measuring $\sim 0.01\%$ THD+N or less, over output levels ranging from 0.5 to 5V_{rms} , from 20Hz - 20kHz . Of course, as with any high gain circuit, layout and lead dress into the circuit are extremely critical to noise, and must be arranged for minimum susceptibility. Supply voltages must be low in noise, and well regulated.

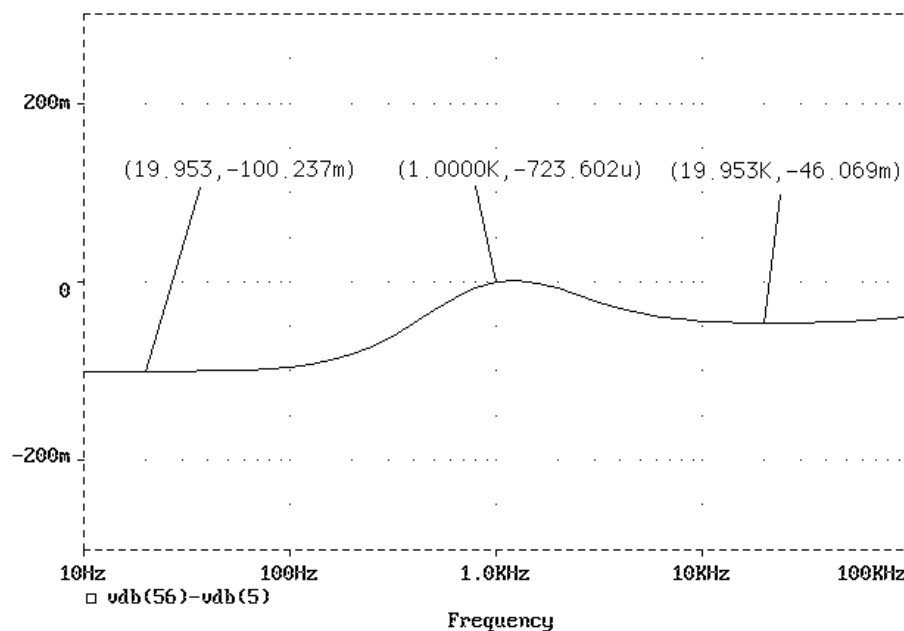


Figure 6-16: *Relative error (B) versus frequency for DC-coupled active feedback RIAA moving coil preamp, gain of 45dB (simulation)*

This exercise has illustrated both the basic design process of the active RIAA equalizer, as well as a convenient SPICE analysis method to optimize the design for best frequency response. It is not suggested that the exact network values shown of the examples are the only ones suitable. To the contrary, great many sets of values can be used with success comparable to that shown above.

This final active equalizer circuit example is the best of the bunch, and has a virtue of being easily adapted for other operating conditions; i.e., higher gain, other networks, etc. For example, note that even lower noise MC operation is possible, by using the $\leq 1\text{nV}/\sqrt{\text{Hz}}$ AD797 for U1, and scaling the N1 RC components further downward. This will have the desirable effect of making R3 lower than 50Ω , which minimizes the R1-R4 network's noise. Note that gains of 50dB or more are also possible, suitable for very low output moving coil cartridges (given suitable attention to worst case U1 offsets).

■ OP AMP APPLICATIONS

Passively Equalized RIAA Preamp Topologies

Another RIAA design approach is the so-called *passively equalized* preamp (see Reference 11). This topology consists of two high quality, wideband gain blocks, separated by a three terminal passive network, N (N can be either network N1 or N2). The gain blocks are assumed very wide in bandwidth, so in essence the preamp's entire frequency response is defined by the passive network, thus the name passively equalized.

A circuit topology useful for such RIAA phono applications is shown in Figure 6-17. This circuit consists of two high-quality wide bandwidth gain blocks, U1 and U2, as discussed above. Selection of these amplifiers and their operating conditions optimizes the preamp for gain, noise, and overload characteristics. The circuit can be set up for either MM or MC operation by simple value changes and op amp selection.

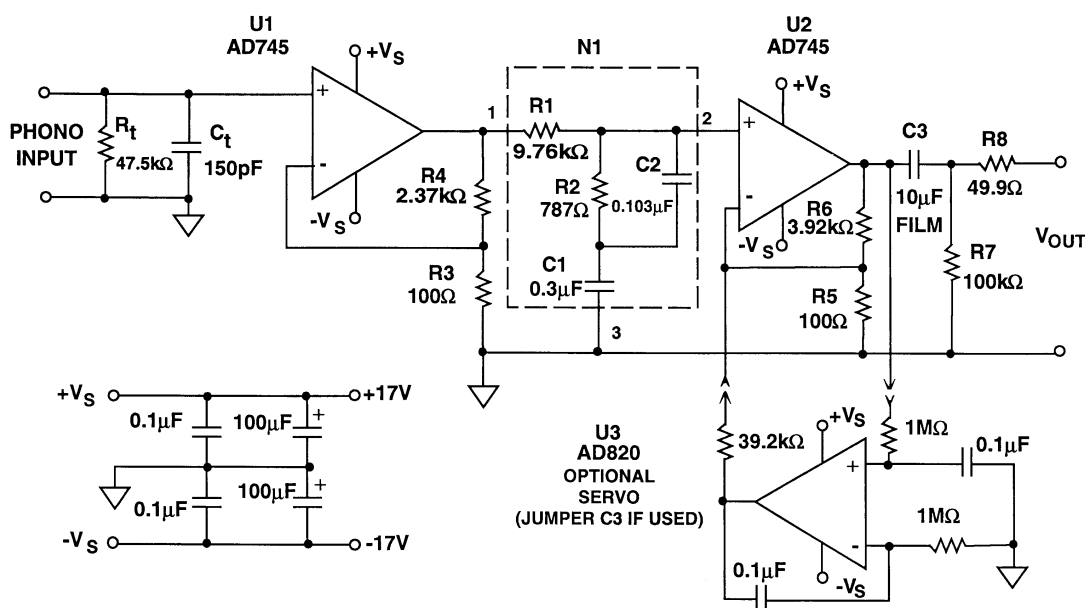


Figure 6-17: A passively equalized RIAA preamp with 40dB gain

The gain stages are set up for the required total gain, via R_4 - R_3 and R_6 - R_5 . In general, the total 1kHz gain of this circuit G is:

$$G = 0.101 \cdot [1 + (R_4/R_3)] \cdot [1 + (R_6/R_5)] \quad \text{Eq. 6-11}$$

The op amp gain blocks could be made identical for purposes of simplicity but are not necessarily so for the following reasons. A preamplifier topology such as this must be carefully optimized for signal-handling capability, both from an overload standpoint and from a low-noise viewpoint. Stage U1 is desirably chosen for a gain sufficiently high that the input-referred noise will be predominantly due to this stage and the cartridge, but yet not so high that it will readily clip at high-level high-frequency inputs. Amplifiers with a $\approx 10V_{rms}$ output capability allow U1 to accept $\approx 400 mV_{rms}$ at high frequencies using $\pm 18V$ supplies, while still operating with useful gain (about 25 times).

The gain of the two blocks are set by R_4 - R_3 and R_6 - R_5 , as defined by Eq. 6-11. The gain values shown yield a 1kHz gain that is the product of the U1-U2 stage gains (24.7 times

40.2), times that of the interstage network N (0.101). This yields an overall 40dB 1kHz gain. Other gains are realized most simply by changes to R_5 or R_3 .

As noted previously, a passively equalized preamplifier such as this must be carefully optimized both from an overload standpoint and from a low noise viewpoint. Stage U1 is desirably chosen for a gain sufficiently high that the input-referred noise will be predominantly due to this stage (and the cartridge, when connected), but yet not so high that it will readily clip at high-level high-frequency inputs. To aid this objective, maximum supply voltage and a high output capability amplifier should be used for U1.

Note that U1 operates at relatively high gain, but it needn't be unity gain stable. Decompensated low noise op amps such as the OP37 and the FET input AD745 will provide best signal/noise ratio here. For other FET-input types, the AD845 as well as the OP17 family types will also yield good performance, but with higher noise levels.

In general, the preceding factors dictate that gain distribution between U1 and U2 be LOW/HIGH from an overload standpoint, but HIGH/LOW from a noise standpoint. Practically, these conflicting requirements can be mitigated by choosing the highest allowable supply voltage for U1, as well as a low noise device. Because of nearly 40dB loss in the network N at 20 kHz, the output overload of the circuit will be noted at high frequencies first. With the gain distribution shown, the circuit allows a 3Vrms undistorted output to 20kHz with $\pm 15V$ supplies, or more with higher supply voltages.

The equalization network N following U1 should use the lowest impedance values practical from the standpoint of low noise, as the noise output at pin 2 of the network is equivalent to the input referred noise of A2. The network of Fig. 6-17 uses the "N1" RC values of R_1 - R_2 - C_1 - C_2 of Figure 6-9a. As noted, scaling can be applied to either network of Figure 6-9 for component selection, as long as the same ratios are maintained.

Noise in amplifier U2 is less critical than U1 at low frequencies, but is still not negligible. A low voltage noise device is very valuable to the U1 and U2 positions, as is a relatively low input current noise. If extremely low noise performance is sought, such as for a moving coil preamp, then the N1 values can be reduced further, and R_3 be lowered for lower noise and additional gain. For example, a 45dB gain preamp could be realized by just dropping R_3 to 56.2Ω , and using an OP37 for U1.

As mentioned before, a low bias current device is appropriate to U1 using bipolar amplifiers. With a 100nA or less bias current device, direct coupling to a moving magnet phono cartridge is practical. For example, the 80nA (maximum) bias current of the OP37 will induce only an additional 80-160 μV input voltage offset at U1 for a typical 1-2k Ω cartridge resistance. For lower DC resistance MC cartridges, this will be much less of course. Similarly, the bias current induced offset voltage of U2, from the 10k Ω DC resistance of R_1 will also be low relative to the amplified offset of U1. As a result, the worst-case overall output DC offset using two AD745s can be held to under 2V for a 40dB gain, allowing a single C_3 coupling capacitor for DC blocking purposes.

■ OP AMP APPLICATIONS

Frequency response of this passively equalized preamp tends to be better than that of the active versions, because of less interaction with the amplifier(s) as compared to the active preamps. It can approach the inherent accuracy of the network components in the audio range, with potentially greater errors at higher frequencies.

Figure 6-18 illustrates this point, in a simulation of the Figure 6-17 circuit using the OP37 models. The midband error is on the order of $\pm 0.02\text{dB}$ with the N1 network composite values. For practical purposes then, the frequency response errors of this circuit will be governed by the tolerances of the network components used within it.

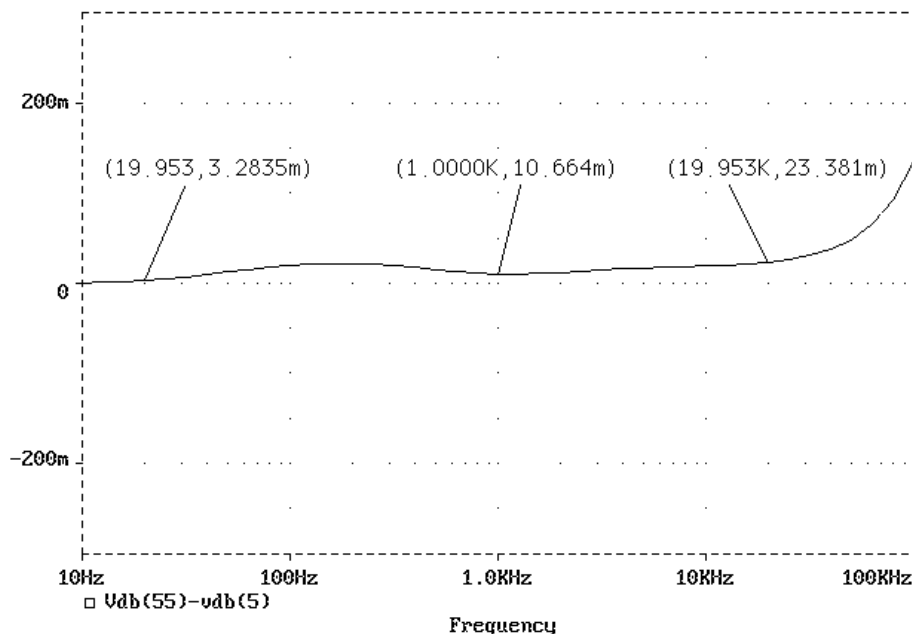


Figure 6-18: Relative error (B) versus frequency for passively equalized RIAA preamp, gain of 40dB (simulation)

This circuit also can be optionally adapted to servo control of the output offset. This is accomplished by deleting coupling capacitor C_3 , substituting a jumper in its place, and using the noninverting servo integrator U3 around stage U2. This is shown as an option within Figure 6-17. A general-purpose noninverting servo can be used for U3, along with a low-offset op amp, such as the AD820, or the OP97.

REFERENCES: RIAA PHONO PREAMPLIFIERS

1. RIAA, "Standard Recording and Reproducing Characteristic, Bulletin E1," November 6, 1978, RIAA, 1 E. 57th St, NY, NY, 10022.
2. IEC, "Publication 98 (1964), Amendment #4," September 1976.
3. F. Bradley, R. McCoy, "Driftless DC Amplifier," **Electronics**, April 1952.
4. G. Korn, T. Korn, **Electronic Analog and Hybrid Computers, 2nd Ed.**, McGraw-Hill, 1972, ISBN 0-07-035363-8.
5. D. Stout, M. Kaufman, **Handbook of Operational Amplifier Circuit Design**, McGraw-Hill, 1976, ISBN 0-07-061797-X.
6. S. Lipshitz, "On RIAA Equalization Networks," **JAES**, Vol. 27 #6, June 1979, pp. 458-481.
7. P. Baxendall, "Comments on 'On RIAA Equalization Networks'," **JAES**, Jan/Feb 1981. See also: S. Lipshitz, "Author's response".
8. Walt Jung, "The PAT-5/WJ-1 Equalization Errors" (Letters), **The Audio Amateur**, issue 3/78, pp. 49-53.
9. S. Lipshitz, W. Jung, "A High Accuracy Inverse RIAA Network," **The Audio Amateur**, issue 1/80, pp. 22-24.
10. Walt Jung, "SPICE Technique Compares Frequency Responses," **EDN**, November 25, 1993, pp. 188 and 190.
11. Walt Jung, "Topology Considerations for RIAA Phono Preamplifiers," **67th AES Convention**, November 1980, preprint #1719.
12. G. Erdi, et al, "Op Amps Tackle Noise- and for Once Noise Loses," **Electronic Design**, December 20, 1980, pp. 65-71.
13. Walt Jung, "Audio Preamplifiers, Line Drivers, and Line Receivers," within Chapter 8 of Walt Kester, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3, pp. 8-1 to 8-100.
14. W. Jung, R. Marsh, "Picking Capacitors," Parts 1&2, **Audio**, February & March 1980.
15. B. Duncan, "With a Strange Device," multiple part series on capacitors, **Hi-Fi News And Record Review**, April-November 1985.
16. B. Duncan, M. Colloms, "Pièce de Résistance, Parts 1-3," **Hi-Fi News And Record Review**, March, April 1987 (Duncan); June 1987 (Colloms).

■ OP AMP APPLICATIONS

Audio Line Level Stages

Audio line level stages represent an intermediate level in dynamic range for practical audio circuits using modern IC devices. Line level amplifying stages generally work with single-ended or balanced input/output signal levels of 1-10V, and at medium levels of power. This section discusses some basic types of audio line stages which are:

■ *Line receivers*— including line receiver stages which accept single-ended or balanced line level signals with maximum noise immunity, and provide scaled outputs for further processing.

■ *Line amplifiers*— including amplifiers which scale a received signal in single-ended form and feature low distortion designs.

■ *Line drivers*— including single-ended and balanced drivers, which are capable of driving appreciable output levels in terms of voltage, swing, current levels, and/or difficult loads, such as capacitive lines.

Some general concepts of line driving and buffer amplifier design have been covered previously, with emphasis on video applications (see References 1-3). Some of the material in this section continues and expands on those themes with audio line-receiver and line-driver discussions in a wide variety of applications. Video applications for line driving and receiving is discussed in detail within section 6-2.

Audio transmission systems, unlike their video counterparts, do *not* use terminated transmission lines as a rule, so long transmission lines usually appear capacitive. Therefore, the concepts of capacitive load isolation are also important to audio drivers. In general, when building practical audio circuits of any type, "housekeeping" rules of layout and bypassing are strongly recommended, particularly so for audio buffer and line driver circuits. They are discussed in further detail in that section.

The function of sending/receiving audio signals between various system components has traditionally involved tradeoffs of one form or another. Fully differential or *balanced* transmission systems are best at rejecting low frequency and RF noise, so they are used for highest performance, and are discussed in some detail following.

A typical audio system block diagram using differential or balanced transmission is shown in Figure 6-19 (opposite). In concept, a balanced transmission system like this could use several input/output coupling schemes within the driver and receiver. Some major points distinguishing coupling method details are discussed briefly below, before addressing actual circuits.

Transformers (see References 4-7) have been a traditional audio line coupling element. They can be used at either input or output stage. They also have well known problems with noise pickup, frequency response, distortion, and operating level. While these problems are soluble to some degree, answers are usually costly. Nevertheless, transformers are unexcelled in notable areas.

The single most outstanding virtue of transformer operation lies in the ability to isolate while transmitting the signal, which shows up in two regards. The first of these is that transformers, which transmit an audio signal between the end terminals of an isolated winding, thus *galvanically isolate* driver and receiver stages. This is accomplished at common mode voltage (ground difference) levels up to the breakdown potential of the windings, allowing signals to be transmitted across very high ground potential differences (tens or hundreds of volts). This feature is one very difficult to achieve with solid state circuitry. Secondly, suitably designed transformers can have very high common mode rejection (CMR) over the audio range, $\geq 100\text{dB}$ in some cases, a factor basically intrinsic to their nature. Some transformer-isolated stages are described in this section.

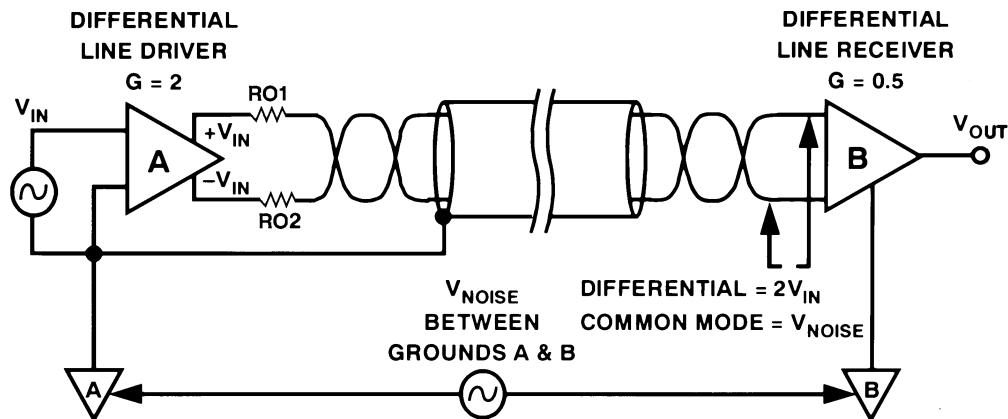


Figure 6-19: An audio balanced transmission system

In practice the general system of Figure 6-19 can use either transformer-based or active stage coupling to the line, at either end. The goal for either approach is to reproduce a final signal V_{OUT} equal to V_{IN} , while rejecting noise between grounds A and B by a factor of 80-100dB. Typically, a unity gain (overall) design uses a balanced line drive of $\pm V_{IN}$, followed by a receiver gain "G" of $\frac{1}{2}$, which maximizes the receiver CM range.

A point worth noting that the $\pm$ voltage drive to the line need *not* be exactly balanced to reap the benefits of balanced transmission. In fact the drive can be asymmetrical to some degree, and the signal will still be received at V_{OUT} with correct amplitude, and with good noise rejection. What *does* need to be provided is two well-balanced line-driving impedances, R_{O1} and R_{O2} . Also, in conjunction with these balanced drive impedances, the associated (+) and (-) receiver input impedances should also be equal. The technical reasons for this will be apparent shortly.

■ OP AMP APPLICATIONS

Audio Line Receivers

A brief review of the topologies and application points of audio line receivers helps in understanding their evolution, and more importantly, how their audio performance differs with topological changes. Figure 6-20 is a diagram of a classic 4 resistor differential amplifier. This general circuit is also known as the most simple instrumentation amplifier form, even though its performance as an in-amp has severe limitations. Within audio applications, this and related circuits are called "line receivers" for the sake of brevity. Various in-amp type topologies have developmental histories dating from the late sixties up to and including today's modern in-amp ICs (see References 8-12 and Chapter 2).

In today's professional audio world, signals by and large get transmitted in balanced mode (Fig. 6-19, again). This fact is simply due to the much greater noise immunity of this method, vis-à-vis the more simple, but highly noise-susceptible single ended method.

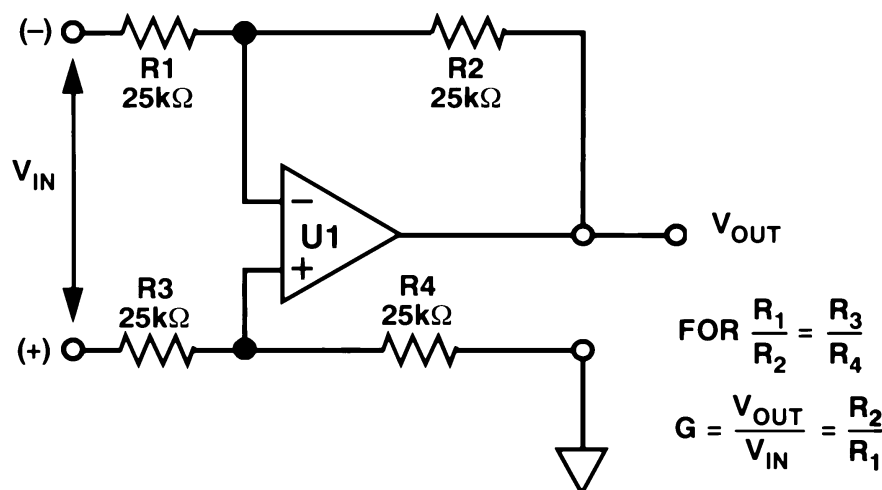


Figure 6-20: A simple line receiver using a 4-resistor differential amplifier

Yet, even within the professional audio world there is no real unanimity on signal driver and receiver circuits for use within balanced circuits—they take on many forms and have differing performance, and a wide variety of circuits find use. Before getting into actual receiver circuitry, it is helpful to take a brief look at some problems impacting circuit performance in terms of common mode (CM) noise susceptibility. This will illustrate how careful hardware choices lowers system cost/size, and maintains excellent performance. Conversely, simple receivers can also be used, for modest performance.

Source-Load Interactions Within Balanced Systems

Some recent attention has focused on the general problem of noise susceptibility in audio system interfacing (see references 13 and 14). The discussions below are concerned with how balanced system drivers and receivers interact fundamentally to produce undesired side effects of noise susceptibility. Suggestions for practical solutions are then offered.

In most simple form, a balanced audio transmission system consists of a differential output driver, an interconnecting cable, and a differential input receiver, such as shown in Fig. 6-19 (again). The driver produces nominally equal and out-of-phase output signals, with some characteristic (and matched) source impedances, R_{OUT1} and R_{OUT2} . As will be

seen, from a noise susceptibility standpoint, it is highly desirable that these two impedances be well balanced, i.e., matched. The driver is connected to the input end of a balanced transmission line, typically a shielded twisted pair. At the opposite end of this line, a differential input receiver receives the balanced signal, and (ideally) rejects the CM noise voltage V_{NOISE} .

The design of both the driver and the receiver has great influence upon how well the overall scheme works in transmitting a noise-free audio signal from driver to receiver. References 14 and 15 discuss different driver and receiver types, active and passive. These papers bring out the inherent degradation in noise sensitivity active receivers can trigger, if they do not have input characteristics which are an appropriate complement to the system driving impedances.

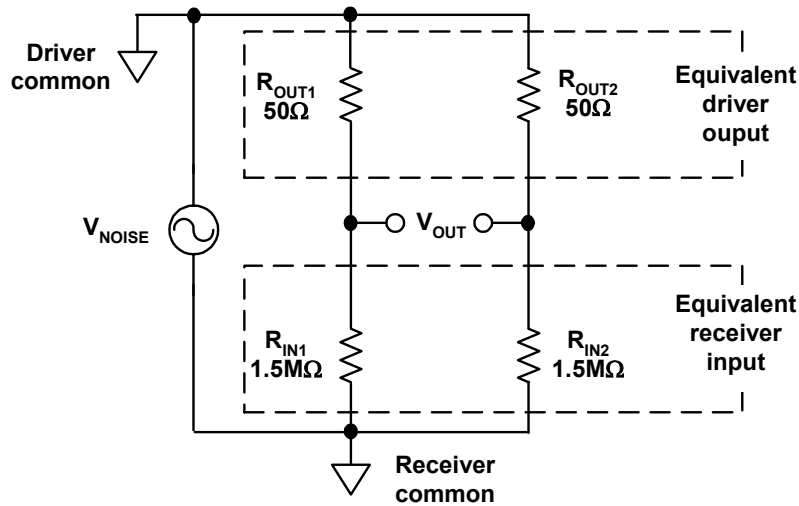


Figure 6-21: A conceptual driver/receiver diagram of a balanced line audio system with key impedances and CM noise

From a noise introduction point of view, the balanced transmission system we're talking about can be analyzed as a bridge circuit, as shown in Figure 6-21 above. Here the two source resistances R_{OUT1} and R_{OUT2} correspond to the output resistances of the differential driver voltage sources. Similarly, input resistances R_{IN1} and R_{IN2} correspond to the input resistances of the differential receiver. V_{OUT} represents the output of this bridge, which is due solely to the bridge mismatching and CM noise V_{NOISE} .

Such a bridge as Fig. 6-21, when *maximized* for output sensitivity, will produce a differential output V_{OUT} which is highest as a function of element unbalance when all four resistances are of the same order.

The following general expression illustrates the intrinsic common mode rejection (CMR) of this bridge as a function of the resistance values and their deviation, K_R :

$$\text{CMR(dB)} = 20 \bullet \log_{10} [(1+(R_{\text{IN}}/R_{\text{OUT}}))/K_R] \quad \text{Eq. 6-12}$$

Some sample calculations with this relationship show that *CMR is a minimum* for a given change in K_R (the total resistor deviation expressed in fractional form) when $R_{\text{IN}} \approx R_{\text{OUT}}$. A CMR minimum is simply another way of saying that the bridge is most sensitive to the

■ OP AMP APPLICATIONS

excitation voltage V_{NOISE} when $R_{\text{IN}} \approx R_{\text{OUT}}$. To place this in perspective, a conventional instrumentation bridge operates thusly, with all four arms nominally equal. This yields the highest sensitivity to the applied voltage (see Chapter 4).

On the other hand, CMR is maximum and bridge sensitivity is *minimized*, when the upper and lower arm resistances differ widely. This improves substantially as R_{IN} becomes $\gg R_{\text{OUT}}$. Or, within an audio system, as the driver R_{OUT} is by design made much less than the receiver R_{IN} . With the example values above, there is a 1/30,000 ratio between the $R_{\text{OUT}}/R_{\text{IN}}$ upper/lower elements. This factor makes relatively high percentage changes in either the upper (or the lower) arm resistances a somewhat harmless phenomenon. In other words, small R_{OUT} or R_{IN} changes will then have little CMR effect upon the output.

For example, taking the Fig. 6-21 values and assuming a 10% change in R_{OUT} , will produce an output which is about 110dB down from the noise voltage V_{NOISE} . By contrast, if all the bridge values were to be equal, the same 10% deviation would produce an output only 26dB down! Note that there are two control point towards this. One can *lower* R_{OUT} , for a given R_{IN} , and increase CMR. Or, one can achieve the same effect by *increasing* R_{IN} , for a given R_{OUT} . This makes the point that *a high ratio between R_{IN} and R_{OUT}* aids in maintaining high CMR, as is shown by Figure 6-22.

$R_{\text{IN}}/R_{\text{OUT}}$	CMR(dB) for $K_R = 0.1$	CMR(dB) for $K_R = 0.01$
10	40.8	60.8
100	60.1	80.1
400	72.1	92.1
1k	80	100
10k	100	120
30k	109.5	129.5
100k	120	140

Figure 6-22: High $R_{\text{IN}}/R_{\text{OUT}}$ minimizes sensitivity to CM noise, bridge imbalance

In a real transmission system, there will be inevitable noise potentials developed between the respective driver and receiver chassis common points, since they are located separately and are powered with different power sources. The resulting noise voltage can be predicted with the aid of Fig. 6-22. As a minimum, a good system should maintain an $R_{\text{IN}}/R_{\text{OUT}}$ of at least 1000, with 10k or more a goal. Under such conditions, with a bridge unbalance of 10%, this will still allow a theoretical CMR of 80 to 100dB (see center column, with cited examples shown in boldface).

As noted, dependent upon the bridge impedance-related sensitivity, some fraction of the CM V_{NOISE} appears as V_{OUT} . The basic process of the conversion of the CM noise voltage into a differential voltage is called *mode conversion*. It is important to understand that mode conversion can only be prevented, not fixed after the fact. Once the noise voltage appears as a differential signal, no receiver can distinguish it from a valid signal.

Finally, it is important to realize that what has been discussed thus far addresses the most basic portion of this system and the impact on CMR. The line receiver circuitry itself obviously also has a big influence, as it determines R_{IN} . This is discussed next.

The Simple Line Receiver

The simple line receiver circuit of Figure 6-23 below uses four matched resistors and an op amp for gain. Such bridge-based difference amplifiers are *critically* dependent upon the resistor ratio matching for good performance, an enormously important point. The amplifier can also be critical, but most practical limitations of this topology arises from *un*-balancing of the R_1 - R_4 bridge (for either/both AC and DC).

The circuit appears somewhat trivial, as the minimum ingredients are four matched film resistors and a good audio op amp. While this works functionally, how well it works in rejecting noise is another thing. C1 and C2 are optional, and can be used to trim HF CMR. Also optional is the use of an in-the-loop unity-gain buffer (more below).

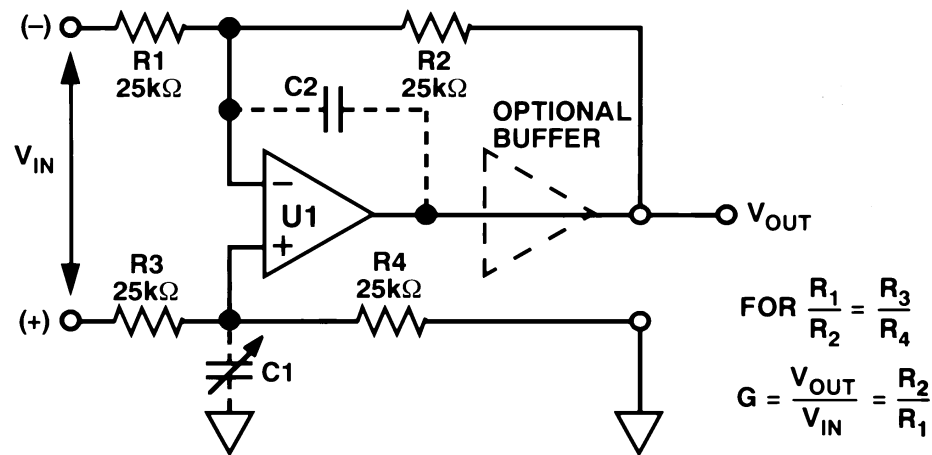


Figure 6-23: A simple line receiver with optional HF trim and buffered output

A main purpose of this circuit and all line receivers is to reject CM noise, as discussed above. But even with a high quality op amp for U1, noise rejection is only as good as the resistor matching. More precisely, the resistor *ratios* R_1/R_2 and R_3/R_4 must match extremely well to reject noise (the absolute values are relatively unimportant).

With care, picking four 1% resistors from a same-vendor, same-batch lot is a step which can yield ratio matching of, say, 0.1% total error, and will achieve a common mode rejection (CMR) of 66dB. Four 1% tolerance resistors with just one off by 1% will yield about 46dB CMR. In general, the worst case CMR of a circuit of this type is:

$$\text{CMR(dB)} = 20 \bullet \log_{10} [(1+(R_2/R_1))/4K_R] \quad \text{Eq. 6-13}$$

where here " K_R " is the *individual resistor tolerance* in fractional form. This form of the expression is most useful for cases using 4 discrete resistors (see Reference 8). More likely, a single component network with a *net matching tolerance* of K_R would be used for this function, in which case the expression then becomes:

$$\text{CMR(dB)} = 20 \bullet \log_{10} [(1+(R_2/R_1))/K_R] \quad \text{Eq. 6-14}$$

In either case, this assumes a significantly higher *amplifier* CMR, such as $\geq 100\text{dB}$).

■ OP AMP APPLICATIONS

Eq. 6-13 shows that the worst case CMR due to tolerance build-up for 4 unselected 1% resistors to be much worse, 34dB in fact. Clearly for high and stable noise rejection, circuits such as these need four single-substrate resistors, made/trimmed simultaneously. Networks using thick-film and thin-film technology are available from companies such as Caddock and Vishay-Ohmtek, in ratio matches of 0.01% or better.

Some simulations may bring this point of critical matching home more clearly. Figure 6-24 shows the effect of various DC matching of a differential amplifier such as like Figure 6-23, with a single resistor mismatch in R_1 of 0.1% (top trace), 0.01% (middle trace) and perfect matching (bottom). Also, this display of CM error vs. frequency indicates a reactive imbalance for the perfect DC-balanced (bottom) case. This is due to an intentional capacitive mismatch in the circuit, representing stray capacitance imbalance.

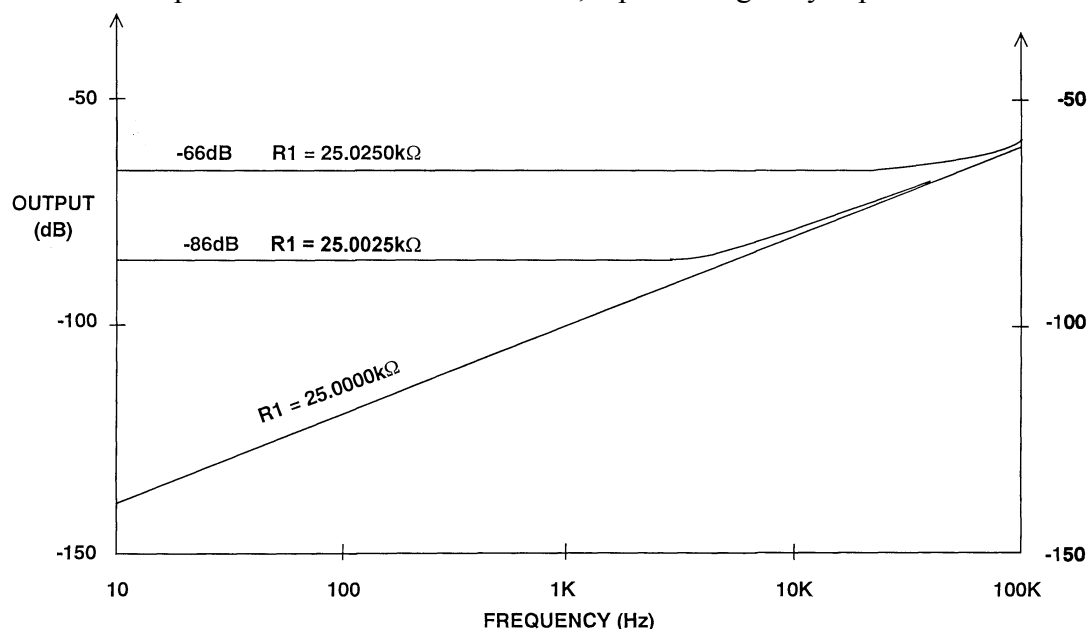


Figure 6-24: Simple line receiver CM rejection vs. frequency for various R_1 trims (simulation)

The effect of AC matching on this differential amplifier using matched resistances, with C_2 matched to C_1 (10pF) results in essentially flat CM error vs. frequency (not shown). In contrast, a 10% capacitive mismatch results in a CM degradation as low as 10kHz. Clearly then, for wideband audio uses, the bridge ratio needs to be maintained for AC as well as DC, to achieve flat CMR versus frequency. Capacitances from the R_2/R_1 and R_4/R_3 nodes need to be balanced. In practice, this is best achieved with very low and/or balanced parasitic capacitances at C_1 - C_2 .

It is worthy of note that this circuit also has a highly desirable side property; that is, it *divides down* the input CM voltage. Thus it is inherently protected against overvoltage. In general, practical line receivers *require* some sort of input protection, for safe use in harsh environments, and to allow CM voltages to exceed the supply rails. The receiver gain-set resistors double in performing this function here. The working CM input range of Figure 6-23 is $(1+(R_3/R_4)) \times V_{CM(U1)}$, and differential input resistance is R_1+R_3 . Circuit gain isn't easily changed, because of the matched R_1 - R_4 ratios.

Implementing the Simple Line Receiver Function

To offer a reasonably high impedance to the line, simple receivers such as these typically use input resistances of 25k Ω or more. When working from 50 Ω sources, this allows a basic R_{IN}/R_{OUT} ratio of 500 (see previous discussion and Fig. 6-22, again). Given a well matched resistor network and low or balanced parasitic capacitances, suggested amplifiers for U1 are the AD711, AD744 (singles), and the AD712, AD746, OP249, OP275 (duals). With 10k Ω -25k Ω resistances, extremely low voltage noise in the amplifier isn't critical. High slew rate (SR) and output drive allows clean high frequency, high amplitude levels, and 600 Ω load capability. If low impedance loads need to be driven, output current of a standard op amp can be boosted with an in-the-loop unity gain buffer, connected between U1 and the load/feedback point. Devices such as the BUF04 or a (follower-connected) AD811 can serve well here (discussed in the later sections).

From an DC and AC trim/balance perspective, the Figure 6-23 topology is most effective with resistors and amplifier made simultaneously in a single monolithic IC. The ADI SSM2141 and SSM2143 are such ICs, characterized as low distortion, high CMR audio line receivers with net gains of unity (SSM2141), and 0.5 (SSM2143). The SSM2141 has resistors as shown in Figure 6-23, while the SSM2143 uses 12k Ω /6k Ω resistors.

In applying circuits of the Figure 6-23 type (or other topologies which resistively load the source), a designer must bear in mind that all *external* resistances added to the four resistances can potentially degrade CMR, unless kept to proportional value increases. To place this in perspective, a 2.5 Ω or 0.01% mismatch can easily occur with wiring, and if not balanced out, this mismatch will degrade the CMR of otherwise perfectly matched 25k Ω resistors to 86dB. These circuits are therefore best fed from balanced, low impedance drive sources, preferably 25 Ω or less.

Other Issues with the Simple Line Receiver

An application point which becomes relevant for large high performance systems with multiple balanced pair lines is the issue of receiver *load balance*. Ideally, an audio line receiver should exhibit equal AC loading at the two inputs. With the simple line receiver of Figure 6-23 (and all similar circuits), this goal isn't met— i.e., the basic circuit does *not* present balanced loading to the two input lines. It is important to note at this point that this is not a function of the devices used to implement the circuit, it is more a function of the architecture itself.

When Figure 6-23 is driven from complementary sources V_{IN} and $-V_{IN}$, the simple line receiver exhibits a property of unbalanced input currents in the R_1 and R_3 legs, due generally to feedback action. For the like values of Figure 6-23, the current in R_1 is 3 times that in R_3 . Thus the inputs load the two input lines differently, as noted.

In large systems with multiple balanced transmission line pairs, the current imbalance in the input lines is potentially serious, as associated fields will not cancel as they do for completely balanced loading. Thus there is potential for crosstalk impairment in such systems using the simple line receiver topology.

■ OP AMP APPLICATIONS

On the other hand, while not optimum from a large system and/or line balance viewpoint, the simple line receiver is nevertheless useful in more modest situations. With resistors R_1 - R_3 relatively high (20k or more), it is adequate for small-scale or confined systems where I/O lines are relatively short, few in number, they are not cabled, and the source impedances are low. In such uses, devices like the SSM2141 and SSM2143 can serve well as efficient, single IC line receiver solutions.

Balanced Line Receivers

For highest performance uses, it is a key point that audio line receivers exhibit equal loading to the source at both inputs— i.e., they should be truly balanced. At least two topologies meet this criteria and are thus suited for professional use in balanced systems.

Balanced Feedback Differential Line Receiver

David Birt of the BBC has analyzed the simple line receiver topology and presented a modified and balanced form, as shown here in Figure 6-25 (see Reference 7). Here U1 uses an 4 resistor network identical to that of Figure 6-23, while a second feedback path from unity gain inverter U2 drives the previously grounded R_4 reference terminal. This has two basic effects overall; the input currents in the R_1 - R_3 input legs become equal in magnitude, or balanced, and the gain of the stage is halved.

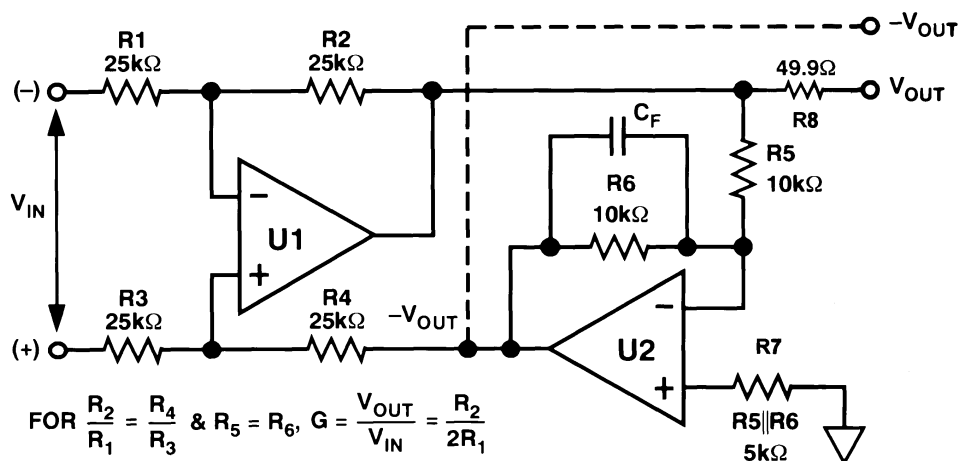


Figure 6-25: *Balanced line receiver using push-pull feedback*

Compared to Figure 6-23, and for like resistor ratios, the Figure 6-25 gain from V_{IN} to V_{OUT} is $\frac{1}{2}$, or a gain of -6dB (0.5) as shown. However it also offers an optional complementary output from U2, $-V_{OUT}$. Like Figure 6-23, the gain of this circuit is not easily changed, as it also involves precise resistor ratios.

Because of the two feedback paths, this circuit holds the inputs of U1 at a null for differential input signals. However CM signals are still seen by U1, and the CM range of the circuit is $(1+(R_3/R_4)) \times V_{CM(U1)}$. Differential input resistance is R_1+R_3 . As can be noted from the figure, the circuit can be broken into a simple line receiver (left), plus an inverter (right). Existing line receivers like Figure 6-23 can be converted to a balanced topology by adding inverter U2. A performance example of this is discussed below.

Alternate Balanced Line Receivers

Other instrumentation amplifier types can achieve the goal of fully balanced input loading, but may not be desirable for other reasons. For example, there are standard in-amp circuits not shown here which use either 2 or 3 amplifiers and have properties of high input impedance, due to the use of non-inverting inputs (see References 8-11). The drawbacks of these topologies as audio line receivers lie in limited gain and CM range. Also, importantly, they require 4 resistors beyond those for gain, just for input overload protection. Since these resistors also influence gain and CMR, they must also be precision ratio matched types. As a net result, workable audio line receivers using these in-amps aren't really highly practical (8 or more matched resistors, plus 2 or 3 op amps).

An "All Inverting" Balanced Line Receiver

Figure 6-26 below is an elegantly attractive topology that seems well suited to audio line receiver use. Using all amplifiers as inverting stages, this circuit can be configured for very high CM voltage range and high input resistance. With the resistor ratios matched as shown, the CMR of this circuit can be better than the others for a given resistor match, since both amplifiers see no CM voltage. The CM range of this circuit is set as $(R_1/R_2) \times V_{OUT(MAX)U1}$. The differential input resistance is $R_1 + R_3$.

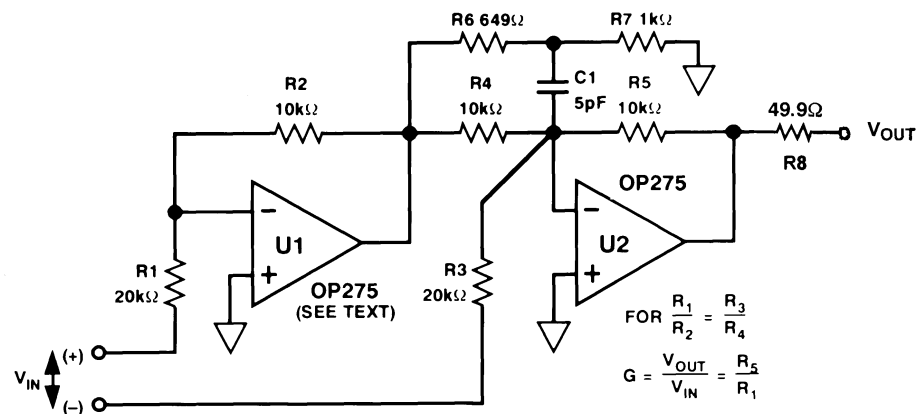


Figure 6-26: "All Inverting" balanced line receiver

The circuit has the unusual and desirable property of single resistor gain adjustment via R_5 , *without* any CMR interaction. Gain can also range from below to above unity, making it flexible in that regard. As shown it is driven with a balanced signal, but note that it can also be driven with single-ended sources at either the (+) or (–) terminal, with no gain interaction from the opposite input port, due to the use of inverting amplifiers. Multiple inputs can be summed, with additional ratio matched input resistor pairs (not shown). In this example gain is set at 0.5, consistent with general line receiver system requirements.

This circuit is also well known in basic form (see References 8-11). However, note that in this configuration, optional phase lead compensation is used to enhance high frequency CMR. A small capacitor shunting R_4 with a value chosen to compensate for the gain-bandwidth of U_1 compensates for the lag through U_1 , and maximizes phase matching of the $\pm$ CM signals at U_2 . However, for op amps with gain bandwidths above a few MHz and practical resistor values, this can result in difficult-to-control small capacitor values.

■ OP AMP APPLICATIONS

The R_6 - R_7 - C_1 tee network reduces the effective value of C_1 , by dividing the applied voltage. A nominal division ratio can be approximated by this expression:

$$K_C = 1/(2\pi BW_{(U1)} R_4 C_1) \quad \text{Eq. 6-15}$$

where K_C is the division ratio of R_6 - R_7 . For this example, with $BW(U1)$ about 5MHz (the closed loop bandwidth of $U1$), K_C is about 0.6, making C_C effectively about 3pF. Circuit parasitics, loading effects and part variations make this inexact; however, once nominal compensation for a given layout and devices is achieved, a 30dB CMR improvement in 10kHz CMR is possible (vis-a-vis no phase compensation). This trim network isn't necessary to the circuit's basic function, but is nevertheless useful for audio applications.

Performance of balanced line receivers

The balanced line receiver configurations of Figure 6-25 and Figure 6-26 were tested for CM performance, with common conditions of $G=0.5$, $V_s=\pm 18V$, and a 10Vrms input sweep, 20Hz-50kHz, filter bandwidth of 80kHz. The Figure 6-25 topology was implemented with an SSM2141 for $U1$, with $U2$ an OP275 inverter, with $C_F=68pF$. Figure 6-26 was implemented with an OP275 and a resistor network matched to 0.005%, with the C_1 network trimmed as shown. These results are shown in Figure 6-27 below.

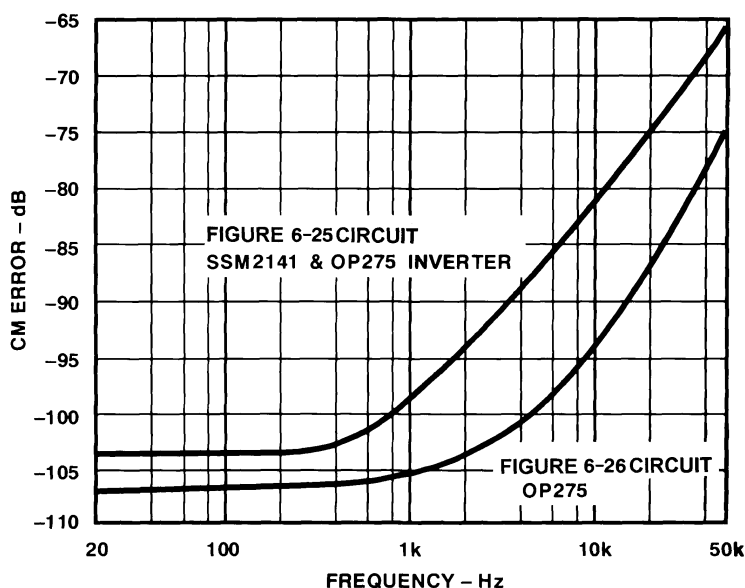


Figure 6-27: Balanced line receivers CM error vs. frequency

Both circuits show excellent results, with $\leq 1kHz$ CM errors of $-100dB$ or lower. The Figure 6-26 topology offers better results at the higher frequencies, perhaps due to the trimming technique used (not applicable to the Figure 6-25 circuit). In the worst case, the CM errors are no poorer than $-80dB$ at 10kHz, still very good for an untrimmed circuit.

THD+N data was taken on both circuits and, while dominated by the noise floor at many levels, there are some differences worth noting between the two. Figure 6-28 shows THD+N performance of the Figure 6-25 SSM2141/OP275 circuit for loading conditions of $100\text{k}\Omega$, successive input sweeps of 1, 2, 5 and 10V_{rms} , and $\pm 18\text{V}$ supplies. The lower level sweeps are noise dominated, while the 5 and 10V sweeps show some distortion rise at high frequencies. Distortion of this circuit also rises with loading of 600Ω (not shown).

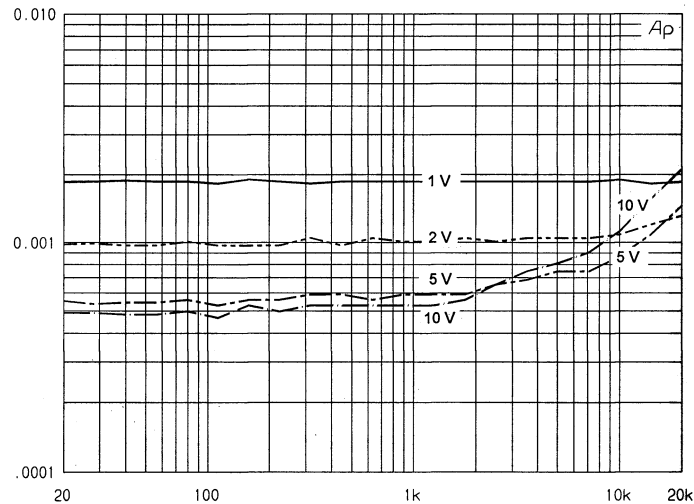


Figure 6-28: *Balanced line receiver of Fig. 6-25, THD+N vs. frequency*

The performance of the Figure 6-26 circuit for similar input drive and power supply conditions is shown in Figure 6-29, and for conditions of 600Ω loading. These data indicate less loading and frequency dependence, due primarily to the OP275's higher slew rate, and greater available output drive into 600Ω loads.

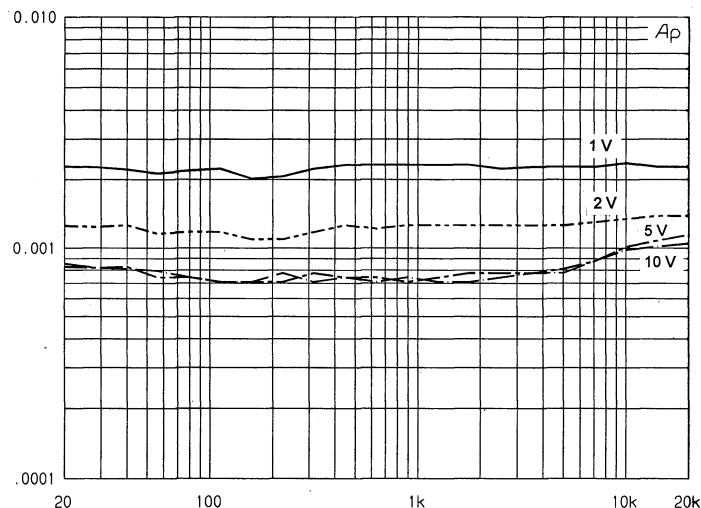


Figure 6-29: *Balanced line receiver of Fig. 6-26, THD+N vs. frequency*

In the all inverting circuit of Fig. 6-26, THD+N performance is much more limited by noise than actual distortion, over frequency. The circuit is a very flexible one, and it can be set up for a variety of other op amps and input resistances, as well as the already mentioned single-resistor gain change operation.

■ OP AMP APPLICATIONS

The line receivers covered just above offer good performance. However, with input resistances on the order of 25kΩ, they still can be subject to CM errors due to driver impedance mismatches. This is not an issue that can be dealt with cleanly, as the designer of a line receiver circuit doesn't necessarily have any pre-knowledge of the worst case driver impedance characteristics. So, to guarantee high CMR performance even in the instance of substantial driver impedance and/or mismatching, there are two possible solutions. One is to make the line receiver circuit input impedance as high as practical, which then allows good CM performance with impedance mismatches on the order of 10% (Fig. 6-22, again). Alternately, a line receiver can utilize a line transformer, which offers high CM rejection and galvanic isolation. Both approaches are discussed next.

A Buffered Input Balanced Line Receiver

The circuit of Figure 6-30 below represents an example of a classic 3 op amp instrumentation amplifier (in-amp) topology, dressed up and optimized for use as an audio line receiver (see Reference 16). The use of FET input stage buffers in amplifiers U1A and U1B allows megohm-level bias resistance to be used for R_{IN1} and R_{IN2} , which greatly de-sensitizes this receiver against loading of the source and CM errors. Optional resistor R_{IN3} terminates the line differentially. Protection resistors R_{P1} and R_{P2} allow over voltages at the two inputs, by limiting amplifier fault currents to safe levels. The input stage can use either dual or single amplifiers, with performance options described below.

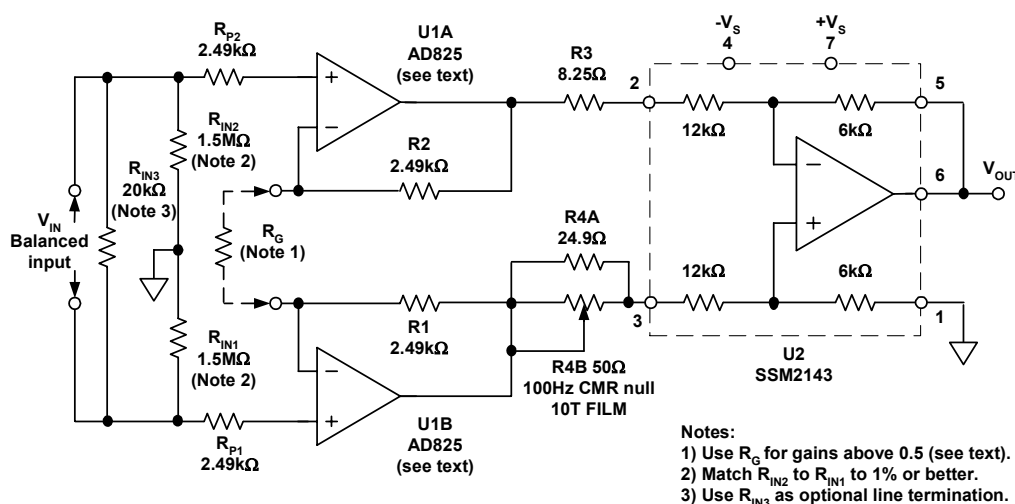


Figure 6-30: A buffered input balanced line receiver

Within the circuit, the *differential* gain of stage U1A, U1B (or G_1) is set by R_1 - R_2 - R_G , as:

$$G_1 = 1 + [(2R_1)/R_G] \quad \text{Eq. 6-16}$$

where $R_1 = R_2$, and R_G is used for high gains. Without R_G , the first stage gain is unity.

While the differential gain of U1 is as noted, the CM gain is unity, since the connection simply passes CM signals to the output. Thus both differential and CM forms of signal are presented to the inputs of stage U2. Note however, because differential and CM signals are scaled differently by U1, there can be a net potential gain in CMR. Practically,

it means that this overall configuration can achieve useful CMR figures higher than the intrinsic CMR of U2, whatever that figure may be.

The U2 stage, a pre-trimmed 4 resistor in amp, suppresses the CM component from U1A/U1B, while amplifying differential signals by a factor of 1/2. For an overall net gain G higher than 0.5 from this line receiver, the value of R_G is:

$$R_G = R_1/(G - 0.5) \quad \text{Eq. 6-17}$$

For net overall gains of 1, 2 and 4 times, the required gain resistance R_G works out to be 4.99k Ω , 1.69k Ω , and 715 Ω , respectively (using closest standard values).

Seasoned analog designers may wonder what's so new about this circuit, as it has been around for more than 30 years in solid state form (Reference 9, again). While true, some refinements here lend it worthwhile audio utility. First, as mentioned, FET input op amps for the U1 stages allow very low bias current, and load the inputs infinitesimally. Source loading will be essentially determined by the 1% resistances used for R_{IN1} and R_{IN2} .

While FET amplifiers are most useful here, a serious selection caveat is in order. The types used for U1A and U1B must *not* be general purpose types prone to sign-reversal, which could possibly come about with combined large signal and CM inputs (see Chapter 7 overvoltage discussions). All of the types tested for Fig. 6-30 have FET input stages, with CM ranges of at least $\pm 10V$ on $\pm 15V$ supplies. Note that any op amp can misbehave if severely over-driven at the input (i.e., beyond the rails). Of the three types tested, the AD825 is the most robust for overload, while the AD845 is less robust, but offers best CM performance. Wideband operation is a virtue, allowing better high frequency performance before degradation sets in. Finally, an FET input structure is less susceptible to RF rectification problems, which can be critically important in an audio line receiver used within an RF environment (see Chapter 7 RFI discussions).

Selection of the U2 device also has a great bearing on CMR. Although there are a number of unity gain 4 resistor in-amps available for the U2 function, the choice here is for less than unity gain (in this case 0.5). To extract the highest possible CMR performance, the U2 network balance is externally trimmed by the R_3 - R_4 resistances. R_{4B} can be either the film trimmer noted, or a selected fixed resistor. The values shown allow a trim range of more than $\pm 0.05\%$, sufficient to trim any SSM2143 part to a null. In the performance data following, the SSM2143 used for U2 reflected such a trim, with a basic low frequency CMR of $\sim 110dB$ for the stage.

Buffered Input Balanced Line Receiver Performance

To demonstrate these concepts, a number of measurements were made on the Fig. 6-30 circuit, using a number of single and dual IC op amps for the U1A and B positions, and an SSM2143 for U2. Although this basic 3 amplifier in amp structure can in principle offer potential gains in CM performance over the intrinsic CMR of U2, this phenomenon is less pronounced at relatively low overall gains as true here (i.e., gains of 1, 2 or 4 times). And, it is also dependent upon the specific U1 and U2 performance. Thus the CMR of both the U1 and U2 stage devices can effect the measured CM performance.

■ OP AMP APPLICATIONS

The test setup used employs an Audio Precision System 1 in a modified crosstalk test mode, where channel A drives the test circuit, which in turn has its output monitored by channel B. This allows a swept narrow band tracking analysis, over a dynamic range of 130dB or more at low frequencies, and a frequency range of 20 Hz to 200 kHz. In the results following, the CM error curves displayed are referenced to a 0dB calibrated output level from U2 of 1Vrms, with the circuit set for a gain of 0.5 (i.e., R_G open). The drive to the circuit was 2Vrms, and power supplies were $\pm 13V$.

Figure 6-31 below shows CM error results for paired (2) single samples of the AD825, the AD845, and the dual AD823. All devices have CM errors at or below a -90dB level below 3kHz, with low frequency errors of the AD845s well below -100dB . The CM corner for all devices is enough to achieve -80dB or better 20kHz CM error.

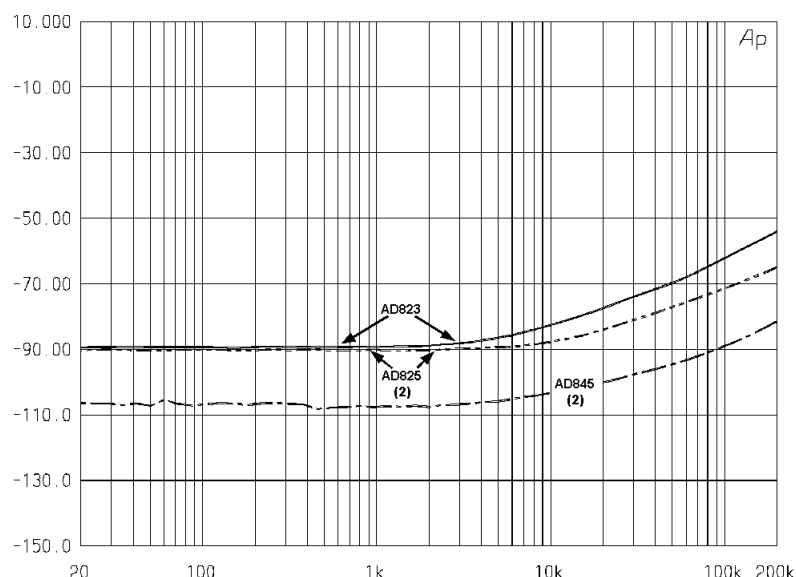


Figure 6-31: CM error (dB) vs. frequency (Hz), for various U1A and U1B devices within the circuit of Fig. 6-30, gain = 0.5.

While these results show generally what the various devices can do in this circuit, the data should not be taken as an absolute indication of future results. Sample-sample variations of a few dB will exist, and this should be taken into account. One point worth noting however is that devices with intrinsically high CMR input stages perform best. An example of this is the AD845, which uses cascode inputs for high CMR, and shows a CMR of 100dB or more to above 20kHz. It does however have less input dynamic range than the AD823 and AD825, because of the bias headroom required for the cascode stage

A criticism directed towards active line receiver circuits such as the SSM2141 and SSM2143 has been their high sensitivity to source resistance mismatches. However, as the discussions above have shown, the problem comes from the relative source/load impedances, and the degree by which these are mis-matched. One can specify a very well-controlled, high CMR receiver such as for example the SSM2141 or SSM2143, and have the as-used CMR degrade simply due to uncontrolled source impedance(s).

In the relatively uncontrolled environment of real world audio system interfacing, source resistance mismatches of a few ohms can be typical. This mismatch level is sufficient to ruin the CMR performance of a simple line receiver, if the receiver uses resistances on the order of 20k, and is fed from a source resistance of 50Ω or more. This can be readily illustrated by a sample calculation using the bridge circuit CMR relationship of Eq. 6-12, and plugging in R_{OUT} resistances of 50 and 55Ω (a 10% mis-match), using an R_{IN} resistance of 20kΩ. This degree of mis-match for 20kΩ loading conditions destroys CMR, as it degrades to 72dB with a 50Ω source mis-matched 10% (see Fig. 6-22, again).

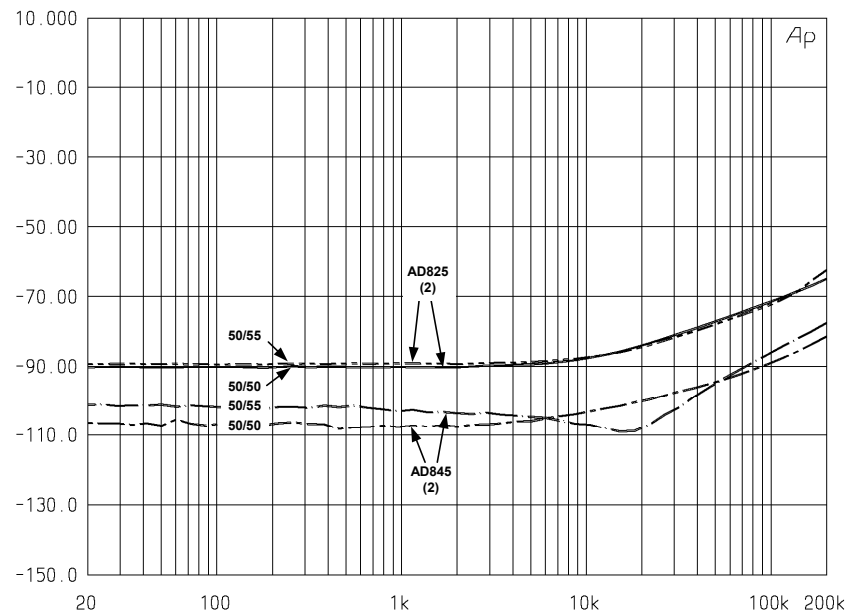


Figure 6-32: CM error (dB) vs. frequency (Hz), for AD825 and AD845 pairs, nominally 50Ω source impedances matched/mis-matched 10%

The buffered topology of Fig. 6-30 directly addresses this issue, as shown in the dual matched/mis-matched source resistance CM plots of Figure 6-32 above. In these tests, the Fig. 6-30 circuit is again exercised with the AD825 and AD845 op amp paired samples at an overall gain of 0.5, and $R_{IN3} = 20k\Omega$. The circuit is fed from a source resistance of 50Ω, and is operated under both matched and mis-matched source resistance conditions. This allows the degradation with mis-matching to be clearly shown as separation between the 50/50 (matched) and 50/55 (mis-matched) paired curves for each device type.

In the AD825 pair curves, the CM degradation is less than 1dB, even for the test condition of the relatively high 10% source resistance mis-match. The AD845 device has better overall CMR than the AD825, which shows up as low frequency errors of better than -100dB with the matched impedances. While mis-matching degrades CMR by a few dB, it is still -100dB or below as high as 20kHz. The AD845 can be used for U1A and U1B in the Fig. 6-30 circuit for highest wideband CMR (do however be careful to note the device dynamic range limits). Considering its superior overload behavior, the AD825 is the best choice, and is thus recommended.

These tests make clear that higher R_{IN} aids in de-sensitizing system CMR degradation against source mismatching. The designer has the option of using even higher input resistance for R_{IN1} and R_{IN2} , to further reduce the source sensitivity.

■ OP AMP APPLICATIONS

Transformer-Input Line Receiver

The classic solution to the CM isolation of audio signals is the *line input* transformer (see References 4,5, again). This device, usually a 1:1 ratio unit, offers galvanic isolation and very high CM voltage breakdown ratings. It is a preferred (or only) solution where true galvanic isolation is a necessity. Transformers are also useful for high and consistent low to middle audio frequency CM performance, both from unit-unit, and also when high immunity to varying differential source resistance is sought. These features do come at some cost however. Quality transformers are pricey, at about 10-20 times a single IC's cost. They also occupy a relatively large package size vis-à-vis a solid state equivalent.

All of the various factors above are the designer's ultimate decision points, dependent upon the exact system requirements. When optimized for high performance, it is not likely that either a completely solid-state or a completely transformer based line receiver solution will be considered either simple, or low in cost. Performance comes with a price.

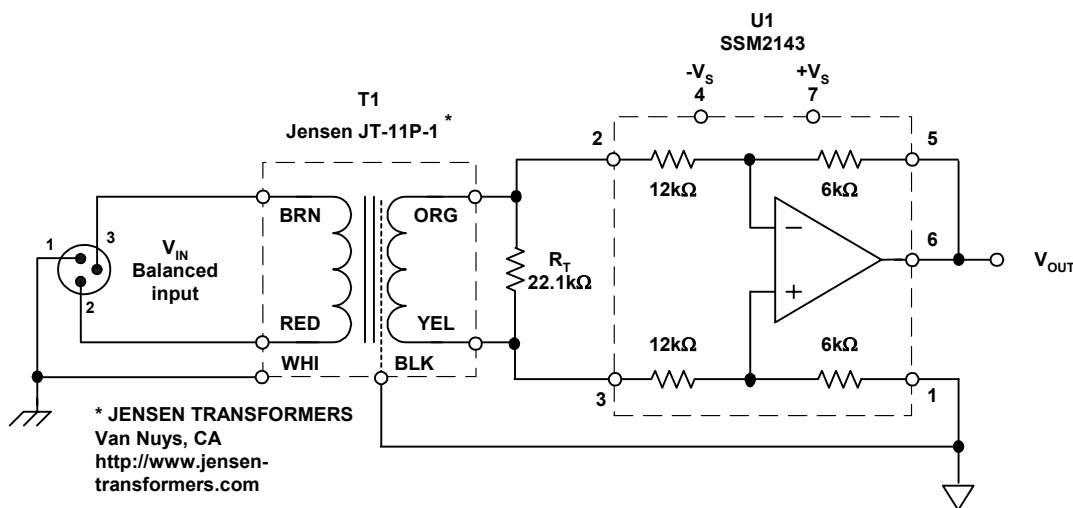


Figure 6-33: A transformer-input line receiver circuit

Interestingly, when a near-ultimate in low frequency CM rejection is required, or completely tweak-free operation is sought, a hybrid line receiver solution may be a good choice. An example of a line transformer buffered by a simple line receiver is shown in Figure 6-33 above. This circuit combines good features of the simple line receiver and the transformer, and offers outstanding performance with attractive simplicity.

The circuit has only three components; T1, the secondary termination resistor, and U1. It can be noted that this combination can operate the transformer in either a balanced mode (just as shown) or in an alternate single-ended mode (with U1-2 grounded). The choice of which mode is used does make a big CMR performance difference, as will be apparent.

The transformer secondary is terminated in a net 10kΩ resistance, which here is comprised of R_T and the 12kΩ input resistances of U1. If other forms of termination are used, or an alternate U1 part, R_T should be adjusted accordingly (see Reference 17). The net gain of the circuit is product of the loaded transformer primary/secondary voltage ratio, and the voltage gain of U1 (0.5). Inasmuch as T1 shows a voltage loss for such loaded conditions, the overall gain of the circuit is approximately 0.35 (or -9dB). If more

gain is desired, an alternate SSM2143 operational mode is possible. The 12k Ω /6k Ω input output resistors can be reversed, which will then allow it to operate at a gain of 2.

The CMR test results for the circuit of Fig. 6-33 are shown in Figure 6.34. Conditions for this test setup are similar to those of the buffered balanced line receiver test, described above. They are referenced to a 1V operating level at the output of U1.

Several important points should be apparent from these results. In the single-ended mode (upper curve), the basic data sheet performance of the JT-11P-1 transformer can be seen. This includes an approximate 60Hz CM error of -107dB . While this simply buffered operating mode does offers excellent low frequency CMR, it can also be noted that this also degrades with rising frequency. At 20kHz the single-ended mode error is just slightly more than -50dB — good, but not superlative. Any of the ICs tested for the Fig. 6-30 circuit better this performance, by about 30dB or more.

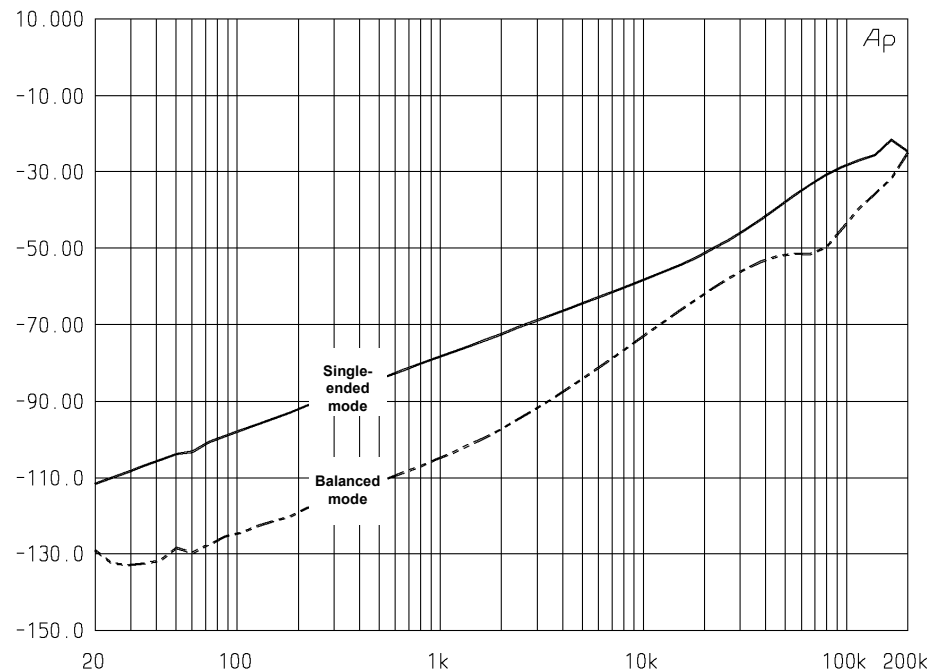


Figure 6-34: CMR errors for Fig. 6-33 transformer-input line receiver circuit, operating in single-ended and balanced modes

On the other hand, by simply letting the SSM2143 operate in a balanced mode, the lower curve CMR performance results. This is clearly a major improvement vis-à-vis the single-ended case, with the low frequency CM error reduced to -130 dB or better, approaching the noise floor of the instrumentation. The errors aren't quite as low at the higher frequencies, with a 20kHz CMR of $\sim 60\text{dB}$.

A point worth noting is the transformer-based line receiver cannot really compete with the buffered balanced receiver of Fig. 6-30 for high frequency CMR. This is because the transformer inter-winding capacitance acts as a high pass filter for CM noise. This has the effect of passing CM noise to the output at the higher frequencies. The degree of severity for this phenomenon will of course vary with the design of the specific transformer.

■ OP AMP APPLICATIONS

Nevertheless, it is readily apparent from the performance data that the preferred method of transformer operation is to operate it in a balanced secondary mode (i.e., as shown in Fig. 6-33), which does mitigate the CMR loss with frequency somewhat.

A Summary of Line Receivers

Both active and passive solutions for minimizing balanced transmission system CM noise have been explored, each with their own set of characteristics. Readers should take these data for the general trends they convey, not any absolute performance levels.

To summarize, the buffered balanced line receiver of Fig. 6-30 offers excellent broadband CMR with low differential source resistance sensitivity, and can be user customized in a variety of ways, including gain, CM input impedance, etc. The solid state line receiver approach here has virtues of the best high frequency CMR in absolute terms, as well as the better CMR vs. frequency flatness. While the example circuit shown works well, optimization for a production role may need some enhancement for worst case minimum CMR. This can be done via careful trim or selection of the U2 circuit, and/or selection of an optimum pair of singles for U1A and U1B. Input dynamic range of the circuit can be optimized by selection of the U1A-U1B pair types, and the supply voltages used. All types tested can be operated at supplies of up to $\pm 18\text{V}$ (maximum), or as low as $\pm 13\text{V}$. The typically used $\pm 15\text{V}$ supplies will provide both excellent performance and high input dynamic range.

The transformer-input approach to a line receiver offers good to superlative low frequency CMR, combined with "no tweak" operation. When the transformer used is combined with an in-amp for balanced mode secondary buffering, as in Fig. 6-33, further CMR reduction is possible over a range of low to middle frequencies. On the downside, there are negatives of cost, size, and eventual CMR degradation with frequency.

REFERENCES: AUDIO LINE RECEIVERS

1. Walt Jung, "Op Amps in Line-Driver and Receiver Circuits, Part 1," **Analog Dialogue**, 26-2, 1992.
2. W. Jung, A. Garcia, "Op Amps in Line-Driver and Receiver Circuits, Part 2," **Analog Dialogue**, 27-1, 1993.
3. Walt Jung, "Applications for Amplifiers in Audio," Ch. 5 within Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, Inc., Norwood, MA, 1992, ISBN 0-916550-10-9.
4. Deane Jensen, "Transformer Application Notes (various)," Jensen Transformers, 7135 Hayvenhurst Avenue, Van Nuys, CA, 91406, (213) 876-0059.
5. Bruce Hofer, "Transformers in Audio Design," **Sound & Video Contractor**, March 15, 1986.
6. Henry Ott, *Noise Reduction Techniques in Electronic Systems*, 2d Ed., Wiley, 1988.
7. David Birt, "Electronically Balanced Analogue- Line Interfaces," **Proceedings of Institute of Acoustics Conference**, Windermere, U.K., Nov. 1990.
8. Robert Demrow, "Narrowing the Margin of Error," **Electronics**, April 15, 1968.
9. Robert Demrow, "Evolution from Operational Amplifier to Data Amplifier," **Analog Devices Application Note**, September, 1968.
10. Walter Borlase, "Application/Analysis of the AD520 Monolithic Data Amplifier," **Analog Devices Application Note**, 1972.
11. Jeff Riskin, "A User's Guide to IC Instrumentation Amplifiers, **Analog Devices AN244**, January, 1978.
12. S. Wurcer, L. Counts, "A Programmable Instrumentation Amplifier for 12-Bit Resolution Systems," **IEEE Journal of Solid State Circuits**, Vol. SC-17 #6, Dec. 1982.
13. N. Muncy, "Noise Susceptibility in Analog and Digital Signal Processing Systems," **JAES**, Vol 43, No 6, June, 1995.
14. B. Whitlock, "Balanced Lines in Audio— Fact, Fiction, and Transformers," **JAES**, Vol 43, No 6, June, 1995.
15. B. Whitlock, "A New Balanced Audio Input Circuit For Maximum Common-Mode Rejection In Real-World Environments," **presented at 101st AES Convention**, November 1996, preprint #4372.
16. Walt Jung, "Practical Circuits for Quiet Audio Transmissions," **Electronic Design Analog Applications Issue**, November 17, 1997, pp. 45-49
17. "JT-11P-1 Line Input Transformer Data Sheet," Jensen Transformers, 7135 Hayvenhurst Avenue, Van Nuys, CA, 91406, (213) 876-0059.

■ OP AMP APPLICATIONS

Audio Buffers and Line Drivers

Audio line drivers and buffer amplifiers can take on a wide variety of forms. These include both single-ended and differential output drivers, as well as transformer isolated drivers. Within these general formats there are also many different performance options, and many of these are covered in this section. Note that a later section of this chapter also discusses buffers for a general context, as for video and instrumentation applications.

Many op amps useful as video drivers and/or buffers do well for audio drivers/buffers, because of the high current output stages necessary for good linearity over video bandwidths (see References 1-4). Some examples of video IC amplifiers that are audio-useful are the AD810, AD811 and AD812, AD815, AD817 and AD826, AD818, AD829, AD845, and AD847. Other types notable for either high or unusually linear output drives or other performance features useful towards audio are the AD797 and the OP275.

Some High Current Buffer Basics

As a preliminary to detailed application discussions, some basic circuit principles germane to high current buffers and drivers should be treated first. With output currents up to 100mA or more, "housekeeping" details of bypassing, grounding and wiring also become important, and must be considered to achieve high performance. These are briefly discussed here in the context of high current audio buffers, using the unity gain buffer circuit of Figure 6-35 below, as a point of departure.

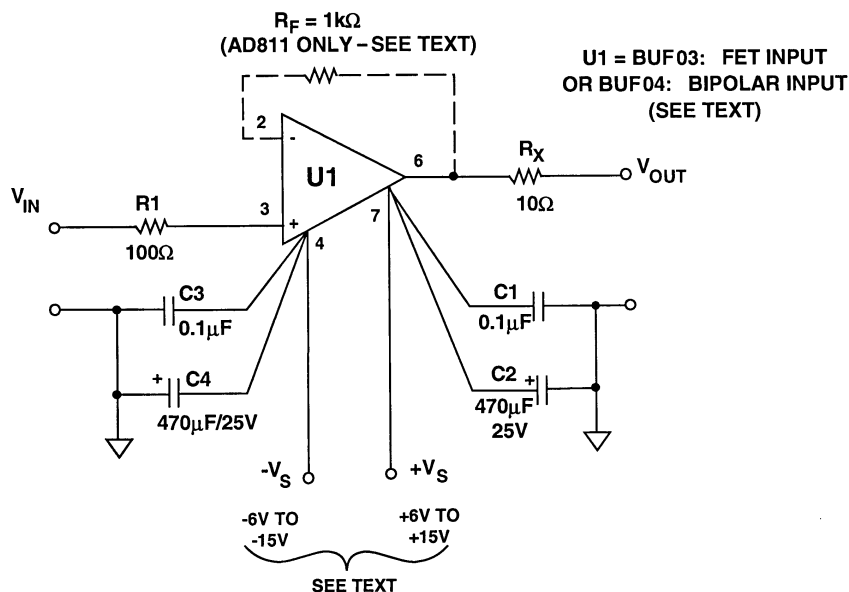


Figure 6-35: A unity-gain, standalone buffer circuit

First, despite which IC is used for U1, close attention should be given to making buffer stages free from parasitic effects, at both input, output, and supplies. Physical construction of buffer-drivers and other high current stages should be in accordance with high speed rules. A heavy copper ground plane is preferred, and circuit layout should be compact, with low capacitance high-Z nodes. Signal and ground runs should be laid out with signal coupling and load current flow in mind (see References 5-7 and Chapter 7).

In addition, the power supplies should be well bypassed close to the high current supply pins. In Figure 6-35 this is indicated by the Kelvin connections of C_1 - C_4 to the U1 $\pm V_s$ pins. This should be used as standard practice for all high current stages, and is intended as a given for all the driver applications of this section.

As a minimum, local low inductance/low ESR RF bypass caps should be used within 0.25" of the device supply pins, shown as C_1 and C_3 . These are preferably 0.1 μF stacked polyester film, or other low inductance capacitor type, preferably films. In addition, for high peak current loads, the high frequency bypasses are paralleled by local, short lead/large value, low ESR electrolytics such as C_2 and C_4 , in a range of 470 μF /25V and up. Note that capacitor ESR reduces in inverse proportion to electrical size and voltage rating, so larger size and/or voltage units help. These capacitors carry transient output currents, and should be aluminum electrolytic types rated for high frequency use, that is switching supply types. Such types tend to have a broad range of lowest high frequency minimum impedance and are thus less likely to cause power line resonance than are tantalum units.

DC power management and dissipation can also be important with buffer ICs. For example, the BUF03 and the AD811 ICs can dissipate fairly large power levels even with light loading, for supplies above $\pm 12V$. This is because the quiescent current of these devices is 15-18mA, relatively independent of operating voltage.

As a conservative general rule of reliability, any IC with a power dissipation above 300mW should not be used without a heat sink. For buffer or driver circuits using this power or more, use the lowest thermal resistance package possible, and add the appropriate heatsink (Thermalloy 2227 for the BUF03 or other TO-99 ICs, or Aavid #5801 for the BUF04, AD811 or other high dissipation 8 pin DIP ICs).

Output resistor R_X in this circuit should be 10 ohms or more, to isolate the buffer from capacitive loading (more on this, elsewhere in this chapter). For an extra safety margin against possible de-stabilization due to capacitive loads, make this resistor as high as feasible from a voltage loss point of view.

The input resistor R_1 is a "bullet-proof" safety item, and can serve two purposes. One is as a parasitic suppression device, which may be required for stability with some amplifiers (not absolutely essential for those here). A subtler feature of this resistance comes about when the buffer is operated within a feedback loop, and is driven from an op amp output. Internally, many buffer ICs have clamping diodes from input to output, and under overload conditions, these diodes act to clamp overdrive. With the inclusion of R_1 , this prevents excess current drive into the buffer IC under this clamping condition.

Because of this stage's very high bandwidth, low phase shift, and low output impedance, fast buffers such as this can be used both "stand alone" just as shown, or as a more conventional "in loop" buffer as well, to minimize loading of a weaker, slower amplifier. The improvement raises the linear output up to $\pm 100mA$ with the AD811 or the BUF04, while maximizing linearity, preserving gain, and lowering distortion.

■ OP AMP APPLICATIONS

Buffer THD+N Performance

Operating in a pure stand-alone mode, THD+N tests on several unity gain buffers are shown in Figure 6-36 below. These tests were for common conditions of 10Vrms output into a 600Ω load, operating from ±18V power supplies.

The BUF03, an open loop design, shows a distortion for these conditions of about 0.15%. The BUF04, a closed loop current feedback design buffer, shows a very low distortion of about 0.004%. The AD811 is also a current feedback amplifier, but it is externally configured as a unity gain follower, with $R_F = 1\text{k}\Omega$. Note that *all current feedback ICs will require such a resistor*, but the value required may vary part to part. The AD811 shows an intermediate distortion level, under 0.01%.

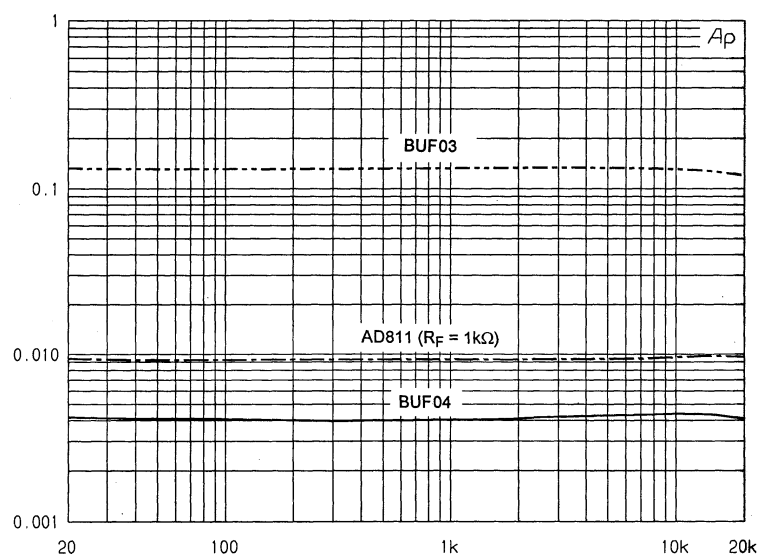


Figure 6-36: THD+N (%) vs. frequency (Hz) for various buffer ICs, for $V_{OUT} = V_{IN} = 10\text{Vrms}$, $R_{LOAD} = 600\Omega$, $V_S \pm 18\text{V}$

As a choice among these types, both the BUF04 and AD811 are capable of more than ±100mA of output, with input currents on the order of 1-2μA. The BUF03 has a lower output current (±70mA), but the advantage of a much lower input current (~200pA).

Dual Amplifier Buffers

In addition to standard operation of the various single op amps as unity-gain buffers, certain high output current *dual* op amp ICs also work exceedingly well as buffers. Using a dual IC to buffer a signal has the advantage of doubling the output drive while using basically the same package size, an obvious benefit. Two design steps allow this to be implemented successfully. The first is the selection of a basically linear single device that also is available as a dual. The second is to devise a method of combining the outputs of the two op amps in a linear fashion, without any side effects.

Among the suitable candidates for this task are the dual version of the AD817, the AD826, as well as the AD811 dual, the AD812. Figure 6-37 below shows a hookup that is useful towards increasing buffer output current to more than 100mA.

Ignoring Q1-Q2 for the moment, the circuit can be seen as a pair of unity-gain followers paralleled at the output, through small value resistors R_3 and R_4 . These resistors provide balanced drive from the U1A and B sections, linearly combining the signals. With the use of the voltage feedback AD826 op amp, the circuit is quite simple, since R_1 and R_2 reduce to zero. If the AD812 current feedback device is used, these two resistors should be 1k Ω (shown dotted). The Q1-Q2 bi-directional clamp circuit is optional, and when used can provide protection against input overdrive, and/or adjustable current limiting via R_5 .

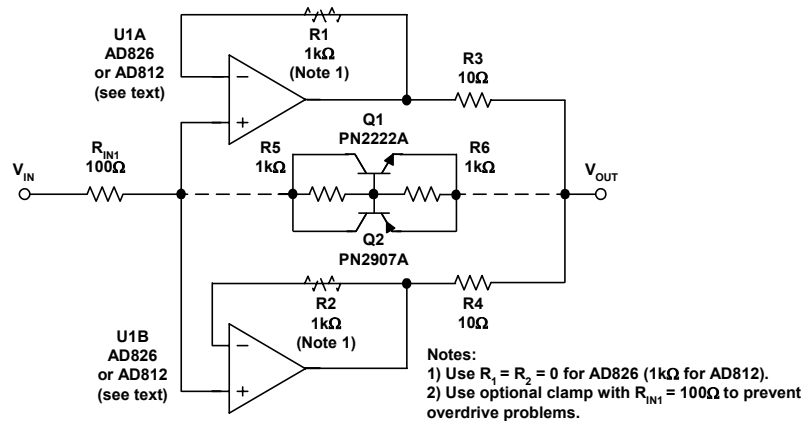


Figure 6-37: Dual op amp buffer circuit raises output current to more than 100mA with low distortion

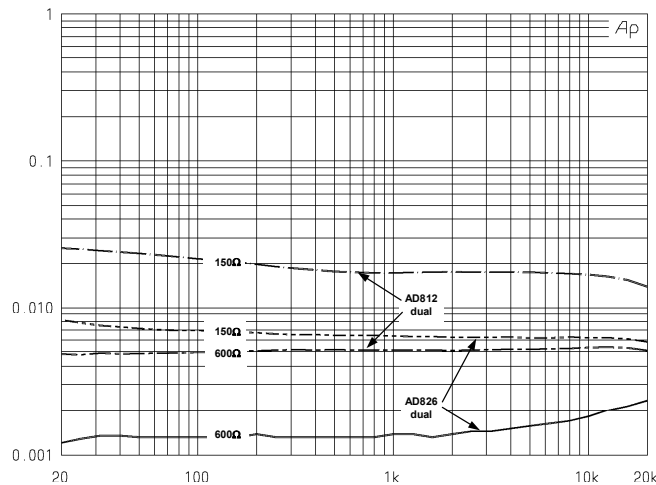


Figure 6-38: THD+N (%) vs. frequency (Hz) for AD826 and AD812 dual buffer ICs, for $V_{OUT} = V_{IN} = 10V_{rms}$, $R_{LOAD} = 150/600\Omega$, $V_S \pm 18V$

This circuit offers excellent performance, as shown in Figure 6-38 above. These THD+N plots show performance with loads of both 600 and 150 Ω , at an output level of 10V_{rms}, from $\pm 18V$ supplies (without clamping active). The AD826 offers the lowest distortion, due to the voltage feedback architecture, with less than 0.01% THD+N, even when driving 150 Ω , which is an approximate 100mA combined peak output.

■ OP AMP APPLICATIONS

Capacitive Loading Issues

Audio driver output stages are typically operated as voltage sources feeding high impedance loads. When connected via long transmission lines between stages, the result is that the driver sees an unterminated line, which can appear highly capacitive. Audio driver stability with capacitive loading can be a difficult design issue, but for good reason—it isn't always an easy thing to achieve. If easy, it may be at the expense of performance or circuit complexity. Fortunately, some standard techniques exist for stabilizing op amp drivers with capacitive loads, and these can be implemented in a reasonably direct fashion. These are covered in detail within the next section of this chapter. The discussions immediately following emphasize driver linearity.

Op Amp Device/Topology Related Distortions

Single-ended audio drivers can be built using a linear, non-inverting gain stage as a starting point. Indeed such a circuit, given appropriate op amp choice and gain scaling, can well serve as a basic audio driver. Topologically, a non-inverting gain stage is preferable, since it loads the signal source less, and, it also adds no sign inversion. However, this configuration is subject to certain distortions, which should be understood in order to extract the best performance in an application. Distortion performance for a number of audio op amps in such a line driver circuit are now discussed, in this context.

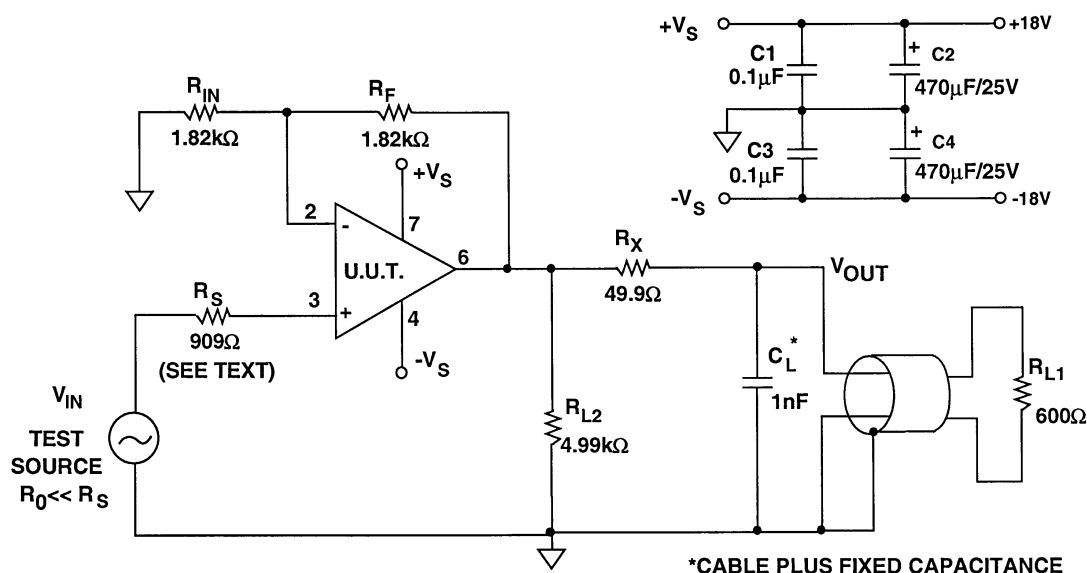


Figure 6-39: Test circuit for audio line driver amplifiers

The circuit of Figure 6-39 above is a test configuration that loads the U.U.T. op amp with 500Ω and 1nF. This is a reasonably stressful test load, which can differentiate the distortion of various devices with outputs of 7Vrms or more. A gain of 2 is used, which subjects the U.U.T. to a relatively high input CM voltage, thus this configuration is sensitive to CM distortion in the amplifier. For the following tests of this section (except as noted to the contrary), $V_S = \pm 18V$, and the analyzer bandwidth is 10Hz-80kHz.

Given amplifiers with sufficient load drive and output stage linearity in this circuit, there can still be non-linear effects due to the CM voltage. This distortion is due to the non-

linear C-V characteristic seen at the two amplifier inputs, and can be minimized by matching the two impedances seen at the respective (+) and (−) inputs (see Reference 8). When this done, the differential component of the error is minimized, and the distortion seen in V_{OUT} falls to a minimum.

This general point is illustrated by Figure 6-40, a family of plots for an OP275 op amp within the circuit of Figure 6-39. The OP275 use junction FET devices in the input stage, which have appreciable (non-linear) capacitance to the substrate. The test is done with various values of source resistance R_S . As noted, distortion is lowest when R_S is equal to the parallel equivalent of R_F and R_{IN} , or in this case, about 910Ω . For either higher or lower values of R_S , distortion rises. Appreciably higher source impedance ($10k\Omega$) can cause the distortion to rise lower in frequency, making performance much worse overall.

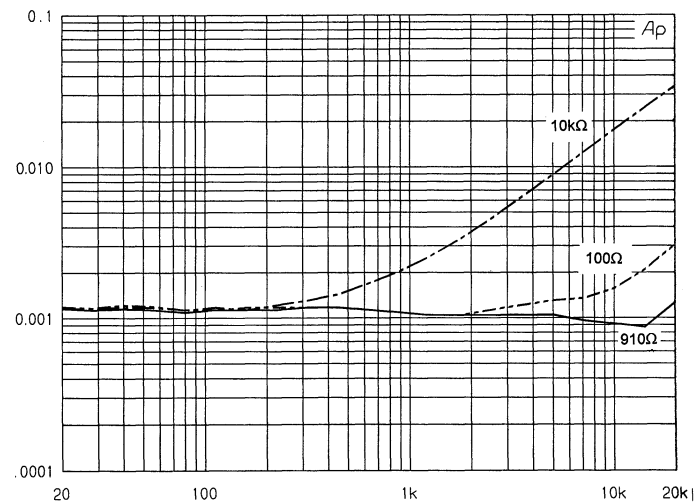


Figure 6-40: Follower mode R_S sensitivity of OP275 bipolar/JFET input op amp—THD+N (%) vs. frequency (Hz), $V_{OUT} = 7V_{rms}$, $R_L = 500\Omega$, $V_S = \pm 18V$

It is therefore suggested that, whenever possible, amplifiers operated as voltage followers should have their source impedances balanced for lowest distortion. Note that the OP275 device is just one example, and its sensitivity to CM distortion effects is not at all unique in this regard.

While the balancing of the two source impedances is most helpful, *lowering the absolute value* can also minimize this distortion. With R_S low, this has the effect of moving the high frequency breakpoint of the distortion rise upwards in the spectrum, where it is less likely to be harmful. The best overall control of this distortion mechanism with an amplifier subject to it is the use of the lowest practical, balanced source impedances.

It is important to understand that virtually all IC op amps, *particularly those using JFET inputs*, as well as discrete JFET and bipolar transistors are subject to non-linear C-V effects, to some degree. In the tests of other amplifiers within the Figure 6-39 circuit, R_S was maintained at 910Ω , so as to minimize the effects of this distortion mechanism.

■ OP AMP APPLICATIONS

With high output, high slew rate linear amplifiers, the distortion generated for these test conditions can parallel that of the test equipment residual, as shown in Figure 6-41. Here the AD817, AD818 and AD845 amplifiers show THD+N which is essentially equal to the residual for these conditions, and appreciably below 0.001%.

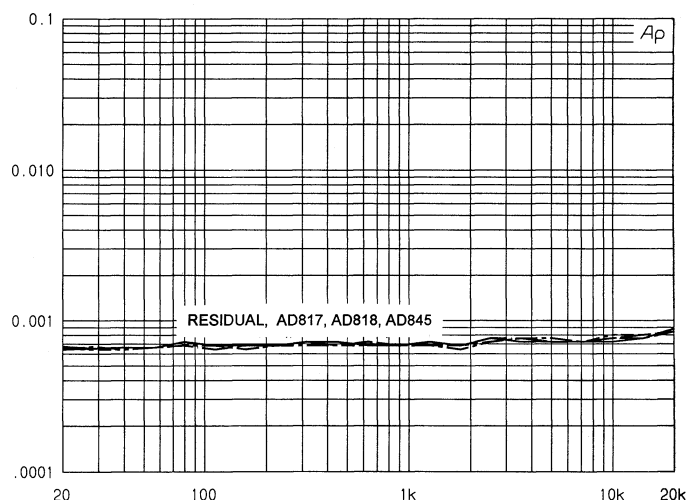


Figure 6-41: A driver group, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 7V_{rms}$, $R_S = 909\Omega$, $R_L = 500\Omega$, $V_S = \pm 18V$

Amplifier types expressly designed for audio use also do well for these THD+N tests, as shown in Figure 6-42. The industry standard 5534 is near or just above the residual level, while the OP275 plot falls just above the 0.001% level and the 5532 is slightly higher.

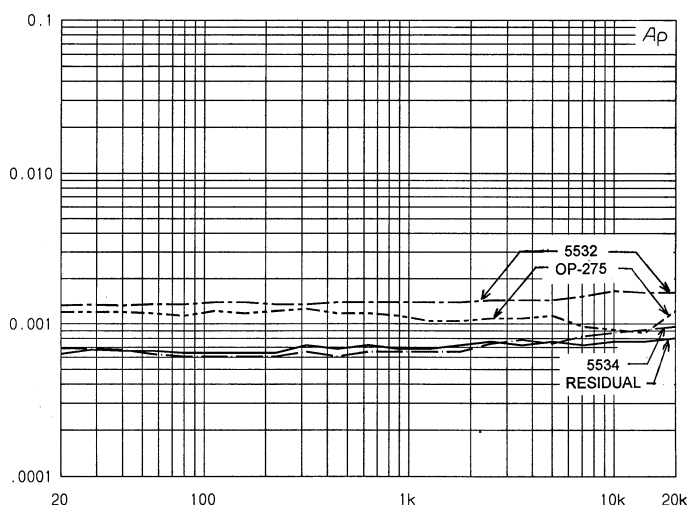


Figure 6-42: B driver group, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 7V_{rms}$, $R_S = 909\Omega$, $R_L = 500\Omega$, $V_S = \pm 18V$

These tests reflect performance of a variety of single amplifiers, as exercised for matched-source test conditions, with medium output loading of 600Ω . Varying test conditions may change the absolute levels of performance. So also may different samples, or in the case of industry standard parts, alternate vendors.

The data of Figs. 6-41 and 6-42 reflect older (but available) op amp devices capable of very high performance in these tests. Several more recent devices also do well for this driver test. Figure 6-43 below shows performance of newer FET input op amps, the AD825, the AD8610 and the AD8065. The AD825 was tested under conditions identical to those of Fig. 6-41 and 6-42. The AD8610 and AD8065 were tested under similar conditions, but with $\pm 13V$ power supplies, reflecting a lower maximum supply rating.

Interestingly, note that the latter two amplifiers still can accommodate more than a 7Vrms output swing, even with the reduced supplies. Under these conditions, the AD8065 distortion is near the test set residual and the AD8610 slightly higher at high frequencies. The AD825 has somewhat higher distortion, but this is almost frequency independent.

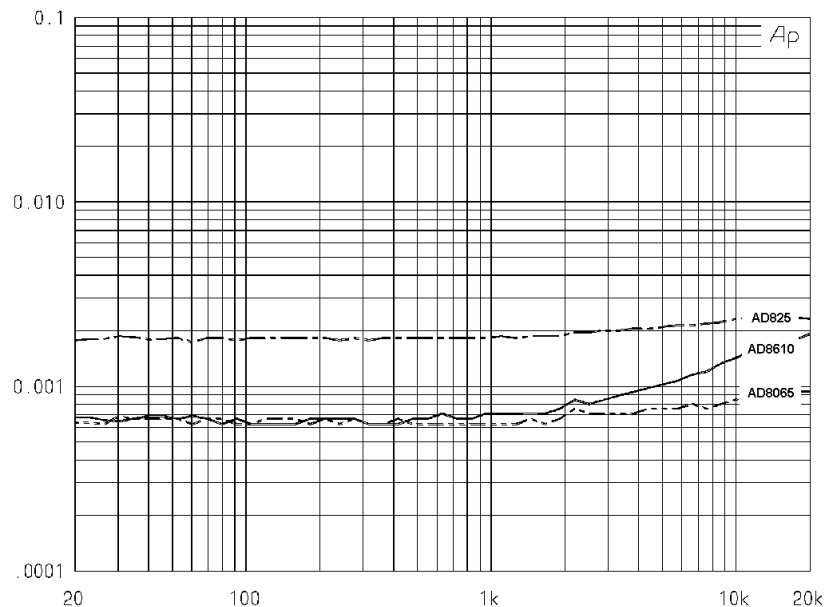


Figure 6-43: *C* driver group, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 7V_{rms}$, $R_S = 909\Omega$, $R_L = 500\Omega$, $V_S = \pm 13V$ or $\pm 18V$

Single-Ended Line Drivers

This section discusses a variety of line driver circuit examples that drive single-ended lines, optimized for different operating environments, supply voltages, and performance.

Consumer Equipment Line Driver

One common driver application is a line output stage for consumer preamps, CD and DVD players, etc. This is typically an economical audio stage with a nominal gain of 5 to 10 times, operating from supplies of $\pm 10V$ to $\pm 18V$, usually with a rated output of 2-3Vrms, and a capability of driving loads of 10k Ω or more.

For simplicity of biasing and minimum output DC offset, AC coupling is used, and the circuit is typically fed from a volume control. For stereo operation, a dual channel device is typically sought for this type application, one which is also optimized for audio uses.

■ OP AMP APPLICATIONS

Such a stage is shown in Figure 6-44 below, and it uses an OP275 dual op amp as the gain element. In this circuit input and feedback resistors R_1 and R_3 are set equal, which makes the nominal DC bias at U1's output close to zero. The U1 bias current flowing in these resistors also serves to polarize coupling capacitors C_1 and C_2 positively, as noted. This bias is due to the sign of the OP275's PNP input stage bias currents, so reverse C_1 - C_2 if an NPN input amplifier is used.

R_2 sets the gain of the stage in conjunction with R_1 . The stage gain is nominally 5 times for the values shown. C_2 sets the low frequency rolloff along with R_2 , which in this case is $\approx 0.3\text{Hz}$. Although this frequency is quite low, it does allow some range of gain increase if desired, simply by lowering R_2 . Output capacitor C_3 must be non-polarized, since the worst case DC at the output of U1 is $\leq 10\text{mV}$ (and can be bipolar). Typically it will be about $\frac{1}{4}$ this, so if a few mV can be tolerated, C_3 can be eliminated.

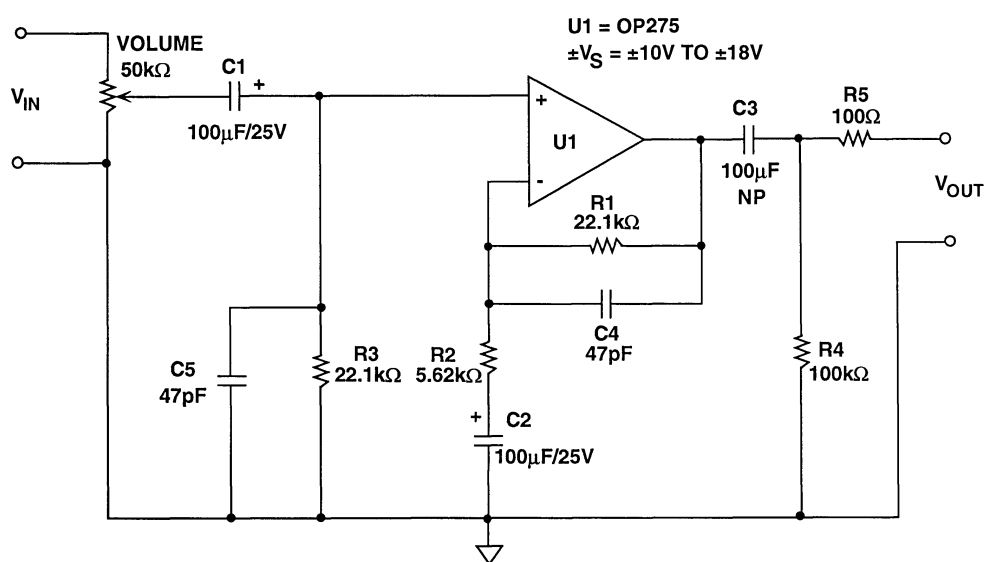


Figure 6-44: Consumer equipment line driver stage

THD+N performance of the stage (not shown) was measured for outputs of 1-3Vrms into a 10k Ω /600pF load, using $\pm 18\text{V}$ supplies with an R_S of 1k Ω . At lower output levels performance is noise limited, measuring less than 0.002%. At the 3V output level a slight increase in high frequency distortion is noted. Although this application is an example where the amplifier $\pm$ source impedances cannot be matched (due to the variations of the volume control), nevertheless the performance is still quite good.

Noise is the limiting factor for lower level signals, so if lower noise is desired, R_2 can be reduced. The ultimate practical limit to noise is the volume control's finite output impedance. This causes higher noise at positions of high output resistance, interacting with the noise current from U1. For example if the effective R_S from the volume control is 10k Ω , a 1.2pA/ $\sqrt{\text{Hz}}$ noise current from U1 will produce an input referred 12nV/ $\sqrt{\text{Hz}}$ noise voltage, from this source alone. The Fig. 6-44 driver is a flexible one, and operates at supplies as low as $\pm 10\text{V}$ with outputs up to 3Vrms, with slight distortion increases. With $\pm 5\text{V}$ supplies up to 2Vrms is available, with higher distortion (but still $\leq 0.01\%$).

Paralleled Output Line Driver

Often a modest increase in output may be needed for a driver, but circumstances may not warrant the use of additional buffer devices. Figure 6-45 shows how a second section of a dual op amp can be used to provide additional load drive.

In this circuit using an OP275 dual op amp, the U1A section is a gain-of-five voltage amplifier, while the U1B section is a voltage follower, used simply to provide additional current to the load. Current sharing is determined by output summer resistors R_4 and R_5 , and the parallel stage drives 600 Ω loads with less distortion than a single OP275 section.

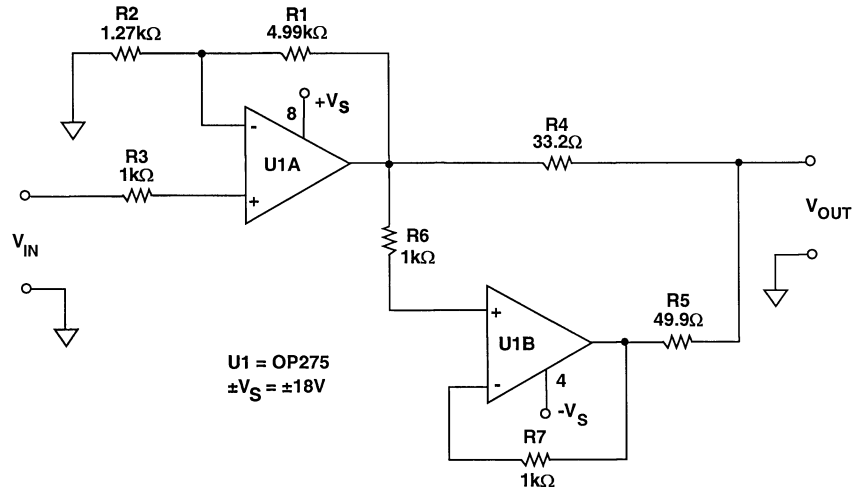


Figure 6-45: Paralleled output dual op amp line driver

THD+N performance data is shown in Figure 6-46 below, with the driver operating from $\pm 18\text{V}$ supplies, and for output levels of 1, 2, 5, and 9Vrms into 600 Ω .

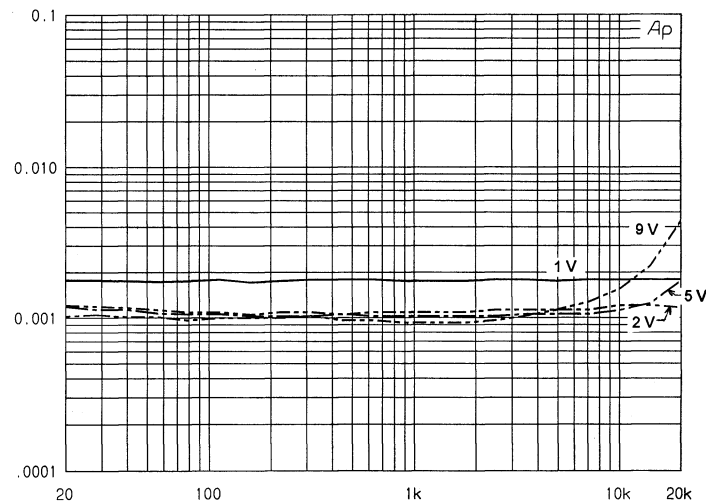


Figure 6-46: Paralleled output dual op amp line driver, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 5, 9\text{Vrms}$, $R_L = 500\Omega$, $V_S = \pm 18\text{V}$

This general scheme can be used with any unity gain stable dual op amp, and also can be adapted for various gain levels, via R_1 - R_2 . For different devices and/or gains, the ratio of R_4 and R_5 may need adjustment, for lowest distortion into the load.

■ OP AMP APPLICATIONS

A Wide Dynamic Range Ultra Low Distortion Driver

Single ended line drivers are simple conceptually, but when pushed to performance limits in dynamic range and distortion, they challenge device choice. The AD797 answers this challenge with its input noise of $\leq 1\text{nV}/\sqrt{\text{Hz}}$ and a distortion canceling output stage. These features allow low and high extremes of dynamic range to be pushed simultaneously.

The AD797 uses a single voltage gain stage, comprised of a folded cascode input combined with a boot strapped current mirror load, allowing the high incremental impedance necessary for a 146dB gain. This buffered single-stage topology is a departure from past devices using multiple stages, with performance benefits in terms of bandwidth, phase margin, settling time, and input noise (see Reference 9).

For standard uses, the AD797 is employed like any 5 pin op amp, such as shown in Figure 6-47 below (neglecting the capacitors for the moment). From the A/B part of the table, relatively low values for resistors R_1 - R_2 are recommended for lowest noise. Selecting these resistors should be done with care, since values $\geq 100\Omega$ will degrade noise performance. Suggested values for gains of $G = 10$ -1000 are noted. The AD797 can drive loads of up to 50mA, and is rated for distortion driving loads of 600Ω .

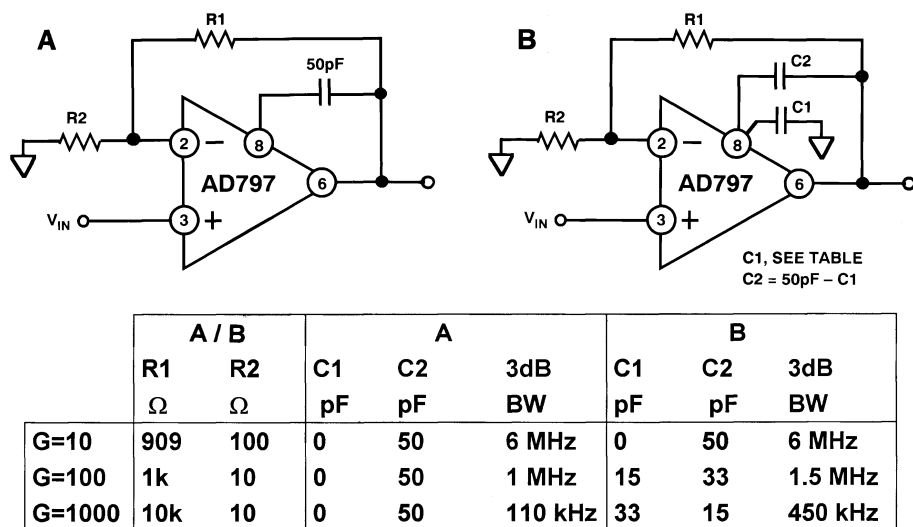


Figure 6-47: Recommended AD797 connections for distortion cancellation and/or bandwidth enhancement

For amplifier applications requiring more top grade performance, optional capacitors C_1 and C_2 can be used. In the Fig. 6-47A configuration, superior performance is realized due to distortion cancellation with the use of a single extra capacitor, enabled simply by adding the 50pF unit as shown. This provides compensation for output stage distortion without effecting the forward gain path, effective over the range of gains noted.

An additional option with the AD797 is the use of *controlled decompensation*, available with the Fig. 6-47B option and the use of capacitors C_1 , C_2 . At gains of 100 or more, adding C_1 as in column B of the table enhances amplifier open loop bandwidth, allowing very high gain-bandwidths to be achieved— 150MHz at $G=100$, and 450MHz at $G=1000$. For high gain operation this extra gain-bandwidth can be very effective.

A family of distortion curves for various AD797 gain configurations driving 600Ω is shown in Figure 6-48 below. As noted, at low frequencies the data is limited by noise, while at high frequencies distortion is measurable, but still extremely low. The distortion for a gain of 10 times at 20kHz for example, is on the order of $\approx 0.0001\%$, implying a dynamic range of about 120dB re $3V_{rms}$, or even more for higher level signals.

An additional point worth making at this point is one regarding the AD797's special distortion cancellation ability. Referring to Fig. 6-47A (again), it should be noted that the 50pF capacitor is connected between pins 8 and 6 of the AD797. Pin 6 is of course the output of the device for standard hookups. However, in special situations, even greater output current may be required, and a unity gain buffer amplifier can be added between pin 6 and the load. For example, one of the buffers of Fig. 6-36 or 6-37 could be used to extend output current to $\geq 100mA$.

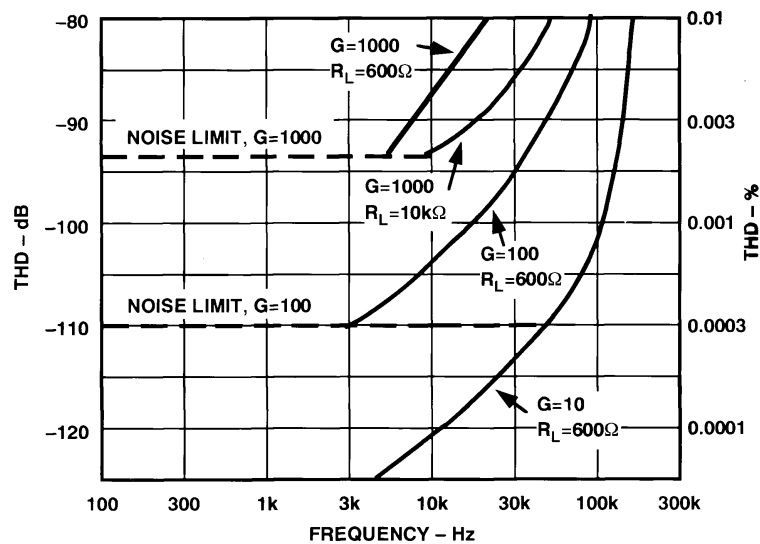


Figure 6-48: THD vs. frequency at $3V_{rms}$ output for AD797 distortion cancellation and/or bandwidth enhancement circuit of Fig. 6-48

The special point worth noting for this situation is that the 50pF distortion cancellation capacitor should then be connected between the AD797 pin 8 and the output of the buffer (not the AD797 pin 6). This step allows the distortion correction to be applied not just to the AD797 internal circuits, but also extends it to include the buffer.

A case in point where this would come to useful purpose lies with the AD797 mic preamp, discussed in some detail earlier in the chapter (Fig. 6-5, again). As mentioned therein, a BUF04 would work well as just such a buffered output option for the AD797 preamp. It would be connected as described above, with the 50pF distortion cancellation capacitor. Details of this are left as a reader exercise (but should even so be obvious).

■ OP AMP APPLICATIONS

Current Boosted Buffered Line Drivers

When load drive capability suitable for less than 600Ω in impedance is required, it is most likely outside the output current and/or linearity rating of even the best op amps. For such cases, a current boosted (buffered) driver stage can be used, allowing loads down to as low as 150Ω (or less) to be driven. Another example would a driver for long audio lines, i.e., lines more than several hundred feet in length.

Figure 6-49 below is a high quality current boosted driver example, using an AD845 at U1 as a gain stage and voltage driver, in concert with a unity voltage gain current booster stage, U2. The overall voltage gain is 5 times as shown, but this is easily modifiable via alternate values for R_1 and R_2 . In any case, for lowest CM distortion effects, input resistor R_3 should be set equal to $R_1 \parallel R_2$ (this assumes a low impedance source for V_{IN}).

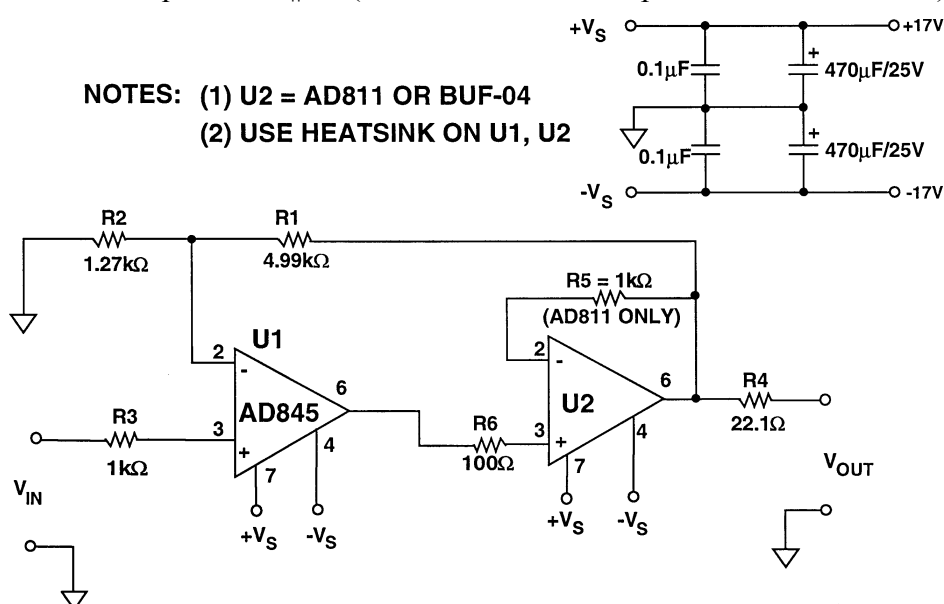


Figure 6-49: Current boosted line driver

The amplifier used for U2 can be either the AD811 op amp, or the BUF04 buffer for simplicity. If the AD811 or similar CFB op amp is used (AD812 etc.), it needs to be configured as a follower, with R_5 connected as shown. Since the BUF04 is internally connected as a follower, it doesn't need the R_5 external feedback resistor.

Because of the high internal dissipation of the AD845 and AD811, these devices *must* be used with a heat sink on supplies of $\pm 17V$. But, such high supplies are only justified for extreme outputs. Supplies of $\pm 12V$ also work, and will eliminate need for a heat sink (with lower maximum outputs). In any case, power supplies should be well bypassed.

A special note is applicable here— Always observe maximum device breakdown voltage ratings within applications. Production versions of this circuit should use supplies of $\pm 17V$ or less, for 36V(max) rated parts. Similarly, 24V(max) rated parts should use supplies of $\pm 12V$ or less. In all cases, use only enough supply voltage to achieve low distortion at the maximum required output swing.

For loads of 150Ω , the output series isolation resistor R_4 is lowered to 22.1Ω to minimize power loss, and to allow levels of $7V_{rms}$ or more. The THD+N data for this circuit is shown in Figure 6-51 below, using an AD811 as the U2 buffer. The test conditions are input sweeps resulting in 1, 2, 4 and $8V_{rms}$ output, using $\pm 18V$ power supplies.

For the AD811 operating as U2, the Figure 6-50 data below shows THD+N dominated by noise and residual distortion at nearly all levels and frequencies driving 150Ω , up to $8V_{rms}$. At this level, a slight distortion rise is noted above $10kHz$, yet it is still $\approx 0.001\%$. With the BUF04 as U2 (not shown) THD+N is comparable at lower output levels, but does show a distortion rise with $8V_{rms}$ output at high frequencies (yet still below 0.01%).

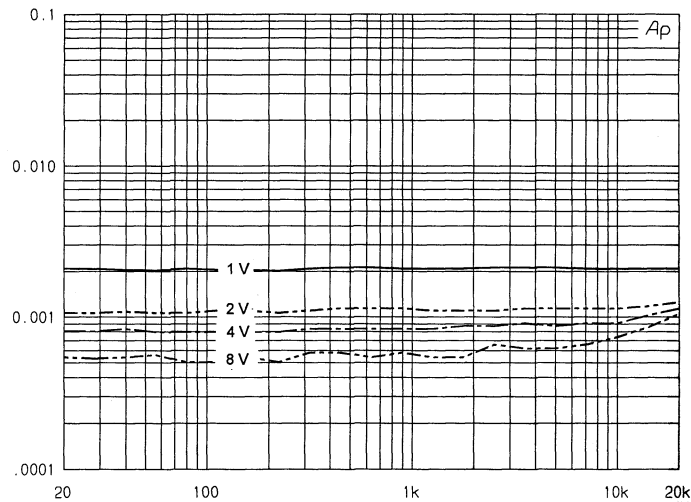


Figure 6-50: Current boosted driver of Fig. 6-50 using AD811 as U2, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 4, 8V_{rms}$, $R_L = 150\Omega$, $V_S = \pm 18V$

There is a power/performance tradeoff involved with the choice between the two mentioned U2 devices which should be understood. The BUF04 has a standby dissipation of about $200mW$ on $\pm 15V$, while the AD811 is more than double this dissipation, at $500mW$. So while the AD811 does yield the lower distortion, it also should be operated more conservatively from a power standpoint. As noted above, only the minimum ($\pm$) supply voltage required to sustain a given output should be used with the circuit in general, and particularly with the AD811 employed at U2.

As for U1 in this circuit, other amplifiers can be used, but only with due caution against poor performance. Quite simply, it is difficult to improve upon the AD845's performance in this application. Three possible candidates would include the "Group C" op amps of Fig. 6-43, operating on suitable power supplies; $\pm 17V$ or less for the AD825, and $\pm 12V$ or less for the AD8610 and AD8065. Of these, the AD8065 would seem to hold the greatest promise, having shown the lowest wideband distortion in the Fig. 6-43 tests.

But, as considered within the buffered driver circuit of Fig. 6-50, the AD8065 will be operated in an even more linear fashion; that is it is operating essentially unloaded at the output. This is a key factor towards highest performance, as it moves the burden of linear load drive to the buffer stage. Further variations of this circuit technique will be reprised later, within other driver applications to be discussed.

■ OP AMP APPLICATIONS

Composite Current Boosted Drivers

Another useful current-boosted circuit technique combines the positive aspects of two different amplifiers into a single composite amp structure, producing a very high performance line driver (see References 10-13 for several variations of this basic circuit). With an FET input IC used as the input stage, DC offset change from source resistance variations of a typical volume control of $\approx 50\text{k}\Omega$ is nil, allowing total direct coupling. As noted previously, with a high current, wide band booster output stage, line impedances down to 150Ω can be driven with excellent linearity.

This type of composite amplifier allows good features of two dissimilar ICs to be exploited; each optimized for the respective input and output tasks. Figure 6-51 shows a low distortion composite amplifier using two op amps ICs with such performance.

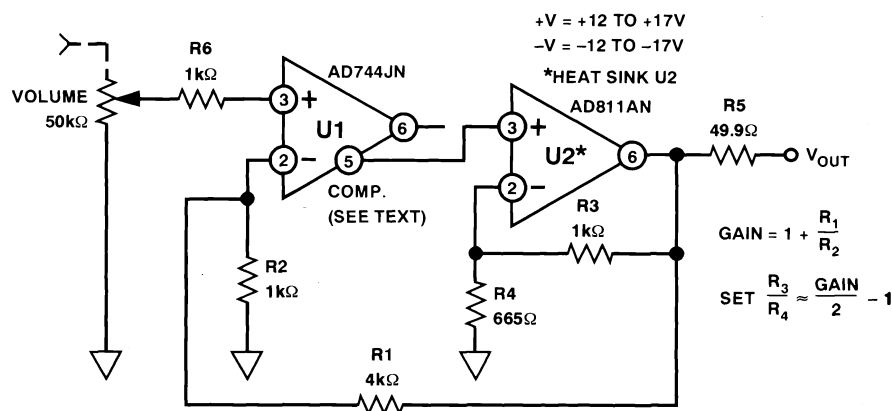


Figure 6-51: Composite current boosted line driver one

A factor here aiding performance is that the U1 AD744 stage operates unloaded, and also that the AD744's compensation pin (5) drives U2. This step (unique to the AD744) removes any possibilities of U1 class AB output stage distortion. Another key point is that the overall gain bandwidth and SR of U1 are boosted by a factor equal to the voltage gain of U2, an AD811 op amp, which itself operates at a voltage gain. These factors enhance this circuit by providing both high and linear load current capability, providing a composite equivalent of an FET input power op amp.

This design operates at an *overall* voltage gain set by R_1 and R_2 (just as a conventional non-inverting amplifier) which in this case is 5 times. Since the circuit also uses a local loop around stage U2, the R_3/R_4 ratio setting the U2 stage gain should be selected as noted. This complements the overall gain set by R_1 and R_2 , and optimizes loop stability.

Also note that the U2's feedback resistor R_3 has a preferred minimum value for stability purposes (again, as is unique to CFB amplifier types). Here with the AD811, a $1\text{k}\Omega$ value suffices, so this value is fixed. R_4 is then chosen for the required U2 stage gain. Further design details are contained in the original references (see References 2 and 10, again).

The composite amplifier performance for a typical audio load of 600Ω , THD+N at output levels of 1, 2, 4 and 8Vrms is shown in Figure 6-52 below, while operating from supplies of $\pm 18V$ for this test. The apparent distortion is noise or residual limited at almost all levels, rising just slightly at the higher frequencies.

Lower impedance loads can also be driven with this circuit, down to 150Ω . Note that for operating supply voltages of more than $\pm 12V$, a clip on heat sink is recommended for U2, as previously discussed for the AD811. Practical versions of this circuit can readily use supplies of $\pm 12V$, and still operate very well.

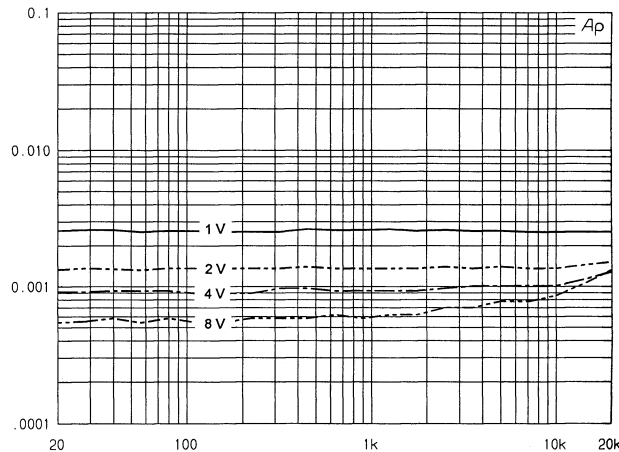


Figure 6-52: Composite current boosted driver one of Fig. 6-51, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 4, 8V_{rms}$, $R_L = 600\Omega$, $V_S = \pm 18V$

The circuit of Fig. 6-51 is a very flexible one, and it can also be adapted a variety of ways. Although the original version shown uses the AD744 compensation pin (5) to drive the output stage U2 device, conventional internally compensated op amps can also be used for U1, and still realize the many features of the architecture.

The ability to adapt the topology to differing devices in single and dual op amp formats allows such dual FET devices as the AD823 to be usefully employed in a stereo realization. Similarly, dual CFB op amps such as the AD812 can be used in the U2 output stage. Thus a complete stereo version of the circuit can be efficiently built, based on only two IC packages.

This topology's flexibility also opens up a diversity of other applications beyond the basic line driver. For example, using a power-packaged dual CFB op amp such as the AD815 for U2, allows very low impedance loads such as headphones to be driven, down to as low as 10Ω (see Reference 12, again).

■ OP AMP APPLICATIONS

The composite current boosted line driver two, shown in Figure 6-53 below, summarizes a number of the above mentioned options, and adds some other features as well.

Similarities within this circuit to the predecessor are resistances R_1 - R_4 , which perform similar functions to the previous version. Overall gain is again calculated via R_1 - R_2 , while output stage gain is set via R_3 , R_4 , etc.

Here, note that an additional pair of resistances, R_C and R_D , form a local feedback path around stage U1. This addition allows the effective open loop bandwidth of U1 as it operates within the overall loop to be increased. For the values shown, using an AD823 for U1, the open loop bandwidth is about 100kHz. This means that the open loop bandwidth of the entire circuit is greater than the audio bandwidth, which means phase errors within the passband will be minimized.

An optional small capacitance (C_F , 10-20pF) can be useful for stabilizing the U1 stage, particularly if it employs a wide bandwidth device such as the AD825. When C_F is used, a like capacitor C_{IN} can also be used, to preserve high frequency impedance matching.

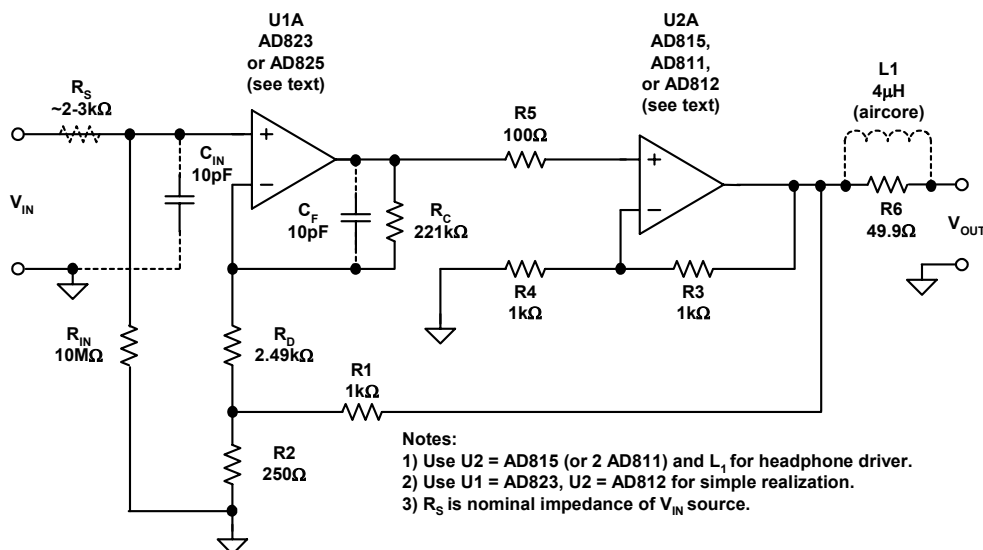


Figure 6-53: Composite current boosted line driver two

The primary input impedance balancing of the circuit is accomplished via resistance R_D , which has a dual role. External resistance R_S is the nominal output resistance of a volume control (typical for a 50kΩ audio taper control at listening level). R_D is chosen to match R_S , and R_C will be approximately 100 times the R_D value when using the AD823.

The necessity of inductor output L1 depends upon whether the circuit is to be used with low impedance loads. For headphones, the L1 choke is necessary to prevent excessive voltage loss from a simple R_6 connection. R_6 is used, in either a headphone or line driver case. If configured as a headphone driver, the circuit should use several square inches of PCB area around U2, to heat sink the AD815 device (see device data sheet). The AD811 and AD812 can also be used to drive higher impedance phones, such as 100Ω or more.

Because of the vast number of options with this circuit, no performance is presented here. However, some insight into headphone driver performance is contained in Reference 12.

Differential Line Drivers

Unlike differential line receivers, a standard circuit topology for differential line *drivers* isn't nearly so clear-cut. A variety of different circuit types for driving audio lines in a balanced mode are discussed in this section, with their contrasts in performance and complexity. The virtues of balanced audio line operation are many. The largest and most obvious advantage is the inherent rejection of inevitable system ground noises, between the driver and receiver equipment locations.

There are also more subtle advantages to balanced line operation. Differential drivers tend to inject less noise onto the power supply rails. Related to this, they also produce inherently less noise onto the ground system, since by definition the return path for a differential signal is not ground. This can be a significant advantage when high currents need to be driven into a long audio line, as it can reduce multiple channel crosstalk. The circuits that follow illustrate a variety of methods for differential line driving.

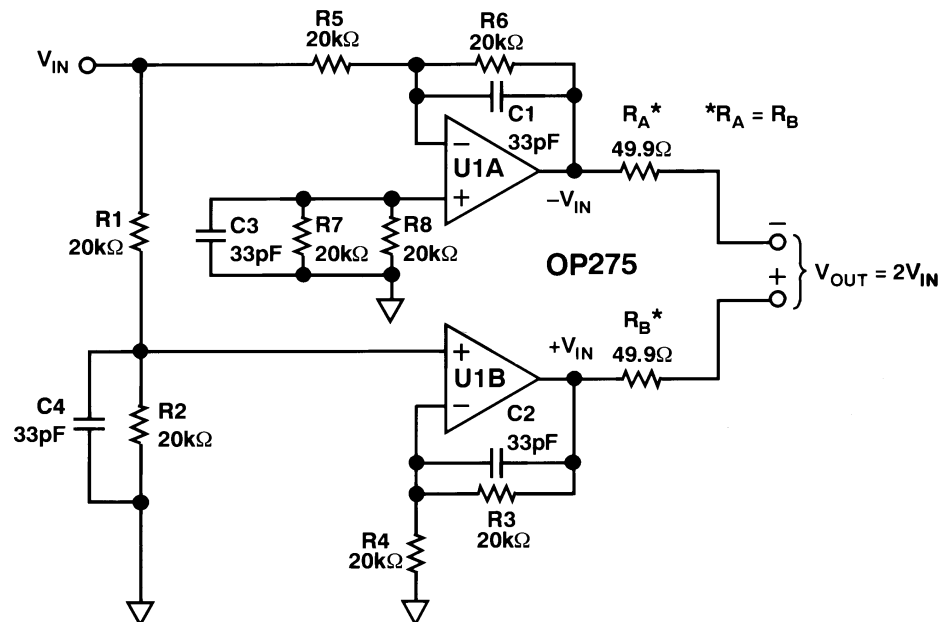


Figure 6-54: An "inverter-follower" differential line driver

"Inverter-Follower" Differential Line Driver

A straightforward approach to developing a differential drive signal of $2V_{IN}$ is to amplify in complementary fashion a single-ended input V_{IN} , with equal gain inverter and follower op amp stages. With op amp gains of ± 1 , this develops outputs $-V_{IN}$ and V_{IN} with respect to common, or $V_{OUT} = 2V_{IN}$ differentially. This "inverter/follower" driver is easily accomplished with a dual op amp such as the OP275, plus an 8x20k film resistor network (or discrete), as shown above, in Figure 6-54. Here U1A provides the gain of -1 channel, while U1B operates at a gain of $+1$. The differential output signal across the balanced output line is $2V_{IN}$, and the differential output impedance is equal to $R_A + R_B$, or 100Ω . The output resistors $R_A + R_B$ should be well matched, for reasons discussed earlier.

Use of like-value gain resistors around the U1 sections makes the respective channel noise gains match, and also makes their purchase easy. In addition, this forces the source

■ OP AMP APPLICATIONS

impedances seen by the op amp $\pm$ inputs to be matched. Capacitors C_1 - C_2 provide a ultrasonic rolloff, and enhance stability into capacitive lines. Overall, this circuit is high in performance for its cost and simplicity. Note that if a resistor network is used for R_1 - R_8 , the entire circuit can be built with only 8 components.

THD+N performance of the Fig. 6-54 circuit operating on $\pm 18V$ supplies is shown in Figure 6-55 below, for a series of successive sweeps resulting in output levels of 1, 2, 5 and 10Vrms across 600 Ω . The distortion in most instances is about 0.001%, and somewhat higher at a 1V output level (noise limited at this level). Maximum output level is about 12Vrms into 600 Ω before clipping (not shown).

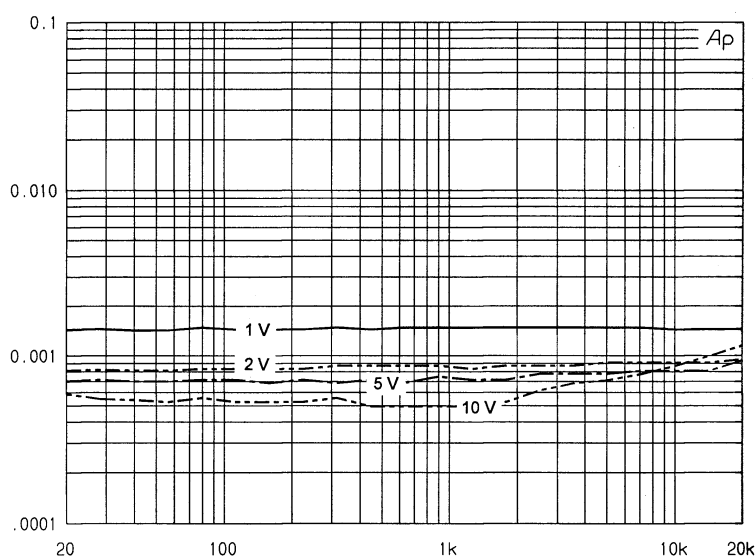


Figure 6-55: Inverter-follower driver of Fig. 6-54, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 5, 10V_{rms}$, $R_L = 600\Omega$, $V_S = \pm 18V$

In system terms, this type of differential line driver can potentially run into application problems, and should be used with some caveats in mind. In reality, this driver circuit uses two mirror-imaged, single-ended drivers, and they produce voltage output signals with respect to the source (V_{IN}) common point.

At the load end of a cable being driven, if the receiver used has a high impedance differential input (such as those discussed in the line receiver section) there is no real problem in application for this driver circuit. However, it should be noted that one side of the differential output from Figure 6-54 *cannot be grounded without side effect*. This is because the source drive V_{OUT} is *not* truly floating, as would be in the case of a transformer winding.

In this sense, the circuit is *pseudo differential*, and it shouldn't be used indiscriminately. Nevertheless, within small and defined systems, it still has the obvious advantage of simplicity, and, as noted, it can achieve high performance. Note also that with the matched source resistances R_A and R_B of 49.9 Ω as shown, nothing will be damaged even if one output is shorted—other than a loss of half the signal! Finally, if balanced high impedance differential loading is used at the receiver, there will be no side effects.

Cross-Coupled Differential Line Driver

A more sophisticated form of differential line driver uses a pair of *cross-coupled* op amps with both positive and negative feedback paths. The general form of this type of circuit is a cross-coupled Howland circuit, after the classic resistor bridge based current pump. The cross-coupled form was described by Pontis in a solid-state transformer emulator for high performance instrumentation (see Reference 13).

Application-wise, this configuration provides maximum flexibility, allowing a differential output signal V_{OUT} to be maintained constant and independent of the load common connections. This means that either side can be shorted to common without loss of signal level, i.e., as can be done with a transformer.

Figure 6-56 below shows the SSM2142 balanced line driver IC in an application. The SSM2142 consists of two Howland circuits A2 and A3, cross-coupled as noted, plus an input buffer (A1). The trimmed multiple resistor array and trio of op amps shown is packaged in an 8 pin miniDIP IC with the pinout noted.

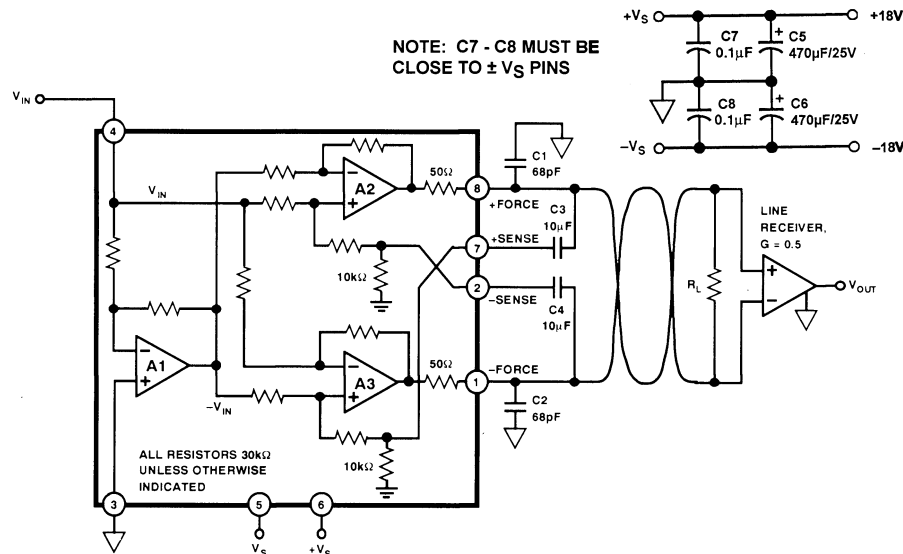


Figure 6-56: SSM2142 cross-coupled differential line driver used within balanced driver/receiver system

The SSM2142 line driver is designed for a single-ended to differential gain of 2 working into a 600Ω load. In the simplest use, it is strapped with the respective output FORCE/SENSE pins tied together (7-8, 1-2). Small film capacitors C_1 - C_2 preload the IC for stability against varying cable lengths. To decouple line dc offsets, the optional capacitors C_3 - C_4 are used as shown, and should be non-polar types, preferably films.

An additional "housekeeping" caveat with the SSM2142 involves the high frequency power supply bypassing. The $0.1\mu\text{F}$ low inductance bypass caps C_7 and C_8 must be within 0.25" of power supply pins 5 and 6, as noted in the figure. If this bypassing is compromised by long lead lengths, excessive THD will be evident.

■ OP AMP APPLICATIONS

In a system application, the SSM2142 is used with a complementary gain of 0.5 receiver, either an SSM2143, or one of the other line receivers discussed previously. The complete hookup of Fig. 6-56 comprises an entire single-ended to differential and back to single-ended transmission system, with noise isolation and a net end-to-end unity gain.

Figure 6-57 shows the THD+N performance of the SSM2142 driver portion of Fig. 6-56, for sweeps yielding output levels of 1, 2, 5 and 10Vrms across 600Ω. While performance is noise limited for the 1V output curve, distortion drops to $\leq 0.001\%$ and near residual for most higher levels, rising only with higher frequencies and the 10V output curve.

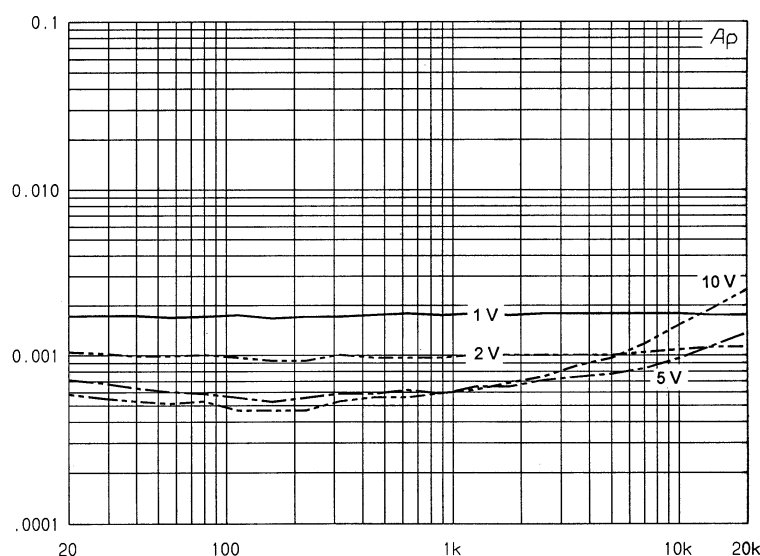


Figure 6-57: SSM2142 driver portion of Fig. 6-56, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 5, 10V_{rms}$, $R_L = 600\Omega$, $V_S = \pm 18V$

These two differential drivers are suited for 600Ω or higher loads, and, within those constraints, perform well.

As should be obvious, these drivers do *not* offer galvanic isolation, which means that in all applications there must be a DC current path between the grounds of the driver and the final receiver. In practice however this isn't necessarily a problem.

The following circuits illustrate differential drivers that do offer galvanic isolation, and can therefore be used with ground potential differences up to several hundred volts (or the actual voltage breakdown rating of the transformer in use).

Transformer Coupled Line Drivers

Transformers provide a unique method of signal coupling, which is one that allows completely isolated common potentials, i.e., *galvanic isolation*. As noted previously in the line receiver section, transformers are not without their technical and practical limitations, but their singular ability to galvanically isolate grounds maintains a place for them in difficult application areas (see References 15 and 16).

Basic Transformer Coupled Line Driver

The circuit of Figure 6-58 below uses some previously described concepts to form a basic low DC offset, high linearity driver using a high quality nickel core output transformer. U1 and U2 form a high current driver, similar to the Fig. 6-49 current boosted driver.

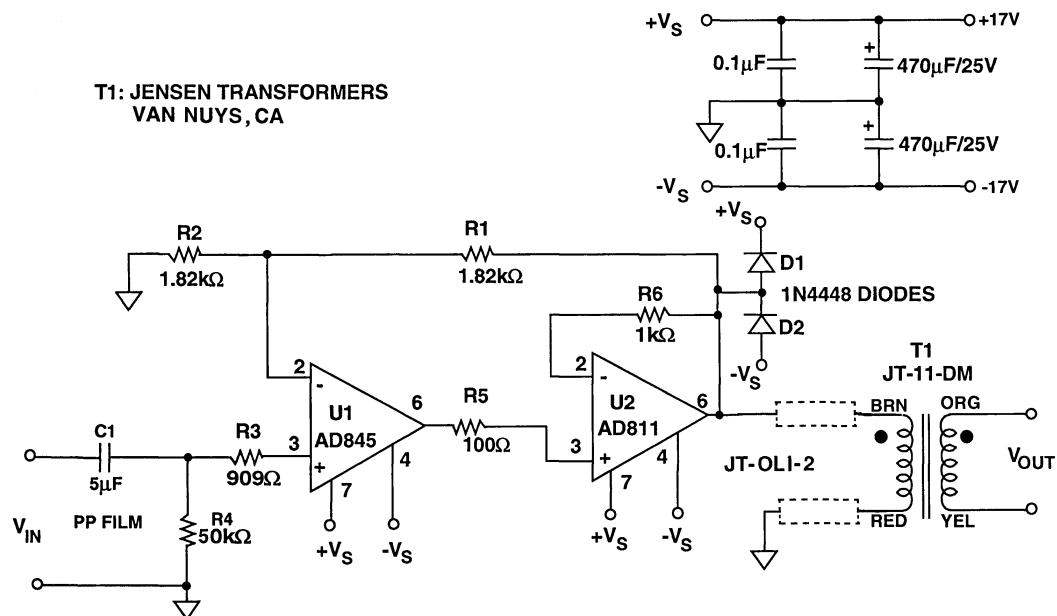


Figure 6-58: A basic transformer coupled line driver

In this circuit U1 is a low offset voltage FET input op amp, for the purpose of holding the DC offset seen at the primary of T1 to a minimum ($\pm 12.5\text{mV}$ maximum as shown, typically less). DC current flowing into the primary winding of a transformer should be minimized, for lowest distortion. C₁, a high quality film capacitor, decouples any DC offset present on V_{IN}, for similar reasons.

The U1-U2 device combination is capable of $\pm 100\text{mA}$ or more of output, which aids greatly in the ability of this circuit to drive low impedances. The buffering of U2 is recommended for long lines, or for the absolute lowest distortion. Although T1 is shown with a 1:1 coupling ratio, other winding configurations are possible with transformer variations, that is step-up or step-down, allowing either 600Ω or 150Ω loads.

As can be noted, the T1 primary isn't driven directly, but is isolated by two series isolation devices, Jensen JT-OLI-2s. Each of these is an LR shunt combination of about 39Ω and 3.7μH. The net impedance offers a very low DCR, and an increasing impedance

■ OP AMP APPLICATIONS

above 1.5MHz for load isolation (see device data sheet and Reference 17). The use of two isolators as shown offers best output CMR rejection for the transformer, but one will also work (with less CMR performance), as will a single 10 Ω resistor.

THD+N performance for this driver-transformer combination is shown in Figure 6-59 below, for supplies of $\pm 18V$ and successive input sweeps, resulting in outputs of 1, 2, 4, and 8Vrms into 600 Ω . These data were taken with a single series resistance of 10 Ω driving T1 (which could be conservative compared to operation with two isolators).

As with the 2x and 5x basic drivers previously described, these data are essentially distortion free above 100Hz. At lower frequencies there is seen a level dependent, inverse-frequency dependent distortion. The measured distortion reaches a maximum at 20Hz with output levels of 8Vrms ($\approx 20dBm$), while at lower levels it is substantially less.

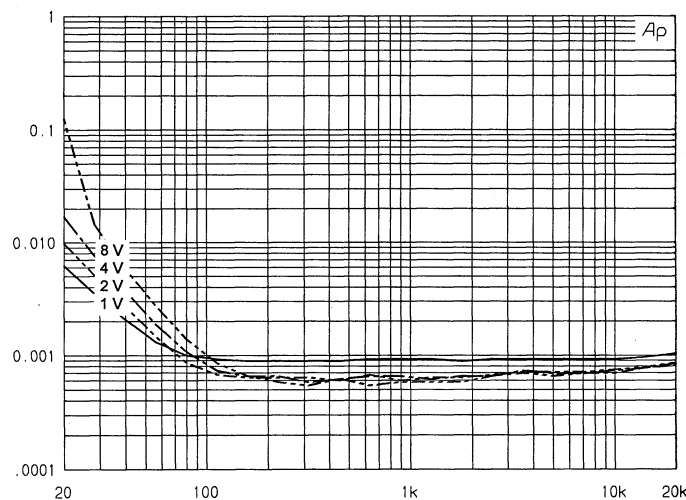


Figure 6-59: Transformer driver of Fig. 6-58, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 1, 2, 4, 8V_{rms}$, $R_L = 600\Omega$, $V_S = \pm 18V$

This distortion phenomenon is basic to audio transformers, to one degree or another. It is lessened (but not totally eliminated) in the higher quality transformer types, such as the nickel-core unit used in the Fig. 6-58 circuit.

In practice, there are some factors that tend to mitigate the seriousness of the low frequency distortion seen in the performance data of Fig. 6-59. First, rarely will maximum audio levels ever be seen at 20Hz. Thus suitably derated operation of T1 will strongly reduce the incidence of this distortion.

However, if the lowest distortion possible independent of level is desired, then some additional effort will need to be expended on making the transformer driver more sophisticated. This can take the form of actively applying feedback around the transformer, so as to lower its non-linearity to negligible levels. This is design approach is discussed with the next driver circuits.

Feedback Transformer Coupled Line Drivers

While non-premium core transformers are more economical than the nickel core types, as a tradeoff they do have much higher distortion. To further complicate the design issue, the distortion characteristics of most transformers varies with level and frequency in complex ways, rising more rapidly at higher levels and lower frequencies. This behavior is even less forgiving than that of the nickel core types, and complicates somewhat the application of audio transformers. While a nickel core transformer has distortion characteristics sufficiently low so as to allow their use without distortion correction (Fig. 6-58, again) the same simply isn't true for other core materials.

A family of distortion curves for another transformer type illustrates this behavior, shown in Figure 6-60 below. This series of plots is for a Lundahl LL1517 silicon iron C core unit, with successive output levels of 0.5, 1, 2 and 5Vrms into a 600Ω load. Individual device samples will vary, but the general pattern is typical of many audio transformers.

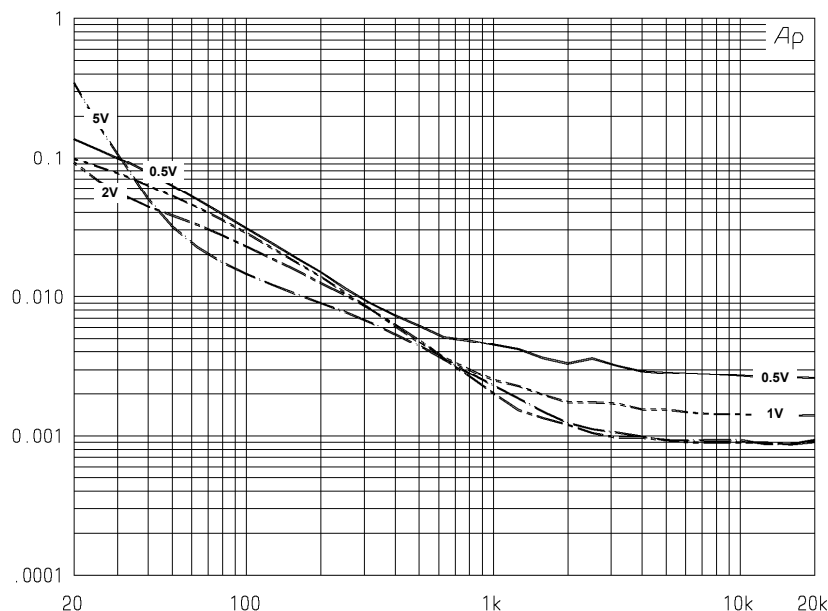


Figure 6-60: Lundahl LL1517 transformer and driver (without feedback), $THD+N$ (%) vs. frequency (Hz), for $V_{OUT} = 0.5, 1, 2, 5V_{rms}$, $R_L = 600\Omega$

Werner Baudisch (see Reference 18) developed a very effective driver technique for minimization of transformer distortion. The technique involves the use of a drive amplifier, connected to the transformer primary in a direct manner. The amplifier uses conventional negative feedback for gain stabilization. In addition, a primary sensing resistance develops a voltage sample proportional to primary current, and the voltage thus derived is also fed back to the amplifier. This second feedback path is positive feedback, so the arrangement is also known as a *mixed feedback* driver (see Reference 19).

This very useful technique of the mixed feedback driver can be used to advantage to integrate a line driver with the transformer primary within a feedback loop, which cancels the bulk of the objectionable distortion. In practice, with careful driver adjustment it is possible to reduce the distortion of the transformer plus driver almost to *that of the driver stage, operating without the transformer*. The beauty of the principle is that the inherent

■ OP AMP APPLICATIONS

floating transformer operation is not lost, and is still effectively applied in a highly linear mode. Due to the action of the mixed feedback, the transformer primary resistance is effectively cancelled, thus appreciably lowering the net secondary output impedance.

The circuit of Figure 6-61 below is a basic, single-ended mixed feedback driver using either a Lundahl LL1582 or LL2811 transformer as T1, and an AD845 or an OP275 as the amplifier. These transformers have two 1:1 primaries, as well as two 1:1 secondaries. As used, both primaries are connected in series, and the T1 net voltage transfer is unity.

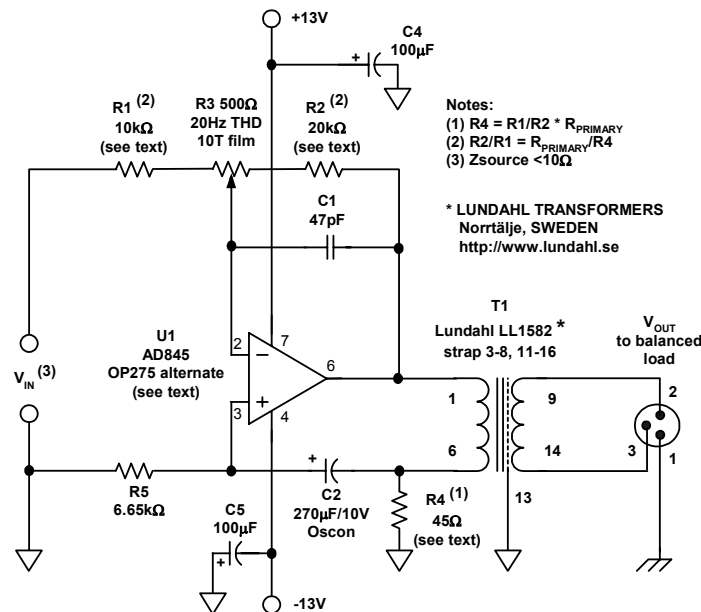


Figure 6-61: A basic single-ended mixed feedback transformer driver

To enable correct mixed feedback operation, two key ratios within the circuit must be set to match. One ratio is between the net T1 primary resistance, R_{PRIMARY} and sample resistor R_4 , and the other is R_2 and R_1 . This relationship is:

$$R_{\text{PRIMARY}}/R_4 = R_2/R_1 \quad \text{Eq. 6-18}$$

It is important to note that R_{PRIMARY} is the total effective DC resistance of T1. As used here, two series 45Ω primaries are used, so R_{PRIMARY} is 90Ω. Gain of the driver circuit is established as in a standard inverter, or the R_2 - R_1 ratio. For a gain of 2x, R_2 is then simply 2 times R_1 , i.e., 20kΩ and 10kΩ. R_4 may then be selected as:

$$R_4 = R_1/R_2 * R_{\text{PRIMARY}} \quad \text{Eq. 6-19}$$

With the R_1/R_2 ratio of 0.5, this makes R_4 simply $\frac{1}{2} R_{\text{PRIMARY}}$, or in this case 45Ω.

Note the value of R_1 is critical, thus the V_{IN} source impedance must be low ($<10\Omega$). This and other subtleties are effective performance keys. One is the sensitivity of the ratio match described by Eq. 6-18. Only when trimmed optimally will the lowest frequency THD be minimum. Thus a multi-turn film trimmer R_3 is used to trim out the various tolerances and the winding resistance of T1. Further, the positive feedback path is AC-coupled via C_2 . This provision prevents DC latchup, should positive feedback override

the negative. However, a simple time constant of say, 8ms (corresponding to 20Hz) is *not sufficient* for lowest low frequency THD. To counteract this, the C_2 - R_5 time constant is set quite long (≈ 1.8 seconds), which enable lowest possible 20Hz THD. With the suggested AD845 for U1, distortion is lowest, as it is also with an Oscon capacitor for C_2 . A larger value ordinary aluminum electrolytic can also be used for C_2 , with a penalty of somewhat high distortion. Alternately, an OP275 can also be used for U1 (see below).

With the AD845 FET input op amp used for U1, the maximum DC at the T1 primary is essentially the amplifier V_{os} times the stage's 3x noise gain, or $\leq 7.5\text{mV}$. Since the AD845 can also dissipate $\approx 250\text{mW}$, the lowest possible supplies help keep the offset change with temperature as low as possible.

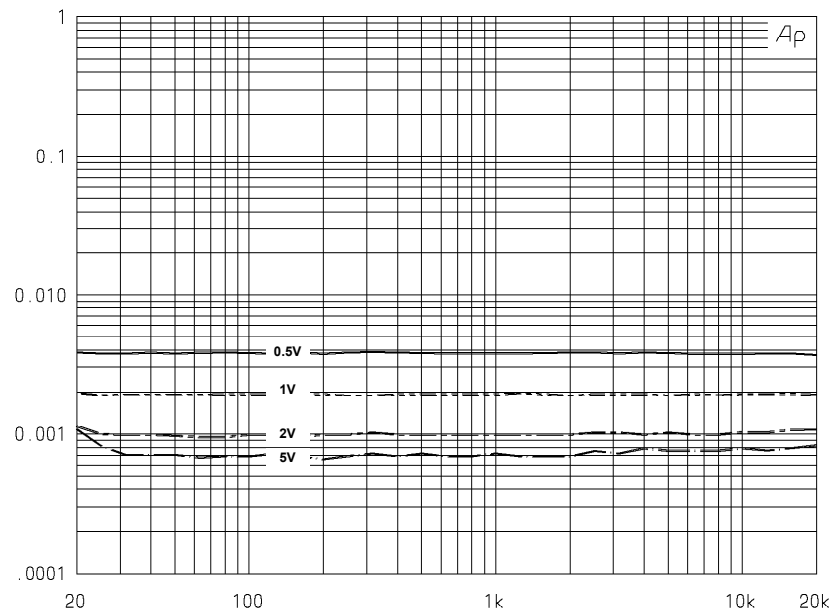


Figure 6-62: Fig. 6-61 driver with Lundahl LL2811 transformer and AD845, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 0.5, 1, 2, 5\text{Vrms}$, $R_L = 600\Omega$

Lab THD+N measurements of Fig. 6-61 were made using an LL2811, a transformer like the LL1582, but without a Faraday shield. The two transformers are very similar, but the LL1582 is recommended for single-ended drive circuits. The performance of this feedback driver is shown in Figure 6-62 above, for successive output levels of 0.5, 1, 2, and 5Vrms into a 600Ω load. Comparison of these data with Fig. 6-60 bears out the utility of the distortion reduction; it is decreased by orders of magnitude. More importantly, the level dependence with decreasing frequency is essentially eliminated.

These data do in fact represent almost an ideal THD+N pattern; the distortion level is flat with frequency, and it decreases with increasing output level. An extremely slight increase in THD+N can just be discerned at 20Hz in the 5V curve. The alternate OP275 for U1 also works well, but does have slightly higher distortion (not shown).

Although directly comparable data is not presented for it, it is worth noting that the LL1517 transformer can also be used with the Fig. 6-61 circuit, with $R_4 = 9.2\Omega$, and the two primaries connected in series. However, some additional data on a circuit quite similar to Fig. 6-61 does reveal a potential limitation for this type of driver.

■ OP AMP APPLICATIONS

Figure 6-63 below shows a set of high level THD+N curves for a mixed feedback driver using an AD8610 op amp for U1, and the LL1517 transformer. Three THD+N sweeps are made, with the lowest THD+N curve representing the best possible null. The other two curves show increased THD+N at low frequencies, for conditions of R_2/R_1 ratio mismatches of 1 and 5%, respectively. This demonstrates how critical a proper null is towards achieving the lowest possible distortion at the low end of the audio band.

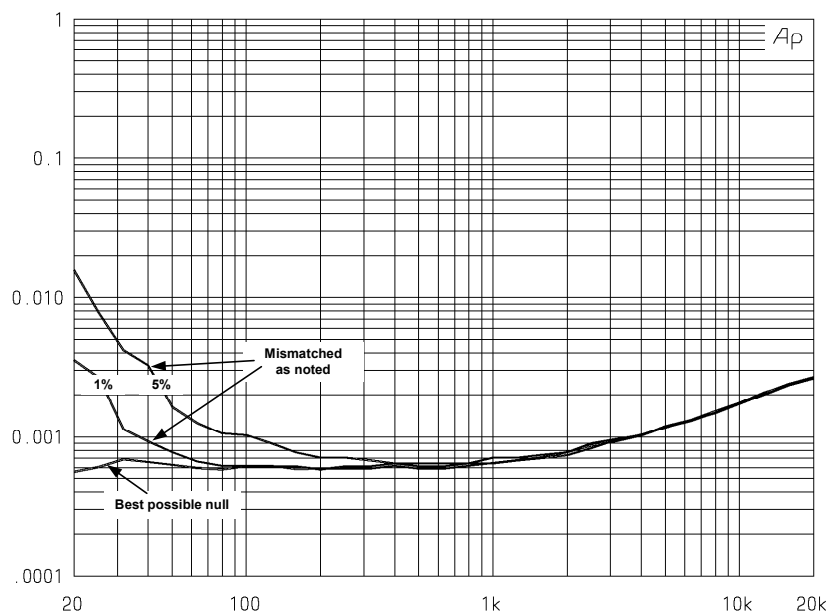


Figure 6-63: Lundahl LL1517 transformer with mixed feedback AD8610 driver, THD+N (%) vs. frequency (Hz) for various null accuracies

It is possible to tweak the ratio via R_3 for an excellent 20Hz high level null at room temperature, and this is recommended to get the most from one of these circuits. But, it must also be remembered that the TC of the T1 copper windings is about 0.39%/°C. So, only a 10°C ambient temperature change would be sufficient to degrade the best null by nearly 5%. The resulting performance would then roughly represent the upper curve of Fig. 6-63— still quite good, but just not quite as good as possible in absolute terms.

For the best and most consistent performance, wide temperature range applications of this type of circuit should therefore employ some means of temperature compensation for the copper winding(s) of T1. One means of achieving this would be to employ a thermally sensitive device to track the copper TC of T1. The net goal should be to hold the R_{PRIMARY}/R_4 ratio constant over temperature. It should also be noted that for this approach to work, it is assumed that the R_2/R_1 ratio is temperature independent. This is possible with the use of close tolerance, low TC metal film resistors, i.e., 50ppm/°C or better (or the use of a low tracking TC network).

It should also be noted that the *output balance* of an audio transformer is a very important factor when designing audio line drivers. Poor transformer balance can lead to mode conversion of CM signals on the output line (see Reference 20). The result is that a spurious differential mode signal can be created due to poor balance. A transformer can attain good balance (i.e., 60dB or better) by the use of sophisticated winding techniques, or the use of a Faraday shield, as is true in the case of the LL1582 and the LL1517.

Transformer drivers can of course also be operated in a balanced drive fashion. This has the advantage of doubling the available drive voltage for given supply voltages, plus lowering the distortion produced. Mixed feedback principles can be extended to a balanced arrangement, which lowers the distortion in the same manner as for the single-ended circuit just described. An example circuit is shown in Figure 6-64, below.

In Fig. 6-64, a U1-U2 low distortion op amp pair drive an LL1517 transformer. U1 is an inverting gain circuit as defined by gain resistors R₁-R₂, which drives the top of T1. Placed in series with the T1 primary, R₃ acts as a current sampling resistor, and develops a correction voltage to drive the second inverter, U2, through R₄.

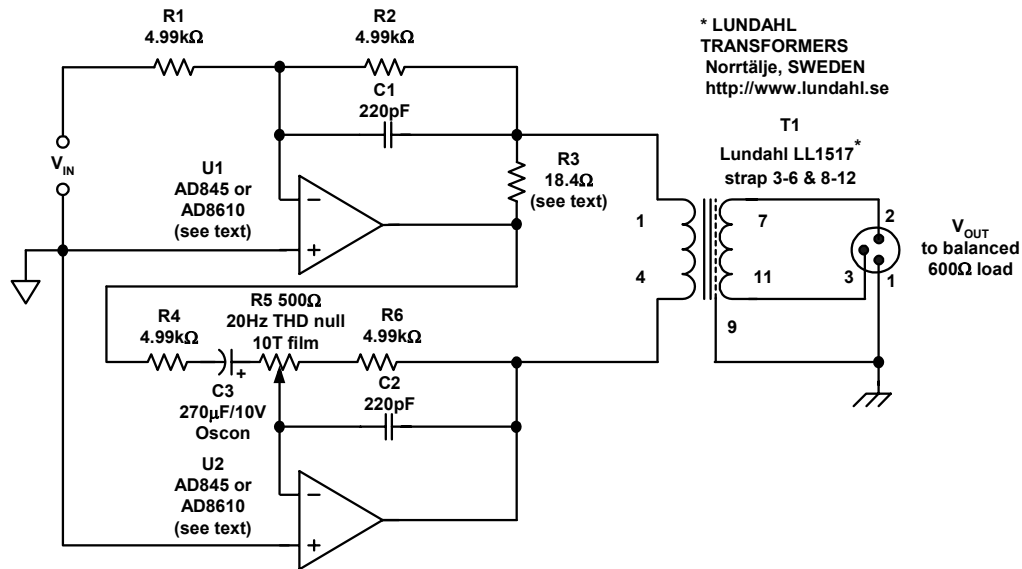


Figure 6-64: A balanced transformer driver circuit that applies mixed feedback principles of distortion minimization

This scheme is adapted from the mixed feedback balanced driver circuit of Arne Offenbergh (see Reference 21). There are however two main differences in this version. One is operation of the U1 stage as an inverter, which eliminates any CM distortion effects in U1, and the second being the ability to easily set the overall driver gain via R₁-R₂. Within this circuit, it should be noted that the resistances R₁-R₂ do *not* affect the distortion null (as they do in the simpler circuit of Fig. 6-61).

The distortion null in this form of the circuit occurs when the ratios R₃/R_{PRIMARY} and R₄/R₆ match. For simplicity, the second inverter gain is set to unity, so R₃ is selected as:

$$R_3 = R_4 / R_6 * R_{PRIMARY} \quad \text{Eq. 6-20}$$

For the R₄-R₆ values shown, R₃ then is simply equal to R_{PRIMARY}, or 18.4Ω when used with the LL1517 transformer with series connected primaries.

As with any of these driver circuits, the exact op amp selection has a great bearing on final performance. Within a circuit using two amplifiers, dual devices are obviously attractive. The distortion testing below discusses amplifier options.

■ OP AMP APPLICATIONS

THD+N performance data for the balanced transformer driver of Fig. 6-64 is shown in Figure 6-65 below, using an LL1517 transformer with successive output levels of 0.5, 1, 2, and 5Vrms into a 600Ω load. For these tests, the supply voltages were ±13V, and the U1-U2 op amp test devices were pairs of either the AD8610 or the AD845 (note— two AD8610 singles are comparable to a single AD8620 dual).

An interesting thing about these plots are the fact that the THD+N is both low and essentially unchanged with frequency, which is again, near ideal. Low frequency nulling of the distortion is almost as critical in this circuit as in the previous, and a slight upturn in THD+N can be seen at 20Hz, for the highest level (5V).

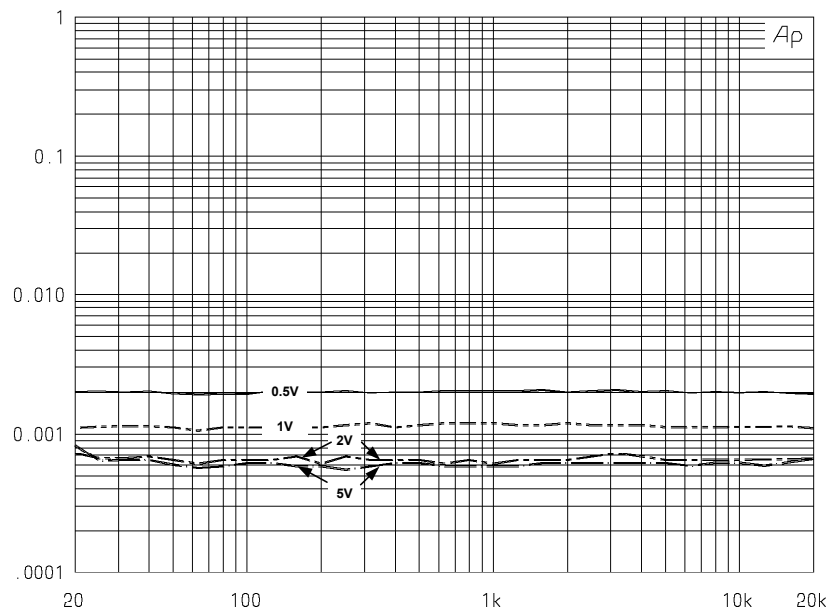


Figure 6-65: Fig. 6-64 balanced driver with Lundahl LL1517 transformer and two AD8610s, THD+N (%) vs. frequency (Hz), for $V_{OUT} = 0.5, 1, 2, 5V_{rms}$, $R_L = 600\Omega$

For the AD8610 devices shown by these data, the wideband THD+N was slightly lower than a comparable test with the AD845 pair (the latter not shown). Both amplifier sets show essentially flat THD+N vs. frequency characteristics.

Because of the balanced drive nature of this circuit, the realization offers lower distortion than the simpler single-ended version of Fig. 6-61, plus a buffering of the distortion null sensitivity against the input source impedance and gain adjustment. It can thus be considered a more robust method of transformer distortion minimization. For these reasons, the balanced form of driver is recommended for professional or other high performance requirements. Note however that similar caveats do apply with regard to stabilizing the distortion null against temperature.

This driver can be used with the LL1517 and many other transformers, with of course an appropriate choice of R_3 . Note that the performance data above reflects use of the op amps operating unbuffered. For very low impedance loads and/or long lines, a pair of the previously described unity gain buffers should be considered, and both the U1 and U2 stages operated with output buffering. This will allow the retention of THD+N performance as is shown in Fig. 6-65 above, but in the face of more difficult loads.

REFERENCES: AUDIO LINE DRIVERS

1. Walt Jung, "Op Amps in Line-Driver and Receiver Circuits, Part 1," **Analog Dialogue**, 26-2, 1992.
2. W. Jung, A. Garcia, "Op Amps in Line-Driver and Receiver Circuits, Part 2," **Analog Dialogue**, 27-1, 1993.
3. Walt Jung, "Walt's Tools & Tips: 'Op Amp Audio: Buffers Part I'" **Electronic Design**, September 1, 1998, pp. 165, 166.
4. Walt Jung, "Walt's Tools & Tips: 'Op Amp Audio: Buffers Part II'" **Electronic Design**, October 1, 1998, pp. 103, 104.
5. Paul Brokaw, "An IC Amplifier User's Guide to Decoupling, Grounding, And Making Things Go Right For a Change," **Analog Devices AN202**.
6. P. Brokaw, J. Barrow, "Grounding for Low and High Frequency Circuits," **Analog Dialogue**, 23-3 (1989).
7. Alan Rich, "Shielding and Guarding," **Analog Dialogue**, 17-1 (1983).
8. Scott Wurcer, "Input Impedance Compensation," discussion within **AD743 data sheet**, Analog Devices.
9. Scott Wurcer, "An Operational Amplifier Architecture With a Single Gain Stage and Distortion Cancellation," presented at **92nd Audio Engineering Society Convention**, March 1992, preprint #3231.
10. W. Jung, S. Wurcer, 'A High Performance Audio Composite Line Stage' within "Applications for Amplifiers in Audio," Ch. 5 in W. Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, Inc., Norwood, MA, 1992, ISBN 0-916550-10-9.
11. Walt Jung, 'High Performance Audio Stages Using Transimpedance Amplifiers', within Gary Galo, "POOGE-5: Rite of Passage for the DAC960," **The Audio Amateur**, issue 2, 1992.
12. Walt Jung, "Composite Line Driver with Low Distortion" **Electronic Design Analog Special Issue**, June 24, 1996, pp. 78.
13. Walt Jung, "Walt's Tools & Tips: 'Op Amp Audio: Minimizing Input errors'" **Electronic Design**, December 14, 1998, pp. 80-82.
14. George Pontis, "Floating a Source Output," **HP Journal**, August 1980.
15. "Transformer Application Notes (various)," Jensen Transformers, 7135 Hayvenhurst Avenue, Van Nuys, CA, 91406, (213) 876-0059.
16. Bruce Hofer, "Transformers in Audio Design," **Sound & Video Contractor**, March 15, 1986.
17. Deane Jensen, "Some Tips on Stabilizing Operational Amplifiers," Jensen Transformers AN-001, 7135 Hayvenhurst Avenue, Van Nuys, CA, 91406, (213) 876-0059.
18. Werner Baudisch, "Schaltungsanordnung mit Verstärker mit Ausgangsübertrager," **German patent DE2901567**, issued July 24, 1980.
19. Per Lundahl, "Mixed Feedback Drive Circuits For Audio Output Transformers," Lundahl Transformers, Norrtälje, Sweden, <http://www.lundahl.se>

■ OP AMP APPLICATIONS

20. Per Lundahl, "Winding Arrangements of Output Transformers," Lundahl Transformers, Norrtälje, Sweden, <http://www.lundahl.se>
21. Arne Offenberg, "Mixed Feedback Balanced Driver Circuit," LL2811 Audio Output Transformer data sheet, Lundahl Transformers, Norrtälje, Sweden, <http://www.lundahl.se>

ACKNOWLEDGEMENTS:

Portions of this section were adapted from Walt Jung, "Audio Preamplifiers, Line Drivers, and Line Receivers," within Chapter 8 of Walt Kester, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3, pp. 8-1 to 8-100.

During the preparation of this material the author received much appreciated helpful comments and other inputs from Per Lundahl of Lundahl Transformers, and from Arne Offenberg of Norway.

SECTION 6-2: BUFFER AMPLIFIERS AND DRIVING CAPACITIVE LOADS

Walt Jung, Walt Kester

Buffer Amplifiers

In the early days of high speed circuits, simple emitter followers were often used as high speed buffers. The term *buffer* was generally accepted to mean a unity-gain, open-loop amplifier. With the availability of matching PNP transistors, a simple emitter follower can be improved, as shown below in Figure 6-66A. This complementary circuit offers first-order cancellation of DC offset voltage, and can achieve bandwidths greater than 100MHz. Typical offset voltages without trimming are usually less than 50mV, even with unmatched discrete transistors. The HOS-100 hybrid amplifier from Analog Devices represented an early implementation of this circuit. This device was a popular building block in early high speed ADCs, DACs, sample-and-holds, and multiplexers.

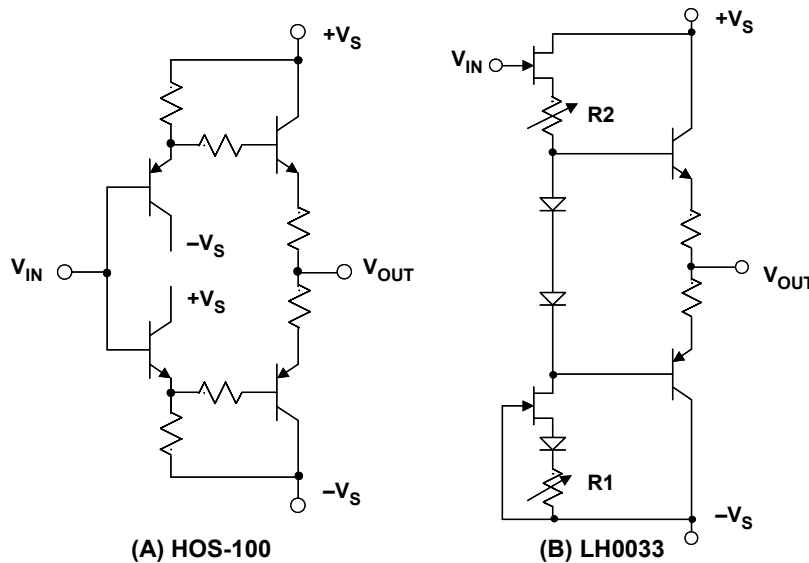


Figure 6-66: Early open-loop hybrid buffer amplifiers: (A) HOS-100 bipolar, (B) LH0033 FET input

If high input impedance is required, a dual FET can be used as an input stage ahead of a complementary emitter follower, as shown in Figure 6-66B. This form of the buffer circuit was implemented by both National Semiconductor Corporation as the LH0033, and by Analog Devices as the ADLH0033.

In the realizations of these hybrid devices, thick film resistors were laser trimmed to minimize input offset voltage. For example, in the Fig. 6-66(B) circuit, $R1$ is first trimmed to set the bias current in the dual matched FET pair, which is from the 2N5911 series of parts. $R2$ is then trimmed to minimize the buffer input-to-output offset voltage.

■ OP AMP APPLICATIONS

Circuits such as these achieved bandwidths of about 100MHz at fairly respectable levels of harmonic distortion, typically better than -60dBc . However, they suffered from DC and AC nonlinearities when driving loads less than 500Ω .

One of the first totally monolithic implementations of these functions was the Precision Monolithics, Inc. BUF03 shown below in Figure 6-67 (see Reference 1). PMI is now a division of Analog Devices. This open-loop IC buffer achieved a bandwidth of about 50MHz for a 2V peak-to-peak signal.

The BUF03 circuit is interesting because it demonstrates techniques that eliminated the requirement for the slow, bandwidth-limited vertical PNP transistors associated with most IC processes available at the time of the design (approximately 1979).

Within the circuit, input transistor J1 is a FET source follower that is biased by an identical FET J2, thereby making the gate-to-source voltage of J1 nominally zero. The output of J1 is applied to emitter follower Q1, and diodes Q5 and Q6 compensate for the combined base-emitter drops of Q1 and Q7/Q9.

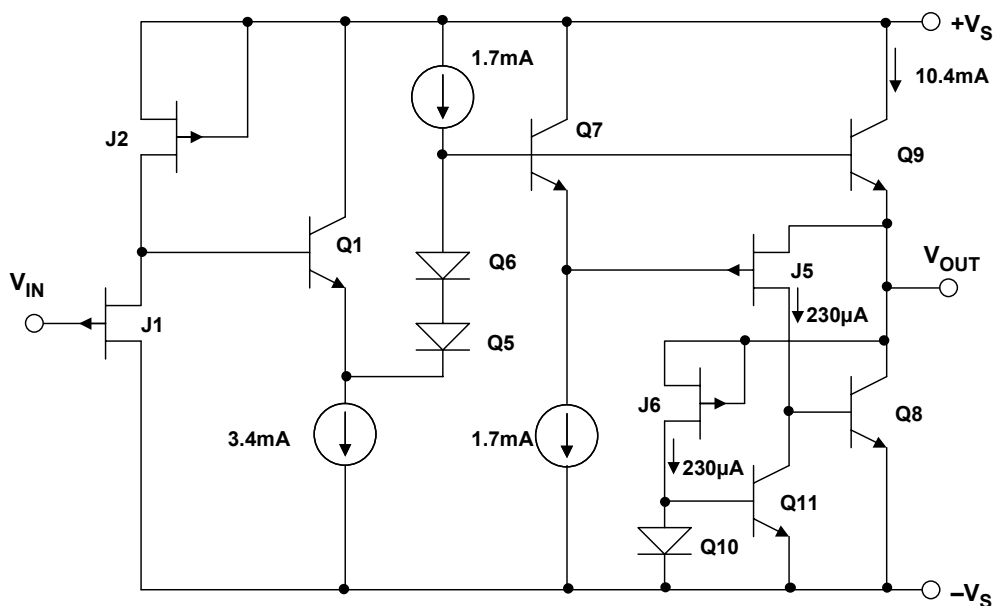


Figure 6-67: BUF03 monolithic open-loop buffer 1979 vintage

The current through Q7 is held at a constant 1.7mA, therefore its V_{BE} is constant. Transistors Q7 and Q9 are scaled such that the current in Q9 is six times that of Q7 for equal V_{BE} drops. If the load current changes, and Q9 is required to source more or less current, its V_{BE} attempts to increase or decrease. This change is applied between the gate and source of J5, which then reduces/increases current in the base of Q8 to maintain the current in Q9 at six times that of Q7. The localized feedback works for load currents up to $\pm 10\text{mA}$. The current in Q9 is therefore held constant at 10.4mA (independent of load) because its V_{BE} drop does not change with either output voltage or load current.

With a $1\text{k}\Omega$ load, and an output voltage of $+10\text{V}$, transistor Q8 must sink 0.2mA, and Q9 supplies 10mA to the load, 0.2mA to J5, and 0.2mA to J6. For an output of -10V , Q8

must sink 20.2mA so that the net current delivered to the load is -10mA . In addition to achieving a bandwidth of approximately 50MHz (2V peak-to-peak output), on-chip zener-zap trimming was used to achieve a DC offset of typically less than 6mV.

One of the problems with all the open-loop buffers discussed thus far is that although high bandwidths can be achieved, the devices discussed don't take advantage of negative feedback. Distortion and DC performance suffer considerably when open-loop buffers are loaded with typical video impedance levels of 50, 75, or 100 Ω . The solution is to use a properly compensated wide bandwidth op amp in a unity-gain follower configuration. In the early days of monolithic op amps, process limitations prevented this, so the open-loop approach provided a popular interim solution.

Today, however, practically all unity-gain-stable voltage or current feedback op amps can be used in a simple follower configuration. Usually, however, the general-purpose op amps are compensated to operate over a wide range of gains and feedback conditions. Therefore, bandwidth suffers somewhat at low gains, especially in the unity-gain non-inverting mode, and additional external compensation is usually required.

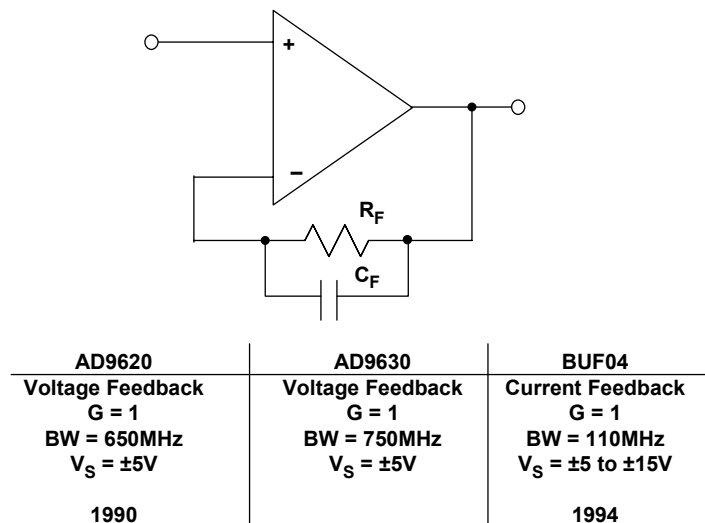


Figure 6-68: Early closed-loop unity-gain monolithic buffers

A practical solution is to compensate the op amp for the desired closed-loop gain, while including the gain setting resistors on-chip, as shown in Figure 6-68 above. Note that this form of op amp, internally configured as a buffer, may typically have no feedback pin. Also, putting the resistors and compensation on-chip also serves to reduce parasitics.

There are a number of op amps optimized in this manner. Roy Gosser's AD9620 (see Reference 2) was probably the earliest monolithic implementation. The AD9620 was a 1990 product release, and achieved a bandwidth of 600MHz using $\pm 5\text{V}$ supplies. It was optimized for unity gain, and used the voltage feedback architecture. A newer design based on similar techniques is the AD9630, which achieves a 750MHz bandwidth.

The BUF04 unity gain buffer (see Reference 3) was released in 1994 and achieves a bandwidth of 120MHz. This device was optimized for large signals and operates on

■ OP AMP APPLICATIONS

supplies from $\pm 5\text{V}$ to $\pm 15\text{V}$. Because of the wide supply range, the BUF04 is useful not only as a standalone unit-gain buffer, but also within a feedback loop with a standard op amp, to boost output (see discussions within "Audio Amplifiers" portion of this chapter).

Closed-loop buffers with a gain of two find wide applications as transmission line drivers, as shown below in Figure 6-69. The internally configured fixed gain of the amplifier compensates for the loss incurred by the source and load termination. Impedances of 50, 75, and 100Ω are popular cable impedances. The AD8074/AD8075 500MHz triple buffers are optimized for gains of 1 and 2, respectively. The dual AD8079A/AD8079B 260MHz buffer is optimized for gains of 2 and 2.2, respectively.

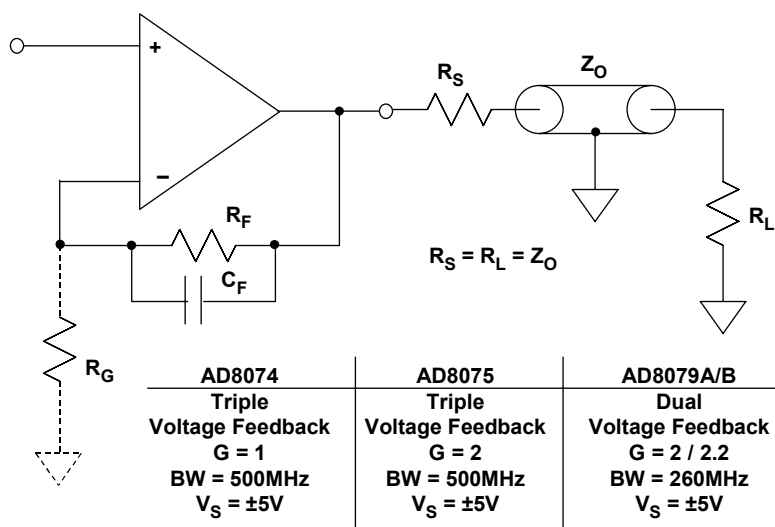


Figure 6-69: Fixed-gain video transmission line drivers

The buffer amplifiers discussed above are all dedicated to either unity or some higher fixed gain setting. As wide bandwidth fixed gain blocks they are simply applied, without the need for additional gain configuration components. They will of course (as with any high-speed amplifier) require supply bypassing components, as well as appropriate layout.

Buffers can also be implemented with almost any unity-gain-stable voltage or current feedback op amp. Examples that come to mind for voltage feedback devices are the single AD817 (or the dual counterpart AD826), or for current feedback devices the AD811, AD8001, AD8015, along with their dual-device cousins (as is applicable). In addition, there are op amps with feature rail-rail outputs as well as operation at low supply voltages—the AD8031/32 and AD8041/42 are examples.

In implementing a high-speed unity-gain buffer with a voltage feedback op amp, there will typically be no resistor required in the feedback loop, which considerably simplifies the circuit. Note that this isn't a 100% hard-and-fast rule however, so always check the device data sheet to be sure. A unity-gain buffer with a current feedback op amp will *always* require a feedback resistor, typically in the range of 500-1000 Ω . So, be sure to use a value appropriate to not only the basic part, but also the specific power supplies in use.

Driving Capacitive Loads

From either a system or signal fidelity point of view, transmission line coupling between stages is best, and is described in some detail in the next section. However, complete transmission line system design may not always be possible or practical. In addition, various other parasitic issues need careful consideration in high performance designs. One such problem parasitic is amplifier *load capacitance*, which potentially comes into play for all wide bandwidth situations that do not use transmission line signal coupling.

A general design rule for wideband linear drivers is that capacitive loading (cap loading) effects should *always* be considered. This is because PC board capacitance can build up quickly, especially for wide and long signal runs over ground planes insulated by a thin, higher K dielectric. For example, a 0.025" PC trace using a G-10 dielectric of 0.03" over a ground plane will run about 22pF/foot (see References 4 and 5). Even relatively small load capacitance (i.e., <100pF) can be troublesome, since while not causing outright oscillation, it can still stretch amplifier settling time to greater than desirable levels for a given accuracy.

The effects of cap loading on high speed amplifier outputs are not simply detrimental, they are actually an anathema to high quality signals. However, before-the-fact designer knowledge still allows high circuit performance by employing various tricks of the trade to combat the capacitive loading. If it is not driven via a transmission line, remote signal circuitry should be checked for capacitive loading very carefully, and characterized as best possible. Drivers which face poorly defined load capacitance should be bulletproofed accordingly with an appropriate design technique from the options list below.

Short of a true matched transmission line system, a number of ways exist to drive a load that is capacitive in nature, while still maintaining amplifier stability.

Custom capacitive load (cap load) compensation includes two possible options, namely a); overcompensation, and b); an intentionally forced-high loop noise gain allowing crossover in a stable region. Both of these steps can be effective in special situations, as they reduce the amplifier's effective closed loop bandwidth, so as to restore stability in the presence of cap loading.

Overcompensation of the amplifier, when possible, reduces amplifier bandwidth so that the additional load capacitance no longer represents a danger to phase margin. As a practical matter however, amplifier compensation nodes to allow this are available on very few of the newer high speed amplifiers.

Nevertheless, there are still useful examples, and one is the AD829, compensated by a single capacitor to AC-common, at pin 5. A more recent and analogous part is the AD8021, which is compensated similarly. In general, almost any amplifier using external compensation can always be over compensated to reduce bandwidth. This will restore stability against cap loads, by lowering the amplifier's unity gain frequency.

■ OP AMP APPLICATIONS

Forcing a high noise gain is shown in Figure 6-70 below, where the left side capacitively loaded amplifier with a noise gain of unity is unstable, due to a $1/\beta$ - open loop rolloff intersection on the Bode diagram in a -12dB/octave rolloff region. For such a case, introducing higher noise gain can restore often stability, so that the critical intersection occurs in a stable -6dB/octave region, as depicted at the right diagram and Bode plot.

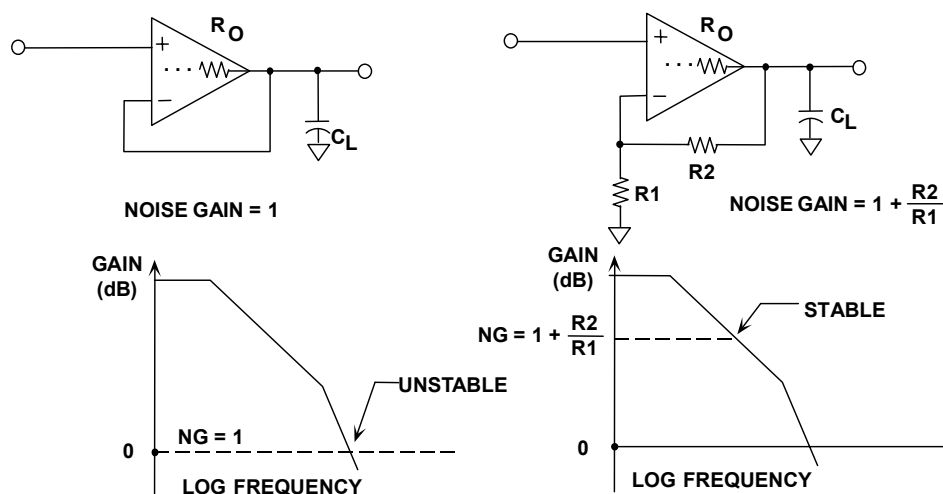


Figure 6-70: Effect of capacitive loading on op amp stability

To enable a higher noise gain (which does not necessarily need to be the same as the stage's *signal gain*), use is made of resistive or RC pads at the amplifier input, as in Figure 6-71 below. This trick is broader in scope than overcompensation, and has the advantage of not requiring access to any internal amplifier nodes. This generally allows use with any amplifier setup, even voltage followers (left) or inverters (right). An extra resistor R_D , is added which works against R_F to force the noise gain of the stage to a level appreciably higher than the signal gain (unity for both cases here).

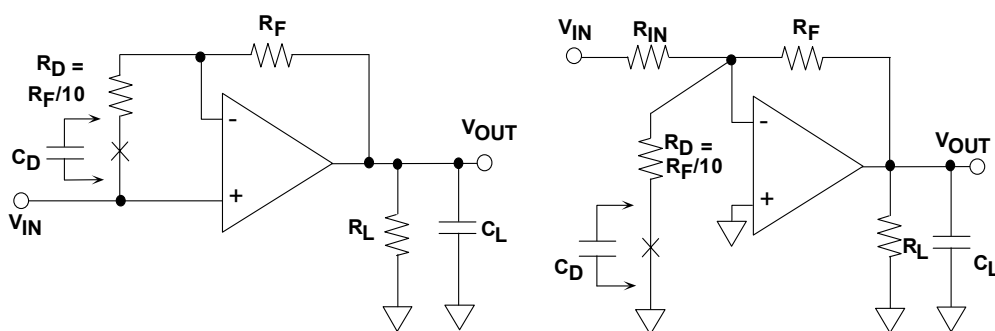


Figure 6-71: Raising noise gain (DC or AC) for follower (A) or inverter (B) stability

Assuming C_L is a value that produces a parasitic pole slightly above or near the amplifier's natural crossover, this loading combination would lead to oscillation due to the excessive phase lag. However with R_D connected, the forced higher amplifier noise gain produces a new $1/\beta$ and open loop rolloff intersection, purposely set about a decade lower in frequency. This is low enough that the extra phase lag from C_L near the amplifier's natural unity-gain crossover is no longer a problem, and stability is restored.

A drawback to this trick is that both the DC offset and input noise of the amplifier are raised by the value of the noise gain, when R_D is DC-connected. But, when C_D is used in series with R_D , the offset voltage of the amplifier is not raised, and the gained-up AC noise components are confined to a frequency region above $1/(2\pi \cdot R_D \cdot C_D)$. A further caution is that this technique can be somewhat tricky when separating these operating DC and AC regions, and should be applied carefully with regard to settling time (see Reference 6). Note that these simplified examples are generic, and in practice the absolute component values should be matched to a specific amplifier.

"Passive" cap load compensation, shown in Figure 6-72 below, is the most simple (and most popular) isolation technique available. It uses a simple "out-of-the-loop" series resistor R_X to isolate the cap load, and can be used with any amplifier, current or voltage feedback, FET or bipolar input.

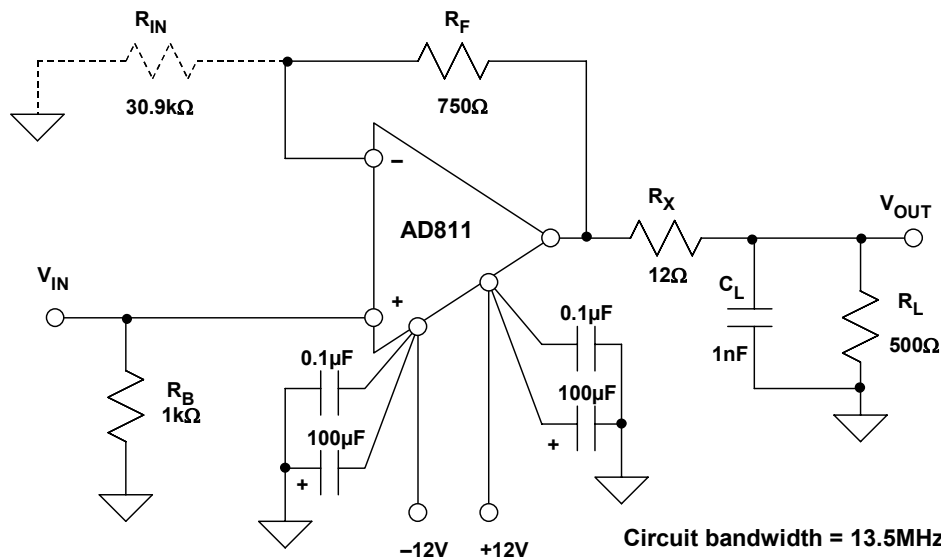


Figure 6-72: Open-loop series resistance isolates capacitive load

As noted, since this technique applies to just about any amplifier, this is a major reason why it is so useful. It is shown here with a current feedback amplifier suitable for high current line driving, the AD811, and it consists of just the simple (passive) series isolation resistor, R_X . This resistor's minimum value for stability will vary from device to device, so the amplifier data sheet should be consulted for other ICs. Generally, information will be provided as to the amount of load capacitance tolerated, and a suggested minimum resistor value for stability purposes.

Drawbacks of this approach are the loss of bandwidth as R_X works against C_L , the loss of voltage swing, a possible lower slew rate limit due to I_{MAX} and C_L , and a gain error due to the R_X - R_L division. The gain error can be optionally compensated with R_{IN} , which is ratioed to R_F as R_L is to R_X . In this example, a $\pm 100\text{mA}$ output from the op amp into C_L can slew V_{OUT} at a rate of $100\text{V}/\mu\text{s}$, far below the intrinsic AD811 slew rate of $2500\text{V}/\mu\text{s}$. Although the drawbacks are serious, this form of cap load compensation is nevertheless useful because of its simplicity. If the amplifier isn't otherwise protected, then an R_X resistor of $50\text{-}100\Omega$ should be used with virtually any amplifier facing capacitive loading. Although a non-inverting amplifier is shown, the technique applies equally to inverters.

■ OP AMP APPLICATIONS

With very high speed amplifiers, or in applications where lowest settling time is critical, even small values of load capacitance can be disruptive to frequency response, but are nevertheless sometimes inescapable. One case in point is an amplifier used for driving ADC inputs. Since high speed ADC inputs quite often look capacitive in nature, this presents an oil/water type problem. In such cases the amplifier *must* be stable driving the capacitance, but it must also preserve its best bandwidth and settling time characteristics. To address this cap load case, R_S and C_L data for a specified settling time is appropriate.

Some applications, in particular those that require driving the relatively high impedance of an ADC, do not have a convenient back termination resistor to dampen the effects of capacitive loading. At high frequencies, an amplifier's output impedance is rising with frequency and acts like an inductance, which in combination with C_L causes peaking or even worse, oscillation. When the bandwidth of an amplifier is an appreciable percentage of device F_t , the situation is complicated by the fact that the loading effects are reflected back into its internal stages. In spite of this, the basic behavior of most very wide bandwidth amplifiers such as the AD8001 is very similar.

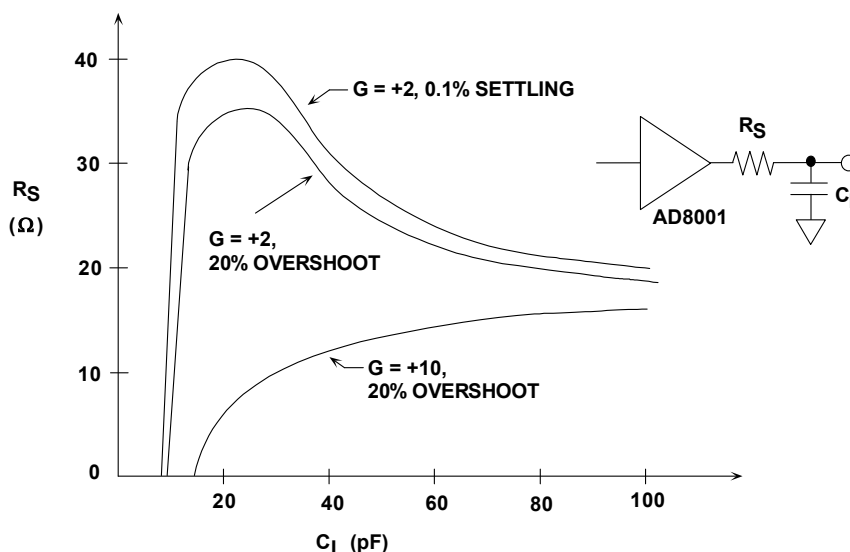


Figure 6-73: AD8001 R_S required for various C_L values

In general, a small damping resistor (R_S) placed in series with C_L will help restore the desired response (see Figure 6-73 above). The best choice for this resistor's value will depend upon the criterion used in determining the desired response. Traditionally, simply stability or an acceptable amount of peaking has been used, but a more strict measure such as 0.1% (or even 0.01%) settling will yield different values. For a given amplifier, a family of R_S - C_L curves exists, such as those of Fig. 6-73. These data will aid in selecting R_S for a given application.

The basic shape of this curve can be easily explained. When C_L is very small, no resistor is necessary. When C_L increases to some threshold value an R_S becomes necessary. Since the frequency at which the damping is required is related to the $R_S \cdot C_L$ time constant, the R_S needed will initially increase rapidly from zero, and then will decrease as C_L is increased further. A relatively strict requirement, such as for 0.1%, settling will generally require a larger R_S for a given C_L , giving a curve falling higher (in terms of R_S) than that

for a less stringent requirement, such as 20% overshoot. For the common gain configuration of +2, these two curves are plotted in the figure for 0.1% settling (upper-most curve) and 20% overshoot (middle curve). It is also worth mentioning that higher closed loop gains lessen the problem dramatically, and will require less R_S for the same performance. The third (lower-most) curve illustrates this, demonstrating a closed loop gain of 10 R_S requirement for 20% overshoot for the AD8001 amplifier. This can be related to the earlier discussion associated with Figure 6-70.

The recommended values for R_S will optimize response, but it is important to note that generally C_L will degrade the maximum bandwidth and settling time performance which is achievable. In the limit, a large $R_S \cdot C_L$ time constant will dominate the response. In any given application, the value for R_S should be taken as a starting point in an optimization process which accounts for board parasitics and other secondary effects.

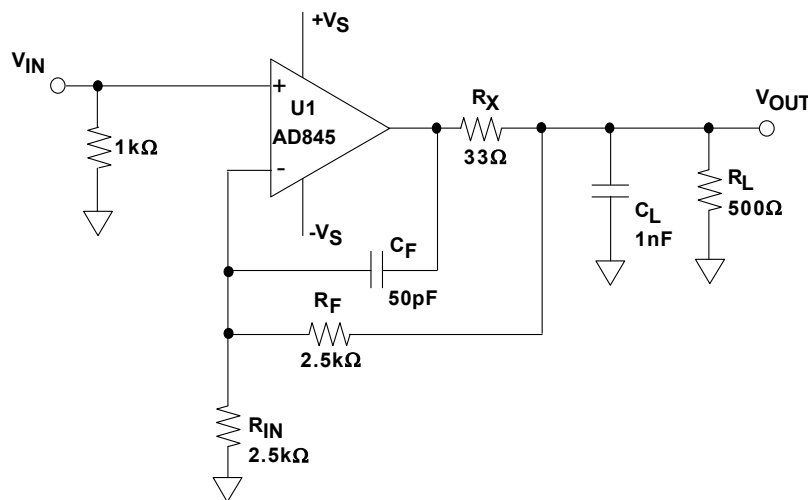


Figure 6-74: Active "in-the-loop" capacitive load compensation corrects for DC and LF gain errors

Active or "in-the-loop" cap load compensation can also be used as shown above in Figure 6-74, and this scheme modifies the passive configuration, providing feedback correction for the DC and low frequency gain error associated with R_X . In contrast to the passive form, active compensation can only be used with voltage feedback amplifiers, because current feedback amplifiers do not allow the integrating connection of C_F .

This circuit returns the DC feedback from the output side of isolation resistor R_X , thus correcting for errors. AC feedback is returned via C_F , which bypasses R_X/R_F at high frequencies. With an appropriate value of C_F (which varies with C_L , for fixed resistances) this stage can be adjusted for a well damped transient response (see References 6 and 7). There is still a bandwidth reduction, a headroom loss, and also (usually) a slew rate reduction, but the DC errors can be very low. A drawback is the need to tune C_F to C_L , as even if this is done well initially, any change to C_L will alter the response away from flat. The circuit as shown is useful for voltage feedback amplifiers only, because capacitor C_F provides integration around U1. It also can be implemented in inverting fashion, by driving the bottom end of R_{IN} , while grounding the op amp (+) input.

■ OP AMP APPLICATIONS

Internal cap load compensation involves the use of an amplifier that has topological provisions for the effects of external cap loading. To the user, this is the most transparent of the various techniques, as it works for any feedback situation, for any value of load capacitance. Drawbacks are that it produces higher distortion than does an otherwise similar amplifier without the network, and the compensation against cap loading is somewhat signal level dependent.

The internal cap load compensated amplifier sounds at first like the best of all possible worlds, since the user need do nothing at all to set it up. Figure 6-75 below is a simplified diagram of an AD817 amplifier with internal cap load compensation. The cap load compensation is the C_F -resistor network, which is highlighted by the dotted area within the unity gain output stage of the amplifier. It is important to note at this point that *this RC network only makes its presence felt for certain load conditions*.

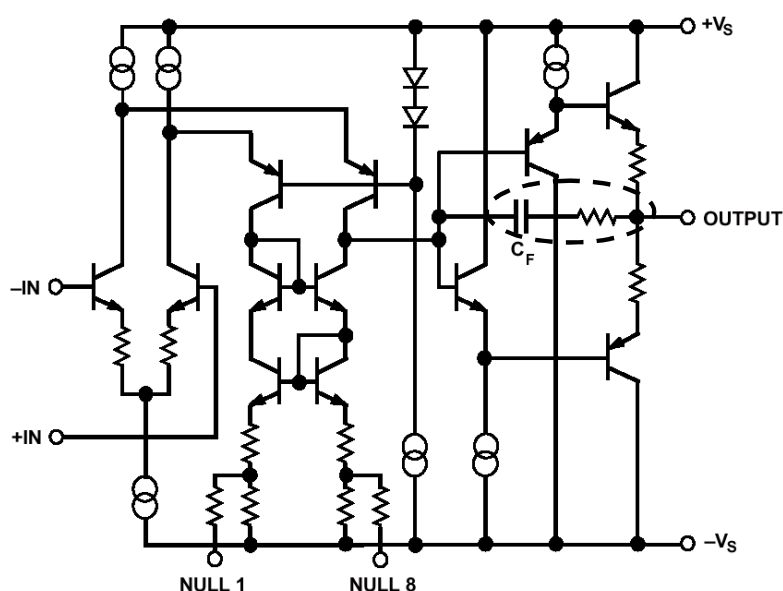


Figure 6-75: AD817 simplified schematic illustrates internal compensation for driving capacitive loads

Under normal (non-capacitive or light resistive) loading, there is limited input/output voltage error across the output stage, so the C_F network then sees a relatively small voltage drop, and has little or no effect on the AD817's high impedance compensation node. However when a capacitor (or other heavy) load is present, the high currents in the output stage produce a voltage difference across the C_F network, which effectively adds capacitance to the compensation node. With this relatively heavy loading, a net larger compensation capacitance results, and reduces the amplifier speed in a manner that is adaptive to the external capacitance, C_L . *As a point of reference, note that it requires 6.3mA peak current to support a 2Vp-p swing across a 100pF load at 10MHz.*

Since this mechanism is resident in the amplifier output stage and it affects the overall compensation characteristics dynamically, it acts independent of the specific external feedback hookup, as well as the external capacitor's size. In other words, it can be transparent to the user in the sense that no specific design conditions need be set to make

it work (other than selecting the right IC). Some amplifiers using internal cap load compensation are the AD817, the AD847, and their dual equivalents, AD826 and AD827.

There are, however, some caveats also associated with this internal compensation scheme. As with the passive compensation techniques, bandwidth decreases as the device slows down to prevent oscillation with higher load currents. Also, this adaptive compensation network has its greatest effect when enough output current flows to produce significant voltage drop across the C_F network. Conversely, at small signal levels, the effect of the network on speed is less, so greater ringing may actually be possible for some circuits, with lower-level outputs.

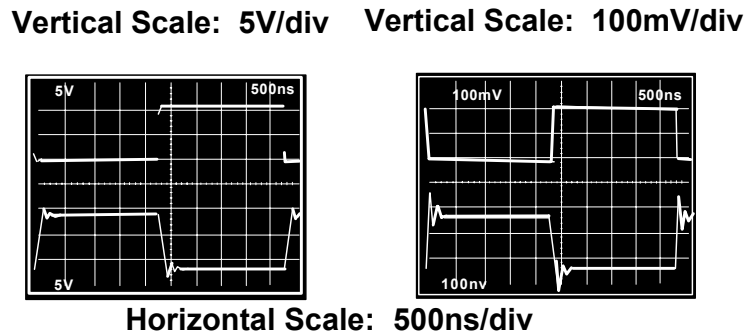


Figure 6-76: *Response of internal cap load compensated amplifier varies with signal level*

The dynamic nature of this internal cap load compensation is illustrated in Figure 6-76, which shows an AD817 unity gain inverter being exercised at both high (left) and low (right) output levels, with common conditions of $V_S = \pm 15V$, $R_L = 1k\Omega$, $C_L = 1nF$, and using $1k\Omega$ input/feedback resistors. In both photos the input signal is on the top trace and the output signal is on the bottom trace, and the time scale is fixed.

In the 10Vp-p output left photo, the output has slowed down appreciably to accommodate the capacitive load, but settling is still relatively clean, with a small percentage of overshoot. For this high level case, the bandwidth reduction due to C_L is most effective. In the right photo, the 200mVp-p output shows greater overshoot and ringing, for the lower level signal. The point is that the performance of the cap load compensated amplifier is signal dependent, but is always stable with any cap load.

Finally, because the circuit is based on a nonlinear principle, the internal network affects distortion performance and load drive ability, and these factors influence amplifier performance in video applications. Though the network's presence does not by any means make devices like the AD817 or AD847 unusable for video, it does not permit the very lowest levels of distortion and differential gain and phase which are achievable with amplifiers without this network, but otherwise comparable.

While the individual techniques for countering cap loading outlined above have various specific tradeoffs as noted, all of the techniques have a common drawback of reducing speed (both bandwidth and slew rate). If these parameters cannot be sacrificed, then a

■ OP AMP APPLICATIONS

matched transmission line system is the solution, and is discussed in more detail later, in the "Video Amplifiers" portion of this chapter.

As for choosing among the cap load compensation schemes, it would seem on the surface that amplifiers using the internal form offer the best possible solution to the problem—just pick the right amplifier and simply forget about it. And indeed, that would seem the "panacea" solution for all cap load situations— if you use the "right" amplifier you never need to think about cap loading again. Could there be more to it?

Yes! The "gotcha" of internal cap load compensation is subtle, and lies in the fact that the dynamic adaptive nature of the compensation mechanism actually can produce higher levels of distortion, vis-à-vis an otherwise similar amplifier, *without* the C_F -resistor network. Like the old saying about no free lunches, if you care about attaining top-notch levels of high frequency AC performance, you should give the issue of whether to use an internally compensated cap load amplifier more serious thought than simply picking a trendy device. For example, the AD818, which is a gain-of-two stable video op amp, offers excellent performance in terms of video gain and phase measurements. It is simply a gain-of-two stable AD817 op amp, but without the internal cap load compensation network. For similar video stage driver applications, the AD817 will not perform as well as the more suitable AD818.

On the other hand, if you have no requirements for the lowest levels of distortion, then such an amplifier as the AD817 could be a very good choice. Such amplifiers are certainly easier to use, and are relatively forgiving about output loading issues.

REFERENCES: BUFFERS AND DRIVING CAP LOADS

1. George Erdi, "A 300V/ μ s Monolithic Voltage Follower," **IEEE Journal of Solid State Circuits**, Vol. SC-14, No. 6, December, 1979, pp. 1059-1065.
2. Royal A. Gosser, "Wideband Transconductance Generator," **US Patent 5,150,074**, Filed May 3, 1991, issued September 22, 1992.
3. Derek F. Bowers, "A 6.8mA Closed-Loop Monolithic Buffer with 120MHz Bandwidth, 4000V/ μ s Slew Rate, and $\pm$ 12V Signal Compatibility," **1994 Bipolar/BiCMOS Circuits and Technology Meeting 1.3**, pp. 23-26.
4. Walt Kester, "Maintaining Transmission Line Impedances on the PC Board," within Chapter 11 of Walt Kester, Editor, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3.
5. William R. Blood, Jr., **MECL System Design Handbook**, (HB205, Rev.1), Motorola Semiconductor Products, Inc., 1988.
6. Joe Buxton, "Careful Design Tames High-Speed Op Amps," **Electronic Design**, April 11, 1991.
7. Walt Jung, "Op Amps in Line-Driver and Receiver Circuits, Part 1," **Analog Dialogue**, Vol. 26-2, 1992.
8. Dave Whitney, Walt Jung, "Applying a High-Performance Video Operational Amplifier," **Analog Dialogue**, 26-1, 1992.

▣ OP AMP APPLICATIONS

NOTES:

SECTION 6-3: VIDEO AMPLIFIERS

Walt Kester

Video Signals and Specifications

Before discussing some video applications for op amps, we will review some basics regarding video signals and specifications. The standard video format is the specification of how the video signal looks from an electrical point of view. Light strikes the surface of an image sensing device within the camera, producing a voltage level corresponding to the amount of light hitting a particular spatial region of the surface. This information is then placed into the standard format and sequenced out of the camera. Along with the actual light and color information, synchronization pulses are added to the signal to allow the receiving device—a television monitor, for instance—to identify where the sequence is in the frame data.

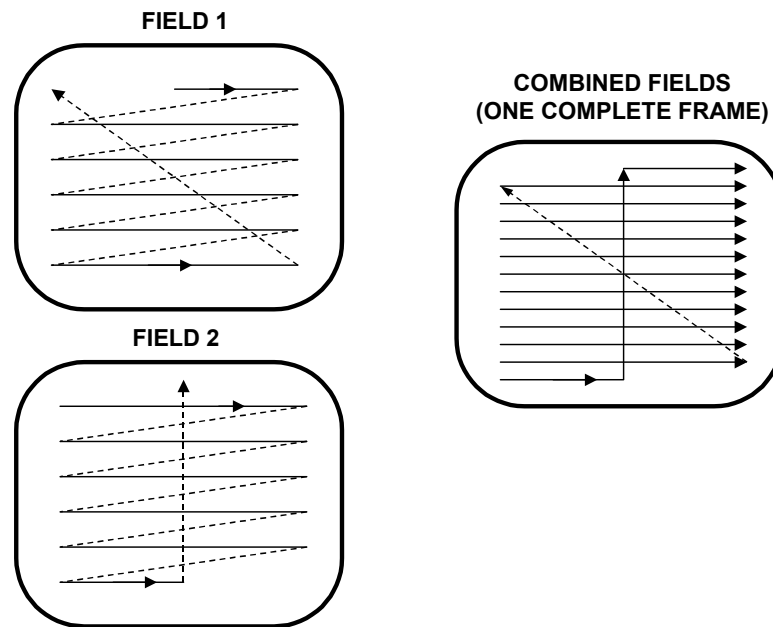


Figure 6-77: Standard broadcast television interlace format

A standard video format image is read out on a line-by-line basis from left to right, top to bottom. A technique called *interlacing* refers to the reading of all even numbered lines, top to bottom, followed by all odd lines as shown in Figure 6-77 above.

The television picture *frame* is thus divided into even and odd *fields*. Interlacing is used to produce an apparent update of the entire frame in half the time that a full update actually occurs. This results in a television image with less apparent flicker. Typical broadcast television frame update rates are 30 and 25Hz, depending upon the line frequency. It should be noted that interlacing is not always required in graphics display systems where the refresh rate is usually greater (typically 60Hz).

■ OP AMP APPLICATIONS

The original black and white, or *monochrome*, television specification in the USA is the EIA RS-170 specification that prescribes all timing and voltage level requirements for standard commercial broadcast video signals. The standard American specification for color signals, NTSC, modifies RS-170 to work with color signals by adding color information to the signal which otherwise contains only brightness information.

A video signal comprises a series of analog television lines. Each line is separated from the next by a synchronization pulse called the *horizontal sync*. The fields of the picture are separated by a longer synchronization pulse, called the *vertical sync*. In the case of a monitor receiving the signal, its electron beam scans the face of the display tube with the brightness of the beam controlled by the amplitude of the video signal. A single line of an NTSC color video signal is shown in Figure 6-78, below.

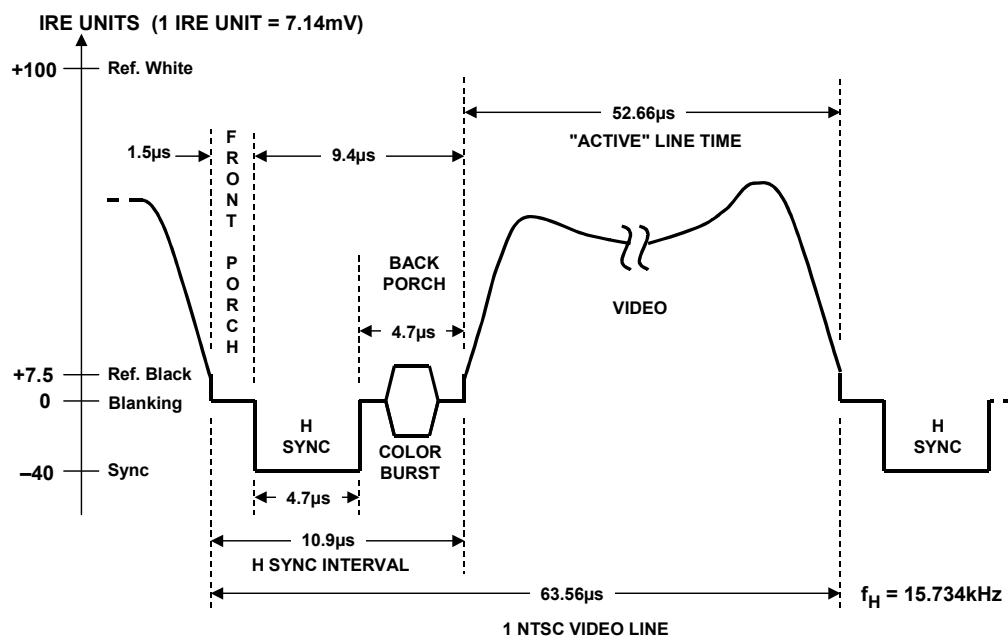


Figure 6-78: NTSC composite color video line

Whenever a horizontal sync pulse is detected, the beam is reset to the left side of the screen and moved down to the next line position. A vertical sync pulse, indicated by a horizontal sync pulse of longer duration, resets the beam to the top left point of the screen to a line centered between the first two lines of the previous scan. This allows the current field to be displayed between the previous one.

A simplified block diagram of the NTSC color processing system is shown in Figure 6-79, below. The three color signals (RGB: red, green, and blue) from the color camera are combined in a *matrix* unit to produce what is called the *luminance* signal (Y) and two color difference signals (I and Q). These *components* are further combined to produce what is called the *composite* color signal.

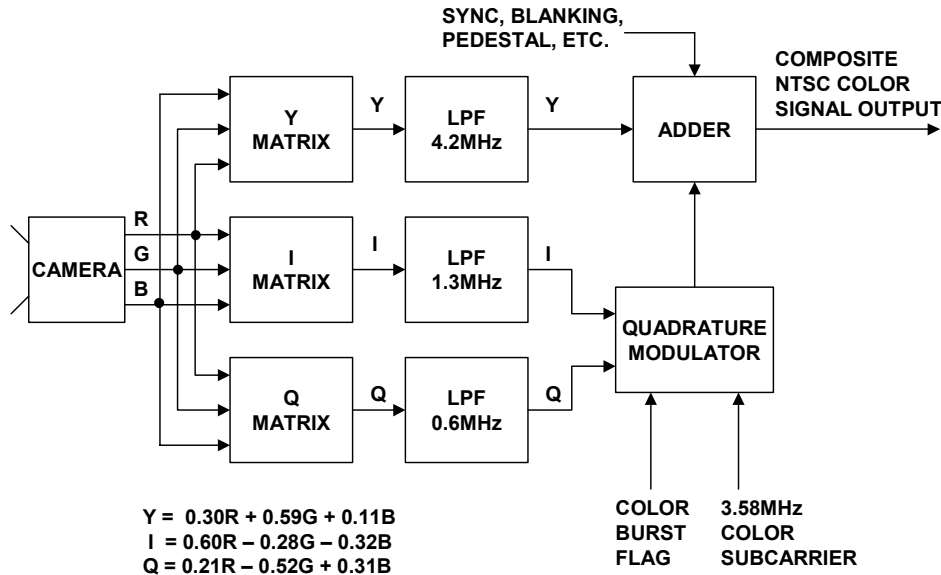


Figure 6-79: Generating the composite NTSC color signal

In the NTSC system (used in the U.S. and Japan), the color subcarrier frequency is 3.58MHz. The PAL system (used in the U.K. and Germany) and SECAM system (used in France) use a 4.43MHz color subcarrier.

	NTSC	PAL
Horizontal Lines	525	625
Color Subcarrier Frequency	3.58MHz	4.43MHz
Frame Frequency	30Hz	25Hz
Field Frequency	60Hz	50Hz
Horizontal Sync Frequency	15.734kHz	15.625kHz

Figure 6-80: NTSC and PAL signal characteristics

In terms of their key frequency differences, a comparison between the NTSC system and the PAL system are given in Figure 6-80, above.

Differential Gain And Phase Specifications

The color (or *chrominance*) information in the composite video signal is contained in the amplitude and phase of the subcarrier. The *intensity* or *saturation* of the color is determined by the amplitude of the subcarrier signal, and the precise color displayed (i.e. red, green, blue, and combinations) is determined by the phase of the subcarrier signal with respect to the phase of the color burst. The chrominance signal modulates the luminance signal which determines the relative blackness or whiteness of the color. To preserve color fidelity, it is important that the amplitude and phase of a constant-amplitude and phase color subcarrier remain constant across the range of black to white.

Any variation of the *amplitude* of the color subcarrier from black to white levels is called *differential gain* (expressed in %), and any variation in phase with respect to the color subcarrier is called *differential phase* (expressed in degrees). Degradations of up several percent of differential gain and several of degrees differential phase are acceptable for home viewing purposes, but individual components in the video signal path (amplifiers, switches, etc.) must meet much tighter specifications. This is because the signal must pass through many circuits from the camera to the home. As a result, individual professional video systems therefore have stringent requirements for differential gain and phase, usually limiting changes to less than 0.1% and 0.1°.

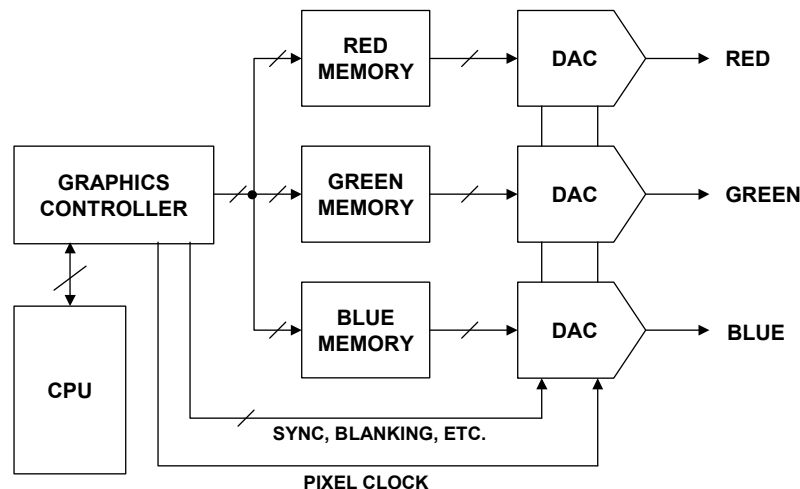


Figure 6-81: Simplified graphics control system for generating RGB signals

These system specifications mandate even more stringent standards for individual components, with the differential gain and differential phase requirements for op amps approaching 0.01% and 0.01°.

Video Formats in Graphics Display Systems

There are several system architectures which may be used to build a graphics display system. The most general approach is illustrated in Figure 6-81, above. It consists of a host microprocessor, a graphics controller, three color memory banks or frame buffer, one for each of the primary colors red, green, and blue, (only one for monochrome systems). The microprocessor provides the image information to the graphics controller. This

information typically includes position and color information. The graphics controller is responsible for interpreting this information and adding the required output signals such as sync, blanking, and memory management signals.

Unlike broadcast video, the horizontal and vertical resolution as well as the refresh rate in a graphics display system can vary widely depending upon the desired performance. The resolution in such a system is defined in terms of *pixels*: the number of horizontal lines (expressed as pixels) and the number of pixels in each line. For instance, a 640×480 monitor has 480 horizontal lines, and each horizontal line is divided into 640 pixels. So a single frame would contain 307,200 pixels. In a color system, each pixel requires RGB intensity data. This data is generally stored as 8 or 10-bit words in the memory.

The memory holds the intensity information for each pixel. The DACs use the words in the memory and information from the memory controller to write the pixel information to the monitor. Special video DACs called "RAMDACs" greatly simplify the storage of the pixel data by using color lookup tables. These DACs also have inputs to facilitate the generation of the sync and blanking signals.

RESOLUTION	PIXEL RATE
640 × 480	25MHz
800 × 600	38MHz
1024 × 768	65MHz
1280 × 1024	105MHz
1500 × 1500	180MHz
2048 × 2048	330MHz

$$\text{Pixel Rate} \approx \text{Vertical Resolution} \times \text{Horizontal Resolution} \times \text{Refresh Rate} \times 1.3$$

Figure 6-82: Typical graphics resolution and pixel rates for 60Hz non-interlaced refresh rate

Figure 6-82 above shows some typical resolutions and pixel rates for common display systems, assuming a 60Hz, non-interlaced refresh rate. Standard computer graphics monitors, like television monitors, use a display technique known as *raster scan*. This technique writes information to the screen line by line, left to right, top to bottom, as has been previously discussed. The monitor must receive a great deal of information to display a complete picture. Not only must the intensity information for each pixel be present in the signal but information must be provided to determine when a new line needs to start (HSYNC) and when a new picture frame should start (VSYNC). The computer industry has generally standardized on formats defined in EIA video standard RS-343A. Unlike broadcast video, the refresh rate can also vary, and interlacing may or may not be utilized. The pixel clock frequency gives a good idea of the settling time and bandwidth requirements for any analog component, such as the DAC, which is placed in the path of the RGB signals. The pixel clock frequency can be estimated by finding the product of the horizontal resolution times the vertical resolution times the refresh rate. An additional 30% should be added, called the retrace factor, to allow for overhead.

Bandwidth Considerations in Video Applications

The bandwidth of an op amp used in a video application must be sufficient so that the video signal is not attenuated or shifted in phase significantly. This generally implies that the bandwidth of the op amp be much greater than that of the maximum video frequency. It is not uncommon to require that amplifiers in the signal path in video equipment such as switchers or special effects generators have 0.1dB bandwidths of 50MHz or greater. High definition television requires even higher 0.1dB bandwidth. Circuit parasitics as well as the load impedance can significantly affect the 0.1dB bandwidth at high frequencies. This implies careful attention to layout, decoupling, and grounding as well as the use of transmission line techniques at the op amp output. It is common to use source and load terminations with high quality 75Ω coaxial cable so that the load presented to the op amp output appears as a 150Ω resistive load. Maintaining accurate control of 0.1dB bandwidth is almost impossible with reactive loads.

Achieving the highest 0.1dB bandwidth flatness is therefore important in many video applications. Voltage feedback op amps can be optimized for maximum 0.1dB bandwidth provided the closed-loop gain and load conditions are known. In video applications, closed-loop gains of +1 and +2 are the most common with a 100Ω or 150Ω output load, representing the impedance of 50Ω or 75Ω source and load terminated cables.

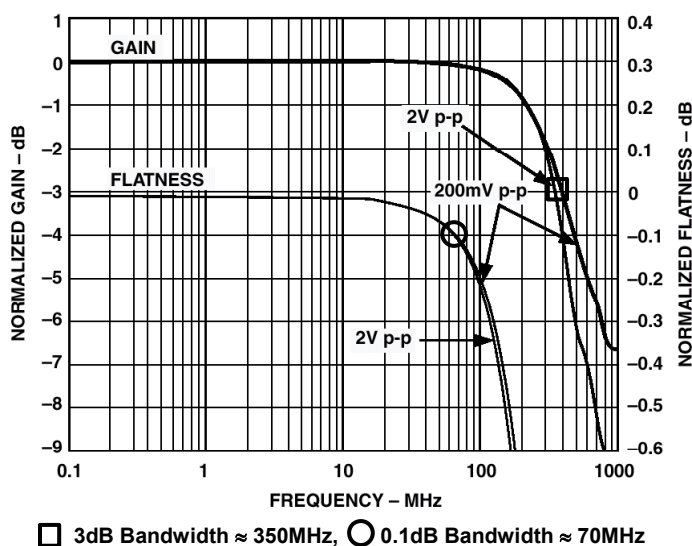


Figure 6-83: AD8075 triple video buffer gain and gain flatness, $G = +2$, $R_L = 150\Omega$

As an example, the AD8074 ($G = +1$) and AD8075 ($G = +2$) are triple video buffers optimized for driving source and load terminated 75Ω cables. These devices use a voltage feedback architecture and have on-chip gain-setting resistors. Figure 6-83 above shows the frequency response of the AD8075 buffer on two vertical scales: 1dB/division and 0.1dB/division. The plots labeled "GAIN" show a 3dB bandwidth of 350MHz (□), and the plots labeled "FLATNESS" show a 0.1dB bandwidth of 70MHz (○). Note that the small-signal (200mVp-p) and large-signal (2Vp-p) bandwidths are approximately equal.

Voltage feedback op amps are optimized for bandwidth flatness by adjusting both the compensation capacitor which sets the dominant pole as well as the external feedback network. However, because of the critical relationship between the feedback resistor and the bandwidth of a current feedback op amp, optimum bandwidth flatness is highly dependent on the feedback resistor value, the resistor parasitics, as well as the op amp package and PCB parasitics. Figure 6-84 below shows the bandwidth flatness (0.1dB/division) plotted versus the feedback resistance for the AD8001 in a non-inverting gain of 2. The 100Ω load resistor represents a source and load terminated 50Ω cable. These plots were made using the AD8001 evaluation board with surface mount resistors.

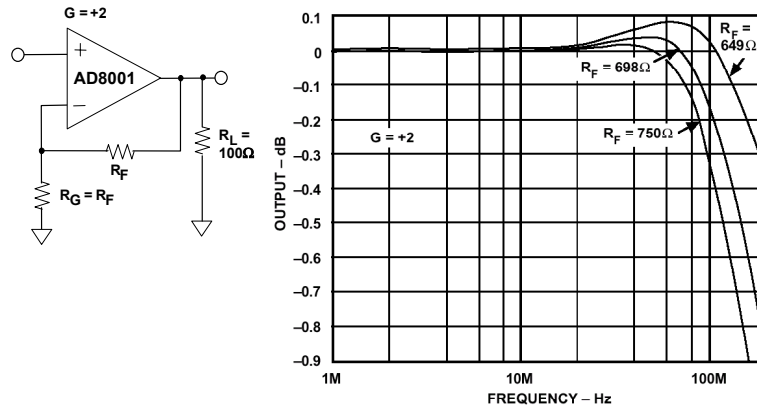


Figure 6-84: AD8001 current feedback op amp bandwidth flatness versus feedback resistor value

It is recommended that once the optimum resistor values have been determined, 1% tolerance values should be used. In addition, resistors of different construction have different associated parasitic capacitance and inductance. Surface mount resistors are an optimum choice, thus leaded components aren't recommended for high frequency use.

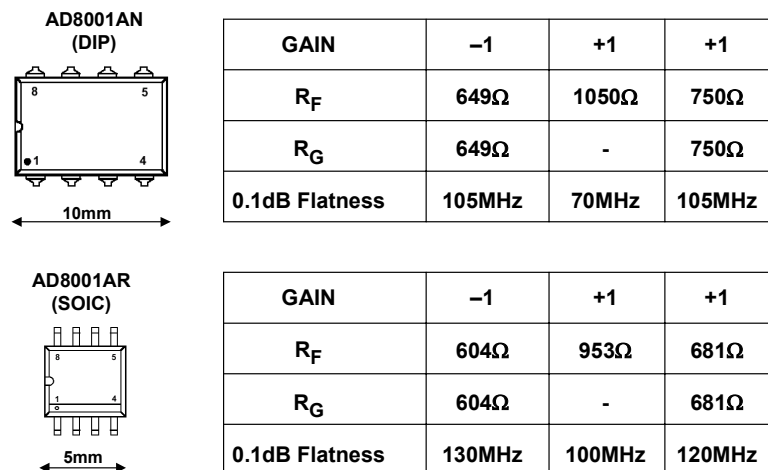


Figure 6-85: Optimum values of R_F and R_G for AD8001 DIP and SOIC packages for maximum 0.1dB bandwidth

Slightly different resistor values may be required to achieve optimum performance of the AD8001 in the DIP versus the SOIC packages (see Figure 6-85 above). The SOIC package exhibits slightly lower parasitic capacitance and inductance than the DIP. The

■ OP AMP APPLICATIONS

data shows the optimum feedback (R_G) and feedforward (R_F) resistors for highest 0.1dB bandwidth for the AD8001 in the DIP and the SOIC packages. As you might suspect, the SOIC package can be optimized for higher 0.1dB bandwidth because of lower parasitics.

As has been discussed, the current feedback op amp is relatively insensitive to capacitance on the inverting input when it is used in the inverting mode (as in an I/V application). This is because the low inverting input impedance is in parallel with the external capacitance and tends to minimize its effect. In the non-inverting mode, however, even a few picofarads of stray inverting input capacitance may cause peaking and instability. Figure 6-86 shows the effects of adding summing junction capacitance to the inverting input of the AD8004 (SOIC package) for $G = +2$. Note that only 1pF of added inverting input capacitance (C_J) causes a significant increase in bandwidth and an increase in peaking. For $G = -2$, however, 5pF of additional inverting input capacitance causes only a small increase in bandwidth and no significant increase in peaking.

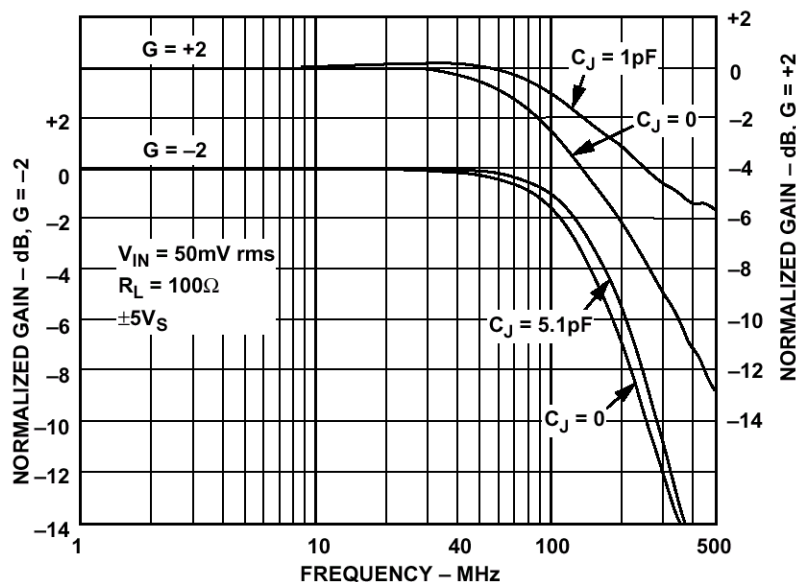


Figure 6-86: AD8004 current feedback op amp sensitivity to inverting input capacitance for $G = +2$, $G = -2$

It should be noted that high-speed voltage feedback op amps are sensitive to stray inverting input capacitance when used in either the inverting or non-inverting mode, because both positive and negative inputs are high impedance.

Video Signal Transmission

High quality video signals are best transmitted over terminated coaxial cable having a controlled characteristic impedance. The characteristic impedance is given by the equation $Z_0 = \sqrt{L/C}$ where L is the distributed inductance per foot, and C is the distributed capacitance per foot. Popular values are 50, 75, and 93 or 100 Ω .

If a length of coaxial cable is properly terminated, it presents a *resistive* load to the driver. If left unterminated, however, it may present a predominately capacitive load to the driver depending on the output frequency. If the length of an unterminated cable is much less than the wavelength of the output frequency of the driver, then the load appears approximately as a lumped capacitance. For instance, at the audio frequency of 20kHz (wavelength $\approx$ 50,000 feet, or 9.5miles), a 5 foot length of unterminated 50 Ω coaxial cable would appear as a lumped capacitance of approximately 150pF (the distributed capacitance of coaxial cable is about 30pF/ft).

- ◆ All interconnections are really transmission lines which have a characteristic impedance (even if not controlled).
- ◆ The characteristic impedance is equal to $\sqrt{L/C}$, where L and C are the distributed inductance and capacitance.
- ◆ Correctly terminated transmission lines have impedances equal to their characteristic impedance.
- ◆ Unterminated transmission lines behave approximately as lumped capacitance if the wavelength of the output frequency is much greater than the length of the cable.
 - Example: At 20kHz (wavelength = 9.5 miles), 5 feet of unterminated 50 Ω cable (30pF/ft) appears like a 150pF load
 - Example: At 100MHz, (wavelength = 10 feet), 5 feet of 50 Ω must be properly terminated to prevent reflections and standing waves.

Figure 6-87: Driving cables

At 100MHz (wavelength $\approx$ 10 feet), however, the unterminated coax must be treated as a transmission line in order to calculate the standing wave pattern and the voltage at the unterminated cable output. Figure 6-87 above summarizes transmission line behavior for different frequencies.

Because of skin effect and wire resistance, coaxial cable exhibits a loss that is a function of frequency. This varies considerably between cable types. For instance at 100MHz the attenuation RG188A/U is 8dB/100ft, RG58/U is 5.5dB/100ft, and RG59/U 3.6dB/100ft (see Reference 4). Skin effect also affects the pulse response of long coaxial cables. The response to a fast pulse will rise sharply for the first 50% of the output swing, then taper off during the remaining portion of the edge. Calculations show that the 10 to 90% waveform risetime is 30 times greater than the 0 to 50% risetime when the cable is skin effect limited (Reference 4, again).

■ OP AMP APPLICATIONS

Transmission Line Driver Lab

It is useful to examine the fidelity of a pulse signal, for conditions of proper/improper transmission line source/load terminations. Some lab experiments were set up to do this.

To illustrate the behavior of a high speed op amp driving a coaxial cable, consider the circuit of Figure 6-88 below. Here the AD8001 drives 5 feet of 50Ω coaxial cable, which is load-end terminated in the characteristic impedance of 50Ω . No termination is used at the amplifier (driving) end. The pulse response is also shown in the figure.

The output of the cable was measured by connecting it directly to the 50Ω input of a 500MHz Tektronix 644A digitizing oscilloscope. The 50Ω resistor termination is actually the input of the scope. However, this 50Ω load is not a perfect line termination, it is lower at high frequencies (due to the scope shunt input capacitance of about 10pF).

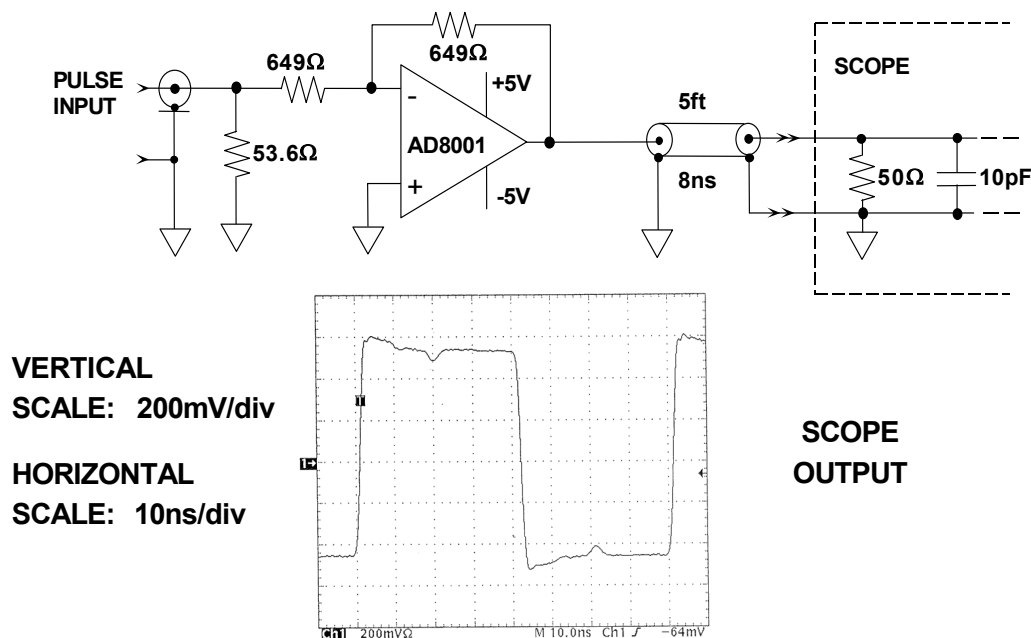


Figure 6-88: Pulse response of AD8001 driving 5 feet of load-only terminated 50Ω coaxial cable

As a consequence some of the positive going pulse edge is reflected out of phase to the source. When this reflection reaches the op amp, it sees the op amp closed-loop output impedance, which, at 100MHz, is approximately 100Ω (higher than line impedance).

Upon arriving at the op amp output, the negative-going reflection from the load is then re-reflected back towards the load, without undergoing another phase reversal. This then accounts for the negative going "blip" seen on the upper plateau of the waveform, which occurs approximately 16ns after the leading edge. This time difference is equal to the round-trip delay of the cable ($2 \cdot 5\text{ft} \cdot 1.6\text{ ns/ft} = 16\text{ns}$). An additional point worth noting is that, in the frequency domain (which is not shown by these tests) the cable mismatch will also cause a loss of bandwidth flatness at the load.

Figure 6-89 below shows a second case, the results of driving the same coaxial cable, but now used with both a 50Ω source-end as well as the 50Ω load-end termination at the scope. It should be noted that this case is the preferred way to drive a transmission line, because a portion of the reflection from the load impedance mismatch is absorbed by the amplifier's source termination resistor of 50Ω . A disadvantage is that there is a gain loss of 6dB, because of the 2/1 voltage division which occurs between the equal value source and load terminations, i.e., $50\Omega/50\Omega$.

However, a major positive attribute of this configuration, with the line impedance matched source and load terminations in conjunction with a low-loss cable, is that *the best bandwidth flatness is ensured*, especially at lower operating frequencies. In addition to this, the amplifier is operated with a near optimum load condition, i.e., into a resistive load. The load in this case is 50Ω plus 50Ω , or 100Ω . In general, it will be twice the impedance of the transmission line in use, i.e., 150Ω for a 75Ω line, etc.

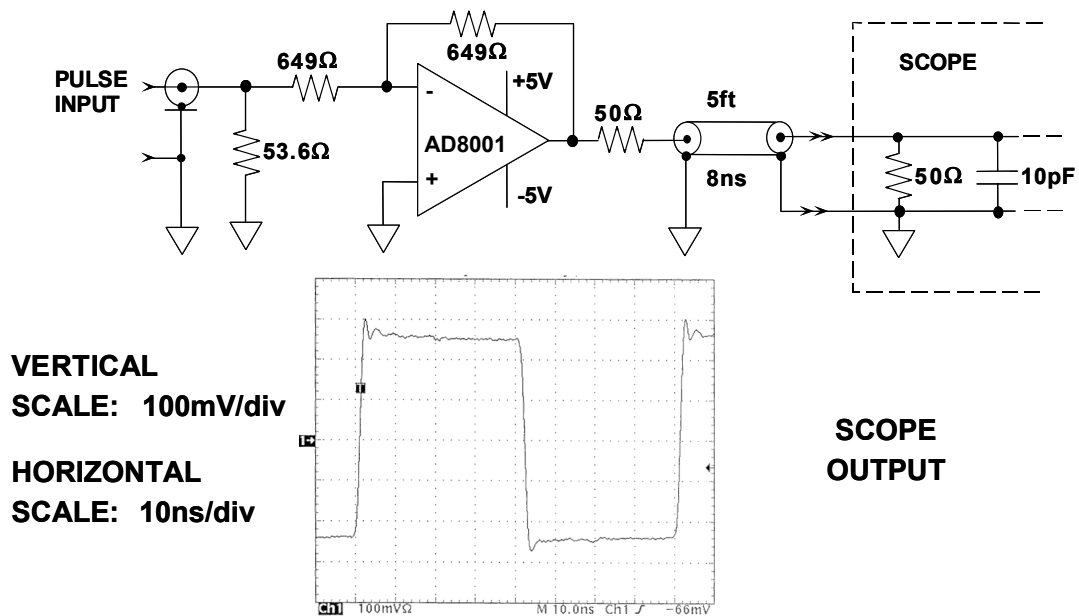


Figure 6-89: Pulse response of AD8001 driving 5 feet of source and load terminated 50Ω coaxial cable

In practice, the gain loss associated with the 2/1 source/load impedance is easily made up, simply by operating the line driver stage at a gain of 2x. Typically, video driver stages are non-inverting to preserve the waveform sign, and operate at a fixed and precise gain of 2 times. Thus they will inherently provide a net signal transfer gain of unity, as measured from input to the final end-of-line load termination (this neglects any associated transmission line losses, and assumes precise resistor ratios for the gain resistors). Another very practical point is that the same driver can be used for a wide variety of transmission lines, simply by changing the value of the source termination resistor.

■ OP AMP APPLICATIONS

Source-end (only) terminations can also be used as shown in Figure 6-90 below, where the op amp is now source terminated by the 50Ω resistor which drives the cable. At the load end, the scope is set for $1M\Omega$ input impedance, which represents an approximate open circuit. The initial leading edge of the pulse at the op amp output sees a 100Ω load (the 50Ω source resistor in series with the 50Ω coax impedance). When the pulse reaches the load, a large portion is reflected in phase, because of the high load impedance, resulting in a full-amplitude pulse at the load. When the reflection reaches the source-end of the cable, it sees the 50Ω source resistance in series with the op amp closed loop output impedance (approximately 100Ω at the frequency represented by the 2ns risetime pulse edge). The re-reflected portion remains in phase, and then appears at the scope input as the positive going "blip", approximately 16ns after the leading edge.

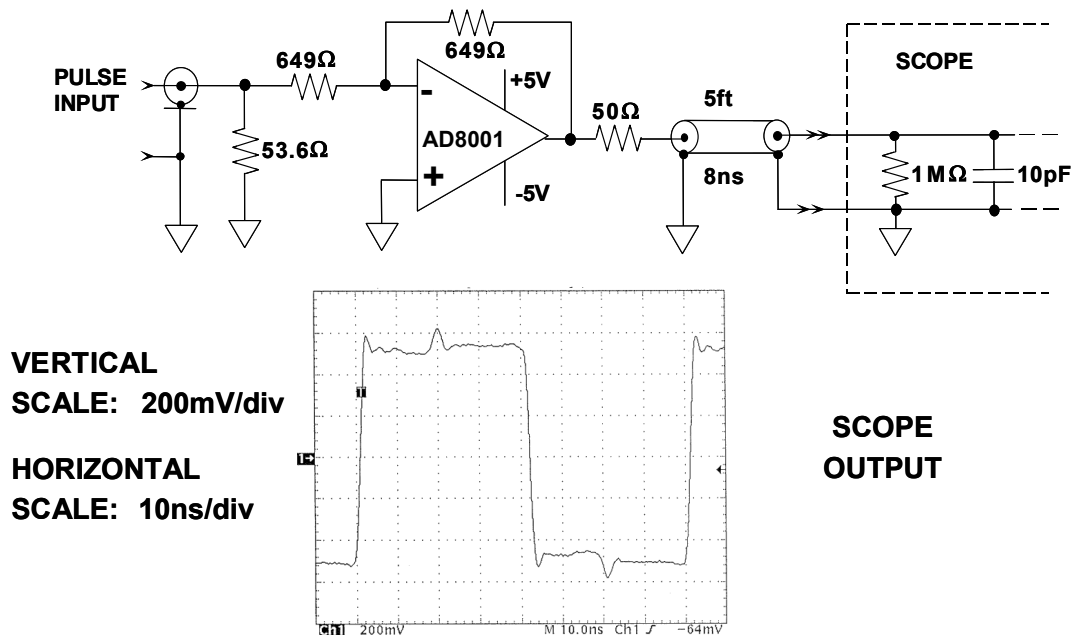


Figure 6-90: Pulse response of AD8001 driving 5 feet of source-only terminated 50Ω coaxial cable

From these experiments, one can easily see that the preferred method for minimum reflections (and therefore maximum bandwidth flatness) is to use both source and load terminations and try to minimize any reactance associated with the load. The experiments represent a worst-case condition, where the frequencies contained in the fast edges are greater than 100MHz . (using the rule-of-thumb that bandwidth = $0.35/\text{risetime}$).

At less demanding video frequencies, either load-only, or source-only terminations may give acceptable results, but the op amp data sheet should always be consulted to determine the op amp's closed-loop output impedance at the maximum frequency of interest; i.e., is it less than the line impedance. A major disadvantage of the source-only termination is that it requires a truly high impedance load (high resistance and minimal parasitic capacitance) for minimum absorption of energy. It also places a burden on the driving amplifier, to maintain the low output impedance at high frequencies.

Now, for a truly worst case, let us replace the 5 feet of coaxial cable with an uncontrolled-impedance cable (one that is largely capacitive with little inductance). Also, let's use a capacitance of 150pF to simulate the cable (corresponding to the total capacitance of 5 feet of coaxial cable, whose distributed capacitance is about 30pF/foot). Figure 6-91 below shows the output of the AD8001 op amp, driving a lumped 160pF capacitance (including the scope input capacitance of 10pF).

Overshoot and severe ringing on the pulse waveform is noted, due to the capacitive loading. This example illustrates the need to use good quality controlled-impedance coaxial cable in the transmission of high frequency signals, even over short distances. Failure to adhere to controlled-impedance lines for signal distribution can result in severe loss of pulse waveform fidelity, and loss of bandwidth flatness in the frequency domain.

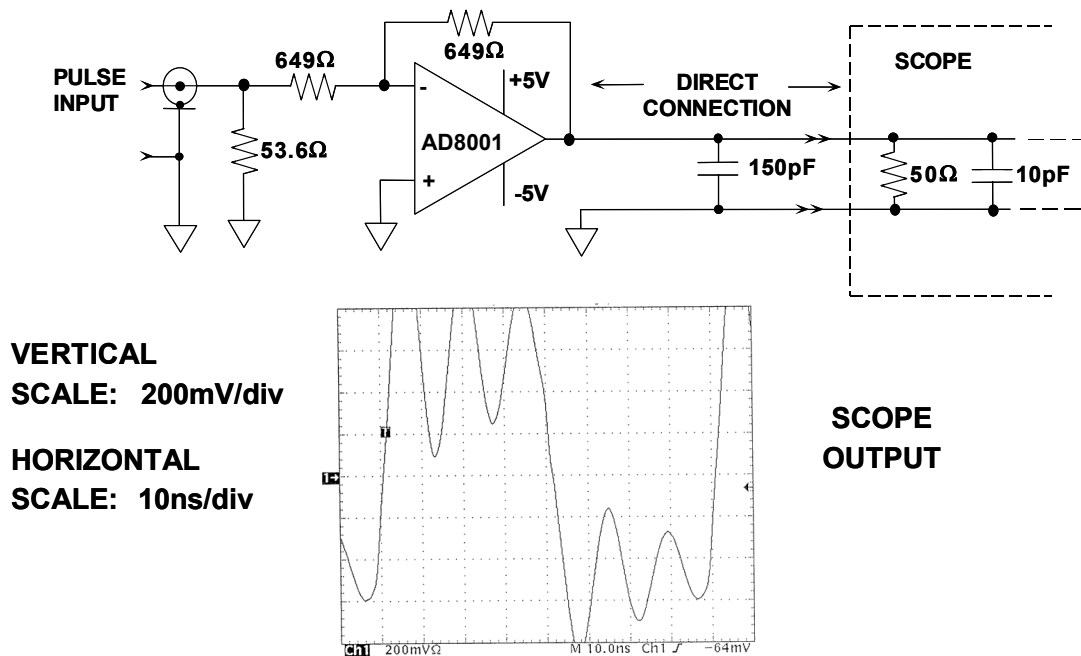


Figure 6-91: Pulse response of AD8001 driving 160pF || 50Ω load

To summarize, transmission line driver circuits should use proper line terminations for best response. The ideal method of line termination is matching line-impedance-value resistances at both source and load end (Fig. 6-89, again). The associated 6dB gain loss is easily made up in the amplifier. Next best is a source-only termination (Fig. 6-90, again), with due care towards maintaining a high impedance at the load end, and a low drive impedance amplifier. This type of termination provides near-full amplitude level at the load end, making the gain of the driver less critical. Load-only termination can also be used (Fig. 6-88, again), but may be more critical of load end parasitic effects and the amplifier performance. It also provides near-full amplitude level at the load end.

Direct drive of uncontrolled load impedances, especially lumped capacitive lines, should be avoided wherever signal fidelity is important (Fig. 6-91, again).

■ OP AMP APPLICATIONS

Video Line Drivers

The AD8047 and AD8048 voltage feedback op amps have been optimized to offer outstanding performance as video line drivers. They utilize the "quad core" g_m stage as previously described for high slew rate and low distortion (see Chapter 1). The AD8048 (optimized for $G = +2$) has a differential gain of 0.01% and a differential phase of 0.02° , making it well suited for HDTV applications.

In the configuration using the AD8048 shown in Figure 6-92 below, the 0.1dB bandwidth is 50MHz for $\pm 5V$ supplies, slew rate is $1000V/\mu s$, and 0.1% settling time is 13ns. The total quiescent current is 6mA ($\pm 5V$), and quiescent power dissipation 60mW.

Performance of this circuit will be optimum with the gain-of-two stable AD8048 op amp, as its parameters have been optimized for this gain. Alternately, if a gain-of-one stable op amp is desired, the AD8047 can be used.

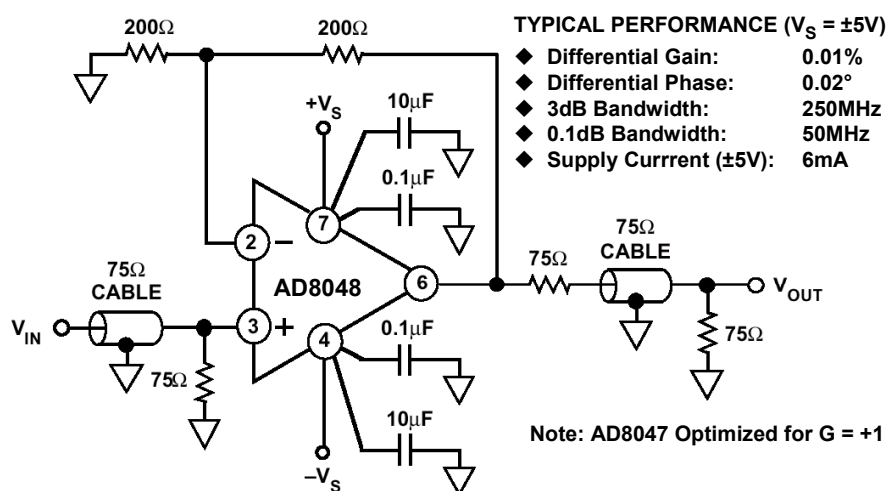


Figure 6-92: High performance video line driver using the AD8048

Note that a very wide variety of both voltage feedback and current feedback devices can be used similarly as a gain-of-two line driver (although the required feedback resistances may vary by device). Examples would be the AD818/828, AD8055/56, AD8057/58, and AD8061/62/63 families of voltage feedback op amps, and the AD811/12/13, AD8001/02, AD8012 families as a partial list. There are differences among all of these devices for applicable supply ranges, single-supply compatibility, etc., so consult device data sheets.

It is often desirable to drive more than one coaxial cable, which represents a DC load of 150Ω to a driver. The typical maximum video signal level is 1V into 75Ω , which represents 2V at the output of the driver, and a current of 13.3mA. Thus a 50mA output current video op amp such as the AD8047 or AD8048 would theoretically be capable of driving three source and load-terminated 75Ω loads. But, there are other important subtle considerations for this application. Differential gain and phase may be degraded for high output currents. Also, the op amp closed-loop output impedance affects crosstalk between the driven output channels. So it is often better to select a video driver fully specified for the required fan-out and load, especially if the fan-out is greater than two.

Video Distribution Amplifier

The AD8010 op amp is optimized for driving multiple video loads in parallel. Video performance of 0.02% differential gain and 0.03% differential phase is maintained, while driving eight 75Ω source and load-terminated video lines. The AD8010 uses the current feedback architecture and has a 0.1dB bandwidth of 60MHz with eight video loads. Typical supply current (neglecting load current) is 15.5mA on ±5V supplies. A typical connection diagram is shown below in Figure 6-93. The AD8010 is offered in three packages: an 8-lead DIP ($\theta_{JA} = 90^{\circ}\text{C/W}$), 16-lead wide body SOIC ($\theta_{JA} = 73^{\circ}\text{C/W}$), and a low thermal resistance, 8-lead SOIC ($\theta_{JA} = 122^{\circ}\text{C/W}$).

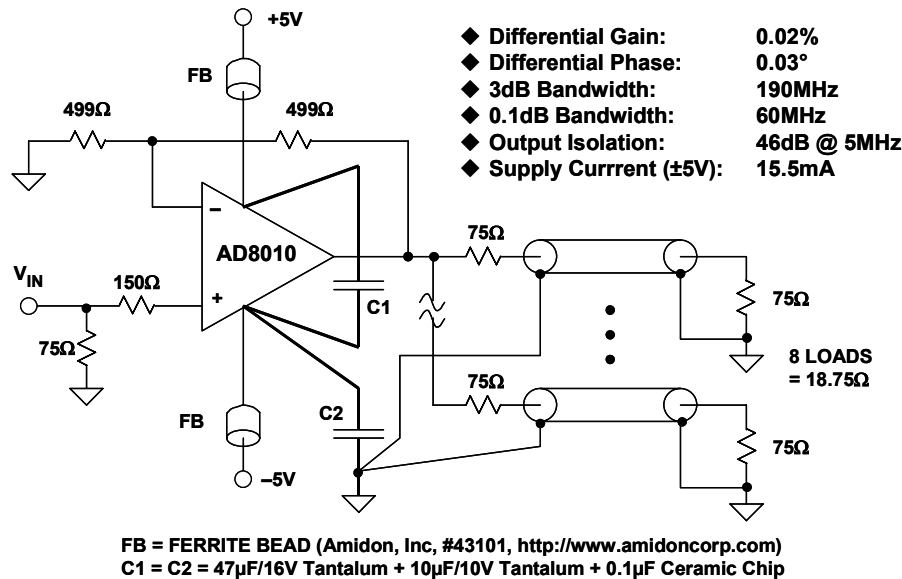


Figure 6-93: The AD8010 video distribution amplifier

The power supply decoupling scheme used for the AD8010 requires special attention. The conventional technique of bypassing each power supply pin individually to ground can have an adverse effect on the differential phase error of the circuit. This is because there is an internal compensation capacitor in the AD8010 that is referenced to the negative supply. The recommended technique shown in Figure 6-93 is to connect three parallel bypass capacitors from the positive supply to the negative supply, and then to bypass the negative supply to ground with a similar set, as shown. For high frequency decoupling, 0.1μF ceramic surface mount capacitors are recommended. The high currents that can flow through the power supply pins require additional large tantalum electrolytic decoupling capacitors. As shown, a 47μF/16V tantalum in parallel with a 10μF/10V tantalum capacitor is desirable. The grounded side of the C2 capacitors bypassing the negative supply should be brought to a single-point output return ground. In addition to the bypass capacitors described above, ferrite beads such as those noted should be placed in series with both positive and negative supplies for further decoupling.

Another important consideration for driving multiple cables is high frequency isolation between the outputs. Due largely to its low output impedance, the AD8010 achieves better than 46dB output-to-output isolation at 5MHz, while driving 75Ω source and load-terminated cables.

Differential Line Drivers/Receivers

There are a number of applications for differential signal drivers and receivers. Among these are analog-digital-converter (ADC) input buffers, where differential operation can provide lower levels of second order distortion for certain converters. Other uses include high frequency bridge excitation, and drivers for balanced transmission twisted pair lines such as in ADSL and HDSL.

The transmission of high quality signals across noisy interfaces (either between individual PC boards or between racks) has always been a challenge to designers. Differential techniques using high common-mode-rejection-ratio (CMRR) instrumentation amplifiers largely solves the problem at low frequencies. Examples of this have already been discussed, under the "Audio Amplifiers" portion of this chapter.

At audio frequencies, transformers, or products such as the SSM2142 balanced line driver and SSM2141/SSM2143 line receivers offer outstanding CMRRs and the ability to transmit low-level signals in the presence of large amounts of noise, as noted. At high frequencies, small bifilar-wound toroid transformers are effective.

In contrast to this, the problem of signal transmission at video frequencies is a more complex one. Transformers suitable for video coupling aren't very effective, because the baseband video signal has low-frequency components down to a few tens of Hz, and an upper bandwidth limit that can be in the tens (hundreds) of MHz. This make a workable video transformer an item extremely difficult to make.

Another point is that video signals are generally processed in single-ended form, and therefore don't adapt easily to balanced transmission line techniques. Related to this, shielded twin-conductor coaxial cable with good bandwidth is usually somewhat bulky and expensive, and has not found great acceptance.

As a result of these factors, designing high bandwidth, low distortion differential video drivers and receivers with high CMRR at high frequencies is an extremely difficult task.

Nevertheless, even in the face of all of the above problems, there are various differential techniques available right now that offer distinct advantages over single-ended methods. Some of these techniques make use of discrete components, while others utilize state-of-the-art video differential amplifiers.

Approaches To Video Differential Driving/Receiving

Two solutions to differential transmission and reception are shown in Figure 6-94 below. One is the ideal case (top), where a balanced differential driver drives a balanced twin-conductor coaxial cable, which then drives a terminated differential line receiver. However, as discussed, this circuit is difficult to implement fully at video frequencies.

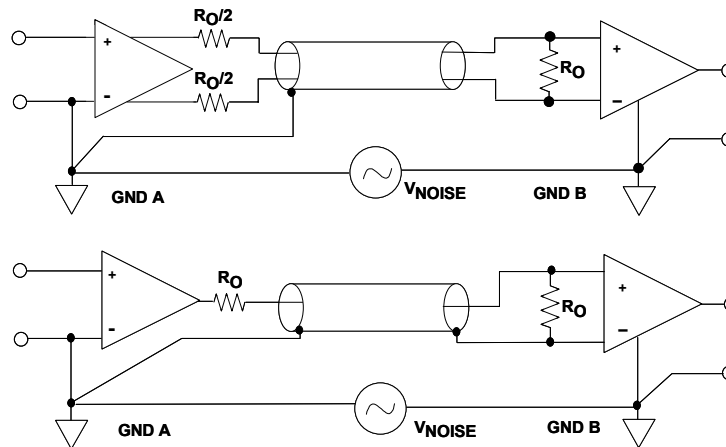


Figure 6-94: Two approaches to differential line driving and receiving

A second, most often used approach uses a single-ended driver driving a source-terminated coaxial cable (bottom), with the cable shield grounded at the transmitter. At the receiver, the coaxial cable is terminated in its characteristic impedance, but the shield is left floating in order to prevent a ground loop between the two systems. Common mode ground noise is rejected by the CMRR of the differential line receiver.

Inverter-Follower Differential Driver

The circuit of Figure 6-95 below is a useful differential driver for high speed 10-12 bit ADCs, differential video lines, and other balanced loads at 1-4V_{rms} output levels.

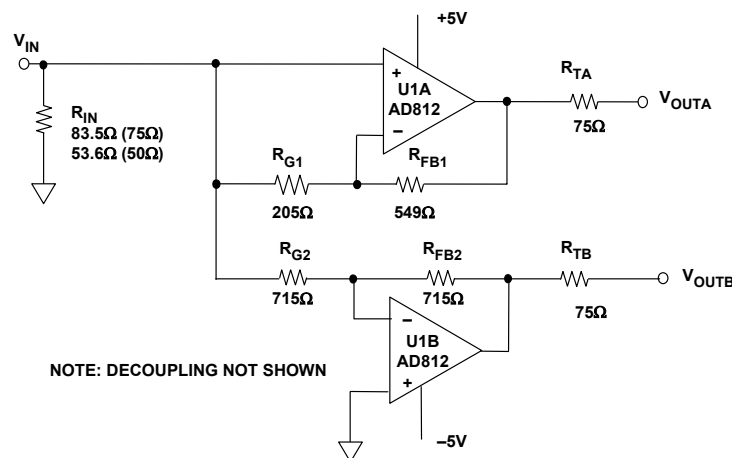


Figure 6-95: Differential driver using an inverter and a follower

It is shown operating from $\pm 5V$ supplies, but it can also be adapted to supplies in the range of ± 5 to $\pm 15V$. When operated directly from $\pm 5V$ as here, it minimizes potential for

■ OP AMP APPLICATIONS

destructive ADC overdrive when higher supply voltage buffers drive a $\pm 5\text{V}$ powered ADC, in addition to also minimizing driver power.

In many of these differential drivers the performance criteria is often high. In addition to low output distortion, the two signals should maintain gain and phase flatness. In this topology, two sections of an AD812 dual current feedback amplifier are used for the channel A and B buffers. This provides inherently better open-loop bandwidth matching than using two singles, where bandwidth varies between different manufacturing lots. The two buffers here operate with precise gains of ± 1 , as defined by their respective feedback and input resistances. Channel B buffer U1B is conventional, and uses a matched pair of 715Ω resistors— an optimum value for the AD812 on $\pm 5\text{V}$ supplies.

In channel A, non-inverting buffer U1A has an inherent signal gain of 1, by virtue of the bootstrapped feedback network R_{FB1} and R_{G1} (see Reference 5). It also has a higher noise gain, for phase matching. Normally a current feedback amplifier operating as a simple unity gain follower would use one (optimum) resistor R_{FB1} , and no gain resistor at all. Here, with input resistor R_{G1} added, a U1A noise gain like that of U1B results. Due to the bootstrap connection of R_{FB1} - R_{G1} , the signal gain is maintained at unity. Given the matched open loop bandwidths of U1A and U1B, similar noise gains in the A-B channels provide closely matched output bandwidths between the driver sides, a distinction which greatly impacts overall matching performance.

In setting up a design for the driver, the effects of resistor gain errors should be considered for R_{G2} - R_{FB2} . Here a worst case 2% mis-match will result in less than 0.2dB gain error between channels A and B. This error can be improved simply by specifying tighter resistor ratio matching, avoiding trimming.

If desired, phase match can be trimmed via R_{G1} , so that the phase of channel A matches that of B. This can be done by using a pair of closely matched (0.1% or better) resistors to sum the A and B channels, as R_{G1} is adjusted for the best null conditions at the sum node. The A-B gain and phase matching is quite effective in this driver; the test results of the circuit as shown 0.04dB and 0.1° between the A and B output signals at 10MHz, when operated into dual 150Ω loads. The 3dB bandwidth of the driver is about 60MHz.

Net input impedance of the circuit is set to a standard line termination value such as 75Ω (or 50Ω), by choosing R_{IN} so that the desired value results when paralleled with R_{G2} . In this example, an R_{IN} value of 83.5Ω provides a standard input impedance of 75Ω when paralleled with 715Ω . For the circuit just as shown, dual voltage feedback amplifier types with sufficiently high speed and low distortion can also be used. This allows greater freedom with regard to resistor values using such devices as the AD826 and AD828.

Gain of the circuit can be changed if desired, but this isn't totally straightforward. An easy step to satisfy diverse gain requirements is to simply use a triple amplifier such as the AD813 or AD8013, with the third channel as a variable gain input buffer. Note that if an amplifier is used with specifications substantially different than the AD812, some adjustment of resistor values may be necessary.

Cross-Coupled Differential Driver

Another differential driver approach uses cross-coupled feedback to get very high CMR and complementary outputs at the same time. In Figure 6-96 below, AD8002 dual current feedback amplifier sections are used as cross-coupled inverters, the outputs are forced equal and opposite, assuring zero output common mode voltage (see Reference 6).

The gain cell that results, U1A and U1B plus cross-coupling resistances R_X , is fundamentally a differential input/output topology. But, it behaves as a voltage feedback amplifier with regard to the feedback port at the U1A (+) node. The V_{IN} to V_{OUT} gain is:

$$G = \frac{V_{OUT}}{V_{IN}} = \frac{2R_2}{R_1} \quad \text{Eq. 6-21}$$

where V_{OUT} is the differential output, equal to $V_{OUTA} - V_{OUTB}$.

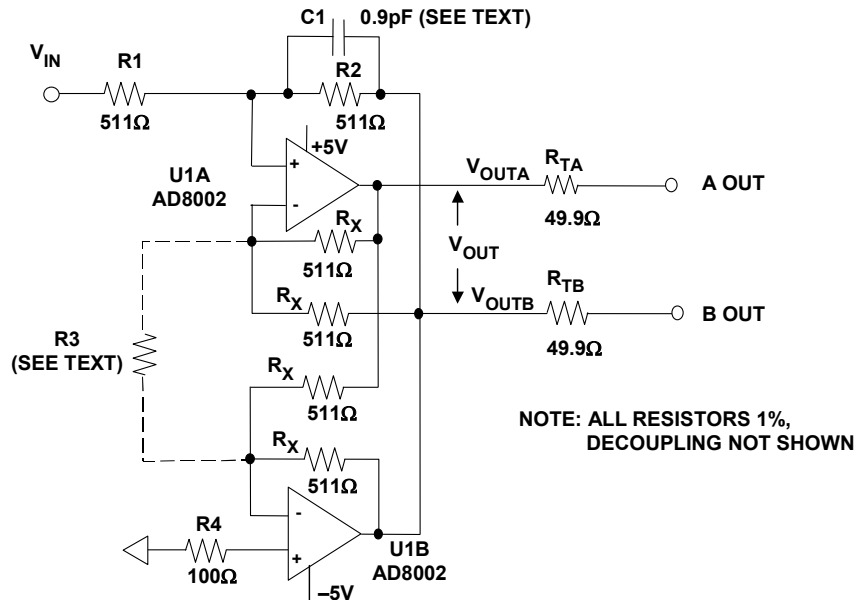


Figure 6-96: Cross-coupled differential driver provides balanced outputs and 250MHz bandwidth

The relationship of Eq. 6-21 may not be obvious, so it can be derived as follows:

Using the conventional inverting op amp gain equation, the input voltage V_{IN} develops an output voltage V_{OUTB} given by:

$$V_{OUTB} = -V_{IN} \frac{R_2}{R_1} \quad \text{Eq. 6-22}$$

Also, $V_{OUTA} = -V_{OUTB}$, because V_{OUTA} is inverted by U1B.

However, $V_{OUT} = V_{OUTA} - V_{OUTB} = -2V_{OUTB}$.

■ OP AMP APPLICATIONS

Therefore,

$$V_{OUT} = -2\left(-V_{IN} \frac{R_2}{R_1}\right) = 2V_{IN} \frac{R_2}{R_1}, \quad \text{Eq. 6-23}$$

and

$$\frac{V_{OUT}}{V_{IN}} = \frac{2R_2}{R_1}. \quad \text{Eq. 6-24}$$

This circuit has some unique benefits. First, the differential voltage gain is set by a single resistor ratio, so there is no necessity for side-side resistor matching with gain changes, as is the case for conventional differential amplifiers (see line receivers, below). Second, because the (overall) circuit emulates a voltage feedback amplifier, these gain resistances are not as restrictive as in the case of a conventional current feedback amplifier. Thus, they are not highly critical as to value as long as the equivalent resistance seen by U1A is reasonably low ($\leq 1\text{k}\Omega$ in this case).

A third and important advantage is that cell bandwidth can be optimized to a desired gain by a single optional resistor, R3, as follows. If for instance, a gain of 20 is desired ($R_2/R_1=10$), the bandwidth would otherwise be reduced by roughly this amount, since without R3, the cell operates with a constant gain-bandwidth product (voltage feedback mode). With R3 present however, advantage can be taken of the AD8002 current feedback amplifier characteristics. Additional internal gain is added by the connection of R3, which, given the appropriate value, effectively raises gain-bandwidth to a level so as to restore the bandwidth which would otherwise be lost by the higher closed loop gain.

In the circuit as shown, no R3 is necessary at the low working gain of 2, since the 511Ω R_X resistors are already optimized for maximum bandwidth. Note that these four matched R_X resistances are somewhat critical, and will change in absolute value with the use of another current feedback amplifier. At higher gain closed loop gains, R3 can be chosen to optimize the working transconductance in the input stages of U1A and U1B, as follows:

$$R_3 \cong \frac{R_X}{(R_2 / R_1) - 1}. \quad \text{Eq. 6-25}$$

As in any high speed inverting feedback amplifier, a small high-Q chip type feedback capacitance, C1, may be needed to optimize flatness of frequency response. In this example, a 0.9pF value was found optimum for minimizing peaking. In general, provision should be made on the PC layout for an NPO chip capacitor in the range of 0.5-2pF. This capacitor is then value selected at board characterization for optimum frequency response.

Performance for the circuit of Fig. 6-96 was examined with a dual trace, 1-500MHz swept frequency response plot, as is shown in Figure 6-97 below. The test output levels were 0dBm into matched 50 Ω loads, through back termination resistances R_{TA} and R_{TB} , as measured at V_{OUTA} and V_{OUTB} .

In this plot the vertical scale is 2dB/div, and it shows the 3dB bandwidth of the driver measuring about 250MHz, with peaking about 0.1dB. The four R_X resistors along with R_{TA} and R_{TB} control low frequency amplitude matching, which was within 0.1dB in the lab tests, using 511 Ω 1% resistor types. For tightest amplitude matching, these resistor ratios can be more closely controlled.

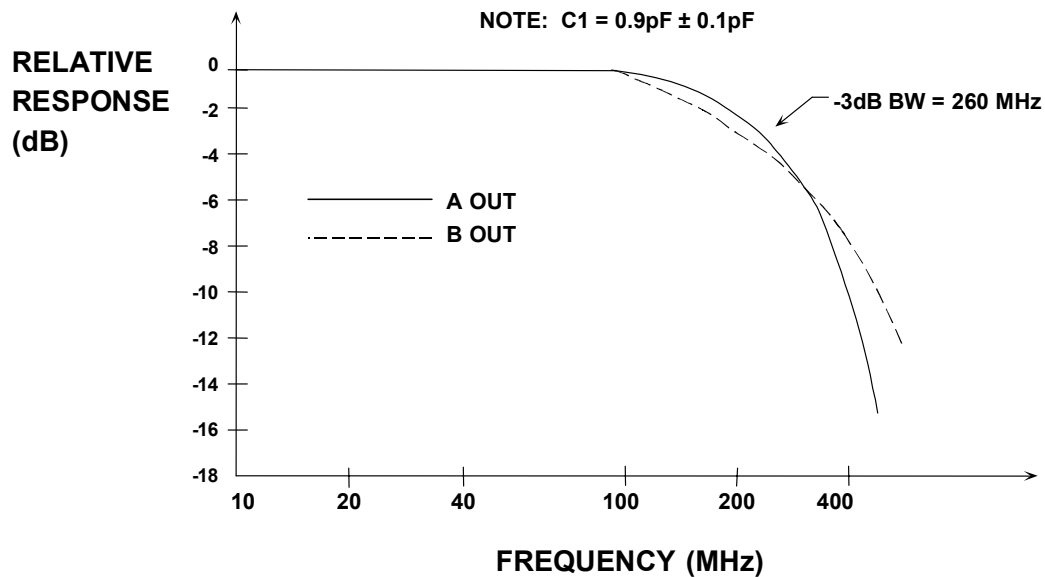


Figure 6-97: Frequency response of AD8002 cross-coupled driver

Due to the very high gain-bandwidths involved with the AD8002, the construction of this circuit should only be undertaken by following RF rules. This includes the use of a heavy ground plane, and the use chip bypass capacitors of zero lead length at the $\pm 5V$ supply pins. For lowest parasitic effect and low inductance, chip style resistors are also recommended for this circuit (see Reference 7). The optimization of C1 has already been noted, above. While a chip style NP0 is good in general for C1, a small film trimmer may also be useful, as it will allow optimizing peaking on an individual circuit basis.

Although this circuit example illustrates wideband a video driver, it should be noted that lower bandwidth applications could also find this push-pull topology useful. An audio frequency application could for example use an AD812 for U1A and U1B, or pair of AD811s. Operating on $\pm 15V$, these will allow a high level of balanced, linear output.

■ OP AMP APPLICATIONS

Fully Integrated Differential Drivers

A block diagram of the new AD813X family of fully differential amplifiers optimized for differential driving is shown in Figure 6-98. Figure 6-98A shows the details of the internal circuit, and Figure 6-98B shows the equivalent circuit. The gain is set by the external R_F and R_G resistors, and the common-mode voltage is set by the voltage on the V_{OCM} pin. The internal common-mode feedback forces the V_{OUT+} and V_{OUT-} outputs to be balanced, i.e., the signals at the two outputs are always equal in amplitude but 180° out of phase per the equation:

$$V_{OCM} = (V_{OUT+} + V_{OUT-}) / 2 . \quad \text{Eq. 6-26}$$

The circuit can be used with either a differential or a single-ended input, and the voltage gain is equal to the ratio of R_F to R_G .

The AD8138 has a 3dB small-signal bandwidth of 320MHz ($G = +1$) and is designed to give low harmonic distortion as an ADC driver (see Chapter 3). The circuit provides excellent output gain and phase matching, and the balanced structure suppresses even-order harmonics.

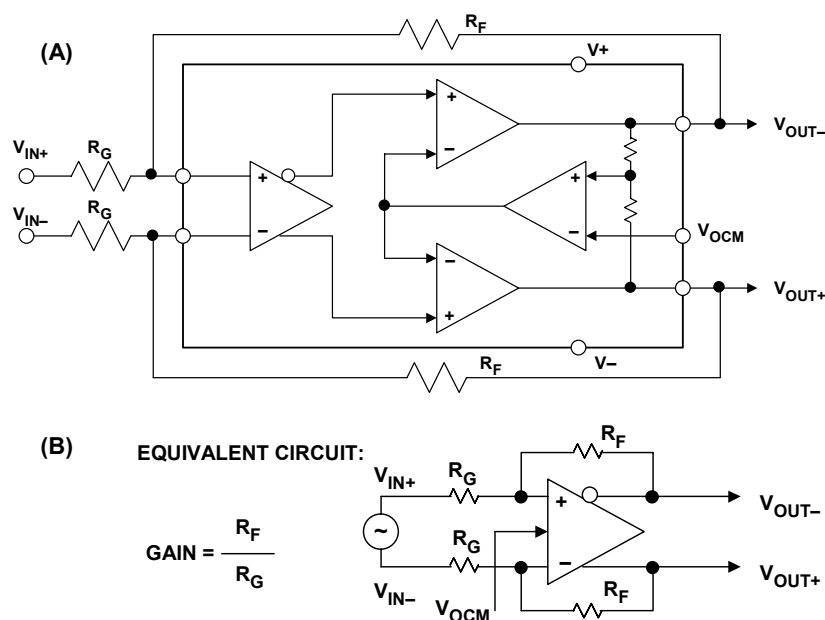


Figure 6-98: AD8138 differential driver amplifier functional schematic (A) and equivalent circuit (B)

It should be noted that the AD8131 differential driver is a sister device to the AD8138 in terms of the function illustrated in Fig. 6-98A, and includes internal gain-set resistors.

A 4-Resistor Differential Line Receiver

Figure 6-99 below shows a low cost, medium performance line receiver using a high speed op amp which is rated for video use. It is actually a standard 4 resistor difference amplifier optimized for high speed, with a differential to single-ended gain of R_2/R_1 . Using low value, DC-accurate, AC-trimmed resistances for R_1 - R_4 and a high speed, high CMR op amp provides the good performance.

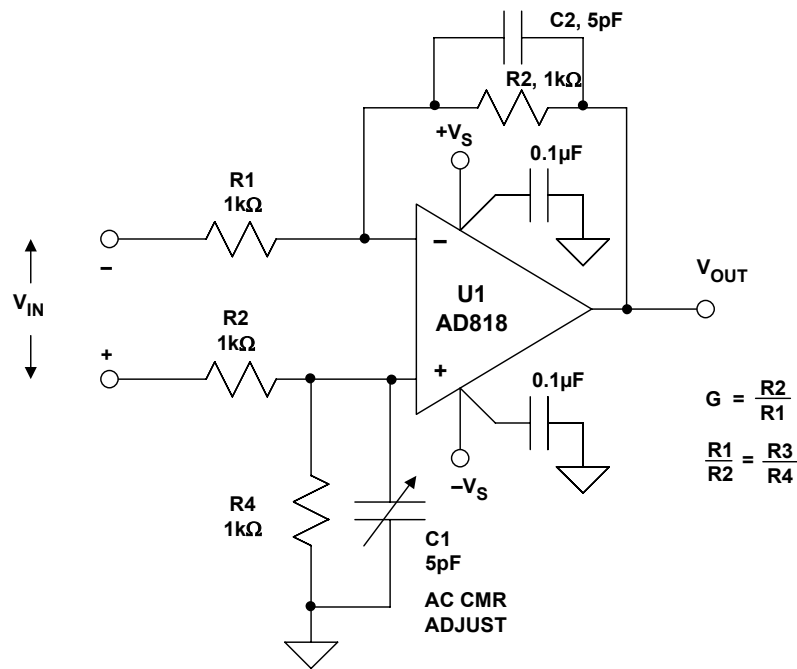


Figure 6-99: Simple video line receiver using the AD818 op amp

Practically speaking however, at low frequencies resistor matching can be more critical to overall CMR than the rated CMR of the op amp. For example, the worst case CMR (in dB) of this circuit due to resistor mismatch is:

$$\text{CMR} = 20 \log_{10} \left(\frac{1 + \frac{R_2}{R_1}}{4Kr} \right). \quad \text{Eq. 6-27}$$

In this expression the term "Kr" is a single resistor tolerance in fractional form (1%=0.01, etc.), and it is assumed the amplifier has significantly higher CMR ($\geq 100\text{dB}$). Using discrete 1% metal films for R_1/R_2 and R_3/R_4 yields a worst case CMR of 34dB, 0.1% types 54dB, etc. Of course 4 random 1% resistors will on the average yield a CMR better than 34dB, but not dramatically so. A single substrate dual matched pair thin film network is preferred, for reasons of best noise rejection and simplicity. One type suitable is the Vishay VTF series part 1005, (see Reference 7) which has a ratio match of 0.1%, which will provide a worst case low frequency CMR of 66dB.

■ OP AMP APPLICATIONS

This circuit has an interesting and desirable side property. Because of the resistors it divides down the input voltage, and the amplifier is protected against overvoltage. This allows CM voltages to exceed $\pm 5V$ supply rails in some cases without hazard. For operation with $\pm 15V$ supplies, inputs should not exceed the supply rails.

At frequencies above 1MHz, the bridge balance is dominated by AC effects, and a C1-C2 capacitive balance trim should be used for best performance. The C1 adjustment is intended to allow this, providing for the cancellation of stray layout capacitance(s) by electrically matching the net C1-C2 values.

Within a given PC layout with low and stable parasitic capacitance, C1 is best adjusted once in 0.5pF increments, for best high frequency CMR. Using designated PC pads, production values then would use the trimmed value. Good AC matching is essential to achieving good high frequency CMR. C1-C2 should be types similar physically, such as NPO ceramic chip capacitors.

While the circuit as shown has unity gain, it can be gain-scaled in discrete steps, as long as the noted resistor ratios are maintained. In practice, this means using taps on a multi-ratio network for gain change, so as to raise both R2 and R4, in identical proportions. There is no other simple way to change gain in this receiver circuit.

Alternately, a scheme for continuous gain control without interaction with CMR is to simply follow this receiver with a scaling amplifier/driver, with adjustable gain. The use of an AD828 amplifier (AD818 dual) allows this, with the addition of only two resistors.

Video gain/phase performance of this stage is dependent upon the device used for U1 and the operating supply voltages. Suitable voltage feedback amplifiers work best at supplies of ± 10 to $\pm 15V$, which maximizes op amp bandwidth. And, while many high speed amplifiers function in this circuit, those expressly designed with very low distortion video operation perform best.

The circuit just as shown can be used with supplies of ± 5 to $\pm 15V$, but lowest NTSC video distortion occurs for supplies of $\pm 10V$ or more using the AD818, where differential gain/differential phase errors are less than 0.01%/0.05°. With the AD818 operating at $\pm 5V$ supplies, the distortion rises somewhat, but the lowest power drain of 70mW occurs. If low distortion and lowest power operation on $\pm 5V$ is important, the use of an AD8055 (or AD8056 section) should be considered for the U1 function; they will dissipate 50mW.

A drawback to this circuit is that it does load a 75 Ω video line to some extent, and so should be used with this loading taken into account. On the plus side, it has wide dynamic range for both signal and CM voltages, plus the inherent overvoltage protection.

Active Feedback Differential Line Receiver

The AD8129/AD8130 differential line receivers, along with their predecessor the AD830, utilize a novel amplifier topology called *active feedback* (see Reference 8). A simplified block diagram of these devices is shown below in Figure 6-100.

The AD830 and the AD8129/AD8130 have two sets of fully differential inputs, available at V_{X1} - V_{X2} and V_{Y1} - V_{Y2} , respectively. Internally, the outputs of the two GM stages are summed and drive a buffer output stage.

In this device the overall feedback loop forces the internal currents I_X and I_Y to be equal. This condition forces the differential voltages V_{X1} - V_{X2} and V_{Y1} - V_{Y2} to be equal and opposite in polarity. Feedback is taken from the output back to one input differential pair, while the other pair is driven directly by an input differential input signal.

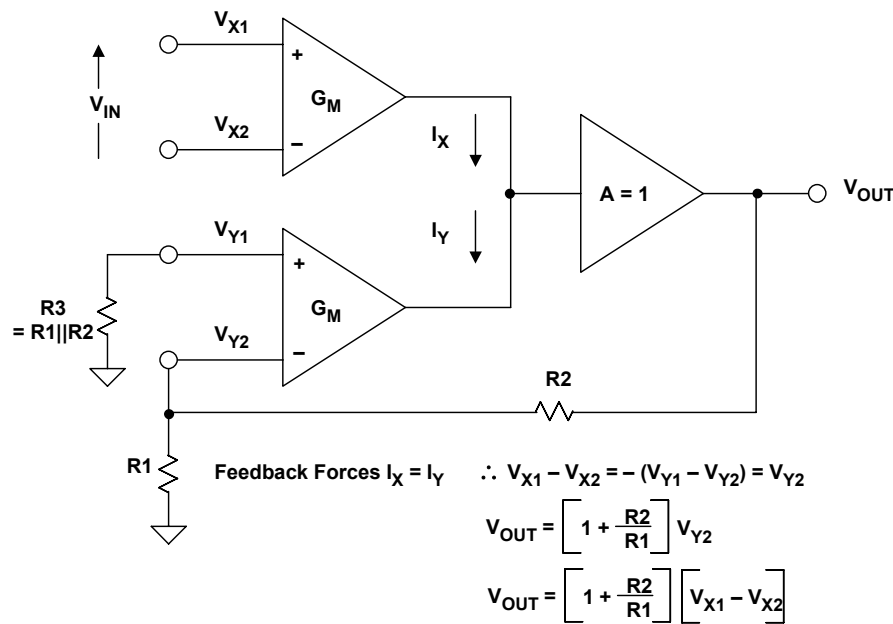


Figure 6-100: The AD830/AD8129/AD8130 active feedback amplifier topology

An important point of this architecture is that high CM rejection is provided by the two differential input pairs, so *CMR is not dependent on resistor bridges* and their associated matching problems. The inherently wideband balanced circuit and the quasi-floating operation of the driven input provide the high CMR, which is typically 100dB at DC.

The general expression for the stage's gain "G" is like a non-inverting op amp, or:

$$G = \frac{V_{OUT}}{V_{IN}} = 1 + \frac{R_2}{R_1} \quad \text{Eq. 6-28}$$

As should be noted, this expression is identical to the gain of a non-inverting op amp stage, with R_2 and R_1 in analogous positions.

■ OP AMP APPLICATIONS

The AD8129 is a low-noise high-gain ($G = 10$ or greater) version of this family, intended for applications with very long cables where signal attenuation is significant. The related AD8130 device is stable at a gain of one. It is used for those applications where lower gains are required, such as a gain-of-two, for driving source and load terminated cables.

The AD8129 and AD8130 have a wide power supply range, from single +5V to $\pm 12V$, allowing wide common-mode and differential-mode voltage ranges. The wide common-mode range enables the driver/receiver pair to operate without isolation transformers in many systems where the ground potential difference between driver and receiver locations is several volts. Both devices include a logic-controlled power-down function.

Both devices have high, balanced input impedances, and achieve 70dB CMR @ 10MHz, providing excellent rejection of high-frequency common-mode signals. Figure 6-101 below shows AD8130 CMR for various supplies. As can be noted, it can be as high as 95dB at 1MHz, an impressive figure considering that no trimming is required.

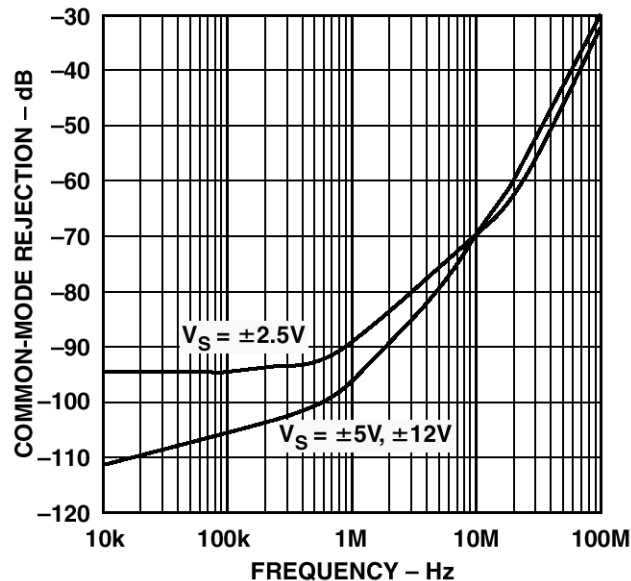


Figure 6-101: AD8130 common-mode rejection versus frequency for $\pm 2.5V$, $\pm 5V$, and $\pm 12V$ supplies

The typical 3dB bandwidth for the AD8129 is 200MHz, while the 0.1dB bandwidth is 30MHz in the SOIC package, and 50MHz in the μ SOIC package. The conditions for these specifications are for $V_S = \pm 5V$ and $G = 10$.

The typical 3dB bandwidth for the AD8130 is 270MHz, and the 0.1dB bandwidth is 45MHz, in either package. The conditions for these specifications are for $V_S = \pm 5V$ and $G = 1$. Typical differential gain and phase specifications for the AD8130 for $G = 2$, $V_S = \pm 5V$, and $R_L = 150\Omega$ are 0.13% and 0.15° , respectively.

A Cable-Tap or Loopthrough Amplifier

Figure 6-102 below shows an example of a video *cable-tap* amplifier (or *loopthrough*) connection where the input signal is tapped from a coax line and applied to one input stage of the AD8130, with the output signal tied to the second input stage. The net gain is unity. Functionally, the input and local grounds are isolated by the CMR of the AD8130, which is typically 70dB at 10MHz. Note that in order to provide a DC path for the input bias currents of the upper stage, there must be a common path between the source and local grounds (shown as Z_{CM}). This impedance is not critical, but must be low enough that 60Hz noise and other voltage components remain within the AD8130's CM range.

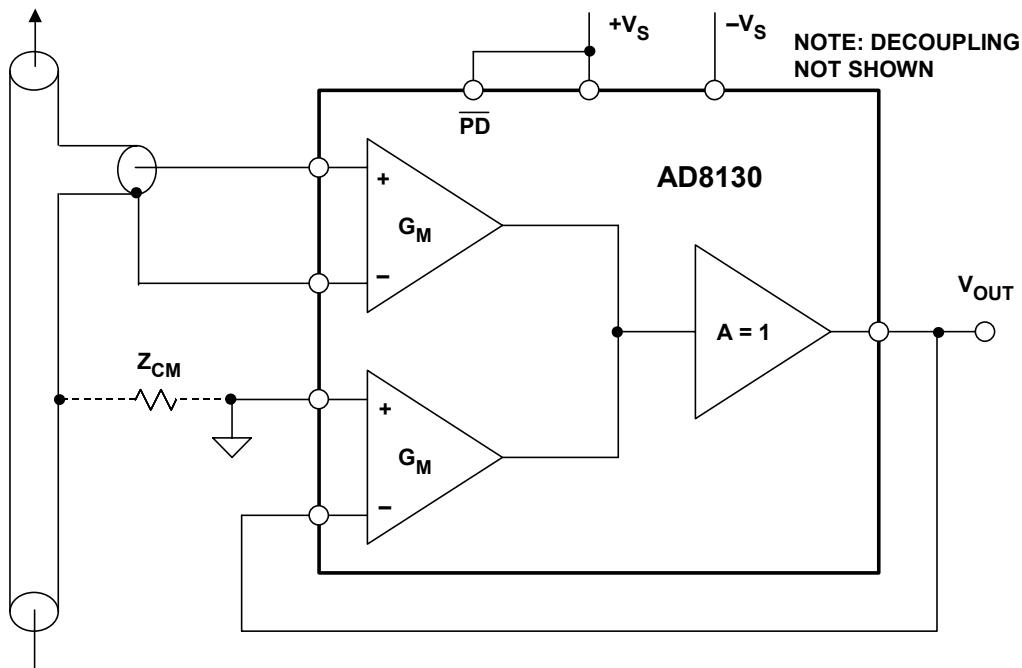


Figure 6-102: Video "cable-tap" amplifier using the AD8130

The circuit is efficient with the simplicity as shown, and requires no gain set resistors, etc. to implement. Normal bypass capacitors and supply decoupling must of course be used, as in any high-speed circuit. Other than the necessary DC path for the two inputs, it has little affect on the video cable it is monitoring, due to the high impedance AD8130 inputs. The circuit just as shown operates on supplies of $\pm 5V$ to $\pm 12V$, but a $\pm 15V$ version can also be implemented by using the AD830 (without the AD8130's power down function).

This circuit can also act as a video repeater, by connecting equal value feedforward and feedback resistors to implement a gain-of-two, for driving a source and load-terminated video cable (i.e., R_2 and R_1 , as in Fig. 6-100, again).

Further application examples of this family of active feedback amplifiers are contained in the "Amplifier Ideas" section of this chapter, plus of course the device data sheets.

High Speed Clamping Amplifiers

There are many situations where it is desirable to *clamp* the output of an op amp, to prevent overdriving following circuitry. Specially designed high speed, fast recovery clamping amplifiers offer an attractive alternative to designing external clamping/protection circuits. The AD8036/AD8037 low distortion, wide bandwidth clamp amplifiers represent a significant breakthrough in this technology. These devices allow the designer to specify a high (V_H) and low (V_L) clamp voltage. The output of the device clamps when the input exceeds either of these two levels. The AD8036/AD8037 offer superior clamping performance compared to competing devices that use output-clamping. Recovery time from overdrive is less than 5ns, and small signal bandwidth is 240MHz (AD8036) and 270MHz (AD8037).

The key to the AD8036 and AD8037's fast, accurate clamp and amplifier performance is their proprietary input clamp architecture. This new design reduces clamp errors by more than 10x over previous output clamp based circuits, as well as substantially increasing the bandwidth, precision, and versatility of the clamp inputs.

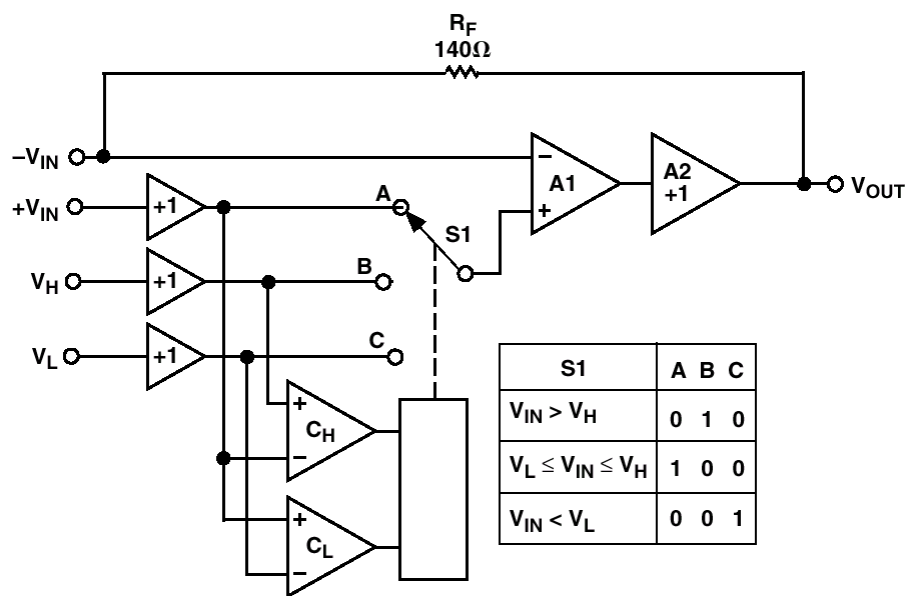


Figure 6-103: AD8036/AD8037 clamp amplifier equivalent circuit

Figure 6-103 above is an idealized block diagram of the AD8036 clamp amplifier, connected as a unity gain voltage follower. The primary signal path comprises A1 (a 1200V/ μ s, 240MHz high voltage gain, differential to single-ended amplifier) and A2 (a $G=+1$ high current gain output buffer). The AD8037 differs from the AD8036 only in that A1 is optimized for closed-loop gains of two or greater.

The input clamp section is comprised of comparators C_H and C_L , which drive switch S1 through a decoder. The unity-gain buffers before the $+V_{IN}$, V_H , and V_L inputs isolate the input pins from the comparators and S1, without reducing bandwidth or precision. The two comparators have about the same bandwidth as A1 (240MHz), so they can keep up with signals within the useful bandwidth of the AD8036. To illustrate the operation of the

input clamp circuit, consider the case where V_H is referenced to +1V, V_L is open, and the AD8036 is set for a gain of +1 by connecting its output back to its inverting input through the recommended 140 Ω feedback resistor. Note that the main signal path always operates closed loop, since the clamping circuit only affects A1's non-inverting input.

If a 0V to +2V voltage ramp is applied to the AD8036's $+V_{IN}$ for the connection, V_{OUT} should track $+V_{IN}$ perfectly up to +1V, then limit at exactly +1V as $+V_{IN}$ continues to +2V. In practice, the AD8036 comes close to this ideal behavior. As the $+V_{IN}$ input voltage ramps from zero to 1V, the output of the high limit comparator C_H starts in the off state, as does the output of C_L . When $+V_{IN}$ just exceeds V_H (practically, by about 18mV), C_H changes state, switching S1 from "A" to "B" reference level. Since the + input of A1 is now connected to V_H , further increases in $+V_{IN}$ have no effect on the AD8036's output. The AD8036 is now operating as a unity-gain buffer for the V_H input, as any variation in V_H , for $V_H > 1V$, will be faithfully produced at V_{OUT} .

AD8036 operation for negative inputs and negative V_L clamp levels is similar, with comparator C_L controlling S1. Since the comparators see the voltage on the $+V_{IN}$ pin as their common reference level, the voltage V_H and V_L are defined as "High" or "Low" with respect to $+V_{IN}$. For example, if V_{IN} is zero volts, V_H is open, and V_L is +1V, comparator C_L will switch S1 to "C", and the AD8036 will buffer the V_L voltage and ignore $+V_{IN}$.

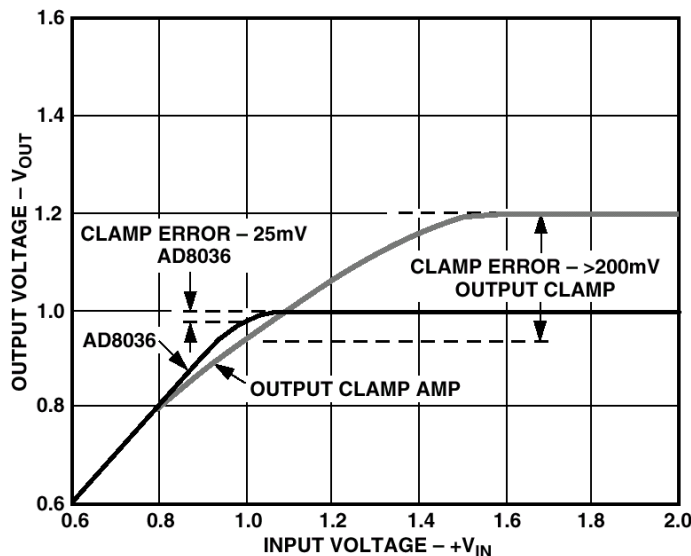


Figure 6-104: Comparison between input and output clamping

The AD8036/AD8037 performance closely matches the described ideal. The comparator's threshold extends from 60mV inside the clamp window defined by the voltages on V_L and V_H to 60mV beyond the window's edge. Switch S1 is implemented with current steering, so that A1's + input makes a continuous transition from say, V_{IN} to V_H as the input voltage traverses the comparator's input threshold from 0.9V to 1.0V for $V_H = 1.0V$.

The practical effect of the non-ideal operation softens the transition from amplification to clamping modes, without compromising the absolute clamp limit set by the input clamping circuit. Figure 6-104 above is a graph of V_{OUT} versus V_{IN} for the AD8036 and a typical *output clamp* amplifier. Both amplifiers are set for $G=+1$ and $V_H = +1V$. The

► OP AMP APPLICATIONS

worst case error between V_{OUT} (ideally clamped) and V_{OUT} (actual) is typically 18mV times the amplifier closed-loop gain. This occurs when V_{IN} equals V_H (or V_L). As V_{IN} goes above and/or below this limit, V_{OUT} will stay within 5mV of the ideal value.

In contrast, the output clamp amplifier's transfer curve typically will show some compression starting at an input of 0.8V, and can have an output voltage as far as 200mV over the clamp limit. In addition, since the output clamp causes the amplifier to operate open-loop in the clamp mode, the amplifier's output impedance will increase, potentially causing additional errors, and the recovery time is significantly longer.

Flash Converter with Clamp Amp Input Protection

Figure 6-105 below shows the AD9002 8-bit, 125MSPS flash converter driven by the AD8037 (240MHz bandwidth) clamping amplifier. The clamp voltages on the AD8037 are set to +0.55 and -0.55V, referenced to the $\pm 0.5V$ input signal, with the twin 806 Ω /100 Ω external resistive dividers. The AD8037 also supplies a gain of two, and an offset of -1V (using the AD780 voltage reference), to match the 0 to -2V input range of the AD9002 flash converter. The output signal is clamped at +0.1V and -2.1V.

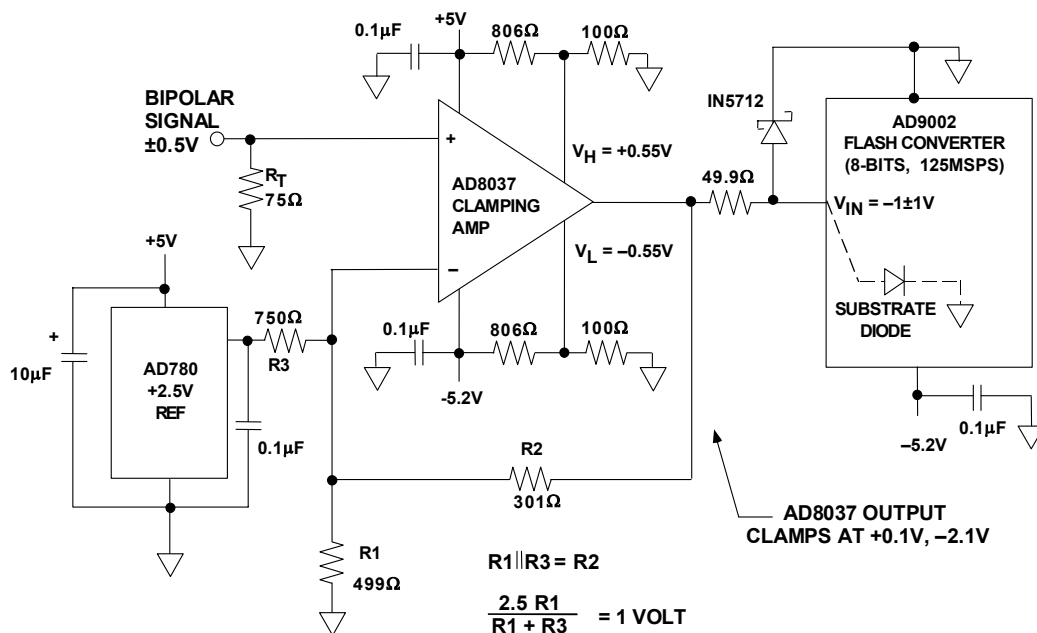


Figure 6-105: AD9002 8-bit, 125MSPS flash converter driven by AD8037 clamp amplifier

This multi-function clamping circuit therefore performs several important functions as well as preventing damage to the flash converter (which would otherwise occur should the input exceed +0.5V, thereby forward biasing the substrate diode). The 1N5712 Schottky diode is a safety-valve device, adding further protection for the flash converter during power-up.

There are multiple criteria that must be met in designing the feedback network around the AD8037. These are a specified gain, and a fixed offset which will enable the output swing of the clamped amplifier to match the target input range of the converter.

The feedback resistor, $R2 = 301\Omega$, is selected for optimum bandwidth per the data sheet recommendation. For a gain of two, the parallel combination of $R1$ and $R3$ must also equal $R2$:

$$\frac{R1 \cdot R3}{R1 + R3} = R2 = 301\Omega , \quad \text{Eq. 6-29}$$

(nearest 1% standard resistor value).

In addition, the Thevenin equivalent output voltage from the AD780 +2.5V reference and the $R3/R1$ divider must be +1V, to provide the required -1V offset at the output of the AD8037. This will cause the output swing of the AD8037 to be biased at -1V when V_{IN} is zero, and to range from 0 to -2V as V_{IN} ranges from -0.5 to 0.5V.

$$\frac{2.5 \cdot R1}{R1 + R3} = 1\text{volt} . \quad \text{Eq. 6-30}$$

Solving these equations yields resistance values of $R1 = 499\Omega$, $R3 = 750\Omega$, using the nearest 1% standard values.

Other input and output voltages ranges can also be accommodated, by appropriate changes in the external resistors.

Further fast clamping op amp application examples are given in Reference 9, and the "Amplifier Ideas" section of this chapter (plus of course the device data sheets).

High Speed Video Multiplexing with Op Amps Utilizing Disable Function

A common video circuit function is the multiplexer, a stage which selects one of "n" video inputs and transmits a buffered version of the selected signal to the output. A number of video op amps (AD810, AD813, AD8013, AD8074/AD8075) have a *disable* mode which, when activated by applying the appropriate control level to a pin on the package, disables the op amp output stage and drops the power to a lower value.

In the case of the AD8013 (triple current-feedback op amp), asserting any one of the disable pins about 1.6V from the negative supply will put the corresponding amplifier into a disabled, powered-down state. In this condition, the amplifier's quiescent current drops to about 0.3mA, its output becomes a high impedance, and there is a high level of isolation from the input to the output. In the case of the gain-of-two line driver, for example, the impedance at the output node will be about equal to the sum of the feedback and feedforward resistors (1.6k Ω) in parallel with about 12pF capacitance. Input-to-output isolation is about 66dB at 5MHz.

Leaving the disable pin disconnected (floating) will leave the corresponding amplifier operational (i.e., enabled). The input impedance of the disable pin is about 40k Ω in parallel with 5pF. When driven to 0V, with the negative supply at $-5V$, about 100 μA flows into the disable pin.

When the disable pins are driven by CMOS logic, on a single +5V supply, the disable and enable times are about 50ns. When operated on dual supplies, level shifting will be required from standard logic outputs to the disable pins.

The AD8013's input stages include protection from the large differential voltages that may be applied when disabled. Internal clamps limit this voltage to about $\pm 3V$. The high input-to-output isolation will be maintained for voltages below this limit.

Wiring the amplifier outputs together as shown in Figure 6-106 (opposite) forms a 3:1 multiplexer with about 50ns switching time between channels. The 0.1dB bandwidth of the circuit is 35MHz, and the OFF channel isolation is 60dB at 10MHz. The simple logic level-shifting circuit shown on the diagram does not significantly affect switching time.

Setting up this amplifier is not entirely straightforward, and some explanation will help with subtleness. The feedback resistor R2 of 845 Ω was chosen first, to allow optimum bandwidth of the AD8013 current feedback op amp. The analogous resistors of the other channels use an identical value, for similar reasons.

Note that when any given channel is ON, it must drive both the termination resistor R_L , and the net dummy resistance, $R_X/2$, where R_X is an equivalent series resistance equal to $R_1 + R_2 + R_3$. To provide a net overall gain of unity, as well as an effective source resistance of 75 Ω , the other resistor values must be as shown. In essence, the Thevenin

equivalent value of $R_X/2$ and R_3 should equal the desired source termination impedance of 75Ω (which it does).

It is also desirable that the ON channels have a net gain of $2x$, as seen behind the 75Ω output impedance. The lower value of R_1 vis-à-vis R_2 , along with the above relationship, allows these mutual criteria to be met.

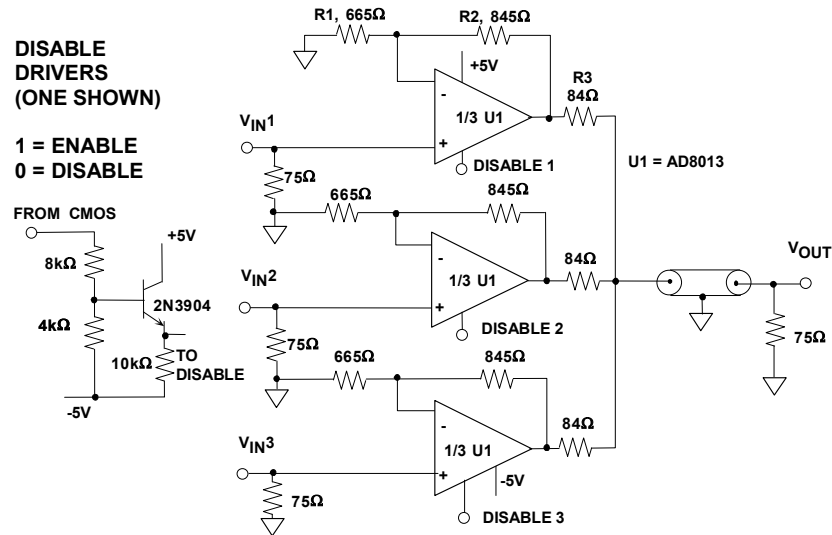


Figure 6-106: AD8013 3:1 video multiplexer switches in 50ns

Configuring two amplifiers of an AD8013 as unity gain followers with the third to set the gain results in a high performance 2:1 multiplexer, as shown in Figure 6-107 below.

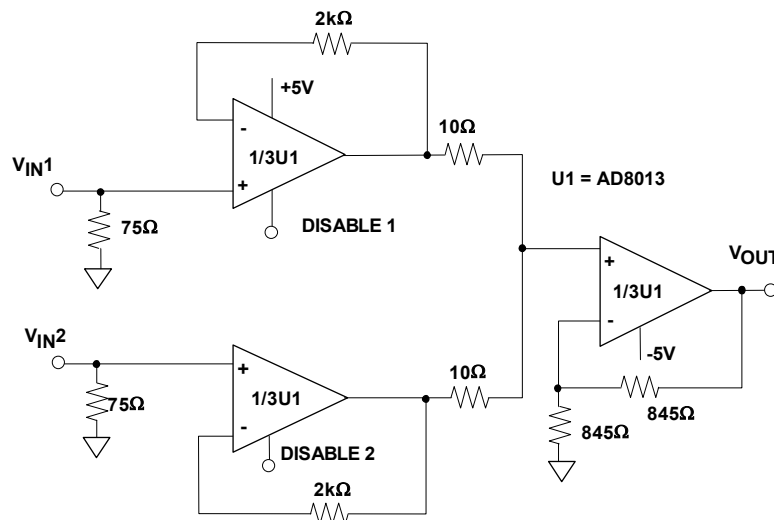


Figure 6-107: 2:1 video multiplexer based on the AD8013

This circuit takes advantage of the low crosstalk between the amplifiers, and achieves an OFF channel isolation of 50dB at 10MHz. The differential gain and phase performance of the circuit is 0.03% and 0.07° , respectively. The output stage operates at a gain of $2x$, and can drive a 75Ω source terminated line if desired.

■ OP AMP APPLICATIONS

Programmable Gain Amplifier using the AD813 Current Feedback Video Op Amp

Closely related to the multiplexers described above is a programmable gain video amplifier, or PGA, as shown in Figure 6-108 below. In the case of the AD813, the individual channels are disabled by pulling the disable pin about 2.5V below the positive supply. This puts the corresponding amplifier in its powered down state. In this condition, the amplifier's quiescent supply current drops to about 0.5mA, its output becomes a high impedance, and there is a high level of isolation between the input and the output. Leaving the disable pin disconnected (floating) will leave the amplifier operational, in the enabled state. When grounded, about 50 μ A flows out of a disable pin when operating on $\pm$ 5V supplies. The switching threshold is such that the disable pins can be driven directly from +5V CMOS logic as shown, with no level shifting (as in the previous example).

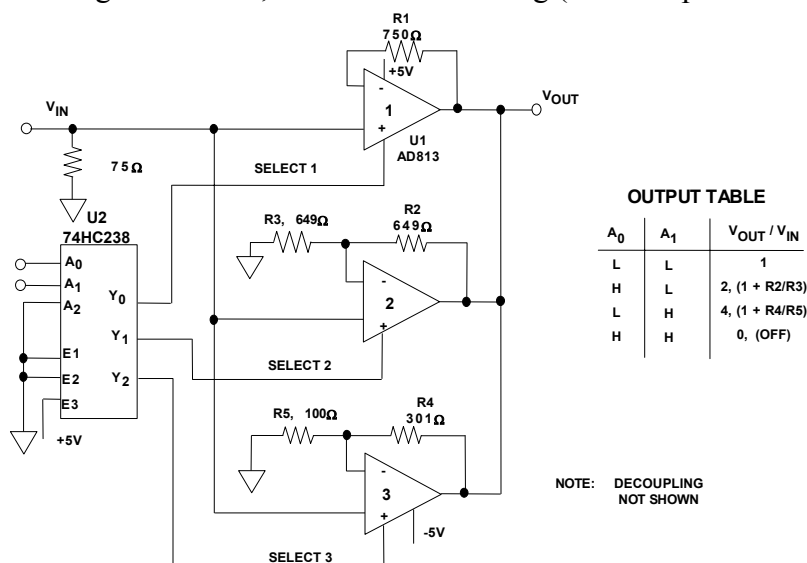


Figure 6-108: Programmable gain video amplifier using the AD813 triple current feedback amplifier

With a two-line digital control input, this circuit can be set up to provide 3 different gain settings. This makes it a useful circuit in various systems which can employ signal normalization or gain ranging prior to A/D conversion, such as CCD systems, ultrasound, etc. The gains can be binary related as here, or they can be arbitrary. An extremely useful feature of the AD813 CFB current feedback amplifier is the fact that the bandwidth does not reduce as gain is increased. Instead, it stays relatively constant as gain is raised. Thus, more useful bandwidth is available at the higher programmed gains than would be true for a fixed gain-bandwidth product VFB amplifier type.

In the circuit, channel 1 of the AD813 is a unity gain channel, channel 2 has a gain of 2, and channel 3 a gain of 4, while the fourth control state is OFF. As is indicated by the table, these gains can easily be varied by adjustment of the R₂/R₃ or R₄/R₅ ratios. For the gain range and values shown, the PGA will be able to maintain a 3dB bandwidth of about 50MHz or more for loading as shown (a high impedance load of 1k Ω or more is assumed). Fine tuning the bandwidth for a given gain setting can be accomplished by lowering the resistor values at the higher gains, as shown in the circuit, where for G=1, R₁=750 Ω , for G=2, R₂=649 Ω , and for G=4, R₄=301 Ω .

Integrated Video Multiplexers and Crosspoint Switches

Traditional CMOS switches and multiplexers suffer from several disadvantages at video frequencies. Their switching time (typically 100ns or so) is not fast enough for today's applications, and they require external buffering in order to drive typical video loads. In addition, the small variation of the CMOS switch "on" resistance with signal level (R_{on} modulation) introduces unwanted distortion in differential gain and phase. Multiplexers based on complementary bipolar technology offer a better solution at video frequencies.

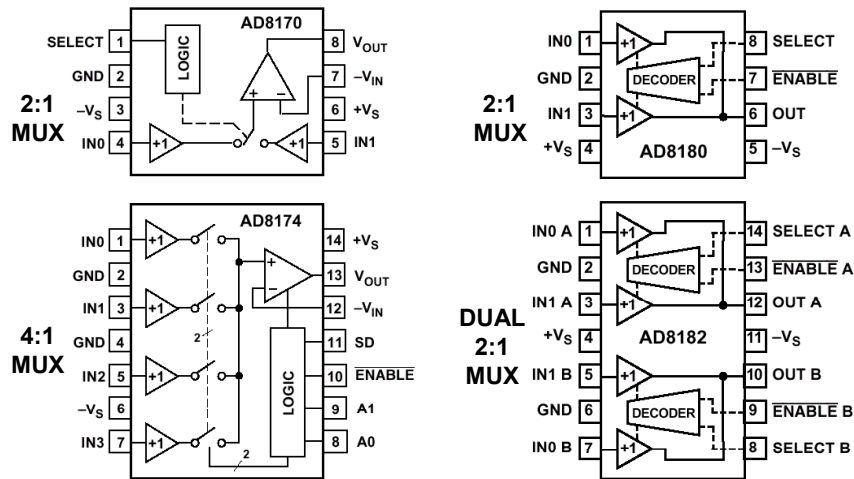


Figure 6-109: AD8170/8174/8180/8182 bipolar video multiplexers

Functional block diagrams of the AD8170/8174/8180/8182 bipolar video multiplexer are shown in Figure 6-109 above. The AD8183/AD8185 video multiplexer is shown in Figure 6-110, below. These devices offer a high degree of flexibility and are ideally suited to video applications, with excellent differential gain and phase specifications. Switching time for all devices in the family is 10ns to 0.1%.

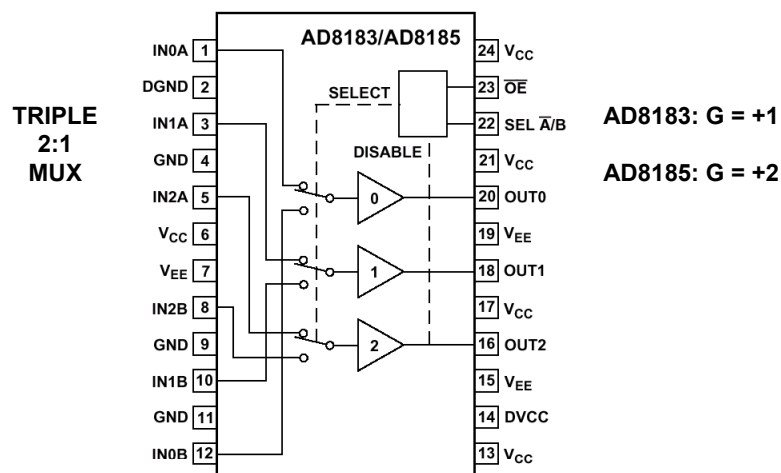


Figure 6-110: AD8183/AD8185 triple 2:1 video multiplexers

The AD8170/8174 series of muxes include an on-chip current feedback op amp output buffer whose gain can be set externally. Off channel isolation and crosstalk are typically greater than 80dB for the entire family.

■ OP AMP APPLICATIONS

Dual RGB Source Video Multiplexer

Figure 6-111 below shows an application circuit for three AD8170 2:1 muxes, where a single RGB monitor is switched between two RGB computer video sources.

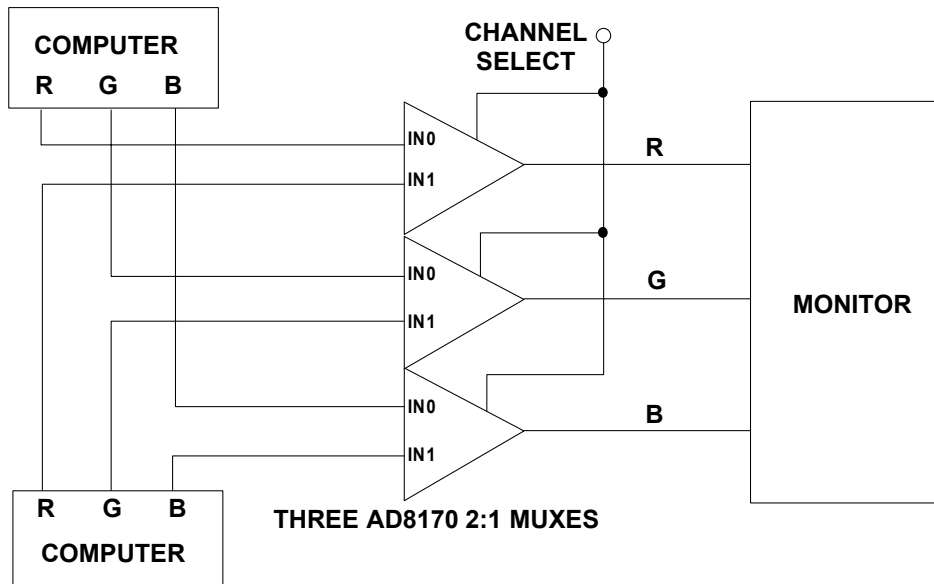


Figure 6-111: Dual source RGB multiplexer using three 2:1 muxes

In this setup, the overall effect is that of a three-pole, double-throw switch. The three video sources constitute the three poles, and either the upper or lower of the video sources constitute the two switch states.

Digitizing RGB Signals With One ADC

The AD8174 4:1 mux is used in Figure 6-112 below, to allow a single high speed ADC to digitize the RGB outputs of a scanner.

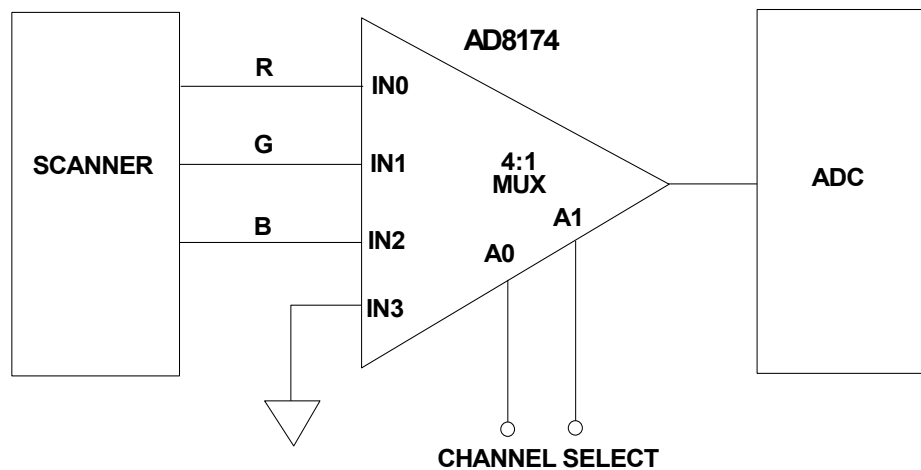


Figure 6-112: Digitizing RGB signals with one ADC and a 4:1 mux

The RGB video signals from the scanner are fed in sequence to the ADC, and digitized in sequence, making efficient use of the scanner data with one ADC.

Figure 6-113 below shows two AD8174 4:1 muxes functionally expanded into an 8:1 multiplexer. The A0 and A1 inputs are conventional, with complemented Enable inputs.

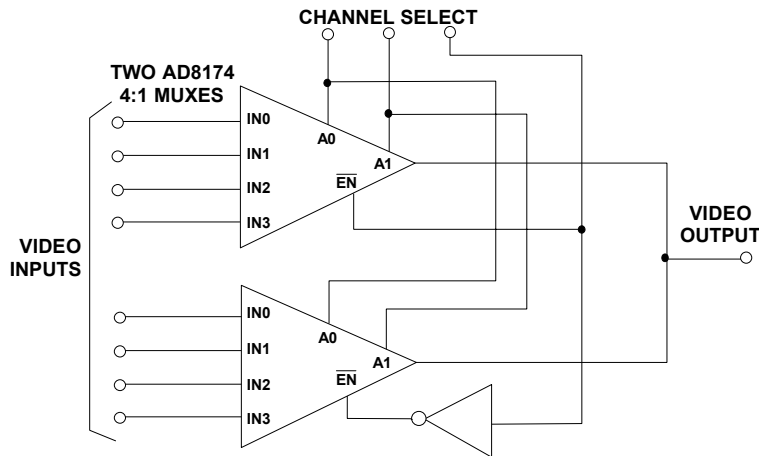


Figure 6-113: Expanding two 4:1 muxes into an 8:1 mux

The AD8116 extends the mux concepts to a fully integrated, 16×16 buffered video crosspoint switch matrix (Figure 6-114). The 3dB bandwidth is greater than 200MHz, and the 0.1dB gain flatness extends to 60MHz. Channel switching time is less than 30ns to 0.1%. Channel-to-channel crosstalk is -70dB measured at 5MHz. Differential gain and phase is 0.01% and 0.01° for a 150Ω load. Total power dissipation is 900mW on ±5V.

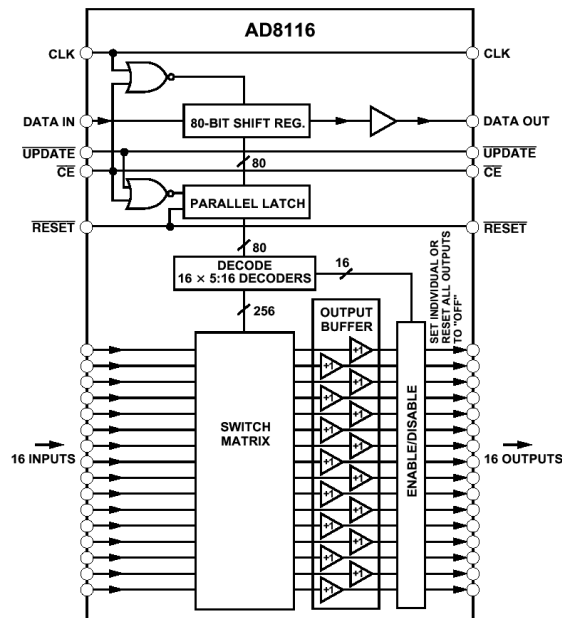


Figure 6-114: AD8116 16×16 200MHz buffered video crosspoint switch

The AD8116 includes output buffers that can be put into a high impedance state for paralleling crosspoint stages so that the off channels do not load the output bus. The channel switching is performed via a serial digital control that can accommodate "daisy chaining" of several devices. The AD8116 package is a 128-pin 14mm×14mm LQFP. Other members of the crosspoint switch family include the AD8110/AD8111 260MHz 16×8 buffered crosspoint switch, the AD8113 audio/video 60MHz 16×16 crosspoint switch, and the AD8114/AD8115 low cost 225MHz 16×16 crosspoint switch.

Single Supply Video Applications

Optimum video performance in terms of differential gain and phase, bandwidth flatness, etc., is generally achieved using dual supplies of $\pm 5\text{V}$ or $\pm 12\text{V}$. In many applications, however, stringent broadcast standards are not required, and single-supply operation may be desirable from a cost and power standpoint. This section illustrates a few op amp single-supply applications. All of the op amps are fully specified for both $\pm 5\text{V}$ and $+5\text{V}$ (and $+3\text{V}$ where the design supports it). Both rail-to-rail and non-rail-to-rail applications are shown (details of rail-to-rail op amp topologies are discussed in Chapter 1).

Single-Supply RGB Buffer

Op amps such as the AD8041/AD8042/ and AD8044 can provide buffering of RGB signals that include ground, while operating from a single $+3\text{V}$ or $+5\text{V}$ supply. The signals that drive an RGB monitor are usually supplied by current output DACs that operate from a single $+5\text{V}$ supply. Examples are triple video DACs such as the ADV7120/21/22 from Analog Devices.

During the horizontal blanking interval, the current output of the DACs goes to zero, and the RGB signals are pulled to ground by the termination resistors. If more than one RGB monitor is desired, it cannot simply be connected in parallel because this would be a mis-termination. Therefore, buffering must be provided before connecting a second monitor.

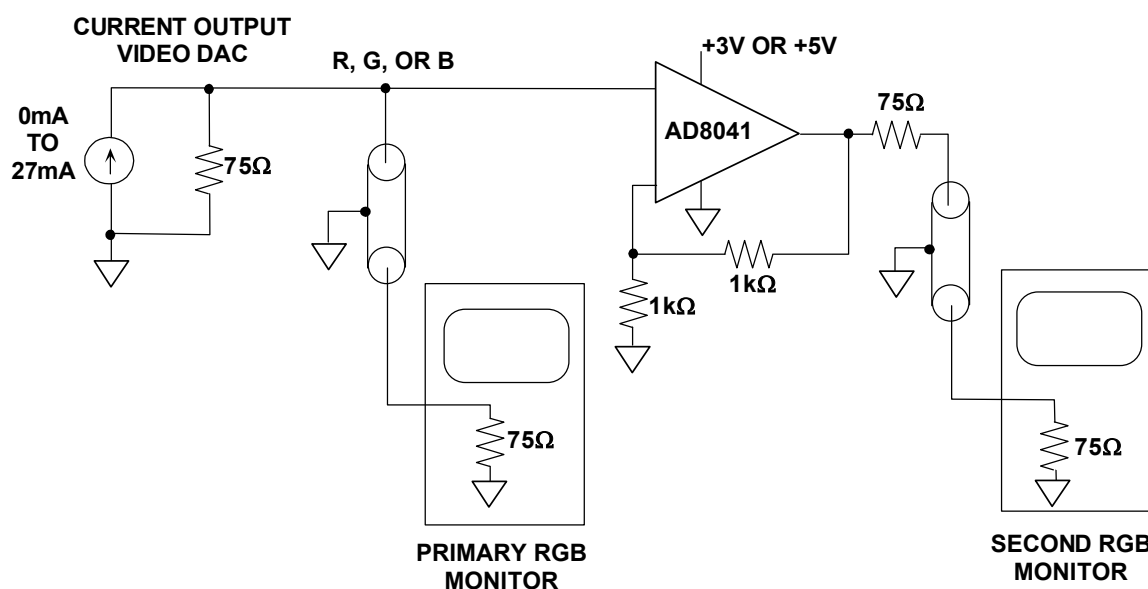


Figure 6-115: Single-supply RGB buffer operates on $+3\text{V}$ or $+5\text{V}$

RGB signals include ground as part of their dynamic output range. Previously a dual supply op amp has been required for this buffering, with sometimes this being the only component requiring a negative supply. This makes it quite inconvenient to incorporate a multiple monitor feature. Figure 6-115 shows a diagram of one channel of a single supply op amp gain-of-two buffer, for driving a second RGB monitor. No current is required when the amplifier output is at ground. The termination resistor at the monitor helps pull the output down at low voltage levels.

Note that the input and output are at ground during the horizontal blanking interval. The RGB signals are specified to output a maximum of 700mV peak. The peak output of the AD8041 is +1.4V, with the termination resistors providing a divide-by-two. All three channels (RGB) signals can be buffered in a like manner with duplication of this circuit. Another possibility is to use three sections of the (similar) quad AD8044 op amp.

Single-Supply Sync Stripper

Some RGB monitors use only three cables total and carry the synchronizing signals and the Green (G) signal on the same cable (Green-with-sync). The sync signals are pulses that go in the negative direction from the blanking level of the G signal.

In some applications, for example prior to digitizing component video signals with ADCs, it is desirable to remove or strip the sync portion from the G signal. Figure 6-116 is a circuit using the AD8041 running on a single +5V supply to perform this function. The signal at V_{IN} is the Green-with-sync signal from an ADV7120, a single supply triple video DAC.

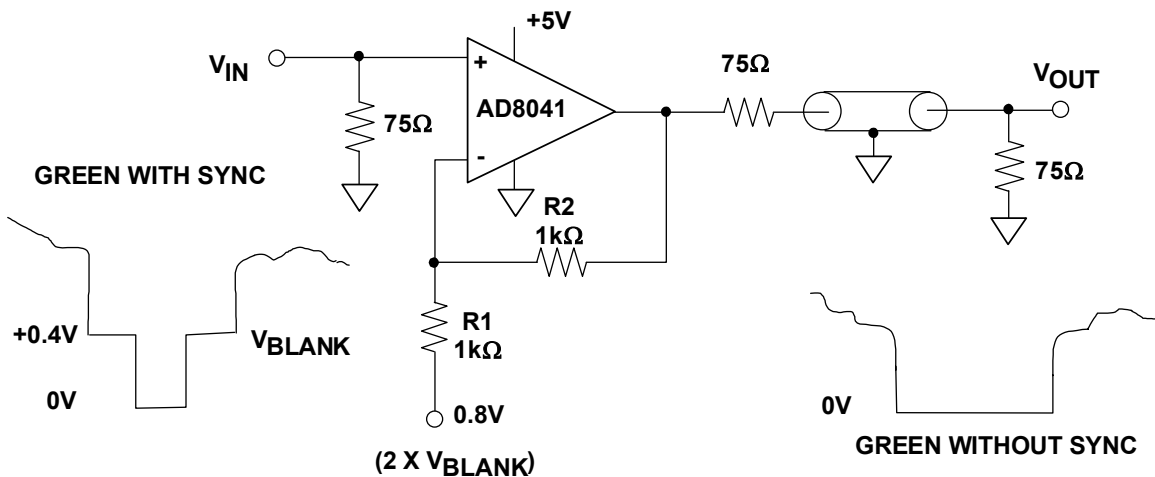


Figure 6-116: Single-supply video sync stripper

Because of the fact that the DAC used is single supply, the lowest level of the sync tip is at ground or slightly above. The AD8041 is set for a gain of two to compensate for the divide-by-two of the output terminations.

In this setup, the op amp used must have a CM capability that includes zero (as is true for the AD8041 family). For voltages *above* $\frac{1}{2}$ the 0.8V reference level applied to R1, the op amp operates as a linear amplifier, going positive from ground level at the output. For inputs *below* the reference level, the op amp saturates, with the output going to ground as used here. The result is that the negative sync tips are removed.

The reference voltage for R1 is twice the DC blanking level of the G signal; normally, this is $2 \times 0.4V = 0.8V$. Alternately, if the blanking level is at ground and the sync tip is negative (as in some dual supply systems), then R1 is tied to ground. The resulting V_{OUT} will have the sync removed, and the blanking level at ground, as noted.

■ OP AMP APPLICATIONS

A Low Distortion, Single-Supply Video Line Driver with Zero-Volt Output

When operated with a single supply, the AD8031 80MHz rail-to-rail voltage feedback op amp has optimum distortion performance when the signal has a common mode level of $V_S/2$, and when there is also about 500mV of headroom to each rail. If this rule is violated, distortion performance suffers. But, if low distortion is required for signals close to ground, a level-shifting emitter follower can be used at the op amp output.

Figure 6-117 below shows an AD8031 op amp, configured as a single supply gain-of-two line driver. With the output driving a back terminated 50 Ω line, the overall gain is unity from V_{IN} to V_{OUT} . In addition to minimizing reflections, the 50 Ω back termination resistor protects the transistor from damage if the cable is short circuited.

The 2N3904 emitter follower inside the feedback loop ensures that the output voltage from the AD8031 always stays about 700mV (or more) above ground, which minimizes distortion. Using this circuit excellent distortion is obtained, even when the output signal swings to within 50mV of ground.

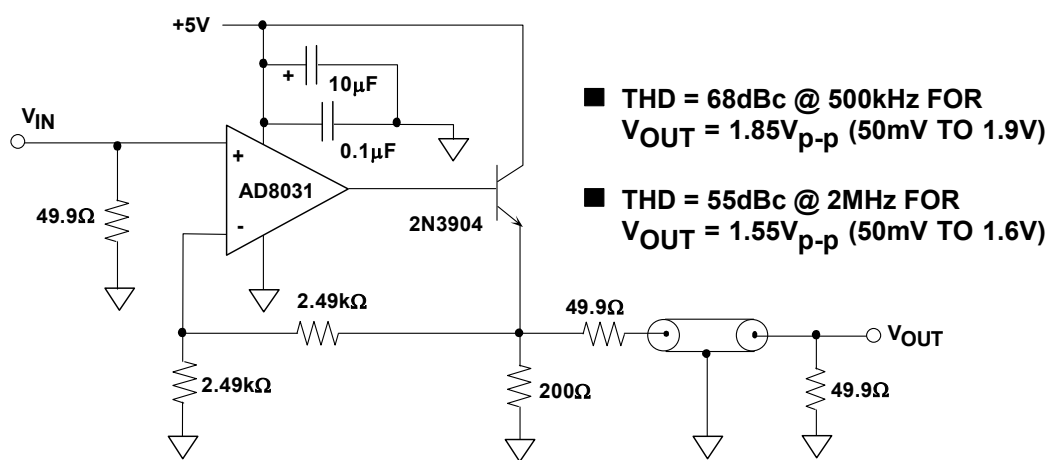


Figure 6-117: Low distortion zero-volt output single-supply line driver using the AD8031

The circuit was tested at 500kHz and 2MHz using a single +5V supply. For the 500kHz signal, THD was 68dBc with a peak-to-peak swing at a V_{OUT} of 1.85V (50mV to +1.9V). This corresponds to a signal at the emitter follower output of 3.7Vp-p (100mV to 3.8V). Data was taken with an output signal of 2MHz, and a THD of 55dBc was measured with a V_{OUT} of 1.55Vp-p (50mV to 1.6V).

This circuit can also be used to drive the analog input of a single supply high speed ADC whose input voltage range is ground-referenced. In this case, the emitter of the external transistor is connected to the ADC input, and the termination resistor is deleted. In this case, peak positive voltage swings of approximately 3.8V are possible before significant distortion begins to occur.

Headroom Considerations in AC-Coupled Single-Supply Video Circuits

The AC coupling of arbitrary waveforms can actually introduce problems that don't exist at all in DC coupled or DC restored systems. These problems have to do with the waveform duty cycle, and are particularly acute with signals that approach the rails, as they can in AC coupled, low supply voltage systems.

In Figure 6-118(A), an example of a 50% duty cycle square wave of about 2Vp-p level is shown, with the signal swing biased symmetrically between the upper and lower clip points of a 5V supply amplifier. Assume that the amplifier has a complementary emitter follower output and can only swing to the limited DC levels as marked, about 1V from either rail. In cases (B) and (C), the duty cycle of the input waveform is adjusted to both low and high duty cycle extremes *while maintaining the same peak-to-peak input level*. At the amplifier output, the waveform is seen to clip either negative or positive, in (B) and (C), respectively.

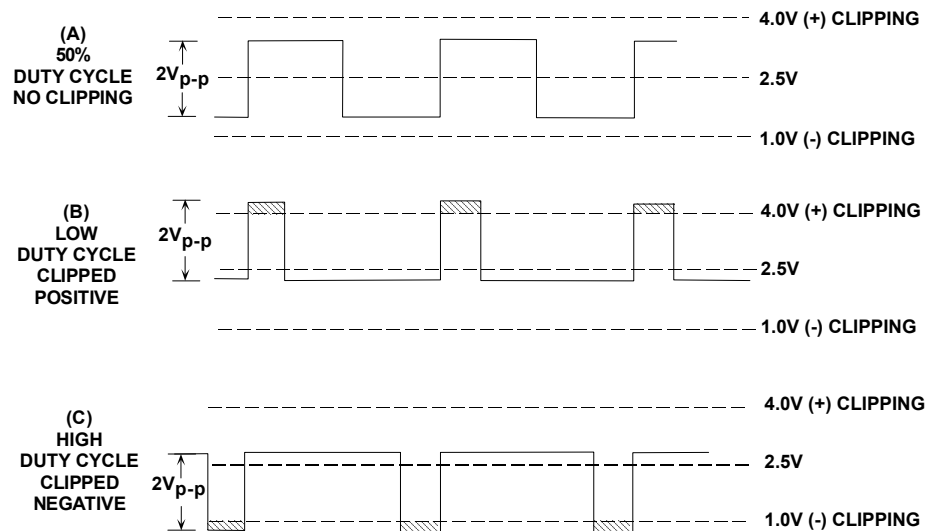


Figure 6-118: Waveform duty cycle taxes headroom in AC coupled single-supply op amps

Since standard video waveforms *do* vary in duty cycle as the scene changes, the point is made that low distortion operation on AC coupled single supply stages must take the duty cycle headroom degradation effect into account. If a stage has a 3Vp-p output swing available before clipping, and it must cleanly reproduce an *arbitrary* waveform, then the maximum allowable amplitude is less than $\frac{1}{2}$ this 3Vp-p swing, that is <1.5Vp-p.

An example of violating this criteria are the 2Vp-p waveforms of Figure 6-118(B) and (C), which clip for both the low and high duty cycles. Note that the criteria set down above is based on avoiding hard clipping, while subtle distortion increases may in fact take place at lower levels. This suggests an even more conservative criterion for lowest distortion operation, such as in composite NTSC video amplifiers.

■ OP AMP APPLICATIONS

Single-supply AC coupled composite video line driver

Figure 6-119 shows a single supply gain-of-two composite video line driver using the AD8041. Since the sync tips of a composite video signal extend below ground, the input must be AC coupled and shifted positively to prevent clipping during negative excursions. The input is terminated in 75Ω and AC coupled via the $47\mu\text{F}$ to a voltage divider that provides the DC bias point to the input. Setting the optimal common-mode bias voltage requires some understanding of the nature of composite video signals and the video performance of the AD8041.

As discussed above, signals of bounded peak-to-peak amplitude that vary in duty cycle require larger dynamic swing capability than their peak-to-peak amplitude after AC coupling. As a worst case, the dynamic signal swing required will approach twice the peak-to-peak value. The two bounding cases are for a duty cycle that is mostly low, but occasionally goes high at a fraction of a percent duty cycle, and vice versa.

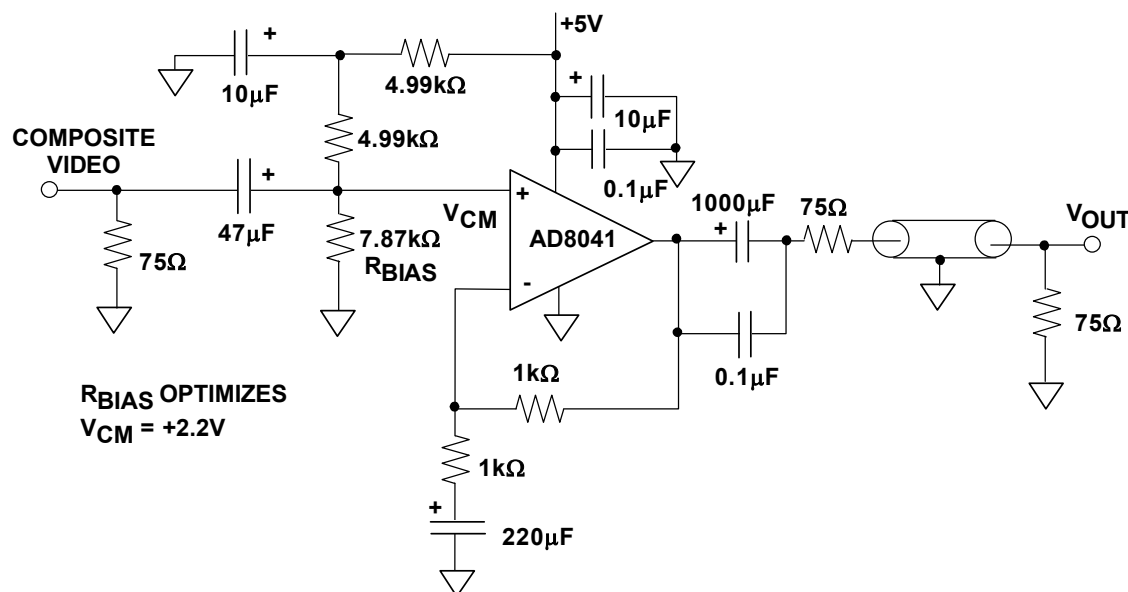


Figure 6-119: Single-supply AC coupled composite video line driver has $\Delta G = 0.06\%$ and $\Delta\phi = 0.06^\circ$

Composite video is not quite this demanding. One bounding extreme is for a signal that is mostly black for an entire frame, but occasionally has a white (full intensity), minimum width spike at least once per frame.

The other extreme is for a video signal that is full white everywhere. The blanking intervals and sync tips of such a signal will have negative going excursions in compliance with composite video specifications. The combination of horizontal and vertical blanking intervals limit such a signal to being at its highest level (white) for about 75% of the time.

As a result of the duty cycle variations between the extremes presented above, a 1Vp-p composite video signal that is multiplied by a gain-of-two requires about 3.2Vp-p of dynamic voltage swing at the output for the op amp, to pass a composite video signal of arbitrary duty cycle without distortion.

The AD8041 device family not only has ample signal swing capability to handle the dynamic range required, but also has excellent differential gain and phase when buffering these signals in an AC coupled configuration.

To test this, the differential gain and phase were measured for the AD8041 while the supplies were varied. As the lower supply is raised to approach the video signal, the first effect is that the sync tips become compressed before the differential gain and phase are adversely affected. Thus, there must be adequate swing in the negative direction to pass the sync tips without compression.

As the upper supply is lowered to approach the video, the differential gain and phase were not significantly adversely affected until the difference between the peak video output and the supply reached 0.6V. Thus, the highest video level should be kept at least 0.6V below the positive supply rail.

Taking the above into account, it was found that the optimal point to bias the non-inverting input was at +2.2V DC. Operating at this point, the worst case differential gain was 0.06% and the differential phase 0.06°.

The AC coupling capacitors used in the circuit may at first glance appear quite large. There is a reason for this. Note that a composite video signal has a lower frequency band edge of 30Hz. The resistances at the various AC coupling points— especially at the output— are quite small. In order to minimize phase shifts and baseline tilt, the large value capacitors shown are required for best waveform reproduction.

For video system performance that is not to be of the highest quality, the value of these capacitors can be reduced by a factor of up to five with only a slight observable change in the picture quality.

■ OP AMP APPLICATIONS

Single-Supply AC Coupled Single-Ended-to-Differential Driver

The circuit shown below in Figure 6-120 provides a flexible solution to differential line driving in a single-supply application and utilizes the dual AD8042. The basic operation of the cross-coupled configuration has been described earlier in this section. The input, V_{IN} , is a single-ended signal that is capacitively coupled into the feedforward resistor, $R1$. The non-inverting inputs of each half of the AD8042 are biased at +2.5V.

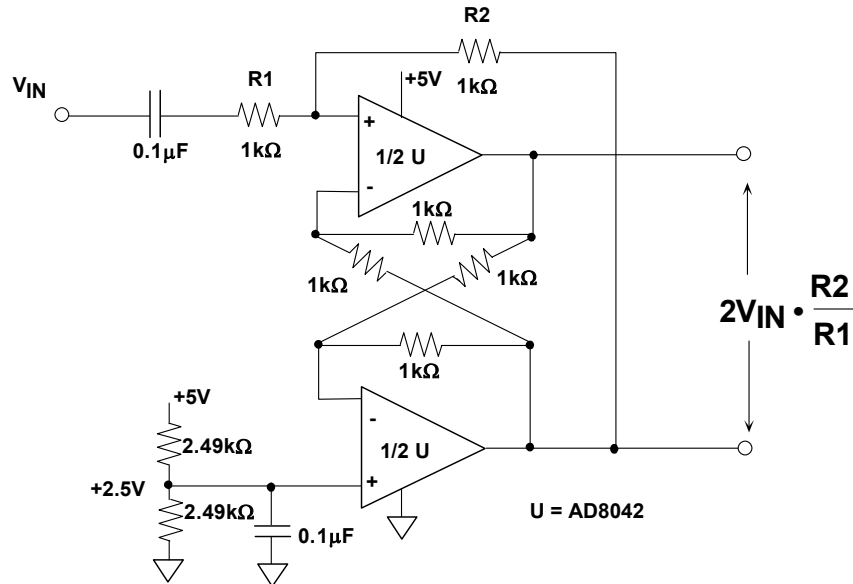


Figure 6-120: Single-supply AC coupled differential driver

The gain from single-ended input to the differential output is equal to $2R2/R1$, as noted in the figure. If desired, this gain can be varied by simply changing one resistor (either $R1$ or $R2$). The input capacitor may need increase, for the processing of low frequency information with low phase shift.

It should also be noted that there is no output coupling capacitor, as none is required for differentially connected loads. The output terminals will be biased at approximately 2.5V.

REFERENCES: VIDEO AMPLIFIERS

1. W. A. Kester, "PCM Signal Codecs for Video Applications," **SMPTE Journal**, No. 88, November 1979, pp. 770-778.
2. "IEEE Standard for Performance Measurements of A/D and D/A Converters for PCM Television Circuits," **IEEE Standard 746-1984**.
3. Walt Kester, "Maintaining Transmission Line Impedances on the PC Board," within Chapter 11 of Walt Kester, Editor, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3.
4. William R. Blood, Jr., **MECL System Design Handbook** (HB205, Rev.1), Motorola Semiconductor Products, Inc., 1988.
5. Dave Whitney, Walt Jung, "Applying a High-Performance Video Operational Amplifier," **Analog Dialogue**, 26-1, 1992.
6. Walt Jung, Scott Wurcer, "Design Video Circuits Using High-Speed Op-Amp Systems," **Electronic Design Analog Applications Issue**, November 7, 1994.
7. Vishay chip resistors and type VTF networks, <http://www.vishay.com>.
8. Walt Kester, "Video Line Receiver Applications Using the AD830 Active Feedback Amplifier Topology," within Chapter 11 of Walt Kester, Editor, **System Application Guide**, Analog Devices, Inc., 1993, ISBN 0-916550-13-3.
9. Peter Checkovich, "Understanding and Using High-Speed Clamping Amplifiers," **Analog Dialogue**, Vol. 29-1, 1995.
10. "Chapters 1, 2, and 4," within Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, Inc., Norwood, MA, 1995, ISBN 0-916550-16-8.

NOTES:

SECTION 6-4: COMMUNICATIONS AMPLIFIERS

Walt Kester

Components used in the signal path in communications systems must have wide dynamic range at high frequencies. Dynamic range is primarily limited by distortion and noise introduced by the active elements in amplifiers, mixers, etc. In the past, amplifiers for communications applications consisted primarily of "gain blocks" with appropriate specifications. Typically such amplifiers are specified for gain, bandwidth, distortion, etc., as a system is designed, and purchased as a self-contained package. This package itself is actually a communications amplifier sub-system.

Today, however, op amps with bandwidths of hundreds of megahertz, low noise, high dynamic range and flexible supply voltages also make popular building blocks in communications systems. They are easily configured for a given gain, and can deliver good performance.

Communications-Specific Specifications

As a necessity, this means that high frequency op amps must be fully specified not only in terms of traditional op amp AC specifications (bandwidth, slew rate, settling time), but also in terms of *communications-specific specifications*. These latter specifications would include performance for harmonic distortion, spurious free dynamic range (SFDR), intermodulation distortion, intercept points (IP2, IP3), noise, and noise figure (NF). Figure 6-121 illustrates these specifications, below.

- ◆ **Noise**
 - Noise referred to input (RTI)
 - Noise referred to output (RTO)
- ◆ **Distortion**
 - Second and third order intercept points (IP2, IP3)
 - Spurious free dynamic range (SFDR)
 - Harmonic distortion
 - Single-tone
 - Multi-tone
 - Out-of-band
 - Multitone Power Ratio (MTPR)
 - Noise Factor (NF), Noise Figure (NF)

Figure 6-121: *Dynamic range specifications in communications systems*

Within this portion of the chapter we will examine these specifications, and how they apply to the amplifiers used in wireless and wired communications systems. In addition, several application specific amplifiers such as variable gain amplifiers (VGAs), CATV drivers, and xDSL drivers will also be discussed.

Distortion Specifications

When a spectrally pure sinewave passes through an amplifier (or other active device), various harmonic distortion products are produced depending upon the nature and the severity of the non-linearity. However, simply measuring harmonic distortion produced by single tone sinewaves of various frequencies does not give all the information required to evaluate the amplifier's potential performance in a communications application. In most communications systems there are a number of channels which are "stacked" in frequency. It is often required that an amplifier be rated in terms of the intermodulation distortion (IMD) produced with two or more specified tones applied.

Intermodulation distortion products are of special interest in the IF and RF area, and a major concern in the design of radio receivers. Rather than simply examining the harmonic distortion or total harmonic distortion (THD) produced by a single tone sinewave input, it is often useful to look at the distortion products produced by two tones.

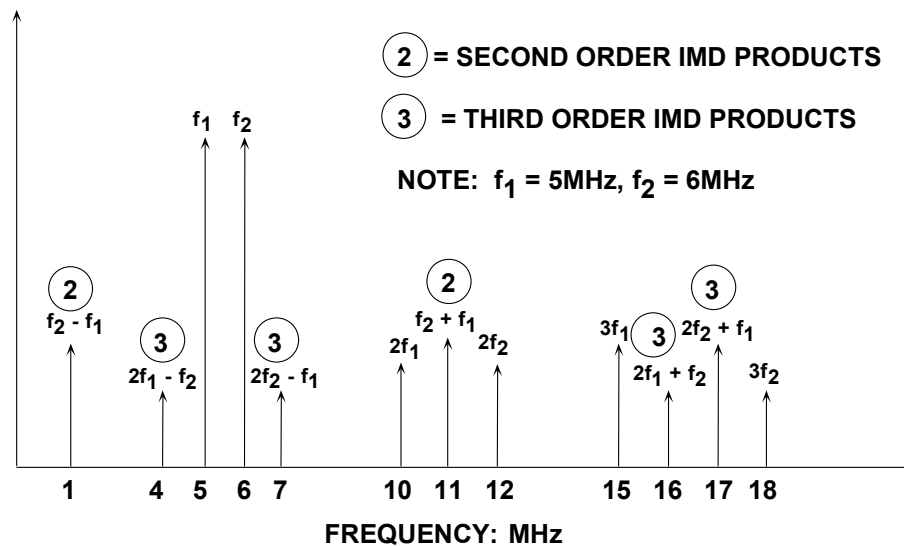


Figure 6-122: Second and third order intermodulation distortion products

As shown in Figure 6-122 above, two tones will produce second and third order intermodulation products. The example shows the second and third order products produced by applying two frequencies, f_1 and f_2 , to a nonlinear device. The second order products located at $f_2 + f_1$ and $f_2 - f_1$ are located far away from the two tones, and may be removed by filtering. The third order products located at $2f_1 + f_2$ and $2f_2 + f_1$ may likewise be filtered. The third order products located at $2f_1 - f_2$ and $2f_2 - f_1$, however, are close to the original tones, and filtering them is difficult.

Third order IMD products are especially troublesome in multi-channel communications systems where the channel separation is constant across the frequency band. Third-order IMD products can mask out small signals in the presence of larger ones.

Third order IMD is often specified in terms of the *third order intercept* point, as is shown by Figure 6-123, below. Two spectrally pure tones are applied to the system. The output signal power in a single tone (in dBm) as well as the relative amplitude of the third-order products (referenced to a single tone) is plotted as a function of input signal power. The fundamental is shown by the *slope = 1* curve in the diagram. If the system non-linearity is approximated by a power series expansion, it can be shown that second-order IMD amplitudes increase 2dB for every 1dB of signal increase, as represented by *slope = 2* curve in the diagram.

Similarly, the third-order IMD amplitudes increase 3dB for every 1dB of signal increase, as indicated by the *slope = 3* plotted line. With a low level two-tone input signal, and two data points, one can draw the second and third order IMD lines as they are shown in Figure 6-123 (using the principle that a point and a slope define a straight line).

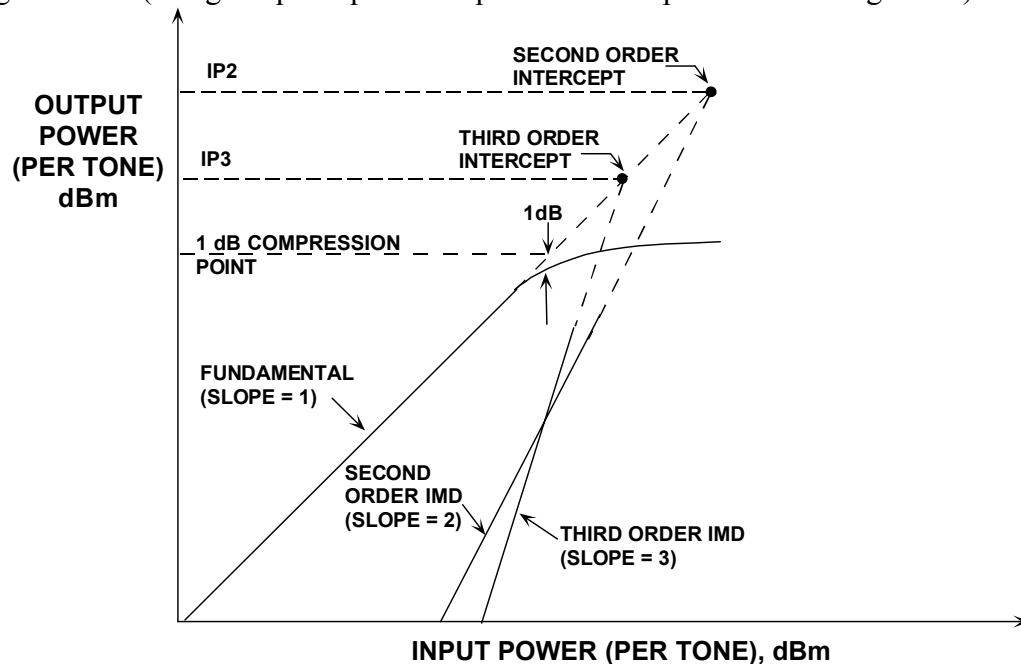


Figure 6-123: Intercept points and 1dB compression point

Once the input reaches a certain level however, the output signal begins to soft-limit, or compress. A parameter of interest here is the *1dB compression point*. This is the point where the output signal is compressed 1dB from an ideal input/output transfer function. This is shown in Figure 6-123 within the region where the ideal slope = 1 line becomes dotted, and the actual response exhibits compression (solid).

Nevertheless, both the second and third-order intercept lines may be extended, to intersect the (dotted) extension of the ideal output signal line. These intersections are called the *second* and *third order intercept points*, respectively, or IP2 and IP3. These power level values are usually referenced to the output power of the device delivered to a matched load (usually, but not necessarily 50Ω) expressed in dBm.

It should be noted that IP2, IP3, and the 1dB compression point are all a function of frequency, and as one would expect, the distortion is worse at higher frequencies.

■ OP AMP APPLICATIONS

For a given frequency, knowing the third order intercept point allows calculation of the approximate level of the third-order IMD products as a function of output signal level. Figure 6-124 below shows the third order intercept value as a function of frequency for a typical wideband low-distortion amplifier.

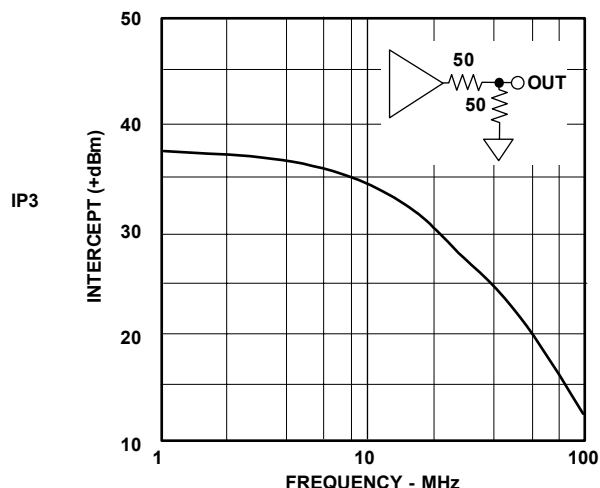


Figure 6-124: Third order intercept point (IP3) versus frequency for a low distortion amplifier

Assume the op amp output signal is 5MHz and 2V peak-to-peak into a 100Ω load (50Ω source and load termination). The voltage into the 50Ω load is therefore 1V peak-to-peak, corresponding to +4dBm. From Fig. 6-124, the value of the third order intercept at 5MHz is 36dBm. The difference between +36dBm and +4dBm is 32dB. This value is then multiplied by 2 to yield 64dB (the value of the third-order intermodulation products referenced to the power in a single tone). Therefore, the intermodulation products should be -64dBc (dB below carrier frequency), or at an output power level of -60dBm.

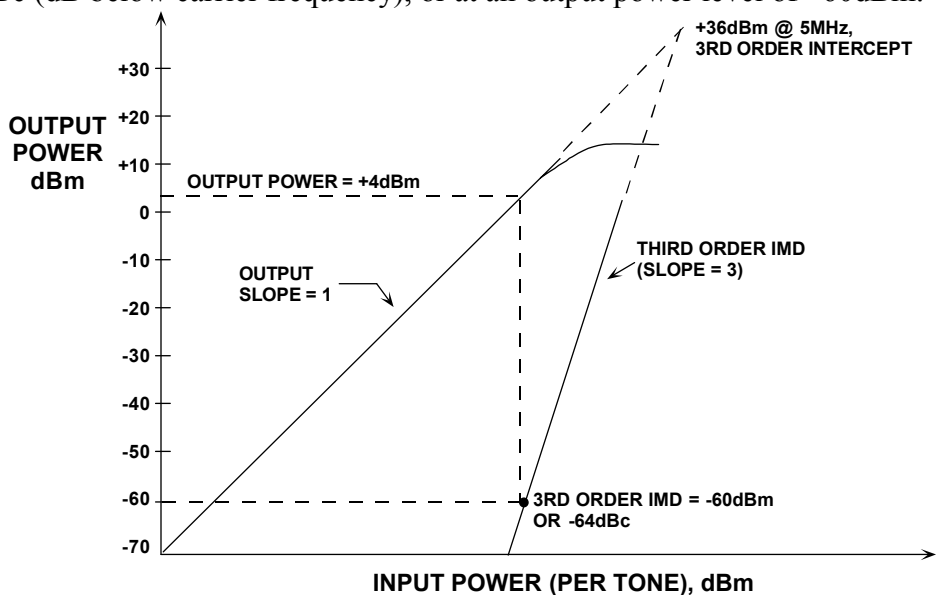


Figure 6-125: Using IP3 to calculate the third-order IMD product amplitude

Figure 6-125 shows the graphical analysis for this example. A similar analysis can be performed for the second-order intermodulation products, using data for IP2.

Another popular specification in communications systems is *spurious free dynamic range*, or SFDR. Figure 6-126 below shows two variations of this specification. Single-tone SFDR (left) is the ratio of the signal (or carrier) to the *worst* spur in the bandwidth of interest. This spur may or may not be harmonically related to the signal. SFDR can be referenced to the signal or carrier level (dBc), or to full scale (dBFS).

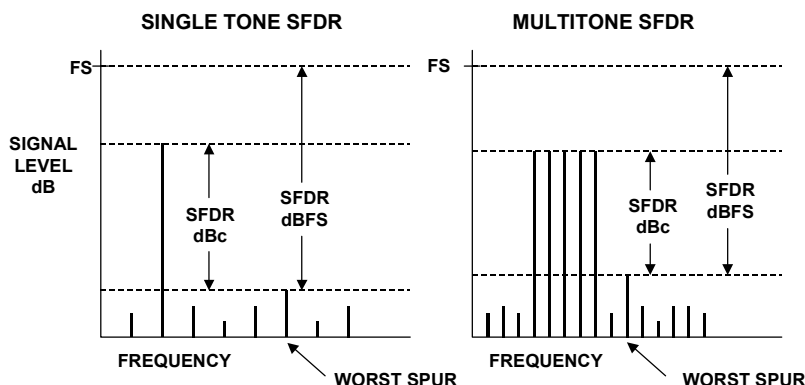


Figure 6-126: Spurious free dynamic range (SFDR) in communications systems

Because most amplifiers are soft limiters, the dBc unit is more often used. However, in systems that have a hard-limiter that precisely defines full scale (such as with ADCs), both dBc and dBFS may be used. It is important to understand that they both describe the worst spur amplitude. SFDR can also be specified for two tones or multitones (right), thereby simulating complex signals that contain multiple carriers and channels.

Multitone power ratio is another way of describing distortion in a multichannel communication system. Figure 6-127 below shows the frequency partitioning in an xDSL system. The QAM signals in the upstream data path are represented by a number of equal amplitude tones, separated equally in frequency. One channel is completely eliminated from the input signal (shown as an empty bin), but intermodulation distortion caused by the system nonlinearity will cause a small signal to appear in that bin.

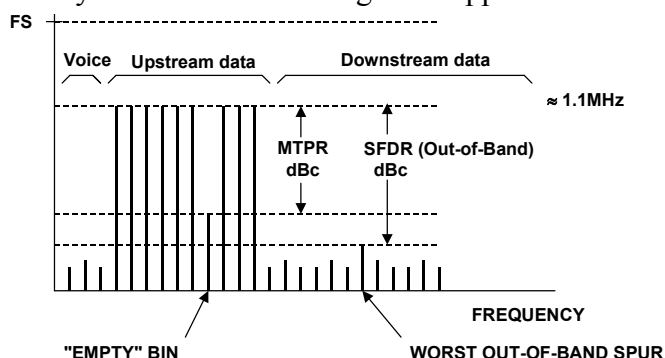


Figure 6-127: Multitone power ratio (MTPR) and out-of-band SFDR in xDSL applications

The ratio of the tone amplitude to the amplitude of the unwanted signal in the empty bin is defined as the multitone power ratio, or MTPR. It is equally important that the amplitude of the intermodulation products caused by the multitone signal (simulating multiple channels) not interfere with signals in either the voice band or the downstream data band. The amplitude of the worst spur produced in these bands to the amplitude of the multitone signal is therefore defined as the *out-of-band SFDR*.

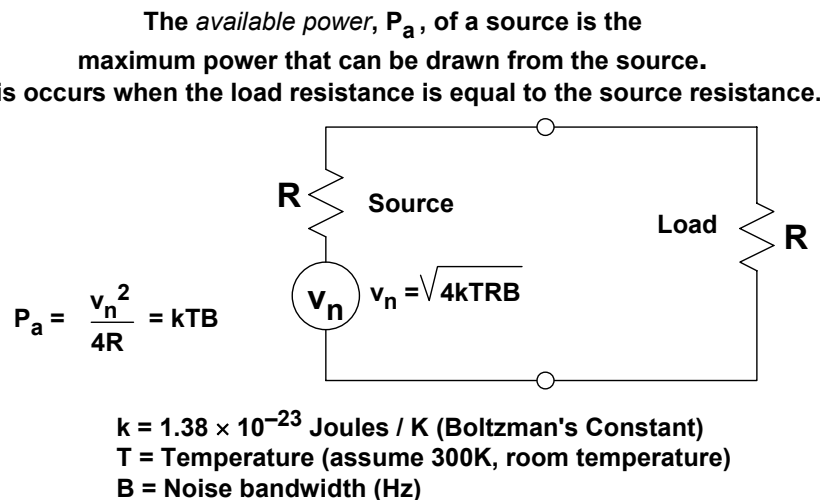
Noise Specifications

Op amp noise is generally specified in terms of input current and voltage noise, as previously discussed in Chapter 1 of this book. In communications systems, however, noise is often specified in terms of *noise figure* (NF)— see Figure 6-128. This can lead to confusion, especially when op amps are used as gain blocks and the noise figure of the op amp is not specified for the specific circuit conditions. In order to understand how to apply noise figure to op amps, we will first review the basic theory behind noise figure.

- ◆ NF is usually specified for matched input/output conditions, but this is not always a system requirement
- ◆ Noise Figure is a popular figure of merit in RF applications: LNAs, Mixers, etc.
- ◆ Difficulties arise when applying NF to op amps. NF is dependent on
 - Impedance levels
 - Feedback network
 - Closed loop gain
- ◆ Other difficulties arise due to different definitions of NF as found in various textbooks
- ◆ We will start with the basics and work up to the op amp issues

Figure 6-128: Noise figure in communications applications

The first concept is that of *available power* from a source. The available power of a source is the maximum power that can be drawn from the source. Figure 6-129 below shows a resistor of value R as the noise source. The thermal noise of this source is $\sqrt{4kTB}$. The maximum noise that can be transferred to an ideal noiseless load occurs when the load resistance is also equal to R .

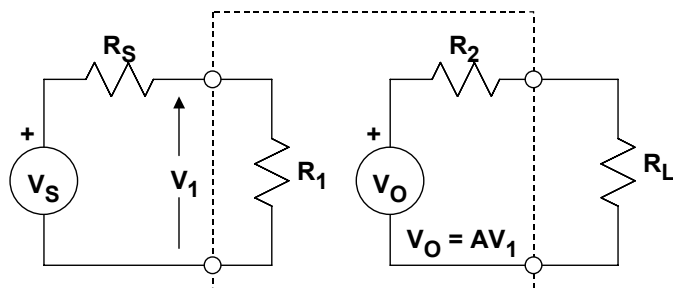


$$P_a (\text{dBm}) = -174\text{dBm} + 10 \log B$$

Figure 6-129: Available noise power from a source

Under these conditions, the maximum available noise power from the source reduces to kTB , where k is Boltzmann's constant, T is the absolute temperature, and B is the noise bandwidth. Note that this power is independent of the value of the source resistance, R .

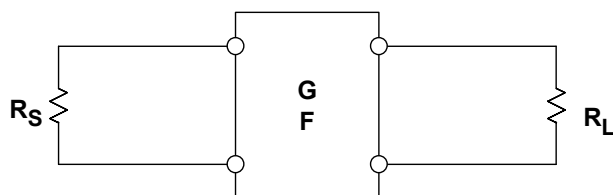
The next important concept is that of *available power gain* of a two-port network, as shown in Figure 6-130 below. The two-port network is driven from a signal source having an impedance. The equations show the available signal power from the source and the available signal power from the output of the network. The available power gain is simply the ratio of the available output power to the available power from the source.



- Available signal power from source = $P_{as} = \frac{V_S^2}{4R_S}$
- Available signal power at output = $P_{ao} = \frac{V_O^2}{4R_2}$
- Available power gain = $G_a = \frac{P_{ao}}{P_{as}} = \frac{V_O^2 R_S}{V_S^2 R_2}$

Figure 6-130: Available power gain of a two-port network

The gain and the noise of a two-port network can now be defined in terms of the available power gain, G , and the noise factor, F , as shown below in Figure 6-131. The noise factor, F , is defined as the ratio of the total available output noise power to the available output noise power due to the source only. For a resistive source, the available noise power from the source is simply kTB , and the output noise power due to the source only is $G \cdot kTB$.



G = Available Power Gain of Network

$$F = \text{Noise Factor} = \frac{\text{Total Available Output Noise Power}}{\text{Available Output Noise Power Due to Source Only}}$$

$$= \frac{\text{Total Available Output Noise Power}}{G \cdot kTB}$$

$$NF = \text{Noise Figure} = 10 \log_{10} F$$

Figure 6-131: Definition of noise factor and noise figure for a two-port noisy network.

Note that the noise factor, F , is expressed as a ratio, and the noise figure, NF , is simply the ratio F expressed in dB. An ideal noiseless two-port network therefore has noise factor $F = 1$, and a noise figure $NF = 0\text{dB}$.

■ OP AMP APPLICATIONS

We can use these same definitions to calculate the NF of an op amp circuit, however it is much easier to work in terms of the square of voltage noise spectral density and current noise spectral density, rather than power or power spectral density (see Figure 6-132 below). Also, unmatched conditions are easier to deal with using this approach. *The noise factor F for an op amp is simply the ratio of the square of the total output noise spectral density to the square of the output noise spectral density due to the source only.* The noise figure $NF = 10 \bullet \log F$.

- ◆ With op amps, it is easier to work with voltage and current noise spectral density, rather than power or power spectral density.
- ◆ Unmatched conditions are more easily dealt with using voltage noise spectral density analysis.
- ◆ Voltage noise spectral densities add using root-sum-squares (RSS).
- ◆ A 1000Ω resistor has a voltage noise spectral density of 4nV/√Hz @ 25°C (300K). (This is good to remember!)
- ◆ The basic definition of Noise Factor and Noise Figure in terms of voltage noise spectral density becomes:

$$\text{Noise Factor} = F = \frac{(\text{Total Output Voltage Noise Spectral Density})^2}{(\text{Output Voltage Noise Spectral Density Due to Source Only})^2}$$

$$\text{Noise Figure} = NF = 10 \log_{10} F$$

Figure 6-132: Noise figure for op amps

In RF or IF gain blocks, the input impedance is defined. However, when using an op amp in the non-inverting mode as a gain block, the input impedance is high (relative to transmission line impedances), and there are several options regarding the input termination which affect the noise figure. These options have been generalized to cover any two-port network with optional input terminations in Figure 6-133 below.

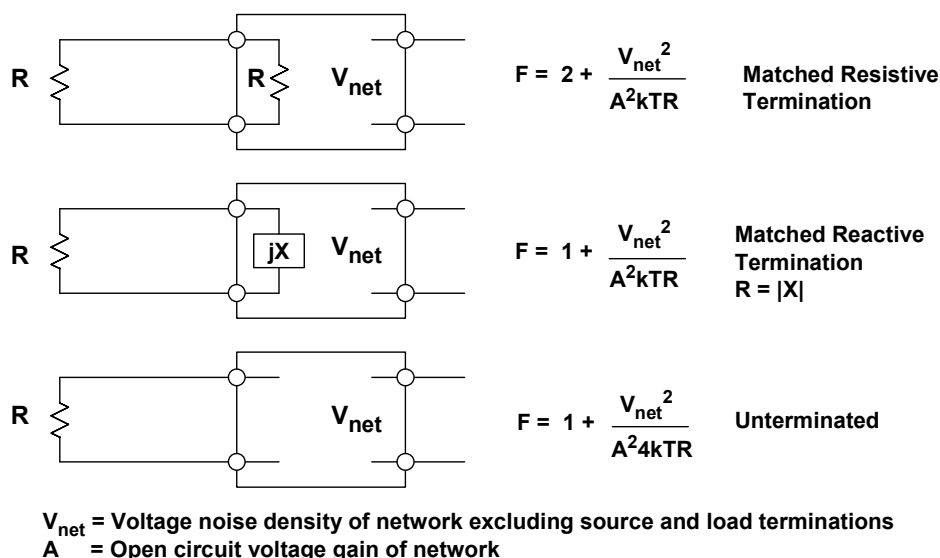


Figure 6-133: Noise factor for resistive, reactive, and unterminated conditions

Assume that the open circuit voltage gain of the network is A and that the total output noise spectral density (excluding that due to the source resistance and the input termination) is equal to V_{net} .

The top diagram of Figure 6-133 (opposite) shows the traditional matched case where the input is resistively terminated to match the source impedance. In this case, the input termination resistor not only attenuates the voltage noise of the source by a factor of 2 but also contributes noise due to its own thermal resistance.

The middle diagram of this figure shows the case of a reactive matched termination. Reactive terminations are often used where the bandwidth is limited but centered on a high frequency carrier. In this case, the source voltage noise is attenuated by a factor of 2, but the reactive termination adds no additional noise of its own to the total output noise.

The bottom diagram in Figure 6-133 shows the case of an unmatched, unterminated input. In this case, the voltage noise of the source is not attenuated, and there is obviously no additional noise due to the input termination because there is no input termination! Although this situation is not likely in a system using RF/IF gain blocks which generally require impedance matching at all interfaces, it is a possibility when using an op amp as the gain block, since the non-inverting configuration input impedance is relatively high.

- ◆ **For a low noise network, adding the matching input termination resistor makes the noise figure 3dB worse. The voltage gain is also reduced by a factor of 2.**
- ◆ **For a high noise network, adding the matching termination resistor makes the noise figure 6dB worse.**
- ◆ **Reactive matched terminations are often used at fixed IF/RF frequencies in LNAs, mixers, etc.**
- ◆ **Using large source and termination resistors decreases noise figure but increases overall circuit noise.**
- ◆ **Noise figures should only be compared at the same impedance level.**

Figure 6-134: Effects of input termination on noise figure

If we assume that the noise of the network, V_{net} , is very small relative to the source noise, then it is obvious that the input termination resistor adds 3dB to the overall noise figure as well as reduces the overall voltage gain by a factor of 2. This is compared to the lowest noise case where there is no input termination. In fact, the lowest possible noise figure for a noiseless network with only an input resistive matched termination is 3dB. Lower noise figures can be obtained only by using matched reactive terminations.

On the other hand, if the noise of the network, V_{net} , is very large with respect to the source noise, then adding the resistive termination increases the overall noise figure by 6dB compared to the unmatched unterminated case.

Summarizing, it is interesting to note that using large source resistances will decrease the noise figure but increase overall circuit noise. This illustrates the important fact that *noise figures can be compared only if they are specified at the same impedance level*. In Figure 6-134 above these effects of amplifier input terminations on overall circuit noise and noise figure are summarized.

■ OP AMP APPLICATIONS

Op amp noise has two components: low frequency noise whose spectral density is inversely proportional to the square root of the frequency, and white noise at medium and high frequencies. The low-frequency noise is known as 1/f noise (the noise *power* obeys a 1/f law—the noise voltage or noise current is proportional to $1/\sqrt{f}$). The frequency at which the 1/f noise spectral density equals the white noise is known as the "1/f Corner Frequency". This is an important figure of merit for op amps, with low values indicating better performance. Values of the 1/f corner frequency vary from a few Hz for the most modern low noise low frequency amplifiers, to several hundreds, or even thousands of Hz for some high-speed op amps.

In most applications of high speed op amps, it is the total output RMS noise that is generally of interest. Because of the high bandwidths, the chief contributor to the output RMS noise is the white noise, and that of the 1/f noise is negligible.

In order to better understand the effects of noise in high speed op amps, we use the classical noise model shown in Figure 6-135 below. This diagram identifies all possible white noise sources, including the external noise in the source and the feedback resistors.

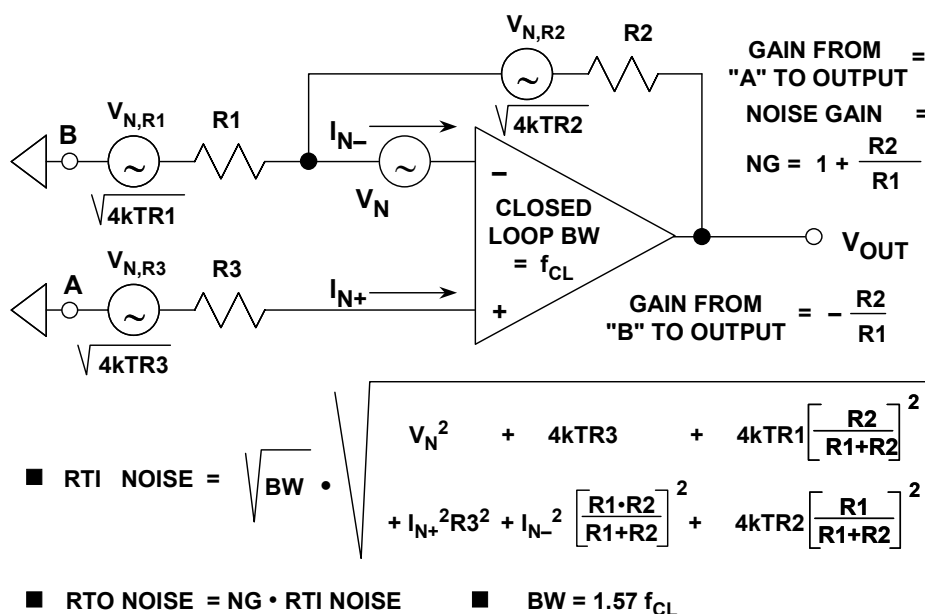


Figure 6-135: Calculating total op amp circuit noise

The equation in the figure allows you to calculate the total output RMS noise over the closed-loop bandwidth of the amplifier. This formula works quite well when the frequency response of the op amp is relatively flat. If there is more than a few dB of high frequency peaking, however, the actual noise will be greater than the predicted because the contribution over the last octave before the 3dB cutoff frequency will dominate.

In most applications, the op amp feedback network is designed so that the bandwidth is relatively flat, and the formula provides a good estimate. Note that BW in the equation is the equivalent *noise bandwidth*, which, for a single-pole system, is obtained by multiplying the closed-loop 3dB bandwidth by 1.57.

Figure 6-136 below is a table which indicates how the individual noise contributors of Fig. 6-135 are referred to the output. After calculating the individual noise spectral densities in this table, they can be squared, added, and then the square root of the sum of the squares yields the RSS value of the output noise spectral density, since all the sources are uncorrelated. This value is multiplied by the square root of the noise bandwidth (noise bandwidth = closed-loop 3dB bandwidth multiplied by a correction factor of 1.57) to obtain the final value for the output RMS noise.

Typical high speed op amps have bandwidths greater than 150MHz or so, and bipolar input stages have input voltage noises ranging from about 2 to 20nV/ $\sqrt{\text{Hz}}$. To put voltage noise in perspective, let's look at the Johnson noise spectral density of a resistor:

$v_n = \sqrt{4kTR \cdot BW}$, where k is Boltzmann's constant, T is the absolute temperature, R is the resistor value, and BW is the equivalent noise bandwidth of interest. (The equivalent noise bandwidth of a single-pole system is 1.57 times the 3dB frequency). Using the formula, a 100 Ω resistor has a noise density of 1.3nV/ $\sqrt{\text{Hz}}$, and a 1000 Ω resistor about 4nV/ $\sqrt{\text{Hz}}$ (values are at room temperature: 27°C, or 300K).

NOISE SOURCE EXPRESSED AS A VOLTAGE	MULTIPLY BY THIS FACTOR TO REFER TO THE OP AMP OUTPUT
Johnson Noise in R3: $\sqrt{(4kTR3)}$	Noise Gain = $1 + R2/R1$
Non-inverting Input Current Noise Flowing in R3: $I_{n+}R3$	Noise Gain = $1 + R2/R1$
Input Voltage Noise: V_n	Noise Gain = $1 + R2/R1$
Johnson Noise in R1: $\sqrt{(4kTR1)}$	$-R2/R1$ (Gain from input of R1, "B", to Output)
Johnson Noise in R2: $\sqrt{(4kTR2)}$	1
Inverting Input Current Noise Flowing in R2: $I_{n-}R2$	1

Figure 6-136: Referring all noise sources to the output

The base-emitter in a bipolar transistor has an equivalent noise voltage source which is due to the "shot noise" of the collector current flowing in the transistor's (noiseless) incremental emitter resistance, r_e . The current noise is proportional to the square root of the collector current, I_c . The emitter resistance, on the other hand, is inversely proportional to the collector current, so *the shot-noise voltage is inversely proportional to the square root of the collector current*.

Voltage noise in FET-input op amps tends to be larger than for bipolar ones, but current noise is extremely low (generally only a few tens of fA/ $\sqrt{\text{Hz}}$) because of the low input bias currents. However, FET-inputs are not generally required for op amp applications requiring bandwidths greater than 100MHz.

■ OP AMP APPLICATIONS

Op amps also have input current noise on each input. For high-speed FET-input op amps, the gate currents are so low that input current noise is almost always negligible (measured in $\text{fA}/\sqrt{\text{Hz}}$).

For a voltage feedback (VFB) op amp, the inverting and non-inverting input current noise are typically equal, and almost always uncorrelated. Typical values for wideband VFB op amps range from $0.5\text{pA}/\sqrt{\text{Hz}}$ to $5\text{pA}/\sqrt{\text{Hz}}$. The input current noise of a bipolar input stage is increased when input bias-current cancellation generators are added, because their current noise is not correlated, and therefore adds (in an RSS manner) to the intrinsic current noise of the bipolar stage.

The input voltage noise in current feedback (CFB) op amps tends to be lower than for VFB op amps having the same approximate bandwidth. This is because the input stage in a CFB op amp is usually operated at a higher current, thereby reducing the emitter resistance and hence the voltage noise. Typical values for CFB op amps range from about 1 to $5\text{nV}/\sqrt{\text{Hz}}$.

The input current noise of CFB op amps tends to be larger than for VFB op amps because of the generally higher bias current levels. The inverting and non-inverting current noise of a CFB is usually different because of the unique input architecture, and are specified separately. In most cases, the inverting input current noise is the larger of the two. Typical input current noise for CFB op amps ranges from 5 to $40\text{pA}/\sqrt{\text{Hz}}$.

The general principle of noise calculation is that uncorrelated noise sources add in a root-sum-squares manner, which means that if a noise source has a contribution to the output noise of a system which is less than 20% of the amplitude of the noise from other noise source in the system, then its contribution to the total system noise will be less than 2% of the total, and that noise source can almost invariably be ignored—in many cases, noise sources smaller than 33% of the largest can be ignored. This can simplify the calculations using the formula, assuming the correct decisions are made regarding the sources to be included and those to be neglected.

The sources which dominate the output noise are highly dependent on the closed-loop gain of the op amp. Notice that for high values of closed loop gain, the op amp voltage noise will tend be the chief contributor to the output noise. At low gains, the effects of the input current noise must also be considered, and may dominate, especially in the case of a CFB op amp.

Feedforward/feedback resistors in high speed op amp circuits may range from less than 100Ω to more than $1\text{k}\Omega$, so it is difficult to generalize about their contribution to the total output noise without knowing the specific values and the closed loop gain. The best way to make the calculations is to write a simple computer program that performs the calculations automatically and includes all noise sources (see Reference 1 for one example). In most high speed applications, the source impedance noise can be neglected for source impedances of 100Ω or less.

Figure 6-137 below shows an example calculation of total output noise for the AD8011 (300MHz, 1mA) CFB op amp. All six possible sources are included in the calculation. The appropriate multiplying factors which reflect the sources to the output are also shown on the diagram. For $G=2$, the close-loop bandwidth of the AD8011 is 180MHz. The correction factor of 1.57 in the final calculation converts this single-pole bandwidth into the circuit's equivalent noise bandwidth.

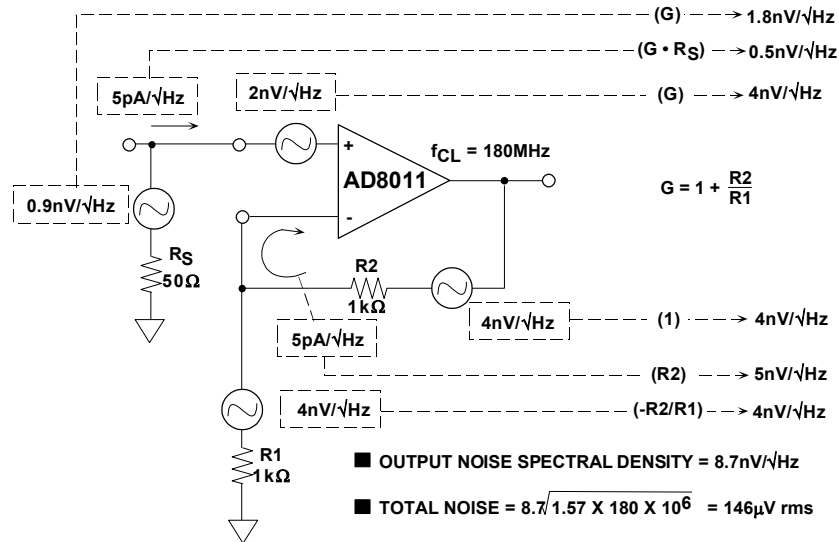


Figure 6-137: AD8011 output noise analysis

Now that the total output noise has been calculated, we can address the issue of noise figure. Figure 6-138 below shows two cases for the AD8011 circuit: the top diagram corresponds to an unterminated input condition, and the bottom diagram corresponds to a terminated input condition.

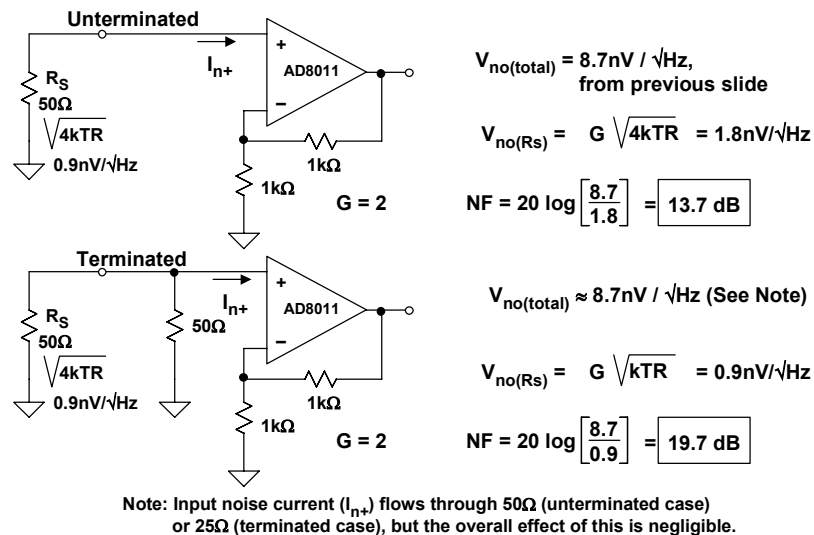


Figure 6-138: AD8011 noise figure for unterminated and terminated input conditions

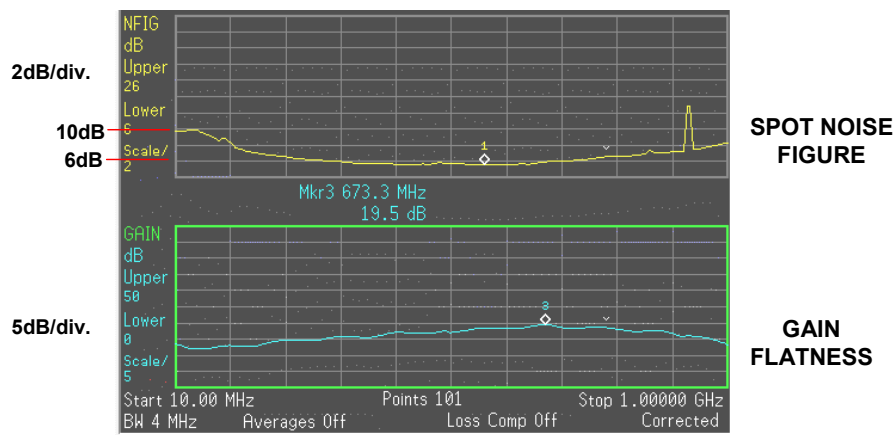
For the unterminated case (top), the total output noise from the previous diagram (i.e., Fig. 6-137) was 8.7nV/√Hz. Note this includes the noise of the 50Ω source. The output

■ OP AMP APPLICATIONS

noise due only to the source is simply the noise gain multiplied by the noise of the source, or $G\sqrt{4kTR_S} = 1.8\text{nV}/\sqrt{\text{Hz}}$. The noise figure is simply $\text{NF} = 20\log(8.7/1.8) = 13.7\text{dB}$.

For the terminated case (bottom), the total output noise is still approximately $8.7\text{nV}/\sqrt{\text{Hz}}$. Note that the input noise current (I_{n+}) now actually flows through 25Ω rather than 50Ω for the unterminated case, but the overall effect of this difference on the total output noise calculation is negligible.

The noise of the source, however, is now $\sqrt{kTR}$ due to the 50Ω divider network, and reflected output, it becomes $G\sqrt{kTR} = 0.9\text{nV}/\sqrt{\text{Hz}}$. The noise figure is calculated as $\text{NF} = 20\log(8.7/0.9) = 19.7\text{dB}$. Notice that the terminated case yields a noise figure which is approximately 6dB worse than the unterminated case.



- Spot Noise Figure shows variations over frequency: 10MHz to 1GHz
- Measurement bandwidth set to 4MHz for above data
- Integration required to get average noise power

Figure 6-139: AD8350 spot noise figure and gain vs. frequency from 10MHz to 1GHz

Finally, it should be noted that noise figure is actually a function of frequency. In Figure 6-139 above is shown the noise figure of the AD8350 measured with a spot noise meter, as a function of frequency over the bandwidth 10MHz to 1GHz. The top curve is the noise figure, and the bottom curve is the closed-loop gain flatness.

In most cases, the approximations such as those used in the example of Figs. 6-135 through 6-137 will give sufficient accuracy, provided the closed-loop bandwidth is relatively flat. However, using the actual spot noise figure may be desirable in high frequency narrowband applications involving specific carrier frequencies.

Variable Gain Amplifiers (VGAs) in Automatic Gain Control (AGC)

Wideband, low distortion variable gain amplifiers find wide applications in communications systems. One example is automatic gain control (AGC) in radio receivers. Typically, the received energy exhibits a large dynamic range due to the variability of the propagation path, requiring dynamic-range compression in the receiver. In this case, the wanted information is in the modulation envelope (whatever the modulation mode), not in the absolute magnitude of the carrier. For example, a 1MHz carrier modulated at 1kHz to a 30% modulation depth would convey the same information, whether the received carrier level is at 0dBm or -120dBm. Some type of automatic gain control (AGC) in the receiver is generally utilized to restore the carrier amplitude to some normalized reference level, in the presence of large input fluctuations. AGC circuits are dynamic-range compressors which respond to some signal metric (often mean amplitude) acquired over an interval corresponding to many periods of the carrier.

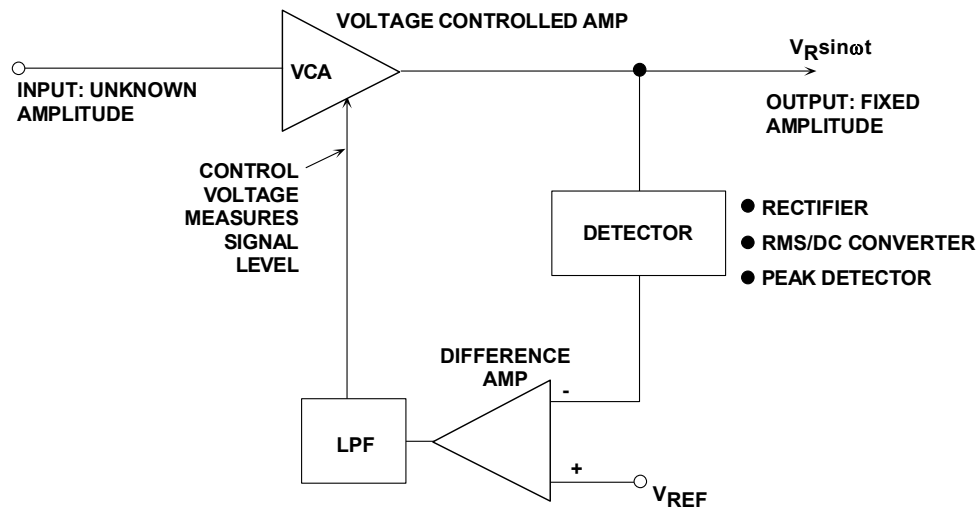


Figure 6-140: A typical automatic gain control (AGC) system

Consequently, they require time to adjust to variations in received signal level. The time required to respond to a sudden increase in signal level can be reduced by using peak detection methods, but with some loss of robustness, since transient noise peaks can now activate the AGC detection circuits. Nonlinear filtering and the concept of "delayed AGC" can be useful in optimizing an AGC system. Many tradeoffs are found in practice; Figure 6-140 above shows a basic AGC system.

It is interesting to note that an AGC loop actually has *two* outputs. The more obvious output is of course the amplitude-stabilized signal. The less obvious output is the control voltage to the VCA. In reality, this voltage is a measure of the average amplitude of the input signal. If the system is precisely scaled, the control voltage may be used as a measure of the input signal, which is sometimes also known as a *received signal strength indicator* (RSSI).

This latter point, given a suitably precise VCA gain control law, allows implementation of a receiving system which is calibrated for incoming signal level.

Voltage Controlled Amplifiers (VCAs)

An analog multiplier can be used as a variable-gain amplifier, as shown in Figure 6-141 below. The control voltage is applied to one input, and the signal to the other. In this configuration, the gain is directly proportional to the control voltage.

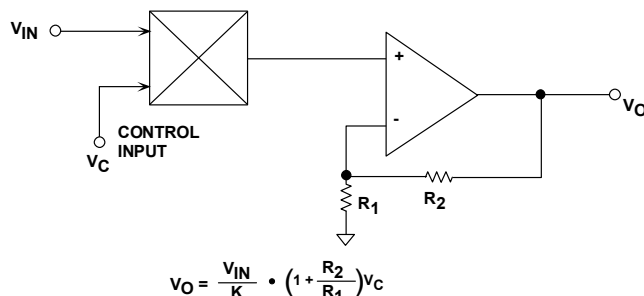


Figure 6-141: Using a multiplier as a voltage-controlled amplifier (VCA)

Most VCAs made with analog multipliers have gain which is *linear in volts* with respect to the control voltage, and they tend to be noisy. There is a demand, however, for a VCA which combines a wide gain range with constant bandwidth and phase, low noise with large signal-handling capabilities, and low distortion with low power consumption, while providing accurate, stable, *linear-in-dB* gain. The AD600, AD602, AD603, AD604, and AD605, and AD8367 achieve these demanding and conflicting objectives with a unique and elegant solution—the X-AMP® (for *exponential amplifier* see Reference 2).

The concept is simple: a fixed-gain amplifier follows a passive, broadband attenuator, with special means to alter voltage-controlled attenuation, as in Figure 6-142 below.

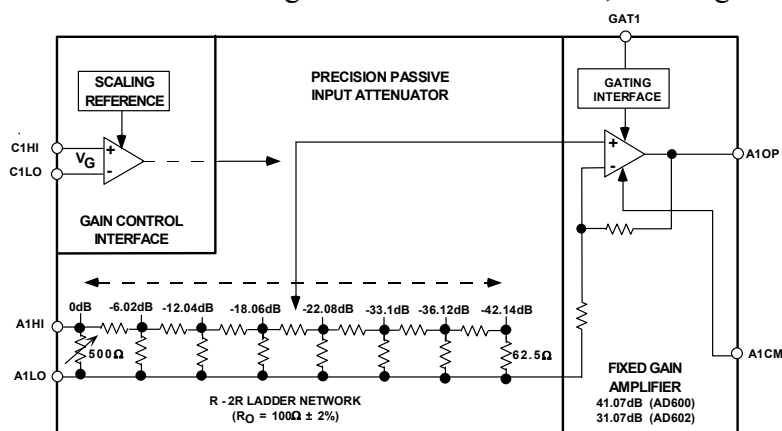


Figure 6-142: Single channel of the dual 30MHz AD600/AD602 X-AMP®

The AD600/AD602 amplifier stage is optimized for low input noise, and negative feedback is used to accurately define its moderately high gain (30-40dB) and minimize distortion. Since amplifier gain is fixed, so also are its AC and transient response characteristics, including distortion and group delay. As its gain is high, the input is never driven beyond a few millivolts, always operating within a small signal response range.

The attenuator network is a 7-section (8-tap) R-2R ladder. The ratio between adjacent taps is exactly 2, or 6.02dB, providing the basis for precise linear-in-dB behavior, while overall attenuation is 42.14dB. As will be shown, the amplifier's input can be connected to any one of these taps, or even *interpolated* between them, with only a small deviation error of about ± 0.2 dB. Overall gain can be varied from the fixed (maximum) gain, to a value 42.14dB less. In the AD600, the fixed gain is 41.07dB (voltage gain of 113); using this choice, the full gain range is -1.07 dB to $+41.07$ dB. The gain is related to control voltage by the relationship $G_{dB} = 32V_G + 20$ where V_G is in volts. For the AD602, the fixed gain is 31.07dB (voltage gain of 35.8), and the gain is given by $G_{dB} = 32V_G + 10$.

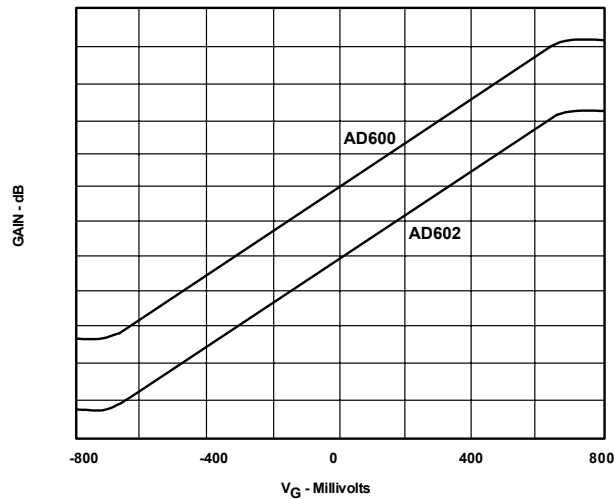


Figure 6-143: Gain of the AD600/AD602 as a function of control voltage

The gain at $V_G = 0$ is laser trimmed to an absolute accuracy of ± 0.2 dB. The gain scaling is determined by an on-chip bandgap reference (shared by both channels), laser trimmed for high accuracy and low temperature coefficient. Figure 6-143 above shows the gain versus the differential control voltage for both the AD600 and the AD602. Deviation from an ideal control law is only a fraction of a dB over a large part of the dynamic range.

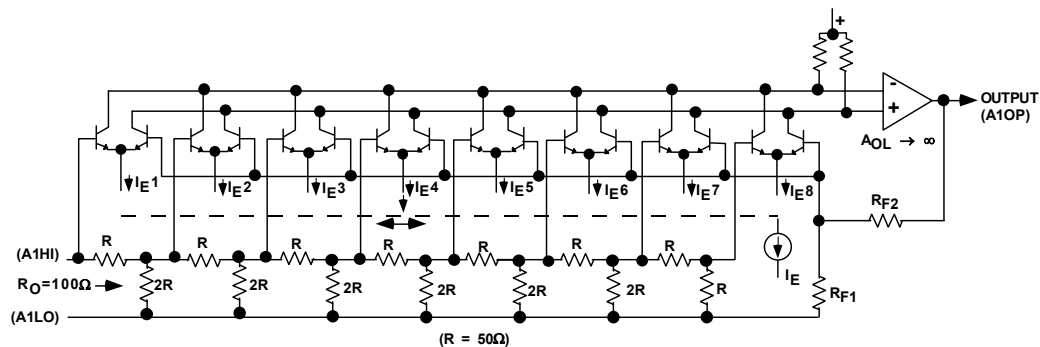


Figure 6-144: Continuous interpolation between taps using current-controlled g_m stages in the X-AMP®

In order to understand the operation of the X-AMP®, consider the simplified diagram shown in Figure 6-144 above. Note that each of the eight taps is connected to an input of one of eight bipolar differential pairs, used as current-controlled transconductance (g_m) stages; the other input of all these g_m stages is connected to the amplifier's gain-

■ OP AMP APPLICATIONS

determining feedback network, R_{F1}/R_{F2} . When emitter bias current I_E is directed to one of the 8 transistor pairs (not shown here), it becomes the complete amplifier input stage.

When I_E is connected to the left-most pair, the signal input is connected directly to the amplifier, giving maximum gain. The distortion is very low, even at high frequencies, due to the careful open-loop design, aided by the negative feedback. If I_E were now to be abruptly switched to the second pair, the overall gain would drop by exactly 6.02dB, and the distortion would remain low, because only one g_m stage remains active.

In reality, the bias current is *gradually* transferred from the first pair to the second. When I_E is equally divided between two g_m stages, both are active, and the situation arises where we have an op amp with two input stages fighting for control of the loop, one getting the full signal, and the other getting a signal exactly half as large.

Analysis shows that the effective gain is reduced, not by 3dB, as one might first expect, but rather by $20\log 1.5$, or 3.52dB. This error, when divided equally over the whole range, would amount to a gain ripple of ± 0.25 dB; however, the interpolation circuit actually generates a Gaussian distribution of bias currents, and a significant fraction of I_E always flows in adjacent stages. This smoothes the gain function and actually lowers the ripple. As I_E moves further to the right, the overall gain progressively drops.

	BANDWIDTH	DISTORTION	NOISE	INPUT Z	SUPPLY
AD600/602	35MHz	-60dBc @ 10MHz	1.4nV/ $\sqrt{\text{Hz}}$	100 Ω	$\pm 5\text{V}$
AD603	90MHz	-60dBc @ 10MHz	1.3nV/ $\sqrt{\text{Hz}}$	100 Ω	$\pm 5\text{V}$
AD604	40MHz	-43dBc @ 10MHz	0.8nV/ $\sqrt{\text{Hz}}$	300k Ω	$\pm 5\text{V}$
AD605	40MHz	-51dBc @ 10MHz	1.8nV/ $\sqrt{\text{Hz}}$	200 Ω	+5V
AD8367	500MHz	IP3 = +31.5dBm @140MHz	NF = 7.8dB @140MHz	200 Ω	+2.7 to +5V

Figure 6-145: X-AMP® family key specifications

The key features of the X-AMP product family are summarized in Figure 6-145 above. Note the other members of the family beyond the described AD600/AD602.

The total input-referred noise of the AD600/AD602 X-AMP® is 1.4nV/ $\sqrt{\text{Hz}}$; only slightly more than the thermal noise of a 100 Ω resistor (1.29nV/ $\sqrt{\text{Hz}}$ at 25°C). The input-referred noise is constant regardless of the attenuator setting, therefore the output noise is always constant and independent of gain.

For the AD600, the amplifier gain is 113 and the output noise spectral density is therefore 1.4nV/ $\sqrt{\text{Hz}}$ × 113, or 158nV/ $\sqrt{\text{Hz}}$. Referred to its maximum output of 2V_{rms}, the signal-to-noise ratio would be 82dB in a 1MHz bandwidth. The corresponding signal-to-noise ratio of the AD602 is 10dB greater, or 92dB.

Digitally Controlled Variable Gain Amplifiers for CATV Upstream Data Line Drivers

Cable modems offer much higher data rates than standard dial-up connections and have become very popular. In addition to receiving data (*downstream*), the cable modem also transmits data (*upstream*). This requires a low distortion digitally controlled variable gain amplifier capable of driving the 75Ω coaxial cable at a nominal level of 1V RMS (+11.2dBm, or 60dBmV). The AD8323 is a member of a family of CATV upstream line drivers suitable for this application. The AD8323 gain is controlled by an 8-bit serial word that determines the desired gain over a 53.5dB range, resulting in gain changes of 0.7526dB/LSB. The AD8323 block diagram is shown Figure 6-146 below.

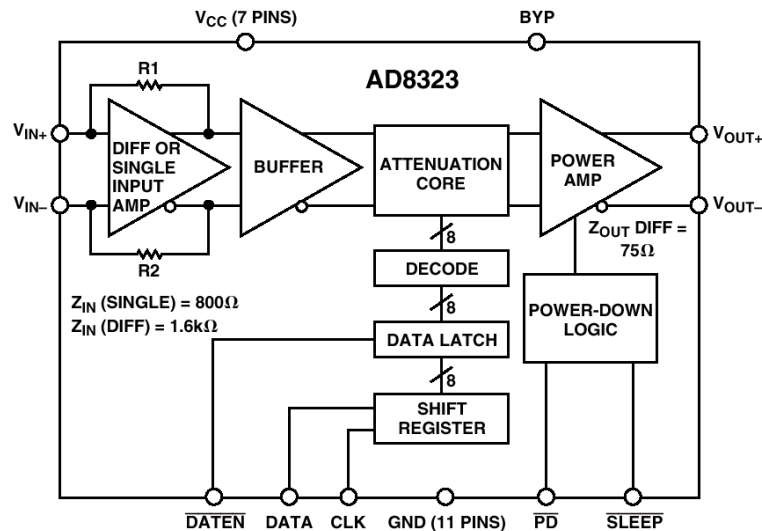


Figure 6-146: AD8323 CATV digitally controlled variable gain amplifier

The AD8323 has a variable attenuator core where the attenuation is digitally controlled from 0dB to -53.5dB. The input buffer has a gain of approximately +27.5dB, therefore the resulting overall gain range is from -26dB to +27.5dB. The AD8323 is composed of four analog functions in the power-up mode. The input amplifier (preamp) can be used either single-ended or differentially. The preamp stage drives a vernier stage that provides the fine tune gain adjustment. The 0.7526dB/LSB resolution is implemented in this stage and provides a total of approximately 5.25dB of attenuation. After the vernier stage, a DAC provides the bulk of the AD8323's attenuation (8-bits, or 48dB).

The signals in the preamp and vernier gain blocks are differential to improve the PSRR and linearity. A differential current is fed from the DAC to the output stage, which amplifies these currents to the appropriate levels necessary to drive the 75Ω load.

A key performance and cost advantage of the AD8323 results from the ability to maintain a constant dynamic output impedance of 75Ω during power-up/ power-down conditions. The output stage utilizes negative feedback to implement a differential 75Ω dynamic output impedance. This eliminates the need for an external 75Ω termination, resulting in twice the effective output voltage when compared to a standard op amp.

■ OP AMP APPLICATIONS

These features allow the AD8323 to operate on a single +5V supply and still deliver the required output power. Distortion performance of -56dBc is achieved with an output level up to 1V RMS ($+11.2\text{dBm}$, or 60dBmV) at a 21MHz bandwidth.

- ◆ Supports cable modem DOCSIS (Data Over Cable Service) standard for upstream path transmission
- ◆ Gain/attenuation programmable in 0.7526dB steps over a 53.5dB range:
 - 26dB to +27.5dB
- ◆ 3-wire SPI digital interface
- ◆ Bandwidth: > 100MHz (all gains)
- ◆ Low distortion @ 1V RMS ($+11.2\text{dBm}$, $+60\text{dBmV}$) output into 75Ω
 - -56dBc SFDR @ 21MHz
 - -55dBc SFDR @ 42MHz
- ◆ Single +5V supply (133mA)
- ◆ Power-down mode (35mA), sleep mode (4mA)
- ◆ 75Ω dynamic output impedance in power-up or power-down modes

Figure 6-147: AD8323 CATV line driver key specifications

The key specifications for the AD8323 are shown in Figure 6-147 above.

xDSL Upstream Data Line Drivers

Various versions of DSL are now used to provide fast internet connections. The upstream data path requires the transmission of $+13\text{dBm}$ discrete multitone (DMT) signals occupying a bandwidth between approximately 144kHz and 500kHz. The DMT signal can have a crest factor as high as 5.3, requiring the line driver to provide peak power of $+27.5\text{dBm}$, which translates into 7.5V peak voltage on the 100Ω telephone line.

DMT modulation appears in the frequency domain as power contained in several individual frequency subbands, sometimes referred to as tones or bins, each of which is uniformly separated in frequency. A quadrature amplitude modulated (QAM) signal occurs at the center of each subband or tone. Difficulties will exist when decoding these subbands if a signal from one subband is corrupted by the signal from other subbands, regardless of whether the corruption comes from an adjacent subband or harmonics of other subbands.

Conventional methods of expressing the output signal integrity of line drivers, such as single-tone harmonic distortion or THD, two-tone intermodulation distortion (IMD), and third order intercept (IP3), become significantly less meaningful when amplifiers are required to process DMT and other heavily modulated waveforms.

A typical ADSL upstream DMT signal can contain as many as 27 carriers (subbands or tones) of QAM signals (as shown in Figure 6-148, opposite). Multitone power ratio (MTTR) is the relative difference between the measured power in a typical subband (at one tone or carrier) versus the power at another subband specifically selected to contain no QAM data. In other words, a selected subband (or tone) remains open and void of intentional power (without a QAM signal), yielding an empty frequency bin. MTPR,

sometimes referred to as the "empty bin test," is typically expressed in dBc and is a key specification for all types of DSL systems.

Another important specification for an xDSL line driver is *out-of-band* SFDR. Spurs produced by distortion of the DMT upstream data can fall in the downstream frequency regions and distort voiceband and downstream data.

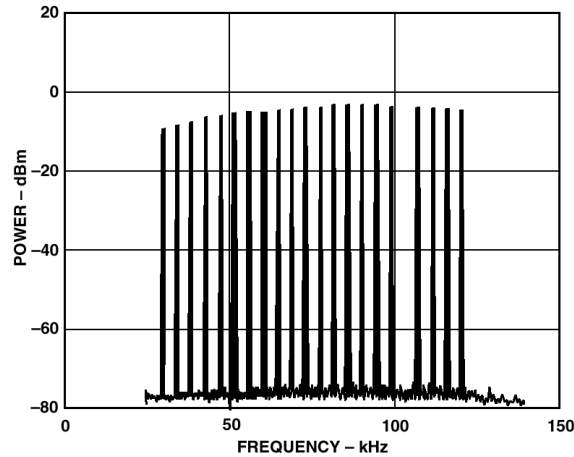


Figure 6-148: Discrete multitone (DMT) signal in the frequency domain

Figure 6-149 below shows an xDSL line driver application circuit based on the AD8018 line driver (one member of a family of Analog Devices' DSL line drivers). The peak DMT signal can be 7.5V on the 100Ω telephone line.

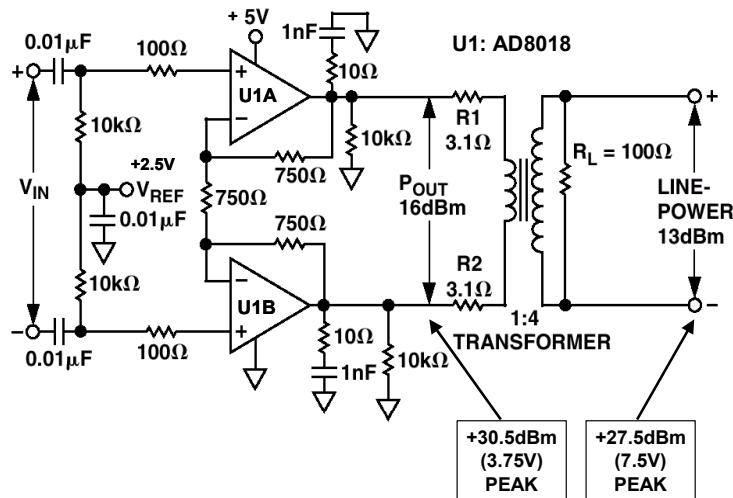


Figure 6-149: AD8018 xDSL upstream data line driver application

Assuming maximum low-distortion output swing available from the AD8018 line driver on a single +5V supply is 4V, taking into account the power lost due to the two 3.1Ω back-termination resistors, a transformer with a 1:4 or greater step-up is needed.

The AD8018 is therefore coupled to the phone line through a step-up transformer with a 1:4 turns ratio. R1 and R2 are back-termination or line-matching resistors, each 3.1Ω. The total differential load presented to the AD8018 output is 12.5Ω, including the

■ OP AMP APPLICATIONS

termination resistors. Even under these conditions, the AD8018 provides low distortion signals to within 0.5V of the power rails.

The transformer circuit presents a complex impedance to the AD8018 output, and therefore for stability, a series R-C network should be connected between each amplifier's output and ground. The recommended values are 10 Ω for the resistor and 1nF for the capacitor to create a low impedance path to ground at frequencies above 16MHz. The 10k Ω output resistors connected to ground are added to improve common-mode stability.

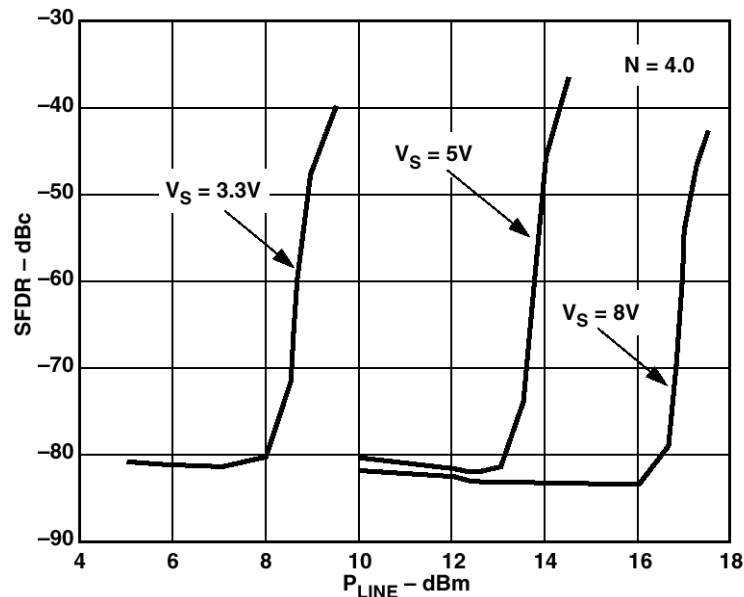


Figure 6-150: Out-of-band SFDR vs. upstream line power, 144kHz to 500kHz

For the AD8018 circuit of Fig. 6-149, the out-of-band SFDR versus upstream line power is shown in Figure 6-150 above for various supply voltages

- ◆ Dual current feedback amplifiers
- ◆ Bandwidth: 130MHz (–3dB)
- ◆ Slew rate: 300V/ μ s
- ◆ Rail-to-rail output stage (swings within 0.5V of rails for $R_L = 5\Omega$)
 - +16dBm into 12.5 Ω load
 - +30.5dBm peak power (3.75V) with +5V supply
- ◆ MTPR: –70dBc (25kHz to 138kHz)
- ◆ Maintains –82dBc out-of-band SFDR, 144kHz to 500kHz, for output power = +16dBm, $R_L = 12.5\Omega$
- ◆ Input voltage noise: 4.5nV/ $\sqrt{\text{Hz}}$ @100kHz
- ◆ Low supply current: 9mA/amplifier (full power mode)
- ◆ Standby mode (4.5mA/amplifier)
- ◆ Shutdown mode (0.3mA/amplifier)

Figure 6-151: AD8018 xDSL line driver key specifications

Some key AD8018 features and specifications are summarized in Figure 6-151, above.

REFERENCES: COMMUNICATIONS AMPLIFIERS

1. Bob Clarke, "Find Op Amp Noise with Spreadsheet", **Electronic Design**, December 13, 1990, or Analog Devices AN253.
2. Barrie Gilbert, "A Low Noise Wideband Variable-Gain Amplifier Using an Interpolated Ladder Attenuator," **IEEE ISSCC Technical Digest**, 1991, pp. 280, 281, 330.

NOTES:

SECTION 6-5: AMPLIFIER IDEAS

Walt Jung, Walt Kester

This section of the chapter features miscellaneous op amp applications, within a format of *amplifier ideas*. They range broadly across the spectrum, illustrating many innovative op amp uses that don't otherwise fit categories. Some of the concepts have been inspired by publication elsewhere. In such cases, an appropriate original reference is given.

High Efficiency Line Driver

Conventional video line drivers use a series or *back-termination* resistor, selected to match the transmission line characteristic impedance. Although simple, this scheme is inherently inefficient, as both load and series termination resistors drop the same voltage. This isn't usually a problem with 1Vp-p video signals operating on high voltage supplies, such as $\pm 12\text{V}$ or $\pm 15\text{V}$. However, with lower voltage supplies, particularly 5V or less, driver headroom is definitely an issue. For such conditions, a conventional driver may simply not be able to accommodate a signal of twice V_{OUT} without distortion.

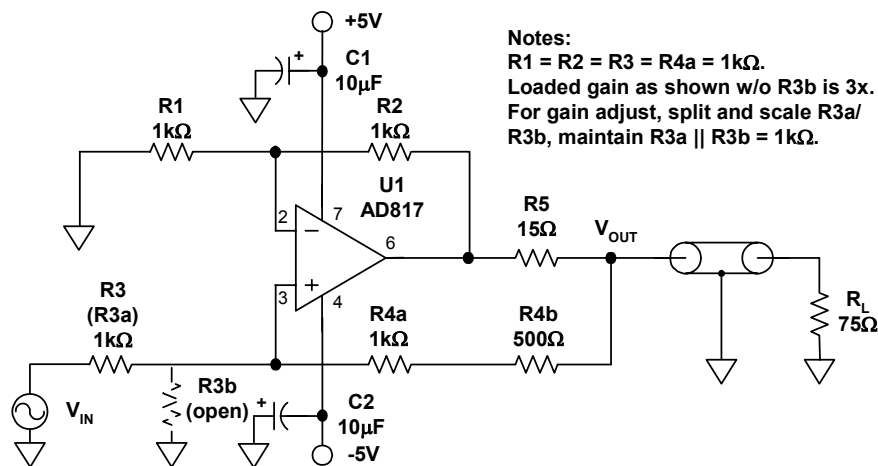


Figure 6-152: A high efficiency video line driver

Figure 6-152 illustrates a solution to this driver efficiency problem. In this line driver (adapted from a circuit by Victor Koren, see Reference 1), a Howland type of feedback configuration is used. This allows the series termination resistor $R5$ to be appreciably smaller, thus dropping less voltage and improving stage efficiency. Both positive and negative loop feedback paths are used around the op amp, $R3$ and $R4$, plus $R1$ and $R2$. An AD817 is chosen for its video characteristics, and line driving capability. The circuit also works with many other op amps, provided they have sufficient output drive.

In this example, a 75Ω line is being driven, and $R5$ is set to 15Ω . With the scaling chosen, this produces $1/5$ the voltage drop of a more conventional 75Ω resistor. For every volt of V_{OUT} , the amplifier needs only to produce 20% more, i.e., 1.2V per V of V_{OUT} . This allows the design to operate easily on 5V or even lower supplies, and still provide undistorted 1Vp-p video signals at V_{OUT} . The $\pm$ feedback paths produce the proper synthesized source impedance when the $R1$ - $R5$ resistors are properly selected.

■ OP AMP APPLICATIONS

Given the desired output impedance R_O , R_5 is related by a scaling factor, so that $R_5 < R_O$. A direct design approach is to simply set R_5 at some fraction of R_O , which then leads to a R_1 through R_4 resistor set that will provide the proper R_O . In this example design, R_5 is set at $1/5 R_O$ as noted earlier, or 15 ohms.

As per the notes of the figure, a major simplifying design step is to make four of the feedback resistors equal, namely R_1 through R_3 and R_{4a} . It also helps further to make these a common, readily available value. This should be a value moderately higher than the target load impedance. In this case, a 1k Ω base value is chosen.

This defines the R_4 value ($R_{4a} + R_{4b}$) as:

$$R_4 = (R_5 \bullet R_1 + R_O \bullet R_2) / (R_O - R_5) \quad \text{Eq. 6-31}$$

R_{4b} is then simply $R_4 - R_{4a}$. The design is further simplified with all of the noted resistors part of a single common array, *including* R_{4b} (which is made from two parallel 1k Ω resistors in this case). Note that R_{4b} won't necessarily be so easily achieved in other design examples. Nevertheless, it is desirable for as many of the R_1 through R_4 resistors as possible be part of a common array, matched to 1% or better.

Gain of the stage just as shown is about $3\times$ with the output loaded. If gain must be adjusted, there is a specific procedure to be followed. This is a necessary condition for proper stage function (for any gain), and is needed to maintain the synthesized R_O . For example, if a unity ($1\times$) overall gain is desired, R_3 can be changed to two resistors, i.e., $R_{3a} = 3\text{k}\Omega$, $R_{3b} = 1.5\text{k}\Omega$. Note that this reduces the drive to the op amp, but it also maintains the same 1k Ω Thevenin impedance for R_3 (where R_3 is the resistance looking back to the input from R_{4a}). Similarly, equal value 2k Ω resistors could be used, which provides a net loaded stage gain of $1.5\times$. Of course, for arbitrary gains, a common array may not be possible, and ordinary 1% metal film types can also be used.

Also related to the above, the driving source at V_{IN} must be a very low impedance with respect to 1k Ω , again so as to maintain the synthesized impedance relations. This is best achieved by use of an R_3 driving source direct from an op amp output. Alternately, if the V_{IN} driving impedance is both fixed and known, it can be subtracted from R_3 .

A more general caveat (which applies to all Howland circuits) is that the design environment must maintain this source driving impedance *for all conditions*, as the circuit itself is *not* open source stable. For example, if R_3 is opened, the positive feedback can override the negative feedback via R_1 and R_2 , and the circuit could latch up.

Finally, although this design illustrates a driver oriented to a video standard of 75 Ω with 1Vp-p signals, there is no reason why the same design principles cannot be applied to other impedances and/or signal levels.

A Simple Wide Bandwidth Noise Generator

While most electronic designs seek noise minimization, there are occasions where a known quantity and/or quality of spectrally flat (white) noise is desirable. One such example is a dither source for enhanced dynamic resolution A/D conversion. For such applications, it is useful to be able to predict the output of a noise generator. It turns out that a carefully chosen decompensated op amp set up to amplify its own input noise is very useful as a wideband noise generator (see Reference 2).

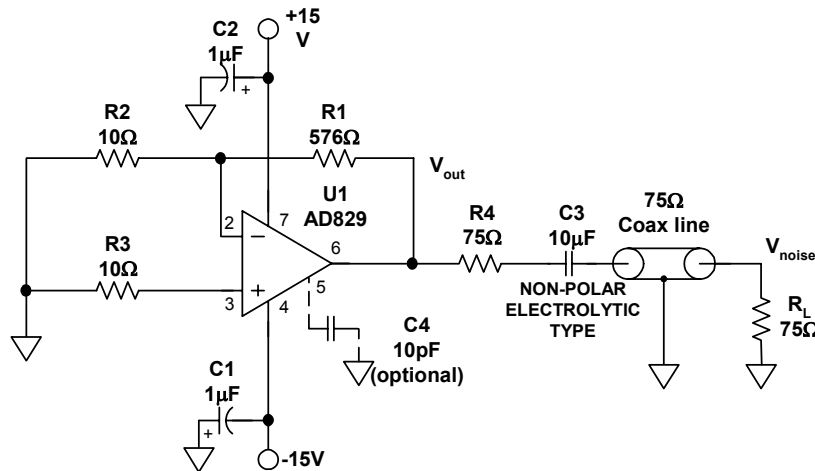


Figure 6-153: A simple wideband noise generator

Figure 6-153 above illustrates this technique, which simply employs the op amp U1 as a fixed gain stage, amplifying its input noise by the stage factor G , where $G = 1 + (R1/R2)$. This process is made easier by some simplifying assumptions, as described below.

By purposely selecting $R2$ and $R3$ values of 10Ω or less, their Johnson noise contribution is forced to be less than the voltage noise of the amplifier. Similarly, the amplifier's current noise components in $R2$ - $R3$, when converted to voltage noise, are also negligible. Thus the dominant circuit noise is reduced to the input voltage noise of U1.

To scale the amplifier noise to a given level of V_{noise} across R_L , select a stage gain which produces a noise density at V_{OUT} which is $2G$ times the typical U1 noise of $1.7\text{nV}/\sqrt{\text{Hz}}$. This will produce a V_{OUT} twice V_{noise} . For example, for a V_{noise} of $50\text{nV}/\sqrt{\text{Hz}}$, using a fixed $R2$ value of 10Ω , the required $R1$ is:

$$R1 = 10 \cdot (((2 \cdot V_{noise}) / 1.7) - 1) \quad \text{Eq. 6-32}$$

Where V_{noise} is in $\text{nV}/\sqrt{\text{Hz}}$, and the 1.7 is the U1 voltage noise ($\text{nV}/\sqrt{\text{Hz}}$). This computes to 576Ω (nearest standard value) for a wideband $50\text{nV}/\sqrt{\text{Hz}}$. Alternately, an audio range noise source of $1000\text{nV}/\sqrt{\text{Hz}}$ with several hundred kHz of bandwidth is achieved with $R1 = 11.8\text{k}\Omega$, and $C3 = 100\mu\text{F}$. By choosing a bipolar-input, voltage feedback U1 device, with a single effective gain stage, a major performance point is achieved. Such an amplifier has a flat, frequency independent input voltage noise response (i.e., a white noise characteristic). Many of the ADI high-speed amplifiers use this topology within a folded cascode architecture.

■ OP AMP APPLICATIONS

In contrast to this, multiple stage, pole-zero compensated amplifiers such as the OP27 (and other similar architectures) can have peaks in the output noise response. This is due to the frequency compensation method used, and the associated gain distributions in the signal path. When picking U1, look for a noise characterization plot which shows flat input referred voltage noise over several decades.

For the AD829 device used, input voltage noise is flat from below 100Hz to more than 10MHz, as is noted below in Figure 6-154. Within the actual circuit, the upper bandwidth limit will be gain/compensation dependent, which can be controlled as described below.

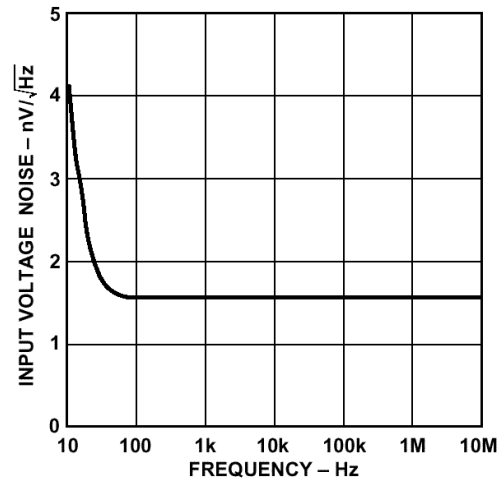


Figure 6-154: AD829 input voltage noise spectral density

The output is coupled through a non-polar capacitor, C3, which removes any amplified DC offset at V_{out} . The C3 value should be large enough to pass the lowest noise frequencies of interest. As shown the response of this network is -3 dB at about 100Hz, but C3 can be changed for other low frequency limits. Source termination resistance R4 allows standard 75Ω cables to be driven, providing distribution to a remote 75Ω load, R_L .

In general, this noise generator's utility is greatest with a decompensated (or externally compensated) op amp, to take advantage of the maximum bandwidth possible. For the AD829, bandwidth is highest with lowest pin 5 capacitance (i.e., no PCB pin 5 pad, or pin 5 cut off). Conversely, C4 can be used to reduce bandwidth, if desired. With minimum pin 5 capacitance, the AD829 gain-bandwidth can be above 500MHz, allowing extended response. In any case, the stage's effective -3 dB bandwidth varies inversely proportional to stage gain G. Some noise variations can be expected from IC sample-sample, so an R1 trim method can be used to set a output calibration level. Alternately, if ultra-wide bandwidth noise isn't required, another op amp to consider is the AD817.

A final note— bipolar input amplifiers such as the AD829 typically use PTAT biasing for the input's differential stage tail current. Since equivalent input noise varies as the square root of this tail current, this can make noise output vary somewhat with temperature. The net effect causes noise to change less than 1dB for a 50°C temperature change.

Single-Supply Half and Full-Wave Rectifier

There are a number of ways to construct half and full-wave rectifiers using combinations of op amps and diodes, but the circuit shown in Figure 6-155 below requires only a dual op amp, two resistors, and operates on a single supply (see Reference 3).

The circuit will work with any single-supply op amp whose inputs can withstand being pulled below ground. The AD820 (single) or AD822 (dual) op amps have N-channel JFET inputs, which allow the input voltage to go to 20V below the negative supply.

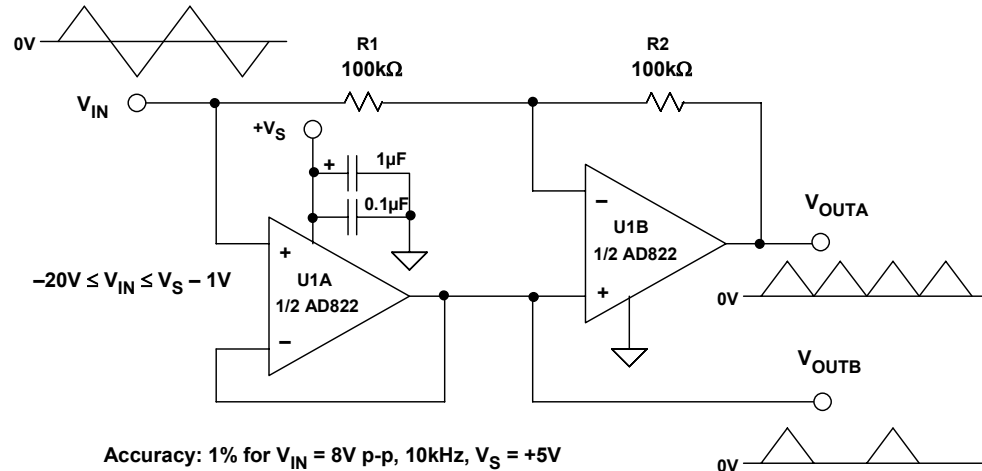


Figure 6-155: Single-supply half and full-wave rectifier uses no diodes

The output stage of these op amps is a complementary bipolar common emitter rail-to-rail stage with an output resistance of approximately 40Ω when sourcing current and 20Ω when sinking current. As a result of this stage, the outputs can go within a few millivolts of the supply rails under light loading.

When the input signal is above ground, unity-gain follower U1A and the loop of the amplifier U1B bootstrap R1. This bootstrapping forces the inputs of U1B to be equal. Thus, no current flows in R1 or R2, and the output V_{OUTA} tracks the input. Conversely, when the input is negative, the output of U1A is forced to zero (saturated). The non-inverting input of U1B sees the ground-level output of U1A, and during this phase operates as a unity-gain inverter, rectifying the negative portion of the input V_{IN} .

The net output at V_{OUTA} is therefore a full-wave rectified version of V_{IN} . In addition, a half-wave rectified version is obtained at the output of U1A (V_{OUTB}) if desired.

The circuit operates with a single power supply of +3 to +20V. The circuit will maintain an accuracy of better than 1% over a 10kHz bandwidth for inputs of 8Vp-p on a +5V supply. The input should not go more than 20V below the negative supply, or closer than 1V to the positive supply. Inputs of ±18V can be rectified using a single +20V supply.

Paralleled Amplifiers Drive Loads Quietly

Paralleling op amps is a method to increase load drive while keeping output impedance low, and also to reduce noise voltage. Figure 6-156A below shows a classic stacked-amplifier circuit. This configuration halves the input voltage noise of a single op amp, and quadruples load drive. However, it does have several weaknesses.

First, you need to set the correct gain for each amplifier *individually*. Second, series resistors must be added to each output, to ensure equal load current distribution among the op amps. Third, the input range can become limited at high gains because of the inherent offset of any of the amplifiers.

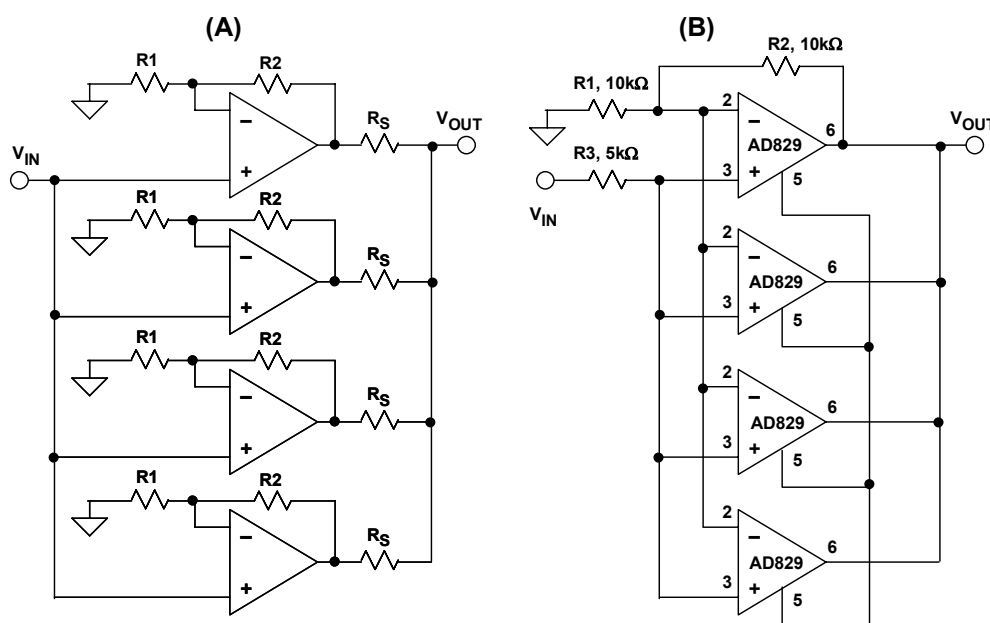


Figure 6-156: Paralleled amplifiers drive loads quietly

The circuit shown in Figure 6-156B also has half the noise voltage of an individual amplifier, and it also quadruples the load drive. But in so doing, it reduces the component count from twelve resistors to three (see Reference 4). In addition, the circuit has a gain-bandwidth product of about 750MHz. Although the topology of Figure 6-156B is generally applicable to all externally compensated amplifiers (i.e., those with a pinned-out high impedance node before the output driver stage— such as the AD844 and AD846), the AD829 op amp is particularly well suited to video and other broadband applications.

Note that in the B circuit, external R_S load sharing resistors aren't required, because the only voltage difference between the individual outputs is due to the slight offset mismatch between the AD829 complementary emitter follower output driver stages, and internal 15Ω emitter resistors ensure equal output current distribution. The closed-loop gain has no effect on this small offset voltage.

The end result of the 4 paralleled output stages in Fig. 6-156B is a composite amplifier with both greater load drive and lower noise, but using only the conventional feedback

components. The circuit in B increases the drive current by a factor of four, similar to A, but with a vast difference in the parts count.

In order to understand how the circuit reduces noise, let the RTI voltage noise of the individual op amps be V_{N1} , V_{N2} , V_{N3} , and V_{N4} , and let the total noise voltage be V_N .

Because all the inputs are connected in parallel, as well as the high impedance nodes, then:

$$(V_N - V_{N1})g_m + (V_N - V_{N2})g_m + (V_N - V_{N3})g_m + (V_N - V_{N4})g_m = 0 \quad \text{Eq. 6-33}$$

$$V_N = \frac{1}{4} (V_{N1} + V_{N2} + V_{N3} + V_{N4}). \quad \text{Eq. 6-34}$$

But because the voltage noise of the amplifiers is uncorrelated, and the noise spectral density for each amplifier is the same:

$$V_N = \frac{1}{4} \sqrt{4(V_{N1})^2} \quad \text{Eq. 6-35}$$

$$V_N = V_{N1} / 2 \quad \text{Eq. 6-36}$$

This result also implies that all uncorrelated parameters such as input offset voltage, input offset voltage drift, CMRR, PSRR, etc., will also approach their true mean values, thus reducing effects arising from the variability of the devices.

The AD829 is flexible and can operate on supply voltages from $\pm 5V$ to $\pm 15V$. Its uncompensated gain-bandwidth product is 750MHz. Nominal output current for rated performance is 20mA, so in the circuit shown, 80mA is available to drive the load.

The input voltage noise of a single AD829 is $1.7nV/\sqrt{Hz}$, so the parallel circuit has an input voltage noise of approximately $0.85nV/\sqrt{Hz}$. In order to take advantage of this low voltage noise, however, the circuit must be driven from a relatively low source impedance, because the input current noise of a single AD829 is $1.5pA/\sqrt{Hz}$. In the parallel circuit, the input current noise is therefore $3pA/\sqrt{Hz}$.

Notice that in the AD829 circuit, $R_3 = R_1 || R_2$ for bias current cancellation. This works because the input bias currents of the AD829 are not internally compensated: they are approximately equal, and of the same sign. If amplifiers with internal bias current compensation or current feedback op amps are used, the input bias currents may not be equal or of the same sign, and R_3 should be made equal to zero.

Power-Down Sequencing Circuit for Multiple Supply Applications

The operating time of battery operated portable equipment can be extended by using power-down techniques. Many new components offer a power-down feature to implement this function. However, there may be times when this feature is not offered and other means must be devised. The solution may also require proper power supply sequencing in multiple supply systems.

Figure 6-157 shows a single-supply op amp powered from +15V driving an single-supply ADC powered from +5V. In many cases, the same +5V can supply the op amp and the ADC, and in that case, there is no sequencing problem. However, in some cases better system performance is obtained by driving the op amp with a higher supply voltage.

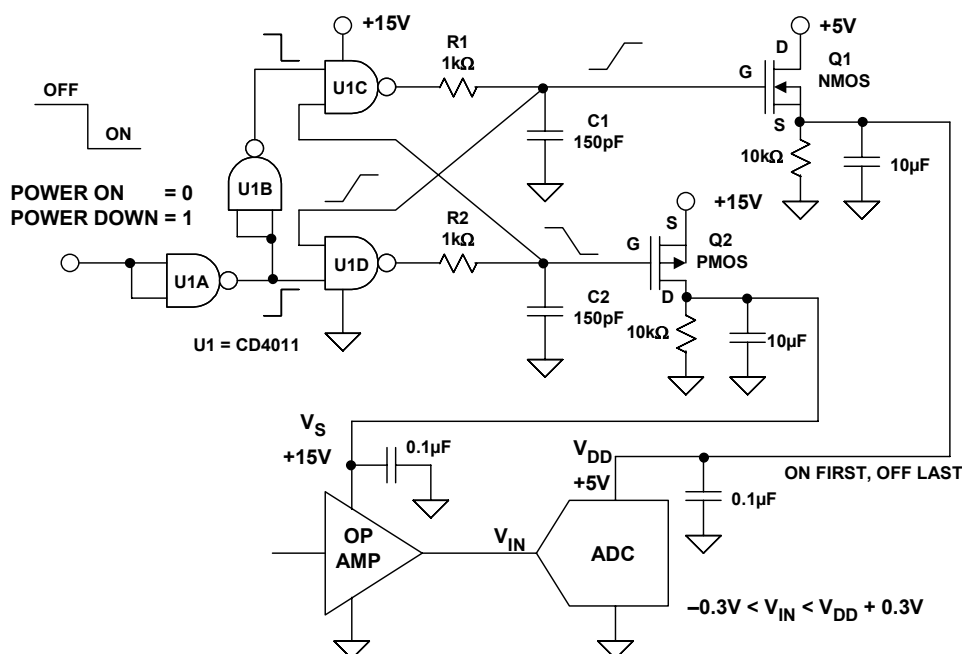


Figure 6-157: Power-down sequencing circuit for multiple supply applications

In the circuit, MOSFETs Q1 and Q2 switch the +5V and the +15V to the devices in the proper sequence. On power-up, the voltage to the ADC must be supplied first; and on power-down, the voltage to the ADC must be removed last. This is to ensure that the V_{IN} input to the ADC is never more than 0.3V above the V_{DD} positive supply or more than 0.3V below the negative supply, thereby preventing damage and possible latch-up.

The MOSFETs, Q1 and Q2 switch the +5V and the +15V to the ADC and the op amp, respectively, in a sequence controlled by two cross-coupled CD4011 CMOS NAND gates (U1C and U1D). The gates are powered from the +15V supply so that sufficient gate drive voltage is available to turn Q1 and Q2 on and off.

To initiate the power-on mode, a logic 0 is applied to the input of U1A, forcing its output high. This forces the output of U1B low, which causes U1C's output to go high. The R1-C1 time constant delays the application of the +15V to the gate of Q1 which ultimately

turns Q1 on with 5V at its source. The delayed output of U1C is also applied to an input of U1D which forces its output low. U1D's output is delayed by the R2-C2 time constant, and ultimately forces the gate of Q2 to zero, which applies +15V to the drain of Q2, and to the op amp V_S supply.

To initiate the power-down mode, a logic 1 is applied to the input of U1A which forces its output to zero, and the output of U1D is forced high, ultimately turning off Q2, the +15V supply. The delayed output of U1D is applied to an input of U1C, the output of U1C goes low, and ultimately the gate of Q1 is forced to zero, turning off the +5V to the V_{DD} input of the ADC.

It is important to note that when system power is applied to the overall circuit, the +5V should come up either before, or simultaneously with the +15V. Similarly, when system power is removed, the +15V should be removed first or simultaneously with the +5V.

This circuit is based on a modification of the one described in Reference 5, where the desired sequencing is the reverse of the one described here. The reverse sequence (+15V turned on before +5V, and +5V turned off before +15V) can be easily achieved by replacing the CD4011 NAND gates with CD4001 NOR gates, reversing the "sense" of the power-on/power-down control input, and swapping the gate drive signals to Q1 and Q2.

Programmable Pulse Generator Using the AD8037 Clamping Amplifier

The AD8036 ($G \geq 1$ stable) and AD8037 ($G \geq 2$ stable) clamp amplifier outputs can be set accurately to a well controlled flat levels determined by the clamping voltages. This, along with wide bandwidth and high slew rate suits them well for numerous applications.

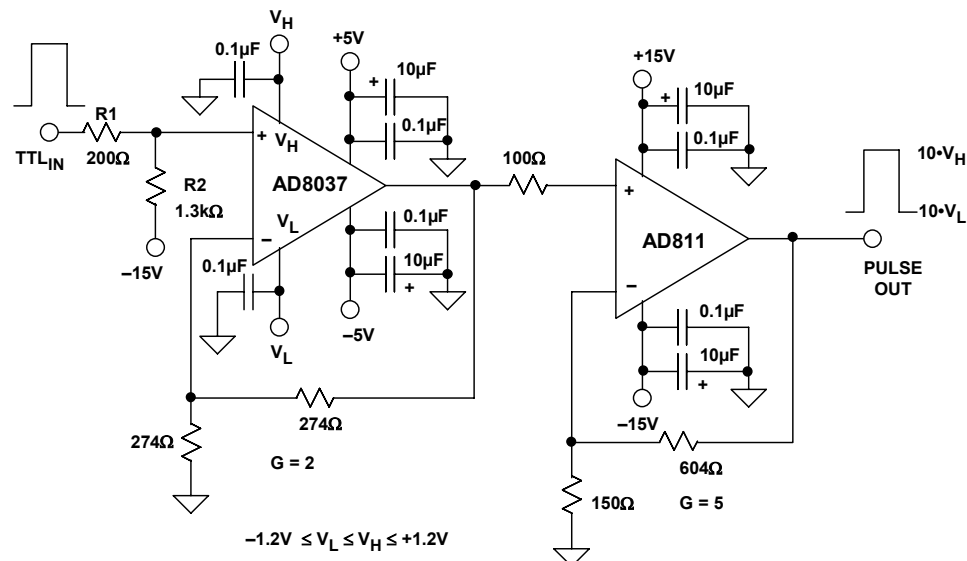


Figure 6-158: Programmable pulse generator using AD8037 clamping amplifier

A basic description of the AD8036/AD8037 operation can be found in the Video Applications section of this chapter. Figure 6-158 above is a diagram of a programmable level pulse generator (see Reference 6).

■ OP AMP APPLICATIONS

The circuit accepts a TTL timing signal for its input and generates pulses at the output up to 24Vp-p with 2500V/ μ s slew rate. The output levels can be programmed to anywhere in the range between -12V to $+12\text{V}$.

The circuit uses an AD8037 operating at a gain of two with an AD811 to boost the output to the $\pm 12\text{V}$ range. The AD811 was chosen for its ability to operate with $\pm 15\text{V}$ supplies and its high slew rate. R1 and R2 level shift the TTL input signal level approximately 2V negative, making it symmetrical above and below ground. This ensures that both the high and low logic levels will be clamped by the AD8037. For well controlled signal levels in the output pulse, the high and low output levels result from the clamping action of the AD8037 and aren't controlled by either the high/ low logic levels passing through a linear amplifier. For good output rise/fall times, logic with high edge speed should be used.

The high logic levels are clamped at 2 times the voltage at V_H , while the low logic levels are clamped at two times the voltage at V_L . The output of the AD8037 is amplified by the AD811 operating at a gain of 5. The overall gain of 10 will cause the high output level to be 10 times the voltage at V_H , and the low output level 10 times the voltage at V_L . For this gain, the clamping levels for a $\pm 12\text{V}$ output pulse are $V_H = +1.2\text{V}$ and $V_L = -1.2\text{V}$.

Full-Wave Rectifier Using the AD8037 Clamping Amplifier

The clamping inputs can be used as additional inputs to the AD8036/AD8037. As such, they have an input bandwidth comparable to the amplifier inputs and lend themselves to some unique functions when they are driven dynamically.

Figure 6-159 (opposite) is a schematic for a full wave rectifier, also called an absolute value generator (Reference 6 again). It works well up to 20MHz and can operate at significantly higher frequencies with some performance degradation. The distortion performance is significantly better than diode-based full-wave rectifiers, especially at high frequencies.

The AD8037 is configured as an inverting amplifier with a gain of unity. The V_{IN} input drives the inverting amplifier and also directly drives V_L , the lower level clamping input. The high level clamping input, V_H , is left floating and plays no role in the circuit.

When the input is negative, the amplifier acts as a unity-gain inverter and outputs a positive signal at the same amplitude as the input, with opposite polarity. V_L is driven negative by V_{IN} , so it performs no clamping action, because the positive output signal is always higher than the negative level driving V_L .

When the input is positive, the output result is the sum of two separate effects. First, the inverting amplifier multiplies the V_{IN} input by -1 , because of the unity-gain inverting configuration. This effectively produces an offset at the output, but with a dynamic level that is equal to -1 times the input. Second, although the positive input is grounded (through 100Ω), the output is clamped at two times the voltage applied to V_L (a positive, dynamic voltage in this case). The factor of two is because the $2\times$ amplifier noise gain.

The sum of these two actions results in an output that is equal to unity times the input signal for positive input signals, as shown in Figure 6-159. Thus, for either positive or negative input signals, the output is unity times the absolute value of the input signal. The circuit can be easily configured to produce the negative absolute value of the input by applying the input to V_H rather than V_L .

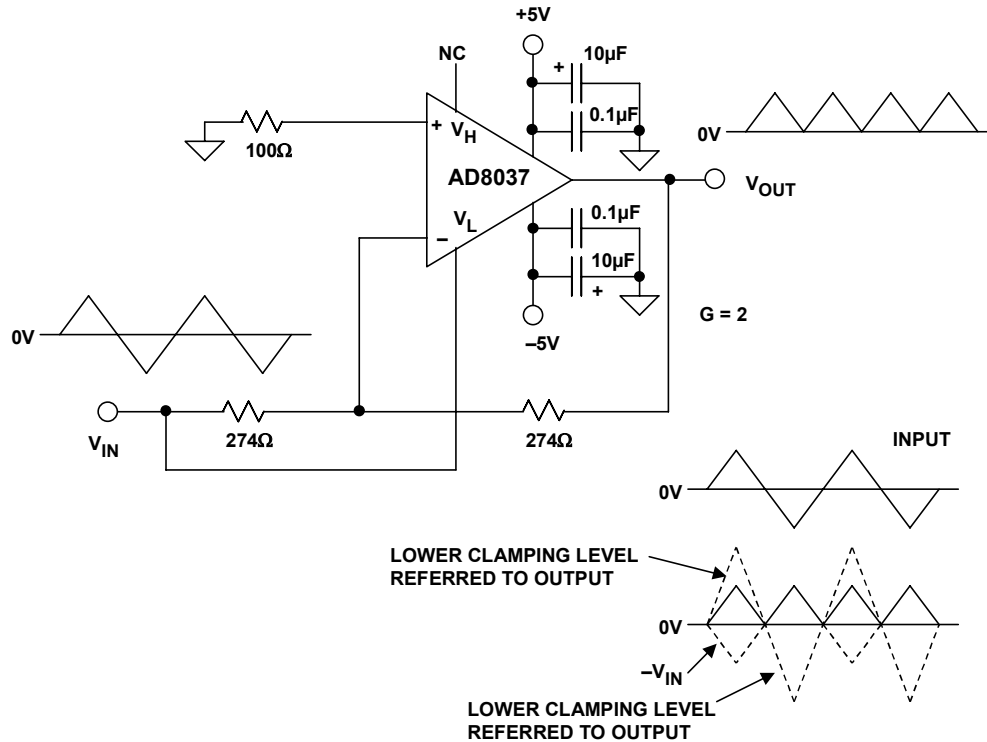


Figure 6-159: Full-wave rectifier using the AD8037 clamping amplifier

The circuit can get to within about 40mV of ground during the time when the input crosses zero. This voltage is fixed over a wide frequency range, and is a result of the switching between the conventional op amp input and the clamp input. However, because there are no diodes to rapidly switch from forward to reverse bias, the performance far exceeds diode-based full wave rectifiers. Signals up to 20MHz can be rectified with minimal distortion.

If desired, the 40mV offset can be removed by adding an offset to the circuit, with little additional complexity. A 27.4kΩ input resistor to the inverting input will have a gain of 0.01, while changing the gain of the circuit by only 1%. A plus or minus 4V DC level (depending on the polarity of the rectifier) fed into this resistor will then compensate for the offset.

Full wave rectifiers are useful in many applications including AM signal detection, high frequency AC voltmeters, and various arithmetic operations.

AD8037 Clamping Amplifier Amplitude Modulator

The AD8037 can also be configured as an amplitude modulator as shown in Figure 6-160 (Reference 6 again). The positive input of the AD8037 is driven with a square wave of sufficient amplitude to produce clamping action at both the high and low levels set by V_H and V_L . This is the higher frequency carrier signal.

The modulation signal is applied to both the input of a unity gain inverting amplifier and to V_L , the lower clamping input. V_H is biased at +0.5V for the example to be discussed but can assume other values.

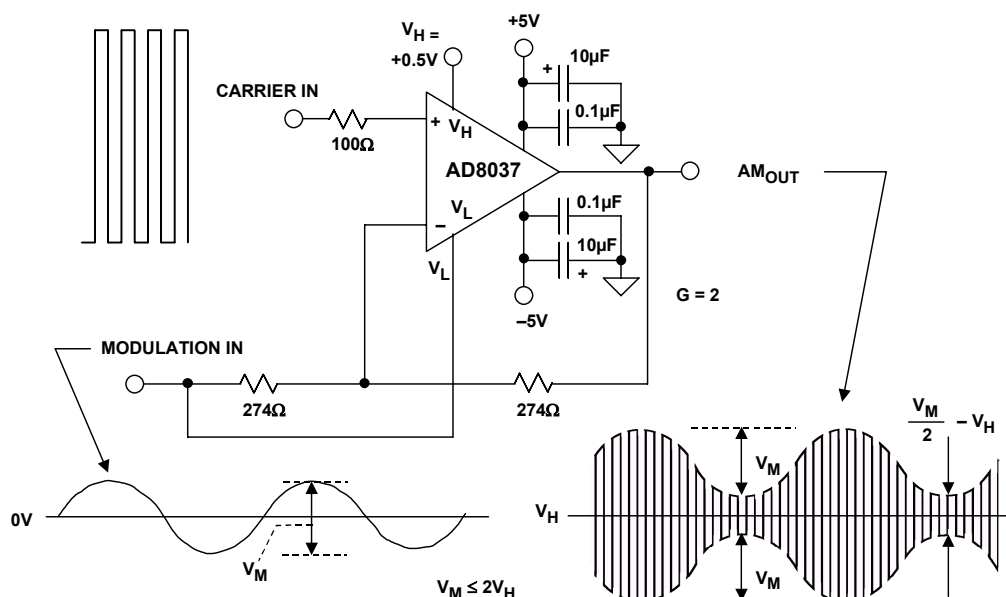


Figure 6-160: AD8037 clamping amplifier amplitude modulator

To understand the circuit operation, it is helpful to first consider a simpler circuit. If both V_H and V_L are DC biased at +0.5V and -0.5V, respectively, and the carrier and modulation inputs driven as above, the output would be a 2Vp-p square wave at the carrier frequency riding on a waveform at the modulating frequency.

The inverting input (modulation signal) is creating a varying offset to the 2Vp-p square wave at the output. Both the high and low levels clamp at twice the input levels on the clamps because the noise gain of the circuit is two.

When V_L is driven by the modulation signal instead of being held at a DC level, a more complicated situation results. The resulting waveform is composed of an upper envelope and a lower envelope with the carrier square wave in between. The upper and lower envelopes are 180° out of phase as in a typical AM waveform.

The upper envelope is produced by the upper clamp level being offset by the waveform applied to the inverting input. This offset is the opposite polarity of the input waveform because of the inverting configuration.

The lower envelope is produced by the sum of two effects. First, it is offset by the waveform applied to the inverting input as in the case of the simpler circuit above. The polarity of this offset is in the same direction as the upper envelope.

Second, the output is driven in the opposite direction of the offset at twice the offset voltage by the modulation signal being applied to V_L . This results from the noise gain being equal to two, and since there is no inversion in this connection, it is opposite in polarity from the offset.

The result at the output for the lower envelope is the sum of these two effects, which produces the lower envelope of an AM waveform. The depth of modulation can be modified by changing the amplitude of the modulation signal. This changes the amplitude of the upper and lower envelope waveforms.

The modulation depth can also be changed by changing the DC bias applied to V_H . In this case, the amplitudes of the upper and lower envelope waveforms stay constant, but the spacing between them changes. This alters the ratio of the envelope amplitude to the amplitude of the overall waveform.

For $V_H = +0.5V$, 100% modulation occurs when the peak-to-peak amplitude of the modulation input $V_M = 1V$. The AM output is always offset by V_H for a bipolar modulation input. In general, for a peak-to-peak modulation amplitude of V_M , the two output modulated envelopes are separated by an amount equal to $V_M/2 - V_H$.

Sync Inserter Using the AD8037 Clamping Amplifier

Video signals typically combine an active video region with both horizontal and vertical blanking intervals during their respective retrace times. A sync signal is required during the blanking intervals in some systems. In RGB systems, the sync is usually inserted on the Green signal. In composite video systems, it is inserted during blanking on the single-channel composite signal, or on the luminance (or Y) signal in an S-video system. Further details on video signals can be found in the Video Applications section of this chapter.

The AD8037 input clamping amplifier can be used to make a video sync inserter that does not require accuracy in the amplitude or shape of the sync pulse (see Reference 7). The circuit shown below in Figure 6-161 uses the AD8037 to create the proper amplitude sync insertion, and the DC levels of the sync pulse do not affect the active video level. The circuit is also non-inverting with a gain-of-two, which allows for driving a back-terminated cable with no loss of amplitude.

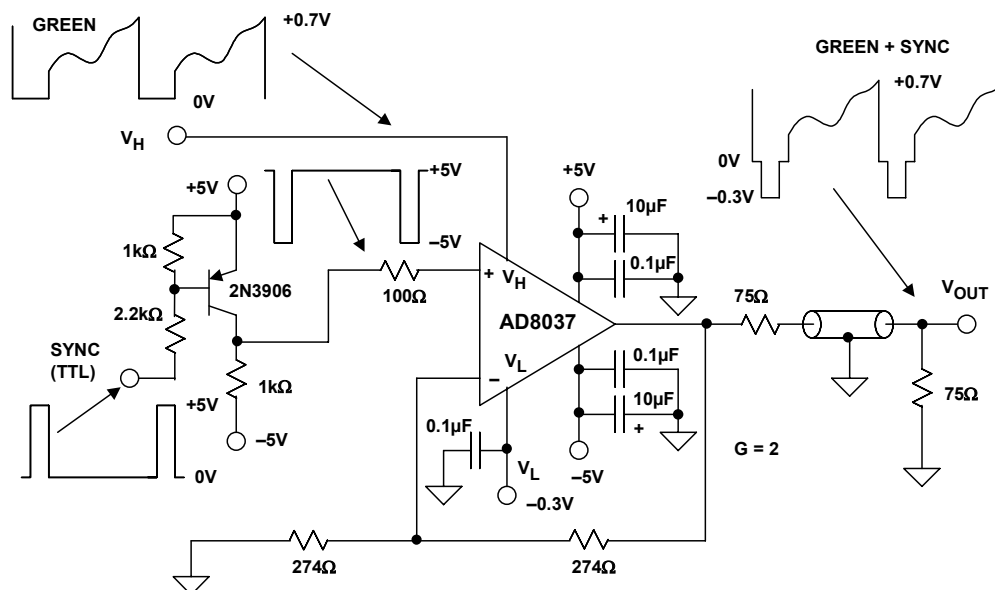


Figure 6-161: Sync inserter using the AD8037 clamping amplifier

The Green video signal is applied to the V_H input of the AD8037. This signal has a blanking level of 0V and an approximate full-scale value of +0.7V. The TTL-level sync pulse is applied to the base of the 2N3906 transistor. The signal at the collector of the 2N3906 are inverted sync pulses with an amplitude of 10Vp-p which are applied to the non-inverting input of the AD8037.

The amplitude of the signal into the non-inverting input of the AD8037 is +5V during the active video portion, and since this is greater than the maximum positive Green signal excursion of +0.7V on the V_H input, the Green signal is passed through to the output of the AD8037 with a gain of +2.

During the blanking interval, the sync signal into the non-inverting input of the AD8037 goes to -5V, and the output of the AD8037 is clamped to a value which is 2 times the DC

level on the V_L input. Nominally, the sync should be -0.3V referenced to a 0V blanking level, and this level is applied to V_L from a voltage reference, or a simple divider.

The high and low levels of the sync pulse generated by the 2N3906 can be relatively loosely defined. The value of the high-level sync input to the non-inverting input of the AD8037 must be higher than the active video signal; the value of the low-level sync input must be lower than the DC voltage on the V_L input. The rising and falling edges of the sync pulse input determine the timing of the inserted sync, but the DC level at the V_L input of the AD8037 will always determine the sync amplitude.

The 2N3906 PNP transistor serves as a level translator and simply provides an appropriate drive signal from a TTL source of positive-going sync. The sync input to the non-inverting input of the AD8037 neither influences the DC level of the output video nor determines the amplitude of the inserted sync.

AD8037 Clamped Amplifier As Piecewise Linear Amplifier

Piecewise linear amplifiers are often implemented using diodes in the feedback loop of an op amp. When the diodes become forward biased, they switch in resistors that alter the closed-loop gain of the amplifier.

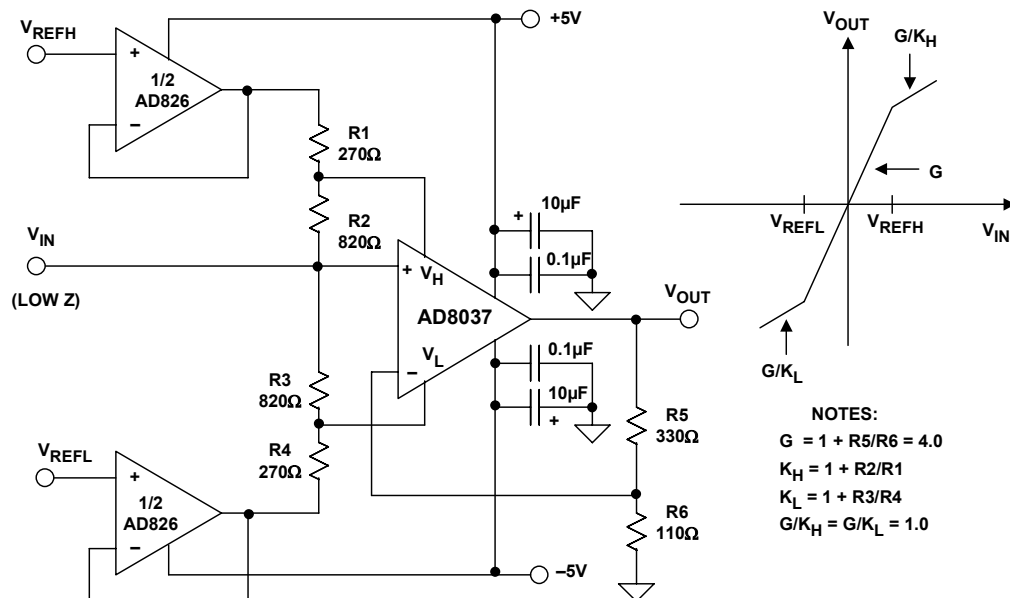


Figure 6-162: Piecewise linear amplifier uses AD8037 clamping amplifier rather than diodes

This approach has three disadvantages. First, the diode's forward voltage drop (even with Schottky diodes) reduces accuracy and speed during the switching region. Second, diode stray capacitance can limit bandwidth. Third, the $2\text{mV}/^\circ\text{C}$ drift of the diode's forward bias voltage introduces errors in the transfer function. The circuit shown in Figure 6-162 avoids these problems by using the fast and accurate clamping function of the AD8037 to set the breakpoints (see Reference 8).

■ OP AMP APPLICATIONS

If the V_{IN} signal applied to the noninverting input of the AD8037 lies between the clamp voltages (set by V_{REFH} and V_{REFL}), the AD8037 works as a standard op amp with a gain = $G = 1 + R5/R6$. If the input signal is greater than the upper clamp voltage, V_H , the amplifier disconnects the input signal, and V_H becomes the noninverting signal input. Likewise, if the signal at the noninverting input of the AD8037 is below the lower clamp voltage, V_L , then the amplifier also disconnects the signal input, and V_L becomes the noninverting signal input.

Figure 6-162 also graphically illustrates the operation of the circuit. When V_{IN} is between V_{REFH} and V_{REFL} , the circuit is a standard noninverting op amp with a gain $G = 1 + R5/R6$. When V_{IN} is greater than V_{REFH} , V_H becomes the noninverting input to the amplifier.

The transfer function from V_{IN} to V_{OUT} comprises two parts under this condition. From V_{IN} to V_H , the signal is attenuated by a factor $K_H = 1 + R2/R1$. From V_H to V_{OUT} , the gain remains $G = 1 + R5/R6$. This leads to an overall gain of G/K_H in this region. The circuit behaves similarly when V_{IN} is below V_{REFL} . The gain in this condition is G/K_L , where $K_L = 1 + R3/R4$.

Careful layout ensures adherence to the desired nonlinear transfer function over a 5MHz bandwidth. The stability of the breakpoints is determined by the tracking of the resistor temperature coefficients, the $10\mu V/^\circ C$ offset voltage drift of the AD8037, and the temperature stability of the reference voltages.

The reference voltages can be generated using precision voltage references or DACs. To maintain accuracy, the reference voltages should be buffered with a fast op amp, such as the dual AD826, to provide a low source impedance throughout the input signal bandwidth.

The analog input voltage, V_{IN} , should also be driven from a low impedance source, such as an op amp, to prevent errors due to the loading effect of the $R1$ - $R2$ - $R3$ - $R4$ network.

Using the AD830 Active Feedback Amplifier as an Integrator

The active feedback amplifier topology used in the AD830/AD8130 can be used to produce a precision voltage-to-current converter, which in turn, makes possible the creation of grounded-capacitor integrators (see Reference 9).

The design discussed here uses the AD830 to deliver a bipolar output current at high impedance (see Figure 6-163 below). Using $R = 1\text{k}\Omega$, the output current is simply equal to V_{IN}/R , or 1mA per volt of input. The maximum output current is limited to $\pm 30\text{mA}$ by the output drive capability of the AD830.

The output resistance is determined by the CMR performance of the AD830. A CMR of 60dB yields an effective output resistance of $1000 \times R$. The output impedance at any frequency can be determined by consulting the CMR data provided in the data sheet. The compliance range on the output is $\pm (V_S - 2V)$ reduced or increased by V_{IN} .

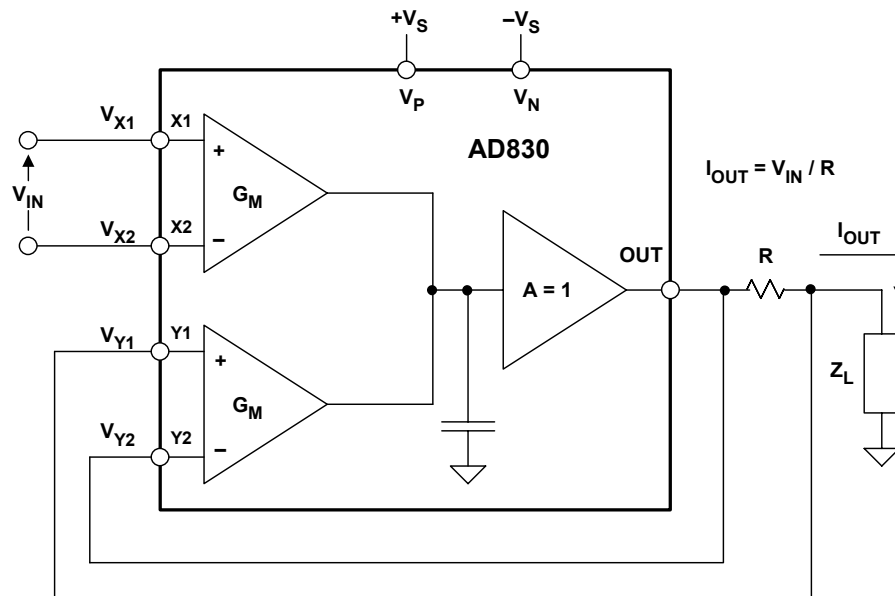


Figure 6-163: Constant current source using the AD830 active feedback amplifier

Figure 6-164A (next page) shows a standard op-amp integrator circuit using the AD825, and Figure 6-164B shows the improved AD830 grounded-capacitor circuit. The DC operating point for testing purposes is determined by R_1 in the AD825 op amp circuit, and by R_1 and C_1 in the AD830 circuit. R_2 and C_2 determine the integrator time constant in both circuits.

If the op amp in Figure 6-164A is assumed to be ideal, i.e., zero output impedance, and infinite input impedance, then the only difference between the two circuit topologies is the finite input resistance of the op amp based integrator as set by R_2 .

■ OP AMP APPLICATIONS

However, in a real op amp, the output resistance is finite and increases with signal frequency as the open-loop gain decreases. This causes the "ground" at the output of the op amp to degrade at high frequencies. The result is a relatively large spike on the output voltage waveform whenever the input switches.

This can be explained as follows. Assume the input V_{IN} switches between $-V_A$ and $+V_A$. When the input is at $-V_A$ long enough for the op amp to settle, then the current in resistor R_2 is $-V_A/R_2$, and the output increases due to C_2 being charged by the op amp. As V_{IN} suddenly switches to $+V_A$, the voltage across C_2 cannot change instantaneously, and neither can the op amp's output because it behaves itself as an integrator. This implies that the change in input voltage will be impressed upon the voltage divider formed by the R_O of the op amp and R_2 . This change in voltage at V_{OUT} will also be coupled by C_2 to the summing node at the inverting input of the op amp.

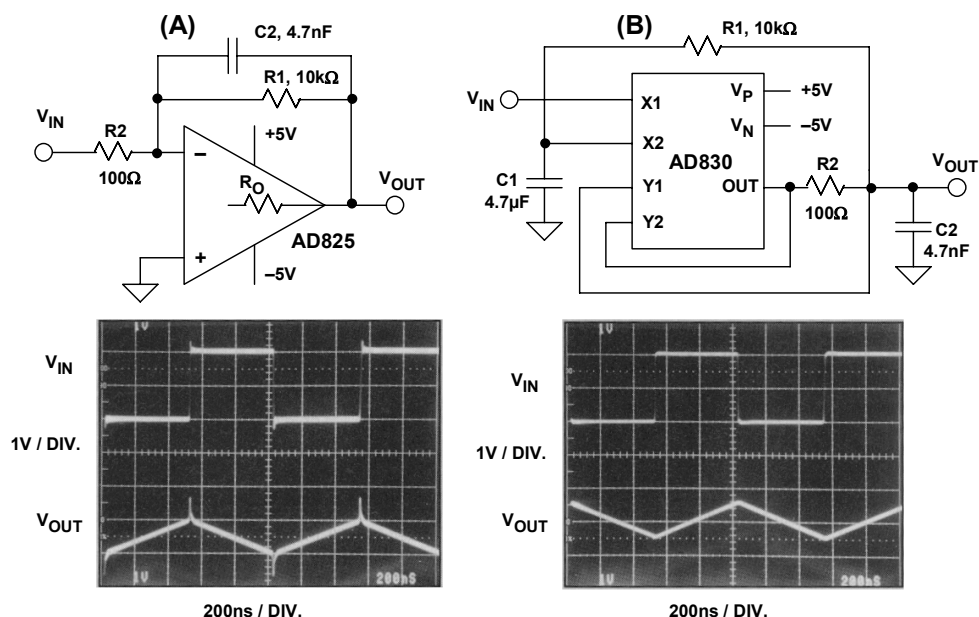


Figure 6-164: Traditional op amp integrator vs. grounded capacitor integrator

If V_{IN} is generated by a source with finite source resistance, this voltage spike will also appear at the input. Only after the amplifier settles will the external components again define the integrator time constant and the circuit function as desired.

It can be seen by comparing the waveforms of (A) and (B) that no spike develops in the output waveform produced by the grounded capacitor integrator using the AD830. This is because the integrating capacitor is connected to a true ground. In addition, the input is completely isolated from the output. Therefore, if an aberration did occur, it would not be coupled back to the driving source.

Various active filter topologies can be realized from this fundamental integrator building block. For example, two such sections can implement a biquad. An example of a simple all-pass filter using the AD830 is described in Reference 10.

Instrumentation Amplifier with 290MHz Gain-Bandwidth

The circuit shown in Figure 6-165 below combines a dual AD828 op amp with the AD830 active feedback difference amplifier to form a high frequency instrumentation amplifier (see Reference 11). The circuit's performance for $\pm 5V$ supplies for gains of 10 and 50 are shown in the figure, along with appropriate component values.

The circuit can be configured for different gains, and will operate on supplies ranging from $\pm 4V$ to $\pm 16.5V$.

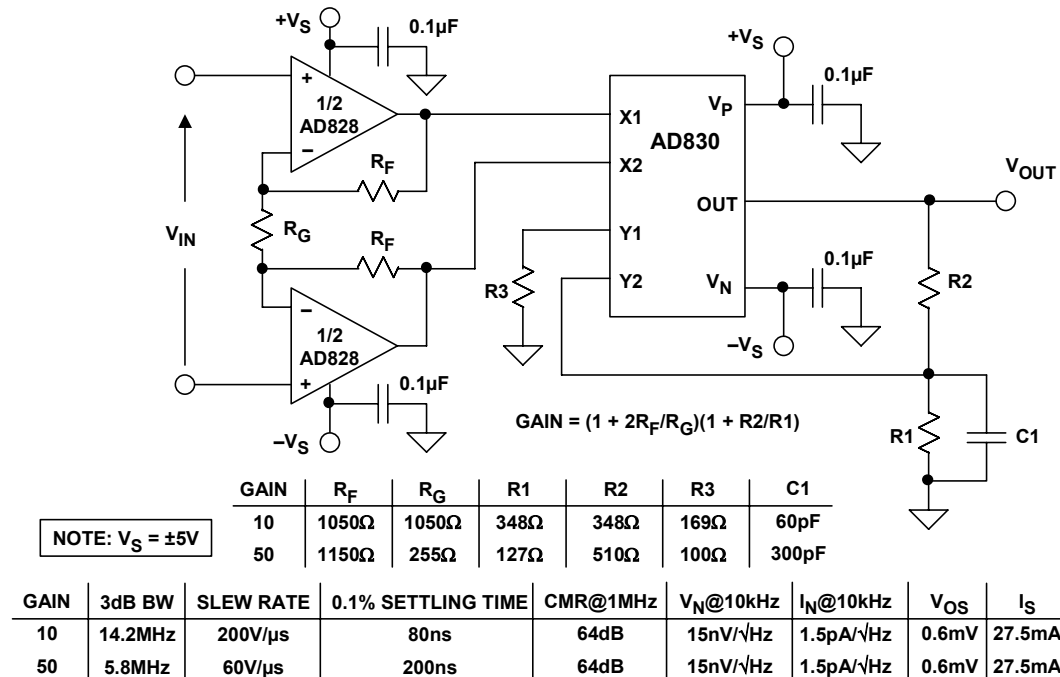


Figure 6-165: Instrumentation amplifier with 290MHz gain-bandwidth

The gain is proportioned between the AD828 stage and the AD830 stage, such that the closed-loop bandwidths of both stages are approximately equal. Under these conditions, a gain-bandwidth of 290MHz is obtained.

The input AD828 stage dominates the effective referred-to-input (RTI) input voltage noise and offset voltage. Capacitor C1 causes gain peaking in the AD830 which compensates for the AD828 input stage rolloff. The optimum value for C1 must be determined experimentally in a prototype or by a careful SPICE evaluation.

Note that R3 is made equal to the parallel combination of R1 and R2 to provide first-order input bias current cancellation at the Y1-Y2 input of the AD830.

■ OP AMP APPLICATIONS

Programmable Gain Amplifier with Arbitrary Attenuation Step Size

The R/2R ladder is a popular resistor topology often used to implement a current or voltage 6dB step attenuator. However, if the resistors are appropriately scaled, the network can be modified to provide any desired attenuation step.

A programmable gain amplifier (PGA) can be made with a attenuating ladder network followed by CMOS multiplexer and a fixed gain amplifier, as in Figure 6-166 below. This circuit has several advantages (see Reference 12).

First, as stated above, the attenuation step size doesn't have to be 6dB. Manipulating the resistor ratios, as described below, can easily change it. Second, the bandwidth of the circuit is always the same, regardless of the attenuation, due to the fact that the op amp buffer operates at a fixed gain. Third, the circuit is flexible, because practically any CMOS multiplexer and op amp can be used. The bandwidth of the circuit is determined primarily by the output op amp. Switching time between gain settings is determined by the multiplexer switching time and the op amp settling time.

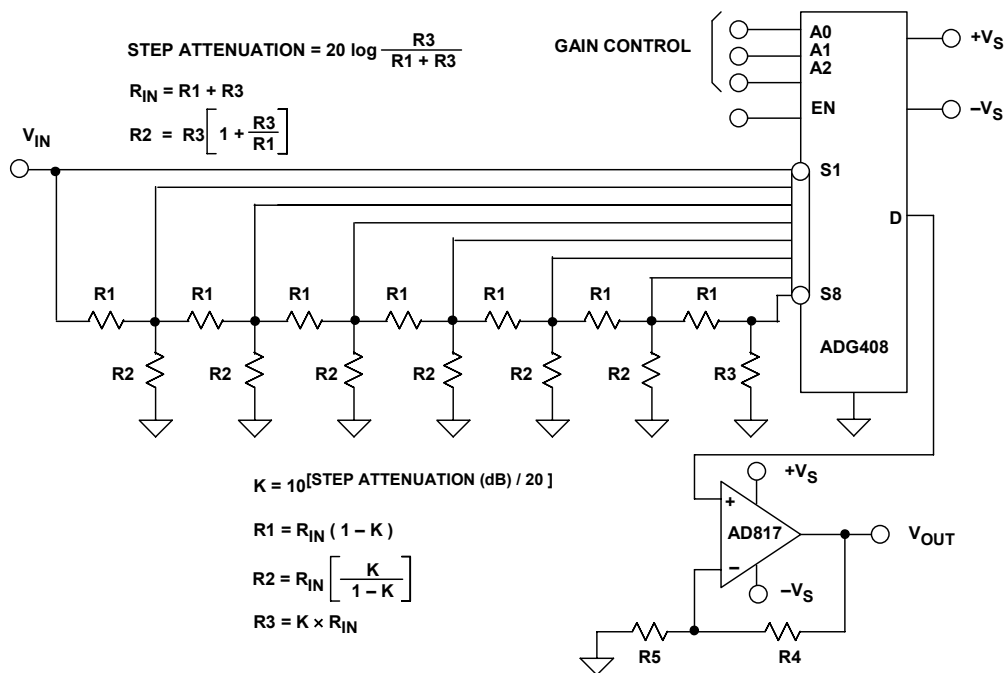


Figure 6-166: Programmable gain amplifier with arbitrary attenuation step size

The resistor ladder as shown uses three different resistor values: R_1 , R_2 , and R_3 . The step attenuation in dB is given by

$$\text{Step Attenuation (dB)} = 20 \log [R_3 / (R_1 + R_3)] \quad \text{Eq. 6-37}$$

Also, the following relationships apply:

$$R_{IN} = R1 + R3 \quad \text{Eq. 6-38}$$

$$R2 = R3 [1 + R3/R1] \quad \text{Eq. 6-39}$$

If $R1 = R3$, then $R2 = 2 \times R1$. In this case, the R-2R network provides 6dB step attenuation.

To determine the resistor values for a specific step attenuation and input resistance, use the formulas as follows:

$$K = 10^{[\text{Step Attenuation (dB)} / 20]} \quad \text{Eq. 6-40}$$

where the step attenuation is entered as a negative number.

Then, the following equations complete the design:

$$R1 = R_{IN} (1 - K) \quad \text{Eq. 6-41}$$

$$R2 = R_{IN} \times K / (1 - K) \quad \text{Eq. 6-42}$$

$$R3 = K \times R_{IN} \quad \text{Eq. 6-43}$$

For example, to implement a resistor ladder with a -1.5dB step attenuation and a 500Ω input impedance: $K = 0.8414$, $R1 = 79.3\Omega$, $R2 = 2653\Omega$, and $R3 = 420.7\Omega$ using the above equations.

The gain of the op amp is equal to $1 + R4/R5$. The overall gain of the PGA is equal to the op amp gain minus the attenuation setting.

Finally, it is interesting to note that the AD60x-series of X-Amps® discussed in the previous "Communications Amplifiers" section of this chapter uses the same basic approach described above. In the AD60x X-Amp® series however, attenuation is continuously variable, because an interpolation circuit rather than a multiplexer is used to connect the individual taps of the network to the op amp input.

■ OP AMP APPLICATIONS

A Wideband In-Amp

Some op amps with provisions for external offset trim can be used in unusually creative ways. In fact, if the two offset null inputs are considered as an additional differential signal input pair, this point becomes more clear. Although designed principally for adjustment of device V_{OS} , the null inputs can often be used for additional signals. An example is the wideband in-amp of Gerstenhaber and Gianino (see Reference 13). In the circuit of Figure 6-167 below, the op amp used is the AD817. Designed for low distortion video circuits, it has a relatively high resistance between the input differential pair emitters, R_E , approximately $1k\Omega$. It also has internal, large-value $8k\Omega$ resistors in series with the V_{OS} nulling terminals at pins 1 and 8, labeled here as R_1 .

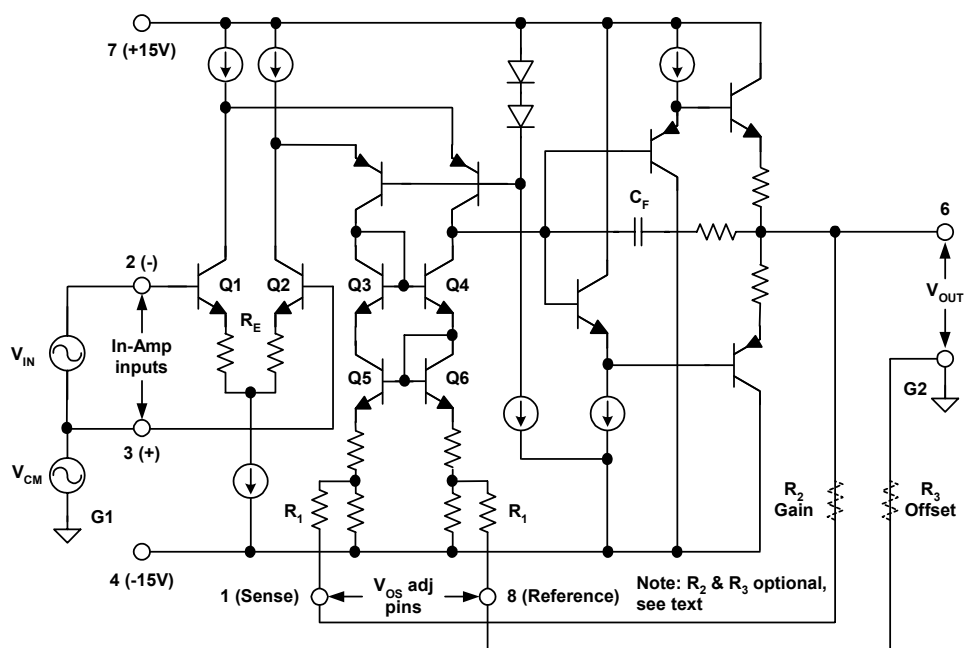


Figure 6-167: An AD817 wideband in-amp configuration

Functioning here as an in-amp, the AD817 is operated unconventionally. No feedback is used to inverting input pin 2. Instead, the V_{IN} differential signal is applied between pins 2 and 3, as noted. Typically there is also an associated CM noise, V_{CM} . Note that there must be a return path between the input ground $G1$ and output ground $G2$, to allow bias current flow (as with standard in-amps). The input differential pair stage $Q1$ and $Q2$ produces an output signal current, driving quad-connected current mirror stage $Q3$, $Q4$, $Q5$, $Q6$. At the bottom of the current mirror is the balancing resistor network, functioning here as a signal current input. The connection from pin 1 to the amplifier output closes a negative feedback loop, to the output at pin 6. A balancing reference input is applied to pin 8, either ground (as shown) or a variable offset voltage. Differential gain of the circuit, G , is:

$$G = V_{OUT}/V_{IN} = R_1/R_E \quad \text{Eq. 6-44}$$

For the values noted, gain is about $8\times$, and bandwidth is 5MHz. CMR is excellent, measuring more than 80dB at 1MHz. Optional trim resistors R_2 and R_3 are used to adjust gain via R_2 , or, alternately, offset via R_3 . The values should be the same, for best CMR.

Negative Resistance Buffer

There is often a requirement for driving a lower load impedance than a given op amp may be capable of meeting. This can be particularly true for precision op amps in general, and, more specifically, with rail-rail output types. The latter class of op amps typically can have an output impedance on the order of several $k\Omega$, which can limit load drive and lower open loop gain when driving low impedances. A straightforward way of addressing this problem is a unity gain buffer, which will work in almost all cases. But ordinary op amps can also be used for the buffer function. An interesting method is to use a second op amp as a negative resistance generator, to synthesize a negative resistance whose value is set equal to the load resistance. When this is carefully done the load disappears, as a parallel connection of R_L and $-R_L$ is infinite. Figure 6-168 below illustrates this technique, in both basic and practical forms.

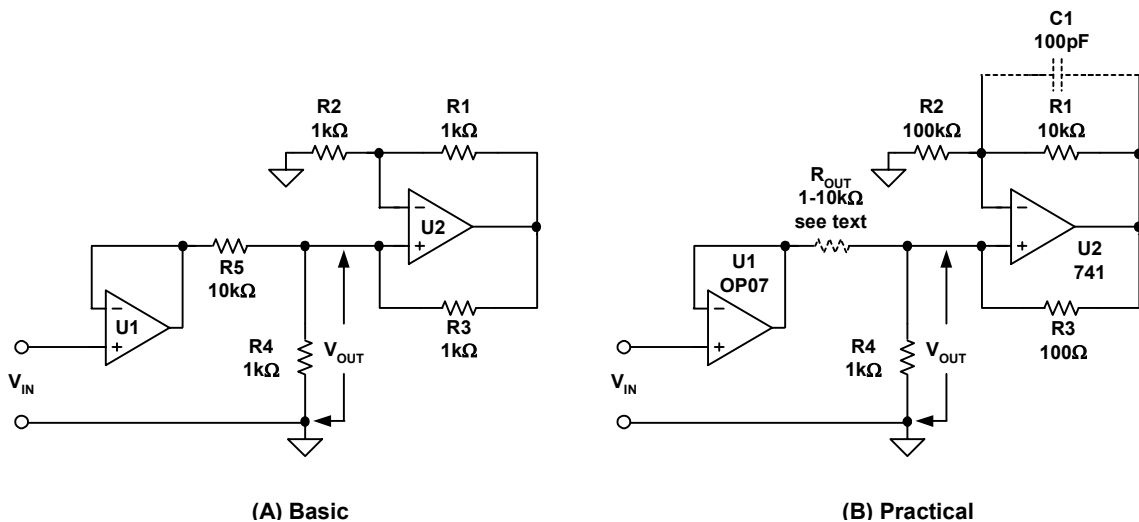


Figure 6-168: Negative resistance buffer circuits

In Fig. 6-168A, a basic form of the circuit is shown, to illustrate the concept. Here op amp U1 is intended to drive load resistor R_4 , but would normally be prevented from doing so with high precision by the output resistance represented by R_5 . But, due to the connection of the U2 stage the voltage V_{OUT} is amplified by a factor of $1 + R_1/R_2$. This amplified voltage is fed back to the V_{OUT} node by R_3 . With the R_1 – R_4 values scaled as shown, this produces a negative resistance of $-R_4$ at the V_{OUT} node. Thus the driving amplifier U1 does not see the real resistance R_4 loading, which can be confirmed by examining the (small) current in R_5 . If operation is not apparent, the circuit can be analyzed by viewing it as a balanced bridge, with ratios of R_1/R_2 matching R_3/R_4 .

But the Fig. 6-168A circuit isn't very efficient, as twice the load voltage must be developed for operation, and double the load current flows in U2. The same principles are employed in the more practical Figure 6-168B version, with the R_1 – R_4 values rescaled to reduce power and to gain headroom in U2, *while maintaining the same ratios*. In a real circuit there is likely no need for R_5 , and U1 can drive the load directly. It does so taking full advantage of the precision characteristics of the U1 type. U2 can be almost any ordinary op amp capable of the load current required.

Cross-Coupled In-Amps Provide Increased CMR

A primary in-amp benefit is the ability to reject CM signals in the process of amplifying a low-level differential signal. While most in-amps perform well below about 100Hz, their CM rejection degrades rapidly with frequency. The circuit below in Figure 6-169 is a composite in-amp with much increased CMR vis-à-vis more conventional hookups (see Reference 15). It consists of three in-amps, with unity-gain connected U1 and U2 cross-coupled at their inputs. In-amp U3 amplifies the difference between V_{O1} and V_{O2} , while rejecting CM signals. The in-amps are AD623s, but the scheme works with other devices.

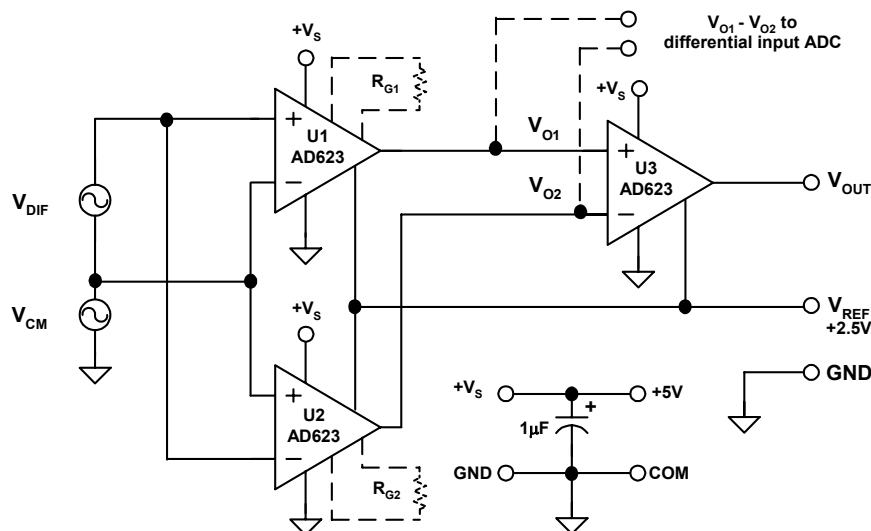


Figure 6-169: Two cross-coupled and similar in-amp devices followed by a third provides much increased CMR with frequency

Because of the fact that U1 and U2 have CM responses that are correlated (by the nature of their design), their output CM errors due to V_{CM} will be similar. For U3, this CM error appears as a CM signal, and is rejected further. Meanwhile, the desired differential signal, equal to $2V_{DIF}$ appears as $V_{O1} - V_{O2}$, and is amplified by U3 at unity gain. Overall gain is $2\times$ as shown, but can be raised by a gain factor programmable by R_{G1} and R_{G2} . Note that due to the fact that the U1 and U2 CM errors correlate, their matching isn't necessary.

A big advantage of this scheme is the extended frequency range over which the composite in-amp has good CMR. For example, at the gain of 2 as shown, CMR as measured at either V_{O1} or V_{O2} will be on the order of 60dB or more at 10kHz. At V_{OUT} however (the output of U3), the CMR is increased to about 85dB, or more than 20dB. The low frequency CMR corner of the composite in-amp is about 6kHz, as opposed to about 500Hz as measured at either V_{O1} or V_{O2} . At higher gains, for example a gain of 100 (as set by R_{G1} and $R_{G2} = 2.05k\Omega$), CMR increases to more than 110dB at low frequencies, and a corner frequency of about 2kHz is noted, while 10kHz CMR is more than 100dB. For these measurements $+V_S$ was 5V, and the V_{REF} applied to all devices was 2.5V.

Although the example shown is single-supply, it is also useful with dual supply in-amps. Another possible mode is to use V_{O1} and V_{O2} to drive a differential input ADC, which eliminates a need for in-amp U3. ADC scaling can be matched via R_G , and the V_{REF} used.

REFERENCES: AMPLIFIER IDEAS

1. Victor Koren, "Line Driver Economically Synthesizes Impedance," **EDN**, January 6, 1994 pp. 79. See also: "Feedback and Amplification," **EDN**, May 26, 1994 pp. 106.
2. Walt Jung, "Simple Wideband Noise Generator," **Electronic Design**, October 1, 1996, pp. 102.
3. Lewis Counts, Mark Murphy, JoAnn Close, "Diode-less Rectifier Takes Rail-to-Rail Input," **EDN**, October 28, 1993.
4. Moshe Gerstenhaber, Mark Murphy, "Paralleled Amplifiers Drive Loads Quietly," **EDN**, April 23, 1992, p. 171.
5. John Wynne, "Simple Circuit Adds Power Down," **Electronic Design**, January 7, 1993, p. 116.
6. "AD8036/AD8037 APPLICATIONS," within **Data Sheet for AD8036/AD8037 Low Distortion, Wide Bandwidth Voltage Feedback Clamp Amps**, <http://www.analog.com>
7. Peter Checkovich, "Clamp Amp Serves as Sync Inserter," **Electronic Design**, October 14, 1996, pp. 132, 134.
8. Brian Harrington, "Piecewise Linear Amplifier Eschews Diodes," **EDN**, October 12, 1995, pp. 112-113.
9. Eberhard Bruner, "Turn Feedback Amp Into Integrator," **Electronic Design**, July 10, 1995, pp. 101-102.
10. Eberhard Bruner, "Simple All-Pass Filter," **Electronic Design**, June 26, 1995, pp. 106-108.
11. Paul Hendricks, "Instrumentation Amplifier Has 290MHz GBW," **EDN**, May 12, 1994, p. 86.
12. Victor Koren, "Programmable-Gain Amp Uses Arbitrary-Attenuation Step Ladder," **Electronic Design**, April 16, 2001, p. 99.
13. Moshe Gerstenhaber, Mike Gianino, "Op Amp Doubles As Instrumentation Amplifier," **EDN**, September 15, 1994, pp. 164.
14. Elliott Simons, "Negative Resistor Cancels Op Amp Load," **EDN**, May 24, 2001, pp. 108.
15. Moshe Gerstenhaber, Chau Tran, "Composite Instrumentation Amp Extends CMRR Frequency Range 10 $\times$," **Electronic Design**, February 4, 2002, pp. 65, 66.

ACKNOWLEDGMENTS:

Helpful comments on this section were received from Victor Koren and Moshe Gerstenhaber, and were much appreciated.

NOTES:

SECTION 6-6: COMPOSITE AMPLIFIERS

Walt Jung

The term "composite op amp" can mean a variety of things. In the most general sense of the word, any additional circuitry at either the input or the output of an op amp could make the combination what is termed a composite amplifier. This can be a valuable thing, as often such enhancements allow new performance levels to be realized from the resultant amplifier.

Some straightforward op amp performance enhancements of this type of have already been treated elsewhere in this book. For example, within the "Buffer Amplifiers" section of this chapter, as well as some of the specialized buffers in the "Audio" section of this chapter are found what could be termed composite op amps. In these examples, a standard output stage buffering design step is to utilize a unity-gain buffer, running on the same supplies as the op amp being buffered. So long as this buffer has sufficient bandwidth, this is an easy and straightforward step—insert the buffer between the op amp and the load, connect the feedback around the op amp plus buffer, and that's it.

A very useful means of increasing op amp performance can be obtained by blending the performance advantages of two ICs, or a standard op amp IC and discrete transistors. Such a combination is known as a *composite amplifier*. In special situations, a well-designed composite amp can often outperform standard op amps. The reason this is true is that the composite amplifier can be optimized for a unique and specialized performance, a combination that may not be available (or practical) in a standard op amp.

However, whenever an input (or output) circuit is added to an op amp that provides additional voltage gain, then the open-loop gain/phase characteristics of the composite op amp may need to be examined for possible stability problems. Note that this applies even when a unity-gain stable op amp is used within the composite, because the additional voltage gain raises the net open-loop gain of the combination. This will be made clearer by some circuit examples that follow.

In this section composite op amp circuits are described which fall into these categories:

- Multiple Op Amp Composite Amplifiers
- Voltage-Boosted Output Composite Amplifiers
- Gain-Boosted Input Composite Amplifiers
- A Nostalgia Composite Op Amp

These sections follow, with one or more circuit examples of each type.

■ OP AMP APPLICATIONS

Multiple Op Amp Composite Amplifiers

The simplest composite amplifier form utilizes two (or more) op amps, merged into a single equivalent composite. This is usually done for reasons of offset control, although in some instances it may be for increased gain capability, more output swing, etc.

Two Op Amp Composite Amplifier

The most flexible composite amplifier version combines two op amps in such a way that both signal inputs are still accessible to an application. A good example of this is the circuit shown below in Figure 6-170 (see Reference 1).

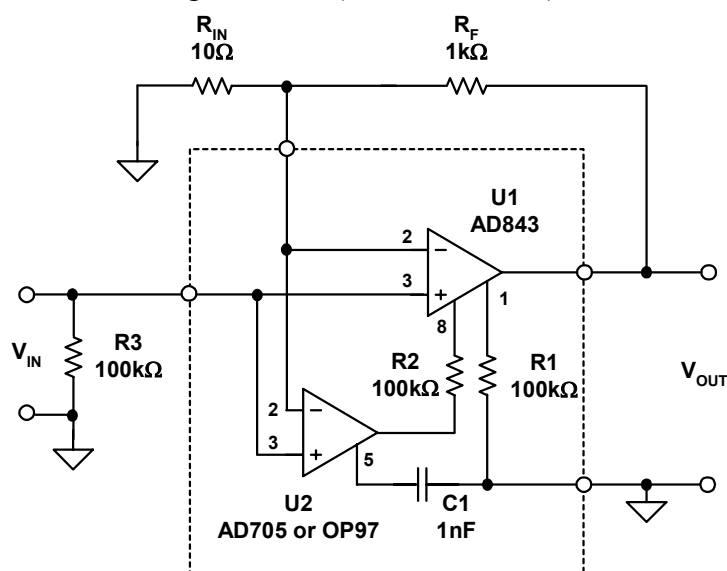


Figure 6-170: Low noise, low drift two op amp composite amplifier

In this circuit, U1 is a high-speed FET input op amp, the AD843. While FET input devices are typically excellent for fast data acquisition applications, their offset and drift is often higher than the best bipolar op amps. By combining the fast AD843 with a low offset and low drift, super- β input device for U2, the best of both worlds is achieved. Offset and drift are reduced essentially to the maximum U2 specification levels—an offset of $60\mu\text{V}$, a drift of $0.6\mu\text{V}/^\circ\text{C}$, and 100pA of bias current (for the OP97E). The composite op amp formed is the dotted box outline, and is applied as a four-pin op amp.

Both the U1 and U2 op amps have inputs connected in parallel, and both amplify the signal. Device U1 drives the load and feedback loop directly. U2 however drives the offset null input of U1, via a $100\text{k}\Omega$ resistor connected to pin 8, R2. R1 provides a complementary resistance at the opposite offset input, pin 1. C1 is used to over-compensate U2, at pin 5. Note that the three just described pins are unique to the AD843, and either the AD705 or the OP97 devices. The U1/U2 signal inputs, output, and power supply pins are all standard, and the circuit operates on conventional $\pm 15\text{V}$ supplies.

The circuit as shown has a non-inverting gain of $101\times$, as determined by R_F and R_{IN} . However, other application are also possible, both inverting and non-inverting in style. A detailed technical analysis of this circuit was presented in Reference 2.

Low Voltage Single-Supply to High Output Voltage Interface

There are numerous cases when an op amp designed for low (or single) supply voltage operation might need to be interfaced into a system operating on higher voltage and/or dual power supplies. An example would be the numerous low voltage chopper-stabilized op amps, which, without some means of easy interfacing, would simply not be available for use on high voltage supplies.

The circuit of Figure 6-171 below shows how a low voltage, single-supply chopper-stabilized amplifier, the AD8551, can be used on a $\pm 15\text{V}$ supply system.

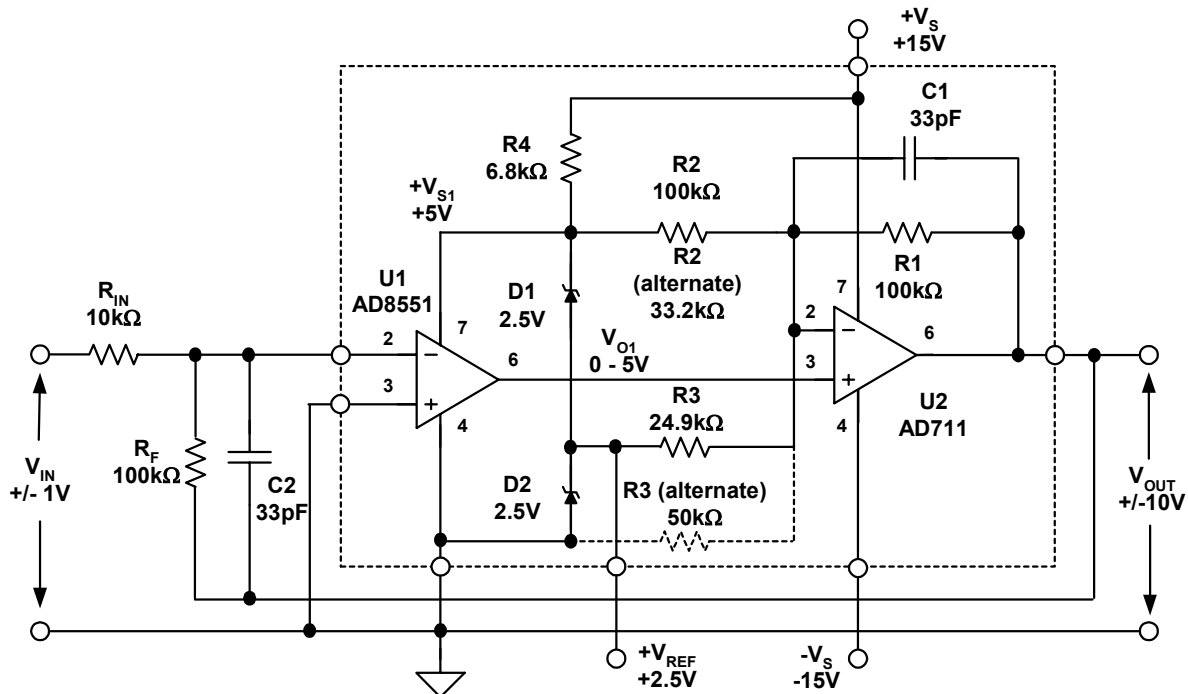


Figure 6-171: Chopper-stabilized 160dB gain, low voltage single-supply to high output voltage composite amplifier

In this circuit, the U1 AD8551 operates as a precision input stage of the composite amplifier, working from a local +5V supply generated from the main +15V rail by reference diodes D1, D2. This satisfies the supply requirements for the U1 stage, with the R4 value selected to supply the required current.

To interface the 0 to 5V output swing of U1 to a $\pm 10\text{V}$ range, the U2 output stage operates as both a level shifter and a gain stage. A nominal gain of $6\times$ is provided, with a DC offset providing the required level shifting. With the R1-R3 resistors and the 5V supply used as a reference, the gain and level-shifting is accomplished. The gain of $6\times$ translates back to a 0.833 to 4.167V positive output swing from U1, a range which even non rail-rail output op amps can most likely accommodate.

A virtue of this circuit is that the output of U1 is not loaded, and thus it operates at its full characteristic gain. For the AD8551, this is typically 145dB. When the additional 15.6dB of the U2 stage is added, the net open-loop gain of the composite amplifier formed is

■ OP AMP APPLICATIONS

more than 160dB. Further, this high gain will be maintained for relatively lower impedance loads, by virtue of the fact that a standard emitter-follower type output stage is used within U2. So, the DC accuracy of this composite amplifier will be very high, and will also be well-maintained over a wide range of loads due to the buffering of stage U2.

The voltage-offsetting network as shown uses two 2.5V reference diodes, which provides a 2.5V V_{REF} output from D2. This is usually a handy asset to have in any +5V supply device system. Alternately, a single 5V reference diode can be used, with the alternate values and connections for R2 and R3 substituted.

To compensate the system for the additional voltage gain of U2, two feedback capacitors are used, C1 and C2. *Note that for anything other than this exact circuit, one or both of these capacitors may require adjustment.* This is best done by applying a low-level square wave to the input such that the final output is on the order of 100mVp-p or less, and verifying that the output step response is well-damped, with minimal overshoot.

C1 ensures stability for stage U2, and C2 provides overall bandwidth control for the main feedback loop. In general, the U2 amplifier should have more bandwidth than the U1 stage. However, the relative DC accuracy of U2 is not at all critical. The AD711 is shown as one possible choice, but many other types can also be used. While not shown for simplicity, conventional supply bypassing of the composite amplifier should be used.

System-wise, this composite amplifier behaves as a $\pm 15V$ powered op amp with an input CM range equal to the specification of the U1 device in use. Overall loop feedback is provided as with any conventional feedback stage, i.e., by R_F , R_{IN} and C2. The circuit is applied by treating the parts within the dotted box as a single op amp, with the external components adjusted to suit a given application. An application caveat is that saturation of U1–U2 should be avoided, due to the longer overload recovery. This can be addressed with a 11-12V back-back clamping network across the feedback impedance.

Although the example hookup shown is a gain-of-10 inverter, other inverting configurations such as integrators and also non-inverting stages are possible. The caveat here is that the CM range of U1 must be observed. However this is likely no handicap; as such an amplifier is most likely to be used with very high gain and low CM input voltages. It could for example be used as a 5V-powered bridge amplifier with a $\pm 10V$ output range.

There are of course any number of other op amp input and output devices that will work within this general setup. A more general-purpose low voltage part for U1 would be the AD8541. Offset voltage will be higher, and gain less, vis-à-vis the AD8551. For optimum dynamic range and linearity, the biasing of the U1 stage output is centered within the total U1 supply voltage V_{S1} , which can be different than 5V if needed. This feature is provided by R1-R3. The absolute values of these resistors aren't critical, but they should be maintained as to their ratio. They can be part of a common 100k Ω array for simplicity.

Voltage-Boosted Output Composite Amplifiers

A number of schemes are useful towards boosting the output swing of standard op amps. This can be either to achieve greater swing (i.e., closer to the rails), or, to develop swings greater than normally possible with standard ICs, i.e., $\approx 40V$ swings. In both cases it may also be desirable to increase load drive to 100mA or more.

Voltage Boosted, Rail-Rail Output Driver

A common requirement in modern system is the rail-rail capable op amp. But all op amps aren't designed with rail-rail outputs, so this may not be possible in all instances. Of course, it makes good sense to utilize standard off-the-shelf rail-rail IC op amps, whenever they meet the application requirements. Nevertheless, it is possible to add an output stage to a standard op amp device that may itself not be rail-rail in function. By using common-emitter (or common-source) discrete transistors external to the op amp, a rail-rail capability is realized. An example designed in this fashion is Figure 6-172 below.

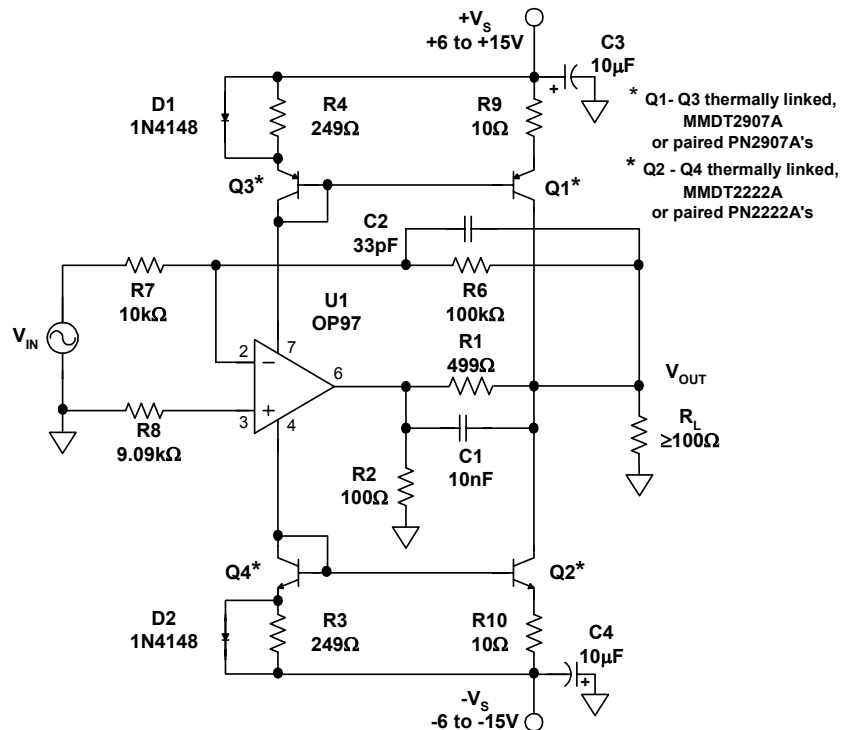


Figure 6-172: Voltage boosted rail-rail output composite op amp

Within this circuit Q1 and Q2 are the complementary buffer transistors that provide the rail-rail output swing. The circuit works as follows: Q1 is driven by the voltage drop across R4, and diode-connected Q3. This voltage is developed from the positive rail supply terminal of U1, so the quiescent bias current of Q1 will be related to the quiescent current of U1. Similarly, Q2 is driven from R3 and Q4, via the negative rail terminal of U1. The Q1-Q3 and Q2-Q4 pairs make up current mirrors, developing a quiescent bias current that flows in Q1-Q2. The U1 quiescent current is about 400μA, and with the resistance values shown, the Q1-Q2 bias current is about 10mA.

■ OP AMP APPLICATIONS

The output stage added to the U1 op amp adds additional voltage gain, and a current gain boost of 25 times, essentially the ratio of R4/R9 and R3/R10. Thus for a 100mA output from Q1-Q2, U1 only supplies 4mA. The swing across R2 is relatively low, allowing operation on low voltage supplies of $\pm 6\text{V}$, or up to $\pm 15\text{V}$.

The simulation data of Figure 6-173 illustrates some salient characteristics of the composite op amp while driving a load of 85Ω . The open loop gain of the circuit is shown by the topmost, or composite gain curve, which indicates a low frequency gain of over 130dB, crossing unity gain at about 630kHz. The intermediate curve is the OP97 op amp gain characteristics. The difference between this and the upper curve is the added gain, which is about 13dB. The lowest curve indicates the closed-loop gain versus frequency characteristics of the composite op amp, which is 20dB in this case, as set by R6 and R7 (as in a standard inverter).

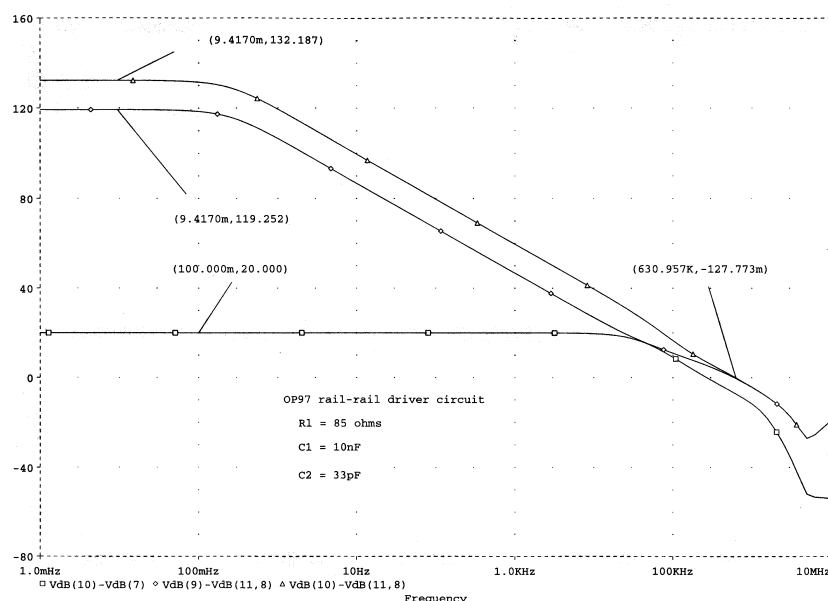


Figure 6-173: Gain (dB) versus frequency characteristics of Fig. 6-172 composite op amp

There are a couple of critical points in setting up this circuit. Bandwidth can be controlled by C1 and C2. C1 reduces the added gain at high frequencies, which can be noticed from the composite gain curve, starting below 100kHz. C2 reduces the closed-loop gain, starting about 50kHz. For greater closed-loop gains, C2 may not be needed at all.

Bias control is achieved by the use of thermal coupling between the dual current mirror transistors. The easiest way to accomplish this is to use packaged dual types, either SOT-363 or SM-8 devices (see References 3 and 4). Alternately, TO92 equivalent PN2222A and PN2907A types can be used, with the two flat sides facing and clipped together.

The circuit as shown drives a 100Ω load to within 2V of the rails, limited by the drop across R9 and R10. Current limiting is provided by a shunt silicon diode across R4 and R3, either with 1N4148 diodes, or diode connected transistors. This limits peak output current to about $\pm 60\text{mA}$. More output current is possible, by adding additional like devices in parallel to Q1 and Q2, with additional 10Ω emitter resistors for each.

A point that should be noted about the booster circuit of Fig. 6-172 is that *the biasing is dependent upon the quiescent current of the op amp*. Thus, this current must be stable within certain bounds, otherwise the idle current in Q1-Q2 could deviate— either too low (causing excess distortion), or too high, causing overheating. So, changing the U1 op amp isn't recommended, unless the biasing loop is re-analyzed for the new device.

Another point is that this type of circuit, which uses the power pins of the op amp for a signal path, *may not model at all in SPICE!* This is due to the fact that many op amp SPICE models do *not* model power supply currents so as to reflect output current— so be forewarned. However, the ADI OP97 model does happen to model these currents correctly, so the reader can easily replicate this circuit with the OP97 (as well as many other ADI models). Discussion of these models can be found in Chapter 7 of this book.

High Voltage Boosted Output Driver

With some subtle but key changes to the basic voltage-boosted composite amplifier of Fig. 6-172, output swing can be extended even higher, more than double the standard $\pm 10\text{V}$ swing for $\pm 15\text{V}$ rails. A basic circuit that does this is shown below in Figure 6-174.

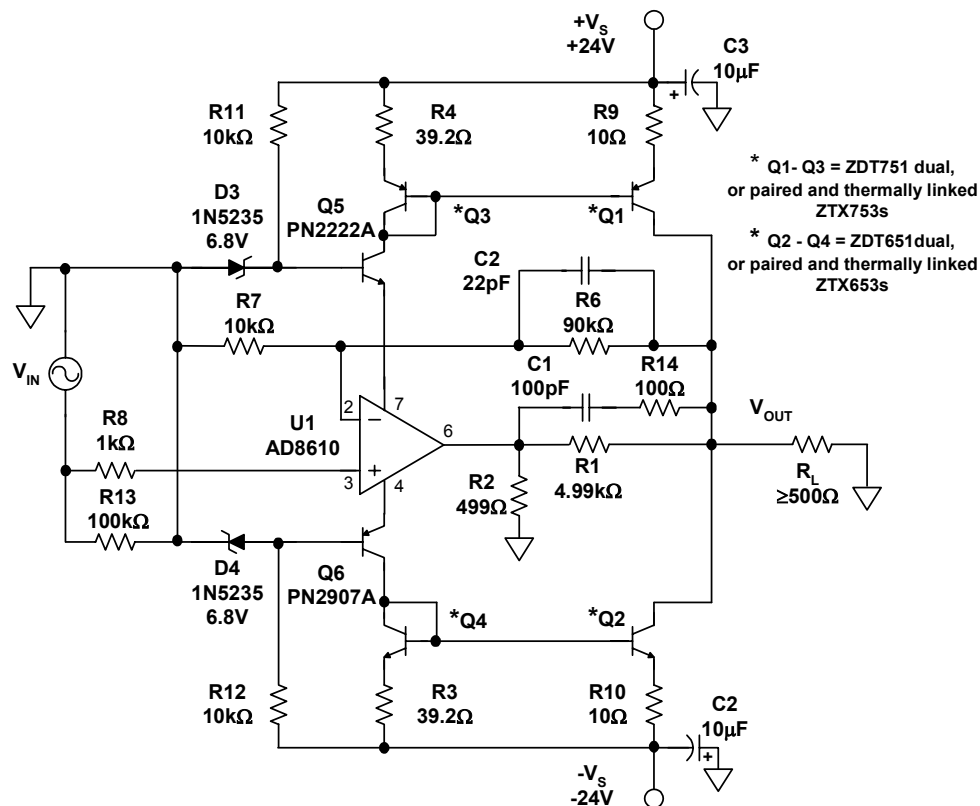


Figure 6-174: High voltage boosted rail-rail output composite op amp

This circuit can readily be recognized as being similar to the lower voltage counterpart of Fig. 6-172. To achieve higher voltage capability, the U1 op amp is operated from a pair of combination level-shift/regulator transistors, Q5 and Q6. These are biased in turn from the D3 and D4 zener diodes at their bases, to $\pm 6.8\text{V}$, respectively. The op amp rails are

■ OP AMP APPLICATIONS

then simply $\pm 6.2\text{V}$, while the main rails $+V_S$ and $-V_S$ can be virtually any potential, as will be ultimately limited by the Q1 and Q2 voltage/power ratings.

The op amp supply current passes through Q5 and Q6, driving the bases of Q1–Q2 as previously. To accommodate the higher $\pm 24\text{V}$ supplies, the standard 2222A and 2907A parts used in Fig. 6-172 are replaced by higher voltage parts, the dual ZDT751 and ZDT651 (see Reference 4). Thermal matching is best maintained by using these dual types, but comparable TO92 pairs can also be used, for Q1-Q3, ZTX753s, and for Q2-Q4, ZTX653s (see References 5 and 6). In any case, a large area PCB land (i.e., 1-2 square inches) should be used to for the Q1–Q2 collectors for heat sinking purposes.

In this new circuit, an AD8610 op amp is used for U1, offering very low offset voltage, and higher speed. The quiescent current of the AD8610 is typically 2.5mA. A 4/1 gain is used in this circuit, as established by R4/R9 and R3/R10. The idle current in Q1-Q2 is therefore about 10mA, leading to a $\approx 240\text{mW}$ dissipation each, on $\pm 24\text{V}$ supplies. This is low enough to not require a heat sink. However, the copper land area described above should be provided on the PCB around Q1-Q2 for heat sink purposes, tied electrically to their collectors. These measures, plus the active current limiting, help protect the output devices against shorts.

This circuit has a novel method of current limiting. As operated on $\pm 6.2\text{V}$, the AD8610 will swing just over $\pm 5\text{V}$. In driving R2 to this limit, $\pm 10\text{mA}$ of current will be delivered to the current mirrors, resulting in a maximum output current four times this, or $\pm 40\text{mA}$. This is just about the DC safe-area limit of the Q1 and Q2 devices as used on $\pm 24\text{V}$. For low impedance loads below 500Ω , the maximum output voltage is a product of the 40mA limit and the load (for example 40mA into 100Ω yields 4V peak). The maximum voltage swing into a 500Ω load is then about $\pm 20\text{V}$, again, as determined by the current limiting. Into higher impedance loads, the swing is proportionally greater, up until the point Q1-Q2 reach their saturation limits.

Although the circuit is quite versatile as shown, there are many other options also possible. Other op amps can be used, but as noted before, the idle current should be taken into account. This is even more critical on higher voltage supplies, as it directly affects the power dissipated in Q1 and Q2.

For higher output currents from Q1-Q2, additional similar transistors can be paralleled, each with individual emitter resistors like R9 and R10. This will be practical for scaling up current by a factor of 2-3 times (assuming one additional package of the ZDT751 and ZDT651 types).

For ampere level current outputs, an additional current gain stage in the form of a complementary emitter follower can be added, driven from the Q1-Q2 collectors, with a 1:1 gain in the current mirror, and appropriate emitter follower biasing. With this step, the circuit will have been converted into a complete power amplifier. Details of this are left as an exercise-for-the-reader. However, a good starting point towards this might be the Alexander power amplifier topology (see Reference 7).

Gain-Boosted Input Composite Amplifiers

One of the most popular configurations used to enhance op amp performance is the gain-boosted input composite op amp. Here, a preamp gain stage is added ahead of a standard IC op amp, allowing greater open-loop gain, lower noise, and other performance enhancements. Another worthy improvement is the thermal isolation between the critical input stage, and the IC output stage that delivers the load current. The preamp can be a matched pair of bipolar transistors (NPN or PNP), or JFETs of either N or P types.

Prototype bipolar transistor gain-boosted input composite amplifier

For illustration of the basics, a prototype example composite amplifier is the two-stage op amp of Fig. 6-175. This circuit uses a matched NPN differential pair as a preamp stage ahead of U1, a standard AD711 type op amp. The preamp stage adds voltage gain to that of U1, making the overall gain higher, thus lowering gain-related errors.

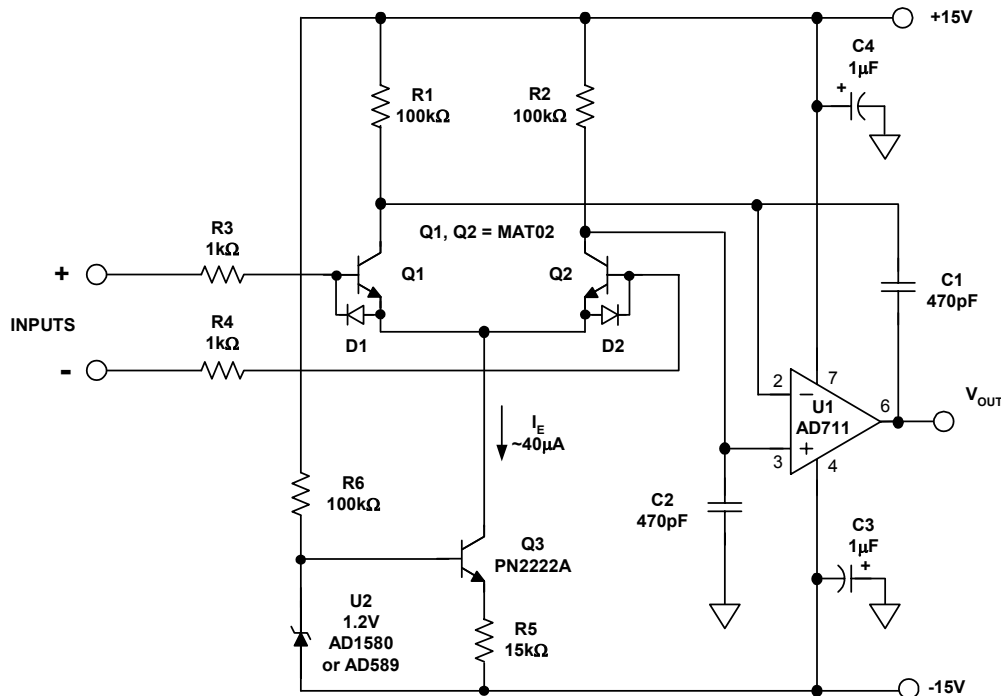


Figure 6-175: Bipolar transistor gain-boosted input composite op amp

Because of the added gain, the relative precision of the output op amp isn't very critical. It can be selected for sufficient output drive, slew rate and bandwidth. Within a given application the composite amplifier has overall feedback around both stages. Note—for this and following circuit examples, the op amp is *uncommitted* (i.e., external feedback).

In this example, a bipolar transistor differential pair, Q1-Q2 is loaded by a stable, matched load resistor pair, R1 and R2 (where $R_1 = R_2 = R_L$). The exact value of these resistors isn't overly critical, but they should match and track well. R1 and R2 are selected to drop 2-3V at $\frac{1}{2} I_E$. For a 2V drop, a suitable R_L is then:

$$R_L = \frac{4}{I_E} \quad \text{Eq. 6-45}$$

■ OP AMP APPLICATIONS

Here, Q1 and Q2 operate at $\approx 20\mu\text{A}$ each, so $100\text{k}\Omega$ values work for R1-R2. Note that in operation, *the second stage op amp must be capable of operating with input CM voltages of 2V below the $+V_{S\text{rail}}$* . This criterion is fine for many PFET input amplifiers such as the AD711, but others should be checked for CM input voltage compatibility. Note that similar preamp stages can also be built with PNP bipolars, or with JFETs, and some of these will be described later.

The DC or low frequency gain of the preamp stage, G1, can be quite high with bipolar transistors, since their g_m is high. G1 can be expressed as:

$$G1 = \frac{R_L I_E}{2V_T} \quad \text{Eq. 6-46}$$

Where $V_T = KT/q (\approx 0.026\text{V at } 27^\circ\text{C})$.

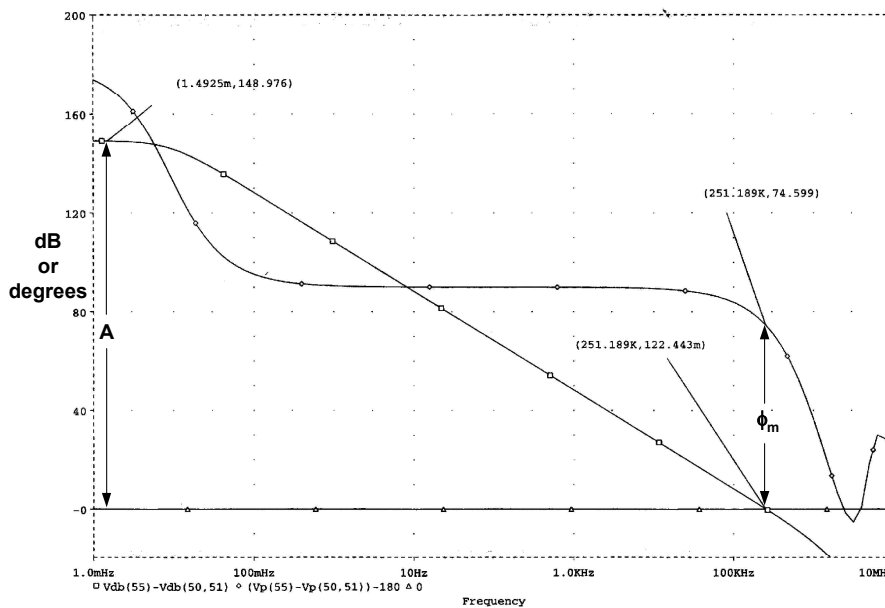


Figure 6-176: Gain/phase versus frequency for Fig. 6-175 composite op amp

In this example, at 27°C , G1 is about 77 times (37.7dB). Overall numeric gain is of course the product of the preamp gain and the U1 op amp gain. The minimum AD711 DC gain is 150,000, so the gain of the composite is more than 11.5 million ($\geq 141\text{dB}$).

As the preamp stage provides additional gain, this extra gain must be phase controlled at high frequencies for unity-gain stability of the composite amplifier with applied feedback. In this circuit, compensation caps C1 and C2 provide this function with U1 connected as a differential integrator.

The unity gain frequency, F_u , can be expressed approximately as:

$$F_u \approx \frac{I_E}{4\pi C_C V_T} \quad \text{Eq. 6-47}$$

where π is 3.14, $C1 = C2 = C_C$.

In the gain and phase versus frequency simulation plot of Figure 6-176 (opposite), the performance of this composite op amp is illustrated. The additional gain of the preamp raises the net DC gain to $\approx 149\text{dB}$, and the unity gain crossover frequency is shown to be $\approx 252\text{kHz}$, both of which generally agree with the estimated figures. The phase margin ϕ_m at 252kHz is about 75° , which is conservative. This op amp should be stable for all closed-loop gains down to unity (in fact, C1 and C2 could possibly be lowered).

Slew rate of the composite op amp can be no higher than the specified SR of output stage U1. For cases where the effective SR is to be lower, it can be estimated as:

$$\text{SR} \approx \frac{I_E}{C_C} \quad \text{Eq. 6-48}$$

Using Eq. 6-47 and 6-48, the chosen values calculate a bandwidth of 260kHz , and an SR of just under $0.1\text{V}/\mu\text{s}$. Actual bandwidths of $236\text{--}238\text{kHz}$ were measured on 4 op amps for U1 (AD711, AD820, LM301A and LF356), while SR was $+0.085$ and $-0.087\text{V}/\mu\text{s}$. As would be expected, the least bandwidth was measured with the lowest bandwidth U1 device, an LM301A. This demonstrates the relative insensitivity to U1 bandwidth.

With high gain input transistors, the bias current can be low. Generally, this will be:

$$I_B = \frac{I_E}{2H_{FE}} \quad \text{Eq. 6-49}$$

Where I_B is the bias current of either Q1 or Q2, and H_{FE} is their DC gain. The MAT02 diodes protect against E-B reverse voltage, while the $1\text{k}\Omega$ resistors limit diode current.

Bias currents of 30nA were measured with a MAT02 for Q1-Q2. Similar results can be obtained with high gain discrete transistors, such as 2N5210s. Offset voltage however, is a different story. Monolithic duals such as the MAT02 will be far superior for offset voltage, with a V_{OS} specification of $50\mu\text{V}$. Non-monolithic packaged duals will also function in this circuit, but with degradation of DC parameters versus a monolithic device such as the MAT02. As can be noted from the numbers quoted above, speed isn't major asset of this amplifier. However, the DC performance is excellent, as noted, placing it in an OP177 class for gain.

The Q1 - Q2 emitter current, I_E , can be established by a variety of means. The most general form is the U2, Q3, and R5 arrangement. This works for a wide range of inputs, and also offers relatively flat gain for a bipolar Q1-Q2 gain stage, since the PTAT current from Q3 compensates the temperature-related gain (Eq. 6-46). For those applications where the input of the amplifier is operating in an inverting mode, a more simple solution would be a resistor of $332\text{k}\Omega$ from the Q1-Q2 emitters to $-V_S$.

Of course, as a practical matter one wouldn't use the complex Fig. 6-175 circuit, if an OP177 (or another standard device) could do the job more simply or inexpensively. Nevertheless, what the above discussion does do is illustrate how one can *tailor* a composite op amp's characteristics, to get exactly what is needed. The composite op amp circuit of Fig. 6-175 could be used with a rail-rail output stage device for U1 (AD820), or with a very high current output stage (AD817, AD825), or any other performance niche not available from standard devices. Examples of these performance options follow.

■ OP AMP APPLICATIONS

Low noise, gain-booster input composite amplifier

One of the more sound reasons for adding a preamp stage before a standard op amp is to lower the effective input noise, to a level lower than that of readily available IC devices. Figure 6-177 below shows how this can be achieved within the same basic topology as described above for the Fig. 6-175 prototype composite. As will be noticed, this circuit is similar to the prototype, with the exception of the added offset trim network, a higher bias level for I_E , and a faster output op amp, U1. Rtrim nulls the offset for best DC accuracy in critical applications. If this isn't necessary, connect R1-R2 as in Fig. 6-175.

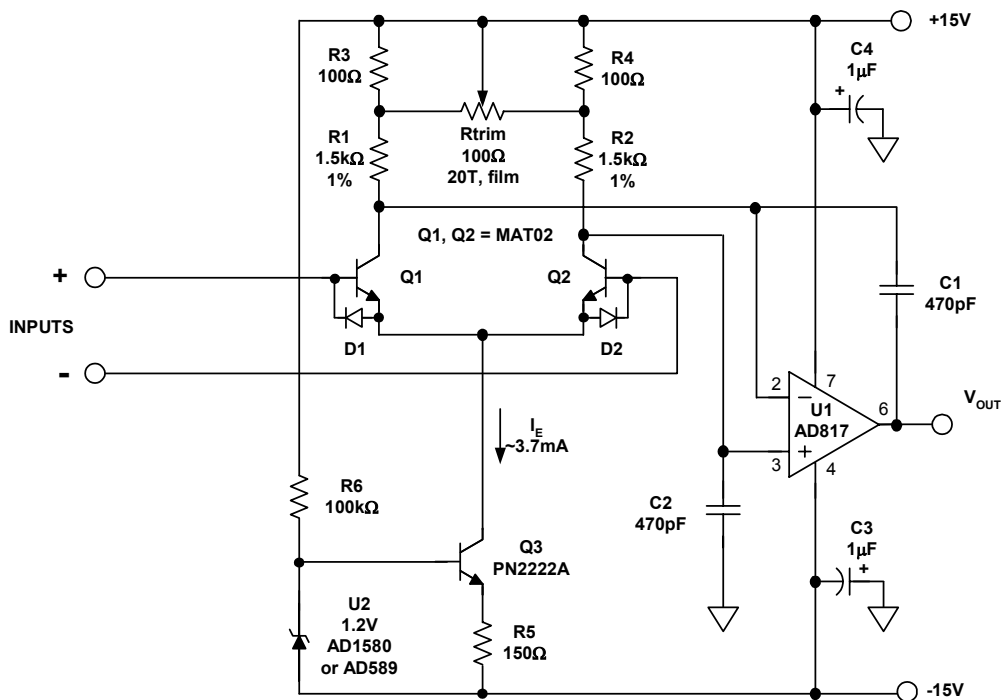


Figure 6-177: Low noise gain-booster input composite op amp

By raising the current level of Q1-Q2 by roughly a factor of 100× compared to the prototype, the effective input noise of Q1-Q2 is lowered dramatically. At the operating current level of 3.7mA, the MAT02 achieves an input voltage noise density $<1\text{nV}/\sqrt{\text{Hz}}$ (see Reference 10). To eliminate their added noise, the series base resistors are dropped.

Both bandwidth and SR are also improved in this circuit, since they are both proportional to I_E . The estimated bandwidth and SR for this circuit are 24MHz and 7.9V/μs, respectively. Measurements show about 21.2MHz for bandwidth, and a SR close 7.6V/μs. Both the actual bandwidth and SR are less than the AD817 specifications of 50MHz and 350V/μs. The circuit as shown is close to unity-gain stable, with 44° of phase margin at the unity gain frequency. Of course, very low noise amplifiers such as this will often be applied at some appreciably higher gain, for example 10, 100 or more. When this is the case, then C1 and C2 can be reduced, allowing greater bandwidth and SR.

An even lower noise op amp can be achieved simply by adding one or more low noise pairs parallel to Q1-Q2, and operating the combination at 6mA of current. See References 11 and 12 for examples to achieve 0.5nV/√Hz or less noise.

JFET transistor gain-boosted input composite amplifier

The circuit of 6-178 below illustrates an alternative compensation method for composite op amps. This technique has the advantage of simplicity, but also the disadvantage of being conditionally stable. This technique goes back to the very earliest days of IC op amps, when discrete or monolithic matched FET pairs were used ahead of a standard IC op amp such as the 741, 709, etc. Further details are contained within References 13 and 14. The example here isn't offered as a practical example, inasmuch as so many superior IC FET op amps are available today. However, it does give insight into this type of compensation, which is applicable either to FET or bipolar input stages.

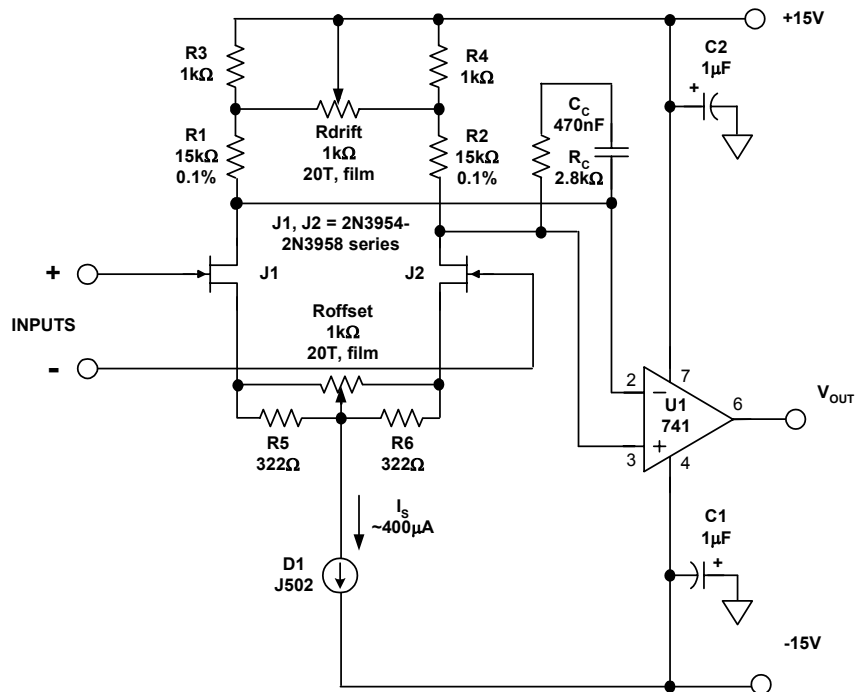


Figure 6-178: JFET transistor gain-boosted input composite amplifier

There are practical reasons why this type of FET input composite amplifier isn't used today. One is that to do it correctly involves many involved trims, another is that it requires a considerable number of parts. Dual FET devices don't come with sub-mV offsets, as do bipolars, so there is the need to trim out offset. Roffset does this, for J1-J2 V_{OS} up to 50mV. For lowest drift, the drain currents should also be trimmed, via Rdrift.

N-channel JFET duals such as the 2N3954 and J401 series are specified for operation at a total I_S of 400μA, or 200μA/side. Their transconductance is much lower than a bipolar; for these conditions; it is typically $\approx 1400\mu S$. Therefore the gain of this preamp will typically be much lower than would a bipolar stage. With matched load resistors, gain is:

$$G1 = \frac{R_L \cdot g_{fs}}{2} \quad \text{Eq. 6-50}$$

where g_{fs} is the specified JFET transconductance at $I_S/2$.

For the conditions shown, G1 works out to be 10.5 (20.4dB). Note— if used, the Roffset network reduces gain somewhat, and Eq. 6-50 *doesn't* take this into account.

■ OP AMP APPLICATIONS

Compensation for this composite amplifier is via the RC network, R_C - C_C . This network reduces the gain of the preamp to unity above the zero frequency, which allows the aggregate open-loop response to then assume that of the U1 amplifier before the unity-gain crossover. It is chosen by setting R_C as:

$$R_C = \frac{4}{g_{fs}} \quad \text{Eq. 6-51}$$

where g_{fs} is again the specified JFET transconductance at $I_S/2$.

In this case, R_C works out to be $2.8\text{k}\Omega$. C_C is then chosen to provide a zero at some frequency that should be a very small fraction of the U1 op amp's unity gain frequency. The importance of this point will be made clearer by various open-loop response shapes.

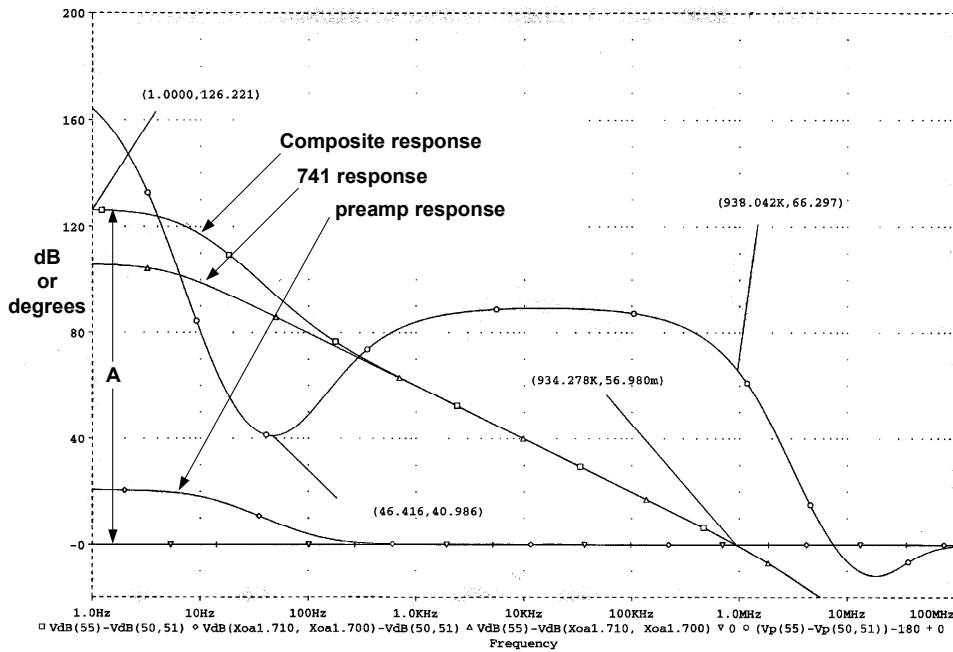


Figure 6-179: Gain/phase versus frequency for Fig. 6-178 composite amplifier

It should be recalled that the classic open-loop response of an unconditionally stable op amp is a constant -6dB/octave for gain, with an associated 90° phase shift. For such a device, any $1/\beta$ closed-loop response that intersects this open-loop response will be stable. For example, the 741 response (Δ), as so marked in Figure 6-179 above, is such a characteristic. But, as can also be noted from Fig. 6-179, the gain/phase response of a composite op amp compensated as in Fig. 6-178 just isn't a simple matter.

In the case of the added preamp stage and the R_C - C_C network compensation, the composite gain response ($\square$) assumes a *multiple-slope response*. Associated with this gain response, note also that *the phase characteristic ($\circ$) varies radically with frequency*. In particular, the phase dip around 46Hz signifies a frequency where a loop closure could be problematic, as the phase margin is only 40° at this point.

Faced with this type of open-loop gain/phase response, a designer needs to be careful in crafting the closed-loop gain configuration. The first step is to decide what level of

closed-loop gain is required by the application. Given that, an ideal $1/\beta$ curve can be drawn on a Bode diagram, to determine the rate-of-closure at the intersection. For optimum stability, it is desirable that this intersection occurs with a relative -6dB/octave between the open-loop gain curve and an ideal $1/\beta$ curve. Note that if such a $1/\beta$ curve were drawn on Fig. 6-179 at a 100dB gain, it would intersect in a -12dB/octave region. This is because C_C is 470nF in this example, which places the phase dip at 46Hz , with a composite gain curve which, as noted, is dropping at a rate greater than 6dB/octave . So, with the proposed gain curve intersecting in this region, stability could be marginal.

On the other hand however, with the R_C - C_C "phase funnies" forced down to a low frequency that corresponds to very high closed-loop gains (i.e., $\approx 100\text{dB}$), the practical potential for instability is minimized. Note also that it doesn't make good sense to build a 100dB gain feedback amplifier based on limited open-loop gain such as Fig. 6-179.

On the other hand, if C_C were smaller, this wouldn't necessarily be the case, because the associated phase dip would then move upward in frequency. This could wreak havoc with loop closures at more practical closed-loop gains. In contrast to this, with C_C sized as shown at 470nF , the phase anomalies are confined to very low frequencies, yet the added DC gain of the preamp is still available. Loop closures at frequencies above $\approx 200\text{Hz}$ (at closed-loop gains of 80dB or less) see a high frequency composite response closely resembling the 741 ($\square$ and Δ , respectively), and should thus be stable.

Watch out for the time domain response!

It should also be noted that there is a more subtle side effect related to the R_C - C_C method of compensation, as illustrated in Fig. 6-178. Simply put, this is the fact that *the time domain response of the resulting composite amplifier will be marred, compared to that of a classic 6dB/octave roll off* (see Reference 15).

So, wherever time domain response is critical, then the more conservative, unconditionally stable compensation method of Fig. 6-175 should be used. A case in point using this method with a FET preamp is the next composite amplifier.

In summary, for the AC performance characteristics as a composite amplifier, the circuit of Fig. 6-178 offers a gain raised higher than that of U1, or by about 20dB (126dB total) using the 741. At high frequencies, the overall gain bandwidth properties of this composite mimics the U1 amplifier, when the R_C - C_C time constant is relatively large.

DC performance limitations

The DC input characteristics of this circuit will be those specified for the J1-J2 pair, with typical room temperature bias currents of 50pA or less. Common-mode rejection will be limited by the J1-J2 specifications, and typically no more than about 80dB , over a limited CM range. This could be improved by cascoding the J1-J2 pair, but again, given the availability of such FET-input IC amplifiers as the AD8610, this would be a questionable design for a precision FET amplifier. In Fig. 6-178, the current source used for I_S is a simple FET current limiter diode (see Reference 16). This offers simple, two-terminal operation, at a current level optimum for the J1-J2 pair.

■ OP AMP APPLICATIONS

Low noise JFET gain-boosted input composite amplifier

An alternative method of executing a JFET input gain-boosted composite is to operate the input differential pair into an output op amp stage that acts as a differential integrator (i.e., similar to the Fig. 6-175 prototype, insofar as the compensation). The circuit of Figure 6-180 below is such an example, one that is also optimized for low noise operation, medium speed, and higher output current.

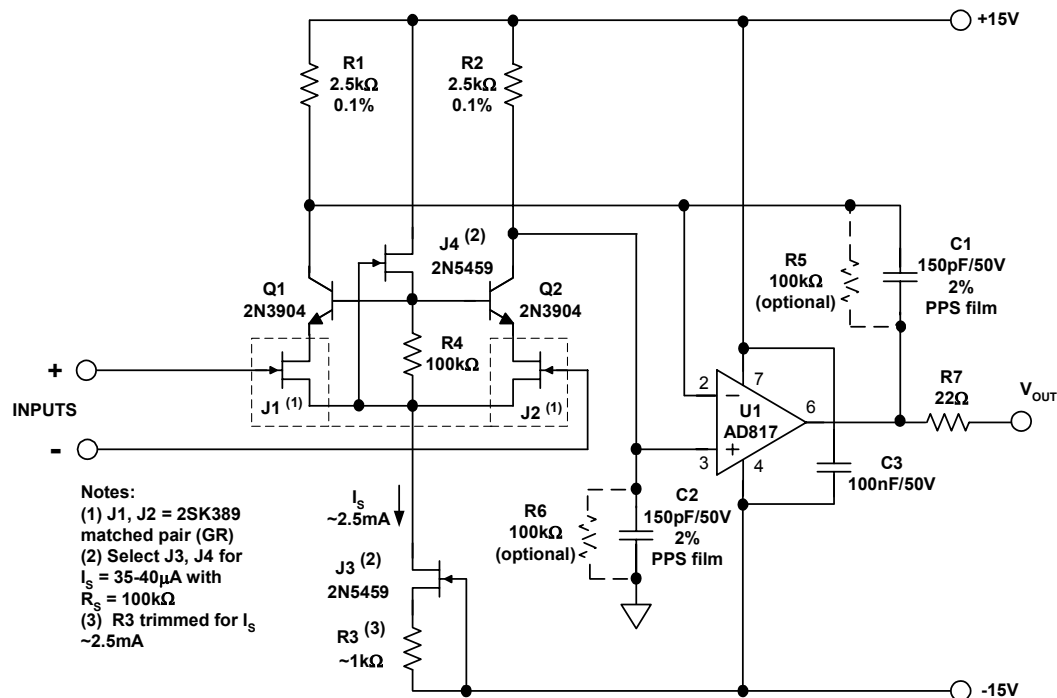


Figure 6-180: Low noise JFET gain-boosted input composite amplifier

The design uses a low noise JFET pair as the gain stage, the 2SK389 device (see Reference 17). Biased for drain currents of more than 2mA, this device pair is capable of achieving an input voltage noise density of less than $1.5\text{nV}/\sqrt{\text{Hz}}$. The basic device is available in three I_{DSS} grades, GR (2.6-6.5mA), BL (6-12mA), and V (12-20mA). The lowest noise will be found with use of the highest I_{DSS} parts, at the expense of course, of supply current. This design example can use any grade, by biasing J1-J2 for the GR parts (at an I_{DSS} of 2.5mA). This still gives good noise performance for an FET-input amplifier (about $1.8\text{nV}/\sqrt{\text{Hz}}$), but at a still reasonable power supply drain.

A byproduct of the large geometry devices of this devices series is a relatively high capacitance. If this factor is not addressed, this large and non-linear capacitance could cause distortion, for applications operating the circuit as a follower. To counteract this, the input stage of J1-J2 is cascoded, by the Q1-Q2 and J4 arrangement. This removes the major degradation of operation due to the J1-J2 capacitance, and it also stabilizes the DC operating points of J1-J2. From the output collectors of Q1-Q2 onward, the amplifier operates generally as the Fig. 6-177 circuit previously described. An AD817 is used for U1, so as to take advantage of its wide bandwidth and high output current.

The unity-gain bandwidth of this circuit is about 15MHz, but the open-loop gain is user selectable, by virtue of optional resistors R5 and R6. With these resistors connected, the composite amplifier open-loop bandwidth is $\approx 10\text{kHz}$, and open-loop gain is about 63dB. These attributes make it well suited for audio applications, for example. Without R5 and R6, the open-loop gain is more like that of a conventional op amp, with a gain of more than 100dB at low frequencies.

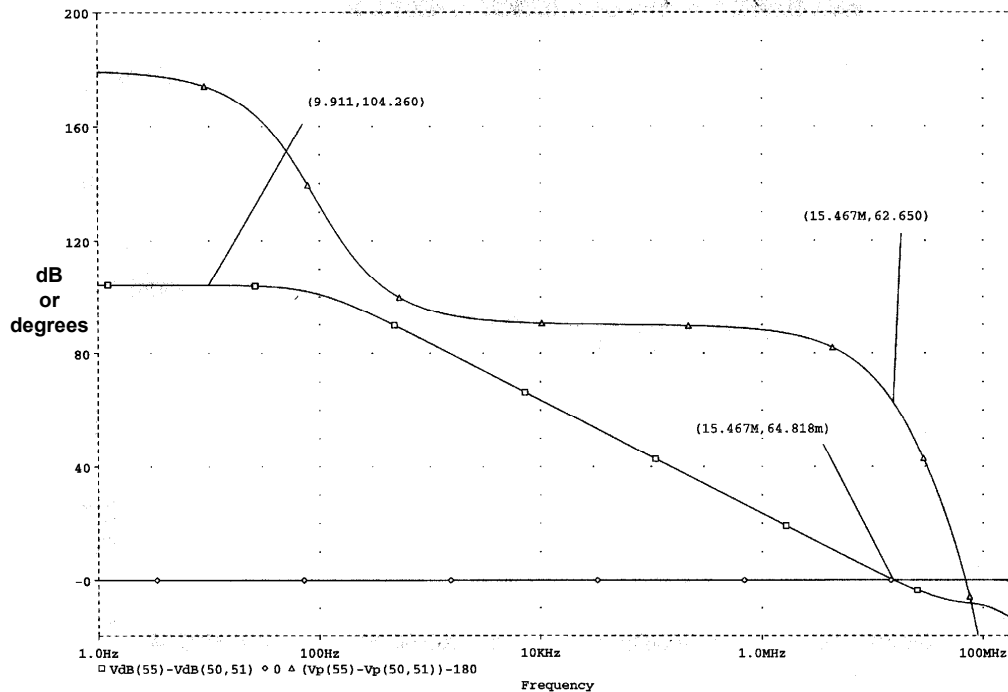


Figure 6-181: Gain/phase versus frequency for Fig. 6-180 composite amplifier

The open-loop response for R5 and R6 open is shown in Figure 6-181 above. In this simulation the load resistance was 600Ω . As can be noted, the response is clean, without phase aberrations. Phase margin at the unity-gain crossover frequency is about 63° , and the low frequency gain is about 104dB.

Although this circuit does have some excellent AC characteristics, it should be noted that it is *not* a general purpose op amp circuit. One reason for this is that the cascode input stage is a two-edged sword. While it reduces capacitance and improves distortion, it also limits the allowable CM input range. The positive swing headroom is limited by roughly the DC drop across R1 and R2, plus that of the cascode, or $3 + 4.5\text{V}$. This means the most positive CM input should be less than about 5V_{peak} , or 3.5V_{rms} . This of course won't be a practical limitation for non-inverting amplifiers with noise gains of $5\times$ or more, or for low-level preamps with high gains of 100 or $1000\times$.

As compensated in Fig. 6-180, the composite op amp should be unity-gain stable. At closed-loop gains appreciably higher than about $5\times$, a reduction of C1-C2 can be considered, which will allow greater bandwidth and SR to be realized. Offset of the J1-J2 pair can be as high as 20mV, so offset trim may be in order for DC-coupled applications. For lower noise, a high I_{DSS} grade for J1-J2 should be used, with I_{S} raised to 5mA or more. R1-R2 will need to be lowered, and C1-C2 raised, in proportion.

"Nostalgia" vacuum tube input/output composite op amp

In keeping with the theme of this book's History chapter, the final composite amplifier design for this section uses venerable vacuum tube devices, which formed the basis of the first ever op amps. Today however, designing a vacuum tube op amp has some advantages, vis-à-vis the early days—transistors didn't exist! Thus the nostalgia op amp shown in Figure 6-182 below incorporates techniques of both today as well as yesteryear.

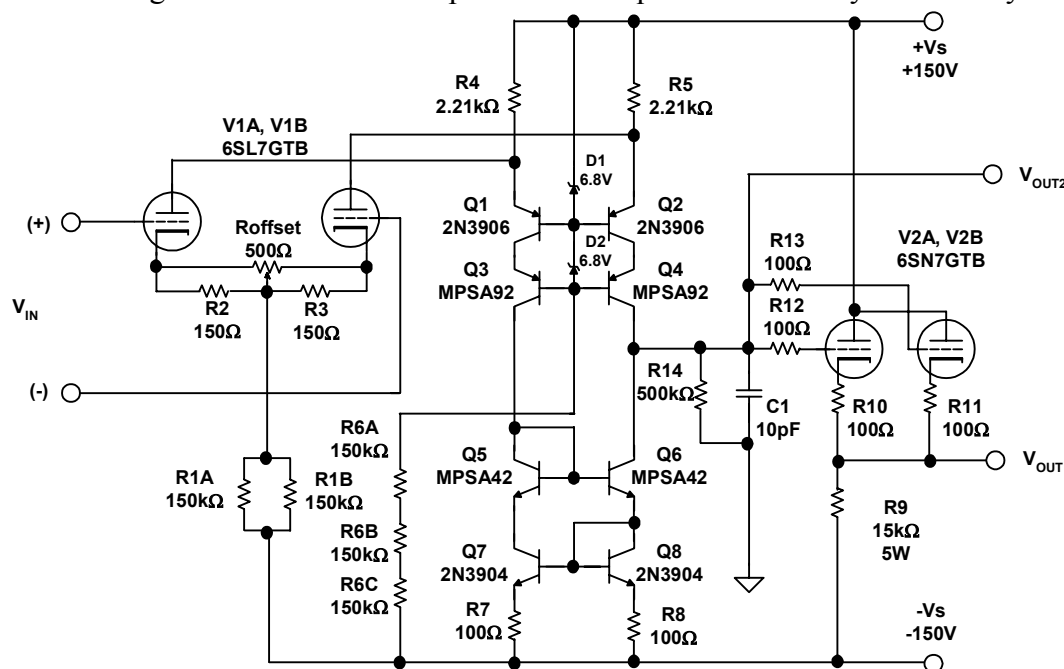


Figure 6-182: "Nostalgia" vacuum tube input/output composite op amp

Note that this particular circuit should be taken as a design exercise rather than a practical example. Yet, it was still a lot of fun to design using available SPICE models (see Reference 18). As such, it offers some insights not available to early op amp designers.

Here V1 is a dual triode input stage, using the high gain 6SL7 (or alternately, the close-cousin 12AX7 miniature). It is operated here as a linear transconductance, long-tailed differential pair. Rather than using conventional plate loading, the output signal current from V1A-V1B is passed to a folded cascode stage Q1-Q4, which is loaded by a linear, high voltage current mirror, Q5-Q8. The transistors of the current mirror are also cascaded, both for higher output impedance as well as for required high voltage capability. A regulated 6.3V heater supply for V1 is suggested, for highest stability.

Voltage gain of this one stage op amp is approximately equal to the V1A-V1B transconductance times the nodal impedance seen at V_{OUT2} . With R14 open this impedance is very high, so gain can also be quite high (≈ 77 dB simulated). With R14 500k, gain is about 53dB. Open-loop bandwidth is established by the shunt capacitance at the high-Z node and R14, and measures about 7MHz gain bandwidth in simulation. A 6SN7 dual cathode follower output stage for V2 allows up to 10mA of load drive.

Laboratory test results for this design are left as an exercise for the interested reader, and feedback is welcome.

REFERENCES: COMPOSITE AMPLIFIERS

1. Moshe Gerstenhaber, Mark Murphy, Scott Wurcer, "Composite Amp Has Low Noise, Drift," **Electronic Design**, January 21, 1993, pp. 62, 63.
2. Paul Brokaw, "Composite Amp Has Low Noise, Drift (Update)," **Electronic Design**, June 5, 2000.
3. Data Sheet for **MMDT2907A Dual PNP Small Signal Surface Mount Transistor**, Rev B-2 and Data Sheet for **MMDT2222A Dual NPN Small Signal Surface Mount Transistor**, Rev C-2, <http://www.diodes.com/products/>
4. Data Sheet for **ZDT751 SM-8 Dual PNP Medium Power Transistors**, August 1997, and Data Sheet for **ZDT651 SM-8 Dual NPN Medium Power Transistors**, August 1997, <http://www.zetex.com>
5. Data Sheet for **ZTX752, ZDX753 PNP Silicon Planar Medium Power Transistors**, July 1994, <http://www.zetex.com>
6. Data Sheet for **ZTX652, ZDX653 NPN Silicon Planar Medium Power Transistors**, July 1994, <http://www.zetex.com>
7. Mark Alexander, "The Alexander Current-Feedback Audio Amplifier," **Analog Devices AN211**. See also: Mark Alexander, "Current Feedback Audio Power Amplifier," **US Patent 5,097,223**, filed May 22, 1990, issued March 17, 1992.
8. Barry Kline, "Enhanced Op Amp Delivers 100Vp-p," **EDN**, September 5, 1985, pp. 309, 311-312.
9. Walter G. Jung, "Chapter 5, Amplifier Circuit Techniques," **IC Array Cookbook**, Hayden Book Company, 1980, ISBN 0-8104-0762-0.
10. Data sheet for **MAT02 Low Noise, Matched Dual Monolithic Transistor**, <http://www.analog.com>
11. Andrew Jenkins, Derek Bowers, "NPN Pairs Yield Ultra-Low-Noise Op Amp," **EDN**, May 3, 1984.
12. Data sheet for **MAT03 Low Noise, Matched Dual PNP Transistor**, <http://www.analog.com>
13. "Choosing and Using N-Channel Dual J-FETs," **Analog Dialogue**, Vol. 4, No. 2, pp. 4-9.
14. "TDN: Temperature Drift Nonlinearity— A New Dual-FET Specification," **Analog Dialogue**, Vol. 6, No. 1, pp. 13-14 .
15. Robert I. Demrow, "Settling Time of Operational Amplifiers," **Analog Dialogue**, Vol. 4, No. 1. See also ADI application note AN359.
16. "The FET Constant-Current Source/Limiter," Siliconix AN103, March 10, 1997, <http://www.vishay.com/brands/siliconix/SSFan.html>
17. Data sheet for **2SK389 Dual FET, Silicon Monolithic N-Channel Junction Type**, <http://www.semicon.toshiba.co.jp/eng/solution/audio/pdf/e001543.pdf>
18. Duncan Munro's SPICE vacuum tube models, <http://www.duncanamps.com>

ACKNOWLEDGMENTS:

Helpful comments on this section were received from Erno Borbely, Steve Bench, and Gary Longrie.

▣ OP AMP APPLICATIONS

NOTES:

- H Op Amp History**
 - 1 Op Amp Basics**
 - 2 Specialty Amplifiers**
 - 3 Using Op Amps with Data Converters**
 - 4 Sensor Signal Conditioning**
 - 5 Analog Filters**
 - 6 Signal Amplifiers**
 - ◆ 7 Hardware and Housekeeping Techniques**
 - 1 Passive Components**
 - 2 PCB Design Issues**
 - 3 Op Amp Power Supply Systems**
 - 4 Op Amp Protection**
 - 5 Thermal Considerations**
 - 6 EMI/RFI Considerations**
 - 7 Simulation, Breadboarding and Prototyping**

▣ OP AMP APPLICATIONS

CHAPTER 7: HARDWARE AND HOUSEKEEPING TECHNIQUES

***Walt Kester, James Bryant, Walt Jung,
Joe Buxton, Wes Freeman***

This chapter, one of the longer of those within the book, deals with topics just as important as all of those basic circuits immediately surrounding the op amp, discussed earlier. The chapter deals with various and sundry circuit/system issues which fall under the guise of system *hardware and housekeeping techniques*. In this context, the hardware and housekeeping may be all those support items surrounding an op amp, excluding the op amp itself. This includes issues of passive components, printed circuit design, power supply systems, protection of op amp devices against overvoltage and thermal effects, EMI/RFI issues, and finally, simulation, breadboarding and prototyping. Some of these topics aren't directly involved in the actual signal path of a design, but they are every bit as important as choosing the correct device and surrounding circuit values.

SECTION 7-1: PASSIVE COMPONENTS

James Bryant, Walt Jung, Walt Kester

Introduction

When designing with op amps and other precision analog devices, it is critical that users avoid the pitfall of poor passive component choice. In fact, the wrong passive component can derail even the best op amp or data converter application. This section includes discussion of some basic traps of choosing passive components for op amp applications.

So, you've spent good money for a precision op amp or data converter, only to find that, when plugged into your board, the device doesn't meet spec. Perhaps the circuit suffers from drift, poor frequency response, and oscillations—or simply doesn't achieve expected accuracy. Well, before you blame the device, you should closely examine your passive components—including capacitors, resistors, potentiometers, and yes, even the printed circuit boards. In these areas, subtle effects of tolerance, temperature, parasitics, aging, and user assembly procedures can unwittingly sink your circuit. And all too often these effects go unspecified (or underspecified) by passive component manufacturers.

In general, if you use data converters having 12 bits or more of resolution, or op amps that cost more than a few dollars, pay very close attention to passive components. Consider the case of a 12-bit DAC, where $\frac{1}{2}$ LSB corresponds to 0.012% of full scale, or only 122 ppm. A host of passive component phenomena can accumulate errors far exceeding this! But, buying the most expensive passive components won't necessarily solve your problems either. Often, a *correct* 25-cent capacitor yields a better-performing, more cost-effective design than a premium-grade part. With a few basics, understanding and analyzing passive components may prove rewarding, albeit not easy.

Capacitors

Most designers are generally familiar with the range of capacitors available. But the mechanisms by which both static and dynamic errors can occur in precision circuit designs using capacitors are sometimes easy to forget, because of the tremendous variety of types available. These include dielectrics of glass, aluminum foil, solid tantalum and tantalum foil, silver mica, ceramic, Teflon, and the film capacitors, including polyester, polycarbonate, polystyrene, and polypropylene types. In addition to the traditional leaded packages, many of these are now also offered in surface mount styles.

Figure 7-1 is a workable model of a non-ideal capacitor. The nominal capacitance, C , is shunted by a resistance R_p , which represents *insulation resistance* or leakage. A second resistance, R_s , —*equivalent series resistance*, or ESR, —appears in series with the capacitor and represents the resistance of the capacitor leads and plates.

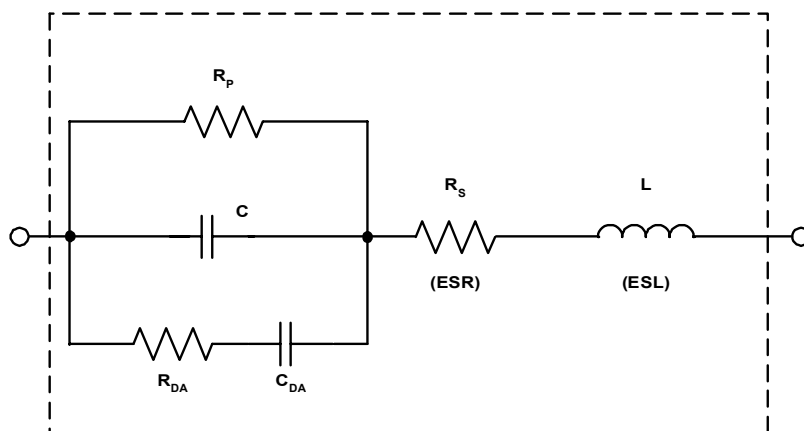


Figure 7-1: A non-ideal capacitor equivalent circuit includes parasitic elements

Note that capacitor phenomena aren't that easy to separate out. The matching of phenomena and models is for convenience in explanation. Inductance, L —the *equivalent series inductance*, or ESL, —models the inductance of the leads and plates. Finally, resistance R_{DA} and capacitance C_{DA} together form a simplified model of a phenomenon known as *dielectric absorption*, or DA. It can ruin fast and slow circuit dynamic performance. In a real capacitor R_{DA} and C_{DA} extend to include multiple parallel sets. These parasitic RC elements can act to degrade timing circuits substantially, and the phenomenon is discussed further below.

Dielectric Absorption

Dielectric absorption, which is also known as "soakage" and sometimes as "dielectric hysteresis" — is perhaps the least understood and potentially most damaging of various capacitor parasitic effects. Upon discharge, most capacitors are reluctant to give up all of their former charge, due to this memory consequence.

Figure 7-2 illustrates this effect. On the left of the diagram, after being charged to the source potential of V volts at time t_0 , the capacitor is shorted by the switch $S1$ at time t_1 , discharging it. At time t_2 , the capacitor is then open-circuited; a residual voltage slowly

builds up across its terminals and reaches a nearly constant value. This error voltage is due to DA, and is shown in the right figure, a time/voltage representation of the charge/discharge/recovery sequence. Note that the recovered voltage error is proportional to both the original charging voltage V , as well as the rated DA for the capacitor in use.

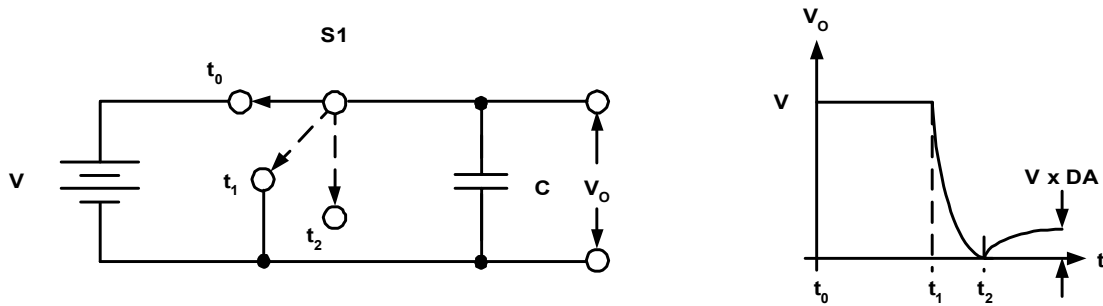


Figure 7-2: A residual open-circuit voltage after charge/discharge characterizes capacitor dielectric absorption

Standard techniques for specifying or measuring dielectric absorption are few and far between. Measured results are usually expressed as the percentage of the original charging voltage that reappears across the capacitor. Typically, the capacitor is charged for a long period, then shorted for a shorter established time. The capacitor is then allowed to recover for a specified period, and the residual voltage is then measured (see Reference 8 for details). While this explanation describes the basic phenomenon, it is important to note that real-world capacitors vary quite widely in their susceptibility to this error, with their rated DA ranging from well below to above 1%, the exact number being a function of the dielectric material used.

In practice, DA makes itself known in a variety of ways. Perhaps an integrator refuses to reset to zero, a voltage-to-frequency converter exhibits unexpected nonlinearity, or a sample-and-hold (SH) exhibits varying errors. This last manifestation can be particularly damaging in a data-acquisition system, where adjacent channels may be at voltages which differ by nearly full scale, as shown below.

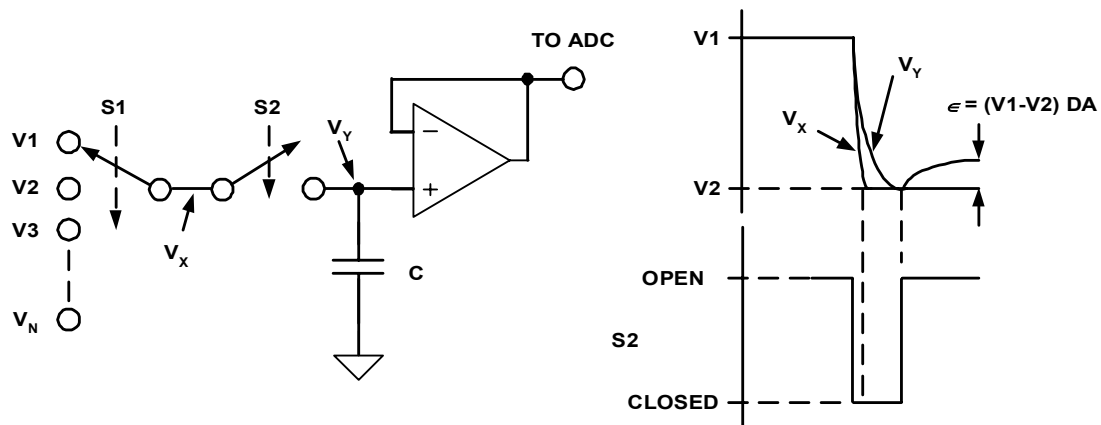


Figure 7-3: Dielectric absorption induces errors in SH applications

Figure 7-3 illustrates the case of DA error in a simple SH. On the left, switches S1 and S2 represent an input multiplexer and SH switch, respectively. The multiplexer output voltage is V_X , and the sampled voltage held on C is V_Y , which is buffered by the op amp for presentation to an ADC. As can be noted by the timing diagram on the right, a DA

■ OP AMP APPLICATIONS

error voltage, ϵ , appears in the hold mode, when the capacitor is effectively open circuit. This voltage is proportional to the difference of voltages V_1 and V_2 , which, if at opposite extremes of the dynamic range, exacerbates the error. As a practical matter, the best solution for good performance in terms of DA in a SH is to use only the best capacitor.

The DA phenomenon is a characteristic of the dielectric material itself, although inferior manufacturing processes or electrode materials can also affect it. DA is specified as a percentage of the charging voltage. It can range from a low of 0.02% for Teflon, polystyrene, and polypropylene capacitors, up to a high of 10% or more for some electrolytics. For some time frames, the DA of polystyrene can be as low as 0.002%.

Common high-K ceramics and polycarbonate capacitor types display typical DA on the order of 0.2%, it should be noted this corresponds to $\frac{1}{2}$ LSB at only 8 bits! Silver mica, glass, and tantalum capacitors typically exhibit even larger DA, ranging from 1.0% to 5.0%, with those of polyester devices failing in the vicinity of 0.5%. As a rule, if the capacitor spec sheet doesn't specifically discuss DA *within your time frame and voltage range*, exercise caution! Another type with lower *specified* DA is likely a better choice.

DA can produce long tails in the transient response of fast-settling circuits, such as those found in high-pass active filters or ac amplifiers. In some devices used for such applications, Figure 7-1's R_{DA} - C_{DA} model of DA can have a time constant of milliseconds. Much longer time constants are also quite usual. In fact, several paralleled R_{DA} - C_{DA} circuit sections with a wide range of time constants can model some devices. In fast-charge, fast-discharge applications, the behavior of the DA mechanism resembles "analog memory"; the capacitor in effect tries to remember its previous voltage.

In some designs, you can compensate for the effects of DA if it is simple and easily characterized, and you are willing to do custom tweaking. In an integrator, for instance, the output signal can be fed back through a suitable compensation network, tailored to cancel the circuit equivalent of the DA by placing a negative impedance effectively in parallel. Such compensation has been shown to improve SH circuit performance by factors of 10 or more (Reference 6).

Capacitor Parasitics and Dissipation Factor

In Figure 7-1, a capacitor's leakage resistance, R_p , the effective series resistance, R_s , and effective series inductance, L , act as parasitic elements, which can degrade an external circuit's performance. The effects of these elements are often lumped together and defined as a dissipation factor, or DF.

A capacitor's leakage is the small current that flows through the dielectric when a voltage is applied. Although modeled as a simple insulation resistance (R_p) in parallel with the capacitor, the leakage actually is nonlinear with voltage. Manufacturers often specify leakage as a megohm-microfarad product, which describes the dielectric's self-discharge time constant, in seconds. It ranges from a low of 1s or less for high-leakage capacitors, such as electrolytic devices, to the 100's of seconds for ceramic capacitors. Glass devices exhibit self-discharge time-constants of 1,000 or more; but the best leakage performance is shown by Teflon and the film devices (polystyrene, polypropylene), with time

constants exceeding 1,000,000 megohm-microfarads. For such a device, external leakage paths—created by surface contamination of the device's case or in the associated wiring or physical assembly—can overshadow the internal dielectric-related leakage.

Effective series inductance, ESL (Figure 7-1, again) arises from the inductance of the capacitor leads and plates, which, particularly at the higher frequencies, can turn a capacitor's normally capacitive reactance into an inductive reactance. Its magnitude strongly depends on construction details within the capacitor. Tubular wrapped-foil devices display significantly more lead inductance than molded radial-lead configurations. Multilayer ceramic and film-type devices typically exhibit the lowest series inductance, while ordinary tantalum and aluminum electrolytics typically exhibit the highest. Consequently, standard electrolytic types, if used alone, usually prove insufficient for *high-speed* local bypassing applications. Note however that there also are more specialized aluminum and tantalum electrolytics available, which may be suitable for higher speed uses. These are the types generally designed for use in switch-mode power supplies, which are covered more completely in a following section.

Manufacturers of capacitors often specify effective series impedance by means of impedance-versus-frequency plots. Not surprisingly, these curves show graphically a predominantly capacitive reactance at low frequencies, with rising impedance at higher frequencies because of the effect of series inductance.

Effective series resistance, ESR (resistor R_S of Figure 7-1), is made up of the resistance of the leads and plates. As noted, many manufacturers lump the effects of ESR, ESL, and leakage into a single parameter called *dissipation factor*, or DF. Dissipation factor measures the basic inefficiency of the capacitor. Manufacturers define it as the ratio of the energy lost to energy stored per cycle by the capacitor. The ratio of ESR to total capacitive reactance— at a specified frequency— approximates the dissipation factor, which turns out to be equivalent to the reciprocal of the figure of merit, Q . Stated as an approximation, $Q \approx 1/DF$ (with DF in numeric terms). For example, a DF of 0.1% is equivalent to a fraction of 0.001; thus the inverse in terms of Q would be 1000.

Dissipation factor often varies as a function of both temperature and frequency. Capacitors with mica and glass dielectrics generally have DF values from 0.03% to 1.0%. For ordinary ceramic devices, DF ranges from a low of 0.1 % to as high as 2.5 % at room temperature. And electrolytics usually exceed even this level. The film capacitors are the best as a group, with DFs of less than 0.1 %. Stable-dielectric ceramics, notably the NP0 (also called COG) types, have DF specs comparable to films (more below).

Tolerance, Temperature, and Other Effects

In general, precision capacitors are expensive and— even then— not necessarily easy to buy. In fact, choice of capacitance is limited both by the range of available values, and also by tolerances. In terms of size, the better performing capacitors in the film families tend to be limited in practical terms to 10 μ F or less (for dual reasons of size and expense). In terms of low value tolerance, $\pm 1\%$ is possible for NP0 ceramic and some film devices, but with possibly unacceptable delivery times. Many film capacitors can be made available with tolerances of less than $\pm 1\%$, but on a special order basis only.

■ OP AMP APPLICATIONS

Most capacitors are sensitive to temperature variations. DF, DA, and capacitance value are all functions of temperature. For some capacitors, these parameters vary approximately linearly with temperature, in others they vary quite nonlinearly. Although it is usually not important for SH applications, an excessively large *temperature coefficient* (TC, measured in ppm/°C) can prove harmful to the performance of precision integrators, voltage-to-frequency converters, and oscillators. NP0 ceramic capacitors, with TCs as low as 30 ppm/°C, are the best for stability, with polystyrene and polypropylene next best, with TCs in the 100-200 ppm/°C range. On the other hand, when capacitance stability is important, one should stay away from types with TCs of more than a few hundred ppm/°C, or in fact any TC which is nonlinear.

A capacitor's maximum working temperature should also be considered, in light of the expected environment. Polystyrene capacitors, for instance, melt near 85°C, compared to Teflon's ability to survive temperatures up to 200°C.

Sensitivity of capacitance and DA to applied voltage, expressed as *voltage coefficient*, can also hurt capacitor performance within a circuit application. Although capacitor manufacturers don't always clearly specify voltage coefficients, the user should always consider the possible effects of such factors. For instance, when maximum voltages are applied, some high-K ceramic devices can experience a decrease in capacitance of 50% or more. This is an inherent distortion producer, making such types unsuitable for signal path filtering, for example, and better suited for supply bypassing. Interestingly, NP0 ceramics, the stable dielectric subset from the wide range of available ceramics, do offer good performance with respect to voltage coefficient.

Similarly, the capacitance, and dissipation factor of many types vary significantly with frequency, mainly as a result of a variation in dielectric constant. In this regard, the better dielectrics are polystyrene, polypropylene, and Teflon.

Assemble Critical Components Last

The designer's worries don't end with the design process. Some common printed circuit assembly techniques can prove ruinous to even the best designs. For instance, some commonly used cleaning solvents can infiltrate certain electrolytic capacitors— those with rubber end caps are particularly susceptible. Even worse, some of the film capacitors, polystyrene in particular, actually melt when contacted by some solvents. Rough handling of the leads can damage still other capacitors, creating random or even intermittent circuit problems. Etched-foil types are particularly delicate in this regard. To avoid these difficulties it may be advisable to mount especially critical components as the last step in the board assembly process— if possible.

Table 7-1 summarizes selection criteria for various capacitor types, arranged roughly in order of decreasing DA performance. In a selection process, the general information of this table should be supplemented by consultation of current vendor's catalog information (see References at end of section).

Table 7-1
Capacitor Comparison Chart

TYPE	TYPICAL DA	ADVANTAGES	DISADVANTAGES
Polystyrene	0.001% to 0.02%	Inexpensive Low DA Good stability (~120ppm/°C)	Damaged by temperature > +85°C Large High inductance Vendors limited
Polypropylene	0.001% to 0.02%	Inexpensive Low DA Stable (~200ppm/°C) Wide range of values	Damaged by temperature > +105°C Large High inductance
Teflon	0.003% to 0.02%	Low DA available Good stability Operational above +125 °C Wide range of values	Expensive Large High inductance
Polycarbonate	0.1%	Good stability Low cost Wide temperature range Wide range of values	Large DA limits to 8-bit applications High inductance
Polyester	0.3% to 0.5%	Moderate stability Low cost Wide temperature range Low inductance (stacked film)	Large DA limits to 8-bit applications High inductance (conventional)
NP0 Ceramic	<0.1%	Small case size Inexpensive, many vendors Good stability (30ppm/°C) 1% values available Low inductance (chip)	DA generally low (may not be specified) Low maximum values ($\leq 10\text{nF}$)
Monolithic Ceramic (High K)	>0.2%	Low inductance (chip) Wide range of values	Poor stability Poor DA High voltage coefficient
Mica	>0.003%	Low loss at HF Low inductance Good stability 1% values available	Quite large Low maximum values ($\leq 10\text{nF}$) Expensive
Aluminum Electrolytic	Very high	Large values High currents High voltages Small size	High leakage Usually polarized Poor stability, accuracy Inductive
Tantalum Electrolytic	Very high	Small size Large values Medium inductance	High leakage Usually polarized Expensive Poor stability, accuracy

■ OP AMP APPLICATIONS

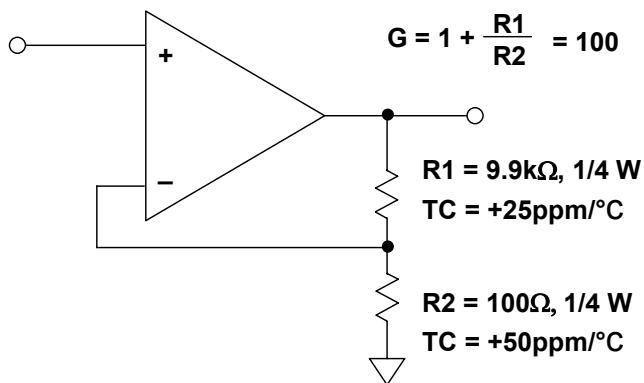
Designers should also consider the natural failure mechanisms of capacitors. Metallized film devices, for instance, often self-heal. They initially fail due to conductive bridges that develop through small perforations in the dielectric film. But, the resulting fault currents can generate sufficient heat to destroy the bridge, thus returning the capacitor to normal operation (at a slightly lower capacitance). Of course, applications in high-impedance circuits may not develop sufficient current to clear the bridge, so the designer must be wary here.

Tantalum capacitors also exhibit a degree, of self-healing, but—unlike film capacitors—the phenomenon depends on the temperature at the fault location rising slowly. Therefore, tantalum capacitors self-heal best in high impedance circuits which limit the surge in current through the capacitor's defect. Use caution therefore, when specifying tantals for high-current applications.

Electrolytic capacitor life often depends on the rate at which capacitor fluids seep through end caps. Epoxy end seals perform better than rubber seals, but an epoxy sealed capacitor can explode under severe reverse-voltage or overvoltage conditions. Finally, *all* polarized capacitors must be protected from exposure to voltages outside their specifications.

Resistors and Potentiometers

Designers have a broad range of resistor technologies to choose from, including carbon composition, carbon film, bulk metal, metal film, and both inductive and non-inductive wire-wound types. As perhaps the most basic—and presumably most trouble-free—of components, resistors are often overlooked as error sources in high performance circuits.



◆ Temperature change of 10°C causes gain change of 250ppm

◆ This is 1LSB in a 12-bit system and a disaster in a 16-bit system

Figure 7-4: Mismatched resistor TCs can induce temperature-related gain errors

Yet, an improperly selected resistor can subvert the accuracy of a 12-bit design by developing errors well in excess of 122 ppm ($\frac{1}{2}$ LSB). When did you last read a resistor data sheet? You'd be surprised what can be learned from an informed review of data.

Consider the simple circuit of Figure 7-4, showing a non-inverting op amp where the 100x gain is set by R1 and R2. The TCs of these two resistors are a somewhat obvious source of error. Assume the op amp gain errors to be negligible, and that the resistors are

perfectly matched to a 99/1 ratio at +25°C. If, as noted, the resistor TCs differ by only 25ppm/°C, the gain of the amplifier changes by 250ppm for a 10°C temperature change. This is about a 1 LSB error in a 12-bit system, and a major disaster in a 16-bit system. Temperature changes, however, can limit the accuracy of the Figure 7-4 amplifier in several ways. In this circuit (as well as many op amp circuits with component-ratio defined gains), the *absolute* TC of the resistors is less important— *as long as they track one another in ratio*. But even so, some resistor types simply aren't suitable for precise work. For example, *carbon composition* units—with TCs of approximately 1,500 ppm/°C, won't work. Even if the TCs could be matched to an unlikely 1%, the resulting 15 ppm/°C differential still proves inadequate— an 8°C shift creates a 120 ppm error.

Many manufacturers offer metal film and bulk metal resistors, with absolute TCs ranging between ±1 and ±100 ppm/°C. Beware, though; TCs can vary a great deal, particularly among discrete resistors from different batches. To avoid this problem, more expensive matched resistor pairs are offered by some manufacturers, with temperature coefficients that track one another to within 2 to 10 ppm/°C. Low-priced thin-film networks have good relative performance and are widely used.

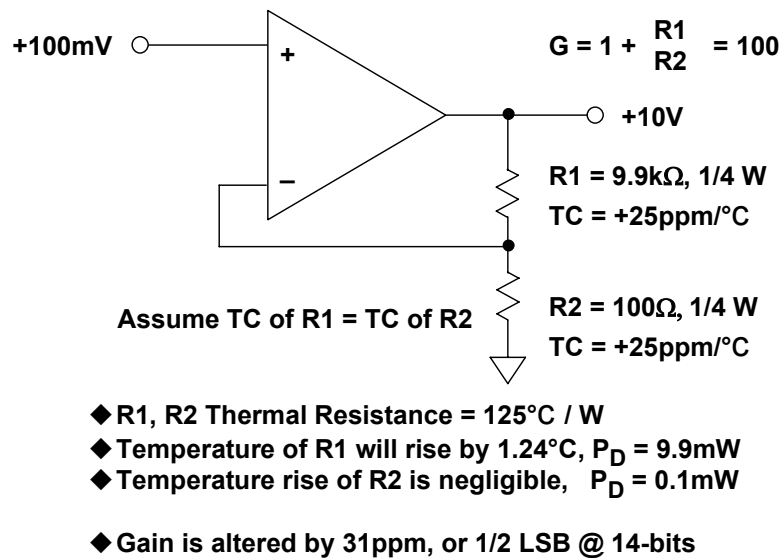


Figure 7-5: Uneven power dissipation between resistors with identical TCs can also introduce temperature-related gain errors

Suppose, as shown in Figure 7-5, R1 and R2 are ¼W resistors with identical 25ppm/°C TCs. Even when the TCs are identical, there can still be significant errors! When the signal input is zero, the resistors dissipate no heat. But, if it is 100mV, there is 9.9V across R1, which then dissipates 9.9mW. It will experience a temperature rise of 1.24°C (due to a 125°C/W ¼W resistor thermal resistance). This 1.24°C rise causes a resistance change of 31ppm, and thus a corresponding gain change. But R2, with only 100mV across it, is only heated a negligible 0.0125°C. The resulting 31 ppm net gain error represents a fullscale error of ½ LSB at 14-bits, and is a disaster for a 16-bit system.

Even worse, the effects of this resistor self-heating also create easily calculable *nonlinearity errors*. In the Figure 7-5 example, with ½ the voltage input, the resulting self-heating error is only 15 ppm. In other words, the stage gain is not constant at ½ and

■ OP AMP APPLICATIONS

full scale (nor is it so at other points), as long as uneven temperature shifts exist between the gain-determining resistors. This is by no means a worst-case example; physically smaller resistors would give worse results, due to higher associated thermal resistance.

These, and similar errors, are avoided by selecting critical resistors that are accurately matched for both value and TC, are well derated for power, and have tight thermal coupling between those resistors where matching is important. This is best achieved by using a resistor network on a single substrate—such a network may either be within an IC, or it may be a separately packaged thin-film resistor network.

When the circuit resistances are very low ($\leq 10\Omega$), *interconnection stability* also becomes important. For example, while often overlooked as an error, the resistance TC of typical copper wire or printed circuit traces can add errors. The TC of copper is typically $\sim 3,900$ ppm/ $^{\circ}\text{C}$. Thus a precision 10Ω , 10 ppm/ $^{\circ}\text{C}$ wirewound resistor with 0. 1Ω of copper interconnect effectively becomes a 10.1Ω resistor with a TC of nearly 50 ppm/ $^{\circ}\text{C}$.

One final consideration applies mainly to designs that see widely varying ambient temperatures: a phenomenon known as *temperature retrace* describes the change in resistance which occurs after a specified number of cycles of exposure to low and high ambients with constant internal dissipation. Temperature retrace can exceed 10 ppm/ $^{\circ}\text{C}$, even for some of the better thin-film components.

- ◆ Closely match resistance TCs.
- ◆ Use resistors with low absolute TCs.
- ◆ Use resistors with low thermal resistance (higher power ratings, larger cases).
- ◆ Tightly couple matched resistors thermally (use standard common-substrate networks).
- ◆ For large ratios consider using stepped attenuators.

Figure 7-6: A number of points are important towards minimizing temperature-related errors in resistors

In summary, to design resistance-based circuits for minimum temperature-related errors, consider the points noted in Figure 7-6 (along with their cost):

Resistor Parasitics

Resistors can exhibit significant levels of parasitic inductance or capacitance, especially at high frequencies. Manufacturers often specify these parasitic effects as a reactance error, in % or ppm, based on the ratio of the difference between the impedance magnitude and the dc resistance, to the resistance, at one or more frequencies.

Wirewound resistors are especially susceptible to difficulties. Although resistor manufacturers offer wirewound components in either normal or noninductively wound form, even noninductively wound resistors create headaches for designers. These

resistors still appear slightly inductive (of the order of 20 μH) for values below 10k Ω . Above 10k Ω the same style resistors actually exhibit 5 pF of shunt capacitance.

These parasitic effects can raise havoc in dynamic circuit applications. Of particular concern are applications using wirewound resistors with values both greater than 10k Ω . Here it isn't uncommon to see peaking, or even oscillation. These effects become more evident at low-kHz frequency ranges.

Even in low-frequency circuit applications, parasitic effects in wirewound resistors can create difficulties. Exponential settling to 1 ppm may take 20 time constants or more. The parasitic effects associated with wirewound resistors can significantly increase net circuit settling time to beyond the length of the basic time constants.

Unacceptable amounts of parasitic reactance are often found even in resistors that aren't wirewound. For instance, some metal-film types have significant interlead capacitance, which shows up at high frequencies. In contrast, when considering this end-end capacitance, carbon resistors do the best at high frequencies.

Thermoelectric Effects

Another more subtle problem with resistors is the *thermocouple effect*, also sometimes referred to as *thermal EMF*. Wherever there is a junction between two different metallic conductors, a thermoelectric voltage results. The thermocouple effect is widely used to measure temperature, as described in detail within Chapter 4. However, in any low level precision op amp circuit it is also a potential source of inaccuracy, since wherever two different conductors meet, a thermocouple is formed (whether we like it or not). In fact, in many cases, it can easily produce the dominant error within an otherwise precision circuit design.

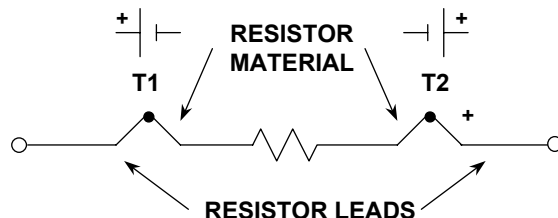
Parasitic thermocouples will cause errors when and if the various junctions forming the parasitic thermocouples are at different temperatures. With two junctions present on each side of the signal being processed within a circuit, by definition we have formed at least one thermocouple pair. If the two junctions of this thermocouple pair are at different temperatures, there will be a net temperature dependent error voltage produced. Conversely, if the two junctions of a parasitic thermocouple pair are kept at an identical temperature, then the net error produced will be zero, as the voltages of the two thermocouples effectively will be canceled.

This is a critically important point, since in practice we cannot avoid connecting dissimilar metals together to build an electronic circuit. But, what we can do is carefully control temperature differentials across the circuit, so such that the undesired thermocouple errors cancel one another.

The effect of such parasitics is very hard to avoid. To understand this, consider a case of making connections *with copper wire only*. In this case, even a junction formed by different copper wire alloys can have a thermoelectric voltage which is a small fraction of 1 $\mu\text{V}/^\circ\text{C}$! And, taking things a step further, even such apparently benign components as resistors contain parasitic thermocouples, with potentially even stronger effects.

■ OP AMP APPLICATIONS

For example, consider the resistor model shown in Figure 7-7. The two connections between the resistor material and the leads form thermocouple junctions, T1 and T2. This thermocouple EMF can be as high as $400\mu\text{V}/^\circ\text{C}$ for some carbon composition resistors, and as low as $0.05\mu\text{V}/^\circ\text{C}$ for specially constructed resistors (see Reference 15). Ordinary metal film resistors (RN-types) are typically about $20\mu\text{V}/^\circ\text{C}$.



TYPICAL RESISTOR THERMOCOUPLE EMFs

◆ CARBON COMPOSITION	$\approx 400 \mu\text{V}/^\circ\text{C}$
◆ METAL FILM	$\approx 20 \mu\text{V}/^\circ\text{C}$
◆ EVENOHM OR MANGANIN WIREWOUND	$\approx 2 \mu\text{V}/^\circ\text{C}$
◆ RCD Components HP-Series	$\approx 0.05 \mu\text{V}/^\circ\text{C}$

Figure 7-7: Every resistor contains two thermocouples, formed between the leads and resistance element

Note that these thermocouple effects are relatively unimportant for ac signals. Even for DC-only signals, they will nicely cancel one another, if, as noted above, the entire resistor is at a uniform temperature. However, if there is significant power dissipation in a resistor, or if its orientation with respect to a heat source is non-symmetrical, this can cause one of its ends to be warmer than the other, causing a net thermocouple error voltage. Using ordinary metal film resistors, an end-to-end temperature differential of 1°C causes a thermocouple voltage of about $20\mu\text{V}$. This error level is quite significant compared to the offset voltage drift of a precision op amp like the OP177, and extremely significant when compared to chopper-stabilized op amps, with their drifts of $<1\mu\text{V}/^\circ\text{C}$.

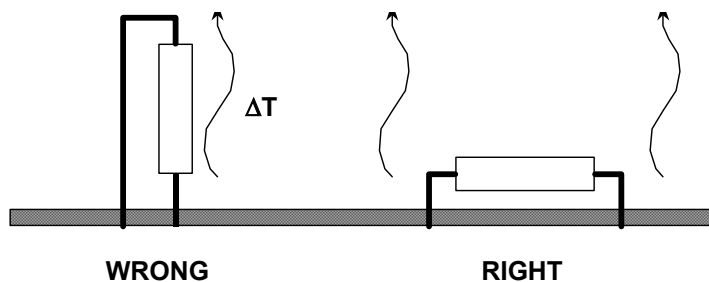


Figure 7-8: The effects of thermocouple emfs generated by resistors can be minimized by orientation that normalizes the end temperatures

Figure 7-8 shows how resistor orientation can make a difference in the net thermocouple voltage. In the left diagram, standing the resistor on end in order to conserve board space will invariably cause a temperature gradient across the resistor, especially if it is dissipating any significant power. In contrast, placing the resistor flat on the PC board as shown at the right will generally eliminate the gradient. An exception might occur, if

there is end-to-end resistor airflow. For such cases, orienting the resistor axis perpendicular to the airflow will minimize this source of error, since this tends to force the resistor ends to the same temperature.

Note that this line of thinking should be extended, to include orientation of resistors on a vertically mounted PC board. In such cases, natural convection air currents tend to flow upward across the board. Again, the resistor thermal axis should be perpendicular to convection, to minimize thermocouple effects. With tiny surface mount resistors, the thermocouple effects can be less problematic, due to tighter thermal coupling between the resistor ends.

In general, designers should strive to avoid thermal gradients on or around critical circuit boards. Often this means thermally isolating components that dissipate significant amounts of power. Thermal turbulence created by large temperature gradients can also result in dynamic noise-like low-frequency errors.

Voltage Sensitivity, Failure Mechanisms, and Aging

Resistors are also plagued by changes in value as a function of applied voltage. The deposited-oxide high-megohm type components are especially sensitive, with voltage coefficients ranging from 1 ppm/V to more than 200 ppm/V. This is another reason to exercise caution in such precision applications as high-voltage dividers.

The normal failure mechanism of a resistor can also create circuit difficulties, if not carefully considered beforehand. For example, carbon-composition resistors fail safely, by turning into open circuits. Consequently, in some applications, these components can play a useful secondary role, as a fuse. Replacing such a resistor with a carbon-film type can possibly lead to trouble, since carbon-films can fail as short circuits. (Metal-film components usually fail as open circuits.)

All resistors tend to change slightly in value with age. Manufacturers specify long-term stability in terms of change— ppm/year. Values of 50 or 75 ppm/year are not uncommon among metal film resistors. For critical applications, metal-film devices should be burned-in for at least one week at rated power. During burn-in, resistance values can shift by up to 100 or 200 ppm. Metal film resistors may need 4-5000 operational hours for full stabilization, especially if deprived of a burn-in period.

Resistor Excess Noise

Most designers have some familiarity with thermal, or Johnson noise, which occurs in resistors. But a less widely recognized secondary noise phenomenon is associated with resistors, and it is called *excess noise*. It can prove particularly troublesome in precision op amp and converter circuits, as it is evident only when current passes through a resistor.

To review briefly, thermal noise results from thermally induced random vibration of charge resistor carriers. Although the average current from the vibrations remains zero, instantaneous charge motions result in an instantaneous voltage across the terminals.

■ OP AMP APPLICATIONS

Excess noise on the other hand, occurs primarily when dc flows in a discontinuous medium—for example the conductive particles of a carbon composition resistor. The current flows unevenly through the compressed carbon granules, creating microscopic particle-to-particle "arcing". This phenomenon gives rise to a $1/f$ noise-power spectrum, in addition to the thermal noise spectrum. In other words, the excess spot noise voltage increases as the inverse square-root of frequency.

Excess noise often surprises the unwary designer. Resistor thermal noise and op amp input noise set the noise floor in typical op amp circuits. Only when voltages appear across input resistors and causes current to flow does the excess noise become a significant—and often dominant—factor. In general, carbon composition resistors generate the most excess noise. As the conductive medium becomes more uniform, excess noise becomes less significant. Carbon film resistors do better, with metal film, wirewound and bulk-metal-film resistors doing better yet.

Manufacturers specify excess noise in terms of a noise index—the number of microvolts of rms noise in the resistor in each decade of frequency per volt of dc drop across the resistor. The index can rise to 10dB (3 microvolts per dc volt per decade of bandwidth) or more. Excess noise is most significant at low frequencies, while above 100 kHz thermal noise predominates.

Potentiometers

Trimming potentiometers (trimpots) can suffer from most of the phenomena that plague fixed resistors. In addition, users must also remain vigilant against some hazards unique to these components.

For instance, many trimpots aren't sealed, and can be severely damaged by board washing solvents, and even by excessive humidity. Vibration—or simply extensive use—can damage the resistive element and wiper terminations. Contact noise, TCs, parasitic effects, and limitations on adjustable range can all hamper trimpot circuit operation. Furthermore, the limited resolution of wirewound types and the hidden limits to resolution in cermet and plastic types (hysteresis, incompatible material TCs, slack) make obtaining and maintaining precise circuit settings anything but an "infinite resolution" process. Given this background, two rules are suggested for the potential trimpot user. Rule 1: Use infinite care and infinitesimal adjustment range to avoid infinite frustration when applying manual trimpots. Rule 2: *Consider the elimination of manual trimming potentiometers altogether, if possible!* A number of digitally addressable potentiometers (RDACs) are now available for direct application in similar circuit functions as classic trimpots (See Reference 17). There are also many low cost multi-channel voltage output DACs expressly designed for system voltage trimming.

Table 7-2 summarizes selection criteria for various fixed resistor types, both in discrete form and as part of networks. In a selection process, the general information of this table should be supplemented by consultation of current vendor's catalog information (see References at end of section).

HARDWARE AND HOUSEKEEPING TECHNIQUES PASSIVE COMPONENTS

Table 7-2
Resistor Comparison Chart

	TYPE	ADVANTAGES	DISADVANTAGES
DISCRETE	Carbon Composition	Lowest Cost High Power/Small Case Size Wide Range of Values	Poor Tolerance (5%) Poor Temperature Coefficient (1500 ppm/°C)
	Wirewound	Excellent Tolerance (0.01%) Excellent TC (1ppm/°C) High Power	Reactance is a Problem Large Case Size Most Expensive
	Metal Film	Good Tolerance (0.1%) Good TC (<1 to 100ppm/°C) Moderate Cost Wide Range of Values Low Voltage Coefficient	Must be Stabilized with Burn-In Low Power
	Bulk Metal or Metal Foil	Excellent Tolerance (to 0.005%) Excellent TC (to <1ppm/°C) Low Reactance Low Voltage Coefficient	Low Power Very Expensive
	High Megohm	Very High Values (10^8 to $10^{14}\Omega$) Only Choice for Some Circuits	High Voltage Coefficient (200ppm/V) Fragile Glass Case (Needs Special Handling) Expensive
NETWORKS	Thick Film	Low Cost High Power Laser-Trimable Readily Available	Fair Matching (0.1%) Poor TC (>100ppm/°C) Poor Tracking TC (10ppm/°C)
	Thin Film	Good Matching (<0.01%) Good TC (<100ppm/°C) Good Tracking TC (2ppm/°C) Moderate Cost Laser-Trimable Low Capacitance Suitable for Hybrid IC Substrate	Often Large Geometry Limited Values and Configurations

■ OP AMP APPLICATIONS

Inductance

Stray Inductance

All conductors are inductive, and at high frequencies, the inductance of even quite short pieces of wire or printed circuit traces may be important. The inductance of a straight wire of length L mm and circular cross-section with radius R mm in free space is given by the first equation shown in Figure 7-9.

The inductance of a strip conductor (an approximation to a PC track) of width W mm and thickness H mm in free space is also given by the second equation in Figure 7-9.

In real systems, these formulas both turn out to be approximate, but they do give some idea of the order of magnitude of inductance involved. They tell us that 1cm of 0.5mm od wire has an inductance of 7.26nH, and 1cm of 0.25mm PC track has an inductance of 9.59nH - these figures are reasonably close to measured results.

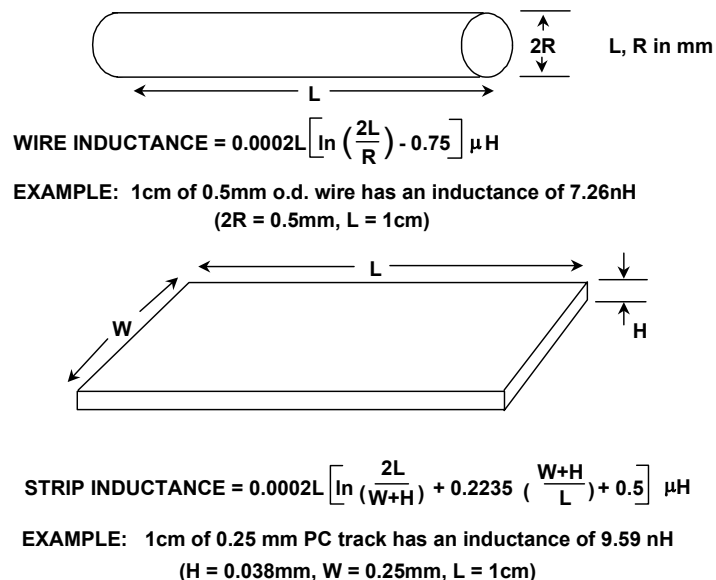


Figure 7-9: Wire and strip inductance calculations

At 10MHz, an inductance of 7.26nH has an impedance of 0.46 ohm, and so can give rise to 1% error in a 50 ohm system.

Mutual Inductance

Another consideration regarding inductance is the separation of outward and return currents. As we shall discuss in more detail later, Kirchoff's Law tells us that current flows in closed paths— there is always an outward and return path. The whole path forms a single-turn inductor.

This principle is illustrated by the contrasting signal trace routing arrangements of Figure 7-10 (opposite). If the area enclosed within the turn is relatively large, as in the upper "nonideal" picture, then the inductance (and hence the AC impedance) will also be large.

On the other hand, if the outward and return paths are closer together, as in the lower "improved" picture, the inductance will be much smaller.

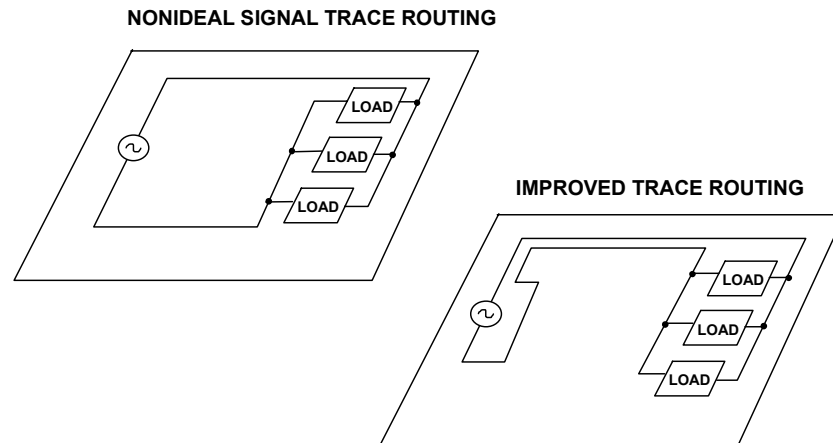


Figure 7-10: Nonideal and improved signal trace routing

Note that the nonideal signal routing case of Figure 7-10 has other drawbacks—the large area enclosed within the conductors produces extensive external magnetic fields, which may interact with other circuits, causing unwanted coupling. Similarly, the large area is more vulnerable to interaction with external magnetic fields, which can induce unwanted signals in the loop.

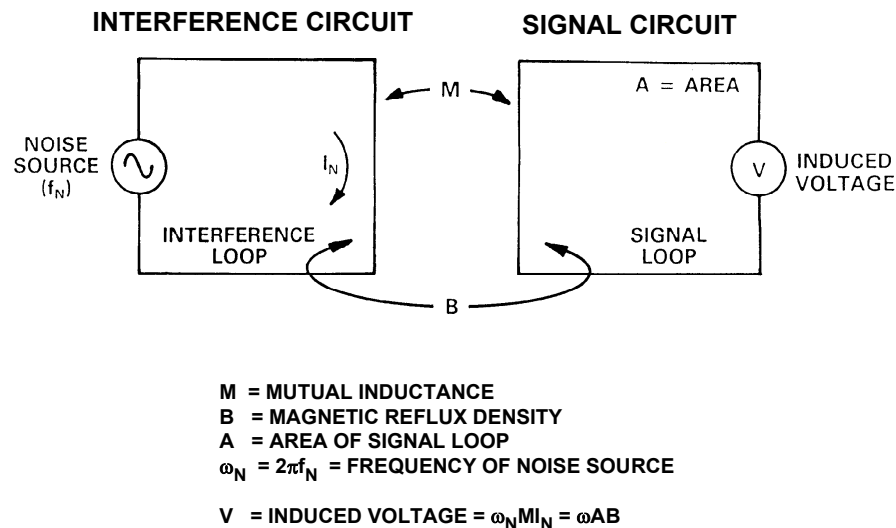


Figure 7-11: Basic principles of inductive coupling

The basic principle is illustrated in Figure 7-11, and is a common mechanism for the transfer of unwanted signals (noise) between two circuits.

As with most other noise sources, as soon as we define the working principle, we can see ways of reducing the effect. In this case, reducing any or all of the terms in the equations in Figure 7-11 reduces the coupling. Reducing the frequency or amplitude of the current causing the interference may be impracticable, but it is frequently possible to reduce the

■ OP AMP APPLICATIONS

mutual inductance between the interfering and interfered with circuits by reducing loop areas on one or both sides and, possibly, increasing the distance between them.

A layout solution is illustrated by Figure 7-12. Here two circuits, shown as Z1 and Z2, are minimized for coupling by keeping each of the loop areas as small as is practical.

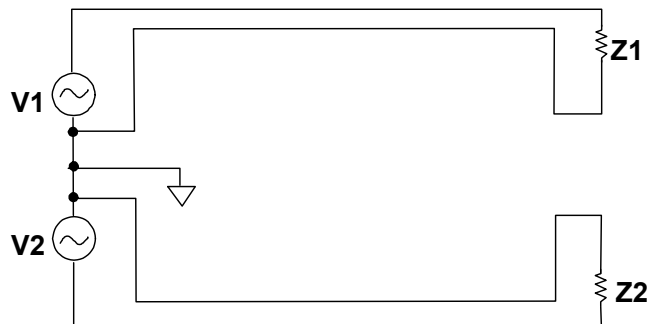


Figure 7-12: Proper signal routing and layout can reduce inductive coupling

As also illustrated in Figure 7-13, mutual inductance can be a problem in signals transmitted on cables. Mutual inductance is high in ribbon cables, especially when a single return is common to several signal circuits (top). Separate, dedicated signal and return lines for each signal circuit reduces the problem (middle). Using a cable with twisted pairs for each signal circuit as in the bottom picture is even better (but is more expensive and often unnecessary).

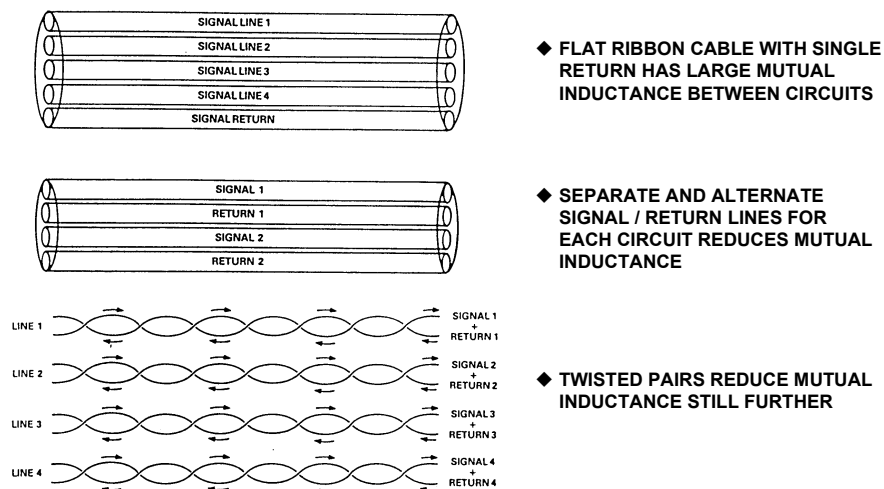


Figure 7-13: Mutual inductance and coupling within signal cabling

Shielding of magnetic fields to reduce mutual inductance is sometimes possible, but is by no means as easy as shielding an electric field with a Faraday shield (following section). HF magnetic fields are blocked by conductive material provided the skin depth in the conductor at the frequency to be screened is much less than the thickness of the conductor, and the screen has no holes (Faraday shields can tolerate small holes, magnetic screens cannot). LF and DC fields may be screened by a shield made of mu-metal sheet. Mu-metal is an alloy having very high permeability, but it is expensive, its magnetic properties are damaged by mechanical stress, and it will saturate if exposed to too high fields. Its use, therefore, should be avoided where possible.

Ringing

An inductor in series or parallel with a capacitor forms a resonant, or "tuned", circuit, whose key feature is that it shows marked change in impedance over a small range of frequency. Just how sharp the effect is depends on the relative Q of the tuned circuit. The effect is widely used to define the frequency response of narrow-band circuitry, but can also be a potential problem source.

If stray inductance and capacitance (which may or may not be stray) in a circuit should form a tuned circuit, then that tuned circuit may be excited by signals in the circuit, and ring at its resonant frequency.

An example is shown in Figure 7-14, where the resonant circuit formed by an inductive power line and its decoupling capacitor may possibly be excited by fast pulse currents drawn by the powered IC.

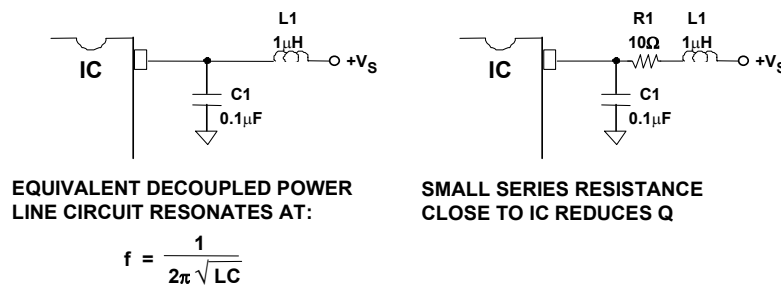


Figure 7-14: Resonant circuit formed by power line decoupling

While normal trace inductance and typical decoupling capacitances of 0.01 – 0.1 μF will resonate well above a few MHz, an example 0.1 μF capacitor and 1 μH of inductance resonates at 500 kHz. Left unchecked, this could present a resonance problem, as shown in the left case. Should an undesired power line resonance be present, the effect may be minimized by lowering the Q of the inductance. This is most easily done by inserting a small resistance (~10 Ω) in the power line close to the IC, as shown in the right case.

Parasitic Effects in Inductors

Although inductance is one of the fundamental properties of an electronic circuit, inductors are far less common as components than are resistors and capacitors. As for precision components, they are even more rare. This is because they are harder to manufacture, less stable, and less physically robust than resistors and capacitors. It is relatively easy to manufacture stable precision inductors with inductances from nH to tens or hundreds of μH, but larger valued devices tend to be less stable, and large.

As we might expect in these circumstances, circuits are designed, where possible, to avoid the use of precision inductors. We find that stable precision inductors are relatively rarely used in precision analog circuitry, except in tuned circuits for high frequency narrow band applications.

Of course, they are widely used in power filters, switching power supplies and other applications where lack of precision is unimportant (more on this in a following section).

■ OP AMP APPLICATIONS

The important features of inductors used in such applications are their current carrying and saturation characteristics, and their Q. If an inductor consists of a coil of wire with an air core, its inductance will be essentially unaffected by the current it is carrying. On the other hand, if it is wound on a core of a magnetic material (magnetic alloy or ferrite), its inductance will be non-linear, since at high currents, the core will start to saturate. The effects of such saturation will reduce the efficiency of the circuitry employing the inductor and is liable to increase noise and harmonic generation.

As mentioned above, inductors and capacitors together form tuned circuits. Since all inductors will also have some stray capacity, all inductors will have a resonant frequency (which will normally be published on their data sheet), and should only be used as precision inductors at frequencies well below this.

Q or "Quality Factor"

The other characteristic of inductors is their Q (or "Quality Factor"), which is the ratio of the reactive impedance to the resistance, as indicated in Figure 7-15.

◆ $Q = 2\pi f L/R$

◆ The Q of an inductor or resonant circuit is a measure of the ratio of its reactance to its resistance.

◆ The resistance is the HF and NOT the DC value.

◆ The 3 dB bandwidth of a single tuned circuit is F_c/Q where F_c is the center frequency.

Figure 7-15: Inductor Q or quality factor

It is rarely possible to calculate the Q of an inductor from its DC resistance, since skin effect (and core losses if the inductor has a magnetic core) ensure that the Q of an inductor at high frequencies is always lower than that predicted from DC values.

Q is also a characteristic of tuned circuits (and of capacitors— but capacitors generally have such high Q values that it may be disregarded, in practice). The Q of a tuned circuit, which is generally very similar to the Q of its inductor (unless it is deliberately lowered by the use of an additional resistor), is a measure of its bandwidth around resonance. LC tuned circuits rarely have Q of much more than 100 (3dB bandwidth of 1%), but ceramic resonators may have a Q of thousands, and quartz crystals tens of thousands.

Don't Overlook Anything

Remember, if your precision op amp or data-converter-based design does not meet specification, try not to overlook anything in your efforts to find the error sources. Analyze both active *and* passive components, trying to identify and challenge any assumptions or preconceived notions that may blind you to the facts. Take nothing for granted.

For example, when not tied down to prevent motion, cable conductors, moving within their surrounding dielectrics, can create significant static charge buildups that cause errors, especially when connected to high-impedance circuits. Rigid cables, or even costly low-noise Teflon-insulated cables, are expensive alternative solutions.

As more and more high-precision op amps become available, and system designs call for higher speed and increased accuracy, a thorough understanding of the error sources described in this section (as well those following) becomes more important.

Some additional discussions of passive components within a succeeding power supply filtering section complements this one. In addition, the very next section on PCB design issues also complements many points within this section. Similar comments apply to the section on EMI/RFI.

REFERENCES: PASSIVE COMPONENTS

1. James E. Buchanan, "Dielectric Absorption— It Can Be a Real Problem In Timing Circuits," **EDN**, January 20, 1977, p 83.
2. Lew Counts and Scott Wurcer, "Instrumentation Amplifier Nears Input Noise Floor," **Electronic Design**, June 10, 1982.
3. W. Doeling, W. Mark, T. Tadewald, and P. Reichenbacher, "Getting Rid of Hook: The Hidden PC-Board Capacitance," **Electronics**, October 12, 1978, p 111-117.
4. Tarlton Fleming, "Data-Acquisition System (DAS) Design Considerations," **WESCON '81 Professional Program Session Record No. 23**.
5. Walter G. Jung and Richard Marsh, "Picking Capacitors, Parts I and II," **Audio**, February and March, 1980.
6. Robert A. Pease, "Understand Capacitor Soakage to Optimize Analog Systems", **EDN**, October 13, 1982, page 125.
7. Andy Rappaport, "Capacitors" **EDN**, October 13, 1982, page 105.
8. Specification MIL-PRF-19978G, Capacitors, Fixed, Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic or Glass Cases), Established and Non-Established Reliability General Specification for, May 27, 1999.
9. Specification MIL-PRF-123B, Capacitors, Fixed, Ceramic Dielectric, (Temperature Stable and General Purpose), High Reliability, General Specification for, August 6, 1990.
10. **Tantalum and Ceramic Surface Mount Capacitor Catalog**, Kemet Electronics Corporation, P.O. Box 5928, Greenville, SC, 29606, 864-963-6300.
11. A general capacitor information resource: <http://www.faradnet.com/>
12. Southern and F-Dyne film capacitors, Southern Electronics, 215 Research Drive, Milford, CT, 06460, 203-876-7488.
13. Wesco film capacitors, Wesco Electrical Company, 201 Munson Street, Greenfield, MA, 01301, 413-774-4358.
14. Doug Grant and Scott Wurcer, "Avoiding Passive Component Pitfalls," **The Best of Analog Dialogue**, Analog Devices, 1991, p. 143-148.
15. RCD Components, Inc., 520 E. Industrial Park Drive, Manchester NH, 03109, 603-669-0054, <http://www.rcd-comp.com>
16. Steve Sockolov and James Wong, "High-Accuracy Analog Needs More Than Op Amps," **Electronic Design**, October 1, 1992, p.53.
17. Selection guide for digital potentiometers:
http://www.analog.com/support/standard_linear/selection_guides/dig_pot.pdf
18. Precision Resistor Co., Inc., 10601 75th St. N., Largo, FL, 33777-1427, 727 541-5771, <http://www.precisionresistor.com>
19. Ohmite Victoreen MAXI-MOX Resistors, 3601 Howard Street, Skokie, IL 60076, 847 675-2600, <http://www.ohmite.com/victoreen/>.

HARDWARE AND HOUSEKEEPING TECHNIQUES PASSIVE COMPONENTS

20. Vishay/Dale Resistors, 2300 Riverside Blvd., Norfolk, NE, 68701-2242, 402 371-0800, <http://www.vishay.com>.
21. Beyschlag Resistor Products, PO Box 1220, D-25732 Heide, Germany, <http://www.beyschlag.com>
22. B. I. & B. Bleaney, **Electricity & Magnetism**, Oxford at the Clarendon Press, 1957, pp 23, 24, & 52.
23. Henry W. Ott, **Noise Reduction Techniques in Electronic Systems**, 2nd Edition, John Wiley, Inc., 1988, ISBN: 0-471-85068-3.
24. G. W. A. Dummer, **Materials for Conductive and Resistive Functions**, Hayden, 1970.

ACKNOWLEDGEMENTS:

Portions of this and the following section were adapted from Doug Grant and Scott Wurcer, "Avoiding Passive Component Pitfalls," originally published in **Analog Dialogue 17-2**, 1983.

NOTES:

SECTION 7-2: PCB DESIGN ISSUES

James Bryant

Printed circuit boards (PCBs) are by far the most common method of assembling modern electronic circuits. Comprised of a sandwich of insulating layer (or layers) and one or more copper conductor patterns, they can introduce various forms of errors into a circuit, particularly if the circuit is operating at either high precision or high speed. PCBs then, act as "unseen" components, wherever they are used in precision circuit designs. Since designers don't always consider the PCB electrical characteristics as additional components of their circuit, overall performance can easily end up worse than predicted. This general topic, manifested in many forms, is the focus of this section.

PCB effects that are harmful to precision circuit performance include leakage resistances, spurious voltage drops in trace foils, vias, and ground planes, the influence of stray capacitance, dielectric absorption (DA), and the related "hook". In addition, the tendency of PCBs to absorb atmospheric moisture, *hygroscopicity*, means that changes in humidity often cause the contributions of some parasitic effects to vary from day to day.

In general, PCB effects can be divided into two broad categories— those that most noticeably affect the static or dc operation of the circuit, and those that most noticeably affect dynamic or ac circuit operation.

Another very broad area of PCB design is the topic of grounding. Grounding is a problem area in itself for all analog designs, and it can be said that implementing a PCB based circuit doesn't change that fact. Fortunately, certain principles of quality grounding, namely the use of ground planes, are intrinsic to the PCB environment. This factor is one of the more significant advantages to PCB based analog designs, and appreciable discussion of this section is focused on this issue.

Some other aspects of grounding that must be managed include the control of spurious ground and signal return voltages that can degrade performance. These voltages can be due to external signal coupling, common currents, or simply excessive IR drops in ground conductors. Proper conductor routing and sizing, as well as differential signal handling and ground isolation techniques enables control of such parasitic voltages.

One final area of grounding to be discussed is grounding appropriate for a mixed-signal, analog/digital environment. Although this isn't the specific overall focus of the book, it is certainly true that interfacing with ADCs (or DACs) is a major task category of op amps, and thus it shouldn't be overlooked. Indeed, the single issue of quality grounding can drive the entire layout philosophy of a high performance mixed signal PCB design— as it well should.

■ OP AMP APPLICATIONS

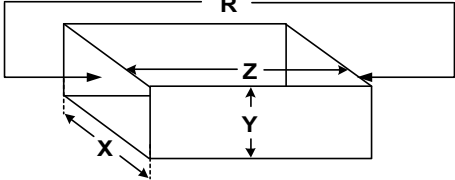
Resistance of Conductors

Every engineer is familiar with resistors— little cylinders with wire or tab ends— although perhaps fewer are aware of their idiosyncrasies, as generally covered in section 7-1. But far too few engineers consider that all the wires and PCB traces with which their systems and circuits are assembled are also resistors. In higher precision systems, even these trace resistances and simple wire interconnections can have degrading effects. Copper is *not* a superconductor— and too many engineers appear to think it is!

Figure 7-16 illustrates a method of calculating the sheet resistance R of a copper square, given the length Z , the width X , and the thickness Y .

$$R = \frac{\rho Z}{XY}$$

ρ = RESISTIVITY



**SHEET RESISTANCE CALCULATION FOR
1 OZ. COPPER CONDUCTOR:**

$$\rho = 1.724 \times 10^{-6} \text{ } \Omega\text{cm}, Y = 0.0036\text{cm}$$
$$R = 0.48 \frac{Z}{X} \text{ m}\Omega$$
$$\frac{Z}{X} = \text{NUMBER OF SQUARES}$$
$$R = \text{SHEET RESISTANCE OF 1 SQUARE (Z=X)} \\ = 0.48\text{m}\Omega/\text{SQUARE}$$

Figure 7-16: Calculation of sheet resistance and linear resistance for standard copper PCB conductors

At 25°C the resistivity of pure copper is 1.724E-6 ohm cm. The thickness of standard 1 ounce PCB copper foil is 0.036mm (0.0014"). Using the relations shown, the resistance of such a standard copper element is therefore 0.48mΩ /square. One can readily calculate the resistance of a linear trace, by effectively "stacking" a series of such squares end-end, to make up the line's length. The line length is Z and the width is X , so the line resistance R is simply a product of Z/X and the resistance of a single square, as noted in the figure.

For a given copper weight and trace width, a resistance/length calculation can be made. For example, the 0.25mm (10 mil) wide traces frequently used in PCB designs equates to a resistance/length of about 19mΩ/cm (48mΩ /inch), which is quite large. Moreover, the temperature coefficient of resistance for copper is about 0.4% /°C around room temperature. This is a factor that shouldn't be ignored, in particular within low impedance precision circuits, where the TC can shift the net impedance over temperature.

As shown in Figure 7-17, PCB trace resistance can be a serious error when conditions aren't favorable. Consider a 16-bit ADC with a $5\text{k}\Omega$ input resistance, driven through 5cm of 0.25mm wide 1 oz. PCB track between it and its signal source. The track resistance of nearly 0.1Ω forms a divider with the $5\text{k}\Omega$ load, creating an error. The resulting voltage drop is a gain error of $0.1/5\text{k}$ ($\sim 0.0019\%$), well over 1LSB (0.0015% for 16 bits).

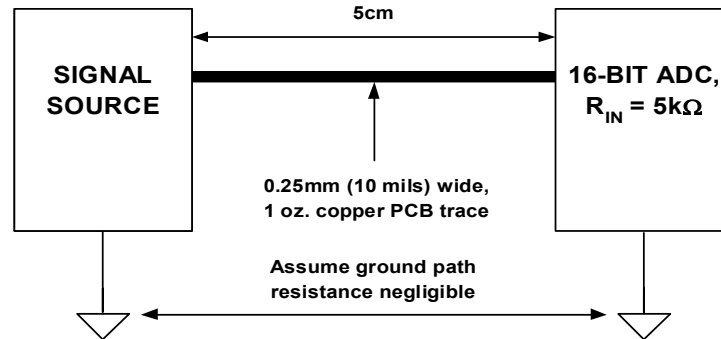


Figure 7-17: Ohm's law predicts >1LSB of error due to drop in PCB conductor

So, when dealing with precision circuits, the point is made that even simple design items such as PCB trace resistance cannot be dealt with casually. There are various solutions that can address this issue, such as wider traces (which may take up excessive space), the use of heavier copper (which may be too expensive), or simply choosing a high impedance converter. But, the most important thing is to think it all through, avoiding any tendency to overlook items appearing innocuous on the surface.

Voltage Drop in Signal Leads - "Kelvin" Feedback

The gain error resulting from resistive voltage drop in PCB signal leads is important only with high precision and/or at high resolutions (the Figure 7-17 example), or where large signal currents flow. Where load impedance is constant and resistive, adjusting overall system gain can compensate for the error. In other circumstances, it may often be removed by the use of "Kelvin" or "voltage sensing" feedback, as shown in Figure 7-18.

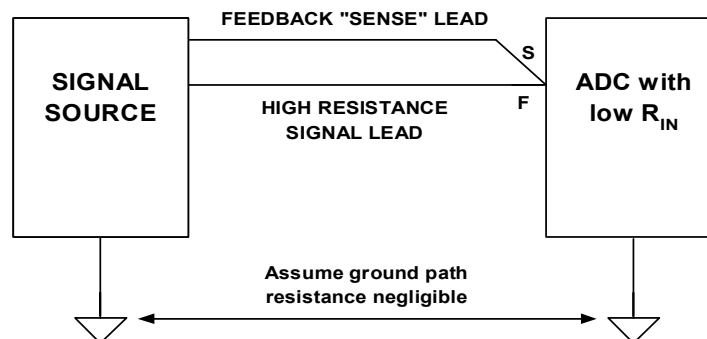


Figure 7-18: Use of a sense connection moves accuracy to the load point

In this modification to the case of Figure 7-17, a long resistive PCB trace is still used to drive the input of a high resolution ADC, with low input impedance. In this case however, the voltage drop in the signal lead does *not* give rise to an error, as feedback is taken directly from the input pin of the ADC, and returned to the driving source. This

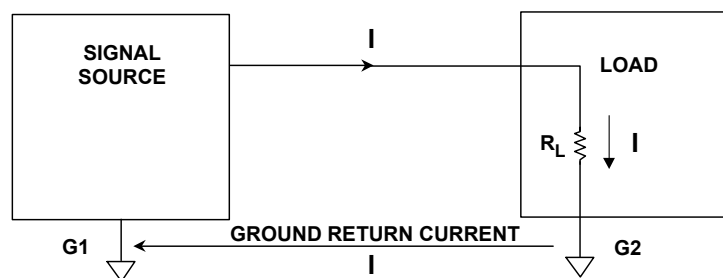
■ OP AMP APPLICATIONS

scheme allows full accuracy to be achieved in the signal presented to the ADC, despite any voltage drop across the signal trace.

The use of separate force (F) and sense (S) connections at the load removes any errors resulting from voltage drops in the force lead, but, of course, may only be used in systems where there is negative feedback. It is also impossible to use such an arrangement to drive two or more loads with equal accuracy, since feedback may only be taken from one point. Also, in this much-simplified system, errors in the common lead source/load path are ignored, the assumption being that ground path voltages are negligible. In many systems this may not necessarily be the case, and additional steps may be needed, as noted below.

Signal Return Currents

Kirchoff's Law tells us that at any point in a circuit the algebraic sum of the currents is zero. This tells us that all currents flow in circles and, particularly, that the return current must always be considered when analyzing a circuit, as is illustrated in Figure 7-19 (see References 7 and 8).



AT ANY POINT IN A CIRCUIT
THE ALGEBRAIC SUM OF THE CURRENTS IS ZERO
OR
WHAT GOES OUT MUST COME BACK
WHICH LEADS TO THE CONCLUSION THAT
ALL VOLTAGES ARE DIFFERENTIAL
(EVEN IF THEY'RE GROUNDED)

Figure 7-19: Kirchoff's law helps in analyzing voltage drops around a complete source/load coupled circuit

In dealing with grounding issues, common human tendencies provide some insight into how the correct thinking about the circuit can be helpful towards analysis. Most engineers readily consider the ground return current "I", *when they are considering a fully differential circuit*.

However, when considering the more usual circuit case, where a single-ended signal is referred to "ground", it is common to assume that all the points on the circuit diagram where ground symbols are found are at the same potential. Unfortunately, this happy circumstance just ain't necessarily so!

This overly optimistic approach is illustrated in Figure 7-20, where, if it really should exist, "infinite ground conductivity" would lead to zero ground voltage difference between source ground G1 and load ground G2. Unfortunately this approach isn't a wise practice, and when dealing with high precision circuits, it can lead to disasters.

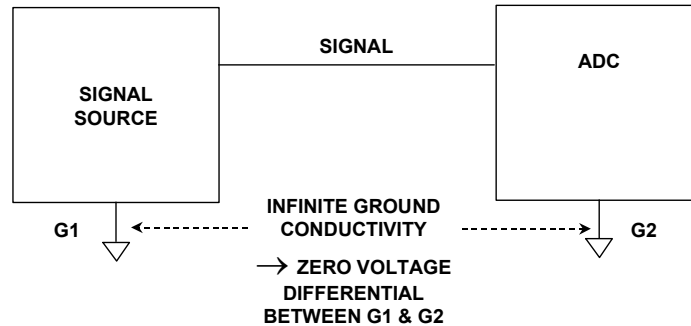


Figure 7-20: Unlike this optimistic diagram, it is unrealistic to assume infinite conductivity between source/load grounds in a real-world system

A more realistic approach to ground conductor integrity includes analysis of the impedance(s) involved, and careful attention to minimizing spurious noise voltages.

Ground Noise & Ground Loops

A more realistic model of a ground system is shown in Figure 7-21. The signal return current flows in the complex impedance existing between ground points G1 and G2 as shown, giving rise to a voltage drop ΔV in this path. But it is important to note that additional *external* currents, such as I_{EXT} , may also flow in this same path. It is critical to understand that such currents may generate uncorrelated noise voltages between G1 and G2 (dependent upon the current magnitude and relative ground impedance).

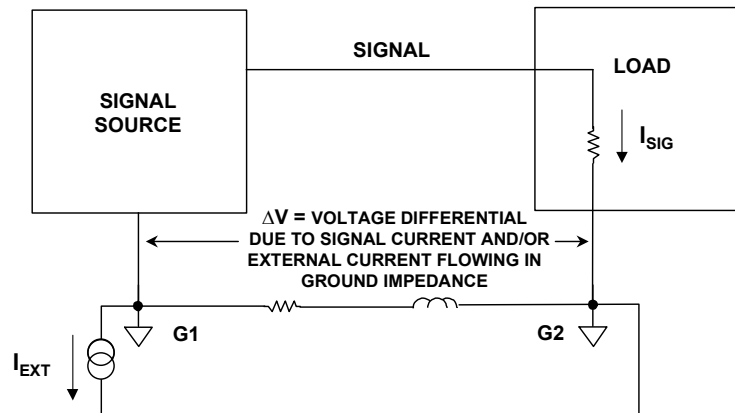


Figure 7-21: A more realistic source-to-load grounding system view includes consideration of the impedance between G1-G2, plus the effect of any non-signal-related currents

Some portion of these undesired voltages may end up being seen at the signal's load end, and they can have the potential to corrupt the signal being transmitted.

■ OP AMP APPLICATIONS

It is evident, of course, that other currents can only flow in the ground impedance, if there is a current path for them. In this case, severe problems can be caused by a high current circuit sharing an *unlooped* ground return with the signal source.

Figure 7-22 shows just such a common ground path, shared by the signal source and a high current circuit, which draws a large and varying current from its supply. This current flows in the common ground return, causing an error voltage ΔV to be developed.

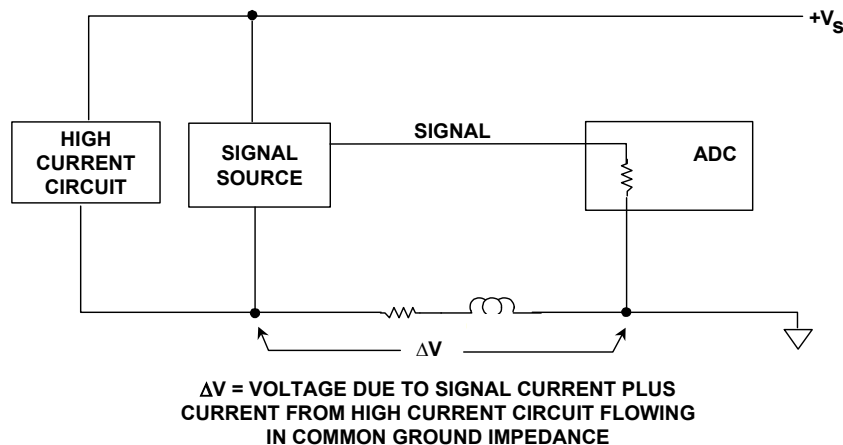


Figure 7-22: Any current flowing through a common ground impedance can cause errors

From Figure 7-23 below, it is also evident that if a ground network contains *loops*, or circular ground conductor patterns (with S1 closed), there is an even greater danger of it being vulnerable to EMFs induced by external magnetic fields. There is also a real danger of ground-current-related signals "escaping" from the high current areas, and causing noise in sensitive circuit regions elsewhere in the system.

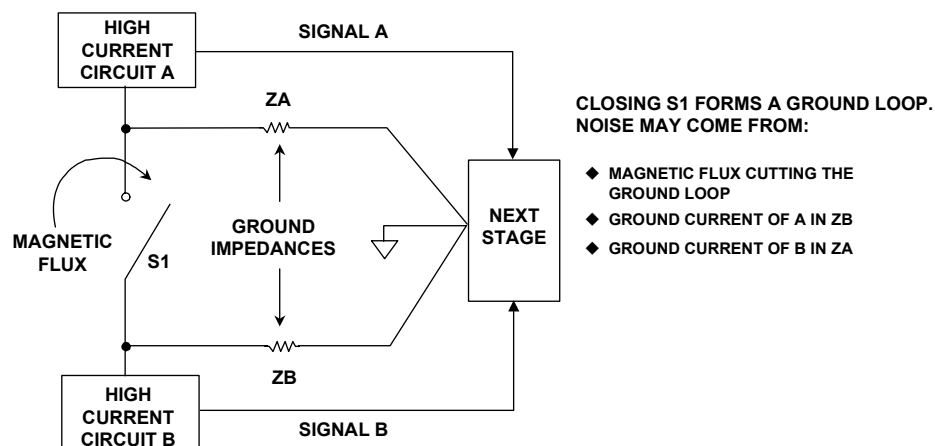


Figure 7-23: A ground loop

For these reasons ground loops are best avoided, by wiring all return paths within the circuit by separate paths back to a common point, i.e., the common ground point towards the mid-right of the diagram. This would be represented by the S1 open condition.

There are a number of possible ways of attacking the ground noise problem, apart from the presently impracticable one of using superconducting grounds. It is rare for any single method to be used to the exclusion of all others, and real systems generally contain a mixture. For descriptive purposes each approach is addressed separately.

Star Grounds

The "star" ground philosophy builds on the theory that there is one single ground point in a circuit to which all voltages are referred. This is known as the *star ground* point. It can be better understood by a visual analogy—the multiple conductors extending radially from the common schematic ground resemble a star. This can be appreciated by regarding Figure 7-23 again, considering many more ground returns from the common point. Note that the star point need not look like a star—it may be a point on a ground plane—but the key feature of the star ground system is that all voltages are measured with respect to a particular point in the ground network, not just to an undefined "ground" (i.e. wherever one can clip a probe). Figure 7-24 succinctly summarizes the philosophy.

- ◆ IF ALL SIGNAL VOLTAGES WITHIN A SYSTEM ARE MEASURED WITH RESPECT TO A SINGLE POINT, THAT POINT IS SAID TO BE THE SYSTEM *STAR GROUND*.

Figure 7-24: The star ground concept

This star grounding philosophy is reasonable theoretically, but can encounter practical difficulties. For example, if we design a star ground system, drawing out all signal paths to minimize signal interaction and the effects of high impedance signal or ground paths, we often find implementation problems. When the power supplies are added to the circuit diagram, they either add unwanted ground paths, or their supply currents flowing in the existing ground paths are sufficiently so large, or noisy (or both) so as to corrupt the signal transmission. This particular problem can often be avoided by having separate power supplies (and thus separate ground returns) for the various circuit portions. For example, separate analog and digital supplies with separate analog and digital grounds, joined at the star point, are common in mixed signal applications.

Separate Analog and Digital Grounds

As a fact of life, digital circuitry is noisy. Saturating logic draws large, fast current spikes from its supply during switching. However, logic stages, with hundreds of millivolts (or more) of noise immunity, usually have little need for high levels of supply decoupling. On the other hand, analog circuitry is quite vulnerable to noise on both power supply rails and grounds. So, it is very sensible to separate analog and digital circuitry, to prevent digital noise from corrupting analog performance. Such separation involves separation of both ground returns *and* power rails, which is inconvenient in a mixed signal system. Nevertheless, if a mixed signal system is to deliver full performance capability, it is often essential to have separate analog and digital grounds, and separate power supplies. The fact that some analog circuitry will "operate" (i.e., function) from a single +5V supply does *not* mean that it may safely be operated from the same noisy +5V supply as the microprocessor and dynamic RAM, the electric fan, and the solenoid jackhammer! What is required is that the analog portion *operate with full performance from such a low*

■ OP AMP APPLICATIONS

voltage supply, not just be functional. This distinction will by necessity require quite careful attention to both the supply rails and the ground interfacing.

Figures 7-25 and 7-26 summarize some analog and digital power supply and grounding concepts which are useful to bear in mind as systems are designed.

- ◆ DIGITAL CIRCUITRY IS *NOISY*
- ◆ ANALOG CIRCUITRY IS *QUIET*
- ◆ CIRCUIT NOISE FROM DIGITAL CIRCUITRY CARRIED BY POWER AND GROUND LEADS CAN CORRUPT PRECISION ANALOG CIRCUITRY
- ◆ IT IS ADVISABLE TO SEPARATE THE POWER AND GROUND OF THE DIGITAL AND ANALOG PARTS OF A SYSTEM
- ◆ ANALOG AND DIGITAL GROUNDS MUST BE JOINED AT *ONE* POINT

Figure 7-25: *Some power supply and ground noise concepts appropriate for mixed signal systems*

Note that, analog and digital ground in a system must be joined at some point, to allow signals to be referred to a common potential. This star point, or analog/digital common point, is chosen so that it does not introduce digital currents into the ground of the analog part of the system—it is often convenient to make the connection at the power supplies.

Note also that many ADCs and DACs have separate *analog ground* (AGND) and *digital ground* (DGND) pins. On the device data sheets, users are often advised to connect these pins together at the package. This seems to conflict with the advice to connect analog and digital ground at the power supplies, and, in systems with more than one converter, with the advice to join the analog and digital ground at a single point.

- ◆ MONOLITHIC AND HYBRID ADCS FREQUENTLY HAVE SEPARATE AGND AND DGND PINS, WHICH MUST BE JOINED TOGETHER AT THE DEVICE.
- ◆ THIS ISN'T DONE TO BE DIFFICULT, BUT BECAUSE BONDWIRE VOLTAGE DROPS ARE TOO LARGE TO ALLOW INTERNAL CONNECTION.
- ◆ THE BEST SOLUTION TO THE GROUNDING PROBLEM ARISING FROM THIS REQUIREMENT IS TO CONNECT BOTH PINS TO SYSTEM "ANALOG GROUND."
- ◆ IT IS LIKELY THAT NEITHER THE DIGITAL NOISE SO INTRODUCED IN THE SYSTEM AGND, NOR THE SLIGHT LOSS OF DIGITAL NOISE IMMUNITY WILL SERIOUSLY AFFECT THE SYSTEM PERFORMANCE.

Figure 7-26: *Treatment of analog and digital grounds with data converters of mixed signal systems*

There is, in fact, no conflict. The labels "analog ground" and "digital ground" on these pins refer to the parts of the converter to which the pins are connected, and not to the system grounds to which they must go. For example, with an ADC, generally these two pins should be joined together and to the *analog* ground of the system. It is not possible to join the two pins within the IC package, because the analog part of the converter

cannot tolerate the voltage drop resulting from the digital current flowing in the bond wire to the chip. But they can be so tied, *externally*.

Figure 7-27 illustrates this concept of ground connections for an ADC. If these pins are connected in this way, the digital noise immunity of the converter is diminished somewhat, by the amount of common-mode noise between the digital and analog system grounds. However, since digital noise immunity is of the order of hundreds or thousands of millivolts, this factor is unlikely to be important.

The analog noise immunity is diminished only by the external digital currents of the converter itself flowing in the analog ground. These currents should be kept quite small, and this can be minimized by ensuring that the converter outputs don't see heavy loads. A good solution towards this is to use a low input current buffer at the ADC output, such as a CMOS buffer-register IC.

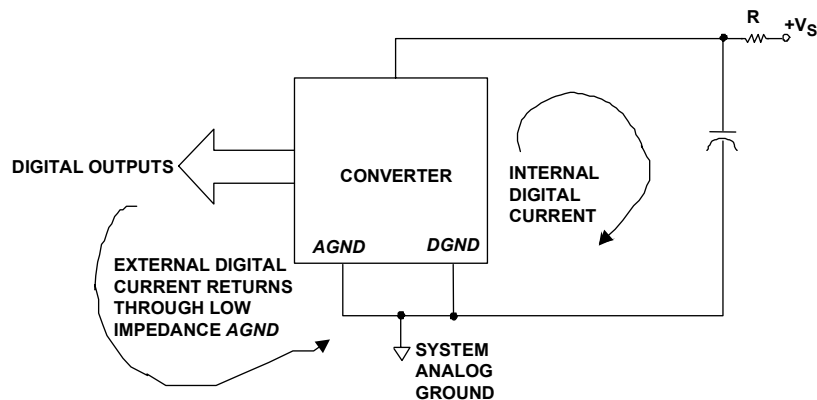


Figure 7-27: Analog (AGND) and digital ground (DGND) pins of a data converter should be returned to system analog ground

If the logic supply to the converter is isolated with a small resistance and decoupled to analog ground with a local $0.1\mu\text{F}$ capacitor, all the fast-edge digital currents of the converter will return to ground through the capacitor, and will not appear in the external ground circuit. If the analog ground impedance is maintained low, as it should be for adequate analog performance, additional noise due to the external digital ground current should rarely present a problem.

Ground planes

Related to the star ground system discussed earlier is the use of a *ground plane*. To implement a ground plane, one side of a double-sided PCB (or one layer of a multi-layer one) is made of continuous copper and used as ground. The theory behind this is that the large amount of metal will have as low a resistance as is possible. It will, because of the large flattened conductor pattern, also have as low an inductance as possible. It then offers the best possible conduction, in terms of minimizing spurious ground difference voltages across the conducting plane.

■ OP AMP APPLICATIONS

Note that ground plane concept can also be extended to include *voltage planes*. A voltage plane offers advantages similar to a ground plane, i.e., a very low impedance conductor, but is dedicated to a one (or more) of the system supply voltages. Thus a system can have more than one voltage plane, as well as a ground plane.

It has been sometimes argued that ground planes shouldn't be used, as they are liable to introduce manufacture and assembly problems. Such an argument may have had limited validity some years ago when PCB adhesives were less well developed, wave-soldering less reliable, and solder resist techniques less well understood, but not today.

A summary of key points related to the construction and operation of ground planes is contained in Figure 7-28.

- ◆ ONE ENTIRE PCB SIDE (OR LAYER) IS A CONTINUOUS GROUNDED CONDUCTOR.
- ◆ THIS GIVES MINIMUM GROUND RESISTANCE AND INDUCTANCE, BUT ISN'T ALWAYS SUFFICIENT TO SOLVE ALL GROUNDING PROBLEMS.
- ◆ BREAKS IN GROUND PLANES CAN IMPROVE OR DEGRADE CIRCUIT PERFORMANCE — THERE IS NO GENERAL RULE.
- ◆ YEARS AGO GROUND PLANES WERE DIFFICULT TO FABRICATE. TODAY THEY AREN'T.
- ◆ MULTI-LAYER, GROUND AND VOLTAGE PLANE PCB DESIGNS ARE STANDARD

Figure 7-28: Characteristics of ground planes

While ground planes solve many ground impedance problems, it should still be understood they aren't a panacea. Even a continuous sheet of copper foil has residual resistance and inductance, and in some circumstances, these can be enough to prevent proper circuit function. Designers should be wary of injecting very high currents in a ground plane, as they can produce voltage drops that interfere with sensitive circuitry.

Skin Effect

At high frequencies, also consider *skin effect*, where inductive effects cause currents to flow only in the outer surface of conductors. Note that this is in contrast to the earlier discussions of this section on dc resistance of conductors.

The skin effect has the consequence of increasing the resistance of a conductor at high frequencies. Note also that this effect is separate from the increase in impedance due to the effects of the self-inductance of conductors as frequency is increased.

Skin effect is quite a complex phenomenon, and detailed calculations are beyond the scope of this discussion. However, a good approximation for copper is that the skin depth in centimeters is $6.61/\sqrt{f}$, (f in Hz).

A summary of the skin effect within a typical PCB conductor foil is shown in Figure 7-29. Note that this copper conductor cross-sectional view assumes looking into the *side* of the conducting trace.

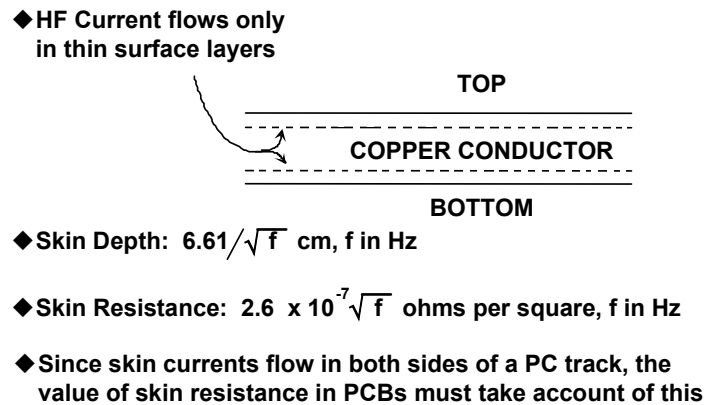


Figure 7-29: Skin depth in a PC conductor

Assuming that skin effects become important when the skin depth is less than 50% of the thickness of the conductor, this tells us that for a typical PC foil, we must be concerned about skin effects at frequencies above approximately 12MHz.

Where skin effect is important, the resistance for copper is $2.6 \times 10^{-7} \sqrt{f}$ Ohms per square, (f in Hz). This formula is invalid if the skin thickness is greater than the conductor thickness (i.e. at DC or LF).

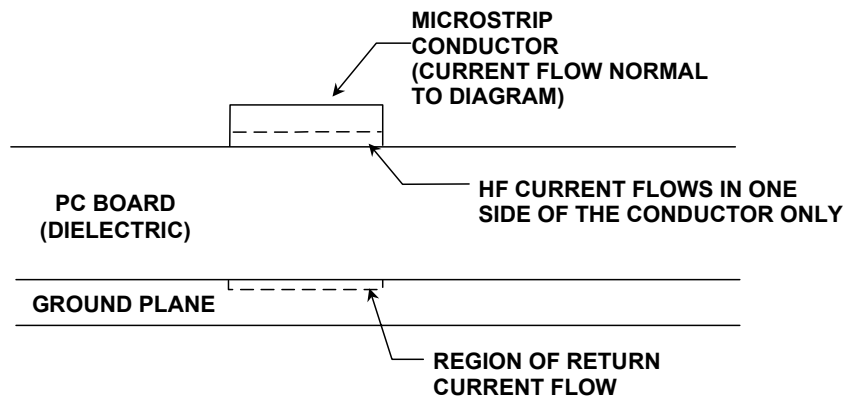


Figure 7-30: Skin effect with PC conductor and ground plane

Figure 7-30 illustrates a case of a PCB conductor with current flow, as separated from the ground plane underneath.

In this diagram, note the (dotted) regions of HF current flow, as reduced by the skin effect. When calculating skin effect in PCBs, it is important to remember that current generally flows in both sides of the PC foil (this is not necessarily the case in microstrip lines, see below), so the resistance per square of PC foil may be half the above value.

■ OP AMP APPLICATIONS

Transmission Lines

We earlier considered the benefits of outward and return signal paths being close together so that inductance is minimized. As shown previously in Figure 7-30, when an HF signal flows in a PC track running over a ground plane, the arrangement functions as a *microstrip* transmission line, and the majority of the return current flows in the ground plane underneath the line.

Figure 7-31 shows the general parameters for a microstrip transmission line, given the conductor width, w , dielectric thickness, h , and the dielectric constant, E_r .

The characteristic impedance of such a microstrip line will depend upon the width of the track and the thickness and dielectric constant of the PCB material. Designs of microstrip lines are covered in more detail within section six of this chapter.

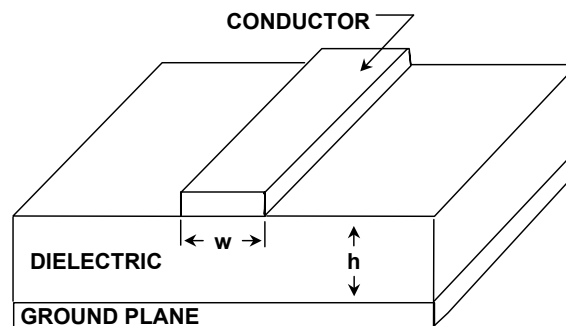


Figure 7-31: A PCB microstrip transmission line is an example of a controlled impedance conductor pair

For most dc and lower frequency applications, the characteristic impedance of PCB traces will be relatively unimportant. Even at frequencies where a track over a ground plane behaves as a transmission line, it is not necessary to worry about its characteristic impedance or proper termination if the free space wavelengths of the frequencies of interest are greater than ten times the length of the line.

However, at VHF and higher frequencies it is possible to use PCB tracks as microstrip lines within properly terminated transmission systems. Typically the microstrip will be designed to match standard coaxial cable impedances, such as 50, 75 or 100 Ω , simplifying interfacing.

Note that if losses in such systems are to be minimized, the PCB material must be chosen for low high frequency losses. This usually means the use of Teflon or some other comparably low-loss PCB material. Often, though, the losses in short lines on cheap glass-fibre board are small enough to be quite acceptable.

Be Careful With Ground Plane Breaks

Wherever there is a break in the ground plane beneath a conductor, the ground plane return current must by necessity flow *around* the break. As a result, both the inductance and the vulnerability of the circuit to external fields are increased. This situation is diagrammed in Figure 7-32, where conductors A and B must cross one another.

Where such a break is made to allow a crossover of two perpendicular conductors, it would be far better if the second signal were carried across both the first and the ground plane by means of a piece of wire. The ground plane then acts as a shield between the two signal conductors, and the two ground return currents, flowing in opposite sides of the ground plane as a result of skin effects, do not interact.

With a multi-layer board, both the crossover and the continuous ground plane can be accommodated without the need for a wire link. Multi-layer PCBs are expensive and harder to trouble-shoot than more simple double-sided boards, but do offer even better shielding and signal routing. The principles involved remain unchanged but the range of layout options is increased.

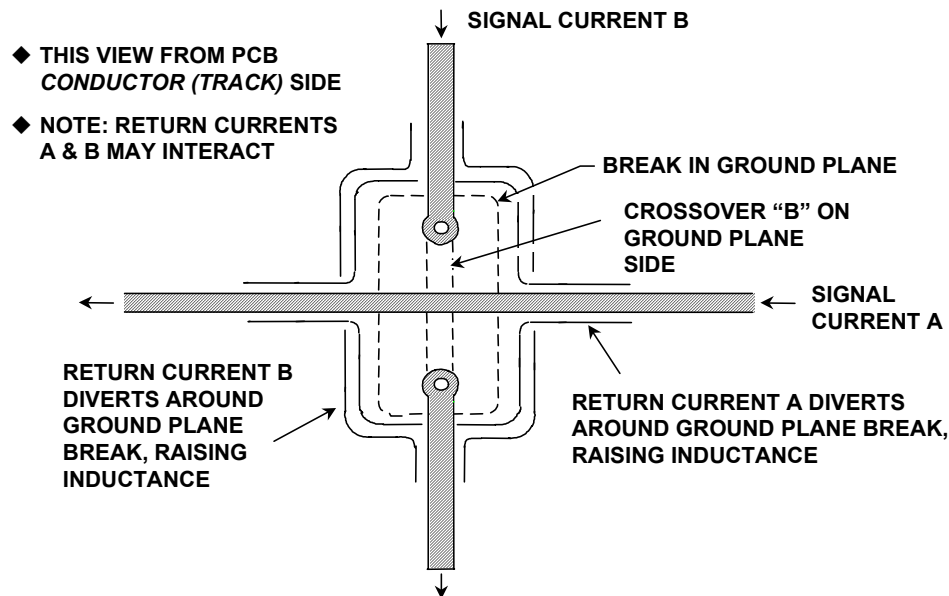


Figure 7-32: A ground plane break raises circuit inductance, and increases vulnerability to external fields

The use of double-sided or multi-layer PCBs with at least one continuous ground plane is undoubtedly one of the most successful design approaches for high performance mixed signal circuitry. Often the impedance of such a ground plane is sufficiently low to permit the use of a single ground plane for both analog and digital parts of the system. However, whether or not this is possible does depend upon the resolution and bandwidth required, and the amount of digital noise present in the system.

OP AMP APPLICATIONS

Ground Isolation Techniques

While the use of ground planes does lower impedance and helps greatly in lowering ground noise, there may still be situations where a prohibitive level of noise exists. In such cases, the use of ground error minimization and isolation techniques can be helpful.

Another illustration of a common-ground impedance coupling problem is shown in Figure 7-33. In this circuit a precision gain-of-100 preamp amplifies a low-level signal V_{IN} , using an AD8551 chopper-stabilized amplifier for best dc accuracy. At the load end, the signal V_{OUT} is measured with respect to G2, the local ground. Because of the small $700\mu\text{A}$ I_{SUPPLY} of the AD8551 flowing between G1 and G2, there is a $7\mu\text{V}$ ground error—about 7 times the typical input offset expected from the op amp!

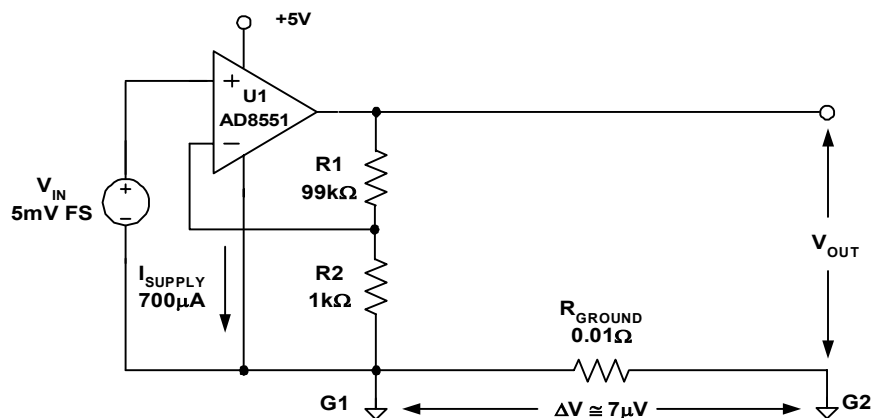


Figure 7-33: Unless care is taken, even small common ground currents can degrade precision amplifier accuracy

This error can be avoided simply by routing the negative supply pin current of the op amp back to star ground G2 as opposed to ground G1, by using a separate trace. This step eliminates the G1-G2 path power supply current, and so minimizes the ground leg voltage error. Note that there will be little error developed in the "hot" V_{OUT} lead, so long as the current drain at the load end is small.

In some cases, there may be simply unavoidable ground voltage differences between a source signal and the load point where it is to be measured. Within the context of this "same-board" discussion, this might require rejecting ground error voltages of several tens-of-mV. Or, should the source signal originate from an "off-board" source, then the magnitude of the common-mode voltages to be rejected can easily rise into a several volt range (or even tens-of-volts).

Fortunately, full signal transmission accuracy can still be accomplished in the face of such high noise voltages, by employing a principle discussed earlier. This is the use of a differential-input, *ground isolation* amplifier. The ground isolation amplifier minimizes the effect of ground error voltages between stages by processing the signal in differential fashion, thereby rejecting CM voltages by a substantial margin (typically 60dB or more).

Two ground isolation amplifier solutions are shown in Figure 7-34. This diagram can alternately employ either the AD629 to handle CM voltages up to $\pm 270\text{V}$, or the AMP03, which is suitable for CM voltages up to $\pm 20\text{V}$.

In the circuit, input voltage V_{IN} is referred to G1, but must be measured with respect to G2. With the use of a high CMR unity-gain difference amplifier, the noise voltage ΔV existing between these two grounds is easily rejected. The AD629 offers a typical CMR of 88dB, while the AMP03 typically achieves 100dB. In the AD629, the high CMV rating is done by a combination of high CM attenuation, followed by differential gain, realizing a net differential gain of unity. The AD629 uses the first listed value resistors noted in the figure for R1-R5. The AMP03 operates as a precision four-resistor differential amplifier, using the 25k Ω value R1-R4 resistors noted. Both devices are complete, one package solutions to the ground-isolation amplifier.

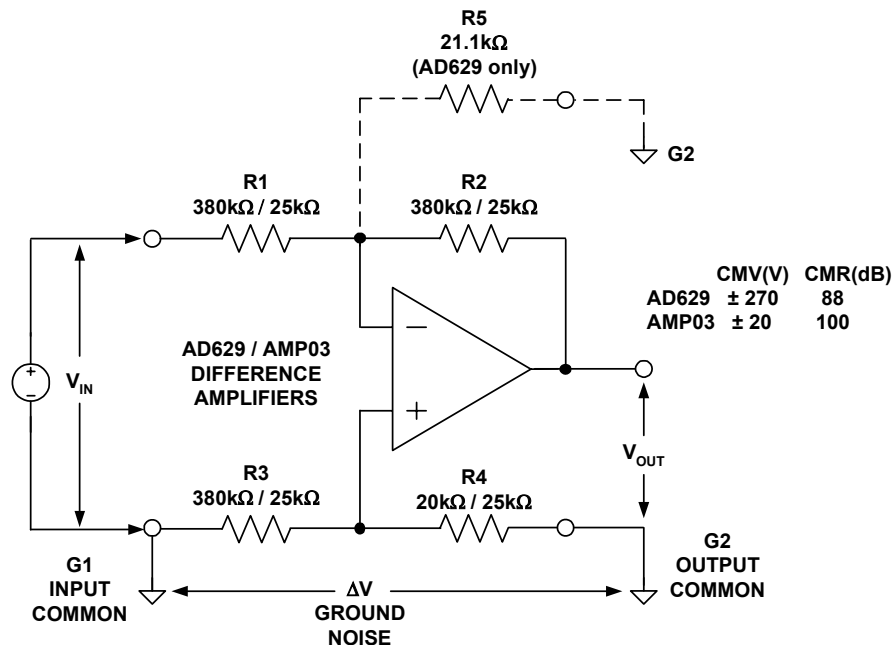


Figure 7-34: A differential input ground isolating amplifier allows high transmission accuracy by rejecting ground noise voltage between source (G1) and measurement (G2) grounds

This scheme allows relative freedom from tightly controlling ground drop voltages, or running additional and/or larger PCB traces to minimize such error voltages. Note that it can be implemented either with the fixed gain difference amplifiers shown, or also with a standard in-amp IC, configured for unity gain. The AD623, for example, also allows single-supply use. In any case, signal polarity is also controllable, by simple reversal of the difference amplifier inputs.

In general terms, transmitting a signal from one point on a PCB to another for measurement or further processing can be optimized by two key interrelated techniques. These are the use of high-impedance, differential signal-handling techniques. The high impedance loading of an in-amp minimizes voltage drops, and differential sensing of the remote voltage minimizes sensitivity to ground noise.

■ OP AMP APPLICATIONS

When the further signal processing is A/D conversion, these transmission criteria can be implemented *without* adding a differential ground isolation amplifier stage. Simply select an ADC which operates differentially. The high input impedance of the ADC minimizes load sensitivity to the PCB wiring resistance. In addition, the differential input feature allows the output of the source to be sensed directly at the source output terminals (even if single-ended). The CMR of the ADC then eliminates sensitivity to noise voltages between the ADC and source grounds.

An illustration of this concept using an ADC with high impedance differential inputs is shown in Figure 7-35. Note that the general concept can be extended to virtually any signal source, driving any load. All loads, even single-ended ones, become differential-input by adding an appropriate differential input stage.

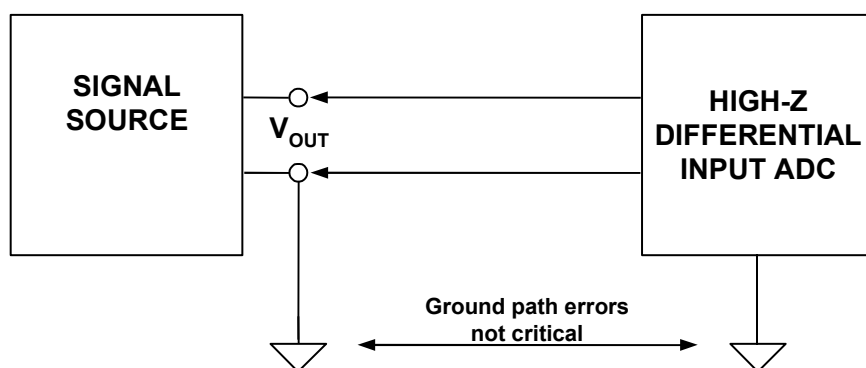


Figure 7-35: A high-impedance differential input ADC also allows high transmission accuracy between source and load

The differential input can be provided by either a fully developed high-Z in-amp, or in many cases it can be a simple subtractor stage op amp, such as Figure 7-34.

Static PCB Effects

Leakage resistance is the dominant static circuit board effect. Contamination of the PCB surface by flux residues, deposited salts, and other debris can create leakage paths between circuit nodes. Even on well-cleaned boards, it is not unusual to find 10 nA or more of leakage to nearby nodes from 15-volt supply rails. Nanoamperes of leakage current into the wrong nodes often cause volts of error at a circuit's output; for example, 10 nA into a 10 megohm resistance causes 0.1 V of error. Unfortunately, the standard op amp pinout places the $-V_S$ supply pin next to the + input, which is often hoped to be at high impedance! To help identify nodes sensitive to the effects of leakage currents ask the simple question: If a spurious current of a few nanoamperes or more were injected into this node, would it matter?

If the circuit is already built, you can localize moisture sensitivity to a suspect node with a classic test. While observing circuit operation, blow on potential trouble spots through a simple soda straw. The straw focuses the breath's moisture, which, with the board's salt content in susceptible portions of the design, disrupts circuit operation upon contact.

There are several means of eliminating simple surface leakage problems. Thorough washing of circuit boards to remove residues helps considerably. A simple procedure includes vigorously brushing the boards with isopropyl alcohol, followed by thorough washing with deionized water and an 85°C bakeout for a few hours. Be careful when selecting board-washing solvents, though. When cleaned with certain solvents, some water-soluble fluxes create salt deposits, exacerbating the leakage problem.

Unfortunately, if a circuit displays sensitivity to leakage, even the most rigorous cleaning can offer only a temporary solution. Problems soon return upon handling, or exposure to foul atmospheres, and high humidity. Some additional means must be sought to stabilize circuit behavior, such as conformal surface coating.

Fortunately, there is an answer to this, namely *guarding*, which offers a fairly reliable and permanent solution to the problem of surface leakage. Well-designed guards can eliminate leakage problems, even for circuits exposed to harsh industrial environments. Two schematics illustrate the basic guarding principle, as applied to typical inverting and non-inverting op amp circuits.

Figure 7-36 illustrates an inverting mode guard application. In this case, the op amp reference input is grounded, so the guard is a grounded ring surrounding all leads to the inverting input, as noted by the dotted line.

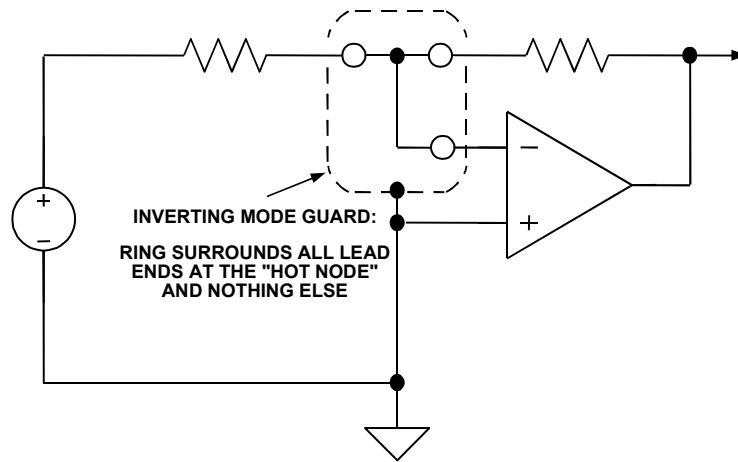


Figure 7-36: *Inverting mode guard encloses all op amp inverting input connections within a grounded guard ring*

Guarding basic principles are simple: *Completely* surround sensitive nodes with conductors that can readily sink stray currents, and maintain the guard conductors at the exact potential of the sensitive node (as otherwise the guard will serve as a leakage source rather than a leakage sink). For example, to keep leakage into a node below 1 pA (assuming 1000-megohm leakage resistance) the guard and guarded node must be within 1 mV. Generally, the low offset of a modern op amp is sufficient to meet this criterion.

There are important caveats to be noted with implementing a true high-quality guard. For traditional through-hole PCB connections, the guard pattern should appear on *both* sides of the circuit board, to be most effective. And, it should also be connected along its length by several vias. Finally, when either justified or required by the system design

■ OP AMP APPLICATIONS

parameters, do make an effort to include guards in the PCB design process from the outset—there is little likelihood that a proper guard can be added as an afterthought.

Figure 7-37 illustrates the case for a non-inverting guard. In this instance the op amp reference input is directly driven by the source, which complicates matters considerably. Again, the guard ring completely surrounds all of the input nodal connections. In this instance however, the guard is driven from the low impedance feedback divider connected to the inverting input.

Usually the guard-to-divider junction will be a direct connection, but in some cases a unity gain buffer might be used at "X" to drive a cable shield, or also to maintain the lowest possible impedance at the guard ring.

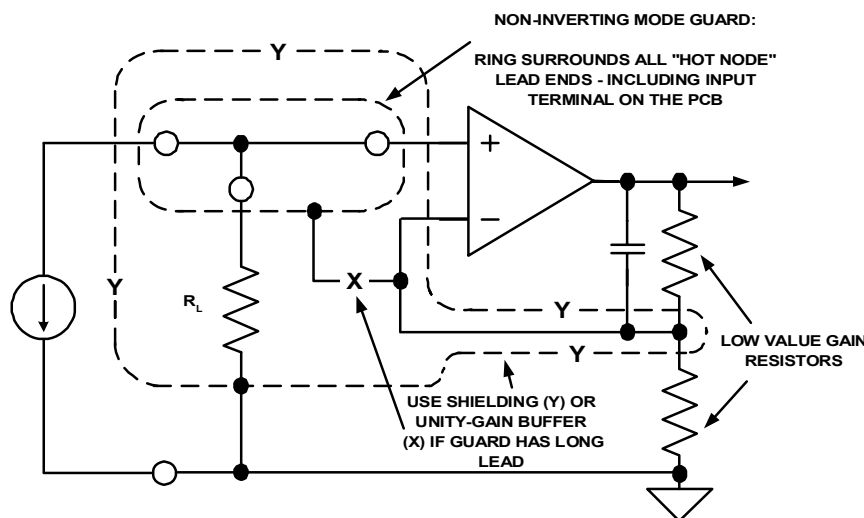


Figure 7-37: Non-inverting mode guard encloses all op amp non-inverting input connections within a low impedance, driven guard ring

In lieu of the buffer, another useful step is to use an additional, directly grounded screen ring, "Y", which surrounds the inner guard and the feedback nodes as shown. This step costs nothing except some added layout time, and will greatly help buffer leakage effects into the higher impedance inner guard ring.

Of course what hasn't been addressed to this point is just how the op amp itself gets connected into these guarded islands without compromising performance. The traditional method using a TO-99 metal can package device was to employ double-sided PCB guard rings, with both op amp inputs terminated within the guarded ring.

The high impedance sensor discussions in Chapter 4 use the above-described method. The section immediately following illustrates how more modern IC packages can be mounted to PCB patterns, and take advantage of guarding and low-leakage operation.

Sample MINIDIP and SOIC op amp PCB guard layouts

Modern assembly practices have favored smaller plastic packages such as 8 pin MINIDIP and SOIC types. Some suggested partial layouts for guard circuits using these packages is shown in the next two figures. While guard traces may also be possible with even more tiny op amp footprints, such as SOT23 etc., the required trace separations become even more confining, challenging the layout designer as well as the manufacturing processes.

For the ADI "N" style MINIDIP package, Figure 7-38 illustrates how guarding can be accomplished for inverting (left) and non-inverting (right) operating modes. This setup would also be applicable to other op amp devices where relatively high voltages occur at pin 1 or 4. Using a standard 8 pin DIP outline, it can be noted that this package's 0.1" pin spacing allows a PC trace (here, the guard trace) to pass between adjacent pins. This is the key to implementing effective DIP package guarding, as it can adequately prevent a leakage path from the $-V_S$ supply at pin 4, or from similar high potentials at pin 1.

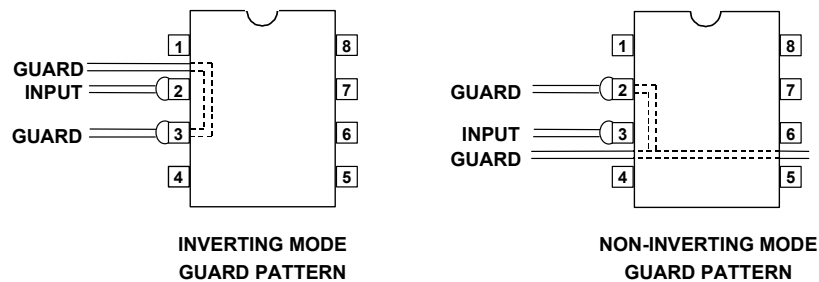


Figure 7-38: PCB guard patterns for inverting and non-inverting mode op amps using 8 pin MINIDIP (N) package

For the left-side inverting mode, note that the pin 3 connected and grounded guard traces surround the op amp inverting input (pin 2), and run parallel to the input trace. This guard would be continued out to and around the source and feedback connections of Figure 7-36 (or other similar circuit), including an input pad in the case of a cable. In the right-side non-inverting mode, the guard voltage is the feedback divider voltage to pin 2. This corresponds to the inverting input node of the amplifier, from Figure 7-37.

Note that in both of the cases of Figure 7-38, the guard physical connections shown are only partial—an actual layout would include all sensitive nodes within the circuit. In both the inverting and the non-inverting modes using the MINIDIP or other through-hole style package, the PCB guard traces should be located on both sides of the board, with top and bottom traces connected with several vias.

■ OP AMP APPLICATIONS

Things become slightly more complicated when using guarding techniques with the SOIC surface mount ("R") package, as the 0.05" pin spacing doesn't easily allow routing of PCB traces between the pins. But, there is still an effective guarding answer, at least for the inverting case. Figure 7-39 shows guards for the ADI "R" style SOIC package.

Note that for many single op amp devices in this SOIC "R" package, pins 1, 5, and 8 are "no connect" pins. For such instances, this means that these locations can be employed in the layout to route guard traces. In the case of the inverting mode (left), the guarding is still completely effective, with the dummy pin 1 and pin 3 serving as the grounded guard trace. This is a fully effective guard without compromise. Also, with SOIC op amps, much of the circuitry around the device will not use through-hole components. So, the guard ring may only be necessary on the op amp PCB side.

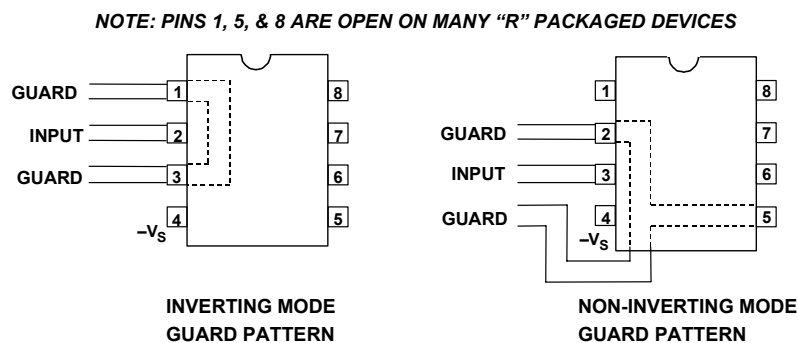


Figure 7-39: PCB guard patterns for inverting and non-inverting mode op amps using 8 pin SOIC (R) package

In the case of the follower stage (right), the guard trace must be routed around the negative supply at pin 4, and thus pin 4 to pin 3 leakage isn't fully guarded. For this reason, a precision high impedance follower stage using an SOIC package op amp isn't generally recommended, as guarding isn't possible for dual supply connected devices.

However, an exception to this caveat does apply to the use of a *single-supply* op amp as a non-inverting stage. For example, if the AD8551 is used, pin 4 becomes ground, and some degree of intrinsic guarding is then established by default.

Dynamic PCB Effects

Although static PCB effects can come and go with changes in humidity or board contamination, problems that most noticeably affect the dynamic performance of a circuit usually remain relatively constant. Short of a new design, washing or any other simple fixes can't fix them. As such, they can permanently and adversely affect a design's specifications and performance. The problems of stray capacitance, linked to lead and component placement, are reasonably well known to most circuit designers. Since lead placement can be permanently dealt with by correct layout, any remaining difficulty is solved by training assembly personnel to orient components or bend leads optimally.

Dielectric absorption (DA), on the other hand, represents a more troublesome and still poorly understood circuit-board phenomenon. Like DA in discrete capacitors, DA in a printed-circuit board can be modeled by a series resistor and capacitor connecting two closely spaced nodes. Its effect is inverse with spacing and linear with length.

As shown in Figure 7-40, the RC model for this effective capacitance ranges from 0.1 to 2.0 pF, with the resistance ranging from 50 to 500 M Ω . Values of 0.5 pF and 100 M Ω are most common. Consequently, circuit-board DA interacts most strongly with high-impedance circuits.

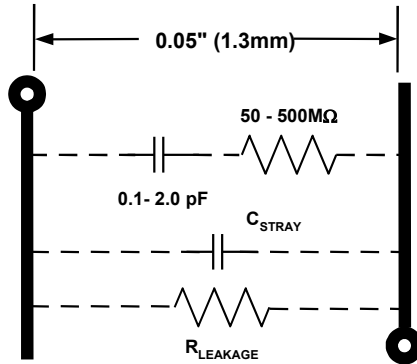


Figure 7-40: DA plagues dynamic response of PCB-based circuits

PCB DA most noticeably influences dynamic circuit response, for example, settling time. Unlike circuit leakage, the effects aren't usually linked to humidity or other environmental conditions, but rather, are a function of the board's dielectric properties. The chemistry involved in producing plated-through holes seems to exacerbate the problem. If your circuits don't meet expected transient response specs, you should consider PCB DA as a possible cause.

Fortunately, there are solutions. As in the case of capacitor DA, external components can be used to compensate for the effect. More importantly, surface guards that totally isolate sensitive nodes from parasitic coupling often eliminate the problem (note that these guards should be duplicated on both sides of the board, in cases of through-hole components). As noted previously, low-loss PCB dielectrics are also available.

PCB "hook", similar if not identical to DA, is characterized by variation in effective circuit-board capacitance with frequency (see Reference 1). In general, it affects high-impedance circuit transient response where board capacitance is an appreciable portion of the total in the circuit. Circuits operating at frequencies below 10 kHz are the most susceptible. As in circuit board DA, the board's chemical makeup very much influences its effects.

Stray Capacitance

When two conductors aren't short-circuited together, or totally screened from each other by a conducting (Faraday) screen, there is a capacitance between them. So, on any PCB, there will be a large number of capacitors associated with any circuit (which may or may not be considered in models of the circuit). Where high frequency performance matters (and even dc and VLF circuits may use devices with high F_t and therefore be vulnerable to HF instability), it is very important to consider the effects of this stray capacitance.

Any basic textbook will provide formulas for the capacitance of parallel wires and other geometric configurations (see References 9 and 10). The example we need consider in this discussion is the parallel plate capacitor, often formed by conductors on opposite sides of a PCB. The basic diagram describing this capacitance is shown in Figure 7-41.

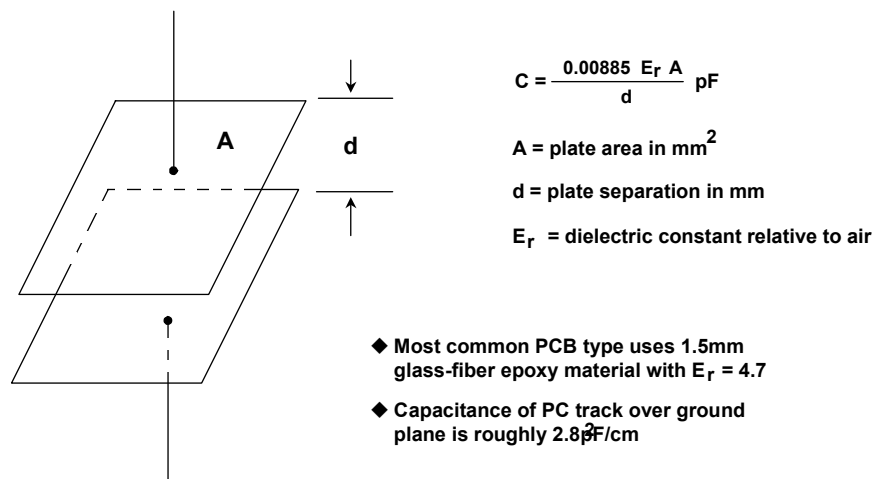


Figure 7-41: Capacitance of two parallel plates

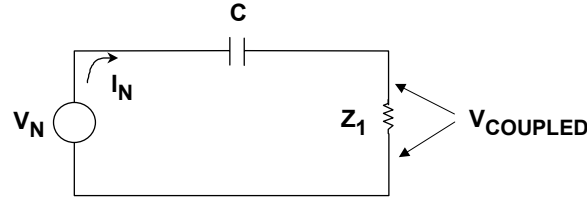
Neglecting edge effects, the capacitance of two parallel plates of area $A \text{ mm}^2$ and separation $d \text{ mm}$ in a medium of dielectric constant E_r relative to air is $0.00885 E_r A/d \text{ pF}$.

From this formula, we can calculate that for general purpose PCB material ($E_r = 4.7$, $d = 1.5\text{mm}$), the capacitance between conductors on opposite sides of the board is just under 3pF/cm^2 . In general, such capacitance will be parasitic, and circuits must be designed so that it does not affect their performance.

While it is possible to use PCB capacitance in place of small discrete capacitors, the dielectric properties of common PCB substrate materials cause such capacitors to behave poorly. They have a rather high temperature coefficient and poor Q at high frequencies, which makes them unsuitable for many applications. Boards made with lower-loss dielectrics such as Teflon are expensive exceptions to this rule.

Capacitive Noise & Faraday Shields

There is a capacitance between any two conductors separated by a dielectric (air or vacuum are dielectrics). If there is a change of voltage on one, there will be a movement of charge on the other. A basic model for this is shown in Figure 7-42.



$$Z_1 = \text{CIRCUIT IMPEDANCE}$$

$$Z_2 = 1/j\omega C$$

$$V_{\text{COUPLED}} = V_N \left(\frac{Z_1}{Z_1 + Z_2} \right)$$

Figure 7-42: Capacitive coupling equivalent circuit model

It is evident that the noise voltage, V_{COUPLED} appearing across Z_1 , may be reduced by several means, all of which reduce noise current in Z_1 . They are reduction of the signal voltage V_N , reduction of the frequency involved, reduction of the capacitance, or reduction of Z_1 itself. Unfortunately however, often none of these circuit parameters can be freely changed, and an alternate method is needed to minimize the interference. The best solution towards reducing the noise coupling effect of C is to insert a grounded conductor, also known as a *Faraday shield*, between the noise source and the affected circuit. This has the desirable effect of reducing Z_1 noise current, thus reducing V_{COUPLED} .

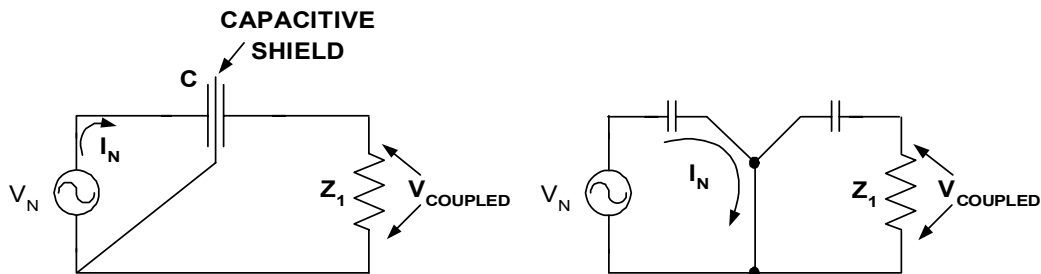


Figure 7-43: An operational model of a Faraday shield

A Faraday shield model is shown by Figure 7-43. In the left picture, the function of the shield is noted by how it effectively divides the coupling capacitance, C . In the right picture the net effect on the coupled voltage across Z_1 is shown. Although the noise current I_N still flows in the shield, most of it is now diverted away from Z_1 . As a result, the coupled noise voltage V_{COUPLED} across Z_1 is reduced.

A Faraday shield is easily implemented and almost always successful. Thus capacitively coupled noise is rarely an intractable problem. However, to be fully effective, a Faraday shield must completely block the electric field between the noise source and the shielded circuit. It must also be connected so that the displacement current returns to its source, without flowing in any part of the circuit where it can introduce conducted noise.

■ OP AMP APPLICATIONS

The Floating Shield Problem

And, it is quite important to note here— *a conductor that is intended to function as a Faraday shield must never be left floating, as this almost always increases capacity and exacerbates the noise problem!*

An example of this "floating shield" problem is seen in side-brazed ceramic IC packages. These DIP packages have a small square conducting Kovar lid soldered onto a metallized rim on the ceramic package top. Package manufacturers offer only two options: the metallized rim may be connected to one of the corner pins of the package, or it may be left unconnected.

Most logic circuits have a ground pin at one of the package corners, and therefore the lid is grounded. Alas, many analog circuits don't have a ground pin at a package corner, and the lid is left floating— acting as an antenna for noise. Such circuits turn out to be far more vulnerable to electric field noise than the same chip in a plastic DIP package, where the chip is completely unshielded.

Whenever practical, it is good practice for the user to ground the lid of any side brazed ceramic IC where the lid is not grounded by the manufacturer, thus implementing an *effective* Faraday shield. This can be done with a wire soldered to the lid (this will not damage the device, as the chip is thermally and electrically isolated from the lid). If soldering to the lid is unacceptable, a grounded phosphor-bronze clip may be used to make the ground connection, or conductive paint from the lid to the ground pin.

A safety note is appropriate at this point. Never attempt to ground such a lid without first verifying that it is unconnected. Occasionally device types are found with the lid connected to a power supply rather than to ground!

A case where a Faraday shield is impracticable is between IC chip bondwires, which has important consequences. The stray capacitance between chip bondwires and associated leadframes is typically $\approx 0.2\text{pF}$, with observed values generally between 0.05 and 0.6pF.)

Buffering ADCs Against Logic Noise

If we have a high resolution data converter (ADC or DAC) connected to a high speed data bus which carries logic noise with a 2-5 V/ns edge rate, this noise is easily connected to the converter analog port via stray capacitance across the device. Whenever the data bus is active, intolerable amounts of noise are capacitively coupled into the analog port, thus seriously degrading performance.

This particular effect is illustrated by the diagram of Figure 7-44, where multiple package capacitors couple noisy edge signals from the data bus into the analog input of an ADC.

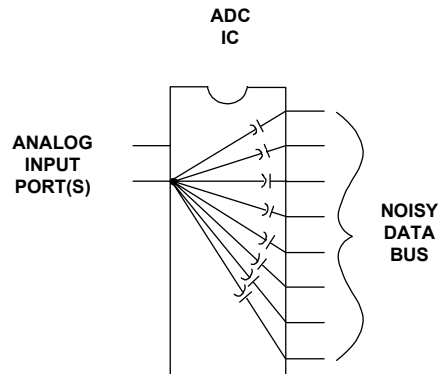
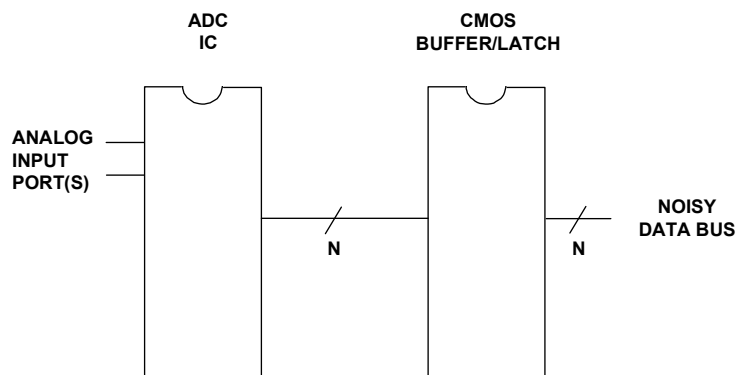


Figure 7-44: A high speed ADC IC sitting on a fast data bus couples digital noise into the analog port, thus limiting performance

Present technology offers no cure for this problem, within the affected IC device itself. The problem also limits performance possible from other broadband monolithic mixed signal ICs with single-chip analog and digital circuits. Fortunately, this coupled noise problem can be simply avoided, by *not* connecting the data bus directly to the converter.



- ◆ THE OUTPUT BUFFER/LATCH ACTS AS A FARADAY SHIELD BETWEEN “N” LINES OF A FAST, NOISY DATA BUS AND A HIGH PERFORMANCE ADC.
- ◆ THIS MEASURE ADDS COST, BOARD AREA, POWER CONSUMPTION, RELIABILITY REDUCTION, DESIGN COMPLEXITY, AND MOST IMPORTANTLY, *IMPROVED PERFORMANCE!*

Figure 7-45: A high speed ADC IC using a CMOS buffer/latch at the output shows enhanced immunity of digital data bus noise

Instead, use a CMOS latched buffer as a converter-to-bus interface, as shown by Figure 7-45. Now the CMOS buffer IC acts as a Faraday shield, and dramatically reduces noise coupling from the digital bus. This solution costs money, occupies board area, reduces reliability (very slightly), consumes power, and it complicates the design— but it does improve the signal-to-noise ratio of the converter! The designer must decide whether it is worthwhile for individual cases, but in general it is highly recommended.

REFERENCES: PCB DESIGN ISSUES

1. W. Doeling, W. Mark, T. Tadewald, and P. Reichenbacher, "Getting Rid of Hook: The Hidden PC-Board Capacitance," **Electronics**, October 12, 1978, p 111-117.
2. Alan Rich, "Shielding and Guarding", **Analog Dialogue**, Vol. 17 N0. 1, 1983, pp. 8.
3. Ralph Morrison, *Grounding and Shielding Techniques in Instrumentation*, 3rd Edition, John Wiley, Inc., 1986, ISBN: 0-471-83805-5.
4. Henry W. Ott, **Noise Reduction Techniques in Electronic Systems**, 2nd Edition, John Wiley, Inc., 1988, ISBN: 0-471-85068-3.
5. Paul Brokaw, "An IC Amplifier User's Guide to Decoupling, Grounding and Making Things Go Right for a Change", **Analog Devices AN202**.
6. Paul Brokaw, "Analog Signal-Handling for High Speed and Accuracy," **Analog Devices AN342**.
7. Paul Brokaw and Jeff Barrow, "Grounding for Low- and High-Frequency Circuits," **Analog Devices AN345**.
8. Jeff Barrow, "Avoiding Ground Problems in High Speed Circuits," **RF Design**, July 1989.
9. B. I. & B. Bleaney, **Electricity & Magnetism**, Oxford at the Clarendon Press, 1957, pp 23, 24, & 52.
10. G. W. A. Dummer, H. Nordenberg, **Fixed and Variable Capacitors**, McGraw-Hill, 1960, pp 11-13.

ACKNOWLEDGEMENTS:

Portions of this section were adapted from Doug Grant and Scott Wurcer, "Avoiding Passive Component Pitfalls", originally published in **Analog Dialogue 17-2**, 1983.

SECTION 7-3: OP AMP POWER SUPPLY SYSTEMS

Walt Jung, Walt Kester

Op amp circuits have traditionally been powered from well-regulated, low noise linear power supplies. This type of power system is typically characterized by medium-to-low power conversion efficiency. Such linear regulators usually excel in terms of self-generated and radiated noise components. If the designer's life were truly simple, it might continue with such familiar designs offering good performance and minimal side effects.

But, the designer's life is hardly so simple. Modern systems may allow using linear regulators, but multiple output levels and/or polarities are often required. There may also be some additional requirements set for efficiency, which may dictate the use of dc-dc conversion techniques, and, unfortunately, their higher associated noise output.

This section addresses power supply design issues for op amp systems, taking into account the regulator types most likely to be used. The primary dc power sources are assumed to be either rectified and smoothed ac sources (i.e., mains derived), a battery stack, or a switching regulator output. The latter example could be fed from either a battery or a mains-derived dc source.

As noted in Figure 7-46, linear mode regulation is generally recommended as an optimum starting point in all instances (first bullet). Nevertheless, in some cases, a degree of hybridization between fully linear and switching mode regulation may be required (second bullet). This could be either for efficiency or other diverse reasons.

- ◆ **High performance analog power systems use *linear regulators*, with primary power derived from:**
 - AC line power
 - Battery power systems
 - DC- DC power conversion systems
- ◆ **Switching regulators should be avoided if at all possible, but if not...**
 - Apply noise control techniques
 - Use quality layout and grounding
 - Be aware of EMI

Figure 7-46: Regulation priorities for op amp power supply systems

Whenever switching-type regulators are involved in powering precision analog circuits, noise control is very likely to be a design issue. Therefore some focus of this section is on minimizing noise when using switching regulators.

Linear IC Regulation

Linear IC voltage regulators have long been standard power system building blocks. After an initial introduction in 5V logic voltage regulator form, they have since expanded into other standard voltage levels spanning from 3 to 24V, handling output currents from as low as 100mA (or less) to as high as 5A (or more). For several good reasons, linear style IC voltage regulators have been valuable system components since the early days. As mentioned above, a basic reason is the relatively low noise characteristic vis-à-vis the switching type of regulator. Others are a low parts count and overall simplicity compared to discrete solutions. But, because of their power losses, these linear regulators have also been known for being relatively inefficient. Early generation devices (of which many are still available) required 2V or more of unregulated input above the regulated output voltage, making them lossy in power terms.

More recently however, linear IC regulators have been developed with more liberal (i.e., lower) limits on minimum input-output voltage. This voltage, known more commonly as *dropout* voltage, has led to what is termed the *Low DropOut* regulator, or more simply, the LDO. Dropout voltage (V_{MIN}) is defined simply as that minimum input-output differential where the regulator undergoes a 2% reduction in output voltage. For example, if a nominal 5.0V LDO output drops to 4.9V (-2%) under conditions of an input-output differential of 0.5V, by this definition the LDO's dropout voltage is 0.5V.

Dropout voltage is extremely critical to a linear regulator's power efficiency. The lower the voltage allowable across a regulator while still maintaining a regulated output, the less power the regulator dissipates as a result. A low regulator dropout voltage is the key to this, as it takes a lower dropout to maintain regulation as the input voltage lowers. In performance terms, the bottom line for LDOs is simply that more useful power is delivered to the load and less heat is generated in the regulator. LDOs are key elements of power systems providing stable voltages from batteries, such as portable computers, cellular phones, etc. This is because they maintain a regulated output down to lower points on the battery's discharge curve. Or, within classic mains-powered raw DC supplies, LDOs allow lower transformer secondary voltages, reducing system shutdowns under brownout conditions, as well as allowing cooler operation.

Some Linear Voltage Regulator Basics

A brief review of three terminal linear IC regulator fundamentals is necessary before understanding the LDO variety. Most (but not all) of the general three terminal regulator types available today are *positive leg*, *series style* regulators. This simply means that they control the regulated voltage output by means of a pass element in series with the positive unregulated input. And, although they are fewer in number, there are also *negative leg* series style regulators, which operate in a fashion complementary to the positive units.

A basic hookup diagram of a three terminal regulator is shown in Figure 7-47. In terms of basic functionality, many standard voltage regulators operate in a series mode, three-terminal form, just as shown here. As can be noted from this figure, the three I/O terminals are V_{IN} , GND (or Common), and V_{OUT} . Note also that this regulator block, in

the absence of any assigned voltage polarity, could in principle be a positive type regulator. Or, it might also be a negative style of voltage regulator—the principle is the same for both—a common terminal, as well as input and output terminals.

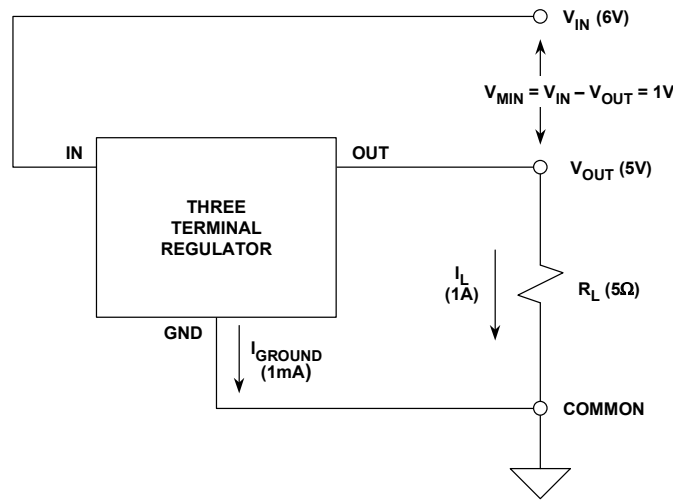


Figure 7-47: A basic three terminal regulator hookup (either positive or negative)

In operation, there are two power components which get dissipated in the regulator, one a function of $V_{IN} - V_{OUT}$ and I_L , plus a second which is a function of V_{IN} and I_{ground} . The first of these is usually dominant. Analysis of the situation will reveal that as the dropout voltage V_{MIN} is reduced, the regulator is able to deliver a higher percentage of the input power to the load, and is thus more efficient, running cooler and saving power. This is the core appeal of the modern LDO type of regulator (see Reference 1).

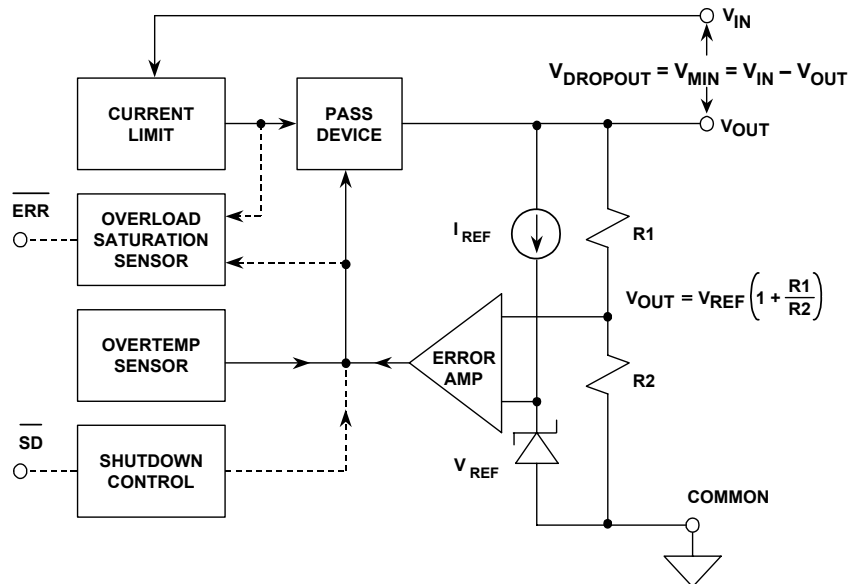


Figure 7-48: Functional diagram of a typical voltage regulator

A more detailed look within a typical regulator block diagram reveals a variety of elements, as is shown in Figure 7-48. Note that all regulators will contain those functional components connected via solid lines. The connections shown dotted indicate options, which might be available when more than three I/O pins are available.

■ OP AMP APPLICATIONS

In operation, a voltage reference block produces a stable voltage V_{REF} , which is almost always a voltage based on the bandgap voltage of silicon, typically $\sim 1.2V$ (see Reference 2). This allows output voltages of 3V or more from supplies as low as 5V. This voltage drives one input of an error amplifier, with the second input connected to the divider, R1-R2. The error amplifier drives the pass device, which in turn controls the output. The resulting regulated voltage is then simply:

$$V_{OUT} = V_{REF} \left(1 + \frac{R1}{R2} \right) \quad \text{Eq. 7-1}$$

Pass Devices

The pass device is a foremost regulator part, and the type chosen here has a major influence on almost all regulator performance issues. Most notable among these is dropout voltage, V_{MIN} . Analysis shows the use of an *inverting* mode pass transistor allows the pass device to be effectively saturated, thus minimizing the associated voltage losses. Therefore this factor makes the two most desirable pass devices for LDO use a PNP bipolar, or a PMOS transistor. These device types achieve the lowest levels of $V_{IN} - V_{OUT}$ required for LDO operation. In contrast, NPN bipolars are poor as pass devices in terms of low dropout, particularly when they are Darlington connected.

Standard fixed-voltage IC regulator architectures illustrate this point on regarding pass devices. For example, the fixed-voltage LM309 5V regulators and family derivatives such as the 7805, 7815 et al, (and their various low and medium current alternates) are poor in terms of dropout voltage. These designs use a Darlington pass connection, not known for low dropout ($\sim 1.5V$ typical), or for low quiescent current ($\sim 5mA$).

$\pm 15V$ regulator using adjustable voltage ICs

Later developments in references and three-terminal regulation techniques led to the development of the *voltage-adjustable* regulator. The original IC to employ this concept was the LM317, a positive regulator. The device produces a fixed reference voltage of 1.25V, appearing between the V_{OUT} and ADJ pins of the IC. External scaling resistors set up the desired output voltage, adjustable in the range of 1.25 – 30V. A complementary device, the LM337, operates in similar fashion, regulating negative voltages.

An application example using standard *adjustable* three terminal regulators to implement a $\pm 15V$ linear power supply is shown in Figure 7-49. This is a circuit as might be used for powering traditional op amp supply rails. It is capable of better line regulation performance than would an otherwise similar circuit, using standard fixed-voltage regulator devices, such as for example 7815 and 7915 ICs. However, in terms of power efficiency it isn't outstanding, due to the use of the chosen ICs, which require 2V or more of headroom for operation.

In the upper portion of this circuit an LM317 adjustable regulator is used, with R2 and R1 chosen to provide a 15V output at the upper output terminal. If desired, R2 can easily be adjusted for other output levels, according to the figure's V_{OUT} expression. Resistor R1 should be left fixed, as it sets the minimum regulator drain of 10mA or more.

In this circuit, capacitors C1 and C2 should be tantalum types, and R1-R2 metal films. C3 is optional, but is highly recommended if the lowest level of output noise is desired. The normally reverse biased diode D1 provides a protective output clamp, for system cases where the output voltage would tend to reverse, if one supply should fail. The circuit operates from a rectified and filtered ac supply at V_{IN} , polarized as shown. The output current is determined by choosing the regulator IC for appropriate current capability.

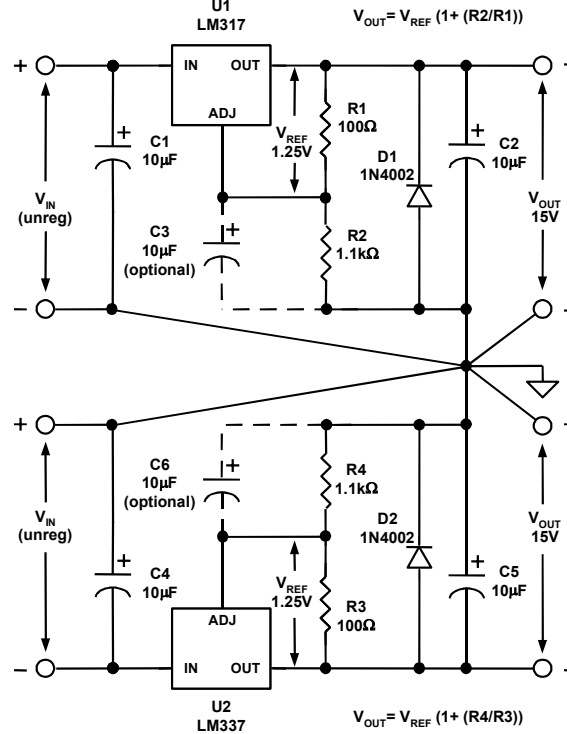


Figure 7-49: A classic $\pm 15V$, 1A linear supply regulator using adjustable voltage regulator ICs

To implement the negative supply portion, the sister device to the LM317 is used, the LM337. The bottom circuit section thus mirrors the operation of the upper, delivering a negative 15V at the lowest output terminal. Programming of the LM337 for output voltage is similar to that of the LM317, but uses resistors R4 and R3. R4 should be used to adjust the voltage, with R3 remaining fixed. C6 is again optional, but is recommended for reasons of lowest noise.

Low Dropout Regulator Architectures

In contrast to traditional three terminal regulators with Darlington or single-NPN pass devices, low dropout regulators employ lower voltage threshold pass devices. This basic operational difference allows them to operate effectively down to a range of 100-200mV in terms of their specified V_{MIN} . In terms of use within a system, this factor can have fairly significant operational advantages.

An effective implementation of some key LDO features is contained in the Analog Devices series of anyCAP™ LDO regulators. Devices of this ADP330X series are so named for their relative insensitivity to the output capacitor, in terms of both its size and ESR. Available in power efficient packages such as the ADI Thermal Coastline (and

OP AMP APPLICATIONS

other thermally enhanced packages), they come in both stand-alone LDO and LDO controller forms (used with an external PMOS FET). They also offer a wide span of fixed output voltages from 1.8 to 5V, with rated current outputs up to 500mA. User-adjustable output voltage versions are also available. A basic simplified diagram for the family is shown schematically in Figure 7-50.

One of the key differences in the ADP330X LDO series is the use of a high gain vertical PNP pass device, Q1, allowing typical dropout voltages for the series to be on the order of 1mV/mA for currents of 200mA or less.

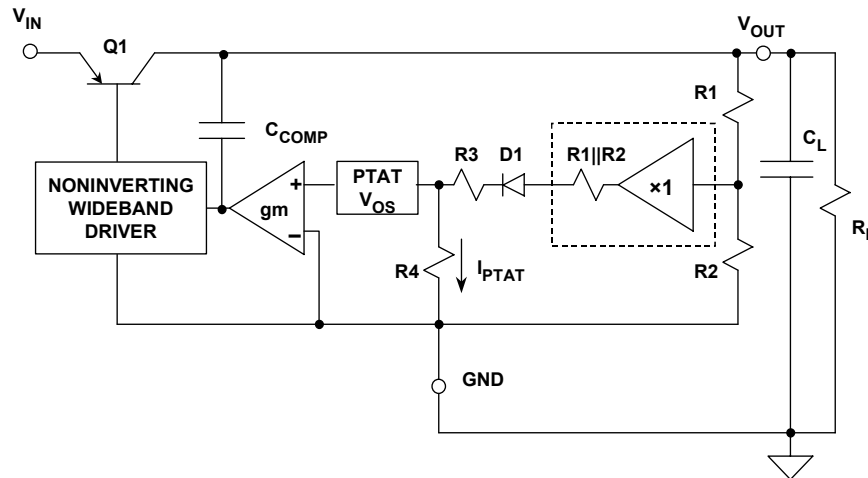


Figure 7-50: The ADP330X anyCAP™ LDO architecture has both dc and ac performance advantages

In circuit operation, V_{REF} is defined as a reference voltage existing at the output of a zero impedance divider of ratio $R1/R2$. In the figure, this is depicted symbolically by the (dotted) unity gain buffer amplifier fed by $R1/R2$, which has an output of V_{REF} . This reference voltage feeds into a series connection of (dotted) $R1||R2$, then actual components $D1$, $R3$, $R4$, etc. The regulator output voltage is:

$$V_{OUT} = V_{REF} \left(1 + \frac{R1}{R2} \right) \quad \text{Eq. 7-2}$$

In the various devices of the ADP330X series, the $R1$ - $R2$ divider is adjusted to produce standard output voltages of 1.8, 2.5, 2.7, 3.0, 3.2, 3.3, and 5.0V. The regulator behaves as if the entire error amplifier has simply an offset voltage of V_{REF} volts, as seen at the output of a conventional $R1$ - $R2$ divider.

While the above described dc performance enhancements of the ADP330X series are worthwhile, more dramatic improvements come in areas of ac-related performance. Capacitive loading and the potential instability it brings is a major deterrent to easy LDO applications. One method of providing some measure of immunity to variation in an amplifier response pole is the use of a frequency compensation technique called *pole splitting*. In the Figure 7-50 circuit, C_{COMP} functions as the pole splitting capacitor, and provides benefits of a buffered, C_L independent single-pole response. As a

result, frequency response is dominated by the regulator's internal compensation, and becomes relatively immune to the value and ESR of load capacitor C_L .

This feature makes the design tolerant of virtually any output capacitor type. C_L , the load capacitor, can be as low as $0.47\mu\text{F}$, and it can also be a multi-layer ceramic capacitor (MLCC) type, allowing a very small physical size for the entire regulation function.

Fixed Voltage, 50/100/200/500 mA LDO Regulators

A basic regulator application diagram common to various fixed voltage devices of the ADP330X device series is shown by Figure 7-51. Operation of the various pins and internal functions is discussed below. To adapt this general diagram to a specific current and voltage requirement, select a basic device for output current from the table in the diagram. Then select the output voltage by the part # suffix, consistent with the table.

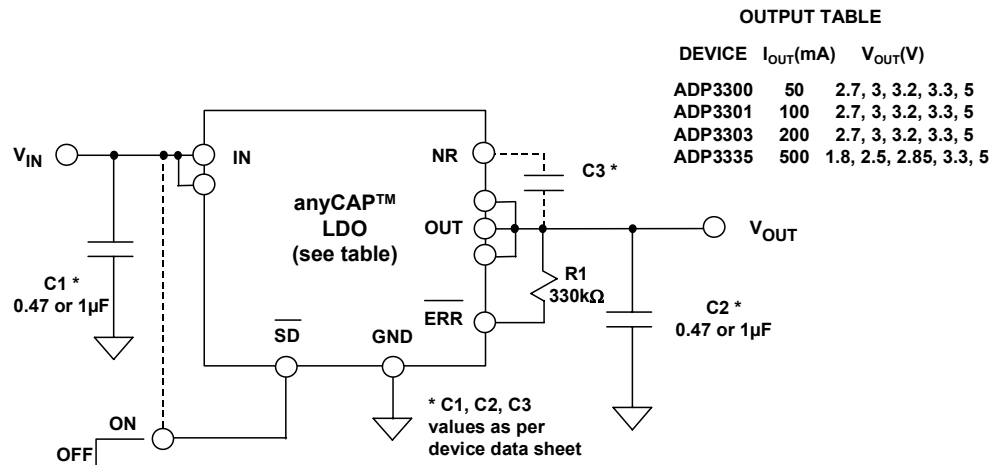


Figure 7-51: A basic LDO regulator hookup useful by device selection from 50 to 500mA, at fixed voltages per table

This circuit is a general one, illustrating common points. For example, the ADP3300 is a 50mA basic LDO regulator device, designed for those fixed output voltages as noted. An actual ADP3300 device ordered would be ADP3300ART-YY, where the "YY" is a voltage designator suffix such as 2.7, 3, 3.2, 3.3, or 5, for the respective table voltages. The "ART" portion of the part number designates the package (SOT23 6-lead). To produce 5V from the circuit, use the ADP3300ART-5. Similar comments apply to the other devices, insofar as part numbering. For example, an ADP3301AR-5 depicts an SO-8 packaged 100mA device, producing 5V output.

In operation, the circuit produces rated output voltage for loads under the max current limit, for input voltages above $V_{OUT} + V_{MIN}$ (where V_{MIN} is the dropout voltage for the specific device used, at rated current). The circuit is ON when the shutdown input is in a HIGH state, either by a logic HIGH control input to the $\overline{\text{SD}}$ pin, or by simply tying this pin to V_{IN} (shown dotted). When $\overline{\text{SD}}$ is LOW or grounded, the regulator shuts down, and draws a minimum quiescent current.

The anyCAP™ regulator devices maintain regulation over a wide range of load, input voltage and temperature conditions. Most devices have a combined error band of $\pm 1.4\%$

OP AMP APPLICATIONS

(or less). When an overload condition is detected, the open collector $\overline{\text{ERR}}$ goes to a LOW state. R1 is a pullup resistor for the $\overline{\text{ERR}}$ output. This resistor can be eliminated if the load provides a pullup current.

C3, connected between the OUT and NR pins, can be used for an optional noise reduction (NR) feature. This is accomplished by bypassing a portion of the internal resistive divider, which reduces output noise ~10 dB. When exercised, only the recommended low leakage capacitors as specific to a particular part should be used.

The C1 input and C2 output capacitors should be selected as either 0.47 or 1 μF values respectively, again, as per the particular device used. For most devices of the series 0.47 μF suffices, but the ADP3335 uses the 1 μF values. Larger capacitors can also be used, and will provide better transient performance.

Heat sinking of device packages with more than 5 pins is enhanced, by use of multiple IN and OUT pins. All of the pins available should therefore be used in the PCB design, to minimize layout thermal resistance.

Adjustable Voltage, 200mA LDO Regulator

In addition to the fixed output voltage LDO devices discussed above, adjustable versions are also available, to realize non-standard voltages. The ADP3331 is one such device, and it is shown in Figure 7-52, configured as a 2.8V output, 200mA LDO application.

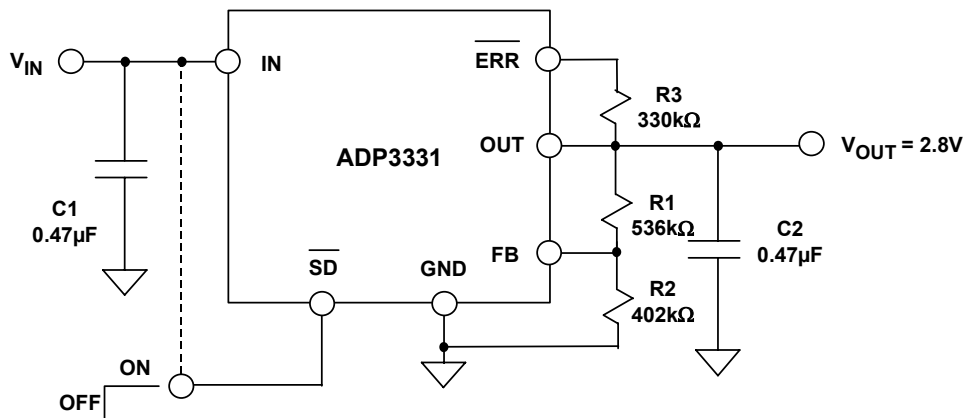


Figure 7-52: An adjustable 200mA LDO regulator set up for a 2.8V output

The ADP3331 is generally similar to other anyCAP™ LDO parts, with two notable exceptions. It has a lower quiescent current (~34 μA when lightly loaded) and most importantly, the output voltage is user-adjustable. As noted in the circuit, R1 and R2 are external precision resistors used to define the regulator operating voltage.

The output of this regulator is V_{OUT} , which is related to feedback pin FB voltage V_{FB} as:

$$V_{\text{OUT}} = V_{\text{FB}} \left(1 + \frac{R1}{R2} \right) \quad \text{Eq. 7-3}$$

where V_{FB} is 1.204V. Resistors R1 and R2 program V_{OUT} , and their parallel equivalent should be kept close to 230k Ω for best stability.

To select R1 and R2, first calculate their ideal values, according to the following two expressions:

$$R1 = 230 \left(\frac{V_{OUT}}{V_{FB}} \right) k\Omega \quad \text{Eq. 7-4}$$

$$R2 = \frac{230}{\left(1 + \frac{V_{FB}}{V_{OUT}} \right)} k\Omega \quad \text{Eq. 7-5}$$

In the example circuit, V_{OUT} is 2.8V, which yields $R1 = 534.9k\Omega$, and $R2 = 403.5k\Omega$. As noted in the figure, closest standard 1% values are used, which provides an output of 2.8093V (perfect resistors assumed). In practice, the resistor tolerances should be added to the $\pm 1.4\%$ tolerance of the ADP3331 for an estimation of overall error.

To complement the above-discussed anyCAP™ series of standalone LDO regulators, there is the LDO *regulator controller*. The regulator controller IC picks up where the standalone regulator stops for either load current or power dissipation, using an external PMOS FET pass device. As such, the current capability of the LDO can be extended to several amps. An LDO regulator controller application is shown later in this discussion.

These application examples above illustrate a subset of the entire anyCAP™ family of LDOs. Further information on this series of standalone and regulator controller LDO devices can be found in the references at the end of the section.

Charge-Pump Voltage Converters

Another method for developing supply voltage for op amp systems employs what is known as a *charge-pump* circuit (also called switched capacitor voltage conversion). Charge-pump voltage converters accomplish energy transfer and voltage conversion using charges stored on capacitors, thus the name, charge-pump.

Using switching techniques, charge-pumps convert supply voltage of one polarity to a higher or lower voltage, or to an alternate polarity (at either higher or lower voltage). This is accomplished with only an array of low resistance switches, a clock for timing, and a few external storage capacitors to hold the charges being transferred in the voltage conversion process. No inductive components are used, thus EMI generation is kept to a minimum. Although relatively high currents are switched internally, the high current switching is localized, and therefore the generated noise is not as great as in inductive type switchers. With due consideration towards component selection, charge-pump converters can be implemented with reasonable noise performance.

The two common charge-pump voltage converters are the *voltage inverter* and the *voltage doubler* circuits. In a voltage inverter, a charge pump capacitor is charged to the input voltage during the first half of the switching cycle. During the second half of the switching cycle the input voltage stored on the charge pump capacitor is inverted, and is applied to an output capacitor and the load. Thus the output voltage is essentially the negative of the input voltage, and the average input current is approximately equal to the

■ OP AMP APPLICATIONS

output current. The switching frequency impacts the size of the external capacitors required, and higher switching frequencies allow the use of smaller capacitors. The duty cycle—defined as the ratio of charge pump charging time to the entire switching cycle time—is usually 50%, which yields optimal transfer efficiency.

A voltage doubler works similarly to the inverter. In this case the pump capacitor accomplishes a voltage doubling function. In the first phase it is charged from the input, but in the second phase of the cycle it appears in series with the output capacitor. Over time, this has the effect of doubling the magnitude of the input voltage across the output capacitor and load. Both the inverter and voltage doubler circuits provide no voltage regulation in basic form. However, techniques exist to add regulation (discussed below).

There are advantages and disadvantages to using charge-pump techniques, compared to inductor-based switching regulators. An obvious key advantage is the elimination of the inductor and the related magnetic design issues. In addition, charge-pump converters typically have relatively low noise and minimal radiated EMI. Application circuits are simple, and usually only two or three external capacitors are required. Because there are no inductors, the final PCB height can generally be made smaller than a comparable inductance-based switching regulator. Charge-pump inverters are also low in cost, compact, and capable of efficiencies greater than 90%. Obviously, current output is limited by the capacitor size and the switch capacity. Typical IC charge-pump inverters have 150mA maximum outputs.

- ◆ **No Inductors!**
- ◆ **Minimal Radiated EMI**
- ◆ **Simple Implementation: 2 External Capacitors
(Plus an Input Capacitor)**
- ◆ **Efficiency > 90% Achievable**
- ◆ **Low Cost, Compact, Low Profile (Height)**
- ◆ **Optimized for Doubling or Inverting Supply Voltage:
-- ADM660 or ADM8660**
- ◆ **Voltage Regulated Output Devices Available:
-- ADP3603/ADP3604/ADP3605/ADP3607**

Figure 7-53: Some general charge-pump characteristics

On their downside, charge-pump converters don't maintain high efficiency for a wide voltage range of input to output, unlike inductive switching regulators. Nevertheless, they are still often suitable for lower current loads where any efficiency disadvantages are a small portion of a larger system power budget. A summary of general charge-pump operating characteristics is shown in Figure 7-53.

An example of charge-pump applicability is the voltage inverter function. Inverters are often useful where a relatively low current negative voltage (i.e., -3V) is required, in addition to a primary positive voltage (such as 5V). This may occur in a single supply system, where only a few high performance parts require the negative voltage. Similarly, voltage doublers are useful in low current applications, where a voltage greater than a primary supply voltage is required.

Unregulated inverter and doubler charge-pumps

Illustrating these principles are a pair of basic charge-pump ICs from Analog Devices, shown in Figure 7-54. The ADM660 is a popular charge-pump IC, and is shown here operating as both a voltage inverter (left) and the doubler (right) functions. Switching frequency of this IC is selectable between 25kHz and 120kHz using the FC input pin. With the FC input is open as shown, the switching frequency is 25kHz; with it connected to the V+ pin, frequency increases to 120kHz. Generally, efficiency is greater when operating at the higher frequency. Only two external electrolytic capacitors are required for operation, C1 and C2 (ESR should be $<200\text{m}\Omega$). The value of these capacitors is flexible. For a 25kHz switching frequency 10 μF tantalum types are recommended; for 120kHz operation 2.2 μF provides comparable performance. Larger values can also be used, and will provide lower output ripple (at the expense of greater size and cost).

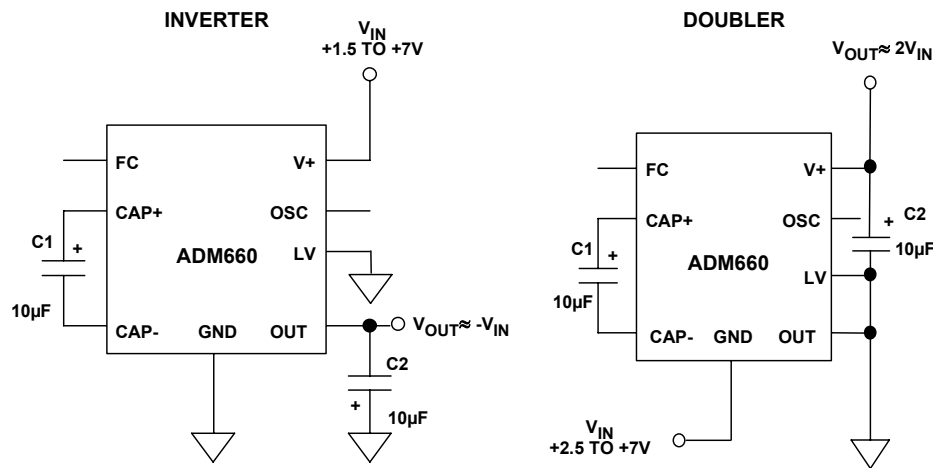


Figure 7-54: ADM660 IC functions as a supply inverter (left) or doubler (right)

These circuits accept V_{IN} inputs over the ranges noted, and deliver a nominal voltage output tracking the input voltage in magnitude, as noted in the output expressions. Although the output voltage is not regulated in these basic designs, it is still relatively low in impedance, due to the nominal 9 Ω resistance of the IC switches.

Efficiency of these circuits using the ADM660/ADM8660 can be 90% or more, for output currents up to 50mA at a 120kHz frequency. The ADM8660 is a device similar to the ADM660, however it is optimized for inverter operation, and includes a shutdown feature which reduces the quiescent current to 5 μA .

Regulated Output Charge-Pump Voltage Converters

Adding regulation to a simple charge-pump voltage converter function greatly enhances its usefulness for most applications. There are several techniques for adding regulation to a charge-pump converter. The most straightforward is to follow the charge-pump inverter/doubler with an LDO regulator. The LDO provides the regulated output, and can also reduce the charge-pump converter's ripple. This approach, however, adds complexity and reduces the available output voltage by the dropout voltage of the LDO (~200mV). These factors may or may not be a disadvantage.

By far the simplest and most effective method for achieving regulation in a charge-pump voltage converter is to simply use a charge-pump design with an internal error amplifier, to control the on-resistance of one of the switches.

This method is used in the ADP3603/3604/3605 voltage inverters, devices offering regulated outputs for positive input voltage ranges. The output is sensed and fed back into the device via a sensing pin, V_{SENSE} . Key features of the series are good output regulation, 3% in the ADP3605, and a high switching frequency of 250kHz, good for both high efficiency and small component size.

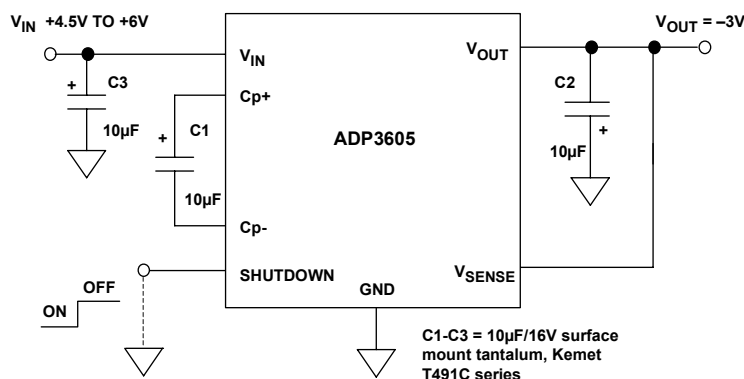


Figure 7-55: ADP3605 5V to -3V, 60mA regulated supply inverter

An example circuit for the ADP3605 IC from this series is shown in Figure 7-55. The application is a 5V to -3V inverter, with the output regulated $\pm 3\%$ for currents up to 60mA. In normal operation, the SHUTDOWN pin is connected to ground (as shown dotted). Alternately, a logic HIGH at this pin shuts the device down to a standby current of 2 μ A.

The 10 μ F capacitors for C1-C3 should have ESRs of less than 150m Ω (4.7 μ F can be used at the expense of slightly higher output ripple voltage). C1 is the most critical of the 3, because of its higher current flow. The tantalum type listed is recommended for lowest output ripple.

With values as shown, typical output ripple voltage ranges up to approximately 60mV as the output current varies over the 60mA range. Although output is regulated for currents up to 60mA, higher currents of up to 100mA are also possible with further voltage deviation, and proportionally greater ripple.

These application examples illustrate a subset of the entire charge-pump IC family. Further information on these devices can be found in the end-of-section references.

Linear post regulator for switching supplies

Another powerful noise reduction option which can be utilized in conjunction with a switching type supply is the option of a *linear post regulator* stage. This is at best an LDO type of regulator, chosen for the desired clean analog voltage level and current. It is preceded by a switching stage, which might be a buck or boost type inductor-based design, or it may also be a charge-pump. The switching converter allows the overall design to be more power-efficient, and the linear post regulator provides clean regulation at the load, reducing the noise of the switcher. This type of regulator can also be termed *hybrid regulation*, since it combines both switching and linear regulation concepts.

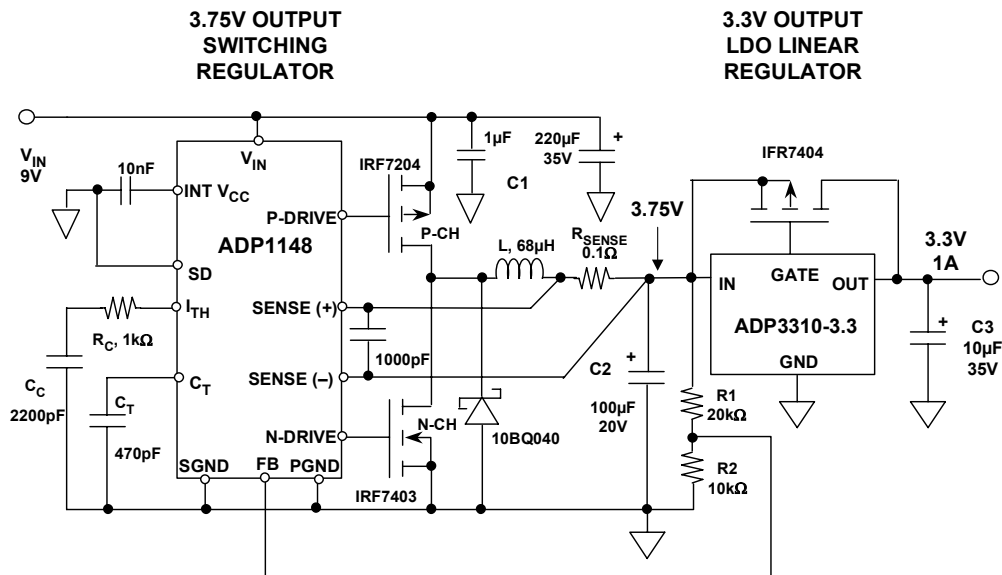


Figure 7-56: A linear post-regulator operating after a switching/linear regulator is capable of low noise, as well as good DC efficiency

An example circuit is shown in Figure 7-56, which features a 3.3V/1A low noise, analog-compatible regulator. It operates from a nominal 9V supply, using a buck or step-down type of switching regulator, as the first stage at the left. The switcher output is set for a few hundred mV above the desired final voltage output, minimizing power in the LDO stage at the right. This feature can eliminate need for a heat sink on the LDO pass device.

In this example the ADP1148 IC switcher is set up for a 3.75V output by R1-R2, but in principle, this voltage can be anything suitable to match the headroom of the companion LDO (within specification limits, of course). In addition, the principle extends to any LDO devices and other current levels, and other switching regulators. The ADP3310-3.3 is a fixed-voltage LDO controller, driving a PMOS FET pass device, with a 3.3V output.

The linear post regulation stage provides both noise-reduction (in this case about 14dB), as well as good dc regulation. To realize best results, good grounding practices must be followed. In tests, noise at the 3.3V output was about 5mVp-p at the 150kHz switcher frequency. Note that the LDO noise rejection for such relatively high frequencies is much less than at 100/120Hz. Note also that C2's ESR will indirectly control the final noise output. The ripple figures given are for a general-purpose C2 part, and can be improved.

■ OP AMP APPLICATIONS

Power Supply Noise Reduction and Filtering

During the last decade or so, switching power supplies have become much more common in electronic systems. As a consequence, they also are being used for analog supplies. Good reasons for the general popularity include their high efficiency, low temperature rise, small size, and light weight.

In spite of these benefits, switchers *do* have drawbacks, most notably high output noise. This noise generally extends over a broad band of frequencies, resulting in both conducted and radiated noise, as well as unwanted electric and magnetic fields. Voltage output noise of switching supplies are short-duration voltage transients, or spikes. Although the fundamental switching frequency can range from 20kHz to 1MHz, the spikes can contain frequency components extending to 100MHz or more. While specifying switching supplies in terms of RMS noise is common vendor practice, as a user you should also specify the *peak* (or p-p) amplitudes of the switching spikes, with the output loading of your system.

This section discusses filter techniques for rendering a switching regulator output *analog ready*, that is sufficiently quiet to power precision op amp and other analog circuitry with relatively small loss of DC terminal voltage. The filter solutions presented are generally applicable to all power supply types incorporating switching element(s) in their energy path. This includes charge-pump as well as other switching type converters and supplies. This section focuses on reducing *conducted type* switching power supply noise with external post filters, as opposed to radiated type noise.

- ◆ **Capacitors**
- ◆ **Inductors**
- ◆ **Ferrites**
- ◆ **Resistors**
- ◆ **Linear Post Regulation**
- ◆ **Proper Layout and Grounding**
- ◆ **Physical Separation!**

Figure 7-57: Tools useful in reducing power supply noise

Tools useful for combating high frequency switcher noise are shown by Figure 7-57. These differ in electrical characteristics as well as practicality towards noise reduction, and are listed roughly in an order of priorities. Of these tools, L and C are the most powerful filter elements, and are the most cost-effective, as well as small in size.

Capacitors

Capacitors are probably the single most important filter component for reducing switching-related noise. As noted in the first section of this chapter, there are many different types of capacitors. It is also quite true that understanding of their individual characteristics is absolutely mandatory to the design of effective and practical power supply filters. There are generally three classes of capacitors useful in 10kHz-100MHz

filters, broadly distinguished as the generic dielectric types; *electrolytic*, *film*, and *ceramic*. These discussions complement earlier ones, focusing on power-related concepts.

With any dielectric, a major potential filter loss element is ESR (equivalent series resistance), the net parasitic resistance of the capacitor. ESR provides an ultimate limit to filter performance, and requires more than casual consideration, because it can vary both with frequency and temperature in some types. Another capacitor loss element is ESL (equivalent series inductance). ESL determines the frequency where the net impedance characteristic switches from capacitive to inductive. This varies from as low as 10kHz in some electrolytics to as high as 100MHz or more in chip ceramic types. Both ESR and ESL are minimized when a leadless package is used. All capacitor types mentioned are available in surface mount packages, preferable for high speed uses.

The *electrolytic* family provides an excellent, cost-effective low-frequency filter component, because of the wide range of values, a high capacitance-to-volume ratio, and a broad range of working voltages. It includes *general purpose aluminum electrolytic* types, available in working voltages from below 10V up to about 500V, and in size from 1 to several thousand μF (with proportional case sizes). All electrolytic capacitors are polarized, and cannot withstand more than a volt or so of reverse bias without damage.

A subset of the general electrolytic family includes *tantalum* types, generally limited to voltages of 100V or less, with capacitance of 500 μF or less (see Reference 7). In a given size, tantalums exhibit a higher capacitance-to-volume ratios than do general purpose electrolytics, and have both a higher frequency range and lower ESR. They are generally more expensive than standard electrolytics, and must be carefully applied with respect to surge and ripple currents.

A subset of aluminum electrolytic capacitors is the *switching* type, designed for handling high pulse currents at frequencies up to several hundred kHz with low losses (see Reference 8). This capacitor type can compete with tantalums in high frequency filtering applications, with the advantage of a broader range of values.

A more specialized high performance aluminum electrolytic capacitor type uses an organic semiconductor electrolyte (see Reference 9). The *OS-CON* capacitors feature appreciably lower ESR and higher frequency range than do other electrolytic types, with an additional feature of minimal low-temperature ESR degradation.

Film capacitors are available in very broad value ranges and an array of dielectrics, including polyester, polycarbonate, polypropylene, and polystyrene. Because of the low dielectric constant of these films, their volumetric efficiency is quite low, and a 10 μF /50V polyester capacitor (for example) is actually a handful. Metalized (as opposed to foil) electrodes does help to reduce size, but even the highest dielectric constant units among film types (polyester, polycarbonate) are still larger than any electrolytic, even using the thinnest films with the lowest voltage ratings (50V). Where film types excel is in their low dielectric losses, a factor which may not necessarily be a practical advantage for filtering switchers. For example, ESR in film capacitors can be as low as 10m Ω or less, and the behavior of films generally is very high in terms of Q. In fact, this can cause problems of spurious resonance in filters, requiring damping components.

OP AMP APPLICATIONS

As typically constructed using wound layers, film capacitors can be inductive, which limits their effectiveness for high frequency filtering. Obviously, only non-inductively made film caps are useful for switching regulator filters. One specific style which is non-inductive is the *stacked-film* type, where the capacitor plates are cut as small overlapping linear sheet sections from a much larger wound drum of dielectric/plate material. This technique offers the low inductance attractiveness of a plate sheet style capacitor with conventional leads (see References 8 and 10). Obviously, minimal lead length should be used for best high frequency effectiveness. Very high current polycarbonate film types are also available, specifically designed for switching power supplies, with a variety of low inductance terminations to minimize ESL (see Reference 11). Dependent upon their electrical and physical size, film capacitors can be useful at frequencies to above 10MHz. At the highest frequencies, only stacked film types should be considered. Leadless surface mount packages are now available for film types, minimizing inductance.

Ceramic is often the capacitor material of choice above a few MHz, due to its compact size, low loss, and availability up to several μF in the high-K dielectric formulations (X7R and Z5U), at voltage ratings up to 200V (see ceramic families of Reference 7).

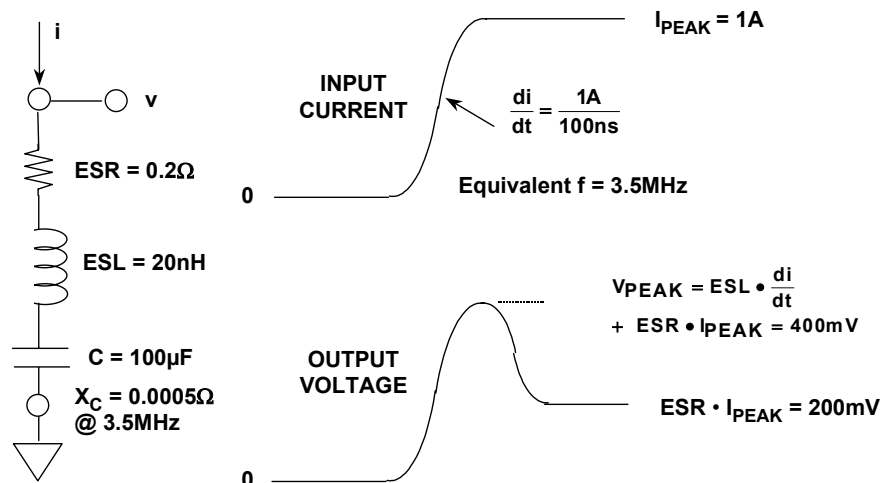


Figure 7-58: Capacitor equivalent circuit and response to input current pulse

Multilayer ceramic "chip caps" are very popular for bypassing and/or filtering at 10MHz or more, simply because their very low inductance design allows near optimum RF bypassing. For smaller values, ceramic chip caps have an operating frequency range to 1GHz. For high frequency applications, a useful selection can be ensured by selecting a value which has a self-resonant frequency *above* the highest frequency of interest.

The capacitor model and waveforms of Figure 7-58 illustrate how the various parasitic model elements become dominant, dependent upon the operating frequency. Assume an input current pulse changing from 0 to 1A in 100ns, as noted in the figure, and consider what voltage will be developed across the capacitor.

The fast-rising edge of the current waveform shown results in an initial voltage peak across the capacitor, which is proportional to the ESL. After the initial transient, the voltage settles down to a longer duration level which is proportional to the ESR of the

capacitor. Thus the ESL determines how effective a filter the capacitor is for the fastest components of the current signal, and the ESR is important for longer time frame components. Note that an overall time frame of a few microseconds (or even less) is relevant here. As things turn out, this means switching frequencies in the 100kHz to 1MHz range. Unfortunately however, this happens to be the region where most electrolytic types begin to perform poorly.

All electrolytics will display impedance curves similar in general shape to that of Figure 7-59. In a practical capacitor, at frequencies below about 10kHz the net impedance seen at the terminals is almost purely capacitive (C region). At intermediate frequencies, the net impedance is determined by ESR, for example about 0.1 to 0.5 Ω at ~125kHz, for several types (ESR region). Above about several hundred kHz to 1MHz these capacitor types become inductive, with net impedance rising (ESL region).

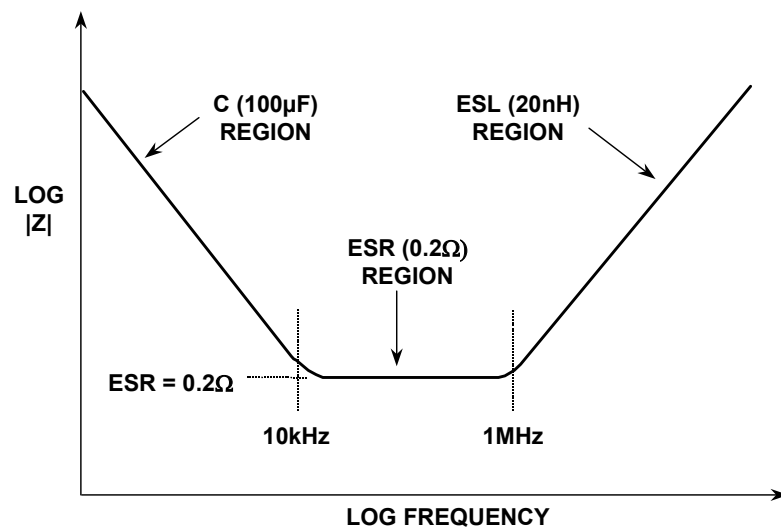


Figure 7-59: Electrolytic capacitor impedance versus frequency

The minimum impedance within the 10kHz – 1MHz range will vary with the magnitude of the capacitor's ESR. This is the primary reason why ESR is the most critical item in determining a given capacitor's effectiveness as a switching supply filter element. Higher up in frequency, the inductive region will vary with ESL (which in turn is also strongly effected by package style). It should go without saying that a wideband impedance plot for a capacitor being considered for a filter application will go a long way towards predicting its potential value, as well as for comparing one type against another.

It should be understood that all real world capacitors have some finite ESR. While it is usually desirable for filter capacitors to possess low ESR, this isn't always so. In some cases, the ESR may actually be helpful in reducing resonance peaks in filters, by supplying "free" damping. For example, in most electrolytic types, a nominally flat broad series resonance region can be noted in an impedance vs. frequency plot. This occurs where $|Z|$ falls to a minimum level, nominally equal to the capacitor's ESR at that frequency. This low Q resonance can generally be noted to cover a relatively wide frequency range of several octaves. Contrasted to the high Q sharp resonances of film and ceramic caps, electrolytic's low Q behavior can be useful in controlling resonant peaks.

■ OP AMP APPLICATIONS

Ferrites

A second important filter element is the inductor, available in various forms. The use of *ferrite* core materials is prevalent in inductors most practical for power supply filtering.

Regarding inductors, ferrites, which are non-conductive ceramics manufactured from the oxides of nickel, zinc, manganese, etc., are extremely useful in power supply filters (see Reference 12). Ferrites can act as either inductors or resistors, dependent upon their construction and the frequency range. At low frequencies (<100kHz), inductive ferrites are useful in low-pass LC filters. At higher frequencies, ferrites become resistive, which can be an important characteristic in high-frequency filters. Again, exact behavior is a function of the specifics. Ferrite impedance depends on material, operating frequency range, DC bias current, number of turns, size, shape, and temperature. Figure 7-60 summarizes a number of ferrite characteristics.

- ◆ **Ferrites Good for Frequencies Above 25kHz**
- ◆ **Many Sizes / Shapes Available Including Leaded "Resistor Style"**
- ◆ **Ferrite Impedance at High Frequencies Primarily Resistive -- Ideal for HF Filtering**
- ◆ **Low DC Loss: Resistance of Wire Passing Through Ferrite is Very Low**
- ◆ **High Saturation Current Versions Available**
- ◆ **Choice Depends Upon:**
 - **Source and Frequency of Interference**
 - **Impedance Required at Interference Frequency**
 - **Environmental: Temperature, AC and DC Field Strength, Size and Space Available**
- ◆ **Always Test the Design!**

Figure 7-60: A summary of ferrite characteristics

Several ferrite manufacturers offer a wide selection of ferrite materials from which to choose, as well as a variety of packaging styles for the finished network (see References 13 and 14). A simple form is the *bead* of ferrite material, a cylinder of the ferrite which is simply slipped over the power supply lead to the decoupled stage. Alternately, the *leaded ferrite bead* is the same bead, pre-mounted on a length of wire and used as a component (see Reference 14). More complex beads offer multiple holes through the cylinder for increased decoupling, plus other variations. Surface mount beads are also available. PSpice models of Fair-Rite ferrites are available, allowing ferrite impedance estimations (see Reference 15). The models match measured rather than theoretical impedances.

A ferrite's impedance is dependent upon a number of inter-dependent variables, and is difficult to quantify analytically, thus selecting the proper ferrite is not straightforward. However, knowing the following system characteristics will make selection easier. First, determine the frequency range of the noise to be filtered. Second, the expected temperature range of the filter should be known, as ferrite impedance varies with temperature. Third, the DC current flowing through the ferrite must be known, to ensure that the ferrite does not saturate. Although models and other analytical tools may prove useful, the general guidelines given above, coupled with actual filter experimentation connected under system load conditions, should lead to a proper ferrite selection.

Card Entry Filter

Using proper component selection, low and high frequency band filters can be designed to smooth a noisy switching supply output so as to produce an *analog ready* supply. It is most practical to do this over two (and sometimes more) stages, each stage optimized for a range of frequencies.

A basic stage can be used to carry the entire load current, and filter noise by 60dB or more up to a 1-10MHz range. Figure 7-61 illustrates this type of filter, which is used as a *card entry filter*, providing broadband filtering for all power entering a PC card.

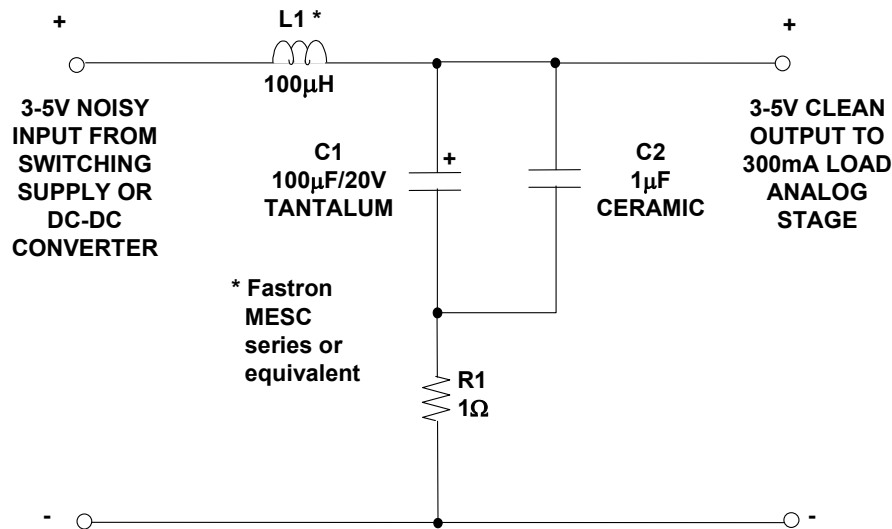


Figure 7-61: A card-entry filter is useful for low-medium frequency power line noise filtering in analog systems

In this filter, L1 and C1 perform the primary filtering, which provides a corner frequency of about 1.6kHz. With the corner thus placed well below typical switching frequencies, the circuit can have good attenuation up to 1MHz, where the typical attenuation is on the order of 60dB. At higher frequencies parasitics limit performance, and a second filter stage will be more useful.

The ultimate level of performance available from this filter will be related to the components used within it. L1 should be derated for the operating current, thus for 300mA loads it is a 1A type. The specified L1 choke has a typical DCR of 0.65Ω, for low drop across the filter (see Reference 16). C1 can be either a tantalum or an aluminum electrolytic, with moderately low ESR. For current levels lower than 300mA, L1 can be proportionally downsized, saving space. The resistor R1 provides damping for the LC filter, to prevent possible ringing. R1 can be reduced or even possibly eliminated, if the ESR of C1 provides a comparable impedance.

While the example shown is a single-supply configuration, obviously the same filter concepts apply for dual supplies.

■ OP AMP APPLICATIONS

Rail bypass/distribution filter

A complement to the card-entry filter is the rail-bypass filter scheme of Figure 7-62. When operating from relatively clean power supplies, the heavy noise filtering of the card entry filter may not be necessary. However, some sort of low frequency bypassing with appreciable energy storage is almost always good, and this is especially true if high currents are being delivered by the stages under power.

In such cases, some lumped low frequency bypassing is appropriate on the card. Although these energy storage filters need not be immediately adjacent to the ICs they serve, they should be within a few inches. This type of bypassing scheme should be considered a minimum for powering any analog circuit. The exact capacitor values aren't critical, and can vary appreciably. The most important thing is to avoid leaving them out!

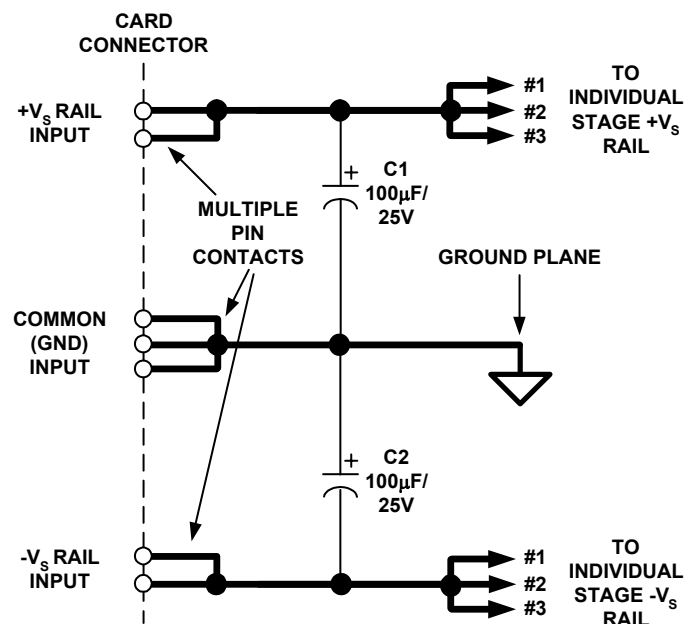


Figure 7-62: Dual-supply low frequency rail bypass/distribution filter

The circuit shown uses C1 and C2 as these bypasses in a dual-rail system. Note that multiple card contacts are recommended for the I/O pins, especially ground connection. From the capacitors outward, supply rail traces are distributed to each stage as shown, in "star" distribution fashion. Note— while this is the optimum method to minimize inter-stage crosstalk, in practice some degree of "daisy chaining" is often difficult to avoid. A prudent designer should therefore carefully consider common supply currents effects in designing these PCB distribution paths.

Wider than normal traces are recommended for these supply rails, especially those carrying appreciable current. If the current levels are in the ampere region, then star-type supply distribution with ultra-wide traces should be considered mandatory. In extreme cases, a dedicated power plane can be used. The impedance of the ground return path is minimized by the use of a ground plane.

Local high frequency bypass/decoupling

At each individual analog stage, further local, high-frequency-only filtering is used. With this technique, used in conjunction with either the card-entry filter or the low frequency bypassing network, such smaller and simpler local filter stages provide optimum high frequency decoupling. *These stages are provided directly at the power pins, of all individual analog stages.*

Figure 7-63 shows this technique, in both correct (left) as well as incorrect example implementations (right). In the left example, a typical $0.1\mu\text{F}$ chip ceramic capacitor goes directly to the opposite PCB side ground plane, by virtue of the via, and on to the IC's GND pin by a second via. In contrast, the less desirable setup at the right adds additional PCB trace inductance in the ground path of the decoupling cap, reducing effectiveness.

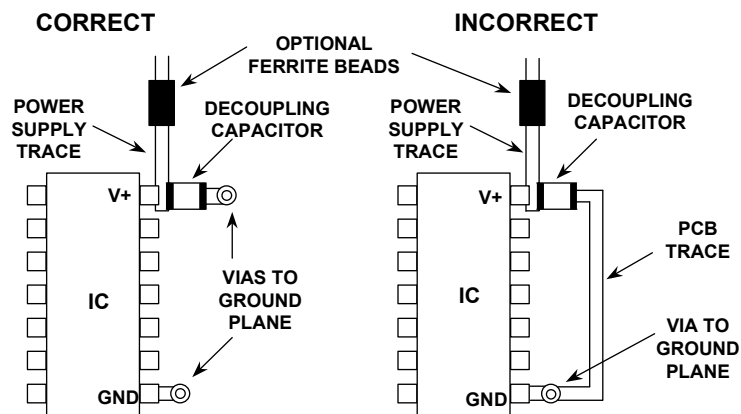


Figure 7-63: Localized high frequency supply filter(s) provides optimum filtering and decoupling via short low-inductance path (ground plane)

The general technique is shown here as suitable for a single-rail power supply, but the concept obviously extends to dual rail systems. Note— if the decoupled IC in question is an op amp, the GND pin shown is the $-V_S$ pin. For dual supply op amp uses, there is no op amp GND pin per se, so the dual decoupling networks should go directly to the ground plane when used, or other local ground.

All high frequency (i.e., $\geq 10\text{MHz}$) ICs should use a bypassing scheme similar to Figure 7-63 for best performance. Trying to operate op amps and other high performance ICs without local bypassing is almost always folly. It *may* be possible in a few circumstances, *if* the circuitry is strictly micropower in nature, and the gain-bandwidth in the kHz range. To put things into an overall perspective however, note that a pair of $0.1\mu\text{F}$ ceramic bypass caps cost less than 25 cents. Hardly a worthy saving compared to the potential grief and lost time of troubleshooting a system without bypassing!

In contrast, the ferrite beads aren't 100% necessary, but they will add extra HF noise isolation and decoupling, which is often desirable. Possible caveats here would be to verify that the beads never saturate, when the op amps are handling high currents.

▣ OP AMP APPLICATIONS

Note that with some ferrites, even before full saturation occurs, some beads can be non-linear, so if a power stage is required to operate with a low distortion output, this should also be lab checked.

Figure 7-64 summarizes the previous points of this section regarding power supply conditioning techniques for op amp circuitry.

- ◆ **Use Proper Layout and Grounding Techniques!**
- ◆ **At HF Local Decoupling at IC Power Pins is Mandatory**
- ◆ **At HF Ground Planes are Mandatory**
- ◆ **External LC Filters Very Effective in Reducing Ripple**
- ◆ **Low ESR/ESL Capacitors Give Best Results**
- ◆ **Parallel Caps Lower ESR/ESL and Increase C**
- ◆ **Linear Post Regulation Effective for Noise Reduction and Best Regulation**
- ◆ **Completely Analytical Approach Difficult**
 - Prototyping Required for Optimum Results*
- ◆ **Once Design is Final, Don't Switch Vendors or Substitute Parts**
 - Without First Verifying Performance Within the Circuit!*

Figure 7-64: *A summary of power supply conditioning techniques for high performance op amp circuitry*

REFERENCES: OP AMP POWER SUPPLY SYSTEMS

1. Walt Jung, "References and Low Dropout Linear Regulators," Section 2 within Walt Kester, Ed., **Practical Design Techniques for Power and Thermal Management**, Analog Devices, Inc., 1998, ISBN 0-916550-19-2.
2. Paul Brokaw, "A Simple Three-Terminal IC Bandgap Voltage Reference," **IEEE Journal of Solid State Circuits**, Vol. SC-9, December, 1974.
3. Frank Goodenough, "Vertical-PNP-Based Monolithic LDO Regulator Sports Advanced Features," **Electronic Design**, May 13, 1996.
4. Frank Goodenough, "Low Dropout Regulators Get Application Specific," **Electronic Design**, May 13, 1996.
5. Walt Kester, Brian Erisman, Gurgit Thandi, "Switched Capacitor Voltage Converters," Section 4 within Walt Kester, Editor, **Practical Design Techniques for Power and Thermal Management**, Analog Devices, Inc., 1998, ISBN 0-916550-19-2.
6. Walt Jung, Walt Kester, Bill Chestnut, "Power Supply Noise Reduction and Filtering," portion of Section 8 within Walt Kester, Editor, **Practical Design Techniques for Power and Thermal Management**, Analog Devices, Inc., 1998, ISBN 0-916550-19-2.
7. **Tantalum Electrolytic and Ceramic Capacitor Families**, Kemet Electronics, Box 5928, Greenville, SC, 29606, (803) 963-6300.
8. Type HFQ Aluminum Electrolytic Capacitor and Type V Stacked Polyester Film Capacitor, Panasonic, 2 Panasonic Way, Secaucus, NJ, 07094, (201) 348-7000.
9. **OS-CON Aluminum Electrolytic Capacitor Technical Book**, Sanyo, 3333 Sanyo Road, Forrest City, AK, 72335, (501) 633-6634.
10. Ian Clelland, "Metallized Polyester Film Capacitor Fills High Frequency Switcher Needs," **PCIM**, June 1992.
11. Type 5MC Metallized Polycarbonate Capacitor, Electronic Concepts, Inc., Box 1278, Eatontown, NJ, 07724, (908) 542-7880.
12. Henry W. Ott, **Noise Reduction Techniques in Electronic Systems, 2nd Edition**, John Wiley, Inc., 1988, ISBN: 0-471-85068-3.
13. **Fair-Rite Linear Ferrites Catalog**, Fair-Rite Products, Box J, Wallkill, NY, 12886, (914) 895-2055.
14. Type EXCEL leaded ferrite bead EMI filter, and Type EXC L leadless ferrite bead, Panasonic, 2 Panasonic Way, Secaucus, NJ, 07094, (201) 348-7000.
15. Steve Hageman, "Use Ferrite Bead Models to Analyze EMI Suppression," **The Design Center Source**, MicroSim Newsletter, January, 1995.
16. "MESC series RFI suppression chokes," FASTRON GmbH, Zum Kaiserblick 25, 83620 Feldkirchen-Westerham, Germany, www.fastron.de

NOTES:

SECTION 7-4: OP AMP PROTECTION

***Walt Jung, Walt Kester, James Bryant, Joe Buxton,
Wes Freeman***

Frequently, op amps and other analog ICs require protection against destructive potentials at their input and output terminals. One basic reason behind this is that these ICs are by nature relatively fragile components. Although designed to be as robust as possible *for normal signals*, there are nevertheless certain application and/or handling conditions where they can see voltage transients beyond their ratings. This situation can occur for either of two instances. The first of these is *in-circuit*, that is, operating within an application circuit. The second instance is *out-of-circuit*, which might be at anytime after receipt from a supplier, but prior to final assembly and mounting of the IC. In either case, under over-voltage conditions, it is a basic fact-of-life that unless the designer limits the fault currents at the input (or possibly output) of the IC, it can be damaged or destroyed.

So, obviously the designer should fully understand all of the fault mechanisms internal to those ICs that may require protection. This then allows design of networks that can protect the in-circuit IC throughout its lifetime, without undue compromise of speed, precision, etc. Or, for the out-of-circuit IC, it can help define proper protective handling procedures until it reaches its final destination. This section of the chapter examines a variety of protection schemes to ensure adequate protection for op amps and other analog ICs for in-circuit applications, as well as for out-of-circuit environments.

In-Circuit Over-Voltage Protection

There are many common cases that stress op amps and other analog ICs at the input, while operating within an application, i.e., in-circuit. Since these ICs must often interface to the outside world, this may entail handling voltages exceeding their absolute maximum ratings. For example, sensors are often placed in environments where a fault condition can expose the circuit to a dangerously high voltage. With the sensor connected to a signal processing amplifier, the input then sees excessive voltages during a fault.

General Input Common Mode Limitations

Whenever an op amp input common mode (CM) voltage goes outside its supply range, the op amp can be damaged, even if the supplies are turned off. Accordingly, the absolute maximum input ratings of almost all op amps limits the greatest applied voltage to a level equal to the positive and negative supply voltage, plus about 0.3V beyond these voltages (i.e., $+V_S + 0.3V$, or $-V_S - 0.3V$). While some exceptions to this general rule might exist it is important to note this: *Most IC op amps require input protection when over-voltage of more than 0.3V beyond the rails occurs.*

A safe operating rule is to always keep the applied op amp CM voltage between the rail limits. Here, "safe" implies prevention of outright IC destruction. As will be seen later, there are also intermediate "danger-zone" CM conditions between the rails with certain op amps, which can invoke dangerous (but not necessarily destructive) behavior.

OP AMP APPLICATIONS

Speaking generally, it is important to note that almost *any* op amp input will break down, given sufficient over-voltage to the positive or negative rail. Under breakdown conditions high and uncontrolled current can flow, so the danger is obvious. The exact breakdown voltage is entirely dependent on the individual op amp input stage. It may be a 0.6V diode drop, or a process-related breakdown of 50V or more. In many cases, over-voltage stress can result in currents over 100mA, which destroys a part almost instantly.

Therefore, unless otherwise stated on the data sheet, op amp input fault current should be limited to $\leq 5\text{mA}$ to avoid damage. This is a conservative rule of thumb, based on metal trace widths in a typical op amp input. Higher levels of current can cause *metal migration*, a cumulative effect, which, if sustained, eventually leads to an open trace. Should a migration situation be present, failure may only appear after a long time due to multiple over-voltages, a very difficult failure to identify. So, even though an amplifier may appear to withstand over-voltage currents well above 5mA for a short time period, it is important to limit the current to 5mA (or preferably less) for long term reliability.

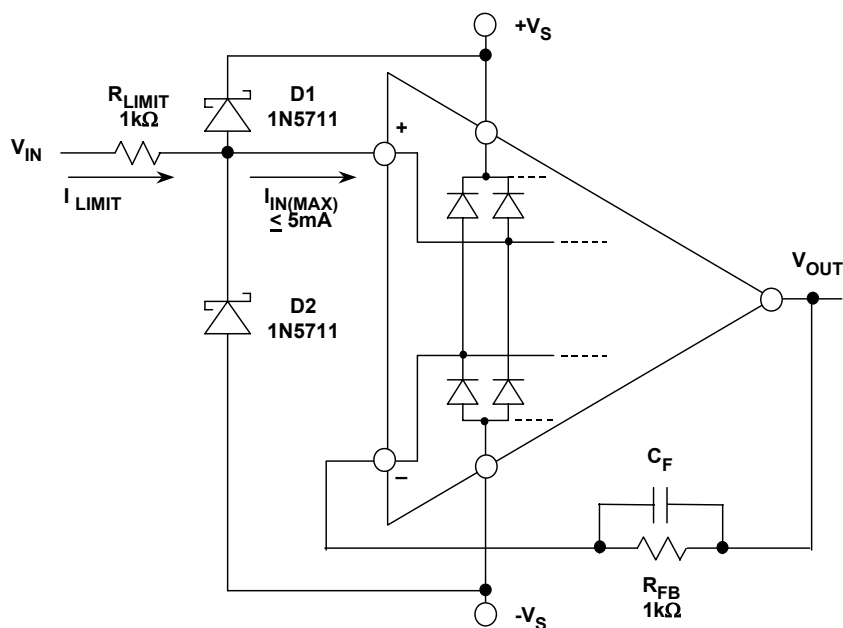


Figure 7-65: A general-purpose op amp CM over-voltage protection network using Schottky clamp diodes with current limit resistance

Figure 7-65 illustrates an external, general-purpose op amp CM protection circuit. The basis of this scheme is the use of Schottky diodes D1 and D2, plus an external current limiting resistor, R_{LIMIT} . With appropriate selection of these parts, input protection for a great many op amps can be ensured. Note that an op amp may also have *internal* protection diodes to the supplies (as shown) which conduct at about 0.6V forward drop above or below the respective rails. In this case however, the external Schottky diodes effectively parallel any internal diodes, so the internal units never reach their threshold. Diverting fault currents externally eliminates potential stress, protecting the op amp.

The external diodes also allow other degrees of freedom, some not so obvious. For example, if fault current is allowed to flow in the op amp, R_{LIMIT} must then be chosen so that the maximum current is no more than 5mA for the worst case V_{IN} . This criterion can

result in rather large R_{LIMIT} values, and the associated increase in noise and offset voltage may not be acceptable. For instance, to protect against a V_{IN} of 100V with the 5mA criterion, R_{LIMIT} must be $\geq 20k\Omega$. However with external Schottky clamping diodes, this allows R_{LIMIT} to be governed by the maximum allowable D1-D2 current, which can be larger than 5mA. However, care must be used here, for at very high currents the Schottky diode drop may exceed 0.6V, possibly activating internal op amp diodes.

It is very useful to keep the R_{LIMIT} value as low as possible, to minimize offset and noise errors. R_{LIMIT} , in series with the op amp input, produces a bias-current-proportional voltage drop. Left uncorrected, this voltage appears as an increase in the circuit's offset voltage. Thus for op amps where the bias currents are moderate and approximately equal (most bipolar types) compensation resistor R_{FB} balances the dc effect, and minimizes this error. For low bias current op amps ($I_b \leq 10nA$, or FET types) it is likely R_{FB} won't be necessary. To minimize noise associated with R_{FB} , bypass it with a capacitor, C_F .

Clamping Diode Leakage

For obvious reasons, it is critical that diodes used for protective clamping at an op amp input have a leakage sufficiently low to not interfere with the bias level of the application.

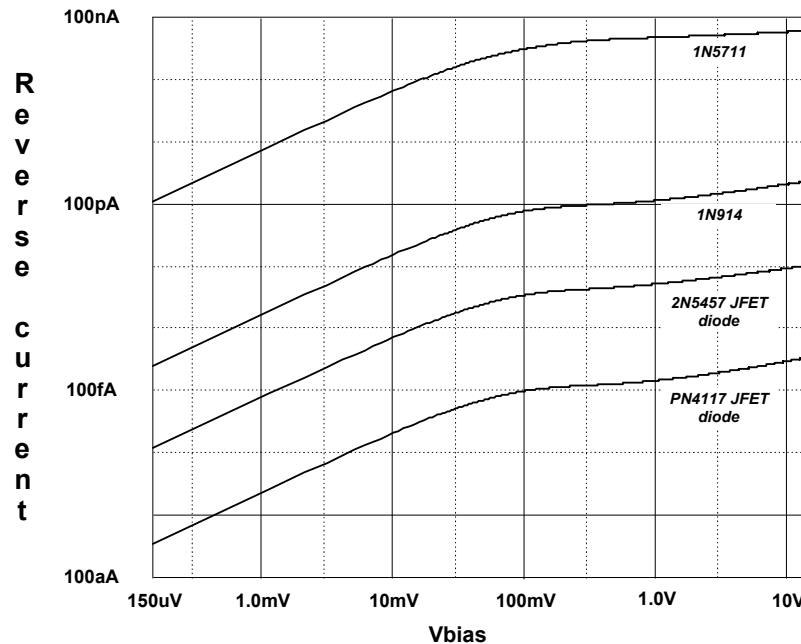


Figure 7-66: Reverse bias current characteristics for diodes useful in protective clamping networks (PSpice simulation)

Figure 7-66 illustrates how some well-known diodes differ in terms of leakage current, as a function of the reverse bias voltage, V_{bias} .

In this chart, a 25°C simulation using PSpice diode models, it is easy to see that not only is the diode type critical, so is the reverse bias. The 1N5711 Schottky type for example, has a leakage of nearly 100nA at a reverse bias of 15V, as it would typically be used with a $\pm 15V$ powered op amp. With this level of leakage, such diodes will only be useful with

OP AMP APPLICATIONS

op amps with bias currents of several μA . For protection of appreciably lower bias current op amps (particularly most FET input devices) much lower leakage is necessary.

As the data of Figure 7-66 shows, not only does selecting a better diode help control leakage current, but operating it at a low bias voltage condition reduces leakage substantially. For example, while an ordinary 1N914 or 1N4148 diode may have 200pA of leakage at 15V, this is reduced to slightly more than a pA with bias controlled to 1mV. But there is a caveat here! When used in a high impedance clamp circuit, glass diodes such as the 1N914/1N4148 families should either be shielded from incident light, or use opaque packages. This is necessary to minimize parasitic photocurrent from the surrounding light, which effectively appears as diode leakage current.

Specialty diodes with much lower leakage are also available, such as diode-connected FET devices characterized as protection diodes (see DPAD series of Reference 2). Within the data of Figure 7-66, the 2N5457 is a general purpose JFET, and the 2N4117/PN4117 family consists of parts designed for low current levels. Other low leakage and specialty diodes are described in References 3 and 4.

A Flexible Voltage Follower Protection Circuit

Of course, it isn't a simple matter to effectively apply protective clamping to op amp inputs, while reducing diode bias level to a sub-mV level.

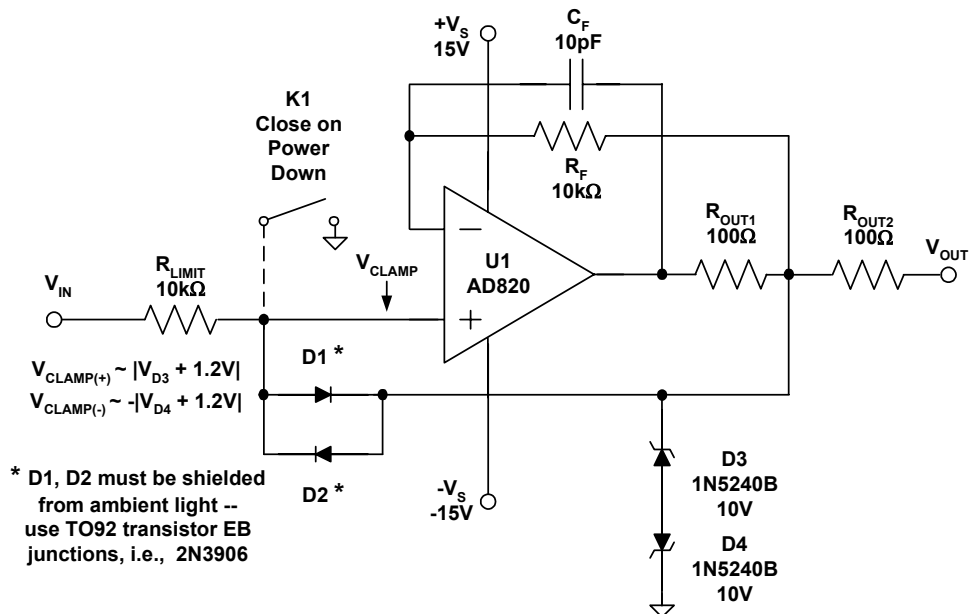


Figure 7-67: Bootstrapping the D1-D2 protection network reduces diode leakage to negligible levels, and is voltage-programmable for clamp level.

The circuit of Figure 7-67 shows low-leakage input clamping and other means used with a follower connected FET op amp, with protection at input and output, for both power on or off conditions.

Disregarding the various diodes momentarily, this circuit is an output-current-limited voltage follower. With the addition of diodes D1-D2 and D3-D4, it has both a voltage-

7.78

limited output, and an over-voltage protected input. Operating below the voltage threshold of output series-connected zener diodes D3-D4, the circuit behaves as a precision voltage follower. Under normal follower operation, that is at input/output voltages $< |V_Z + 0.6|$ volts (where V_Z is the breakdown voltage of D3 or D4), diodes D1-D2 see only the combined offset and CM voltage errors of U1 as bias voltage. This reduces the D1-D2 leakage to very low levels, consistent with the pA level bias current of a FET input op amp. Note that D1-D2 *must* be prevented from photo-conduction, and one direct means of this is to use opaque package diodes, such as the 2N3906 EB junctions discussed by Pease (see References 3 and 4). If 1N914s are used they must be light shielded. In either case, bootstrapping greatly reduces the effective D1-D2 leakage.

For input/output voltage levels greater than $V_Z + 0.6V$, zener diodes D3-D4 breakdown. This action clamps both the V_{OUT} output node and the V_{CLAMP} node via D1-D2. The input of the op amp is clamped to either polarity of the two input levels of V_{CLAMP} , as indicated within the figure. Under clamp conditions, input voltage V_{IN} can rise to levels beyond the supply rails of U1 without harm, with excess current limited by R_{LIMIT} . If sustained high-level ($\sim 100V$) inputs will be applied, R_{LIMIT} should be rated as a 1-2W (or fusible) type.

This circuit has very good DC characteristics, due to the fact that the clamping network is bootstrapped. This produces very low input/output errors below the V_{CLAMP} threshold (consistent with the op amp specifications, of course). Note that this bootstrapping has ac benefits as well, as it reduces the D1-D2 capacitance seen by the source. While the $\sim 100pF$ capacitance of D3-D4 might cause a loading problem with some op amps, this is mitigated by the isolating effect of R_{OUT1} , plus the feedback compensation of C_F . Both R_{OUT1} and R_{OUT2} protect the op amp output.

The input voltage clamping level is also programmable, and is set by the choice of zener voltage V_Z . This voltage plus 1.2V should be greater than the maximum input, but below the rail voltage, as summarized in the figure. The example uses $10V \pm 5\%$ zener diodes, so input clamping typically will occur at $\pm 11.2V$, allowing $\pm 10V$ swings.

An important caveat to the above is that it applies for *power on* conditions. With *power off*, D1-D4 still clamp to the noted levels, but this now produces a condition whereby the U1 input and output voltage can exceed the rails.

Note that this could be dangerous, for a given U1 device. If so, an optional and simple means towards providing a lower, safe clamping level for power off conditions is to use a relay at the V_{CLAMP} node. The contacts are open with power applied, and closed with power absent. With attention paid to an overall PCB layout, this can preserve a pA level bias current of FET op amps used for U1.

CM Over-Voltage Protection Using CMOS Channel Protectors

A much simpler alternative for over-voltage protection is the CMOS *channel protector*. A channel protector is a device in series with the signal path, for example preceding an op amp input. It provides over-voltage protection by dynamically altering its resistance under fault conditions. Functionally, it has the distinct advantage of affording protection for sensitive components from voltage transients, whether the power supplies are present or not. Representative devices are the ADG465/ADG466/ADG467, which are channel protectors with single, triple, and octal channel options. Because this form of protection works whether supplies are present or not, the devices are ideal for use in applications where input over-voltages are common, or where correct power sequencing can't always be guaranteed. One such example is within hot-insertion rack systems.

An application of a channel protector for over-voltage protection of a precision buffer circuit is shown in Figure 7-68. A single channel device, the ADG465 at U2, is used here at the input of the U1 precision op amp buffer, an OP777.

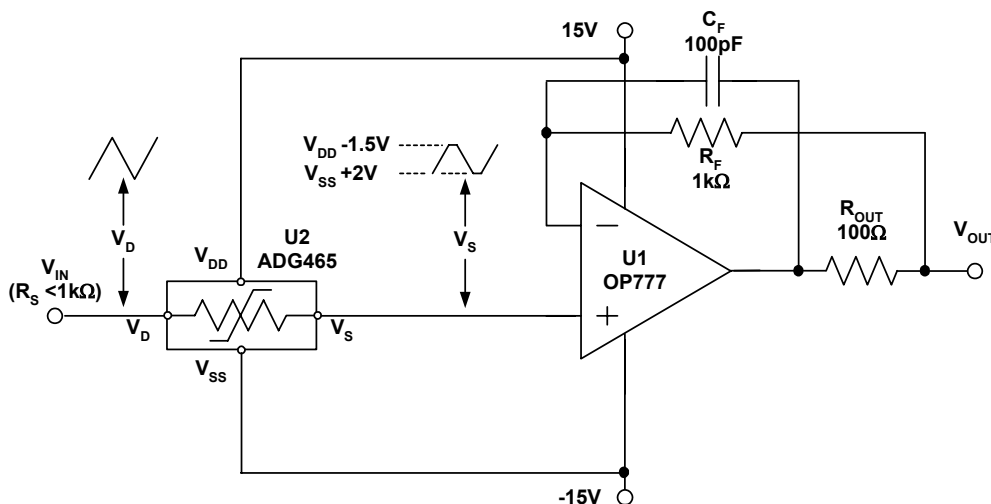


Figure 7-68: Using an ADG465 channel protector IC with a precision buffer offers great simplicity of protection and fail-safe operation during power off.

In operation, a channel protector behaves just like a series resistor of 60 Ω to 80 Ω in normal operation (i.e., non-fault conditions). Consisting of a series connection of multiple P and N MOSFETs, the protector dynamically adjusts channel resistance according to the voltage seen at the V_D terminal. Normal conduction occurs with V_D more than a threshold level above or below the rails, i.e., $(V_{SS} + 2V) < V_D < (V_{DD} - 1.5V)$. For fault conditions the analog input voltage exceeds this range, causing one of the series MOSFETs to switch off, thus raising the channel resistance to a high level. This clamps the V_S output at one extreme range, either $V_{SS} + 2V$ or $V_{DD} - 1.5V$, as shown in Figure 7-68.

A major channel protector advantage is the fact that both circuit and signal source protection are provided, in the event of over-voltage or power loss. Although shown here operating from op amp $\pm 15V$ supplies, these channel protectors can handle total supplies of up to 40V. They also can withstand over-voltage inputs from $V_{SS} - 20V$ to $V_{DD} + 20V$ with power on (or $\pm 35V$ in the circuit shown). With power off ($V_{DD} = V_{SS} = 0V$),

maximum input voltage is $\pm 35\text{V}$. Maximum room temperature channel leakage is 1nA , making them suitable for op amps and in-amps with bias currents of several nA and up.

Related to the ADG46X series of channel protectors are several *fault-protected multiplexers*, for example the ADG508F/509F, and the ADG438F/439F families. Both the channel protectors and the fault-protected multiplexers are low power devices, and even under fault conditions, their supply current is limited to sub microampere levels. A further advantage of the fault-protected multiplexer devices is that they retain proper channel isolation, even for input conditions of one channel seeing an over-voltage, that is the remaining channels still function.

CM Over-Voltage Protection Using High CM Voltage In-Amp

The ultimate simplicity for analog channel over-voltage protection is achieved with resistive input attenuation ahead of a precision op amp. This combination equates to a high voltage capable in-amp, such as the AD629, which is able to linearly process differential signals riding upon CM voltages of up to $\pm 270\text{V}$. Further, and most important to over-voltage protection considerations, the on-chip resistors afford protection for either common mode or differential voltages of up to $\pm 500\text{V}$. All of this is achieved by virtue of a precision laser-trimmed thin-film resistor array and op amp, as shown in Figure 7-69.

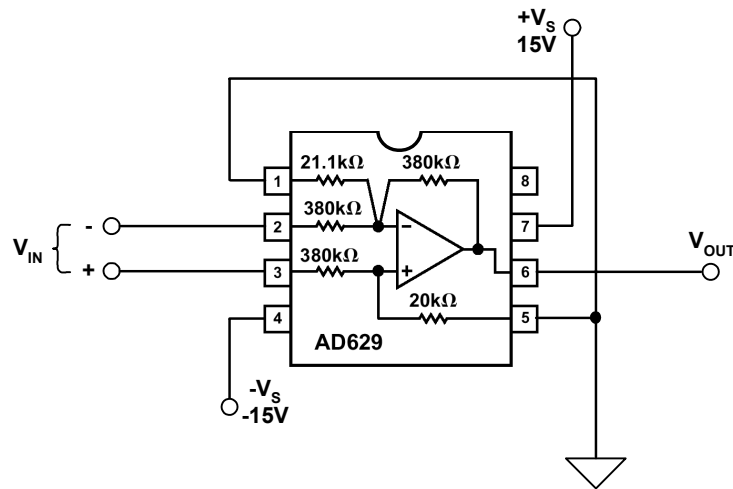


Figure 7-69: The AD629 high voltage in-amp IC offers $\pm 500\text{V}$ input over-voltage protection, one-component simplicity, and fail-safe power off operation

Examination of this topology shows that the resistive network around the AD629's precision op amp acts to divide down the applied CM voltage at V_{IN} by a factor of 20/1. The AD629 simultaneously processes the input differential mode signal V_{IN} to a single-ended output referred to a local ground, at a gain factor of unity. Gain errors are no more than ± 0.03 or 0.05% , while offset voltage is no more than 0.5 or 1mV (grade dependent). The AD629 operates over a supply range of ± 2.5 to $\pm 18\text{V}$.

These factors combine to make the AD629 a simple, one-component choice for the protection of off-card analog inputs that can potentially see dangerous transient voltages. Due to the relatively high resistor values used, protection of the device is also inherent with no power applied, since the input resistors safely limit fault currents. In addition, it

■ OP AMP APPLICATIONS

offers those operating advantages inherent to an in-amp: high CMR (86dB minimum at 500Hz), excellent overall dc precision, and the flexibility of simple polarity changes.

On the flip side of performance issues, several factors make the AD629's output noise and drift relatively high, if compared to a lower gain in-amp configuration such as the AMP03. These are the Johnson noise of the high value resistors, and the high noise gain of the topology (21x). These factors raise the op amp noise and drift along with the resistor noise by a factor higher than typical. Of course, whether or not this is an issue relevant to an individual application will require evaluation on a case-by-case basis.

Inverting mode op amp protection schemes

There are some special cases of over-voltage protection requirements that don't fit into the more general CM protection schemes above. Figure 7-70 is one such example, a low bias current FET input op amp I/V converter.

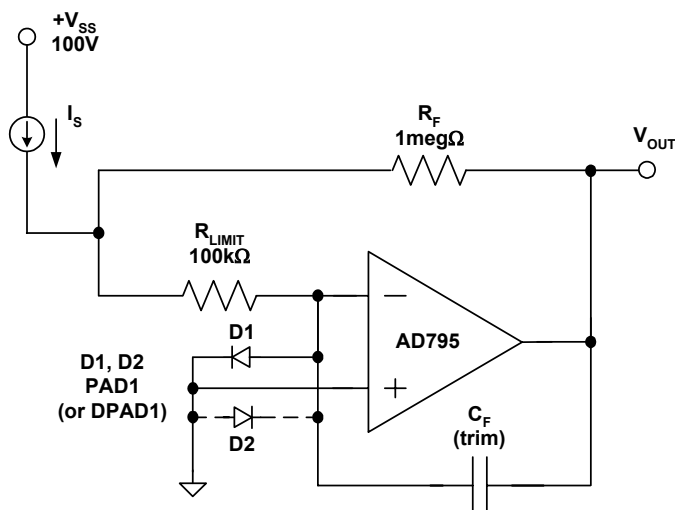


Figure 7-70: A low bias current FET input op amp I/V converter with over-voltage protection network R_{LIMIT} and $D1$

In this circuit the AD795 1pA bias current op amp is used as a precision inverter. Some current-source nature signals can originate from a high voltage potential, such as the 100V V_{SS} level shown. As such, they have the potential of developing fault voltage levels beyond the op amp rails, producing fault current into the op amp well above safe levels. To prevent this, protection resistor R_{LIMIT} is used inside the feedback loop as shown, along with voltage clamp $D1$ ($D2$).

For normal signal condition (i.e., $I_S \leq 10\mu A$) the op amp's inverting node is very close to ground, with just a tiny voltage drop across R_{LIMIT} . Normal I/V conversion takes place, with gain set by R_F . For protection, $D1$ is a special low leakage diode, clamping any excess voltage at the (-) node to $\sim 0.6V$, thus protecting the op amp. The value of R_{LIMIT} is chosen to allow a 1mA max current under fault conditions. Bootstrapping the $D1$ (and/or $D2$) clamp diodes as shown minimizes the normal operating voltage across the inverting node, keeping the diode leakage low (see Figure 7-66, again). Note that for a positive source voltage as shown, only positive clamping is needed, so just one diode suffices.

Only the lowest leakage diodes ($\leq 1\text{pA}$) such as the PAD1 (or the DPAD1 dual) should be used in this circuit. As noted previously, any clamping diode used here should either be shielded from light (or use opaque packaging), to minimize photocurrent from ambient light. Even so, the diode(s) will increase the net input current and shunt capacitance, and feedback compensation C_F will likely be necessary to control response peaking. C_F should be a very low leakage type. Also, with the use of very low input bias current devices such as the AD795, it isn't possible to use the same level of internal protection circuitry as with other ADI op amps. This factor makes the AD795 more sensitive to handling, so ESD precautions should be taken.

Amplifier Output Voltage Phase-Reversal

As alluded to above, there are "gray-area" op amp groups that have anomalous CM voltage zones, falling between the supply rails. As such, protection for these devices cannot be guaranteed by simply ensuring that the inputs stay between the rails—they must additionally stay *entirely* within their rated CM range, for consistent behavior.

Peculiar to some op amps, this misbehavior phenomenon is called *output voltage phase-reversal*. It is seen when one or both of the op amp inputs exceeds their allowable input CM voltage range. Note that the inputs may still be well within the extremes of rail voltage, but simply below one specified CM limit. Typically, this is towards the negative range. Phase-reversal is most often associated with JFET and/or BiFET amplifiers, but some bipolar single-supply amplifiers are also susceptible to it.

The Figure 7-71 waveforms illustrates this general phenomenon, with an overdriven voltage follower input on the left, and the resulting output phase-reversal at the right.

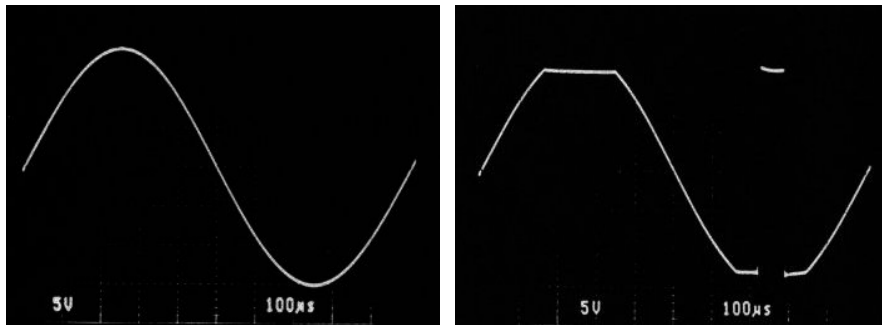


Figure 7-71: An illustration of input overdriving waveform (left) and the resulting output phase-reversal (right), using a JFET input op amp

While the specific details of the internal mechanism may vary with individual op amps, it suffices to say that the output phase-reversal occurs when a critical section of the amplifier front end saturates, causing the input-output sign relationship to temporarily reverse. Under this condition, when the CM range is exceeded, the negative going input waveform in Figure 7-71 (left) does not continue going more negative in the output waveform, Figure 7-71 (right). Instead, the input-output relationship *phase-reverses*, with the output suddenly going positive, i.e., the spike. It is important to note that this is *not* a latching form of phase-reversal, as the output will once again continue to properly track

■ OP AMP APPLICATIONS

the input, when the input returns to the CM range. In Figure 7-71, this can be seen in the continuance of the output sine wave, after the positive-going phase-reversal spike settles.

In most applications, this output voltage phase-reversal does no harm to the op amp, nor to the circuit where it is used. Indeed, since it is triggered when the CM limit is exceeded, non-inverting stages with appreciable signal gain never see it, since their applied CM voltage is too small.

Note that with inverting applications the output phase-reversal problem is non-existent, as the CM range isn't exercised. So, although a number of (mostly older) op amps suffer from phase-reversal, it still is rarely a serious problem in system design.

Nevertheless, when and if a phase-reversal susceptible amplifier used in a servo loop application sees excess CM voltage, the effect can be disastrous— it goes **Bang!** So, the best advice is to be forewarned.

An Output Phase-Reversal Do-it-Yourself Test

Since output phase-reversal may not always be fully described on a data sheet, it is quite useful to test for it. This is easily done in the lab, by driving a questionable op amp as a unity-gain follower, from a source impedance (R_{LIMIT}) of $\sim 1k\Omega$. It is helpful to make this a variable, 1-100k Ω range resistance.

With a low resistance setting (1k Ω), while bringing the driving signal level slowly up towards the rail limits, observe the amplifier output. If a phase-reversal mechanism is present, when the CM limit of the op amp is exceeded, the output will suddenly reverse (see Figure 7-71, right, again). If there is no phase-reversal present in an amplifier, the output waveform will simply clip at the limits of its swing. It may prove helpful to have a well-behaved op amp available for this test, to serve as a performance reference. One such device is the AD8610.

Note that in general, some care should be used with this test. Without a series current-limit resistor, if the generator impedance is too low (or level too high), it could possibly damage an internal junction of the op amp under test. So, obviously, caution is best for such cases.

Once a suitable R_{LIMIT} resistance value is found, well-behaved op amps will simply show a smooth, bipolar range, clipped output waveform when overdriven. This clipping will appear more like the *upper (positive swing)* portion of the waveform within Fig. 7-71, right (again).

Fixes For Output Phase-Reversal

An op amp manufacturer might not always give the R_{LIMIT} resistance value appropriate to prevent output phase-reversal. But, the value can be determined empirically with the driving method mentioned above. Most often, the R_{LIMIT} resistor value providing protection against phase-reversal will also safely limit fault current through any input CM clamping diodes. If in doubt, a nominal value of $1k\Omega$ is a good starting point for testing.

Typically, FET input op amps will need only the current limiting series resistor for protection, but bipolar input devices are best protected with this same limiting resistor, *along with a Schottky diode* (i.e., R_{LIMIT} and D2, of Figure 7-65, again).

For a more detailed description of the output voltage phase-reversal effect, see References 7 and 8. Figure 7-72 summarizes a number of the key points relating to output voltage phase-reversal.

Alternately, any of the several previously mentioned CM clamping schemes can be used to prevent output phase-reversal, by setting the clamp voltage to be less than the amplifier CM range limit where phase-reversal occurs. For example, Figure 7-67 would operate to prevent phase-reversal in FET amplifiers susceptible to it, if the negative clamp limit is set so that $V_{CLAMP(-)}$ never exceeds the typical negative CM range of $-11V$ on a $-15V$ rail.

- ◆ **Non-Latching Inversion of Transfer Function, Triggered by Exceeding Common Mode Limit**
- ◆ **Sometimes Occurs in FET and Bipolar (Single-Supply) Op Amps**
- ◆ **Doesn't Harm Amplifier... but *Disastrous* for Servo Systems!**
- ◆ **Not Usually Specified on Data Sheet, so Amplifier Must be Checked**
- ◆ **Easily Prevented:**
 - **All op amps: Limit applied CM voltage by clamping or other means**
 - **BiFETs: Add series input resistance, R_{LIMIT}**
 - **Bipolars: R_{LIMIT} and Schottky clamp diode to rail**

Figure 7-72: A summary of key points regarding output phase-reversal in FET and bipolar input op amps

For validation of this or any of the previous over-voltage protection schemes, the circuit should be verified on a number of op amps, over a range of conditions as suitable to the final application environment.

■ OP AMP APPLICATIONS

Input Differential Protection

The discussions thus far have been on over-voltage common-mode conditions, which is typically associated with forward biasing of PN junctions inherent in the structure of the input stage. There is another equally important aspect of protection against over-voltage, which is that due to excess *differential* voltages. Excessive differential voltage, when applied to certain op amps, can lead to degradation of their operating characteristics.

This degradation is brought about by *reverse junction breakdown*, a second case of undesirable input stage conduction, occurring under conditions of *differential* over-voltage. However, in the case of reverse breakdown of a PN junction, the problem can be more subtle in nature. It is illustrated by the partial op amp input stage in Figure 7-73.

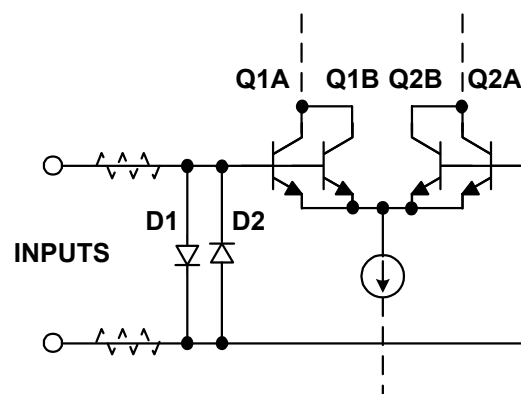


Figure 7-73: An op amp input stage with D1-D2 input differential over-voltage protection network

This circuit, applicable to a low noise op amp such as the OP27, is also typical of many others using low noise bipolar transistors for differential pair Q1-Q2. In the absence of any protection, it can be shown that voltages above about 7V between the two inputs will cause a reverse junction breakdown of either Q2 or Q1 (dependent upon relative polarity). Note that, in cases of E-B breakdown, even small reverse currents can cause degradation in both transistor gain and noise (see Reference 6). After E-B breakdown occurs, op amp parameters such as the bias currents and noise may well be out of specification. This is usually permanent, and it can occur gradually and quite subtly, particularly if triggered by transients. For these reasons, virtually all low-noise op amps, whether NPN or PNP based, utilize protection diodes such as D1-D2 across the inputs. These diodes conduct for applied voltages greater than $\pm 0.6\text{V}$, protecting the transistors.

The dotted series resistors function as current limiters (protection for the protection diodes) but aren't used in all cases. For example, the AD797 doesn't have the resistors, simply because they would degrade the part's specified noise of $1\text{nV}/\sqrt{\text{Hz}}$. Note—when the resistors are absent internally, some means of external current limiting must be provided, when and if differential over-voltage conditions do occur. Obviously, this is a tradeoff situation, so the confidence of full protection must be weighed against the noise degradation. Note that an application circuit itself may provide sufficient resistance in the op amp inputs, such that additional resistance isn't needed.

In applying a low noise bipolar input stage op amp, first check the chosen part's data sheet for internal protection. When necessary, protection diodes D1-D2, if not internal to the op amp, should be added to guarantee prevention of Q1-Q2 E-B breakdown. If differential transients of more than 5V can be seen by the op amp in the application, the diodes are in order. Ordinary low capacitance diodes will suffice, such as the 1N4148 family. Add current limiting resistors as necessary, to limit diode current to safe levels.

Other IC device junctions, such as base-collector and JFET gate-source junctions don't exhibit the same degradation in performance upon break down, and for these the input current should be limited to 5mA, unless the data sheet specifies a different value.

Protecting In-Amps Against Over-Voltage

From a protection standpoint, instrumentation amplifiers (in-amps) are similar in many ways to op amps. Like op amps, their absolute maximum ratings must be observed for both common and differential mode input voltages.

A much simplified schematic of the AD620 in-amp is shown in Figure 7-74, showing the input differential transistors and their associated protection parts.

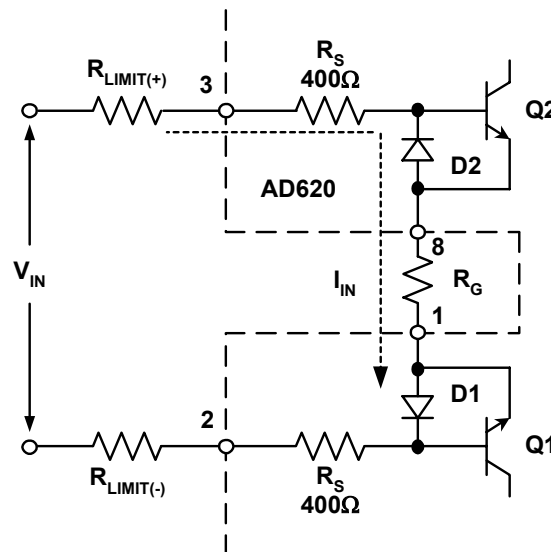


Figure 7-74: The AD620 in-amp input internally uses D1-D2 and series resistors R_S for protection (additional protection can be added externally)

An important point, unique to the AD620 device, is the fact that the 400Ω internal R_S protection resistors are *thin-film* types. Therefore these resistors don't show symptoms of diode-like conduction to the IC substrate (as would be the case were they diffused resistors). Practically, this means that the input ends of these resistors (pins 3 and 2) can go above or below the supplies. Differential fault currents will be limited by the combination of twice the internal R_S plus the external gain resistance, R_G . Excess applied CM voltages will show current limited by R_S .

In more detail, it can be noted that input transistors Q1 and Q2 have protection diodes D1 and D2 across their base-emitter junctions, to prevent reverse breakdown. For differential voltages, analysis of shows that a fault current, I_{IN} , flows through the external R_{LIMIT}

OP AMP APPLICATIONS

resistors (if present), the internal R_S resistors, the gain-setting resistor R_G , and two diode drops (Q2, D1). For the AD620 topology, R_G varies inversely with gain, and a worst case (lowest resistance) occurs with the maximum gain of 1000, when R_G is 49.9Ω . Therefore the lowest total internal path series resistance is about 850Ω .

For the AD620, any combination of CM and differential input voltages should be limited to levels that limit the input fault current to 20mA, maximum. A purely differential voltage of 17V would result in this current level, for the lowest resistance case. For CM voltages which may go beyond either rail, an internal diode not shown in Fig. 7-74 conducts, effectively clamping the driven input to either $+V_S$ or $-V_S$ at the R_S inner end. For this over-voltage CM condition, the 400Ω value of R_S and the excess voltage beyond the rail determines the current level. If for example V_{IN} is 23V with $+V_S$ at 15V, 8V appears across R_S , and the 20mA current rating is reached. Higher fault voltages can be dealt with by adding R_{LIMIT} resistance, to maintain fault current at 20mA or less.

A more generalized external voltage protection circuit for an in-amp like the AD620 is shown in Figure 7-75.

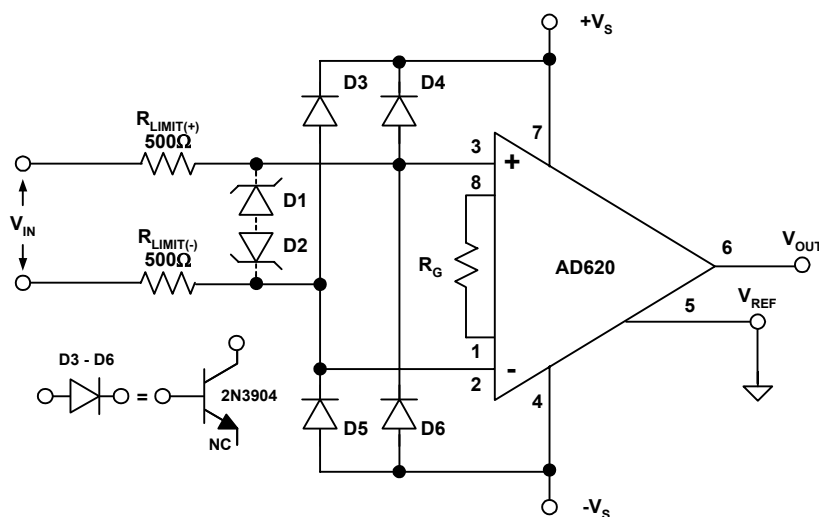


Figure 7-75: A generalized diode protection circuit for the AD620 and other in-amps uses D3-D6 for CM clamping and series resistors R_{LIMIT} for protection

In this circuit, low-leakage diodes D3-D6 are used as CM clamps. Since the in-amp bias current may be only 1nA or so (for the AD620), a low-leakage diode type is mandatory. As can be noted from the topology, diode bootstrapping isn't possible with this configuration.

It should be noted that not only must the diodes have basically low leakage, they must also maintain low leakage at the highest expected temperature. This suggests either FET type diodes (see Fig. 7-66, again), or the transistor C-B types shown. The R_{LIMIT} resistors are chosen to limit the maximum diode current under fault conditions. If additional differential protection is used, either back-back zener or Transzorb clamps can be used, shown as D1-D2. If this is done, leakage of these diodes should be carefully considered.

The protection scheme of Figure 7-75, while effective using appropriate parts, has the downside of being busy for components. A much more simple in-amp protection using fault protected devices is shown in Figure 7-76. Although shown with an AD620, this circuit is useful with many other dual-supply in-amps with bias currents of 1nA or more. It uses two-thirds of a triple ADG466 channel-protector for the in-amp differential inputs.

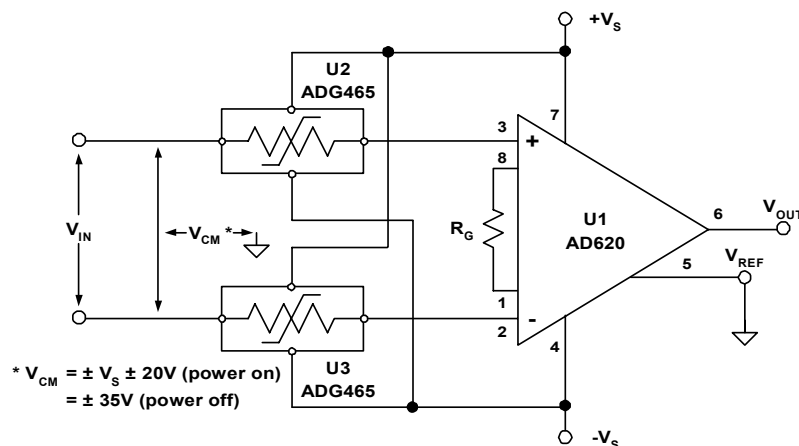


Figure 7-76: A channel protector device (or fault-protected multiplexer) provides protection for dual-supply in-amps with a minimum of extra parts

Because the nature of a channel protection device is to turn off as V_{IN} approaches either rail, the scheme of Figure 7-76 doesn't function with rail sensing single-supply in-amps. If near-rail operation and protection is required in an in-amp application, an alternative method is necessary. Many single-supply in-amps are topologically similar to the two-amplifier in-amp circuit which is shown within the dotted box of Figure 7-77.

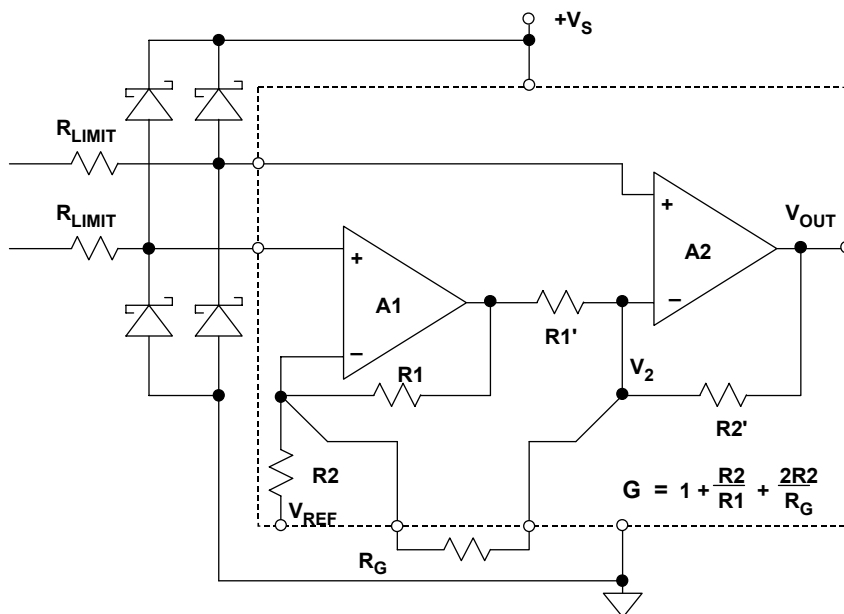


Figure 7-77: Single-supply in-amps may or may not require external protection in the form of resistors and clamp diodes— if so, they can be added as shown

In terms of the necessity for externally added protection components, a given in-amp may or may not require them. Each case needs to be considered individually. For example,

■ OP AMP APPLICATIONS

some in-amps have clamp diodes as shown, but *internal to the device*. The AD623 is such a part, but it lacks the series resistors, which can be added externally when and if necessary. Note that this approach allows the R_{LIMIT} value to be optimized for protection, with negligible impact on noise for those applications not needing the protection.

Also, some in-amp devices have both internal protection resistors *and* clamping diodes, an example here is the AD627. In this device, the internal protection is adequate for transients up to 40V beyond the supplies (a 20mA fault current in the internal resistors). For over-voltage levels higher than this, external R_{LIMIT} resistors can be added.

The use of the Schottky diodes as shown at the two inputs is an option for in-amp protection. If no clamping is specifically provided internally, then they are applicable. Their use is generally similar to the op amp protection case of Figure 7-65, with comparable caveats as far as leakage. Note that in many cases, due to internal protection networks of modern in-amps, these diodes just won't be necessary. But again, there aren't hard rules on this, so always check the data sheet before finalizing an application.

To summarize, Figure 7-78 reviews the major points of the in-circuit over-voltage issues discussed in this section.

◆ INPUT VOLTAGES MUST NOT EXCEED ABSOLUTE MAXIMUM RATINGS

(Usually Specified With Respect to Supply Voltages)

◆ Requires $V_{IN(CM)}$ Stay Within a Range Extending to $\pm 0.3V$ Beyond Rails

($-V_S - 0.3V \geq V_{IN} \leq +V_S + 0.3V$)

◆ IC Input Stage Fault Currents *Must* Be Limited

($\leq 5mA$ Unless Otherwise Specified)

◆ Avoid Reverse-Bias Breakdown in Input Stage Junctions!

◆ Differential and Common Mode Ratings Often Differ

◆ No Two Amplifiers are Exactly the Same

◆ Watch Out for Output Phase-Reversal in JFET and SS Bipolar Op Amps!

◆ Some ICs Contain *Internal* Input Protection

- Diode Voltage Clamps, Current Limiting Resistors (or both)
- Absolute Maximum Ratings Must Still Be Observed

Figure 7-78: A summary of in-circuit over-voltage points

If these varied over-voltage precautions for op amps and in-amps seem complex, yes indeed, they are! Whenever op amp (or in-amp) inputs (and outputs) go outside equipment boundaries, dangerous or destructive things can happen to them. Obviously, these potentially hazardous situations should be anticipated, for highest reliability.

Fortunately, most applications are contained entirely within the equipment, and usually see inputs and outputs to/from other ICs on the same power system. Therefore clamping and protection schemes typically aren't necessary for these cases.

Out-of-Circuit Over-Voltage Protection

Linear ICs such as op amps and in-amps must also be protected prior to the time that they are mounted to a printed circuit board, that is an *out-of-circuit* state. In such a condition, ICs are completely at the mercy of their environment as to what stressful voltage surges they may see. Most often the harmful voltage surges come from *electrostatic discharge*, or, as more commonly referenced, ESD. This is a single, fast, high current transfer of electrostatic charge resulting from one of two conditions. These conditions are:

- 1) *Direct contact transfer between two objects at different potentials (sometimes called contact discharge)*
- 2) *A high electrostatic field between two objects when they are in close proximity (sometimes called air discharge)*

The prime sources of static electricity are mostly insulators and are typically synthetic materials, e.g., vinyl or plastic work surfaces, insulated shoes, finished wood chairs, Scotch tape, bubble pack, soldering irons with ungrounded tips, etc. Voltage levels generated by these sources can be extremely high since their charge is not readily distributed over their surfaces or conducted to other objects.

- ◆ **Walking Across a Carpet**
1000V - 1500V
- ◆ **Walking Across a Vinyl Floor**
150V - 250V
- ◆ **Handling Material Protected by Clear Plastic Covers**
400V - 600V
- ◆ **Handling Polyethylene Bags**
1000V - 2000V
- ◆ **Pouring Polyurethane Foam Into a Box**
1200V - 1500V
- ◆ **Note: Above Assumes 60% RH. For Low RH (30%),
Voltages Can Be > 10 Times**

Figure 7-79: ESD voltages generated by various ordinary circumstances

The generation of static electricity caused by rubbing two substances together is called the *triboelectric* effect. Some common examples of ordinary acts producing significant ESD voltages are shown in Figure 7-79.

ICs can be damaged by the high voltages and high peak currents generated by ESD. Precision analog circuits, often featuring very low bias currents, are more susceptible to damage than common digital circuits, because traditional input-protection structures which protect against ESD damage increase input leakage—and thus can't be used.

▣ OP AMP APPLICATIONS

For the design engineer or technician, the most common manifestation of ESD damage is a catastrophic failure of the IC. However, exposure to ESD can also cause increased leakage or degrade other parameters. If a device appears to not meet a data sheet specification during evaluation, the possibility of ESD damage should be considered. Figure 7-80 outlines some relevant points on ESD induced failures.

◆ ESD Failure Mechanisms:

- Dielectric or junction damage
- Surface charge accumulation
- Conductor fusing

◆ ESD Damage Can Cause:

- Increased leakage
- Degradation in performance
- Functional failures of ICs

◆ ESD Damage is often Cumulative:

- For example, each ESD "zap" may increase junction damage until, finally, the device fails.

Figure 7-80: Understanding ESD damage

All ESD-sensitive devices are shipped in protective packaging. ICs are usually contained in either conductive foam or antistatic shipping tubes, and the container is then sealed in a static-dissipative plastic bag. The sealed bag is marked with a distinctive sticker, such as in Figure 7-81, which outlines the appropriate handling procedures.

All static sensitive devices are sealed in protective packaging and marked with special handling instructions

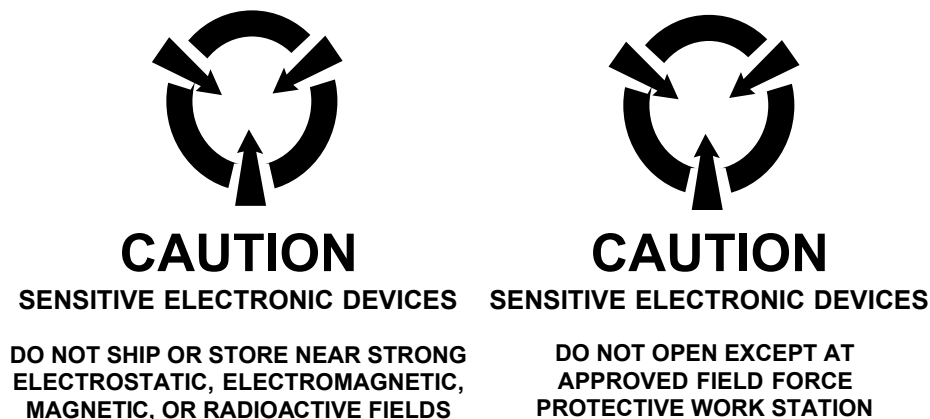


Figure 7-81: Recognizing ESD-sensitive devices by package and labeling

The presence of outside package notices such as those shown in Figure 7-81 is notice to the user that device handling procedures appropriate for ESD protection are necessary.

HARDWARE AND HOUSEKEEPING TECHNIQUES OP AMP PROTECTION

In addition, data sheets for ESD-sensitive ICs generally have a bold statement to that effect, as shown in Figure 7-82.

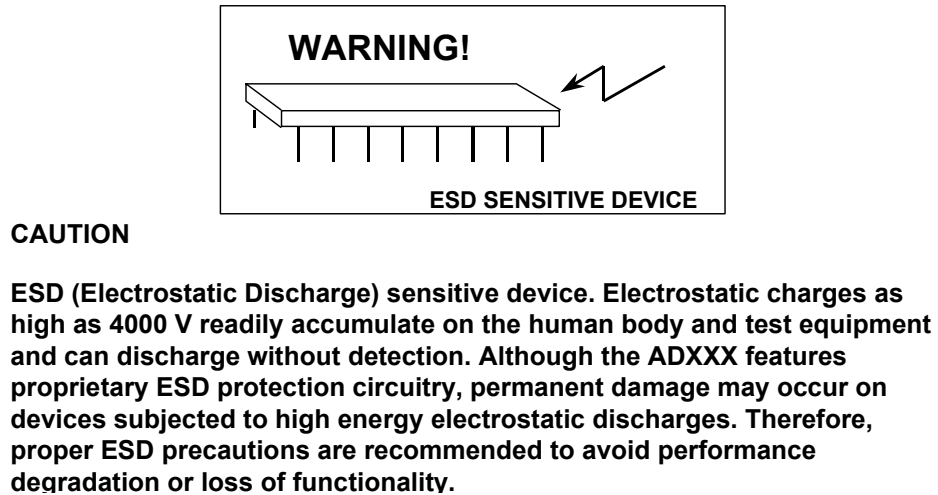


Figure 7-82: ESD data sheet statement for linear ICs

Once ESD-sensitive devices are identified, protection is relatively easy. Obviously, keeping ICs in their original protective packages as long as possible is a first step. A second step is discharging potentially damaging ESD sources before IC damage occurs. Discharging such voltages can be done quickly and safely, through a high impedance.

A key component required for ESD-safe IC handling is a workbench with a static-dissipative surface, shown in the workstation of Figure 7-83. The surface is connected to ground through a $1\text{M}\Omega$ resistor, which dissipates any static charge, while protecting the user from electrical ground fault shock hazards. If existing bench tops are nonconductive, a static-dissipative mat should be added, along with the discharge resistor.

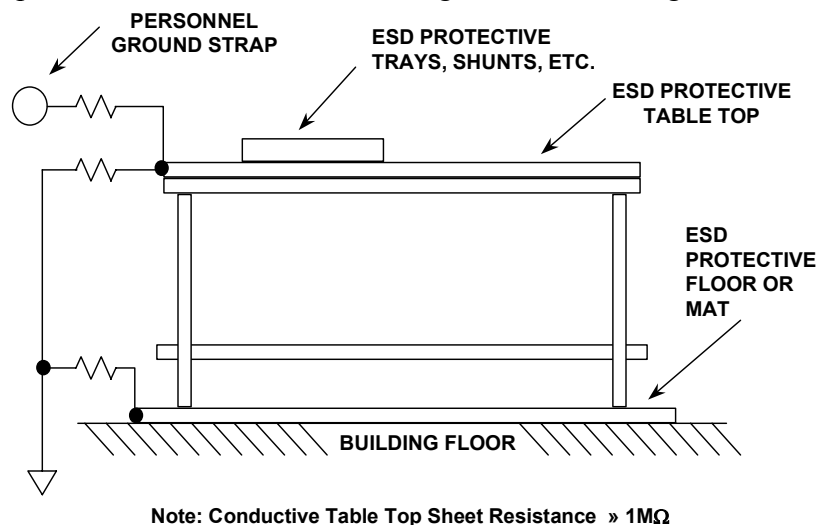


Figure 7-83: A workstation environment suitable for handling ESD-sensitive ICs

Note that the surface of the workbench has a moderately high sheet resistance. It is neither necessary nor desirable to use a low-resistance surface (such as a sheet of copper-clad PC board) for the work surface. Remember, a high peak current may flow if a charged IC is discharged through a low impedance. This is precisely what happens when

■ OP AMP APPLICATIONS

a charged IC contacts a grounded copper clad board. When the same charged IC is placed on the high impedance surface of Figure 7-83 however, the peak current isn't high enough to damage the device.

Several personnel handling techniques are keys to minimizing ESD-related damage. At the workstation, a conductive wrist strap is recommended while handling ESD-sensitive devices. The wrist strap ensures that normal tasks, such as peeling tape from packages, won't cause IC damage. Again, a $1\text{M}\Omega$ resistor, from the wrist strap to ground, is required for safety. When building prototype breadboards or assembling PC boards which contain ESD-sensitive ICs, all passive components should be inserted and soldered before the ICs. This minimizes the ESD exposure of the sensitive devices. The soldering iron must, of course, have a grounded tip.

Protecting ICs from ESD requires the participation of both the IC manufacturer and the customer. IC manufacturers have a vested interest in providing the highest possible level of ESD protection for their products. IC circuit designers, process engineers, packaging specialists and others are constantly looking for new and improved circuit designs, processes, and packaging methods to withstand or shunt ESD energy.

ANALOG DEVICES:

- **Circuit Design and Fabrication -**
 - ↓ **Design and manufacture products with the highest level of ESD protection consistent with required analog and digital performance.**
- **Pack and Ship -**
 - ↓ **Pack in static dissipative material. Mark packages with ESD warning.**

CUSTOMERS:

- **Incoming Inspection -**
 - ↓ **Inspect at grounded workstation. Minimize handling.**
- **Inventory Control -**
 - ↓ **Store in original ESD-safe packaging. Minimize handling.**
- **Manufacturing -**
 - ↓ **Deliver to work area in original ESD-safe packaging. Open packages only at grounded workstation. Package subassemblies in static dissipative packaging.**
- **Pack and Ship -**
 - ↓ **Pack in static dissipative material if required. Replacement or optional boards may require special attention.**

Figure 7-84: *ESD protection requires a partner relationship between ADI and the end customer with control at key points*

A complete ESD protection plan, however, requires more than building ESD protection into ICs. The users of ICs must also provide their employees with the necessary knowledge of and training in ESD handling procedures, so that protection can be built in at all key points along the way, as outlined in Figure 7-84.

Special care should be taken when breadboarding and evaluating ICs. The effects of ESD damage can be cumulative, so repeated mishandling of a device can eventually cause a failure. Inserting and removing ICs from a test socket, storing devices during evaluation, and adding or removing external components on the breadboard should all be done while

observing proper ESD precautions. Again, if a device fails during a prototype system development, repeated ESD stress may be the cause.

The key word to remember with respect to ESD is *prevention*. There is no way to undo ESD damage, or to compensate for its effects.

ESD Models and Testing

Some applications have higher sensitivity to ESD than others. ICs which are located on a PC board surrounded by other circuits are generally much less susceptible to ESD damage than circuits which must interface with other PC boards or the outside world. These ICs are generally not specified or guaranteed to meet any particular ESD specification (with the exception of MIL-STD-883 Method 3015 classified devices). A good example of an ESD-sensitive interface is the RS-232 interface port ICs on a computer, which can easily be exposed to excess voltages. In order to guarantee ESD performance for such devices, the test methods and limits must be specified.

A host of test waveforms and specifications have been developed to evaluate the susceptibility of devices to ESD. The three most prominent of these waveforms currently in use for semiconductor or discrete devices are: The Human Body Model (HBM), the Machine Model (MM), and the Charged Device Model (CDM). Each of these models represents a fundamentally different ESD event, consequently, correlation between the test results for these models is minimal.

- ◆ **IEC1000-4 Electromagnetic Compatibility EMC**
- ◆ **IEC1000-4-1 Overview of Immunity Tests**
- ◆ **IEC1000-4-2 Electrostatic Discharge Immunity (ESD)**
- ◆ **IEC1000-4-3 Radiated Radio-Frequency Electromagnetic Field Immunity**
- ◆ **IEC1000-4-4 Electrical Fast Transients (EFT)**
- ◆ **IEC1000-4-5 Lightning Surges**
- ◆ **IEC1000-4-6 Conducted Radio Frequency Disturbances above 9kHz**
- ◆ **Compliance Marking: **

Figure 7-85: *A listing of the IEC standards applicable to ESD specifications and testing procedures*

Since 1996, all electronic equipment sold to or within the European Community must meet Electromechanical Compatibility (EMC) levels as defined in specification IEC1000-4-x. Note that this does not apply to individual ICs, *but to the end equipment*. These standards are defined along with test methods in the various IEC1000 specifications, and are listed in Figure 7-85.

■ OP AMP APPLICATIONS

IEC1000-4-2 specifies compliance testing using two coupling methods, *contact discharge* and *air-gap discharge*.

Contact discharge calls for a direct connection to the unit being tested. Air-gap discharge uses a higher test voltage, but does not make direct contact with the unit under test. With air discharge, the discharge gun is moved toward the unit under test, developing an arc across the air gap, hence the term air discharge. This method is influenced by humidity, temperature, barometric pressure, distance and rate of closure of the discharge gun. The contact-discharge method, while less realistic, is more repeatable and is gaining acceptance in preference to the air-gap method.

Although very little energy is contained within an ESD pulse, the extremely fast risetime coupled with high voltages can cause failures in unprotected ICs. Catastrophic destruction can occur immediately as a result of arcing or heating. Even if catastrophic failure does not occur immediately, the device may suffer from parametric degradation, which may result in degraded performance. The cumulative effects of continuous exposure can eventually lead to complete failure.

I-O lines are particularly vulnerable to ESD damage. Simply touching or plugging in an I-O cable can result in a static discharge that can damage or completely destroy the interface product connected to the I-O port (such as RS-232 line drivers and receivers).

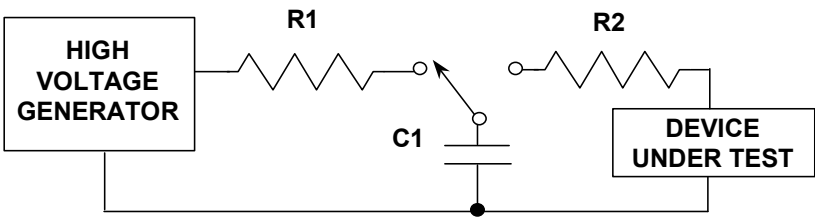
Traditional ESD test methods such as MIL-STD-883B Method 3015.7 do not fully test a product's susceptibility to this type of discharge. This test was intended to test a product's susceptibility to ESD damage during handling. Each pin is tested with respect to all other pins. There are some important differences between the MIL-STD-883B Method 3015.7 test and the IEC test, noted as follows:

- 1) *The IEC test is much more stringent in terms of discharge energy. The peak current injected is over four times greater.*
- 2) *The current risetime is significantly faster in the IEC test.*
- 3) *The IEC test is carried out while power is applied to the device.*

It is possible that ESD discharge could induce latch-up in the device under test. This test is therefore more representative of a real-world I-O discharge where the equipment is operating normally with power applied. For maximum confidence, however, both tests should be performed on interface devices, thus ensuring maximum protection both during handling, and later, during field service.

HARDWARE AND HOUSEKEEPING TECHNIQUES
OP AMP PROTECTION

A comparison of the test circuit values for the IEC1000-4-2 model versus the MIL-STD-883B Method 3015.7 Human Body Model is shown in Figure 7-86.



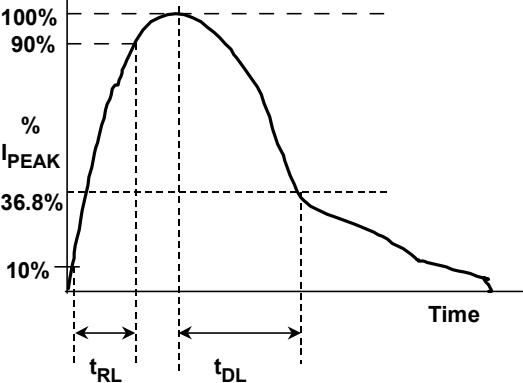
ESD TEST METHOD	R2	C1
Human Body Model MIL STD 883B Method 3015.7	1.5k Ω	100pF
IEC 1000-4-2	330 Ω	150pF

NOTE: CONTACT DISCHARGE VOLTAGE SPEC FOR IEC 1000-4-2 IS $\pm 8\text{kV}$

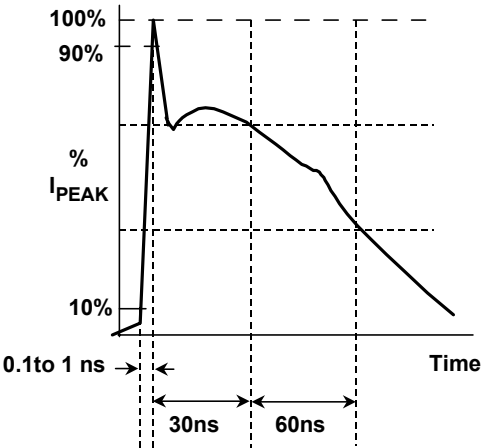
Figure 7-86: ESD test circuits and values

The ESD waveforms for the MIL-STD-883B, METHOD 3015.7 and IEC 1000-4-2 tests are compared in Figure 7-87, left and right, respectively.

**HUMAN BODY MODEL
MIL-STD-883B, METHOD 3015.7**



IEC 1000-4-2



- ◆ Voltage : 8 kV
- ◆ Peak Current :
 - MIL-883B, Method 3015.7 HBM : 5 A
 - IEC 1000-4-2 : 25 A

Figure 7-87: ESD test waveforms

Suitable ESD-protection design measures are relatively easy to incorporate, and most of the over-voltage protection methods already discussed in this section will help. Additional protection can also be obtained. For RS-232 and RS-485 drivers and receivers, the ADMXXX-E series is supplied with guaranteed 15kV (HBM) ESD specifications.

▣ OP AMP APPLICATIONS

For more general uses, the addition of TransZorbs at appropriate places in a system can provide protection against ESD (see References).

Figure 7-88 summarizes the major points about ESD prevention, from both an out-of-circuit as well as an in-circuit perspective.

- ◆ **Observe all Absolute Maximum Ratings on Data Sheet!**
- ◆ **Read ADI AN-397 (See Reference 16)**
- ◆ **Purchase ESD-Specified Digital Interface Devices**
 - **ADMXXX-E Series of RS-232 / RS-485 Drivers / Receivers**
(See Reference 18)
- ◆ **Follow General Over-voltage Protection Recommendations**
 - **Add Series Resistance to Limit Currents**
 - **Add Zeners or Transient Voltage Suppressors (TVS) for Extra Protection**
(See Reference 19)

Figure 7-88: A summary of ESD points

REFERENCES: OP AMP PROTECTION

1. Walt Kester, Wes Freeman, Joe Buxton, "Overvoltage Protection," portion of Section 10 within Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, Inc., 1999, ISBN 0-916550-20-6.
2. Siliconix PAD/JPAD/SSTPAD series Low leakage Pico-Amp Diodes, Vishay/Siliconix, <http://www.vishay.com/brands/siliconix/SSFsglld.html>
3. Bob Pease, "Bounding, Clamping Techniques Improve Circuit Performance," **EDN**, November 10, 1983, p. 277.
4. Bob Pease, "Understanding Diodes and Their Problems," Chapter 6 within **Troubleshooting Analog Circuits**, Butterworth-Heinemann, 1991, ISBN 0-7506-9184-0.
5. Dov Kurz, Avner Cohen, "Bootstrapping Reduces Amplifier Input Capacitance," **EDN**, March 20, 1978.
6. C.D. Motchenbacher, J.A. Connelly, Chapter 5, within **Low Noise Electronic System Design**, John Wiley, 1993.
7. Adolfo Garcia, "Operational Amplifier Output Voltage Phase Reversal," Section 11, pp. 1-10 within Walt Kester, Editor, **1992 Amplifier Applications Guide**, Analog Devices, Inc., Norwood, MA, 1992, ISBN 0-916550-10-9.
8. Adolfo Garcia, Wes Freeman, "Overvoltage Effects On Analog Integrated Circuits," Section 7 within Walt Kester, Editor, **Practical Analog Design Techniques**, Analog Devices, Inc., Norwood, MA, 1995, ISBN 0-916550-16-8.
9. Charles Kitchen, Lew Counts, **A Designer's Guide to Instrumentation Amplifiers**, Analog Devices, Inc., 2000.
10. Walt Kester, Wes Freeman, James Bryant, "Electrostatic Discharge," portion of Section 10 within Walt Kester, Editor, **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, Inc., 1999, ISBN 0-916550-20-6.
11. MIL-STD-883 Method 3015, "Electrostatic Discharge Sensitivity Classification." Available from Standardization Document Order Desk, 700 Robbins Ave., Building #4, Section D, Philadelphia, PA, 19111-5094.
12. EIAJ ED-4701 Test Method C-111, "Electrostatic Discharges." Available from the Japan Electronics Bureau, 250 W 34th St., New York NY 10119, Attn.: Tomoko.
13. ESD Association Standard S5.2 for "Electrostatic Discharge (ESD) Sensitivity Testing -Machine Model (MM)- Component Level." Available from the ESD Association, Inc., 200 Liberty Plaza, Rome, NY 13440.
14. ESD Association Draft Standard DS5.3 for "Electrostatic Discharge (ESD) Sensitivity Testing - Charged Device Model (CDM) Component Testing." Available from the ESD Association, Inc., 200 Liberty Plaza, Rome, NY 13440.
15. **ESD Prevention Manual**, Analog Devices, Inc.
16. Niall Lyne, "Electrically Induced Damage to Standard Linear Integrated Circuits: The Most Common Causes and the Associated Fixes to Prevent Reoccurrence," **Analog Devices AN397**.

▣ OP AMP APPLICATIONS

17. Mike Bryne, "How to Reliably Protect CMOS Circuits Against Power Supply Overvolutaging," **Analog Devices AN311**.
18. Data sheet for **ADM3311E RS-232 Port Transceiver**, Analog Devices, Inc., <http://www.analog.com>
19. TransZorbs are available from General Semiconductor, Inc., 10 Melville Park Road, Melville, NY, 11747-3113, (631) 847-3000, <http://www.gensemi.com/product/categories/tvs/tvs.htm>

SECTION 7-5: THERMAL CONSIDERATIONS

Walt Jung

For reliability reasons, op amp systems handling appreciable power are increasingly called upon to observe *thermal management*. All semiconductors have some specified safe upper limit for junction temperature (T_J), usually on the order of 150°C (sometimes 175°C). Like maximum power supply voltages, maximum junction temperature is a worst case limitation which shouldn't be exceeded. In conservative designs, it won't be approached by less than an ample safety margin. Note that this is critical, since semiconductor lifetime is inversely related to operating junction temperature. Simply put, the cooler op amps are, the more they can approach their maximum life.

This limitation of power and temperature is basic, and is illustrated by a typical data sheet statement as in Figure 7-89. In this case it is for the AD8017AR, an 8-pin SOIC device.

The maximum power that can be safely dissipated by the AD8017 is limited by the associated rise in junction temperature. The maximum safe junction temperature for plastic encapsulated device is determined by the glass transition temperature of the plastic, approximately +150°C. Temporarily exceeding this limit may cause a shift in parametric performance due to a change in the stresses exerted on the die by the package. Exceeding a junction temperature of +175°C for an extended period can result in device failure.

Figure 7-89: Maximum power dissipation data sheet statement for the AD8017AR, an ADI thermally enhanced SOIC packaged device

Tied to these statements are certain conditions of operation, such as the power dissipated by the device, and the package mounting specifics to the printed circuit board (PCB). In the case of the AD8017AR, the part is rated for 1.3W of power at an ambient of 25°C. This assumes operation of the 8-lead SOIC package on a two-layer PCB with about 4 inches² (~2500 mm²) of 2 oz. copper for heat sinking purposes. Predicting safe operation for the device under other conditions is covered below.

Thermal basics

The symbol θ is generally used to denote *thermal resistance*. Thermal resistance is in units of °C/watt (°C/W). Unless otherwise specified, it defines the resistance heat encounters transferring from a hot IC junction to the ambient air. It might also be expressed more specifically as θ_{JA} , for *thermal resistance, junction-to-ambient*. θ_{JC} and θ_{CA} are two additional θ forms used, and are further explained below.

In general, a device with a thermal resistance θ equal to 100°C/W will exhibit a temperature differential of 100°C for a power dissipation of 1W, as measured between two reference points. Note that this is a linear relationship, so 1W of dissipation in this part will produce a 100°C differential (and so on, for other powers). For the AD8017AR example, θ is about 95°C/W, so 1.3W of dissipation produces about a 124°C junction-to-ambient temperature differential. It is of course this rise in temperature that is used to predict the internal temperature, in order to judge the thermal reliability of a design. With

■ OP AMP APPLICATIONS

the ambient at 25°C, this allows an internal junction temperature of about 150°C. In practice most ambient temperatures are above 25°C, so less power can then be handled.

For any power dissipation P (in watts), one can calculate the effective temperature differential (ΔT) in °C as:

$$\Delta T = P \times \theta \quad \text{Eq. 7-6}$$

where θ is the total applicable thermal resistance.

Figure 7-90 summarizes a number of basic thermal relationships.

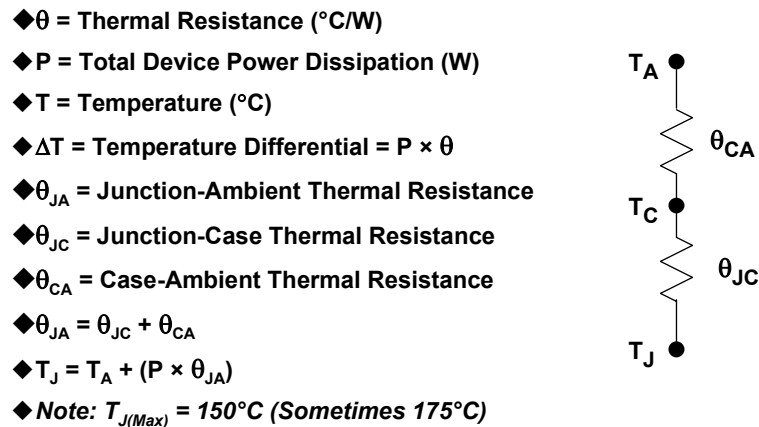


Figure 7-90: Basic thermal relationships

Note that series thermal resistances, such as the two shown at the right, model the total thermal resistance path a device may see. Therefore the total θ for calculation purposes is the sum, i.e., $\theta_{JA} = \theta_{JC} + \theta_{CA}$. Given the ambient temperature T_A , P , and θ , then T_J can be calculated. As the relationships signify, to maintain a low T_J , either θ or the power being dissipated (or both) must be kept low. A low ΔT is the key to extending semiconductor lifetimes, as it leads to lower maximum junction temperatures.

In ICs, one temperature reference point is always the device junction, taken to mean the hottest spot inside the chip operating within a given package. The other relevant reference point will be either T_C , the case of the device, or T_A , that of the surrounding air. This then leads in turn to the above mentioned individual thermal resistances, θ_{JC} and θ_{JA} .

Taking the most simple case first, θ_{JA} is the thermal resistance of a given device measured between its *junction* and the *ambient* air. This thermal resistance is most often used with small, relatively low power ICs such as op amps, which often dissipate 1W or less. Generally, θ_{JA} figures typical of op amps and other small devices are on the order of 90-100°C/W for a plastic 8 pin DIP package, as well as the better SOIC packages.

It should be clearly understood that these thermal resistances are *highly* package dependent, as different materials have different degrees of thermal conductivity. As a general rule of thumb, thermal resistance of conductors is analogous to electrical resistances, that is copper is the best, followed by aluminum, steel, and so on. Thus copper lead frame packages offer the highest performance, i.e., the lowest θ .

Heat sinking

By definition, a *heat sink* is an added low thermal resistance device attached to an IC to aid heat removal. A heat sink has additional thermal resistance of its own, θ_{CA} , rated in $^{\circ}\text{C}/\text{W}$. However, most current op amp packages don't easily lend themselves to heat sink attachment (exceptions are older TO99 metal can types). Devices meant for heat sink attachment will often be noted by a θ_{JC} dramatically lower than the θ_{JA} . In this case θ will be composed of more than one component. Thermal impedances add, making a net calculation relatively simple. For example, to compute a net θ_{JA} given a relevant θ_{JC} , the thermal resistance of the heat sink, θ_{CA} , or *case to ambient* is added to the θ_{JC} as:

$$\theta_{JA} = \theta_{JC} + \theta_{CA} \quad \text{Eq. 7-7}$$

and the result is the θ_{JA} for that specific circumstance.

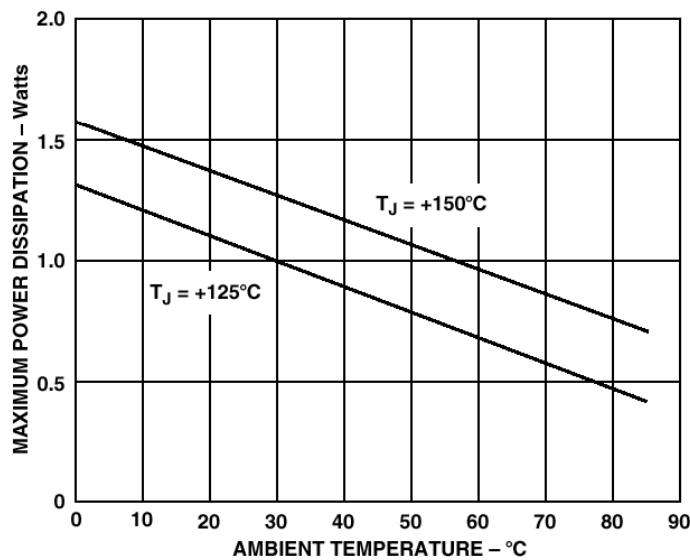


Figure 7-91: Thermal rating curves for AD8017AR op amp

More generally however, modern op amps *don't* use commercially available heat sinks. Instead, when significant power needs to be dissipated, such as $\geq 1\text{W}$, low thermal resistance copper PCB traces are used as the heat sink. In such cases, the most useful form of manufacturer data for this heat sinking are the boundary conditions of a sample PCB layout, and the resulting θ_{JA} for those conditions. This is in fact the type of specific information supplied for the AD8017AR, as mentioned earlier. Applying this approach, example data illustrating thermal relationships for such conditions is shown by Figure 7-91. These data apply for an AD8017AR mounted to a heat sink with an area of about 4 square inches on a 2 layer, 2 ounce copper PCB.

These curves indicate the maximum power dissipation vs. temperature characteristic for the AD8017, for maximum junction temperatures of both 150°C and 125°C . Such curves are often referred to as *derating* curves, since allowable power decreases with ambient temperature.

With the AD8017AR, the proprietary ADI *Thermal Coastline* IC package is used, which allows additional power to be dissipated with no increase in the SO-8 package size. For a

OP AMP APPLICATIONS

$T_{J(max)}$ of 150°C , the upper curve shows the allowable power in this package, which is 1.3W at an ambient of 25°C . If a more conservative $T_{J(max)}$ of 125°C is used, the lower of the two curves applies.

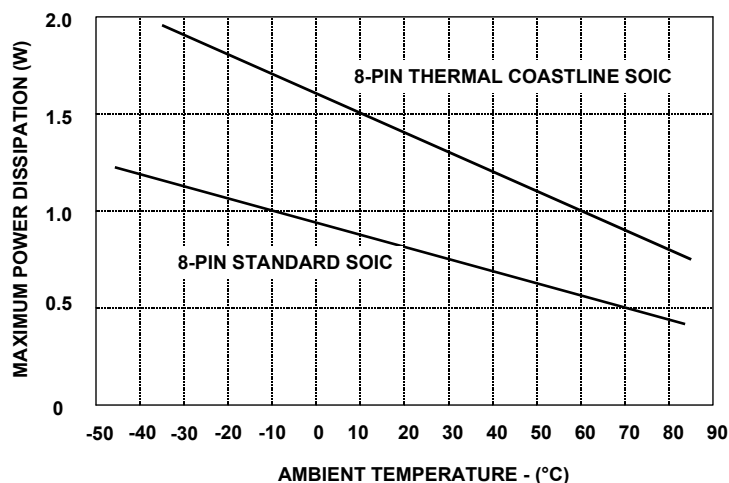


Figure 7-92: Thermal rating curves for standard (lower) and ADI Thermal Coastline (upper) 8-pin SOIC packages

A performance comparison for an 8-pin standard SOIC and the ADI Thermal Coastline version is shown in Figure 7-92. Note that the Thermal Coastline provides an allowable dissipation at 25°C of 1.3W , whereas a standard package allows only 0.8W . In the Thermal Coastline heat transfer is increased, accounting for the package's lower θ_{JA} .

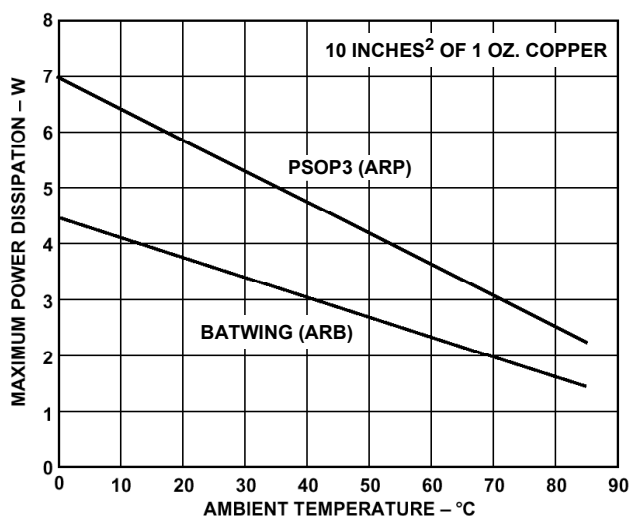


Figure 7-93: Thermal characteristic curves for the AD8016 BATWING (lower) and PSOP3 (upper) packages, for $T_{J(max)}$ equal to 125°C

Even higher power dissipation is possible, with the use of IC packages better able to transfer heat from chip to PCB. An example is the AD8016 device, available with two package options rated for 5.5 and 3.5W at 25°C , respectively, as shown in Figure 7-93.

Taking the higher rated power option, the AD8016ARP PSOP3 package, when used with a 10 inch^2 1 oz. heat sink plane, the combination is able to handle up to 3W of power at

an ambient of 70°C, as noted by the upper curve. This corresponds to a θ_{JA} of 18°C/W, which in this case applies for a maximum junction temperature of 125°C.

The reason the PSOP3 version of the AD8016 is so better able to handle power lies with the use of a large area copper slug. Internally, the IC die rest directly on this slug, with the bottom surface exposed as shown in Figure 7-94. The intent is that this surface be soldered directly to a copper plane of the PCB, thereby extending the heat sinking.

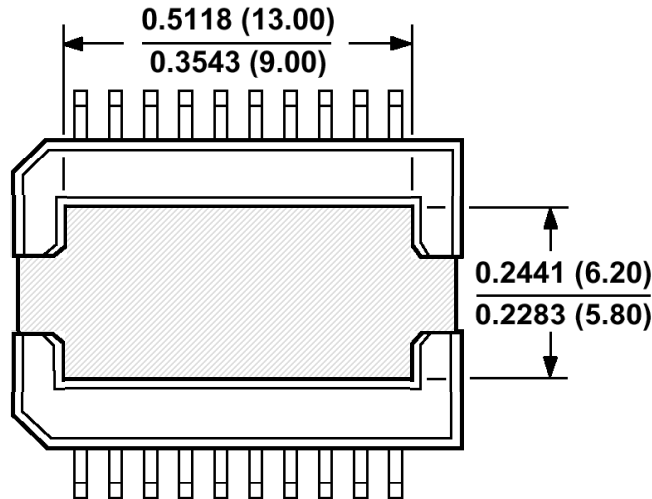


Figure 7-94: Bottom view of AD8016 20-lead PSOP3 package showing copper slug for aid in heat transfer (central grayed area)

Both of AD8016 package options are characterized for both still and moving air, but the thermal information given above applies *without* the use of directed airflow. Therefore, adding additional airflow lowers thermal resistance further (see Reference 2).

For reliable, low thermal resistance designs with op amps, several design *Do's and Don'ts* are listed below. Consider all of these points, as may be practical.

- 1) *Do use as large an area of copper as possible for a PCB heat sink, up to the point of diminishing returns.*
- 2) *In conjunction with 1), do use multiple (outside) PCB layers, connected together with multiple vias.*
- 3) *Do use as heavy copper as is practical (2 oz. or more preferred).*
- 4) *Do provide sufficient natural ventilation inlets and outlets within the system, to allow heat to freely move away from hot PCB surfaces.*
- 5) *Do orient power-dissipating PCB planes vertically, for convection-aided airflow across heat sink areas.*
- 6) *Do consider the use of external power buffer stages, for precision op amp applications.*

■ OP AMP APPLICATIONS

- 7) *Do consider the use of forced air, for situations where several watts must be dissipated in a confined space.*
- 8) *Don't use solder mask planes over heat dissipating traces.*
- 9) *Don't use excessive supply voltages on ICs delivering power.*

For the most part, these points are obvious. However, one that could use some elaboration is number 9. Whenever an application requires only modest *voltage* swings (such as for example standard video, 2Vp-p) a wide supply voltage range can often be used. But, as the data of Figure 7-95 indicates, operation of an op amp driver on higher supply voltages produces a large IC dissipation, even though the load power is constant.

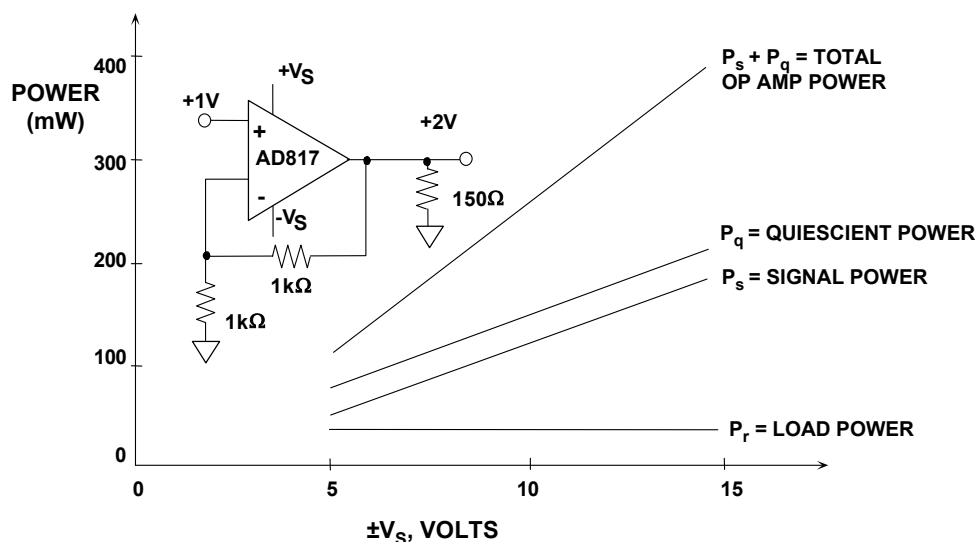


Figure 7-95: *Power dissipated in video op amp driver for various supply voltages with low voltage output swing*

In such cases, as long as the distortion performance of the application doesn't suffer, it can be advantageous to operate the IC on lower supplies, say $\pm 5\text{V}$, as opposed to $\pm 15\text{V}$. The above example data was calculated on a dc basis, which will generally tax the driver more in terms of power than a sine wave or a noise-like waveform, such as a DMT signal (see Reference 2). The general principles still hold for these ac waveforms, i.e., the op amp power dissipation is high when load current is high and the voltage low.

While there is ample opportunity for high power handling with the thermally enhanced packages described above for the AD8016 and AD8107, the increasingly popular smaller IC packages actually move in an opposite direction. Without question, it is true that today's smaller packages do noticeably sacrifice thermal performance. But, it must be understood that this is done in the interest of realizing a smaller size for the packaged op amp, and, ultimately, a much greater final PCB density for the overall system.

These points are illustrated by the thermal ratings for the AD8057 and AD8058 family of single and dual op amp devices, as is shown in Figure 7-96. The AD8057 and AD8058 op

HARDWARE AND HOUSEKEEPING TECHNIQUES

THERMAL CONSIDERATIONS

amps are available in three different packages. These are the SOT-23-5, and the 8-pin μ SOIC, along with standard SOIC.

As the data shows, as the package size becomes smaller and smaller, much less power is capable of being removed. Since the lead frame is the only heatsinking possible with such tiny packages, their thermal performance is thus reduced. The θ_{JA} for the packages mentioned is 240, 200, and 160°C/W, respectively. Note this is more of a *package* than *device* limitation. Other ICs with the same packages have similar characteristics.

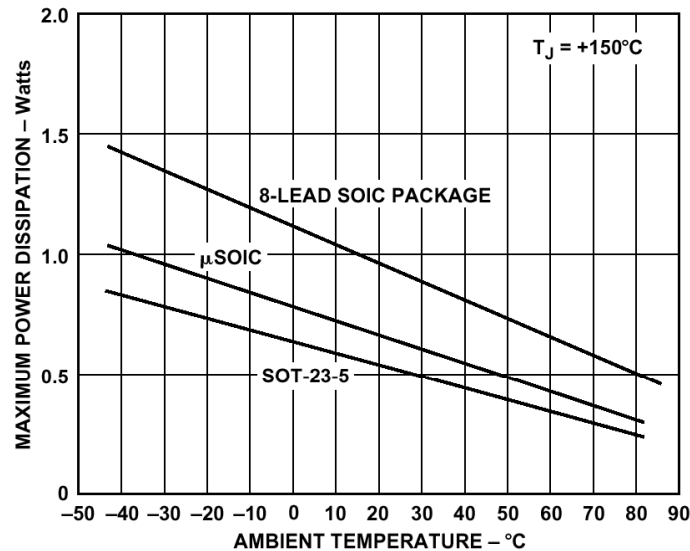


Figure 7-96: Comparative thermal performance for several AD8057/58 op amp package options

These discussions on the thermal application issues of op amps haven't dealt with the classic techniques of using clip-on (or bolt-on) type heat sinks. They also have not addressed the use of forced air cooling, generally considered only when tens of watts must be handled. These omissions are mainly because these approaches are seldom possible or practical with today's op amp packages.

The more general discussions within References 4-7 can be consulted for this and other supplementary information.

REFERENCES: THERMAL CONSIDERATIONS

1. Data sheet for **AD8017 Dual High Output Current, High Speed Amplifier**, Analog Devices, Inc., <http://www.analog.com>
2. Data sheet for **AD8016 Low Power, High Output Current, xDSL Line Driver**, Analog Devices, Inc., <http://www.analog.com/>.
3. "Power Consideration Discussions," data sheet for **AD815 High Output Current Differential Driver**, Analog Devices, Inc., <http://www.analog.com/>.
4. Walt Jung, Walt Kester, "Thermal Management," portion of Section 8 within Walt Kester, Editor, **Practical Design Techniques for Power and Thermal Management**, Analog Devices, Inc., 1998, ISBN 0-916550-19-2.
5. General Catalog, **AAVID Thermal Technologies, Inc.**, One Kool Path, Laconia, NH, 03246, (603) 528-3400.
6. Seri Lee, "How to Select a Heat Sink," **Aavid Thermal Technologies**, <http://www.aavid.com>
7. Seri Lee, "Optimum Design and Selection of Heat Sinks," **11th IEEE SEMI-THERM™ Symposium**, 1995, <http://www.aavid.com>

SECTION 7-6: EMI/RFI CONSIDERATIONS

James Bryant, Walt Jung, Walt Kester

Analog circuit performance is often affected adversely by high frequency signals from nearby electrical activity. And, equipment containing your analog circuitry may also adversely affect systems external to it. Reference 1 (page 4) describes this complementary transmission of undesirable high frequency signals from or into local equipment as per an IEC50 definition. These corresponding aspects of broad arena of *electromagnetic compatibility*, better known as EMC, are:

- ◆ *It describes the ability of electrical and electronic systems to operate without interfering with other systems...*
- ◆ *It also describes the ability of such systems to operate as intended within a specified electromagnetic environment.*

So, complete EMC assurance would indicate that the equipment under design should neither produce spurious signals, nor should it be vulnerable to out-of-band external signals (i.e., those outside its intended frequency range). It is the latter class of EMC problem to which analog equipment most often falls prey. And, it is the graceful handling of these spurious signals that are emphasized within this section.

The externally produced electrical activity may generate noise, and is referred to either as electromagnetic interference (EMI), or radio frequency interference (RFI). In this section, we will refer to EMI in terms of both electromagnetic and radio frequency interference. One of the more challenging tasks of the analog designer is the control of equipment against undesired operation due to EMI. It is important to note that in this context, *EMI and or RFI is almost always detrimental*. Once given entrance into your equipment, it can and will degrade its operation, quite often considerably.

This section is oriented heavily towards minimizing undesirable analog circuit operation due to the *receipt* of EMI/RFI. Misbehavior of this sort is also known as EMI or RFI *susceptibility*, indicating a tendency towards anomalous equipment behavior when exposed to EMI/RFI. There is of course a complementary EMC issue, namely with regard to spurious *emissions*. However, since analog circuits typically involve fewer of pulsed, high speed, high current signal edges that give rise to such spurious signals (compared to high speed logic, for example), this aspect of EMC isn't as heavily treated here. Nevertheless, the reader should bear in mind that it can be important, particularly if the analog circuitry is part of a mixed-signal environment along with high speed logic.

Since all of these various EMC design points can be critical, *the end-of-chapter references are strongly recommended for supplementary study*. Indeed, for a thorough, fully competent design with respect to EMI, RFI and EMC, the designer will need to become intimately acquainted with one or more of these references (see References 1- 6). As for the material following, it is best viewed as an introduction to this extremely broad but increasingly important topic.

■ OP AMP APPLICATIONS

EMI/RFI mechanisms

To understand and properly control EMI and RFI, it is helpful to first segregate it into manageable portions. Thus it is useful to remember that when EMI/RFI problems do occur, they can be fundamentally broken down into a *Source*, a *Path*, and a *Receiver*. As a systems designer, you have under your direct control the receiver part of this landscape, and perhaps some portion of the path. But seldom will the designer have control over the actual source.

EMI noise sources

There are countless ways in which undesired noise can couple into an analog circuit to ruin its accuracy. Some of the many examples of these noise sources are listed in Figure 7-97.

- ◆ **EMI/RFI noise sources can couple from *anywhere***
- ◆ **Some common sources of externally generated noise:**
 - Radio and TV Broadcasts
 - Mobile Radio Communications
 - Cellular Telephones
 - Vehicular Ignition
 - Lightning
 - Utility Power Lines
 - Electric Motors
 - Computers
 - Garage Door Openers
 - Telemetry Equipment

Figure 7-97: Some common EMI noise sources

Since little control is possible over these sources of EMI, the next best management tool one can exercise over them is to recognize and understand the possible paths by which they couple into the equipment under design.

EMI coupling paths

The EMI coupling paths are actually very few in terms of basic number. Three very general paths are by:

1. *Interference due to conduction (common-impedance)*
2. *Interference due to capacitive or inductive coupling (near-field interference)*
3. *Electromagnetic radiation (far-field interference)*

Noise Coupling Mechanisms

EMI energy may enter wherever there is an impedance mismatch or discontinuity in a system. In general this occurs at the interface where cables carrying sensitive analog signals are connected to PC boards, and through power supply leads. Improperly connected cables or poor supply filtering schemes are often perfect conduits for interference.

Conducted noise may also be encountered when two or more currents share a common path (impedance). This common path is often a high impedance "ground" connection. If two

circuits share this path, noise currents from one will produce noise voltages in the other. Steps may be taken to identify potential sources of this interference (see References 1 and 2, plus section 2 of this chapter).

Figure 7-98 shows some of the general ways noise can enter a circuit from external sources.

- ◆ **Impedance mismatches and discontinuities**
- ◆ **Common-mode impedance mismatches → Differential Signals**
- ◆ **Capacitively Coupled (Electric Field Interference)**
 - **$dV/dt \rightarrow$ Mutual Capacitance $\rightarrow$ Noise Current**
(Example: 1V/ns produces 1mA/pF)
- ◆ **Inductively Coupled (Magnetic Field)**
 - **$di/dt \rightarrow$ Mutual Inductance $\rightarrow$ Noise Voltage**
(Example: 1mA/ns produces 1mV/nH)

Figure 7-98: How EMI finds paths into equipment

There is a capacitance between any two conductors separated by a dielectric (air and vacuum are dielectrics, as well as all solid or liquid insulators). If there is a change of voltage on one conductor there will be change of charge on the other, and a *displacement current* will flow in the dielectric. Where either the capacitance or the dV/dT is high, noise is easily coupled. For example, a 1V/ns rate-of-change gives rise to displacement currents of 1 mA/pF.

If changing magnetic flux from current flowing in one circuit threads another circuit, it will induce an emf in the second circuit. Such *mutual inductance* can be a troublesome source of noise coupling from circuits with high values of dI/dT . As an example, a mutual inductance of 1nH and a changing current of 1A/ns will induce an emf of 1V.

Reducing Common-Impedance Noise

Steps to be taken to eliminate or reduce noise due to the conduction path sharing of impedances, or *common-impedance noise* are outlined in Figure 7-99.

- ◆ **Common-impedance noise**
 - **Decouple op amp power leads at LF and HF**
 - **Reduce common-impedance**
 - **Eliminate shared paths**
- ◆ **Techniques**
 - **Low impedance electrolytic (LF) and local low inductance (HF) bypasses**
 - **Use ground and power planes**
 - **Optimize system design**

Figure 7-99: Some solutions to common-impedance noise

These methods should be applied in conjunction with all of the related techniques discussed earlier within section 2 of this chapter.

Power supply rails feeding several circuits are good common-impedance examples. Real world power sources may exhibit low output impedance, or may they not— especially over frequency. Furthermore, PCB traces used to distribute power are both inductive and resistive,

■ OP AMP APPLICATIONS

and may also form a ground loop. The use of power and ground planes also reduces the power distribution impedance. These dedicated conductor layers in a PCB are continuous (ideally, that is) and as such, offer the lowest practical resistance and inductance.

In some applications where low-level signals encounter high levels of common-impedance noise it will not be possible to prevent interference and the system architecture may need to be changed. Possible changes include:

1. *Transmitting signals in differential form*
2. *Amplifying signals to higher levels for improved S/N*
3. *Converting signals into currents for transmission*
4. *Converting signals directly into digital form*

Noise Induced by Near-Field Interference

Crosstalk is the second most common form of interference. In the vicinity of the noise source, i.e., near-field, interference is not transmitted as an electromagnetic wave, and the term crosstalk may apply to either inductively or capacitively coupled signals.

Reducing Capacitance-Coupled Noise

Capacitively-coupled noise may be reduced by reducing the coupling capacity (by increasing conductor separation), but is most easily cured by shielding. A conductive and grounded shield (known as a *Faraday shield*) between the signal source and the affected node will eliminate this noise, by routing the displacement current directly to ground.

- ◆ **Reduce Level of High dV/dt Noise Sources**
- ◆ **Use Proper Grounding Schemes for Cable Shields**
- ◆ **Reduce Stray Capacitance**
 - **Equalize Input Lead Lengths**
 - **Keep Traces Short**
 - **Use Signal-Ground Signal-Routing Schemes**
- ◆ **Use Grounded Conductive Faraday Shields to Protect Against Electric Fields**

Figure 7-100: Methods to reduce capacitance-coupled noise

With the use of such shields, it is important to note that it is always *essential* that a Faraday shield be grounded. A floating or open-circuit shield almost invariably increases capacitively-coupled noise. For a brief review of this shielding, consult Section 2 of this chapter again, and see References 2 and 3 at the end of this section.

Methods to eliminate capacitance-coupled interference are summarized in Figure 7-100.

Reducing Magnetically-Coupled Noise

Methods to eliminate interference caused by magnetic fields are summarized in Figure 7-101.

- ◆ **Careful Routing of Wiring**
- ◆ **Use Conductive Screens for HF Magnetic Shields**
- ◆ **Use High Permeability Shields for LF Magnetic Fields (mu-Metal)**
- ◆ **Reduce Loop Area of Receiver**
 - **Twisted Pair Wiring**
 - **Physical Wire Placement**
 - **Orientation of Circuit to Interference**
- ◆ **Reduce Noise Sources**
 - **Twisted Pair Wiring**
 - **Driven Shields**

Figure 7-101: Methods to reduce magnetically-coupled noise

To illustrate the effect of magnetically-coupled noise, consider a circuit with a closed-loop area of $A \text{ cm}^2$ operating in a magnetic field with an rms flux density value of B gauss. The noise voltage V_n induced in this circuit can be expressed by the following equation:

$$V_n = 2 \pi f B A \cos\theta \times 10^{-8} \text{ V} \quad \text{Eq. 7-8}$$

In this equation, f represents the frequency of the magnetic field, and θ represents the angle of the magnetic field B to the circuit with loop area A . Magnetic field coupling can be reduced by reducing the circuit loop area, the magnetic field intensity, or the angle of incidence. Reducing circuit loop area requires arranging the circuit conductors closer together. Twisting the conductors together reduces the loop net area. This has the effect of canceling magnetic field pickup, because the sum of positive and negative incremental loop areas is ideally equal to zero. Reducing the magnetic field directly may be difficult. However, since magnetic field intensity is inversely proportional to the cube of the distance from the source, physically moving the affected circuit away from the magnetic field has a very great effect in reducing the induced noise voltage. Finally, if the circuit is placed perpendicular to the magnetic field, pickup is minimized. If the circuit's conductors are in parallel to the magnetic field the induced noise is maximized because the angle of incidence is zero.

There are also techniques that can be used to reduce the amount of magnetic-field interference, *at its source*. In the previous paragraph, the conductors of the receiver circuit were twisted together, to cancel the induced magnetic field along the wires. The same principle can be used on the source wiring. If the source of the magnetic field is large currents flowing through nearby conductors, these wires can be twisted together to reduce the net magnetic field.

Shields and cans are not nearly as effective against magnetic fields as against electric fields, but can be useful on occasion. At low frequencies magnetic shields using high-permeability material such as Mu-metal can provide modest attenuation of magnetic fields. At high frequencies simple conductive shields are quite effective provided that the thickness of the shield is greater than the skin depth of the conductor used (at the frequency involved).

Note—copper skin depth is $6.6/\sqrt{f}$ cm, with f in Hz.

Passive Components: Your Arsenal Against EMI

Passive components, such as resistors, capacitors, and inductors, are powerful tools for reducing externally induced interference when used properly.

Simple RC networks make efficient and inexpensive one-pole, low-pass filters. Incoming noise is converted to heat and dissipated in the resistor. But note that a fixed resistor does produce thermal noise of its own. Also, when used in the input circuit of an op amp or in-amp, such resistor(s) can generate input-bias-current induced offset voltage. While matching the two resistors will minimize the dc offset, the noise will remain. Figure 7-102 summarizes some popular low-pass filters for minimizing EMI.

LP Filter Type	ADVANTAGE	DISADVANTAGE
RC Section	Simple Inexpensive	Resistor Thermal Noise $I_B \times R$ Drop $\rightarrow$ Offset Single-Pole Cutoff
LC Section (Bifilar)	Very Low Noise at LF Very Low IR Drop Inexpensive Two-Pole Cutoff	Medium Complexity Nonlinear Core Effects Possible
π Section (C-L-C)	Very Low Noise at LF Very Low IR Drop Pre-packaged Filters Multiple-Pole Cutoff	Most Complex Nonlinear Core Effects Possible Expensive

Figure 7-102: Using passive components within filters to combat EMI

In applications where signal and return conductors aren't well-coupled magnetically, a common-mode (CM) choke can be used to increase their mutual inductance. Note that these comments apply mostly to in-amps, which naturally receive a balanced input signal (whereas op amps are inherently unbalanced inputs— unless one constructs an in-amp with them). A CM choke can be simply constructed by winding several turns of the differential signal conductors together through a high-permeability (> 2000) ferrite bead. The magnetic properties of the ferrite allow differential-mode currents to pass unimpeded while suppressing CM currents.

Capacitors can also be used before and after the choke, to provide additional CM and differential-mode filtering, respectively. Such a CM choke is cheap and produces very low thermal noise and bias current-induced offsets, due to the wire's low DCR. However, there is a field around the core. A metallic shield surrounding the core may be necessary to prevent coupling with other circuits. Also, note that high-current levels should be avoided in the core as they may saturate the ferrite.

The third method for passive filtering takes the form of packaged π -networks (C-L-C). These packaged filters are completely self-contained and include feedthrough capacitors at the input and the output as well as a shield to prevent the inductor's magnetic field from radiating noise. These more expensive networks offer high levels of attenuation and wide operating frequency ranges, but the filters must be selected so that for the operating current levels involved the ferrite doesn't saturate.

Reducing System Susceptibility to EMI

The general examples discussed above and the techniques illustrated earlier in this section outline the procedures that can be used to reduce or eliminate EMI/RFI. Considered on a *system* basis, a summary of possible measures is given in Figure 7-103.

- ◆ **Always Assume That Interference Exists!**
- ◆ **Use Conducting Enclosures Against Electric and HF Magnetic Fields**
- ◆ **Use mu-Metal Enclosures Against LF Magnetic Fields**
- ◆ **Implement Cable Shields Effectively**
- ◆ **Use Feedthrough Capacitors and Packaged PI Filters**

Figure 7-103: Reducing system EMI/RFI susceptibility

Other examples of filtering techniques useful against EMI are illustrated later in this section, under "Reducing RFI rectification within op amp and in-amp circuits".

The section immediately below further details shielding principles.

A Review of Shielding Concepts

The concepts of shielding effectiveness presented next are background material. Interested readers should consult References 4-9 cited at the end of the section for more detailed information.

Applying the concepts of shielding effectively requires an understanding of the source of the interference, the environment surrounding the source, and the distance between the source and point of observation (the receiver). If the circuit is operating close to the source (in the *near*, or induction-field), then the field characteristics are determined by the source. If the circuit is remotely located (in the *far*, or radiation-field), then the field characteristics are determined by the transmission medium.

A circuit operates in a near-field if its distance from the source of the interference is less than the wavelength (λ) of the interference divided by 2π , or $\lambda/2\pi$. If the distance between the circuit and the source of the interference is larger than this quantity, then the circuit operates in the far field. For instance, the interference caused by a 1ns pulse edge has an upper bandwidth of approximately 350MHz. The wavelength of a 350MHz signal is approximately 32 inches (the speed of light is approximately 12"/ns). Dividing the wavelength by 2π yields a distance of approximately 5 inches, the boundary between near- and far-field. If a circuit is within 5 inches of a 350MHz interference source, then the circuit operates in the near-field of the interference. If the distance is greater than 5 inches, the circuit operates in the far-field of the interference.

Regardless of the type of interference, there is a characteristic impedance associated with it. The characteristic, or wave impedance of a field is determined by the ratio of its electric (or E-) field to its magnetic (or H-) field. In the far field, the ratio of the electric field to the magnetic field is the characteristic (wave impedance) of free space, given by $Z_0 = 377\Omega$. In the near field, the wave-impedance is determined by the nature of the

▣ OP AMP APPLICATIONS

interference and its distance from the source. If the interference source is high-current and low-voltage (for example, a loop antenna or a power-line transformer), the field is predominately magnetic and exhibits a wave impedance which is less than 377Ω . If the source is low-current and high-voltage (for example, a rod antenna or a high-speed digital switching circuit), then the field is predominately electric and exhibits a wave impedance which is greater than 377Ω .

Conductive enclosures can be used to shield sensitive circuits from the effects of these external fields. These materials present an *impedance mismatch* to the incident interference, because the impedance of the shield is lower than the wave impedance of the incident field. The effectiveness of the conductive shield depends on two things: First is the loss due to the *reflection* of the incident wave off the shielding material. Second is the loss due to the *absorption* of the transmitted wave *within* the shielding material. The amount of reflection loss depends upon the type of interference and its wave impedance. The amount of absorption loss, however, is independent of the type of interference. It is the same for near- and far-field radiation, as well as for electric or magnetic fields.

Reflection loss at the interface between two media depends on the difference in the characteristic impedances of the two media. For electric fields, reflection loss depends on the frequency of the interference and the shielding material. This loss can be expressed in dB, and is given by:

$$R_e(\text{dB}) = 322 + 10\log_{10}\left[\frac{\sigma_r}{\mu_r f^3 r^2}\right] \quad \text{Eq. 7-9}$$

where σ_r = relative conductivity of the shielding material, in Siemens per meter;
 μ_r = relative permeability of the shielding material, in Henries per meter;
 f = frequency of the interference, and
 r = distance from source of the interference, in meters

For magnetic fields, the loss depends also on the shielding material and the frequency of the interference. Reflection loss for magnetic fields is given by:

$$R_m(\text{dB}) = 14.6 + 10\log_{10}\left[\frac{fr^2\sigma_r}{\mu_r}\right] \quad \text{Eq. 7-10}$$

and, for plane waves ($r > \lambda/2\pi$), the reflection loss is given by:

$$R_{pw}(\text{dB}) = 168 + 10\log_{10}\left[\frac{\sigma_r}{\mu_r f}\right] \quad \text{Eq. 7-11}$$

Absorption is the second loss mechanism in shielding materials. Wave attenuation due to absorption is given by:

$$A(\text{dB}) = 3.34 t\sqrt{\sigma_r\mu_r f} \quad \text{Eq. 7-12}$$

HARDWARE AND HOUSEKEEPING TECHNIQUES EMI/RFI CONSIDERATIONS

where t = thickness of the shield material, in inches. This expression is valid for plane waves, electric and magnetic fields. Since the intensity of a transmitted field decreases exponentially relative to the thickness of the shielding material, the absorption loss in a shield one skin-depth (δ) thick is 9dB. Since absorption loss is proportional to thickness and inversely proportional to skin depth, increasing the thickness of the shielding material improves shielding effectiveness at high frequencies.

Reflection loss for plane waves in the far field decreases with increasing frequency because the shield impedance, Z_s , increases with frequency. Absorption loss, on the other hand, increases with frequency because skin depth decreases. For electric fields and plane waves, the primary shielding mechanism is reflection loss, and at high frequencies, the mechanism is absorption loss.

Thus for high-frequency interference signals, lightweight, easily worked high conductivity materials such as copper or aluminum can provide adequate shielding. At low frequencies however, both reflection and absorption loss to magnetic fields is low. It is thus very difficult to shield circuits from low-frequency magnetic fields. In these applications, high-permeability materials that exhibit low-reluctance provide the best protection. These low-reluctance materials provide a magnetic shunt path that diverts the magnetic field away from the protected circuit.

To summarize the characteristics of metallic materials commonly used for shielded purposes: Use high conductivity metals for HF interference, and high permeability metals for LF interference.

A properly shielded enclosure is very effective at preventing external interference from disrupting its contents as well as confining any internally-generated interference. However, in the real world, openings in the shield are often required to accommodate adjustment knobs, switches, connectors, or to provide ventilation. Unfortunately, these openings may compromise shielding effectiveness by providing paths for high-frequency interference to enter the instrument.

The longest dimension (not the total area) of an opening is used to evaluate the ability of external fields to enter the enclosure, because the openings behave as slot antennas. Equation Eq. 7-13 can be used to calculate the shielding effectiveness, or the susceptibility to EMI leakage or penetration, of an opening in an enclosure:

$$\text{Shielding Effectiveness (dB)} = 20 \log_{10} \left(\frac{\lambda}{2 \cdot L} \right) \quad \text{Eq. 7-13}$$

where λ = wavelength of the interference and
 L = maximum dimension of the opening

Maximum radiation of EMI through an opening occurs when the longest dimension of the opening is equal to one half-wavelength of the interference frequency (0dB shielding effectiveness). A rule-of-thumb is to keep the longest dimension less than 1/20 wavelength of the interference signal, as this provides 20dB shielding effectiveness.

■ OP AMP APPLICATIONS

Furthermore, a few small openings on each side of an enclosure is preferred over many openings on one side. This is because the openings on different sides radiate energy in different directions, and as a result, shielding effectiveness is not compromised. If openings and seams cannot be avoided, then conductive gaskets, screens, and paints alone or in combination should be used judiciously to limit the longest dimension of any opening to less than $1/20$ wavelength. Any cables, wires, connectors, indicators, or control shafts penetrating the enclosure should have circumferential metallic shields physically bonded to the enclosure at the point of entry. In those applications where unshielded cables/wires are used, then filters are recommended at the shield entry point.

General Points on Cables and Shields

Although covered in detail elsewhere, it is worth noting that the improper use of cables and their shields can be a significant contributor to both radiated and conducted interference. Rather than developing an entire treatise on these issues, the interested reader should consult References 2, 3, 5, and 6 for background.

As shown in Figure 7-104, proper cable/enclosure shielding confines sensitive circuitry and signals *entirely within the shield*, with no compromise to shielding effectiveness.

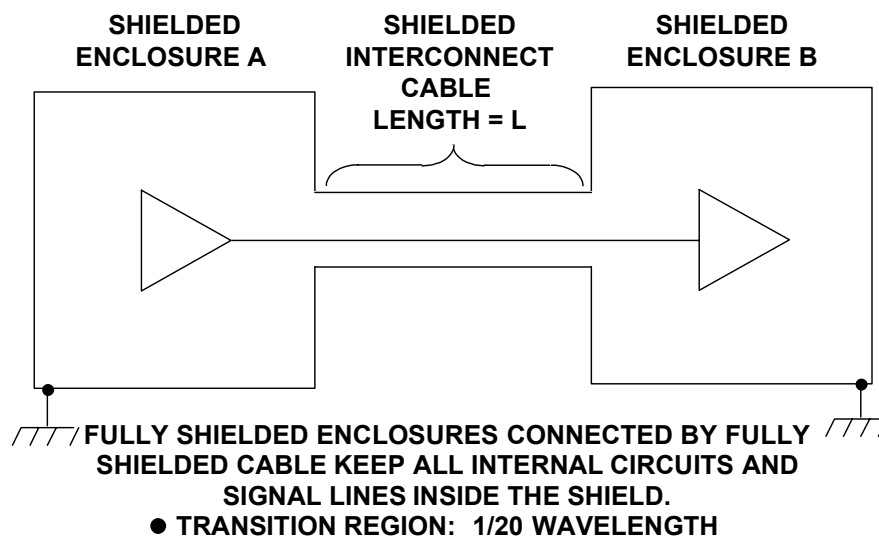


Figure 7-104: Shielded interconnect cables are either electrically long or short, depending upon the operating frequency

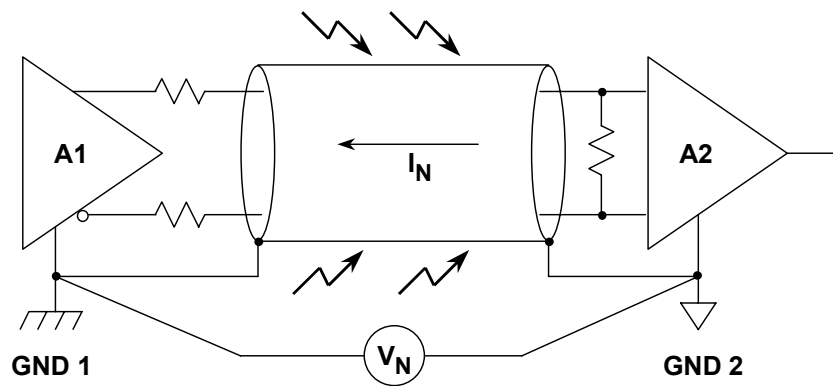
As can be noted by this diagram, the enclosures and the shield must be grounded properly, otherwise they can act as an antenna, thereby making the radiated and conducted interference problem worse (rather than better).

Depending on the type of interference (pickup/radiated, low/high frequency), proper cable shielding is implemented differently and is very dependent on the length of the cable. The first step is to determine whether the length of the cable is *electrically short* or *electrically long* at the frequency of concern. A cable is considered electrically short if the length of the cable is less than $1/20$ wavelength of the highest frequency of the interference. Otherwise it is considered to be electrically long.

For example, at 50/60Hz, an electrically short cable is any cable length less than 150 miles, where the primary coupling mechanism for these low frequency electric fields is capacitive. As such, for any cable length less than 150 miles, the amplitude of the interference will be the same over the entire length of the cable.

In applications where the length of the cable is electrically long, or protection against high-frequency interference is required, then the preferred method is to connect the cable shield to low-impedance points, *at both ends*. As will be seen shortly, this can be a direct connection at the driving end, and a capacitive connection at the receiver. If left ungrounded, unterminated transmission lines effects can cause reflections and standing waves along the cable. At frequencies of 10MHz and above, circumferential (360°) shield bonds and metal connectors are required to main low-impedance connections to ground.

In summary, for protection against low-frequency (<1MHz), electric-field interference, grounding the shield at one end is acceptable. For high-frequency interference (>1MHz), the preferred method is grounding the shield at both ends, using 360° circumferential bonds between the shield and the connector, and maintaining metal-to-metal continuity between the connectors and the enclosure.



- ◆ V_N Causes Current in Shield (Usually 50/60Hz)
- ◆ Differential Error Voltage is Produced at Input of A2 Unless:
 - A1 Output is Perfectly Balanced and
 - A2 Input is Perfectly Balanced and
 - Cable is Perfectly Balanced

Figure 7-105: Ground loops in shielded twisted pair cable can cause errors

However in practice, there is a caveat involved with directly grounding the shield at both ends. When this is done, it creates a low frequency ground loop, shown in Figure 7-105.

Whenever two systems A1 and A2 are remote from each other, there is usually a difference in the ground potentials at each system, i.e., V_N . The frequency of this potential difference is generally the line frequency (50 or 60Hz) and multiples thereof. But, if the shield is directly grounded at both ends as shown, noise current I_N flows in the shield. In a perfectly balanced system, the common-mode rejection of the system is infinite, and this current flow produces no differential error at the receiver A2. However, perfect balance is never achieved in the driver, its impedance, the cable, or the receiver,

■ OP AMP APPLICATIONS

so a certain portion of the shield current will appear as a differential noise signal, at the input of A2. The following illustrate correct shield grounding for various examples.

As noted above, cable shields are subject to both low and high frequency interference. Good design practice requires that the shield be grounded at both ends if the cable is electrically long to the interference frequency, as is usually the case with RF interference.

Figure 7-106 (below) shows a remote passive RTD sensor connected to a bridge and conditioning circuit by a shielded cable. The proper grounding method is shown in the upper part of the figure, where the shield is grounded at the receiving end.

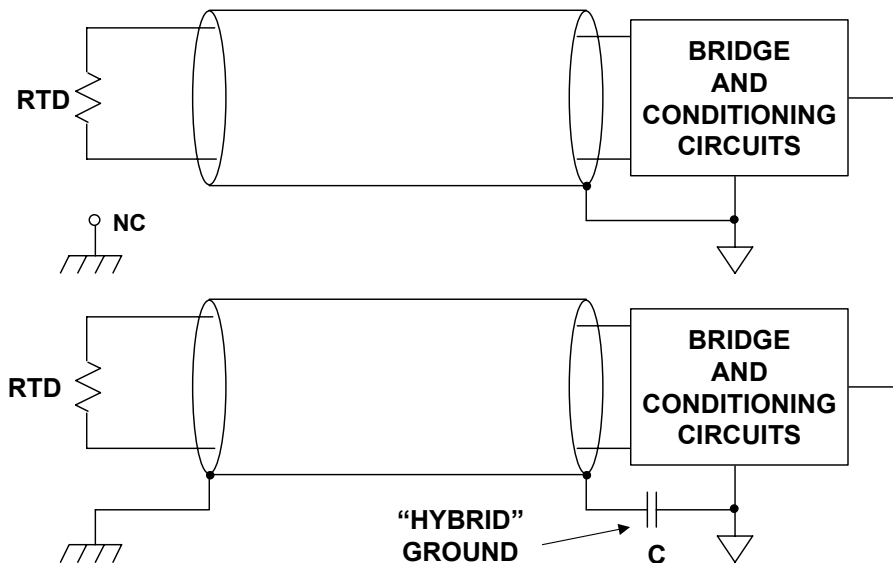


Figure 7-106: Hybrid grounding of shielded cable with passive sensor

However, safety considerations may require that the remote end of the shield also be grounded. If this is the case, the receiving end can be grounded with a low inductance ceramic capacitor ($0.01\mu\text{F}$ to $0.1\mu\text{F}$), still providing high frequency grounding. The capacitor acts as a ground to RF signals on the shield but blocks low frequency line current to flow in the shield. This technique is often referred to as a *hybrid ground*.

A case of an active remote sensor and/or other electronics is shown Figure 7-107 (opposite). In both of the two situations, a hybrid ground is also appropriate, either for the balanced (upper) or the single-ended (lower) driver case. In both instances the capacitor "C" breaks the low frequency ground loop, providing effective RF grounding of the shielded cable at the A2 receiving end at the right side of the diagram.

There are also some more subtle points that should be made with regard to the source termination resistances used, R_S . In both the balanced as well as the single-ended drive cases, the driving signal seen on the balanced line originates from a net impedance of R_S , which is split between the two twisted pair legs as twice $R_S/2$. In the upper case of a fully differential drive, this is straightforward, with an $R_S/2$ valued resistor connected in series with the complementary outputs from A1.

In the bottom case of the single-ended driver, note that there are still two $R_S/2$ resistors used, one in series with both legs. Here the grounded dummy return leg resistor provides an impedance-balanced ground connection drive to the differential line, aiding in overall system noise immunity. Note that this implementation is only useful for those applications with a balanced receiver at A2, as shown.

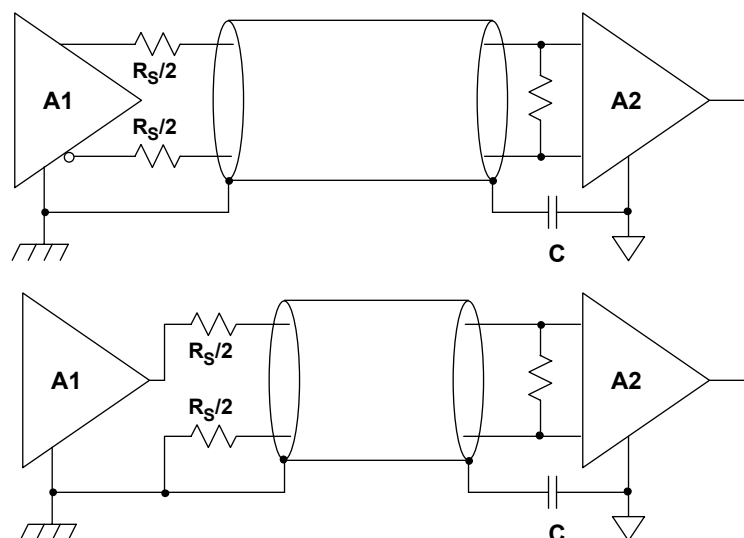


Figure 7-107: *Impedance-balanced drive of balanced shielded cable aids noise-immunity with either balanced or single-ended source signals*

Coaxial cables are different from shielded twisted pair cables in that the signal return current path is through the shield. For this reason, the ideal situation is to ground the shield at the driving end and allow the shield to float at the differential receiver (A2) as shown in the upper portion of Figure 7-108 (below). For this technique to work, however, the receiver must be a differential type with good high frequency CM rejection.

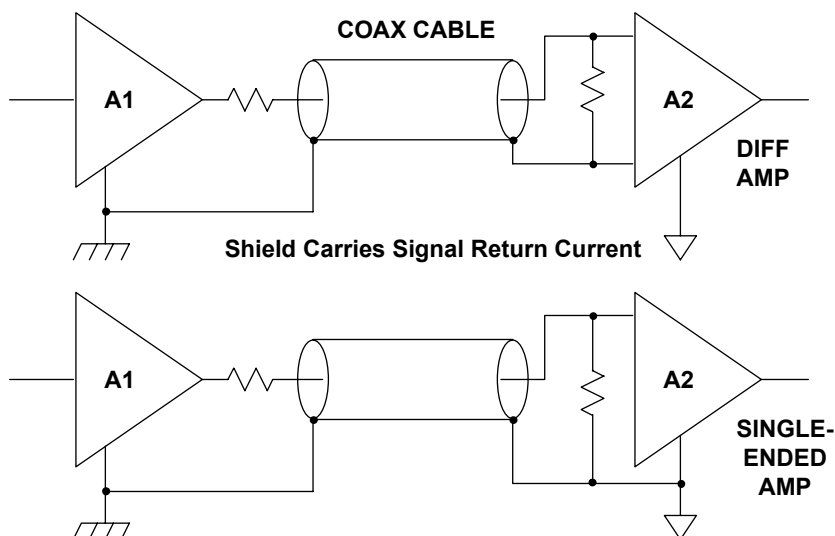


Figure 7-108: *Coaxial cables can use either balanced or single-ended receivers*

However, the receiver may be a single-ended type, such as typical of a standard single op amp type circuit. This is true for the bottom example of Figure 7-108, so there is no choice but to ground the coaxial cable shield at both ends for this case.

Input-stage RFI Rectification Sensitivity

A well-known but poorly understood phenomenon in analog integrated circuits is *RFI rectification*, specifically as it occurs in op amps and in-amps. While amplifying very small signals these devices can rectify large-amplitude, out-of-band HF signals, i.e., RFI. As a result, dc errors appear at the output in addition to the desired signal. The undesired HF signals can enter sensitive analog circuits by various means. Conductors leading into and out of the circuit provide a path for interference coupling into a circuit. These conductors pick up noise through capacitive, inductive, or radiation coupling, as discussed earlier. The spurious signals appear at the amplifier inputs, along with the desired signal. The spurious signals can be several tens of mV in amplitude however, which causes problems. Simply stated, it cannot be assumed that a sensitive, low-bandwidth dc amplifier will always reject out-of-band spurious signals. While this would be the case for a simple linear low pass filter, op amp and in-amp devices actually rectify high-level HF signals, leading to non-linearities and anomalous offsets. Methods of analysis for as well as the prevention of RFI rectification are discussed in this section.

Background: Op amp and In-Amp RFI Rectification Sensitivity Tests

Just about all in-amp and op amp input stages use emitter-coupled BJT or source-coupled FET differential pairs of some type. Depending on the device operating current, the interfering frequency and its relative amplitude, these differential pairs can behave as high-frequency detectors. As will be shown, the detection process produces spectral components at the harmonics of the interference, as well at dc! It is the detected dc component of the interference that shifts amplifier bias levels, leading to inaccuracies.

The effect of RFI rectification within op amps and in-amps can be evaluated with relatively simple test circuits, as described for the *RFI Rectification Test Configuration* (see page 1-38 of Reference 10). In these tests, an op amp or in-amp is configured for a gain of -100 (op amp), or 100 (in-amp), with dc output measured after a 100-Hz low-pass filter, preventing interference from other signals. A 100MHz, $20\text{mV}_{\text{p-p}}$ signal is the test stimulus, chosen to be well above test device frequency limits. In operation, the test evaluates dc output shift observed under stimulus presence. While an ideal dc shift for this measurement would be zero, the actual dc shift of a given part indicates the relative RFI rectification sensitivity. Devices using both BJT and FET technologies can be tested by this method, as can devices operating at either low or high supply current levels.

In the original op amp test device set of Reference 10, some FET-input devices (OP80, OP42, OP249 and AD845) exhibited no observable shift in their output voltages, while several others showed shifts of less than $10\mu\text{V}$ referred to the input. Of the BJT-input op amps, the amount of shift decreased with increasing device supply current. Only two devices showed no observable output voltage shift (AD797 and AD827), while others showed shifts of less than $10\mu\text{V}$ referred to the input (OP200 and OP297). For other op amps, it is to be expected that similar patterns would be shown under such testing.

From these tests, some generalizations on RFI rectification can be made. First, device susceptibility appears to be inversely proportional to supply current; that is, devices biased at low quiescent supply currents exhibit greatest output voltage shift. Second, ICs

with FET-input stages appeared to be less susceptible to rectification than those with BJTs. Note that these points are independent of whether the device is an op amp or an in-amp. In practice this means that the lower power op amps *or* in-amps will tend to be more susceptible to RFI rectification effects. And, FET-input op amps (or in-amps) will tend to be *less* susceptible to RFI, especially those operating at higher currents.

Based on these data and from the fundamental differences between BJTs and FETs, we can summarize what we know. Bipolar transistor action is controlled by a forward-biased p-n junction (the base-emitter junction) whose I-V characteristic is exponential and quite nonlinear. FET behavior, on the other hand, is controlled by voltages applied to a reverse-biased p-n junction diode (the gate-source junction). The I-V characteristic of FETs is a square-law, and thus it is inherently more linear than that of BJTs.

For the case of the lower supply current devices, transistors in the circuit are biased well below their peak f_T collector currents. Although the ICs may be constructed on processes whose device f_T s can reach hundreds of MHz, charge transit times increase, when transistors are operated at low current levels. The impedance levels used also make RFI rectification in these devices worse. In low-power op amps, impedances are on the order of hundreds to thousands of k Ω s, whereas in moderate supply-current designs impedances might be no more than just a few k Ω . Combined, these factors tend to degrade a low-power device's RFI rectification sensitivity.

- ◆ **BJT input devices *rectify readily***
 - **Forward-biased B-E junction**
 - **Exponential I-V Transfer Characteristic**
- ◆ **FET input devices *less sensitive to rectifying***
 - **Reversed-biased p-n junction**
 - **Square-law I-V Transfer Characteristic**
- ◆ **Low I_{supply} devices versus High I_{supply} devices**
 - **Low I_{supply} $\Rightarrow$ *Higher* rectification sensitivity**
 - **High I_{supply} $\Rightarrow$ *Lower* rectification sensitivity**

Figure 7-109: *Some general observations on op amp and in-amp input stage RFI rectification sensitivity*

Figure 7-109 summarizes these general observations on RFI rectification sensitivity, and is applicable to both op amps and in-amps.

An Analytical Approach: BJT RFI Rectification

While lab experiments can demonstrate that BJT-input devices exhibit greater RFI rectification sensitivity than comparable devices with FET inputs, a more analytical approach can also be taken to explain this phenomenon.

RF circuit designers have long known that p-n junction diodes are efficient rectifiers because of their nonlinear I-V characteristics. A spectral analysis of a BJT transistor current output for a HF sinewave input reveals that, as the device is biased closer to its "knee," nonlinearity increases. This, in turn, makes its use as a detector more efficient.

■ OP AMP APPLICATIONS

This is especially true in low-power op amps, where input transistors are biased at very low collector currents.

A rectification analysis for the collector current of a BJT has been presented in Reference 10, and will not be repeated here except for the important conclusions. These results reveal that the original quadratic second-order term can be simplified into a frequency-dependent term, $\Delta i_C(AC)$, at twice the input frequency and a dc term, $\Delta i_C(DC)$. The latter component can be expressed as noted in Eq. 7-14, the final form for the rectified dc term:

$$\Delta i_C(DC) = \left(\frac{V_X}{V_T} \right)^2 \cdot \frac{I_C}{4} \quad \text{Eq. 7-14}$$

This expression shows that the dc component of the second-order term is directly proportional to the *square* of the HF noise amplitude V_X , and, also, to I_C , the quiescent collector current of the transistor. To illustrate this point on rectification, note that the change in dc collector current of a bipolar transistor operating at an I_C of 1mA with a spurious 10mV_{peak} high-frequency signal impinging upon it will be about 38uA.

Reducing the amount of rectified collector current is a matter of reducing the quiescent current, or the magnitude of the interference. Since the op amp and in-amp input stages seldom provide adjustable quiescent collector currents, reducing the level of interfering noise V_X is by far the best (and almost always the only) solution. For example, reducing the amplitude of the interference by a factor of 2, down to 5mV_{peak} produces a net 4 to 1 reduction in the rectified collector current. Obviously, this illustrates the importance of keeping spurious HF signals away from RFI sensitive amplifier inputs.

An Analytical Approach: FET RFI Rectification

A rectification analysis for the drain current of a JFET has also been presented in Reference 10, and isn't repeated here. A similar approach was used for the rectification analysis of a FET's drain current as a function of a small voltage V_X , applied to its gate. The results of evaluating the second-order rectified term for the FET's drain current are summarized in Eq. 7-15. Like the BJT, an FET's second-order term has an ac and a dc component. The simplified expression for the dc term of the rectified drain current is given here, where the rectified dc drain current is directly proportional to the square of the amplitude of V_X , the spurious signal. However, Eq. 7-15 also reveals a very important difference between the *degree* of the rectification produced by FETs relative to BJTs.

$$\Delta i_D(DC) = \left(\frac{V_X}{V_P} \right)^2 \cdot \frac{I_{DSS}}{2} \quad \text{Eq. 7-15}$$

Whereas in a BJT the change in collector current has a direct relationship to its quiescent collector current level, the change in a JFET's drain current is proportional to its drain current at zero gate-source voltage, I_{DSS} , and inversely proportional to the square of its channel pinch-off voltage, V_P — parameters that are geometry and process dependent. Typically, JFETs used in the input stages of in-amps and op amps are biased with their

quiescent current of $\sim 0.5 \cdot I_{DSS}$. Therefore, the change in a JFET's drain current is independent of its quiescent drain current; hence, independent of the operating point.

A quantitative comparison of second-order rectified dc terms between BJTs and FETs is illustrated in Figure 7-110. In this example, a bipolar transistor with a unit emitter area of $576\mu\text{m}^2$ is compared to a unit-area JFET designed for an I_{DSS} of $20\mu\text{A}$ and a pinch-off voltage of 2V. Each device is biased at $10\mu\text{A}$ and operated at $T_A = 25^\circ\text{C}$.

The important result is that, under identical quiescent current levels, the change in collector current in bipolar transistors is about 1500 times greater than the change in a JFET's drain current. This explains why FET-input amplifiers behave with less sensitivity to large amplitude HF stimulus. As a result, they offer more RFI rectification immunity.

◆ BJT: Emitter area = $576\mu\text{m}^2$ $I_C = 10\mu\text{A}$ $V_T = 25.68\text{mV @ } 25^\circ\text{C}$ $\Delta i_C = \left(\frac{V_X}{V_T} \right)^2 \cdot \frac{I_C}{4}$ $= \frac{V_X^2}{264}$	◆ JFET: $I_{DSS} = 20\mu\text{A (Z/L=1)}$ $V_P = 2\text{V}$ $I_D = 10\mu\text{A}$ $\Delta i_D = \left(\frac{V_X}{V_P} \right)^2 \cdot \frac{I_{DSS}}{2}$ $= \frac{V_X^2}{400 \times 10^3}$
---	---

◆ **Conclusion: BJTs ~1500 more sensitive than JFETs!**

Figure 7-110: Relative sensitivity comparison - BJT versus JFET

What all this boils down to is this: Since a user has virtually no access to the amplifier's internal circuitry, the prevention of IC circuit performance degradation due to RFI is left essentially to those means which are external to the ICs.

As the analysis above shows, regardless of the amplifier type, *RFI rectification is directly proportional to the square of the interfering signal's amplitude*. Therefore, to minimize RFI rectification in precision amplifiers, the level of interference must be reduced or eliminated, *prior to the stage*. The most direct way to reduce or eliminate the unwanted noise is by proper filtering.

This topic is covered in the section immediately following.

Reducing RFI rectification within op amp and in-amp circuits

EMI and RFI can seriously affect the dc performance of high accuracy analog circuits. Because of their relatively low bandwidth, precision op amps and in-amps simply won't accurately amplify RF signals in the MHz range. However, if these out-of-band signals are allowed to couple into a precision amplifier through either its input, output, or power supply pins, they can be internally rectified by various amplifier junctions, ultimately causing an undesirable dc offset at the output. The previous theoretical discussion of this phenomenon has shown its basic mechanisms. The logical next step is to show how proper filtering can minimize or eliminate these errors.

Elsewhere in this chapter we have discussed how proper supply decoupling minimizes RFI on IC power pins. Further discussion is required with respect to the amplifier inputs and outputs, *at the device level*. It is assumed at this point that system level EMI/RFI approaches have already been implemented, such as an RFI-tight enclosure, properly grounded shields, power rail filtering, etc. The steps following can be considered as circuit-level EMI/RFI prevention.

Op amp inputs

The best way to prevent input stage rectification is to use a low-pass filter located close to the op amp input as shown in Figure 7-111. In the case of the inverting op amp at the left, filter capacitor C is placed between equal-value resistors R1-R2. This results in a simple corner frequency expression, as shown in the figure. At very low frequencies or dc, the closed loop gain of the circuit is $-R3/(R1+R2)$. Note that C cannot be connected directly to the inverting input of the op amp, since that would cause instability. The filter bandwidth can be chosen at least 100 times the signal bandwidth to minimize signal loss.

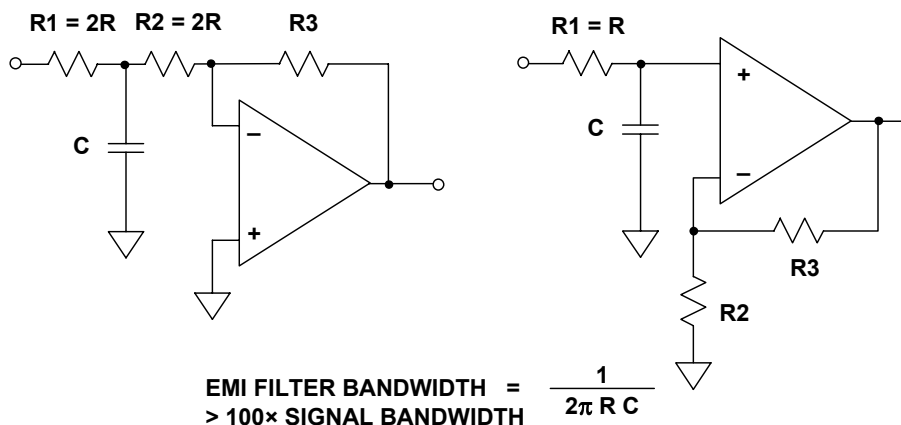


Figure 7-111: Simple EMI/RFI noise filters for op amp circuits

For the non-inverting case on the right, capacitor C can be connected directly to the op amp input as shown, and an input resistor with a value "R" yields the same corner frequency as the inverting case. In both cases low inductance chip-style capacitors should be used, such as NP0 ceramics. The capacitor should in any case be free of losses or voltage coefficient problems, which limits it to either the NP0 mentioned, or a film type.

It should be noted that a ferrite bead can be used instead of R1, however ferrite bead impedance is not well controlled and is generally no greater than 100Ω at 10MHz to 100MHz. This requires a large value capacitor to attenuate lower frequencies.

In-amp inputs

Precision in-amps are particularly sensitive to dc offset errors due to the presence of CM EMI/RFI. This is very much like the problem in op amps. And, as is true with op amps, the sensitivity to EMI/RFI is more acute with the lower power in-amp devices.

A general-purpose approach to proper filtering for device level application of in-amps is shown in Figure 7-112. In this circuit the in-amp could in practice be any one of a number of devices. The relatively complex balanced RC filter preceding the in-amp performs all of the high frequency filtering. The in-amp would be programmed for the gain required in the application, via its gain-set resistance (not shown).

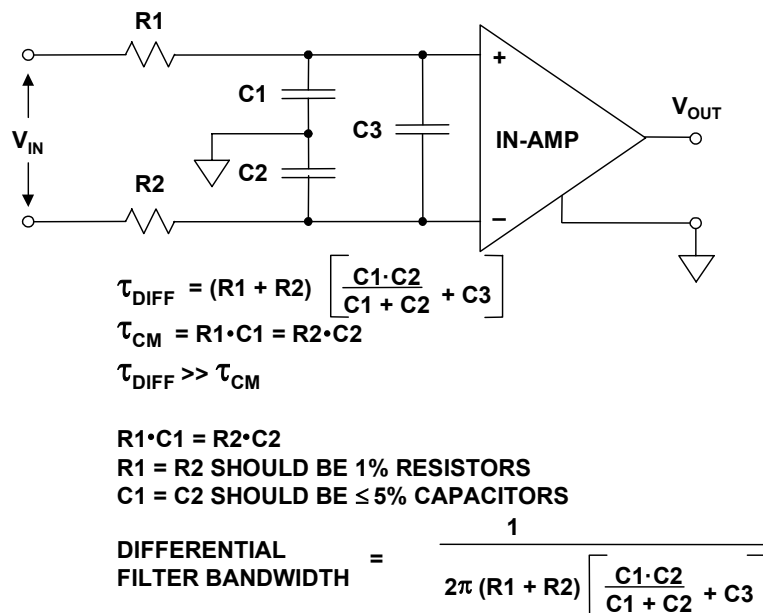


Figure 7-112: A general-purpose common-mode/differential-mode RC EMI/RFI filter for in-amps

Within the filter, note that fully balanced filtering is provided for both CM (R1-C1 and R2-C2) as well as differential mode (DM) signals (R1+R2, and C3 || the series connection of C1-C2). If R1-R2 and C1-C2 aren't well matched, some of the input common-mode signal at V_{IN} will be converted to a differential mode signal at the in-amp inputs. For this reason, C1 and C2 should be matched to within at least 5% of each other. Also, R1 and R2 should be 1% metal film resistors, so as to aid this matching. It is assumed that the source resistances seen at the V_{IN} terminals are low with respect to R1-R2, and matched. In this type of filter, C3 should be chosen much larger than C1 or C2 (C3 ≥ C1, C2), in order to suppress spurious differential signals due to CM⇒DM conversion resulting from mismatch of the R1-C1 and R2-C2 time constants.

OP AMP APPLICATIONS

The overall filter bandwidth should be at least 100 times the input signal bandwidth. Physically, the filter components should be symmetrically mounted on a PC board with a large area ground plane and placed close to the in-amp inputs for optimum performance.

Figure 7-113 shows a family of these filters, as suited to a range of different in-amps. The RC components should be tailored to the different in-amp devices, as per the table. These filter components are selected for a reasonable balance of low EMI/RFI sensitivity and a low increase in noise (vis-à-vis that of the related in-amp, without the filter).

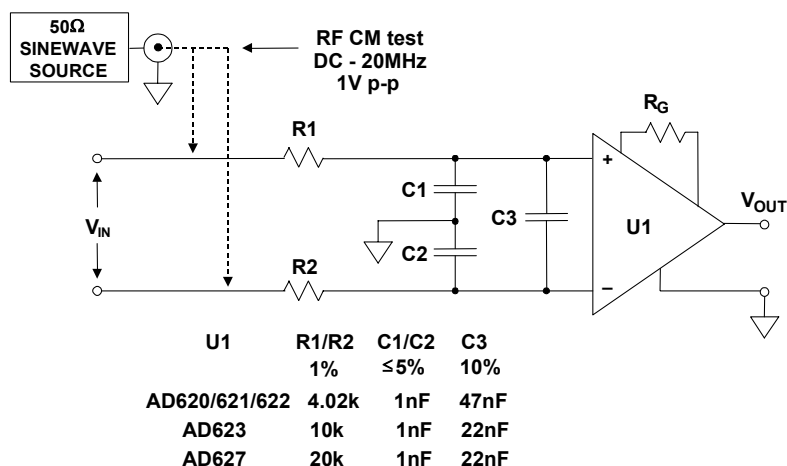


Figure 7-113: Flexible common-mode and differential-mode RC EMI/RFI filters are useful with the AD620 series, the AD623, AD627, and other in-amps

To test the EMI/RFI sensitivity of the configuration, a 1V_{p-p} CM signal can be applied to the input resistors, as noted. With a typically used in-amp such as the AD620 working at a gain of 1000, the maximum RTI input offset voltage shift observed was 1.5μV over the 20MHz range. In the AD620 filter example, the differential bandwidth is about 400 Hz.

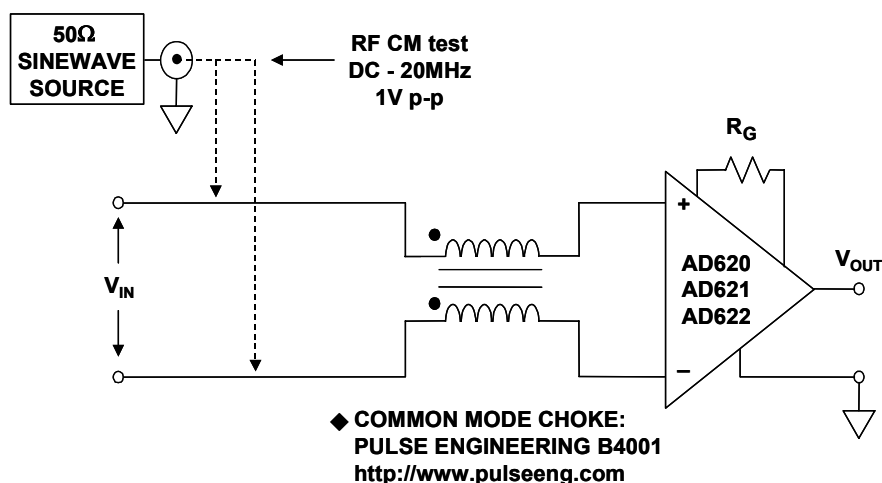


Figure 7-114: For simplicity as well as lowest noise EMI/RFI filter operation, a common-mode choke is useful with the AD620 series in-amp devices

Common-mode chokes offer a simple, one-component EMI/RFI protection alternative to the passive RC filters, as shown in Figure 7-114.

In addition to being a low component count approach, choke-based filters offer low noise, by dispensing with the resistances. Selecting the proper common-mode choke is critical, however. The choke used in the circuit of Figure 7-114 is a Pulse Engineering B4001. The maximum RTI offset shift measured from DC to 20MHz at $G = 1000$ was $4.5\mu\text{V}$. Either an off-the-shelf choke such as the B4001 can be used for this filter, or, alternately one can be constructed. Since balance of the windings is important, bifilar wire is suggested. The core material must of course operate over the expected frequency band. Note that, unlike the Figure 7-113 family of RC filters, a choke-only filter offers no differential filtration. Differential mode filtering can be optionally added, with a second stage following the choke, by adding the R1-C3-R2 connections of Figure 7-112.

For further information on in-amp EMI/RFI filtering, see References 10, and 12 - 15.

Amplifier outputs and EMI/RFI

In addition to filtering the input and power pins, amplifier *outputs* also need to be protected from EMI/RFI, especially if they must drive long lengths of cable, which act as antennas. RF signals received on an output line can couple back into the amplifier input where it is rectified, and appears again on the output as an offset shift.

A resistor and/or ferrite bead, or both, in series with the output is the simplest and least expensive output filter, as shown in Figure 7-115 (upper circuit).

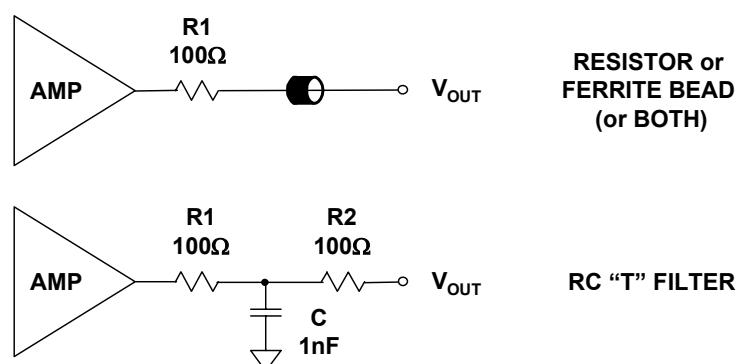


Figure 7-115: *Op amp and in-amp outputs should be protected against EMI/RFI, particularly if they drive long cables.*

Adding a resistor-capacitor-resistor "T" circuit as shown in Figure 7-115 (lower circuit) improves this filter with just slightly more complexity. The output resistor and capacitor divert most of the high frequency energy away from the amplifier, making this configuration useful even with low power active devices. Of course, the time constant of the filter parts must be chosen carefully, to minimize any degradation of the desired output signal. In this case the RC components are chosen for an approximate 3MHz signal bandwidth, suitable for instrumentation or other low bandwidth stages.

Printed Circuit Board Design for EMI/RFI Protection

This section summarizes general points on EMI/RFI with respect to the printed circuit board (PCB) layout. It complements earlier chapter discussions on general PCB design techniques.

When a PCB design has not been optimized in terms of EMI/RFI, system performance can be compromised. This is true not only for signal-path performance, but also for the system's susceptibility to EMI, plus the degree of EMI radiated by the system. Failure to implement sound PCB layout techniques will very likely lead to system/instrument EMC failures.

To summarize earlier points of this section, a real-world PCB layout may allow multiple paths through which high-frequency noise can couple/radiate into and/or out of the circuit. This is especially true for digital circuitry, operating at high *edge rates*. It is the rapid changes of logic state ($1 \Rightarrow 0$ or $0 \Rightarrow 1$), i.e., the edge rate which contains the HF energy which can easily radiate as EMI. While similar points are applicable to precision high-speed analog or mixed analog/digital circuits, logic devices are by far the worst potential EMI offenders. Identifying critical circuits and paths helps in designing the PCB for both low emissions and susceptibility to radiated and conducted external and internal noise sources.

Choose logic devices carefully!

Logic family speaking, a key point in minimizing system noise problems is to *choose devices no faster than actually required by the application*. Many designers assume that faster is always better—fast logic is better than slow, high bandwidth amplifiers better than low bandwidth ones, and fast DACs and ADCs are better, even if the speed isn't required by the system. Unfortunately, faster is *not* better, and actually may be worse for EMI concerns.

Many fast DACs and ADCs have digital inputs and outputs with edge rates in the 1ns/V region. Because of this wide bandwidth, the sampling clock and the digital inputs can respond to any form of high frequency noise, even glitches as narrow as 1 to 3ns. These high speed data converters and amplifiers are thus easy prey for the high frequency noise of microprocessors, digital signal processors, motors, switching regulators, hand-held radios, electric jackhammers, etc. With some of these high-speed devices, a small amount of input/output filtering may be required to desensitize the circuit from its EMI/RFI environment. A ferrite bead just before the local decoupling capacitor is very effective in filtering high frequency noise on supply lines. Of course, with circuits requiring bipolar supplies, this technique should be applied to both positive and negative supply lines.

To help reduce emissions generated by extremely fast moving digital signals at DAC inputs or ADC outputs, a small resistor or ferrite bead may be required at each digital input/output.

Design PCBs thoughtfully

Once the system's critical paths and circuits have been identified, the next step in implementing sound PCB layout is to partition the printed circuit board according to circuit function. This involves the appropriate use of power, ground, and signal planes. Good PCB layouts also isolate critical analog paths from sources of high interference (I/O lines and connectors, for example). High frequency circuits (analog and digital) should be separated from low frequency ones. Furthermore, automatic signal routing

CAD layout software should be used with extreme caution. Critical signal paths should be routed by hand, to avoid undesired coupling and/or emissions.

Properly designed multilayer PCBs can reduce EMI emissions and increase immunity to RF fields, by a factor of 10 or more, compared to double-sided boards. A multilayer board allows a complete layer to be used for the ground plane, whereas the ground plane side of a double-sided board is often disrupted with signal crossovers, etc. If the system has separate analog and digital ground and power planes, the analog ground plane should be underneath the analog power plane, and similarly, the digital ground plane should be underneath the digital power plane. There should be no overlap between analog and digital ground planes, nor analog and digital power planes.

Designing Controlled Impedances Traces on PCBs

A variety of trace geometries are possible with controlled impedance designs, and they may be either integral to or allied to the PCB pattern. In the discussions below, the basic patterns follow those of the IPC, as described in standard 2141 (see Reference 16).

Note that the figures below use the term "ground plane". It should be understood that this plane is in fact a large area, low impedance *reference* plane. In practice it may actually be either a ground plane or a power plane, both of which are assumed to be at zero ac potential.

The first of these is the simple wire-over-a-plane form of transmission line, also called a *wire microstrip*. A cross-sectional view is shown in Figure 7-116. This type of transmission line might be a signal wire used within a breadboard, for example. It is composed simply of a discrete insulated wire spaced a fixed distance over a ground plane. The dielectric would be either the insulation wall of the wire, or a combination of this insulation and air.

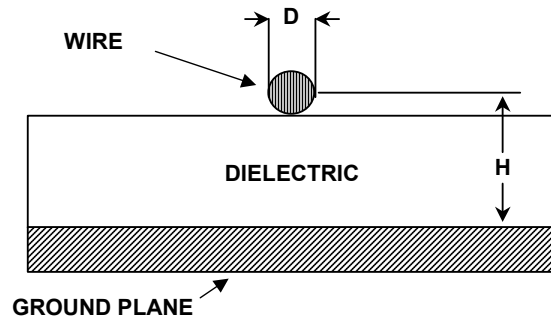


Figure 7-116: A wire microstrip transmission line with defined impedance is formed by an insulated wire spaced from a ground plane

The impedance of this line in ohms can be estimated with Eq. 7-16. Here D is the conductor diameter, H the wire spacing above the plane, and ϵ_r the dielectric constant.

$$Z_o(\Omega) = \frac{60}{\sqrt{\epsilon_r}} \ln \left[\frac{4H}{D} \right] \quad \text{Eq. 7-16}$$

For patterns integral to the PCB, there are a variety of geometric models from which to choose, single-ended and differential. These are covered in some detail within IPC standard 2141 (see Reference 16), but information on two popular examples is shown here.

■ OP AMP APPLICATIONS

Before beginning any PCB-based transmission line design, it should be understood that there are abundant equations, all claiming to cover such designs. In this context, "Which of these are accurate?" is an extremely pertinent question. The unfortunate answer is, *none are perfectly so!* All of the existing equations are approximations, and thus accurate to varying degrees, depending upon specifics. The best known and most widely quoted equations are those of Reference 16, but even these come with application caveats.

Reference 17 has evaluated the Reference 16 equations for various geometric patterns against test PCB samples, finding that predicted accuracy varies according to target impedance. Reference 18 also evaluates the Reference 16 equations, offering an alternative and even more complex set (see Reference 19). The equations quoted below are from Reference 16, and are offered here as a starting point for a design, subject to further analysis, testing and design verification. The bottom line is, study carefully, and take PCB trace impedance equations with a proper dose of salt.

Microstrip PCB transmission lines

For a simple two-sided PCB design where one side is a ground plane, a signal trace on the other side can be designed for controlled impedance. This geometry is known as a *surface microstrip*, or more simply, *microstrip*.

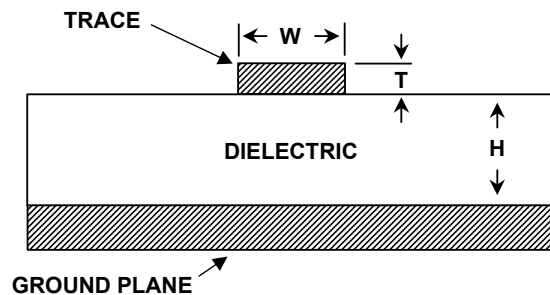


Figure 7-117: A microstrip transmission line with defined impedance is formed by a PCB trace of appropriate geometry, spaced from a ground plane

A cross-sectional view of a two-layer PCB illustrates this microstrip geometry as shown in Figure 7-117.

For a given PCB laminate and copper weight, note that all parameters will be predetermined except for W , the width of the signal trace. Eq. 7-17 can then be used to design a PCB trace to match the impedance required by the circuit. For the signal trace of width W and thickness T , separated by distance H from a ground (or power) plane by a PCB dielectric with dielectric constant ϵ_r , the characteristic impedance is:

$$Z_o(\Omega) = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left[\frac{5.98H}{(0.8W + T)} \right] \quad \text{Eq. 7-17}$$

Note that in these expressions, measurements are in common dimensions (mils).

These transmission lines will have not only a characteristic impedance, but also capacitance. This can be calculated in terms of pF/in as shown in Eq. 7-18.

$$C_o (\text{pF/in}) = \frac{0.67(\epsilon_r + 1.41)}{\ln[5.98H/(0.8W + T)]} \quad \text{Eq. 7-18}$$

As an example including these calculations, a 2-layer board might use 20-mil wide (W), 1 ounce (T=1.4) copper traces separated by 10-mil (H) FR-4 ($\epsilon_r=4.0$) dielectric material. The resulting impedance for this microstrip would be about 50Ω. For other standard impedances, for example the 75Ω video standard, adjust "W" to about 8.3 mils.

Some Microstrip rules-of-thumb

This example touches an interesting and quite handy point. Reference 17 discusses a useful rule of thumb pertaining to microstrip PCB impedance. For a case of dielectric constant of 4.0 (FR-4), it turns out that when W/H is 2/1, the resulting impedance will be close to 50Ω (as in the first example, with W=20 mils).

Careful readers will note that Eq. 7-17 predicts Z_o to be about 46Ω, generally consistent with accuracy quoted in Reference 17 (>5%). The IPC microstrip equation is most accurate between 50 and 100Ω, but is substantially less so for lower (or higher) impedances. Reference 20 gives tabular results of various PCB industry impedance calculator tools.

The propagation delay of the microstrip line can also be calculated, as per Eq. 7-19. This is the one-way transit time for a microstrip signal trace. Interestingly, for a given geometry model, *the delay constant in ns/ft is a function only of the dielectric constant, and not the trace dimensions* (see Reference 21). Note that this is quite a convenient situation. It means that, with a given PCB laminate (and given ϵ_r), the propagation delay constant is fixed for various impedance lines.

$$t_{pd} (\text{ns/ft}) = 1.017\sqrt{0.475\epsilon_r + 0.67} \quad \text{Eq. 7-19}$$

This delay constant can also be expressed in terms of ps/in, a form which will be more practical for smaller PCBs. This is:

$$t_{pd} (\text{ps/in}) = 85\sqrt{0.475\epsilon_r + 0.67} \quad \text{Eq. 7-20}$$

Thus for an example PCB dielectric constant of 4.0, it can be noted that a microstrip's delay constant is about 1.63ns/ft, or 136ps/in. These two additional rules-of-thumb can be useful in designing the timing of signals across PCB trace runs.

Symmetric Stripline PCB transmission lines

A method of PCB design preferred from many viewpoints is a multi-layer PCB. This arrangement *embeds* the signal trace between a power and a ground plane, as shown in the cross-sectional view of Figure 7-118. The low-impedance ac-ground planes and the embedded signal trace form a *symmetric stripline* transmission line.

As can be noted from the figure, the return current path for a high frequency signal trace is located directly above and below the signal trace on the ground/power planes. The high frequency signal is thus contained entirely inside the PCB, minimizing emissions, and providing natural shielding against incoming spurious signals.

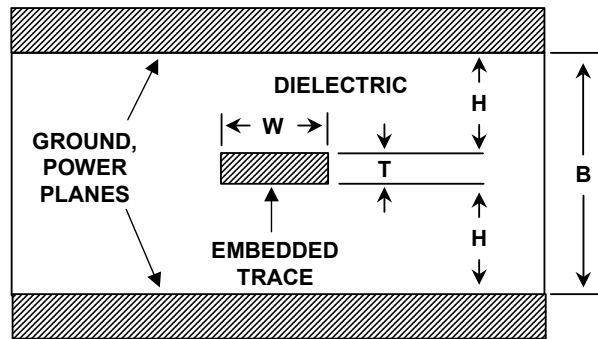


Figure 7-118: A symmetric stripline transmission line with defined impedance is formed by a PCB trace of appropriate geometry embedded between equally spaced ground and/or power planes

The characteristic impedance of this arrangement is again dependent upon geometry and the ϵ_r of the PCB dielectric. An expression for Z_0 of the stripline transmission line is:

$$Z_0 (\Omega) = \frac{60}{\sqrt{\epsilon_r}} \ln \left[\frac{1.9(B)}{(0.8W + T)} \right] \quad \text{Eq. 7-21}$$

Here, all dimensions are again in mils, and B is the spacing between the two planes. In this symmetric geometry, note that B is also equal to $2H + T$. Reference 17 indicates that the accuracy of this Reference 16 equation is typically on the order of 6%.

Another handy rule-of-thumb for the symmetric stripline in an $\epsilon_r = 4.0$ case is to make B a multiple of W, in the range of 2 to 2.2. This will result in an stripline impedance of about 50Ω . Of course this rule is based on a further approximation, by neglecting T. Nevertheless, it is still useful for ballpark estimates.

The symmetric stripline also has a characteristic capacitance, which can be calculated in terms of pF/in as shown in Eq. 7-22.

$$C_0 (\text{pF/in}) = \frac{1.41(\epsilon_r)}{\ln[3.81H/(0.8W + T)]} \quad \text{Eq. 7-22}$$

The propagation delay of the symmetric stripline is shown in eq. 7-23.

$$t_{pd}(\text{ns/ft}) = 1.017\sqrt{\epsilon_r} \quad \text{Eq. 7-23}$$

or, in terms of ps:

$$t_{pd}(\text{ps/in}) = 85\sqrt{\epsilon_r} \quad \text{Eq. 7-24}$$

For a PCB dielectric constant of 4.0, it can be noted that the symmetric stripline's delay constant is almost exactly 2ns/ft, or 170ps/in.

Some Pros and Cons of Embedding Traces

The above discussions allow the design of PCB traces of defined impedance, either on a surface layer or embedded between layers. There of course are many other considerations beyond these impedance issues.

Embedded signals do have one major and obvious disadvantage—the debugging of the hidden circuit traces is difficult to impossible. Some of the pros and cons of embedded signal traces are summarized in Figure 7-119.

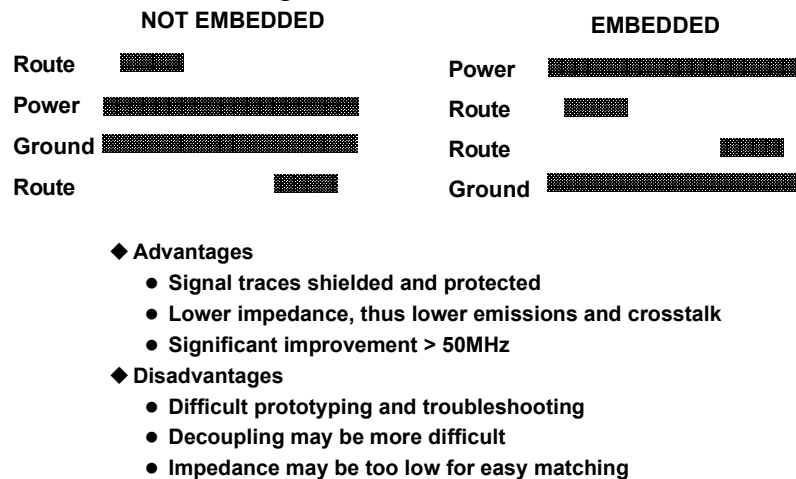


Figure 7-119: *The pros and cons of not embedding vs. the embedding of signal traces in multi-layer PCB designs*

Multi-layer PCBs can be designed *without* the use of embedded traces, as is shown in the left-most cross-sectional example. This embedded case could be considered as a doubled two-layer PCB design (i.e., four copper layers overall). The routed traces at the top form a microstrip with the power plane, while the traces at the bottom form a microstrip with the ground plane. In this example, the signal traces of both outer layers are readily accessible for measurement and troubleshooting purposes. But, the arrangement does nothing to take advantage of the shielding properties of the planes.

This non embedded arrangement will have greater emissions and susceptibility to external signals, vis-a-vis the embedded case at the right, which uses the embedding, and does take full advantage of the planes. As in many other engineering efforts, the decision

■ OP AMP APPLICATIONS

of embedded vs. not-embedded for the PCB design becomes a tradeoff, in this case one of reduced emissions vs. ease of testing.

Transmission Line Termination Rule-of-Thumb

Much has been written about terminating PCB traces in their characteristic impedance, to avoid signal reflections. A good rule-of-thumb to determine when this is necessary is as follows: *Terminate the transmission line in its characteristic impedance when the one-way propagation delay of the PCB track is equal to or greater than one-half the applied signal rise/fall time (whichever edge is faster)*. For example, a 2 inch microstrip line over an $\epsilon_r=4.0$ dielectric would have a delay of $\sim 270\text{ps}$. Using the above rule strictly, termination would be appropriate whenever the signal rise time is $< \sim 500\text{ps}$. A more conservative rule is to use a 2 inch (PCB track length)/nanosecond (rise/fall time) rule. If the signal trace exceeds this trace-length/speed criterion, then termination should be used.

For example, PCB tracks for high-speed logic with rise/fall time of 5ns should be terminated in their characteristic impedance if the track length is equal to or greater than 10 inches (where measured length *includes* meanders).

In the analog domain, it is important to note that this same 2 inch/nanosecond rule of thumb should also be used with op amps and other circuits, to determine the need for transmission line techniques. For instance, if an amplifier must output a maximum frequency of f_{max} , then the equivalent risetime t_r is related to this f_{max} . This limiting risetime, t_r , can be calculated as:

$$t_r = 0.35/f_{\text{max}} \quad \text{Eq. 7-25}$$

The maximum PCB track length is then calculated by multiplying t_r by 2 inch/nanosecond. For example, a maximum frequency of 100MHz corresponds to a risetime of 3.5ns, so a 7 inch or more track carrying this signal should be treated as a transmission line.

The best ways to keep sensitive analog circuits from being affected by fast logic are to physically separate the two by the PCB layout, and to use no faster logic family than is dictated by system requirements. In some cases, this may require the use of several logic families in a system. An alternative is to use series resistance or ferrite beads to slow down the logic transitions where highest speed isn't required.

A general method of doing this is to use a series R at a logic driver output, and a shunt C at a CMOS gate input. The series resistance and the net input capacitance of the gate form a lowpass filter. Typical CMOS input capacitance is 10pF. Locate the series resistor close to the driving gate, adding an additional small capacitance, as needed. The resistor minimizes transient switching currents, and may also eliminate the necessity for transmission line techniques. The value of the resistor should be chosen such that the rise and fall times at the receiving gate are fast enough to meet system requirement, but no faster. Also, make sure that the resistor is not so large that the logic levels at the receiver are out of specification because of the source and sink current which must flow through the resistor. Use of CMOS logic will simplify this, since the input currents are so low.

REFERENCES: EMI/RFI CONSIDERATIONS

1. Tim Williams, **EMC for Product Designers**, 2nd Ed., Newnes, Oxford, 1996, ISBN: 0 7506 2466 3.
2. Henry Ott, **Noise Reduction Techniques In Electronic Systems**, 2nd Ed., John Wiley & Sons, New York, 1988, ISBN 0-471-85068-3.
3. Mark Montrose, **EMC and the Printed Circuit Board**, IEEE Press, 1999, ISBN 0-7803-4703-X.
4. Ralph Morrison, **Grounding And Shielding Techniques in Instrumentation**, 3rd Ed., John Wiley & Sons, New York, 1986, ISBN 0-471-83805-5.
5. Daryl Gerke and William Kimmel, "Designer's Guide to Electromagnetic Compatibility," **EDN**, January 20, 1994.
6. **Designing for EMC** (Workshop Notes), Kimmel Gerke Associates, Ltd., 1994.
7. Daryl Gerke and William Kimmel, "EMI and Circuit Components," **EDN**, September 1, 2000.
8. Alan Rich, "Understanding Interference-Type Noise," **Analog Dialogue**, Vol. 16, No. 3, 1982, pp. 16-19 (also available as application note AN346).
9. Alan Rich, "Shielding and Guarding," **Analog Dialogue**, Vol. 17, No. 1, 1983, pp. 8-13 (also available as application note AN347).
10. James Wong, Joe Buxton, Adolfo Garcia, James Bryant, "Filtering and Protection Against EMI/RFI" and "Input Stage RFI Rectification Sensitivity", Chapter 1, pg. 21-55 of **Systems Application Guide**, 1993, Analog Devices, Inc., Norwood, MA, ISBN 0-916550-13-3.
11. Adolfo Garcia, "EMI/RFI Considerations", Chapter 7, pg 69-88 of **High Speed Design Techniques**, 1996, Analog Devices, Inc., Norwood, MA, 1993, ISBN 0-916550-17-6.
12. Walt Kester, Walt Jung, Chuck Kitchen, "Preventing RFI Rectification", Chapter 10, pg 10.39-10.43 of **Practical Design Techniques for Sensor Signal Conditioning**, Analog Devices, Inc., Norwood, MA, 1999, ISBN 0-916550-20-6.
13. Charles Kitchen, Lew Counts, **A Designer's Guide to Instrumentation Amplifiers**, Analog Devices, Inc., 2000.
14. **B4001 and B4003 common-mode chokes**, Pulse Engineering, Inc., 12220 World Trade Drive, San Diego, CA, 92128, 619-674-8100, <http://www.pulseeng.com>
15. **Understanding Common Mode Noise**, Pulse Engineering, Inc., 12220 World Trade Drive, San Diego, CA, 92128, 619-674-8100, <http://www.pulseeng.com>
16. Standard IPC-2141, "Controlled Impedance Circuit Boards and High Speed Logic Design," 1996, Institute for Interconnection and Packaging Electronic Circuits, 2215 Sanders Road, Northbrook, IL, 60062-6135, (847) 509-9700, <http://www.ipc.org>
17. Eric Bogatin, "Verifying the Accuracy of 2D Field Solvers for Characteristic Impedance Calculation," Ansoft Seminar, October 11, 2000, <http://www.bogatinterprises.com>
18. Andrew Burkhardt, Christopher Gregg, Alan Staniforth, "Calculation of PCB Track Impedance," Technical Paper S-19-5, presented at the IPC Printed Circuits Expo '99 Conference, March 14-18, 1999

▣ OP AMP APPLICATIONS

19. Brian C. Wadell, **Transmission Line Design Handbook**, Artech House, Norwood, MA, 1991, ISBN: 0-89006-436-9
20. Eric Bogatin, "No Myths Allowed, "Impedance Calculations", " a Chip Center column", November 1, 1999, <http://www.chipcenter.com/signalintegrity>
21. William R. Blood, Jr., **MECL System Design Handbook** (HB205/D, Rev. 1A May 1988), ON Semiconductor, August, 2000.
22. Paul Brokaw, "An IC Amplifier User Guide To Decoupling, Grounding, And Making Things Go Right For A Change," **Analog Devices AN202**.

Some useful EMC and signal integrity related URLs:

Eric Bogatin website, <http://www.bogatinenterprises.com>

Chip Center's "Signal Integrity" page, <http://www.chipcenter.com/signalintegrity>

Kimmel Gerke Associates website, <http://www.emiguru.com>

Henry Ott website, <http://www.hottconsultants.com>

IEEE EMC website, <http://www.ewh.ieee.org/soc/emcs>

Mark Montrose website, <http://www.montrosecompliance.com/index.html>

Tim Williams website, <http://www.elmac.co.uk>

ACKNOWLEDGMENTS:

Eric Bogatin made helpful comments on this section, which were very much appreciated.

SECTION 7-7: SIMULATION, BREADBOARDING AND PROTOTYPING

Joe Buxton, Walt Kester, James Bryant, Walt Jung

In this final section of the chapter, the practical aspects of assembling hardware for op amp and other analog functions are brought to play. Various experimental techniques are useful towards verifying the integrity of a design. These include electronic *analog circuit simulation* programs, to be used with (but not to the exclusion of) the allied lab processes of *breadboarding* and *prototyping*.

Analog Circuit Simulation

In the past decade, circuit simulation has taken on an increasingly important role within analog circuit design. The most popular simulation tool for this is SPICE, which is available in multiple forms for various computer platforms (see References 1 and 2). However, to achieve meaningful simulation results, designers need accurate models of many system components. The most critical of these are realistic models for ICs, the active devices that drive modern designs. In the early 1990's, Analog Devices developed an advanced op amp SPICE model, which is in fact still in use today (see References 3 and 4). Within this innovative open amplifier architecture, gain and phase response can be fully modeled, enabling designers to accurately predict ac, dc, and transient performance behavior. This modeling methodology has also been extended to include other devices such as in-amps, voltage references, and analog multipliers.

- ◆ Understand Realistic Simulation Goals
- ◆ Evaluate Available Models Accordingly
- ◆ Know the Capabilities for Each Competing Op Amp Model
- ◆ Following Simulation, *Breadboarding is Always Desirable and Necessary*

Figure 7-120: *Used wisely, simulation is a powerful design tool*

Figure 7-120 lists some major SPICE simulation objectives. The popularity of SPICE simulation has led to many op amp macromodel releases, which (ideally) software-mimic amplifier electrical performance. With numerous models available, several confusions are possible. There may be uncertainty as to what is/isn't modeled, plus a basic question of *model accuracy*. All of these points are important, in order to place confidence in simulation results. So, *verification* of a model is important, checking it by comparison to the actual device performance conditions, before trusting it for serious designs.

Of course, a successful first design step using an accurate op amp model by itself doesn't necessarily guarantee totally valid simulations. A simulation based on incomplete information has limited value. All parts of a target circuit should be modeled, including the surrounding passive components, various parasitic effects, and temperature changes. Then, the circuit needs to be verified in the lab, by breadboarding and prototyping. A breadboard circuit is a quickly executed mockup of a circuit design using a semi-permanent lab platform, i.e., one which is less than final physical form. It is intended to show real performance, but without the total physical environment. A good breadboard can often reveal behavior not predicted by SPICE, either because of an incomplete

■ OP AMP APPLICATIONS

model, external circuit parasitics, or numerous other reasons. However, by using SPICE along with intelligent breadboarding techniques, a circuit can be efficiently and quickly designed with reasonably good assurance of working properly on a prototype version, or even a final PCB. The following prototype phase is just one step removed from a final PCB, and may in fact be an actual test PCB, with nearly all design components incorporated, and with close to full performance.

The breadboard/prototype design steps are closely allied to simulation, usually following it in the overall design process. These are more fully discussed in subsequent sections.

Macromodel vs. Micromodel

The distinction between *macromodel* and *micromodel* is often unclear. A micromodel uses the actual *transistor level* and other SPICE models of an IC device, with all active and passive parts fully characterized according to the manufacturing process. In differentiating this type of model from a macromodel, some authors use the term *device level model* to describe the resulting overall op amp model (see Reference 5). Typically, a micromodel is used in the actual design process of an IC.

	METHODOLOGY	ADVANTAGES	DISADVANTAGES
MACROMODEL	Ideal Elements Model Device Behavior	Fast Simulation Time, Easily Modified	May Not Model All Characteristics
MICROMODEL	Fully Characterized Transistor Level Circuit	Most Complete Model	Slow Simulation, Convergence Difficulty, Non-Availability

Figure 7-121: Differentiating the Macromodel and Micromodel

A macromodel goes another route in emulating op amp performance. Taking into consideration final device performance, it uses ideal native SPICE elements to model observed behavior—as many as necessary. In developing a macromodel, a real device is measured in terms of lab and data sheet performance, and the macromodel is adjusted to match this behavior. Some aspects of performance may be sacrificed in doing this. Figure 7-121 compares the major pros and cons between macromodels and micromodels.

There are advantages and disadvantages to both approaches. A micromodel can give a complete and accurate model of op amp circuit behavior under almost all conditions. But, because of a large number of transistors and diodes with non-linear junctions, simulation time is very long. Of course, manufacturers are also reluctant to release such models, since they contain proprietary information. And, even though all transistors may be included, this isn't a guarantee of total accuracy, as the transistor models themselves don't cover all operational regions precisely. Furthermore, with a high node count, SPICE can have convergence difficulties, causing a failed simulation. This point would make a micromodel virtually useless for multiple amplifier active filters, for example.

On the other hand, a carefully developed macromodel can produce both accurate results and simulation time savings. In more advanced macromodels such as the ADSpice model

described below, transient and ac device performance can be closely replicated. Op amp non-linear behavior can also be included, such as output voltage and current swing limits.

However, because these macromodels are still simplifications of real devices, all non-linearities aren't modeled. For example, not all ADSpice models include common mode input voltage range, or noise (while more recent ones do). Typically, in model development parameters are optimized as may be critical to the intended application—for example, ac and transient response. Including every possible characteristic could lead to cumbersome macromodels that may even have convergence problems. Thus, ADSpice macromodels include those op amp behavior characteristics critical to intended performance for normal operating conditions, but not necessarily all non-linear behavior.

The ADSpice Op Amp Macromodels

The basic ADSpice model was developed as an op amp macromodeling advance, and as an improved design tool for more accurate application circuit simulations. Since being introduced in 1990, it has become a standard op amp macromodel topology, as evidenced by industry adoption of the frequency shaping concepts (see References 6 and 7).

Prior to about 1990, a dominant op amp model architecture was the Boyle model (see Reference 8). This macromodel, developed in the early 70's, cannot accurately model higher speed amplifiers. The primary reason for this is that it has limited frequency shaping ability—only two poles and no zeroes. In contrast, the ADSpice model topology has a flexible and open architecture, allowing virtually unlimited pole and zero frequency shaping stages to be cascaded. This key difference provides much more accurate ac and transient response, vis-à-vis the more simplistic Boyle model topology.

An ADSpice model is comprised of three main portions, described as follows. The first of these is a combined input and gain stage, which will include transistor models as appropriate to the device being modeled (NPN or PNP bipolar, JFET, MOSFET, etc.). Next are the synthetic pole and zero stages, which are comprised of ideal SPICE native elements. There may be only a few of these or there may be many, dependent on the complexity of the op amp's frequency response. Finally, there is an output stage, which couples the first two sections to the outside world.

Before describing these sections in detail, it is important to realize that many variations upon what is shown below do in fact exist. This is due to not just differences from one op amp model to another, but also to evolutionary topology developments in op amp hardware, which in turn has led to corresponding modeling changes. For example, modern op amps often include either rail-rail output or input stages, or both. Consequently more recent developments in the ADSpice models have addressed these issues, along with corresponding model developments.

Furthermore, although the Boyle model and the original ADSpice models were designed to support *voltage feedback* op amp topologies, subsequent additions have added *current feedback* amplifier topologies. In fact, Reference 9 describes an ADSpice current feedback macromodel which appeared just shortly after the voltage feedback model of Reference 3. These current feedback macromodels are discussed in more detail below.

Input and Gain/Pole stages

A basic ADSpice voltage feedback op amp macromodel input stage is shown in Figure 7-122. As noted, it uses what are (typically) the only transistors in the entire model, in this example the Q1-Q2 NPN pair, to the left on the diagram. These are needed to properly model an op amp's differential input stage characteristics. A basic tenet of this model topology is that this stage is designed for unity gain, by the proper choice of Q1-Q2 operating current and gain-setting resistors R3-R4 and R5-R6.

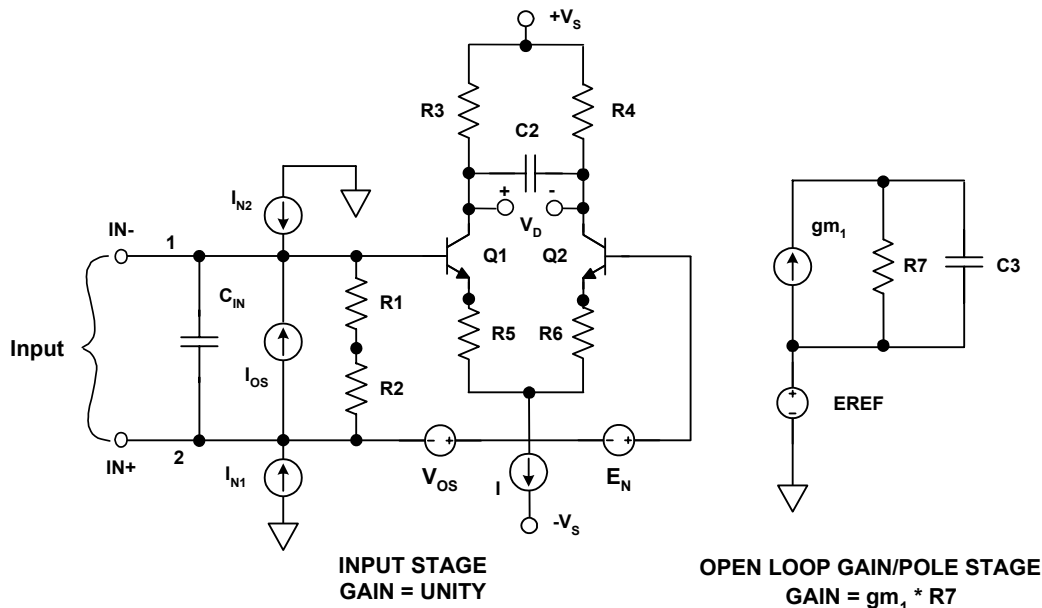


Figure 7-122: Input and gain/pole stages of ADSpice macromodel

Although this example uses NPN transistors, the input stage is easily modified to use PNP bipolars, JFET, or MOSFET devices. The rest of the input stage uses simple SPICE elements such as resistors, capacitors, and controlled sources.

The open-loop gain vs. frequency characteristics of the modeled op amp is provided by the gain stage, to the right in the diagram. Here controlled source gm_1 senses the differential collector voltage V_D from the input stage, converting this voltage to a proportional current. The gm_1 output current flows in load resistor R_7 , producing a single ended voltage referenced to an internal voltage, E_{REF} . Typically, this voltage is derived as a supply voltage midpoint, and is used throughout the model.

By simply making the gm_1 - R_7 product equal to the specified gain of the op amp, this stage produces the entire open-loop gain of the macromodel. This design factor means that all other model stages operate at unity gain, a feature leading to significant flexibility in adding and deleting subsequent stages. This approach allows the quick synthesis of the complex ac characteristics typical of high performance, high speed op amps. In addition, this stage also provides the dominant pole of the amplifier's ac response. The open-loop pole frequency is set by selection of capacitor C_3 , as noted in the diagram.

Frequency Shaping Stages

Following the gain stage of the macromodel is a variable but unlimited number of pole and/or zero stages, which in combination provide frequency response shaping. Typical topologies for these stages are as shown in the Figure 7-123 diagram. The stages can be either a single pole or a single zero, or combined pole/zero or zero/pole stages. All such stages have a dc transfer gain of unity, and a given amplifier type can have all or just a few of these stages, as may be require to synthesize its response.

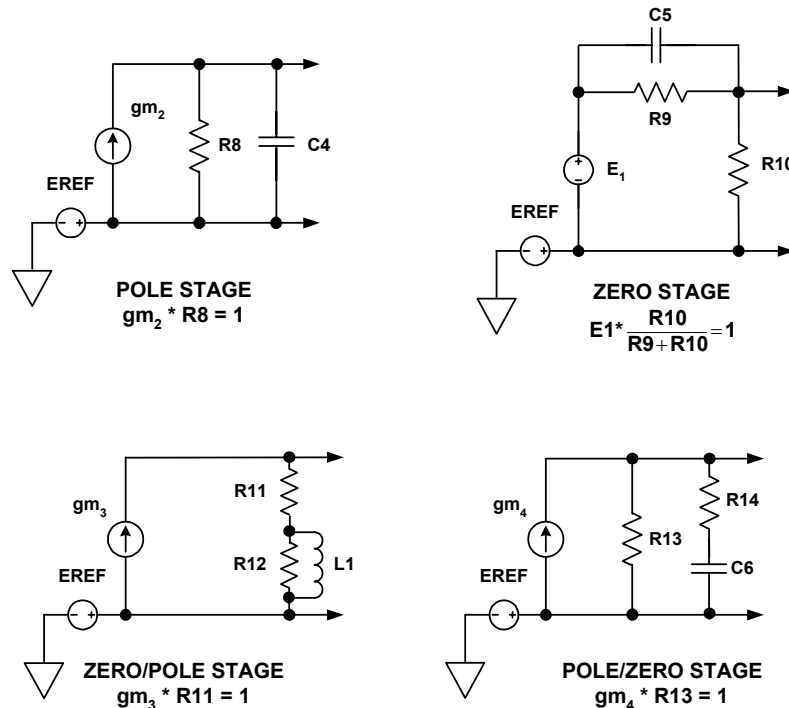


Figure 7-123: The frequency shaping stages possible within the ADSpice model

The pole or zero frequency is set by the combination of the resistor(s) and capacitor, or resistor(s) and inductor, as may be the case. Because an infinite number of values are possible in SPICE, choice of RC values is somewhat arbitrary, and a wide range work. Early ADSpice models used relatively high values, while later ones employ lower values to reduce noise (described in more detail later). In all instances, it is assumed that each stage provides zero loading to the driving stage. The stages shown reflect no particular op amp, but example principles can be found within the OP27 model (see Reference 10).

Because all of these frequency-shaping stages are dc-coupled and have unity gain, any number of them can be added or deleted, with no affect on the model's low frequency response. Most importantly, the high frequency gain and phase response can be precisely tailored to match a real amplifier's response. The benefits of this frequency-shaping flexibility are especially apparent in performance comparisons of the ADSpice model closed loop pulse response and stability analysis, versus that of a more simplistic model. This point is demonstrated by a later example.

Macromodel output stages

A general form of the output stage for the ADSpice model, shown in Figure 7-124, models a number of important op amp characteristics. The Thevenin equivalent resistance of R_{O1} and R_{O2} mimics the op amp's dc open loop output impedance, while inductor L_O models the rise in impedance at high frequencies. A unity gain characteristic for the stage is set by the g_7 - R_{O1} and g_8 - R_{O2} products.

Additionally, output load current is correctly reflected in the supply currents. This feature is a significant improvement over the Boyle model, because the power consumption of the loaded circuit can be analyzed accurately. Furthermore, circuits using the op amp supply currents as part of the signal path can also be correctly simulated. The output stage shown is not intended to reflect any particular op amp, but close similarity is found within the AD817 model (see Reference 11).

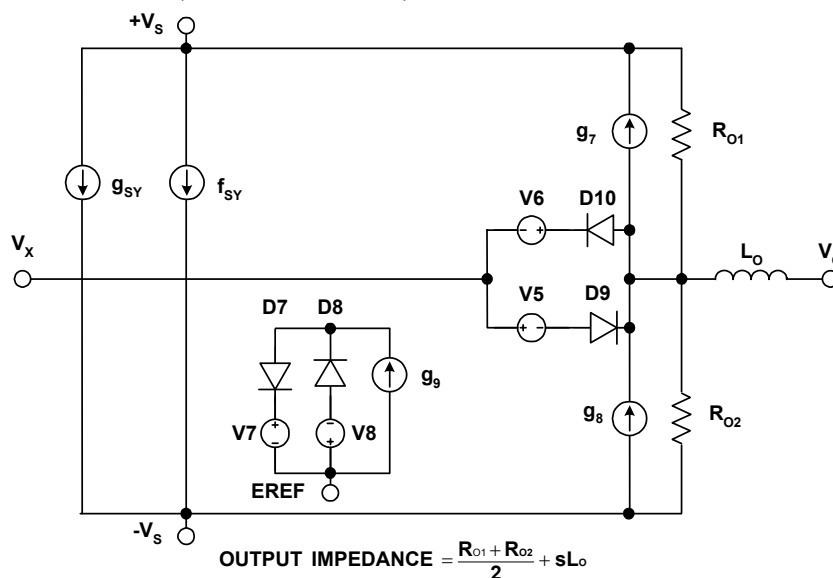


Figure 7-124: General-purpose macromodel output stage

With the recent advent of numerous rail-rail output stage op amps, a number of customized model topologies have been developed. This expands the ADSpice library to include rail-rail model behavior, matching op amp architectures using P and N MOSFET devices, as well as bipolar devices. Characteristically, a rail-rail output stage includes several key differentiating performance points. First and foremost is the ability to swing the op amp output to within a few mV of both supplies. A second point is the fact that such an output stage has a voltage gain greater than one, and a third is the relatively high output impedance (high as contrasted to traditional emitter follower outputs).

Examples of several modeling approaches to rail-rail output stages are found in the ADI SPICE macromodel library. Reference 12 employs CMOS devices to realize a rail-rail output, while Reference 13 uses bipolar devices to the same end. The macromodels of References 14 and 15 use synthesis techniques to model rail-rail outputs. References 16-18 utilize combinations of selected discrete device models and synthesis techniques, to realize rail-rail output operation for both op amp and in-amp devices.

In addition to rail-rail output operation, many modern op amps also feature rail-rail *input stages*. Such stages essentially duplicate, for example, an NPN-based differential stage with a complementary PNP stage, both stages operating in parallel. This allows the op amp to provide a CM range that includes both supply rails. This performance feature can also be accomplished within CMOS op amps, using both a P and N type MOS differential pairs. Model examples reflecting rail-rail input stages include References 13, 14, and 17.

Model transient response

The performance advantage of the multiple pole/zero stages is readily demonstrated in a transient pulse response test, as in Figure 7-125. This figure compares an actual OP249 op amp, the ADSpice model, and the Boyle model. It reveals the improved execution resulting from the unlimited number of poles and zeros in this model.

The difference is easily apparent from this transient analysis plot for a unity gain follower circuit. An OP249 amplifier was used, with the output connected to the inverting input, and a 260pF capacitive load.

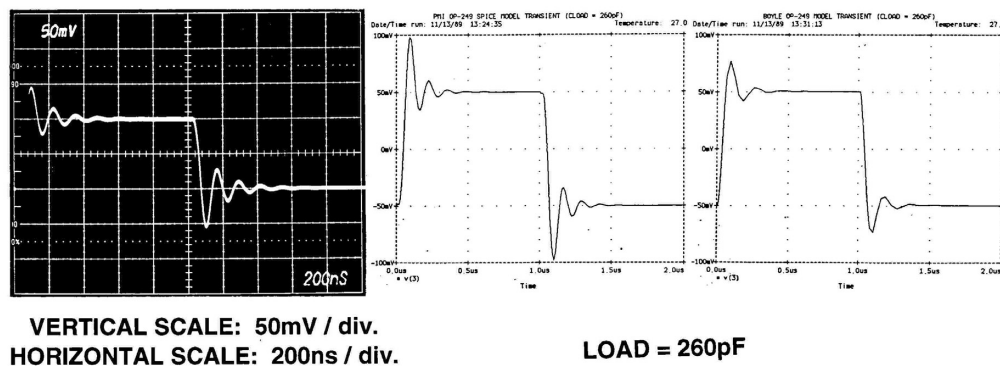


Figure 7-125: A pulse response comparison of an OP249 follower (left) model favors the ADSpice model in terms of fidelity (center), but not the Boyle (right)

As can be noted, this results in ringing, as seen in the op amp response (left). Note that the ADSpice model accurately predicts the amount of overshoot and frequency of the damped ringing (center). In contrast, the Boyle model (right) predicts about half the overshoot and significantly less ringing.

The Noise Model

An important enhancement to the ADSpice model is the ability to realistically model noise performance of an op amp. The capability to model a circuit's noise in SPICE can be appreciated by anyone who has tried to analyze noise by hand. A complete analysis is an involved and tedious task that involves adding all the individual noise contributions from all active devices and all resistors, and referring them to the input.

To aid this task, the ADSpice model was enhanced to include noise generators that accurately mimic the broadband and 1/f noise of an actual op amp. Conceptually, this involves first making an existing model noiseless, and then adding discrete noise generators, so as to emulate the target device. As noted earlier, all ADI models aren't

OP AMP APPLICATIONS

necessarily designed for this noise-accurate performance. Selected device models are designed for noise however, when their typical uses include low noise applications.

The first step is an exercise in scaling down the model internal impedances. For example, by reducing the resistances in the pole/zero stages from a base resistance of $1\text{E}6\Omega$ to 1Ω , total noise is reduced dramatically, as figure 7-126 illustrates.

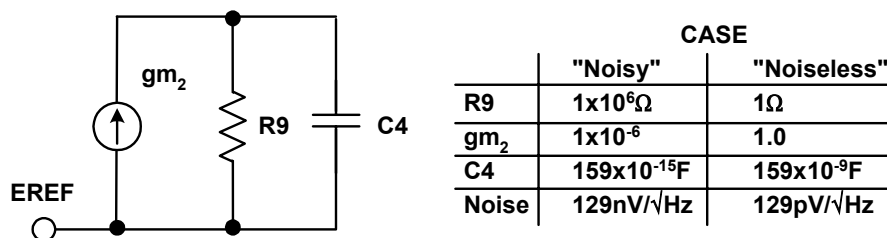


Figure 7-126: Towards achieving low noise operation, a first design step is the reduction of pole/zero cell impedances to low values

For the "Noisy" column of the table, the noise from the pole stage shown with a large R9 resistor value is $129 \text{nV}/\sqrt{\text{Hz}}$. But when this resistor is scaled down by a factor of 10^6 , to 1Ω , as in the "Noiseless" column, stage noise is $129 \text{pV}/\sqrt{\text{Hz}}$. Note also that transconductance and capacitance values are also scaled by the same factor, maintaining the same gain and pole frequency. To make the model's input stage noiseless, it is operated at a high current and with reduced load resistances, making noise contributions negligible. Extending these techniques to the entire model renders it essentially noiseless.

Once global noise reduction is achieved, independent noise sources are added, one for voltage noise and two for current noise. The basic noise source topology used is like Figure 7-127, and it can be set up to produce both voltage and current noise outputs.

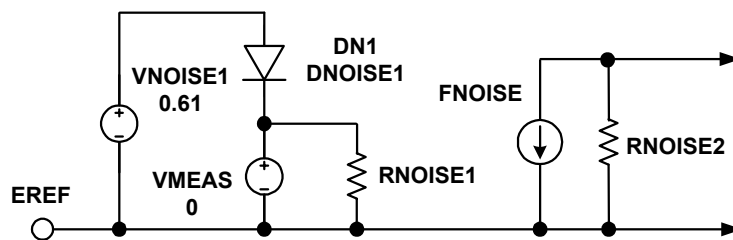


Figure 7-127: A basic SPICE noise generator is formed with diodes, resistors, and controlled sources

Note that, within SPICE, semiconductor models can generate $1/f$ (flicker) noise. The noise generators use diodes such as DN1 to produce this portion of the noise, modeling the $1/f$ noise of the op amp. By properly specifying diode model parameters and bias voltage VNOISE1, the $1/f$ noise is tailored to match the op amp. The noise current from DN1 passes through a zero voltage source. Here VMEAS is being used as a measurement device, combining the $1/f$ noise from DN1 and the broadband noise from RNOISE1.

RNOISE1 is selected for a value providing an appropriate broadband noise. The combined noise current in VMEAS is monitored by FNOISE, and appears as a voltage across RNOISE2. This voltage is then injected in series with one amplifier input via a

controlled voltage source, such as E_N of Figure 7-122 (see again). Either FNOISE or a controlled voltage source coefficient can be used for overall noise voltage scaling.

Current noise generation is similar to the above, except that the RNOISE2 voltage producing resistor isn't used, and two current-controlled sources drive the amplifier inputs. With all noise generators symmetrical about ground, dc errors aren't introduced.

Current Feedback Amplifier Models

As noted previously, a new model topology was developed for current feedback amplifiers, to accommodate their unique input stage structure (see Reference 9). The model uses a topology as shown in Figure 7-128 for the input and gain stages. The remaining model portions (not shown) contain multiple pole/zero stages and the output stage, and are essentially the same as voltage feedback amplifiers, described above.

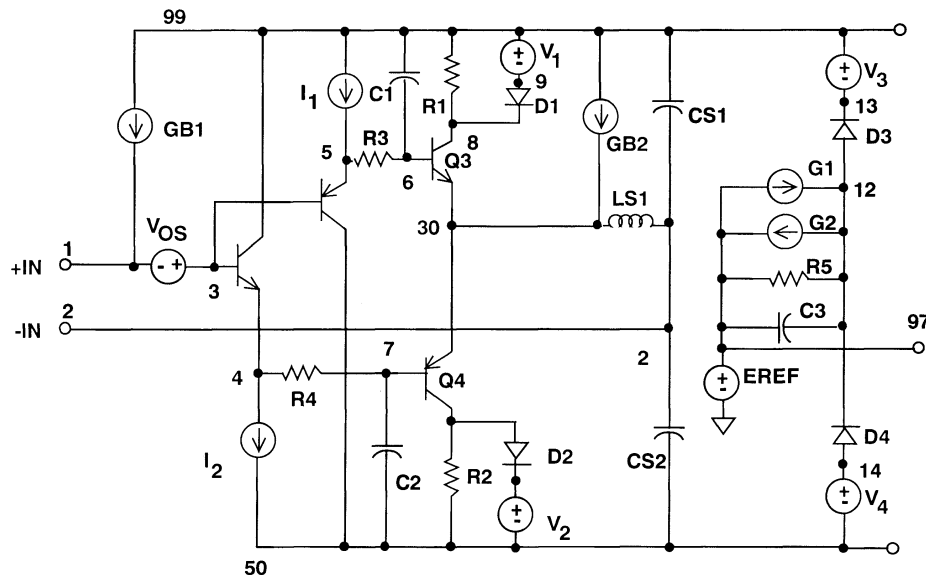


Figure 7-128: Input and gain stages of current feedback op amp macromodel

The four bipolar transistor input stage resembles actual current feedback amplifiers, with a high impedance non-inverting input (+IN) and a low impedance inverting input (-IN). In current feedback amplifiers, the maximum slew rate is very high, because dynamic slew current isn't limited to a differential pair tail current (as in voltage feedback op amps). In current feedback op amp designs, much larger amounts of error current can flow in the inverting input, as developed by the feedback network. Internally, this current flows in either Q3 or Q4, and charges compensation capacitor C3 via current mirrors.

The current mirrors of the ADSpice model are actually voltage controlled current sources in the gain stage, G1 and G2. They sense voltage drops across input stage resistors R1 and R2, and translating this into a C3 charging current. By making the value of G1 and G2 equal to the R1- R2 reciprocal, the slew currents will be identical. By clamping the R1-R2 voltage drops via D1-V1 and D2-V2, the maximum current is limited, which thus sets the highest slew rate. Open loop gain or transresistance of the model is set by R5, and the open loop pole frequency by C3-R5 (as described previously, Figure 7-122

OP AMP APPLICATIONS

again). The output from across R5-C3 (node 12) drives the model's succeeding frequency shaping stages, and EREF is again an internal reference voltage.

One of the unique properties of current feedback amplifiers is that bandwidth is a function of the feedback resistor and the internal compensation capacitor, C3. The lower the feedback resistor, the greater the bandwidth, until a practical lower limit is reached, i.e., the value at which the part oscillates. As the model includes a low impedance inverting input, it accurately mimics real part behavior as R_F is altered. Figure 7-129 compares the ADSpice model to the actual device for an AD811 video amplifier. As shown, the model accurately predicts the gain roll off at the much lower frequency for the $1\text{k}\Omega$ feedback resistor as opposed to the 500Ω resistor.

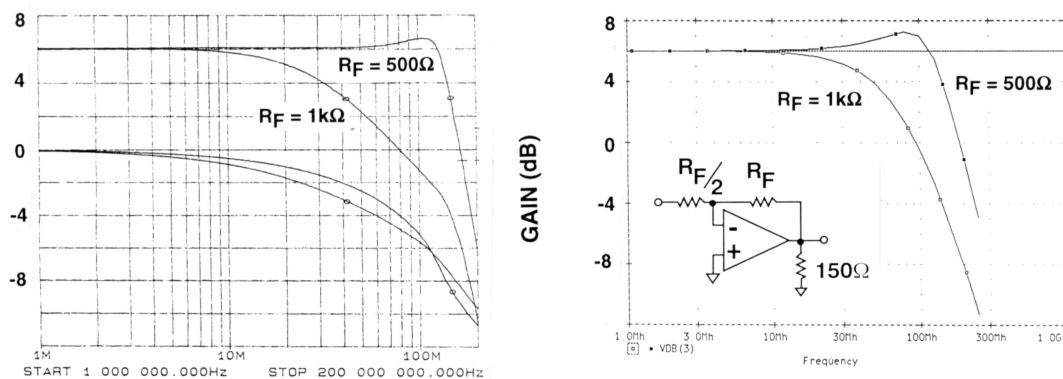


Figure 7-129: Comparison of a real AD811 current feedback op amp (left) with macromodel (right) shows similar characteristics as feedback resistance is varied

The current feedback amplifier input and gain stage is an enhancement to the ADSpice model that increases flexibility in modeling different op amp devices, and provides a net increase in design cycle speed.

Simulation Must Not Replace Breadboarding!

No matter how accurate your models are, or how much confidence you have with simulations, *SPICE analysis alone should never totally replace breadboarding*. As part of a layout and the actual devices existing within a real world PCB assembly, there are second and third order effects which can easily become relevant to performance. By and large, SPICE will never ever "know" of such things, unless you explicitly enter them into the SPICE netlist. However, this may be either difficult or outright impossible. You may not even be aware of some things before a PCB is built and tested within the final system— spurious signal coupling, the effects of crosstalk, the inevitable parasitic capacitance, inductance, and resistance— on and on goes the list. Let's face it, it is all but impossible to include all of these effects in a simulation. Even if generally aware of their existence, you simply won't have any data whatsoever on the magnitudes involved without actually building a PCB, and operating it under the intended conditions.

Furthermore, remember the fact that no macromodel includes all op amp characteristics. For example, exceeding the input voltage range can cause non-linear behavior in an op

amp, which is not necessarily included in its model. Because of such effects that a simulation might not predict, it is necessary to breadboard the circuit.

Even with models as comprehensive as those of the ADSpice library, external effects can easily cause a circuit to work improperly. As noted, PCB parasitics can significantly alter the frequency performance in high speed designs. Such parasitics are easily overlooked in a SPICE simulation, but a breadboard will reveal the problems.

Ultimately, simulation and breadboarding should be used together to maximize the design efficiency. Figure 7-130 summarizes these pro and con points of analog simulations.

- ◆ **Understand What's Real (Hardware), and What Isn't (SPICE)**
- ◆ **Use Breadboarding/Prototyping as Final Design Verification**
- ◆ **Be Aware of Non-Modeled Op Amp Characteristics**
- ◆ **Pay Attention to PCB Parasitics Impacting Circuit Behavior**

Figure 7-130: Some analog simulation caveats

Obviously, the designer needs to be wary of what SPICE can/cannot do, and the necessity of closely allying simulations with breadboarding and prototyping.

Simulation is a Tool to be Used *Wisely*

It must be remembered that while simulation is an extremely powerful tool, it must be used wisely to realize its full benefits. This includes knowing models well, understanding PCB and other parasitic effects, and anticipating the results. For example, consider a simple differential amplifier comprised of an op amp and four equal resistors, to be analyzed for common mode rejection ratio (CMRR) performance. At low frequencies, CMRR will be dominated by resistor mismatch, while at higher frequencies it is dominated by op amp CMRR performance. However, a SPICE simulation will only show this if the external resistors are realistically mismatched, and the op amp model used also properly treats not only dc CMRR, but also CMRR reduction at higher frequencies. If these critically important points are overlooked in the analysis, then an optimistic result will show excellent CMRR performance over the entire circuit bandwidth.

Unfortunately, this is simply wrong. Alternatively, substituting into the netlist resistors mismatched by their specified tolerances as well as an ADSpice model (which *does* have CMRR frequency effects modeled) the end results will be quite different. CMRR performance at low frequencies will be limited by resistor mismatch errors, and it will degrade at higher frequencies, as would a real op amp device with CMRR versus frequency effects.

Know the Models

Using various dc and ac tests, any op amp macromodel can be checked for accuracy and functional completeness. Specialized test simulations can also be devised for other op amp parameters important for a particular analysis. All this is critically important, as knowing a model's capabilities ahead of time can help prevent many headaches later.

■ OP AMP APPLICATIONS

Understand PCB Parasitics

Even if the model passes all preliminary tests, caution still should be exercised. As noted, PCB parasitics can have significant impact on a circuit's performance. This is especially true for high speed circuits. A few picofarads of capacitance on the input node can make the difference between a stable circuit and one that oscillates. Thus, these effects need to be carefully considered when simulating the circuit to achieve meaningful results.

To illustrate the impact of PCB parasitics, the simple voltage follower circuit of Figure 7-131 (left) was built twice. The first time this was on a carefully laid out PCB, and the second time on a component plug-in type of prototype board. An AD847 op amp is used because of its 50 MHz bandwidth, which makes the parasitic effects much more critical (smaller C values will have a greater effect on results).

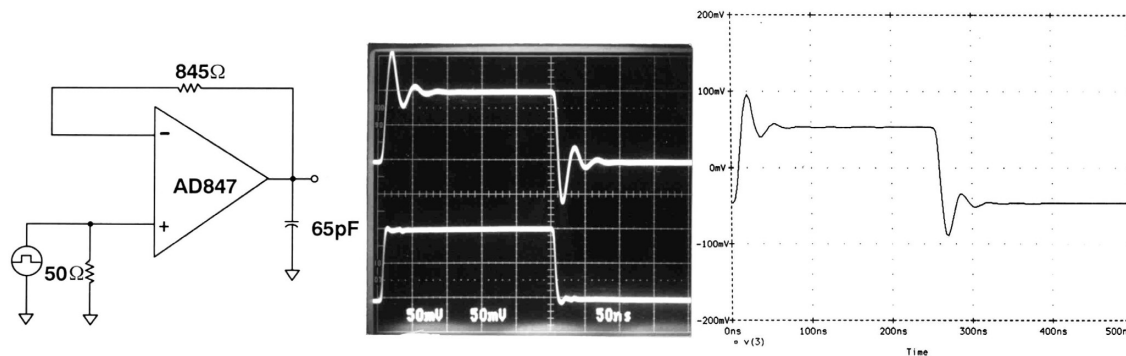


Figure 7-131: *With care and low parasitic effects in the PCB layout, results of lab testing (center) and simulation (right) can converge*

As the results above indicate, this circuit executed on a properly laid out PCB has a clean response with minor overshoot and ringing (center picture). The SPICE model results also closely agree with the real part, showing a corresponding simulation (right picture).

On the other hand, the same circuit built on the plug-in prototype board shows distinctly different results. In general it shows much worse performance, due to the relatively high nodal capacitances around the op amp inputs, which degrade the square wave response to severe ringing, much less than full capability of the part.

This is shown in Figure 7-132 (opposite) in the center and right pictures, respectively. The voltage follower circuit on the left shows the additional capacitances as inherent to the prototype board. With this test circuit and corresponding analysis, there was (initially) no agreement between the poor lab test, and the parallel SPICE test. However, when the relevant PCB parasitic capacitances are included in the SPICE file, then the simulation results do agree with the real circuit, as noted in the right picture.

This example illustrates several key points. One, PCB parasitics can easily make a high speed circuit behave much differently from a simplistic SPICE analysis. Secondly, when the SPICE netlist is adjusted to more reasonably reflect the parasitic elements of a PCB, then the simulation results do compare with the actual lab test. Finally, a point that should be obvious, a clean PCB layout with minimal parasitics is critically important to high

speed designs. To put this in a broader perspective, op amps of today are capable of operating to 1GHz or more!

Another interesting point is that the simulation can be used as a rough measure of the PCB layout design. If the simulation, without any parasitics, agrees with the PCB, then there is a reasonable assurance that PCB is well laid out.

Parasitic PCB elements are not the only area that may cause differences between the simulation and the breadboard. A circuit may exhibit non-linear behavior during power-on that will cause a device to lock-up. Or, a device may oscillate due to insufficient power supply decoupling or lead inductance. SPICE circuits need *no* bypassing, *but real world ones always do!* It is, practically speaking, impossible to anticipate all normal or abnormal operating conditions to which an amplifier might be subjected.

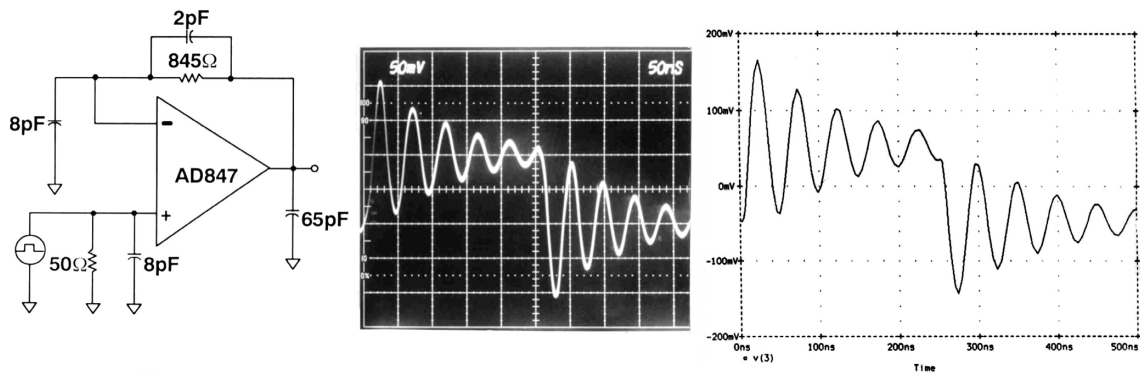


Figure 7-132: Without low parasitics, lab testing results (center) and parallel simulation (right) still show convergence—with a poorly damped response

Thus, it is always important that circuits be breadboarded and thoroughly checked in the lab. Careful forethought in these stages of design helps minimize any unknown problems from showing up when the final PCBs are manufactured.

Simulation Speeds the Design Cycle

Simulation is very effective in the initial design phase, to try out different ideas and circuit configurations. When a circuit topology has been decided upon and tested in SPICE, then a breadboard can be built. If the simulation was done carefully, the breadboard has good likelihood of working correctly without significant modifications.

When the simulation and the actual results correlate, then the circuit can be easily altered in SPICE to perform many different types of analysis. For example, it is much easier to try to optimize the circuit while working within in SPICE, as opposed to repeatedly modifying a breadboard. Quick substitutions of the op amps and components can be made in SPICE and the results immediately viewed.

Worst-case and sensitivity analyses are also done in SPICE much easier than on paper, and with multiple SPICE runs, the sensitivity to a certain parameter can be determined. Consider for example an analysis of a multi-stage active filter, for all possible combinations of component values. This is a nightmare if not impossible either by hand

■ OP AMP APPLICATIONS

or in the lab, but valid results for response extremes can be obtained relatively easily via a SPICE Monte Carlo option, providing greater design confidence.

Some general SPICE-related points are useful towards an overall healthy perspective on this, as shown in Figure 7-133.

- ◆ **Quickly Check Circuit Ideas**
- ◆ **Eases Circuit Optimization**
- ◆ **Allows Component Alteration for Worst Case and Sensitivity Analyses**
- ◆ **Allows Quick Comparison of Different Op Amps**

Figure 7-133: *Some useful points on using SPICE simulations*

While simulation cannot reasonably be allowed to replace breadboarding, the two can and should be used together, to increase the efficiency of a design cycle.

SPICE support

A variety of industry vendors offer SPICE analysis packages for various computer platforms, including the PC. The first of these and among the most popular is PSpice®, a commercial program which now includes allied packages for both schematic capture and PCB layout (see Reference 19). In addition, many vendors also offer low or no cost limited capability student versions of their SPICE programs.

Model support

The ADSpice model library is available in several different forms. Included within it are models of several IC device types, in addition to the op amps discussed above. These are for in-amps, analog multipliers, voltage references, analog switches, analog multiplexers, matched transistors, and buffers. Individual op amp models are available as listings on many data sheets. Electronic ASCII text files of the model library are found from either the ADI website (see References), the Analog Devices Literature Center via 1-800-ANALOGD (1-800-262-5643), or on the ADI support CD.

ACKNOWLEDGEMENTS:

There have been numerous model authors of SPICE macromodels for the ADI library. These include Derek Bowers, Eberhard Brunner, Joe Buxton, Vic Chang, Bob Day, Wes Freeman, Adolfo Garcia, Antonio Germano, John Hayes, John McDonald, Troy Murphy, Al Neves, Steve Reine, Bill Tolley, Tim Watkins, and James Wong.

Breadboard and Prototyping Techniques

A basic principle of a breadboard or a prototype structure is that it is a *temporary* one, designed to test the performance of an electronic circuit or system. By definition it must therefore be easy to modify, particularly so for a breadboard.

There are many commercial prototyping systems, but unfortunately for the analog designer, almost all of them are designed for prototyping *digital* systems. In such environments, noise immunities are hundreds of millivolts or more. Prototyping methods commonly used include non copper-clad Matrix board, Vectorboard, wire-wrap, and plug-in breadboard systems. Quite simply, these all are unsuitable for high performance or high frequency analog prototyping, because of their excessively high parasitic resistance, inductance, and capacitance levels. Even the use of standard IC sockets is inadvisable in many prototyping applications (more on this, below).

- ◆ **Always Use a Ground Plane for Precision or High Frequency Circuits**
- ◆ **Minimize Parasitic Resistance, Capacitance, and Inductance**
- ◆ **If Sockets Are Required, Use "Pin Sockets" ("Cage Jacks")**
- ◆ **Pay Equal Attention to Signal Routing, Component Placement, Grounding, and Decoupling in Both the Prototype and the Final Design**
- ◆ **Popular Prototyping Techniques:**
 - **Freehand "Deadbug" Using Point-to-point Wiring**
 - **"Solder-mount"**
 - **Milled PC Board From CAD Layout**
 - **Multilayer Boards: Double-sided With Additional Point-to-point Wiring**

Figure 7-134: A summary of analog prototyping system key points

Figure 7-134 summarizes a number of key points on selecting a useful analog breadboard and/or prototyping system, which are further discussed below.

One of the more important considerations in selecting a prototyping method is the requirement for a large-area ground plane. This is required for high frequency circuits as well as low speed precision circuits, especially when prototyping circuits involving ADCs or DACs. The differentiation between *high-speed* and *high-precision* mixed-signal circuits is difficult to make. For example, 16+ bit ADCs (and DACs) may operate on high speed clocks (>10MHz) with rise and fall times of less than a few nanoseconds, while the effective throughput rate of the converters may be less than 100kSPS. Successful prototyping of these circuits requires that equal (and thorough) attention be given to good high-speed and high-precision circuit techniques.

Deadbug prototyping

A simple technique for analog prototyping uses a solid copper-clad board as a ground plane (see References 20 and 21). In this method, the ground pins of the ICs are soldered directly to the plane, and the other components are wired together above it. This allows

■ OP AMP APPLICATIONS

HF decoupling paths to be very short indeed. All lead lengths should be as short as possible, and signal routing should separate high-level and low-level signals. Connection wires should be located close to the surface of the board to minimize the possibility of stray inductive coupling. In most cases, 18-gauge or larger insulated wire should be used. Parallel runs should not be "bundled" because of possible coupling. Ideally the layout (at least the relative placement of the components on the board) should be similar to the layout to be used on the final PCB. This approach is often referred to as *deadbug* prototyping, because the ICs are often mounted upside down with their leads up in the air (with the exception of the ground pins, which are bent over and soldered directly to the ground plane). The upside-down ICs look like deceased insects, hence the name.

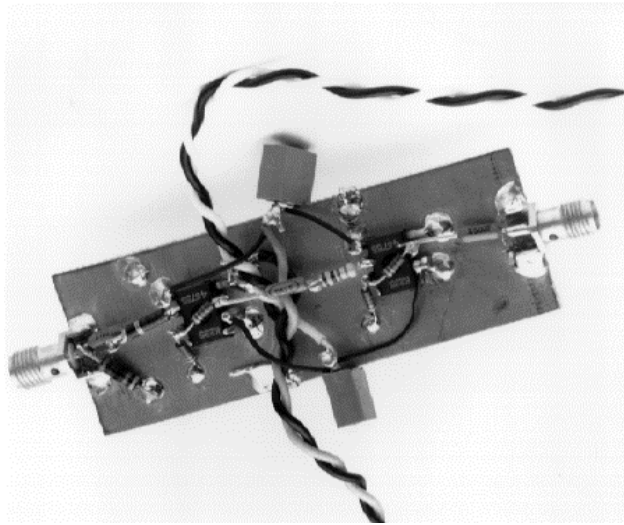


Figure 7-135: A "deadbug" analog breadboard

Figure 7-135 shows a hand-wired "deadbug" analog breadboard. This circuit uses two high speed op amps, and in fact gives excellent performance in spite of its lack of esthetic appeal. The IC op amps are mounted upside down on the copper board with the leads bent over. The signals are connected with short point-to-point wiring. The characteristic impedance of a wire over a ground plane is about 120Ω , although this may vary as much as $\pm 40\%$ depending on the distance from the plane. The decoupling capacitors are connected directly from the op amp power pins to the copper-clad ground plane. When working at frequencies of several hundred MHz, it is a good idea to use only one side of the board for ground. Many people drill holes in the board and connect the sides together by soldering short pieces of wire. If care isn't taken, this may result in unexpected ground loops between the two sides of the board, especially at RF frequencies.

Pieces of copper-clad board may be soldered at right angles to the main ground plane to provide screening, or circuitry may be constructed on both sides of the board (with through-hole connections) with the board itself providing screening. For this, the board will need corner standoffs to protect underside components from being crushed.

When the components of a breadboard of this type are wired point-to-point in the air (a type of construction strongly advocated by Bob Pease (see Reference 21) and sometimes known as "bird's nest" construction) there is always the risk of the circuitry being crushed and resulting short-circuits. Also, if the circuitry rises high above the ground plane, the screening effect of the ground plane is diminished, and interaction between different parts

of the circuit is more likely. Nevertheless, the technique is very practical and widely used because the circuit may easily be modified (this of course assumes the person doing the modifications is adept with soldering techniques).

Another prototype breadboard variation is shown in Figure 7-136. Here the single-sided copper-clad board has pre-drilled holes on 0.1" centers (see Reference 22). Power busses are used at the top and bottom of the board. The decoupling capacitors are used on the power pins of each IC. Because of the loss of copper area due to the pre-drilled holes, this technique does not provide as low a ground impedance as a completely covered copper-clad board of Figure 7-135, so be forewarned.

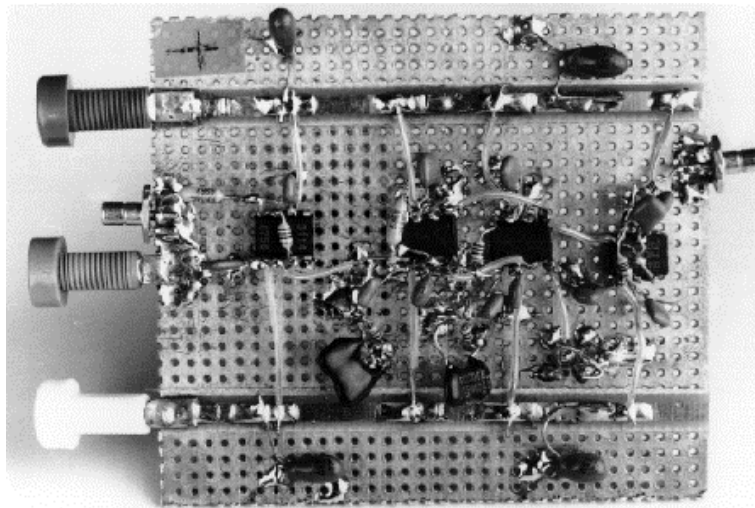


Figure 7-136: *"A deadbug" prototype using 0.1" pre-drilled single-sided, copper-clad printed board material*

In a variation of this technique, the ICs and other components are mounted on the non-copper-clad side of the board. The holes are used as vias, and the point-to-point wiring is done on the copper-clad side of the board. Note that the copper surrounding each hole used for a via must be drilled out, to prevent shorting. This approach requires that all IC pins be on 0.1" centers. For low frequency circuits, low profile sockets can be used, and the socket pins then will allow easy point-to-point wiring.

Solder-Mount Prototyping

There is a commercial breadboarding system which has most of the advantages of the above techniques (robust ground, screening, ease of circuit alteration, low capacitance and low inductance) and several additional advantages: it is rigid, components are close to the ground plane, and where necessary, node capacitances and line impedances can be calculated easily. This system is made by Wainwright Instruments and is available in Europe as "Mini-Mount" and in the USA (where the trademark "Mini-Mount" is the property of another company) as "Solder-Mount" (see References 23 and 24).

Solder-Mount consists of small pieces of PCB with etched patterns on one side and contact adhesive on the other. These pieces are stuck to the ground plane, and components are soldered to them. They are available in a wide variety of patterns,

■ OP AMP APPLICATIONS

including ready-made pads for IC packages of all sizes from 8-pin SOICs to 64-pin DILs, strips with solder pads at intervals (which intervals range from 0.040" to 0.25", the range includes strips with 0.1" pad spacing which may be used to mount DIL devices), strips with conductors of the correct width to form microstrip transmission lines (50Ω , 60Ω , 75Ω or 100Ω) when mounted on the ground plane, and a variety of pads for mounting various other components. Self-adhesive tinned copper strips and rectangles (LO-PADS) are also available as tie-points for connections. They have a relatively high capacitance to ground and therefore serve as low-inductance decoupling capacitors. They come in sheet form and may be cut with a knife or scissors.

The main advantage of Solder-Mount construction over "bird's nest" or "deadbug" is that the resulting circuit is far more rigid, and, if desired, may be made far smaller (the latest Solder-Mounts are for surface-mount devices and allow the construction of breadboards scarcely larger than the final PCB, although it is generally more convenient if the prototype is somewhat larger). Solder-Mount is sufficiently durable that it may be used for small quantity production as well as prototyping.

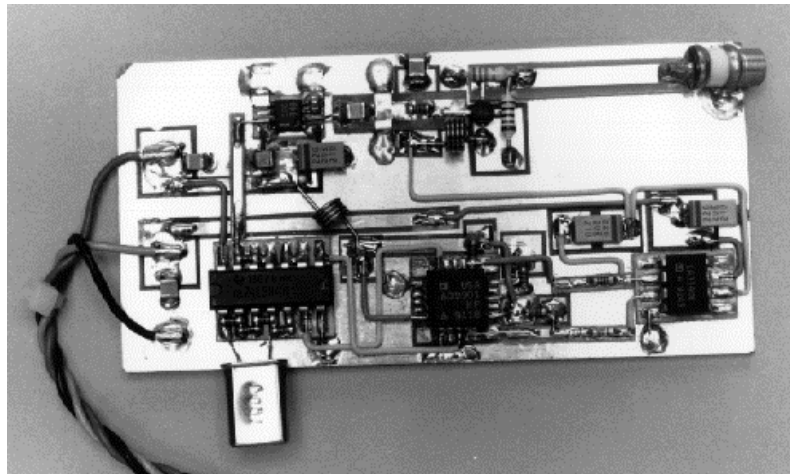


Figure 7-137: A "Solder-Mount" constructed prototype board

Figure 7-137 shows an example of a 2.5GHz phase-locked-loop prototype, built with Solder-Mount techniques. While this is a high speed circuit, the method is equally suitable for the construction of high resolution low frequency analog circuitry.

A particularly convenient feature of Solder-Mount at VHF is the relative ease with which transmission lines can be formed. As noted earlier, if a conductor runs over a ground plane, it forms a microstrip transmission line. The Solder-Mount components include strips which form microstrip lines when mounted on a ground plane (they are available with impedances of 50Ω , 60Ω , 75Ω , and 100Ω). These strips may be used as transmission lines for impedance matching, or alternately, more simply as power buses. Note that glass fiber/epoxy PCB is somewhat lossy at VHF/ UHF, but losses will probably be tolerable if microstrip runs are short.

Milled PCB Prototyping

Both "deadbug" and "Solder-Mount" prototypes become tedious for complex analog circuits, and larger circuits are better prototyped using more formal layout techniques.

There is a prototyping approach that is but one step removed from conventional PCB construction, described as follows. This is to actually lay out a double-sided board, using conventional CAD techniques. PC-based software layout packages offer ease of layout as well as schematic capture to verify connections (see References 25 and 26). Although most layout software has some degree of auto-routing capability, this feature is best left to digital designs. The analog traces and component placements should be done by hand, following the rules discussed elsewhere in this chapter. After the board layout is complete, the software verifies the connections per the schematic diagram net list.

Many designers find that they can make use of CAD techniques to lay out simple boards. The result is a pattern-generation tape (or Gerber file) which would normally be sent to a PCB manufacturing facility where the final board is made.

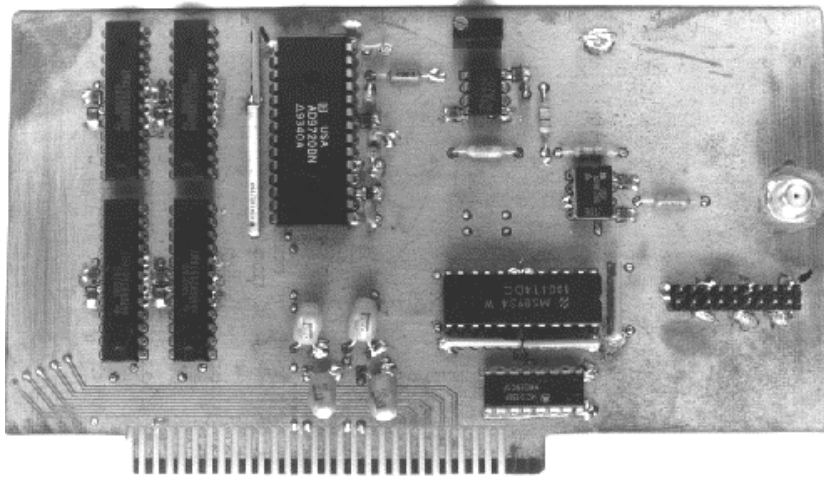


Figure 7-138: A milled circuit construction prototype board (top view)

Rather than use a PCB manufacturer, however, automatic drilling and milling machines are available which accept the PG tape directly (see References 27 and 28). An example of such a prototype circuit board is shown in Figure 7-138 (top view).

These systems produce either single or double-sided circuit boards directly, by drilling all holes and using a milling technique to remove conductive copper, thus creating the required insulation paths, and finally, the finished prototype circuit board. The result can be a board functionally quite similar to a final manufactured double-sided PCB.

However, it should be noted that a chief caveat of this method is that there is no "plated-through" hole capability. Because of this, any conductive "vias" required between the two layers of the board must be manually wired and soldered on both sides.

■ OP AMP APPLICATIONS

Minimum trace widths of 25 mils (1 mil = 0.001") and 12 mil spacing between traces are standard, although smaller trace widths can be achieved with care. The minimum spacing between lines is dictated by the size of the milling bit used, typically 10 to 12 mils.

A bottom-side view of this same milled prototype circuit board is shown in Figure 7-139. The accessible nature of the copper pattern allows access to the traces for modifications.

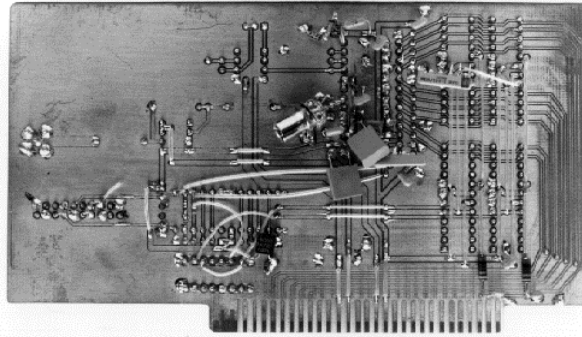


Figure 7-139: A milled circuit construction prototype board (bottom view)

Perhaps the greatest single advantage of the milled circuit type of prototype circuit board is that it approaches the format of the final PCB design most closely. By its very nature however, it is basically limited to only single or double-sided boards.

Beware of Sockets!

IC sockets can degrade the performance of high speed or high precision analog ICs. Although they make prototyping easier, even *low-profile* sockets often introduce enough parasitic capacitance and inductance to degrade the performance of a high speed circuit. If sockets must be used, a socket made of individual *pin sockets* (sometimes called *cage jacks*) mounted in the ground plane board may be acceptable, as in Figure 7-140.

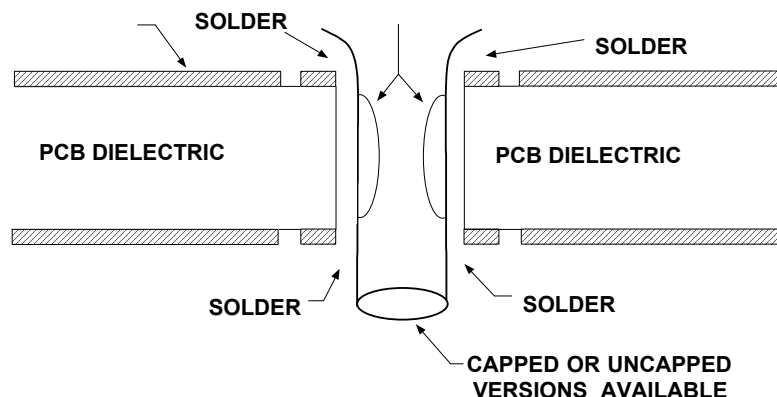


Figure 7-140: When necessary, use pin sockets for minimal parasitic effects

To use this technique, clear the copper (on both sides of the board) for about 0.5mm around each ungrounded pin socket. Then solder the grounded socket pins to ground, on both sides of the board.

Both capped and uncapped versions of these pin sockets are available (AMP part numbers 5-330808-3, and 5-330808-6, respectively). The pin sockets protrude through the board far enough to allow point-to-point wiring interconnections.

Because of the spring-loaded gold-plated contacts within the pin socket, there is good electrical and mechanical connection to the IC pins. Multiple insertions, however, may degrade the performance of the pin socket, so this factor should be kept in mind.

Note also that the uncapped versions allow the IC pins to extend out the bottom of the socket. This feature leads to an additional useful function. Once a prototype using the pin sockets is working and no further changes are to be made the IC pins can be soldered directly to the bottom of the socket. This establishes a rugged, permanent connection.

Some Additional Prototyping Points

The prototyping techniques discussed so far have been limited to single or double-sided PCBs. Multilayer PCBs do not easily lend themselves to standard prototyping techniques. If multilayer board prototyping is required, one side of a double-sided board can be used for ground and the other side for power and signals. Point-to-point wiring can be used for additional runs which would normally be placed on the additional layers provided by a multi-layer board. However, it is difficult to control the impedance of the point-to-point wiring runs, and the high frequency performance of a circuit prototyped in this manner may differ significantly from the final multilayer board.

Other difficulties in prototyping may occur with op amps or other linear devices having bandwidths greater than a few hundred megahertz. Small variations in parasitic capacitance ($<1\text{pF}$) between the prototype and the final board can cause subtle differences in bandwidth and settling time.

Sometimes, prototyping is done with DIP packages, when the final production package is an SOIC. *This is not recommended!* At high frequencies, small package-related parasitic differences can account for different performance, between prototype and final PCB. To minimize this effect, always prototype with the final packages.

■ OP AMP APPLICATIONS

Evaluation Boards

Most manufacturers of analog ICs provide *evaluation boards* usually at a nominal cost. These boards allow customers to evaluate ICs without constructing their own prototypes. Regardless of the product, the manufacturer has taken proper precautions regarding grounding, layout, and decoupling to ensure optimum device performance. Where applicable, the evaluation PCB artwork is usually made available free of charge, should a customer wish to copy the layout directly or make modifications to suit an application.

General Purpose Op Amp Evaluation Boards

Evaluation boards can be either dedicated to a particular IC, or they can also be general purpose. With op amps the most universal linear IC, it is logical that evaluation boards be developed for them, to aid easy applications. However, it is also important that a good quality evaluation board avoid the parasitic effects discussed above. An example is the general purpose dual amplifier evaluation board of in figure 7-141 (see Reference 29).

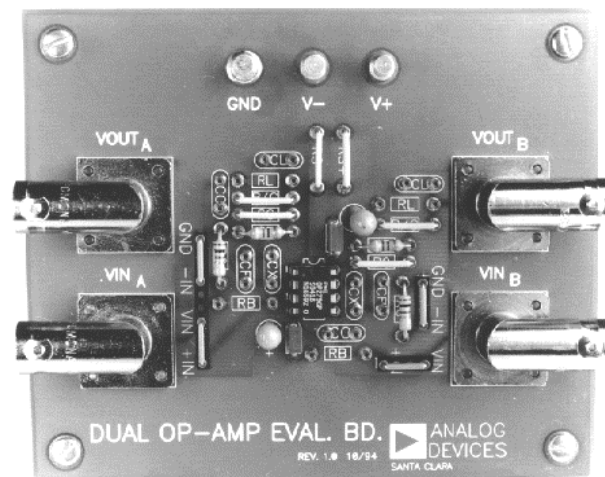


Figure 7-141: *A general purpose op amp evaluation board allows fast, easy configuration of low frequency op amp circuits*

This board uses pin sockets for any standard dual op amp pinout device, and a flexible set of component jumper locations allows it to be setup for inverting or non-inverting amplifiers. Various gains can be configured by choice of the component values, in either ac or dc-coupled configurations.

The card design provides signal coupling via BNC connectors at input and output. It also uses external lab power supplies, which are wired to the lug terminals at the top. The card does however contain local supply voltage decoupling and bypassing components.

These general purpose boards are intended for medium to high precision uses at frequencies below 10MHz, with moderate op amp input currents. For higher operating speeds, a dedicated, device-specific evaluation board is likely to be a better choice.

Dedicated Op Amp Evaluation Boards

In high speed/high precision ICs, special attention must be given to power supply decoupling. For example, fast slewing signals into relatively low impedance loads produce high speed transient currents at the power supply pins of an op amp. The transient currents produce corresponding voltages across any parasitic impedance that may exist in the power supply traces. These voltages, in turn, may couple to the amplifier output, because of the op amp's finite power supply rejection at high frequencies.

The AD8001 high speed current-feedback amplifier is a case in point, and a dedicated evaluation board is available for it. A bottom side view of this SOIC board is shown in Figure 7-142. A triple decoupling scheme was chosen, to ensure a low impedance ground path at all transient frequencies. Highest frequency transients are shunted to ground by dual 1000pF/0.01 μ F ceramic chip capacitors, located as close to the power supply pins as possible to minimize series inductance and resistance. With these surface mount components, there is minimum stray inductance and resistance in the ground plane path. Lower frequency transient currents are shunted by the larger 10 μ F tantalum capacitors.

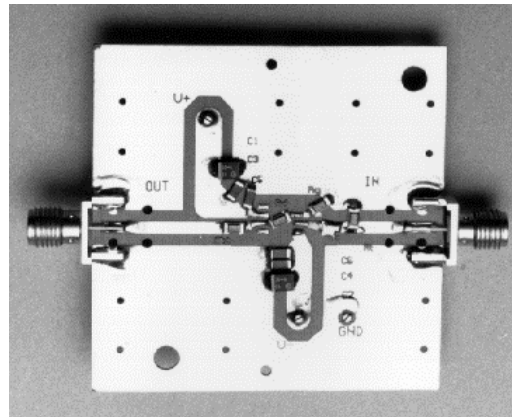


Figure 7-142: A high speed op amp such as the AD8001 requires a dedicated evaluation board with suitable ground planes and decoupling (bottom view)

The input and output signal traces of this board are 50 Ω microstrip transmission lines, as can be noted towards the right and left. Gain-set resistors are chip-style film resistors, which have low parasitic inductance. These can be seen in the center of the photo, mounted at a slight diagonal.

Note also that there is considerable continuous ground plane area on both sides of the PCB. Plated-through holes connect the top and bottom side ground planes at several points, in order to maintain lowest possible impedance and best high frequency ground continuity.

Input and output connections to the card are provided via the SMA connectors as shown, which terminate the input/output signal transmission lines. The board's power connection from external lab supplies is made via solder terminals, which are seen at the ends of the broad supply line traces.

▣ OP AMP APPLICATIONS

Some of these points are more easily seen in a topside view of the same card, which is shown in Figure 7-143. This AD8001 evaluation board is a non-inverting signal gain stage, optimized for lowest parasitic capacitance. The cutaway area around the SOIC outline of the AD8001 provides lowest stray capacitance, as can be noted in this view.

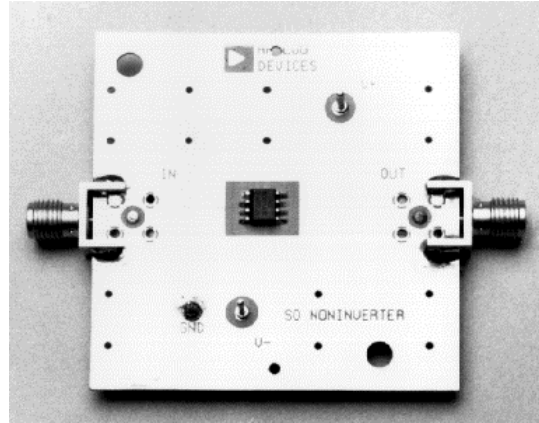


Figure 7-143: *The AD8001 evaluation board uses a large area ground plane as well as minimal parasitic capacitance (top view)*

In this view is also seen the virtually continuous ground plane and the multiple vias, connecting the top/bottom planes.

Summary

In summary, good analog designers utilize as many tools as possible to ensure that the final system design performs correctly. The first step is the intelligent use of IC op amp and other macromodels, where available, to simulate the circuit. The second step is the construction of a prototype board to further verify the design, and to validate the simulation. The final PCB layout should then be based on the prototype layout as much as possible, with careful attention to parasitic effects.

REFERENCES: SIMULATION, BREADBOARDING AND PROTOTYPING

1. L. W. Nagel, "SPICE2: A Computer Program to Simulate Semiconductor Circuits," May 1975, UCB/ERL M75/520, Univ. of California, Berkeley, CA, 94720.
2. A. Vladimirescu, K. Zhang, A. R. Newton, D. O. Pederson, "SPICE Version 2G User's Guide," August 1981, Department of Electrical Engineering and Computer Sciences, Univ. of California, Berkeley, CA, 94720.
3. Mark Alexander, Derek Bowers, "SPICE-Compatible Op Amp Macromodels," **EDN**, February 15, 1990 and March 1, 1990 (available as Analog Devices, Inc. AN138).
4. Joe Buxton, "Analog Circuit Simulation," Chapter 13 of **Amplifier Applications Guide**, 1992, Analog Devices, Inc., Norwood, MA, ISBN 0-916550-10-9.
5. Andrei Vladimirescu, **The SPICE Book**, John Wiley & Sons, New York, 1994, ISBN 0-471-60926-9.
6. "Development of an Extensive SPICE Macromodel for "Current-Feedback" Amplifiers," National Semiconductor AN-840, July 1992.
7. David Hindi, "A SPICE Compatible Macromodel for CMOS Operational Amplifiers," National Semiconductor AN-856, September 1992.
8. Boyle, et al, "Macromodelling of Integrated Circuit Operational Amplifiers," **IEEE Journal of Solid State Circuits**, Vol. SC-9, no.6, December 1974.
9. Derek Bowers, Mark Alexander, Joe Buxton, "A Comprehensive Simulation Macromodel for 'Current Feedback' Operational Amplifiers," **IEE Proceedings**, Vol. 137, Pt. G, # 2, April 1990.
10. Joe Buxton, "OP27 op amp macromodel, Rev B," Analog Devices, Inc., SPICE model library, December 1990, <http://www.analog.com>
11. Antonio Germano, "AD817 op amp macromodel, Rev A," Analog Devices, Inc., SPICE model library, November 1992, <http://www.analog.com>
12. Antonio Germano, "OP295 op amp macromodel, Rev B," Analog Devices, Inc., SPICE model library, February 1995, <http://www.analog.com>
13. Antonio Germano, "OP284 op amp macromodel, Rev B," Analog Devices, Inc., SPICE model library, November 1995, <http://www.analog.com>
14. Steve Reine, "AD8031A op amp macromodel, Rev C," Analog Devices, Inc., SPICE model library, August 1996, <http://www.analog.com>
15. Steve Reine, "AD823AN op amp macromodel, Rev C," Analog Devices, Inc., SPICE model library, April 1997, <http://www.analog.com>
16. John Hayes, "AD8051 op amp macromodel, Rev 0," Analog Devices, Inc., SPICE model library, September 1998, <http://www.analog.com>
17. Troy Murphy, "AD8552 op amp macromodel, Rev 1.0," Analog Devices, Inc., SPICE model library, July, 1999, <http://www.analog.com>
18. John Hayes, Tim Watkins, "AD623 in-amp macromodel, Rev B," Analog Devices, Inc., SPICE model library, September 2000, <http://www.analog.com>

■ OP AMP APPLICATIONS

19. PSpice® Simulation software, 1 (888) 671-9500, <http://www.pspice.com>
20. Jim Williams, "High Speed Amplifier Techniques," Linear Technology AN-47, August, 1991.
21. Robert A. Pease, **Troubleshooting Analog Circuits**, Butterworth-Heinemann, 1991, ISBN 0-7506-9184-0.
22. Vector Electronic Company, 12460 Gladstone Ave., Sylmar, CA 91342, Tel. 818-365-9661.
23. Wainwright Instruments Inc., 69 Madison Ave., Telford, PA, 18969-1829, (215) 723-4333.
24. Wainwright Instruments GmbH, Widdersberger Strasse 14, DW-8138 Andechs-Frieding, Germany. +49-8152-3162.
25. PADS Software, Advanced CAM Technologies, Inc., 16450 Los Gatos Blvd., Suite 110, Los Gatos, CA 95032, <http://www.ecam.com/>
26. ACCEL Technologies, Inc., 17140 Bernardo Center Drive, Suite 100, San Diego, CA 92128, <http://www.acceltech.com/>
27. LPKF Laser & Electronics, 28220 SW Boberg Rd., Wilsonville, OR 97020, 800-345-LPKF or (503) 454-4200, <http://www.lpkfcadcam.com>
28. T-Tech, Inc., 5591-B New Peachtree Road, Atlanta, GA, 34341, 800 370-1530 or (770) 455-0676, <http://www.T-Tech.com>
29. Adolfo Garcia, "Evaluation Boards for Single, Dual and Quad Operational Amplifiers," **Analog Devices AN398**, January 1996.